Computing the Riemann Hypothesis

Volume I

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Volume I

Representations of Ξ, Thorin Measures
and the Reciprocal Ξ Function

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Computing the Riemann Hypothesis, Volume I: Representations of Ξ*, Thorin Measures and the Reciprocal* Ξ *Function*

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Cover: the Thorin density of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ at eight resolutions, sharpening onto the zeros.

Source, code and data: `https://github.com/VadimSokolov/computing_rh`

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Preface

The Riemann Hypothesis can be stated as the positivity of a measure. The reciprocal Xi clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is the Laplace transform of a generalised gamma convolution exactly when every nontrivial zero lies on the critical line, and then its Thorin measure is the discrete measure $\sum_{\gamma>0}\delta_{\gamma^2}$ on the squared ordinates. This book asks what computation can say about that positivity condition.

Its answer has three parts. First, the Thorin measure is computable at every positive resolution, with explicit error bounds, from the Pólya kernel alone, with no primes and no list of zeros. Second, the distance between the computed measure and the measure demanded by RH can be measured exactly, and the signature an off line zero would leave can be calibrated. Third, a finite computation could refute RH by certifying a zero off the line, but neither a computation of the zeros up to a finite height nor a finite set of positivity tests on the heat trace can prove it. A last question is how much of the positivity can be proved. Moved from the centre of the critical strip to the basepoint one, the clock has a positive Thorin measure with no hypothesis. Chapters 12 to 14 follow each construction of the book back across the strip as far as it holds unconditionally; below the basepoint one, zero density estimates still show that the zeros to the right of the basepoint, the only possible source of negativity, carry a vanishing fraction of the mass of the density.

The book grew out of the first author's programme on generalised gamma convolutions and the Pólya density, collected in the monograph *On Hilbert's 8th Problem* [282], and out of a joint paper computing the Thorin measure directly [293]. The first author's arXiv paper of the same title is a separate work, which we cite by version: [279] is its version 24, of 2026, and [280] its version 5, of 2017. Chapters 10 and 11 incorporate Part IX of that monograph [282], the first author's work on the reciprocal zeta function, written as the manuscript [288], and Chapters 12 to 14 a paper of the authors on the reciprocal Xi function at the basepoint one and across the critical strip, together with notes on Riesz's criterion and on almost generalised gamma convolutions. Sections 15.1 and 15.2 draw on a manuscript of the first author on the Thorin measures of the Dedekind eta function [285], Chapter 20 on a note of the first author on ferromagnetic random variables, Chapter 19 on a manuscript of the first author on the Pólya density as a maximum order statistic [287], Chapter 6 on a working report of

the authors [294], Chapter 22 on a chapter of the first author on Pólya and the zeros of entire functions, and Chapters 15.12 to 16 on five further chapters of the first author, on Yoshida's Hermitian forms, on the compact windows of Zhu [369], on three families of generalised gamma convolutions, on deconvolution as a damped Weil criterion, and on Dirichlet L functions and the Linnik constant, and Chapters 18 and 23 on two more chapters of the first author, on friable approximants of Ξ after Miyatake and the cosh ladder and on the work of Haseo Ki read through the Thorin programme, and Chapter 5 on a chapter and two notes of Nick Polson on hitting times and the reciprocal ξ function [141, 151, 150], and Chapter 19.1.7 on a chapter of Nick Polson on extreme values, hitting times and a path to RH [135], Chapter 15 on a chapter of Nick Polson on Weil positivity and the Thorin condition [148], Chapter 20 on a note of Nick Polson on the identity of Comtet and Majumdar [133], Chapter 17.1.4 on two notes of Nick Polson on the Thorin truncation of $L(s, \chi_4)$ and on the Hermite coefficients of Ξ [146, 140], and Chapter 17 on two notes of Nick Polson on the secant dilates and the Möbius function, Chapter 10 on a manuscript of Nick Polson on the level one measure and the backward Cauchy flow [143], Chapter 11 on his notes on the complexity and acceleration of that flow, Chapter 12 on a manuscript of his on the explicit formula as a deconvolution [134], Chapter 7 on his notes on Cauchy deconvolution, Chapter 13 on his note on the deconvolution from level one to the critical line, and Chapter 14 on his note on the criterion of Balazard, Saias and Yor. The core evaluator runs in ball arithmetic. The code and data of every computation are in the book's GitHub repository. No proof of RH is claimed.

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Part I

Foundations: Ξ, its representations and the reciprocal Ξ function

Chapter 1

Introduction: the Riemann Hypothesis as the positivity of a measure

The book rests on one equivalence, between the Riemann Hypothesis and the positivity of the Thorin measure of a function built from the completed zeta function.

1.1 The Thorin condition and what this book shows

Write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ for the completed zeta function, whose zeros are the nontrivial zeros ρ of ζ. The Riemann Hypothesis (RH) asserts that every ρ has real part $\frac{1}{2}$. Put

$$G(u) = \frac{\xi(\frac{1}{2} + \sqrt{u})}{\xi(\frac{1}{2})}.$$

Since $\xi(\frac{1}{2} + s)$ is even in s, G is entire of order one half in u, and Hadamard's theorem gives

$$G(u) = \prod_k \Big(1 + \frac{u}{a_k}\Big), \qquad a_k = -(\rho_k - \tfrac{1}{2})^2,$$

one factor for each pair $\{\rho, 1-\rho\}$ (derived as (2.1.1) in Section 2.1). A zero on the line has $a_k = \gamma_k^2 > 0$; a zero off the line has a_k complex. The reciprocal $1/G$ is the *clock*. By the theory of generalised gamma convolutions (GGC) of Thorin and Bondesson [344, 345, 39], a function $\exp\{-\int \log(1+u/z)\,U(dz)\}$ with U a positive measure is the Laplace transform of a positive random variable, and the Thorin measure U is unique. The rigidity theorem of [282] (Proposition 2.1.1) says that the clock is the Laplace transform of a GGC if and only if RH holds, and that its Thorin measure is then

$$U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}, \tag{1.1.1}$$

with clock variable $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ for independent standard exponentials e_γ.

We call the requirement that the Thorin measure of the clock be positive the *Thorin condition*. Each of its equivalent forms is a sign condition. The logarithmic derivative $G'/G(u) = \sum_k (u+a_k)^{-1}$ must be a Stieltjes transform. The heat trace $W(t) = \sum_k e^{-a_k t}$ must be completely monotone. The Thorin density $\rho_\varepsilon(\theta) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$ must be nonnegative at every resolution $\varepsilon > 0$, which is Lagarias's criterion [198]; for $\varepsilon \geq \frac{1}{2}$, where the basepoint is at least one, it is positive with no hypothesis (Section 12.1). Weil's explicit formula functional must be positive on positive definite test functions [357]. The clock must be hyperbolically completely monotone (Corollary 9.1.3). All are computable at a finite resolution on a finite window (Section 4.1.3).

1.1.1 What computation can and cannot say

Platt and Trudgian verified RH below height 3×10^{12} [277]; Appendix 1.5 gives the history, with landmarks in Table 1.4. None of it proves RH, for a structural reason: every computation sees finitely many zeros, and there are functions with the order, type and symmetry of ξ that pass a given finite set of heat trace tests yet have zeros off the line just above any sufficiently large height R for which RH has been verified up to $R + 50$ (Theorem 1.4.1 [279, Theorem D]). The Thorin condition gives computation a different role: instead of locating zeros one at a time, it asks for the sign of a measure that can be computed at every positive resolution without knowing where a single zero is.

Three evaluators of ξ appear throughout. The first is Pólya's integral $\Xi(\theta) = \int \Phi(u) \cos(\theta u)\, du$, with the positive, doubly exponentially decaying Pólya kernel Φ built from the Jacobi theta function. It is exact and needs no primes and no zeros, and since Φ is analytic in a strip the trapezoid rule is spectrally accurate; the cost is the cancellation $e^{-\pi\theta/4}$, which the tilted contour of Section 2.1 largely removes. The second is the Riemann Siegel formula and its multi point descendant, the Odlyzko Schönhage algorithm [253], the tools of choice at large height. The third is the explicit formula, which evaluates the same objects from the primes and the gamma factor; the sine squared lemma is its Thorin form.

What ties them together is that the Thorin measure at resolution ε is both the Poisson smoothing of the zeros and the sine squared image of the primes. The first description makes it computable from any evaluator of ξ off the line; the second explains why the primes alone cannot compute it near the line. In Selberg's smoothed explicit formula the prime cutoff x plays the part of the resolution: if no zero has real part greater than $\frac{1}{2}$, the error at resolution ε is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so k correct digits need $x \approx 10^{k/\varepsilon}$, with primes up to x^2 (Section 5.1.1). Figure 1.1 shows the first description at work.

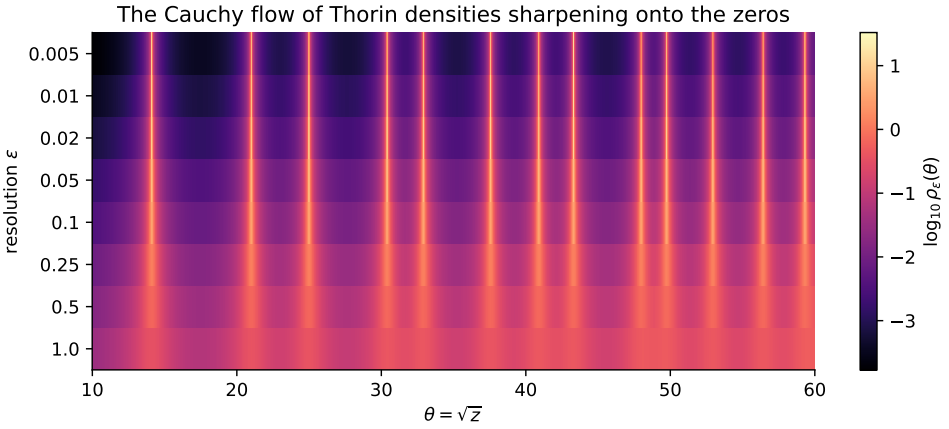


Figure 1.1: The Thorin density $\rho_\varepsilon(\theta)$ on $10 \le \theta \le 60$, computed from the Pólya density at eight resolutions from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top), on a logarithmic colour scale. As the resolution sharpens, the density concentrates on bright vertical lines at the ordinates of the zeros, all of which lie on the critical line in this window: the Cauchy flow converging to the RH measure $\sum \delta_{\gamma^2}$.

1.1.2 Plan of the book and its principal results

Two ideas from Sections 2.1 and 4.1 carry the book. The first is that the Thorin measure at resolution ε is the Poisson smoothing of the zeros, converging as $\varepsilon \downarrow 0$ to the atoms on the line. The second is that the hierarchy of infinitely divisible laws becomes a hierarchy of positivity conditions on one function, the heat trace, whose lower rungs are theorems and whose top rung is RH.

Chapter 2 evaluates ξ and ξ' in the complex plane by the Pólya route. The core is Chapters 3 to 8: Chapter 3 measures the distance from the computed Thorin measure to U_{RH} and computes the signature of an off line zero, Chapter 9 treats weak convergence and hyperbolic complete monotonicity, and Chapter 8 climbs the hierarchy of positivity on the heat trace. Chapters 10 and 11 study the reciprocal zeta function in the variable u, and Chapter 18 treats the clock as a random variable.

Chapters 12 to 14 move the basepoint. The shifted clock T_α has Laplace transform $\xi(\alpha)/\xi(\alpha + \sqrt{u})$, and $T_{1/2}$ is the clock; these chapters carry both ideas from the basepoint one, the resolution $\varepsilon = \frac{1}{2}$, into the critical strip. Chapter 12 works out the basepoint one and constructs T_α as a hitting time for every $\alpha > \frac{1}{2}$. Chapter 13 extends the hierarchy to symmetric laws on the line and reads zero free half planes as completeness statements after Wiener, Salem and Riesz. Chapter 14 reads the hierarchy through Sato and Bondesson, uses zero density estimates to bound how far each T_α can be from a GGC, and closes with Pakes's work on size

biasing, the Lambert law and ladders of self decomposable laws.

Chapter 25 assembles the results into a path on which every step is marked proved, computed, announced or open. The remaining chapters read the programme against particular objects and literatures. Chapter 20: RH as the statement that the Pólya variable is ferromagnetic (its moment generating function vanishes only on the imaginary axis), with provably ferromagnetic laws within total variation 6.1×10^{-6} of it with no hypothesis and 2×10^{-6} by a computer assisted proof. Chapter 19: the Pólya density as the density of a maximum of independent exponentials. Chapter 6: the Thorin phase diagram of the de Bruijn Newman flow, the set of (t,α) at which the shifted clock of ξ_t is a GGC, with the finite part of one of its points certified below height 160. Chapter 22: Pólya's papers on zeros of entire functions, ending with what the unrefereed zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [257] would give. Chapter 15.12: Yoshida's Hermitian forms for the explicit formula. Chapter 16: the compact windows of Zhu [369]. Chapter 8: the three one parameter GGC families containing the clock (Esscher tilt, basepoint family, heat flow) and the deconvolution theorem. Chapter 9: the deconvolution as a damped Weil criterion. Chapter 16: Dedekind zeta and Dirichlet L functions and the Linnik constant. Chapter 18: the friable approximants of Ξ (Riemann's theta series restricted to P friable integers) as a test of computational criteria, an identity of Miyatake read probabilistically, and the bottom of the cosh ladder against the quasi Riemann abscissa. Chapter 23: the work of Haseo Ki and coauthors on the heat flow and repeated differentiation, and the fact that no property surviving a Gaussian tilt of the Pólya density can imply RH. Chapter 5: whether RH can be obtained from the limiting behaviour of hitting times, with a universal stable limit for the prime side hitting times and a rigidity theorem for factorisations of the clock law. Chapter 19.1.7: the Pólya density as a superposition of extreme value laws and three hitting time routes to RH, with the place where each stops. Chapter 15: Weil's criterion on Gaussian test functions as the Thorin condition, with archimedean and prime parts both of extreme value form. Chapter 20: the identity of Comtet and Majumdar, negative on the arithmetic side and positive for the uniform law. Chapter 17.1.4: truncations of the Biane, Pitman and Yor lattice for $L(s,\chi_4)$, the Gram parity, and the Hermite coefficients of Ξ. Chapter 17: the question of Lai and Vere-Jones on completeness of dilates of the hyperbolic secant, answered in weighted L^2. Chapter 10: the Thorin measure at level one in closed form, and the Cauchy semigroup run backward as a positive linear inverse problem solved by EM. Chapter 11: the cost of that computation and its acceleration. Chapter 12: Weil's explicit formula as the deconvolution at level one half, and recovery of zeros from the level one density by the matrix pencil, above the verified range as well. Chapter 7: Cauchy deconvolution as an ill posed problem, and the facts the book uses. Chapter 13: deconvolution from level one to the critical line by particle filters, contour moments and zero free regions from positivity. Chapter 14: the Balazard, Saias and Yor criterion at every level

and height, solved by Monte Carlo.

Tables 1.1, 1.2 and 1.3 collect the principal results of Chapters 2 to 14 with the section where each is developed. A result is marked proved when that section proves it or cites a proof, and computed when it rests on a computation; the section states how far the computation is certified.

Result	Status	Where
Pólya route: ξ and ξ' from two fixed polynomials in ball arithmetic, about 60 correct digits at height 200 at 420 bits, 0.7 ms per point	computed	Section 2.1
Tilted contour: 25 digits of $\rho_{0.05}$ at height 5000 against `mpmath`, in 28 ms per point; to leading exponential order the loss to cancellation falls from about $0.341\,\theta$ digits to about $\delta\theta/\log 10$	proved and computed	Section 2.1
The Lehmer pair near height 7005 resolved by the tilted Pólya route at 34 ms per point	computed	Section 2.2.3.1
$N(1000) = 649$ from the Thorin mass at resolution $\frac{1}{2}$ and the argument along a horizontal segment	computed	Section 2.2.3.2
Raw total variation to U_{RH} maximal; matched total variation at the floor $3.7 \times 10^{-9}\varepsilon$ of the tail model; cell distance $5.44\,\varepsilon$	computed	Section 3.1
Mass defect $-(\varepsilon/\pi)\Xi'(\Theta)/\Xi(\Theta)$ to first order; at $\Theta = 200$ the observed slope agrees with it to five digits at $\varepsilon = 0.005$	proved and computed	Section 3.1
Signatures of an off line pair in total variation and negative mass, exact for an omitted pair and for the negative mass, to leading order for a displaced pair	proved	Section 3.2
Zero density $\nu(\frac{1}{2} + \varepsilon, 200) = 0$ at eight resolutions, from the argument principle identity of Proposition 3.3.2	computed	Section 3.3
Prime side of the sine squared lemma: when no zero has real part greater than $\frac{1}{2}$, Selberg's smoothed sum computes ρ_ε with relative error at most $(x^{-\varepsilon} + x^{-2\varepsilon})/(\varepsilon \log x)$, so with error of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$ at the peaks, the single zero law; tested at $\varepsilon = \frac{1}{4}$	proved and computed	Section 5.1
The Thorin measures converge vaguely to the measure of the zeros on the line, whether or not RH holds	proved	Section 9.1
The clock is HCM if and only if RH holds, and the shifted clock at any basepoint $\alpha \geq 1$ is HCM unconditionally; the 216 normalised derivatives of a finite HCM test are all positive, certified in ball arithmetic	proved and computed	Section 9.1
Bosch type threshold: $1/G(x^{1/\beta})$ is HCM if and only if $\lvert\beta\rvert \geq 1/(1-\alpha_\xi)$, with α_ξ set by the largest angle of the zeros; unconditionally $1/G(x^p)$ is HCM for $\lvert p\rvert \leq 1 - 1/(\pi T_0)$, where $T_0 = 3 \times 10^{12}$ is the verified height, in particular for $\lvert p\rvert \leq 1 - 1.1 \times 10^{-13}$	proved	Section 9.3

The clock is infinitely divisible and self decomposable with no hypothesis, and $(-1)^k W^{(k)} > 0$ for every $k \leq 1.16 \times 10^{25}$, that is, the Williamson measures, inverse gamma smoothings of the Thorin measure, are positive to that order; the two proofs use the verification of RH below heights 1000 and 3×10^{12} respectively [277]	proved	Section 8.1

Table 1.1: Principal results of Chapters 2 to 8.

Result	Status	Where
The normalised reciprocal zeta function is infinitely divisible with no hypothesis, and it is not self decomposable, not a GGC and not in the Bondesson class	proved and computed	Section 11.1
Conjecture S, that the symmetrised truncations of the Biane, Pitman and Yor lattice have their zeros in the strip on the line, holds for the sizes $n \leq 24$ and $n = 32, 36$ above explicit heights and numerically below them; for infinitely many n it would imply RH	proved and computed	Section 17.1
Law of the clock by Talbot inversion, with no zero as input; its hazard tends to γ_1^2, reached to eleven digits at $t = 0.25$	computed	Section 5.2
Kent approximants with 1700 zeros and a moment matched Weyl tail: relative error 3.5×10^{-23} at $u = 1$ and 7.0×10^{-13} at $u = 10^4$	computed	Section 5.2
The clock is not the first passage time of Brownian motion to an independent random level; the boundary from the Volterra equation, solved in three zones, reproduces its law to Monte Carlo accuracy (Kolmogorov distance 0.0020, against 0.0059 for the tangent boundary)	proved and computed	Section 3.1
Keiper Li coefficients $\lambda_1, \ldots, \lambda_{100}$ from the Pólya density, λ_1 to 25 digits	computed	Section 23.6
Closest pair below height 200 collides at de Bruijn Newman time -0.288; the Lehmer pair near 7005 at -7.11×10^{-4}	computed	Section 4.1
For $L(s, \chi_{12})$ and $L(s, \Delta)$ the sine squared lemma has no pole piece but a signed prime piece	proved	Section 15.1
The eta kernel of $L(s, \chi_{12})$: GRH below height 100 from the zero density identity and the 67 sign changes	computed	Section 15.1
For an imaginary quadratic field the Pólya kernel of ζ_K is positive exactly when an explicit theta sum is positive on a finite interval, which gives $L(\frac{1}{2}, \chi_d) > 0$ for $d = -3, -4, -7, -8$; it takes negative values for every $d \equiv 5 \pmod{24}$	proved	Section 15.1
For the Dirichlet beta function $L(s, \chi_4)$, whose kernel is the cube of the eta function: GRH below height 100; a positive Thorin density at every basepoint $\alpha \geq 1$, which the absolute values of the prime weights certify only for $\alpha > 2.2823$; the density as the difference of those of $\xi_{\mathbb{Q}(i)}$ and ξ, and as the derivative of the argument of the completed function on the line	proved and computed	Section 15.1.7

Result	Status	Where
Nonnegative deconvolution of $\rho_{0.1}$ on $[15, 55]$, with the zeros outside $[5, 65]$ as background, recovers every zero in the window to within 10^{-5}	computed	Section 10.8
For Pólya's Bessel function $K_{i\theta/2}(2\pi)$ the index side clock is the hitting time of an explicit diffusion, its Thorin atoms are the squared zeros and its trace is completely monotone; its de Bruijn Newman constant is finite, by Newman's theorem, although none of its real zeros below height 90 collide down to $t = -60$	proved and computed	Section 18.1

Table 1.2: Principal results of Chapters 10 to 18.

Result	Status	Where
T_1, with Laplace transform $\xi(1)/\xi(1+\sqrt{u})$, is a GGC with positive Thorin density, smallest at the origin, by a computation of the zeros below height 2000 in ball arithmetic and explicit zero counts above it, so that T_1 is a positive stable one half variable plus an independent GGC; its law has infinite mean and a stable one half tail, and the Lévy form of $\xi(1+s)$, which converges by the prime number theorem, gives its Thorin density from the primes	proved and computed	Section 12.1
Below the basepoint one, $v_\alpha(\theta)/(\alpha - \frac{1}{2})$ is nonincreasing in α away from the zeros off the line; the Thorin density is smallest at the origin, and only there, for every $\alpha \in (\frac{1}{2}, 7.35]$ on $\lvert\theta\rvert \le 3\times 10^{12} - 1.21$, for $L(s, \chi_4)$ and $L(s, \chi_{12})$ at every height for every $\alpha \ge 1$, and at every height for every $\alpha \in (\frac{1}{2}, 1)$ exactly when RH holds; for every $\alpha > \frac{1}{2}$ it is positive on $\lvert\theta\rvert \le 3 \times 10^{12} - \frac{1}{2}$	proved and computed	Sections 12.1.6, 14.3
Every T_α, $\alpha > \frac{1}{2}$, exists as the tilted centre clock run through the inverse local time of a sticky Brownian motion, and it is a GGC if and only if no zero has real part greater than α	proved	Section 12.2
For real u and $\alpha \ge \frac{1}{2}$, $\xi(\alpha)/\xi(\alpha + \lvert u\rvert)$ is the characteristic function of an infinitely divisible normal scale mixture; for $\frac{1}{2} < \alpha < 1$ it is an extended GGC, and the translates of every tilted Pólya kernel $\Phi_{\alpha'}$ with $\alpha' > \alpha$ are complete in L^1, exactly when no zero has real part greater than α	proved	Section 13.1
The complete monotonicity conditions of the Poisson trace $K_{1/2}$, which together are RH, hold for every order $n < T_0(\pi - y)$ at the point y; $K_{1/2}$ and the trace of Pólya's Bessel function differ by $\frac{9}{8} + O(y \log \frac{1}{y})$ as $y \to 0$	proved and computed	Section 13.1.5
$1/\zeta(2k)$ is, up to an elementary factor, a value of the Laplace transform of the centre clock and of T_1; for $\frac{1}{2} \le \alpha < 1$ the bound $O(x^{\alpha/2+\delta})$ on Riesz's function, for every $\delta > 0$, holds exactly when every $T_{\alpha'}$ with $\alpha' > \alpha$ is a GGC	proved	Section 13.2

Result	Status	Where
The constant of $\sum_{n\le x}\mu(n)^2/n$ is an elementary term plus $\mathbb{E}V_1 - 2\,\mathbb{E}V_2$, two tilted Pólya means; RH holds exactly when the squarefree count smoothed by an exponential or a Gamma(2) law, or the count in mean square, has error $O(x^{1/4+\varepsilon})$; the error is $\Omega_\pm(x^{1/4})$; Gamma(2) smoothing writes it as an average of the Mertens function under a positive law, and two is the largest shape for which this holds	proved	Section 13.3
The centre clock is driven by a subordinator with Lévy density $-W'$ and is unimodal, as is T_α for $\alpha \ge 1$; the tilted Pólya variable is quasi infinitely divisible for every $\alpha > 1$ but not at $\alpha = 1$	proved	Section 14.1
For $L(s,\chi_{12})$ the Thorin measure at the basepoint one is positive, and its Lévy form there has no pole to compensate and prime atoms of both signs	proved	Section 15.2
For $\frac12 < \alpha < 1$, in the density $\rho_{\alpha-1/2}$ the Cauchy kernels of the zeros to the right of α carry at heights $\theta \le T$ at most $T^{e(\alpha)-1+o(1)}$ times the mass of the kernels of the other zeros, with $e(\alpha) < 1$, by the zero density estimates of Ingham, of Guth and Maynard and of Huxley [163, 103, 159]	proved	Section 14.3
With no hypothesis, the size biased centre clock is the clock plus an independent variable with density $W/\mathbb{E}T$, and RH holds exactly when this variable is a mixture of exponential laws, with weight $\gamma^{-2}/\mathbb{E}T$ at the rate γ^2, about a fifth at the lowest zero; the Lambert law, with Laplace transform $\mathcal{W}(u)/u$, is a GGC whose Thorin measure is the image of the uniform law on $(0,\pi)$ under $b \mapsto b\,e^{-b\cot b}/\sin b$, by a proof of which only one step fails for the clock, and that step is RH	proved and computed	Sections 14.4.2, 14.4.3
The centre clock lies in Urbanik's class L_2 of multiply self decomposable laws and not in L_3; it is not the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter	proved and computed	Section 14.4.4
A property of even kernels that also holds for the damped Pólya kernel $e^{tv^2/4}\Phi(v)$ for some $t < 0$ cannot by itself imply RH, by the theorem of Rodgers and Tao; the GGC property of the variable of Biane, Pitman and Yor fails for the damped kernel at every $t < 0$ tested and, provably, for small $\lvert t\rvert$, so the filter does not exclude it there	proved and computed	Section 14.4.5

Table 1.3: Principal results of Chapters 12 to 14. All are unconditional, and several use verified computations: the verification of RH below height 1000, through Theorem 8.1.1, or below the height $T_0 = 3 \times 10^{12}$, Platt's verification of GRH for small moduli, or computations in ball arithmetic.

1.1.3 Conventions and how to read this book

The clock is a function of $u = s^2$; earlier papers of the programme used $s^2/2$, which rescales Thorin measures by $z \mapsto z/2$. Thorin measures live in the variable

z, and $\theta = \sqrt{z}$ is the chart in which the Thorin measure at the basepoint $\frac{1}{2} + \varepsilon$ has density $\rho_\varepsilon(\theta) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$; ε is the resolution. The Pólya kernel Φ is in Titchmarsh's normalisation, $\Xi(\theta) = \int_{\mathbb{R}} \Phi(u) \cos(\theta u)\, du$. $N(T)$ counts zeros with $0 < \operatorname{Im} \rho \le T$, and $\nu(\sigma, T)$ those with, in addition, $\operatorname{Re} \rho > \sigma$. Total variation of a signed measure is the norm $|\mu|(E)$ on the stated window. Chapters 10 and 11, and Sections 9.2 and 9.3, write s for u, and the Kent approximants of Section 5.2 write w; each declares its variables, and Appendix A lists the notation with the place of definition. Numerical values are quoted to the accuracy at which they have been checked.

Readers mainly interested in computation can read Sections 2.1, 2.1 and 2.3 and go straight to Chapters 3 to 8. Readers interested in probability can start with Sections 4.1, 5.2 and 3.1 and continue with Chapters 12 to 14, which build on Sections 2.1 and 4.1 and use Section 3.3 for the zero count. Chapters 10 and 11, the most technical part, can be read after Chapter 1 and the statements of Theorems 8.1.1 and 8.1.6. The conclusions are in Sections 25.1 and 25.2.6.

1.2 The two volumes

This volume is the theory, and its companion, Volume II, is the computation. Part I gives every representation of Ξ the two volumes use: the analytic ones in Chapter 2, the probabilistic ones in Chapter 3, the classes of infinitely divisible laws in which they live (generalised gamma convolutions, their extended form on the line, the Bondesson hierarchy and hyperbolic complete monotonicity) in Chapter 4, and the reciprocal Ξ function, whose Thorin measure is the subject of both volumes, in Chapter 5. Parts II to V follow the Thorin condition through the reciprocal zeta function, the basepoints and the critical strip, other L functions, and the probabilistic and statistical mechanical readings. Volume II computes the objects defined here, from the Pólya kernel and in ball arithmetic, and its central part runs the Cauchy flow backwards: the Thorin measure at level θ is the Cauchy smoothing of the zeros, and the question whether that smoothing can be undone down to the critical line is RH.

References into Volume II are printed with their numbers there.

1.3 Conventions

Every statement carries one of six words. *Proved*: a proof or a pointer to one is given. *Computed*: a script of the book produced the number, in floating point at the stated precision. The scripts and their data are not named in the text: they are in the book's repository, `https://github.com/VadimSokolov/computing_rh`, where `code/README.md` lists them by chapter. *Certified*: the number was computed in ball arithmetic and the enclosure is rigorous. *Conditional*: the statement assumes RH, GRH, or the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October

2026 and not refereed; the hypothesis is named each time. *Heuristic* and *open* mean what they say. No statement of either volume proves RH.

One density appears throughout under the names of the chapters that use it. With $\alpha = \theta = \sigma = \frac{1}{2} + \varepsilon$,

$$\rho_\varepsilon(t) = \frac{1}{\pi}\, v_\alpha(t), \qquad v_\alpha(t) = \operatorname{Re} \frac{\xi'}{\xi}(\alpha + it), \qquad u_\theta(v) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\theta + iv) = m_\theta(v),$$

so that ρ_ε (the Thorin density at resolution ε), v_α/π (the density at the basepoint α) and u_θ (the density at level θ) are one function in three parametrisations. This volume writes basepoints α and the bracket v_α; Volume II writes levels θ and the density u_θ. We write $b_\alpha = v_\alpha(0) = \frac{\xi'}{\xi}(\alpha)$.

1.4 The clearest path to the Riemann Hypothesis

The two volumes test many routes to RH and none of them closes here. Read together, they also show which route is clearest, what any proof along it must contain, and where it stops. We set this out here, before the details, so that each later chapter can be read against it.

The descent in level For $\alpha > \frac{1}{2}$ the shifted clock $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of a law with no hypothesis on the zeros (Theorem 12.2.5, which uses the verification of RH below height 1000), and the density v_α/π is the Cauchy smoothing of the zeros at widths $\alpha - \beta$ (Proposition 2.1.3); it is the Thorin density of that law whenever α lies to the right of every zero (Theorem 2.1.2). Two theorems turn RH into a single monotone question about this family.

1. Positivity is a zero free half plane (Theorem 11.9.1 of Volume II): for $\frac{1}{2} \le \theta < 1$, the density at level θ is a positive Thorin measure exactly when ζ has no zero with $\operatorname{Re} s > \theta$. At $\theta = \frac{1}{2}$ the equivalence with $\operatorname{Re}(\xi'/\xi) > 0$ on $\operatorname{Re} s > \frac{1}{2}$ is Lagarias's theorem.

2. The minimum sits at the origin (Theorem 12.1.18): RH holds if and only if $v_\alpha(t) > b_\alpha$ for every $\alpha \in (\frac{1}{2}, 1)$ and every $t \neq 0$, and the same statement for $L(s, \chi_4)$ and $L(s, \chi_{12})$ is GRH for those characters.

At the top of the strip the second statement is a theorem: $v_1(t) > v_1(0)$ for every $t \neq 0$, with no hypothesis (Proposition 12.1.11), so the clock at the basepoint one splits as a positive stable law of index $\frac{1}{2}$ plus a GGC. For the two Dirichlet characters it holds for every $\alpha \ge \frac{1}{2} + \frac{1}{\sqrt{2}}$. RH is the statement that the minimum at the origin, which holds at $\alpha = 1$, persists as α descends to $\frac{1}{2}$. In the language of Volume II, the Cauchy flow from the level one density, which the Euler product makes computable, can be run backwards to the critical line through positive measures (Theorem 8.2.2 of Volume II).

Three constraints on any proof The volumes prove three facts that rule out whole classes of argument, and a route is worth pursuing only if it survives all three.

1. It cannot be finite. A perturbation of ξ with the same zeros below a large height and four zeros off the line above it has an infinitely divisible, self decomposable reciprocal clock whose heat trace is strictly log convex and agrees with that of ξ to the constant term (Theorem 1.4.1 of Volume II); finite positivity tests on the heat trace, such as its leading Hankel blocks and its derivative signs to order 10^{17}, pass for it. A computation can refute RH; it cannot prove it.

2. It must use the Euler product. The Davenport and Heilbronn function has a functional equation of the same shape, a positive trace and an infinitely divisible reciprocal law, and a zero off the line near $0.8085 + 85.6993\,i$ (Theorem 14.4.3 of Volume II). Any argument that uses only the functional equation, positivity of a trace and infinite divisibility proves a false statement for it. Ki's approximation theorems point the same way (Chapter 23).

3. It must not rest on a property that survives a Gaussian tilt. If a class of kernels is closed under $K \mapsto Ke^{-bu^2}$ and all its members have Fourier transforms with only real zeros, the Pólya kernel is not in it (Proposition 23.2.3), because the de Bruijn Newman constant is not negative. Log concavity of the kernel and every other upper bound on $(\log K)''$ fall here.

The descent in level survives the three. It is an analytic statement for all heights at once; its input at level one is the Euler product, whose prime sum fixes the density on the line $\operatorname{Re} s = 1$, and it fails for the Davenport and Heilbronn function; and it is a statement about where the zeros are, not a convexity property of the kernel.

Where it stops The first step below the top is already the quasi Riemann hypothesis. Positivity at any level $\theta < 1$ is a zero free half plane $\operatorname{Re} s > \theta$ (statement 1 above), and no such half plane is known: the regions of Vinogradov and Korobov narrow towards the line $\operatorname{Re} s = 1$ as the height grows, and they are the best available. The density at level one is a smooth gamma part minus a compensated prime sum (Chapter 10 of Volume II); pushing its minimum downward is a statement about the size of that prime sum against the gamma part, which is, in Fourier variables, the error term $\psi(x) - x = O(x^{\theta+\varepsilon})$. Every other route in the two volumes, through the heat trace, Weil positivity in windows, hitting times, ferromagnetic approximations, truncations or friable approximants, either reduces to this step or meets one of the three constraints, and the chapters say which.

Two things would change the picture. If the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 is confirmed, the Thorin property holds at every basepoint above $\frac{7}{8}$ (Section 17.7), and the open step becomes the descent from $\frac{7}{8}$ to $\frac{1}{2}$,

a problem of the same kind and no easier in principle. And the de Bruijn Newman flow gives the only known analytic control below level one: at time $t = 0.2$ every zero is real [277], and the Thorin phase diagram of Volume II (Chapter 6) shows that what separates that theorem from RH is the existence of an analytic tail at positive time and the lack of a known one at time zero, not anything a computed density shows.

The clearest path is therefore to prove a zero free half plane by a positivity argument that uses the Euler product, and to carry it down to the line. It is honest to say that this is the hard core of the problem and not a way around it: the theorems of these volumes make the core precise, compute everything around it, and show that the remaining step is analytic. The penultimate chapter of this volume (Sections 24.1 to 24.9) assembles this route from the whole book, in six equivalent forms, compares it with curves over finite fields, where the same route is a proof, and states the missing step as a single positivity problem.

1.5 Exercises

On Section 2.1.

1.1. Show from the Hadamard product (2.1.1) that $\sum_k a_k^{-1} = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2}))$, and compute it to ten digits from the Pólya density by differentiating (2.1.4) twice under the integral.

1.2. Prove that Φ is even by applying the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1 + 2\omega(x))$ to $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and differentiating twice.

1.3. Show that if X has the Pólya density then $\mathbb{E}X^2 = 2\sum_k a_k^{-1}$, and deduce that the Pólya law has variance 0.0462.

1.4. Verify the first row of Table 2.1 by summing $e^{-\gamma^2 t}$ over the first forty zeros, and estimate the error of truncation using $N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi e}$.

1.5. Show that an off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ contributes $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, and find the first time at which that contribution is negative.

1.6. Show that at a basepoint $\alpha > 1$ the Thorin density $v_\alpha(\sqrt{z})/(2\pi\sqrt{z})$ is bounded below by a positive multiple of $z^{-1/2}\log(2+z)$ for large z. Use Proposition 2.1.3 with the Riemann von Mangoldt formula, or, more simply, $v_\alpha(r) = \mathrm{Re}[\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2}) + \frac{\zeta'}{\zeta}(s)]$ at $s = \alpha + ir$ with Stirling's formula.

1.7. Let $\delta = 0.49$, $\gamma_0 = 10.5$ and

$$F(s) = \frac{\sinh(\pi(s-\frac{1}{2}))}{\pi(s-\frac{1}{2})}\Big(1 - \frac{(s-\frac{1}{2})^2}{(\delta+i\gamma_0)^2}\Big)\Big(1 - \frac{(s-\frac{1}{2})^2}{(\delta-i\gamma_0)^2}\Big).$$

Show that F is entire of order one, positive on the real axis and symmetric under $s \mapsto 1-s$, with zeros $\frac{1}{2} \pm ik$ $(k \geq 1)$ and $\frac{1}{2} \pm \delta \pm i\gamma_0$. Show that $\mathrm{Re}(F'/F)(0.6+ir) > 0$ for every real r, although F vanishes at $0.99+10.5i$. (Put $\varepsilon = 0.1$. The zeros $\frac{1}{2} \pm \delta + i\gamma_0$ together contribute $2\varepsilon(t^2-\delta^2+\varepsilon^2)/((t^2+(\delta-\varepsilon)^2)(t^2+(\delta+\varepsilon)^2))$ at $t = r-\gamma_0$, and their conjugates the same expression at $t = r+\gamma_0$; it is at least $-2\varepsilon/(\delta^2-\varepsilon^2)$ and is nonnegative for $t^2 \geq \delta^2-\varepsilon^2$; the zeros on the line contribute $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) - \varepsilon/(\varepsilon^2+r^2) > 0$, and $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) \geq \pi\tanh(\pi\varepsilon)$.)

On Section 4.1.

1.8. Show that the Gamma(a,b) law, with shape a and rate b, has Thorin measure $a\delta_b$, and that the sum of independent Gamma(a_j,b_j) variables has Thorin measure $\sum_j a_j\delta_{b_j}$.

1.9. Derive the Thorin measure of sech $\sqrt{u}$ from the product $\cosh x = \prod_{k\geq 1}(1+4x^2/((2k-1)^2\pi^2))$.

1.10. Prove that $\int_0^\infty \log(1+u/z)\,(2\pi\sqrt{z})^{-1}dz = \sqrt{u}$ (integrate by parts), and deduce the entry for the positive stable law of index $\frac{1}{2}$ in Table 4.3.

1.11. Show that $e^{-t}(1+\frac{1}{2t})$ is completely monotone while $k(t) = e^{-t}(t+\frac{1}{2})$ is not monotone, and conclude that the infinitely divisible law with Lévy density $e^{-t}(1+\frac{1}{2t})$ and no drift is in BO but not in SD.

1.12. For the sinh clock, derive $\mu_\varepsilon^{(0)}((0,\Theta^2]) = m+\pi^{-1}\arctan(\varepsilon/\Theta)$ at $\Theta = m+\frac{1}{2}$ from the periodicity of $\sum_{k\in\mathbb{Z}} P_\varepsilon(\theta-k)$, and check the first order value $0.00630\,\varepsilon$ of the mass defect at $\Theta = 50.5$.

1.13. Show that the HCM property of $\varphi(u) = a/(a+u)$ reduces to complete monotonicity of $a^2/(a^2+auw+u^2)$ in w, and deduce that finite products of such functions are HCM.

1.14. Show that the right side of (4.1.1) extends to a holomorphic nonvanishing function on $\mathbb{C}\setminus(-\infty,0]$. Deduce that if an entire function F with $F(0)=1$ satisfies $1/F(u) = \mathbb{E}e^{-uH}$ for $0<u<1$, with H a GGC, then F has no zeros in $\mathbb{C}\setminus(-\infty,0]$.

Chapter 2

Analytic representations of the Riemann Ξ function

2.1 The Xi function, the Pólya density and the clock

Most of the material is classical or taken from [282].

2.1.1 The completed zeta function and the clock

The completed zeta function ξ of Section 1.1 is entire of order one and maximal type, satisfies $\xi(s) = \xi(1-s)$ and $\overline{\xi(\bar{s})} = \xi(s)$, and is positive on the real axis. On the critical line $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ is real and even. The zeros $\rho = \beta + i\gamma$ of ξ lie in the open strip $0 < \operatorname{Re} s < 1$ and are symmetric under $\rho \mapsto 1-\rho$ and $\rho \mapsto \bar{\rho}$.

Hadamard's theorem applied to G, entire of order $\frac{1}{2}$, gives, with $u = s^2$,

$$\frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \prod_k \Big(1 - \frac{s^2}{(\rho_k - \frac{1}{2})^2}\Big) = \prod_k \Big(1 + \frac{s^2}{a_k}\Big), \tag{2.1.1}$$

with $\sum |a_k|^{-1} < \infty$. Logarithmic differentiation gives the partial fraction expansion

$$\frac{\xi'}{\xi}(s) = \sum_\rho \frac{1}{s-\rho}, \tag{2.1.2}$$

in which each ρ is grouped with $1-\rho$ (as differentiating (2.1.1) produces) or with $\bar{\rho}$. Grouped either way the series converges absolutely, since each group is $O(\gamma^{-2})$ for fixed s and $\sum_\rho \gamma^{-2} < \infty$; ungrouped it converges only when summed symmetrically in $\operatorname{Im}\rho$.

Passing to $u = s^2$ folds each pair $\pm(\rho_k - \frac{1}{2})$ of zeros of $\xi(\frac{1}{2}+s)$ into the single zero $-a_k$ of G, so the product has genus zero. Since ξ has no real zeros, a_k is real only when ρ_k lies on the critical line, and then $a_k = \gamma_k^2 > 0$. RH is therefore the statement that every zero of G lies on the negative real axis. The reciprocal of a genus zero product with negative zeros is a product of gamma Laplace transforms, and this is how generalised gamma convolutions enter.

A random variable $H \geq 0$ is a generalised gamma convolution (GGC) if $\mathbb{E}e^{-uH} = \exp\{-du - \int_{(0,\infty)} \log(1+u/z)\, U(dz)\}$ for $u > 0$, with a drift $d \geq 0$ and a positive measure U, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1} U(dz) < \infty$. The drift and the Thorin measure are unique, and the Laplace transform of a GGC extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$.

Unconditionally, $1/G$ is the Laplace transform of a probability law on $[0,\infty)$ ([282, Theorem A], Theorem 8.1.1). A random variable T with $\mathbb{E}e^{-uT} = 1/G(u)$ for $u \geq 0$ is the clock variable. Its mean and variance are $\mathbb{E}T = \sum_k a_k^{-1} = G'(0) = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2})) = 0.0231050$ and $\operatorname{Var} T = \sum_k a_k^{-2} = 3.71726 \times 10^{-5}$; these constants, and $\xi(\frac{1}{2}) = 0.4971207781883141$, recur throughout the book.

Proposition 2.1.1 (Rigidity, [282, Proposition 3.4]). *The clock $1/G$ is the Laplace transform of a GGC if and only if RH holds, and then its Thorin measure is U_{RH} of (1.1.1) and the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ.*

If $1/G$ is the Laplace transform of a GGC with drift d and Thorin measure U, then $d + \int U(dz)/(u+z) = G'/G(u) = \sum_k (u+a_k)^{-1}$ for $u > 0$. The left side is holomorphic on $\mathbb{C}\setminus(-\infty,0]$, so every pole $-a_k$ of the right side lies on the negative real axis. Hence every a_k is positive, which is RH, and uniqueness of the Stieltjes representation forces $d = 0$ and $U = \sum_k \delta_{a_k}$. Conversely, under RH, (2.1.1) writes $1/G$ as a product of gamma Laplace transforms.

2.1.2 The Pólya kernel and the Pólya density

Riemann's continuation writes ξ as a Mellin transform of a combination of $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and its derivatives, using the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1+2\omega(x))$, $x > 0$. The substitution $x = e^{2u}$ turns it into a Fourier transform; in Titchmarsh's normalisation [348, Section 10.1],

$$\begin{aligned} \Phi(u) &= 2\sum_{n\geq 1} \left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\right) \exp(-\pi n^2 e^{2u}), \\ \Xi(\theta) &= \int_{\mathbb{R}} \Phi(u) \cos(\theta u)\, du. \end{aligned} \tag{2.1.3}$$

Here u is the variable of the kernel, not the argument of the clock. Pólya studied this representation [302], and Φ is the Pólya kernel.

The kernel is even and strictly positive, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2} \exp(-\pi e^{2|u|})$ as $|u| \to \infty$. Evenness is the functional equation read through the theta transformation (Exercise 1.2). For $u \geq 0$ every term of (2.1.3) is positive, because $2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2} = \pi n^2 e^{5u/2}(2\pi n^2 e^{2u} - 3)$ and $2\pi n^2 e^{2u} \geq 2\pi > 3$; positivity for $u < 0$ follows by evenness. Term n is $n^{-1/2}$ times the first term translated left by $\log n$, so for $u \geq 0$ the first term dominates, while for $u < 0$ the terms

are of order $n^{-1/2}$ and cancel almost completely; computations therefore evaluate $\Phi(|u|)$.

Each term of (2.1.3) is entire in u, and since $\operatorname{Re} e^{2u} = e^{2x}\cos 2y$ for $u = x+iy$, the series converges locally uniformly in the strip $|\operatorname{Im} u| < \pi/4$, so Φ is analytic there; the boundary lines $\operatorname{Im} u = \pm\pi/4$ are a natural boundary inherited from the theta function [313, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The strip of analyticity makes the trapezoid rule spectrally accurate and permits the tilted contour of Section 2.1.3. Figure 2.1 shows $|\Phi|$ on lines inside the strip.

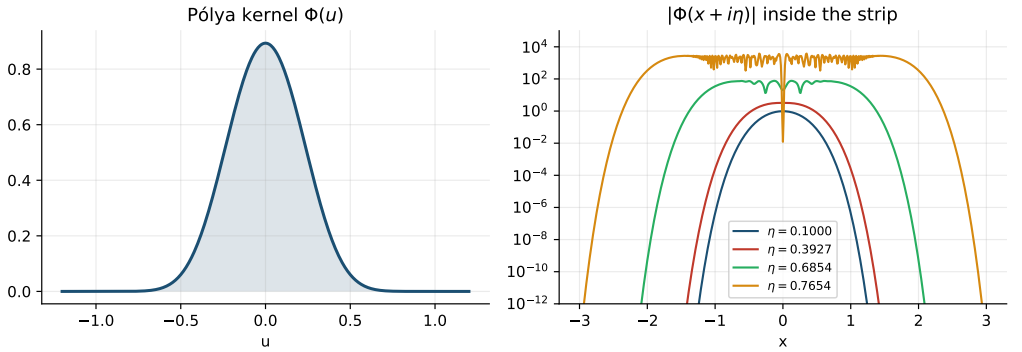


Figure 2.1: Left: the Pólya kernel $\Phi(u)$ on the real line. Right: $|\Phi(x+i\eta)|$, on a logarithmic scale, on the lines $\eta = 0.1000$, 0.3927, 0.6854 and 0.7654 inside the strip of analyticity, that is at distances 0.6854, 0.3927, 0.1 and 0.02 below its edge $\eta = \pi/4$ (the tilts of Section 2.1.3). As the line approaches the natural boundary $\eta = \pi/4$ the kernel grows and spreads (compare Table 2.1), and it still decays doubly exponentially in $|x|$.

Since $\Xi(0) = \xi(\frac{1}{2}) = \int_{\mathbb{R}} \Phi$, the Pólya density $p = \Phi/\Xi(0)$ is the law of a symmetric random variable X with all exponential moments, and

$$\frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \mathbb{E}\, e^{sX} = \mathbb{E}\cosh(sX), \qquad s \in \mathbb{C}. \tag{2.1.4}$$

The clock is $1/\mathbb{E}\cosh(\sqrt{u}\,X)$, and at $u = \theta^2$ it forms with the characteristic function of X a van Dantzig pair, a pair (φ_1, φ_2) of characteristic functions with $\varphi_2(\theta) = 1/\varphi_1(i\theta)$, after a problem of van Dantzig studied by Lukacs [228]. Here $\varphi_1(\theta) = \Xi(\theta)/\Xi(0)$ is the characteristic function of X, and $1/\varphi_1(i\theta) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\theta) = \mathbb{E}e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with Z standard normal and independent of T [282, Theorem A]. G. Letac asked whether φ_1 is the first member of a van Dantzig pair (private communication), and Theorem 8.1.1 shows that it is, unconditionally.

Two classical pairs show the shape to expect: $(\cos\theta, 1/\cosh\theta)$ and $(\sin\theta/\theta, \theta/\sinh\theta)$. In both the first member has only real zeros and the second is

the product of the reciprocal factors: $1/\cosh\theta = \prod_{k\geq 1}(1+\theta^2/((k-\frac{1}{2})\pi)^2)^{-1}$ is the characteristic function of $\sum_{k\geq 1} L_k/((k-\frac{1}{2})\pi)$, where the L_k are independent standard Laplace variables, with density $\frac{1}{2}e^{-|x|}$ and characteristic function $1/(1+\theta^2)$, and likewise $\theta/\sinh\theta$ is that of $\sum_{k\geq 1} L_k/(k\pi)$.

The Xi pair has the classical shape exactly when RH holds. If RH holds, $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 2.1.1), so $Z\sqrt{2T}$ has the law of $\sum_{\gamma>0} L_\gamma/\gamma$, the sum that Section 9.2 calls $\log W$. Conversely, if $Z\sqrt{2T}$ has the law of $\sum_k L_k/c_k$ for real c_k with $\sum_k c_k^{-2} < \infty$, then $\xi(\frac{1}{2}+\theta)/\xi(\frac{1}{2}) = \prod_k(1+\theta^2/c_k^2)$ for real θ, hence for complex θ, and every zero of ξ lies on the critical line. Konstantopoulos, Patie and Sarkar state the same equivalence in the language of the Laguerre Pólya class and of Pólya frequency functions [192, Theorem 2.3 and Corollary 2.4].

The Pólya variable has a probabilistic construction. Let $S_2 = \sum_{k\geq 1} \frac{2}{\pi^2 k^2}(e_k + e'_k)$ with independent standard exponentials e_k, e'_k; it has the law of the sum of two independent hitting times of level 1 by a three dimensional Bessel process started at 0. By the theorem of Biane, Pitman and Yor [33], going back to Williams [363], $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$. With $Y = \frac{1}{2}\log(\pi S_2/2)$ this reads $\mathbb{E}e^{sY} = 2\xi(s)$, so X is Y tilted by $e^{Y/2}$: $\mathbb{E}f(X) = \mathbb{E}[f(Y)e^{Y/2}]/\mathbb{E}e^{Y/2}$. Tilting Y by $e^{\alpha Y}$ in place of $e^{Y/2}$ gives, at every real basepoint α and with no hypothesis on the zeros, a variable V_α with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$ and $V_{1/2} = X$ (Section 12.1.1).

2.1.3 The heat trace: spectral and arithmetic forms

The heat trace of the zeros is

$$W(t) = \sum_k e^{-a_k t} = \tfrac{1}{2}\sum_\rho e^{(\rho-\frac{1}{2})^2 t}, \qquad t > 0,$$

which converges absolutely for every $t > 0$ wherever the zeros lie, since $\operatorname{Re} a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > \gamma_k^2 - \frac{1}{4}$ for $\rho_k = \beta_k + i\gamma_k$. Integrating $e^{-(u+a_k)t}$ term by term, and using Frullani's integral $\log(1+u/a) = \int_0^\infty (1-e^{-ut})e^{-at}t^{-1}dt$ for $\operatorname{Re} a > 0$, gives, unconditionally for $u > 0$ [282, Theorem 4.3],

$$\frac{G'}{G}(u) = \int_0^\infty e^{-ut} W(t)\, dt, \qquad \log G(u) = \int_0^\infty \big(1-e^{-ut}\big)\frac{W(t)}{t}\, dt.$$

The second formula shows that $W(t)/t$ is the only possible Lévy density of the clock variable T.

The classes of infinitely divisible laws of Section 4.1.2 become sign conditions on W [282, Theorem 4.6]: T is infinitely divisible if and only if $W \geq 0$, self decomposable if and only if W is nonincreasing, and a GGC if and only if W is completely monotone, which by Proposition 2.1.1 happens if and only if RH holds. The first two conditions hold unconditionally, in the strict forms $W > 0$ and $W' < 0$ on $(0,\infty)$ (Theorem 8.1.1, [282, Theorem A]), by a proof that uses the verification of RH below height 1000.

For even test functions h holomorphic and rapidly decreasing in a strip $|\operatorname{Im} r| \le \frac12 + c$ with $c > 0$, Weil's explicit formula [357], [167, Theorem 5.12], reads

$$\sum_{\rho} h(\gamma_\rho) = h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\big(\operatorname{Re}\psi(\tfrac14 + \tfrac{ir}{2}) - \log\pi\big)dr \\ - 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\, g(\log n),$$

where $\rho = \frac12 + i\gamma_\rho$, $g(x) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r)e^{-irx}dr$ and $\psi = \Gamma'/\Gamma$ is the digamma function. The Gaussian $h(r) = e^{-r^2 t}$ has $g(x) = e^{-x^2/4t}/(2\sqrt{\pi t})$ and $\sum_\rho h(\gamma_\rho) = 2W(t)$, which gives the arithmetic form [282, Theorem 4.4]

$$W(t) = e^{t/4} + \frac{1}{4\pi}\int_{\mathbb{R}} e^{-r^2 t} m(r)\, dr - \frac{1}{2\sqrt{\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}} e^{-(\log n)^2/4t}, \\ m(r) = \operatorname{Re}\psi\big(\tfrac14 + \tfrac{ir}{2}\big) - \log\pi. \tag{2.1.5}$$

The three terms of (2.1.5) are the polar, archimedean and prime contributions. The polar term $e^{t/4}$ comes from $h(\pm\frac{i}{2})$, the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, which the factor $s(s-1)$ of ξ removes. The prime term is $\sum \Lambda(n) n^{-1/2}$ smoothed by a Gaussian of standard deviation $\sqrt{2t}$ in $\log n$.

Table 2.1 reproduces [282, Table 4.1]; the two sides agree to fifteen digits. The polar and archimedean terms are of order one while $W(t)$ decays like $e^{-\gamma_1^2 t}$ with $\gamma_1^2 = 199.79$, so an estimate of W from (2.1.5) must survive a cancellation whose depth grows linearly in t. Section 8.2.2 extends the comparison and gives that depth explicitly.

t	$W(t)$ from zeros	polar + archimedean − prime	archimedean	prime
0.01	0.149698011253780	0.149698011253780	−0.8527967	8.3995×10^{-6}
0.02	$1.85412917313195 \times 10^{-2}$	$1.85412917313195 \times 10^{-2}$	−0.9840612	0.0024101
0.05	$4.58783504857357 \times 10^{-5}$	$4.58783504857357 \times 10^{-5}$	−0.9546186	0.0579140
0.10	$2.10479979582235 \times 10^{-9}$	$2.10479979582235 \times 10^{-9}$	−0.8625096	0.1628055
0.20	$4.43018218022364 \times 10^{-18}$	$4.43018218022364 \times 10^{-18}$	−0.7497538	0.3015173

Table 2.1: The heat trace from the zeros and from the explicit formula, after [282, Table 4.1]. At $t = 1$ the cancellation between the arithmetic terms reaches about eighty seven digits.

2.1.4 The Thorin density at a shifted basepoint

The Thorin density of the theorem below is defined at every basepoint $\alpha > \frac12$, whatever the zeros do, and is the starting point of the computations in Chapters 3

to 8. For a real basepoint α write $v_\alpha(r) = \operatorname{Re}(\xi'/\xi)(\alpha + ir)$, and let U_α be the signed measure on $(0,\infty)$ given by

$$U_\alpha(dz) = \frac{v_\alpha(\sqrt{z})}{2\pi\sqrt{z}}\,dz. \tag{2.1.6}$$

At $\alpha = \frac12 + \varepsilon$, where $\varepsilon > 0$ is the resolution, the chart $\theta = \sqrt{z}$ gives U_α the density

$$\rho_\varepsilon(\theta) = \frac{1}{\pi}\,v_{1/2+\varepsilon}(\theta) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}\Big(\frac12 + \varepsilon + i\theta\Big), \tag{2.1.7}$$

so that $U_\alpha(dz) = \rho_\varepsilon(\theta)\,d\theta$. The family $\varepsilon \mapsto \rho_\varepsilon$ is the Cauchy flow of [282, Chapter 8].

Theorem 2.1.2 ([282, Theorem 3.7 and Remark 3.10]). *If $\alpha > \operatorname{Re}\rho$ for every zero, in particular for every $\alpha \ge 1$, then $u \mapsto \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of a GGC T_α with Thorin density* (2.1.6)*, and $v_\alpha > 0$. It is the Laplace transform of a GGC if and only if $\alpha \ge \sup_\rho \operatorname{Re}\rho$.*

No basepoint $\alpha < 1$ is known unconditionally to satisfy the hypothesis; the unrefereed zero free half plane $\operatorname{Re} s > \frac78$ announced in October 2026 would give every $\alpha > \frac78$ (Section 25.1.2). At every basepoint $\alpha > \frac12$ the shifted clock is still the Laplace transform of a probability law, with no hypothesis on the zeros (Theorem 12.2.5); the zeros decide whether that law is a GGC. Let $P_a(x) = a/(\pi(a^2 + x^2))$ be the Poisson kernel, with $P_{-a} = -P_a$. Write μ_ε for the signed measure $U_{1/2+\varepsilon}$ and for its image in the chart $\theta = \sqrt{z}$, with density (2.1.7). By Theorem 2.1.2, μ_ε is the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac12 + \varepsilon > \operatorname{Re}\rho$ for every zero, in particular for every $\varepsilon \ge \frac12$; in general we call it the Thorin measure at resolution ε.

Proposition 2.1.3 (Poisson structure). *For $\varepsilon > 0$ and $\theta \ge 0$ such that $\frac12 + \varepsilon + i\theta$ is not a zero of ξ,*

$$\begin{aligned}\rho_\varepsilon(\theta) = {} & \sum_{\gamma>0,\ \operatorname{Re}\rho=1/2} \big[P_\varepsilon(\theta-\gamma) + P_\varepsilon(\theta+\gamma)\big] \\ & + \sum_{\text{off line } \rho=\frac12\pm\delta+i\gamma,\ \gamma>0} \Big(\big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta-\gamma) + \big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta+\gamma)\Big),\end{aligned} \tag{2.1.8}$$

where the second sum counts each off line pair $\{\frac12 + \delta + i\gamma, \frac12 - \delta + i\gamma\}$, $0 < \delta < \frac12$, once. Equivalently, with $\alpha = \frac12 + \varepsilon$, $\rho_\varepsilon(\theta)$ is the sum of $P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)$ over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$. In particular, under RH, $\mu_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ restricted to $\theta > 0$, where $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma>0}(\delta_\gamma + \delta_{-\gamma})$.

Proof. Take real parts in (2.1.2) at $s = \alpha + i\theta$ with $\alpha = \frac12 + \varepsilon$: a zero $\rho = \beta + i\gamma$ contributes $\pi P_{\alpha-\beta}(\theta-\gamma)$ to $v_\alpha(\theta)$, and $\alpha - \beta = \varepsilon \mp \delta$ when $\beta = \frac12 \pm \delta$. Dividing by

π and grouping each zero ρ with $\bar\rho$ gives the second form, and collecting the zeros ρ, $\bar\rho$, $1-\rho$ and $1-\bar\rho$ gives (2.1.8). The series converges absolutely, and uniformly for θ in compact sets avoiding the zeros, since each term is $O(\gamma^{-2})$ there and $\sum_\rho \gamma^{-2} < \infty$. □

An off line pair contributes kernels of widths $\varepsilon+\delta$ and $\varepsilon-\delta$, and the second is negative once $\varepsilon < \delta$, that is, once the line $\operatorname{Re} s = \frac12+\varepsilon$ lies left of the zero $\frac12+\delta+i\gamma$. If every zero lies left of the line, every kernel is positive and so is v_α; a negative value of v_α therefore proves a zero to the right of α.

Remark 2.1.4 (Positivity on one line). Theorem 2.1.2 concerns the analytic continuation of the shifted clock, not the sign of v_α on the single line $\operatorname{Re} s = \alpha$. Once a zero ρ lies right of the line, the shifted clock is not a GGC, since $\xi(\alpha+\sqrt{u})$ vanishes at $u = (\rho-\alpha)^2 \in \mathbb{C}\setminus(-\infty,0]$, but the density need not change sign, because the positive kernels of the other zeros can outweigh the negative one. Exercise 1.7 gives an entire function with the order, symmetry and reality of ξ whose density is positive on a line with a zero to its right; for ξ itself it is not known, without assuming RH, whether positivity on one line excludes zeros to its right. Positivity on every line to the right of $\operatorname{Re} s = \sigma_0$ does exclude zeros there: if $v_\alpha \ge 0$ for every $\alpha > \sigma_0$, then no zero has $\operatorname{Re}\rho > \sigma_0$, because $\operatorname{Re}(\xi'/\xi)(\rho-b) \sim -m/b$ as $b \downarrow 0$ at a zero ρ of multiplicity m. With $\sigma_0 = \frac12$ this is Lagarias's criterion [198]: RH holds if and only if $v_\alpha \ge 0$ for every $\alpha > \frac12$.

Integrated over $\theta > 0$, a line zero contributes mass one, which concentrates at $\theta = \gamma$ as $\varepsilon \downarrow 0$. An off line pair contributes mass two when $\varepsilon > \delta$, and when $\varepsilon < \delta$ a signed measure of mass zero whose total variation tends to zero as $\varepsilon \downarrow 0$. These limits can be summed over the zeros: if f is continuous with support in $[a,b] \subset (0,\infty)$, then for $0 < \varepsilon \le 1$ each term of (2.1.8) with $\gamma > b+1$ integrates f to a number of modulus at most $C_f\gamma^{-2}$, where C_f depends only on f, because the widths c in (2.1.8) satisfy $|c| < \varepsilon + \frac12 \le \frac32$, $|P_c(x)| \le |c|/(\pi x^2)$, and $|\theta \pm \gamma| \ge \gamma - b \ge \gamma/(b+1)$ for $\theta \in [a,b]$; as $\sum_\rho \gamma^{-2} < \infty$, dominated convergence applies to the sum over the zeros. Hence $\mu_\varepsilon \to U_{\text{line}} = \sum_{\text{line zeros}} \delta_{\gamma^2}$ vaguely, whatever the truth of RH [279, Proposition 9.5].

2.1.5 The sine squared lemma: the Thorin density from the primes

The sine squared lemma [279, Lemma 8.1] produces the same Thorin density from the Lévy side, where the primes appear as atoms. For $x > 0$ and $s \ge 0$,

$$e^{-sx} + sx - 1 = x\int_0^\infty \log\Big(1+\frac{s^2}{z}\Big)\frac{\sin^2(x\sqrt{z}/2)}{\pi\sqrt{z}}\,dz. \tag{2.1.9}$$

For the proof, $z = \theta^2$ turns the right side into $\frac{x}{\pi}\int_0^\infty \log(1+s^2/\theta^2)(1-\cos x\theta)\,d\theta$. Both sides vanish at $s = 0$, and their derivatives in s agree because $\int_0^\infty (1-\cos x\theta)\,2s(\theta^2+s^2)^{-1}d\theta = \pi(1-e^{-sx})$.

α	θ	sine squared, primes $\le 2\times10^6$	Pólya route	error of prime side
3.0	14.1347	0.561447238789	0.561447238789	1.1×10^{-13}
3.0	50	1.120967014740	1.120967015289	5.5×10^{-10}
2.0	14.1347	0.766398694727	0.766399169607	4.7×10^{-7}
2.0	50	1.224475714058	1.224476216908	5.0×10^{-7}
1.5	14.1347	1.065762504228	1.067140080666	1.4×10^{-3}
1.5	50	1.385288214458	1.386706213218	1.4×10^{-3}

Table 2.2: The bracket $v_\alpha(\theta)$ of the sine squared lemma computed from the canonical measure and from the Pólya density; $\theta = 14.1347$ is the first ordinate γ_1, rounded. The Pólya values agree with an independent evaluation of $\mathrm{Re}(\xi'/\xi)$ through Γ and ζ to about 10^{-15}.

Integrating (2.1.9) against a measure turns Lévy representations into Thorin representations. A Lévy representation $\log f(\alpha+s)/f(\alpha) = cs + \int (e^{-sx} + sx - 1)L(dx)$ for $s \ge 0$, with c real and L a real measure on $(0,\infty)$ such that $\int \min(x,x^2)\,|L|(dx) < \infty$, becomes a Thorin representation with density $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\,xL(dx)]$, and the bracket equals $\mathrm{Re}(f'/f)(\alpha + i\sqrt{z})$ [279, Theorem 8.2]. For $f = \xi$ at $\alpha > 1$ the input is the signed canonical measure $xL(dx)$, with atoms $\Lambda(n)n^{-\alpha}$ at $\log n$, an archimedean density and a pole density [282, Theorem 8.1].

Table 2.2 compares the bracket computed from the canonical measure with primes truncated at $N = 2\times10^6$ and $v_\alpha(\theta)$ computed by the Pólya route of Section 2.1. The difference is the prime tail $\sum_{n>N}\Lambda(n)n^{-\alpha}(1-\cos(\theta\log n))$, of order $N^{1-\alpha}/(\alpha-1)$, except at $\alpha = 3$, $\theta = 50$, where it comes from the double precision quadrature of the archimedean integral.

Chapter 3

Probabilistic representations of Ξ

3.1 Brownian boundaries and Williams' structure

Under RH with simple zeros, Kent's theorem makes the clock variable T the hitting time of a level by a diffusion, but it describes that diffusion only through its spectrum, the set $\{\gamma^2\}$ that is to be found (Section 5.2.3). We look instead for a representation by Brownian motion itself. Unconditionally, T is not a Brownian first passage time to an independent random level for any constant drift (Section 3.1.1). Across a curved boundary it is one: such a boundary exists by a theorem of Ekström and Janson, and we compute it (Section 3.1.2). Williams' representation of the Pólya kernel through Bessel hitting times explains the tilted contour of Section 2.1.3, but it transfers no positivity to the clock (Section 3.1.3).

3.1.1 The clock is not a first passage time to a random level

The first passage time of Brownian motion with drift $c \geq 0$ to a level $a > 0$ has the inverse Gaussian law, with transform $\exp\{a(c - \sqrt{c^2 + 2u})\}$. Mixing over an independent random level with law F gives the transforms $\int \exp\{a(c - \sqrt{c^2 + 2u})\}F(da)$.

Proposition 3.1.1. *For no drift $c \geq 0$ and no probability law F on $(0, \infty)$ is the clock variable T equal in law to the first passage time of Brownian motion with drift c to an independent level $A \sim F$.*

Proof. On the parabola $u = \frac{1}{2}(-q^2 + 2icq)$, $q \in \mathbb{R}$, we have $c^2 + 2u = (c + iq)^2$, so $\operatorname{Re}\sqrt{c^2 + 2u} = c$ and every mixture transform has modulus at most one there. Suppose T had such a law, so that $1/G(u) = \mathbb{E}e^{-uT}$ and the mixture transform agree for $u \geq 0$. The mixture transform is analytic and bounded by one on $D = \{u : \operatorname{Re}\sqrt{c^2 + 2u} > c\}$ and continuous up to its boundary, the parabola; D is the image of the half plane $\operatorname{Re} v > c$ under $v \mapsto (v^2 - c^2)/2$, so it is connected and contains the right half plane. By the identity theorem $1/G$ coincides with the mixture transform on D, so it has no pole in D and is bounded by one there, and

by continuity $|1/G| \le 1$ on the parabola away from the zeros of $\xi(\frac{1}{2}+\sqrt{u})$. For the clock, however, $\sqrt{u} = (c+iq)/\sqrt{2}+O(1/q)$ as $q \to \infty$ (for $c = 0$ the parabola is the negative axis, approached from above, and $\sqrt{u} = i|q|/\sqrt{2}$), so $|\xi(\frac{1}{2}+\sqrt{u})|$ decays like $\exp(-\pi|q|/(4\sqrt{2}))$ times a power, by Stirling's formula and the polynomial growth of ζ in vertical strips. Hence $|\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})|$ grows exponentially along the parabola, and the transforms cannot agree. □

The proof covers the nonincreasing lines $a - ct$, $c \ge 0$; an increasing line, with $c < 0$, gives a passage time that is infinite with probability $1 - \mathbb{E}e^{2cA} > 0$, which the clock is not. Expanding $G(u) = 1 + m_1 u + m_2 u^2 + \cdots$ with $m_1 = \mathbb{E}T$ and $m_1^2 - 2m_2 = \operatorname{Var} T$ gives $|G(u)|^2 = 1 - q^2(m_1 - c^2 \operatorname{Var} T) + O(q^3)$ on the parabola, so the clock transform already exceeds one at small q whenever

$$c < c_* = \sqrt{\mathbb{E}T/\operatorname{Var} T} = \sqrt{0.0231050/3.71726 \times 10^{-5}} = 24.93.$$

3.1.2 A curved Brownian boundary for the clock

Finding a boundary whose first passage time has a prescribed law is the inverse first passage problem [57, Theorem 1]. By the theorems of Ekström and Janson [81, Theorems 2.3, 8.2 and 9.2], stated in Section 12.2.4, it has a solution for the clock, the exact boundary. Since the density of the clock is continuous and positive on $(0, \infty)$ (Exercise 18.10), by the same theorems the exact boundary is continuous on $(0, \infty)$ and is the unique continuous solution b with $\liminf_{t \downarrow 0} b(t) \ge 0$ of the Volterra equation (3.1.1). The solution of the discretised equation is the Volterra boundary.

3.1.2.1 The Roberts Shortland tangent boundary

Roberts and Shortland [317] approximate the hazard of a Brownian first passage across a smooth boundary b by the hazard of the tangent line at the current time, for which the Bachelier Lévy formula is exact:

$$h_{\tan}(t) = \frac{\alpha(2\pi t^3)^{-1/2} \exp\{-(\alpha+\beta t)^2/2t\}}{\mathcal{N}\big((\alpha+\beta t)/\sqrt{t}\big) - e^{-2\alpha\beta}\mathcal{N}\big((\beta t - \alpha)/\sqrt{t}\big)},$$
$$\beta = b'(t),\ \alpha = b(t) - tb'(t),$$

where $\mathcal{N}$ is the standard normal distribution function and $\overline{\mathcal{N}} = 1 - \mathcal{N}$. We use the formula in reverse: setting $h_{\tan}(t) = h(t)$, the computed hazard of the clock, gives an ordinary differential equation for b, which we integrate by fourth order Runge Kutta with step 10^{-4} from $t_0 = 0.011$. The starting value $b(t_0) = 0.4023$ is the constant level with the crossing mass of the clock, $2\overline{\mathcal{N}}(b(t_0)/\sqrt{t_0}) = P(T \le t_0) = 1.25 \times 10^{-4}$.

The boundary falls steeply at first, with slope -87.5 at t_0, and then becomes asymptotically linear with slope tending to $-\sqrt{2}\,\gamma_1 = -19.99$, since an inverse Gaussian hazard tends to $\beta^2/2$.

3.1.2.2 The boundary from the Volterra equation

The Volterra equation determines the exact boundary ([81, Theorems 9.1 and 9.2]), and solving its discretisation for the clock needs three zones: small times, the bulk and the right tail.

For a continuous boundary b with $b(0+) > 0$, the first passage density f of Brownian motion started at 0 satisfies the Volterra equation of the first kind [270]

$$\begin{aligned} P\big(B_t \geq b(t)\big) &= \int_0^t f(s)\, P\big(B_t \geq b(t) \mid B_s = b(s)\big)\, ds, \\ \text{that is} \quad \overline{\mathcal{N}}\Big(\frac{b(t)}{\sqrt{t}}\Big) &= \int_0^t f(s)\, \overline{\mathcal{N}}\Big(\frac{b(t)-b(s)}{\sqrt{t-s}}\Big) ds, \end{aligned} \tag{3.1.1}$$

In the inverse problem f is the density of the clock computed in Section 5.2, and we solve for b one time step at a time: once b is known at the earlier grid points, the discretised equation at time t is a scalar equation for $b(t)$. We discretise with 2900 steps on $[0.009, 0.3]$, midpoint quadrature in s, and a scalar root search at each step.

Before $t_0 = 0.009$ the clock has crossed with probability only $P(T \leq t_0) = 1.42 \times 10^{-7}$. Its density is super flat: it vanishes to all orders at $t = 0$, like $\exp(-(1+o(1))\log^2(1/t)/(16t))$, as the growth $\log G(u) \sim \frac{1}{4}\sqrt{u}\log u$ at infinity dictates; de Bruijn's exponential Tauberian theorem [34, Theorem 4.12.9] gives this rate for $P(T \leq t)$, and the density has the same rate because the clock is unimodal (Corollary 14.1.2). So the exact boundary is unbounded as $t \downarrow 0$: a path above $b(t)$ at time t has crossed by then, so $\overline{\mathcal{N}}(b(t)/\sqrt{t}) \leq P(T \leq t)$, which forces $b(t) \geq (1+o(1))\log(1/t)/(2\sqrt{2})$. In this first zone we replace the exact boundary by a constant level, fixed by the reflection principle, $2\overline{\mathcal{N}}(b/\sqrt{t_0}) = P(T \leq t_0)$, which gives $b(t_0) = 0.4992$. The Volterra boundary then falls from 0.4992 to 0.3908 at $t = 0.011$. The tangent boundary starts there at 0.4023, the constant level with the same crossing mass by time 0.011; a boundary that decreases on $[0, 0.011]$ with that crossing mass must end below this level, so the gap is no defect of the tangent approximation.

In the bulk, $0.011 \leq t \leq 0.05$, the Volterra and tangent boundaries agree closely: 0.0721 against 0.0710 at $t = 0.02$, and -0.5894 against -0.5954 at $t = 0.05$. There is no cancellation: both sides of (3.1.1) are of order one.

In the right tail the equation cancels. Solved in the form (3.1.1), the computation fails at $t = 0.0937$, where the clock has survival probability 3.8×10^{-7}: both sides of the equation are within that amount of $P(T \leq t) \approx 1$. In general the absolute error of the quadrature must stay below $P(T > t)$, which is 5×10^{-25} at $t = 0.3$, beyond double precision. The cure is to subtract the equation from $P(T \leq t) = \int_0^t f$, which gives the complementary form

$$\mathcal{N}\Big(\frac{b(t)}{\sqrt{t}}\Big) - P(T > t) = \int_0^t f(s)\, \mathcal{N}\Big(\frac{b(t)-b(s)}{\sqrt{t-s}}\Big) ds,$$

in which every term is small and computed without cancellation, provided the survival function is known to relative accuracy; the spectral expansion of Section 5.2 supplies it to full working precision even where $P(T > t)$ is of order 10^{-25}. With the switch made where $P(T > t)$ falls below one half, the solution runs smoothly to $t = 0.3$.

In this zone the boundary is nearly linear with slope $\beta = -\sqrt{2}\gamma_1$, but a linear boundary does not suffice: its crossing time has survival function of order $t^{-3/2}e^{-\beta^2 t/2}$, while that of the clock is asymptotic to a multiple of $e^{-\gamma_1^2 t}$. A logarithmic correction removes the power: for $b(t) = \beta t + a + \kappa \log t$ the Gaussian factor $e^{-b(t)^2/2t}$ acquires $t^{-\kappa\beta}$, and $\kappa\beta = -\frac{3}{2}$ cancels the power. So the right tail boundary is

$$b(t) \approx \beta t + a + \frac{3}{2|\beta|}\log t, \qquad \beta = -\sqrt{2}\,\gamma_1, \qquad \frac{3}{2|\beta|} = 0.0750.$$

Fitting a on $[0.15, 0.3]$ gives $a = 0.578$ with largest residual 0.0095. The slope of the Volterra boundary over $[0.2, 0.3]$ is -19.77, between the slope -19.99 without the logarithm and the slope $-19.99 + 0.075/0.25 = -19.69$ of the asymptote at $t = 0.25$.

We simulate $n = 2 \times 10^5$ Brownian paths against both boundaries with step 2×10^{-5} and a Brownian bridge crossing correction; sampling noise alone gives a Kolmogorov distance of about $0.87/\sqrt{n} \approx 0.002$. For the Volterra boundary the Kolmogorov distance to the law of the clock is 0.0020, at the level of the Monte Carlo noise, and the mean crossing time is 0.023085 against $\mathbb{E}T = 0.0231050$. For the tangent boundary the distance is 0.0059 and the mean 0.023021 (Figure 3.1). The right tail itself cannot be checked by plain simulation, since only about one path in ten million survives past $t = 0.1$.

Wald's identities constrain the exact boundary as a whole. Since $\mathbb{E}T$ is finite, the first passage time H across it satisfies $\mathbb{E}B_H = 0$ and $\mathbb{E}B_H^2 = \mathbb{E}H$, and since $B_H = b(H)$ and H has the law of the clock, $\int_0^\infty b(t)f(t)\,dt = 0$ and $\int_0^\infty b(t)^2 f(t)\,dt = \mathbb{E}T$.

3.1.3 Williams' Gamma(2) structure: the tilt, a benchmark and its limits

Williams [363] and Biane, Pitman and Yor [33] showed that the variable $S_2 = \sum_{n\geq 1} \frac{2}{\pi^2 n^2}\Gamma_n$ of Section 5.2.2 is a sum of two independent hitting times of level one by the three dimensional Bessel process started at 0, and that $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$, so that the Pólya variable is $Y = \frac{1}{2}\log(\pi S_2/2)$ tilted by $e^{Y/2} = (\pi S_2/2)^{1/4}$ (Section 2.1; [282, Section 16.2.1]). The Thorin measure of S_2 has atoms of mass two at the points $\pi^2 n^2/2$, the Gamma(2) shape of a sum of two copies.

First, the tilt. Write y for the variable of the Pólya kernel, so that the value y of Y corresponds to the value e^{2y} of $\pi S_2/2$. Moving the line of integration to

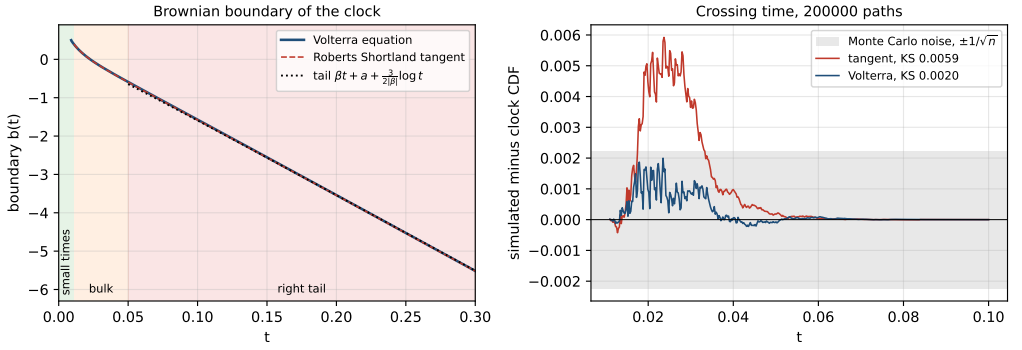


Figure 3.1: Left: the Brownian boundary of the clock computed from the Volterra equation, the Roberts Shortland tangent boundary, and the right tail asymptote with its logarithmic correction, over the three zones. Right: the simulated distribution function of the crossing time minus that of the clock, for both boundaries; the grey band is $\pm 1/\sqrt{n}$, twice the largest standard error of a simulated distribution function.

$\operatorname{Im} y = \eta$, as in Proposition 2.1.1, rotates e^{2y} by the angle 2η. The density of S_2, a theta series, continues analytically to the right half plane, whose boundary, the natural boundary of the theta function [313, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], is reached exactly at $\eta = \pi/4$. So the admissible tilts of Proposition 2.1.1 are the rotations of Williams' variable within the half plane of analyticity of its density, and the saving $e^{-\eta\theta}$ is the factor gained by the rotation.

Second, a benchmark. Since $\pi^2 S_2/2 = \sum_n \Gamma_n/n^2$, the Thorin measure of $\pi^2 S_2/2$ is $2\sum_n \delta_{n^2}$, and in the square root variable its Cauchy flow $P_\varepsilon * 2\sum_n \delta_n$ is explicit [282, Section 16.4]. On it the distances of Section 3.1 separate the effects of the method from those of the arithmetic, with no separate computation: $\pi^2 S_2/2$ is the sum of two independent copies of the sinh clock of Section 4.1.4, so its distances follow from the closed forms there (Section 25.2.3).

Third, the limits. Williams' atoms are the Dirichlet eigenvalues $\pi^2 n^2/2$ of an interval and live on the Laplace side of the kernel; the zeros of ξ live on the Mellin side of the same variable and reappear as unit atoms only on the Laplace side of the reciprocal. The Gamma(2) structure gives positivity and an exact sampler [76] on the kernel side, but transfers no positivity to the Thorin measure of the clock. In particular it does not give a positive Gamma(2) scale mixture representation of the Pólya density, a route that [282, Section 12.6] excludes.

Fourth, Maślanka's series. Maślanka [132] expanded $(\sigma - 1)\zeta(\sigma) = \sum_{k\geq 0} A_k\,(1-\frac{\sigma}{2})_k/k!$, with $A_k = \sum_{j=0}^{k}(-1)^j\binom{k}{j}(2j+1)\zeta(2j+2)$, and Báez Duarte [11] proved it rigorously. The coefficients are Laguerre moments of Williams' variable [152].

Proposition 3.1.2. *Let* $W = \pi^2 S_2/2 = \sum_{n\ge1} \Gamma_n/n^2$, *so that* $\mathbb{E}e^{-\lambda W} = (\pi\sqrt{\lambda}/\sinh \pi\sqrt{\lambda})^2$, *and let* $L_k^{(1)}$ *be the generalised Laguerre polynomials.*

(i) $A_k = \mathbb{E}[W L_k^{(1)}(W)]/(2(k+1))$.

(ii) $\sum_{k\ge0} A_k y^k = \big(1 - \mathbb{E}e^{-Wy/(1-y)}\big)/(2y)$ *for* $0 < |y| < 1$.

(iii) For $0 < \operatorname{Re}\sigma < 2$,

$$(\sigma-1)\zeta(\sigma) = \frac{\sin(\pi\sigma/2)}{2\pi} \int_0^\infty r^{-\sigma/2-1}\big(1 - \mathbb{E}e^{-rW}\big)\, dr.$$

For real $0 < \sigma < 2$ *this equals* $\mathbb{E}\big[(1-\mathbb{E}[e^{-WR} \mid \mathcal{B}])/(2(1-\mathcal{B}))\big]$, *with* $\mathcal{B}$ *a beta variable of parameters* $\frac{\sigma}{2}$ *and* $1-\frac{\sigma}{2}$, *independent of* W, *and* $R = (1-\mathcal{B})/\mathcal{B}$; *since* $\mathbb{E}(1-\mathcal{B})^k = (1-\frac{\sigma}{2})_k/k!$, *expanding (ii) at* $y = 1-\mathcal{B}$ *gives Maślanka's series.*

Proof. At $s = 2j+2$ the identity $2\xi(s) = \mathbb{E}[(W/\pi)^{s/2}]$ reads $(2j+1)\zeta(2j+2) = \mathbb{E}W^{j+1}/(2(j+1)!)$, and $\binom{k}{j}/(j+1)! = \binom{k+1}{k-j}/((k+1)\,j!)$ turns $\sum_j (-1)^j \binom{k}{j} W^{j+1}/(j+1)!$ into $W L_k^{(1)}(W)/(k+1)$, which is (i). For (ii), $\sum_k L_k^{(1)}(w)\, y^{k+1}/(k+1) = \int_0^y (1-t)^{-2} e^{-wt/(1-t)} dt = (1 - e^{-wy/(1-y)})/w$, and the bound $|L_k^{(1)}(w)| \le (k+1)e^{w/2}$ for $w \ge 0$, with $\mathbb{E}\, W e^{W/2} < \infty$ because $\mathbb{E}e^{tW} = \prod_n (1 - t/n^2)^{-2}$ is finite for $t < 1$, allows the exchange of sum and expectation for $|y| < 1$. For (iii) put $z = \sigma/2$. Fubini and $\int_0^\infty r^{-z-1}(1 - e^{-rw})\, dr = w^z \Gamma(1-z)/z$ for $0 < \operatorname{Re} z < 1$, with $\mathbb{E}W^z = 2\pi^z \xi(2z)$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$, give $\pi^z \xi(2z)/\Gamma(z+1) = (2z-1)\zeta(2z)$. The change of variables $r = (1-b)/b$ carries the beta density times $db/(2(1-b))$ to $\frac{\sin \pi z}{2\pi} r^{-z-1} dr$, and the expansion in powers of $1-\mathcal{B}$ is Maślanka's series, proved rigorously in [11]. □

So Maślanka's coefficients determine the law of W and give ξ with no zeros and no primes, but ξ enters through the Mellin transform of W, and for the reason in the third point they transfer no positivity to the Thorin measure of the clock.

Chapter 4

Infinite divisibility, GGC, EGGC and the Bondesson hierarchy

4.1 Generalised gamma convolutions and the Thorin condition

The Thorin condition, equivalent to RH by Proposition 2.1.1, is that the clock $1/G$ be the Laplace transform of a generalised gamma convolution.

4.1.1 The definition and its three readings

As in Section 2.1, a random variable $H \geq 0$, or its law, is a generalised gamma convolution (GGC) if

$$\mathbb{E}e^{-uH} = \exp\Big\{ -du - \int_{(0,\infty)} \log\Big(1 + \frac{u}{z}\Big) U(dz) \Big\}, \qquad u > 0, \tag{4.1.1}$$

with drift $d \geq 0$ and a positive Radon measure U on $(0,\infty)$, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1} U(dz) < \infty$ [344, 345], [39, p. 29]. The drift and the Thorin measure are unique [39, p. 30]. Since $\log(1 + u/z)$ is the Laplace exponent of the exponential law with rate z, a discrete $U = \sum_j c_j \delta_{z_j}$ makes H the sum of d and independent gamma variables with shapes c_j and rates z_j; the class is the closure of such sums under weak limits (Thorin's closure theorem, Section 9.1.1).

The first reading is through the Lévy measure. Frullani's integral $\log(1 + u/z) = \int_0^\infty (1 - e^{-ut}) e^{-zt} t^{-1} dt$ turns (4.1.1) into

$$-\log \mathbb{E}e^{-uH} = du + \int_0^\infty \big(1 - e^{-ut}\big) \frac{k(t)}{t}\, dt, \qquad k(t) = \int e^{-zt}\, U(dz),$$

so H is infinitely divisible with Lévy density $k(t)/t$ and completely monotone canonical function k. Conversely, by Bernstein's theorem, an infinitely divisible law on $[0,\infty)$ with drift d and Lévy density $k(t)/t$, k completely monotone, is a

GGC whose Thorin measure is the measure representing k [168, Section 1.2]. For the clock, the unconditional formula for $\log G$ in Section 2.1 gives $k = W$, so the Thorin condition is the complete monotonicity of the heat trace.

The second reading is through the Stieltjes transform. Differentiating the logarithm of (4.1.1) gives

$$f_H(u) = -\frac{d}{du}\log \mathbb{E}e^{-uH} = d + \int \frac{U(dz)}{u+z},$$

a nonnegative constant plus the Stieltjes transform of a positive measure, holomorphic on $\mathbb{C}\setminus(-\infty,0]$. At distance $\eta>0$ below the cut, $\pi^{-1}\operatorname{Im} f_H(-x-i\eta) = \int P_\eta(x-z)\,U(dz)$ is the Poisson smoothing of U, with P_η the Poisson kernel of Section 2.1, and letting $\eta\downarrow 0$ recovers U by Stieltjes inversion: for continuity points $a<b$ of U,

$$U\big((a,b]\big) = \lim_{\eta\downarrow 0}\frac{1}{\pi}\int_a^b \operatorname{Im} f_H(-x-i\eta)\,dx. \tag{4.1.2}$$

For the clock, $G'/G = -\frac{d}{du}\log(1/G)$ takes the place of f_H whether or not RH holds. The chain rule on $G(s^2) = \xi(\frac12+s)/\xi(\frac12)$ gives $(\xi'/\xi)(\frac12+s) = 2s\,(G'/G)(s^2)$. At $s=\varepsilon+i\theta$ we have $\bar s^2 = -x-i\eta$ with $x=\theta^2-\varepsilon^2$ and $\eta = 2\varepsilon\theta$, and since G is real on the real axis, (2.1.7) becomes

$$\rho_\varepsilon(\theta) = 2\theta\cdot\frac{1}{\pi}\operatorname{Im}\frac{G'}{G}(-x-i\eta) + \frac{2\varepsilon}{\pi}\operatorname{Re}\frac{G'}{G}(s^2).$$

The first term is the integrand of (4.1.2) at distance $\eta = 2\varepsilon\theta$ from the cut times the Jacobian $dx/d\theta = 2\theta$; the second carries the factor ε. The Thorin density at resolution ε is thus a regularised Stieltjes inversion along the parabola $u=(\varepsilon-i\theta)^2$ rather than a horizontal line.

Under RH the logarithm gives the counting function directly: for $y>0$ not of the form γ^2, $U_{\mathrm{RH}}((0,y]) = -\pi^{-1}\operatorname{Im}\log G(-y-i0) = N(\sqrt{y})$, with $\log G$ continued from $(0,\infty)$ through the lower half plane, since the factor $1+u/\gamma^2$ of G has argument $-\pi$ at $u=-y-i0$ when $\gamma^2<y$ and 0 otherwise.

The third reading is as a product: for discrete $U=\sum_j c_j\delta_{z_j}$, (4.1.1) is $e^{-du}\prod_j(1+u/z_j)^{-c_j}$. Conversely, if F is entire of genus zero with $F(0)=1$ and all its zeros real and negative, then $1/F$ is the Laplace transform of a GGC with $d=0$ and a unit atom at $-u_j$ for each zero u_j of F, counted with multiplicity. For the clock $F=G$, the product is (2.1.1), and the Thorin condition says that every a_k is positive.

4.1.2 The Bondesson hierarchy and the clock

Between the infinitely divisible laws on $[0,\infty)$ (ID) and the GGCs lie two classes defined by conditions on the Lévy density $k(t)/t$: the self decomposable laws (SD),

for which k is nonincreasing, and the Bondesson class (BO), for which $k(t)/t$ is completely monotone. The GGCs are the laws for which k itself is completely monotone, and inside them lie the laws with a hyperbolically completely monotone density (HCM, defined below):

$$\text{HCM} \subset \text{GGC} \subset \text{SD} \subset \text{ID}, \qquad \text{GGC} \subset \text{BO} \subset \text{ID},$$

and SD and BO are not nested. The Lévy density $e^{-t}(1+\frac{1}{2t})$ is completely monotone, but $k(t) = e^{-t}(t+\frac{1}{2})$ increases on $(0, \frac{1}{2})$, so the law is in BO and not in SD [279, Section 2]; the Lévy density $t^{-1}\mathbf{1}_{t<1}$ gives a law in SD and not in BO, since a completely monotone function that vanishes on $(1,\infty)$ vanishes identically.

A positive continuous function φ on $(0,\infty)$ is hyperbolically completely monotone if for every $u > 0$ the function $\varphi(uv)\varphi(u/v)$ is completely monotone in $w = v + v^{-1}$, and a law is HCM if it has an HCM density. By Bondesson's theorem [39, Theorem 6.1.1], a function φ with $\varphi(0+) = 1$ is the Laplace transform of a GGC if and only if it is HCM, so the Thorin condition becomes a family of sign conditions on real derivatives. An HCM law is a GGC, so the HCM property of a law is stronger than that of its Laplace transform, strictly: the GGC of Theorem 9.3.20 has an HCM Laplace transform and a density that is not HCM.

The extended generalised gamma convolutions (EGGC), introduced by Thorin [346], are the weak limits of convolutions of gamma laws and their reflections [39, Section 7.1]. A symmetric law is an EGGC if and only if it is the law of $Z\sqrt{2H}$ with Z standard normal and H an independent GGC [39, Theorem 7.3.1]. Since $\mathbb{E}e^{i\theta Z\sqrt{2T}} = \mathbb{E}e^{-\theta^2 T}$ determines the law of T, the variable $Z\sqrt{2T}$ of the van Dantzig pair of Section 2.1 is an EGGC if and only if T is a GGC, that is, by Proposition 2.1.1, if and only if RH holds. With no hypothesis $Z\sqrt{2T}$ is self decomposable: subordinating the Lévy measure $W(v)\,dv/v$ of T to the heat kernel gives the Lévy density $k(|x|)/|x|$ with $k(r) = \int_0^\infty e^{-1/4v}(4\pi v)^{-1/2}W(r^2 v)\,dv/v = \sum_j e^{-r\sqrt{a_j}}$ (the first passage identity for each term, with the principal root), and k decreases because W does (Theorem 8.1.1). The z distributions of Barndorff-Nielsen, Kent and Sørensen [19] are the lattice analogue of this family of variance mixtures.

Table 4.1 records the status of the sign conditions on W that the classes become for the clock; Section 1.4 explains why no finite number of derivative conditions reaches the last row.

Urbanik's classes of multiply self decomposable laws give conditions on W of another kind: with L_0 the class of self decomposable laws and L_n that of the self decomposable laws whose driving process has its law in L_{n-1}, the clock lies in L_n exactly when $(-t\frac{d}{dt})^j W \geq 0$ for $j \leq n+1$ (Section 14.4.4), and for $n \geq 1$ this does not follow from the GGC property, since the exponential law is a GGC whose canonical function e^{-t} has $(-t\frac{d}{dt})^2 e^{-t} = -t(1-t)e^{-t} < 0$ for $t < 1$. For the clock these conditions hold for $n = 2$ and fail for $n = 3$, with no hypothesis (Theorem 14.4.8), while membership of every L_n would imply RH, since $\bigcap_n L_n$

class or property	condition on W	status for the clock
ID	$W \geq 0$	proved, with the zeros verified to height 1000 (Theorem 8.1.1)
SD	W nonincreasing	proved, with the zeros verified to height 1000 (Theorem 8.1.1)
strict log convexity	$WW'' > W'^2$ and $W'W''' > W''^2$	proved, with the zeros verified to height 1000 (Section 8.1)
derivative signs to order 1.16×10^{25}	$(-1)^n W^{(n)} > 0$ for $n \leq 1.16 \times 10^{25}$	proved, with the zeros verified to height 3×10^{12} (Theorem 8.1.13)
complete monotonicity near the origin	$(-1)^n W^{(n)}(t) > 0$ for every n and $t \leq 1.29$	proved, with the zeros verified to height 10^{12} (Theorem 8.1.9)
BO	$W(t)/t$ completely monotone	equivalent to RH (Section 15.1.4); open
GGC	W completely monotone	equivalent to RH; open

Table 4.1: The Bondesson hierarchy on the clock, whose Lévy density is $W(t)/t$.

lies inside the GGCs [29].

4.1.3 Diagnostics for the Thorin condition

RH can be stated as a GGC identity on a real interval. The right side of (4.1.1) extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$ (Exercise 1.14), so RH holds if and only if there is a GGC H with $\xi(\frac{1}{2})/\xi(\frac{1}{2}+s) = \mathbb{E}e^{-s^2 H}$ for real $0 < s < 1$ alone [279, Proposition 3.3].

As it stands the identity is useless to a computation, because analytic continuation from an interval is unstable. A unit Thorin atom at z_0 multiplies the transform by $(1 + u/z_0)^{-1}$, which differs from one by less than $1/z_0$ on $(0, 1)$, so transforms that agree to high accuracy on $(0, 1)$ can have very different Thorin measures at large z. Riesz's criterion meets the same obstruction: its numbers $1/\zeta(2k)$ are, up to explicit factors, values of the clock at the real points $(2k - \frac{1}{2})^2$, and they bear on the zeros only through the analytic interpolation of $k \mapsto 1/\zeta(2k)$ into the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$ (Section 13.2.6). The information about the zeros enters through the analyticity the representation forces, and a computation must supply it otherwise: through values of ξ and ξ' at complex arguments (Chapter 2) or through sign conditions on derivatives of every order. Each characterisation of Sections 4.1.1 and 4.1.2 yields such a test; Table 4.2 lists them with what a failure proves.

The diagnostics differ in how strongly they weight the higher zeros. Those built from W see a zero at height γ through $e^{-\gamma^2 t}$: in $(-1)^n W^{(n)}(t) = \sum_j a_j^n e^{-a_j t}$

diagnostic	computes	a failure proves
pole test	poles of G'/G off $(-\infty, 0]$	not GGC; for the clock, a zero off the line
Thorin density at resolution ε	$\rho_\varepsilon = \pi^{-1}\operatorname{Re}(\xi'/\xi)$ at $\frac{1}{2}+\varepsilon+i\theta$	a zero with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ (Proposition 2.1.3, [198])
zero density scan	$M_\varepsilon(T)+\tau_\varepsilon(T)$ against $N(T)$ (Proposition 3.3.2)	$\frac{1}{2}[N(T)-M_\varepsilon(T)-\tau_\varepsilon(T)]$ zeros with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ below height T
heat trace signs	$(-1)^n W^{(n)}(t)$	not GGC; finitely many orders are consistent with off line zeros (Section 1.4)
Hankel blocks	$\det[(-1)^{i+j}W^{(i+j)}(t)]$	not GGC (Weil's criterion at one scale, Section 8.1)
HCM test	$(-1)^n\partial_w^n[\varphi(uv)\varphi(u/v)]$	not GGC (Section 9.1)
Bernstein test	$(-1)^{n-1}L^{(n)}$, $L=\log G$	not ID, which Theorem 8.1.1 excludes for the clock
Stieltjes moments	Hankel matrices $[m_{i+j}]$ and $[m_{i+j+1}]$, $m_n=\sum_k a_k^{-n-1}$	not GGC

Table 4.2: Diagnostics for the Thorin condition. The pole test, the Thorin density and the zero density scan locate a failure in height; the others detect it only globally.

it enters with weight $(\gamma/\gamma_1)^{2n}e^{-(\gamma^2-\gamma_1^2)t}$ relative to the first zero. These tests certify easily and discriminate poorly. The Bernstein test on $L(u)=\log G(u)$, computed for $u>\frac{1}{4}$, where the Dirichlet series of $\log\zeta(\frac{1}{2}+\sqrt{u})$ converges, is better conditioned [282, Section 18.2.2]: since $(-1)^{n-1}L^{(n)}(u)=(n-1)!\sum_k(u+a_k)^{-n}$, a zero at height γ enters with relative weight of order $(\gamma_1/\gamma)^{2n}$. It too sees the higher zeros only through subdominant terms, and for the clock it cannot fail: by the formula for $\log G$ in Section 2.1, $(-1)^{n-1}L^{(n)}(u)=\int_0^\infty t^{n-1}e^{-ut}W(t)\,dt$, which is positive because $W>0$ (Theorem 8.1.1). Read at $u=0$ through (2.1.4), which gives $\log\mathbb{E}e^{hX}=\log G(h^2)$, the same formula makes the even cumulants of the Pólya variable X alternate in sign with no hypothesis: $(-1)^{n-1}\kappa_{2n}(X)=\frac{(2n)!}{n!}\int_0^\infty t^{n-1}W(t)\,dt>0$ for every $n\geq 1$. Under RH this is the sign rule that Newman proved for laws of Lee Yang type [241]. Complete monotonicity of L' is weaker than the Thorin condition, which asks that L' be the Stieltjes transform $\int U(dz)/(u+z)$ of a positive measure U.

The Thorin density and the zero density scan built from it are local in the ordinate without a search for zeros: by Proposition 2.1.3, a zero at height γ contributes about $1/(\pi\varepsilon)$ to ρ_ε at $\theta=\gamma$ and about $\varepsilon/(\pi d^2)$ at distance $d\gg\varepsilon$.

4.1.4 Explicit Thorin measures and the sinh benchmark

Table 4.3 lists GGCs with explicit Thorin measures, in the variable u of the clock: models for the clock, test cases for code, and building blocks of the Vere-Jones representation of Section 10.1 (Theorem 10.1.8 and Table 10.1).

Laplace transform	Thorin measure	law
$(1+u/z_0)^{-a}$	$a\,\delta_{z_0}$	Gamma(a, z_0)
$\pi\sqrt{u}/\sinh(\pi\sqrt{u})$	$\sum_{k\geq 1}\delta_{k^2}$	$2K^2$, K Kolmogorov
$(\sqrt{u}/\sinh\sqrt{u})^2$	$2\sum_{k\geq 1}\delta_{\pi^2k^2}$	$S_2/2$ (Williams)
$\operatorname{sech}\sqrt{u}$	$\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$	exit time of $\sqrt{2}B$ from $(-1,1)$
$(\operatorname{sech}\sqrt{u})^{1/2}$	$\frac{1}{2}\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$	Meixner time
$e^{-c\sqrt{u}}$	$c\,(2\pi\sqrt{z})^{-1}dz$	positive stable, index $\frac{1}{2}$
$(1+\sqrt{u}/c)^{-1}$	$c\,\{2\pi\sqrt{z}(z+c^2)\}^{-1}dz$	E^2S/c^2
$e^{x(\alpha-\sqrt{\alpha^2+2u})}$	$x\{\pi\sqrt{2z-\alpha^2}\}^{-1}\mathbf{1}_{z>\alpha^2/2}dz$	inverse Gaussian
$\exp\{1-x\sqrt{u}-e^{-x\sqrt{u}}\}$	$x\sin^2(x\sqrt{z}/2)(\pi\sqrt{z})^{-1}dz$	one compensated jump, by (2.1.9)
$\mathcal{W}(u)/u$	image of db/π on $(0,\pi)$ under $r(b)=b\,e^{-b\cot b}/\sin b$	Lambert law (Theorem 14.4.3)
$K_0(2\pi)/K_{\sqrt{u}}(2\pi)$	$\sum_{n\geq 1}\delta_{\nu_n^2}$, ν_n the zeros of $\nu\mapsto K_{i\nu}(2\pi)$	hitting time of a diffusion (Theorem 18.1.10)
$\xi(\alpha)/\xi(\alpha+\sqrt{u})$, $\alpha\geq 1$	$v_\alpha(\sqrt{z})(2\pi\sqrt{z})^{-1}dz$	shifted clock (Theorem 2.1.2)
$\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$	$\sum_{\gamma>0}\delta_{\gamma^2}$ if and only if RH holds	the clock (Proposition 2.1.1)

Table 4.3: GGC Laplace transforms in the variable u, their Thorin measures in z, and the corresponding laws.

In the table, K is the Kolmogorov variable; S_2 is as in Section 2.1; B is a standard Brownian motion; the Meixner time H has $\mathbb{E}e^{-uH}=(\operatorname{sech}\sqrt{u})^{1/2}$, so that $Z\sqrt{2H}$, with Z standard normal and independent of H, has the Meixner law of Section 10.2, whose time in Proposition 10.2.10 is $2H$; and S is the positive stable variable with $\mathbb{E}e^{-uS}=e^{-\sqrt{u}}$, independent of the standard exponential variable E. Rows two to five are the classical hyperbolic laws [33]. At $u=-t^2$ the reciprocals of rows two and four become $\sin(\pi t)/(\pi t)$ and $\cos t$, the characteristic functions of the uniform law on $(-\pi,\pi)$ and of a symmetric sign. Like Ξ both have infinitely many real zeros: real zeros do not obstruct the Thorin property, they are required by it. For the clock, unlike these models, the positivity of the points a_k is the conclusion to be proved [279, Section 10].

The sinh clock $1/G_0$, with $G_0(u)=\sinh(\pi\sqrt{u})/(\pi\sqrt{u})=\prod_{k\geq 1}(1+u/k^2)$, is the second row of the table. It has the structure of the clock under RH, a GGC with unit Thorin atoms, but its ordinates are the integers and no arithmetic enters. Its

Thorin density at resolution ε, the density in $\theta = \sqrt{z}$ of a measure $\mu_\varepsilon^{(0)}$, is

$$\rho_\varepsilon^{(0)}(\theta) = \frac{1}{\pi} \operatorname{Re}\Big[\pi \coth\big(\pi(\varepsilon + i\theta)\big) - \frac{1}{\varepsilon + i\theta}\Big] = \sum_{k \neq 0} P_\varepsilon(\theta - k),$$

exactly the Poisson smoothing of the atoms and their reflections, so the matched total variation of Section 3.1 vanishes identically. As for ξ (Section 3.1), the cumulative mass is π^{-1} times the change in the argument of $\sinh(\pi s)/(\pi s)$ along $s = \varepsilon + i\theta$, and at a half integer $\Theta = m + \frac{1}{2}$ the continuous argument of $\sinh(\pi(\varepsilon + i\theta))$ has grown from 0 to $\pi\Theta$ and that of $\varepsilon + i\theta$ to $\frac{\pi}{2} - \arctan(\varepsilon/\Theta)$, so that

$$\mu_\varepsilon^{(0)}\big((0, \Theta^2]\big) = m + \frac{1}{\pi} \arctan \frac{\varepsilon}{\Theta}.$$

The mass defect $\mu_\varepsilon^{(0)}((0, \Theta^2]) - m$ is therefore $\varepsilon/(\pi\Theta) + O(\varepsilon^3)$. On the window $0 < \theta \leq 50.5$, which holds 50 atoms, the defect is $0.00630\,\varepsilon$ to first order and 3.151×10^{-3} at $\varepsilon = 0.5$.

The cell distance, the sum over cells C of $|\mu_\varepsilon^{(0)}(C) - 1|$ as in Section 3.1, is also explicit. We use the cells $(k - \frac{1}{2}, k + \frac{1}{2}]$, $k = 1, \ldots, 50$, and add the mass of the strip $(0, \frac{1}{2}]$, which lies in no cell, as unmatched. For the full lattice $\mathbb{Z}$ each cell would have mass exactly one. The sinh clock lacks the atom at 0, so each of its cells is short by the mass that P_ε puts there, and the strip carries $\pi^{-1} \arctan 2\varepsilon$; Section 8.3 computes these masses. Summing,

$$d_{\mathrm{cell}}^{(0)}(\varepsilon) = \frac{1}{\pi}\Big[2 \arctan 2\varepsilon - \arctan \frac{\varepsilon}{50.5}\Big] = \frac{4 - 1/50.5}{\pi}\,\varepsilon + O(\varepsilon^3),$$

which is 0.4968, 0.1250, 0.0631, 0.01267 and 0.00633 at $\varepsilon = 0.5$, 0.1, 0.05, 0.01 and 0.005, and $1.267\,\varepsilon$ in the limit. All of it is due to the atom missing at 0.

For the 79 zeros below height 200 the constant is 5.44 (Section 3.1), with a different origin: the zeros are irregularly spaced, so the mass leaking out of a cell is no longer replaced by mass leaking in from its neighbours, and every cell contributes; Section 8.3 reads the constant as a statistic of the spacings. Figure 4.1 divides both distances by the number of atoms, which gives $1.267\,\varepsilon/50 = 0.0253\,\varepsilon$ for the sinh clock and $5.44\,\varepsilon/79 = 0.069\,\varepsilon$ for the zeros. Their ratio, nearly three, depends on the window, since for the lattice the distance per atom falls like $1/N$ as the window grows to N atoms. Williams' Thorin measure $2\sum_k \delta_{\pi^2 k^2}$ of Table 4.3 needs no separate benchmark: it is the same lattice rescaled by π, with atoms of mass two, so its Thorin density at resolution ε is $\frac{2}{\pi}\rho_{\varepsilon/\pi}^{(0)}(\theta/\pi)$ and its distances follow by scaling from the closed forms for the sinh clock.

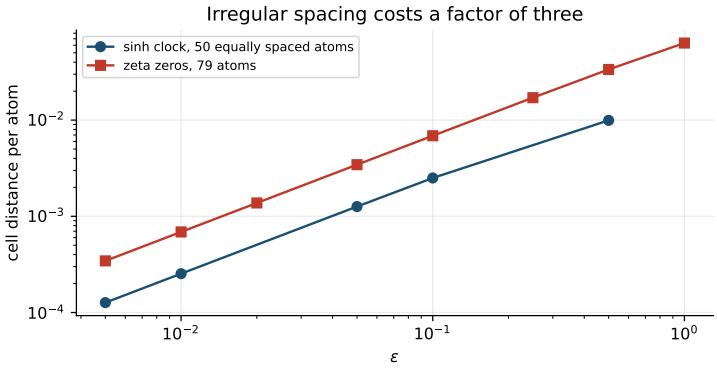


Figure 4.1: Cell distance divided by the number of atoms in the window, for the sinh clock (50 atoms, cells $(k-\frac{1}{2}, k+\frac{1}{2}]$) and for the 79 zeros below height 200 (Voronoi cells of Section 3.1). Both vanish linearly in ε. For the sinh clock the undivided distance stays below $4\varepsilon/\pi$ whatever the number of atoms, so the gap between the curves depends on the window, and the factor of three in the title of the figure holds for these two windows only.

Chapter 5

The reciprocal Ξ function

The reciprocal of the normalised ξ function is the Laplace transform of a probability law, the law of the clock T of Section 2.1. Under the Riemann Hypothesis that law is the hitting time of a Krein string and the explosion time of a pure birth chain, and its density is Kent's eigenfunction expansion over the zeros. We ask whether the argument can be run backwards, so that RH follows from the limiting behaviour of hitting times, and show where the content of such an argument must lie. We test approximating laws built with no assumption on the zeros (Jensen laws, Gram smoothed zeros, truncated Euler products, boundaries of Brownian motion), prove a universal stable limit for the prime side hitting times and a rigidity theorem for factorisations of the clock law, and study the density of the clock directly: its residue expansion, its log concavity and the failure of hyperbolic complete monotonicity. No step proves RH.

The material is based on [141, 151, 150]; its numbers were recomputed for this book, and results no independent computation reached are marked unchecked. Throughout, $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ as in Section 2.1, the clock T has law μ and Laplace transform $1/G$, the canonical function is the heat trace W (Chapter 8), and the de Bruijn and Newman kernel is Φ_{dBN} of Section 4.1, not the kernel Φ of (2.1.3). Results proved elsewhere are cited, not restated: the self decomposability of μ and the equivalence of the GGC property with RH (Theorem 8.1.1, Proposition 2.1.1), the spectral expansion and the Kent approximants (Section 5.2), and the closure of the GGC class under weak limits (Corollary 9.1.2).

5.1 Further readings of the sine squared lemma

We follow the sine squared lemma (2.1.9) in two directions.

5.1.1 The prime side near the critical line

For $\alpha > 1$ the integrated lemma is the argument of ζ. In the chart $\theta = \sqrt{z}$ a unit of $xL(dx)$ at x contributes $\pi^{-1}(1-\cos\theta x)$ to the density, and $\pi^{-1}\int_0^{\Theta}(1-\cos\theta x)\,d\theta =$

$\pi^{-1}(\Theta - \sin(\Theta x)/x)$, so the prime atoms $\Lambda(n)n^{-\alpha}$ at $x = \log n$ return a term linear in Θ plus π^{-1} times

$$-\sum_{n\ge 2} \frac{\Lambda(n)}{n^{\alpha}} \frac{\sin(\Theta \log n)}{\log n} = \arg \zeta(\alpha + i\Theta),$$

on the branch with $\arg \zeta(\alpha) = 0$, since $\log \zeta(s) = \sum_n \Lambda(n) n^{-s}/\log n$ for $\operatorname{Re} s > 1$. Divided by π, this is $S_{\alpha-1/2}(\Theta)$ of Section 3.3.1. The series converges absolutely for $\alpha > 1$ and diverges for $\alpha < 1$, so the function S of Turing's method (Section 2.2.1), at $\alpha = \frac{1}{2}$, is reached only through a smoothing.

Selberg [330] truncates the Dirichlet series of ζ'/ζ smoothly and accounts for the truncation exactly by a sum over the zeros [348, Section 14.20]. With $\Lambda_x(n) = \Lambda(n)$ for $n \le x$ and $\Lambda_x(n) = \Lambda(n)\log(x^2/n)/\log x$ for $x < n \le x^2$, and for $x > 1$ and s different from 1, from every zero ρ and from every $-2q$,

$$\begin{aligned} \frac{\zeta'}{\zeta}(s) = &- \sum_{n\le x^2} \frac{\Lambda_x(n)}{n^s} + \frac{x^{2(1-s)} - x^{1-s}}{(1-s)^2 \log x} \\ &+ \frac{1}{\log x} \sum_{q\ge 1} \frac{x^{-2q-s} - x^{-2(2q+s)}}{(2q+s)^2} + \frac{1}{\log x} \sum_{\rho} \frac{x^{\rho-s} - x^{2(\rho-s)}}{(s-\rho)^2}. \end{aligned} \tag{5.1.1}$$

At $s = \alpha + i\theta$ with $\alpha = \frac{1}{2} + \varepsilon$, a zero $\rho = \beta + i\gamma$ with $\beta < \alpha$ contributes to the last sum a term of modulus about $x^{\beta-\alpha}/(|s-\rho|^2 \log x)$. For a line zero at its own ordinate the term is $(x^{-\varepsilon} - x^{-2\varepsilon})/(\varepsilon^2 \log x)$, a fraction $x^{-\varepsilon}(1 - x^{-\varepsilon})/(\varepsilon \log x)$ of the contribution $1/\varepsilon$ of that zero to v_α; we call this the single zero law. The same computation bounds the whole sum: if no zero has real part greater than $\frac{1}{2}$, then $|x^{\rho-s}| = x^{-\varepsilon}$ and $|x^{2(\rho-s)}| = x^{-2\varepsilon}$ for every zero, and every zero contributes $\operatorname{Re}(1/(s-\rho)) = \varepsilon/|s-\rho|^2$ to $v_\alpha(\theta)$, so that $\sum_\rho |s-\rho|^{-2} = v_\alpha(\theta)/\varepsilon$ by Proposition 2.1.3. Hence the bracket v_α computed from Selberg's smoothed prime sum has error at most

$$\frac{x^{-\varepsilon} + x^{-2\varepsilon}}{\varepsilon \log x}\, v_\alpha(\theta),$$

which is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$ at a peak, where v_α is about $1/\varepsilon$, and k correct digits at an ordinate need $x^{\varepsilon}\, \varepsilon \log x \approx 10^k$, that is

$$x \approx 10^{k/\varepsilon}$$

to leading order, with primes up to x^2. A zero with $\frac{1}{2} < \beta < \alpha$ gives a term that decays like $x^{\beta-\alpha}$, more slowly than $x^{-\varepsilon}$, and one with $\beta > \alpha$ a term that grows like $x^{2(\beta-\alpha)}$, since then $x^{2(\rho-s)}$ dominates. With $B = \sup_\rho \operatorname{Re}\rho$ the error is $O(x^{B-\alpha}/\log x)$ when $B < \alpha$, and when $B > \alpha$ it reaches $x^{2(B-\alpha)+o(1)}$ along a sequence of cutoffs, so that the prime side diverges as $x \to \infty$ if a zero lies to the right of the line $\operatorname{Re} s = \alpha$. Selberg's abscissa $\sigma_{x,t} = \frac{1}{2} + 2\max(\beta - \frac{1}{2}, 2/\log x)$, the

maximum over the zeros near height t, accounts for this; when none of these zeros lies right of $\frac12+2/\log x$ it equals $\frac12+4/\log x$, where $x^{-\varepsilon}=e^{-4}$ [330]. Section 12.1 meets the same dichotomy without smoothing.

The terms of the zeros right of the line $\operatorname{Re} s=\alpha$ cannot cancel one another. Suppose that $B>\alpha$, write $E_x(s)$ for the last term of (5.1.1), the sum over the zeros divided by $\log x$, and put $y=\log x$, $\lambda_\rho=\rho-s$ and $R(y)=y\operatorname{Re}E_{e^y}(s)$. Bounding each term by its modulus gives $|R(y)|\le\sum_\rho|s-\rho|^{-2}\bigl(e^{(B-\alpha)y}+e^{2(B-\alpha)y}\bigr)$, so for $\operatorname{Re}w>2(B-\alpha)$ the Laplace transform $\int_0^\infty e^{-wy}R(y)\,dy$ may be taken term by term, and it is

$$\frac12\sum_\rho\Bigl[\frac{1}{\lambda_\rho^2}\Bigl(\frac{1}{w-\lambda_\rho}-\frac{1}{w-2\lambda_\rho}\Bigr)+\frac{1}{\bar\lambda_\rho^2}\Bigl(\frac{1}{w-\bar\lambda_\rho}-\frac{1}{w-2\bar\lambda_\rho}\Bigr)\Bigr].$$

The series converges locally uniformly away from its poles, since $\sum_\rho|\lambda_\rho|^{-2}<\infty$ and $|\lambda_\rho|\to\infty$, and so defines a meromorphic function in the whole plane. A zero ρ_0 with $\operatorname{Re}\rho_0>(B+\alpha)/2$ gives a pole at $2\lambda_{\rho_0}$, to the right of every pole λ_ρ and $\bar\lambda_\rho$; each fraction with a pole there contributes $-\frac12\lambda_{\rho_0}^{-2}$ to the residue, which is therefore not zero. If $R(y)=O(e^{\eta y})$, the Laplace integral converges on $\operatorname{Re}w>\eta$ and continues the series holomorphically there, so $\eta\ge2(\operatorname{Re}\rho_0-\alpha)$. Letting $\operatorname{Re}\rho_0$ tend to B, and using the bound on $|R|$, gives $\limsup_{x\to\infty}\log|\operatorname{Re}E_x(s)|/\log x=2(B-\alpha)$.

The cutoff interpolates between the zeros and the primes. For $x<\sqrt2$ the prime sum is empty, and as $x\downarrow1$ each remaining term tends to its partial fraction, $1/(1-s)$, $1/(s+2q)$ or $1/(s-\rho)$. The limits cannot be taken inside the sums separately: the sum over q grows like $\frac12\log(1/\log x)$ and the real part of the sum over the zeros falls like $-\frac12\log(1/\log x)$, and only their total, $\zeta'/\zeta(s)-(x^{2(1-s)}-x^{1-s})/((1-s)^2\log x)$, has a limit, $\zeta'/\zeta(s)+1/(s-1)$. Adding the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ turns this limit into $\xi'/\xi(s)$, whose real part divided by π is the density $\rho_\varepsilon(\theta)$, the sum of the kernels of Proposition 2.1.3.

Figure 5.1 tests the formula at $\varepsilon=0.25$ ($\alpha=\frac34$) on 501 points of $10\le\theta\le60$, against the Pólya values of $v_{3/4}$. With the primes alone the maximal relative error is 66%, 20% and 7% for $x=30$, 300 and 3000, attained where $v_{3/4}$ is small, and the maximal absolute errors 1.14, 0.53 and 0.23, attained at peaks over zeros, follow the single zero law, which predicts 1.15, 0.51 and 0.23; the relative errors stay below the bound $(x^{-\varepsilon}+x^{-2\varepsilon})/(\varepsilon\log x)$, which is 0.72, 0.21 and 0.077. Both errors decrease faster than $x^{-\varepsilon}$. Adding the sum over the first 1700 zeros reduces the error below 1.6×10^{-7}, the residual being the zeros above height 2197. At $\varepsilon=0.005$ ten digits would need $x\approx10^{2000}$, about e^{4600}, to leading order (about e^{4000} by the single zero law), while the cost of one Pólya evaluation does not depend on ε (Section 1.3.2). The sine squared lemma identifies the Thorin density, but near the line the primes compute it only at a cost exponential in $1/\varepsilon$.

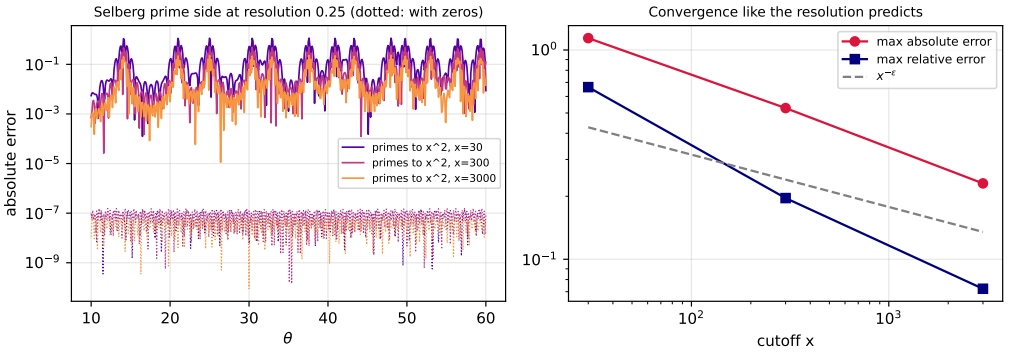


Figure 5.1: Left: absolute error of Selberg's smoothed prime side of $v_{3/4}$, at resolution 0.25, for three cutoffs, and (dotted) with the sum over the first 1700 zeros added. Right: the maximal absolute and relative errors against the cutoff, with $x^{-\varepsilon}$ for comparison.

5.1.2 Positivity: Brownian subordination and Weil's criterion

The lemma cannot produce a Brownian first passage time to a random level from a positive input. Brownian motion with variance two per unit time first reaches a level $x > 0$ at a time with Laplace transform $e^{-x\sqrt{u}}$, so its first passage time T_Y to an independent level $Y \geq 0$ has $\mathbb{E}e^{-uT_Y} = \mathbb{E}e^{-Y\sqrt{u}}$. The exponent $\log f(\alpha+s)/f(\alpha) = cs + \int (e^{-sx} + sx - 1)L(dx)$ to which the lemma applies (Section 2.1) would have to equal $-\log \mathbb{E}e^{-sY}$, which is concave in s by Hölder's inequality, whereas $L \geq 0$, $L \neq 0$ makes it strictly convex, with second derivative $\int x^2 e^{-sx} L(dx) > 0$. With $c \geq 0$ the lemma gives a GGC, but not a first passage time. Conversely, an infinitely divisible level with $\mathbb{E}e^{-sY} = \exp(-bs - \int (1-e^{-sx})\Pi(dx))$ enters the lemma with $c = b + \int x\,\Pi(dx)$ and the negative input $L = -\Pi$, and T_Y receives the signed density $(2\pi\sqrt{z})^{-1}[b + \int x\cos(x\sqrt{z})\,\Pi(dx)]$. For a Poisson level ($\Pi = \delta_1$, $b = 0$) this is $\cos(\sqrt{z})/(2\pi\sqrt{z})$, and T_Y is not a GGC: it has an atom e^{-1} at 0. A positive input and a random level coexist only for pure drift, with transform $e^{-c\sqrt{u}}$ and Thorin density $c/(2\pi\sqrt{z})$.

The reciprocal direction does work: for $Y \geq 0$ infinitely divisible with finite mean m, $T_m \stackrel{d}{=} T_Y + R$, with T_m the first passage time to the level m and R a GGC independent of T_Y, which for the clock is Proposition 12.1.7 and for a Barnes beta law is Proposition 12.1.9.

For ξ there is no random level at any basepoint. By Stirling's formula and the polynomial growth of ζ in vertical strips, $|\xi(\alpha)/\xi(\alpha+it)|$ is at least $e^{\pi|t|/4}$ divided by a power of $|t|$, while $|\mathbb{E}e^{-itY}| \leq 1$; so $s \mapsto \xi(\alpha)/\xi(\alpha+s)$ is never a Laplace transform in s, although for $\alpha \geq 1$ it is one in $u = s^2$ (Theorem 2.1.2). For $\alpha > 1$ the input of ξ is signed, and the positivity of U_α is created by the zeros through the Poisson kernels of Proposition 2.1.3, not inherited from a positive

input (Section 1.3.3); compare [282, Remark 3.8]. At $\alpha = \frac{1}{2}$ the obstruction is the drift zero case of Proposition 3.1.1, whose Brownian motion has variance one, and the curved boundaries of Section 3.1 take the place of the straight boundary with a random intercept. The obstruction concerns first passage alone (Section 12.2).

The same kernel puts Weil's criterion in variogram form. Section 8.2 defines the Thorin functional U on the class $\mathcal{T}$ of functions $\varphi(z) = h(\sqrt{z})$ with h even, holomorphic on a strip $|\operatorname{Im} r| \le \frac{1}{2} + c$ for some $c > 0$, and $O((1+|r|)^{-2-c})$ there. With $g(x) = \frac{1}{2\pi} \int h(r) e^{-irx}\, dr$,

$$\int_0^\infty \varphi(z)\, \frac{2\sin^2(x\sqrt{z}/2)}{2\pi\sqrt{z}}\, dz = g(0) - g(x), \tag{5.1.2}$$

both sides being $\pi^{-1} \int_0^\infty h(r)(1-\cos xr)\, dr$. If $h \ge 0$ on the real line, g is positive definite and $g(0) - g(x) \ge 0$ is a variogram. Weil's criterion [357] states that RH holds if and only if $\langle U, \varphi \rangle \ge 0$ for every such φ, and by the explicit formula U evaluates φ at every zero. Formally, the lemma writes $\langle U, \varphi \rangle$ as the pairing of variograms with the canonical measure of (1.3.1) taken at the centre, the measure ν of Section 8.2, whose archimedean density and prime atoms are nonnegative and whose only negative input is the pole density $-e^{x/2}dx$. So RH says that on every variogram of a positive definite function the pole mass is dominated by the prime and archimedean mass. Both masses are infinite, and the absolutely convergent form of Section 8.2 makes the statement exact, the divergent constants cancelling through $\xi'(\frac{1}{2}) = 0$ (Section 1.3.3).

The linear programming formulation of [282, Sections 18.2.8 and 18.2.9] works on this form, with the variograms as feasible directions: in a finite dimensional space of test functions the regularised domination is a feasibility problem with dual certificates, which Section 25.2.4 proposes to solve at increasing dimension. Weil's criterion concerns U, which sees every zero, whereas the limit $\alpha \downarrow \frac{1}{2}$ of the Thorin measures is U_{line}, which pairs nonnegatively with every such φ whatever the zeros do (Section 8.2).

5.2 The clock as a random variable

The clock variable T of Section 2.1, with $\mathbb{E}e^{-uT} = 1/G(u)$, exists unconditionally and is self decomposable (Theorem 8.1.1). Under RH it is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables e_γ (Proposition 2.1.1), and if the zeros are also simple, the theorems of Kent and Krein make it the hitting time of a level by a diffusion whose Dirichlet spectrum is $\{\gamma^2\}$ [184], [282, Section 16.3].

5.2.1 The law of the clock and its decay rate

The Pólya route gives $\mathbb{E}e^{-uT} = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ at every complex u other than its poles, untilted for $|\operatorname{Im}\sqrt{u}| < 120$ and tilted above (Section 2.1.3). Talbot's

contour inversion [341] in `mpmath` [237] then gives the density f of T and its survival function $\bar F(t) = P(T > t)$ with no reference to the zeros.

Under RH, with simple zeros, the density also has the spectral expansion [184]

$$f(t) = \sum_k R_k e^{-\gamma_k^2 t}, \qquad P(T>t) = \sum_k \frac{R_k}{\gamma_k^2} e^{-\gamma_k^2 t}, \qquad R_k = \frac{2i\gamma_k\,\xi(\frac12)}{\xi'(\frac12 + i\gamma_k)},$$

whose coefficients are the residues of the transform at its poles $u = -\gamma_k^2$. The first step of the proof of Theorem 9.3.22 derives it by a contour shift, with $R_k = -2\gamma_k \Xi(0)/\Xi'(\gamma_k)$ in the role of c_n there; the Pólya route supplies the ordinates, as Newton roots, and the values of ξ' at them. The residues alternate in sign (Exercise 18.1) and grow like $e^{\pi\gamma_k/4}$ up to powers of γ_k and the factor $1/|\zeta'(\rho_k)|$, because Stirling's formula puts the factor $e^{-\pi\gamma_k/4}$ into $|\Xi'(\gamma_k)| = |\xi'(\rho_k)|$; thus $R_1 = 1.0164 \times 10^4$ and $R_2 = -1.1775 \times 10^6$. The series converges for $t > 0$ but cancels heavily at small t, where Talbot's contour is the better tool (Exercise 18.2). Computed with 50 digits from the 79 zeros below 200, the spectral expansion agrees with the Talbot values to 24 digits for $0.010 \le t \le 0.12$. Beyond that range the Talbot values lose relative accuracy as $\bar F(t)$ falls, down to six digits at $t = 0.3$, and the spectral expansion is the accurate one (Figure 5.2).

The primes give a second expansion of f, with no hypothesis. Write $\xi = \eta\zeta$ with the gamma factor $\eta(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, and put $\phi_\eta(u) = \xi(\frac12)/\eta(\frac12 + \sqrt{u})$.

Proposition 5.2.1. *Let $c > \frac14$, let g_1 be the inverse Laplace transform of ϕ_η along $\operatorname{Re} u = c$, and let $q_\lambda(x) = \frac{\lambda}{2\sqrt{\pi}} x^{-3/2} e^{-\lambda^2/4x}$ for $x > 0$, both extended by 0 to $x \le 0$. Then for every $t > 0$*

$$f(t) = \sum_{n\ge1} \frac{\mu(n)}{\sqrt{n}}\, g_n(t), \qquad g_n = g_1 * q_{\log n}, \qquad |g_n(t)| \le \sup_{0<x\le t} |g_1(x)|\ \operatorname{erfc}\Big(\frac{\log n}{2\sqrt{t}}\Big),$$

so the series converges absolutely, faster than any power of n. Moreover $g_1(t) = 2\xi(\frac12)e^{t/4} + O(e^{\varepsilon t})$ as $t \to \infty$ for every $\varepsilon > 0$.

Proof. On $\operatorname{Re} u = c$ we have $\operatorname{Re}\sqrt{u} \ge \sqrt{c} > \frac12$, so the series (10.1.1) for $1/\zeta(\frac12 + \sqrt{u})$ converges absolutely there, with the sum of the absolute values of its terms at most $\zeta(\frac12 + \sqrt{c})$. By Stirling's formula $|\phi_\eta(u)|$ decays faster than any power of $|u|$ on vertical lines in $\operatorname{Re} u > 0$, uniformly on compact ranges of $\operatorname{Re} u$; hence $|g_1(x)| \le C_c e^{cx}$, the Laplace transform of g_1 is ϕ_η on $\operatorname{Re} u > c$, and f, whose transform $\phi_\eta(u)/\zeta(\frac12 + \sqrt{u})$ is integrable on the line, is its Bromwich integral along $\operatorname{Re} u = c$. Dominated convergence inverts the series term by term. The term of n is $\phi_\eta(u)e^{-\sqrt{u}\log n}$, and $e^{-\lambda\sqrt{u}}$ is the transform of q_λ by (10.1.2), which gives the convolution; the bound follows from $\int_0^t q_\lambda = \operatorname{erfc}(\lambda/2\sqrt{t})$. Finally ϕ_η is analytic off $(-\infty, 0]$ except for a simple pole at $u = \frac14$, where $\eta(\frac12 + \sqrt{u}) = \frac12(u - \frac14) + O((u-\frac14)^2)$, so the residue is $2\xi(\frac12)$; moving the contour to $\operatorname{Re} u = \varepsilon$ gives the asymptotic of g_1. □

So g_1 is not a density: it grows like $2\xi(\frac{1}{2})e^{t/4}$. The same contour shift gives $g_n(t) = 2\xi(\frac{1}{2})n^{-1/2}e^{t/4} + O_n(e^{\varepsilon t})$, so for fixed N the partial sum over $n \le N$ is $2\xi(\frac{1}{2})e^{t/4}\sum_{n\le N}\mu(n)/n + O_N(e^{\varepsilon t})$ and grows, while f decays like $e^{-\gamma_1^2 t}$. For large t the series rests on the cancellation $\sum_n \mu(n)/n = 0$, the prime number theorem; for small t it converges very fast. Computed by Talbot inversion at 30 digits, the terms with $n \le 5$ ($n \le 7$ at $t = 0.04$) agree with the Talbot value of f to at least 20 digits at $t = 0.008, 0.012, 0.016, 0.02, 0.03, 0.04$. Relative to f, the term of the prime 2 is 1×10^{-26} at $t = 0.008$, 3×10^{-9} at $t = 0.02$ and 2.8 percent at $t = 0.04$. The two expansions are complementary: for relative accuracy 10^{-8} the spectral expansion needs 50 zeros at $t = 0.008$, 10 at $t = 0.02$ and 2 at $t = 0.06$ (among $1, 2, 5, 10, 20, 50, 120$ zeros), while the Möbius series needs only $n = 1$ up to $t = 0.02$, and $n \le 3$ at $t = 0.04$.

The law is unimodal with no hypothesis (Corollary 14.1.2) and sharply concentrated. Its mean and variance are given in Section 2.1, and its median is 0.0220. At the other end, with no hypothesis, $P(T \le y) \le \exp\{-(1 + o(1))(\log\frac{1}{y})^2/16y\}$ as $y \downarrow 0$, so the density vanishes to infinite order at the origin: Chernoff's bound $P(T \le y) \le e^{uy}/G(u)$, with $\log G(u) \sim \frac{\sqrt{u}}{4}\log u$ by Stirling's formula for ξ, is optimised at $\sqrt{u} \approx \log(1/y)/4y$.

Proposition 5.2.2. *RH holds if and only if the density f of T is a Pólya frequency function of infinite order, that is* $\det[f(x_i - y_j)]_{i,j\le r} \ge 0$ *for every r and all $x_1 < \cdots < x_r$, $y_1 < \cdots < y_r$.*

Proof. Under RH, f is the density of $\sum_\gamma e_\gamma/\gamma^2$ (Proposition 2.1.1), exponential densities are Pólya frequency functions, and the class is closed under convolution and weak limits. Conversely, by Schoenberg's theorem [328] the bilateral Laplace transform of an integrable Pólya frequency function of infinite order is $1/\Psi$ with $\Psi(u) = Ce^{-cu^2+bu}\prod_k(1 + \lambda_k u)e^{-\lambda_k u}$, $c \ge 0$, λ_k real. Here the transform is $1/G$, so $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ has only real zeros. Its zeros are $u = -a_k$ with $a_k = -(\rho - \frac{1}{2})^2$, which is real only if $\rho - \frac{1}{2}$ is real or purely imaginary, and ξ has no real zeros, so every ρ is on the critical line. □

This is the one sided form of Proposition 22.5.1. Under RH with simple zeros, assuming that the spectral expansion converges absolutely for $t > 0$ and may be differentiated termwise, the opposite signs of R_1 and R_2 make the upper tail of f log concave: $f(t) = R_1 e^{-\gamma_1^2 t}(1 - \varepsilon(t)) + O(e^{-\gamma_3^2 t})$ with $\varepsilon = |R_2/R_1|e^{-\Delta t}$ and $\Delta = \gamma_2^2 - \gamma_1^2$, so $(\log f)''(t) = -\Delta^2\varepsilon/(1 - \varepsilon)^2 + O(e^{-(\gamma_3^2-\gamma_1^2)t}) < 0$ for large t. On 91 logarithmically spaced points of $[0.006, 0.08]$, which carry all but 6×10^{-6} of the mass, a floating point Talbot computation at 25 digits (no interval arithmetic) gives $(\log f)'' < 0$ with maximum -0.026, the first two orders of hyperbolic complete monotonicity of f. These are necessary conditions for Proposition 5.2.2 tested in the bulk, not a proof.

The hazard $h = f/\bar{F}$ tends to the principal eigenvalue γ_1^2. Under RH with

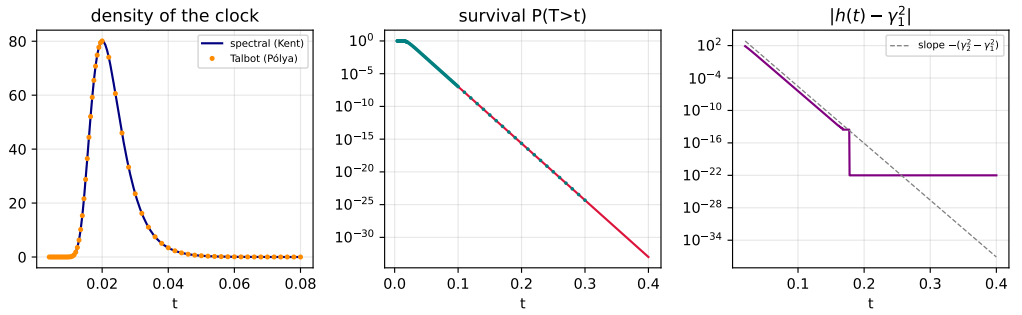


Figure 5.2: The law of the clock variable computed from the Pólya density by Talbot inversion (points) and by the spectral expansion with Pólya residues (lines): density, survival function, and the distance of the hazard from γ_1^2, whose logarithm has slope $-(\gamma_2^2-\gamma_1^2)$ (dashed). Beyond $t\approx 0.17$ the last curve reflects the double precision of the plot, not the hazard.

simple zeros $h(t)-\gamma_1^2=O(e^{-(\gamma_2^2-\gamma_1^2)t})$ (Exercise 18.3). This is the quasi stationary behaviour of a hitting time; Roberts [316, Section 2] obtains approximations of Brownian boundary hitting laws from approximations of the conditioned diffusion. Talbot inversion shows the limit with no zero as input: $h(0.05)=199.72$, $h(0.1)=199.7904544$ and $h(0.25)=199.79045483$, the last equal to γ_1^2 to the eleven digits shown. The rate of approach is the spectral gap: a fit of $\log|h(t)-\gamma_1^2|$ on $[0.05,0.12]$ has slope -242.12, against $-(\gamma_2^2-\gamma_1^2)=-242.14$.

The clock also generates a natural exponential family in the sense of Barndorff-Nielsen. The cumulant generating function $K(\theta)=\log\mathbb{E}e^{\theta T}=\log\{\xi(\frac12)/\xi(\frac12+\sqrt{-\theta})\}$ is finite exactly for $\theta<\gamma_1^2$ and $K'(\theta)\to\infty$ as $\theta\uparrow\gamma_1^2$, so the family is regular and steep with natural parameter space $(-\infty,\gamma_1^2)$, whose boundary is the first zero; under RH the tilted laws are $\sum_\gamma e_\gamma/(\gamma^2-\theta)$. The saddlepoint approximation of Barndorff-Nielsen and Cox [18], $\hat f(y)=\exp\{K(\hat\theta)-\hat\theta y\}/\sqrt{2\pi K''(\hat\theta)}$ with $K'(\hat\theta)=y$, needs ξ only on the real axis $s\ge\frac12$ (for $y<\mathbb{E}T$) and on the segment of the critical line from $\frac12$ to $\frac12+i\gamma_1$ (for $y>\mathbb{E}T$), so it never passes the first zero. It is not normalised (it integrates to 0.98492) and agrees with the Talbot density to about 0.5 percent in the lower tail and within 2.5 percent on $[0.011,0.035]$ (Table 5.1); in the upper tail the ratio tends to $e/\sqrt{2\pi}=1.0844$, the error of the approximation for an exponential density, because the tail is $R_1e^{-\gamma_1^2y}$. As $y\downarrow0$ the exponent $K(\hat\theta)-\hat\theta y$ is $-(1+o(1))(\log\frac1y)^2/16y$, the Chernoff exponent above, but slowly: at $y=0.006$ the leading term gives -272 against -33.0 for the exact $\log f$, while the saddlepoint, which keeps all of ξ, is within 0.5 percent of f. The tail formula of Lugannani and Rice [221], built from the same $\hat\theta$, $K(\hat\theta)$ and $K''(\hat\theta)$, is within 1.4 percent of the Talbot values of $P(T\le y)$ and within 2.4 percent of those of $P(T>y)$ at the sampled y.

y	$\hat{\theta}$	$f(y)$ (Talbot)	$\hat{f}(y)$	ratio
0.0060	-15909.8	5.165×10^{-15}	5.138×10^{-15}	0.9949
0.0120	-1674.1	1.7018	1.6953	0.9962
0.0201	-114.5	80.1084	78.5287	0.9803
0.0239	19.4	61.4522	59.9432	0.9754
0.0401	148.8	3.3491	3.4507	1.0303
0.0800	182.8	0.0012	0.0013	1.0762

Table 5.1: The saddlepoint density of the clock against the Talbot inversion of its Laplace transform, both from ξ alone.

terms	γ_1^2	γ_2^2	γ_3^2	γ_4^2
2	199.86679	427.28		
3	199.79031	442.129	618.93	
4	199.790455	441.9255	625.63	907.24
true	199.7904548	441.9261506	625.5429969	925.6730873

Table 5.2: Prony extraction of the atoms from $P(T > t)$ on $t \in [0.03, 0.108]$. Identical results for relative noise 10^{-16}, 10^{-30} and 10^{-45}: the error is model truncation, not noise.

5.2.2 The cost of the zeros in time and in samples

We fit sums of p exponentials to $P(T > t)$ by Prony's method, from 40 values at $t = 0.03, 0.032, \ldots, 0.108$ carrying relative noise from 10^{-16} down to 10^{-45} (Table 5.2). The error of the fitted atoms comes from the neglected terms, of relative size $e^{-(\gamma_{p+1}^2 - \gamma_1^2)t_0}$ on a window starting at t_0, up to a factor set by the residues. Resolving the kth atom therefore needs data accurate to about $(\gamma_k^2 - \gamma_1^2)t_0 / \log 10$ digits, unbounded in k.

Sampling meets the same cancellation in another currency. By the identity $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Section 2.1, with $S_2 = \sum_{k \geq 1} \frac{2}{\pi^2 k^2} \Gamma_k$ for independent Gamma$(2,1)$ variables Γ_k, the variable $Y = \frac{1}{2}\log(\pi S_2/2)$ satisfies

$$2\,\Xi(z) = \mathbb{E}\big[e^{Y/2}\cos(zY)\big].$$

Since $\mathbb{E}e^Y = 2\xi(1) = 1$, the variable has finite variance and a sample mean estimates $\Xi(z)$ at every z. Table 5.3 compares the estimates from $n = 10^6$ copies of S_2 with $\Xi(z)$, a check of the identity on the critical line; Table 12.1 checks it at real s (Section 12.1.1).

The standard error settles at the standard deviation of $e^{Y/2}\cos(zY)/2$ over $\sqrt{n}$, while $|\Xi(z)|$ decays like $e^{-\pi z/4}$, so the sample size needed to resolve the sign of Ξ grows like $e^{\pi z/2}$. Even the first sign change, at $z = 14.13$, needs about 10^6 samples: between the first two zeros $|\Xi|$ is at most 8.0×10^{-4} (at $z = 15.59$), and

z	Monte Carlo	standard error	$\Xi(z)$	n with standard error $\lvert\Xi(z)\rvert$
0	0.497061	5.4×10^{-5}	0.497121	1.2×10^{-2}
2	0.453136	7.3×10^{-5}	0.453100	2.6×10^{-2}
5	0.275888	2.4×10^{-4}	0.275550	0.77
8	0.103756	3.4×10^{-4}	0.103303	11
10	0.038401	3.5×10^{-4}	0.037968	86
20	0.000314	3.5×10^{-4}	-3.67×10^{-5}	9.3×10^{7}
30	0.000289	3.5×10^{-4}	-1.50×10^{-8}	5.5×10^{14}

Table 5.3: Monte Carlo estimates of $\Xi(z)$ from Williams' representation with 10^6 samples. The last column is the sample size at which the standard error equals $\lvert\Xi(z)\rvert$.

resolving it at two standard errors takes about 8×10^5 samples. Sampling is the right tool for the law of the Pólya variable, its moments and the Bayesian inverse problems of [282, Section 19.2], and the wrong tool for the zeros: Devroye's exact samplers for theta type laws [76] make each sample cheap, but do not change the growth $e^{\pi z/2}$.

5.2.3 Kent approximants with a Weyl tail

In the rest of this section w is the variable u of the clock. Kent's theorem [184] identifies the first hitting time of a level by a diffusion started at a reflecting or entrance end of its interval with the GGC $\prod_n (1 + w/\lambda_n)^{-1}$, where the λ_n are the Dirichlet eigenvalues of the interval up to that level. Krein's inverse spectral theory [78] builds a string, and so a generalised diffusion, from any increasing sequence $\lambda_n > 0$ with $\sum 1/\lambda_n < \infty$. The GGCs that arise as such hitting times are therefore those without drift whose Thorin measure consists of unit atoms at distinct points. By Proposition 2.1.1 the clock is one of them exactly when RH holds and the zeros are simple, and its diffusion then has Dirichlet spectrum $\{\gamma^2\}$.

The approximants of the clock that Kent's theorem suggests are the transforms $1/G_N(w)$ of the partial products $G_N(w) = \prod_{k \le N}(1 + w/\gamma_k^2)$, each the transform of the hitting time of a finite string with N point masses. They are admissible in the closure criterion of Corollary 9.1.2, are the approximating sequence, under RH, in the proof of [282, Proposition 18.10], and converge slowly. With $\tau_1(N) = \sum_{k>N} \gamma_k^{-2}$, the relative error $G(w)/G_N(w) - 1$ of the clock transform lies, under RH, between $w\tau_1(N)$ and $e^{w\tau_1(N)} - 1$.

We keep the verified zeros as atoms and replace the others by their average density. By the Riemann von Mangoldt formula $N(t) = \vartheta(t)/\pi + 1 + S(t)$ of Section 2.2, the zeros above the cutoff Θ_N have density $\vartheta'(\gamma)/\pi \approx \frac{1}{2\pi} \log \frac{\gamma}{2\pi}$,

which is positive for $\gamma > 6.3$. So

$$G_N^{\mathrm{W}}(w) = \prod_{k \le N} \Big(1 + \frac{w}{\gamma_k^2}\Big) \exp\Big\{ \int_{\Theta_N}^{\infty} \log\Big(1 + \frac{w}{\gamma^2}\Big) \frac{\vartheta'(\gamma)}{\pi}\, d\gamma \Big\},$$

with W for Weyl, has reciprocal equal to the Laplace transform of a GGC with N unit atoms and the continuous Thorin density $\vartheta'(\sqrt{z})/(2\pi\sqrt{z})$ on $z > \Theta_N^2$. For Θ_N we take the Gram point g_{N-1} of Section 2.2.1, where $\vartheta(\Theta_N)/\pi + 1 = N$; for the values of N used here it lies between γ_N and γ_{N+1}, so $S(\Theta_N) = 0$. Table 5.4 compares the two approximants with the clock computed from ξ.

N	$w = 1$	$w = 100$	$w = 1000$	$w = 10^4$
1	1.8×10^{-2} / 1.3×10^{-4}	4.8 / 1.0×10^{-2}	$\gg 1$ / 3.3×10^{-2}	$\gg 1$ / 4.3×10^{-2}
20	7.2×10^{-3} / 4.3×10^{-7}	1.04 / 4.2×10^{-5}	$\gg 1$ / 3.7×10^{-4}	$\gg 1$ / 1.5×10^{-3}
79	3.6×10^{-3} / 1.5×10^{-7}	0.43 / 1.5×10^{-5}	34 / 1.5×10^{-4}	$\gg 1$ / 1.2×10^{-3}
300	1.6×10^{-3} / 7.4×10^{-9}	0.17 / 7.4×10^{-7}	3.95 / 7.4×10^{-6}	$\gg 1$ / 7.2×10^{-5}
1700	5.0×10^{-4} / 2.9×10^{-11}	0.051 / 2.9×10^{-9}	0.64 / 2.9×10^{-8}	142 / 2.9×10^{-7}

Table 5.4: Relative error of the clock transform: partial product over N zeros, $|G/G_N - 1|$, and with the Weyl tail, $|G_N^{\mathrm{W}}/G - 1|$. The two measures of error agree to first order, and $\gg 1$ marks an error above 10^3. The partial product values agree with [282, Section 18.2.1] where both are given.

At the four points $w = 1, 100, 1000, 10^4$ the table exhibits an explicit positive Thorin measure whose Thorin transform $\int \log(1 + w/z)\, U(dz)$ matches $\log G(w)$ to within 3×10^{-7}. This is a finite instance of the weak Thorin condition of [282, Section 18.2.3], which asks for such a measure, to within any $\eta > 0$, at every finite set of points, and which is equivalent to RH by the closure criterion of Corollary 9.1.2. A finite instance proves nothing: agreement for $w \le 10^4$ cannot detect off line zeros above the verified height, which change $\log G$ there by far less than the error. The error of the Weyl tail grows linearly in w, and to first order it is a weighted average of S beyond the cutoff,

$$\log G(w) - \log G_N^{\mathrm{W}}(w) \approx w\big(\tau_1(N) - \tau_1^{\mathrm{W}}(N)\big) = 2w \int_{\Theta_N}^{\infty} \frac{S(\gamma)}{\gamma^3}\, d\gamma,$$

where $\tau_1^{\mathrm{W}}(N) = \int_{\Theta_N}^{\infty} \gamma^{-2} \vartheta'(\gamma) \pi^{-1} d\gamma$ is the Weyl estimate of $\tau_1(N)$ and the identity is an integration by parts that uses $S(\Theta_N) = 0$.

5.2.4 Moment matching and the error at large w

The error of the Weyl tail comes from the first order term of $\log(1 + w/\gamma^2) = w/\gamma^2 - w^2/(2\gamma^4) + \cdots$. For w small compared with Θ_N^2 the error of $\log G_N^{\mathrm{W}}$ is approximately w times $\tau_1(N) - \tau_1^{\mathrm{W}}(N)$, and the second order term is $-w^2/2$ times the corresponding difference for $\tau_2(N) = \sum_{k>N} \gamma_k^{-4}$. Both true tail sums

are available without any zero beyond γ_N, because the power sums $p_m = \sum_k a_k^{-m}$, with $a_k = -(\rho_k - \frac12)^2$ as in Section 2.1 and equal to $\sum_{\gamma>0} \gamma^{-2m}$ under RH, are Taylor coefficients of ξ at the centre: if $\xi(\frac12 + s)/\xi(\frac12) = 1 + c_2 s^2 + c_4 s^4 + \cdots$, then

$$p_1 = \sum_k a_k^{-1} = c_2 = 0.023104993115418970 7889,$$
$$p_2 = \sum_k a_k^{-2} = c_2^2 - 2c_4 = 0.000037172599285269 6862,$$

and $\tau_m(N) = p_m - \sum_{k\le N} \gamma_k^{-2m}$. Here $p_1 = \mathbb{E}T$ and $p_2 = \operatorname{Var} T$. We replace the Weyl density $\vartheta'(\gamma)/\pi$ on $[\Theta_N, \infty)$ by $\lambda\,\vartheta'(\gamma)/\pi$ on $[\Theta'_N, \infty)$, and choose the two parameters λ and Θ'_N so that the tail reproduces $\tau_1(N)$ and $\tau_2(N)$ exactly. The tail density stays positive, so the approximant is still a GGC, and the adjustments are small: for $N = 79$ the cutoff moves from 199.649 to 199.675 and $\lambda = 1.000059$.

N	tail	$w=1$	$w=10^2$	$w=10^4$	$w=10^5$	$w=10^6$	$w=10^7$
79	Weyl	1.5×10^{-7}	1.5×10^{-5}	1.2×10^{-3}	4.3×10^{-3}	5.8×10^{-3}	6.0×10^{-3}
	matched	2.6×10^{-17}	2.6×10^{-11}	1.8×10^{-5}	3.1×10^{-3}	6.1×10^{-2}	0.47
300	Weyl	7.4×10^{-9}	7.4×10^{-7}	7.2×10^{-5}	5.5×10^{-4}	1.7×10^{-3}	2.1×10^{-3}
	matched	2.3×10^{-20}	2.3×10^{-14}	2.2×10^{-8}	1.5×10^{-5}	1.8×10^{-3}	2.8×10^{-2}
1700	Weyl	2.9×10^{-11}	2.9×10^{-9}	2.9×10^{-7}	2.8×10^{-6}	2.4×10^{-5}	9.3×10^{-5}
	matched	3.5×10^{-23}	1.1×10^{-17}	7.0×10^{-13}	3.6×10^{-10}	2.5×10^{-7}	5.0×10^{-5}

Table 5.5: Relative error $|G_{\mathrm{a}}/G - 1|$ of the approximant G_{a} of G with the Weyl tail and with the moment matched tail. The cutoffs are $\Theta_N^2 = 3.99\times10^4$, 2.95×10^5 and 4.83×10^6 for $N = 79$, 300 and 1700.

Table 5.5 shows two regimes. For w below Θ_N^2 moment matching gains ten to twelve digits at $w = 1$ and about two as w approaches Θ_N^2. Well beyond Θ_N^2 the gain disappears, and at $w = 10^7$ with $N = 79$ the matched tail is worse than the plain Weyl tail. For $w \gg \Theta_N^2$ the matched tail differs from the Weyl tail mainly by $(\lambda-1)\int_{\Theta'_N}^{\infty} \log(1 + w/\gamma^2)\,\vartheta'(\gamma)\pi^{-1} d\gamma$, which grows like $(\lambda-1)\frac{\sqrt{w}}{2}\log\frac{\sqrt{w}}{2\pi}$, while the error of the Weyl tail tends to a constant.

Under RH, since $S(\Theta_N) = 0$ and no ordinate lies in $(\gamma_N, \Theta_N]$, the true tail sum minus its Weyl estimate is a Stieltjes integral against S, and an integration by parts gives

$$\log G_N^{\mathrm{W}}(w) - \log G(w) = -\int_{\Theta_N}^{\infty} \log\Big(1 + \frac{w}{\gamma^2}\Big)\, dS(\gamma) = -\int_{\Theta_N}^{\infty} \frac{2w\,S(\gamma)}{\gamma(\gamma^2 + w)}\, d\gamma$$
$$\longrightarrow\ -2\int_{\Theta_N}^{\infty} \frac{S(\gamma)}{\gamma}\, d\gamma \qquad (w \to \infty).$$

The limit is $-\ell_N$, with $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\, d\gamma$ as in Exercise 18.5. It is finite because $\int_0^t S(u)\, du = O(\log t)$ (Littlewood; Section 2.2.1), and in general it is not

zero. The measured errors of the Weyl tail level off at about 6×10^{-3} for $N = 79$ and 2×10^{-3} for $N = 300$.

The constant is no obstruction of principle. It depends only on $\gamma_1, \dots, \gamma_N$ and on ξ along the real axis, and Stirling's formula gives ℓ_N in closed form (Exercise 18.5). It is also a floor only for the Weyl density. Adding to ϑ'/π a continuous ω with compact support in (Θ_N, ∞) and $\int \omega = 0$ changes the limit by $-2\int \omega(\gamma) \log \gamma \, d\gamma$, and two wide bumps of opposite sign placed far apart give this any value while the density stays positive, so a suitable positive tail removes the constant.

5.3 The law and its density

Let $u_\rho = (\rho - \frac{1}{2})^2$ for each zero $\rho = \frac{1}{2} + \delta + i\gamma$ of ζ in the upper half plane, so that $G(u) = \prod_{\gamma>0}(1 - u/u_\rho)$, and under RH $u_\rho = -\gamma^2$. With no hypothesis $1/G$ is the Laplace transform of the self decomposable law μ of the clock T, with canonical function $W(x) = \sum_{\gamma>0} e^{u_\rho x}$; μ is a GGC if and only if RH holds, and then $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Theorem 8.1.1, Proposition 2.1.1). The mean of T is $p_1 = \sum_{\gamma>0} \gamma^{-2} = 0.0231049931$.

The spectral expansion of the density f of μ is in Section 5.2.1; here its residues are written $w_\gamma = 1/G'(-\gamma^2) = R_k$, and the point is what holds with no hypothesis on the zeros.

5.3.1 The partial fraction series diverges, the density series converges

If the zeros are simple, $1/G$ has poles u_ρ with residues $1/G'(u_\rho)$, and the Mittag Leffler expansion $1/G(u) = \sum_\rho 1/(G'(u_\rho)(u - u_\rho))$ does not converge. Under RH, $G'(-\gamma^2) = \xi'(\frac{1}{2} + i\gamma)/(2i\gamma\,\xi(\frac{1}{2}))$, and the gamma factor of ξ gives $|w_\gamma| \asymp \gamma^{-3/4} e^{\pi\gamma/4}/|\zeta'(\rho)|$, so the weights alternate in sign and are enormous: $w_{\gamma_1} = 1.0164 \times 10^4$, $w_{\gamma_2} = -1.1775 \times 10^6$, $w_{\gamma_5} = 8.0078 \times 10^9$, and $w_{\gamma_{120}} = -3.909 \times 10^{89}$ at $\gamma_{120} = 269.970$, where $\log|w_{\gamma_{120}}| = 206.3$ against $\pi\gamma/4 = 212.0$ (computed, code/ht_residues.py and code/ht_resid.py). Neither $\sum w_\gamma/(u + \gamma^2)$ nor $\sum w_\gamma/\gamma^2$ converges. Multiplying by e^{uy} before summing cures this, since $e^{-\gamma^2 y}$ beats $e^{\pi\gamma/4}$ for every $y > 0$.

Theorem 5.3.1 (Grouped residue expansion). *There are radii $R_n \to \infty$ with $R_n \in [n, 2n]$ such that for every $y > 0$*

$$f(y) = \lim_{n\to\infty} \sum_{|u_\rho|<R_n} \operatorname*{Res}_{u=u_\rho} \frac{e^{uy}}{G(u)}, \tag{5.3.1}$$

with no hypothesis on the zeros. A simple zero contributes $e^{u_\rho y}/G'(u_\rho)$, a zero of multiplicity m contributes $e^{u_\rho y} P_\rho(y)$ with P_ρ a polynomial of degree $m - 1$, and

a pair of zeros $\frac12 \pm \delta + i\gamma$ *off the line has conjugate values* u_ρ *and contributes* $2\operatorname{Re}\big(e^{u_\rho y}/G'(u_\rho)\big)$. *For* y *in a compact subset of* $(0,\infty)$ *the error after the* n*th group is at most* $R_n \exp(-\frac12 R_n y + C\sqrt{R_n}\log R_n) + \exp(-c\sqrt{R_n}\log R_n)$ *for constants* $c, C > 0$.

Sketch. Write $f(y) = (2\pi i)^{-1}\int_{1-i\infty}^{1+i\infty} e^{uy} G(u)^{-1}\,\mathrm{d}u$, which converges absolutely because $\log|G(1+it)|$ grows like $c\sqrt{|t|}\log|t|$ with $c>0$. Close the contour by an arc of the circle $|u| = R$. The function G has order one half and $\log\max_{|u|=r}|G| \sim \frac14\sqrt{r}\log r$, so the minimum modulus theorem for functions of finite order [207] produces $R_n \in [n, 2n]$ with $\log|G(u)| \ge -C\sqrt{R_n}\log R_n$ on the whole circle $|u| = R_n$. On the part of the arc in $\operatorname{Re} u \le -R_n/2$ the factor $|e^{uy}| \le e^{-R_n y/2}$ dominates. On the rest of the arc $\sqrt{u}$ has argument at most $\pi/3$ and $\operatorname{Re}(\frac12 + \sqrt{u}) \ge \frac12\sqrt{R_n}$, so $\zeta(\frac12 + \sqrt{u})$ is bounded away from zero, Stirling's formula gives $\log|G(u)| \ge c\sqrt{R_n}\log R_n$, and $|e^{uy}| \le e^{y}$. Both contributions tend to zero, and the residue theorem gives (5.3.1). □

So the expansion holds for every $y > 0$, multiple and off line zeros included, provided the residues are summed in the order of $|u_\rho|$ and grouped between the radii R_n. Only absolute convergence needs a hypothesis.

Theorem 5.3.2 (Residue expansion). *Assume that the zeros of* ζ *are simple and that* $|\xi'(\rho)| \ge e^{-C|\gamma|}$ *for some constant* C *and all zeros. Then the series*

$$f(y) = \sum_{\gamma>0} \frac{e^{u_\rho y}}{G'(u_\rho)}, \qquad y > 0, \tag{5.3.2}$$

converges absolutely and locally uniformly. Under RH it reads $f(y) = \sum_{\gamma>0} w_\gamma e^{-\gamma^2 y}$ *with* $w_\gamma = 1/G'(-\gamma^2) = \gamma^2 \prod_{\gamma'\ne\gamma}(1 - \gamma^2/\gamma'^2)^{-1}$.

Proof. The hypothesis gives $|1/G'(u_\rho)| \le 2|\gamma||\xi(\frac12)|e^{C|\gamma|}$, against $|e^{u_\rho y}| \le e^{-(\gamma^2 - 1/4)y}$. The product form of w_γ is the derivative of the Hadamard product at $-\gamma^2$. □

The hypothesis, a lower bound for $|\zeta'(\rho)|$ times the gamma factor, is expected to hold with room to spare (Gonek's conjectures predict polynomial bounds for $1/|\zeta'(\rho)|$ on the line) but is not known.

The weights of (5.3.2) are those of the classical density of a sum of independent exponentials with rates λ_k, so under RH Theorem 5.3.2 writes out the density of $\sum_\gamma e_\gamma/\gamma^2$. The series cannot be integrated term by term, since $\sum w_\gamma/\gamma^2$ diverges: the mass $\int f = 1$ and the mean $\int yf = p_1$ are properties of the sum, not of the terms. With the first 120 zeros the series reproduces the density at $y \ge 0.008$ to the digits of Table 5.6, its integral over $[0.008, \infty)$ is 0.9999999995 (so 5×10^{-10} of the mass lies below 0.008, where the series is not used), and its first moment over $[0.008, \infty)$ is 0.0231049931114 against $p_1 = 0.0231049931154$, although the

zeros used have $\sum \gamma^{-2} = 0.0203057$: the residues carry the whole function G, not just the zeros summed (computed, code/ht_resid.py). The positivity of f, unconditional because μ is a probability law, is a massive cancellation: near $y = 0.008$ the largest single term exceeds $f(y)$ by a factor of about $e^{28.5}$ (the term -7.83×10^6 at $\gamma = 49.77$ against $f(0.008) = 3.42 \times 10^{-6}$, computed).

y	0.010	0.015	0.020	0.025	0.030	0.040	0.050	0.060	0.100
$f(y)$	0.0244	28.765	80.104	53.175	23.433	3.4137	0.4660	0.0632	2.14×10^{-5}

Table 5.6: The density of μ from the residue expansion with the first 120 zeros (computed, 110 digits, agreeing with 140 digits to 10^{-103}).

Proposition 5.3.3 (Tail of the density). *Let $\gamma_1 < \gamma_2 < \gamma_3$ be the first three ordinates and $c_3 = \frac{1}{2}(\gamma_2^2 + \gamma_3^2)$. With no hypothesis,*

$$f(y) = w_1 e^{-\gamma_1^2 y} + w_2 e^{-\gamma_2^2 y} + O\big(e^{-c_3 y}\big), \qquad y \to \infty,$$

with the same bound for each derivative, where $w_1 = 1.01635 \times 10^4 > 0$ and $w_2 = -1.17754 \times 10^6 < 0$.

Proof. Shift the Bromwich line to $\operatorname{Re} u = -c_3$. Only the poles $-\gamma_1^2$ and $-\gamma_2^2$ are crossed: a zero with u_ρ to the right of $-c_3$ has $\gamma^2 - \delta^2 < c_3$, so $\gamma < 23.2$, and every such zero is known to lie on the line and to be simple. On $\operatorname{Re} u = -c_3$ the function $|G|$ is bounded below and grows like $e^{c\sqrt{|t|}\log|t|}$, so the shifted integral is $O(e^{-c_3 y})$, and differentiating under the integral gives the derivatives. □

5.3.2 Density and canonical function

The density (5.3.1) and the canonical function $W(x) = \sum e^{u_\rho x}$ are sums over the same exponentials, with weights $1/G'(u_\rho)$ and 1, tied by Steutel's equation for infinitely divisible laws on $[0, \infty)$,

$$y\, f(y) = \int_0^y f(y - x) W(x)\, \mathrm{d}x, \tag{5.3.3}$$

which is the Laplace transform of the identity $-\varphi' = \varphi\, G'/G$ for $\varphi = 1/G$, with $G'/G = \int e^{-ux} W(x)\, \mathrm{d}x$. The GGC property is a statement about W (complete monotonicity, whose weights are all one); the residue expansion moves the question to f, whose weights alternate.

5.4 Kent's formula

Kent [184] showed that the first passage time of a generalised diffusion, equivalently of a Krein string, from its entrance point to a level is a sum $\sum_n e_n/\lambda_n$ of

independent exponentials, where the λ_n are the Dirichlet eigenvalues of the string, and that its density has the eigenfunction expansion

$$f(t) = \sum_n \lambda_n \prod_{m \neq n} \frac{\lambda_m}{\lambda_m - \lambda_n} e^{-\lambda_n t}. \tag{5.4.1}$$

With $G(u) = \prod(1 + u/\lambda_m)$ the weight is $1/G'(-\lambda_n)$, so (5.4.1) is the residue expansion. The theorem is used in Section 5.2.3 and Corollary 12.2.3.

Proposition 5.4.1 (Kent's formula and the residue expansion). *If RH holds and the zeros are simple,* (5.3.2) *is Kent's expansion* (5.4.1) *with $\lambda_n = \gamma_n^2$, and f is the density of the hitting time of a Krein string with Dirichlet spectrum $\{\gamma_n^2\}$. Without these hypotheses the grouped expansion* (5.3.1) *is the formal continuation of Kent's formula to complex eigenvalues: an off line pair contributes $2\,\mathrm{Re}\left(e^{u_\rho y}/G'(u_\rho)\right)$, a multiple zero contributes a polynomial times an exponential, and no string has such a spectrum, so the formula survives while its probabilistic meaning does not.*

Proof. The weights agree by the product formula of Theorem 5.3.2. The existence of the string for an increasing sequence λ_n with $\sum 1/\lambda_n < \infty$ is Krein's inverse spectral theorem [78], as in Section 5.2.3. The spectrum of a string is real, positive and simple. □

5.4.1 A check: Brownian exit from an interval

The exit time of standard Brownian motion from $(-1, 1)$, started at 0, has Laplace transform $1/\cosh\sqrt{2u}$, eigenvalues $\lambda_k = (k + \frac{1}{2})^2\pi^2/2$ and residues $(-1)^k\pi(k+\frac{1}{2})$, which gives the theta series $f(t) = \sum_k (-1)^k \pi(k+\frac{1}{2}) e^{-(k+\frac{1}{2})^2\pi^2 t/2}$. Numerically it has mass 1 and mean 1 (to 40 digits, the theta series and its dual series agreeing to 4×10^{-61}), and its Laplace transform at $u = 1$ equals $1/\cosh\sqrt{2} = 0.459098131085425$ to fifteen digits. The truncated Kent products $\lambda_k \prod_{m<N, m\neq k} \lambda_m/(\lambda_m - \lambda_k)$ converge to the residues, slowly for large k (computed, code/ht_kent.py):

k	$N = 50$	$N = 400$	residue
0	1.5630	1.5698	1.5708
1	−4.5050	−4.6860	−4.7124
5	−9.4241	−16.0202	−17.2788

5.4.2 What is special about ξ

For the exit time the weights grow like $\sqrt{\lambda_k}$: the function $\cosh\sqrt{2u}$ has exponential type in $\sqrt{u}$ and lies in Krein's class, so that $\sum |w_k|/\lambda_k^2 < \infty$ and the partial fraction series of $1/\cosh\sqrt{2u}$ converges, although $\sum w_k/\lambda_k$ converges only conditionally.

For ξ the weights grow like $e^{\pi\sqrt{\lambda}/4}$ up to the factor $1/|\zeta'(\rho)|$. Krein's class K consists of the real entire functions E with simple real zeros a_k whose reciprocal has the expansion

$$\frac{1}{E(z)} = c + \sum_k c_k\Big(\frac{1}{z-a_k} + \frac{1}{a_k}\Big), \qquad \sum_k \frac{|c_k|}{a_k^2} < \infty,$$

and Krein's theorem says that every function in K has exponential type and lies in the Cartwright class [131].

Proposition 5.4.2 (Not in Krein's class). *The function $\xi(\frac{1}{2}+it)$ does not belong to Krein's class, whether or not RH holds. Moreover, under RH the partial fraction series of $1/G$ over the points $u_\rho = -\gamma_n^2$ does not converge absolutely.*

Proof. Along the imaginary direction of t, $\log|\xi(\frac{1}{2}+it)|$ grows like $\frac{1}{2}|t|\log|t|$, so $\xi(\frac{1}{2}+it)$ has order one and infinite type, hence is not of exponential type, and Krein's theorem excludes it from K. For $1/G$ the argument is different, since G has exponential type zero. Under RH absolute convergence needs $\sum|w_\gamma|/\gamma^4 < \infty$, but $|w_\gamma| \asymp \gamma^{-3/4}e^{\pi\gamma/4}/|\zeta'(\rho)|$ and $|\zeta'(\frac{1}{2}+i\gamma)| \ll \gamma^{1/4}\log^2\gamma$ by the convexity bound, so the terms tend to infinity. □

The residue expansion of the density is the natural substitute for the partial fraction expansion that Krein's theorem forbids: multiplying by e^{uy} is the regularisation a function of infinite type requires. Under RH the truncations G_N have finitely many zeros and are trivially in K; the divergence appears only in the limit, as the weights $1/G_N'(u_\rho)$ grow to the residues (Table 5.7). The theory of Ostrovskii on the class K, and the work of Kamynin, Ostrovskii and Vishnyakova on zeros of entire functions that are characteristic functions, concern the class to which $\xi(\frac{1}{2}+it) = \mathbb{E}\cos(tV)$ belongs; we state no result from it.

5.5 Processes behind the reciprocal xi law

Under RH there are two processes whose hitting times have law μ.

1. The Krein string of Proposition 5.4.1 (for simple zeros), a generalised diffusion whose time to reach the end has law $\sum e_\gamma/\gamma^2$.

2. A pure birth chain that leaves its nth state at rate γ_n^2 (with multiplicities counted as repeated rates). Since $\sum\gamma^{-2} < \infty$ the chain explodes in finite time, and its explosion time has law μ.

Conversely, any process with nonnegative speed measure, or any family of GGC laws, that produces $1/G$ proves RH. This is the probabilistic form of the Hilbert and Pólya idea; its operator theoretic counterpart, de Branges spaces and canonical systems attached to ζ and to Dirichlet L functions, has been studied by Lagarias [130].

Proposition 5.5.1 (Reduction to approximating hitting times). *RH holds if and only if there are GGC laws, for instance hitting times of strings, converging weakly to μ.*

Proof. Under RH the laws of $\sum_{n\le N} e_n/\gamma_n^2$ converge to μ. Conversely, a weak limit of GGC laws is a GGC (Corollary 9.1.2, Bondesson's continuity theorem [39]), and μ is a GGC only under RH. □

The limit costs nothing: the entire content of a hitting time proof is the construction of approximating processes whose speed measures remain nonnegative.

5.6 The candidate families

De Bruijn and Newman. Replacing Riemann's kernel by $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ gives laws μ_t that are GGC exactly when $t \ge \Lambda$ (Corollary 9.2.5), and hitting times of strings when moreover the zeros are simple, which is so for $t > \Lambda$. Rodgers and Tao proved $\Lambda \ge 0$ [318], and $\Lambda \le 0.2$ is known [277]. RH is $\Lambda = 0$: the strings would have to exist for every $t > 0$ and fail for every $t < 0$, so any positivity preserving evolution in t would have to be sharp at $t = 0$.

Basepoint. For $\alpha \ge 1$ the law $\mu^{(\alpha)}$ with Laplace transform $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ is a GGC with Lorentzian Thorin density $\pi^{-1}\operatorname{Re}\xi'/\xi(\alpha + i\tau)$, the hitting time of a diffusion with continuous spectrum, and as $\alpha \downarrow \frac{1}{2}$ the spectrum sharpens onto the atoms γ^2. Survival of these laws down to α is the quasi Riemann hypothesis at α (Theorem 11.1.14, Chapter 12, Section 23.1).

Truncated Hadamard products. These are hitting times exactly when the zeros used lie on the line (Proposition 5.7.1); the construction is circular.

Riemann's integral. Since $G(u) = \mathbb{E}\cosh(\sqrt{u}\,V)$ with V symmetric and density proportional to $\Phi_{\mathrm{dBN}}(|v|/2)$, the function $\varphi = 1/G$ is the harmonic mean, over the random half width $V/\sqrt{2}$, of the transforms $\operatorname{sech}(\sqrt{u}\,V)$ of Brownian exit times. Each is a GGC, and a harmonic mean of GGC transforms need not be one. Jensen's inequality gives $\varphi(u) \le \mathbb{E}\operatorname{sech}(\sqrt{u}\,V)$, a scale mixture of GGC transforms (the exit time from $(-a, a)$ has Thorin atoms at $(k + \frac{1}{2})^2\pi^2/(2a^2)$), and the reciprocal of a mixture is not a mixture of reciprocals, so this route gives bounds and not the GGC property.

Tilted balance. For $\lambda > \frac{1}{4}$, $Y_\lambda + T_{\log N}$ equals an explicit archimedean law, where Y_λ is the Esscher tilt of μ and $T_{\log N}$ the first passage of drifted Brownian motion to the random level $\log N$, with N zeta distributed [279], and the limit $\lambda \downarrow \frac{1}{4}$ meets the pole of ζ.

Truncated Euler products. For every $\alpha > 0$ and every P there is a GGC Y_P with $T_{\mathbb{E}\log N_P} \overset{d}{=} Y_P + T_{\log N_P}$ (Proposition 21.3.1). It is the one family built from the primes that exists unconditionally at $\alpha = \frac{1}{2}$, and it escapes as $P \to \infty$.

5.7 Truncations, total positivity and the HCM route

5.7.1 Bondesson's finite approximations

Bondesson's construction of GGC laws starts from finite convolutions of exponentials and passes to weak limits [39]. For $N \geq 1$ order the zeros by height, never splitting a pair $\frac{1}{2} \pm \delta + i\gamma$, and put

$$G_N(u) = \prod_{n \leq N} \Big(1 - \frac{u}{u_{\rho_n}}\Big), \qquad \varphi_N = 1/G_N.$$

Then φ_N is the Laplace transform of a signed measure μ_N of total mass one, a probability law when the first N zeros lie on the line, with density the finite residue sum $f_N(y) = \sum_{n \leq N} e^{u_{\rho_n} y}/G_N'(u_{\rho_n})$ (for simple zeros).

Proposition 5.7.1 (Truncations). *(i) $\varphi_N \to \varphi$ pointwise on $[0,\infty)$, and for each fixed zero the finite weight converges to the residue, $1/G_N'(u_\rho) \to 1/G'(u_\rho)$, with correction factor $\prod_{n>N}(1 - u_\rho/u_{\rho_n})^{-1} \approx \exp\big(+u_\rho \sum_{n>N} u_{\rho_n}^{-1}\big)$, which under RH equals $\exp(\gamma^2 \sum_{n>N} \gamma_n^{-2}) > 1$.*

(ii) μ_N is a GGC if and only if the first N zeros lie on the line; in particular it is a GGC, unconditionally, for every $N \leq N_0(H)$, the number of zeros below the verified height $H = 3 \cdot 10^{12}$ [277].

(iii) RH holds if and only if every μ_N is a GGC, and then μ is a GGC by Bondesson's continuity theorem (Corollary 9.1.2).

(iv) Under RH, for $u \geq 0$, $0 \leq \log G(u) - \log G_N(u) \leq u\,\tau_1(N)$ with $\tau_1(N) = \sum_{n>N} \gamma_n^{-2} \approx (\log(T_N/2\pi) + 1)/(2\pi T_N)$, where T_N is the height of the Nth zero.

Proof. (i) is the convergence of the Hadamard product and of its derivative at u_ρ; the sign follows from $\prod(1 - x_n)^{-1} \approx \exp(\sum x_n)$ with $x_n = u_\rho/u_{\rho_n}$, which is positive under RH. (ii) A finite product $\prod(1 - u/u_n)^{-1}$ is a GGC Laplace transform exactly when every u_n is real and negative, and $u_\rho = (\delta + i\gamma)^2$ is real and negative exactly when $\delta = 0$; when a pair is not split, G_N is real. (iii) If some zero is off the line, μ_N fails to be a GGC for all large N; conversely use (ii) and Bondesson's theorem. (iv) Under RH $\log G(u) - \log G_N(u) = \sum_{n>N} \log(1 + u/\gamma_n^2) \leq u\tau_1(N)$ for $u \geq 0$. $\square$

The approximants are explicit GGC laws, unconditionally, up to height $3 \cdot 10^{12}$ (the relative error of the clock transform is bounded in Section 5.2.3). But Bondesson's closure theorem needs all N, and (iii) only restates RH: the method converts RH into the statement that no truncation ever leaves the GGC class, and gives no mechanism to prevent it.

	$N = 10$	$N = 30$	$N = 60$	$N = 120$
first zero, raw	1476	3118	4447	5808
first zero, corrected	9970	10132	10155	10161
second zero, corrected	-1.068×10^6	-1.160×10^6	-1.173×10^6	-1.176×10^6
fifth zero, corrected	4.15×10^9	7.28×10^9	7.80×10^9	7.96×10^9

Table 5.7: Finite truncation weights $1/G'_N(u_\rho)$ against the residues $w_\gamma = 1.0164 \times 10^4$, -1.1775×10^6 and 8.0078×10^9 at the first, second and fifth zeros. The corrected rows multiply the raw weight by the factor of Proposition 5.7.1(i), with $\sum_{n>N} \gamma_n^{-2}$ taken from the exact total p_1; for the first zero and $N = 120$ the factor is $\exp(199.79 \times 0.0027993) = 1.7494$ and the corrected weight agrees with the residue to 0.025 per cent (computed, code/ht_resid.py).

5.7.2 Pólya frequency functions

By Proposition 5.2.2, RH holds if and only if the density f of μ is a Pólya frequency function of infinite order, so that RH sits at the top of a ladder $\mathrm{PF}_2 \Leftarrow \mathrm{PF}_3 \Leftarrow \cdots \Leftarrow \mathrm{PF}_\infty \iff$ RH of determinant inequalities on the explicit function (5.3.1), alongside the ladder of derivatives of W (Section 8.1). The lowest rung, PF_2, is log concavity of f, which self decomposability does not settle: it gives unimodality (Corollary 14.1.2) only. If an off line zero exists, Schoenberg's theorem [328] says some rung fails, and the heuristic of the Gaussian window argument suggests that it fails only at an order that grows with the height of that zero.

5.7.3 Log concavity

The rung PF_2 splits into three ranges of y.

Theorem 5.7.2 (Log concavity in the tail). *There is Y such that f is strictly log concave on $[Y, \infty)$, with no hypothesis.*

Proof. By Proposition 5.3.3, $f(y) = w_1 e^{-\gamma_1^2 y}(1 + \varepsilon(y) + \eta(y))$ with $\varepsilon(y) = (w_2/w_1)e^{-\Delta y}$, $\Delta = \gamma_2^2 - \gamma_1^2 = 242.1$, and η, together with its first two derivatives, of size $O(e^{-(c_3 - \gamma_1^2)y})$. Then $(\log f)'' = \Delta^2 \varepsilon(1 + o(1)) + O(e^{-(c_3 - \gamma_1^2)y})$. Since $w_2/w_1 < 0$ the first term is negative, and since $c_3 - \gamma_1^2 = 333.9 > \Delta$ it dominates the second for large y. □

The sign of w_2/w_1 is the whole point: the negative second residue bends $\log f$ downwards, as at every pair of consecutive rates in a sum of exponentials. On the numerical grid of step 0.001 in $[0.008, 0.2]$ one finds $(\log f)'' < 0$ with minimum -2.59×10^6 at $y = 0.008$ and value -6.3×10^{-15} at $y = 0.2$ (computed, code/ht_resid.py; the value -4.8×10^{-5} near $y = 0.106$ is the two term tail value, not a grid computation). That this holds on all of $[0.008, \infty)$ is Theorem 5.7.5, a computation based on an explicit remainder bound that needs no hypothesis on the zeros but one numerical input that is not enclosed, so its status is computed.

Near the origin the density is governed by the archimedean growth of ξ. Write $\psi = \log G$ and $\Lambda_f(y) = \sup_{u>0}(\psi(u) - uy)$. Chernoff's bound gives $\log P(T \le y) \le -\Lambda_f(y)$ (Section 5.2), and the saddle point approximation

$$\log f(y) \approx -\Lambda_f(y) - \tfrac{1}{2}\log\big(2\pi|\psi''(u_y)|\big), \qquad \psi'(u_y) = y, \tag{5.7.1}$$

gives $(\log f)'' \approx -1/|\psi''(u_y)|$, which is large and negative: since $\psi(u) \sim \frac{1}{4}\sqrt{u}\log u$, one finds $\sqrt{u_y} \approx \log(1/y)/4y$ and $\Lambda_f''(y)$ of order $\log^2(1/y)/y^3$.

Proposition 5.7.3 (Log concavity near the origin). *If the saddle point approximation* (5.7.1) *holds as $y \to 0$ together with two derivatives, then f is strictly log concave on $(0, y_0]$ for some $y_0 > 0$.*

The hypothesis is the usual one for densities of infinitely divisible laws with regularly varying exponent. Making it rigorous with derivatives needs control of ψ in a sector around the positive axis, which the growth of ξ right of the critical line supplies but which is not written out here; the status is conditional. Table 5.8 shows the approximation against $\log f$ at five points as reported in the source; these values were not reproduced and are unchecked. What remains of log concavity, unlike RH, is a problem about a fixed analytic function on a bounded range, and it is expected to be true.

y	$\log f(y)$	saddle (5.7.1)	$\log P(T \le y)$	$-\Lambda_f(y)$	$\Lambda_f''(y)$
0.008	-12.5846	-12.5857	-21.40	-19.06	2.62×10^6
0.010	-3.7121	-3.7144	-11.82	-9.75	1.09×10^6
0.012	0.5749	0.5710	-6.91	-5.09	5.19×10^5
0.015	3.3592	3.3511	-3.30	-1.84	2.01×10^5
0.020	4.3833	4.3636	-1.07	-0.16	5.38×10^4

Table 5.8: The lower tail of μ and the saddle point approximation, as printed in the source (unchecked: not reproduced). The Chernoff exponent $-\Lambda_f(y)$ bounds $\log P(T \le y)$ from above, as it must.

5.7.4 A certificate for log concavity on $[0.008, \infty)$

The grouped expansion has an explicit error once the Bromwich line passes between two consecutive poles. Let $N = 120$ and $c = \frac{1}{2}(\gamma_{120}^2 + \gamma_{121}^2) = 73296.53$,

with $\gamma_{120} = 269.970$ and $\gamma_{121} = 271.494$ (computed).

Proposition 5.7.4 (Explicit truncation). *With no hypothesis, for every $y > 0$ and $j \geq 0$,*

$$\left| f^{(j)}(y) - \sum_{n \leq 120} w_n (-\gamma_n^2)^j e^{-\gamma_n^2 y} \right| \leq M_j e^{-cy}, \qquad M_j = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|-c+it|^j}{|G(-c+it)|} \, \mathrm{d}t.$$

Numerically $\log M_0 = 206.26$ and $M_{j+1}/M_j = 1.0000185\,c$ for $j = 0, 1, 2$ (computed).

Proof. Shift the Bromwich line to $\operatorname{Re} u = -c$. The poles crossed are those with $\operatorname{Re} u_\rho = \delta^2 - \gamma^2 > -c$, which forces $\gamma < \sqrt{c + \frac{1}{4}} < 270.8$; these are the first 120 zeros, which are simple and on the line. On $\operatorname{Re} u = -c$ the function $1/|G|$ is integrable, because $\operatorname{Re} \sqrt{u}$ grows like $|t|^{1/2}$ and with it $|\xi(\frac{1}{2} + \sqrt{u})|$. Differentiating under the integral gives the derivatives. □

The bound uses nothing about zeros above height 270.8, and it is minute: with $\log M_j = 206.26, 217.46, 228.67, 239.87, 251.07$ for $j \leq 4$, $M_j e^{-cy} \leq e^{-330}$ for $y \geq 0.008$. The constants M_j, $j \leq 8$, were enclosed by ball arithmetic with the integral cut at $|t| = 2^{20}$ and the zeros and the derivative of ζ taken from Arb; $M_0 = 3.78781207977 \times 10^{89}$ is enclosed to a relative 1.6×10^{-12}. The integrand at $|t| = 2^{20}$ is 10^{-572} of its peak, but the tail beyond is not bounded analytically, and the aliasing of the circle quadrature for ζ' was checked at two sizes and not certified. So the M_j are computed, and not proved.

Two devices make log concavity checkable. The first is an exact identity for sums of exponentials: if $f = \sum_k w_k e^{-\lambda_k y}$ then

$$f f'' - f'^2 = \frac{1}{2} \sum_{j,k} w_j w_k (\lambda_j - \lambda_k)^2 e^{-(\lambda_j + \lambda_k) y}. \tag{5.7.2}$$

The slowest pair term is $(1, 2)$, with negative coefficient $w_1 w_2 (\lambda_2 - \lambda_1)^2$. Once it dominates the sum of the absolute values of the others, which then persists, $g = f f'' - f'^2$ is negative for good. With the truncation bound added, the ratio of the other terms to the leading one is 2.83 at $y = 0.02$, 1.39 at 0.022, 0.622 at 0.025, 0.033 at 0.04 and 2.2×10^{-5} at 0.08 (computed), and in ball arithmetic it is below one and decreasing on the grid $0.025, 0.030, \ldots, 0.3$, with the enclosure $0.621855669841 \pm 2 \times 10^{-13}$ at $y = 0.025$. Hence $g < 0$ on $[0.025, \infty)$, given the M_j.

The second device is a Taylor model for the window below 0.025. At a point y_i compute $g^{(m)}(y_i)$ for $m < K = 6$ from (5.7.2) or by Leibniz's rule, and bound $|g^{(K)}|$ on $[y_i, y_i + h]$ by replacing every $f^{(j)}$ with $A_j(y_i) = \sum_{n \leq 120} |w_n| \lambda_n^j e^{-\lambda_n y_i} + M_j e^{-c y_i}$, which dominates $|f^{(j)}|$ on the whole step because each term decreases in y. If the Taylor polynomial of g plus remainder and truncation errors is negative on

$[y_i, y_i+h]$, then $g < 0$ there; take the largest such h and move on. In ball arithmetic at 400 bits, with the range $[0.008, 0.08]$ split into seven sub windows (five of them in $[0.008, 0.012]$), every sub window was covered. Table 5.9 lists the windows; the step counts depend on the rule that chooses h, which the source does not give, so those of the check (a greedy doubling rule) are printed and the source's 25856 steps (its parts sum to 25859), 73 and 74 are not reproduced.

window	Taylor steps	method
$[0.008, 0.012]$	29984	Taylor model, $K = 6$, 400 bits
$[0.012, 0.020]$	88	Taylor model, $K = 6$, 400 bits
$[0.020, 0.080]$	90	Taylor model, $K = 6$, 400 bits
$[0.025, \infty)$	none	pair identity (5.7.2) with Proposition 5.7.4

Table 5.9: Windows on which $(\log f)'' < 0$ was verified, given the constants M_j (computed, code/ht_lcert.py). The step count near 0.008 is large because of the cancellation in (5.3.1).

THEOREM 5.7.5 (Log concavity, computed). *Assume the values of the constants M_j, $j \le 8$, of Proposition 5.7.4, which are computed by quadrature. Then, with no hypothesis on the zeros, the density f of μ is strictly log concave on $[Y_*, \infty)$ with $Y_* = 0.008$.*

Its status is computed until the M_j are enclosed with a bound on the tail of the integral. The range it leaves open, $(0, Y_*)$, carries mass $P(T < Y_*) = 5 \times 10^{-10}$ (computed), and there the saddle point approximation (5.7.1) would give log concavity with $\Lambda_f''(y)$ of order 10^6 and growing (Proposition 5.7.3, conditional). Closing it rigorously needs either more zeros, so that c grows and the window can be pushed down (at $y = 0.005$ the 120 term sum with its truncation balls still encloses $g = -6.5 \times 10^{-40}$ away from zero, with ball radius 7.6×10^{-49} against a cancellation of 6.3×10^{35} in f, but no Taylor model was run below 0.008), or a rigorous saddle point bound. Neither involves the location of the zeros. Log concavity of f is the strong unimodality of μ, by the theorem of Ibragimov [115].

5.7.5 Off line zeros are invisible to the low rungs

The pair of zeros $\frac{1}{2} \pm \delta + i\gamma$ contributes $2|G'(u_\rho)|^{-1} e^{(\delta^2-\gamma^2)y} \cos(2\delta\gamma y - \arg G'(u_\rho))$ to the density, while u_ρ makes the angle $\theta_\rho = 2\arctan(\delta/\gamma)$ with the negative axis. Over the range $y \lesssim \gamma^{-2}$ where it is not exponentially negligible its phase moves by only about $2\delta/\gamma$, and Schoenberg's theorem requires exactly that u_ρ stay near the negative axis. A determinant of order k resolves the density to roughly k derivatives, so the first failing rung is expected, heuristically, to be of order γ/δ at best, and larger once the lower zeros are allowed to smooth the pair. For a zero above the verified height this is at least of order 10^{12}. Heuristically, then,

the first failing rung is none that can be computed. In an experiment reported in the source, nine toy densities were built from the residue formula with the first $m-1$ zeros on the line and the mth replaced by an off line pair (at $\gamma_m = 30.4$, 43.3, 67.1 and $\delta = 0.45$, 0.25, 0.1); all were positive, log concave on $(0, 0.2]$ and had nonnegative Toeplitz determinants of orders three and four. This experiment was not reproduced here and is unchecked. Numerical confirmation of low rungs, for either ladder, carries no information about zeros above the verified height.

5.7.6 The HCM route

Bondesson calls a density f on $(0,\infty)$ hyperbolically completely monotone (HCM) if, for each $u>0$, $f(uv)f(u/v)$ is a completely monotone function of $v+1/v$. HCM densities are GGC densities, and the class is closed under products, quotients and powers $|q| \geq 1$ of independent variables [39] (Chapter 9). HCM is sufficient for the GGC property, not necessary, so it gives a route to RH in one direction only.

Proposition 5.7.6 (HCM route). *If the density f of μ is HCM, then μ is a GGC and RH holds.*

The kth rung of this route is $(-1)^k \partial_w^k [f(uv)f(u/v)] \geq 0$ with $w = v + 1/v$. Bondesson's representation $f(x) = Cx^{\beta-1}h_1(x)h_2(1/x)$, with h_1, h_2 Laplace transforms of GGC laws, gives a consequence that is easy to test: an HCM density extends to a holomorphic function on $\mathbb{C} \setminus (-\infty, 0]$ with no zeros there, since $h_1(x)$ and $h_2(1/x)$ have the form $\exp(-ax - \int \log(1 + x/t)U(\,\mathrm{d}t))$ and are holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$.

Theorem 5.7.7 (No HCM). *(i) No density of a sum of $n \geq 2$ independent exponential variables with distinct rates is HCM (proved).*

(ii) With the first 120 *terms of* (5.3.2)*, the continuation of f to $\operatorname{Re} z > 0$ vanishes at*

$$z_* = 0.013831874754506 + 0.020001801712034\,i$$

(computed). If the omitted terms are controlled by an explicit remainder bound on a sector containing z_*, this proves that f is not HCM.

Proof. (i) The density is an exponential polynomial $\sum_{k \leq n} c_k e^{-\lambda_k z}$ with distinct real λ_k and nonzero c_k. It has at most $n-1$ real zeros, by the Descartes rule for exponential sums, and infinitely many zeros altogether, since an entire function of order one with finitely many zeros is a polynomial times an exponential, which a sum of two or more exponentials with distinct rates is not. So it has nonreal zeros, which lie in $\mathbb{C} \setminus (-\infty, 0]$, and no HCM density has any. For $e^{-x} - e^{-2x}$ they are the points $2\pi i k$, $k \neq 0$.

(ii) Newton's method on the 120 term sum at 50 digits gives the stated z_*, floating point and not enclosed; the largest term of the normalised sum $f(z)e^{\gamma_1^2 z}/w_1$ at z_* has modulus 5.4, and the normalised sum is 1.9×10^{-12} at the printed

fifteen digits (the root to forty digits was not tested). The argument principle on the box $0.012 \le \operatorname{Re} z \le 0.15$, $0 < \operatorname{Im} z \le 0.05$ gives winding number 5 for the 120 term sum, and 1 on a box of half width 10^{-3} around z_* (floating point, code/ht_resid.py). □

Part (ii) is a computation on the 120 term sum, not a proof, for four reasons. The series (5.3.2) presupposes RH and simple zeros, and its convergence for $\operatorname{Re} z > 0$ needs the hypothesis of Theorem 5.3.2. The termwise estimate for the omitted zeros uses residues that are not controlled. The remainder bound of Proposition 5.7.4 does not apply: it is for real y only, because on $\operatorname{Re} u = -c$ the factor $|e^{uz}| = e^{-c \operatorname{Re} z - t \operatorname{Im} z}$ grows with $|t|$ while $1/|G(-c+it)|$ decays only like $\exp(-c'\sqrt{|t|}\log|t|)$, so the integral diverges for every nonreal z. And the argument principle needs that bound on the boundary of the box. A proof is within reach and needs no hypothesis on the zeros: bend the contour of the Bromwich integral at $-c$ into the two rays $\arg(u+c) = \pm(\pi/2+\eta)$ with η a little above $\arg z_* = 55.3°$, where the poles are real up to $|u| = 9 \times 10^{24}$ and within angle $1/\gamma$ of the negative axis above, the rays cross none of them, and $1/|G|$ decays on them, which gives an explicit remainder bound on a sector containing z_*. If RH fails then f is not HCM at all, because HCM densities are GGC, so only the case of RH needs the computation. Under RH, simple zeros and two hypotheses on the zeros, Theorem 9.3.22 proves that the density is not HCM.

For the two exponential density the failure of the HCM ladder appears only at high order. With $w = 2.1$, the normalised derivative $(-1)^k \partial_w^k [f(uv)f(u/v)]/f(uv)f(u/v)$ of $f = e^{-x} - e^{-2x}$ at $u = 5$ is positive for $k \le 14$ and equals -1.21×10^9 at $k = 15$ (stable between 300 and 450 digits). At $u = 6$ and $u = 7$ the first negative order is 14 and 18, and for $u = 3, 4, 8, 10$ none appears up to order 40 (computed, code/ht_hcm2exp.py). Low order evidence for an HCM property of f would have been misleading. Since the computed zero indicates that the HCM class is unavailable even under RH, Proposition 5.7.6, while true, is vacuous for this law as far as the computed zero shows: the HCM route to RH appears closed, provided the bound of the bent contour is supplied.

5.8 What the reciprocal Xi function proves

Every route in this chapter ends in an equivalence with RH whose unconditional part stops exactly where classical knowledge stops. We record the frontier for each, add one quantitative statement about the canonical function, and identify the single inequality at which arithmetic would have to enter.

5.8.1 Partial complete monotonicity

The law μ is a GGC if and only if its canonical function $\Theta(x) = \sum_{\gamma>0} e^{s_\rho x}$ is completely monotone, that is $(-1)^n \Theta^{(n)}(x) \geq 0$ for all n and x. Each zero on the line contributes $\gamma^{2n} e^{-\gamma^2 x} > 0$, an off line pair an oscillating term; the verified zeros give a quantitative piece of the condition.

Theorem 5.8.1. *Let $T = 3 \cdot 10^{12}$, so that every zero with $0 < \gamma \leq T$ is on the line [277]. For every $x \geq 1/T$ and every integer*

$$0 \leq n \leq \frac{(T^2 - \gamma_1^2 - 1)\, x - 5}{\log(T^2/\gamma_1^2)},$$

we have $(-1)^n \Theta^{(n)}(x) > 0$. Numerically the bound is $n \leq 1.7 \cdot 10^{23} x$. RH holds if and only if the bound can be replaced by $n < \infty$ for every $x > 0$. Status: proved, with the verification height T as input; the constants are computed.

Proof. The zeros with $\gamma \leq T$ contribute $\sum \gamma^{2n} e^{-\gamma^2 x} \geq \gamma_1^{2n} e^{-\gamma_1^2 x}$. A zero with $\gamma > T$ has $s_\rho = (\delta + i\gamma)^2$ with $|\delta| < \frac{1}{2}$, so its contribution has modulus at most $g(\gamma) = (\gamma^2 + \frac{1}{4})^n e^{-(\gamma^2 - \frac{1}{4})x}$. Under the bound on n we have $n \leq T^2 x/52$, so g is decreasing for $\gamma \geq T$ and $g(T + j + 1)/g(T + j) \leq e^{-1.9Tx} \leq e^{-1.9}$. There are at most $2\log(T + j + 1)$ zeros with $T + j < \gamma \leq T + j + 1$, by an explicit bound for the argument of ζ [108], so the tail is at most $4\log(T + 1) g(T)/(1 - e^{-1.9}) < 136\, g(T) < e^5 g(T)$. The bound on n gives $n \log((T^2 + \frac{1}{4})/\gamma_1^2) + 5 < (T^2 - \frac{1}{4} - \gamma_1^2)x$, which makes this smaller than $\gamma_1^{2n} e^{-\gamma_1^2 x}$. □

At scale x the nth Bernstein condition weighs the zeros by $\gamma^{2n} e^{-\gamma^2 x}$, which peaks at $\gamma^2 = n/x$. The theorem says that the conditions which look only below height T hold, and no more: it is zero verification written as complete monotonicity, with the same height to order trade off as the Jensen degree ($d \approx \gamma^2$, a heuristic of Section 21.1) and the first order boundary response (δ^2/γ^4, Remark 21.7.3).

5.8.2 The frontier

Table 5.10 lists the forms of RH that the reciprocal Xi law provides, the classical statement each is equivalent to, and how far each is known unconditionally.

Two entries carry the diagnosis. First, the base point row: Lagarias proved that RH is equivalent to $\operatorname{Re} \xi'/\xi(s) > 0$ throughout $\operatorname{Re} s > \frac{1}{2}$ [198], and the Thorin density of $\mu^{(\alpha)}$ is exactly $\pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + it)$ whenever $\mu^{(\alpha)}$ is a GGC (Section 21.8, Theorem 2.1.2). So the base point descent is Lagarias' criterion, and pushing it below $\alpha = 1$ at a single value is the quasi Riemann hypothesis there: the unrefereed announced half plane $\operatorname{Re} s > \frac{7}{8}$ of [257] (Section 25.1.2), open for every $\alpha < \frac{7}{8}$; before it the best known were the zero free regions of de la Vallée Poussin and of Vinogradov and Korobov, which shrink to $\operatorname{Re} s = 1$ as

Reciprocal Xi form	Classical form	Known unconditionally
μ is a GGC	RH	Theorem 5.8.1: Bernstein order $1.7 \cdot 10^{23} x$
density of μ is PF_∞	RH (Schoenberg [328])	log concave on $[0.008, \infty)$ (computed, Section 21.14)
$\mu^{(\alpha)}$ GGC for all $\alpha > \frac{1}{2}$	$\operatorname{Re} \xi'/\xi > 0$ on $\operatorname{Re} s > \frac{1}{2}$ (Lagarias [198])	$\alpha \geq 1$ only
$\mu^{(\alpha)}$ GGC at one $\alpha < 1$	quasi RH at α	announced for $\alpha \geq \frac{7}{8}$ [257], open below
de Bruijn laws μ_t GGC	$t \geq \Lambda$	$t \geq 0.2$ [277]
Jensen laws μ_d exist	Jensen and Pólya criterion	$d \leq 2 \cdot 10^{17}$ [97]
hybrid laws $\hat{\mu}^{[N]}$ GGC	zeros up to γ_N on the line	$N \leq 1.2 \cdot 10^{13}$
GGC limit of prime built laws	Weil type positivity at the centre	the scaling of Theorem 21.3.4 erases the zeros; Theorem 5.8.4 for $\theta < \beta < 2 - \theta$ under QRH(θ)
boundary, Brownian, extremal representations	none	exist for every law

Table 5.10: Equivalent forms of RH from the reciprocal Xi law and their unconditional frontier. The Jensen bound is Chassé's, as quoted in [97].

the height grows. Second, the last row: the probabilistic representations exist for every law and carry no information. Every unconditional entry in the right column except the base point, quasi Riemann hypothesis and prime built rows also holds for Davenport and Heilbronn below the height of their first off line zero, apart from the log concavity row, which has been computed for μ only and is not claimed for them. The base point and prime built rows rest on the zero free half plane $\operatorname{Re} s \geq 1$ and on the prime sum, which come from the Euler product and fail for Davenport and Heilbronn, whose function has zeros with real part greater than 1 [72]. So the entries that survive for Davenport and Heilbronn cannot be extended to RH without the Euler product, and the entries that use the Euler product are the ones that fail for it.

5.8.3 The quasi Riemann hypothesis and the prime sum

Write QRH(α), $\frac{1}{2} \leq \alpha < 1$, for the statement that every zero has real part at most α. It was announced for $\alpha = \frac{7}{8}$ in October 2026, with a formal proof in Lean not yet refereed [257], and is open for every $\alpha < \frac{7}{8}$. It is exactly a statement about the prime sum, and exactly one about the base point hitting times.

Proposition 5.8.2. *For $\frac{1}{2} \leq \alpha < 1$ the following are equivalent.*

1. QRH(α).

2. $\psi(x) = x + O(x^{\alpha+\varepsilon})$ *for every* $\varepsilon > 0$.
3. *The Dirichlet series* $D(s) = \sum_{n\geq 1}(\Lambda(n) - 1)n^{-s}$ *converges for* $\operatorname{Re} s > \alpha$, *so that* $-\zeta'/\zeta = \zeta + D$ *there.*
4. $\operatorname{Re} \xi'/\xi(s) > 0$ *for* $\operatorname{Re} s > \alpha$.
5. *The base point laws* $\mu^{(\beta)}$ *are GGCs, that is diffusion hitting times, for every* $\beta > \alpha$.

Status: proved (the equivalence of (4) and (5) with (1) is Theorem 2.1.2 and Remark 2.1.4).

Proof. (1) and (2) are equivalent by the explicit formula. (2) gives $\sum_{n\leq x}(\Lambda(n) - 1) = \psi(x) - \lfloor x \rfloor = O(x^{\alpha+\varepsilon})$, and partial summation gives (3). Under (3), $-\zeta'/\zeta = \zeta + D$ is analytic in $\operatorname{Re} s > \alpha$ apart from the pole at 1, so ζ has no zeros there, which is (1). Under (1), the zeros pair as ρ and $1 - \bar{\rho}$ with real parts in $[1 - \alpha, \alpha]$, so each term $\operatorname{Re} 1/(s-\rho)$ of $\operatorname{Re} \xi'/\xi(s)$ is positive when $\operatorname{Re} s > \alpha$, which is (4). By Theorem 2.1.2, $\mu^{(\beta)}$ is a GGC if and only if $\beta \geq \sup_\rho \operatorname{Re} \rho$, which gives (1) if and only if (5); and (4) is Lagarias' criterion on the half plane, Remark 2.1.4, whose last step shows that positivity of $\operatorname{Re} \xi'/\xi$ on every line to the right of α excludes zeros there. □

So the quasi Riemann hypothesis does control the prime sum below 1: under QRH(α), a smoothed explicit formula expresses the Thorin density of $\mu^{(\beta)}$ at height t, for $\beta > \alpha$, as the archimedean term plus a prime sum $\operatorname{Re} \sum_n \Lambda(n) n^{-\beta-it} w(n/X)$ minus a smooth main term in $X^{1-\beta-it}$, with an error $O(X^{\alpha-\beta} \log t)$ from the zeros (compare Proposition 12.1.13). The base point hitting times are then built from the primes up to X to within that error. Two facts stop this going further.

First, positivity cannot be carried downward in β. In the variable $w = \sqrt{s}$, moving the base point is a translation, and if $w \mapsto 1/\xi(\beta + w)$ were the Laplace transform of a positive measure in w, a translation would be an exponential tilt of that measure and positivity would pass from β to $\beta - \Delta$ for free. It never is (for the one sided case see Remark 12.1.8; the lemma adds the two sided case).

Lemma 5.8.3. *For no real* β *is* $w \mapsto 1/\xi(\beta + w)$ *the one or two sided Laplace transform of a positive measure. Status: proved.*

Proof. By Landau's theorem the abscissa of convergence of such a transform is a singular point on the real axis [359]. The function $1/\xi(\beta + w)$ is meromorphic with poles only at $w = \rho - \beta$, none of them real, since $\xi(\sigma) > 0$ for real σ. A finite abscissa would therefore have to be a regular point, and an infinite one would make the transform entire, contradicting the poles. □

This is why every positive representation in this chapter lives in the variable $s = w^2$, where the poles $-\gamma^2$ are real and negative, and why no translation argument in β is available. Second, any step from QRH(α) to QRH($\alpha - \varepsilon$) is new information about zeros in the strip $\alpha - \varepsilon < \operatorname{Re} s \le \alpha$. Davenport and Heilbronn satisfy all the structural properties used here and have zeros in such strips, so the information cannot come from structure; it must come from the Euler product, through the prime sum above.

5.8.4 Prime built hitting times under the quasi Riemann hypothesis

Suppose QRH(θ) holds for some $\theta < 1$; the case $\theta = \frac{7}{8}$ is the one announced in [257], so the statements below are conditional on that announcement for $\theta = \frac{7}{8}$. Then the base point laws above θ can be built from the primes, and this is the correct renormalisation of the Euler hitting times of Theorem 21.3.4. Let $w(u) = (1-u)_+^3$, with Mellin transform $W(z) = 3!\,\Gamma(z)/\Gamma(z+4) = 6/(z(z+1)(z+2)(z+3))$, and for $\beta > \theta$ define

$$d_X(t) = \frac{1}{\pi}\Big(\operatorname{Re} A(\beta + it) - \operatorname{Re} \sum_{n \le X} \frac{\Lambda(n)}{n^{\beta+it}} \Big(1 - \frac{n}{X}\Big)^3 + \operatorname{Re}\, X^{1-\beta-it} W(1-\beta-it) \Big),$$

which uses only the primes up to X and the gamma function. Shifting the contour in $\frac{1}{2\pi i}\int (-\zeta'/\zeta)(s+z)X^z W(z)\,\mathrm{d}z$ gives the explicit formula

$$\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\beta + it) = d_X(t) - \frac{1}{\pi}\operatorname{Re}\sum_{\rho} X^{\rho-\beta-it} W(\rho - \beta - it) + O\big(X^{-1}\log(|t|+2)\big).$$

The poles of W at $z = -1, -2, -3$, with residues -3, 3, -1, produce the remainder, which is of order X^{-1} and not the $X^{-2-\beta}$ of the first trivial zero alone. The formula is used only for $\theta < \beta < 2 - \theta$ under QRH(θ): for $2 - \theta \le \beta \le 1 + \theta$ a zero with $\beta_\rho = \beta - 1$ meets a pole of W (for $\beta > 1 + \theta$ nothing collides and the X^{-1} remainder dominates), and the remainder $-3X^{-1}(-\zeta'/\zeta)(s-1)$ has poles in the critical strip, so the bound is not uniform.

Theorem 5.8.4. *Assume* QRH(θ) *and let* $\theta < \beta < 2 - \theta$. *Then* $|d_X(t) - \pi^{-1}\operatorname{Re}\,\xi'/\xi(\beta + it)| \ll_\beta X^{\theta-\beta}\log(t+2)$ *uniformly in* t, *and there is* $X_0(\beta, \theta)$ *such that for* $X \ge X_0$ *the function* d_X *is nonnegative. The GGC laws* $\mu_X^{(\beta)}$ *with Thorin density* d_X *in the variable* $z = t^2$ *are then hitting times built from the primes up to* X, *and* $\mu_X^{(\beta)} \Rightarrow \mu^{(\beta)}$ *as* $X \to \infty$. *Conversely, if for some* β *and every* t *the values* $d_X(t)$ *are nonnegative for all large* X, *then* QRH(β) *holds. Status: proved, conditional on* QRH(θ) *for the first part. The unconditional range* $1 < \beta < 2$ *is not covered, see the discussion after the proof.*

Proof. Under QRH(θ), $|X^{\rho-\beta-it}| \le X^{\theta-\beta}$ for every zero, and $|W(\rho-\beta-it)| \ll (1+|\gamma-t|)^{-4}$, so the zero sum is $\ll X^{\theta-\beta}\log(t+2)$, and the remainder $O(X^{-1}\log(|t|+2))$ is no larger because $\beta < 2-\theta \le 1+\theta$. By Proposition 5.8.2 the exact density is positive, and it is continuous, so it has a positive minimum on any compact set. For large t, $\pi^{-1}\operatorname{Re}\xi'/\xi(\beta+it) = \frac{1}{2\pi}\log(t/2\pi) - \pi^{-1}\operatorname{Re}(-\zeta'/\zeta)(\beta+it) + O(1/t)$ and $\zeta'/\zeta(\beta+it) = o(\log t)$ under QRH(θ), so the exact density is at least $c\log t$ for large t, and the error is a factor $X^{\theta-\beta}$ smaller, which gives X_0 (no appeal to the smoothed counting function of zeros is needed). The Laplace exponents converge because $d_X \to \pi^{-1}\operatorname{Re}\xi'/\xi$ uniformly against $\log(1+s/t^2)$.

For the converse we cannot use the pointwise limit, since a zero with $\beta_\rho > \beta$ makes the term $X^{\rho-\beta-it}W(\rho-\beta-it)$ grow like $X^{\beta_\rho-\beta}$. Fix t that is not an ordinate of a zero and not half the difference of two ordinates, and put $s = \beta+it$. For $\operatorname{Re}u$ large, $\int_1^\infty \sum_{n\le X}\Lambda(n)n^{-s}(1-n/X)^3X^{-u-1}\mathrm{d}X = (-\zeta'/\zeta)(s+u)W(u)$, because $\int_0^1(1-v)^3v^{u-1}\mathrm{d}v = W(u)$. So the Mellin transform $M(u) = \int_1^\infty \pi d_X(t)X^{-u-1}\mathrm{d}X$ is a combination of $\operatorname{Re}A(s)/u$, of $(\zeta'/\zeta)(s+u)W(u)$ and its conjugate counterpart at $\bar{s}$, and of $W(1-s)/(u+s-1)$ with its conjugate; the pole at $u = 1-s$ of $\zeta'/\zeta(s+u)$ cancels against the main term. It is meromorphic in $\operatorname{Re}u > 0$ with poles only at $u = \rho-s$ and $u = \rho-\bar{s}$, none of them real, and every zero with $\beta_\rho > \beta$ gives a pole with $\operatorname{Re}u = \beta_\rho-\beta > 0$ (the choice of t keeps the poles from cancelling). If $d_X(t) \ge 0$ for $X \ge X_0$, the transform of the tail $X \ge X_0$ has a real singularity at its abscissa of convergence by Landau's theorem [359], as in Lemma 5.8.3, while M is regular at every real $u > 0$; so the abscissa is at most 0, M is analytic in $\operatorname{Re}u > 0$, and no zero has real part above β. Since t ranges over a set of full measure, this is QRH(β). □

So under QRH($\frac{7}{8}$) the reciprocal Xi programme has what the heuristic Remark 21.5.1 asks for, at every base point $\frac{7}{8} < \beta < \frac{9}{8}$: GGC hitting times built from the primes and the gamma function, converging to the base point law. The renormalisation that makes this work is the subtraction of the smooth main term $X^{1-s}W(1-s)$, the contribution of the smoothed prime number theorem, rather than the central limit scaling of Theorem 21.3.4, which erases the zeros. Whether the same holds unconditionally for $1 < \beta < 2$ is left open: the two singular terms at $\beta_\rho = \beta-1$ may cancel in the sum, but the proof above does not show it. What QRH($\frac{7}{8}$) does not give is anything at $\beta \le \frac{7}{8}$, and in particular at $\frac{1}{2}$: there the error bound $X^{\theta-\beta}\log(t+2)$ of Theorem 5.8.4 does not decay.

We computed d_X with $X = 10^2, \dots, 10^5$ on $0 < t < 2000$. The error against the exact density is 0.34, 0.11, 0.038, 0.011 at $\beta = 1$, and 0.90, 0.44, 0.22, 0.10 at $\beta = 0.8$ (computed): it decays like $X^{1/2-\beta}$, the rate set by the actual zeros, which lie on the line as far as they have been computed. The densities are nonnegative on the window for $X \ge 10^3$ at $\beta = 1, 0.9, 0.8, 0.7$, with minimum close to the exact minimum (0.0073 against 0.0074 at $\beta = 1$). At $\beta = 0.6$ they are still negative at $X = 10^5$, with minimum -0.16 near $t = 1164$, against an exact minimum of

0.0015. The obstacle is visible: when the kernel width $\beta - \frac{1}{2}$ is smaller than the zero spacing, the exact density between consecutive zeros is tiny, and the error must be pushed below it. Near the centre this costs a power of X that grows without bound as $\beta \downarrow \frac{1}{2}$.

Other consequences of QRH($\frac{7}{8}$) in this chapter's terms are weaker than what is already known. De Bruijn's theorem turns zeros in the strip $|\operatorname{Re}\rho - \frac{1}{2}| \le \frac{3}{8}$ into $\Lambda \le 2(\frac{3}{8})^2 = \frac{9}{32} = 0.28$, which is worse than the proven 0.2 [277], so it improves neither the de Bruijn laws nor the Jensen degree heuristic $D(T) = T^2/(2\Lambda_0)$ of Section 21.1. It removes the defect D_β of Proposition 14.3.3 for $\beta > \frac{7}{8}$, and it gives $\psi(x) = x + O(x^{7/8+\varepsilon})$, a power saving in the prime number theorem that follows from the announcement [257] if it stands.

5.8.5 Zero density as an almost GGC statement

What is known unconditionally is QRH on average: zero density estimates, which Section 14.3.6 and Theorem 14.3.2 turn into a statement about base point hitting times. Fix $\beta > \frac{1}{2}$ and split the Thorin density $\pi^{-1} \operatorname{Re} \xi'/\xi(\beta + it)$ into the Poisson kernels of the zeros with $\operatorname{Re}\rho < \beta$, which are positive, and those with $\operatorname{Re}\rho > \beta$, which are negative, with total masses $m_\beta^+(T)$ and $m_\beta^-(T)$ on $[0, T]$. The book's theorem bounds $m_\beta^-(T)/m_\beta^+(T) \le T^{e(\beta)-1+o(1)}$ with $e(\beta) < 1$ from Ingham's density estimate [163] and from Guth and Maynard [103, Theorem 1.2]. Selberg's estimate $N(\sigma, T) \ll T^{1-(\sigma-1/2)/4} \log T$ [330] gives the weaker relative size $T^{-(\beta-1/2)/4}$ for every $\beta > \frac{1}{2}$, and $N(\sigma, T) \ll T^{30(1-\sigma)/13+o(1)}$ gives $T^{30(1-\beta)/13-1+o(1)}$, which is the stronger of the two once $\beta > 0.575$. Both are beaten by Ingham's exponent on $(\frac{1}{2}, \frac{7}{10})$, and Guth and Maynard's $15(1-\sigma)/(3+5\sigma)$ is sharper on $[\frac{7}{10}, \frac{4}{5}]$. The zero count satisfies $\nu_-[0, T] \ll N(\beta, 2T) + 1$ in the notation above, with ν_- the negative part of the Thorin measure, because a kernel centred at height $\gamma > 2T$ puts mass $O(1/(\gamma - T))$ times its width into $[0, T]$, which sums to $O(1)$ over the sparse set of zeros to the right of β.

What this does not give is a factor. The negative kernels are not the Thorin density of a second hitting time G_- with $\mu^{(\beta)} * G_- = G_+$: the negative Poisson kernel of a zero to the right of β is the Thorin density of the reflected zero, so the product of the two exponentials is analytic in the slit plane, while $\xi(\beta)/\xi(\beta + \sqrt{s})$ has poles there. The correct statement is the canonical factorisation of Proposition 14.3.3: $\mu^{(\beta)}$ has Laplace transform $\varphi_\beta^+ D_\beta$, where φ_β^+ is the Laplace transform of the GGC T_β^+ whose Thorin density is the positive part above and D_β is the defect over the zeros to the right of β, and the book shows that T_β^+ is not the sum of a variable with law $\mu^{(\beta)}$ and an independent nonnegative variable. Density estimates say that the defect D_β is sparse; proving it empty is QRH(β), which no density estimate can do, because density estimates allow a sparse set of exceptions.

5.8.6 Where arithmetic would enter

Writing $\xi'/\xi = A + \zeta'/\zeta$ with A the archimedean part, the base point Thorin density is

$$\frac{1}{\pi}\Big(\operatorname{Re} A(\alpha+it) - \operatorname{Re}\sum_{n\ge 2}\frac{\Lambda(n)}{n^{\alpha+it}}\Big),$$

$$\operatorname{Re} A(\alpha+it) = \tfrac{1}{2}\log\frac{t}{2\pi} + O(t^{-1}),$$

with the Dirichlet series read as $-\zeta'/\zeta(\alpha+it)$ when $\alpha \le 1$. For $\alpha > 1$ the series converges absolutely and is bounded by $-\zeta'/\zeta(\alpha)$, so positivity for large t is automatic. That the base point laws are GGCs for $\alpha \ge 1$ comes from a different fact, which is also arithmetic: every term $\operatorname{Re} 1/(s-\rho)$ of $\operatorname{Re}\,\xi'/\xi$ is positive because no zero has real part ≥ 1 (Theorem 2.1.2). For $\alpha < 1$ positivity at every t requires the prime sum to stay below $\frac{1}{2}\log(t/2\pi)$ in real part, uniformly in t, with no absolute convergence available. That uniform inequality, a positivity statement about $\sum \Lambda(n)n^{-\alpha-it}$ in the critical strip, is the arithmetic input that the routes of this chapter would need, in the form of Lagarias' criterion. Davenport and Heilbronn fail it, because their function has no prime sum: its logarithmic derivative has no Euler product expansion at all.

Notes. The factorisation of the base point law and the status of QRH($\frac{7}{8}$) are those of Sections 14.3 and 25.1; the statements of Biane, Pitman and Yor, of Pitman and Yor, of James, Roynette and Yor, of Roberts and Shortland and of Kent are those of the cited papers.

5.9 Exercises

On Section 5.1.

5.1. Show that $\frac{1}{\pi}\int_0^{\Theta}(1-\cos\theta x)\,d\theta = \frac{1}{\pi}(\Theta - \sin(\Theta x)/x)$ and, for $\alpha > 1$, derive the Dirichlet series for $\arg\zeta(\alpha+i\Theta)$ from the integrated sine squared lemma.

5.2. Implement Selberg's formula at $s = \frac{3}{4} + i\theta$ with $x = 100$ and check that adding the zero sum over the first hundred zeros reduces the error to the size of the omitted zeros.

5.3. Estimate, from the single zero law, the cutoff x needed for six correct digits of $v_{0.55}(\theta)$ at an ordinate $\theta = \gamma$ from the primes, and compare with the cost of the Pólya route at the same point.

5.4. Show that $g(0) - g(x) \ge 0$ for every real positive definite function g, and that $g(0) - g(x)$ is conditionally negative definite.

Chapter 6

Riemann's continuation and the generalised gamma convolution question

This chapter asks how close Riemann's own analytic continuation of 1859 [370] comes to proving that $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u}) = 1/G(u)$ is the Laplace transform of a generalised gamma convolution, the Thorin condition of Section 1.1. Riemann's continuation writes ξ as the Mellin transform of a combination of the theta function and its derivatives. Read probabilistically, it is a signed mixture of Gamma(2) and exponential densities over the rates πn^2. It is also the density of Williams' variable, which is a convolution of Gamma(2) laws and so a GGC (Section 3.1.3). The question is whether the closure properties of the class GGC carry this structure through to the clock. We take the two facts that suggest it and test each against what has been proved. We then treat three further devices: the normal scale mixture of the logistic law due to Stefanski and to Andrews and Mallows, and the extreme value law behind it; the recovery of the Thorin measure from the heat trace through the sine squared lemma (2.1.9); and the de Bruijn tilt $e^{\lambda u^2}$ of the kernel and its counterpart $e^{-|u|}$.

Main results. (1) Riemann's kernel, as the density of $V = \frac{\pi}{2}S_2$, is $\sum_n [4g_2 - 6g_1](\cdot\,; \pi n^2)$, where $g_k(\cdot\,;\lambda)$ is the Gamma(k) density of rate λ (Proposition 6.1.1; proved and computed). (2) Neither starting fact holds as it is usually stated. A scale mixture of Gamma(2) laws is infinitely divisible but need not be a GGC. Harmonic means do not preserve the class: the natural candidate statement is false (Proposition 11.1.7), and we found no harmonic mean closure in the sources we read. Neither link from V or the Pólya variable X to $1/G$ is one of the closure operations that are theorems, and X^2 and $|X|$ are not GGC (Proposition 6.2.1; proved). (3) The Gumbel and logistic laws are EGGCs whose Thorin measures are the integers. For ξ the same shape is the normal scale mixture $Z\sqrt{2T}$, which exists with no hypothesis; it is an EGGC exactly under RH (Section 6.3). (4) The chain from the heat trace W through the sine squared lemma gives a Thorin density m that is positive with no hypothesis (Theorem 6.4.1; proved). It is the Thorin density of the positive half of the reciprocal law, not of the clock. RH is equivalent

to $w \mapsto \pi m(\sqrt{w})/w$ being a Stieltjes function, and then its representing measure is $z^{-1/2}U_{\mathrm{RH}}(dz)$. So positivity cannot be read off m: the RH content is in the analytic continuation. (5) The de Bruijn tilt is a normal mixture of Esscher tilts, that is a Gaussian average of basepoint moves. The tilt $e^{-|u|}$ is a positive mixture of Gaussian dampings, the direction that Proposition 23.2.3 excludes. Its Fourier transform has only finitely many real zeros (Proposition 6.5.1; proved), and a floating point scan with an interpolation bound finds none on $[0, 80]$ (computed, not certified).

Against the path of Section 1.4, every step of Riemann's continuation is unconditional except one. That step passes from a Mellin transform, or an arithmetic mean of transforms, to the reciprocal of a Laplace transform: a harmonic mean, which no closure theorem covers. Making it a GGC is RH (Proposition 2.1.1). The chapter therefore gives no new route. It locates the gap exactly in the second constraint of Section 1.4: nothing in Riemann's continuation uses the Euler product beyond the Dirichlet series of ω, and Section 6.6 explains why the harmonic mean step is where the Davenport and Heilbronn obstruction bites.

6.1 Riemann's continuation as a Gamma(2) structure

Riemann's second proof of the functional equation [370], [80, Section 1.7] starts from $\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s) = \int_0^\infty \omega(x)x^{s/2-1}dx$, with $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$, for $\operatorname{Re} s > 1$. It uses the theta transformation $1 + 2\omega(1/x) = \sqrt{x}(1 + 2\omega(x))$ to continue. Remark 15.1.6 gives the form we need: with $D = -x\,d/dx$, the function $[2D(D - \frac{1}{2})\omega](x)$ decays faster than any power at both ends, and

$$\begin{aligned} \xi(s) &= \int_0^\infty \big[2D(D-\tfrac{1}{2})\omega\big](x)\, x^{s/2-1}dx \quad (s \in \mathbb{C}), \\ \Phi(u) &= 2e^{u/2}\big[2D(D-\tfrac{1}{2})\omega\big](e^{2u}). \end{aligned} \tag{6.1.1}$$

Since $D(D - \frac{1}{2})e^{-\lambda x} = (\lambda^2 x^2 - \frac{3}{2}\lambda x)e^{-\lambda x}$, the bracket is $\sum_n (2\pi^2 n^4 x^2 - 3\pi n^2 x)e^{-\pi n^2 x}$. We write $g_k(x;\lambda) = \lambda^k x^{k-1}e^{-\lambda x}/\Gamma(k)$ for the Gamma(k) density of rate λ, and $V = \frac{\pi}{2}S_2$ for Williams' variable in the normalisation of Section 2.1, so that $2\xi(s) = \mathbb{E}V^{s/2}$.

Proposition 6.1.1. *(i) (Williams; Biane, Pitman and Yor.) $V = \sum_{n\geq 1} \Gamma_n/(\pi n^2)$ with independent Gamma(2) variables Γ_n. Hence V is a GGC with Thorin measure $2\sum_n \delta_{\pi n^2}$.*

(ii) The density of V is

$$f_V(x) = \frac{2}{x}\big[2D(D-\tfrac{1}{2})\omega\big](x) = \sum_{n\geq 1}\Big[4\, g_2(x;\pi n^2) - 6\, g_1(x;\pi n^2)\Big], \qquad x > 0. \tag{6.1.2}$$

This is a signed mixture of Gamma(2) and exponential densities over the rates πn^2. Each term has mass -2, and the masses sum to 1 under zeta regularisation, since $-2\zeta(0) = 1$.

(iii) For $\operatorname{Re} s > 1$ *the Mellin transform can be taken term by term:*

$$\mathbb{E}V^{s/2} = \sum_{n\geq 1} (\pi n^2)^{-s/2} \Big(4\Gamma(2+\tfrac{s}{2}) - 6\Gamma(1+\tfrac{s}{2}) \Big) = s(s-1)\pi^{-s/2}\Gamma(\tfrac{s}{2})\zeta(s) = 2\xi(s).$$

(iv) V is not a scale mixture of Gamma(a) laws for any $a > 0$ (Theorem 14.2.10). Its density is positive, and the negative terms of (6.1.2) *cannot be absorbed into a positive mixture.*

Proof. (i) is [363, 33], as recalled in Section 3.1.3 and Table 4.3: the Laplace transform of V is $\prod_n (1 + u/(\pi n^2))^{-2}$. (ii) The density of S_2 is the theta series given in Section 3.1.3, and $f_V(x) = 2\sum_n (2\pi^2 n^4 x - 3\pi n^2) e^{-\pi n^2 x}$ (Section 15.1.4.3). Moreover $4g_2(x;\lambda) - 6g_1(x;\lambda) = (4\lambda^2 x - 6\lambda)e^{-\lambda x}$, and $\int (4g_2 - 6g_1) = -2$. The regularised sum is a formal identity, since $\sum_n 1$ diverges. (iii) For $\operatorname{Re} s > 1$ the series of absolute values converges, $\int_0^\infty x^{s/2} g_k(x;\lambda)dx = \lambda^{-s/2}\Gamma(k + \frac{s}{2})/\Gamma(k)$, and $4\Gamma(2+\frac{s}{2}) - 6\Gamma(1+\frac{s}{2}) = (2s-2)\Gamma(1+\frac{s}{2}) = s(s-1)\Gamma(\frac{s}{2})$. (iv) is Theorem 14.2.10. □

The identities were checked at 40 and 60 digits (computed). The series in (iii) agrees with $2\xi(s)$ to twenty digits at $s = 3$, $4 + 2i$ and 6. Quadrature of $x^{s/2} f_V(x)$ with 120 terms of (6.1.2) reproduces $2\xi(s)$ to eighteen digits at $s = 0$, 1 and $\frac{1}{2} + 3i$, beyond the range where (iii) holds term by term. The minimum of f_V on the grid $0.03 \leq x \leq 5$, of step 0.01, is $2.8 \times 10^{-39} > 0$ (supported by the 60 digit run only, since at 40 digits this value is below the cancellation noise), at the left end where f_V vanishes to infinite order.

On the logarithmic scale $y = \frac{1}{2}\log x$, a scale change of x is a translation of y. So (6.1.2) is a signed *location* mixture, over the lattice $-\frac{1}{2}\log(\pi n^2)$, of log gamma densities of shapes two and one. Size biasing by $x^{1/4}$, which passes from V to the Pólya variable $X = \frac{1}{2}\log V$ tilted by $e^{X/2}$ (Section 2.1), raises the shapes to $\frac{9}{4}$ and $\frac{5}{4}$. This is the Gumbel decomposition of Theorem 19.17.1, in the normalisation of Rodgers and Tao. This is the sense in which Riemann's kernel is a signed location scale mixture of Gamma(2) and Gamma(1) laws. Two Gamma(2) structures coexist. Williams' convolution (i) is positive and a GGC, and it carries the functional equation. Riemann's mixture (ii) is signed, and it carries the Dirichlet series. Remark 15.1.6 shows that the signs come from the operator $2D(D - \frac{1}{2})$ that removes the poles of ζ.

6.2 The two starting facts, checked

6.2.1 Scale mixtures of Gamma(2) laws

Steutel conjectured, and Kristiansen proved [194], that every scale mixture $\Gamma_2 Y$, with Γ_2 of the Gamma(2) law and $Y \geq 0$ independent of it, is infinitely divisible. It need not be a GGC. If Y takes two distinct positive values, the Laplace transform of $\Gamma_2 Y$ vanishes at non real points, and a GGC transform has no zeros off $(-\infty, 0]$ (Example 11.1.25; Section 15.1.4.3 gives the general discrete case). The correct sufficient conditions that we could verify are the following.

1. If Y is a GGC, then $\Gamma_2 Y$ is a GGC, because products of independent GGC variables are GGC (Bondesson [40]; Section 11.1.4).

2. If Y has an HCM density, then $\Gamma_2 Y$ and Γ_2 / Y are GGC. More generally, the independent product or quotient of a GGC by an HCM variable is a GGC (Bondesson [39, Theorem 6.2.1], as quoted by Jedidi and Simon [371]).

3. If Y is hyperbolically monotone of order two, then $\Gamma_2 Y$ and Γ_2 / Y are GGC, according to the abstract of [372], which we did not read further.

These conditions are sufficient, not necessary: for U uniform on $(0, 1)$, $\Gamma_2 U$ is exponential, a GGC, while U is neither a GGC nor HCM.

6.2.2 Harmonic means

The second starting fact would be closure under harmonic means. In the form

$$\text{(H)} \qquad V_1, V_2 \text{ independent GGC} \implies 2V_1V_2/(V_1 + V_2) \text{ GGC}$$

it is false: $2E_1E_2/(E_1+E_2)$ with independent standard exponentials is not a GGC (Proposition 11.1.7). On the transform side, which is the side the clock needs, the harmonic mean of the GGC transforms $(1 + s)^{-2}$ and $(1 + 2s)^{-2}$ is not even a Laplace transform of a positive measure (Example 11.1.24). For HCM variables V_1, V_2 the reciprocals $1/V_i$ are HCM [371], so $1/V_1 + 1/V_2$ is a GGC, but passing back to its reciprocal would need a closure of GGC under $x \mapsto 1/x$, which we found in none of the sources. We found no harmonic mean closure in [371, 372, 373, 40].

The operations that are theorems are: independent sums, positive scaling and weak limits (Section 4.1); independent products of GGC variables [40]; independent products and quotients with HCM variables [39, 371]. Bondesson conjectured closure under qth powers for $q > 1$ [39, 374]. Two 2026 preprints announce a proof [373, 374]; according to their abstracts the second calls itself a candidate proof. We read neither proof, and we use neither result.

6.2.3 What the operations can and cannot reach

The clock is linked to V and to X in two ways, both unconditional. First, $G(s^2) = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2}) = \mathbb{E}e^{sX}$ with $X = \frac{1}{2}\log V$ tilted by $e^{X/2}$ (2.1.4): a Mellin transform of V. Second, $1/G(u) = 1/\mathbb{E}\cosh(\sqrt{u}\,X) = 1/\mathbb{E}[1/\operatorname{sech}(\sqrt{u}X)]$ (Proposition 11.1.23). This is the harmonic mean over the Pólya density of the GGC transforms $\operatorname{sech}(\sqrt{u}\,x)$, whose Thorin atoms are at $\pi^2(k-\frac{1}{2})^2/x^2$. Neither link is a closure operation. The first is not one because a Mellin transform does not act on Laplace exponents. The second fails because of the counterexamples above. The arithmetic mean $\mathbb{E}\operatorname{sech}(\sqrt{u}X)$, which bounds $1/G(u)$ from above by Jensen's inequality, is the Laplace transform of $\frac{1}{2}X^2C$, with C the exit time of Brownian motion from $(-1,1)$. One might hope to make it a GGC by Bondesson's product theorem. The next proposition shows that the factorisation $\frac{1}{2}X^2\cdot C$ does not meet its hypotheses.

Proposition 6.2.1. *A GGC whose Laplace transform extends to an entire function is a constant. In particular neither X^2 nor $|X|$ is a GGC, for the Pólya variable X. So Bondesson's product theorem does not apply to the factorisation $\frac{1}{2}X^2\cdot C$ of the law whose Laplace transform is the arithmetic mean $\mathbb{E}\operatorname{sech}(\sqrt{u}X)$.*

Proof. Let $\varphi(u) = \exp\{-du - \int \log(1+u/z)U(dz)\}$ be entire. Then $f = -\varphi'/\varphi = d + \int U(dz)/(u+z)$ on $\mathbb{C}\setminus(-\infty,0]$ extends to a meromorphic function on $\mathbb{C}$. Its poles are the zeros of φ, with residue $-m$ at a zero of order m. By Stieltjes inversion (4.1.2), $U((a,b])$ is the limit of $\frac{1}{\pi}\int_a^b \operatorname{Im} f(-x-i\eta)\,dx$. For a meromorphic f, real on $(0,\infty)$, this limit picks up only the poles of f on $(-\infty,0)$, so U is a sum of atoms. An atom of mass $c>0$ at z_0 gives f the residue $+c$ at $-z_0$, against the residue $-m<0$ that a zero of φ gives. Hence $U=0$ and $\varphi(u) = e^{-du}$. For X, the bound $\Phi(v) \le Ce^{9v/2}\exp(-\pi e^{2v})$ for $v \ge 0$ (the proof of Theorem 14.2.10) makes $\mathbb{E}e^{tX^2}$ and $\mathbb{E}e^{t|X|}$ finite for every real t. So the Laplace transforms of X^2 and $|X|$ are entire, and these variables are not constant. □

Whether $\mathbb{E}\operatorname{sech}(\sqrt{u}X)$ is a GGC transform by some other argument is open (Problem 6.1). If it were, this would not bear on RH, since the clock is the harmonic and not the arithmetic mean. The clock itself is not $\Gamma_a Y$ for any $a>0$ and independent $Y \ge 0$ (Theorem 11.1.26). So neither starting fact can be applied with the clock as the outer variable.

6.3 Stefanski's logistic mixture and the extreme value law

Andrews and Mallows [3] and Stefanski [336] represent the logistic law, with characteristic function $\pi t/\sinh \pi t$, as a normal scale mixture. In the language of Section 4.1.2 the mixing variable is the sinh clock H_0, with $\mathbb{E}e^{-uH_0} = \pi\sqrt{u}/\sinh\pi\sqrt{u}$, a GGC with Thorin measure $\sum_k \delta_{k^2}$ and the law of $2K^2$ for the Kolmogorov variable K (Table 4.3). So the logistic law is $Z\sqrt{2H_0} = 2KZ$, and it is an EGGC [39,

Theorem 7.3.1]. The extreme value law enters twice. The logistic law is the difference of two independent Gumbel variables, because $\Gamma(1-it)\Gamma(1+it) = \pi t/\sinh \pi t$. And the Gumbel variable $\mathcal{G} = -\log E$, E standard exponential, satisfies

$$\mathbb{E}e^{-s(\mathcal{G}-\gamma_E)} = e^{\gamma_E s}\Gamma(1+s) = \prod_{k\geq 1} \frac{e^{s/k}}{1+s/k}, \qquad \mathcal{G} - \gamma_E \stackrel{d}{=} \sum_{k\geq 1} \frac{E_k - 1}{k}, \tag{6.3.1}$$

with γ_E Euler's constant and E_k independent standard exponentials, by the Weierstrass product of $1/\Gamma$. This is a compensated GGC with Thorin measure $\sum_k \delta_k$ on the positive half line, an EGGC carried by one side. In Section 19.20 it is the Gumbel law less Euler's constant. Both identities were checked to eighteen digits (computed): the three expressions for the logistic characteristic function at $t = 0.7$, 2 and 5, and (6.3.1) at $s = \frac{1}{2}$ and 3.

The template for ξ replaces the integers by the ordinates, and Section 13.1 carries it out. With no hypothesis, $\xi(\frac{1}{2})/\xi(\frac{1}{2}+|t|)$ is the characteristic function of the normal scale mixture $Y_{1/2} = Z\sqrt{2T}$, which is infinitely divisible and self decomposable (Proposition 13.1.1). It is an EGGC if and only if RH holds (Theorem 13.1.2). Under RH, $Y_{1/2} = A - A'$ with $A \stackrel{d}{=} \sum_k (E_k - 1)/\gamma_k$, the analogue of (6.3.1) (Proposition 13.1.5). Stefanski's representation therefore gives the right shape and no positivity. The normal scale mixture exists for ξ unconditionally, as it does for the logistic law, and the EGGC property of its mixing variable is exactly the Thorin condition. The logistic case is solved because the zeros of sinh are known by inspection, and the points k^2 they give are positive. The extreme value structure of the Pólya kernel itself (Theorem 19.17.1) is a sum of Gumbel pieces over n, each with an explicit Thorin form. A sum of laws is not a product of transforms, and Section 19.20 records that this is where that route stops.

6.4 From the heat trace to a Thorin measure by the sine squared lemma

The Thorin measure U of the clock, if it exists, is the measure that represents W by Bernstein's theorem, $W(t) = \int e^{-zt}U(dz)$ (Section 4.1.1). The sine squared lemma (2.1.9) does something different. It turns a Lévy representation in the variable s into a Thorin representation in s^2, and for ξ at basepoints $\alpha > 1$ it produces v_α from the primes (Section 2.1). At the centre $\alpha = \frac{1}{2}$ the function $\log \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$ has no Lévy form in s, because it is even in s. The lemma applies instead to one half of it. Write $a_k = -(\rho_k - \frac{1}{2})^2$ as in Section 1.1, let $c_k = \sqrt{a_k}$ with the principal root, and let

$$K(x) = \sum_k e^{-c_k x}, \qquad P(s) = \prod_k \Big(1 + \frac{s}{c_k}\Big)e^{-s/c_k}.$$

If $\rho_k = \frac{1}{2} + \delta + i\gamma$, then $c_k = \gamma - i\delta$, so $\operatorname{Re} c_k = \gamma > 0$ whether or not RH holds. The subordination identity of Section 4.1.2 gives $K(r) =$

$\int_0^\infty e^{-1/4v}(4\pi v)^{-1/2}W(r^2v)\,dv/v$, so $K>0$ with no hypothesis, because $W>0$ (Theorem 8.1.1; also Proposition 14.4.1). The function K is the trace $K_{1/2}$ of Section 13.1.5.

THEOREM 6.4.1. *With no hypothesis on the zeros, let*

$$m(\theta)=\frac{1}{\pi}\int_0^\infty \big(1-\cos x\theta\big)K(x)\,dx=\frac{1}{\pi}\sum_k \frac{\theta^2}{c_k(c_k^2+\theta^2)},\qquad \theta>0,$$

with the sum taken over conjugate pairs $c_k,\bar c_k$ together.

(i) $m(\theta)>0$ for every $\theta>0$, and $u\mapsto P(\sqrt{u})$ is the Laplace transform of a GGC whose Thorin measure has density m in the chart $\theta=\sqrt{z}$:

$$-\log P(s)=\int_0^\infty \log\Big(1+\frac{s^2}{\theta^2}\Big)m(\theta)\,d\theta,\qquad s\ge 0.$$

Moreover $P(s)P(-s)=\xi(\frac12+is)/\xi(\frac12)$, and $1/P(s)=\mathbb{E}e^{-sA}$ for the law A of Proposition 13.1.5.

(ii) The function $M(w)=\pi m(\sqrt{w})/w=\sum_k\big(c_k(w+a_k)\big)^{-1}$ extends to a meromorphic function on $\mathbb{C}$, with simple poles at the points $-a_k$ of simple zeros and residues $1/c_k$ there.

(iii) RH holds if and only if M is a Stieltjes function, that is $M(w)=\int\nu(dz)/(w+z)$ for $w>0$ with a positive measure ν. In that case $\nu=\sum_{\gamma>0}\gamma^{-1}\delta_{\gamma^2}$, which is $z^{-1/2}U_{\mathrm{RH}}(dz)$.

Proof. (i) Since $\operatorname{Re}c_k>0$, $\frac{1}{\pi}\int_0^\infty(1-\cos x\theta)e^{-c_kx}dx=\frac{1}{\pi}\big(\frac{1}{c_k}-\frac{c_k}{c_k^2+\theta^2}\big)$, which gives the series. The sum over k of absolute values converges, because each term is $O(\theta^2\gamma_k^{-3})$ for $\gamma_k\ge 2\theta$ and $O(\gamma_k^{-1})$ otherwise. Positivity is immediate from the integral, since $K>0$ and $1-\cos x\theta\ge 0$ is not identically zero. In the chart θ, (2.1.9) reads $e^{-sx}+sx-1=\frac{2x}{\pi}\int_0^\infty\log(1+s^2/\theta^2)\sin^2(x\theta/2)\,d\theta$. Integrating against $K(x)\,dx/x$, using Tonelli (everything is positive) and $2\sin^2(x\theta/2)=1-\cos x\theta$, gives

$$\int_0^\infty\big(e^{-sx}+sx-1\big)\frac{K(x)}{x}\,dx=\int_0^\infty\log\Big(1+\frac{s^2}{\theta^2}\Big)m(\theta)\,d\theta.$$

By Frullani's integral the left side is $\sum_k[s/c_k-\log(1+s/c_k)]=-\log P(s)$ for $s\ge 0$. The exchange of sum and integral is justified because each term is $O(s^2/|c_k|^2)$ and $\sum|c_k|^{-2}<\infty$. The same identity, with $K(x)\,dx/x$ as Lévy measure, is the Laplace exponent of A in Proposition 13.1.5. The function m is $O(\theta^2)$ at 0, and it grows at most like $(\log\theta)^2$ at infinity, by the zero count. So the Thorin measure $m(\sqrt{z})\,dz/(2\sqrt{z})$ satisfies the integrability condition of (4.1.1), and the representation is that of a GGC in $u=s^2$. Finally $(1+s/c)(1-s/c)=1+(-s^2)/a$, and $\prod_k(1+(-s^2)/a_k)=G(-s^2)=\xi(\frac12+is)/\xi(\frac12)$ by (2.1.1). The exponential

factors cancel in pairs. (ii) This follows from $c_k^2 = a_k$ and the absolute convergence of $\sum_k |c_k|^{-1}|w + a_k|^{-1}$ on compact sets avoiding the points $-a_k$; the terms are $O(\gamma_k^{-3})$. (iii) Under RH, $a_k = \gamma_k^2 > 0$ and $c_k = \gamma_k$, so $M(w) = \sum \gamma^{-1}(w + \gamma^2)^{-1}$, which is the Stieltjes transform of $\sum \gamma^{-1}\delta_{\gamma^2}$. Conversely, a Stieltjes function is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, so every pole $-a_k$ of M lies on $(-\infty, 0]$. Since M is a convergent sum of simple fractions with distinct poles, no pole cancels. Hence every a_k is real and positive, which is RH by Section 2.1. □

So the heat trace yields, through subordination and the sine squared lemma, a positive Thorin density with no hypothesis. It is the Thorin density of the positive half of the reciprocal law, in the square root variable, and it smooths every zero by the kernel $\theta^2/(\pi c(c^2+\theta^2))$. It is not the Thorin measure of the clock. The positivity of m carries no information about RH, because it holds whatever the zeros do. The information sits where Proposition 2.1.1 puts it: in the analytic continuation of $m(\sqrt{w})/w$ across the negative axis, which must have its poles on that axis. To recover U from W one must perform this continuation (Stieltjes inversion of M, then $U(dz) = \sqrt{z}\,\nu(dz)$), and, heuristically, it is exactly as hard as locating the zeros. The off line part of m is also invisible numerically. For a hypothetical off line pair with $\gamma = 1000$, $\delta = 0.4$, or $\gamma = 100$, $\delta = 0.45$, its contribution to m is positive on the grid $0 < \theta < 3000$ of step 0.1, with minimum 6.4×10^{-12} and 6.4×10^{-9} respectively. Its pole in M sits at $-a = -999999.84 + 800i$ and $-9999.80 + 90i$ (computed). With the first 400 zeros, the integral and the series for m agree to fifteen digits at $\theta = 5$, γ_1 and 30. With one zero, $\int_0^\infty \log(1 + s^2/\theta^2)m(\theta)\,d\theta$ and $-\log P(s)$ agree to nineteen digits at $s = 1$ and 10 (computed, 30 digits).

Remark 6.4.2. At basepoints $\alpha > 1$ the lemma acts on the Lévy form of $\log \xi(\alpha + s)/\xi(\alpha)$, whose canonical measure carries the primes. It gives v_α directly, as the Thorin density of the clock T_α (Theorem 2.1.2, [279, Theorem 8.2]). For $\frac{1}{2} < \alpha < 1$ the Lévy form exists in a compensated form only when no zero lies to the right of α (Theorem 13.1.2). The theorem above is the unconditional counterpart at $\alpha = \frac{1}{2}$. It buys unconditional positivity by splitting $G(-s^2) = P(s)P(-s)$, and the splitting is what removes the RH content.

6.5 The de Bruijn tilt and the tilt by $e^{-|u|}$

Pólya's universal factors multiply the kernel. Every factor of the form $\mathbb{E}\cosh(uB)$, with B symmetric, acts as a vertical average $\int_{\mathbb{R}} \Phi(u)\mathbb{E}\cosh(uB)e^{izu}\,du = \mathbb{E}\,\Xi(z + iB)$ (Section 18.14.4). The de Bruijn factor is of this kind, with B normal; the damping e^{-bu^2} and the factor $e^{-|u|}$ are mixtures of cosines instead. Let N be standard normal and τ standard exponential.

Proposition 6.5.1. *(i) For $\lambda > 0$, $e^{\lambda u^2} = \mathbb{E}\cosh(\sqrt{2\lambda}\,Nu)$, and for every*

$s, z \in \mathbb{C}$

$$\int_{\mathbb{R}} \Phi(u) e^{\lambda u^2} \cosh(su)\, du = \mathbb{E}\, \xi\big(\tfrac{1}{2} + s + \sqrt{2\lambda} N\big),$$

$$\int_{\mathbb{R}} \Phi(u) e^{\lambda u^2} e^{izu}\, du = \mathbb{E}\, \Xi\big(z + i\sqrt{2\lambda} N\big).$$

So the de Bruijn tilt is a normal scale mixture of Esscher tilts. On the Laplace side it is a Gaussian average of the basepoint moves $\alpha = \frac{1}{2} + \sqrt{2\lambda} N$.

(ii) *For* $b > 0$, $e^{-bu^2} = \mathbb{E} \cos(\sqrt{2b}\, Nu)$, *and* $\int_{\mathbb{R}} \Phi(u) e^{-bu^2} e^{izu}\, du = \mathbb{E}\, \Xi(z + \sqrt{2b} N)$, *a horizontal Gaussian smoothing of* Ξ.

(iii) $e^{-|u|} = \mathbb{E}\big[(\pi\tau)^{-1/2} e^{-u^2/(4\tau)}\big]$. *So* $e^{-|u|}\Phi$ *is a positive mixture, over* $b = 1/(4\tau)$, *of the damped kernels of (ii), and*

$$F(z) = \int_{\mathbb{R}} \Phi(u) e^{-|u|} e^{izu}\, du = \frac{1}{\pi} \int_{\mathbb{R}} \frac{\Xi(z - y)}{1 + y^2}\, dy.$$

(iv) $F(x) = 2\Phi(0)/x^2 + O(x^{-4})$ *as* $x \to \infty$ *along the real axis. So* F *has only finitely many real zeros.*

Proof. (i) $\mathbb{E} e^{cN} = e^{c^2/2}$ gives $\mathbb{E} \cosh(\sqrt{2\lambda} Nu) = e^{\lambda u^2}$. By (2.1.4), $\int \Phi(u) e^{wu} du = \xi(\frac{1}{2} + w)$ for every $w \in \mathbb{C}$. Fubini applies because Φ decays doubly exponentially, so $\int \Phi(u) \mathbb{E} e^{|\sqrt{2\lambda} Nu| + |su|} du < \infty$. This gives $\mathbb{E} \frac{1}{2}[\xi(\frac{1}{2} + s + \sqrt{2\lambda} N) + \xi(\frac{1}{2} - s + \sqrt{2\lambda} N)]$, and the symmetry of N and the evenness of $\xi(\frac{1}{2} + w)$ identify the two halves. The second identity is the first at $s = iz$. (ii) is the same computation with $\mathbb{E} e^{icN} = e^{-c^2/2}$. (iii) The integral $\int_0^\infty (\pi t)^{-1/2} e^{-t - u^2/4t} dt = e^{-|u|}$ is classical, and the convolution theorem with $\int_{\mathbb{R}} e^{-|u|} e^{iyu} du = 2/(1 + y^2)$ gives F. (iv) The kernel $k(u) = e^{-|u|}\Phi(u)$ is even and smooth on $[0, \infty)$, with all derivatives decaying doubly exponentially, and $k'(0+) = -\Phi(0)$ since $\Phi'(0) = 0$. So $F(x) = 2 \int_0^\infty k(u) \cos(xu)\, du$. Two integrations by parts give $-2k'(0+)/x^2$, and two more give a remainder $2k'''(0+)/x^4 + o(x^{-4})$. Since $\Phi(0) > 0$, $F(x) > 0$ for all large real x, and F is entire, so its real zeros in any bounded interval are finite in number. □

The identities were checked (computed). At $\lambda = 0.05$ and $z = 10$ the kernel side and the Gaussian vertical average agree to fifteen digits, both equal to 0.037505051645606. The mixture in (iii) agrees with $e^{-|u|}$ to eighteen digits at $u = 0.3$ and 2. With $\Phi(0) = 0.893393800934246888$, a scan of F at step $\frac{1}{4}$ on $[0, 80]$ finds no sign change. Since $|F''(x)| \le 2 \int_0^\infty u^2 e^{-u} \Phi(u)\, du = 0.0166$, the linear interpolant between grid points is within 1.30×10^{-4} of F, below the minimum 2.816×10^{-4} of F on the grid, attained at $x = 80$; so F has no zero on $[0, 80]$ (computed in floating point, not certified). There Ξ has 21 zeros, the last near 79.34. The ratio $F(x)x^2/(2\Phi(0))$ is 1.0363 at $x = 40$ and 1.0087 at $x = 80$, consistent with a correction of order x^{-2} in the ratio.

The two tilts therefore behave in opposite ways. The de Bruijn tilt is a positive normal mixture of cosh factors, and $\cosh(bu)$ is a universal factor (Section 22.6). The theorem that keeps real zeros forward in time is de Bruijn's, which uses the product structure of the factor and not the mixture. The GGC property of the reciprocal R_t of the tilted function holds exactly for λ at or above one quarter of the de Bruijn Newman constant in the normalisation of Section 4.1.1, unconditionally for $\lambda \geq 0.05$ (Theorem 17.1.16, [277]). It holds at $\lambda = 0$ exactly under RH, with $\Lambda \geq 0$ by Rodgers and Tao [318]. On the Laplace side, (i) writes the tilted function as an arithmetic mean of shifted ξ, so the tilted clock is again a harmonic mean, of the basepoint transforms. It meets the obstruction of Section 6.2, so no closure theorem proves the property at any λ from this representation. Mixing over basepoints gives no positivity of its own.

The tilt by $e^{-|u|}$ is a positive mixture of the Gaussian dampings in (ii), which run the de Bruijn Newman flow backwards. A class of kernels stable under them cannot contain Φ if all its members have only real zeros (Proposition 23.2.3). The tilt $e^{-|u|}$ does more. Its corner at the origin produces the x^{-2} tail of (iv), which removes all but finitely many real zeros; on $[0, 80]$ the scan sees none. On the Laplace side, $e^{-|u|}$ is not of the form $\mathbb{E}\cosh(uB)$ for any symmetric B, since such a factor is at least one; it corresponds to no basepoint move and no clock. So tilting by $e^{-|u|}$ does not behave like the de Bruijn tilt. It is a scale mixture of normal densities, not of normal moment generating functions, and it destroys the structure that the Thorin condition needs.

6.6 Where each step needs RH

Table 6.1 collects the steps of the chapter with their status.

Read against Section 1.4, the analysis stops at the same place by every route. Riemann's continuation uses the theta transformation and the Dirichlet series of ω. It uses no multiplicative structure. The harmonic mean step, the one that would need a closure theorem, has the same form for any Dirichlet series with a functional equation of this shape. Section 21.8 shows that it holds in this form for the Davenport and Heilbronn function. There the harmonic mean has the complex pole coming from the off line zero near $0.8085 + 85.6993\,i$ and is not a GGC. So any closure theorem strong enough to make the harmonic mean a GGC for ξ would prove a false statement for that function, unless it used the Euler product: the second constraint of Section 1.4. The third constraint excludes properties stable under the dampings of Proposition 6.5.1(ii) and (iii).

6.7 Open problems

Problem 6.1. Is $u \mapsto \mathbb{E}\,\mathrm{sech}(\sqrt{u}\,X)$, the Laplace transform of $\frac{1}{2}X^2C$, a GGC transform? By Proposition 6.2.1, Bondesson's product theorem does not apply to

step	status	where
f_V is Riemann's signed Gamma(2) and exponential mixture; $\mathbb{E}V^{s/2} = 2\xi(s)$	proved, computed	Proposition 6.1.1
V is a GGC (Williams' Gamma(2) convolution)	proved	[363, 33]
V, $\lvert X\rvert$, X^2 are not Gamma(a) scale mixtures; $\lvert X\rvert$, X^2 not GGC	proved	Theorem 14.2.10, Proposition 6.2.1
$1/G$ is the harmonic mean of $\operatorname{sech}(\sqrt{u}\,x)$ over the Pólya density	proved	Proposition 11.1.23
harmonic means preserve GGC	false in general	Proposition 11.1.7
$1/G$ is a GGC transform	equivalent to RH	Proposition 2.1.1
$Z\sqrt{2T}$ is a normal scale mixture; an EGGC	proved; equivalent to RH	Section 13.1
$m > 0$; $P(\sqrt{u})$ is a GGC transform	proved	Theorem 6.4.1
$\pi m(\sqrt{w})/w$ is a Stieltjes function	equivalent to RH	Theorem 6.4.1
de Bruijn tilt keeps the GGC property	proved for $\lambda \geq 0.05$; at $\lambda = 0$ equivalent to RH	Theorem 17.1.16
$e^{-\lvert u\rvert}$ tilt has finitely many real zeros	proved, computed	Proposition 6.5.1

Table 6.1: Riemann's continuation and the GGC question: the status of each step.

the factorisation $\frac{1}{2}X^2 \cdot C$. An answer would not bear on RH directly, but it would show whether the arithmetic mean side of Jensen's inequality keeps the Thorin property.

Problem 6.2. Find a class $\mathcal{C}$ of mixing laws for which the harmonic mean $1/\mathbb{E}[1/\varphi_X(u)]$ of GGC transforms φ_x is again a GGC transform, and which is not already explained by Pólya's theorems on real zeros. Any such class containing the Pólya density must use the Euler product. Otherwise the Davenport and Heilbronn function refutes it (Section 6.6).

Problem 6.3. The density m of Theorem 6.4.1 is positive with no hypothesis. Its continuation $M(w)$ has the poles $-a_k$. Is there a quantitative statement, in the spirit of Theorem 13.1.3(3), that bounds $\operatorname{Im} M(-x - i\eta)$ from below on $x > 0$, for some $\eta > 0$, in terms of computable data and a zero free region? Such a bound is a positivity statement at a level below one, so by statement 1 of Section 1.4 it would be at least as strong as a zero free half plane.

PROBLEM 6.4. For which even weights w with $w(0) = 1$, other than $\mathbb{E}\cosh(uB)$, does $w\Phi$ have a Fourier transform with only real zeros under RH? Proposition 6.5.1(iv) shows that the weight $e^{-|u|}$ fails, and the same argument covers every weight that is smooth on $[0, \infty)$ with $w'(0+) \neq 0$. Is smoothness at the origin together with $\log w$ convex sufficient?

Does the topic give a path to RH? No. Riemann's continuation supplies a positive Gamma(2) convolution, Williams' variable, and a signed Gamma(2) mixture, Riemann's kernel. Every closure operation that is a theorem preserves the first, and none is known to reach the clock. The connecting step is a harmonic mean of GGC transforms, and making it a GGC is equivalent to RH. Stefanski's mixture gives the shape that RH predicts, not a proof of it. The sine squared route from the heat trace gives a positive Thorin density with no hypothesis, but for the positive half of the reciprocal law, and RH reappears as the location of the poles of its continuation. The de Bruijn tilt is a normal mixture of basepoint moves, and the tilt by $e^{-|u|}$ is a mixture of the dampings that cannot carry a proof. What the chapter shows is where the single nonclosure step sits. It is the same step that the descent in level of Section 1.4 faces, and it needs the Euler product.

Chapter 7

The reciprocal Ξ function as an extended generalised gamma convolution

This chapter reads $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$, the reciprocal of Ξ on the imaginary axis, as the characteristic function of a normal scale mixture and asks what the z distributions of Barndorff-Nielsen, Kent and Sørensen [19] add to that reading. Section 13.1.1 proved that this function is, with no hypothesis, the characteristic function of $Y = Z\sqrt{2T}$, where Z is standard normal and T is the clock, and Theorem 13.1.2 that Y is an extended generalised gamma convolution (EGGC) exactly when RH holds. We do not repeat those results. We add four things: the z laws as the explicit model of the same structure, with their mixing laws and their Fourier duality with the Meixner laws; the archimedean factor of ξ written through one z law; the de Bruijn Newman flow as a one parameter family of laws, EGGC exactly for $t \geq \Lambda$, that starts, after scaling, at a z law and ends at Y; and the impossibility, short of RH, of an unconditional EGGC family converging to the law of Y, together with the best unconditional substitute.

The main results are these. The symmetric z law $z(c,c)$ is the EGGC $\sum_k L_k/(c+k)$, and its density and that of the Meixner law of order c are, up to constants, each the characteristic function of the other (Proposition 7.2.1, proved). The pair formed by the characteristic function of $z(c,c)$ and the reciprocal of its moment generating function is a van Dantzig pair exactly when $c \geq \frac{1}{2}$; the gamma factor of ζ sits at $c = \frac{1}{4}$, where it is not (Proposition 7.2.2, proved). On the critical line the archimedean factor of Ξ^2 is the characteristic function of $X/2$ with $X \sim z(\frac{1}{4},\frac{1}{4})$, and after it is removed the arithmetic remainder of $Y + Y'$ has negative mean, so it is not a law on $[0,\infty)$ (Proposition 7.3.1, proved and computed). The mixing law that makes Y an EGGC is the GGC with Thorin measure $\sum_{\gamma>0} \delta_{\gamma^2}$, and its existence is RH (Theorem 13.1.2 and Proposition 2.1.1). Along the heat flow, Y_t is an EGGC exactly when $t \geq \Lambda$, so unconditionally for $t \geq 0.2$, and Y_t/u_t tends to a scaled $z(\frac{1}{2},\frac{1}{2})$ law as $t \to \infty$ (Theorem 7.6.1, proved). Finally the truncated families over verified zeros are EGGC laws with no hypothesis and approximate Y to within $v^4 \times 2.3 \times 10^{-37}$ in

the logarithm of the characteristic function for $|v| \le 2 \times 10^{12}$ (Proposition 7.7.2, proved).

Against the path of Section 1.4 the chapter is a change of coordinates and not a new route. The z laws make the archimedean half of the problem completely explicit, and they show that the arithmetic half enters as a deconvolution, the same obstruction that Theorem 10.2.15 meets. The heat flow gives EGGC laws arbitrarily close in time to Y only down to $t = \Lambda$, and Rodgers and Tao put $\Lambda \ge 0$ [318], so the last step is the closed condition $\Lambda \le 0$, which is RH.

7.1 The z distributions as normal variance mean mixtures

For $a, b > 0$ the z distribution $z(a, b)$ has density

$$f_{a,b}(x) = \frac{e^{ax}}{B(a,b)\,(1+e^x)^{a+b}}, \qquad x \in \mathbb{R}. \tag{7.1.1}$$

It is the law of $\log(G_a/G_b)$ for independent gamma variables of shapes a and b; Fisher's z, $\frac{1}{2}\log F$, is a scaled and shifted case. We have read only the title and the bibliographic record of [19] (International Statistical Review 50, 1982, 145 to 159), and not its text. Its title says that the z laws are normal variance mean mixtures; the mixing law below is derived here, and we make no claim that it is written in this form there.

Proposition 7.1.1. *Let $X \sim z(a, b)$.*

(i) For $-a < \operatorname{Re} s < b$, $\mathbb{E}e^{sX} = B(a+s, b-s)/B(a,b) = \Gamma(a+s)\Gamma(b-s)/(\Gamma(a)\Gamma(b))$.

(ii) For the same s,

$$\mathbb{E}e^{sX} = \prod_{k\ge 0} \frac{(a+k)(b+k)}{(a+k+s)(b+k-s)}.$$

(iii) $X \overset{d}{=} \mu W + \sqrt{W}\,Z$ with $\mu = \frac{a-b}{2}$, Z standard normal and W independent of Z with

$$W = \sum_{k\ge 0} \frac{2E_k}{(a+k)(b+k)},$$

the E_k independent standard exponential variables. So W is a GGC with unit Thorin atoms at the points $\frac{1}{2}(a+k)(b+k)$, and X is a normal variance mean mixture.

(iv) In the symmetric case $a = b = c$, $X \overset{d}{=} \sum_{k\ge 0} L_k/(c+k)$ with independent standard Laplace variables L_k, and X is a symmetric EGGC with Thorin measure $\sum_{k\ge 0} \delta_{c+k}$ in the chart θ; its characteristic function is $|\Gamma(c+iv)|^2/\Gamma(c)^2$.

Proof. (i) The substitution $y = e^x/(1+e^x)$ turns $\int e^{sx} f_{a,b}(x)\,dx$ into $B(a,b)^{-1}\int_0^1 y^{a+s-1}(1-y)^{b-s-1}\,dy$. (ii) By the Weierstrass product $1/\Gamma(z) = ze^{\gamma_E z}\prod_{n\ge1}(1+z/n)e^{-z/n}$, the exponential factors of $\Gamma(a+s)/\Gamma(a)$ and $\Gamma(b-s)/\Gamma(b)$ are $e^{\mp\gamma_E s}\prod_n e^{\pm s/n}$ and cancel in the product, and what remains is the displayed product, which converges absolutely because its factors are $1+O(k^{-2})$. (iii) Expanding, $(a+k+s)(b+k-s) = (a+k)(b+k) - 2\mu s - s^2$, so each factor of (ii) is $(1-\lambda_k)^{-1}$ with $\lambda_k = 2(\mu s + s^2/2)/((a+k)(b+k))$, the moment generating function of $2E_k/((a+k)(b+k))$ at $\mu s + \frac12 s^2$. Hence $\mathbb{E}e^{sX} = \mathbb{E}e^{W(\mu s+s^2/2)} = \mathbb{E}e^{s(\mu W+\sqrt{W}Z)}$ for real s near 0, and two laws whose moment generating functions agree on an interval are equal. (iv) With $\mu = 0$, $\mathbb{E}e^{ivX} = \mathbb{E}e^{-v^2W/2} = \prod_k(1+v^2/(c+k)^2)^{-1}$, the characteristic function of $\sum_k L_k/(c+k)$, and it equals $\Gamma(c+iv)\Gamma(c-iv)/\Gamma(c)^2$ by (i) at $s = iv$. The EGGC statement is the case $H = W/2$ of the criterion of Section 4.1.2. □

A computation checks (i), (ii) and (iii) at $a = 0.7$, $b = 1.9$, $s = 0.4$: quadrature of the density, the product and the mixture all give 0.675340919623505, with relative differences below 10^{-31} (computed, 30 digits).

The symmetric z law is thus the arithmetic progression case of a structure the book meets repeatedly: a sum $\sum_k L_k/c_k$ of scaled Laplace variables, whose characteristic function $\prod_k(1+v^2/c_k^2)^{-1}$ is the reciprocal of an even entire function of genus zero in v^2 with only imaginary zeros $\pm ic_k$. The classical pairs of Section 2.1 are the cases $c_k = (k-\frac12)\pi$ and $c_k = k\pi$, and under RH the law of Y is the case $c_k = \gamma_k$ (Theorem 13.1.2). What distinguishes the z case is that the product closes in gamma functions, so everything can be computed and nothing is conditional.

7.2 Meixner duality and van Dantzig pairs

The Meixner law of order $c > 0$, which Section 10.2.3 uses at $c = \frac14$, has the density

$$m_c(x) = \frac{2^{2c}\,|\Gamma(c+ix)|^2}{2\pi\,\Gamma(2c)}, \qquad x \in \mathbb{R},$$

and characteristic function $\cosh(u/2)^{-2c}$, by the Fourier integral quoted before (10.2.3). In the normalisation of (10.2.3), $p(\tau) = \frac12 m_{1/4}(\tau/2)$ and $\int p(\tau)e^{iv\tau}d\tau = \cosh(v)^{-1/2}$.

Proposition 7.2.1 (Meixner duality). *For $c > 0$, the density of $z(c,c)$ is $f_{c,c}(x) = (2\cosh\frac{x}{2})^{-2c}/B(c,c)$, a constant multiple of the characteristic function of the Meixner law of order c, and the characteristic function of $z(c,c)$ is $|\Gamma(c+iv)|^2/\Gamma(c)^2$, a constant multiple of the Meixner density $m_c(v)$. Both laws are symmetric EGGC laws: $z(c,c)$ with Thorin measure $\sum_{k\ge0}\delta_{c+k}$ and the Meixner law with Thorin measure $2c\sum_{k\ge0}\delta_{(2k+1)\pi}$, both in the chart θ. For $c = \frac12$ the two coincide up to scale with the hyperbolic secant law.*

Proof. $e^{cx}/(1+e^x)^{2c} = (e^{-x/2}+e^{x/2})^{-2c}$ gives the density. The characteristic function is Proposition 7.1.1(iv). For the Meixner law, $\cosh(u/2) = \prod_{k\ge0}(1+u^2/((2k+1)\pi)^2)$, so $\cosh(u/2)^{-2c}$ is $\exp\{-\int \log(1+u^2/\theta^2)\, m(d\theta)\}$ with $m = 2c\sum_k \delta_{(2k+1)\pi}$, a positive measure, which makes the law a symmetric EGGC (Section 13.1.3); at $c = \frac14$ this is Proposition 10.2.8(i) in the variable $v = u/2$. At $c = \frac12$, $f_{1/2,1/2}(x) = (2\pi\cosh\frac{x}{2})^{-1}$ and $m_{1/2}(x) = \operatorname{sech}(\pi x)$. □

A computation checks that $m_{0.7}$ has mass one and that its characteristic function at $u = 1.1$ is 0.817206548790001, equal to $\cosh(0.55)^{-1.4}$ and to the product over $(2k+1)\pi$; and that the characteristic function of $z(\frac14,\frac14)$ at $v = 0.8$ is 0.044640877021, equal to $|\Gamma(\frac14+0.8i)|^2/\Gamma(\frac14)^2$ and to $2^{3/2}\pi^{3/2}p(1.6)/\Gamma(\frac14)^2$ (computed; the agreement is to 3×10^{-16}, the accuracy of the oscillatory quadrature).

The duality explains a remark of Section 10.2.3. There the Meixner density p is the archimedean part of Ξ^2 on the critical line. By Proposition 7.2.1 the same function of v is the characteristic function of a z law, so the archimedean part of Ξ^2 is at once a density and a characteristic function, and the two readings are exchanged by the Fourier transform.

The van Dantzig pairs of Section 2.1 pair a characteristic function φ_1 with $\varphi_2(\theta) = 1/\varphi_1(i\theta)$. The z laws give a family in which one can see when such a pair exists.

Proposition 7.2.2. *Let $c > 0$ and $\varphi_2(v) = |\Gamma(c+iv)|^2/\Gamma(c)^2$, the characteristic function of $z(c,c)$. There is a characteristic function φ_1 with $\varphi_2(\theta) = 1/\varphi_1(i\theta)$ for all real θ if and only if $c \ge \frac12$. Then $\varphi_1(v) = \Gamma(c)^2/(\Gamma(c+v)\Gamma(c-v))$; for $c > \frac12$ it is the characteristic function of 2Θ, where Θ has density proportional to $\cos^{2c-2}\theta$ on $(-\frac\pi2,\frac\pi2)$, and for $c = \frac12$ it is $\cos(\pi v)$, the characteristic function of a symmetric sign times π.*

Proof. If φ_1 exists, then $\varphi_1(i\theta) = \Gamma(c)^2/(\Gamma(c+i\theta)\Gamma(c-i\theta))$ is finite for every real θ, so the law of φ_1 has all exponential moments and φ_1 is entire; the right side is entire in $i\theta$ as well, so $\varphi_1(w) = \Gamma(c)^2/(\Gamma(c+w)\Gamma(c-w))$ for every complex w. For real $v > c$, the reflection formula $\Gamma(c-v)\Gamma(1-c+v) = \pi/\sin(\pi(c-v))$ gives

$$\varphi_1(v) = \frac{\Gamma(c)^2}{\pi}\,\frac{\Gamma(v+1-c)}{\Gamma(v+c)}\,\sin\big(\pi(c-v)\big),$$

and $\Gamma(v+1-c)/\Gamma(v+c) \sim v^{1-2c}$. For $c < \frac12$ this is unbounded along $v = c+n+\frac12$, so φ_1 is not a characteristic function. For $c > \frac12$, Cauchy's beta integral $\int_{-\pi/2}^{\pi/2}\cos^\mu\theta\, e^{i\nu\theta}d\theta = \pi\Gamma(\mu+1)/(2^\mu\Gamma(1+\frac{\mu+\nu}{2})\Gamma(1+\frac{\mu-\nu}{2}))$, valid for $\mu > -1$ [256, Section 5.12], at $\mu = 2c-2$ and $\nu = 2v$ and divided by its value at $\nu = 0$, gives $\mathbb{E}e^{2iv\Theta} = \varphi_1(v)$. For $c = \frac12$, $\varphi_1(v) = \pi/(\Gamma(\frac12+v)\Gamma(\frac12-v)) = \cos\pi v$. □

At $c = 0.8$, $v = 0.7$ the quadrature of $\mathbb{E}e^{2iv\Theta}$ and the gamma ratio agree to 10^{-18}; at $c = \frac14$ the candidate φ_1 takes the values 9.358, −9.589, 29.59, −29.66 at $v = 10$, 10.5, 100, 100.5, growing like $v^{1/2}$ as the proof says (computed).

Remark 7.2.3. The threshold $c = \frac{1}{2}$ is the hyperbolic secant law, the case $(\cos\theta, 1/\cosh\theta)$ of Section 2.1 up to scale. The gamma factor $\Gamma(\frac{1}{4} + \frac{s}{2})$ of ξ corresponds to $c = \frac{1}{4}$, below the threshold: its z law is an EGGC and the second member of no pair, in the same direction as Proposition 10.2.10(iii), which says that the dual Meixner density at the same parameter is not a Pólya frequency function (we do not derive either statement from the other). The Ξ pair, $(\Xi(v)/\Xi(0),\ \xi(\frac{1}{2})/\xi(\frac{1}{2}+v))$, exists with no hypothesis (Theorem 8.1.1). So the van Dantzig property of the full Ξ pair is not inherited from its archimedean factor; it is produced by the zeta factor and the polynomial factor together.

7.3 The archimedean factor of ξ through a z law

Let $X \sim z(\frac{1}{4}, \frac{1}{4})$, so that by Proposition 7.1.1, $X/2 = \sum_{k\geq 0} L_k/c_k$ with $c_k = 2k + \frac{1}{2}$, the constants of Theorem 10.2.1, and $\mathrm{Var}(X/2) = 2\sum_k c_k^{-2} = \frac{1}{2}\psi'(\frac{1}{4}) = 8.59866457725$ (computed). Put $H_\infty = \sum_k E_k/c_k^2$, a GGC with unit Thorin atoms at c_k^2 and mean $\frac{1}{4}\psi'(\frac{1}{4}) = 4.299332289$, so that $X/2 \stackrel{d}{=} Z\sqrt{2H_\infty}$. Let Y' be an independent copy of $Y = Z\sqrt{2T}$, and $\tilde{T}$ the corresponding copy of T.

Proposition 7.3.1. *(i) For real $|s| < \frac{1}{2}$,*

$$\frac{\xi(\frac{1}{2}+s)^2}{\xi(\frac{1}{2})^2} = (1-4s^2)^2\,\mathbb{E}e^{sX/2}\,\frac{\zeta(\frac{1}{2}+s)\zeta(\frac{1}{2}-s)}{\zeta(\frac{1}{2})^2},$$

and on the critical line, for real v,

$$\frac{\Xi(v)^2}{\Xi(0)^2} = (1+4v^2)^2\,\mathbb{E}e^{ivX/2}\,\frac{|\zeta(\frac{1}{2}+iv)|^2}{\zeta(\frac{1}{2})^2}.$$

(ii) Let $D_0 = T + \tilde{T} - H_\infty$, with T, $\tilde{T}$ and H_∞ independent. For $0 \leq w < \frac{1}{4}$,

$$\mathbb{E}e^{-wD_0} = \frac{\zeta(\frac{1}{2})^2}{(1-4w)^2\,\zeta(\frac{1}{2}+\sqrt{w})\,\zeta(\frac{1}{2}-\sqrt{w})}.$$

(iii) $\mathbb{E}D_0 = 2\mathbb{E}T - \frac{1}{4}\psi'(\frac{1}{4}) = -8 + (\log\zeta)''(\frac{1}{2}) = -4.2531223024$. Consequently there is no law Q on $[0,\infty)$, independent of H_∞, with $T + \tilde{T} \stackrel{d}{=} H_\infty + Q$, and no random variable R independent of X with $Y + Y' \stackrel{d}{=} X/2 + R$.

Proof. (i) By definition $\xi(\frac{1}{2}+s) = \frac{1}{2}(s^2 - \frac{1}{4})\pi^{-1/4-s/2}\Gamma(\frac{1}{4}+\frac{s}{2})\zeta(\frac{1}{2}+s)$. Multiply by the same expression at $-s$, which is $\xi(\frac{1}{2}-s) = \xi(\frac{1}{2}+s)$, divide by the value at $s = 0$, and use $\Gamma(\frac{1}{4}+\frac{s}{2})\Gamma(\frac{1}{4}-\frac{s}{2})/\Gamma(\frac{1}{4})^2 = \mathbb{E}e^{sX/2}$, which is Proposition 7.1.1(i) at $a = b = \frac{1}{4}$ and argument $s/2$. The second display is the first at $s = iv$, both sides being analytic in the strip $|\operatorname{Re} s| < \frac{1}{2}$, and $\Xi(v) = \xi(\frac{1}{2}+iv)$, $\overline{\zeta(\frac{1}{2}+iv)} = \zeta(\frac{1}{2}-iv)$.

(ii) $\mathbb{E}e^{-w(T+\tilde{T})} = \xi(\frac{1}{2})^2/\xi(\frac{1}{2}+\sqrt{w})^2$ and $\mathbb{E}e^{wH_\infty} = \prod_k (1-w/c_k^2)^{-1} = \mathbb{E}e^{\sqrt{w}X/2}$ for $w < c_0^2 = \frac{1}{4}$; insert (i) at $s = \sqrt{w}$. (iii) The first expression for the mean is $\mathbb{E}T + \mathbb{E}\tilde{T} - \mathbb{E}H_\infty$. For the second, the logarithm of the right side of (ii) is a constant minus $2\log(1-4w)$ minus $g(\sqrt{w})$, with $g(x) = \log\zeta(\frac{1}{2}+x) + \log\zeta(\frac{1}{2}-x)$ even and $g''(0) = 2(\log\zeta)''(\frac{1}{2})$, so its derivative at $w=0$ is $8-(\log\zeta)''(\frac{1}{2})$, and the mean is minus this. If $T+\tilde{T} \stackrel{d}{=} H_\infty + Q$, then $\mathbb{E}Q = \mathbb{E}D_0 < 0$, which is impossible for $Q \geq 0$. If $Y + Y' \stackrel{d}{=} X/2 + R$, then $R \in L^2$, because $X/2 + R \in L^2$ with $X/2$ independent of R, and $\operatorname{Var} R = 4\mathbb{E}T - \frac{1}{2}\psi'(\frac{1}{4}) < 0$, since $4\mathbb{E}T = 0.0924$. □

A computation checks (i) at $s = 0.3$ (both sides 1.00416725669921) and at $v = 5$ (both sides 0.255983368484996, in the form $|\zeta(\frac{1}{2}+5i)/\zeta(\frac{1}{2})|^2$), (ii) at $w = 0.1$ (both sides 1.70964296601762), and the two expressions for the mean in (iii), each to better than 10^{-30} (computed).

So the archimedean factor of ξ^2 on the real axis is the moment generating function of a z law, and at $w = s^2$ it is $\mathbb{E}e^{wH_\infty}$, with H_∞ the GGC with Thorin atoms c_k^2. The factor $(1-4s^2)^2$ is the polar part, and the rest is arithmetic. Part (iii) is the z law form of the obstruction met by Theorem 10.2.15 and Proposition 10.2.16: the archimedean law is far wider than Y, with variance 8.6 against 0.046, so the zeta factor does not add an independent component to the archimedean one but removes most of it. The EGGC question for Y is therefore not a question about a product of two EGGC laws, and no factorisation through the z law can make it one. Only the polar and zeta factors together carry the information, and the GGC property of their combination is RH.

7.4 Which mixing law, and why it is RH

For real u, $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u) = \mathbb{E}e^{-u^2T}$ with no hypothesis (Proposition 13.1.1). A symmetric law is an EGGC exactly when it is $Z\sqrt{2H}$ with H a GGC (Section 4.1.2), and the mixing law is determined by the characteristic function, since $v \mapsto \mathbb{E}e^{-v^2H}$ is the Laplace transform of H on $[0,\infty)$. So the only candidate mixing law is the clock itself: Y is an EGGC if and only if T is a GGC, and by Proposition 2.1.1 that holds if and only if RH holds, the Thorin measure being then $\sum_{\gamma>0}\delta_{\gamma^2}$ and $Y \stackrel{d}{=} \sum_{\gamma>0} L_\gamma/\gamma$ (Theorem 13.1.2).

In the language of Section 7.1, RH says that Y is a generalised symmetric z law, $\sum_k L_k/c_k$ with $c_k = \gamma_k$ in place of $c+k$. The comparison shows what changes. For $z(c,c)$ the points $c+k$ are given and the product $\prod_k(1+v^2/(c+k)^2)$ is recognised as $\Gamma(c)^2/|\Gamma(c+iv)|^2$; for Y the function $\xi(\frac{1}{2}+v)/\xi(\frac{1}{2})$ is given and the positivity of the points $a_k = -(\rho_k - \frac{1}{2})^2$ is the question. The z law provides no mechanism for the second direction.

Remark 7.4.1. Three weaker mixing statements hold with no hypothesis, and none implies RH. T is infinitely divisible and self decomposable (Theorem 8.1.1),

so Y is a self decomposable normal scale mixture (Result 4.1.2). T lies in Urbanik's class L_2 and not in L_3 (Theorem 14.4.8). And the floor $e^{-c\sqrt{u}}/G(u)$ adds to T a positive $\frac{1}{2}$ stable component, with Thorin density $c/(2\pi\sqrt{t})$, which together with the archimedean factor gives the Thorin density of Theorem 10.2.12 (Theorem 10.2.15); in the variable $v = \sqrt{u}$ this multiplies the characteristic function of Y by the Cauchy factor $e^{-c|v|}$.

7.5 The 2 sin squared lemma on the z laws

The lemma (2.1.9) converts a Lévy representation in $|u|$ into a Thorin representation in u^2, as in the derivation of (13.1.2). For the z laws and the gamma function it works with positive input.

Proposition 7.5.1. *(i) The law $z(c,c)$ has Lévy density $k_c(|x|)/|x|$ with $k_c(r) = e^{-cr}/(1-e^{-r}) = \sum_{n\ge 0} e^{-(c+n)r}$, completely monotone, and the sine squared image of k_c, $\frac{1}{\pi}\int_0^\infty 2\sin^2(x\theta/2)\,k_c(x)\,dx$, is $\frac{1}{\pi}\sum_n \theta^2/((c+n)((c+n)^2+\theta^2))$.*

(ii) For $a > 0$ and real u,

$$e^{\psi(a)|u|}\,\frac{\Gamma(a)}{\Gamma(a+|u|)} = \exp\Big\{-\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big)\big(\operatorname{Re}\psi(a+i\theta)-\psi(a)\big)\,d\theta\Big\},$$

and $\operatorname{Re}\psi(a+i\theta)-\psi(a) > 0$ for $\theta \neq 0$. So the left side is the characteristic function of a symmetric EGGC, and $\Gamma(a)/\Gamma(a+|u|)$ is the product of $e^{-\psi(a)|u|}$, a Cauchy factor when $\psi(a) > 0$ (that is $a > 1.4616$), and that EGGC.

Proof. (i) Malmsten's formula $\log\Gamma(c+s) - \log\Gamma(c) - s\psi(c) = \int_0^\infty (e^{-sx} - 1 + sx)\frac{e^{-cx}}{x(1-e^{-x})}dx$ for $\operatorname{Re} s > -c$ [256, Section 5.9], applied to both gamma factors of Proposition 7.1.1(i) with $a = b = c$, puts $\log \mathbb{E}e^{ivX}$ in Lévy Khintchine form with Lévy density $k_c(|x|)/|x|$, the linear terms cancelling. The image is computed termwise from $\int_0^\infty (1-\cos x\theta)e^{-\kappa x}dx = \theta^2/(\kappa(\kappa^2+\theta^2))$. (ii) Malmsten's formula is a Lévy representation of $\log\Gamma(a+s)/\Gamma(a)$ with drift $\psi(a)$ and canonical measure $xL(dx) = e^{-ax}(1-e^{-x})^{-1}dx \ge 0$. Integrating (2.1.9) against L and using $\int_0^\infty (1-\cos x\theta)\frac{e^{-ax}}{1-e^{-x}}dx = \operatorname{Re}\psi(a+i\theta) - \psi(a)$, from $\psi(z) - \psi(a) = \int_0^\infty \frac{e^{-ax}-e^{-zx}}{1-e^{-x}}dx$, gives the display at $s = |u|$ in the chart θ, with the exchange of integrals justified by positivity (Tonelli). The integrand of the last formula is positive for $\theta \neq 0$. The EGGC statement is the criterion of Section 13.1.3. □

The integral representation of $\log\Gamma$ used in the proof was checked at $c = 0.3$, $s = 1.7$: both sides are 4.85849318292215, to 3×10^{-51} at 50 digits (computed).

At $u = 2$ the two sides of (ii) agree to 10^{-17} at $a = \frac{1}{4}$ (value $6.81130761057 \times 10^{-4}$) and at $a = 3$ (value 0.527641617963) (computed, adaptive quadrature at 30

digits). At $a = \frac{1}{4}$ and with $|u|/2$ in place of $|u|$, (ii) is Theorem 10.2.12 on the real axis.

For the gamma factor the lemma therefore converts a positive Lévy measure into a positive Thorin density, with no hypothesis. For the full ξ at the centre the input is the signed primal measure (10.2.5), whose sine squared image is positive exactly under RH (Proposition 10.2.16). Proposition 7.3.1(iii) is the same fact on the side of laws: the positive archimedean input is overwhelmed by the negative polar input $-2\cosh(x/2)\,dx$, and the primes have to restore positivity.

7.6 The heat flow as the natural EGGC family

Let H_t, $\xi_t(s) = 8H_t(-2i(s-\frac{1}{2}))$ and $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ be as in Section 4.1, so that $\xi_0 = \xi$ and $R_0 = 1/G$. Since $\xi_t(\frac{1}{2}+v) = 8H_t(-2iv) = 8H_t(2iv)$, for real v

$$R_t(v^2) = \frac{\xi_t(\frac{1}{2})}{\xi_t(\frac{1}{2}+v)} = \frac{\mathbb{E}\,\xi(\frac{1}{2}+\sqrt{t/2}\,Z)}{\mathbb{E}\,\xi(\frac{1}{2}+v+\sqrt{t/2}\,Z)} \qquad (t>0), \tag{7.6.1}$$

the second form by Proposition 4.1.18.

Theorem 7.6.1. *(i) For every real t, $H_t(iy) > 0$ for real y. If R_t is the Laplace transform of a law T_t on $[0,\infty)$, then $v \mapsto R_t(v^2)$ is the characteristic function of the symmetric law $Y_t = Z\sqrt{2T_t}$.*

(ii) For $t \geq \Lambda$ the law T_t exists, and $v \mapsto R_t(v^2)$ is the characteristic function of a symmetric EGGC if and only if $t \geq \Lambda$, and then $Y_t \overset{d}{=} \sum_k L_k/\theta_k(t)$ over the positive zeros $\frac{1}{2} \pm i\theta_k(t)$ of ξ_t. In particular Y_t is an EGGC for every $t \geq 0.2$, with no hypothesis, and $Y_0 = Y$ is an EGGC if and only if RH holds.

(iii) With u_t the mode of Proposition 4.1.19, $Y_t/u_t \Rightarrow \frac{2}{\pi}X_{1/2}$ as $t \to \infty$, where $X_{1/2} \sim z(\frac{1}{2},\frac{1}{2})$; the limit has characteristic function $1/\cosh(2v)$.

Proof. (i) $H_t(iy) = \int_0^\infty e^{tw^2}\Phi_{\mathrm{dBN}}(w)\cosh(yw)\,dw > 0$ because $\Phi_{\mathrm{dBN}} > 0$ (Section 4.1), and $\mathbb{E}e^{ivZ\sqrt{2T_t}} = \mathbb{E}e^{-v^2T_t} = R_t(v^2)$. (ii) For $t \geq \Lambda$, Theorem 17.1.16 gives $R_t(u) = \prod_k (1+u/\theta_k(t)^2)^{-1}$, a GGC transform, so $T_t = \sum_k E_k/\theta_k(t)^2$ and $Y_t = Z\sqrt{2T_t}$ is the EGGC $\sum_k L_k/\theta_k(t)$. Conversely, if $R_t(v^2)$ is the characteristic function of an EGGC $Z\sqrt{2H}$ with H a GGC, then $\mathbb{E}e^{-v^2H} = R_t(v^2)$, so R_t is a GGC transform; it is then holomorphic and nonvanishing on $\mathbb{C}\setminus(-\infty,0]$, so $H_t(2i\sqrt{u})$ has no zero there, and every zero z of H_t, being a zero of $H_t(2i\sqrt{u})$ at $u = -z^2/4$, has $-z^2/4 \in (-\infty,0]$, that is z real; so $t \geq \Lambda$ by the definition of Λ. The bound $\Lambda \leq 0.2$ is [277]. At $t = 0$ the statement is Theorem 13.1.2. (iii) By Proposition 4.1.19, $T_t/u_t^2 \Rightarrow 2C$ with $\mathbb{E}e^{-\lambda\,2C} = 1/\cosh(2\sqrt{\lambda})$,

so $\mathbb{E}e^{ivY_t/u_t} = \mathbb{E}e^{-v^2T_t/u_t^2} \to 1/\cosh(2v)$. By Proposition 7.2.1, $X_{1/2}$ has characteristic function $|\Gamma(\frac{1}{2}+iv)|^2/\pi = 1/\cosh(\pi v)$, so $\frac{2}{\pi}X_{1/2}$ has $1/\cosh(2v)$, and Lévy's continuity theorem applies. □

The heat flow is therefore a path of laws that runs, after scaling, from a z law at $t = \infty$ to the law of Y at $t = 0$, and stays inside the EGGC class exactly on $t \geq \Lambda$. It is natural in three senses. It is the only one of the three families of Chapter 8 that changes the zeros and not only the basepoint or a tilt; by (7.6.1) it acts on the mixing law through a Gaussian smoothing of ξ on the real axis, which keeps the reciprocal a GGC transform whenever it is one (Theorem 4.1.20); and its Thorin atoms are the zeros of an entire function of the same type as ξ, so that every member is again a symmetric generalised z law $\sum_k L_k/\theta_k(t)$. It is close to Y where it is known to be an EGGC: from (7.6.1), $R_{0.2}(1)/R_0(1)-1 = -9.57\times10^{-5}$ (computed by adaptive quadrature of the normal averages at 30 digits, in agreement with the value -9.6×10^{-5} of Section 4.1.8).

What the 2 sin squared lemma contributes For each $t \geq \Lambda$ the Thorin measure of T_t in the chart θ is $\sum_k \delta_{\theta_k(t)}$, and by the computation of Proposition 10.2.16(ii) it is the sine squared image of the formal primal measure $-2\sum_k \cos(\theta_k(t)x)\,dx$ of $\log \xi_t(\frac{1}{2}+s)$. The lemma preserves positivity, so a positive primal measure would give the EGGC property at once. At $t = 0$ the primal measure has the explicit form (10.2.5), with a negative polar part, and it is not positive. For $t \neq 0$ no prime side is known (Section 17.1, after Theorem 17.1.16), because no Euler product for H_t is known; so for $t > 0$ the lemma gives no independent route to positivity, and the EGGC property along the flow rests on the reality of the zeros of H_t, proved by de Bruijn's theorem and its refinements [73, 277]. What the lemma does contribute is the identification at every level: the Thorin density of each member is the sine squared image of a Lévy density, and for the z laws and the gamma factor that Lévy density is positive (Proposition 7.5.1).

Modulated normal laws The laws Y_t are normal laws whose variance is modulated by an independent random factor $2T_t$. Romanowski's modulated normal laws of geodesy are, according to the abstracts of McLane [375] and of Platt and Paprzycki [376], a family that tends to the normal law as its modulating parameter grows and that can be approximated by mixtures of normal laws; we have not read Romanowski's papers and use the phrase only in the generic sense of a normal scale mixture. In that sense the flow modulates the normal law by the clock, and the modulation is a GGC exactly when $t \geq \Lambda$.

7.7 An unconditional approximating family

The task of finding an unconditional family of EGGC laws that converges to the law of Y is, as stated, equivalent to RH.

Proposition 7.7.1. *If there are symmetric EGGC laws Y_n with $Y_n \Rightarrow Y$, then RH holds. Conversely, under RH the constant family $Y_n = Y$ is such a family.*

Proof. Write $Y_n \overset{d}{=} Z\sqrt{2H_n}$ with H_n a GGC. Then $\mathbb{E}e^{-v^2 H_n} \to \mathbb{E}e^{-v^2 T}$ for every real v, so the Laplace transforms of H_n converge on $[0,\infty)$ to that of T, which tends to one at the origin, and $H_n \Rightarrow T$ by the continuity theorem for Laplace transforms. The GGC class is closed under convergence in law (Section 9.1.1), so T is a GGC, and Proposition 2.1.1 gives RH; this is Corollary 9.1.2 in the variable $v = \sqrt{u}$. The converse is Theorem 13.1.2. □

Every unconditional family must therefore stop short of Y. The heat flow stops at $t = 0.2$; the basepoint family Y_α of Theorem 13.1.3 stops at $\alpha = 1$, or at $\alpha = \frac{7}{8}$ if the announced and unrefereed zero free half plane $\operatorname{Re} s > \frac{7}{8}$ is confirmed [257, 258, 223] (conditional); and the truncations over verified zeros stop at the verification height. The last family is the closest of the three in the topology of characteristic functions (Proposition 7.7.2).

Proposition 7.7.2. *Let $T_0 = 3 \times 10^{12}$, the height below which every zero lies on the critical line [277], and for $2 \le T' \le T_0$ put*

$$m(T') = \sum_{k:\ \gamma_k > T'} \frac{1}{a_k}, \qquad Y^{(T')} = \sqrt{2m(T')}\, Z_0 + \sum_{0<\gamma\le T'} \frac{L_\gamma}{\gamma},$$

with $a_k = -(\rho_k - \frac12)^2$ summed over the zeros with $\gamma_k > T'$, Z_0 standard normal and the L_γ standard Laplace, all independent. Then $m(T') > 0$, $Y^{(T')}$ is a symmetric EGGC with no hypothesis, $\operatorname{Var} Y^{(T')} = \operatorname{Var} Y = 2\mathbb{E}T$, *and for real v with $v^2 \le T'^2/2$,*

$$\left| \log \mathbb{E}e^{ivY} - \log \mathbb{E}e^{ivY^{(T')}} \right| \le v^4\, \epsilon(T'), \qquad \epsilon(T') = \frac{2}{\pi}\Big(\frac{\log T'}{3T'^3} + \frac{1}{9T'^3}\Big).$$

Proof. The zeros with $\gamma_k > T'$ come in groups $\rho, 1-\bar\rho$ whose values a_k are complex conjugates, so $m(T')$ is real; and $\operatorname{Re}(1/a_k) = (\gamma^2 - \delta^2)/(\gamma^2+\delta^2)^2 > 0$ for $a_k = (\gamma - i\delta)^2$ with $|\delta| < \frac12 < \gamma$, so $m(T') > 0$; the series converges absolutely since $|a_k| \ge \gamma_k^2$. Below T_0 every a_k equals $\gamma_k^2 > 0$. So $H = m(T') + \sum_{0<\gamma\le T'} E_\gamma/\gamma^2$ is a GGC with drift $m(T')$ and finitely many atoms, and $Y^{(T')} = Z\sqrt{2H}$ in law, an EGGC by the criterion of Section 4.1.2. Its variance is $2\mathbb{E}H = 2\sum_k a_k^{-1} = 2\mathbb{E}T$. For the bound, $-\log \mathbb{E}e^{ivY} = \log G(v^2) = \sum_k \log(1 + v^2/a_k)$, and $-\log \mathbb{E}e^{ivY^{(T')}} = \sum_{\gamma_k \le T'} \log(1 + v^2/\gamma_k^2) + v^2 m(T')$, so the difference is $\sum_{\gamma_k > T'} \big(\log(1+x_k) - x_k\big)$ with $x_k = v^2/a_k$. Since $|a_k| \ge \gamma_k^2 > T'^2$ and $v^2 \le T'^2/2$, $|x_k| \le \frac12$, and then

$|\log(1+x)-x| \le \sum_{n\ge 2}|x|^n/n \le |x|^2$. So the difference is at most $v^4\sum_{\gamma>T'}\gamma^{-4}$ in absolute value, summing over the zeros with $\gamma > T'$ counted once each. By partial summation this sum is at most $4\int_{T'}^{\infty} N(t)t^{-5}dt$, and $N(t) \le \frac{t}{2\pi}\log t$ for $t \ge 2$ (proof of Theorem 13.1.3), so it is at most $\frac{2}{\pi}\int_{T'}^{\infty} t^{-4}\log t\,dt = \epsilon(T')$. □

At $T' = T_0$, $\epsilon(T_0) = 2.28\times 10^{-37}$: with no hypothesis, the law of Y is within $v^4 \times 2.3\times 10^{-37}$ of an EGGC in the logarithm of its characteristic function for $|v| \le 2\times 10^{12}$. A test at a height where the zeros are at hand: with the first 300 ordinates, $T' = \gamma_{300} = 541.8474371$ and $m(T') = \mathbb{E}T - \sum_{\gamma\le T'}\gamma^{-2} = 0.0015995195$, the observed error is 4.96×10^{-7} at $v = 5$ and 1.27×10^{-4} at $v = 20$, against the bounds 5.53×10^{-6} and 1.42×10^{-3} (computed; $\mathbb{E}T = \xi''(\frac12)/(2\xi(\frac12)) = 0.0231049931154$, as in Section 2.1). The same ordinates, with the omitted zeros replaced by their exact first moment $m(T')$ and a Weyl estimate of their second, reproduce $\Xi(0)/\Xi(5) = 1.80410373065$ to 3×10^{-8} (computed; the value $2\mathbb{E}T = 0.046210$ is not an independent check, since $m(T')$ is defined from $\mathbb{E}T$).

The Gaussian component $\sqrt{2m(T')}Z_0$ is the second moment of the missing factors and is what Section 5.2.3 calls the Weyl tail at first order. Every member is a generalised z law plus a normal law. As T' increases beyond T_0 the construction gives EGGC laws exactly as long as $a_k = \gamma_k^2$ for the zeros below T', that is as long as they stay on the line, and the family converges to Y; Proposition 7.7.1 says that this convergence through EGGC laws is RH, and Proposition 7.7.2 says how small the unproved part is in the topology of characteristic functions. It is not small in the sense that matters: by Section 1.4 and Theorem 1.4.1, functions with zeros off the line above T_0 give laws just as close.

A tilted family The shifted zeros give a second check of the same structure, conditional on RH: for $|\beta| < \gamma_1$ and $u \ge 0$, under RH,

$$\frac{\Xi(\beta)}{\Xi(\sqrt{\beta^2-u})} = \prod_{\gamma>0}\Big(1+\frac{u}{\gamma^2-\beta^2}\Big)^{-1},$$

the Laplace transform at u of the clock tilted by $e^{\beta^2 T}$, whose atoms are $\gamma^2-\beta^2$. With $\beta = 5$, $u = 3$ and 300 ordinates, the tail treated in the same way, both sides are 0.93034098634 to 3.4×10^{-10} (computed). This is the Esscher tilt of Chapter 8; it moves every atom by the same amount and is uninformative for the same reason (Section 8.1.1).

7.8 Open problems

Problem 7.1. Read [19] and determine whether its mixing law for $z(a,b)$ is the GGC of Proposition 7.1.1(iii), and whether its results on the asymmetric laws, $\mu \ne 0$, have an analogue for the tilted Pólya variables V_α of Section 2.1, whose characteristic functions $\xi(\alpha+i\theta)/\xi(\alpha)$ are never infinitely divisible (Section 13.1).

Problem 7.2. Proposition 7.2.2 decides the van Dantzig property for the arithmetic progressions $c_k = c + k$. Characterise the positive sequences (c_k) with $\sum c_k^{-2} < \infty$ for which $\sum_k L_k/c_k$ is the second member of a van Dantzig pair. Under RH the ordinates γ_k are such a sequence, with first member $\Xi(v)/\Xi(0)$; the condition is that $\prod_k (1 - v^2/c_k^2)$ be a characteristic function, so it is a condition on the regularity of the sequence and not on its density alone.

Problem 7.3. Is there a one parameter family of symmetric laws, each an EGGC with no hypothesis, joining $\frac{2}{\pi} X_{1/2}$ to Y other than by the heat flow, along which the EGGC property can be followed by a computation that does not reduce to the reality of the zeros of H_t? Proposition 7.7.1 implies that such a family cannot reach Y unconditionally, so the question is which property of the family degenerates at the endpoint.

Problem 7.4. Is the Lévy density $K_t(|y|)/|y|$ of Y_t, with $K_t(y) = \sum_k e^{-\theta_k(t)y}$ for $t \geq \Lambda$, monotone in t for each fixed y? Problem 4.2 asks the same question for the stochastic order of T_t; for Y_t it would say that the flow moves Lévy mass monotonically between scales.

Problem 7.5. The variable D_0 of Proposition 7.3.1(ii) has negative mean, so $T + \tilde{T}$ does not contain H_∞ as an independent summand. For $0 \leq \lambda \leq 1$ let $H^{(\lambda)}$ be the GGC with Thorin atoms of mass λ at the points c_k^2. For which λ is $\mathbb{E}e^{-w(T+\tilde{T})}/\mathbb{E}e^{-wH^{(\lambda)}}$ the Laplace transform of a law on $[0, \infty)$? The mean gives the necessary condition $\lambda \leq 2\mathbb{E}T/\mathbb{E}H_\infty = 0.01075$, and at $\lambda = 0$ the answer is yes (Theorem 8.1.1). Is the set of such λ an interval $[0, \lambda^*]$ with $\lambda^* > 0$, and what is λ^*?

Does the topic give a path to RH? No, not by itself. The z laws give an explicit and unconditional model of the EGGC structure, and they make the archimedean factor of ξ a z law, but the arithmetic factor enters as a deconvolution of that law (Proposition 7.3.1(iii)), so positivity has to come from the primes and the pole together, which is the step that Section 1.4 identifies as a zero free half plane. The heat flow is the natural EGGC family, with a z law at one end and Y at the other, but it is known to stay in the EGGC class only for $t \geq 0.2$, and closing the gap to $t = 0$ is $\Lambda \leq 0$, which is RH. Any unconditional family of EGGC laws converging to the law of Y would prove RH (Proposition 7.7.1), and the best unconditional families stop at the verification height, at a distance that no finite computation can make zero.

Chapter 8

The heat trace and the hierarchy of positivity

Chapter 9 worked at the top of the Bondesson hierarchy; here we climb it from the bottom. Every rung is a sign condition on the heat trace $W(t)$ of Section 2.1, the pairing of the Thorin functional with e^{-tz} (not the Mellin variable W of Section 9.3). The lower rungs are unconditional theorems, proved with a finite verification of the zeros; the top rung is RH.

8.1 The heat trace and the hierarchy of positivity

The rungs of the hierarchy proved unconditionally are infinite divisibility and self decomposability of the clock, log convexity of W, positive definite Hankel blocks up to size 3×10^7, the signs of all derivatives, equivalently the positivity of the Williamson measures, up to order 1.16×10^{25}, and near the origin the signs of every derivative: W is completely monotone on $(0, 1.29]$. One scheme proves them all, with a reach set by the height to which RH has been verified; no verification reaches the top, where Section 1.4 begins. Results stated without a citation are proved here; the others are from [282, Chapters 4 and 5] and [279].

8.1.1 The heat trace and the first rungs

By Section 2.1, unconditionally,

$$\log G(u) = \int_0^\infty \bigl(1 - e^{-ut}\bigr) \frac{W(t)}{t}\, dt, \qquad W(t) = \sum_j e^{-a_j t},$$

where $a_j = -(\rho_j - \frac{1}{2})^2$ and the sum runs over one zero of each pair $\{\rho, 1-\rho\}$. So $W(t)/t$ is the only candidate for the Lévy density of the clock, and the classes of Section 4.1 become conditions on W: nonnegativity for infinite divisibility, monotonicity for self decomposability, complete monotonicity for the GGC property.

The function W is analytic in $\operatorname{Re} t > 0$, real on the positive axis, and differentiable term by term,

$$c_k(t) = (-1)^k W^{(k)}(t) = \sum_j a_j^k e^{-a_j t}.$$

Under RH every $c_k(t)$ is a sum of positive terms. A quadruple of zeros $\frac12 \pm \delta \pm i\gamma$, $0 < \delta < \frac12$, instead contributes the exponents $a = -(\delta + i\gamma)^2 = \gamma^2 - \delta^2 - 2i\delta\gamma$ and $\bar a$, hence the oscillating term $2|a|^k e^{-\operatorname{Re} a\, t}\cos(2\delta\gamma t - k\phi)$, with $|a| = \gamma^2 + \delta^2$ and $\tan\phi = 2\delta\gamma/(\gamma^2 - \delta^2)$. Its amplitude exceeds the contribution $2\gamma^{2k}e^{-\gamma^2 t}$ of the same two zeros placed at $\frac12 + i\gamma$ by the factor $(1+\delta^2/\gamma^2)^k e^{\delta^2 t} \le \exp\{k\delta^2/\gamma^2 + \delta^2 t\}$ [279, Proposition 11.1].

As $t \downarrow 0$ the archimedean term of (2.1.5) dominates. With ϑ the Riemann Siegel theta function, since $\operatorname{Re}\psi(\frac14 + \frac{iu}{2}) - \log\pi = 2\vartheta'(u)$, the explicit formula gives, with no hypothesis,

$$W(t) = e^{t/4} + \frac{1}{\pi}\int_0^\infty e^{-u^2 t}\vartheta'(u)\,du - \frac{1}{\sqrt{4\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt n}\,e^{-(\log n)^2/4t},$$

and the prime terms are $O(t^{-1/2}e^{-(\log 2)^2/4t})$, smaller than every power of t. Mellin's method for the integral [256, Section 2.5] gives, for every N and as $t \downarrow 0$,

$$W(t) = \frac{\log\frac1t - \gamma_E - 4\log 2 - 2\log\pi}{8\sqrt{\pi t}} + \sum_{n=0}^{N} d_n t^n + \sum_{k=1}^{N} b_k t^{k-\frac12} + O\big(t^{N+\frac12}\big),$$

with $d_n = (1 - \frac{E_{2n}}{8})\frac{4^{-n}}{n!}$ and $b_k = \frac{(1-2k)\Gamma(\frac12 - k)}{2\pi}\beta_k$, where γ_E is Euler's constant, E_{2n} are the Euler numbers and $\beta_k = \frac{(1-2^{1-2k})|B_{2k}|}{4k(2k-1)}$ are the coefficients of Stirling's series $\vartheta(u) \sim \frac{u}{2}\log\frac{u}{2\pi} - \frac{u}{2} - \frac{\pi}{8} + \sum_k \beta_k u^{1-2k}$. The terms $b_k t^{k-\frac12}$ come from the expansion of ϑ' at infinity, and the terms $d_n t^n$ from the Taylor series of $e^{-u^2 t}$ through the Hurwitz value $\zeta(-2n, \frac14) = E_{2n}/4^{2n+1}$, together with the coefficients $4^{-n}/n!$ of $e^{t/4}$. After the leading term the series is $\frac78 + \frac{\sqrt t}{48\sqrt\pi} + \frac{9}{32}t - \frac{7t^{3/2}}{2880\sqrt\pi} + \frac{3}{256}t^2 + \frac{31t^{5/2}}{60480\sqrt\pi} + \frac{69}{3072}t^3 + \cdots$, and it diverges; $\frac78 = d_0$ is the constant of the Riemann von Mangoldt formula. The error of the two term expansion (the leading term and $\frac78$) is $b_1\sqrt t + d_1 t + O(t^{3/2})$, which is 6.5×10^{-4}, 1.5×10^{-4} and 4.0×10^{-5} at $t = 10^{-3}, 10^{-4}, 10^{-5}$ [279, Corollary 5.5]; with the terms through $t^{5/2}$ the remainder agrees with $\frac{69}{3072}t^3$ to the order of the next term, $t^{7/2}$, in a 40 digit computation for $10^{-6} \le t \le 10^{-3}$, and at $t = 10^{-2}$ it is -8.377×10^{-6}, the prime 2. Consequently $\int_\tau^\infty W(t)\,dt/t = \frac{\log\frac1\tau - \gamma_E - 4\log 2 - 2\log\pi - 2}{4\sqrt{\pi\tau}} + \frac78\log\frac1\tau + O(1)$ as $\tau \downarrow 0$, so the clock has infinitely many jumps. As $t \to \infty$ the first zero dominates, $W(t) = e^{-\gamma_1^2 t}(1 + o(1))$ with $\gamma_1^2 = 199.7904548$. In between, the arithmetic terms cancel to a depth that grows linearly in t (Table 2.1).

THEOREM 8.1.1 ([282, Theorem A]). *Unconditionally, that is, using only the verification of RH below height* 1000 *[277], $W(t) > 0$ and $W'(t) < 0$ for every $t > 0$. Consequently the clock is the Laplace transform of an infinitely divisible, self decomposable law with Lévy density $W(t)/t$, and $(\Xi(\theta)/\Xi(0),\ \xi(\frac12)/\xi(\frac12+\theta))$ is a van Dantzig pair in the sense of Section 2.1.*

Proof. We follow [282, Theorem A] with explicit constants. Let $\tau = 1/20000$, a constant of this section only, $\ell = -\frac12\log t$, and $s_k(t) = \Gamma(k+\frac12)t^{-k-\frac12}/(4\pi)$, which is $(-1)^k$ times the kth derivative of $s_0(t) = 1/(4\sqrt{\pi t})$. It suffices to show $c_0, c_1 > 0$, for $t \le \tau$ from (2.1.5) and for $t \ge \tau$ from the zeros.

Let A and P be the archimedean and prime terms of (2.1.5). The series $\psi(z) = -\gamma_E + \sum_{n\ge0}\left(\frac{1}{n+1} - \frac{1}{n+z}\right)$ [256, (5.7.6)] shows that m increases on $[0,\infty)$, from $m(0) = \psi(\frac14) - \log\pi = -\gamma_E - \frac{\pi}{2} - 3\log 2 - \log\pi = -5.372$. Binet's formula $\psi(z) = \log z - \frac{1}{2z} - 2\int_0^\infty v\,dv/\big((v^2+z^2)(e^{2\pi v}-1)\big)$ [256, (5.9.15)] at $z = \frac14 + \frac{ir}{2}$, with the integral split at $v = r/4$, gives $|m(r) - \log\frac{r}{2\pi}| \le \frac{9}{8r^2} + 10^{-10} < 0.003$ for $r \ge 20$. Hence $|m| \le 6$ on $[0,20]$ and $\int_0^{20}|m(r) - \log\frac{r}{2\pi}|\,dr \le 120 + 4\pi + 20\log\frac{10}{\pi} - 20 < 136$. Since $\frac{1}{2\pi}\int_0^\infty r^{2k}e^{-r^2t}\log\frac{r}{2\pi}\,dr = s_k(t)(\ell+\alpha_k)$, with $\alpha_0 = -\frac12\gamma_E - \log 4\pi = -2.8196$ and $\alpha_1 = \alpha_0 + 1$, it follows that

$$\Big|\frac{(-1)^k A^{(k)}(t)}{s_k(t)} - \ell - \alpha_k\Big| \le \frac{272\cdot 20^{2k}\,t^{k+\frac12}}{\Gamma(k+\frac12)} + 0.003 \qquad (k = 0, 1),$$

and the right side, which increases with t, is below 1.089 for $k = 0$ and 0.047 for $k = 1$ at $t = \tau$. For the prime term put $y_n = (\log n)^2/4t$ and $q = (\log 2)/8t$. Both e^{-y_n} and $|y_n - \frac12|e^{-y_n}$ are at most $e^{-y_n/2}$, which is at most n^{-q} because $(\log n)^2 \ge (\log 2)\log n$, and $\Lambda(n) \le \log n < \sqrt{n}$; so $|P^{(k)}(t)| \le (2\sqrt{\pi})^{-1}t^{-k-\frac12}\sum_{n\ge2} n^{-q} < t^{-k-\frac12}2^{-q}$, that is $|P^{(k)}(t)| < t^{-k-\frac12}e^{-\kappa/t}$ for $k = 0, 1$ and $t \le \tau$, where $\kappa = (\log 2)^2/8$ is another constant of this section. Since $\kappa/\tau > 1201$, this is below $10^{-500}s_k(t)$. The polar term adds $e^{t/4}$ to c_0 and subtracts $\frac14 e^{t/4} < 10^{-5}s_1(t)$ from c_1. So for $t \le \tau$

$$c_0(t) \ge s_0(t)\,(\ell - 3.92), \qquad c_1(t) \ge s_1(t)\,(\ell - 1.87),$$

and both are positive, because $\ell \ge \frac12\log 20000 > 4.95$.

For $t \ge \tau$ we use the zeros. Those below height 1000 lie on the line [277] and contribute $\gamma^{2k}e^{-\gamma^2 t} > 0$ to c_k, and by the explicit bound for $N(T)$ of [108] one of them lies in $(20, 50]$ and contributes at least $400^k e^{-2500t}$. A zero above height 1000, wherever it lies, has $\operatorname{Re} a \ge \gamma^2 - \frac14$ and $|a| \le 2\gamma^2$. The same bound gives $N(r) \le r^2$ for $r \ge 1000$, and integration by parts against dN bounds the total modulus of the contributions of these zeros to c_0 and c_1 by

$$H_0(t) = e^{-(10^6-\frac14)t}\big(10^6 + t^{-1}\big), \qquad H_1(t) = 2e^{-(10^6-\frac14)t}\big(10^{12} + 2\cdot 10^6 t^{-1} + 2t^{-2}\big).$$

At $t = \tau$ these are below 2×10^{-16} and 4.1×10^{-10}, while $e^{-2500\tau} > 0.88$, and $e^{2500t} H_k(t)$ decreases with t. So $c_k(t) \geq 400^k e^{-2500t} - H_k(t) > 0$ for $t \geq \tau$ and $k = 0, 1$.

Thus $W > 0$ and $W' < 0$ on $(0, \infty)$. As $0 < W(t) \leq \sum_j e^{-t \operatorname{Re} a_j}$, the measure $W(t)\, dt/t$ satisfies $\int_0^\infty (1 \wedge t)\, W(t)\, dt/t \leq \sum_j 1/\operatorname{Re} a_j < \infty$, so it is a Lévy measure, and the formula for $\log G$ above makes the clock infinitely divisible with this Lévy measure. A law on the half line whose Lévy density is $W(t)/t$ with W nonincreasing is self decomposable [325, Corollary 15.11]. Finally, $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \theta) = \mathbb{E} e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with Z standard normal and independent of T, and with the characteristic function $\Xi(\theta)/\Xi(0)$ of the variable X of Section 2.1 it forms a van Dantzig pair. Every numerical inequality of this proof is evaluated in ball arithmetic. □

Tilting the clock by e^{-T} gives $\xi(\frac{3}{2})/\xi(\frac{1}{2} + \sqrt{1+s}) = G(1)/G(1+s) = \mathbb{E} e^{-sT^{(1)}}$, with $T^{(1)} \geq 0$ infinitely divisible, so unconditionally $h_k = \xi(\frac{3}{2})/\xi(\frac{1}{2} + \sqrt{1+k}) = \mathbb{E}(e^{-T^{(1)}})^k$ is a Hausdorff moment sequence.

The proof needs far fewer zeros. With the same bounds it goes through with $\tau = 1/8300$ and the zeros below height 400. If the archimedean error is bounded by computation instead, from $\int_0^5 r^{2k} |m(r) - \log \frac{r}{2\pi}|\, dr < 0.836$ and 0.404 for $k = 0, 1$ and $r^2 |m(r) - \log \frac{r}{2\pi}| < 0.05$ for $r \geq 5$, it goes through with $\tau = 1/500$ and height 100. If in addition the polar term $e^{t/4}$ is kept in c_0, the prime terms with $n \leq 30$ are taken exactly, and the lower bound for $t \geq \tau$ comes from the first zero, in $(14.1347, 14.1348)$, rather than from a zero in $(20, 50]$, it goes through with $\tau = 1/50$ and height 25, below which there are only two zeros. Every numerical inequality of the three versions is evaluated in ball arithmetic.

The same scheme, which we call the two range method, gives more. With the prime term bounded by $100\, t^{-k-\frac{1}{2}} e^{-\kappa/t}$ for $k \leq 3$ and a second zero in $(100, 200]$, it shows that W and $-W'$ are strictly log convex [279, Theorem 7.3], that is $c_0 c_2 > c_1^2$ and $c_1 c_3 > c_2^2$ for every $t > 0$, and that $(-1)^k W^{(k)} > 0$ for $k \leq 3$.

For the first two rungs the explicit formula can be avoided, because where the size of an off line quadruple cannot be bounded its phase fixes its sign [291]. Suppose that the zeros with ordinate in $(0, H]$ lie on the line, with $H \geq 100$, and let $0 \leq k \leq H/8$. A quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ then has $\gamma > H$, and since $0 < \phi < 2\delta/\gamma < 1/\gamma$ its phase satisfies $|2\delta\gamma t - k\phi| < \gamma t + 1$, so its term in $c_k(t)$ is positive unless $\gamma t > \frac{\pi}{2} - 1 > 0.57$. When $\gamma t > 0.57$ the term is at most $2(\gamma^2 + \frac{1}{4})^k e^{-(\gamma^2 - \frac{1}{4})t}$ in modulus, and since $(\gamma^2 + \frac{1}{4})/\gamma_1^2 \leq \gamma^2/99.9$, $\gamma^2 - \frac{1}{4} - \gamma_1^2 \geq \gamma^2/2$ and $\gamma^2 t/2 > 0.285\gamma$, it is at most $2(\gamma^2/99.9)^k e^{-0.285\gamma}$ times the term $\gamma_1^{2k} e^{-\gamma_1^2 t}$ of the first zero. A quadruple has two zeros of ordinate γ, so at every $t > 0$

$$c_k(t) \geq \big(1 - S_k(H)\big) \gamma_1^{2k} e^{-\gamma_1^2 t}, \qquad S_k(H) = \sum_{\gamma > H} \Big(\frac{\gamma^2}{99.9}\Big)^k e^{-0.285\gamma},$$

the sum running over the zeros above height H, with multiplicity. The explicit bound for $N(T)$ of [108] gives $N(T+1)-N(T) < 2\log T$ for $T \geq 100$, and summing over unit intervals gives $S_0(100) < 10^{-10.5}$ and $S_1(100) < 10^{-8.5}$. As Hutchinson placed the zeros below height 300 on the line in 1925 [158], this proves Theorem 8.1.1 without the explicit formula. For larger k the factor $(\gamma^2/99.9)^k$ must be absorbed by $e^{-0.285\gamma}$ at $\gamma = H$, so the reachable order, about $0.285H/\log(H^2/99.9)$, stays below the height: the verification below 3×10^{12} [277] gives every $k \leq 10^{10}$, with $S_k(H) < 10^{-1.4\times 10^{11}}$. The two range method reaches orders proportional to H^2 (Theorems 8.1.6 and 8.1.13). These bounds are evaluated in ball arithmetic.

Positivity of W also limits how the clock factorises. By Mase's theorem [232, Theorems 1 and 2], an infinitely divisible law without Gaussian component whose Lévy measure has a density that is positive on the whole half line, or that has mass on both half lines, has an indecomposable factor, hence a factor that is not infinitely divisible, so it lies outside Linnik's class I_0 [216] of the infinitely divisible laws all of whose factors are infinitely divisible. Since $W > 0$, this applies unconditionally to the clock variable T, to $Z\sqrt{2T}$ and to every nondegenerate GGC [279, Proposition 12.1].

Proposition 8.1.2 (Truncation error). *Let the zeros be simple, so that the a_j are distinct, and let r_k and $r_k^{(K)}$ be the residues at $u = -a_k$, $k \leq K$, of $1/G(u) = \prod_j a_j/(a_j+u)$ and of its product over $j \leq K$. Then $r_k = a_k \prod_{j\neq k} a_j/(a_j - a_k)$ and, when $|a_j| > |a_k|$ for $j > K$,*

$$\log \frac{r_k^{(K)}}{r_k} = \sum_{j>K} \log\Bigl(1 - \frac{a_k}{a_j}\Bigr) = -a_k \sum_{j>K} a_j^{-1} - \frac{a_k^2}{2} \sum_{j>K} a_j^{-2} - \cdots .$$

Under RH $a_j = \gamma_j^2$ and $\sum_{j>K} \gamma_j^{-2} \sim \log\gamma_K/(2\pi\gamma_K)$, so the first term is $-\gamma_k^2 \log\gamma_K/(2\pi\gamma_K)$: the truncated coefficient is close to the residue for $\gamma_k \leq (\gamma_K/\log\gamma_K)^{1/2}$, and not beyond.

Proof. The product converges absolutely, and its pole at $-a_k$ is simple, with residue a_k times the other factors at $u = -a_k$. The ratio of the two residues is the tail product $\prod_{j>K}(a_j - a_k)/a_j$. For the sum, $\sum_{j>K}\gamma_j^{-2} = \int_{\gamma_K}^{\infty} T^{-2}\,dN(T)$ with $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, which is $(\log(\gamma_K/2\pi)+1)/(2\pi\gamma_K)$ to leading order. □

8.1.2 Hankel blocks and derivative signs

Fix $t > 0$ and form the Hankel matrices $[c_{i+j}(t)]$ and $[c_{i+j+1}(t)]$, $i, j \geq 0$.

Theorem 8.1.3 ([279, Theorem 7.1]). *For any fixed $t > 0$ the following are equivalent: both infinite Hankel matrices are positive semidefinite; Weil's quadratic form is positive semidefinite on the span of $r^m e^{-r^2 t/2}$, $m = 0, 1, 2, \ldots$; RH.*

Under RH the derivatives at one time are the moments of the positive measure $e^{-tz}U(dz)$ on the half line, so both matrices are positive semidefinite. Conversely, if both Hankel matrices are positive semidefinite, Stieltjes' theorem makes the $c_k(t)$ the moments of a positive measure μ on the half line, and since $\sum_k c_k(t)(-s)^k/k! = W(t+s)$ for $|s| < t$, $W(t+s) = \int e^{-sx}\mu(dx)$ there. Both sides are analytic in $\operatorname{Re} s > -t$, the right side because $\int e^{|s|x}\mu(dx) = \sum_k c_k(t)|s|^k/k! = W(t-|s|) < \infty$ for $|s| < t$, so the identity theorem extends the equality to $\operatorname{Re} s > -t$, and W is completely monotone, which is RH. Through the explicit formula, Weil's form on the span of $r^m e^{-r^2t/2}$ gives the same two matrices, even m the first and odd m the second.

Theorem 8.1.3 tests RH on the real axis; the next theorem tests it on a vertical line, where W is a sum of oscillating terms.

THEOREM 8.1.4. *RH holds if and only if $v \mapsto W(\sigma+iv)$ is bounded on $\mathbb{R}$, for one, equivalently every, $\sigma > 0$.*

Proof. Write $s_j = -a_j = (\rho_j - \frac{1}{2})^2 = x_j + iy_j$, $\rho_j = \frac{1}{2} + \delta_j + i\gamma_j$, so that $x_j = \delta_j^2 - \gamma_j^2$, $y_j = 2\delta_j\gamma_j$, $|\delta_j| < \frac{1}{2}$, $|y_j| \le |\gamma_j|$ and $\gamma_j \neq 0$. Only three facts on the zeros are used: $|\delta_j| < \frac{1}{2}$, $\gamma_j \neq 0$ (ξ has no zeros in $(0,1)$) and $N(T) = O(T\log T)$, so that $\sum_j e^{-c\gamma_j^2} < \infty$ for every $c > 0$ and only finitely many $|\gamma_j|$ lie below any height. Put $F(v) = W(\sigma+iv) = \sum_j c_j e^{is_jv}$, $c_j = e^{s_j\sigma}$, $C = \sum_j |c_j| \le e^{\sigma/4}\sum_j e^{-\sigma\gamma_j^2} < \infty$. Since $|e^{is_jv}| = e^{-y_jv} \le e^{|\gamma_j v|}$, the series converges absolutely and locally uniformly in v.

Under RH every $s_j = -\gamma_j^2$ is real, so $|F(v)| \le W(\sigma)$ for every $\sigma > 0$. Conversely let $|F| \le M$ on $\mathbb{R}$ for one σ, and suppose RH fails. For $\omega \in \mathbb{R}$ and complex ε with $\operatorname{Re}\varepsilon > 0$ put $L(\varepsilon) = \int_{\mathbb{R}} F(v)e^{-\varepsilon v^2 - i\omega v}\,dv$.

Step 1. On $\operatorname{Re}\varepsilon \ge \eta > 0$ the integrand is entire in ε and bounded by $Me^{-\eta v^2}$, so L is continuous by dominated convergence and holomorphic by Morera's theorem, with Fubini over triangles justified by the same bound. For real $\varepsilon > 0$, $|L(\varepsilon)| \le M\sqrt{\pi/\varepsilon}$.

Step 2. For $\varepsilon > 0$ and $b \in \mathbb{C}$, $\int_{\mathbb{R}} e^{-\varepsilon v^2 + bv}\,dv = \sqrt{\pi/\varepsilon}\,e^{b^2/4\varepsilon}$: both sides are entire in b and agree for real b by completing the square. With $b = i(s_j - \omega)$ this gives $\int_{\mathbb{R}} e^{is_jv - \varepsilon v^2 - i\omega v}\,dv = \sqrt{\pi/\varepsilon}\,e^{-(s_j-\omega)^2/4\varepsilon}$. The sum over j may be taken under the integral (Fubini and Tonelli) if $\sum_j |c_j| \int_{\mathbb{R}} e^{-y_jv - \varepsilon v^2}\,dv = \sqrt{\pi/\varepsilon}\sum_j e^{\sigma\delta_j^2 - \sigma\gamma_j^2 + \delta_j^2\gamma_j^2/\varepsilon} < \infty$. For real $\varepsilon \ge 1/(2\sigma)$ we have $\delta_j^2/\varepsilon \le \sigma/2$, each term is at most $e^{\sigma/4}e^{-\sigma\gamma_j^2/2}$, and the sum is finite. Hence $L(\varepsilon) = \sqrt{\pi/\varepsilon}\,\Psi_\varepsilon(\omega)$ for $\varepsilon \ge 1/(2\sigma)$, where $\Psi_\varepsilon(\omega) = \sum_j c_j e^{-(s_j-\omega)^2/4\varepsilon}$ and the root is principal.

Step 3. The series $\Psi_\varepsilon(\omega)$ converges locally uniformly on $\operatorname{Re}\varepsilon > 0$. Let K be compact there. Then $u = 1/(4\varepsilon)$ stays in a compact set with $\operatorname{Re} u \ge u_0 > 0$ and $|\operatorname{Im} u| \le u_1$. Write $(s_j-\omega)^2 = X + iY$ with $X = (x_j-\omega)^2 - y_j^2$ and $Y = 2(x_j-\omega)y_j$.

For $\gamma_j^2 \geq \frac{1}{4} + |\omega|$ we have $|x_j - \omega| \geq \gamma_j^2 - \frac{1}{4} - |\omega|$, so $X \geq (\gamma_j^2 - \frac{1}{4} - |\omega|)^2 - \gamma_j^2$ and $|Y| \leq 2|\gamma_j|(\gamma_j^2 + \frac{1}{4} + |\omega|)$, of orders γ_j^4 and $|\gamma_j|^3$. Hence there is $G = G(K, \omega)$ such that $X > 0$ and $u_0 X - u_1|Y| \geq 0$ for $|\gamma_j| \geq G$, and then $|e^{-(X+iY)u}| = e^{-(X \operatorname{Re} u - Y \operatorname{Im} u)} \leq e^{-(u_0 X - u_1|Y|)} \leq 1$, so the jth term is at most $|c_j|$. Only finitely many j have $|\gamma_j| < G$, and $\sum_j |c_j| = C$, so the Weierstrass test applies and Ψ is holomorphic on $\operatorname{Re} \varepsilon > 0$. Now L and $\sqrt{\pi/\varepsilon}\, \Psi_\varepsilon(\omega)$ are holomorphic on the connected half plane and agree on $[1/(2\sigma), \infty)$, so by the identity theorem they agree throughout. With Step 1,

$$|\Psi_\varepsilon(\omega)| \leq M \qquad (\varepsilon > 0,\ \omega \in \mathbb{R}). \tag{$*$}$$

Step 4. Put $t = 1/(4\varepsilon)$ and $q_j(\omega) = y_j^2 - (x_j - \omega)^2$. Then $|e^{-(s_j - \omega)^2 t}| = e^{q_j t}$ and its phase is $e^{-i\phi_j t}$, $\phi_j = 2(x_j - \omega)y_j$. For fixed ω and level l only finitely many j have $q_j(\omega) \geq l$, because $q_j \leq \gamma_j^2 - (\gamma_j^2 - \frac{1}{4} - |\omega|)^2 \to -\infty$ as $|\gamma_j| \to \infty$.

Step 5. Take a zero off the line, with $s_0 = x_0 + iy_0$, $y_0 = 2\delta\gamma \neq 0$. On the open interval $I_0 = (x_0 - |y_0|, x_0 + |y_0|)$, $q_0 > 0$. For two values $s \neq s'$ in $\{s_j\}$ with $(x, |y|) \neq (x', |y'|)$, the difference $q - q' = y^2 - y'^2 - x^2 + x'^2 + 2\omega(x - x')$ is a nonconstant affine function of ω if $x \neq x'$ and a nonzero constant if $x = x'$, so it vanishes for at most one ω. Let E be the countable set of these points together with $\{x_j\}$, and fix $\omega \in I_0 \setminus E$. By Step 4, $Q = \max_j q_j(\omega)$ is attained, $Q \geq q_0 > 0$, and by the choice of ω every maximiser has the same $(x, |y|)$, so $s_j \in \{s^*, \bar{s}^*\}$ for a fixed $s^* = x^* + iy^*$. Now $Q > 0$ forces $y^* \neq 0$, and $\omega \neq x^*$ forces $\phi := 2(x^* - \omega)y^* \neq 0$. Only finitely many j have $q_j \geq 0$, so $Q' = \max\{0,\, q_j(\omega) : s_j \notin \{s^*, \bar{s}^*\}\}$ is a maximum over a finite set and $Q' < Q$.

Step 6. Let $m_1 \geq 1$ and $m_2 \geq 0$ be the numbers of j with $s_j = s^*$ and $s_j = \bar{s}^*$, and $A = e^{s^*\sigma} \neq 0$. The terms with $\bar{s}^*$ have the same q, the phase $e^{+i\phi t}$ and the coefficient $\bar{A}$. Hence $\Psi_\varepsilon(\omega) = e^{Qt} g(t) + r(t)$ with $g(t) = m_1 A e^{-i\phi t} + m_2 \bar{A} e^{i\phi t}$ and $|r(t)| \leq \sum_j |c_j| e^{Q't} = C e^{Q't}$ for $t > 0$. The function g is continuous and periodic with period $2\pi/|\phi|$, and not identically zero, since its Fourier coefficient at the frequency $-\phi \neq \phi$ is $m_1 A \neq 0$. So $\kappa = \max |g| > 0$ is attained in every period, at points $t_k \to \infty$, and with $\varepsilon_k = 1/(4t_k)$, using $Q > 0$ for the growth and $Q > Q'$ for the remainder,

$$|\Psi_{\varepsilon_k}(\omega)| \geq \kappa e^{Qt_k} - C e^{Q't_k} \to \infty,$$

contradicting $(*)$. So RH holds, and since RH gives boundedness for every σ, the two readings agree. □

THEOREM 8.1.5 ([279, Theorem 7.4]). *If all zeros below $T_N = 20000N\sqrt{1 + \log N}$ are on the line, the leading $N \times N$ blocks are positive definite at every $t > 0$. The verification to height 3×10^{12} [277] gives this for every $N \leq 3 \times 10^7$.*

Table 8.1 shows normalised leading minors recomputed in sixty digit arithmetic from the first 1700 zeros, stored in double precision; they agree with every digit printed in [279, Numerical observation 7.2].

t	c_0c_2/c_1^2	$N=2$	$N=3$	$N=4$	$N=5$	block
0.01	1.1346548	0.119	2.45×10^{-3}	1.05×10^{-5}	1.04×10^{-8}	even
0.01		0.182	5.87×10^{-3}	3.30×10^{-5}	4.57×10^{-8}	odd
0.05	1.0000081	8.11×10^{-6}	1.77×10^{-14}	5.35×10^{-28}	2.20×10^{-44}	even
0.05		1.79×10^{-5}	1.22×10^{-13}	1.71×10^{-26}	3.81×10^{-42}	odd

Table 8.1: Normalised leading Hankel minors of the heat trace, $\det[c_{i+j}]_{i,j<N}/\prod_{i<N}c_{2i}$ for the even block and $\det[c_{i+j+1}]_{i,j<N}/\prod_{i<N}c_{2i+1}$ for the odd block, from the first 1700 zeros in sixty digit arithmetic.

At $t=0.05$ (Figure 8.1, left) the first exponents dominate and the others enter only through $e^{-(a_j-a_1)t}$, so the matrix is nearly of rank one. Computed from the zeros, the minors are sums of positive terms by the Cauchy Binet formula $\det[c_{i+j}]_{i,j<N}=\sum_{|S|=N}\prod_{l\in S}e^{-a_lt}\prod_{l<l',\ l,l'\in S}(a_l-a_{l'})^2$ over the sets S of N zeros, so what is lost is conditioning, not sign. The derivative signs (Figure 8.1, right) reach much further, because a sign needs only that the verified terms dominate, not that a small determinant stay positive.

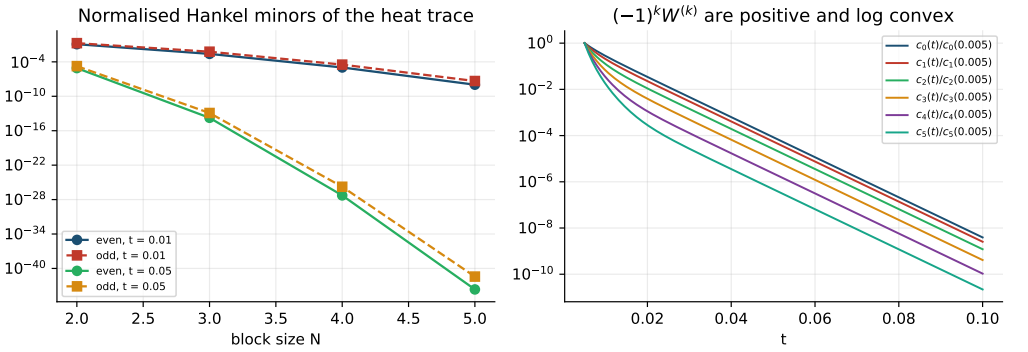


Figure 8.1: Left: normalised Hankel minors of the heat trace at two times; positivity persists but falls below any fixed precision as t grows. Right: the first six $(-1)^kW^{(k)}(t)$, divided by their values at $t=0.005$, from 1700 zeros.

THEOREM 8.1.6 ([279, Theorem 7.10]). *Unconditionally, that is, using only the verification of RH below height* 10^{12} *[277],* $(-1)^kW^{(k)}(t)>0$ *for every* $t>0$ *and every integer* $0\le k\le 10^{17}$.

The proof splits at τ. Near the origin, Cauchy's estimate on a circle about t of radius close to t, such as $(1-\frac{1}{k+1})t$, on which $\mathrm{Re}(1/z)\ge 1/(2t)$ and $|z|\ge t/(k+1)$, bounds the kth derivative of the prime term by a multiple of $(k+1)\Gamma(k+\frac{1}{2})t^{-k-\frac{1}{2}}e^{-\kappa/t}$. Against $s_k(t)$ this loses a factor of order k and gains $e^{-\kappa/t}\le e^{-1201}$, and for $k\ge 1$ the archimedean term dominates by a margin of

more than $2.9\,s_k(t)$, the case $k = 0$ being Theorem 8.1.1. Away from the origin a zero in $(20, 50]$ dominates all zeros above the verified height $H = 10^{12}$, wherever they lie, provided $tH^2 \geq 2(k+1)$ and a logarithmic condition hold; Section 8.1.6 explains why these allow orders up to about 10^{18}.

The Stieltjes moments of Table 4.2 give a Hankel test at the other end of the heat trace, which also returns the zeros [291]. They are the moments of W, $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_j a_j^{-n-1}$, and since $\log G(u) = \sum_j \log(1+u/a_j) = \sum_{n\geq 1}(-1)^{n+1} m_{n-1}u^n/n$ they follow from the Taylor coefficients of $G(u) = \Xi(i\sqrt{u})/\Xi(0) = \sum_k M_k u^k/((2k)!\,M_0)$, where $M_k = \int_{\mathbb{R}} v^{2k}\Phi(v)\,dv$; no zero enters. Under RH the m_n are the moments of the positive measure $\nu = \sum_j a_j^{-1}\delta_{1/a_j}$, so every Hankel matrix $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is positive definite. Conversely, if all of them are, Hamburger's theorem makes the m_n the moments of a positive measure, of compact support because $|m_n| \leq \sum_j |a_j|^{-n-1}$ grows at most geometrically, and then $\sum_n m_n w^n = \sum_j 1/(a_j - w)$ has all its poles on the real axis: every a_j is real, and positive because ξ has no zeros in $(0,1)$, which is RH. The shifted matrices $\mathcal{H}'_N = [m_{i+j+1}]_{i,j<N}$ give the same test, and either family suffices: if every $\mathcal{H}'_N$ is positive definite, Hamburger's theorem applied to (m_{n+1}) and $\sum_n m_n w^n = m_0 + w\sum_n m_{n+1}w^n$ give the same conclusion. Positivity is tested by the Cholesky factorisation $\mathcal{H}_N = LL^\top$, whose factor gives more: $J_N = L^{-1}\mathcal{H}'_N L^{-\top}$ is the Jacobi matrix of ν, tridiagonal, and its eigenvalues are the nodes of the N point Gauss rule for ν. They approximate the largest atoms $1/a_j$ first, so their inverse square roots return the lowest ordinates (Table 8.2). With $M_0, \ldots, M_{100}$ computed at 1000 digits by the trapezoid rule, every $\mathcal{H}_N$ up to order 50 is positive definite.

N	zeros	error of γ_1	last error	$\det\mathcal{H}_N/\det\mathcal{H}_{N-1}$	digits
10	2	4.4×10^{-18}	1.3×10^{-9}	2.0×10^{-61}	60
20	7	1.6×10^{-49}	3.1×10^{-7}	1.2×10^{-138}	140
30	12	8.6×10^{-87}	2.4×10^{-7}	1.0×10^{-221}	230
40	17	6.1×10^{-128}	5.4×10^{-8}	1.3×10^{-308}	310
50	22	5.3×10^{-172}	5.4×10^{-8}	1.4×10^{-398}	400

Table 8.2: The lowest ordinates from the Stieltjes moments $m_0, \ldots, m_{2N-1}$, computed from the Taylor coefficients of Ξ at the centre: the number returned by the Jacobi matrix of order N to within 10^{-6}, the errors of the first and of the last of them, the last Cholesky pivot of $\mathcal{H}_N$, and the least number of digits, in steps of 10, of the Taylor coefficients and of the arithmetic from which on the same ordinates are returned.

This is Gauss quadrature, equivalently Padé approximation of G'/G at the origin, and Table 8.2 shows its two limits. The first is precision: the computation must resolve the last Cholesky pivot, which falls from 10^{-61} at $N = 10$ to 10^{-398} at $N = 50$, so the precision grows linearly, by about 17 digits for each further

zero. The second is reach. An off line quadruple at height γ enters m_n with weight of order γ^{-2n-2}, against γ_1^{-2n-2} for the first zero, so a block of order N sees essentially only the zeros that its Gauss rule resolves; Section 4.1.3 shows this along the de Bruijn Newman flow, where the order 30 test registers the collision of the pair just above the zeros it resolves, and not a collision higher up. The table is computed.

The Hankel matrices also count the zeros off the line. For a real polynomial this is a theorem attributed to Hermite [114]: the Hankel form of the power sums of the roots has as many negative squares as the polynomial has distinct pairs of nonreal roots. Let ι_N and ι'_N be the numbers of negative eigenvalues of $\mathcal{H}_N$ and $\mathcal{H}'_N$.

THEOREM 8.1.7. *Let $K \in \{0,1,2,\dots\} \cup \{\infty\}$ be the number of distinct zeros ρ of ξ with $\operatorname{Re}\rho > \frac12$ and $\operatorname{Im}\rho > 0$. Then ι_N and ι'_N are nondecreasing in N, $\iota_N \le K$ and $\iota'_N \le K$ for every N, and $\iota_N \to K$ and $\iota'_N \to K$ as $N \to \infty$. In particular RH holds if and only if $\iota_N = 0$ for every N, and if it fails, $\iota_N \ge 1$ from some N on; the same holds for ι'_N.*

Proof. Put $b_k = 1/a_k$, listed with multiplicity, so that $m_n = \sum_k b_k^{n+1}$ and $\sum_k |b_k| < \infty$. For a real vector $x = (x_0, \dots, x_{N-1})$ and $P(w) = \sum_i x_i w^i$,

$$x^\top \mathcal{H}_N x = \sum_k b_k P(b_k)^2, \qquad x^\top \mathcal{H}'_N x = \sum_k b_k^2 P(b_k)^2,$$

the sums converging absolutely. A zero on the line gives $b_k = \gamma_k^{-2} > 0$ and nonnegative terms. A zero $\rho = \frac12 + \delta + i\gamma$ with $\delta > 0$ and $\gamma > 0$, of multiplicity μ, gives $a = -(\delta + i\gamma)^2$ with $\operatorname{Im} a = -2\delta\gamma < 0$, hence $b = 1/a$ with $\operatorname{Im} b > 0$; its mirror image $1 - \bar\rho$ gives $\bar a$ and $\bar b$ with the same multiplicity, and since P has real coefficients the two contribute $2\mu \operatorname{Re}\big(bP(b)^2\big)$ to the first form. With $u = \operatorname{Re} P(b)$ and $v = \operatorname{Im} P(b)$ this is $2\mu\big(\operatorname{Re} b\,(u^2 - v^2) - 2\operatorname{Im} b\,uv\big)$, a quadratic form in (u, v) whose matrix has determinant $-4\mu^2|b|^2 < 0$, so it has exactly one negative square, and composed with the linear map $x \mapsto (u, v)$ it has at most one. The terms of the zeros on the line add up to a positive semidefinite form, so $\iota_N \le K$. In the second form the weight is b^2, and $2\mu \operatorname{Re}\big(b^2 P(b)^2\big)$ is again a form in (u, v) of determinant $-4\mu^2|b|^4 < 0$, so $\iota'_N \le K$. Since $\mathcal{H}_N$ is the leading principal submatrix of $\mathcal{H}_{N+1}$, Cauchy's interlacing theorem shows that ι_N does not decrease, and the same holds for ι'_N.

For the lower bound let $\mathcal{B}$ be the closure of $\{b_k\}$. It is a countable compact set, symmetric under conjugation, whose only possible accumulation point is 0, and a countable compact set does not separate the plane, so $\mathbb{C} \setminus \mathcal{B}$ is connected. Fix a finite $m \le K$ and distinct atoms $\omega_1, \dots, \omega_m$ with $\operatorname{Im}\omega_j > 0$, of multiplicities μ_j; they are isolated points of $\mathcal{B}$. On $\mathbb{C}^m = \mathbb{R}^{2m}$ the form $q(y) = \sum_{j \le m} 2\mu_j \operatorname{Re}(\omega_j y_j^2)$ is a sum of m forms in the pairs $(\operatorname{Re} y_j, \operatorname{Im} y_j)$, each with one negative square, so there are an m dimensional real subspace E and $\lambda_0 > 0$ with $q(y) \le -\lambda_0 |y|^2$ on

E. Let $y^{(1)}, \dots, y^{(m)}$ be a basis of E and $\lambda_1 > 0$ such that $|\sum_i \alpha_i y^{(i)}|^2 \geq \lambda_1 |\alpha|^2$ for $\alpha \in \mathbb{R}^m$. Take open discs D_j about ω_j, and their mirror images D_j^* about $\bar\omega_j$, whose closures are disjoint and meet $\mathcal{B}$ only in their centres, and an open set $O \supset \mathcal{B} \setminus \{\omega_1, \bar\omega_1, \dots, \omega_m, \bar\omega_m\}$, symmetric under conjugation and disjoint from the closed discs, which exists because that set is compact. Let f_i be $y_j^{(i)}$ on D_j, $\overline{y_j^{(i)}}$ on D_j^* and 0 on O. Each f_i is holomorphic on a neighbourhood of $\mathcal{B}$, so by Runge's theorem there are polynomials $\tilde P_i$ with $|\tilde P_i - f_i| < \varepsilon$ on $\mathcal{B}$, and since $\mathcal{B}$ and f_i are symmetric under conjugation, $P_i(w) = \frac12\big(\tilde P_i(w) + \overline{\tilde P_i(\bar w)}\big)$ has real coefficients and satisfies the same bound. For $\alpha \in \mathbb{R}^m$ and $P = \sum_i \alpha_i P_i$ we have $|P(b_k)| \leq \varepsilon\sqrt{m}\,|\alpha|$ at every atom other than the ω_j and $\bar\omega_j$, and $P(\omega_j) = \sum_i \alpha_i y_j^{(i)} + \eta_j$ with $|\eta_j| \leq \varepsilon\sqrt{m}\,|\alpha|$. The atoms ω_j and $\bar\omega_j$ contribute $\sum_j 2\mu_j \operatorname{Re}\big(\omega_j P(\omega_j)^2\big)$ to the first form, so for $\varepsilon \leq 1$ and a constant C that depends on neither ε nor α,

$$\sum_k b_k P(b_k)^2 \leq q\Big(\sum_i \alpha_i y^{(i)}\Big) + C\varepsilon|\alpha|^2 \leq -(\lambda_0\lambda_1 - C\varepsilon)|\alpha|^2.$$

For $\varepsilon < \lambda_0\lambda_1/C$ the right side is negative for $\alpha \neq 0$, so the coefficient vectors of $P_1, \dots, P_m$ span an m dimensional space on which the form of $\mathcal{H}_N$ is negative definite as soon as N exceeds their degrees, and $\iota_N \geq m$ for those N. For $\mathcal{H}'_N$ the same argument applies with ω_j^2 in place of ω_j in q and the weights b_k^2 in the sum. Since $m \leq K$ is arbitrary, $\iota_N \to K$ and $\iota'_N \to K$. □

In determinant form the theorem is classical. For a real entire function of order one with zeros z_j, Grommer [100] proved that all its zeros are real if and only if the determinants $D_n = \det(s_{i+j})_{i,j<n}$ of the power sums $s_k = \sum_j z_j^{-k-2}$ are positive for every $n \geq 1$, and Chebotarev [353] that if the sequence $D_{n-1}D_n$, $n \geq 1$, with $D_0 = 1$, has exactly m negative terms, the function has m distinct pairs of complex conjugate zeros; Baricz and Štampach state both in this form [17, Theorem 10]. For Ξ, whose zeros are the square roots of the a_k, $s_{2l} = 2m_l$ and $s_{2l+1} = 0$, so after the even indices are put first $(s_{i+j})_{i,j<n}$ is block diagonal with the blocks $2\mathcal{H}_{\lceil n/2\rceil}$ and $2\mathcal{H}'_{\lfloor n/2\rfloor}$, whose negative indices add up to $\iota_{\lceil n/2\rceil} + \iota'_{\lfloor n/2\rfloor}$; this tends to $2K$, and each zero ρ counted by K gives two pairs of complex conjugate zeros of Ξ, as Chebotarev's count requires. Counting by eigenvalues, Theorem 8.1.7 needs no determinant to be nonzero, allows $K = \infty$ and holds for each family separately. Along the de Bruijn Newman flow the proof applies at every time to the zeros of H_t, none of which is purely imaginary (Section 4.1.3).

8.1.3 Every order near the origin

Let $s_k(t) = \Gamma(k + \frac12)t^{-k-\frac12}/(4\pi)$ as in the proof of Theorem 8.1.1, let $E_k(t)$ be the average of m against the weight $r^{2k}e^{-r^2t}$ on $[0,\infty)$, and let $F_k(x) = {}_1F_1(k + \frac12; \frac12; -x^2)$. Expanding the cosine in its power series gives

$\frac{1}{2\pi}\int_{\mathbb{R}} r^{2k}e^{-r^2t}\cos(ry)\,dr = 2s_k(t)F_k(y/2\sqrt{t})$, whose case $k=0$ is the kernel $e^{-y^2/4t}/(2\sqrt{\pi t})$ of the prime term of (2.1.5). Differentiating (2.1.5) k times term by term, which the Gaussian factors allow, gives with no hypothesis on the zeros

$$\frac{c_k(t)}{s_k(t)} = E_k(t) - 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}F_k(x_n) + \frac{(-\frac14)^k e^{t/4}}{s_k(t)}, \qquad x_n = \frac{\log n}{2\sqrt{t}}. \tag{8.1.1}$$

The archimedean term $E_k(t)$ grows with k.

LEMMA 8.1.8. (a) *For every integer $k\ge 0$ and every $x\ge 0$, $|F_k(x)|\le B(x)e^{-x^2/2}$, where $B(x)=\max\{2.62,\ (\pi(2ex^2+\frac12))^{1/4}\}$.*

(b) *Let $0<\varepsilon<1$, and let $k\ge 1$ and $x\ge 0$ satisfy $e^2x^2\le 4\varepsilon k$. Then $|F_k(x)|\le C_\varepsilon(k)e^{-x^2/2}$, where $C_\varepsilon(k)=\sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}+2\sqrt{k+0.37}\,e^{-\varepsilon k}$.*

Proof. Taking $t=1$ and $y=2x$ above, $F_k(x)=\Gamma(k+\frac12)^{-1}\int_{\mathbb{R}} r^{2k}e^{-r^2+2ixr}\,dr$. Put $\mu=x^2/4$. The integrand is entire and decays in every horizontal strip, so the path can be moved to $\operatorname{Im} r = x/2$, where $|r^{2k}|=(\sigma^2+\mu)^k$, $|e^{-r^2}|=e^{-\sigma^2+\mu}$ and $|e^{2ixr}|=e^{-4\mu}$, with $\sigma=\operatorname{Re} r$. Expanding $(\sigma^2+\mu)^k$ by the binomial theorem gives, for every k and x,

$$|F_k(x)| \le e^{-3\mu}\sum_{j=0}^{k}\frac{\mu^j}{j!}P_{k,j}, \qquad P_{k,j}=\frac{\Gamma(k+1)}{\Gamma(k+\frac12)}\,\frac{\Gamma(k-j+\frac12)}{\Gamma(k-j+1)}.$$

Kershaw's inequalities $\sqrt{y+\frac14}\le \Gamma(y+1)/\Gamma(y+\frac12)\le\sqrt{y+0.37}$, proved for $y>0$ in [186] and true at $y=0$, where the ratio is $1/\sqrt{\pi}$, give $P_{k,j}\le\sqrt{(k+0.37)/(k-j+\frac14)}$, which is less than $\sqrt{2}$ for $j\le k/2$ and at most $2\sqrt{k+0.37}$ for every j. For (a), suppose first that $k\ge 2ex^2$. Since $k/2\ge 4e\mu$, Chernoff's bound $\sum_{j\ge a}\mu^j/j!\le(e\mu/a)^a$ for the Poisson law, valid for $a>\mu$, gives $\sum_{j\ge k/2}\mu^j/j!\le(2e\mu/k)^{k/2}\le 2^{-k}$, and $2\sqrt{k+0.37}\,2^{-k}\le\sqrt{1.37}<1.2$ for $k\ge 1$. Hence $|F_k(x)|\le e^{-3\mu}(\sqrt{2}\,e^{\mu}+1.2)\le(\sqrt{2}+1.2)e^{-x^2/2}$, and $\sqrt{2}+1.2<2.62$; for $k=0$ the assumption forces $x=0$, where $F_0=1$.

Suppose now that $k<2ex^2$. Kummer's transformation and the Hermite case of ${}_1F_1$ [256, (13.2.39), (13.6.16)] give $F_k(x)=e^{-x^2}{}_1F_1(-k;\frac12;x^2)=(-1)^k\frac{k!}{(2k)!}e^{-x^2}H_{2k}(x)$, and the bound $e^{-x^2/2}|H_n(x)|\le(2^n n!)^{1/2}$ [256, (18.14.9)] gives

$$|F_k(x)| \le \Big(\frac{4^k(k!)^2}{(2k)!}\Big)^{1/2}e^{-x^2/2} \le \big(\pi(k+\tfrac12)\big)^{1/4}e^{-x^2/2} < \big(\pi(2ex^2+\tfrac12)\big)^{1/4}e^{-x^2/2},$$

since $4^k(k!)^2/(2k)! = \sqrt{\pi}\,\Gamma(k+1)/\Gamma(k+\frac12)\le\sqrt{\pi(k+\frac12)}$ by Kershaw's upper bound.

For (b), split the sum at $j = \varepsilon k$ instead. For $j \le \varepsilon k$, $P_{k,j} \le \sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}$, and these j contribute at most that times e^{μ}. For $j > \varepsilon k$, $P_{k,j} \le 2\sqrt{k+0.37}$, and since $\varepsilon k \ge e^2\mu > \mu$, Chernoff's bound gives $\sum_{j\ge\varepsilon k}\mu^j/j! \le (e\mu/\varepsilon k)^{\varepsilon k} \le e^{-\varepsilon k}$. Hence $|F_k(x)| \le e^{-3\mu}\big(\sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}\,e^{\mu} + 2\sqrt{k+0.37}\,e^{-\varepsilon k}\big) \le C_\varepsilon(k)e^{-x^2/2}$. □

THEOREM 8.1.9. (a) *For every integer $k \ge 1$ and every $t \in (0, 0.019]$, $(-1)^k W^{(k)}(t) > 0$. The proof uses no information on the zeros.*

(b) *Unconditionally, that is, using only the verification of RH below height 10^{12} [277], W is completely monotone on $(0, 1.29]$: $(-1)^k W^{(k)}(t) > 0$ for every $t \in (0, 1.29]$ and every integer $k \ge 0$.*

Proof. By (8.1.1) and Lemma 8.1.8(a), for every $k \ge 0$ and $t > 0$,

$$\frac{c_k(t)}{s_k(t)} \ge E_k(t) - S_B(t) - Q_k(t),$$

where

$$S_B(t) = 2\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}B(x_n)e^{-(\log n)^2/8t}, \qquad Q_k(t) = \frac{4\pi e^{t/4}t^{k+\frac12}}{4^k\,\Gamma(k+\frac12)}.$$

Above the order $K = 10^{17}$ part (b) of the lemma does better. Take $\varepsilon = 10^{-6}$. The first term of $C_\varepsilon(k)$ decreases in k, because $\frac14 < 0.37(1-\varepsilon)$, and the second decreases for $k > 1/(2\varepsilon)$, so $C_\varepsilon(k) \le C = C_\varepsilon(K+1) < 1 + 6\times10^{-7}$ for $k > K$, and $e^2x^2 \le 4\varepsilon k$ holds for $x \le 2\sqrt{\varepsilon(K+1)}/e$, which exceeds 2.3×10^5. For $t \ge 0.019$ every prime power $n \le 2^{24}$ has $x_n < 61$. So for $k > K$ and $t \ge 0.019$ the bound holds with S_B replaced by S_C, the sum S_B with $B(x_n)$ replaced by C for $n \le 2^{24}$.

The three terms are monotone. Since m increases on $[0,\infty)$, and the weights $r^{2k}e^{-r^2t}$ have likelihood ratios r^2 from k to $k+1$ and $e^{r^2(t'-t)}$ from t' to $t < t'$, both increasing in r, $E_k(t)$ increases in k and decreases in t. $Q_k(t)$ increases in t, and for $t \le 2$ it decreases in k, the ratio of consecutive terms being $t/(4k+2)$. $B(x)e^{-x^2/2}$ decreases on $(0,\infty)$: the second branch of B increases and is 1.78 at $x = 2^{-1/2}$, so $B = 2.62$ below that point, and above it the logarithmic derivative of the second branch times $e^{-x^2/2}$ is less than $1/(2x) - x < 0$; since x_n decreases in t, S_B and S_C increase in t. So for $k \ge k_0$ and $t \le T \le 2$ each bound is at least its value at (k_0, T), and it suffices to prove $E_1(0.019) - S_B(0.019) - Q_1(0.019) > 0$, which gives (a), and $E_{K+1}(1.29) - S_C(1.29) - Q_{K+1}(1.29) > 0$, which gives the orders $k > K$ on $[0.019, 1.29]$; with (a) for $t \le 0.019$ and Theorem 8.1.6 for the orders $k \le K$, this gives (b).

For the first, lower sums over $[0, 64]$, which the monotonicity of m allows, and the positivity of m beyond 64 give $E_1(0.019) > 0.15924$; the prime powers $n \le 30000$ and the tail bound below give $S_B(0.019) < 0.13628$; and $Q_1(0.019) <$

0.00933. For the second, the weight $r^{2K+2}e^{-r^2t}$ makes $v = r^2t$ a gamma variable of shape $\lambda = K + \frac{3}{2}$, and Chernoff's bound for the gamma law, $\Pr(v < a) \le (a/\lambda)^\lambda e^{\lambda-a}$ for $a < \lambda$, gives $\Pr(v < a) < e^{-1000}$ for $a = (1 - 2^{-13})^2\lambda$. As m increases, $E_{K+1}(t) \ge m(r_1) - (m(r_1) - m(0))e^{-1000}$ with $r_1 = \sqrt{a/t}$, which gives $E_{K+1}(1.29) > 17.6066$; the prime powers $n \le 2^{24}$ and the tail bound below give $S_C(1.29) < 17.5848$; and $Q_{K+1}(1.29) < e^{-1000}$. In both cases the prime powers above the cutoff n_0, which is 30000 and 2^{24}, add less than $2\big(2.62 + 2.08(2\sqrt{t})^{-1/2}\big)n_0^{1-q}/(q-1)$ with $q = \log n_0/8t$, because $x_n \ge x_{n_0} \ge 1$ and $e^{-(\log n)^2/8t} \le n^{-q}$ for $n > n_0$, $B(x) \le 2.62 + 2.08\sqrt{x}$ for $x \ge 1$, and $\log n$ and $(\log n)^{3/2}$ are at most $\sqrt{n}$ for $n \ge e^{10}$; this is below 10^{-299} and 5×10^{-4}. Every numerical inequality of this proof and of the proof of Lemma 8.1.8 is evaluated in ball arithmetic. □

In floating point the bound $E_1 - S_B - Q_1$ of (a) vanishes at $t = 0.0193$ and E_1 itself at $t = 0.0261$, so no bound on the size of the prime terms carries (a) much further. Part (b) goes further because $E_k(t) = \frac{1}{2}(\psi(k+\frac{1}{2}) - \log t) - \log 2\pi + O(t/k)$ grows like $\frac{1}{2}\log k$, while at such orders part (b) of the lemma bounds the prime terms at almost no cost: the bound $E_{K+1} - S_C$, computed in the same way, vanishes at $t = 1.2912$, while $E_{K+1} - S_B$ vanishes at $t = 0.578$.

The ray of Section 8.1.5 shows what the proof uses. On the ray $t = k/u^2$, $e^{-f_0}c_k(t)$ is to leading order the Gaussian smoothing $\mathcal{G}(u,t)$ of the zeros (Proposition 8.1.10), $E_k(t)$ is close to $\log(u/2\pi)$, and for large k the factor $F_k(x_n)$ is close to $e^{-(\log n)^2/8t}\cos(u\log n)$. The proof is therefore the phase free bound of Section 25.2.6, made uniform in the order; with C replaced by one, S_C is twice the sum S of that section, up to the prime powers above 2^{24}. Above the order 10^{17} uniformity costs a factor below $1+6\times10^{-7}$, so 1.2912, the root of $E_{K+1}(t) = 2S(t)$, is as far as this argument carries (b) with $K = 10^{17}$, since it bounds the prime terms by one envelope for all orders above K and the archimedean term by its value at $K+1$. A phase free bound that treats the orders separately is not covered by this limit. At low orders it costs the factor $B \ge 2.62$ of part (a), and part (a) is not sharp: on a grid of $0.5 \le x \le 20$ and $k \le 1200$ the ratio $|F_k(x)|e^{x^2/2}/B(x)$ is at most 0.44 (computed). The ray also says where (b) looks. To leading order, complete monotonicity on $(0,T]$ asks for positive Gaussian smoothings at resolutions $1/(2\sqrt{t}) \ge 1/(2\sqrt{T})$, which is 0.44 for $T = 1.29$, finer than the 0.46 at which the quadruples of Table 1.2, at distance $\frac{1}{4}$ from the line, first make the smoothing negative, at $t = 1.18$. There is no conflict, because the orders above 10^{17} at $t \le 1.29$ lie on the rays at heights $u = \sqrt{k/t} > 2.78 \times 10^8$, where the time $t_0(u)$ of Section 25.2.6, below which the phase free bound makes the smoothing positive whatever the zeros, exceeds 1.29. The quadruples lie below height 2000, at heights where t_0 is at most 0.511, and there (b) rests on the verified zeros, through Theorem 8.1.6.

8.1.4 Williamson measures: the Thorin measure smoothed by gamma laws

Complete monotonicity is the limit of k monotonicity, which asks only that the first k derivatives alternate in sign; by Williamson's theorem [364] a k monotone function that vanishes at infinity is a positive mixture of the truncated powers $(1-tz/k)_+^{k-1}$. For the heat trace the mixing measure is explicit,

$$W(t)=\int_0^\infty \Big(1-\frac{tz}{k}\Big)_+^{k-1} U_k(dz), \qquad U_k(dz)=\frac{k^k}{(k-1)!}z^{-k-1}c_k\Big(\frac{k}{z}\Big)dz,$$

with U_k the Williamson measure of order k. It is positive for every $k \le 10^{17}$ by Theorem 8.1.6 [279, Corollary 7.13], and for every $k \le 1.16\times 10^{25}$ by Theorem 8.1.13. RH holds if and only if $U_k \ge 0$ for infinitely many k, since then W is k monotone for infinitely many k, hence completely monotone.

The summands of U_k are inverse gamma laws, by the identity, for $k \ge 1$ and $c \ge 0$,

$$e^{-c}=\mathbb{E}\Big(1-\frac{c}{\Gamma_k}\Big)_+^{k-1}, \qquad \Gamma_k \sim \mathrm{Gamma}(k,1), \tag{8.1.2}$$

since $\int_c^\infty(\omega-c)^{k-1}e^{-\omega}d\omega/\Gamma(k)=e^{-c}$. With $c=a_jt$ and $z=ka_j/\Gamma_k$, so that $1-a_jt/\Gamma_k=1-tz/k$, the identity shows that U_k is the sum over the zeros of the densities $(ka_j)^k z^{-k-1}e^{-ka_j/z}/(k-1)!$. For real a_j this is the inverse gamma law of ka_j/Γ_k. Under RH, therefore, writing $\mathcal{L}(Y)$ for the law of Y,

$$U_k=\sum_\gamma \mathcal{L}\Big(\gamma^2\,\frac{k}{\Gamma_k}\Big),$$

a sum of probability laws: the Thorin measure with each atom γ^2 spread over a relative width of about $1/\sqrt{k}$. As $k\to\infty$, Γ_k/k concentrates at one and the atoms reappear, the passage from Williamson's theorem to Bernstein's; in statistics, Polson, Scott and Windle used k monotone mixtures of this kind to build samplers for the Bayesian bridge [278, Section 2.2, Theorem 2.1] and noted the same limit [278, Section 3].

The Williamson measures form a second resolution family for the Thorin measure, beside the Poisson family ρ_ε of the Cauchy flow (Section 2.1). In the chart $\theta=\sqrt{z}$ the gamma smoothing spreads the atom at γ over a width of about $\gamma/(2\sqrt{k})$, proportional to the height, while the Poisson smoothing spreads every atom over the width ε, and the inverse gamma kernel has tails far lighter than the x^{-2} tails of the Cauchy kernel. In Figure 8.2, where the widths match at $\theta=30$, the Williamson densities fall by many orders of magnitude between neighbouring zeros and the Poisson densities by only one to four.

The Poisson smoothing of the zeros is $\pi^{-1}\operatorname{Re}(\xi'/\xi)$ off the line and is computable from ξ alone at every resolution, which is why ρ_ε is the central object of

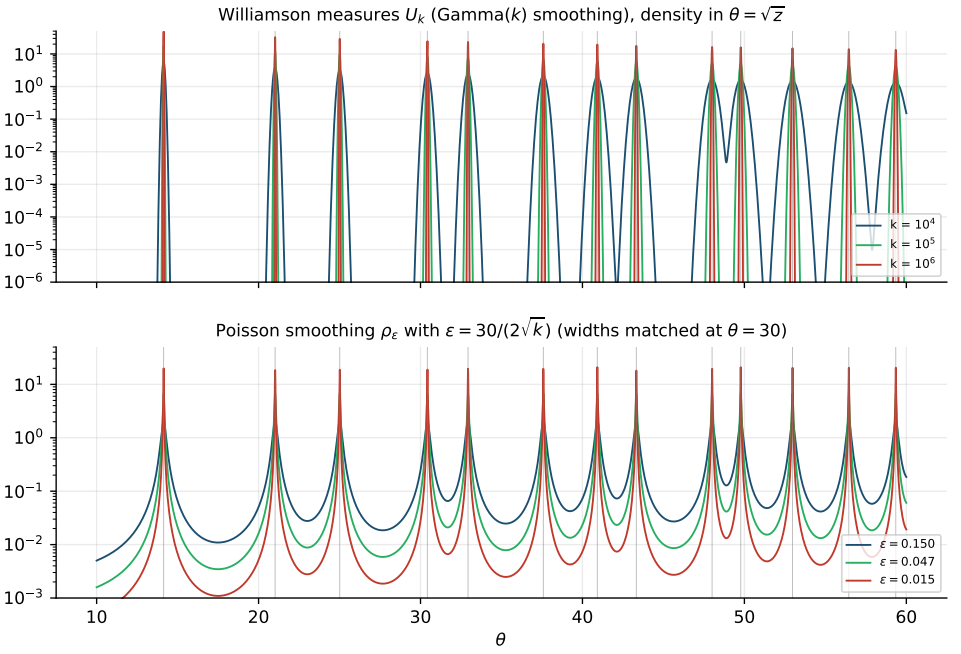


Figure 8.2: Top: the Williamson measures U_k of the heat trace for $k = 10^4$, 10^5, 10^6, computed from 1700 zeros, as densities in $\theta = \sqrt{z}$. Bottom: the Poisson smoothed Thorin densities ρ_ε with $\varepsilon = 30/(2\sqrt{k})$, so that the widths match at $\theta = 30$.

Chapter 3 and Section 9.1; the price is the heavy tail, which makes positivity at one resolution a statement about all zeros at once. The gamma smoothing is local, and an off line pair contributes to U_k a signed, oscillating term near its height.

8.1.5 Along a ray: the Gaussian smoothing of the zeros

Fix a height u and move along the ray $t = k/u^2$ of the (k, t) plane, on which the weight $x^k e^{-xt}$ of $c_k(t)$ peaks at $x = u^2$ and the density of U_k is taken at $z = u^2$. Write each zero as $\rho = \frac{1}{2} + \delta + i\gamma$ and put $\tilde{\gamma} = \gamma - i\delta$, so that $a = \tilde{\gamma}^2$; as in every sum of this section, the zero is taken with $\gamma > 0$ from each pair $\{\rho, 1-\rho\}$. The Gaussian smoothing of the zeros at resolution $1/(2\sqrt{t})$ is

$$\mathcal{G}(u,t) = \sum e^{-2t(\tilde{\gamma}-u)^2}.$$

A zero on the line contributes a Gaussian bump at its ordinate. The zeros $\frac{1}{2} \pm \delta + i\gamma$ of a quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ have conjugate values of $\tilde{\gamma}$, so $\mathcal{G}$ is real, and together they contribute $2e^{2t(\delta^2 - (\gamma-u)^2)} \cos(4t\delta(\gamma - u))$, a bump amplified by $e^{2t\delta^2}$ and modulated in u with period $\pi/(2t\delta)$. At $u = 0$, $\mathcal{G}(0,t) = W(2t)$.

PROPOSITION 8.1.10. *Let $u \geq 2$, $t > 0$ and $k = u^2 t$, not necessarily an integer, with a^k the principal power in $c_k(t) = \sum a^k e^{-at}$, and let $f_0 = k(\log u^2 - 1)$. Then*

$$e^{-f_0} c_k(t) = \sum_{|\tilde\gamma - u| \leq u/2} e^{-2t(\tilde\gamma - u)^2}\big(1 + \varepsilon_\rho\big) + R, \qquad |\varepsilon_\rho| \leq \exp\Big\{\frac{4t}{3u}|\tilde\gamma - u|^3\Big\} - 1,$$

where, with $\ell_\pm = u \pm \frac12\sqrt{u^2-1}$, $H(r) = u^2 \log \frac{r^2+1/4}{u^2} - r^2 + \frac14 + u^2$ and $\eta = 1 - u^2/(\ell_+^2 + \frac14) > 0$,

$$|R| \leq N(\ell_-) e^{tH(\ell_-)} + \Big(\ell_+^2 + \frac{1}{t\eta}\Big) e^{tH(\ell_+)}.$$

No hypothesis on the zeros is used.

Proof. Put $\tilde\gamma = u + v$ and $w = v/u$. Since $\operatorname{Re}\tilde\gamma > 0$ and $|\arg\tilde\gamma| < \frac{\pi}{4}$, the principal logarithm gives $e^{-f_0} a^k e^{-at} = e^E$ with

$$E = tu^2\big(2\log(1+w) - 2w - w^2\big) = -2tv^2 + 2tu^2 \sum_{m \geq 3} \frac{(-1)^{m+1} w^m}{m},$$

and for $|w| \leq \frac12$ the series has modulus at most $|w|^3/(3(1-|w|)) \leq \frac23 |w|^3$. This gives $\varepsilon_\rho = e^{E + 2tv^2} - 1$ and its bound, since $|e^z - 1| \leq e^{|z|} - 1$. For a zero with $|v| > u/2$, $|v|^2 = (\gamma - u)^2 + \delta^2$ and $\delta^2 < \frac14$ give $|\gamma - u| > \frac12\sqrt{u^2-1}$, so that $\gamma < \ell_-$ or $\gamma > \ell_+$, and $|e^E| = \exp t\big(u^2 \log(|a|/u^2) - \operatorname{Re} a + u^2\big) \leq e^{tH(\gamma)}$, because $|a| = \gamma^2 + \delta^2$ and $\operatorname{Re} a = \gamma^2 - \delta^2$. The function H increases up to $\sqrt{u^2 - \frac14}$, which lies between ℓ_- and ℓ_+, so each of the at most $N(\ell_-)$ zeros below ℓ_- contributes at most $e^{tH(\ell_-)}$. For $r \geq \ell_+$, $\log(1+x) \leq x$ gives $H(r) \leq H(\ell_+) - \eta(r^2 - \ell_+^2)$. The explicit Riemann von Mangoldt formula [108, Corollary 1.2] gives $N(r) \leq r^2$ for every r, and partial summation then gives $\sum_{\gamma > \ell_+} e^{-q(\gamma^2 - \ell_+^2)} \leq \ell_+^2 + 1/q$ for $q > 0$; take $q = t\eta$. □

For $u \geq 10$ the ratios $H(\ell_-)/u^2$ and $H(\ell_+)/u^2$ decrease in u, from -0.616 and -0.431 at $u = 10$, and $\eta > 0.55$, so that $|R| \leq (3u^2 + 2/t)e^{-0.43k}$. The term of order $1/u$ of ε_ρ is $\frac{2t}{3u}(\tilde\gamma - u)^3$. Along the ray, therefore, $e^{-f_0} c_k(t)$ is $\mathcal{G}(u,t)$ up to an exponentially small remainder and, in each term, a relative error at most $e^{4t|\tilde\gamma - u|^3/3u} - 1$, which is small when $t|\tilde\gamma - u|^3$ is small against u for the terms that carry the sum; for the zeros within a few multiples of $1/\sqrt{t}$ of u it is of order $1/(\sqrt{t}\,u)$. It is not small in general: at a fixed height, as t grows, the term of the nearest zero dominates, and its relative error grows with t. For an integer k the density of U_k in the chart $\theta = \sqrt{z}$ is exactly $\sqrt{2t/\pi}\, e^{-\lambda_k} e^{-f_0} c_k(t)$ at $\theta = u$, where $t = k/u^2$ and $\lambda_k \in (\frac{1}{12k+1}, \frac{1}{12k})$ is the remainder in Stirling's formula for $k!$ [315]. The density of U_k at θ is therefore about $\sqrt{2t/\pi}\,\mathcal{G}(\theta, t)$ with $t = k/\theta^2$: the zeros smoothed by normal densities of standard deviation $1/(2\sqrt{t}) = \theta/(2\sqrt{k})$, which is the width found above.

PROPOSITION 8.1.11. *RH holds if and only if $\mathcal{G}(u,t) \geq 0$ for every real u and every $t > 0$.*

Proof. Under RH every term is positive. Suppose that $\frac{1}{2} + d + ic$ is a zero with $d > 0$, and let $|u - c| < \frac{d}{2}$. The term of a zero has modulus $e^{2t\phi}$ with $\phi = \delta^2 - (\gamma - u)^2$. For the zero $\frac{1}{2} + d + ic$, $\phi > \frac{3}{4}d^2$; for a zero with $|\gamma - c| > 1$, $\phi < \frac{1}{4} - \frac{9}{16} < 0$. Group the finitely many zeros with $|\gamma - c| \leq 1$ by their type $(\gamma, |\delta|)$. Two types at the same ordinate have exponents ϕ differing by a nonzero constant, and two at different ordinates have exponents differing by a nonconstant affine function of u, so outside finitely many points one type has the largest ϕ. Fix such a u_*, different from every ordinate, and let (γ_*, d_*) be the maximising type and ϕ_* its exponent. Then $\phi_* > \frac{3}{4}d^2$, so $d_* > 0$, and the type consists of $\nu \geq 2$ zeros counted with multiplicity, which contribute $\nu e^{2t\phi_*}\cos(4td_*(\gamma_* - u_*))$. Along $t_j = (2j+1)\pi/(4d_*|\gamma_* - u_*|)$ this is $-\nu e^{2t_j\phi_*}$. The other n zeros with $|\gamma - c| \leq 1$ contribute at most $ne^{2t_j(\phi_* - \Delta)}$, with $\Delta > 0$ the gap to the next exponent, and the zeros with $|\gamma - c| > 1$ at most $32\log(c+2)e^{-5t_j/8}$ once $t_j \geq 1$, since by the explicit Riemann von Mangoldt formula every interval of heights $[y, y+1]$ holds at most $4\log(y+3)$ zeros. Hence

$$e^{-2t_j\phi_*}\mathcal{G}(u_*, t_j) \leq -\nu + ne^{-2t_j\Delta} + 32\log(c+2)e^{-(2\phi_* + 5/8)t_j},$$

which is negative for all large j. □

PROPOSITION 8.1.12. *For $\frac{1}{2} < \alpha < 1$ let $\eta = \alpha - \frac{1}{2}$ and $\mathcal{G}_\alpha(u,t) = \operatorname{Re}\sum e^{-2t(\tilde{\gamma} + i\eta - u)^2}$, over the zeros with $\gamma > 0$ as for $\mathcal{G}$, which is $\mathcal{G}$ at $\eta = 0$. Under RH $\mathcal{G}_\alpha(u,t) < 0$ for some u and t: there is no Gaussian criterion on the line $\operatorname{Re} s = \alpha$.*

Proof. Under RH $\tilde{\gamma} = \gamma$, and the term of γ is $e^{2t\eta^2}e^{-2t(\gamma-u)^2}\cos\big(4t\eta(\gamma - u)\big)$. Take $u = \gamma_1 + \pi/(4t\eta)$, where γ_1 is the lowest ordinate and a simple zero, and let $g > 0$ be the gap to the next ordinate. The term of γ_1 is $-e^{2t\eta^2 - \pi^2/(8t\eta^2)}$. For $t \geq \pi/(2\eta g)$ every other γ has $\gamma - u \geq (\gamma - \gamma_1)/2$, so the other terms total at most $e^{2t\eta^2}\sum_{\gamma > \gamma_1} e^{-t(\gamma-\gamma_1)^2/2} = o(e^{2t\eta^2})$. Hence $\mathcal{G}_\alpha(u,t) = -e^{2t\eta^2}(1 + o(1)) < 0$ as $t \to \infty$. □

Proposition 8.1.11 restricts Weil's criterion to the modulated Gaussians. For $f(x) = e^{-x^2/4t - iux}$ the transform $\hat{f}(r) = \int f(x)e^{irx}dx = \sqrt{4\pi t}\, e^{-t(r-u)^2}$ gives, summed over all zeros $\rho = \frac{1}{2} + i\gamma_\rho$,

$$\sum_\rho \hat{f}(\gamma_\rho)\overline{\hat{f}(\bar{\gamma}_\rho)} = 4\pi t\big(\mathcal{G}(u,t) + \mathcal{G}(-u,t)\big),$$

where $\mathcal{G}(-u,t)$ comes from the zeros $1 - \rho$ and is at most $e^{t/2}((u+14)^2 + \frac{1}{2t})e^{-2t(u+14)^2}$ in modulus for $u \geq 0$, too little to change the sign in the proof.

At $u = 0$ the family gives the first rung, $\mathcal{G}(0,t) = W(2t) > 0$ (Theorem 8.1.1). Since the Gaussian is entire, Weil's formula of Section 2.1 applies to $h(r) = e^{-2t(r-u)^2} + e^{-2t(r+u)^2}$, for which $\sum_\rho h(\gamma_\rho) = 2(\mathcal{G}(u,t) + \mathcal{G}(-u,t))$, and gives $\mathcal{G}$ from the primes:

$$\begin{aligned} \mathcal{G}(u,t) + \mathcal{G}(-u,t) &= 2e^{t/2-2tu^2}\cos 2tu + A(u,t) - P(u,t), \\ A(u,t) &= \frac{1}{4\pi}\int_{\mathbb{R}} \bigl(e^{-2t(r-u)^2} + e^{-2t(r+u)^2}\bigr) m(r)\,dr, \\ P(u,t) &= \frac{1}{\sqrt{2\pi t}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-(\log n)^2/8t}\cos(u\log n), \end{aligned} \tag{8.1.3}$$

with m as in (2.1.5).

Table 8.3 computes both sides. The prime sum converges at every resolution, more slowly the finer the resolution: since Chebyshev's function $\psi(x) = \sum_{n\leq x}\Lambda(n)$ grows like x, the weight $n^{-1/2}e^{-(\log n)^2/8t}$ acts, as a density in $\log n$, like $e^{t/2}$ times a Gaussian with centre $2t$ and standard deviation $2\sqrt{t}$. At $t = 3$ (last row) the rest of the prime sum beyond 10^9 is nearly all main term: with the integral $(2\pi t)^{-1/2}\int_N^\infty x^{-1/2}e^{-(\log x)^2/8t}\cos(u\log x)\,dx$ that the prime number theorem predicts for it, the difference at $N = 10^9$ falls to 3×10^{-10}. Where $\mathcal{G}$ is small the two arithmetic terms cancel, the more deeply the finer the resolution, as for W in Table 2.1; these are the points where an off line quadruple would first make $\mathcal{G}$ negative (Section 1.4).

u	t	$\mathcal{G}(u,t)$	$A(u,t)$	$P(u,t)$	$\lvert\mathcal{G} - A + P_N\rvert$		
					$N = 10^4$	$N = 10^6$	$N = 10^9$
100	1	0.0960845325	0.5519917891	0.4559072567	2.3×10^{-6}	1.2×10^{-10}	9.6×10^{-19}
1977.22	1.75	1.9835406644	0.8672562847	−1.1162843797	1.2×10^{-3}	1.3×10^{-7}	2.2×10^{-13}
1000	2	0.8452691893	0.7150931047	−0.1301760846	5.8×10^{-4}	5.8×10^{-6}	1.9×10^{-11}
100	3	0.0003053958	0.3186935678	0.3183870527	2.0×10^{-3}	5.4×10^{-4}	1.1×10^{-6}

Table 8.3: The Gaussian smoothing of the zeros from the zeros and from the primes, by (8.1.3). $\mathcal{G}(u,t)$ is summed over the zeros below height 2197.26, the rest adding less than 10^{-70000}; the polar term and $\mathcal{G}(-u,t)$ are below 10^{-8000}. P_N is P with the sum cut at $n \leq N$, and the P column is P_{10^9}. The bound on the rest of the prime sum that ignores its phases, from $\psi(x) < 1.03883x$ [320, Theorem 12], is 1.6×10^{-20}, 2.1×10^{-10}, 1.0×10^{-8} and 1.0×10^{-4} at $N = 10^9$ for the four rows; in the first row the difference at $N = 10^9$ is the rounding error of the sums, in extended precision.

8.1.6 What a verification height buys

Chapters 10 and 11 reach further rungs by the same method. Adding to the clock variable an independent positive stable variable of index $\frac{1}{2}$, that is, multiplying the

clock by $e^{-c\sqrt{u}}$ with $c = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2}) = 2.6860917$, gives the floor $e^{-c\sqrt{u}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ of Section 10.2, whose canonical function, y times its Lévy density, is $k(y) = W(y)+c/(2\sqrt{\pi y})$. By Theorem 8.1.1, $k > W > 0$, so the floor is infinitely divisible unconditionally; Section 10.2 proves this independently, and proves that the floor is a GGC if and only if RH holds (Theorem 10.2.15). Weil's functional is positive on the cone of test functions completely monotone in r^2 (Theorem 17.1.15); this is the first rung integrated against a positive weight, $\sum_\rho \hat{h}(\gamma_\rho) = 2\int_0^\infty W(y)g(y)\,dy$ with $g \geq 0$, and for it Section 17.1 proves $W > 0$ afresh.

Every rung is reached by the same mechanism: near the origin the archimedean term dominates the primes, and away from it a known positive term, usually a verified low zero, dominates the unverified high zeros. Only the second limits the order (Theorem 8.1.6); the first limits the time: it gives every order $k \geq 1$ up to $t = 0.019$, and every order above 10^{17} up to $t = 1.29$ (Theorem 8.1.9).

THEOREM 8.1.13. *Let $H = 3\times10^{12}$, the height to which RH is verified [277]. Then $(-1)^k W^{(k)}(t) > 0$ for every $t \geq 1.29$ and every integer k with $0 \leq k \leq t(H-23)^2$. Consequently $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every integer k with $0 \leq k \leq 1.29(H-23)^2$, and $1.29(H-23)^2 > 1.16\times10^{25}$.*

Proof. Fix $t \geq 1.29$ and k, and put $u = \sqrt{k/t}$, $D = H - u \geq 23$ and $f(v) = v^{2k}e^{-tv^2}$. As in Section 8.1.5, write each zero as $\frac{1}{2}+\delta+i\gamma$ and put $\tilde{\gamma} = \gamma - i\delta$, so that $c_k(t) = \sum f(\tilde{\gamma})$ over one zero of each pair $\{\rho, 1-\rho\}$, taken with $\gamma > 0$. On $(0,\infty)$, $(\log f)'' = -2k/v^2 - 2t$ is at most $-2t$, and at least $-4t$ on $[u,\infty)$, while $(\log f)'(u) = 0$; hence $f(v) \leq f(u)e^{-t(v-u)^2}$ for every $v > 0$, and $f(v) \geq f(u)e^{-2t(v-u)^2}$ for $v \geq u$. The zeros below height H lie on the line and contribute $f(\gamma) > 0$. One of them lies in $[u, u+15]$: for $u \leq 1000$ because $\gamma_1 < 15$ and consecutive ordinates below 1015 differ by less than 7, and for $u \geq 1000$ because the explicit Riemann von Mangoldt formula of [108, (1.7) and Corollary 1.4] gives $N(u+15) - N(u) > 4$. That zero contributes at least $f(u)e^{-450t}$. A zero above H, wherever it lies, has $|\delta| < \frac{1}{2}$ and $|\tilde{\gamma}| \geq \gamma > u$, so its term has modulus $f(|\tilde{\gamma}|)e^{2t\delta^2} \leq e^{t/2}f(u)e^{-t(\gamma-u)^2}$. Partial summation with $N(r) \leq r^2$, and $\int_D^\infty e^{-tv^2}dv \leq e^{-tD^2}/(2tD)$, give

$$\sum_{\gamma>H} e^{-t(\gamma-u)^2} \leq e^{-tD^2}\Big(H^2 + \frac{1}{t} + \frac{H}{tD}\Big),$$

and therefore

$$\frac{c_k(t)}{f(u)} \geq e^{-450t} - e^{t/2-tD^2}\Big(H^2 + \frac{1}{t} + \frac{H}{tD}\Big) > 0,$$

since $t(D^2 - 450.5) \geq 1.29\cdot 78.5 > 101$ while $\log(H^2 + 1 + H/23) < 57.5$. For the second statement, Theorem 8.1.9(b) gives every order for $t \leq 1.29$, and for $t > 1.29$ every $k \leq 1.29(H-23)^2$ satisfies $k \leq t(H-23)^2$. The zeros below height

1015 are computed in ball arithmetic, and every numerical inequality of the proof is evaluated. □

For the floor, $c/(2\sqrt{\pi y})$ is completely monotone, so Theorem 8.1.13 gives $(-1)^n k^{(n)} > 0$ for every $n \le 1.16 \times 10^{25}$; Theorem 10.2.17 proves $n \le 10^{10}$ independently, with a Cauchy circle of radius $y/2$, which loses a factor 2^n and makes its reach grow only linearly in H. The same circle applied to W itself [145] bounds the prime terms of (2.1.5) at order k by a quantity that stays below the archimedean term only for $t \lesssim c_P/k$ up to a constant, where $c_P = \frac{1}{6}(\log 2)^2$ is the constant of the factor $e^{-c_P/t}$ that the circle leaves to the prime 2, while the zeros above H stay below the first zero only for $t \gtrsim 2k\log(H/\gamma_1)/H^2$. Bounding the factor 2^k that the circle loses by e^k, the two ranges meet at an order of about $H\big(c_P/(2\log(H/\gamma_1))\big)^{1/2}$, which is 1.1754×10^{11} for $H = 3 \times 10^{12}$ (heuristic); the exact factor $\log 2$ in place of 1 multiplies this by $(\log 2)^{-1/2}$. This limits the circle bound, not the derivative signs, and is far below the $1.29(H-23)^2$ of Theorem 8.1.13.

Problem 8.1. At the verified height $H = 3\times 10^{12}$, Theorem 8.1.16 gives the signs of the derivatives of W at every $t > 0$ of every order $k \le 1.70(H-23)^2$. Can this range be increased to $k \le cH^2$ with an explicit $c > 1.70$? In the quadruple model of Section 1.4.1, a quadruple just above height H first changes a sign at an order of about $H^2 t^*$, with t^* set by the gaps around it, which suggests a scale to investigate (heuristic; the model does not show that such a failure occurs for the zeros of ζ).

8.1.7 A compound Poisson defect of the off line zeros

Group the exponents a_k into the real ones, $a_k = \gamma^2 > 0$, of zeros on the line, and the conjugate pairs $\{a, \bar{a}\}$ of nonreal exponents, one for each off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$, for which $a = A - iB$ with $A = \gamma^2 - \delta^2$ and $B = 2\delta\gamma$. Each pair contributes $2e^{-At}\cos(Bt)$ to W, and $\cos\theta = 1 - 2\sin^2(\theta/2)$ splits this into a Laplace transform part and a nonnegative remainder.

Theorem 8.1.14. *Unconditionally, $W(t) = \int_{(0,\infty)} e^{-tz}\, U_+(dz) - R(t)$, where U_+ is a Thorin measure, $R \ge 0$, and*

$$U_+ = \sum_{a_k \text{ real}} \delta_{a_k} + 2\sum_{\text{pairs}} \delta_A, \qquad R(t) = 4\sum_{\text{pairs}} e^{-At}\sin^2\Big(\frac{Bt}{2}\Big).$$

Let T_+ be the GGC with Thorin measure U_+, and Y the infinitely divisible variable with Lévy density $R(t)/t$, independent of T. Then

$$T + Y \overset{d}{=} T_+, \qquad \mathbb{E}e^{-uY} = \prod_{\text{pairs}} \frac{(1+u/a)(1+u/\bar{a})}{(1+u/A)^2}.$$

The defect Y is compound Poisson, with rate and jump density

$$\lambda = \int_0^\infty \frac{R(t)}{t}\,dt = \sum_{pairs} \log\Big(1+\frac{B^2}{A^2}\Big), \qquad \frac{R(t)}{\lambda t},$$

and $\lambda < 10^{-12}$, so $\Pr(Y=0) = e^{-\lambda} > 1-10^{-12}$. The following are equivalent: $R \equiv 0$; $Y = 0$; RH. In that case $U_+ = U_{\mathrm{RH}}$ and $T_+ = T$.

Proof. The decomposition is the identity applied pair by pair; U_+ is a Thorin measure because $A \ge \frac12|a|$, since $|a| = \gamma^2+\delta^2$ and $\gamma > 14$, and $\sum_k |a_k|^{-1} < \infty$. By Tonelli and the Frullani type integral $\int_0^\infty e^{-At}\sin^2(Bt/2)\,t^{-1}dt = \frac14\log(1+B^2/A^2)$, the function $R(t)/t$ is integrable on $(0,\infty)$ with integral λ, which is finite by the bound below, and a Lévy measure of finite mass is compound Poisson. For the bound let $H = 3\times10^{12}$. Every pair has $\gamma > H$, and $\delta < \frac12$ gives $B^2/A^2 = 4\delta^2\gamma^2/(\gamma^2-\delta^2)^2 < \gamma^2/(\gamma^2-\frac14)^2 < (1+10^{-24})\gamma^{-2}$. The two zeros $\frac12\pm\delta+i\gamma$ of a pair have the same ordinate, so $\lambda < \frac12(1+10^{-24})\sum_{\gamma>H}\gamma^{-2}$, the sum running over all zeros of ordinate above H. Partial summation with the explicit bound for $N(T)$ of [108, Corollary 1.2], exactly as for ε_H in the proof of Proposition 19.7.2, gives $\sum_{\gamma>H}\gamma^{-2} < 1.4798\times10^{-12}$, hence $\lambda < 7.4\times10^{-13}$. This bound is evaluated in ball arithmetic.

By the formula for $\log G$ at the start of this section and Frullani's integral $\log(1+u/z) = \int_0^\infty (1-e^{-ut})e^{-zt}t^{-1}dt$,

$$\log G(u) = \int_0^\infty \big(1-e^{-ut}\big)\frac{W(t)}{t}\,dt = \int \log\Big(1+\frac{u}{z}\Big)U_+(dz) - \int_0^\infty \big(1-e^{-ut}\big)\frac{R(t)}{t}\,dt,$$

that is, $\mathbb{E}e^{-uT}\,\mathbb{E}e^{-uY} = \mathbb{E}e^{-uT_+}$, and the product formula follows from $\mathbb{E}e^{-uT_+} = \prod_{a_k\ \mathrm{real}}(1+u/a_k)^{-1}\prod_{\mathrm{pairs}}(1+u/A)^{-2}$ and (2.1.1). For the equivalences: every term of R is nonnegative, and a pair contributes a positive term outside the discrete set $\{t>0 : Bt \in 2\pi\mathbb{Z}\}$, so $R\equiv0$ if and only if there is no pair, that is, every a_k is real, which happens only for zeros on the line; and $Y=0$ if and only if $\lambda=0$ if and only if $R\equiv0$. Under RH, $U_+ = \sum_k \delta_{a_k} = U_{\mathrm{RH}}$ and $T_+ = T$ by Proposition 2.1.1. □

Remark 8.1.15. Unconditionally, therefore, the clock plus an independent compound Poisson variable of rate below 10^{-12} is a GGC, and the law of the clock is within total variation 10^{-12} of it, since $T = T+Y$ on $\{Y=0\}$.

8.1.8 The prime side of the ladder

The proofs of Theorems 8.1.9 and 8.1.13 stop at 1.29 and 1.16×10^{25} because the proof of Theorem 8.1.9 uses Lemma 8.1.8(b) with $K = 10^{17}$, for which 1.2912 is the root of $E_{K+1}(t) = 2S(t)$. Theorem 8.1.13 supplies every order up to 1.16×10^{25}, so the lemma can be used with that order in the place of 10^{17}. The root of

$E_{K+1}(t) = 2S(t)$ then moves to about 1.719 in floating point, and the argument of both theorems runs at $t = 1.70$.

THEOREM 8.1.16. *W is completely monotone on $(0, 1.70]$, and $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every integer k with $0 \le k \le 1.70(H-23)^2$, where $H = 3\times10^{12}$; and $1.70(H-23)^2 > 1.52 \times 10^{25}$.*

Proof. Let $K = 1.16 \times 10^{25}$. By Theorem 8.1.13, $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every $k \le K$. For $k > K$ and $t \le 0.019$, Theorem 8.1.9(a) applies. For $k > K$ and $t \in [0.019, 1.70]$ the proof of Theorem 8.1.9(b) applies with K in the place of 10^{17} and 1.70 in the place of 1.29, and with the cutoff $n_0 = 10^9$ in the place of 2^{24}: Lemma 8.1.8(b) with $\varepsilon = 10^{-6}$ gives $C_\varepsilon(k) \le C_\varepsilon(K+1) < 1 + 6 \times 10^{-7}$ for $k > K$, and the condition $e^2x^2 \le 4\varepsilon k$ holds for $x \le 2\sqrt{\varepsilon(K+1)}/e$, which exceeds 2.5×10^9, which covers every prime power $n \le 10^9$ at $t \ge 0.019$ (there $x_n < 76$). By the monotonicity in k and t it suffices to show $E_{K+1}(1.70) - S_C(1.70) - Q_{K+1}(1.70) > 0$, with S_C the sum S_B with $B(x_n)$ replaced by C for $n \le 10^9$. Chernoff's bound for the gamma law with $a = (1 - 2^{-13})^2\lambda$ gives $E_{K+1}(1.70) > 26.7532$; the prime powers $n \le 10^9$ give $S_C(1.70) < 26.2968$, with the tail bound of that proof, which is 2.9×10^{-4} at $q = \log n_0/8t = 1.52$; and $Q_{K+1}(1.70) < e^{-1000}$. The margin exceeds 0.4564. So the orders $k > K$ have the right signs on $(0, 1.70]$ and, with the orders $k \le K$, W is completely monotone there.

For $t > 1.70$ and $k \le 1.70(H-23)^2$ the proof of Theorem 8.1.13 applies with 1.70 in the place of 1.29, since $k \le t(H-23)^2$ and the only place where the lower bound on t enters is $t(D^2 - 450.5) \ge 1.70 \cdot 78.5 > 133$, against $\log(H^2 + 1 + H/23) < 57.5$. Finally $1.70(H-23)^2 > 1.52 \times 10^{25}$. Every numerical inequality of the argument for $k > K$ is evaluated in ball arithmetic, with the prime powers $n \le 10^9$ (50 851 223 of them): all of them hold. □

In floating point the root of $E_{K+1}(t) = 2S(t)$ for $K = 1.52 \times 10^{25}$ is about 1.724, which is the time $t_0 = 1.72$ of the phase free bound at the verified height (Section 25.2.6); 1.70 is the safe round number.

By Proposition 8.1.10, on the ray $t = k/u^2$ the rung is $\mathcal{G}(u,t)$ up to the errors of that proposition, and (8.1.3) writes $\mathcal{G}(u,t) + \mathcal{G}(-u,t)$ as a polar term plus $A(u,t) - P(u,t)$, where $A(u,t)$ is close to $\log(u/2\pi)/(2\sqrt{2\pi t})$ and $P(u,t) = (2\pi t)^{-1/2}\Pi(u;t)$, with $\Pi(T;x) = \sum_{n\ge2} \Lambda(n) n^{-1/2} e^{-(\log n)^2/8x} \cos(T \log n)$ the smoothed prime sum. To leading order the rung at height T and scale x is therefore positive when $2\Pi(T;x) < \log(T/2\pi)$, the comparison for which Section 25.2.6 gives the phase free time $t_0(T)$, which agrees to five digits with $x_c(T)$, the root of $M(x) = \log(T/2\pi)$ with $M(x) = 2\sum_{n\ge2} \Lambda(n) n^{-1/2} e^{-(\log n)^2/8x}$ (twice the sum $S(x)$ of that section). Below $x_c(T)$ it holds whatever the phases; above it, it is a statement about cancellation among the primes, which Proposition 8.1.11 makes an exact criterion for RH, with A and $\mathcal{G}$ in place of the leading terms.

The comparison can be evaluated far beyond the verified height, since the cost depends on T only through the precision of $T \log n$ modulo 2π. The table gives

$2\Pi(T;x)$ against $\log(T/2\pi)$, with the prime powers $n \le N$, $N = \lfloor e^{\sqrt{8xD}} \rfloor$ for the tolerance $D = 22$ at $x \le 1$, $D = 20$ at $x = 1.7$, $D = 16$ at $x = 2.5$ and $D = 14$ at $x = 3$, and the phases in ball arithmetic at 300 bits. The weight of the omitted prime powers is about $e^{\sqrt{2xD}-D}$ times a factor of order one, not e^{-D}; the prime number theorem estimate of the sum of their absolute values is below 1.5×10^{-2} in every row. The values are not certified, and the leading term $\log(T/2\pi)$ stands for the archimedean term $A(T,x)$, with a relative error $O(T^{-2})$. The condition holds in every row; the scales $x = 1.7$, 2.5 and 3 at $T = 10^{12}$ are above $x_c(10^{12}) = 1.679$ (computed), where the weights alone no longer decide.

T	x	N	$2\Pi(T;x)$	$\log(T/2\pi)$	$\pi/(4x)$
10^{12}	0.5	1.2×10^4	-0.187	25.79	1.57
10^{12}	1.0	5.8×10^5	-0.374	25.79	0.79
10^{12}	1.7	1.5×10^7	-0.142	25.79	0.46
10^{12}	2.5	5.9×10^7	0.630	25.79	0.31
10^{12}	3.0	9.1×10^7	1.279	25.79	0.26
3×10^{12}	1.7	1.5×10^7	-0.266	26.89	0.46
3×10^{12}	3.0	9.1×10^7	-2.713	26.89	0.26
10^{20}	1.0	5.8×10^5	-0.720	44.21	0.79
10^{30}	1.0	5.8×10^5	-0.077	67.24	0.79
10^{50}	1.0	5.8×10^5	0.881	113.29	0.79

The last column is the product $\delta\Delta$ of the displacement and the gap of an isolated off line pair at which, in the local model of Proposition 1.4.4 and to leading order only, a failure of the comparison at the scale x would first be visible, since $t^*_{\mathrm{loc}} = \omega^*/(2\delta\Delta)$ with ω^* just above $\pi/2$ (Section 1.4.4). The failure sits at one offset of the window in the height, so one value of T excludes nothing; a scan in T over about Δ is needed. In this reading the prime side resolves only isolated zeros well off the line, and a pair with $\delta\Delta \ll 1$, such as a Lehmer type near miss, needs a scale $x \gg 1$ at which the cutoff N is out of reach. The table and the values of M and x_c quoted above are computed.

8.2 The Thorin functional from the arithmetic

The explicit formula gives the Thorin measure from the gamma factor and the primes without reference to any zero [282, Chapter 7].

8.2.1 The Thorin functional and the explicit formula

The Thorin measure cannot be introduced as a measure without assuming the conclusion. Bernstein's theorem would supply a positive measure U with $W(t) = \int e^{-tz} U(dz)$ if W were completely monotone, but that is RH; and $\sum_k \delta_{a_k}$ is a measure on $[0,\infty)$ only if every a_k is real and nonnegative, which is RH again, since an atom at a nonreal a_k can be paired only with functions defined off the

real axis. We therefore define U as a linear functional on a class of test functions large enough to determine it and small enough that the arithmetic side converges.

Let $\mathcal{T}$ be the set of functions $\varphi(z) = h(\sqrt{z})$ with h even, holomorphic in $|\operatorname{Im} r| \le \frac12 + c$ for some $c > 0$ and $O((1+|r|)^{-2-c})$ there, and let $g(x) = \frac{1}{2\pi}\int h(r)e^{-irx}dr$. Then φ is holomorphic on the image $\{r^2 : |\operatorname{Im} r| < \frac12 + c\}$, which contains $-\frac14$ and every a_k. Define

$$\begin{aligned} \langle U, \varphi\rangle &= \varphi\Big(-\frac14\Big) + \frac{1}{4\pi}\int_{\mathbb{R}} \varphi(r^2)\, m(r)\, dr - \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}}\, g(\log n), \\ m(r) &= \operatorname{Re}\psi\Big(\frac14 + \frac{ir}{2}\Big) - \log\pi, \end{aligned} \tag{8.2.1}$$

where $\psi = \Gamma'/\Gamma$ and Λ is the von Mangoldt function; no zero appears. Weil's explicit formula [357], [167, Theorem 5.12] identifies the functional: $\langle U, \varphi\rangle = \sum_k \varphi(a_k)$ for every $\varphi \in \mathcal{T}$, the sum running over the exponents $a_k = -(\rho_k - \frac12)^2$ of Section 2.1, one for each pair $\{\rho, 1-\rho\}$, and in particular $\langle U, e^{-t\cdot}\rangle = W(t)$ [279, Proposition 9.4].

The term $\varphi(-\frac14) = h(\frac{i}{2})$ comes from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, which the factor $s(s-1)$ of ξ removes; it is the only evaluation of φ off $[0,\infty)$. The integral is the contribution of the gamma factor: with the Riemann Siegel theta function $\vartheta(r) = \operatorname{Im}\log\Gamma(\frac14 + \frac{ir}{2}) - \frac{r}{2}\log\pi$ we have $m = 2\vartheta'$, so the integral is $\pi^{-1}\int_0^\infty h\, d\vartheta$, the integral of h against the smooth part of the zero counting function. The prime sum is the fluctuating part; it converges absolutely against the weight $n^{-1/2}$ because the half width $\frac12 + c$ of the strip makes $g(x) = O(e^{-(1/2+c)|x|})$.

The sine squared lemma (2.1.9) rewrites the functional in terms of variograms. By (5.1.2) its kernel carries each $\varphi \in \mathcal{T}$ to $g(0) - g(x)$, a variogram when $h \ge 0$, and the functional takes the absolutely convergent form [279, Proposition 9.2]

$$\begin{aligned} \langle U, \varphi\rangle = \int_0^\infty \frac{e^{-x/2}}{e^{2x}-1}\big(g(0) - g(x)\big)\, dx - C_\Gamma g(0) \\ + \int_0^\infty e^{x/2} g(x)\, dx - \sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}} g(\log n), \end{aligned}$$

where $C_\Gamma = \frac12\log\pi - \frac12\psi(\frac14) - 2 = 0.6860917096$ is the archimedean constant of Section 1.3.3. To derive this form, split $\varphi(-\frac14) = \int_0^\infty (e^{x/2} + e^{-x/2}) g(x)\, dx$; the half with $e^{-x/2}$ combines with the gamma integral into the first two terms.

It regularises a formal pairing: formally $\langle U, \varphi\rangle = \int_0^\infty \big(g(0) - g(x)\big)\, \nu(dx)$, where ν is the canonical measure of (1.3.1) at $\alpha = \frac12$,

$$\nu(dx) = \frac{e^{-x/2}\, dx}{e^{2x}-1} - e^{x/2}\, dx + \sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\, \delta_{\log n}(dx),$$

but the pole and prime parts of this pairing diverge through their $g(0)$ terms. Part by part, the absolutely convergent form exceeds the formal pairing by multiples of $g(0)$: by $-C_\Gamma$ for the gamma factor, by $\int_0^\infty e^{x/2}dx$ for the pole and by $-\sum_n \Lambda(n)n^{-1/2}$ for the primes. Continued analytically as in Section 1.3.3, the last two are -2 and $\zeta'(\frac12)/\zeta(\frac12)$, and the three constants cancel by the identity $C_\Gamma + 2 - \zeta'(\frac12)/\zeta(\frac12) = 0$ of that section, which is equivalent to $\xi'(\frac12) = 0$. The only negative part of ν is the pole density $-e^{x/2}dx$, and Section 5.1.2 reads Weil's criterion from this sign pattern. At the basepoint one the pole density is $-dx$, and no analytic continuation is needed there: the prime atoms compensated by the pole density have a convergent sine squared integral, and its convergence is the prime number theorem (Proposition 12.1.3).

The functional is not in general the limit of the Thorin measures as the basepoint approaches the line. As $\alpha \downarrow \frac12$ the measures U_α of (2.1.6) converge vaguely to $U_{\text{line}} = \sum \delta_{\gamma^2}$, the sum running over the zeros on the line, whatever the truth of RH (Section 2.1 and [279, Proposition 9.5]), whereas unconditionally

$$U = U_{\text{line}} + \sum_{a_k \notin (0,\infty)} \delta_{a_k} \quad \text{on } \mathcal{T},$$

the sum running over the nonreal exponents. These come in conjugate pairs $(\gamma - id)^2$ and $(\gamma + id)^2$, one pair for each off line pair $\frac12 \pm d + i\gamma$ of Proposition 2.1.3 (written $\frac12 \pm \delta + i\gamma$ there), so $U = U_{\text{line}}$ exactly when RH holds [279, Proposition 9.6]. The functional evaluates φ at both exponents of an off line pair, while in (2.1.8) the pair contributes kernels $P_{\varepsilon+d} + P_{\varepsilon-d}$ of total mass zero once $\varepsilon < d$, which vanish in the limit.

8.2.2 The heat trace from the primes

For $\varphi(z) = e^{-tz}$, that is $h(r) = e^{-tr^2}$ and $g(x) = e^{-x^2/4t}/(2\sqrt{\pi t})$, the functional is the arithmetic form (2.1.5) of the heat trace, $W = \Pi + A - P$, with the polar term $\Pi = e^{t/4}$, the archimedean term A and the prime term P described in Section 2.1. Table 8.4 extends the comparison of Table 2.1 to smaller and larger t, with the three terms in sixty digit arithmetic (P over $n \le 2 \times 10^5$) against the sum over the first 1700 zeros; the two agree to every digit shown, and the last column is the number of decimal digits lost to cancellation, $\log_{10}\{(\Pi + |A| + P)/W\}$.

The cancellation follows a simple law. For $t \ge 0.005$ the archimedean term is negative (Exercise 8.8), so $\Pi + |A| + P = 2e^{t/4} - W$ and the digits lost are $\log_{10}(2e^{t/4}/W - 1)$. Since $W = e^{-\gamma_1^2 t}\big(1 + O(e^{-(\gamma_2^2 - \gamma_1^2)t})\big)$, this is

$$\log_{10} 2 + \frac{(\gamma_1^2 + \frac14)\, t}{\log 10} \approx 0.30 + 86.88\, t$$

up to an exponentially small error, below 2×10^{-5} for $t \ge 0.05$.

t	polar Π	archimedean A	prime P
0.002	1.000500	0.782881	2.6×10^{-26}
0.005	1.001251	−0.464059	7.2×10^{-11}
0.01	1.002503	−0.852797	8.4×10^{-6}
0.02	1.005013	−0.984061	0.002410
0.05	1.012578	−0.954619	0.057914
0.1	1.025315	−0.862510	0.162806
0.15	1.038212	−0.797634	0.240578
0.2	1.051271	−0.749754	0.301517
0.3	1.077884	−0.681852	0.396032

t	$\Pi + A - P$	W from zeros	digits lost
0.002	1.78338120537913	1.78338120537913	0.0
0.005	0.537191964091041	0.537191964091041	0.4
0.01	0.14969801125378	0.14969801125378	1.1
0.02	0.0185412917313195	0.0185412917313195	2.0
0.05	$4.58783504857357 \times 10^{-5}$	$4.58783504857357 \times 10^{-5}$	4.6
0.1	$2.10479979582235 \times 10^{-9}$	$2.10479979582235 \times 10^{-9}$	9.0
0.15	$9.65642094572633 \times 10^{-14}$	$9.65642094572633 \times 10^{-14}$	13.3
0.2	$4.43018218022364 \times 10^{-18}$	$4.43018218022364 \times 10^{-18}$	17.7
0.3	$9.32464654810622 \times 10^{-27}$	$9.32464654810622 \times 10^{-27}$	26.4

Table 8.4: The heat trace from the explicit formula and from the zeros. For $t \geq 0.05$ the digits lost are $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t/\log 10 \approx 0.30 + 86.88\,t$.

The cancellation costs working precision and nothing more: 26.4 digits at $t = 0.3$, well within sixty digit arithmetic. The zeros give W cheaply at moderate t, the primes only near $t = 0$. Section 14.1 finds the same profile for the Lévy density $-W'$ of the subordinator that drives the clock.

The arithmetic side is cheap where g is concentrated near $x = 0$, so that only small n contribute: at small t for the heat kernel, and only at large resolution for the Poisson kernel $h(r) = P_\varepsilon(\theta - r) + P_\varepsilon(\theta + r)$. That kernel is not in $\mathcal{T}$, but for $\varepsilon > \frac{1}{2}$ its transform $g(x) = \pi^{-1} e^{-\varepsilon|x|} \cos \theta x$ makes the prime sum converge absolutely; for smaller ε the prime side is Selberg's smoothed sum (Section 5.1.1). The spectral side is cheap where the test function is concentrated at the low zeros: at large t, and at small ε on a window. In the proofs of the lower rungs in Section 8.1 the two forms meet at $t = 1/20000$, where the inequalities are checked in interval arithmetic. The Thorin condition requires positivity at every scale, and neither side is cheap at every scale. The Pólya route of Section 2.1 is cheap at every scale, but only on a window of finite height, and that restriction is the barrier of Section 1.4.

8.3 Spacings, irregularity and the cell distance

On a window of finite height the errors of the Pólya route, measured by the cell distance $d_{\mathrm{cell}}(\varepsilon)$ of Section 3.1, are set by the spacings of the zeros, which follow the random matrix prediction already for the first 1700 zeros.

8.3.1 The cell distance to first order

We use the cells $C_k = (b_{k-1}, b_k]$ of Section 3.1.2, with $\Theta = 200$ and $K = N(\Theta) = 79$, and the first order formula for $\mu_\varepsilon(C_k) - 1$ derived there, whose remainder is $O(\varepsilon^3)$ (Remark 3.1.3); Exercise 8.10 gives its leading term. It needs only that the zeros below Θ be simple and on the line.

Under RH the formula becomes a sum over the ordinates. By Proposition 2.1.3, $\rho_\varepsilon(\theta) = \sum_j [P_\varepsilon(\theta - \gamma_j) + P_\varepsilon(\theta + \gamma_j)]$ over all positive ordinates, and the kernel $P_\varepsilon(\cdot - \gamma)$ has distribution function $F_\varepsilon(y - \gamma)$, where $F_\varepsilon(y) = \frac{1}{2} + \frac{1}{\pi}\arctan(y/\varepsilon) = \mathbf{1}_{y \ge 0} - \varepsilon/(\pi y) + O(\varepsilon^3/|y|^3)$ as $\varepsilon/|y| \to 0$. No ordinate lies on a cell boundary, so

$$\mu_\varepsilon(C_k) - 1 = -\frac{\varepsilon}{\pi} \sum_j \Big[\frac{1}{b_k - \gamma_j} - \frac{1}{b_{k-1} - \gamma_j} + \frac{1}{b_k + \gamma_j} - \frac{1}{b_{k-1} + \gamma_j} \Big] + O(\varepsilon^3),$$

and by (2.1.1) with $s = i\theta$, $\Xi'/\Xi(\theta) = \sum_j [(\theta - \gamma_j)^{-1} + (\theta + \gamma_j)^{-1}]$ under RH, so the two expansions agree. Summing absolute values,

$$d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\, \varepsilon + O(\varepsilon^3),$$
$$\kappa_{\mathrm{cell}} = \frac{1}{\pi} \sum_k \Big| \sum_j \Big[\frac{1}{b_k - \gamma_j} - \frac{1}{b_{k-1} - \gamma_j} + \frac{1}{b_k + \gamma_j} - \frac{1}{b_{k-1} + \gamma_j} \Big] \Big|.$$

Truncated to the first 1700 ordinates, the zero sum gives 5.415, against $\kappa_{\mathrm{cell}} = 5.43665$ from the form in Ξ'/Ξ and the observed slope $d_{\mathrm{cell}}(0.005)/0.005 = 5.43664$; the difference of 0.4 per cent is the contribution of the zeros above γ_{1700}. In the truncated formula the cell errors range from $-0.225\,\varepsilon$ to $+0.250\,\varepsilon$ (Figure 8.3), with the two largest positive values at the end cells: at the last because Θ is not a midpoint between ordinates, and at the first because below b_1 there is only γ_1 and then the reflections $-\gamma_j$, which do not balance the zeros above b_1.

The atoms $\theta = k \ge 1$ of the sinh benchmark of Section 4.1 and their reflections form the lattice $\mathbb{Z}$ without the point 0. For the full lattice every cell $(k - \frac{1}{2}, k + \frac{1}{2}]$ would have mass exactly one, by periodicity, so the deficit of each cell is the mass that the missing kernel at 0 would have put in it; the atom free interval $(0, \frac{1}{2}]$, which the benchmark counts as a cell of mass zero, receives the mass $\frac{1}{2}$ of the full lattice less that of the missing kernel:

$$\mu_\varepsilon\big((k - \tfrac{1}{2}, k + \tfrac{1}{2}]\big) = 1 - \frac{1}{\pi}\arctan\frac{\varepsilon}{\varepsilon^2 + k^2 - \frac{1}{4}}, \qquad \mu_\varepsilon\big((0, \tfrac{1}{2}]\big) = \frac{1}{\pi}\arctan(2\varepsilon).$$

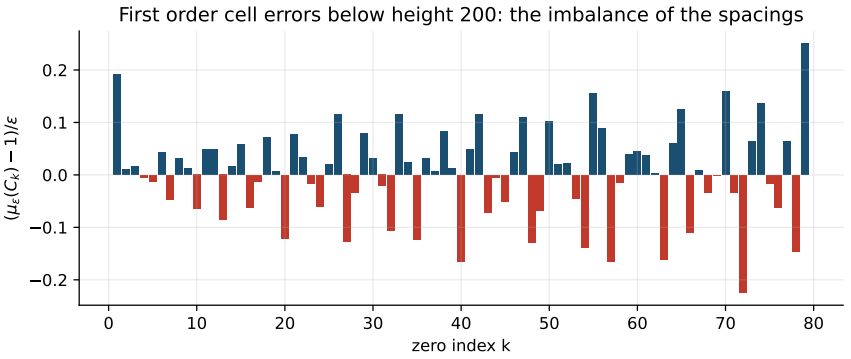


Figure 8.3: First order cell errors $(\mu_\varepsilon(C_k)-1)/\varepsilon$ for the 79 cells below height 200, computed from the first 1700 ordinates.

So the interior cells do not balance: each loses $\varepsilon/(\pi(k^2-\frac14))+O(\varepsilon^3)$, while $(0,\frac12]$ gains $2\varepsilon/\pi+O(\varepsilon^3)$. As $\sum_{k\le N}(k^2-\frac14)^{-1}=2-1/(N+\frac12)$, the cell distance of N atoms is $(4-1/(N+\frac12))\,\varepsilon/\pi+O(\varepsilon^3)$, which is $1.26694\,\varepsilon$ for $N=50$, the constant 1.267 of Section 4.1. With the convention used for the zeros, a first cell $(0,\frac32]$, it would be $(\frac43-1/(N+\frac12))/\pi=0.418$. For the zeros no cell is balanced, and the errors in Figure 8.3 do not decay along the window.

The constant vanishes exactly when the sum of $1/(b-x)$ over the ordinates and their reflections vanishes at every cell boundary b, as it does when these points are symmetric about every boundary. The lattice $\gamma_j=(j-\frac12)h$, whose reflections complete $(\mathbb{Z}+\frac12)h$, has every cell error zero, whereas the lattice of the benchmark does not. For the zeros every ordinate contributes to a cell error, with a weight that decays like $|C_k|/(\gamma_j-b_k)^2$. The two ordinates nearest an interior boundary are symmetric about it, so the asymmetry comes first from the next gaps on either side, and it is small when those gaps are equal.

8.3.2 Spacings of the zeros and the Thorin density

At height γ the mean spacing of the zeros is $\Delta=2\pi/\log(\gamma/2\pi)$, as in Section 1.4.2. We unfold by $\tilde\gamma=\vartheta(\gamma)/\pi$. The 1699 normalised spacings $\tilde\gamma_{k+1}-\tilde\gamma_k$ of the first 1700 zeros, which lie below height 2198, have variance 0.146 and minimum 0.089. Figure 8.4 compares their histogram with the Wigner surmise for the Gaussian unitary ensemble, $p_{\mathrm W}(s)=\frac{32}{\pi^2}s^2e^{-4s^2/\pi}$, and with the Poisson law e^{-s} of independent points. The histogram already follows the surmise, including the quadratic repulsion at small spacings that Montgomery's pair correlation conjecture predicts [235, pp. 183 and 184] and that Odlyzko observed at large height [252]. The variance is below the surmise value $3\pi/8-1=0.178$; the spacing distribution approaches the unitary one only slowly with height [235, p. 185], [35, Section 3].

Repulsion affects the Thorin density in two ways. First, it makes very small



Figure 8.4: The 1699 unfolded spacings of the first 1700 zeros against the GUE Wigner surmise and the Poisson law.

gaps rare but not absent. The smallest normalised spacing among the first 1700 zeros, 0.089, separates the ordinates 1977.17 and 1977.27, the pair that Section 1.4 moves off the line in Figure 1.1. By the Cauchy kernel criterion of Exercise 2.11, an isolated pair of ordinates at distance g in θ gives two peaks of ρ_ε only for $\varepsilon < \sqrt{3}g/2$. The resolution needed to separate every pair on a window is set by the smallest gap, which repulsion controls: if the N normalised spacings were independent, with density $p_{\mathrm{W}}(s) \propto s^2$ near zero, the smallest would be of order $N^{-1/3}$, which is 0.084 for $N = 1699$, close to the observed 0.089.

Second, repulsion sets the scale of the local visibility condition of Section 1.4.2: to first order in ε, an off line zero at distance d from the line makes ρ_ε negative at its ordinate if and only if $2/d^2$ exceeds $\sum_j \delta_j^{-2}$, where δ_j are the distances from its ordinate to the other ordinates. The sum is dominated by the nearest neighbours, and repulsion makes it typically of order Δ^{-2}, because $\mathbb{E}s^{-2}$ is finite for the surmise, where it equals $8/\pi$, and infinite for the Poisson law. With neighbours on a lattice of spacing Δ, the density at the ordinate stays positive at small resolution once d exceeds the threshold $(\sqrt{6}/\pi)\Delta \approx 0.78\,\Delta$ of that section. Since $d < \frac{1}{2}$, a zero hidden in this way can exist only where $\Delta < 0.64$, that is above height about 1.1×10^5. Hidden here means hidden at the ordinate: beside it the density can be negative even when the condition predicts a positive value there (Exercise 8.14).

The behaviour of the cell constant at large height is open (Section 25.2). In unfolded units the cell errors depend mainly on the nearby ordinates, so the sine kernel process of the unitary ensemble would predict their mean and fluctuations per cell; we have not computed that prediction. Computing the cell errors with Odlyzko's zeros near zero number 10^{12} [252], 10^{21} or 10^{22} [254, 255], at heights about 2.7×10^{11}, 1.4×10^{20} and 1.4×10^{21}, would compare the Thorin density with

the spacing statistics there; it needs the ordinates, which define the cells, and the hybrid evaluator of Section 2.2.2 for Ξ'/Ξ at the cell boundaries. Section 14.3.5 proposes the same comparison at large height for the positivity margins of the density, its values at the midpoints between consecutive ordinates.

8.4 Exercises

On Section 8.1.

8.1. Show that $c_0c_2 - c_1^2 = \frac{1}{2}\sum_{j,l}(a_j - a_l)^2 e^{-(a_j+a_l)t}$ and deduce that log convexity of W holds under RH.

8.2. Recompute the first row of Table 8.1 with the first 100 zeros and explain why no printed digit changes.

8.3. Show that for a real exponent a the summand of the Williamson measure U_k belonging to a is the inverse gamma density of ka/Γ_k with $\Gamma_k \sim \mathrm{Gamma}(k,1)$, and compute its variance for $k > 2$.

8.4. Let $\widetilde{W}(t) = e^{-t} + 2e^{-10t}\cos t$, the heat trace of the three exponents 1 and $10 \pm i$. Show that $\widetilde{W} > 0$ and $\widetilde{W}' < 0$ on $(0,\infty)$, using $\cos t > 0$ and $\sin t \geq 0$ for $t \leq 1$ and $2\sqrt{101}\,e^{-9} < 1$ for $t \geq 1$, so that the first two rungs of Theorem 8.1.1 hold. With $\theta = \arctan\frac{1}{10}$, show that $\frac{10}{101} < \theta < \frac{1}{10}$, hence $\frac{2\pi}{3} < 24\theta < \pi$, and deduce that $(-1)^{24}\widetilde{W}^{(24)}(0) = 1 + 2\cdot 101^{12}\cos 24\theta < 0$, so that $\widetilde{W}$ is not completely monotone. Then check numerically that the first order k at which $(-1)^k\widetilde{W}^{(k)}(t) < 0$ for some $t > 0$ is 12, near $t = 2.87$, and compare with the first failures of Table 1.2.

8.5. Show that the term of order $1/u$ of ε_ρ in Proposition 8.1.10 is $\frac{2t}{3u}(\tilde\gamma - u)^3$. For $u = 2$, $t = 1$ and $k = 4$, show that no zero lies in the disc $|\tilde\gamma - u| \leq u/2$, so that $c_4(1) = e^{f_0}R$ is carried entirely by the remainder, and compare the bound on R with the contribution of the first zero.

8.6. Show that $m(0) > -6$ and, comparing the series for ψ with an integral, that $m(60) > 1$. Deduce that for $t \leq 10^{-6}$ the archimedean term of (2.1.5) is at least $1/(4\sqrt{\pi t}) - 210/\pi > 55$ and the prime term at most $(\pi t)^{-1/2}e^{-(\log 2)^2/4t} < 0.01$. With the argument of the proof of Theorem 8.1.1 for $t \geq \tau$, taken at height 10^4 and from $t = 10^{-6}$, conclude that $W > 0$ on $(0,\infty)$ follows from the verification of RH below height 10^4 by estimates that can all be checked by hand.

On Section 8.2.

8.7. Verify the arithmetic form of $W(t)$ at $t = 0.01$ with the prime powers up to 10^4, and explain why so few suffice by showing that only $n = 2$ and $n = 3$ affect the first fifteen digits.

8.8. Show that m increases on $[0, \infty)$ and vanishes at a single point r_*, and deduce that $e^{r_*^2 t} A(t)$ decreases in t, so that A changes sign at most once; with Table 8.4, conclude that $A < 0$ for $t \geq 0.005$. Then show that for $t \geq 0.05$ the number of digits lost in the table equals $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t / \log 10$ up to an exponentially small error, and identify the source of each term.

8.9. Show that the regularised constants of the gamma, pole and prime parts of the variogram form sum to zero if and only if $\xi'(\frac{1}{2}) = 0$.

On Section 8.3.

8.10. Let $\ell = \Xi'/\Xi$. Show that $\xi(\frac{1}{2} + \varepsilon + ib) = \Xi(b - i\varepsilon)$, and deduce from the Taylor series of $\log \Xi$ about b that $\mu_\varepsilon(C_k) - 1 = -\frac{\varepsilon}{\pi}[\ell(b_k) - \ell(b_{k-1})] + \frac{\varepsilon^3}{6\pi}[\ell''(b_k) - \ell''(b_{k-1})] + O(\varepsilon^5)$. Under RH, check that the term in ε^3 agrees with the expansion of F_ε to third order.

8.11. Prove the arctangent formulas for the cell masses of the sinh benchmark, and deduce that its cell distance for N atoms is $(4 - 1/(N + \frac{1}{2}))\,\varepsilon/\pi + O(\varepsilon^3)$, and $(\frac{4}{3} - 1/(N + \frac{1}{2}))\,\varepsilon/\pi + O(\varepsilon^3)$ when the first cell is $(0, \frac{3}{2}]$.

8.12. Show that the Wigner surmise p_{W} has mean 1 and variance $3\pi/8 - 1$, and that $\mathbb{E} s^{-2} = 8/\pi$ under it, while $\mathbb{E} s^{-2} = \infty$ for the Poisson law.

8.13. Show that $\sum_{m \neq 0} (m\Delta)^{-2} = \pi^2/(3\Delta^2)$, deduce the lattice threshold $(\sqrt{6}/\pi)\Delta$ of the visibility condition, and find the height above which the threshold is less than $\frac{1}{2}$.

8.14. In a local model, put an off line pair at distance $d = 0.3$ from the line at the ordinate γ and zeros on the line at $\gamma + 0.2 + m$ and $\gamma - 4 - m$ for $m \geq 0$, and ignore the reflected kernels. At $\varepsilon = 0.005$, show that $2/d^2 < \sum_j \delta_j^{-2}$, so that the visibility condition predicts a positive density at γ, and check that $\rho_\varepsilon(\gamma) > 0$ while $\rho_\varepsilon(\gamma - 0.079) < 0$.

Chapter 9

Weak limits, HCM and a Bosch type threshold

Sections 9.2 and 9.3 bring in Bosch's theorem on powers of the half Cauchy law. The first explains why his result cannot be transferred as an input, and the second transfers his method instead and proves a threshold theorem for powers of the reciprocal Xi function, in which the angle of the zeros replaces Bosch's drift parameter and RH is the extreme case.

9.1 The limit of vanishing resolution

Neither closure under weak limits nor Bondesson's theorem reaches RH.

9.1.1 Weak limits of Thorin measures

Theorem 9.1.1 (Thorin [344, 345]; Bondesson [39, Theorem 3.1.5]; see also [282, Theorem 18.9]). *The class of GGC laws is closed under convergence in law.*

Corollary 9.1.2. *RH holds if and only if there are GGC variables X_N with $\mathbb{E}e^{-uX_N} \to 1/G(u)$ for all u in a set with an accumulation point in $(0,\infty)$ [282, Propositions 18.10 and 18.14].*

Proof. The transforms are analytic and bounded by one on $\operatorname{Re} u > 0$, so by Vitali's theorem they converge locally uniformly there, and the limit is $1/G$ on $(0,\infty)$. Since $1/G(u) \to 1$ as $u \downarrow 0$, the X_N converge in law to a proper limit with transform $1/G$, which is a GGC by the closure theorem, and Proposition 2.1.1 gives RH. Conversely, under RH the clock variable T is a GGC with transform $1/G$; take $X_N = T$. □

The Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 2.1 is the obvious candidate. Its measures μ_ε are the Thorin measures of $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ at $\alpha = \frac{1}{2}+\varepsilon$ whenever these are GGC transforms (Theorem 2.1.2), the transforms converge to $1/G$ pointwise as $\varepsilon \downarrow 0$, and the measures converge vaguely. The flow fails for two reasons.

First, the limit is the wrong measure. As shown after Proposition 2.1.3, $\mu_\varepsilon \to U_{\mathrm{line}}$ vaguely whatever the truth of RH [279, Proposition 9.5], because the contribution of an off line pair tends to zero in total variation (Proposition 3.2.1(ii)); the limit omits the off line zeros. It is the Thorin measure of the clock only under RH, since the Thorin functional U of Section 8.2 counts every zero and $U = U_{\mathrm{line}}$ exactly when RH holds (Section 8.2.1 and [282, Proposition 7.6]). Positivity of the vague limit carries no information.

Second, the closure theorem needs GGC laws along the sequence. The flow supplies a law at every resolution with no hypothesis, that of the shifted clock T_α of Theorem 12.2.5 at $\alpha = \frac{1}{2} + \varepsilon$, but at small ε a GGC law only under the hypothesis to be proved: the transform at basepoint $\frac{1}{2} + \varepsilon$ is a GGC transform if and only if no zero lies to the right of the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ (Theorem 2.1.2). Positivity of ρ_ε at that one ε is necessary for this but is not known to imply it, since near a cluster of line zeros the negative kernel of an off line pair can be outweighed by the kernels of its neighbours (Exercise 1.7 and Section 1.4.2). Positivity of $\rho_{\varepsilon'}$ for every $\varepsilon' > \varepsilon$ does imply it, as shown in Remark 2.1.4; for $\varepsilon = 0$ this is Lagarias's criterion [198], which is RH. Weak closure adds the endpoint $\varepsilon = 0$ to a family already known to consist of GGCs; it cannot show that the family consists of GGCs [282, Remark 3.10].

On the window $0 \le \theta \le 200$ the Pólya route computes $\min \rho_\varepsilon = 0.0147\,\varepsilon > 0$ at each of the eight resolutions of Table 3.1, down to $\varepsilon = 0.005$ (Section 1.3.1); with the verification of RH to height 3×10^{12} this is a theorem, since for every $\varepsilon \le 6.85$ the minimum of ρ_ε over $[0, 3 \times 10^{12} - 1.21]$ is $\rho_\varepsilon(0)$ (Theorem 12.1.18 and Remark 12.1.19), and Section 25.2.2 lists what would make the Pólya computation itself a certificate.

Weak convergence becomes useful only with approximants whose positivity is proved independently of ξ, such as Beurling generalised prime systems whose zeta functions provably have only real ordinates, finite Krein strings, and the partial products over verified zeros completed by the Weyl tail; the last two are the Kent approximants of Section 5.2.3, without and with the Weyl tail. For such approximants the distances of Section 3.1 measure the approximation. Verification also removes factors from the question: if the first N zeros are verified to lie on the line, then $1/G_N$, with $G_N(u) = \prod_{k \le N}(1 + u/\gamma_k^2)$, is a GGC transform with no hypothesis, and RH holds if and only if the tail factor G_N/G is a GGC transform as well [282, Corollary 18.11].

9.1.2 The HCM form of the Thorin condition

Bondesson's theorem (Section 4.1.2; [39, Theorem 6.1.1], see also [168]) and the results of Section 2.1 give the following.

Corollary 9.1.3. *(i) For every $\alpha \ge 1$ the function $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is HCM, unconditionally. (ii) The reciprocal Xi function $\varphi_{1/2}(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$,*

which is the clock $1/G$, is HCM if and only if RH holds.

Proof. (i) is Theorem 2.1.2 with Bondesson's theorem, and (ii) is Proposition 2.1.1 with Bondesson's theorem. □

By the Hadamard product (2.1.1), with $a_k = -(\rho_k - \frac{1}{2})^2$ and $w = v + 1/v$,

$$\varphi_{1/2}(uv)\,\varphi_{1/2}(u/v) = \prod_k \frac{a_k^2}{a_k^2 + a_k\,u\,w + u^2} = \prod_k \frac{a_k/u}{w + c_k(u)}, \qquad c_k(u) = \frac{a_k}{u} + \frac{u}{a_k}. \tag{9.1.1}$$

The numbers $c_k(u)$ are local to this subsection (not the derivatives $c_k(t)$ of Section 8.1). An off line pair, with values a and $\bar a$ of a_k, contributes $(|a|^2/u^2)/((w + c)(w + \bar c))$ with $c = c(u)$ complex, and $1/((w + c)(w + \bar c))$ is the Laplace transform of the signed function $e^{-t\,\mathrm{Re}\,c}\sin(t\,\mathrm{Im}\,c)/\,\mathrm{Im}\,c$. At a single u the remaining factors may smooth this to a positive function; Bondesson's theorem guarantees failure only across all u. At $u = |a|$ both values of c equal $2\cos\arg a$, and the pair contributes $(w + 2\cos\arg a)^{-2}$, which is completely monotone on $[2, \infty)$. Complete monotonicity in w at finitely many u is therefore necessary for HCM but not sufficient.

We test this necessary condition. From 60 digit values of ξ at real arguments, computed by (2.1.2) without cancellation, numerical differentiation gives the normalised derivatives $d_k = (-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\,\varphi_\alpha(u/v)$ for $k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{2.05, 3, 5, 10, 20, 50\}$, at $\alpha = \frac{1}{2}$ and at the control $\alpha = 1$. All 216 values are positive (Figure 9.1 shows $w = 3$ and $w = 20$), and the signs are certified: from the Taylor coefficients of H_u in w, computed in ball arithmetic [169] from $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, and so independently of the Pólya route, every d_k is enclosed in a ball of relative radius below 10^{-70} that lies in $(0, \infty)$. Under RH, (9.1.1) gives $H_u(w + t)/H_u(w) = \mathbb{E}e^{-tY}$ with $Y = \sum_k e_k/(w + c_k(u))$ and independent standard exponentials e_k, so $d_k = \mathbb{E}Y^k$ is of order d_1^k. The smallest value, 2.5×10^{-10} at $\alpha = \frac{1}{2}$, $u = 1$, $w = 50$, $k = 6$, is $2.54\,d_1^6$ with $d_1 = 0.0215$: it reflects the scale of Y, to which many zeros contribute, not closeness to a sign change. The test is consistent with (ii) and cannot decide it (Section 1.4, Table 1.3, and [282, Chapter 11]).

Under RH the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 2.1.1), and if moreover the zeros are simple and satisfy two further hypotheses, its density is not HCM (Theorem 9.3.22); for the Cauchy analogue of the clock variable this holds unconditionally (Theorem 9.3.20).

9.2 Bosch's half Cauchy theorem and HCM of the reciprocal Xi function

By Corollary 9.1.3(ii) the clock $\varphi = 1/G$ is HCM if and only if RH holds, the HCM form of the reciprocal Xi programme of [279]. In this section and in Section 9.3

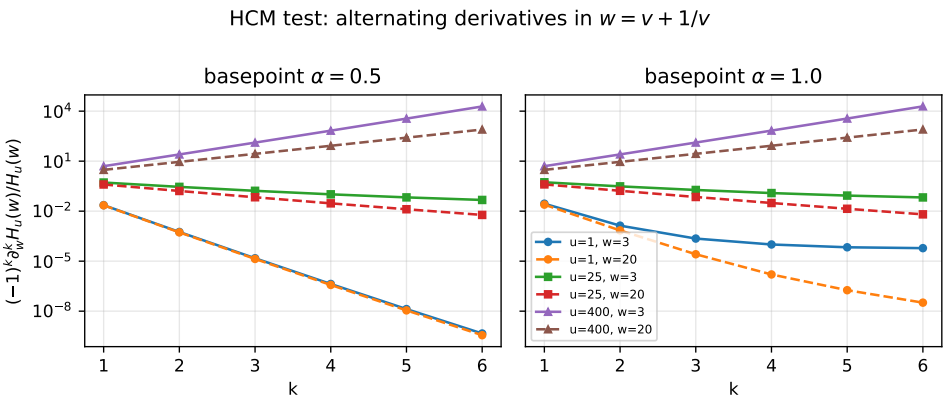


Figure 9.1: Normalised alternating derivatives $(-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\varphi_\alpha(u/v)$ for $1 \le k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{3, 20\}$, computed from the Pólya density, at the centre $\alpha = \frac{1}{2}$ and at the unconditional basepoint $\alpha = 1$.

the clock is a function of s, the variable u of Section 2.1: $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is the function $\varphi_{1/2}$ of Corollary 9.1.3. A sharp HCM theorem for a nearby, exactly solvable function might transfer to φ; the candidate is Bosch's theorem on powers of the half Cauchy law [41].

9.2.1 HCM and Bosch's theorem at the half Cauchy point

Bosch's theorem concerns densities, so we recall HCM (Section 4.1.2) in the form that applies to them: a continuous function $f : (0, \infty) \to [0, \infty)$ is HCM if for every $u > 0$ the map $v \mapsto f(uv)f(u/v)$, the pair function of f at u, is a completely monotone function of $w = v + v^{-1}$ on $[2, \infty)$, and a positive random variable is HCM if it has an HCM density. In logarithmic coordinates $u = e^a$, $v = e^y$ the condition reads: $y \mapsto f(e^{a+y})f(e^{a-y})$ is completely monotone in $w = 2\cosh y$. We use the following facts, all due to Bondesson; (F1), (F2), (F5) and (F6) are also stated in Section 4.1.2.

(F1) If X_1, X_2 are independent and HCM then X_1X_2 and X_1/X_2 are HCM [39, Theorem 5.1.1].

(F2) If X is HCM then X^q is HCM for $|q| \ge 1$ [39, pp. 68 and 69]. The mechanism is that for $|p| \le 1$ the function $v^p + v^{-p}$ is a Bernstein function of $v + v^{-1}$, so if h is an HCM function then so is $x \mapsto h(x^p)$ for $|p| \le 1$; if X has density f, the density of X^q is $|q|^{-1}x^{1/q-1}f(x^{1/q})$, and the power factor is HCM by (F3).

(F3) Pointwise products of HCM functions are HCM, and powers x^r, $r \in \mathbb{R}$, are HCM [39, p. 68].

(F4) Pointwise limits of HCM functions, when finite and positive, are HCM [39, p. 68 and Theorem 5.1.3].

(F5) Every HCM law is a GGC [39, Theorem 5.1.2].

(F6) A probability law on $[0,\infty)$ is GGC if and only if its Laplace transform is an HCM function [39, Theorem 6.1.1].

(F7) A function f that is not identically zero is HCM if and only if

$$\begin{aligned} f(x) &= K x^{r-1} h_1(x) h_2(1/x), \\ h_j(x) &= \exp\Big(-b_j x + \int_{(0,\infty)} \log\frac{1+t}{x+t}\,\Gamma_j(dt)\Big), \end{aligned} \tag{9.2.1}$$

with $K>0$, $r\in\mathbb{R}$, $b_j\ge 0$ and nonnegative measures Γ_j satisfying $\int(1+t)^{-1}\Gamma_j(dt)<\infty$ [39, (5.2.3) and Theorem 5.3.1]. In particular every HCM function that is not identically zero extends to an analytic and zero free function on $\mathbb{C}\setminus(-\infty,0]$ [39, p. 68].

THEOREM 9.2.1 (Bosch [41, Theorem 1.2]). *Let $T_\alpha = (Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with $Z_\alpha, \tilde{Z}_\alpha$ independent positive α stable (here α is a stability index, not a basepoint, and T_α is not the variable of Theorem 2.1.2). Then T_α has density $\sin(\pi\alpha)/(\pi\alpha(t^2+2t\cos\pi\alpha+1))$, a half Cauchy law with drift, and the density of T_α^β is HCM if and only if $\alpha\le 1/2$ and $|\beta|\ge 1/(1-\alpha)$.*

At $\alpha=1/2$ we have $T_{1/2}\overset{d}{=}|C|$ with density $2/(\pi(1+t^2))$. The density of $|C|^\beta$ is

$$f_\beta(x) = \frac{2}{\pi|\beta|}\,\frac{x^{1/\beta-1}}{1+x^{2/\beta}}, \qquad x>0.$$

With $x=e^{a\pm y}$ and $q=2/|\beta|$ a short computation gives

$$f_\beta(e^{a+y})f_\beta(e^{a-y}) = \frac{4}{\pi^2\beta^2}\,e^{-2a}\,\frac{1}{2\cosh(qa)+2\cosh(qy)}. \tag{9.2.2}$$

So at $\alpha=1/2$ Bosch's theorem says that $1/(c+\cosh(qy))$ is completely monotone in $\cosh y$ for every $c\ge 1$ exactly when $q\le 1$; here $c=\cosh(qa)$, which ranges over $[1,\infty)$.

PROPOSITION 9.2.2. *The half Cauchy density is not HCM.*

Proof. Take $\beta=1$, so $q=2$ in (9.2.2). Since $\cosh(2a)+\cosh(2y)=2\sinh^2 a+w^2/2$ with $w=2\cosh y$, the pair function is proportional to $h(w)=1/(c^2+w^2)$ with $c=2|\sinh a|>0$ for $a\ne 0$. For $x>0$,

$$h(2+x) = \int_0^\infty e^{-xt}\,e^{-2t}\,\frac{\sin(ct)}{c}\,dt.$$

If h were completely monotone on $(2,\infty)$, Bernstein's theorem and uniqueness of the Laplace transform would make the signed density $e^{-2t}c^{-1}\sin(ct)$ nonnegative. □

PROPOSITION 9.2.3. *The density of $|C|^\beta$ is HCM for $|\beta| \geq 2$.*

Proof. For $q = 2/|\beta| \leq 1$ the function $\cosh(qy)$ is a Bernstein function of $\cosh y$ by (F2), and $x \mapsto 1/(c+x)$ is completely monotone for $c > 0$, so the composition in (9.2.2) is completely monotone. □

At $\alpha = 1/2$ sufficiency is elementary; the deep part of Bosch's paper is $\alpha < 1/2$.

9.2.2 The half Cauchy law and the Riemann Xi function

On the Cauchy side everything is explicit. The logarithm of $|C|$ has density $g_C(y) = 1/(\pi\cosh y)$ and characteristic function $\operatorname{sech}(\pi t/2)$. The Weierstrass product

$$\operatorname{sech}\Big(\frac{\pi t}{2}\Big) = \prod_{k\geq 1}\Big(1 + \frac{t^2}{(2k-1)^2}\Big)^{-1}$$

says $\log|C| \overset{d}{=} \sum_{k\geq 1} L_k/(2k-1)$ with L_k independent standard Laplace variables. The Mellin transform is $\mathbb{E}|C|^z = \sec(\pi z/2)$ for $|\operatorname{Re} z| < 1$, the reciprocal of $\cos(\pi z/2)$, a function of the Laguerre Pólya class with real zeros $\pm(2k-1)$; these odd integers are the frequencies of the half Cauchy law. The Cauchy reciprocal Xi function

$$\varphi_C(s) = \mathbb{E}|C|^{i\sqrt{s}} = \operatorname{sech}\Big(\frac{\pi\sqrt{s}}{2}\Big) = \prod_{k\geq 1}\Big(1 + \frac{s}{(2k-1)^2}\Big)^{-1}$$

is the Laplace transform of the GGC $\sum_k e_k/(2k-1)^2$, with independent standard exponentials e_k and Thorin measure $\sum_k \delta_{(2k-1)^2}$, hence HCM by (F6). The checks below that name no source used mpmath [237] at 20 to 30 digits. Bourgade, Fujita and Yor [43] derive Euler's formulae for $\zeta(2n)$ from products of independent Cauchy variables, whose Mellin transforms are powers of $\sec(\pi z/2)$ (Section 13.2).

On the Riemann side the frequencies are the zeros γ of $\Xi(z) = \xi(\frac{1}{2} + iz)$. By (2.1.1), whose product converges absolutely because $\sum|\gamma|^{-2} < \infty$, unconditionally

$$\Xi(z) = \Xi(0)\prod_{\gamma}\Big(1 - \frac{z^2}{\gamma^2}\Big),$$

the product over zeros γ of Ξ with $\operatorname{Re}\gamma > 0$, counted with multiplicity; none is purely imaginary because $\xi(\sigma) > 0$ for real σ. Hence

$$\varphi(s) = \frac{\Xi(0)}{\Xi(i\sqrt{s})} = \frac{\xi(1/2)}{\xi(1/2+\sqrt{s})} = \prod_{\gamma}\Big(1 + \frac{s}{\gamma^2}\Big)^{-1}.$$

Under RH define, in the spirit of Biane, Pitman and Yor [33], $\log W = \sum_\gamma L_\gamma/\gamma$ with independent standard Laplace variables L_γ. Then $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1 \approx 14.1347$, the characteristic function of $\log W$ is $\xi(1/2)/\xi(1/2+t)$, and $\varphi(s) = \mathbb{E}W^{i\sqrt{s}}$. The variable W, not the heat trace $W(t)$, exists without RH: $\log W$ has the law of $Z\sqrt{2T}$ in the van Dantzig pair of Section 2.1, with Z standard normal independent of the clock variable T. Only the series for $\log W$ needs RH. Biane, Pitman and Yor represent 2ξ itself, not its reciprocal, as a Mellin transform (Sections 2.1 and 12.1).

	half Cauchy	Riemann
entire function	$\cos(\pi z/2)$	$\Xi(z)/\Xi(0)$
frequencies	$2k-1$	γ
log variable	$\sum_k L_k/(2k-1)$	$\sum_\gamma L_\gamma/\gamma$
Mellin transform	$\sec(\pi z/2)$	$\Xi(0)/\Xi(z)$
reciprocal Xi	$\operatorname{sech}(\pi\sqrt{s}/2)$	$\xi(1/2)/\xi(1/2+\sqrt{s})$
Thorin measure	$\sum_k \delta_{(2k-1)^2}$	$\sum_\gamma \delta_{\gamma^2}$
exit time	Brownian motion on $(-1,1)$	Krein string with spectrum $\{\gamma^2\}$
density of the power	HCM iff $\lvert\beta\rvert \geq 2$ (Bosch)	HCM only if $\lvert\beta\rvert \geq \gamma_2 - \gamma_1$

Table 9.1: The dictionary. In the Riemann column the Mellin transform (Section 2.1 and Theorem 9.3.25) and the reciprocal Xi function are unconditional, the log variable and the Thorin measure presume RH, and the exit time, the hitting time of Kent's theorem (Section 5.2.3), presumes RH and simple zeros. The last entry is the necessary condition of Theorem 9.3.27, which assumes RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$; sufficiency is open (Question 9.3.28).

On the Riemann side the analogue of g_C is the density of $\log W$. The function $t \mapsto \xi(1/2)/\xi(1/2+t)$ is defined for all real t, since ξ has no real zeros, and decays like $\exp(-\frac{1}{2}|t|\log|t|)$, so its Fourier inverse

$$g(y) = \frac{1}{\pi}\int_0^\infty \frac{\xi(1/2)}{\xi(1/2+t)}\cos(ty)\,dt$$

is an even function with $\int g = 1$, entire because the integrand decays faster than any exponential; $g > 0$ unconditionally (Theorem 9.3.25). Under RH, with simple zeros and $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$, the residues at the poles $\pm i\gamma_n$ of the integrand give, for $|y| > u_0$ with the constant $u_0 \geq \pi/4$ of Theorem 9.3.27,

$$g(y) = \sum_n \frac{\gamma_n}{2} A_n e^{-\gamma_n|y|}, \qquad A_n = \prod_{m\neq n}\Big(1-\frac{\gamma_n^2}{\gamma_m^2}\Big)^{-1}, \tag{9.2.3}$$

the analogue of $g_C(y) = (2/\pi)\sum_k (-1)^{k-1}e^{-(2k-1)|y|}$, which converges for every $y \neq 0$ because its coefficients are bounded. The Riemann coefficients $A_1 \approx 50.9$, $A_2 \approx -2664.6$, $A_3 \approx 31420$ (computed) alternate in sign, as the Cauchy coefficients do, but grow roughly like $e^{\pi\gamma_n/4}$ (Theorem 9.3.27), so (9.2.3) diverges for $|y| < \pi/4$.

9.2.3 The reciprocal Xi function is HCM if and only if RH holds

Bosch's method works with the singularities of a function and with its logarithmic derivative, so we state Corollary 9.1.3(ii) in four forms; (iii) is Proposition 2.1.1, and since $\psi = G'/G$, (iv) is the Stieltjes form of the Thorin condition (Section 1.1). A Stieltjes function is a function $a/s + b + \int_{(0,\infty)} \sigma(dt)/(s+t)$ with $a, b \geq 0$ and σ a nonnegative measure satisfying $\int (1+t)^{-1}\sigma(dt) < \infty$ [327, Definition 2.1].

Theorem 9.2.4. *The following are equivalent.*

(i) The Riemann Hypothesis.

(ii) $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$ *is HCM on* $(0,\infty)$.

(iii) φ *is the Laplace transform of a GGC, namely* $\sum_\gamma e_\gamma/\gamma^2$.

(iv) $\psi(s) = -\varphi'(s)/\varphi(s) = \dfrac{1}{2\sqrt{s}}\dfrac{\xi'}{\xi}\big(\frac{1}{2}+\sqrt{s}\big)$ *is a Stieltjes function.*

Proof. (i) implies (iii): each factor $(1 + s/\gamma^2)^{-1}$ is the Laplace transform of an exponential variable with rate $\gamma^2 > 0$, and the product converges because $\sum \gamma^{-2} < \infty$; the sum $\sum_\gamma e_\gamma/\gamma^2$ converges almost surely. (iii) implies (ii) by (F6).

(ii) implies (i): by (F7), φ extends analytically to $\mathbb{C} \setminus (-\infty, 0]$. Suppose a zero γ has $\operatorname{Im}\gamma \neq 0$. Since $\operatorname{Re}\gamma > 0$ the point $s_0 = -\gamma^2$ has $\operatorname{Im} s_0 = -2\operatorname{Re}\gamma \operatorname{Im}\gamma \neq 0$, and its principal square root is $\pm i\gamma$, so $\xi(1/2+\sqrt{s_0}) = \Xi(\pm\gamma) = 0$ by evenness. Then φ has a pole at a nonreal point of $\mathbb{C} \setminus (-\infty, 0]$, a contradiction.

(i) is equivalent to (iv): under RH, $\psi(s) = \sum_\gamma 1/(s+\gamma^2)$, a Stieltjes function with measure $\sum_\gamma \delta_{\gamma^2}$. Conversely a Stieltjes function is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and an off line zero produces the nonreal pole s_0 of ψ found above. □

Condition (iv) is a companion of Lagarias's criterion [198], and it is the pole test of Table 4.2 in analytic form.

The de Bruijn Newman flow embeds φ in a family of the same kind. Let Φ_{dBN} and $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be as in Section 4.1, so that $H_0(z) = \frac{1}{8}\Xi(z/2)$. Put $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$, the function R_t of Theorem 17.1.16, so that $\varphi_0 = \varphi$. The function H_t is even, real on $\mathbb{R}$ and entire of order one, as in the proof of Theorem 17.1.16, so its Hadamard product in z^2 converges. It is positive on the imaginary axis, $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, since $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u) > 0$; hence $\varphi_t > 0$ on $(0,\infty)$ and no zero of H_t is purely imaginary. With these facts the proof of Theorem 9.2.4 applies verbatim.

Corollary 9.2.5. φ_t *is HCM if and only if* $t \geq \Lambda$, *the de Bruijn Newman constant. Hence* φ_t *is HCM for* $t \geq 0.22$ *[311],* φ_t *is not HCM for* $t < 0$ *[318], and RH holds if and only if* φ_t *is HCM for every* $t > 0$.

Proof. The zeros of H_t are real if and only if $t \geq \Lambda$ (de Bruijn [73], Newman [242]). If RH holds, H_0 has only real zeros, so $\Lambda \leq 0$ and φ_t is HCM for every $t > 0$. Conversely, $\varphi_t \to \varphi_0$ pointwise as $t \downarrow 0$, so if every φ_t with $t > 0$ is HCM then φ is HCM by (F4), and Theorem 9.2.4 gives RH. □

The bound $\Lambda \leq 0.2$ of Platt and Trudgian [277] (Section 4.1) lowers the unconditional threshold from 0.22 to 0.2. The reciprocal Xi function, if HCM at all, is HCM with no room to spare, since the property fails for every $t < 0$: the Riemann counterpart of the attained boundary $|\beta| = 1/(1-\alpha)$ in Bosch's theorem, on which Bosch can compute because his kernel is explicit.

9.2.4 What Bosch's theorem can and cannot supply

Bosch's theorem cannot be the input to a proof of Theorem 9.2.4, for an elementary reason.

Proposition 9.2.6. *For any positive numbers c_n with $\sum_n c_n^{-1} < \infty$ the function $\prod_n (1 + s/c_n)^{-1}$ is HCM. If one c_n is replaced by a pair $c, \bar{c}$ with c nonreal, the product is still positive on $(0, \infty)$ but is not HCM.*

Proof. The first claim is the argument for (i) implies (ii) in Theorem 9.2.4. For the second, $(1 + s/c)(1 + s/\bar{c}) = |1 + s/c|^2 > 0$ for real s, and the product has a pole at the nonreal point $-c$, contradicting (F7). □

Heuristically, the HCM closure operations carry real frequencies to real frequencies and create none: products and quotients of independent variables multiply the Mellin transforms, and powers rescale the frequencies. Bosch's theorem lives in the class where the frequency set $\{2k-1\}$ is given, so any deduction of Theorem 9.2.4 from it must import the location of the γ from elsewhere.

Nor can the Riemann statement be proved factor by factor: in $\xi(1/2 + \sqrt{s}) = \frac{1}{2}(s - \frac{1}{4})\, \pi^{-(1/2+\sqrt{s})/2}\, \Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\, \zeta(\frac{1}{2} + \sqrt{s})$ the reciprocal factors are not individually HCM. The reciprocal of $\pi^{-(1/2+\sqrt{s})/2}$ is $\pi^{1/4}\pi^{\sqrt{s}/2}$, and the pair function of $\pi^{\sqrt{s}/2}$ is $\exp(\frac{1}{2}\log\pi\, \sqrt{u}\, \sqrt{w+2})$, which increases in w. The factor $1/\zeta(1/2+\sqrt{s})$ changes sign at $s = 1/4$, since $\zeta < 0$ on $(1/2, 1)$, and HCM functions are positive. The positive combination $1/((s - \frac{1}{4})\zeta(\frac{1}{2} + \sqrt{s}))$ is continuous at $s = 1/4$, and by the argument of Theorem 9.2.4 any HCM statement for it would again imply RH. It is the function to which the sech expansion of $1/\zeta(\frac{1}{2} + \sqrt{s})$ in Theorem 10.1.5 should be applied.

Bosch's theorem is a density statement, and HCM of a density differs from HCM of a Laplace transform: the density of $|C|$ is not HCM, yet φ_C is. On the Cauchy side the two levels are linked only because sech is its own Fourier transform up to scaling, so that the density g_C of $\log|C|$ and the transform φ_C are the same function in different coordinates. On the Riemann side there is no

such self duality: the density level object is g in (9.2.3), the Fourier inverse of $\Xi(0)/\Xi(i\,\cdot)$, and the function level object is φ.

Question 9.2.7. The variable W, with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, exists unconditionally (Section 9.2.2 and Theorem 9.3.25). For which β is the density of W^β HCM? On the Cauchy side the answer is $|\beta| \geq 2$.

Question 9.2.8. Does HCM of the density of W^β for a single β imply RH?

Theorem 9.3.27 answers Question 9.2.7 in part: under RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$, the density of W^β can be HCM only if $|\beta| \geq \gamma_2 - \gamma_1 = 6.8873$, and Question 9.3.28 asks whether this is also sufficient. At the Cauchy point the same rule gives $3 - 1 = 2$, Bosch's threshold.

One might hope that an HCM variable has a logarithm with an EGGC law (Section 4.1.2), which would force the singularities of $\mathbb{E}W^z$, the zeros of Ξ, onto the real axis. This is false in general. The generalised inverse Gaussian law with density proportional to $x^{-1}e^{-(x+1/x)}$ is HCM by (F7), yet its Mellin transform at $z = it$, $K_{it}(2)/K_0(2)$, has real zeros in t, the first at $t \approx 4.4255$ (by a theorem of Pólya [302], every zero of $t \mapsto K_{it}(2)$ is real). So the characteristic function of its logarithm vanishes, the logarithm is not infinitely divisible, and density level HCM does not by itself give RH.

Two things do transfer. First, the dictionary of Table 9.1 is a solved model in which every quantity is explicit, and it tests any proposed argument: a proof that φ is HCM which does not use the values of the frequencies would also prove HCM for $\cos(\pi z/2)$ with frequencies moved off the axis, which is false by Proposition 9.2.6. Second, Corollary 9.2.5 gives a target: to show that φ_t is HCM for all $t > 0$, which is equivalent to $\Lambda \leq 0$, that is, by Rodgers and Tao [318], to $\Lambda = 0$.

9.3 A Bosch type theorem for the reciprocal Xi function

The observation that closes Section 9.2 becomes a theorem through an elasticity criterion (Theorem 9.3.2). Applied to $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ it gives a Bosch type theorem for $\varphi(x^{1/\beta})$ (Theorem 9.3.5), and the same elasticity identifies the Thorin measure of $\varphi(s^p)$ (Theorem 9.3.12); the positivity of its density is where the Riemann problem now sits.

The notation is that of Section 9.2; in particular W is the Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace. The numerical checks confirm formulas and constants and prove nothing.

9.3.1 The elasticity criterion

We use HCM as defined in Section 9.2.1, with the facts (F1) to (F7) listed there, above all Bondesson's representation (9.2.1). Throughout, $\Omega = \mathbb{C} \setminus (-\infty, 0]$, $\mathbb{C}^+ = \{\operatorname{Im} x > 0\}$, and a Pick function is an analytic map $\mathbb{C}^+ \to \overline{\mathbb{C}^+}$. For a GGC with

Laplace transform h, drift $b \geq 0$ and Thorin measure U, in the sense of (4.1.1), $-h'/h = b + \int U(dt)/(x+t)$ is a Stieltjes function. We also need two facts about complete Bernstein functions [327, Theorem 6.2]: a function f on $(0,\infty)$ is complete Bernstein if and only if $f \geq 0$ on $(0,\infty)$ and f continues to a Pick function analytic on Ω, and then $f(x)/x$ is a Stieltjes function.

The elasticity criterion restates (9.2.1) as a Pick function condition, of the kind through which Bosch verifies HCM [41, Section 2.2].

DEFINITION 9.3.1. For h analytic and zero free on Ω and positive on $(0,\infty)$, let $\log h$ be the branch that is real on $(0,\infty)$ and define the elasticity $\varepsilon_h(x) = -xh'(x)/h(x)$. In logarithmic coordinates $x = e^z$ one has $\varepsilon_h(e^z) = -\frac{d}{dz}\log h(e^z)$.

THEOREM 9.3.2. *Let $h > 0$ on $(0,\infty)$. Then h is HCM if and only if h continues to a zero free analytic function on Ω and ε_h is a Pick function, that is* $\operatorname{Im}\varepsilon_h(x) \geq 0$ *for* $\operatorname{Im} x > 0$.

Proof. Necessity. By (9.2.1), h is zero free and analytic on Ω, and its elasticity is

$$\varepsilon_h(x) = 1 - r + b_1 x + \int \frac{x}{x+t}\,\Gamma_1(dt) - \frac{b_2}{x} - \int \frac{\Gamma_2(dt)}{1+tx}.$$

For $\operatorname{Im} x > 0$ each term has nonnegative imaginary part, for instance $\operatorname{Im}\frac{x}{x+t} = \frac{t\operatorname{Im} x}{|x+t|^2}$.

Sufficiency. Fix $u > 0$. The principal branch $y(w) = \operatorname{arccosh}(w/2)$ maps $\mathbb{C}\setminus(-\infty,2]$ conformally onto the half strip $S = \{\operatorname{Re} y > 0,\ |\operatorname{Im} y| < \pi\}$, and $w = 2\cosh y$. For $y \in S$ both ue^{y} and ue^{-y} lie in Ω, so

$$F(w) = -\log h\big(ue^{y(w)}\big) - \log h\big(ue^{-y(w)}\big)$$

is analytic on $\mathbb{C}\setminus(-\infty,2]$ and real on $(2,\infty)$, and $e^{-F(w)}$ is the pair function. Let $\operatorname{Im} w > 0$, so $y = \varrho + i\theta$ with $\varrho > 0$ and $\theta \in (0,\pi)$. Put $r_1 = ue^{\varrho} > r_2 = ue^{-\varrho}$. Since $\log h(\bar{x}) = \overline{\log h(x)}$,

$$\operatorname{Im} F(w) = -\operatorname{Im}\log h(r_1 e^{i\theta}) + \operatorname{Im}\log h(r_2 e^{i\theta}).$$

Along the ray, $\partial_r \operatorname{Im}\log h(re^{i\theta}) = \operatorname{Im}\big(e^{i\theta}h'/h\big) = -r^{-1}\operatorname{Im}\varepsilon_h(re^{i\theta}) \leq 0$, so $\operatorname{Im} F(w) \geq 0$. Thus F is a Pick function that continues analytically across $(2,\infty)$, where it is real. By the Nevanlinna representation $F(w) = a + bw + \int(\frac{1}{t-w} - \frac{t}{1+t^2})\sigma(dt)$ with $b \geq 0$ and $\sigma \geq 0$ carried by $(-\infty,2]$. Hence for $w > 2$,

$$F'(w) = b + \int_{(-\infty,2]} \frac{\sigma(dt)}{(w-t)^2} = b + \int_{(-\infty,2]} \int_0^\infty \lambda e^{-\lambda(w-t)}\,d\lambda\,\sigma(dt),$$

which is completely monotone. A function whose derivative is completely monotone is a Bernstein function up to an additive constant, and e^{-F} is then completely monotone. So every pair function is completely monotone and h is HCM. □

The two conditions play different roles. Analyticity without zeros is global and rigid: one zero or pole of h off the negative axis destroys HCM, however small its effect on the real line. The Pick condition is local and quantitative, and can fail along a ray without any singularity.

9.3.2 Bosch's theorem from the criterion

Let T_α, $0 < \alpha < 1$, be Bosch's variable of Section 9.2.1. Let $c = \cos \pi\alpha$ and $q = 1/|\beta|$, half the q of (9.2.2). Since $T_\alpha \overset{d}{=} 1/T_\alpha$ we may take $\beta > 0$. The density of T_α^β is proportional to

$$h(x) = \frac{x^{q-1}}{D(x)}, \qquad D(x) = x^{2q} + 2cx^q + 1 = (x^q + e^{i\pi\alpha})(x^q + e^{-i\pi\alpha}).$$

Theorem 9.3.3 (Bosch [41, Theorem 1.2]). *h is HCM if and only if $\alpha \le 1/2$ and $q \le 1 - \alpha$.*

Proof. Zeros. For $x \in \Omega$ the value x^q ranges over the sector $|\arg| < q\pi$. The zeros of D satisfy $x^q = e^{\pm i(1-\alpha)\pi}$, so D is zero free on Ω if and only if $q \le 1 - \alpha$.

Elasticity. For $x = re^{i\theta}$ with $0 < \theta < \pi$ write $x^q = Re^{i\omega}$ with $R = r^q$ and $\omega = q\theta \in (0, q\pi)$. Then $\varepsilon_h = 1 - q + xD'(x)/D(x)$ and $\operatorname{Im} \varepsilon_h = qR\,\partial_R \arg D$. A direct computation of $\operatorname{Im}(\partial_R D \cdot \overline{D})$ gives

$$\operatorname{Im} \varepsilon_h(x) = \frac{2qR \sin\omega \left(c(1 + R^2) + 2R\cos\omega\right)}{|D(x)|^2}. \tag{9.3.1}$$

If $\alpha \le 1/2$ and $q \le 1 - \alpha$ then $c \ge 0$ and $\cos\omega \ge \cos((1-\alpha)\pi) = -c$, so, using $1 + R^2 \ge 2R$, the bracket is at least $2R(c + \cos\omega) \ge 0$. Theorem 9.3.2 gives HCM. If $\alpha > 1/2$ then $c < 0$ and the bracket tends to $c < 0$ as $R \to 0$, so ε_h is not Pick. If $q > 1 - \alpha$ then h has poles in Ω. In both cases Theorem 9.3.2 rules out HCM. □

In terms of β, the theorem says that the density of T_α^β is HCM if and only if $\alpha \le 1/2$ and $|\beta| \ge 1/(1-\alpha)$. For the half Cauchy variable, $\alpha = 1/2$, the threshold is $|\beta| \ge 2$; there $c = 0$ and (9.3.1) reduces to $\operatorname{Im} \varepsilon_h = 4qR^2 \sin\omega\cos\omega/|D|^2$, which is nonnegative for every $\omega \in (0, q\pi)$ exactly when $q \le 1/2$. So the half Cauchy threshold is the condition $\omega \le \pi/2$, equivalently that the zeros $x = e^{\pm i\pi/(2q)}$ of $1 + x^{2q}$ stay on or beyond the cut.

9.3.3 The Riemann analogue

As in Section 9.2.2, γ runs over the zeros of $\Xi(z) = \xi(1/2 + iz)$ with $\operatorname{Re}\gamma > 0$, counted with multiplicity; the nonreal ones come in conjugate pairs $\gamma, \bar\gamma$.

By (2.1.1), unconditionally,

$$\varphi(s)=\frac{\xi(1/2)}{\xi(1/2+\sqrt{s})}=\prod_{\gamma}\Big(1+\frac{s}{\gamma^2}\Big)^{-1},\qquad \varepsilon_\varphi(s)=\frac{\sqrt{s}}{2}\,\frac{\xi'}{\xi}\Big(\frac12+\sqrt{s}\Big)=\sum_{\gamma}\frac{s}{s+\gamma^2},$$

both absolutely convergent since $\sum|\gamma|^{-2}<\infty$. Write $\gamma=\varrho e^{i\delta}$. Since the zeros of ξ lie in the open critical strip, $|\delta|<\arctan(1/(2\operatorname{Re}\gamma))$, and since $\operatorname{Re}\gamma>14$, $|\delta|<\pi/4$. Define

$$\delta_*=\max_{\gamma}|\arg\gamma|,\qquad \alpha_\xi=\frac{2\delta_*}{\pi},\qquad p_*=1-\alpha_\xi.$$

The maximum is attained if some zero is off the line, because for every positive angle only finitely many γ have $|\arg\gamma|$ larger than it. RH is the statement $\delta_*=0$.

Lemma 9.3.4. *Let $\gamma=\varrho e^{i\delta}$ with $0\le\delta<\pi/4$ and $s=re^{i\theta}$ with $0<\theta<\pi-2\delta$. Then*

$$\operatorname{Im}\Big(\frac{s}{s+\gamma^2}+\frac{s}{s+\bar\gamma^2}\Big)\ge 0,$$

with strict inequality. For $\delta=0$ the single term $\operatorname{Im}\frac{s}{s+\gamma^2}=\frac{\gamma^2 r\sin\theta}{|s+\gamma^2|^2}$ is positive for $0<\theta<\pi$.

Proof. For $c\in\mathbb{C}$, $\operatorname{Im}\frac{s}{s+c}=\operatorname{Im}\big(-\frac{c}{s+c}\big)=\frac{\operatorname{Im}(\bar c s)}{|s+c|^2}$. With $c=\varrho^2e^{\pm2i\delta}$ this is $\varrho^2 r\sin(\theta\mp2\delta)/|s+c|^2$. The sum is

$$\varrho^2 r\Big(\frac{\sin(\theta-2\delta)}{A_-}+\frac{\sin(\theta+2\delta)}{A_+}\Big),\qquad A_\mp=r^2+\varrho^4+2r\varrho^2\cos(\theta\mp2\delta).$$

Since $\theta+2\delta<\pi$ the second sine is positive. If $\theta\ge2\delta$ both terms are nonnegative. If $\theta<2\delta$ then $|\theta-2\delta|<\theta+2\delta$, so $A_->A_+$, and $\sin(\theta+2\delta)-\sin(2\delta-\theta)=2\cos(2\delta)\sin\theta>0$; the positive term has the larger numerator and the smaller denominator. □

Theorem 9.3.5. *For $p\in\mathbb{R}$ the function $x\mapsto\varphi(x^p)$ is HCM if and only if $|p|\le p_*=1-\alpha_\xi$. Equivalently, $\varphi(x^{1/\beta})$ is HCM if and only if $|\beta|\ge1/(1-\alpha_\xi)$.*

Proof. Since $h(x)$ is HCM if and only if $h(1/x)$ is, take $p>0$. Let $h_p(x)=\varphi(x^p)$, positive on $(0,\infty)$ because $\xi>0$ on $\mathbb{R}$. Its singularities are the poles $x^p=-\gamma^2=\varrho^2e^{\pm i(\pi-2|\delta|)}$, and x^p ranges over $|\arg|<p\pi$ for $x\in\Omega$. Hence h_p is zero free and analytic on Ω if and only if $p\pi\le\pi-2|\delta|$ for every γ, that is $p\le p_*$. When $p\le p_*$,

$$\varepsilon_{h_p}(x)=p\,\varepsilon_\varphi(x^p)=p\sum_{\gamma}\frac{x^p}{x^p+\gamma^2},$$

and for $\operatorname{Im}x>0$ the point $s=x^p$ has $0<\arg s<p\pi\le\pi-2\delta_*$. Grouping conjugate pairs, Lemma 9.3.4 makes every term of $\operatorname{Im}\varepsilon_{h_p}$ nonnegative. Theorem 9.3.2 gives HCM. When $p>p_*$ the pole coming from a zero with $|\arg\gamma|=\delta_*$ lies in Ω, and Theorem 9.3.2 rules out HCM. □

COROLLARY 9.3.6. *The following are equivalent: (i) RH; (ii) φ is HCM; (iii) $\varphi(x^p)$ is HCM for every $|p|<1$; (iv) ε_φ is a Pick function; (v)* $\operatorname{Im}\big[z\,(\xi'/\xi)(\frac12+z)\big]\ge 0$ *for all z with* $\operatorname{Re} z>0$ *and* $\operatorname{Im} z>0$.

Proof. By Theorem 9.3.5, each of (i), (ii) and (iii) says that $p_*=1$; (i) if and only if (ii) is also Theorem 9.2.4. (iv) is the elasticity form of (ii) through Theorem 9.3.2. (v) is (iv) with $s=z^2$, which maps the open first quadrant onto $\mathbb{C}^+$. □

Criterion (v) is an angular companion to Lagarias's criterion $\operatorname{Re}(\xi'/\xi)(\frac12+z)>0$ for $\operatorname{Re} z>0$ [198].

COROLLARY 9.3.7 (unconditional). *RH has been verified for all zeros with* $0<\operatorname{Re}\gamma\le T_0=3\times10^{12}$ *[277]. Hence* $\delta_*<\arctan(1/(2T_0))$ *and*

$$\varphi(x^p)\ \textit{is HCM for all}\ |p|\le 1-\frac{1}{\pi T_0},\qquad 1-\frac{1}{\pi T_0}>1-1.1\times10^{-13}.$$

Unconditionally, $\operatorname{Im}\varepsilon_\varphi\ge0$ *on the sector* $0<\arg s<\pi-1/T_0$.

Proof. Any off line zero has $\operatorname{Re}\gamma>T_0$ and $|\operatorname{Im}\gamma|<1/2$, so $|\arg\gamma|<1/(2T_0)$ and $2\delta_*/\pi<1/(\pi T_0)$. The sector statement is Lemma 9.3.4 applied termwise. □

Corollary 9.3.7 is a statement about the elasticity $\varepsilon_\varphi(s)=s\psi(s)$, where $\psi=-\varphi'/\varphi=\sum_\gamma(s+\gamma^2)^{-1}$ is the function of the Stieltjes form (iv) of Theorem 9.2.4.

LEMMA 9.3.8. *Let $c=c_1+ic_2$ with $c_2>0$, and let $s=x+iy$ with $y>0$ and $s\ne c$. Then*

$$\operatorname{Im}\Big(\frac{1}{s-c}+\frac{1}{s-\bar c}\Big)=-\frac{2y\,\big(|s-c_1|^2-c_2^2\big)}{|s-c|^2\,|s-\bar c|^2},$$

so the imaginary part is positive on the open disc $|s-c_1|<c_2$, whose vertical diameter is the segment from $\bar c$ to c, zero on its boundary and negative outside it.

Proof. With $d=(x-c_1)^2$, $\operatorname{Im}\frac{1}{s-c}=-\frac{y-c_2}{d+(y-c_2)^2}$ and $\operatorname{Im}\frac{1}{s-\bar c}=-\frac{y+c_2}{d+(y+c_2)^2}$. Over the common denominator the numerator of the sum is

$$-(y-c_2)\big(d+(y+c_2)^2\big)-(y+c_2)\big(d+(y-c_2)^2\big)=-2y\,(d+y^2-c_2^2),$$

and $d+y^2=|s-c_1|^2$. □

A nonreal zero $\gamma=a+ib$ of Ξ, with $a>0$ and, after conjugation, $b>0$, contributes to ψ the pair of terms $1/(s-c)+1/(s-\bar c)$ with $c=-\bar\gamma^2=b^2-a^2+2iab$. Let

$$D_\gamma=\big\{s:\ |s-(b^2-a^2)|\le 2ab\big\}$$

be the closed disc with diameter from $-\gamma^2$ to $-\bar\gamma^2$. By Lemma 9.3.8 the pair has positive imaginary part only at interior points of D_γ, and negative imaginary part at every point of the upper half plane outside D_γ.

THEOREM 9.3.9 (unconditional). *Let $T_0 = 3 \times 10^{12}$, so that every zero γ with $0 < \operatorname{Re}\gamma \le T_0$ is real [277], and let*

$$\mathcal{E}_{T_0} = \Big\{ x + iy : \ x < -T_0^2 + T_0 + \tfrac{1}{4}, \ y^2 < \tfrac{1}{2} - x \Big\}.$$

Split $\psi = S_{\text{on}} + R_{\text{off}}$, where $S_{\text{on}}(s) = \sum_{\gamma \text{ real}} (s+\gamma^2)^{-1}$ runs over the zeros on the critical line and R_{off} is the same sum over the nonreal γ.

(i) S_{on} is a Stieltjes function, with measure $U_{\text{line}} = \sum_{\gamma \text{ real}} \delta_{\gamma^2}$.

(ii) R_{off} is meromorphic on $\mathbb{C}$ and nonnegative on $[0,\infty)$. For every nonreal γ the disc D_γ, and with it the poles $-\gamma^2$ and $-\bar\gamma^2$ of R_{off} on its boundary, lies in $\mathcal{E}_{T_0}$, and $\operatorname{Im} R_{\text{off}} \le 0$ at every point of the upper half plane outside these discs.

(iii) $\psi > 0$ on $[0,\infty)$, and $\operatorname{Im}\psi(s) < 0$ at every $s = x + iy$ with $y > 0$ outside $\mathcal{E}_{T_0}$.

Proof. Since $\sum |\gamma|^{-2} < \infty$, the series for ψ, S_{on} and R_{off} converge absolutely and locally uniformly off the points $-\gamma^2$, so R_{off} is meromorphic on $\mathbb{C}$, and (i) holds because $\sum_{\gamma \text{ real}} (1+\gamma^2)^{-1} < \infty$. A real γ contributes $-y/|s+\gamma^2|^2 < 0$ to $\operatorname{Im}\psi(x+iy)$ for $y > 0$, and $(x+\gamma^2)^{-1} > 0$ to $\psi(x)$ for $x \ge 0$.

A nonreal zero is $\gamma = a + ib$ with $a > T_0$ and, after conjugation, $0 < b < \frac{1}{2}$, because ξ vanishes at $\frac{1}{2} \pm b + ia$. With $c = b^2 - a^2 + 2iab$ as above, its pair of terms equals $2(x - c_1)/((x-c_1)^2 + c_2^2) > 0$ at $x \ge 0$, since $c_1 = b^2 - a^2 < 0$, and its imaginary part is negative at every point of the upper half plane outside D_γ. It remains to show that $D_\gamma \subset \mathcal{E}_{T_0}$. A point $x + iy$ of D_γ has

$$x \le b^2 - a^2 + 2ab = 2b^2 - (a-b)^2 < -T_0^2 + T_0 + \tfrac{1}{4},$$

because the middle term increases in b and, for $a > b$, decreases in a, and equals $-T_0^2 + T_0 + \frac{1}{4}$ at $a = T_0$, $b = \frac{1}{2}$; in particular $x < 0$. With $e = x - (b^2 - a^2)$, so that $a^2 = b^2 - x + e$, it also has

$$y^2 \le 4a^2b^2 - e^2 = 8b^4 - 4b^2x - (e - 2b^2)^2 \le 4b^2(2b^2 - x) < 2b^2 - x < \tfrac{1}{2} - x,$$

because $4b^2 < 1$ and $2b^2 - x > 0$. Hence $D_\gamma \subset \mathcal{E}_{T_0}$, and (ii) follows from the signs of the pairs. At a point of the upper half plane outside $\mathcal{E}_{T_0}$ every pair of terms has negative imaginary part, and so has the term of every real γ; with the signs on $[0,\infty)$, since γ_1 is real, this gives (iii). □

Both bounds that define $\mathcal{E}_{T_0}$ are sharp for zeros with $a > T_0$ and $0 < b < \frac{1}{2}$. The rightmost point $2b^2 - (a-b)^2$ of D_γ tends to $-T_0^2 + T_0 + \frac{1}{4}$ as $b \uparrow \frac{1}{2}$ and $a \downarrow T_0$. For $x < \frac{3}{4} - T_0^2$ the supremum of y^2 over the points $x + iy$ of the discs is $\frac{1}{2} - x$: the bound $8b^4 - 4b^2x$ of the proof is attained, at $e = 2b^2$, by the disc

with $a^2 = 3b^2 - x$, which has $a > T_0$ once b is close to $\frac{1}{2}$, and it tends to $\frac{1}{2} - x$ as $b \uparrow \frac{1}{2}$. At the tip $x_0 = -T_0^2 + T_0 + \frac{1}{4}$ the height of $\mathcal{E}_{T_0}$ is exactly $T_0 - \frac{1}{2}$, since $\frac{1}{2} - x_0 = (T_0 - \frac{1}{2})^2$, so the ray from the origin to the corner $x_0 + i(T_0 - \frac{1}{2})$ makes the angle $\theta_0 = \arctan \frac{T_0 - 1/2}{T_0^2 - T_0 - 1/4} = 3.33333 \times 10^{-13}$ with the negative axis, which exceeds $1/T_0$ by the factor $1 + 1.7 \times 10^{-13}$. Further out the angle decreases like $|x|^{-1/2}$.

Corollary 9.3.7 gives the sign of the elasticity $s\psi(s)$ outside a sector of opening $1/T_0$ beside the cut; Theorem 9.3.9 gives the sign of ψ itself outside a much thinner parabola (Figure 9.2), through the exact region of Lemma 9.3.8. It does not make ψ a Stieltjes function off $\mathcal{E}_{T_0}$, a global property; it says that if the Stieltjes form fails, it fails through poles of ψ in $\mathcal{E}_{T_0}$, from zeros above the verification height. Like the corollary, it brings no new information about the zeros, and nothing in it limits the number of poles in $\mathcal{E}_{T_0}$ (Section 14.3.7).

Remark 9.3.10 (the right end). The right end of $\mathcal{E}_{T_0}$ cannot be moved to $-(T_0 - \frac{1}{2})^2$, the value the argument would give if the rightmost point of D_γ were $-(a-b)^2$: it is $2b^2 - (a-b)^2$, and the extra $2b^2$ reaches into the strip $-(T_0 - \frac{1}{2})^2 \le x < -(T_0 - \frac{1}{2})^2 + \frac{1}{2}$. Whether $\operatorname{Im}\psi$ is positive at such points depends on unknown zeros, which the termwise argument cannot exclude.

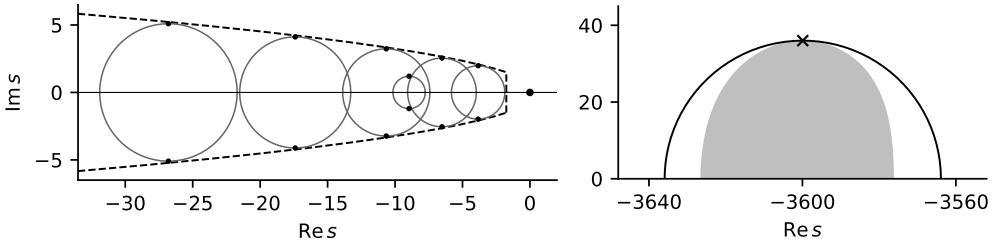


Figure 9.2: Left: the sliver $\mathcal{E}_{T_0}$ of Theorem 9.3.9 for $T_0 = 2$ (dashed) and the discs D_γ of six hypothetical zeros $a + ib$ with $a > 2$ and $0 < b < \frac{1}{2}$, with the poles $-\gamma^2$ and $-\bar{\gamma}^2$ (dots) at the ends of their vertical diameters; the dot at 0 marks the origin. Right: for the toy reciprocal Xi function with the first 200 zeros of ζ and the pair $60 \pm 0.3i$, the set where $\operatorname{Im}\psi > 0$ (shaded) on a grid of 481×300 points, which lies inside the upper half of D_γ (solid); the cross is the pole $-\bar{\gamma}^2 = -3599.91 + 36i$.

The threshold p_* of Theorem 9.3.5 can be followed along the de Bruijn Newman flow. With H_t and $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$ as in Section 9.2.3, so that $\varphi_0 = \varphi$, let $\delta_*(t)$ be the largest angle $|\arg z|$ over the zeros z of H_t with $\operatorname{Re} z > 0$, and $p_*(t) = 1 - 2\delta_*(t)/\pi$. The proof of Theorem 9.3.5 uses only the product over the

zeros, positivity on $(0,\infty)$ and the bound $|\arg z| < \pi/4$ required by Lemma 9.3.4, so whenever the zeros of H_t satisfy that bound, $\varphi_t(x^p)$ is HCM if and only if $|p| \le p_*(t)$. By de Bruijn [73] and Newman [242], $p_*(t) = 1$ exactly for $t \ge \Lambda$, and by Rodgers and Tao [318] $p_*(t) < 1$ for every $t < 0$.

Question 9.3.11. Describe $p_*(t) = 1 - \frac{2}{\pi}\delta_*(t)$ for $t < 0$. Assume RH; how fast does $1 - p_*(t) \to 0$ as $t \uparrow 0$?

Heuristically, the off line zeros of H_t for small $|t|$ come from nearly coincident pairs of real zeros: by the isolated pair estimate of Section 4.1, a pair at distance D_0 collides near $t = -D_0^2/8$ and then has imaginary parts of order $\sqrt{|t|}$, at heights where the gap is of order $\sqrt{|t|}$. Since $1 - p_*(t)$ is governed by $\max \operatorname{Im} z / \operatorname{Re} z$, the answer depends on the lowest height at which such a pair has collided, and so on the small gap statistics of the zeros. Both ingredients of an estimate are missing: de Bruijn's bound on the imaginary parts of the zeros runs only forward in time, from H_0 to H_t with $t > 0$, and no lower bound is known for the heights of the pairs that have collided by time t.

9.3.4 The Thorin measure of $\varphi(s^p)$ and Bosch's variable

By Bondesson's theorem [39, Theorem 6.1.1], a function with value 1 at 0 is HCM exactly when it is the Laplace transform of a GGC, so Theorem 9.3.5 already makes $\varphi(s^p)$ a GGC transform for $0 < p \le p_*$.

Theorem 9.3.12. *Let $0 < p \le p_*$. Then $\varphi(s^p) = \mathbb{E}e^{-sX_p}$ for a GGC variable X_p with no drift. If $p < p_*$, its Thorin measure has the density*

$$u_p(\tau) = \frac{p}{\pi\tau}\,\operatorname{Im}\,\varepsilon_\varphi\big(\tau^p e^{ip\pi}\big) = \frac{p}{2\pi\tau}\,\operatorname{Im}\Big[z\,\frac{\xi'}{\xi}\Big(\frac{1}{2}+z\Big)\Big]_{z=\tau^{p/2}e^{ip\pi/2}} \ge 0, \qquad \tau > 0, \tag{9.3.2}$$

which in terms of the zeros is

$$u_p(\tau) = \frac{p}{\pi\tau}\sum_\gamma \operatorname{Im}\frac{\tau^p e^{ip\pi}}{\tau^p e^{ip\pi} + \gamma^2}. \tag{9.3.3}$$

If $p = p_*$, its Thorin measure is $u_{p_*}(\tau)\,d\tau$ plus a unit atom at $|\gamma|^{2/p_*}$ for every zero γ with $\arg\gamma = -\delta_*$, counted with multiplicity, that is, one atom for each conjugate pair of zeros at the extreme angle; when $p_* = 1$ there is an atom at γ^2 for every zero, and $u_1 = 0$.

Proof. Let $h = h_p$ and $\varepsilon_h(x) = p\,\varepsilon_\varphi(x^p)$. By the proof of Theorem 9.3.5, ε_h is analytic on Ω and Pick. On $(0,\infty)$ it is nonnegative: a real zero contributes $x^p/(x^p+\gamma^2) > 0$ and a pair contributes $2\operatorname{Re}\frac{s}{s+\varrho^2 e^{2i\delta}} = \frac{2s(s+\varrho^2\cos 2\delta)}{|s+\varrho^2 e^{2i\delta}|^2} > 0$. Hence ε_h is complete Bernstein and $\varepsilon_h(x)/x = -(\log h)'(x)$ is a Stieltjes function, $a/x + b + \int U(dt)/(x+t)$. Here $a = \varepsilon_h(0+) = 0$ and $b = \lim_{x\to\infty}\varepsilon_h(x)/x = 0$ because $\varepsilon_\varphi(s) = O(\sqrt{s}\log s)$. Integrating from 0 with $h(0) = 1$ gives $h(x) =$

$\exp(-\int \log(1+x/t)U(dt))$, a GGC transform. For $p < p_*$ the function ε_h is continuous up to the upper edge of the cut, and Stieltjes inversion at $x = \tau e^{i\pi}$ gives $U(d\tau) = u_p(\tau)d\tau$ with $u_p(\tau) = \frac{1}{\pi\tau}\operatorname{Im}\varepsilon_h(\tau e^{i\pi})$, which is (9.3.2); (9.3.3) follows from the Hadamard form of ε_φ. For $p = p_*$ the same inversion applies away from the points $x_0 = \tau_0 e^{i\pi}$ with $\tau_0 = |\gamma|^{2/p_*}$ and $\arg\gamma = -\delta_*$, at which $x_0^{p_*} = -\gamma^2$. Near such a point the term of γ in ε_h is $p_* x^{p_*}/(x^{p_*}+\gamma^2) = x_0/(x-x_0) + O(1)$, so $\varepsilon_h(x)/x = 1/(x+\tau_0) + O(1)$, and inversion gives a unit atom at τ_0. On the ray itself this term equals $p_*\tau^{p_*}/(\tau^{p_*} - |\gamma|^2)$, which is real, so u_{p_*} does not see the atom. □

Corollary 9.3.13. *Let $L_p = Z_p/\tilde{Z}_p$ be the ratio of independent positive p stable variables, whose density $\frac{\sin p\pi}{\pi}\frac{t^{p-1}}{t^{2p}+2t^p\cos p\pi+1}$ is Lamperti's [201]; its power L_p^p is Bosch's variable T_α with $\alpha = p$. Under RH the Thorin measure of X_p is*

$$U_p = p\sum_\gamma \mathcal{L}\big(\gamma^{2/p}L_p\big),$$

and $X_p \overset{d}{=} S_p\, X^{1/p}$, where $X = \sum_\gamma e_\gamma/\gamma^2$ is the clock variable of Proposition 2.1.1, with independent standard exponential variables e_γ, and S_p is a positive p stable variable independent of X with $\mathbb{E}e^{-\lambda S_p} = e^{-\lambda^p}$. At $p = 1/2$, $L_{1/2} \overset{d}{=} C^2$ for a standard Cauchy variable C, and $U_{1/2} = \frac{1}{2}\sum_\gamma \mathcal{L}(\gamma^4 C^2)$. As $p \uparrow 1$, $L_p \to 1$ and $U_p \to \sum_\gamma \delta_{\gamma^2}$ vaguely, the Thorin measure of φ itself.

Proof. For real γ the term of (9.3.3) equals $\frac{p}{\pi}\frac{\gamma^2\tau^{p-1}\sin p\pi}{\tau^{2p}+2\gamma^2\tau^p\cos p\pi+\gamma^4}$; substituting $\tau = \gamma^{2/p}t$ turns it into p times Lamperti's density in t. The subordination identity is $\mathbb{E}e^{-sS_pY} = \mathbb{E}e^{-s^pY^p}$ with $Y = X^{1/p}$. □

Bosch's variable is therefore the kernel of the Thorin measure of $\varphi(s^p)$. Formula (9.3.2) makes sense whatever the zeros, and RH is the statement that it stays nonnegative as p increases to 1 (Question 9.3.16 and the argument after it).

Corollary 9.3.14. *RH holds if and only if X_p exists, that is $\varphi(s^p)$ is a GGC transform, for every $p \in (0,1)$. Unconditionally X_p exists for $0 < p \le 1 - 1/(\pi T_0)$, in particular for $0 < p \le 1 - 1.1\times 10^{-13}$, with the Thorin density* (9.3.2).

Proof. If RH fails, $p_* < 1$, and for $p \in (p_*, 1)$ the function $\varphi(s^p)$ has a pole in Ω, which no GGC transform can have, since $\exp(-bs - \int \log(1+s/t)U(dt))$ is analytic and zero free on Ω. The rest is Theorem 9.3.12 with Corollary 9.3.7, which gives $1 - 1/(\pi T_0) < p_*$. □

Question 9.3.15. Construct $X_{1/2}$ from Brownian motion, Bessel processes or the Biane, Pitman and Yor variables [33] without reference to the zeros. A construction that survived $p \uparrow 1$ would prove RH by Corollary 9.3.14.

9.3.5 Positivity of the Thorin density from the primes

The method reads the threshold off the zeros: the elasticity has nonnegative imaginary part termwise on the sector $0 < \arg s < \pi - 2\delta_*$, and nothing is learnt about δ_*. A proof of RH along these lines must show $u_p \geq 0$ for every $p < 1$ directly from (9.3.2), for instance from the Pólya kernel, from the prime sum of Remark 9.3.17 or from a probabilistic model; this is the Thorin measure form of the reciprocal Xi programme of [279]. It cannot be done factor by factor: only the combination $(s-\frac{1}{4})\zeta(\frac{1}{2}+\sqrt{s})$ can enter an HCM or Pick argument (Section 9.2.4).

Question 9.3.16. Prove $u_p \geq 0$ for every $p \in (0,1)$ directly from (9.3.2), without locating the zeros.

For all $p \in (0,1)$ this is equivalent to RH. Under RH it holds by Theorem 9.3.12. Conversely, suppose RH fails, let $\gamma = \varrho e^{i\delta_*}$ be a zero of maximal angle, and let $p \in (p_*, 1)$; formula (9.3.2) still defines u_p, except at isolated poles, although it is no longer a Thorin density. At $\tau = \varrho^{2/p}$ the term of $\bar\gamma$ in (9.3.3) equals $-\frac{p}{2\pi\tau}\cot\frac{\pi(p-p_*)}{2}$, which tends to $-\infty$ as $p \downarrow p_*$. The sum of the other terms stays bounded, because the other points $-\gamma'^2$ keep a positive distance from $-\bar\gamma^2$ and the tail of the sum is $O(\sum|\gamma'|^{-2})$. So u_p takes negative values for p slightly above p_*.

Along the ray write $\vartheta = p\pi/2 \in (0,\pi/2)$, $t = \tau^{p/2}$, $z = te^{i\vartheta}$ and $w = \frac{1}{2}+z$, so that $u_p(\tau) = \frac{p}{2\pi\tau}\operatorname{Im}[z\,\frac{\xi'}{\xi}(w)]$ by (9.3.2), and split $\xi'/\xi = \Gamma_\xi + \zeta'/\zeta$, where

$$\Gamma_\xi(w) = \frac{1}{w} + \frac{1}{w-1} - \frac{\log\pi}{2} + \frac{1}{2}\psi\Big(\frac{w}{2}\Big)$$

is the logarithmic derivative of the factor $\frac{1}{2}w(w-1)\pi^{-w/2}\Gamma(w/2)$ of ξ, and $\psi = \Gamma'/\Gamma$ is the digamma function, not the function ψ of Theorem 9.2.4. The ray crosses the critical strip for $t < t_e = 1/(2\cos\vartheta)$ and then enters $\operatorname{Re} w > 1$.

Remark 9.3.17 (primes). For $t > t_e$, that is $\tau > (2\cos(p\pi/2))^{-2/p}$, we have $\operatorname{Re} w > 1$ and

$$u_p(\tau) = \frac{p}{2\pi\tau}\operatorname{Im}\Big[z\Big(\frac{1}{w}+\frac{1}{w-1}-\frac{\log\pi}{2}+\frac{1}{2}\psi\Big(\frac{w}{2}\Big)\Big)\Big] - \frac{p}{2\pi\tau}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\operatorname{Im}\big(z\,n^{-z}\big),$$

with Λ the von Mangoldt function.

Proposition 9.3.18. *Let* $\operatorname{Re} w > 1$ *and* $\sigma = \operatorname{Re} w$. *Then*

$$\frac{2\pi\tau}{p}\,u_p(\tau) \;\geq\; m_p(t) := \operatorname{Im}\Big[z\Big(\frac{1}{2w}+\frac{1}{w-1}+\frac{1}{2}\log\frac{w}{2\pi}\Big)\Big] - \frac{t}{6\sigma^2} - t\Big(-\frac{\zeta'}{\zeta}\Big)(\sigma). \tag{9.3.4}$$

Every term of m_p *is explicit, no zero of* ζ *enters, and* $m_p(t) = \frac{t}{2}\sin\vartheta\log t\,(1+o(1))$ *as* $t\to\infty$. *Consequently for each* $p\in(0,1)$ *there is* $\tau_1(p) < \infty$ *with* $u_p(\tau) > 0$ *for all* $\tau > \tau_1(p)$.

Proof. Binet's formula [256, (5.9.13)] gives, for $\operatorname{Re} x > 0$, $\psi(x) = \log x - \frac{1}{2x} - \int_0^\infty b(u)e^{-xu}du$ with $b(u) = \frac{1}{e^u-1} - \frac{1}{u} + \frac{1}{2} \in [0, u/12]$. Hence $\psi(x) = \log x - \frac{1}{2x} - R$ with $|R| \le \frac{1}{12(\operatorname{Re} x)^2}$. With $x = w/2$ this turns $\Gamma_\xi(w)$ into $\frac{1}{2w} + \frac{1}{w-1} + \frac{1}{2}\log\frac{w}{2\pi} - \frac{R}{2}$ with $|R| \le \frac{1}{3\sigma^2}$, so $\operatorname{Im}[z\Gamma_\xi(w)]$ is at least the first term of m_p minus $t/(6\sigma^2)$. For the prime part, $\operatorname{Im}[z\,\zeta'/\zeta(w)] = -\sum \Lambda(n)\operatorname{Im}(zn^{-w}) \ge -t\sum \Lambda(n)n^{-\sigma}$. The asymptotic follows from $\operatorname{Im}[z\log z] = t(\sin\vartheta\log t + \vartheta\cos\vartheta)$ and $-\zeta'/\zeta(\sigma) \to 0$ as $\sigma \to \infty$. □

Table 9.2 gives the point beyond which $m_p > 0$, on a logarithmic grid of 50 points per decade with $-\zeta'/\zeta(\sigma)$ evaluated directly: t_1 is the last grid point at which $m_p \le 0$, so the sign change lies within a factor $10^{1/50}$ above it. These values are computed, not proved: nothing controls m_p between grid points or beyond the last one.

The prime term of m_p can be made elementary, because $-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$ for every $\sigma > 1$. Delange [74, Théorème] proves the sharper bound $-\zeta'/\zeta(\sigma) < 1/(\sigma-1) - 1/(2\sigma^2)$. Since $-\zeta'/\zeta(\sigma)$ enters m_p with the factor $-t$, replacing it by $1/(\sigma-1)$ lowers m_p, and (9.3.4) still holds.

p	strip exit $t_e = \frac{1}{2\cos\vartheta}$	t_1	t_1/t_e	$\tau_1 = t_1^{2/p}$
0.5	0.707	4.07	5.75	274
0.9	3.20	11.1	3.47	210
0.99	31.8	69.6	2.19	5.29×10^3
0.999	318	553	1.74	3.10×10^5
0.9999	3183	4818	1.51	2.33×10^7

Table 9.2: Where the lower bound (9.3.4) becomes positive. On a logarithmic grid from t_e to about $10^{14}t_e$, $m_p(t) > 0$ at every grid point $t > t_1$; the ray reaches t_1 at $\tau_1 = t_1^{2/p}$. The entries are computed, not proved.

Remark 9.3.19. The ratio t_1/t_e tends to 1. Near $\vartheta = \pi/2$ write $\sigma = 1 + x$, so $t = t_e(1 + 2x)$. The dominant terms of m_p/t are $\frac{1}{2}\log\frac{t}{2\pi}$ and $-\frac{1}{x}$, so positivity begins at $x \approx 2/\log(t_e/2\pi)$, that is $t_1(p) = t_e(p)\big(1 + \frac{4+o(1)}{\log(t_e/2\pi)}\big)$.

So Question 9.3.16 reduces, for each p, to the segment $|z| \le t_1(p)$ of the ray, asymptotically its part inside the critical strip, of length $t_e = 1/(2\cos(p\pi/2)) \approx 1/(\pi(1-p))$, so heights and exponents correspond through $1 - p \approx 1/(\pi T)$, as in Corollary 9.3.7.

9.3.6 The density level and natural boundaries

A variable with an HCM density is a GGC and so has an HCM Laplace transform, by (F5) and (F6). Under RH the clock variable $X = \sum_\gamma e_\gamma/\gamma^2$ is the GGC with

transform φ. Its Cauchy counterpart is $X_C = \sum_k e_k/(2k-1)^2$, with transform $\operatorname{sech}(\pi\sqrt{s}/2)$.

THEOREM 9.3.20. *The density of X_C is*

$$f_C(t) = \frac{4}{\pi}\sum_{k\ge 1}(-1)^{k-1}(2k-1)e^{-(2k-1)^2 t} = \frac{4}{\pi}\,\eta\Big(\frac{4it}{\pi}\Big)^3, \qquad t>0,$$

with η the Dedekind eta function. The line $\operatorname{Re} t = 0$ is a natural boundary of f_C. Hence X_C is a GGC whose Laplace transform is HCM, but the density of X_C is not HCM.

Proof. The residue of $\operatorname{sech}(\pi\sqrt{s}/2)$ at $s=-(2k-1)^2$ is $\frac{4}{\pi}(-1)^{k-1}(2k-1)$, which gives the series once the Bromwich contour is moved to the left across the circles $|s| = 4k^2$: for $s = 4k^2 e^{i\theta}$, $|\cosh(\pi\sqrt{s}/2)|^2 = \sinh^2(\pi k\cos\frac{\theta}{2}) + \cos^2(\pi k \sin\frac{\theta}{2})$ is at least $\frac{1}{4}$ for every θ and at least $\sinh^2(\pi k/2)$ for $|\theta| \le 2\pi/3$, while $|e^{st}| \le e^{-2k^2 t}$ for $2\pi/3 \le |\theta| \le \pi$, so for $t>0$ the integrals over the arcs of these circles to the left of the Bromwich line tend to 0. Jacobi's identity $\eta(z)^3 = \sum_{n\ge 0}(-1)^n(2n+1)e^{\pi i(2n+1)^2 z/4}$ [106, Theorem 357], at $z = 4it/\pi$, where $e^{\pi i(2k-1)^2 z/4} = e^{-(2k-1)^2 t}$, gives the eta form. As a modular form η has the real axis as natural boundary [313, Chapter 11, Section 1.4, and Chapter 1, Section 4.2], and the real axis in z is the line $\operatorname{Re} t = 0$. An HCM density continues analytically to Ω by (9.2.1), so f_C is not HCM. □

QUESTION 9.3.21 (density level). Assume RH and that all zeros are simple. Does the density $f_X(t) = \sum_n A_n \gamma_n^2 e^{-\gamma_n^2 t}$ of X, with A_n as in (9.2.3), have $\operatorname{Re} t = 0$ as a natural boundary? If so, X is not HCM.

Pólya's gap theorem for Dirichlet series [300, 306] says that if the exponents μ_n of a Dirichlet series $\sum_n c_n e^{-\mu_n t}$ have density $d = \lim n/\mu_n$ and satisfy $\inf_n(\mu_{n+1} - \mu_n) > 0$, then every segment of length greater than $2\pi d$ on the line of convergence carries a singularity. When $\limsup \log n/\mu_n = 0$ the abscissa of convergence equals $\limsup_n \log|c_n|/\mu_n$ [229].

THEOREM 9.3.22. *Assume RH and that all zeros are simple, and assume*

(G) $\inf_n(\gamma_{n+1}^2 - \gamma_n^2) > 0$;

(D) $\log\big(1/|\zeta'(\frac{1}{2} + i\gamma_n)|\big) = o(\gamma_n^2)$.

Then for $t>0$

$$f_X(t) = \sum_n c_n e^{-\gamma_n^2 t}, \qquad c_n = A_n\gamma_n^2 = -\frac{2\gamma_n\,\Xi(0)}{\Xi'(\gamma_n)},$$

the series has abscissa of convergence 0, and $\operatorname{Re} t = 0$ is a natural boundary of f_X. In particular the density of X is not HCM.

Proof. Residue expansion. The residue of $\varphi(s) = \Xi(0)/\Xi(i\sqrt{s})$ at $s = -\gamma_n^2$ is c_n as stated. Write f_X as a Bromwich integral and move the contour to the left across the arcs of the circles $|s| = T_k^2$ that lie to the left of the Bromwich line, where, for large k, $T_k \in [k, k+1]$ is chosen at distance at least $c/\log k$ from every ordinate, with a small absolute constant $c > 0$; this is possible because $[k-1, k+2]$ contains $O(\log k)$ ordinates [108]. Then $\zeta'/\zeta(\sigma + iy) = O(\log^2 k)$ uniformly in $-1 \le \sigma \le 2$ and $|y - T_k| \le c/(2\log k)$ [348, Theorem 9.6(A) and Section 14.4], and integration from $\sigma = 2$ gives $1/|\zeta(\sigma + iy)| \le \exp(O(\log^2 k))$ there. On the part of such an arc within angle $O(1/T_k)$ of the negative axis, $w = \frac{1}{2} + \sqrt{s}$ has $\frac{1}{2} \le \operatorname{Re} w \le 2$ and $|\operatorname{Im} w| = \sqrt{T_k^2 - (\operatorname{Re} w - \frac{1}{2})^2} = T_k + O(1/T_k)$, so for large k it lies in this band or in its reflection in the real axis, and Stirling's formula gives $1/|\Xi(i\sqrt{s})| \le \exp(O(T_k \log T_k))$, while $|e^{st}| \le e^{-tT_k^2/2}$. On the rest of the arc $\operatorname{Re} w \ge 2$, and Stirling's formula gives $|e^{st}\varphi(s)| \le \exp\bigl(tT_k^2 \cos\theta - \frac{T_k}{2}\cos\frac{\theta}{2}\log T_k + O(T_k)\bigr)$ for $s = T_k^2 e^{i\theta}$. For $t > 0$ both bounds tend to zero faster than any power of T_k, so the arcs contribute nothing in the limit and f_X equals the residue series.

Abscissa. The exponents γ_n^2 satisfy $\log n/\gamma_n^2 \to 0$ and have density zero. By Stirling, $|\Xi'(\gamma_n)| = |\xi'(\rho_n)| = \gamma_n^{O(1)} e^{-\pi\gamma_n/4}|\zeta'(\rho_n)|$, so $\log|c_n| = \frac{\pi}{4}\gamma_n + O(\log\gamma_n) + \log(1/|\zeta'(\rho_n)|)$. Since $|\zeta'(\rho_n)| \le \gamma_n^{O(1)}$ always, (D) gives $\log|c_n|/\gamma_n^2 \to 0$, so the abscissa of convergence is 0.

Natural boundary. By (G) and density zero, Pólya's theorem places a singularity in every segment of the imaginary axis. An HCM density continues analytically to Ω, which contains the imaginary axis minus the origin. □

The same argument, with $\mu_k = (2k-1)^2$, gaps $8k$ and $c_k = \frac{4}{\pi}(-1)^{k-1}(2k-1)$, gives a second proof of Theorem 9.3.20 that does not use modularity: by Pólya's gap theorem [300, 306] the line $\operatorname{Re} t = 0$ is a natural boundary of f_C.

Remark 9.3.23 (status of the hypotheses). Hypothesis (D) is far weaker than the conjectured polynomial lower bounds for $|\zeta'(\rho)|$. Hypothesis (G) asks $\gamma_{n+1} - \gamma_n \ge c_0/(\gamma_n + \gamma_{n+1})$ for some $c_0 > 0$, a gap of order $1/\gamma$ against a mean spacing of order $1/\log\gamma$. Under the GUE model the expected number of violations is $\sum_n (\log\gamma_n/\gamma_n)^3 < \infty$, so by the Borel Cantelli heuristic (G) fails at most finitely often, and then holds with a smaller c_0 if the zeros are simple.

Numerically, for the first 200 zeros the coefficients c_n computed from $\Xi'(\gamma_n)$ alternate in sign without exception; $\log|c_n|/\gamma_n^2$ equals 0.0462, 0.0144, 0.00524, 0.00322, 0.00195 at $n = 1, 10, 50, 100, 200$, tracking $\pi/(4\gamma_n)$; and the smallest gap $\gamma_{n+1}^2 - \gamma_n^2$ is 159, at $n = 4$. The c_n are the residues of the spectral expansion of the clock density in Section 5.2, written R_k there, and Theorem 9.3.22 describes, under its hypotheses, why that expansion cannot be continued past $t = 0$.

For small p the density of X_p escapes this obstruction. The transform $\varphi(s^p)$ is analytic on the wider sector $|\arg s| < (\pi - 2\delta_*)/p$ and, by Stirling's formula applied to $\xi(\frac{1}{2} + z)$ for $|\arg z| < \pi/2$, decays along every ray strictly inside it.

The inverse Laplace transform then continues the density of X_p to the sector $|\arg t| < (\pi - 2\delta_*)/p - \pi/2$, which contains Ω when $p \le 2p_*/3$.

Question 9.3.24. Determine $\{p : X_p \text{ is HCM}\}$, and the same set for the Cauchy family $X_{C,p}$ with transform $\operatorname{sech}(\pi s^{p/2}/2)$.

Two facts constrain the answer. Under RH, $X_p \overset{d}{=} S_p X^{1/p}$, and positive p stable variables are HCM exactly for $p \le 1/2$, a conjecture of Bondesson [39, Sections 5.7, 5.8 and 7.2] proved by Bosch and Simon [42, p. 32]. By (F1) and (F2), HCM of X would give HCM of X_p for $p \le 1/2$; but Theorem 9.3.22 shows that X is not HCM under (G) and (D), so this route is closed unless those hypotheses fail. The elasticity criterion reduces Question 9.3.24 to the absence of zeros of the density f of X_p in Ω and the Pick property of $-tf'(t)/f(t)$; by the continuation argument before Question 9.3.24, f is at least analytic on Ω for small p. A numerical scan of this elasticity for $X_{C,1/2}$ would be the first test.

9.3.7 The Mellin variable W and the first gap

Question 9.2.7 asks for which β the density of W^β is HCM; on the Cauchy side the answer is $|\beta| \ge 2$. Let $\chi(t) = \xi(\frac12)/\xi(\frac12 + t) = \prod_\gamma (1 + t^2/\gamma^2)^{-1}$ and let $g(y) = \frac{1}{\pi}\int_0^\infty \chi(t)\cos(ty)\,dt$ be the function g of Section 9.2.2, which is entire because $\chi(t) = \exp(-\frac12 |t| \log|t|(1 + o(1)))$. By the van Dantzig pair of Section 2.1, χ is a characteristic function without RH: the clock variable T exists unconditionally [282, Theorem A], and for Z standard normal independent of T, $\mathbb{E}e^{itZ\sqrt{2T}} = \mathbb{E}e^{-t^2 T} = \chi(t)$. So g is the density of $Z\sqrt{2T}$, and $W = \exp(Z\sqrt{2T})$ exists without any hypothesis. By the elasticity criterion, HCM of W^β becomes a sign condition on $-g'/g$ in a strip (Proposition 9.3.26), and since g is entire no natural boundary stands in the way.

9.3.7.1 A lower bound for g

Theorem 9.3.25 proves the existence of W a second time, independently of [282, Theorem A].

Theorem 9.3.25. *Unconditionally, by the verification of RH to height T_0 [277], $g(y) \ge 0.9998\, g_R(y) > 0$ for all real y, where g_R is the density of $\sum_{\gamma\ \mathrm{real}} L_\gamma/\gamma$. Hence χ is a characteristic function and there is a positive random variable W with $\mathbb{E}W^{it} = \chi(t)$ and $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1$.*

Proof. Split $\chi = \chi_R \chi_{\mathrm{off}}$ into the products over real zeros and over conjugate pairs of nonreal zeros. For a pair $\gamma = \varrho e^{i\delta} = a + ib$ with $a, b > 0$, the factor $(1 + t^2/\gamma^2)^{-1}(1 + t^2/\bar\gamma^2)^{-1}$ is the Fourier transform of the even kernel

$$k_\gamma(x) = \frac{\varrho}{2\sin 2\delta}\, e^{-a|x|} \sin(b|x| + \delta),$$

of total mass one, positive for $|x| < (\pi-\delta)/b$ and bounded by $\frac{\varrho}{2\sin 2\delta}e^{-a|x|}$. List the pairs as $\gamma^{(1)},\gamma^{(2)},\dots$ in order of $\operatorname{Re}\gamma$, write $k_j = k_{\gamma^{(j)}}$, and let χ_m be the product of the first m pair factors, $\nu_m = k_1 * \cdots * k_m$ and $g_m = g_R * \nu_m$, the function with Fourier transform $\chi_R\chi_m$. We bound g_m uniformly in m and let m grow at the end.

Comparison. $g_R = \ell * \tilde g_R$, where $\ell(z) = \frac{\gamma_1}{2}e^{-\gamma_1|z|}$ is the density of L_{γ_1}/γ_1 and $\tilde g_R$ is the density of the sum over the other real zeros, so $e^{-\gamma_1|x|}g_R(y) \le g_R(y-x) \le e^{\gamma_1|x|}g_R(y)$. Writing $\nu_m = \nu_m^+ - \nu_m^-$,

$$g_m(y) \ge g_R(y)\Big(\int e^{-\gamma_1|x|}\nu_m(x)dx - 2\int \cosh(\gamma_1 x)\,\nu_m^-(x)dx\Big) =: g_R(y)(A_m - 2B_m).$$

The term A_m. By Parseval, $A_m = \frac{1}{\pi}\int \chi_m(t)\frac{\gamma_1}{\gamma_1^2+t^2}dt = \mathbb{E}\,\chi_m(\gamma_1 C)$ with C a standard Cauchy variable. For $s \ge 0$ each pair factor is $|1+s/\gamma^2|^{-2} \ge (1+s/\varrho^2)^{-2} \ge e^{-2s/\varrho^2}$, so $A_m \ge \mathbb{E}e^{-\kappa C^2} = e^{\kappa}\operatorname{erfc}(\sqrt{\kappa})$ with $\kappa = 2\gamma_1^2\Sigma_{\text{off}}$ and $\Sigma_{\text{off}} = \sum_{\text{off}}|\gamma|^{-2}$. All nonreal zeros have $\operatorname{Re}\gamma > T_0 = 3\times 10^{12}$ [277], so with $N(T) \le T\log T$ for $T \ge T_0$, $\Sigma_{\text{off}} \le 2\int_{T_0}^{\infty} N(T)T^{-3}dT \le 2(\log T_0+1)/T_0 < 2\times 10^{-11}$, $\kappa < 8\times 10^{-9}$ and $A_m > 0.99989$.

The term B_m. For even signed measures $\int\cosh(\gamma_1 x)\,d(|\mu| * |\nu|) = \int\cosh(\gamma_1 x)d|\mu|\cdot\int\cosh(\gamma_1 x)d|\nu|$ exactly, and $(\mu*\nu)^- \le \mu^+*\nu^- + \mu^-*\nu^+$. Hence, with $M(\cdot) = \int\cosh(\gamma_1 x)\,d(\cdot)$ and since $M(|k_\gamma|) \ge M(k_\gamma) = |1-\gamma_1^2/\gamma^2|^{-2} \ge 1$,

$$B_m \le \sum_{j\le m} M(k_j^-)\prod_{i\le m,\ i\ne j} M(|k_i|) \le SP, \qquad S = \sum_{\gamma} M(k_\gamma^-), \quad P = \prod_{\gamma} M(|k_\gamma|),$$

with the sum and the product over all pairs. Here k_γ^- lives on $|x| \ge (\pi-\delta)/b$ and $M(k_\gamma^-) \le \frac{\varrho}{\sin 2\delta\,(a-\gamma_1)}\exp\big(-(a-\gamma_1)\frac{\pi-\delta}{b}\big)$. As $\varrho/\sin 2\delta = \varrho^3/(2ab)$, this bound is $\frac{\varrho^3}{2a(a-\gamma_1)}\,b^{-1}e^{-K/b}$ with $K = (a-\gamma_1)(\pi-\delta)$. For fixed a it increases with b: ϱ increases, K decreases because $\delta = \arctan(b/a)$ increases, and $b \mapsto b^{-1}e^{-K/b}$ increases on $(0,K)$. So, as $b < \frac12$, it is at most

$$F(a) = \frac{(a^2+\frac14)^{3/2}}{a(a-\gamma_1)}\exp\Big(-2(a-\gamma_1)\big(\pi - \arctan\tfrac{1}{2a}\big)\Big),$$

and $F(T_0) < \exp(-1.88\times 10^{13})$, while $-(\log F)' > 6$ on $[T_0,\infty)$. Every pair has $a > T_0$, so Stieltjes integration against $N(T) \le T\log T$ gives

$$S \le F(T_0)\Big(T_0\log T_0 + \frac{\log T_0+1}{6} + \frac{1}{36T_0}\Big) < \exp(-1.8\times 10^{13}).$$

Moreover $M(|k_\gamma|) = M(k_\gamma) + 2M(k_\gamma^-)$ and $M(k_\gamma) \le \exp(3\gamma_1^2/\varrho^2)$, so $\log P \le 3\gamma_1^2\Sigma_{\text{off}} + 2S < 1.2\times 10^{-8}$. So $A_m - 2B_m > 0.9998$ for every m.

Limit. For real t each pair factor lies in $(0,1]$, and χ_R is integrable because it contains the factor $(1+t^2/\gamma_1^2)^{-1}$. By dominated convergence in the Fourier

inversion integral, $g_m \to g$ uniformly on $\mathbb{R}$, so $g \geq 0.9998\, g_R$. The comparison also gives $g_m(y) \leq g_R(y) \int e^{\gamma_1|x|}|\nu_m(x)|dx \leq 2P\, g_R(y)$, since $e^{\gamma_1|x|} \leq 2\cosh(\gamma_1 x)$ and $|\nu_m| \leq |k_1| * \cdots * |k_m|$, so $g \leq 2P\, g_R$. Hence g is integrable, its Fourier transform is χ, and $\chi(0) = 1$ makes g a probability density. Finally $\int e^{cy} g_R(y)dy = \prod_{\gamma\ \mathrm{real}}(1 - c^2/\gamma^2)^{-1} < \infty$ for $|c| < \gamma_1$, so $\int e^{c|y|} g(y)dy < \infty$ there, $\mathbb{E}W^z = \int e^{zy} g(y)dy$ is analytic on $|\operatorname{Re} z| < \gamma_1$, and it equals $\Xi(0)/\Xi(z)$ there because the two agree on the imaginary axis. □

The proof uses only that the zeros with $0 < \operatorname{Re}\gamma \leq T_0$ are real; the pair kernel formula and its constants were also checked numerically.

9.3.7.2 Which powers of W are HCM

For real $\beta \neq 0$ the density of W^β is $x^{-1} g(\log x/\beta)/|\beta|$. With $y = \log x$, its elasticity is $1 - \frac{1}{\beta}\frac{g'}{g}(y/\beta)$, and $x \in \mathbb{C}^+$ corresponds to $0 < \operatorname{Im} y < \pi$; for $\beta < 0$ the point $u = y/\beta$ then lies in the lower half of the strip $|\operatorname{Im} u| < \pi/|\beta|$, where the factor $\frac{1}{\beta} < 0$ reverses the sign condition, and $g(\bar{u}) = \overline{g(u)}$, as g is real on $\mathbb{R}$, moves it to the upper half. Theorem 9.3.2 therefore gives the following.

Proposition 9.3.26. *Let $\beta \in \mathbb{R} \setminus \{0\}$. Then W^β is HCM if and only if g has no zeros in the strip $|\operatorname{Im} u| < \pi/|\beta|$ and $\operatorname{Im}\big(-g'(u)/g(u)\big) \geq 0$ for $0 < \operatorname{Im} u < \pi/|\beta|$.*

As $\operatorname{Im} u \downarrow 0$ the condition requires $(\log g)'' \leq 0$ on $\mathbb{R}$; numerically $(\log g)'' \leq -2.2$ on $[0,1]$. On the Cauchy side $g_C = 1/(\pi\cosh u)$, $-g_C'/g_C = \tanh u$, which is Pick on $0 < \operatorname{Im} u < \pi/2$ with a pole at $i\pi/2$; Proposition 9.3.26 gives $|\beta| \geq 2$, Bosch's theorem once more.

Theorem 9.3.27. *Assume RH, simple zeros, and $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$. Then $u_0 = \frac{\pi}{4} + \limsup_n \gamma_n^{-1} \log(1/|\zeta'(\rho_n)|)$ is finite, and for $\operatorname{Re} u > u_0$*

$$g(u) = -\Xi(0) \sum_n \frac{e^{-\gamma_n u}}{\Xi'(\gamma_n)} = \sum_n b_n e^{-\gamma_n u}, \qquad b_1 > 0 > b_2,$$

and if W^β is HCM then $|\beta| \geq \gamma_2 - \gamma_1 = 6.8873\ldots$.

Proof. The expansion is the residue computation for the Fourier integral $g(u) = \frac{1}{2\pi}\int \chi(t) e^{-itu}\, dt$ closed in the lower half plane. We move the line of integration to $\operatorname{Im} t = -T_k$, where $T_k \in [k, k+1]$ are heights at which $1/\zeta(\sigma \pm iT_k) = T_k^{O(1)}$ uniformly in $-1 \leq \sigma \leq 2$ [348, Theorem 9.7]; by Stirling's formula and the functional equation the moved integral is $O(e^{(\pi/4 - \operatorname{Re} u)T_k + O(\log T_k)})$, which tends to 0 for $\operatorname{Re} u > \pi/4$, so there $g(u)$ is the limit of the sums of the residue terms over $\gamma_n < T_k$. By Stirling, $\log|b_n| = \frac{\pi}{4}\gamma_n + \log(1/|\zeta'(\rho_n)|) + O(\log\gamma_n)$, so the series converges absolutely for $\operatorname{Re} u > u_0$, and $u_0 \geq \pi/4$ because $|\zeta'(\rho_n)| \leq \gamma_n^{O(1)}$, so its

sum is $g(u)$ there (conjecturally $u_0 = \pi/4$); the signs come from $b_n = \frac{\gamma_n}{2} A_n$. Put $\Delta = \gamma_2 - \gamma_1$. As $\operatorname{Re} u \to \infty$ with $\operatorname{Im} u = v$ fixed,

$$\operatorname{Im}\Big(-\frac{g'}{g}\Big)(x+iv) = \Delta\Big|\frac{b_2}{b_1}\Big| e^{-\Delta x}\sin(\Delta v)\,\big(1+O(e^{-cx})\big), \qquad c>0.$$

For $\pi/\Delta < v < 2\pi/\Delta$ this is negative for large x. If $|\beta| < \Delta$ the strip $0 < \operatorname{Im} u < \pi/|\beta|$ contains such v, and Proposition 9.3.26 fails. □

Question 9.3.28 (first gap conjecture). Is W^β HCM exactly when $|\beta| \ge \gamma_2 - \gamma_1$? On the Cauchy side the analogous statement, with frequencies $2k-1$, is Bosch's theorem at $\alpha = 1/2$: $3 - 1 = 2$.

With b_n computed from $\Xi'(\gamma_n)$ ($b_1 = 359.52$, $b_2 = -28007$, $b_3 = 392918$) and g computed by quadrature near the origin, a scan of 4840 points with $0 \le x \le 6$ and $0 < v < 0.99\, v_*$, where $v_* = \pi/\Delta = 0.45614$, found no value of $\operatorname{Im}(-g'/g)$ below -1.3×10^{-15}, which is rounding; in 40 digit arithmetic the minimum over $1.1 \le x \le 9.1$ at $v = 0.999\, v_*$ is $+1.0 \times 10^{-27}$. Just above the threshold the sign flips: the minimum is -6.5×10^{-8} at $v = 1.001\, v_*$, so numerically the threshold is sharp to 0.1 per cent.

Sufficiency requires the Pick property of $-g'/g$ in the full strip $0 < \operatorname{Im} u < \pi/\Delta$, which near the origin is a statement about the Pólya type kernel g that does not reduce to the zeros.

9.3.7.3 Solvable models

To see whether the first gap rule is a coincidence of the half Cauchy case, we test it on an explicit family. The logistic variable $\log W$ has characteristic function $\pi t/\sinh(\pi t) = \prod_{k\ge1}(1+t^2/k^2)^{-1}$. Removing a finite set R of frequencies multiplies the characteristic function by $\prod_{k\in R}(1+t^2/k^2)$, which gives $g = \prod_{k\in R}(1-D^2/k^2)L$ with $D = d/dy$ and $L(y) = \frac{1}{4}\operatorname{sech}^2(y/2)$. For each model we locate numerically the largest strip on which $\operatorname{Im}(-g'/g) \ge 0$; as in Proposition 9.3.26, its width h_* gives the HCM threshold $\beta_* = \pi/h_*$ (Table 9.3).

When the first gap is at least every later gap, the computed threshold equals the first gap to the resolution of the computation; otherwise it lies strictly between the first and the largest gap, and depends on where the large gap sits. The asymptotic argument of Theorem 9.3.27 explains the lower bound in all cases. For the Riemann zeros the first gap $\gamma_2 - \gamma_1 = 6.8873$ exceeds every later gap among the first 1000 zeros (the next largest is $\gamma_4 - \gamma_3 = 5.414$). The explicit form of the Riemann von Mangoldt formula of Hasanalizade, Shen and Wong, $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac{7}{8}| \le 2.5167 + \frac{1}{50T}$ for $e \le T \le X_0 = 30\,610\,046\,000$ and $\le 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T}$ for $T > X_0$ [108, (1.7), (2.7) and Corollary 1.4], shows that every interval $(T, T+6.887]$ with $T \ge 1000$ contains an ordinate: the lower bound it gives for the number of ordinates there exceeds

frequencies	gaps	first gap	largest gap	$\beta_* = \pi/h_*$
$1, 2, 3, \ldots$	$1, 1, 1, \ldots$	1	1	1.000
$1, 3, 5, \ldots$ (half Cauchy)	$2, 2, 2, \ldots$	2	2	2 (Bosch)
$1, 3, 4, 5, \ldots$	$2, 1, 1, \ldots$	2	2	2.000
$1, 3, 5, 6, \ldots$	$2, 2, 1, \ldots$	2	2	2.000
$1, 3, 4, 6, \ldots$	$2, 1, 2, 1, \ldots$	2	2	2.000
$1, 4, 5, 6, \ldots$	$3, 1, 1, \ldots$	3	3	3.000
$1, 4, 5, 7, \ldots$	$3, 1, 2, 1, \ldots$	3	3	3.000
$1, 4, 7, 8, \ldots$	$3, 3, 1, \ldots$	3	3	3.000
$1, 5, 6, 7, \ldots$	$4, 1, 1, \ldots$	4	4	4.000
$1, 2, 4, 5, \ldots$	$1, 2, 1, \ldots$	1	2	1.269
$1, 2, 3, 6, \ldots$	$1, 1, 3, 1, \ldots$	1	3	1.317
$1, 2, 5, 6, \ldots$	$1, 3, 1, \ldots$	1	3	1.483

Table 9.3: HCM threshold for powers of W in logistic models with frequencies removed, computed as $\beta_* = \pi/h_*$ from the width h_* of the largest strip on which $\mathrm{Im}(-g'/g) \geq 0$. The half Cauchy row is Bosch's theorem, not a computation.

0.52 at $T = 1000$, increases with T while $T + 6.887 \leq X_0$, and exceeds 11 beyond. So every gap that starts above height 1000 is at most 6.887, and every gap that starts below it is between two of the first 650 zeros, since $N(1000) = 649$: the first gap exceeds every later gap. This refines Question 9.3.28.

Question 9.3.29 (first gap conjecture, refined). Let $Y = \sum_n L_n/\lambda_n$ with independent standard Laplace variables L_n and $0 < \lambda_1 < \lambda_2 < \cdots$, and let $W = e^Y$. If the characteristic function of Y decays faster than every exponential, or if the first gap $\lambda_2 - \lambda_1$ is at least every later gap, is the HCM threshold of W^β exactly $\lambda_2 - \lambda_1$? In particular, is it $\gamma_2 - \gamma_1$ for the Riemann variable?

9.3.8 Where the difficulty lies

Bosch's theorem cannot be the input to a proof that the reciprocal Xi function is HCM, because for a product over the zeros HCM is equivalent to the reality of the zeros (Proposition 9.2.6), and every GGC criterion in the books of Sato and Bondesson is, at the centre, equivalent to RH (Section 14.1). Bosch's method does transfer: at each of three levels it isolates one concrete object on which the question turns.

At the level of transforms the object is the Thorin density u_p of (9.3.2). Its positivity for every $p < 1$ is equivalent to RH; the primes prove it on each ray beyond a point that is asymptotically where the ray leaves the critical strip, leaving a segment of length about $1/(\pi(1-p))$. The first direction is to prove positivity on that segment from the explicit formula for ξ'/ξ rather than from the zeros;

computation alone extends the unconditional range $p \le 1 - 1/(\pi T_0)$ only as fast as the verification height T_0 grows.

At the level of the Mellin variable the object is the sign of $\operatorname{Im}(-g'/g)$ in the strip $0 < \operatorname{Im} u < \pi/(\gamma_2 - \gamma_1)$. The second direction is to prove the first gap conjecture, first in the logistic models of Table 9.3, where every quantity is elementary, and then for W, whose threshold the scan after Question 9.3.28 already locates to 0.1 per cent. Question 9.2.8 remains open.

At the level of densities the object is the natural boundary. The third direction is to establish it for the clock variable without the hypotheses (G) and (D) of Theorem 9.3.22, which would show what replaces, on the Riemann side, the modularity that produces the natural boundary on the Cauchy side.

Section 8.1 approaches the Thorin functional of the clock from the other end of Bondesson's hierarchy, through its Laplace transform, the heat trace $W(t)$.

9.4 Exercises

On Section 9.1.

9.1. Show that if X_N are GGCs converging in law to a proper limit, the Thorin measures converge vaguely on $(0, \infty)$, and use the laws Gamma(n, n) of Section 4.1.1, which converge to the point mass at one, to show that Thorin mass escaping to infinity can reappear in the limit as drift.

9.2. For the sinh clock of Section 4.1.4 with one atom k^2 replaced by the conjugate pair $a, \bar{a}$, $a = -(\delta + ik)^2$ with $0 < \delta < \frac{1}{2}$, as in Proposition 2.1.3, construct the Thorin densities at resolution ε and show that their vague limit is the sinh measure with the atom k^2 removed.

9.3. Verify that $\varphi_{1/2}(uv)\varphi_{1/2}(u/v) = \prod_k a_k^2/(a_k^2 + a_k uw + u^2)$ for the clock, and that at $u = |a|$ an off line pair contributes the completely monotone factor $(w + 2\cos\arg a)^{-2}$.

9.4. Compute $(-1)^k \partial_w^k H_u(w)/H_u(w)$ for $k \le 6$ at $u = 1$ and $w = 3$ from the Pólya density and compare with Figure 9.1.

Part II

The reciprocal zeta function

Chapter 10

Thorin type forms of the reciprocal zeta function

Up to the factor $1-4s$ of the pole and a constant, $1/\zeta(\frac{1}{2}+\sqrt{s})$ is the reciprocal Xi function times the gamma factor; unlike ξ it has a Dirichlet series and an Euler product, from which we read its Thorin type measures.

Sections 10.1 and 10.2 compute the Thorin measure of the gamma factor, the Thorin measures of the normalised reciprocal zeta function and of the reciprocal Xi function under RH, and, unconditionally, the formal Thorin measures of these two functions in the Vere-Jones form of Section 10.1.3. After the stable tilt $e^{-c\sqrt{s}}$, $c=\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$, the reciprocal of the gamma factor is a generalised gamma convolution; multiplying the normalised reciprocal zeta function by it gives the floor $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ of the reciprocal Xi function (Section 10.2.4), in which the zeros are the only possible source of negativity.

The infinite divisibility of the floor, as proved here, rests on the positivity of the canonical function of the normalised reciprocal zeta function (Theorem 11.1.1, Chapter 11). It also follows from the infinite divisibility of the clock (Theorem 8.1.1); the floor adds a splitting of its canonical function into an arithmetic part and a completely monotone archimedean part. The floor is a generalised gamma convolution exactly when RH holds, and the derivatives of its canonical function alternate in sign up to order 10^{10} by the methods of this chapter and up to order 1.16×10^{25} by the estimates of Section 8.1.

10.1 Thorin type forms of the reciprocal zeta function

Under RH the Thorin measure of the normalised reciprocal zeta function is dM, with $M-1=S(\sqrt{t})$ the argument function. Its heat trace, the canonical function w of Theorem 10.1.11, is explicit in the primes, is an absolutely convergent series of Jacobi theta functions, and gives a criterion for RH (Theorem 10.1.15).

10.1.1 The question and the three functions

Write $\sqrt{s}$ for the principal square root on the cut plane $\mathbb{C} \setminus (-\infty, 0]$, so that $\operatorname{Re}\sqrt{s} > 0$ there, and evaluate ζ at $z = \frac{1}{2} + \sqrt{s}$. A zero $\rho = \frac{1}{2} + i\gamma$ then corresponds to $\sqrt{s} = i\gamma$, that is to $s = -\gamma^2$ on the cut, where an atom of a Thorin measure at $t = \gamma^2$ would put its singularity. The question of this chapter and of Chapter 11 is whether the zeros, the primes and the gamma factor assemble into a Thorin measure, and which one. Unless stated otherwise, in both chapters s is the variable u of Sections 2.1 and 4.1, t is the variable z of their Thorin measures, x is the time of the heat trace $W(x)$, τ is the ordinate, and z is the argument of ζ and ξ.

The Möbius series and the Euler product converge absolutely to the right of the parabola $x = \frac{1}{4} - y^2$, on

$$H_{1/2} = \{s : \operatorname{Re}\sqrt{s} > \tfrac{1}{2}\} = \{x + iy : \ x > \tfrac{1}{4} - y^2\},$$

and there

$$\frac{1}{\zeta(\frac{1}{2} + \sqrt{s})} = \sum_{n \geq 1} \frac{\mu(n)}{\sqrt{n}}\, e^{-\sqrt{s}\log n} = \prod_{p} \Big(1 - p^{-1/2} e^{-\sqrt{s}\log p}\Big). \tag{10.1.1}$$

Every term is built from $e^{-\lambda\sqrt{s}}$, which is completely monotone in s: it is the Laplace transform of the positive stable law of index $\frac{1}{2}$,

$$e^{-\lambda\sqrt{s}} = \int_0^\infty e^{-sx}\, \frac{\lambda}{2\sqrt{\pi}}\, x^{-3/2} e^{-\lambda^2/4x}\, dx. \tag{10.1.2}$$

The Thorin measure is read from the logarithm. Recall from (4.1.1) that a generalised gamma convolution (GGC) has Laplace transform

$$f(s) = \exp\Big(-as - \int_{(0,\infty)} \log\big(1 + \tfrac{s}{t}\big)\, U(dt)\Big), \tag{10.1.3}$$

with $a \geq 0$ and $U \geq 0$ the Thorin measure; equivalently $-f'/f(s) = a + \int U(dt)/(s+t)$ is a Stieltjes function [39]. We allow U to be signed, with $\int_{(0,1)} |\log t|\, |U|(dt) + \int_{[1,\infty)} t^{-1} |U|(dt) < \infty$ so that the integral converges, and then call (10.1.3) a Thorin type representation.

Three functions of s run through both chapters. The function $1/\zeta(\frac{1}{2} + \sqrt{s})$ vanishes at $s = \frac{1}{4}$, the pole of ζ, and equals $1/\zeta(\frac{1}{2}) < 0$ at $s = 0$, so it is not the Laplace transform of a probability law. Both defects are removed by the normalised reciprocal zeta function

$$\varphi(s) = \frac{\zeta(\frac{1}{2})}{(1 - 4s)\, \zeta(\frac{1}{2} + \sqrt{s})}, \qquad \varphi(0) = 1. \tag{10.1.4}$$

The other two are the gamma factor $G(s) = \Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$ and the reciprocal Xi function $R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, and $\varphi = RG$ (Proposition 10.1.9). In both chapters G is the gamma factor, not the function G of Section 2.1, whose reciprocal is R, except in Section 11.1.3, where G is a Laguerre Pólya function, and Section 11.1.4, where G is the function of Section 2.1; and $\theta(\tau) = \operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{i\tau}{2}) - \frac{\tau}{2}\log\pi$ is the Riemann Siegel theta function, written ϑ elsewhere in the book (Section 2.2), so that $\theta'(\tau) = \frac{1}{2}\operatorname{Re}\psi(\frac{1}{4} + \frac{i\tau}{2}) - \frac{1}{2}\log\pi$ [80, Section 6.5]. Table 10.1 lists the Thorin measures of the three functions and of the building blocks used below. In it $N(T)$ counts the zeros with $0 < \gamma \le T$, $S(T) = \pi^{-1}\arg\zeta(\frac{1}{2} + iT)$ is the argument function, $c_n = 2n + \frac{1}{2}$, $c = -\theta'(0)$ is the constant of (10.1.6), and U_λ is the sawtooth measure of Lemma 10.1.7.

Function	Thorin measure	Sign
φ	$dM(t)$, $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ (RH)	signed
G	$-\dfrac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}dt = \dfrac{1}{2\pi\sqrt{t}}\Big\{b + \sum_{n\ge0}\Big(\dfrac{c_n}{t + c_n^2} - \dfrac{1}{2n+2}\Big)\Big\}dt$	signed
R	$\sum_{\gamma>0}\delta_{\gamma^2}$ (RH)	positive
$2/(1 + e^{-\lambda\sqrt{s}})$	$-U_\lambda = \sum_k \delta_{((2k+1)\pi/\lambda)^2} - \dfrac{\lambda}{4\pi\sqrt{t}}dt$	signed
$(1 + \sqrt{s}/\kappa)^{-1}$	$\dfrac{\kappa}{2\pi\sqrt{t}\,(t + \kappa^2)}dt$	positive
$\operatorname{sech}(\lambda\sqrt{s})$	$\sum_k \delta_{((k+\frac{1}{2})\pi/\lambda)^2}$	positive
$e^{-c\sqrt{s}}/G$	$\dfrac{\theta'(\sqrt{t}) - \theta'(0)}{2\pi\sqrt{t}}dt$	positive

Table 10.1: Thorin measures of φ, G and R and of the building blocks used in this chapter and in Chapter 11, with the convention $\log f(s) = -\int \log(1 + s/t)\,U(dt)$ after normalising $f(0) = 1$. Here $b = \frac{1}{2}(\gamma_E + \log\pi)$, $\kappa > 0$ and $c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$. The rows for φ and R are Theorem 10.1.10, the row for G is Theorem 10.2.1, the fourth row is Lemma 10.1.7, the fifth and sixth appear in Table 4.3, and the last is Theorem 10.2.12.

10.1.2 Stieltjes inversion and the Euler factors

Most Thorin measures of this chapter and of Chapter 11 are read from the boundary values of $-F'/F$ on $(-\infty, 0]$ by Lemma 10.1.1.

Lemma 10.1.1. *Let f be analytic on $\mathbb{C} \setminus (-\infty, 0]$ with $f(\bar{s}) = \overline{f(s)}$, $f(s) \to 0$ as $|s| \to \infty$ uniformly in $\arg s$, $|f(s)| \le C|s|^{-\alpha}$ near 0 for some $\alpha < 1$, and such that f extends continuously from the upper half plane to the points of $(-\infty, 0)$ apart from finitely many points $-t_0$, at each of which $f(s) - r/(s + t_0)$ is bounded near $-t_0$ in $\mathbb{C} \setminus (-\infty, 0]$ for some real r; $f(-t + i0)$ denotes the values of this extension*

and $f(-t-i0)=\overline{f(-t+i0)}$. *Then*

$$f(s)=\int_{(0,\infty)}\frac{m(dt)}{s+t},\qquad m(dt)=-\frac{1}{\pi}\operatorname{Im} f(-t+i0)\,dt,$$

where the integral is the limit of the integrals over $(0,T]$ *as* $T\to\infty$, *absolutely convergent when* $\int_{(1,\infty)}t^{-1}|m|(dt)<\infty$, *and where each of the points* $-t_0$, *which we call a simple pole of* f, *contributes the atom* $r\,\delta_{t_0}$ *to* m.

Proof. Fix $s\in\mathbb{C}\backslash(-\infty,0]$ and integrate $f(z)/(z-s)$ over a keyhole contour around $(-\infty,0]$, which by Cauchy's formula gives $2\pi i\,f(s)$; the large circle vanishes by the decay at infinity and the small circle by the bound at the origin. By the continuity of the extension, the integrals along the two sides of the keyhole converge to the integrals of the boundary values as the keyhole closes onto the cut, after subtracting $r/(z+t_0)$ near each pole $-t_0$, where the remainder is bounded. The two sides of the cut combine to $\frac{1}{2\pi i}\{f(-t-i0)-f(-t+i0)\}=-\frac{1}{\pi}\operatorname{Im} f(-t+i0)$. At a simple pole the contour follows small semicircles above and below the cut, which together contribute $r/(s+t_0)$. □

The lemma excludes functions with infinitely many poles on the cut, such as $-R'/R$ under RH, which has a pole at every $-\gamma^2$; these need a direct argument from Hadamard's product, as in the proof of Theorem 10.1.10.

For $f(s)=\lambda e^{-\lambda\sqrt{s}}/(2\sqrt{s})=-\frac{d}{ds}e^{-\lambda\sqrt{s}}$ one has $\sqrt{-t+i0}=i\sqrt{t}$, and the lemma gives the density $\frac{\lambda}{2\pi\sqrt{t}}\cos(\lambda\sqrt{t})$, so that

$$e^{-\lambda\sqrt{s}}=1-\int_0^\infty\log\bigl(1+\tfrac{s}{t}\bigr)\frac{\lambda\cos(\lambda\sqrt{t})}{2\pi\sqrt{t}}\,dt.$$

This is not the Thorin measure of $e^{-\lambda\sqrt{s}}$, which is the positive density $\lambda/(2\pi\sqrt{t})$ (Table 4.3), but the Thorin type density contributed by an exponential sitting in an exponent, as in the logarithm of every Euler factor of (10.1.1) and in $\log\zeta(\frac{1}{2}+\sqrt{s})$.

Proposition 10.1.2. *For* $0<q<1$ *and* $\lambda>0$ *the Euler factor* $E(s)=1-qe^{-\lambda\sqrt{s}}$ *is analytic and zero free on* $\mathbb{C}\setminus(-\infty,0]$, *and*

$$\log\frac{E(s)}{1-q}=-\int_0^\infty\log\bigl(1+\tfrac{s}{t}\bigr)\,u(t)\,dt,$$

$$u(t)=-\frac{q\lambda}{2\pi\sqrt{t}}\,\frac{\cos(\lambda\sqrt{t})-q}{1-2q\cos(\lambda\sqrt{t})+q^2}=-\frac{\lambda}{2\pi\sqrt{t}}\sum_{k\ge1}q^k\cos(k\lambda\sqrt{t}).$$

Proof. On $\mathbb{C}\setminus(-\infty,0]$ we have $|e^{\lambda\sqrt{s}}|>1>q$, so E has no zeros. Put $f=-E'/E=-\frac{q\lambda}{2\sqrt{s}}(e^{\lambda\sqrt{s}}-q)^{-1}$, which is $O(|s|^{-1/2})$ at 0 and decays at infinity. At $\sqrt{s}=i\tau$, $\tau=\sqrt{t}$, we get $f=\frac{iq\lambda}{2\tau}(e^{i\lambda\tau}-q)^{-1}$ and $\operatorname{Im} f=\frac{q\lambda}{2\tau}\,\frac{\cos\lambda\tau-q}{1-2q\cos\lambda\tau+q^2}$. Lemma 10.1.1 gives u; integrating $-E'/E=\int u(t)\,dt/(s+t)$ from 0 to s gives the logarithm, and the series is the Poisson kernel expansion. □

With $q = p^{-1/2}$ and $\lambda = \log p$, summing u over p formally gives

$$u_\zeta(t) = -\frac{1}{2\pi\sqrt{t}} \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}} \cos(\sqrt{t}\log n) \quad \text{“}=\text{”} \quad \frac{1}{2\pi\sqrt{t}} \operatorname{Re} \frac{\zeta'}{\zeta}\big(\tfrac12 + i\sqrt{t}\big), \tag{10.1.5}$$

a series that converges only in the sense of distributions. Since $\frac{d}{d\tau}\arg\zeta(\frac12+i\tau) = \operatorname{Re}\frac{\zeta'}{\zeta}(\frac12+i\tau)$, the right side is formally the density of $dS(\sqrt{t})$; Theorem 10.1.10 makes this precise.

10.1.3 The Lai and Vere-Jones identity and sawtooth measures

We trade each exponential for a hyperbolic secant, whose Thorin measure is a lattice of unit atoms, at the price of a Möbius inversion. Lai and Vere-Jones [200] pose the completeness of the dilates $\operatorname{sech}(\theta x)$, $\theta \ge 0$, as an open problem for the hyperbolic secant law. Möbius inversion of the alternating exponential series of the secant expands e^{-x} in the dilates $\operatorname{sech}(mx)$ (Proposition 10.1.3, the Lai and Vere-Jones identity); the expansions of $1/\zeta$ and $\log\zeta$ built from it form the Vere-Jones representation. Here χ_4 is the nontrivial character modulo 4.

Proposition 10.1.3. *For* $\operatorname{Re} x > 0$,

$$\tfrac12 \operatorname{sech} x = \sum_{m \text{ odd}} \chi_4(m)\, e^{-mx}, \qquad e^{-x} = \tfrac12 \sum_{m\ge1} \mu(m)\chi_4(m)\, \operatorname{sech}(mx).$$

Proof. The first is the geometric series $e^{-x}/(1+e^{-2x})$. Since χ_4 is completely multiplicative, $\sum_m \mu(m)\chi_4(m)\sum_k \chi_4(k)e^{-mkx} = \sum_r e^{-rx}\sum_{mk=r}\mu(m)\chi_4(m)\chi_4(k) = e^{-x}$, the rearrangement being justified by $\sum_{m,k} e^{-mk\operatorname{Re}x} < \infty$ for $\operatorname{Re}x > 0$. □

Lemma 10.1.4. *For* $\lambda > 0$, $\operatorname{sech}(\lambda\sqrt{s}) = \prod_{k\ge0}\big(1 + s/\tau_k\big)^{-1}$ *with* $\tau_k = \big((k+\frac12)\pi/\lambda\big)^2$, *written* $\tau_k(\lambda)$ *when the dependence on* λ *matters. It is the Laplace transform of* $T_\lambda = \sum_k \tau_k^{-1} e_k$ *with* e_k *independent standard exponentials, the exit time of* $\sqrt{2}B$ *from* $(-\lambda,\lambda)$, *and it is a GGC with atomic Thorin measure* $\sum_k \delta_{\tau_k}$; *see [33] and Table 4.3.*

Theorem 10.1.5. *For* $s \in H_{1/2}$,

$$\frac{1}{\zeta(\frac12+\sqrt{s})} = 1 + \sum_{N\ge2} A_N \operatorname{sech}(\sqrt{s}\log N) = 1 + \sum_{N\ge2} A_N\, \mathbb{E}\, e^{-sT_{\log N}},$$

where $A_N = \frac12\,\mu(m)\chi_4(m)\mu(n)\,n^{-1/2}$ *if* $N = n^m$ *with* $n \ge 2$ *squarefree and* m *odd, and* $A_N = 0$ *otherwise. In particular* $A_N = \mu(N)/(2\sqrt{N})$ *for squarefree* N.

Proof. Keep the term $n = 1$ of (10.1.1) and apply Proposition 10.1.3 with $x = \sqrt{s}\log n$ to each $n \ge 2$. The pair (n,m) with n squarefree and $n^m = N$ is unique, since all exponents of N must equal m. Absolute convergence follows from $\sum_n n^{-1/2}\sum_m n^{-m\operatorname{Re}\sqrt{s}} < \infty$ for $\operatorname{Re}\sqrt{s} > \frac12$. □

Each $\operatorname{sech}(\sqrt{s}\log N)$ is a solved model, by which we mean a function whose Thorin measure is explicit. Its unit atoms $\tau_k(\log N)$, the pseudo zeros, sit at the squares of the odd multiples of $\pi/(2\log N)$, a lattice of spacing $\pi/\log N$ in $\sqrt{t}$. The coefficients A_N carry the signs of the Möbius function, so positivity is now a question about the coefficients alone.

Theorem 10.1.5 is additive, but a mixture of GGCs need not be a GGC, and the Thorin measure is defined through the logarithm. To reach it we replace sech by $\log(1+e^{-x})$, whose coefficients have Dirichlet series Dirichlet's eta function $\eta(1+v) = (1-2^{-v})\zeta(1+v)$, not the Dedekind eta function of Sections 15.1 and 15.2.

Proposition 10.1.6. *Define d_n by $\sum_n d_n n^{-v} = 1/\eta(1+v)$, that is $d_n = \sum_{2^j m=n} \mu(m)/m$. Then $|d_n| \le 1$ and for $\operatorname{Re} x > 0$*

$$e^{-x} = \sum_{m\ge 1} d_m \log\bigl(1+e^{-mx}\bigr).$$

Proof. Expand $\log(1+e^{-y}) = \sum_k (-1)^{k+1} e^{-ky}/k$. The coefficients $(-1)^{k+1}/k$ have Dirichlet series $\eta(1+v)$, so the Dirichlet convolution with d is $\delta_{r,1}$. If $2^\ell \| n$, a squarefree $m = n/2^j$ requires $j \in \{\ell-1, \ell\}$, so $d_n = \mu(n)/n$ for odd n and $d_n = \mu(k)(2^\ell - 2^{\ell-1})/n = \mu(k)/(2k)$ for $n = 2^\ell k$ with $\ell \ge 1$ and k odd. In particular $|d_n| \le 1$ and $\sum_n |d_n|/n < \infty$. □

Lemma 10.1.7. *For $\lambda > 0$ and $s \in \mathbb{C} \setminus (-\infty, 0]$,*

$$\log\bigl(1+e^{-\lambda\sqrt{s}}\bigr) = \log 2 - \int \log\bigl(1+\tfrac{s}{t}\bigr)\, U_\lambda(dt), \qquad U_\lambda(dt) = \frac{\lambda}{4\pi\sqrt{t}}\, dt - \sum_{k\ge 0} \delta_{\sigma_k(\lambda)},$$

with $\sigma_k(\lambda) = \bigl((2k+1)\pi/\lambda\bigr)^2$. Equivalently $U_\lambda(dt) = -d_t \Sigma_\lambda(\sqrt{t})$, where $\Sigma_\lambda(\tau) = \#\{k \ge 0 : (2k+1)\pi/\lambda \le \tau\} - \lambda\tau/2\pi$ is a sawtooth with $|\Sigma_\lambda| \le \frac12$.

Proof. Write $1 + e^{-y} = 2e^{-y/2}\cosh(y/2)$ with $y = \lambda\sqrt{s}$. Lemma 10.1.4 gives $\log\cosh(\lambda\sqrt{s}/2) = \sum_k \log(1 + s/\sigma_k)$, and $\int_0^\infty \log(1+s/t)\, t^{-1/2} dt = 2\pi\sqrt{s}$ gives $\lambda\sqrt{s}/2 = \int \log(1+s/t)\frac{\lambda}{4\pi\sqrt{t}} dt$. □

The reciprocal of each block has Thorin measure $-U_\lambda$: unit atoms at the pseudo zeros, from $\operatorname{sech}(y/2)$, minus their mean density $\lambda/(4\pi\sqrt{t})$, from $e^{y/2}$. So Σ_λ is a toy model of $S(\tau)$, and $1 + e^{-\lambda\sqrt{s}}$ of a zeta function with all its zeros on the line.

Theorem 10.1.8. *Put $a_n = \Lambda(n)/(\sqrt{n}\log n)$ and $B_N = \sum_{n^m = N} d_m a_n$. For $s \in H_{1/2}$,*

$$\log\zeta\bigl(\tfrac12 + \sqrt{s}\bigr) = \sum_{N\ge 2} B_N \log\bigl(1 + N^{-\sqrt{s}}\bigr)$$

and

$$\frac{d}{ds}\log\zeta\big(\tfrac12+\sqrt{s}\big)=\sum_{N\ge2}B_N\Big[\sum_{k\ge0}\frac{1}{s+\sigma_k(\log N)}-\frac{\log N}{4\sqrt{s}}\Big].$$

The first values are $B_p=p^{-1/2}$, $B_4=\frac14+\frac{1}{2\sqrt2}$, $B_8=-\frac{1}{6\sqrt2}$, $B_9=\frac16+\frac{1}{2\sqrt3}$.

Proof. For $\operatorname{Re}z>1$, $\log\zeta(z)=\sum_n a_n n^{-(z-1/2)}$. Apply Proposition 10.1.6 with $x=\sqrt{s}\log n$ and regroup by $N=n^m$; since $|d_m|\le1$ the double series converges absolutely on $H_{1/2}$. Differentiate with Lemma 10.1.7. □

Formally the Thorin measure of $1/\zeta(\frac12+\sqrt{s})$ is $-\sum_N B_N U_{\log N}=\sum_N B_N\,d\Sigma_{\log N}(\sqrt{t})$, a superposition of sawtooth measures. It does not converge as a series of measures, but its Laplace transforms do; Proposition 10.1.13 and Corollary 10.1.14 make this exact.

10.1.4 The normalised reciprocal and its Thorin measure

The factorisation $\varphi=RG$, with $\xi(z)=\frac12 z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ as in Section 2.1, separates the zeros from the archimedean part.

Proposition 10.1.9. *The function φ is meromorphic on $\mathbb{C}\setminus(-\infty,0]$ and*

$$\varphi(s)=R(s)\,G(s),\qquad R(s)=\frac{\xi(\frac12)}{\xi(\frac12+\sqrt{s})},\qquad G(s)=\frac{\Gamma(\frac14+\frac{\sqrt{s}}{2})\,\pi^{-\sqrt{s}/2}}{\Gamma(\frac14)}.$$

It is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$ if and only if the Riemann Hypothesis holds. On $(0,\infty)$ it is analytic and positive, it is continuous at 0 with $\varphi(0)=1$, and $-\varphi'/\varphi$ is analytic in a neighbourhood of $(0,\infty)$.

Proof. With $z=\frac12+\sqrt{s}$ we have $z(z-1)=s-\frac14$, so $(1-4s)\zeta(z)=-8\,\xi(z)\pi^{z/2}/\Gamma(z/2)$; evaluating at $s=0$ fixes the constant. The gamma factor G is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$ since $\operatorname{Re}(\frac14+\frac{\sqrt{s}}{2})>0$, so φ has no zeros in $\mathbb{C}\setminus(-\infty,0]$, and its poles there are those of R, namely the points $\sqrt{s}=\rho-\frac12$ with $\operatorname{Re}\rho>\frac12$. By the functional equation there are none if and only if RH holds. On the real axis $\zeta(\frac12+x)$ is finite and nonzero for $x>0$, $x\ne\frac12$, and $(1-4s)\zeta(\frac12+\sqrt{s})\to-4$ as $s\to\frac14$. Since $\zeta(\frac12+x)<0$ for $0<x<\frac12$ and $\zeta(\frac12+x)>0$ for $x>\frac12$, while $1-4s$ changes sign at $s=\frac14$ and $\zeta(\frac12)<0$, the function φ is positive on $(0,\infty)$. □

Theorem 10.1.10. *Assume RH and let $N(T)$ count the zeros $\frac12+i\gamma$ with $0<\gamma\le T$, with multiplicity, as do the sums over γ below. Then for $s\in\mathbb{C}\setminus(-\infty,0]$*

$$\log\varphi(s)=-\int_{(0,\infty)}\log\big(1+\tfrac{s}{t}\big)\,dM(t),\qquad M(t)=N(\sqrt{t})-\frac{\theta(\sqrt{t})}{\pi},$$

that is

$$dM(t) = \sum_{\gamma>0} \delta_{\gamma^2} - \frac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}\,dt, \qquad -\frac{\varphi'}{\varphi}(s) = \int \frac{dM(t)}{s+t}.$$

By the Riemann von Mangoldt formula $M(t) = 1 + S(\sqrt{t})$ *for* $\sqrt{t}$ *not an ordinate, with* $S(0^+) = -1$.

Proof. Under RH, Hadamard's product gives $R(s) = \prod_{\gamma>0}(1 + s/\gamma^2)^{-1}$, a GGC with Thorin measure $\sum \delta_{\gamma^2}$, and $-R'/R(s) = \sum_{\gamma>0} 1/(s+\gamma^2)$. For G put $g = -G'/G = -\frac{1}{4\sqrt{s}}\{\psi(\frac{1}{4} + \frac{\sqrt{s}}{2}) - \log\pi\}$, analytic on $\mathbb{C} \setminus (-\infty, 0]$, $O(|s|^{-1/2})$ at 0 and $O(|s|^{-1/2}\log|s|)$ at infinity. At $\sqrt{s} = i\tau$, $g = \frac{i}{4\tau}\{\psi(\frac{1}{4} + \frac{i\tau}{2}) - \log\pi\}$, so $\operatorname{Im} g = \theta'(\tau)/(2\tau)$, and Lemma 10.1.1 gives the density $-\theta'(\sqrt{t})/(2\pi\sqrt{t})$. Add the two Stieltjes representations and integrate from 0 to s; Fubini applies because $|dM|$ integrates $\min(1, 1/t)(1 + |\log t|)$. Finally $N(T) = \theta(T)/\pi + 1 + S(T)$ [348, Section 9.3]. □

Under RH the Thorin measure of φ consists of the unit atoms at the squared ordinates, the Thorin measure of R (a GGC exactly when RH holds, Proposition 2.1.1), and the density $-\theta'(\sqrt{t})/(2\pi\sqrt{t})$ of the gamma factor, positive for $t < \tau_0^2$ and negative beyond, where $\tau_0 \approx 6.28984$ is the minimum point of θ. Between consecutive ordinates, all of which exceed τ_0, the function M decreases; hence dM is not positive, and φ is not a GGC even under RH. Theorem 11.1.3 proves more, unconditionally.

Near the origin $M(t) \sim c\sqrt{t}/\pi$, with

$$c = -\theta'(0) = \frac{\log\pi - \psi(\frac{1}{4})}{2} = \frac{\zeta'(\frac{1}{2})}{\zeta(\frac{1}{2})} \approx 2.68609, \tag{10.1.6}$$

the last equality from $\xi'(\frac{1}{2}) = 0$. The function M rises to $-\theta(\tau_0)/\pi \approx 1.12394$ at $t = \tau_0^2$, falls to ≈ 0.55025 just before γ_1^2, jumps by one, and thereafter is a staircase with mean 1 whose continuous parts decrease (Figure 10.1; Table 17.7 gives these constants to more digits). As $s \to 0$, $\varphi(s) = 1 - c\sqrt{s} + O(s)$.

10.1.5 Lévy density, theta functions and a criterion for RH

Under RH, integrating by parts in Theorem 10.1.10 and using $s/(s+t) = \int_0^\infty (1 - e^{-sx})te^{-tx}dx$ turns the Thorin type representation of φ into a Lévy Khintchine one,

$$\log\varphi(s) = -\int_0^\infty \bigl(1 - e^{-sx}\bigr)\frac{w(x)}{x}\,dx,$$
$$w(x) = \int e^{-tx}\,dM(t), \qquad \frac{w(x)}{x} = \int_0^\infty e^{-tx}M(t)\,dt.$$

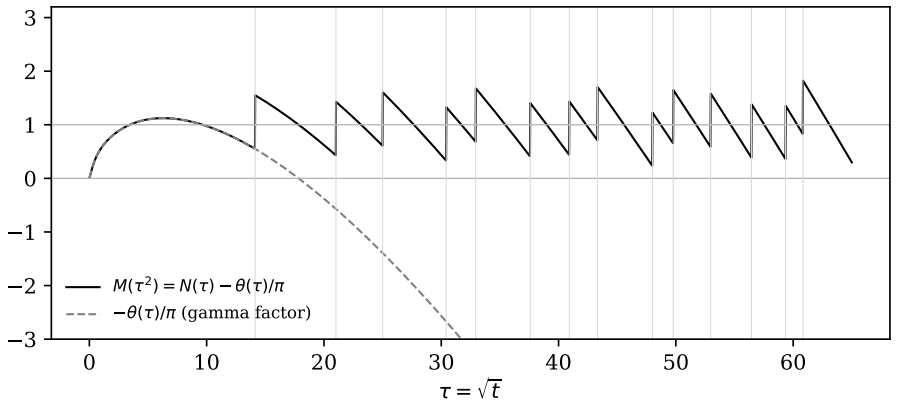


Figure 10.1: The function $M(\tau^2) = N(\tau) - \theta(\tau)/\pi = 1 + S(\tau)$ against $\tau = \sqrt{t}$, with the zero ordinates marked; under RH it is the distribution function of the Thorin measure of φ. The dashed curve is the gamma factor contribution $-\theta(\tau)/\pi$ alone.

THEOREM 10.1.11. *Define, unconditionally,*

$$w(x) = e^{x/4} - P(x), \qquad P(x) = \frac{1}{\sqrt{4\pi x}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x}, \qquad x > 0.$$

Then $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)\, dx$ *for* $s > \frac{1}{4}$, *and for every* $x > 0$

$$\begin{aligned} w(x) &= W(x) - W_0(x), \qquad W(x) = \tfrac{1}{2} \sum_{\rho} e^{x(\rho - \frac{1}{2})^2}, \\ W_0(x) &= \frac{1}{\pi} \int_0^\infty e^{-x\tau^2} \theta'(\tau)\, d\tau, \end{aligned} \tag{10.1.7}$$

the sum running over all nontrivial zeros with multiplicity. Here W *is the heat trace of Section 2.1 in the variable* x*; under RH,* $W(x) = \sum_{\gamma > 0} e^{-x\gamma^2}$ *and* $w(x) = \int e^{-tx} dM(t)$. *The function* w *satisfies* $w(0^+) = 1$ *and* $w(x) \sim c/(2\sqrt{\pi x})$ *as* $x \to \infty$.

Proof. By logarithmic differentiation, $-\varphi'/\varphi = \frac{1}{s - 1/4} + \frac{1}{2\sqrt{s}} \frac{\zeta'}{\zeta}(\frac{1}{2} + \sqrt{s})$. The first term is the transform of $e^{x/4}$; for the second use $\zeta'/\zeta = -\sum \Lambda(n) n^{-z}$ and $\frac{1}{2\sqrt{s}} e^{-a\sqrt{s}} = \int_0^\infty e^{-sx} (4\pi x)^{-1/2} e^{-a^2/4x} dx$, with Tonelli justified on $s > \frac{1}{4}$. Identity (10.1.7) is Weil's explicit formula for the even test function $h(r) = e^{-xr^2}$, whose Fourier partner is $g(u) = (4\pi x)^{-1/2} e^{-u^2/4x}$: the pole contributes $h(\pm \frac{i}{2}) = e^{x/4}$, the archimedean place contributes W_0, and the primes contribute P. It is (2.1.5) of Section 2.1 with $t = x$. Under RH, Theorem 10.1.10 gives the same Laplace

transform for $\int e^{-tx}dM(t)$ and uniqueness identifies the two. The limit $w(0^+)=1$ holds because $P(x)=O(x^{-1/2}e^{-(\log 2)^2/4x})$ as $x\to 0$. The behaviour at infinity needs no hypothesis: every term of W has modulus at most $e^{-x(\gamma^2-1/4)}$ with $\gamma>14$, so W decays exponentially, while $W_0(x)\sim -c/(2\sqrt{\pi x})$ by Lemma 10.1.12 below. □

We call w the canonical function of φ: φ is infinitely divisible exactly when $w\ge 0$, and then $w(x)/x$ is its Lévy density. The proof of Theorem 11.1.1 uses the prime side $e^{x/4}-P(x)$ for small x, finitely many verified zeros for large x, and the following formula for W_0.

Lemma 10.1.12. *For $x>0$,*

$$W_0(x)=\frac{F(x)}{4\sqrt{\pi x}},\qquad F(x)=2\theta'(0)+\int_0^\infty \frac{e^{-u/4}\bigl(1-e^{-u^2/16x}\bigr)}{1-e^{-u}}\,du.$$

The function F is strictly decreasing, $F(0^+)=+\infty$ and $F(\infty)=2\theta'(0)=\psi(\frac14)-\log\pi<0$. Hence W_0 has exactly one zero x_*, with $W_0>0$ on $(0,x_*)$ and $W_0<0$ on (x_*,∞).

Proof. Insert $\psi(z)=\int_0^\infty\bigl(e^{-u}/u-e^{-zu}/(1-e^{-u})\bigr)du$ with $z=\frac14+\frac{i\tau}{2}$ into θ', and use $\int_0^\infty e^{-x\tau^2}\cos(\tau u/2)\,d\tau=\frac12\sqrt{\pi/x}\,e^{-u^2/16x}$; the interchange is justified after writing the ψ integrand as $e^{-u}/u-e^{-u/4}/(1-e^{-u})$ plus $e^{-u/4}(1-\cos(\tau u/2))/(1-e^{-u})$, both absolutely integrable against $e^{-x\tau^2}$. The terms free of τ reassemble into $\psi(\frac14)-\log\pi$. The integrand of F is positive and decreasing in x; as $x\to 0$ it increases to $e^{-u/4}/(1-e^{-u})$, which is not integrable at 0. □

For the sawtooth measure of Lemma 10.1.7 define its heat trace

$$h_\lambda(x)=-\int e^{-tx}\,U_\lambda(dt)=\sum_{k\ge0}e^{-x\sigma_k(\lambda)}-\frac{\lambda}{4\sqrt{\pi x}}=\frac{\lambda}{2\sqrt{\pi x}}\sum_{j\ge1}(-1)^j e^{-j^2\lambda^2/4x},$$

the last form by Poisson summation. The first form sums over the pseudo zeros minus their smooth count, the second over the multiples $j\lambda$: this Jacobi theta identity is the explicit formula of the toy zeta function $1+e^{-\lambda\sqrt{s}}$.

Proposition 10.1.13. *For every $x>0$, with absolutely convergent sums,*

$$w(x)=e^{x/4}+\sum_{N\ge2}B_N\,h_{\log N}(x).$$

Proof. Substitute the theta form of h_λ and regroup the triple sum over (n,m,j) by $r=mj$. The coefficient of $e^{-r^2(\log n)^2/4x}$ is

$$-(4\pi x)^{-1/2}a_n\log n\sum_{mj=r}m\,d_m(-1)^{j+1},$$

and the inner sum is $\delta_{r,1}$ because its Dirichlet series is $\eta(v)^{-1}\cdot\eta(v)$. Since $a_n \log n = \Lambda(n)/\sqrt{n}$ the sum equals $-P(x)$. The bound $|m\, d_m| \le m$ against Gaussian decay gives absolute convergence. □

Proposition 10.1.13 is the Vere-Jones form of Theorem 10.1.11. Unconditionally, w is the termwise Laplace transform of the formal series $\delta_{-1/4} + \sum_N B_N\{\sum_k \delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}dt\}$ of sech atom lattices, with all the arithmetic in the weights B_N; under RH it is also the Laplace transform of dM. The atom at $-\frac{1}{4}$, from the pole of ζ, lies outside $(0,\infty)$, so under RH the sawtooth series must cancel it. Two consequences follow; the first makes the sawtooth expansion of S rigorous after Gaussian averaging.

Corollary 10.1.14. *Assume RH. For every $x > 0$,*

$$\int_0^\infty S(\tau)\, 2\tau e^{-x\tau^2}\, d\tau = \frac{e^{x/4}-1}{x} + \sum_{N\ge 2} B_N \int_0^\infty \Sigma_{\log N}(\tau)\, 2\tau e^{-x\tau^2}\, d\tau,$$

and each integral on the right equals $h_{\log N}(x)/x$.

Proof. By Theorem 10.1.11,

$$\int_0^\infty e^{-tx}(1 + S(\sqrt{t}))dt = w(x)/x,$$

and $\int_0^\infty e^{-tx}\Sigma_\lambda(\sqrt{t})\, dt = h_\lambda(x)/x$ by parts since $\Sigma_\lambda(0) = 0$. Subtract $\int e^{-tx}dt = 1/x$ and apply Proposition 10.1.13; substitute $t = \tau^2$. □

The second consequence is a criterion for RH.

Theorem 10.1.15. *The following are equivalent.*

(i) The Riemann Hypothesis.

(ii) There is a real signed Borel measure m on $[0,\infty)$ with $\int |m|(dt)/(1+t) < \infty$ and $w(x) = \int e^{-tx} m(dt)$ for all $x > 0$.

In that case $m = dM$.

Proof. If RH holds take $m = dM$; the atoms satisfy $\sum(1+\gamma^2)^{-1} < \infty$ and the density is $O(t^{-1/2}(1+t)^{-1}\log(2+t))$ after division by $1+t$. Conversely, given m, put $J(s) = \int m(dt)/(s+t)$, analytic on $\mathbb{C}\setminus(-\infty,0]$. For $s > \frac{1}{4}$ Fubini gives $\int_0^\infty e^{-sx} w(x)dx = J(s)$, so $-\varphi'/\varphi = J$ on $(\frac{1}{4},\infty)$ and hence on all of $\mathbb{C}\setminus(-\infty,0]$ by analytic continuation. A zero or pole of φ in $\mathbb{C}\setminus(-\infty,0]$ would be a pole of $-\varphi'/\varphi$, so φ has none, and Proposition 10.1.9 gives RH. Uniqueness of the Stieltjes transform gives $m = dM$. □

10.1.6 Across the critical strip

Fix σ and evaluate ζ on the line $\operatorname{Re} z = \sigma$ through $z = \sigma + \sqrt{s}$. A zero $\rho = \beta + i\gamma$ then sits at $s = (\rho - \sigma)^2$: if $\beta > \sigma$ it gives a pole of $1/\zeta(\sigma + \sqrt{s})$ in $\mathbb{C} \setminus (-\infty, 0]$, if $\beta = \sigma$ a point of the cut, and if $\beta < \sigma$ no singularity, only a Poisson kernel of width $\sigma - \beta$ in the Thorin density. On each line we ask whether a normalised reciprocal of ζ is infinitely divisible, and whether the completed function $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is a GGC.

To the right of $\operatorname{Re} z = 1$ the first question has a classical answer, in the variable τ. The Dirichlet series converges, $\zeta(\sigma + i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau \log N_\sigma}$ for the zeta distribution $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and the Lévy measure of $\log N_\sigma$ is the prime measure $\sum_n \Lambda(n) n^{-\sigma} (\log n)^{-1} \delta_{\log n}$. The Vere-Jones representation writes this measure as a superposition of the alternating lattices $\sum_j (-1)^{j+1} j^{-1} \delta_{jn \log p}$, the signed Lévy measures of fair coins, one for each prime power p^n, with weight the necklace polynomial $b_n(p^{-\sigma})$ (Theorem 11.1.8). At $\sigma = 0$ the weight $b_n(1)$ is 1 when n is a power of 2 and 0 otherwise, and the identity becomes Euler's binary product, with every block a solved model of unit multiplicity; for $\sigma > 0$ the weights are fractional and some are negative.

Inside the strip $|\zeta(\sigma + i\tau)|$ is unbounded and the characteristic function picture breaks down, but the Vere-Jones representation survives in the variable s. For $0 < \sigma < 1$ put $\varphi_\sigma(s) = \zeta(\sigma)/\{(1 - s/(1-\sigma)^2)\zeta(\sigma + \sqrt{s})\}$, which is φ at $\sigma = \frac{1}{2}$. For $s > (1-\sigma)^2$, $-\varphi'_\sigma/\varphi_\sigma$ is the Laplace transform of a function w_σ, given by the theta series of Proposition 10.1.13 with $e^{(1-\sigma)^2 x}$ in place of $e^{x/4}$ and the weights $b_n(p^{-\sigma})$ in place of $B_{p^n} = b_n(p^{-1/2})$ (Theorem 11.1.12). When $w_\sigma \geq 0$ this is a Lévy representation, $\log \varphi_\sigma(s) = -\int (1 - e^{-sx}) w_\sigma(x) x^{-1} dx$. Theorem 11.1.1 proves $w_{1/2} = w > 0$, and numerically $w_\sigma > 0$ on nine lines from $\sigma = 0.05$ to $\sigma = 0.99$, for $10^{-4} \leq x \leq 10^3$ (Section 11.1); whether infinite divisibility holds on every line of the strip is open (Problem 17.8).

The zeros enter through the Thorin property (Proposition 11.1.13 and Theorem 11.1.14). Chapters 12 to 14 return to R_σ, written there $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$; these are not the normalised functions φ_σ of this section.

10.2 The gamma factor, the Meixner law and the sine squared lemma

On the critical line $|G|^2$ is proportional to the density of a Meixner hypergeometric law, whose characteristic function is a GGC transform in the square of its variable and whose orthogonal polynomials satisfy a local Riemann Hypothesis. The sine squared lemma turns the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$; it shows that after a stable tilt the reciprocals of $\zeta(\beta + \sqrt{s})$, $\beta > 1$, and of the gamma factor are GGC transforms, and how positivity is lost at the centre.

10.2.1 The gamma factor in Vere-Jones form

Recall the constants $c_n = 2n + \frac{1}{2}$, $n \geq 0$, and $b = \frac{1}{2}(\gamma_E + \log \pi) \approx 0.860973$, with γ_E Euler's constant.

Theorem 10.2.1. *For $s \in \mathbb{C} \setminus (-\infty, 0]$,*

$$G(s) = e^{-b\sqrt{s}} \prod_{n\geq 0} \frac{e^{\sqrt{s}/(2n+2)}}{1 + \sqrt{s}/c_n},$$

and the Thorin measure of G is

$$U_G(dt) = \frac{1}{2\pi\sqrt{t}} \Big\{ b + \sum_{n\geq 0} \Big(\frac{c_n}{t + c_n^2} - \frac{1}{2n+2} \Big) \Big\} dt = -\frac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}\, dt.$$

Each factor $(1+\sqrt{s}/\kappa)^{-1}$ is the Laplace transform of a GGC, with positive Thorin density $q_\kappa(t) = \dfrac{\kappa}{2\pi\sqrt{t}\,(t+\kappa^2)}$, a Poisson kernel in $\sqrt{t}$; each factor $e^{a\sqrt{s}}$ with $a > 0$ has the negative Thorin density $-a/(2\pi\sqrt{t})$.

Proof. Weierstrass gives $\log\Gamma(z_0 + v) - \log\Gamma(z_0) = -\gamma_E v - \log(1 + v/z_0) + \sum_{n\geq 1}\{v/n - \log(1 + v/(n + z_0))\}$. Take $z_0 = \frac{1}{4}$, $v = \sqrt{s}/2$, note $v/(n + \frac{1}{4}) = \sqrt{s}/c_n$, and shift the index of v/n. For q_κ: $\frac{d}{ds}\log(1 + \sqrt{s}/\kappa) = \frac{1}{2\sqrt{s}(\sqrt{s}+\kappa)}$, and Lemma 10.1.1 with $\sqrt{-t + i0} = i\sqrt{t}$ gives the density $\frac{\kappa}{2\pi\sqrt{t}(t+\kappa^2)}$. For $e^{a\sqrt{s}}$ use $\int_0^\infty \log(1 + s/t)t^{-1/2}dt = 2\pi\sqrt{s}$. To sum the Thorin densities of the factors, write the bracket with index n in U_G as $(\frac{1}{c_n} - \frac{1}{2n+2}) - \frac{t}{c_n(t+c_n^2)}$: both parts are nonnegative, and over n the first sums to a constant and the second to $O(\log(2 + t))$, so Tonelli's theorem applies to each part against $\log(1 + s/t)\frac{dt}{2\pi\sqrt{t}}$ for $s > 0$, and analytic continuation extends the sum to $\mathbb{C} \setminus (-\infty, 0]$; the same split justifies the termwise integration against e^{-tx} in Proposition 10.2.2. The identification with θ' is the partial fraction series $\operatorname{Re}\psi(\frac{1}{4} + \frac{i\tau}{2}) = -\gamma_E + \sum_{n\geq 0}\{\frac{1}{n+1} - \frac{2c_n}{c_n^2+\tau^2}\}$. □

Formally, the gamma factor is a renormalised infinite convolution of GGC laws. Write X_φ and X_R for random variables with Laplace transforms φ and R; both laws exist and are infinitely divisible, by Theorem 11.1.1 and by Theorem 8.1.1 [282, Theorem A]. Write Y_κ for the variable with $\mathbb{E}e^{-sY_\kappa} = \kappa/(\kappa + \sqrt{s})$, and L_a for the positive stable variable of index $\frac{1}{2}$ with $\mathbb{E}e^{-sL_a} = e^{-a\sqrt{s}}$. Theorem 10.2.1 and Proposition 10.1.9 then say formally

$$X_\varphi \stackrel{d}{=} X_R + L_b + \sum_{n\geq 0} \big(Y_{c_n} \ominus L_{1/(2n+2)} \big), \tag{10.2.1}$$

where $A \ominus B$ stands for a formal variable whose Laplace transform is the quotient of those of A and B. The sum of the Y_{c_n} diverges almost surely because $\sum 1/c_n = \infty$,

and the stable deconvolutions are the compensators that make the product of the Laplace transforms converge.

The Thorin density of G changes sign at τ_0 (Figure 10.2), so the Thorin density of $1/G$ is negative near the origin.

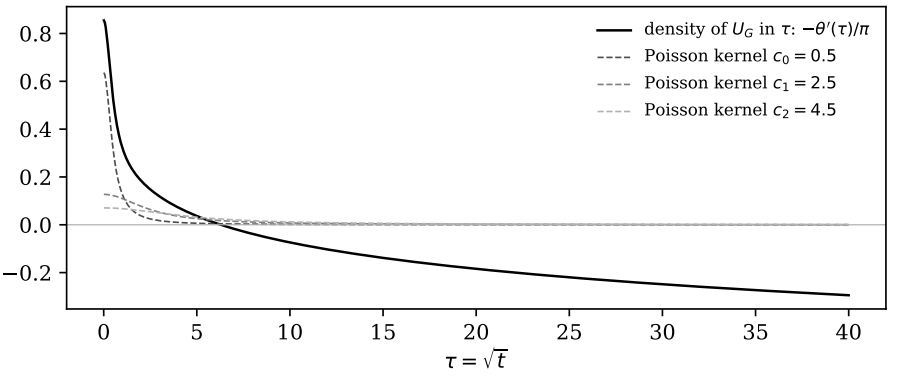


Figure 10.2: The Thorin density of the gamma factor in the variable $\tau = \sqrt{t}$, namely $-\theta'(\tau)/\pi$, together with the first three Poisson kernels $c_n/(\pi(c_n^2 + \tau^2))$ of Theorem 10.2.1. The density is positive for $\tau < \tau_0 \approx 6.29$ and negative beyond, where the flat compensators outweigh the kernels.

Proposition 10.2.2. *For $x > 0$,*

$$W_0(x) = -\int e^{-tx}\, U_G(dt) = -\frac{b}{2\sqrt{\pi x}} + \sum_{n\geq 0}\Big\{\frac{1}{(4n+4)\sqrt{\pi x}} - \frac{1}{2}e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

Proof. $\int_0^\infty e^{-tx} q_\kappa(t)\,dt = \frac{\kappa}{\pi}\int_0^\infty \frac{e^{-xv^2}}{v^2+\kappa^2}dv = \frac{1}{2}e^{\kappa^2 x}\operatorname{erfc}(\kappa\sqrt{x})$ and $\int_0^\infty e^{-tx}\frac{dt}{2\pi\sqrt{t}} = \frac{1}{2\sqrt{\pi x}}$. □

Theorem 10.2.3. *For every $x > 0$, with absolutely convergent sums,*

$$W(x) = e^{x/4} + \sum_{N\geq 2} B_N\, h_{\log N}(x) - \frac{b}{2\sqrt{\pi x}} + \sum_{n\geq 0}\Big\{\frac{1}{(4n+4)\sqrt{\pi x}} - \frac{1}{2}e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

Equivalently, $W(x)$ is the Laplace transform, taken term by term, of the formal

series

$$U_R = \delta_{-1/4} + \sum_{N\ge 2} B_N\Big\{\sum_{k\ge 0}\delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}\,dt\Big\} - \frac{1}{2\pi\sqrt{t}}\Big\{b + \sum_{n\ge 0}\Big(\frac{c_n}{t+c_n^2} - \frac{1}{2n+2}\Big)\Big\}dt,$$

the formal Thorin measure of $R(s) = \xi(\frac12)/\xi(\frac12+\sqrt{s})$. *Under RH,* R *is a GGC, and its Thorin measure* $\sum_{\gamma>0}\delta_{\gamma^2} = dN(\sqrt{t})$, *where* $N(T)$ *counts the zeros with* $0<\gamma\le T$, *is a positive measure whose Laplace transform is also* W.

Proof. $W = w + W_0$ by (10.1.7); substitute Proposition 10.1.13 for w and Proposition 10.2.2 for W_0. Term by term, $\delta_{-1/4}$ has Laplace transform $e^{x/4}$, the bracket with index N has transform $h_{\log N}(x)$ by the definition of h_λ, and the last density is $-U_G$, whose transform is $W_0(x)$ by Proposition 10.2.2. Under RH, Hadamard's product gives $R(s) = \prod_{\gamma>0}(1+s/\gamma^2)^{-1}$, as in the proof of Theorem 10.1.10, and $W(x) = \sum_{\gamma>0} e^{-x\gamma^2}$ by Theorem 10.1.11. □

In U_R the gamma factor gives negative Poisson kernels of widths c_n in the variable $\sqrt{t}$, with poles at $\sqrt{t} = \pm ic_n$, each compensated by a positive flat density. RH is the statement that the atom at $-\frac14$, the sech lattices of the primes and the gamma factor terms combine into unit atoms at the γ^2 and nothing else. They combine only through Laplace transforms: every partial sum of U_R carries the atom at $-\frac14$ and $\sum_{\gamma>0}\delta_{\gamma^2}$ does not, so, like the series after Proposition 10.1.13, U_R does not converge to that measure as a series of measures. By Theorem 10.1.15, R has a Thorin type representation only if RH holds, since adding U_G to its measure would give a measure m as in condition (ii) there.

On the cut the gamma factor has modulus and phase

$$G(-\tau^2+i0) = \frac{|\Gamma(\frac14+\frac{i\tau}{2})|}{\Gamma(\frac14)}\,e^{i\theta(\tau)}.$$

The phase is $-\pi$ times the distribution function of the Thorin measure: $U_G((0,\tau^2]) = -\theta(\tau)/\pi$, since $\operatorname{Im}\log(1+s/t) = \pi$ for $t<\tau^2$ at $s=-\tau^2+i0$. The modulus is the logarithmic potential of the same measure,

$$\log\frac{|\Gamma(\frac14+\frac{i\tau}{2})|}{\Gamma(\frac14)} = -\int \log\Big|1-\frac{\tau^2}{t}\Big|\,U_G(dt), \tag{10.2.2}$$

the integral converging absolutely.

The square of the modulus is proportional to the density of a Meixner hypergeometric law, with parameter $a=\frac14$; Lai and Vere-Jones [200] study this

family and its case $a = \frac{1}{2}$, the hyperbolic secant law. Since $\int |\Gamma(a+ix)|^2 e^{iux} dx = 2\pi\Gamma(2a)(2\cosh\frac{u}{2})^{-2a}$,

$$p(\tau) = \frac{|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2}{2^{3/2}\pi^{3/2}} \quad \text{is a probability density with} \quad \int p(\tau)e^{iv\tau}d\tau = \operatorname{sech}(v)^{1/2}. \tag{10.2.3}$$

In this section p denotes this Meixner density. Since $\Xi(\tau) = -\frac{1}{2}(\tau^2+\frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|Z(\tau)$, with Hardy's function $Z(\tau) = e^{i\theta(\tau)}\zeta(\frac{1}{2}+i\tau)$ of Section 2.2, the Meixner density is the archimedean part of Ξ^2 on the critical line:

$$\Xi(\tau)^2 = \tfrac{1}{4}\big(\tau^2+\tfrac{1}{4}\big)^2\, 2^{3/2}\pi\, p(\tau)\, Z(\tau)^2.$$

10.2.2 From Lévy to Thorin: the sine squared lemma

For a function $f(0)/f(\sqrt{s})$ we call f its primal function. The sine squared lemma of [280, Lemma 1] converts the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$; it applies to the Lévy measures produced by the Vere-Jones representation, sums of atoms at the points $k\log N$.

Lemma 10.2.4 ([280, Lemma 1]). *Suppose that for some measure μ on $(0,\infty)$ and some α with $\int_0^\infty (1\wedge x)e^{-\alpha x}\,\mu(dx) < \infty$,*

$$\log\frac{f(\alpha+s)}{f(\alpha)} - s\frac{f'(\alpha)}{f(\alpha)} = \int_0^\infty \big(e^{-sx}-1+sx\big)e^{-\alpha x}\,\frac{\mu(dx)}{x}, \qquad s\ge 0.$$

Then, for $s \ge 0$,

$$\log\frac{f(\alpha)}{f(\alpha+s)} + s\frac{f'(\alpha)}{f(\alpha)} = -\int_0^\infty \big(1-e^{-\frac{1}{2}s^2t}\big)\frac{\nu_\alpha(t)}{t}\,dt,$$

$$\nu_\alpha(t) = \frac{1}{\sqrt{2\pi}}\int_0^\infty e^{-tz}\Big\{\int_0^\infty 2\sin^2\big(x\sqrt{z/2}\big)e^{-\alpha x}\,\mu(dx)\Big\}\frac{dz}{\sqrt{\pi z}}.$$

The map $\mu \mapsto \nu_\alpha$ is linear and positivity preserving.

We give the proof by subordination, whose two stages, a Lévy density in s and then a Thorin density, reappear in Section 10.2.4.

Lemma 10.2.5 (Lemma 10.2.4 in the scaling $s = u^2$). *Let $\tilde\mu$ be a signed measure on $(0,\infty)$ with $\int(1\wedge x)|\tilde\mu|(dx) < \infty$, and suppose*

$$\log\frac{f(u)}{f(0)} - u\frac{f'(0)}{f(0)} = \int_0^\infty \big(e^{-ux}-1+ux\big)\frac{\tilde\mu(dx)}{x}, \qquad u\ge 0.$$

Then for $s \in \mathbb{C}\setminus(-\infty,0]$

$$\log\frac{f(0)}{f(\sqrt{s})} + \sqrt{s}\,\frac{f'(0)}{f(0)} = -\int_0^\infty \log\Big(1+\frac{s}{t}\Big)U(t)\,dt,$$

$$U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2\Big(\frac{x\sqrt{t}}{2}\Big)\tilde\mu(dx).$$

If $\tilde\mu \ge 0$ then $U \ge 0$, and $s \mapsto f(0)e^{\sqrt{s}f'(0)/f(0)}/f(\sqrt{s})$ is the Laplace transform of a GGC. Since $\sqrt{s} = \int \log(1+s/t)\,\frac{dt}{2\pi\sqrt{t}}$, the Thorin measure of $f(0)/f(\sqrt{s})$ itself is $U(t)\,dt + \frac{f'(0)}{f(0)}\frac{dt}{2\pi\sqrt{t}}$.

Proof. Subordination by the first passage law (10.1.2) gives, for $x, u > 0$,

$$e^{-ux} - 1 + ux = \int_0^\infty \bigl(1 - e^{-u^2 y}\bigr)\bigl(1 - e^{-x^2/4y}\bigr)\frac{x}{\sqrt{4\pi y^3}}\,dy,$$

because $\int_0^\infty (1 - e^{-u^2 y})\frac{x}{\sqrt{4\pi y^3}}dy = ux$. Integrating against $\tilde\mu(dx)/x$ gives a Lévy form in $s = u^2$ with Lévy density $k(y)/y$, where $k(y) = (4\pi y)^{-1/2}\int(1 - e^{-x^2/4y})\tilde\mu(dx)$ is the canonical function. Finally $(1 - e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1 - \cos(x\sqrt{t})\}\frac{dt}{\sqrt{\pi t}}$, so $k(y) = \int e^{-yt}U(t)dt$, and Frullani turns $\int(1 - e^{-sy})k(y)\frac{dy}{y}$ into $\int \log(1 + s/t)U(t)dt$. All kernels in these steps are nonnegative, so by Tonelli's theorem the steps hold with $|\tilde\mu|$ in place of $\tilde\mu$, and there every integral is finite, since $0 \le (e^{-ux} - 1 + ux)/x \le \min(\frac{u^2 x}{2}, u)$ and $\int(1 \wedge x)|\tilde\mu|(dx) < \infty$; this justifies each interchange for the signed $\tilde\mu$. The analytic continuation to $\mathbb{C} \setminus (-\infty, 0]$ is immediate. □

Lemma 10.2.5 is (2.1.9) of Section 2.1 integrated against $\tilde\mu(dx)/x$; the letters s and u are exchanged between the two sections. We call $\tilde\mu$ the primal Lévy measure of $f(0)/f(\sqrt{s})$. Positivity comes from the kernel: the signed cosine density that Lemma 10.1.1 gives for a single exponential in an exponent is $-\cos = 2\sin^2 - 1$, and Lemma 10.2.5 separates the positive part $2\sin^2$ from the flat part -1, which is absorbed into the linear tilt $u\,f'(0)/f(0)$, a stable law of index $\frac{1}{2}$.

10.2.2.1 Zeta to the right of the line, the Vere-Jones blocks and the centre

Proposition 10.2.6. *For $\beta > 1$ put $\tilde\mu_\beta = \sum_{n\ge 2}\Lambda(n)n^{-\beta}\delta_{\log n}$. Then*

$$\frac{\zeta(\beta)}{\zeta(\beta+\sqrt{s})}\,e^{\sqrt{s}\,\zeta'(\beta)/\zeta(\beta)} = \exp\Bigl(-\int_0^\infty \log\Bigl(1 + \frac{s}{t}\Bigr)U_\beta(t)\,dt\Bigr),$$

$$U_\beta(t) = \frac{1}{2\pi\sqrt{t}}\sum_{n\ge 2}\frac{\Lambda(n)}{n^\beta}\,2\sin^2\Bigl(\frac{\sqrt{t}\log n}{2}\Bigr) \ge 0,$$

so the left side is the Laplace transform of a GGC. Adding the tilt, the Thorin measure of $\zeta(\beta)/\zeta(\beta+\sqrt{s})$ is $-\frac{1}{2\pi\sqrt{t}}\sum\Lambda(n)n^{-\beta}\cos(\sqrt{t}\log n)\,dt$, which is the Euler product density (10.1.5) at β in place of $\frac{1}{2}$.

Proof. $\log\zeta(\beta+u) = \sum\Lambda(n)n^{-\beta}e^{-u\log n}/\log n$, so the hypothesis of Lemma 10.2.5 holds with $\tilde\mu_\beta$, which has finite mass for $\beta > 1$. □

The tilt is needed: without it the Thorin density of $\zeta(\beta)/\zeta(\beta+\sqrt{s})$ is negative near $t=0$. For ξ no tilt is needed, since $s\mapsto \xi(\alpha)/\xi(\alpha+\sqrt{s})$ is a GGC transform for every $\alpha\geq 1$ (Theorem 2.1.2); at $\alpha=1$, Section 12.1 obtains its Thorin density from a Lévy form of $\log\xi(1+u)$ by the same lemma.

Apply the lemma to one block $f_\lambda(u)=1+e^{-\lambda u}$ of Theorem 10.1.8, shifted by $\alpha>0$. Since $\log(1+e^{-\lambda(\alpha+u)})=\sum_k(-1)^{k+1}q^k e^{-k\lambda u}/k$ with $q=e^{-\lambda\alpha}$, the primal Lévy measure is the alternating lattice $\tilde\mu_{\lambda,\alpha}=\lambda\sum_{k\geq1}(-1)^{k+1}q^k\delta_{k\lambda}$.

Proposition 10.2.7. *For $\alpha>0$ the sine squared transform of $\tilde\mu_{\lambda,\alpha}$ is positive:*

$$U_{\lambda,\alpha}(t)=\frac{\lambda}{2\pi\sqrt{t}}\,\frac{q(1-q)\big(1-\cos\lambda\sqrt{t}\big)}{(1+q)\big(1+2q\cos\lambda\sqrt{t}+q^2\big)}\geq 0,$$

so $\frac{1+e^{-\lambda\alpha}}{1+e^{-\lambda(\alpha+\sqrt{s})}}\,e^{-\sqrt{s}\lambda q/(1+q)}$ is the Laplace transform of a GGC. As $\alpha\downarrow 0$, $U_{\lambda,\alpha}(t)\,dt\to\sum_{k\geq0}\delta_{\sigma_k(\lambda)}$ vaguely, the tilt density tends to $-\frac{\lambda}{4\pi\sqrt{t}}$, and the Thorin measure of $2/(1+e^{-\lambda\sqrt{s}})$ is recovered as $\sum_k\delta_{\sigma_k(\lambda)}-\frac{\lambda}{4\pi\sqrt{t}}dt=-U_\lambda$, the sawtooth measure of Lemma 10.1.7.

Proof. $\sum_k(-1)^{k+1}q^k=\frac{q}{1+q}$ and $\sum_k(-1)^{k+1}q^k\cos k\omega=\frac{q\cos\omega+q^2}{1+2q\cos\omega+q^2}$; their difference is $\frac{q(1-q)(1-\cos\omega)}{(1+q)(1+2q\cos\omega+q^2)}\geq 0$. As $q\uparrow 1$ this is a Poisson kernel in $\omega=\lambda\sqrt{t}$ concentrating at $\omega=(2k+1)\pi$ with mass π per period, which becomes $\sum_k\delta_{\sigma_k(\lambda)}$ after the change of variable. The tilt is $f'(\alpha)/f(\alpha)=-\lambda q/(1+q)\to-\lambda/2$. □

The negative part of the sawtooth measure is entirely the tilt. Summing the blocks with the weights B_N gives $\log\zeta(\frac12+\alpha+\sqrt{s})$, and the sum of Thorin densities $\sum_N B_N U_{\log N,\alpha}$ converges absolutely for $\alpha>\frac12$, where it equals $U_{1/2+\alpha}$ of Proposition 10.2.6. Positivity is visible termwise in the prime form but not blockwise in the Vere-Jones form, since infinitely many weights are negative: $B_{p^3}=\frac13(p^{-3/2}-p^{-1/2})<0$ for every prime p.

At $\beta=\frac12$ the prime Lévy measure $\tilde\mu_{1/2}=\sum\Lambda(n)n^{-1/2}\delta_{\log n}$ has infinite mass, and both the image and the tilt in Lemma 10.2.5 diverge. The heat trace makes their cancellation exact. Since $\int_0^\infty e^{-xt}\,2\sin^2(\frac{\sqrt{t}L}{2})\frac{dt}{2\pi\sqrt{t}}=\frac{1-e^{-L^2/4x}}{2\sqrt{\pi x}}$, the heat trace of the sine squared image is formally

$$\frac{1}{2\sqrt{\pi x}}\sum_{n\geq2}\frac{\Lambda(n)}{\sqrt{n}}\big(1-e^{-(\log n)^2/4x}\big)=\frac{1}{2\sqrt{\pi x}}\sum_{n\geq2}\frac{\Lambda(n)}{\sqrt{n}}-P(x).$$

Replace the divergent constant by its analytic continuation $\sum\Lambda(n)n^{-1/2}\to-\zeta'(\frac12)/\zeta(\frac12)=-c$, and add the tilt $\frac{c}{2\sqrt{\pi x}}$ (the heat trace of $\frac{c}{2\pi\sqrt{t}}dt$) and the heat trace $e^{x/4}$ of the pole atom $\delta_{-1/4}$. The result is $e^{x/4}-P(x)=w(x)$, in agreement with Theorem 10.1.11. So at the centre Lemma 10.2.5 still produces dM, but only after a regularisation that subtracts an infinite positive constant from a positive density; that is where the sign of dM is lost.

10.2.2.2 The inverse direction: cosine Lévy forms

The Frullani identity behind Lemma 10.2.5 also runs from a Thorin measure in s to a symmetric Lévy measure in $u = \sqrt{s}$: for $\beta > 0$ and real u, $\int_0^\infty 2\sin^2(\frac{ux}{2})e^{-\beta x}\frac{dx}{x} = \frac{1}{2}\log(1 + u^2/\beta^2)$, hence

$$\int \log\Big(1 + \frac{s}{t}\Big)U(dt) = \int_0^\infty 2\sin^2\Big(\frac{ux}{2}\Big)K_U(x)\,\frac{dx}{x}, \qquad K_U(x) = 2\int e^{-\sqrt{t}\,x}\,U(dt). \tag{10.2.4}$$

If $U \geq 0$ then $u \mapsto \exp(-\int \log(1 + u^2/t)U(dt))$ is the characteristic function of a symmetric infinitely divisible law with Lévy density $K_U(|x|)/(2|x|)$. Section 13.1 applies this dictionary to $\xi(\alpha)/\xi(\alpha + |u|)$, which is the characteristic function of a normal scale mixture at every basepoint α.

Proposition 10.2.8. *Under* (10.2.4)*:*

(i) the Meixner law (10.2.3) *has Lévy density* $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$ *on* $\mathbb{R}$*, and its characteristic function, as a function of* v^2*, is a GGC transform with Thorin measure* $\frac{1}{2}\sum_{k\geq 0}\delta_{((k+1/2)\pi)^2}$*, since* $\operatorname{sech}(v)^{1/2} = \prod_k (1 + v^2/((k+\frac{1}{2})\pi)^2)^{-1/2}$*;*

(ii) the sawtooth measure has $K_{U_\lambda}(x) = \dfrac{\lambda}{\pi x} - \dfrac{1}{\sinh(\pi x/\lambda)}$*;*

(iii) under RH, the Thorin measure $U = \sum_{\gamma>0}\delta_{\gamma^2}$ *of* R *has* $K_U(x) = 2\sum_{\gamma>0} e^{-\gamma x}$*, twice* $e^{-ix/2}V(ix)$ *for Cramér's function* $V(z) = \sum_{\gamma>0} e^{\rho z}$ *[66], and* $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ *is the characteristic function of a sum of independent Laplace variables of scales* $1/\gamma$*.*

Proposition 10.2.9. *For every* $x > 0$ *the series* $\sum_N B_N K_{U_{\log N}}(x)$ *diverges.*

Proof. B_N vanishes unless N is a prime power. Grouping the powers of a fixed p, $\sum_k B_{p^k}/k = \sum_{j,m} d_m\, p^{-j/2}/(j^2 m) = \frac{12}{\pi^2}p^{-1/2} + O(p^{-1})$, using $\sum_m d_m/m = 1/\eta(2) = 12/\pi^2$. Since $K_{U_\lambda}(x) = \frac{\pi x}{6\lambda} + O((x/\lambda)^3)$, the group for p equals $\frac{2x}{\pi}\frac{p^{-1/2}}{\log p}(1 + o(1))$, and $\sum_p p^{-1/2}/\log p = \infty$. The grouping changes the order of summation, but in the order of increasing N the partial sums are dominated by the prime terms $B_p K_{U_{\log p}}(x) \sim \frac{\pi x}{6}\frac{p^{-1/2}}{\log p}$, so the series diverges in that order as well. □

The inverse direction fails over the primes for the same reason that Lemma 10.2.5 fails at the centre: without Gaussian damping the weights $p^{-1/2}$ are not summable. The Poisson kernel $e^{-\sqrt{t}\,x}$ is the stable $\frac{1}{2}$ average of the heat kernel, $e^{-\sqrt{t}\,x} = \int_0^\infty e^{-ty}\frac{x}{2\sqrt{\pi}}y^{-3/2}e^{-x^2/4y}dy$, and the average extends over $y \gtrsim (\log N)^2$, where the block heat trace $h_{\log N}(y)$ has lost its Gaussian factor $e^{-(\log N)^2/4y}$ and is close to $-\log N/(4\sqrt{\pi y})$. Under the heat kernel alone each

block heat trace is a theta function with Gaussian decay in $\log N$, and the series converges (Proposition 10.1.13, Theorem 10.2.3).

10.2.3 The Meixner law and its orthogonal polynomials

Let Y have the density p of (10.2.3), so that $\mathbb{E}e^{ivY} = \operatorname{sech}(v)^{1/2}$.

Proposition 10.2.10. *(i) $Y \stackrel{d}{=} B(T)$, where B is a standard Brownian motion and T is independent with $\mathbb{E}e^{-\lambda T} = \operatorname{sech}(\sqrt{2\lambda})^{1/2}$, the variable $C_{1/2}$ of Biane, Pitman and Yor [33], so that Y is the value at time $\frac{1}{2}$ of the process $\widehat{C}_t = B(C_t)$ of Pitman and Yor [274, (1) and (4)]. The time $T = \sum_{k\ge 0} \frac{8\,\Gamma_k}{(2k+1)^2\pi^2}$, with Γ_k independent gamma variables of shape $\frac{1}{2}$ and unit scale, is a GGC with Thorin measure $\frac{1}{2}\sum_k \delta_{(k+1/2)^2\pi^2/2}$.*

(ii) Y is infinitely divisible with Lévy density $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$, one half of the Lévy density $1/(2|x|\sinh(\pi|x|/2))$ of the process $\widehat{C}$ [274, Table 5]; Y has a Meixner law [329, 99]. As $|\tau| \to \infty$, $p(\tau) \sim \pi^{-1/2}|\tau|^{-1/2}e^{-\pi|\tau|/2}$, matching the singularities of $\operatorname{sech}(v)^{1/2}$ at $v = \pm i\pi/2$.

(iii) p is not a Pólya frequency function of infinite order.

Proof. (i) $\mathbb{E}e^{ivB(T)} = \mathbb{E}e^{-v^2T/2} = \operatorname{sech}(v)^{1/2}$, and $\operatorname{sech}(\sqrt{2\lambda})^{1/2} = \prod_k (1+2\lambda/((k+\frac{1}{2})\pi)^2)^{-1/2}$. (ii) is Proposition 10.2.8(i); the tail follows from Stirling, $|\Gamma(\frac{1}{4}+ix)|^2 \sim 2\pi|x|^{-1/2}e^{-\pi|x|}$ with $x = \tau/2$. (iii) By Schoenberg's theorem [328, Theorem 1] the bilateral Laplace transform of a PF_∞ density is the reciprocal of an entire function of Laguerre Pólya class, so the reciprocal of its characteristic function is entire. Here $1/\mathbb{E}e^{ivY} = \cosh(v)^{1/2}$, which has branch points at the simple zeros $v = i(k+\frac{1}{2})\pi$ of cosh and is not entire. □

The time T of (i), which is not the clock variable of Section 2.1, is twice the Meixner time of Table 4.3. The exponent $\frac{1}{2}$ in $\operatorname{sech}(v)^{1/2}$ is the Meixner parameter $2a = \frac{1}{2}$, forced by the factor $\Gamma(z/2)$ of ξ at $z = \frac{1}{2}$. For $2a$ a positive integer the Meixner law is PF_∞; the zeta function sits at a value where it is not, although its characteristic function is still a GGC transform in v^2.

Define the measure ν on $(0,\infty)$ by

$$\nu(dy) = \frac{e^{y/2}}{\sinh y}\,dy = 2\sum_{k\ge 0} e^{-c_k y}\,dy,$$

with $c_k = 2k + \frac{1}{2}$ as in Theorem 10.2.1.

PROPOSITION 10.2.11. *The function* $\Phi_\Gamma(\lambda) = \psi(\frac14+\frac{\lambda}{2}) - \psi(\frac14) = \int_0^\infty (1-e^{-\lambda y})\,\nu(dy)$ *is a Bernstein function. Its boundary values on the imaginary axis are*

$$\operatorname{Re}\Phi_\Gamma(i\tau) = 2\big(\theta'(\tau)-\theta'(0)\big) = \int_0^\infty (1-\cos\tau y)\,\nu(dy),$$

$$\operatorname{Im}\Phi_\Gamma(i\tau) = -\frac{d}{d\tau}\log p(\tau) = \int_0^\infty \sin(\tau y)\,\nu(dy).$$

Consequently $2\theta(\tau)+\tau\log\pi$ *and* $\log p(\tau)$ *are, up to constants, the imaginary and real parts of* $2\log\Gamma(\frac14+\frac{i\tau}{2})$*, the boundary value of a function analytic in the right half plane. So* $2\theta(\tau)+\tau\log\pi$*, which is* 2π *times the smooth part* θ/π *of the zero counting function plus a linear term, is the harmonic conjugate of the Meixner log density. The smooth heat trace is the heat trace of* ν*:*

$$W_0(x) = \frac{\theta'(0)}{2\sqrt{\pi x}} + \frac{1}{4\sqrt{\pi x}}\int_0^\infty \big(1-e^{-y^2/4x}\big)\,\nu(dy),$$

which is Lemma 10.1.12 after the substitution $u=2y$*; the function* F *of that lemma decreases because* ν *is positive.*

Proof. Substitute $u=2y$ in $\psi(z)=\int_0^\infty (e^{-u}/u - e^{-zu}/(1-e^{-u}))du$ with $z=\frac14+\frac{\lambda}{2}$; note $2e^{-y/2}/(1-e^{-2y}) = e^{y/2}/\sinh y$. Taking $\lambda=i\tau$ gives the boundary values, with $\theta' = \frac12\operatorname{Re}\psi(\frac14+\frac{i\tau}{2}) - \frac12\log\pi$ and $\frac{d}{d\tau}\log|\Gamma(\frac14+\frac{i\tau}{2})|^2 = -\operatorname{Im}\psi(\frac14+\frac{i\tau}{2})$. For W_0 integrate $\theta'(\tau)=\theta'(0)+\frac12\int(1-\cos\tau y)\nu(dy)$ against $\frac{1}{\pi}e^{-x\tau^2}$. □

THEOREM 10.2.12. *Let* $c=-\theta'(0)=\frac12(\log\pi-\psi(\frac14))\approx 2.68609$ *be the constant of* (10.1.6)*. Then*

$$\frac{e^{-c\sqrt{s}}}{G(s)} = \frac{\Gamma(\frac14)\,\pi^{\sqrt{s}/2}\,e^{-c\sqrt{s}}}{\Gamma(\frac14+\frac{\sqrt{s}}{2})} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{t}\Big)U_\Gamma(t)\,dt\Big),$$

with

$$U_\Gamma(t) = \frac{\theta'(\sqrt{t})-\theta'(0)}{2\pi\sqrt{t}} = \frac{1}{2\pi\sqrt{t}}\sum_{k\ge0}\frac{t}{c_k\,(c_k^2+t)} \ge 0.$$

So $e^{-c\sqrt{s}}/G(s)$ *is the Laplace transform of a GGC, and its Thorin distribution function is* $(\theta(\sqrt{t})-\theta'(0)\sqrt{t})/\pi$*, a convex increasing function of* $\sqrt{t}$*.*

Proof. With $f(u)=\Gamma(\frac14+\frac{u}{2})\pi^{-u/2}$ we have $f'(0)/f(0) = \frac12\psi(\frac14) - \frac12\log\pi = \theta'(0) = -c$, and $\frac{d}{du}[\log f(u) - uf'(0)/f(0)] = \frac12\Phi_\Gamma(u) = \frac12\int(1-e^{-uy})\nu(dy)$. Integrating from 0, $\log f(u)/f(0) - uf'(0)/f(0) = \int(e^{-uy}-1+uy)\frac{\tilde\mu(dy)}{y}$ with $\tilde\mu = \frac12\nu \ge 0$. Lemma 10.2.5 gives $U_\Gamma(t) = \frac{1}{2\pi\sqrt{t}}\int 2\sin^2(\frac{y\sqrt{t}}{2})\frac12\nu(dy) = \frac{1}{2\pi\sqrt{t}}\sum_k\int_0^\infty(1-\cos y\sqrt{t})e^{-c_k y}dy$, which is the series stated, and Proposition 10.2.11 identifies it with θ'. □

Multiplying $1/G$ by $e^{-c\sqrt{s}}$ adds the Thorin density $\frac{c}{2\pi\sqrt{t}}$, and $U_\Gamma = -U_G + \frac{c}{2\pi\sqrt{t}}$ (the identity $c-b = \sum_k (\frac{1}{c_k} - \frac{1}{2k+2})$) has no negative part: the positive counterpart of Theorem 10.2.1.

10.2.3.1 Meixner Pollaczek polynomials and the local Riemann Hypothesis

Let $\psi_{2m}(x) = H_{2m}(\sqrt{2\pi}\,x)e^{-\pi x^2}$ be the even Hermite functions, eigenfunctions of the Fourier transform $\hat{f}(y) = \int f(x)e^{-2\pi ixy}dx$ with eigenvalue $(-1)^m$. Define the Mellin transform $\mathcal{M}f(s) = \int_0^\infty x^{s-1}f(x)dx$ and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. From here to the end of Section 10.2.3, s is the Mellin variable, the argument of ζ, and not the variable of the clock.

THEOREM 10.2.13. *For each $m \geq 0$, $Q_m(s) = \mathcal{M}\psi_{2m}(s)/\Gamma_{\mathbb{R}}(s)$ is a polynomial of degree m satisfying $Q_m(1-s) = (-1)^m Q_m(s)$, and*

$$Q_m(\tfrac{1}{2} + i\tau) = \kappa_m\, P_m^{(1/4)}\big(\tfrac{\tau}{2}; \tfrac{\pi}{2}\big),$$

where $P_m^{(\lambda)}(x;\phi)$ are the Meixner Pollaczek polynomials and $\kappa_m \neq 0$. These are the orthogonal polynomials of the Meixner law p. Consequently all zeros of Q_m lie on $\operatorname{Re} s = \frac{1}{2}$ and are simple: the Mellin transform of every even Hermite function satisfies the Riemann Hypothesis.

Proof. Expanding $H_{2m}(\sqrt{2\pi}\,x) = \sum_j h_j x^{2j}$ gives

$$\mathcal{M}\psi_{2m}(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\sum_j h_j \pi^{-j}(s/2)_j$$

with Pochhammer symbols, a polynomial of degree m times $\Gamma_{\mathbb{R}}(s)$. Tate's local functional equation [343, Theorem 2.4.1 and Section 2.5], $\mathcal{M}f(s)/\Gamma_{\mathbb{R}}(s) = \mathcal{M}\hat{f}(1-s)/\Gamma_{\mathbb{R}}(1-s)$ for even f, applied to the eigenfunction ψ_{2m} gives $Q_m(1-s) = (-1)^m Q_m(s)$, so $i^{-m}Q_m(\frac{1}{2}+i\tau)$ is a real polynomial $R_m(\tau)$. By Mellin Plancherel,

$$\int_0^\infty \psi_{2m}\psi_{2n}\,dx = \frac{1}{2\pi}\int_{\mathbb{R}} Q_m\overline{Q_n}\,|\Gamma_{\mathbb{R}}(\tfrac{1}{2}+i\tau)|^2 d\tau = \sqrt{2}\,i^{m-n}\int_{\mathbb{R}} R_m(\tau)R_n(\tau)\,p(\tau)\,d\tau,$$

since $|\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)|^2 = \pi^{-1/2}|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2 = 2^{3/2}\pi\,p(\tau)$. The left side vanishes for $m \neq n$, so (R_m) is the orthogonal polynomial sequence of p, which with $x = \tau/2$ is the Meixner Pollaczek family with $\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$ [191, Section 9.7]. Orthogonal polynomials of a measure with infinite support have real simple zeros. □

This is the archimedean local Riemann Hypothesis of Bump, Choi, Kurlberg and Vaaler [50], seen through the Meixner law; we make no priority claim for this route (see Section 10.2.3.2 for the literature).

10.2.3.2 Zeta in Meixner coordinates

Kuznetsov [196, 197] expands Ξ and Dirichlet L functions in Meixner Pollaczek polynomials; Inoue [165] studies the symmetric expansion of the completed zeta function, the multiplicative structure of its coefficients and the zeros of the approximating polynomials; Romik [319] expands Ξ itself in the symmetric Meixner Pollaczek basis $P_n^{(3/4)}(x;\frac{\pi}{2})$, together with the Hermite and continuous Hahn bases, and derives coefficient asymptotics; and Candelpergher [53] gives related expansions whose polynomials have their roots on the line. The expansions below are for comparison with the Thorin picture and are not claimed as new.

The map $\mathcal{U}f(\tau) = \mathcal{M}f(\frac12+i\tau)/\Gamma_{\mathbb{R}}(\frac12+i\tau)$ is, by the Mellin Plancherel identity in the proof of Theorem 10.2.13, a unitary map from $L^2(0,\infty)$ onto $L^2(\sqrt{2}\,p(\tau)d\tau)$, carrying the even Hermite functions to the Meixner Pollaczek polynomials. Dilations become multipliers, $\mathcal{U}[f(n\,\cdot)] = n^{-1/2-i\tau}\mathcal{U}f$. The function $g(x) = (2\pi^2x^4 - 3\pi x^2)e^{-\pi x^2} = \frac12(xD)(xD+1)e^{-\pi x^2}$, with $D = d/dx$, has $\mathcal{U}g = \frac14 s(s-1)$, a Meixner Pollaczek combination of degree two, and its dilation sum Ω is, up to the factor $\frac12 x^{-1/2}$, the Pólya kernel Φ of Section 2.1 in the variable $x = e^u$:

$$\Omega(x) = \sum_{n\geq1} g(nx) = \tfrac12 x^{-1/2}\Phi(\log x), \qquad \mathcal{U}\Omega = \tfrac14 s(s-1)\zeta(s)\big|_{s=\frac12+i\tau}.$$

So in Meixner coordinates $\frac14 s(s-1)\zeta(s)$ on the critical line is the image of a function in $L^2(0,\infty)$ and has a convergent Meixner Pollaczek expansion

$$\tfrac14 s(s-1)\zeta(s) = \sum_{m\geq0} a_{2m}\,Q_m(s), \qquad a_{2m} = \frac{\langle\Omega,\psi_{2m}\rangle}{\langle\psi_{2m},\psi_{2m}\rangle},$$

in $L^2(p)$. Each Q_m satisfies the local Riemann Hypothesis; we ask whether the partial sums do.

The coefficients decay faster than geometrically, but the polynomials grow and the expansion converges slowly as $|\tau|$ grows: near the first zero $\frac12 + 14.1347i$ the partial sums are still far from zero at $N = 24$. The partial sums $\sum_{m\leq N} a_{2m}Q_m(s)$ mix polynomials of both parities under $s \mapsto 1-s$, so they have no reason to vanish on the critical line, and for $N = 12$ and $N = 24$ they do not: all their zeros lie off the line. So the asymmetric expansion cannot test whether the local Riemann Hypothesis survives the superposition; the symmetric one can.

The symmetric choice is the real even function $F(\tau) = (\tau^2+\frac14)Z(\tau)$, proportional to $\Xi(\tau)/|\Gamma(\frac14+\frac{i\tau}{2})|$, expanded in the even Meixner Pollaczek polynomials $P_{2m}^{(1/4)}(\tau/2;\frac{\pi}{2})$, normalised as in [191, Section 9.7]. Its truncations, the orthogonal projections of F in $L^2(p)$ onto the even polynomials of degree at most $2M$, are real even polynomials. The coefficients alternate in sign from $m = 3$ on, and from $m = 4$ on their modulus decreases slowly, to about 0.01 at $m = 20$. The slow decay reflects the polynomial growth of F against the exponential decay of p: the

truncations converge in $L^2(p)$ but approximate F poorly at the ordinates. The real zeros of the truncations do not track the ordinates: the zero nearest 14.13 is 14.50, 14.32, 14.16, 13.73, 13.54, 14.84 for the degrees 38, 40, 42, 50, 60, 80, and there are eight to ten real zeros in $[0, 40]$ against six ordinates. Neither expansion suggests a route from the local Riemann Hypothesis to the global one; such a route must use the arithmetic of the full dilation sum, not only the orthogonality of its archimedean pieces.

The factorisation of Ξ^2 in Section 10.2.1 induces one of the autocorrelation $\Phi * \Phi$ of the Pólya kernel, whose Fourier transform is Ξ^2.

PROPOSITION 10.2.14. *The integral* $\mathcal{C}(v) = \int p(\tau)\,|\zeta(\frac{1}{2} + i\tau)|^2 e^{iv\tau}\,d\tau$ *converges absolutely, and*

$$(\Phi * \Phi)(v) = 2^{-3/2}\Big(\frac{d^2}{dv^2} - \frac{1}{4}\Big)^2 \mathcal{C}(v).$$

Formally $\mathcal{C} = \operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$, *with*

$$\mathcal{A} = \sum_{m,n \geq 1} \frac{\delta_{\log(m/n)}}{\sqrt{mn}},$$

where $\operatorname{sech}(v)^{1/2}$ *is the characteristic function of the Meixner law and* $\mathcal{A}$, *whose Fourier transform is formally* $|\zeta(\frac{1}{2} + i\tau)|^2$, *is the autocorrelation of the measure* $\sum_n n^{-1/2}\delta_{\log n}$. *The measure* $\mathcal{A}$ *is positive and positive definite, and the series for* $\mathcal{C}$ *diverges: truncated at* $m, n \leq X$ *it equals* $X \int \operatorname{sech}(v-r)^{1/2} e^{-|r|/2}\,dr + O(X^{1/2})$. *This growth comes from the pole of* ζ, *and the factor* $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ *does not remove it, since* $(\frac{d^2}{dv^2} - \frac{1}{4})e^{-|v|/2} = -\delta_0$.

Proof. Since Φ is even, $\int (\Phi * \Phi)(v) e^{i\tau v} dv = \Xi(\tau)^2 = \frac{1}{4}(\tau^2 + \frac{1}{4})^2 \pi^{-1/2} |\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2 |\zeta(\frac{1}{2} + i\tau)|^2$, and $\pi^{-1/2}|\Gamma|^2 = 2^{3/2}\pi\, p$. Fourier inversion divides by 2π, the factor $(\tau^2 + \frac{1}{4})^2$ becomes $(\frac{d^2}{dv^2} - \frac{1}{4})^2$, and the inverse transform of the even function $p|\zeta|^2$ is $\mathcal{C}/(2\pi)$; the constant is $\frac{1}{4}\,2^{3/2}\pi/(2\pi) = 2^{-3/2}$. The integral $\mathcal{C}$ converges absolutely because $p(\tau) \sim \pi^{-1/2}|\tau|^{-1/2} e^{-\pi|\tau|/2}$ (Proposition 10.2.10) and $\zeta(\frac{1}{2} + i\tau)$ grows at most polynomially. Formally $|\zeta|^2 = |\sum n^{-1/2-i\tau}|^2 = \sum_{m,n} (mn)^{-1/2} e^{i\tau \log(m/n)}$, and $\int p(\tau) e^{i\tau(v+r)} d\tau = \operatorname{sech}(v + r)^{1/2}$, which gives $\mathcal{C} = \operatorname{sech}^{1/2} * \mathcal{A}$. For the truncation put $D_X(\tau) = \sum_{n \leq X} n^{-1/2-i\tau}$; the truncated series is exactly $\int p(\tau) |D_X(\tau)|^2 e^{iv\tau} d\tau$. For $|\tau| \leq X$, $D_X(\tau) = \zeta(\frac{1}{2} + i\tau) + X^{1/2-i\tau}/(\frac{1}{2} - i\tau) + O(X^{-1/2})$ [348, Theorem 4.11], and for $|\tau| > X$ the bound $|D_X|^2 \leq 4X$ meets the exponential decay of p. Since $\int e^{-|r|/2} e^{i\tau r}\,dr = (\frac{1}{4} + \tau^2)^{-1}$, the squared pole term contributes $X \int p(\tau)(\frac{1}{4} + \tau^2)^{-1} e^{iv\tau} d\tau = X \int \operatorname{sech}(v - r)^{1/2} e^{-|r|/2}\,dr$, up to an exponentially small error from $|\tau| > X$; the remaining terms contribute $O(X^{1/2})$ uniformly in v. Finally $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ turns $X \int \operatorname{sech}(v - r)^{1/2} e^{-|r|/2}\,dr$ into $-X(\frac{d^2}{dv^2} - \frac{1}{4}) \operatorname{sech}(v)^{1/2}$, which is not zero. □

Both factors of $\mathcal{C}$ are positive, the Meixner factor being a GGC transform in v^2 and $\mathcal{A}$ a positive measure, but convolution with a positive measure does not preserve the GGC property. The lattice sum also needs a counterterm, the pole term of the approximate functional equation, which comes from the same pole of ζ that produced the atom at $-\frac{1}{4}$ in Proposition 10.1.13. Problem 17.6 of Section 17.1 asks for the regularised form of $\operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$ and whether it has a Thorin or Lee and Yang structure in v.

10.2.4 An infinitely divisible floor for the reciprocal Xi function

Multiplying $\varphi = RG$ by the GGC transform $e^{-c\sqrt{s}}/G$ of Theorem 10.2.12 gives $e^{-c\sqrt{s}}R$, which we call the floor of the reciprocal Xi function.

Theorem 10.2.15. *Unconditionally, $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density*

$$\frac{W(x)+c/(2\sqrt{\pi x})}{x}, \qquad W(x)=\tfrac{1}{2}\sum_{\rho} e^{x(\rho-\frac{1}{2})^2},$$

and $W(x)+c/(2\sqrt{\pi x})>0$ for all $x>0$. The following are equivalent: (i) RH; (ii) this law is a GGC; (iii) $W(x)+c/(2\sqrt{\pi x})$ is completely monotone; (iv) W is completely monotone. The proof uses $w>0$, which is Theorem 11.1.1 of Section 11.1 and rests on the verification of RH to height 50, an explicit bound for $N(T)$ [108] and the enclosures E1 to E3.

Proof. $e^{-c\sqrt{s}}R=\varphi\cdot(e^{-c\sqrt{s}}/G)$ is the product of the transform of an infinitely divisible law (Theorem 11.1.1) and a GGC (Theorem 10.2.12), so it is infinitely divisible, and its Lévy density is the sum $\{w(x)+\int e^{-xt}U_\Gamma(t)dt\}/x$. By Proposition 10.2.2 and Theorem 10.2.12, $\int e^{-xt}U_\Gamma(t)dt = W_0(x)+\frac{c}{2\sqrt{\pi x}}$, and $w+W_0=W$ by (10.1.7). Positivity holds because $w>0$ and $U_\Gamma\ge 0$. For the equivalences: GGC means the Lévy density times x is completely monotone, so (ii)$\Leftrightarrow$(iii). Under RH $W=\sum e^{-x\gamma^2}$ is completely monotone, so (i)$\Rightarrow$(iv)$\Rightarrow$(iii). If (ii) holds, the transform is analytic on $\mathbb{C}\setminus(-\infty,0]$ and agrees with $e^{-c\sqrt{s}}R$ on $(0,\infty)$, so by uniqueness of analytic continuation R has no poles in $\mathbb{C}\setminus(-\infty,0]$; since R has no zeros and G is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$, $\varphi=RG$ is analytic and zero free there, and Proposition 10.1.9 gives RH. □

Infinite divisibility alone is not new: by Theorem 8.1.1 [282, Theorem A], $W>0$ and $W'<0$ on $(0,\infty)$, so R itself is infinitely divisible and self decomposable, and so is its product with the stable factor $e^{-c\sqrt{s}}$; Section 14.1 draws the consequences of self decomposability. What the floor adds is the splitting of its canonical function $k=W+c/(2\sqrt{\pi x})$ as $k=w+\int e^{-xt}U_\Gamma(t)\,dt$, an arithmetic part plus a completely monotone archimedean part contributed by the Meixner law (Theorem 10.2.12). In law the floor adds to X_R an independent first passage

time L_c of $\sqrt{2}B$ to the level c. Everything that could fail lies in w, and RH is the upgrade of k from positivity, the lowest rung of the ladder of Section 8.1, to complete monotonicity, its top.

10.2.4.1 The sine squared lemma at the floor

As in the proof of Lemma 10.2.5, from here to the end of the section x is the variable of the primal Lévy measure, and the time of the heat trace is written y.

The floor is $e^{-c\sqrt{s}}R(s) = f(0)/f(\sqrt{s})$ with primal function $f(u) = e^{cu}\xi(\frac{1}{2}+u)$, and $f'(0)/f(0) = c$ because $\xi'(\frac{1}{2}) = 0$. The tilt contributes the linear term cu to $\log f$, which cancels in $\log f(u)/f(0) - uf'(0)/f(0) = \log \xi(\frac{1}{2}+u)/\xi(\frac{1}{2})$, so the primal Lévy measure of the floor is that of ξ at the centre. Collecting the three factors of $\xi(\frac{1}{2}+u) = \frac{1}{2}(u^2-\frac{1}{4})\,\pi^{-1/4} f_\Gamma(u)\,\zeta(\frac{1}{2}+u)$, where $f_\Gamma(u) = \pi^{-u/2}\Gamma(\frac{1}{4}+\frac{u}{2})$, gives formally

$$\tilde{\mu}_\xi(dx) = -2\cosh\!\left(\tfrac{x}{2}\right) dx + \sum_{k\geq 0} e^{-c_k x}\, dx + \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, \delta_{\log n}(dx). \tag{10.2.5}$$

The first term comes from $u^2 - \frac{1}{4} = (u+\frac{1}{2})(u-\frac{1}{2})$, each linear factor $1+u/a$ contributing $-e^{-ax}dx$ by Frullani (with $a = -\frac{1}{2}$ by continuation); the second is $\frac{1}{2}\nu$ (Proposition 10.2.11 and the proof of Theorem 10.2.12); the third is the prime measure of Proposition 10.2.6 at $\beta = \frac{1}{2}$. By the explicit formula, as distributions on $x > 0$,

$$\tilde{\mu}_\xi(dx) = -\sum_\rho e^{(\rho-\frac{1}{2})x}\, dx, \qquad \text{which under RH is} \qquad -2\sum_{\gamma>0}\cos(\gamma x)\, dx. \tag{10.2.6}$$

Proposition 10.2.16. *Apply the two stages of Lemma 10.2.5 to $\tilde{\mu}_\xi$, regularising $\int_0^\infty \cos(\gamma x)dx = 0$ and the growing term of* (10.2.5) *by analytic continuation.*

(i) Subordination stage: $(4\pi y)^{-1/2}\int_0^\infty (1-e^{-x^2/4y})\,\tilde{\mu}_\xi(dx) = W(y)$, the heat trace over the zeros. With the tilt, the canonical function of the floor is $k(y) = W(y) + c/(2\sqrt{\pi y})$, as in Theorem 10.2.15, and its positivity is proved.

(ii) Bernstein stage: $U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2(\frac{x\sqrt{t}}{2})\,\tilde{\mu}_\xi(dx)$ equals $\sum_{\gamma>0}\delta_{\gamma^2}$ under RH, and the Thorin measure of the floor is $U + \frac{c}{2\pi\sqrt{t}}dt$.

The two stages are linked by $W(y) = \int_0^\infty e^{-yt}U(t)\,dt$. The floor is a GGC exactly when $U + \frac{c}{2\pi\sqrt{t}}\,dt \geq 0$, that is, when k is completely monotone; by Theorem 10.2.15 this happens exactly under RH, and then $U = \sum_{\gamma>0}\delta_{\gamma^2} \geq 0$.

Proof. (i) For $\tilde{\mu} = -2\cos(\gamma x)dx$,

$$\int_0^\infty (1-e^{-x^2/4y})(-2\cos\gamma x)\,dx = 2\int_0^\infty e^{-x^2/4y}\cos\gamma x\,dx = \sqrt{4\pi y}\,e^{-\gamma^2 y}.$$

Without RH the same computation with $e^{(\rho-\frac12)x}$ gives $e^{y(\rho-\frac12)^2}$, so the result is W in general. (ii) $\int_0^\infty (1-\cos x\sqrt{t})(-2\cos\gamma x)\,dx = \pi\,\delta(\sqrt{t}-\gamma)$, and $\frac{1}{2\pi\sqrt{t}}\pi\,\delta(\sqrt{t}-\gamma) = \delta(t-\gamma^2)$. The link is the identity $(1-e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1-\cos x\sqrt{t}\}\frac{dt}{\sqrt{\pi t}}$ used in the proof of Lemma 10.2.5. □

The lemma preserves positivity, so a positive $\tilde{\mu}_\xi$ would make the floor a GGC and give RH. But $\tilde{\mu}_\xi$ is not positive: its pole part $-2\cosh(x/2)\,dx$ is negative, and at large x it cancels the mean density $e^{x/2}dx$ of the prime atoms, leaving the oscillating sum over zeros (10.2.6). At the centre the lemma turns the question into one about a signed input whose sine squared image is positive if and only if RH holds.

10.2.4.2 Complete monotonicity to finite order

Positivity at stage (i) of Proposition 10.2.16 is Theorem 10.2.15; positivity at stage (ii) is complete monotonicity of $k(y) = W(y) + c/(2\sqrt{\pi y})$, that is $(-1)^n k^{(n)} \geq 0$ for every n.

THEOREM 10.2.17. *Let $k(y) = W(y) + c/(2\sqrt{\pi y})$. For every $n \leq 10^{10}$ and every $y > 0$,*

$$(-1)^n\, k^{(n)}(y) > 0.$$

Consequently, by Williamson's theorem, for each $1 \leq n \leq 10^{10}$ there is a positive measure Ω_n on $(0,\infty)$ with $k(y) = \int (1-ty)_+^{n-1}\,\Omega_n(dt)$. For $n \geq 1$ the proof uses the verification of RH to height $T_0 = 3\times 10^{12}$ [277], an explicit bound for $N(T)$ [108] and one interval enclosure (E4, Section 17.1.3), and no other information about zeros.

Proof. The case $n = 0$ is the positivity in Theorem 10.2.15. Fix $1 \leq n \leq 10^{10}$ and put $y_n = 0.05/n$.

Small y. Write $k = w + (W_0 + \frac{c}{2\sqrt{\pi y}})$. By Theorem 10.2.12 the bracket is $\int e^{-ty}U_\Gamma(t)\,dt$ with $U_\Gamma \geq 0$, so $(-1)^n$ times its nth derivative is $L_n(y) = \int t^n e^{-ty}U_\Gamma(t)dt$, decreasing in y. Since $\theta'(\tau) - \theta'(0) \geq \frac{2\tau^2}{\frac14+\tau^2} \geq 1$ for $\tau \geq \frac12$ (the $k = 0$ term of the series in Theorem 10.2.12),

$$L_n(y) = \frac{1}{\pi}\int_0^\infty \tau^{2n}e^{-y\tau^2}\big(\theta'(\tau)-\theta'(0)\big)d\tau \geq \frac{\Gamma(n+\frac12)}{2\pi}\,y^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)}.$$

On the other side $(-1)^n w^{(n)} = (-1)^n 4^{-n}e^{y/4} - (-1)^n P^{(n)}$. P is analytic in $\operatorname{Re} y > 0$, and on the circle $|z-y| = y/2$ one has $|z| \geq y/2$ and $\operatorname{Re}(1/z) \geq 2/(3y)$, so Cauchy's estimate gives

$$\begin{aligned}|P^{(n)}(y)| &\leq n!\,(2/y)^n\,(2\pi y)^{-1/2}\sum_m \Lambda(m)m^{-1/2}e^{-(\log m)^2/6y}\\ &\leq C_0\,n!\,(2/y)^n\,(2\pi y)^{-1/2}e^{-(\log 2)^2/6y},\end{aligned}$$

where $C_0 = 0.54993$ bounds $e^{(\log 2)^2/6y}\sum_m \Lambda(m)m^{-1/2}e^{-(\log m)^2/6y}$ for every $y \le 0.05$, each term being increasing in y (enclosure E4). Divide by the leading term of the lower bound for L_n and use Gautschi's inequality $n!/\Gamma(n+\frac12) < \sqrt{n+1}$ [256, (5.6.4)]. For $y \le y_n$ we have $(\log 2)^2/6y \ge 1.6015\,n$, so the ratio is at most

$$r_n = C_0\sqrt{2\pi}\sqrt{n+1}\,e^{(\log 2 - 1.6015)n},$$

which decreases in n, since $r_{n+1}/r_n \le \sqrt{3/2}\,e^{\log 2 - 1.6015} < \frac12$, and satisfies $r_n \le r_1 < 0.786$ (enclosure E4). Hence, for $y \le y_n$,

$$(-1)^n k^{(n)}(y) \ge (1-r_n)\frac{\Gamma(n+\frac12)}{2\pi}y_n^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)} - 4^{-n}e^{y/4}.$$

The first term is at least $0.214\,\Gamma(\frac32)\,20^{3/2}/(2\pi) > 2.6$, since $\Gamma(n+\frac12)(20n)^{n+\frac12}$ increases in n, and the two negative terms total less than 0.27. So $(-1)^n k^{(n)} > 0$ on $(0, y_n]$ for every $n \ge 1$.

Large y. Here $(-1)^n k^{(n)} = (-1)^n W^{(n)} + \frac{c\,\Gamma(n+\frac12)}{2\pi}y^{-n-\frac12}$. Zeros with $0 < \gamma \le T_0$ lie on the line and contribute $\gamma^{2n}e^{-y\gamma^2} > 0$. Any other zero has $\gamma > T_0$ and contributes at most $(\gamma+1)^{2n}e^{-y(\gamma^2-\frac14)}$ in modulus. Let $y \ge y_n$. Since $n/y \le 20n^2 < T_0^2$, the function $(\gamma+1)^{2n}e^{-y\gamma^2}$ decreases for $\gamma \ge T_0$, so by $N(t+1)-N(t) \le 2\log(t+1)$, which follows from [108] as in the proof of Theorem 11.1.1, the zeros with $T_0 + j < \gamma \le T_0 + j + 1$ contribute at most $b_j = 2\log(T_0+j+1)\,(T_0+j+1)^{2n}e^{-y((T_0+j)^2-\frac14)}$. For $n \le 10^{10}$ we have $b_{j+1}/b_j \le \exp\bigl(\frac{2n}{T_0} + \frac{1}{T_0\log T_0} - \frac{2T_0+1}{20n}\bigr) < 10^{-13}$, so the tail is at most $2b_0 = 4\log(T_0+1)\,(T_0+1)^{2n}e^{-y(T_0^2-\frac14)}$. Its ratio to the positive term $\frac{c\,\Gamma(n+\frac12)}{2\pi}y^{-n-\frac12}$ is a constant multiple of $y^{n+\frac12}e^{-y(T_0^2-\frac14)}$, which decreases for $y \ge y_n$ because $T_0^2 - \frac14 > 20n(n+\frac12)$. At $y = y_n$, using $\Gamma(n+\frac12) \ge \Gamma(\frac32)$ and $y_n < 1$, the logarithm of the ratio is at most

$$\log\frac{8\pi\log(T_0+1)}{c\,\Gamma(\frac32)} + 2n\log(T_0+1) - \frac{T_0^2-\frac14}{20n} < -4.4\times10^{13},$$

because the left side increases with n, and at $n = 10^{10}$ its first term is below 6 while $y_n(T_0^2-\frac14) \approx 4.5\times10^{13}$ exceeds $2n\log(T_0+1) < 5.8\times10^{11}$ by a wide margin. So the tail is smaller than the positive term.

Since $k(\infty) = 0$, Williamson's theorem [364] gives Ω_n. □

On the large y branch, $y > 0.05/n$, the bound of the proof lets the positive term $c\,\Gamma(n+\frac12)y^{-n-\frac12}/(2\pi)$ contributed by the tilt dominate the zeros above T_0 whenever $y_n(T_0^2-\frac14) > 2n\log(T_0+1) + 6$. This sufficient condition holds while $n < 8.8\times10^{10}$, and we state the safe round number 10^{10}.

Section 8.1.6 compares the reach of Theorem 10.2.17 with that of Theorems 8.1.6 and 8.1.13.

Chapter 11

Infinite divisibility, necklaces and solved models

Section 11.1 proves with no hypothesis the positivity of the canonical function w of Theorem 10.1.11, which the proof of Theorem 10.2.15 used, so that φ itself is infinitely divisible, and places φ in the Bondesson hierarchy: infinitely divisible, but not self decomposable, not in the Bondesson class and not a generalised gamma convolution. It then identifies the Vere-Jones weights as necklace polynomials, extends the Lévy representation across the critical strip, and compares the solved models with the theory of Pólya and Cardon. The truncations of this lattice are studied in Volume II.

The coordinates of Chapter 10 (Section 10.1.1) hold here too: in the variable s of the clock a zero $\frac{1}{2} + i\gamma$ sits at $s = -\gamma^2$ on the cut.

11.1 Infinite divisibility, necklaces and solved models

By Theorem 10.1.11, $-\varphi'/\varphi$ is the Laplace transform of the canonical function $w = e^{x/4} - P = W - W_0$.

11.1.1 Infinite divisibility and the class of φ

Since the candidate Lévy density of φ is $w(x)/x$, infinite divisibility is a question about the sign of w; recall $w(0^+) = 1$ and $w(x) \sim c/(2\sqrt{\pi x})$ as $x \to \infty$, with c the constant of (10.1.6). The proof uses the verification of RH to height 50 [9, 277], the explicit Riemann von Mangoldt bound of Hasanalizade, Shen and Wong [108], and three interval enclosures, $F(0.004) < -0.111$ for the function F of Lemma 10.1.12, $P(0.004) < 2 \times 10^{-13}$ and $E(0.004) < 0.0011$ for the sum E defined in the proof, computed (enclosures E1 to E3).

Theorem 11.1.1. *$w(x) > 0$ for every $x > 0$. Consequently, unconditionally, φ is the Laplace transform of an infinitely divisible probability law on $(0, \infty)$ with zero*

drift and Lévy density $w(x)/x$:

$$\frac{\zeta(\frac12)}{(1-4s)\,\zeta(\frac12+\sqrt{s})} = \exp\Big(-\int_0^\infty \big(1-e^{-sx}\big)\frac{w(x)}{x}\,dx\Big), \qquad \operatorname{Re} s > 0.$$

Proof. Fix $x_1 = 0.004$.

Small x. Each term $x^{-1/2}e^{-(\log n)^2/4x}$ increases in x while $(\log n)^2/4x > \frac12$, which holds for $x \le x_1$ and $n \ge 2$. Hence $P(x) \le P(x_1) < 2\times 10^{-13}$ (enclosure E2: the term $n=2$ in interval arithmetic, and $n \ge 3$ through $(\log n)^2/4x_1 \ge \frac{\log 3}{4x_1}\log n$), and $w(x) = e^{x/4} - P(x) > 0$ on $(0, x_1]$.

Large x. By Lemma 10.1.12 and $F(x_1) < 0$, for $x \ge x_1$ we have $-W_0(x) \ge |F(x_1)|/(4\sqrt{\pi x})$. Here $F(x_1) < -0.111$ by enclosure E1. In W the zeros with $0 < \gamma \le 50$ lie on the line [9] and contribute $e^{-x\gamma^2} > 0$. Any other zero has $\gamma > 50$ and contributes at most $e^{-x(\gamma^2-1/4)}$ in modulus. The explicit Riemann von Mangoldt formula $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78| \le E(T)$ for $T \ge e$, with $E(T) = 2.5167 + \frac{1}{50T}$ for $T \le X_0 = 30\,610\,046\,000$ and $E(T) = 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T} < 0.3\log T$ for $T > X_0$ [108, (1.7), (2.7) and Corollary 1.4], gives $N(t+1) - N(t) \le 2\log(t+1)$ for $t \ge 50$: the main term grows by at most $\frac{1}{2\pi}\log\frac{t+1}{2\pi}$, and while $t+1 \le X_0$ the two error terms add at most $5.0334 + \frac{1}{25t}$, together less than 5.37 at $t = 50$ against $2\log 51 = 7.86$, with a gap that widens as t grows, since its derivative exceeds $(2 - \frac{1}{2\pi})/(t+1)$; for $t+1 > X_0$ the error terms add at most $0.6\log(t+1)$, and $\frac{1}{2\pi} + 0.6 < 2$. So all such zeros together contribute at most

$$E(x) = \sum_{j\ge 0} 2\log(51+j)\, e^{-x((50+j)^2 - \frac14)}.$$

Enclosure E3 gives $E(x_1) < 0.0011$, against $|F(x_1)|/(4\sqrt{\pi x_1}) > 0.247$. Since $(50+j)^2 - \frac14 > 1/(2x_1)$ for every j, $\sqrt{x}\,E(x)$ decreases on $[x_1, \infty)$, so $E(x) < |F(x_1)|/(4\sqrt{\pi x}) \le -W_0(x)$ for all $x \ge x_1$, and $w = W - W_0 > 0$ on $[x_1, \infty)$.

Representation. By Theorem 10.1.11, $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)dx$ for $s > \frac14$. Since $w \ge 0$, Landau's theorem applies [359, Chapter II, Theorem 5b]: if the Laplace transform of a nonnegative function has abscissa of convergence σ_c, then the transform cannot be continued analytically to a neighbourhood of σ_c. Here $-\varphi'/\varphi$ is analytic near every point of $(0,\infty)$ by Proposition 10.1.9. Hence $\sigma_c \le 0$, the integral converges for $\operatorname{Re} s > 0$, and equals the continuation of $-\varphi'/\varphi$ there. Integrating over $[a, s]$ for real $0 < a < s$ with Tonelli, and letting $a \downarrow 0$ with $\varphi(a) \to 1$ and monotone convergence, gives the formula for real $s > 0$; then $\int \min(1,x)\, w(x) x^{-1} dx < \infty$ because the left side is finite, and analytic continuation gives the formula for $\operatorname{Re} s > 0$. □

Theorem 11.1.1 is the main unconditional positive result of Chapters 10 and 11, and its proof has a wide margin (Figure 11.1). The only place where w comes close to a sign change is at infinity, where it decays like $x^{-1/2}$ and the sign is fixed by $\theta'(0) < 0$.

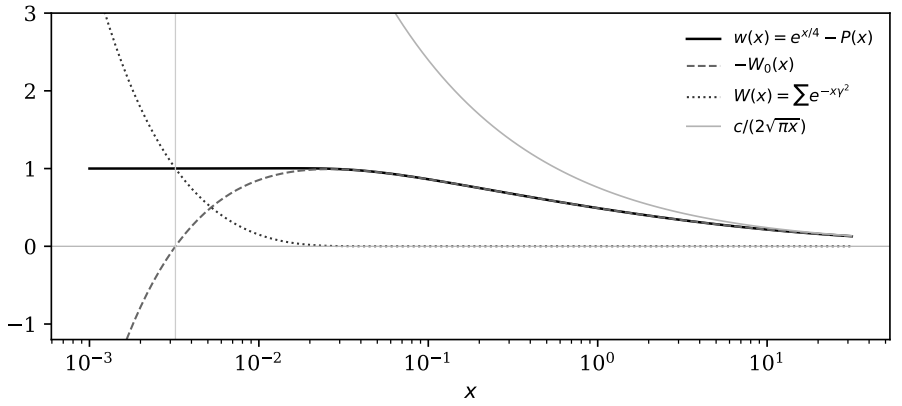


Figure 11.1: The function $w = W - W_0$ on a logarithmic scale, with its two components: the heat trace W over the zeros and the negative smooth part $-W_0$. The vertical line is $x_* \approx 0.00322$, where W_0 changes sign. To the left, w differs from $e^{x/4}$ by less than 2×10^{-13}; to the right, $-W_0 > 0$, and from $x_1 = 0.004$ on it dominates any contribution of off line zeros to W. The faint curve is the asymptote $c/(2\sqrt{\pi x})$.

In arithmetic terms, Theorem 11.1.1 is the Gaussian smoothed Chebyshev bound

$$\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x} < \sqrt{4\pi x}\; e^{x/4} \qquad (x > 0).$$

The right side is the same smoothing applied to dy in place of $d\psi(y)$. To see what the primes alone give, write $\psi(y) = y + \Delta(y)$ and put $I(x) = \frac{1}{2} e^{x/4} \operatorname{erfc}(\frac{\sqrt{x}}{2})$.

Proposition 11.1.2. *For $x > 0$,*

$$w(x) = I(x) - Q_\Delta(x), \qquad Q_\Delta(x) = \frac{1}{\sqrt{4\pi x}} \int_{1^-}^{\infty} y^{-1/2} e^{-(\log y)^2/4x}\, d\Delta(y),$$

with $I(x) \sim (\pi x)^{-1/2}$ and $w(x) \sim \frac{c}{2}(\pi x)^{-1/2}$ as $x \to \infty$. Hence $Q_\Delta(x) \sim -(\frac{c}{2} - 1)(\pi x)^{-1/2}$ with $\frac{c}{2} - 1 \approx 0.343$. On the other hand, any bound $|\Delta(y)| \leq A\sqrt{y} \log^2 y$ for $y \geq 2$, including the one that RH implies, yields only $|Q_\Delta(x)| \leq A\,(x/2 + O(\sqrt{x}))$ after integration by parts.

Proof. With $d\Delta = d\psi - dy$, the main term is $(4\pi x)^{-1/2} \int_1^\infty y^{-1/2} e^{-(\log y)^2/4x} dy = (4\pi x)^{-1/2} \int_0^\infty e^{u/2 - u^2/4x} du = \frac{1}{2} e^{x/4}(1 + \operatorname{erf} \frac{\sqrt{x}}{2})$, and $e^{x/4}$ minus this is I. The asymptotics of w are Theorem 10.1.11. For the bound, integrate by parts and use $\frac{d}{dy}\{y^{-1/2} e^{-(\log y)^2/4x}\} = -y^{-3/2} e^{-(\log y)^2/4x}(\frac{1}{2} + \frac{\log y}{2x})$ and $\int_0^\infty u^2 e^{-u^2/4x}(\frac{1}{2} + \frac{u}{2x}) du = \frac{\sqrt{\pi}}{2}(2\sqrt{x})^3 \cdot \frac{1}{4} + 4x$; the boundary term at 1^- and the range $1 \leq y < 2$ contribute $O(x^{-1/2})$. □

In this Gaussian weighting the error term Δ of the prime number theorem has a definite negative bias at large x, $Q_\Delta(x) \approx -0.343(\pi x)^{-1/2}$, which is $I+W_0$ up to an exponentially small term, and $w>0$ is the statement that $Q_\Delta(x) < I(x)$ for all x. Pointwise bounds on the oscillating Δ cannot give this at large x: the margin there is of order $x^{-1/2}$, while a pointwise bound controls Q_Δ only up to order x; the function $D(y) = -A\sqrt{y}\log^2 y$, in place of Δ, gives $Q_D(x) = -A\big(x/2 + 2\sqrt{x/\pi}\big)$ by the computation in the proof, so the bound of Proposition 11.1.2 cannot be improved using the size of Δ alone. Mean square information on $\Delta(y)/\sqrt{y}$ in the logarithmic scale, of the kind Cramér obtained under RH [67], controls Gaussian smoothed quantities much better, and we make no claim that such information is insufficient.

11.1.1.1 The place of φ in the Bondesson hierarchy

Recall from Section 4.1.2 the classes of laws on $[0,\infty)$ defined through the Lévy density $k(x)/x$: ID, SD (k nonincreasing), BO ($k(x)/x$ completely monotone) and GGC (k completely monotone). They are nested as GGC $\subset$ SD $\subset$ ID and GGC $\subset$ BO $\subset$ ID [39, 340].

THEOREM 11.1.3. *$\varphi \in$ ID but $\varphi \notin$ SD, $\varphi \notin$ BO, and φ is not a mixture of exponentials. In particular φ is not a generalised gamma convolution.*

Proof. ID is Theorem 11.1.1.

SD. By Theorem 10.1.11, $k = w = e^{x/4} - P$, and P together with every one of its derivatives vanishes at 0^+, so $w'(0^+) = \frac14 > 0$ and w strictly increases at the origin.

BO. Membership means that the Lévy density $w(x)/x$ equals $\int e^{-tx} m(dt)$ for some $m \geq 0$. Then $-\varphi'/\varphi(s) = \int_0^\infty x e^{-sx} \int e^{-tx} m(dt)\,dx = \int m(dt)/(s+t)^2$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, so φ has no zeros or poles there and RH follows from Proposition 10.1.9. Under RH, Theorem 10.1.11 gives $w(x) = \int e^{-tx} dM(t)$, with $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ as in Theorem 10.1.10, and since $M(0^+) = 0$ an integration by parts gives $w(x)/x = \int e^{-tx} M(t)\,dt$. Uniqueness of Laplace transforms gives $M\,dt = m \geq 0$, that is $S(\tau) \geq -1$ for almost all τ. But by Selberg's central limit theorem [330], $S(\tau) < -1$ for a positive proportion of $\tau \in [T, 2T]$ once T is large, so $M < 0$ on a set of positive measure. Numerically, for $527.71 \leq \tau \leq 527.90$, just below $\gamma_{289} = 527.9036\ldots$, ball arithmetic gives $N(\tau) = 288$ and $-1.143 < S(\tau) < -1.008$, so $M < 0$ there, and $S(\gamma_{289}^-) = -1.14548\ldots$.

Mixtures of exponentials. The law is a mixture of exponentials exactly when it has a completely monotone density, that is when φ is a Stieltjes function, which requires $\operatorname{Im}\varphi(-t+i0)$ to have one sign. With $\zeta(\frac12 + i\tau) = Z(\tau)e^{-i\theta(\tau)}$ we get $\operatorname{Im}\varphi(-\tau^2+i0) = \zeta(\frac12)\sin\theta(\tau)/\{(1+4\tau^2)Z(\tau)\}$. On $(0,\gamma_1)$, Z has no zero while θ decreases through $-\pi$ at $\tau_\pi = 3.43622\ldots$, so $\sin\theta$ changes sign.

GGC. Since GGC is contained in SD and in BO, either clause excludes it. $\square$

The theorem is unconditional, and its failures have two causes, neither involving a zero off the line. Self decomposability and the GGC property fail because of the pole: the factor $1 - 4s$ that removes it puts $e^{x/4}$ into w, so the canonical function increases near the origin. The Bondesson property fails because the negative Thorin density of the gamma factor meets the fluctuations of the zeros about their smooth count, measured by S, which has large negative excursions: the first zero with $S(\gamma^-) < -1$ is $\gamma_{127} = 282.4651$, where $S(\gamma_{127}^-) = -1.0063$, and the most negative value among the first 460 zeros is at γ_{289} (the seven such zeros are listed after Theorem 17.1.3, and these values of S are enclosed in ball arithmetic by the program cited in the proof). This failure would occur even under RH, and it recurs infinitely often since S is unbounded below [330, 352]. Both causes disappear for R, which is infinitely divisible and self decomposable with no hypothesis (Theorem 8.1.1, [282, Theorem A]); what Theorem 11.1.1 adds there is the splitting of the floor (Theorem 10.2.15) into the arithmetic part w and a completely monotone archimedean part.

11.1.1.2 The density of the law

Let X_φ be, as in Section 10.2, a variable with $\mathbb{E}e^{-sX_\varphi} = \varphi(s)$. Differentiating the representation of Theorem 11.1.1 gives $-\varphi'(s) = \varphi(s)\int_0^\infty e^{-sx}w(x)\,dx$, which says that the measure $x\Pr(X_\varphi \in dx)$ is the convolution of the law of X_φ with $w(x)\,dx$. Since $\varphi(s) \to 0$ as $s \to \infty$, the law has no atom at 0, so it is absolutely continuous, and its density f satisfies Steutel's equation [340, Proposition III.4.16]

$$x\,f(x) = \int_0^x f(x-y)\,w(y)\,dy, \qquad x > 0. \tag{11.1.1}$$

The right side is continuous in x by dominated convergence, since f is integrable and w is bounded and continuous on $[0,\infty)$, so f may be taken continuous on $(0,\infty)$. Near the origin, $w(y) = 1 + O(y)$ gives $f(x) = \bar f(x)(1 + O(x))$ with $\bar f(x) = x^{-1}\int_0^x f$, and $\bar f(x) > 0$ for every $x > 0$, since otherwise $s\varphi(s) \to 0$, whereas $s\varphi(s) \to -\zeta(\frac12)/4 > 0$. As $\bar f' = (f - \bar f)/x$, this gives $(\log \bar f)' = O(1)$ near 0, so $\bar f$ and f have a common positive limit $f(0^+)$, and dominated convergence in $s\varphi(s) = \int_0^\infty e^{-y} f(y/s)\,dy$, f being bounded, gives $f(0^+) = \lim_{s\to\infty} s\varphi(s) = -\zeta(\frac12)/4 \approx 0.36509$.

At infinity the same equation gives $f(x) \sim w(x)/x \sim \frac{c}{2\sqrt{\pi}}x^{-3/2} \approx 0.75773\,x^{-3/2}$, in three steps. First, $xf(x) \le \sup w$. Second, split the integral in (11.1.1), written as $\int_0^x f(u)w(x-u)\,du$, at $u = x/2$: the part $u \le x/2$ is at most $\sup_{y\ge x/2} w(y) = O(x^{-1/2})$, and the part $u \ge x/2$ is at most $\sup_{[x/2,x]} f \cdot \int_0^{x/2} w = O(x^{-1})\,O(x^{1/2})$, so $f(x) = O(x^{-3/2})$. Third, with this bound the part $u \ge x/2$ is $O(x^{-1}) = o(w(x))$, while the part $u \le x/2$ is $w(x)(1 + o(1))$ by dominated convergence, since $w(x-u)/w(x) \to 1$ for fixed u

and stays below 2 for $u \le x/2$ and large x. Hence $xf(x) = w(x)(1+o(1))$. It follows that $\Pr(X_\varphi > x) \sim \frac{c}{\sqrt{\pi}} x^{-1/2}$ and $\mathbb{E}X_\varphi = \infty$, and since $\varphi(s) = 1 - c\sqrt{s} + O(s)$ gives $\varphi(s/n^2)^n \to e^{-c\sqrt{s}}$, the variable X_φ lies in the domain of attraction of the positive stable law of index $\frac12$ that generates the building blocks (10.1.2).

THEOREM 11.1.4. *$f(0^+) = -\zeta(\frac12)/4$ and $f'(0^+) = f(0^+)\,w'(0^+) = -\zeta(\frac12)/16 \approx 0.09127 > 0$. In particular f is not monotone, so X_φ is not a mixture of uniform laws on intervals $[0,a]$, and f is not log convex and not completely monotone.*

Proof. The value of $f(0^+)$ was found after (11.1.1). Since P and all its derivatives vanish at 0^+, w is continuously differentiable on $[0,\infty)$ with $w(0^+) = 1$ and $w'(0^+) = \frac14$. Write (11.1.1) as $xf(x) = \int_0^x f(u)\,w(x-u)\,du$. Since f is continuous on $(0,\infty)$ and bounded near 0, the right side is differentiable, with derivative $f(x)w(0^+) + \int_0^x f(u)w'(x-u)\,du$, so f is differentiable on $(0,\infty)$ and

$$f'(x) = \frac1x \int_0^x f(u)\,w'(x-u)\,du \longrightarrow f(0^+)\,w'(0^+) \qquad (x \downarrow 0),$$

because f and w' are continuous at 0. A nonincreasing density is required for a mixture of uniforms by Khintchine's theorem, and log convex or completely monotone densities are nonincreasing. □

PROPOSITION 11.1.5. *$w' > 0$ on $(0, 0.004]$ and $w' < 0$ on $[4.52, \infty)$. On $[0.004, 4.6]$ a grid evaluation of w' (computed, not proved) finds exactly one sign change, at $x_m = 0.015851\ldots$, where $w(x_m) = 1.003409\ldots$.*

Proof. On $(0, 0.004]$, $w' = \frac14 e^{x/4} - P'(x)$ and $0 \le P'(x) \le P'(0.004) < 2 \times 10^{-9}$, since each prime term of P' increases in x there. For $x \ge 0.004$ write $w' = W' - W_0'$. The zeros on the line contribute $-\sum \gamma^2 e^{-x\gamma^2} < 0$ to W'. Every zero below height 3×10^{12} lies on the line [277], so an off line zero contributes at most $(\gamma^2 + \frac14)e^{-x(\gamma^2 - 1/4)}$ with $\gamma > 3 \times 10^{12}$. With $N(t+1) - N(t) \le 2\log(t+1)$ from the proof of Theorem 11.1.1, these contributions together are less than $e^{-10^{22}} \gamma_1^2 e^{-x\gamma_1^2}$ for $x \ge 0.004$; hence $W' < 0$. By Proposition 10.2.11, with $\nu(dy) = 2\sum_{k\ge0} e^{-(2k+\frac12)y}\,dy$ the measure defined there,

$$-W_0'(x) = \frac{D(x)}{4\sqrt{\pi}\,x^{3/2}}, \qquad D(x) = \theta'(0) + \frac12 \int_0^\infty g\Big(\frac{y^2}{4x}\Big)\nu(dy),$$
$$g(u) = 1 - (1-2u)e^{-u}.$$

Since $g(0) = 0$, $g'(0) = 3$, g is concave on $[0, \frac52]$ and $g \le 1 + 2e^{-3/2} < 3u$ beyond, $g(u) \le 3u$ for all $u \ge 0$. With $\int y^2 \nu(dy) = 4\sum_{k\ge0}(2k + \frac12)^{-3} < 32.34$ this gives $D(x) \le \theta'(0) + 12.13/x < 0$ for $x \ge 4.52$, and then both parts of w' are negative. □

The computed statement rests on an evaluation of w' on $[0.004, 4.6]$ from the first 300 zeros and the integral for W_0', on a grid of spacing 10^{-4} up to 0.05 and 5×10^{-3} beyond. The prime and zero evaluations of w' agree to 10^{-11} at $x = 0.01$.

Numerically f is also unimodal: it rises from 0.365089 to its maximum 0.366883 at the mode $x = 0.024046$ and then decreases to 0 like $0.75773\,x^{-3/2}$, so its mode lies to the right of the maximum of w. Talbot and de Hoog inversion agree to twelve digits at every point used. Unimodality of f remains open (Section 17.1.4). Since X_φ is infinitely divisible but not self decomposable, Yamazato's theorem that every self decomposable law is unimodal [366], [325, Theorem 53.1] does not apply. Section 14.1 applies the theorem to the law of X_R, which is self decomposable, and to the shifted clocks T_α of Theorem 2.1.2 with $\alpha \geq 1$ (Corollary 14.1.2).

11.1.1.3 Closure operations and the harmonic mean

GGC is closed under independent sums and weak limits, and, by a theorem of Bondesson [40], under independent products, so Theorem 11.1.3 gives at once the following.

COROLLARY 11.1.6. *X_φ is not a sum or a product of independent GGC variables, nor a weak limit of such.*

The reciprocal in φ suggests a fourth operation, the harmonic mean. Closure under it, which would have widened Corollary 11.1.6, is the statement

(H) if V_1 and V_2 are independent GGC variables, then so is $2V_1V_2/(V_1+V_2)$.

PROPOSITION 11.1.7. *(H) is false. If E_1 and E_2 are independent standard exponentials, then $H = 2E_1E_2/(E_1+E_2)$ has density*

$$f_H(h) = e^{-h}\int_0^\infty \Big(1+\frac{h}{2u}\Big)e^{-u-h^2/4u}\,du, \qquad h>0,$$

and H is not a GGC.

Proof. For $h>0$, $2E_1E_2 > h(E_1+E_2)$ holds if and only if $E_1, E_2 > \frac{h}{2}$ and $(2E_1-h)(2E_2-h) > h^2$: both factors cannot be negative, since $E_1, E_2 < \frac{h}{2}$ forces $|2E_1-h|, |2E_2-h| < h$. Substituting $E_1 = \frac{h}{2}+u$, $E_2 = \frac{h}{2}+v$ gives

$$\Pr(H>h) = e^{-h}\iint_{uv>h^2/4} e^{-u-v}\,du\,dv = e^{-h}\int_0^\infty e^{-u-h^2/4u}\,du,$$

so differentiating under the integral, which is justified by the bound $\frac{h}{2u}e^{-u-h^2/4u} \leq u^{-1/2}e^{-u}\sup_{y>0} ye^{-y^2}$, locally uniform in h, gives f_H. In terms of the modified Bessel functions of the second kind the two integrals are $h\,\mathrm{K}_1(h)$ and $h\{\mathrm{K}_0(h)+\mathrm{K}_1(h)\}$, by $\int_0^\infty u^{-\nu-1}e^{-u-h^2/4u}\,du = 2(h/2)^{-\nu}\mathrm{K}_\nu(h)$ [256, (10.32.10)]. Splitting the integral at $u=h$ shows that f_H is bounded with $f_H(0^+)=1$.

Suppose H were a GGC with drift a and Thorin measure U of total mass β. Since $f_H > 0$ on $(0,\infty)$ and a GGC satisfies $H \geq a$ almost surely, $a = 0$. Since f_H is bounded, $s\,\mathbb{E}e^{-sH} = \int_0^\infty e^{-y} f_H(y/s)\,dy \to f_H(0^+) = 1$, so $-\log \mathbb{E}e^{-sH} = \log s + o(1)$, while Fatou applied to $\int \log(1 + s/t)\,U(dt)$ gives $\liminf_{s\to\infty}(-\log \mathbb{E}e^{-sH})/\log s \geq \beta$. Hence $\beta \leq 1$.

Now let V be any GGC with zero drift and $\beta \leq 1$, and $\mathcal{L}(s) = \mathbb{E}e^{-sV}$. For $\operatorname{Im} s > 0$ each $1 + s/t$ lies in the upper half plane, so $\operatorname{Im} \log \mathcal{L}(s) = -\int \arg(1 + s/t)\,U(dt) \in [-\pi\beta, 0) \subseteq [-\pi, 0)$, and $\mathcal{L} > 0$ on $(0,\infty)$: $\mathcal{L}$ is a Stieltjes function [327, Corollary 7.4]. In its representation $\mathcal{L}(s) = b_0 + b_1/s + \int (s+r)^{-1} m(dr)$ [327, Definition 2.1] we have $b_0 = \lim_{s\to\infty} \mathcal{L}(s) = 0$ and $b_1 = 0$, since $\mathcal{L}(0^+) = 1$. So $\mathcal{L}(s) = \int_0^\infty e^{-sx}\big(\int e^{-rx} m(dr)\big)dx$ and the density of V is $\int e^{-rx} m(dr)$, completely monotone and in particular nonincreasing.

But f_H is not nonincreasing: $f_H(h) - 1 \sim h \log(1/h)$ as $h \downarrow 0$, and f_H attains 1.11125 at $h = 0.09765$ against $f_H(0^+) = 1$. So H is not a GGC. □

There is also a structural obstruction. The factorisation of φ into R and the gamma factor is a factorisation of Laplace transforms, that is a convolution of laws, while (H) acts on the variables themselves. The gamma factor is not a Laplace transform at all, since it tends to infinity with s, and (10.2.1) shows that it enters only through deconvolution by stable laws. The one place where (H) could have acted is the positive part: R is GGC if and only if RH holds, so an unconditional representation $X_R \stackrel{d}{=} 2V_1V_2/(V_1 + V_2)$ with V_1, V_2 independent GGC variables built from the primes and the gamma factor would have proved RH under (H). Proposition 11.1.7 removes the implication, and we know of no candidate representation.

11.1.2 The Vere-Jones weights and the Lévy representation across the strip

Throughout this subsection σ is the real part of the argument z of ζ.

11.1.2.1 Necklace polynomials and the sign rule

Let $M_k(q) = \frac{1}{k}\sum_{d|k} \mu(k/d)\, q^d$ be the necklace polynomial [234].

Theorem 11.1.8. *$B_N = 0$ unless $N = p^n$ is a prime power, and then*

$$B_{p^n} = b_n\big(p^{-1/2}\big), \qquad b_n(q) = \sum_{i\geq 0,\ 2^i|n} M_{n/2^i}(q).$$

Equivalently, for $|y| < 1$ and $0 < q \leq 1$,

$$\frac{1}{1 - qy} = \prod_{n\geq 1} \big(1 + y^n\big)^{b_n(q)}, \tag{11.1.2}$$

so that $\zeta(\frac{1}{2}+u)=\prod_p\prod_n(1+p^{-nu})^{b_n(p^{-1/2})}$ *for* $\operatorname{Re}u>\frac{1}{2}$. *At* $q=1$, $b_n(1)=1$ *if* n *is a power of* 2 *and* 0 *otherwise, and* (11.1.2) *becomes Euler's binary identity* $\frac{1}{1-y}=\prod_{j\ge0}(1+y^{2^j})$.

Proof. From the definition of B_N in Theorem 10.1.8, $B_{p^n}=\sum_{jm=n}d_m\,a_{p^j}$ with $a_{p^j}=q^j/j$. Proposition 10.1.6 gives $y^j=\sum_m d_m\log(1+y^{mj})$; multiplying by q^j/j and summing over j gives $\sum_n b_n(q)\log(1+y^n)=\sum_j q^jy^j/j=-\log(1-qy)$, which is (11.1.2). The cyclotomic identity $\frac{1}{1-qy}=\prod_k(1-y^k)^{-M_k(q)}$ and $\log(1+y^n)=\log(1-y^{2n})-\log(1-y^n)$ give, comparing coefficients of $\log(1-y^n)$, the recursion $b_n=M_n+b_{n/2}$ for even n and $b_n=M_n$ for odd n, whose solution is the stated sum. At $q=1$, $M_k(1)=\frac{1}{k}\sum_{d\mid k}\mu(k/d)=\delta_{k,1}$. □

The identity (11.1.2) is a 2 adic variant of the cyclotomic identity of Metropolis and Rota [234]; we have not found it stated in this form. Its exponents include $b_2=\frac{q^2+q}{2}$, $b_3=\frac{q^3-q}{3}$, $b_4=\frac{q^4}{4}+\frac{q^2}{4}+\frac{q}{2}$, $b_6=\frac{q^6+q^3-q^2-q}{6}$, which give B_4, B_8 and B_9 of Theorem 10.1.8 and, at $p=2$, B_{16} and B_{64}.

Proposition 11.1.9. *For every prime* p, *with* $q=p^{-1/2}$: $b_{2^a}(q)>0$ *for all* $a\ge1$, *and for odd* $n>1$, $b_n(q)\neq0$ *and the sign of* $b_n(q)$ *is* $(-1)^{\omega(n)}$, *where* $\omega(n)$ *is the number of distinct prime factors of* n.

Proof. For $n=2^a$, $b_n=q+\sum_{j=1}^{a}\frac{q^{2^j}-q^{2^{j-1}}}{2^j}=\frac{q}{2}+\frac{q^2}{4}+\cdots+\frac{q^{2^{a-1}}}{2^a}+\frac{q^{2^a}}{2^a}>0$. For odd n, $b_n=M_n$. Put $\ell=\operatorname{rad}(n)$ and $\tilde q=q^{n/\ell}\le q\le2^{-1/2}$; then $nM_n(q)=\sum_{k\mid\ell}\mu(\ell/k)\tilde q^k=(-1)^{\omega(n)}\big(\tilde q+\sum_{k\mid\ell,k>1}\pm\tilde q^k\big)$, and the divisors $k>1$ of the odd squarefree ℓ are distinct odd integers at least 3, so $\sum_{k>1}\tilde q^k<\sum_{m\ \mathrm{odd}\ge3}\tilde q^m=\tilde q\cdot\frac{\tilde q^2}{1-\tilde q^2}\le\tilde q$. □

For even n other than powers of 2 the sign depends on how the 2 adic parts combine. Among $2\le n\le60$ there are 42 negative exponents for each of $p=2,3,5,7$.

The weights B_N and d_m also govern the truncation of the sawtooth expansion of S. The sawtooth Σ_λ of Lemma 10.1.7 has the Fourier series $\Sigma_\lambda(\tau)=\sum_{j\ge1}(-1)^j\frac{\sin(j\lambda\tau)}{\pi j}$, so the formal series $\sum_N B_N\Sigma_{\log N}(\tau)$ is a double series in (N,j) whose frequencies are $j\log N$.

Proposition 11.1.10. *Truncate the double series by frequency, keeping the terms with* $j\log N\le L$. *Then*

$$\sum_{\substack{N,j\\ j\log N\le L}}B_N\,(-1)^j\frac{\sin(j\tau\log N)}{\pi j}=-\frac{1}{\pi}\sum_{n\le e^L}\frac{\Lambda(n)}{\sqrt{n}\log n}\sin(\tau\log n),$$

the classical truncated series for $S(\tau)$. *Truncation by* $N\le N_0$ *instead is not equivalent: it keeps the harmonics* $j\log N>\log N_0$ *of each retained sawtooth without the terms of larger* N *that cancel them.*

Proof. With $N = n^m$, the frequency is $r \log n$ with $r = mj$, and the constraint $j \log N \le L$ is $r \log n \le L$, which does not separate m from j. The coefficient of $\sin(r\tau \log n)$ is $-\frac{a_n}{\pi} \sum_{mj=r} d_m \frac{(-1)^{j+1}}{j}$, and $\sum_{mj=r} d_m (-1)^{j+1} j^{-1}$ has Dirichlet series $\eta(1+v)^{-1}\eta(1+v) = 1$, so it is $\delta_{r,1}$. □

So, truncated consistently, the sawtooth expansion is the classical series, and truncating it by the index N is worse. Numerically, on a uniform grid of step 10^{-3} in $\tau \in [10, 50]$ and with the Cesàro weight $1 - \log N / \log N_0$, the root mean square error against $S(\tau)$ is 0.228, 0.223, 0.218 for $N_0 = 10^3, 10^4, 10^5$, against 0.146, 0.128, 0.115 for the classical series with the same weight, the root mean square of S itself being 0.320.

11.1.2.2 The cyclotomic form, the prime number theorem and a trace formula

The cyclotomic identity gives a second factorisation, into the blocks $1 - N^{-u}$:

$$\frac{1}{\zeta(\frac{1}{2}+u)} = \prod_p \prod_{k \ge 1} \bigl(1 - p^{-ku}\bigr)^{M_k(p^{-1/2})}, \qquad \operatorname{Re} u > \tfrac{1}{2}.$$

Each block $1 - e^{-\lambda u} = 2e^{-\lambda u/2} \sinh(\lambda u/2)$ is again a solved model, now with a sinh lattice: with $u = \sqrt{s}$, $\frac{\sinh(\lambda\sqrt{s}/2)}{\lambda\sqrt{s}/2} = \prod_{m \ge 1} \bigl(1 + \frac{s}{(2\pi m/\lambda)^2}\bigr)$, a reciprocal GGC with unit atoms at $(2\pi m/\lambda)^2$. The sinh blocks are the local zeta factors on the line $\sigma = 0$; unlike the sech blocks they vanish at $u = 0$, and normalising the product at $u \to 0$ requires two identities.

Proposition 11.1.11. *For $0 < q < 1$, summing over k in increasing order,*

$$\sum_{k \ge 1} M_k(q) = 0, \qquad \sum_{k \ge 1} M_k(q) \log k = \log(1 - q).$$

Both follow from the prime number theorem in the forms $\sum_m \mu(m)/m = 0$ and $\sum_m \mu(m) \log m / m = -1$.

Proof. $\sum_{k \le K} M_k(q) = \sum_d \frac{q^d}{d} \sum_{m \le K/d} \frac{\mu(m)}{m}$. The inner sums are bounded by 1 in absolute value and tend to 0, so dominated convergence gives 0. Similarly $\sum_{k \le K} M_k(q) \log k = \sum_d \frac{q^d}{d} \sum_{m \le K/d} \frac{\mu(m) \log(dm)}{m} \to \sum_d \frac{q^d}{d}(-1) = \log(1-q)$, using $\sum \mu(m)/m = 0$ for the $\log d$ part. □

Heuristically, the two identities are what make the cyclotomic product of solved models converge at the edge $u \to 0$: $\prod_k (1 - y^k)^{-M_k(q)}$ behaves near $y = 1$ like $(1-y)^{-\sum M_k} \prod_k k^{-M_k}$, and the finite limit $\frac{1}{1-q}$ requires both.

In heat trace form the Vere-Jones representation becomes a trace formula. Proposition 10.1.13 and Poisson summation give, for every $x > 0$,

$$\sum_{\gamma > 0} e^{-x\gamma^2} - W_0(x) = e^{x/4} + \sum_p \sum_{n \ge 1} b_n(p^{-1/2}) \Bigl[\sum_{k \ge 0} e^{-x\left(\frac{(2k+1)\pi}{n \log p}\right)^2} - \frac{n \log p}{4\sqrt{\pi x}} \Bigr], \quad (11.1.3)$$

where the left side is written under RH and is $W(x) - W_0(x)$ in general, and the double series converges absolutely once each bracket is written in its theta form. The left side is the heat trace of the zeros net of its smooth part; the right side is the pole plus the heat traces of the arithmetic progressions $\frac{(2k+1)\pi}{n \log p}$ of pseudo zeros, each net of its own mean. In heat trace the zeros of ζ are a signed superposition of arithmetic progressions with spacings $2\pi/(n \log p)$. Heuristically, constructive interference of the progressions places the mass at the ordinates γ, and the negative weights of Proposition 11.1.9 remove the mass from the other points of the progressions.

The trace formula does not give positivity of the Thorin measure of R, and the obstruction is explicit: the weights $b_n(p^{-1/2})$ are not nonnegative integers, and the passage from the binary product at $q = 1$ to the critical line at $q = p^{-1/2}$ is a deformation of the exponents in which integrality and positivity are lost together. A proof of RH along these lines would have to show that the interference of the progressions in (11.1.3), with exactly these weights, produces a positive measure; Section 17.1.4 poses the deformation from $q = 1$ to $q = p^{-1/2}$ as a problem.

11.1.2.3 The Lévy and Thorin representations across the strip

For $\sigma > 1$ the Vere-Jones representation is a statement about the classical zeta distribution (Section 10.1.6): $\zeta(\sigma + i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau \log N_\sigma}$ where $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and $\log N_\sigma$ is compound Poisson with Lévy measure $\nu_\sigma = \sum_n \frac{\Lambda(n)}{n^\sigma \log n} \delta_{\log n}$. Theorem 11.1.8 decomposes this Lévy measure as

$$\nu_\sigma = \sum_p \sum_{n \geq 1} b_n(p^{-\sigma})\, \nu^{\mathrm{alt}}_{n \log p}, \qquad \nu^{\mathrm{alt}}_\lambda = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k}\, \delta_{k\lambda},$$

where $\nu^{\mathrm{alt}}_\lambda$ is the formal Lévy measure of the fair coin λB, $B \in \{0, 1\}$, whose characteristic function $\frac{1}{2}(1 + e^{-i\lambda\tau})$ is the sech block. For $\sigma > 1$ there is also a positive decomposition: by $\frac{1}{1-qy} = \prod_j (1 + q^{2^j} y^{2^j})$, the exponent of p in N_σ has independent binary digits with $\Pr(\text{digit } j = 1) = \frac{p^{-2^j \sigma}}{1 + p^{-2^j \sigma}}$. The Vere-Jones form keeps the coins fair and moves $p^{-\sigma}$ into the multiplicities; the binary digit form keeps unit multiplicities and biases the coins.

Inside the strip neither form survives as a characteristic function (Section 10.1.6); the heat kernel supplies the replacement. For $0 < \sigma < 1$ put

$$\varphi_\sigma(s) = \frac{\zeta(\sigma)}{(1 - s/(1-\sigma)^2)\, \zeta(\sigma + \sqrt{s})}, \qquad \varphi_{1/2} = \varphi.$$

THEOREM 11.1.12. *For* $0 < \sigma < 1$ *and* $s > (1 - \sigma)^2$, $-\varphi'_\sigma/\varphi_\sigma(s) =$

$\int_0^\infty e^{-sx} w_\sigma(x)\,dx$, where w_σ has the three equivalent forms

$$\begin{aligned} w_\sigma(x) &= e^{(1-\sigma)^2 x} - \frac{1}{\sqrt{4\pi x}} \sum_{n\ge 2} \frac{\Lambda(n)}{n^\sigma} e^{-(\log n)^2/4x} \\ &= e^{(1-\sigma)^2 x} + \sum_p \sum_{n\ge 1} b_n(p^{-\sigma})\, h_{n\log p}(x) \\ &= \tfrac12 e^{(1-\sigma)^2 x} \operatorname{erfc}\big((1-\sigma)\sqrt{x}\big) + \sum_{\gamma>0} \operatorname{Re}\Big[e^{a_\gamma^2 x} \operatorname{erfc}(a_\gamma \sqrt{x})\Big] \\ &\qquad + \frac{\gamma_E + \log\pi}{4\sqrt{\pi x}} + \sum_{k\ge 0} \Big[\tfrac12 e^{a_k^2 x} \operatorname{erfc}(a_k\sqrt{x}) - \frac{1}{4(k+1)\sqrt{\pi x}}\Big], \end{aligned}$$

with h_λ the Jacobi theta function of Proposition 10.1.13, $a_\gamma = \sigma - \frac12 - i\gamma$ (the last form written under RH; in general γ runs over $-i(\rho - \frac12)$), and $a_k = 2k + 2 + \sigma$. As $x \to \infty$, $w_\sigma(x) \sim \frac{\zeta'(\sigma)/\zeta(\sigma)}{2\sqrt{\pi x}}$, with $\zeta'(\sigma)/\zeta(\sigma) > 0$ on $(0,1)$. If $w_\sigma \ge 0$, then φ_σ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density $w_\sigma(x)/x$:

$$\log \varphi_\sigma(s) = -\int_0^\infty \big(1 - e^{-sx}\big) \frac{w_\sigma(x)}{x}\, dx, \qquad \operatorname{Re} s > 0.$$

Proof. The first form is the argument of Theorem 10.1.11 with the pole now at $s = (1-\sigma)^2$. For the second, regroup $\sum_n b_n(q)\, n \log p \sum_j (-1)^j e^{-(jn\log p)^2/4x}$ by $r = jn$: with $nb_n = \sum_{ab=n} a\, d_a q^b$ the coefficient of $e^{-(r\log p)^2/4x}$ is $\sum_{b|r} q^b \sum_{ja=r/b} a\, d_a (-1)^j = -q^r$, since the inner sum is $-\delta_{r/b,1}$ (Proposition 10.1.13); this is the prime p part of the first form. For the third, insert the Hadamard product of ζ'/ζ into $-\varphi_\sigma'/\varphi_\sigma = \frac{1}{s-(1-\sigma)^2} + \frac{1}{2\sqrt{s}} \frac{\zeta'}{\zeta}(\sigma + \sqrt{s})$; the pole terms combine to $\frac{1}{2\sqrt{s}(\sqrt{s}+1-\sigma)}$, and each $\frac{1}{\sqrt{s}(\sqrt{s}+a)}$ is the transform of $e^{a^2 x} \operatorname{erfc}(a\sqrt{x})$. At $\sigma = \frac12$ this reduces to $w = W - W_0$ with Proposition 10.2.2.

The asymptotic follows from the third form term by term. Uniformly for $|\arg a| \le \frac{3\pi}{4} - \epsilon$ and $|a|\sqrt{x} \ge 1$,

$$\tfrac12 e^{a^2 x} \operatorname{erfc}(a\sqrt{x}) = \frac{1}{2a\sqrt{\pi x}} + O\big(|a|^{-3} x^{-3/2}\big);$$

every a in the third form has $|\arg a| < \frac{\pi}{2} + \frac{1}{14}$, and $\sum |a|^{-3}$ converges. So $w_\sigma(x) = C_\sigma/(2\sqrt{\pi x}) + O(x^{-3/2})$ with

$$C_\sigma = \frac{1}{1-\sigma} + \sum_\rho \frac{1}{\sigma - \rho} + \frac{\gamma_E + \log\pi}{2} + \sum_{k\ge 0} \Big(\frac{1}{2k+2+\sigma} - \frac{1}{2k+2}\Big),$$

the sum over ρ taken in conjugate pairs, and $C_\sigma = \zeta'(\sigma)/\zeta(\sigma)$ by the Hadamard product of ξ. Infinite divisibility follows from $w_\sigma \ge 0$ by Landau's theorem as in Theorem 11.1.1, since φ_σ is analytic near $(0,\infty)$: $\zeta(\sigma + x) \ne 0$ for $x > 0$ and the pole at $x = 1 - \sigma$ is cancelled. □

Numerically w_σ is positive on every line we tried: we evaluated it on 300 points of $x \in [10^{-4}, 10^3]$, from the prime form for $x \le 0.02$ and the zero form beyond, for $\sigma = 0.05, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 0.9, 0.99$. In every case the minimum is attained at the largest x, where $2\sqrt{\pi x}\, w_\sigma(x)$ is close to $\zeta'(\sigma)/\zeta(\sigma)$ up to $\sigma = 0.75$ and falls short of it as $\sigma \to 1$, where the pole makes the approach slow and the asymptotic has not been reached at $x = 10^3$. Positivity on every vertical line of the strip is open (Problem 17.8).

For the Thorin representation two families have to be distinguished: φ_σ, which carries the gamma factor and the pole, and the completed function $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$. For $\sigma > \frac{1}{2}$ the second is, with no hypothesis, the Laplace transform of an infinitely divisible law obtained by tilting and subordinating the law of X_R (Theorem 12.2.5 of Section 12.2 and the proof of Proposition 13.1.1 of Section 13.1, where R_σ is written φ_α). The first never has the Thorin property.

PROPOSITION 11.1.13. *For every $\sigma \in (0,1)$, φ_σ is not the Laplace transform of a GGC, and if it is infinitely divisible it is not self decomposable.*

Proof. We argue as for Theorem 11.1.3. By Theorem 11.1.12, $-\varphi_\sigma'/\varphi_\sigma(s) = \int_0^\infty e^{-sx} w_\sigma(x)\, dx$ for $s > (1-\sigma)^2$, with

$$w_\sigma = e^{(1-\sigma)^2 x} - P_\sigma, \qquad P_\sigma(x) = \frac{1}{\sqrt{4\pi x}} \sum_{n \ge 2} \frac{\Lambda(n)}{n^\sigma} e^{-(\log n)^2/4x}.$$

Every derivative of P_σ vanishes at 0^+, so $w_\sigma(x) = 1 + (1-\sigma)^2 x + o(x)$ and w_σ strictly increases near the origin. If φ_σ were a GGC transform, then $-\varphi_\sigma'/\varphi_\sigma = a + \int U(dt)/(s+t)$ with $a \ge 0$ and $U \ge 0$ satisfying the Thorin integrability conditions, so that $k_U(x) = \int e^{-tx} U(dt)$ is integrable near 0 and, by Tonelli, $\int U(dt)/(s+t) = \int_0^\infty e^{-sx} k_U(x)\, dx$ for $s > (1-\sigma)^2$. Letting $s \to \infty$ gives $a = 0$, and uniqueness of the Laplace transform gives $w_\sigma = k_U$, which is completely monotone and so nonincreasing, a contradiction. If φ_σ is infinitely divisible, the same uniqueness identifies its Lévy density as $w_\sigma(x)/x$, and self decomposability would require w_σ to be nonincreasing. □

As on the critical line, the obstruction is the factor $1 - s/(1-\sigma)^2$ that removes the pole, not the zeros. Beyond it, for $\frac{1}{2} < \sigma < 1$ with $\sup_\rho \operatorname{Re}\rho < \sigma$, the Thorin density $\frac{1}{2\pi\sqrt{t}} \operatorname{Re} \frac{\zeta'}{\zeta}(\sigma + i\sqrt{t})$ given by Lemma 10.1.1 also takes negative values on every line we examined (the minimum of $\operatorname{Re} \frac{\zeta'}{\zeta}(\sigma + i\tau)$ over $\tau \in (0, 60]$ is -1.049, -0.995, -0.867, -0.769, -0.721 at $\sigma = 0.55, 0.6, 0.75, 0.9, 0.99$), but the proposition does not need this.

The completed functions R_σ carry the zeros. Theorem 11.1.14 is Theorem 2.1.2 of Section 2.1 in the variable of this chapter.

THEOREM 11.1.14. *Let $\sigma \ge \frac{1}{2}$. Then $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is the Laplace transform of a GGC if and only if ζ has no zero ρ with $\operatorname{Re}\rho > \sigma$. In that case its*

Thorin measure is the Poisson smoothed zero density

$$U_\sigma(dt) = \frac{1}{2\pi\sqrt{t}} \sum_\rho \frac{\sigma - \beta_\rho}{(\sigma - \beta_\rho)^2 + (\sqrt{t} - \gamma_\rho)^2}\, dt,$$

the sum running over all nontrivial zeros $\rho = \beta_\rho + i\gamma_\rho$, counted with multiplicity, of both signs of γ_ρ, and taken symmetrically; each zero on $\operatorname{Re}\rho = \sigma$ with $\gamma_\rho > 0$ contributes a unit atom at γ_ρ^2 instead of its kernel. Consequently R_σ is a GGC for every $\sigma \geq 1$, unconditionally, and

$$\Theta := \sup_\rho \operatorname{Re}\rho = \inf\{\sigma \geq \tfrac{1}{2} : \ R_\sigma \text{ is a GGC}\}.$$

RH is the statement $\Theta = \frac{1}{2}$, that is, that $R_{1/2} = R$ is a GGC.

Proof. A zero with $\operatorname{Re}\rho > \sigma$ gives $\sqrt{s} = \rho - \sigma$ on the principal branch and hence a pole of R_σ in $\mathbb{C} \setminus (-\infty, 0]$, so R_σ is not a GGC. Suppose there is none. Write the zeros with positive ordinate, counted with multiplicity, as $\rho = \beta + i\gamma$, put $a = \sigma - \beta \geq 0$, and let $\mathcal{P}_a(x) = a/\{\pi(a^2 + x^2)\}$, for $a > 0$, be the Cauchy density of scale a (the Poisson kernel P_a of Section 2.1). Hadamard's product [70, Chapter 12], with each ρ paired with $\bar\rho$, gives $\xi(z) = \xi(0)\prod_{\gamma>0}(1 - z/\rho)(1 - z/\bar\rho)$, absolutely convergent since $\sum_\rho |\rho|^{-2} < \infty$, and hence for $s > 0$ and $q = \sqrt{s}$

$$R_\sigma(s) = \prod_{\gamma>0} \frac{a^2 + \gamma^2}{(q + a)^2 + \gamma^2}.$$

For $a > 0$ let $U_{a,\gamma}(dt) = \{\mathcal{P}_a(\sqrt{t} - \gamma) + \mathcal{P}_a(\sqrt{t} + \gamma)\}\, dt/(2\sqrt{t})$, and let $U_{0,\gamma} = \delta_{\gamma^2}$. The substitution $t = \tau^2$, the identity $1/(s + \tau^2) = (\pi/q)\mathcal{P}_q(\tau)$ and $\mathcal{P}_a * \mathcal{P}_q = \mathcal{P}_{a+q}$ give

$$\int_0^\infty \frac{U_{a,\gamma}(dt)}{s + t} = \frac{q + a}{q\{(q + a)^2 + \gamma^2\}} = \frac{d}{ds} \log\{(q + a)^2 + \gamma^2\},$$

which also holds for $a = 0$. Integrating over $(0, s]$ with Tonelli, $\int \log(1 + s/t)\, U_{a,\gamma}(dt) = \log\{((q + a)^2 + \gamma^2)/(a^2 + \gamma^2)\}$, which is nonnegative and at most $(s + 2\sigma\sqrt{s})/\gamma^2$. Since $\sum_{\gamma>0} \gamma^{-2} < \infty$, Tonelli's theorem shows that $U_\sigma = \sum_{\gamma>0} U_{a,\gamma}$ is a positive measure with $\int \log(1 + s/t)\, U_\sigma(dt) = -\log R_\sigma(s) < \infty$, which gives the Thorin integrability conditions, and that $R_\sigma(s) = \exp\{-\int \log(1 + s/t)\, U_\sigma(dt)\}$ for $s > 0$. So R_σ is the Laplace transform of a GGC with Thorin measure U_σ, whose density is the stated sum, grouped by the pairs $\rho, \bar\rho$, and each zero on $\operatorname{Re}\rho = \sigma$ with $\gamma_\rho > 0$ gives the atom $\delta_{\gamma_\rho^2}$. For $\sigma \geq 1$ the hypothesis holds by the Euler product.

For $\sigma > \Theta$ the density also follows from Lemma 10.1.1: then $-R_\sigma'/R_\sigma(s) = \frac{1}{2\sqrt{s}} \frac{\xi'}{\xi}(\sigma + \sqrt{s})$ is $O(|s|^{-1/2})$ near 0, $O(|s|^{-1/2} \log|s|)$ at infinity uniformly on $\mathbb{C} \setminus (-\infty, 0]$ (by the partial fraction formula for ζ'/ζ [70, Chapter 15], since every zero

lies at distance at least $\sigma-\Theta$ from the half plane $\operatorname{Re} z \geq \sigma$), and continuous up to the cut away from 0, and at $\sqrt{s}=i\tau$ the lemma gives the density $\frac{1}{2\pi\tau}\operatorname{Re}\frac{\xi'}{\xi}(\sigma+i\tau)=\frac{1}{2\pi\tau}\sum_\rho \operatorname{Re}\frac{1}{\sigma+i\tau-\rho}$, summed symmetrically. □

The quasi Riemann Hypothesis $\Theta<1$, announced in October 2026 with $\Theta\leq\frac{7}{8}$ (Section 25.1.2), is the statement that R_σ is a GGC for some $\sigma<1$; it is step 7 of the path to RH in Section 25.1 (Table 25.1), and short of it Section 14.3 shows with no hypothesis that for $\frac{1}{2}<\sigma<1$, in the density $\rho_{\sigma-1/2}$ of (2.1.7), the Cauchy kernels of the zeros to the right of σ, if there are any, carry below height T a vanishing fraction of the mass of the kernels of the other zeros as $T\to\infty$. Numerically $\min_{0<\tau\leq 70}\operatorname{Re}\frac{\xi'}{\xi}(\sigma+i\tau)$ is 4.6×10^{-5}, 0.0046, 0.0116, 0.0231 at $\sigma=0.501, 0.6, 0.75, 1$, attained as $\tau\to0$.

11.1.3 Pólya and Cardon theory for the solved models

Pólya's theory of Fourier transforms with only real zeros, and Cardon's extension, are built from the objects the Vere-Jones step produces: cosines, hyperbolic cosines and their products. The solved models are special cases of that theory; ζ is not, because its multiplicities are neither integers nor nonnegative. Throughout, Φ is the Pólya kernel and $\Xi(\tau)=\int\Phi(t)\cos(\tau t)\,dt$, as in (2.1.3) with the variable written t; Φ is even, analytic in the strip $|\operatorname{Im} t|<\pi/4$, and has a natural boundary on the lines $\operatorname{Im} t=\pm\pi/4$ [313, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The Pólya variable X of Section 2.1 has density $\Phi/\Xi(0)$ and moment generating function $\mathbb{E}e^{hX}=\xi(\frac{1}{2}+h)/\xi(\frac{1}{2})$, by (2.1.4). In this subsection G denotes a function of the Laguerre Pólya class, not the gamma factor.

11.1.3.1 Solved models and the Laguerre Pólya class

On the critical line the block $1+N^{-\sqrt{s}}$ becomes

$$1+N^{-i\tau}=2e^{-i\tau\ell/2}\cos\Big(\tfrac{\ell\tau}{2}\Big),\qquad \ell=\log N,$$

and $\cos(\ell\tau/2)=\mathbb{E}e^{i\tau(\ell/2)\varepsilon}$ with $\varepsilon=\pm1$ equally likely. A product of blocks with nonnegative integer multiplicities m_N is, up to the phase, the characteristic function of the Rademacher sum $\sum_N\sum_{j\leq m_N}\frac{\log N}{2}\varepsilon_{N,j}$, and all its zeros are real, lying on the lattices $(2k+1)\pi/\log N$. Cardon's theorem on exponential sums [54] extends this: if $a_1,\dots,a_n$ and $b_1,\dots,b_n$ are positive and G is a real entire function of genus 0 or 1 with only real zeros, then

$$\sum_{\varepsilon\in\{\pm1\}^n} G\Big(i\sum_k a_k\varepsilon_k\Big)\exp\Big(iz\sum_k b_k\varepsilon_k\Big),$$

with the same signs in both sums, has only real zeros.

The function ζ is not of this form: formally $\zeta(\frac{1}{2}+i\tau)=\prod_N(1+N^{-i\tau})^{B_N}$, but the exponents B_N are not integers, and some are negative: by Theorem 11.1.8, at prime cubes $B_{p^3}=\frac{1}{3}p^{-3/2}-\frac{1}{3}p^{-1/2}<0$.

In the kernel variable the solved models act as multipliers. By Pólya's Hilfssatz II [302, Hilfssatz II], if $a>0$, b is real and G is a real entire function of genus 0 or 1 with only real zeros, at least one of them, then $e^{ib}G(z+ia)+e^{-ib}G(z-ia)$ has only real zeros. With $G=\Xi$ this says that replacing $\Phi(t)$ by $\Phi(t)\cosh(at)$, a solved model in the variable t, cannot create nonreal zeros: if Ξ has only real zeros, so does the transform of the new kernel. For $a\geq\frac{1}{2}$ the conclusion needs no hypothesis: the transform of $\Phi(t)\cosh(at)$ is $\frac{1}{2}(\xi(s+a)+\xi(s-a))$ with $s=\frac{1}{2}+iz$, and Lagarias proved that for $a\geq\frac{1}{2}$ its zeros all lie on the critical line and are simple [199]. Products of such factors converge to Gaussians, $\prod_{k\leq n}\cosh(ct/\sqrt{n})\to e^{c^2t^2/2}$, and this is de Bruijn's route to the Gaussian multipliers $e^{\lambda t^2}$ [73]. In the variable t, multiplying $\Phi(t)$ by $e^{\lambda t^2/4}$ is the de Bruijn Newman flow at time λ (Section 4.1), the flow of Theorem 17.1.16, so the multiplier $e^{\lambda t^2}$ is the flow at time 4λ. The solved models are thus the elementary steps of the de Bruijn Newman flow, and the bound $\Lambda\leq 0.2$ [277] says that the flow at time 0.2, applied to Φ or, in the normalisation of Rodgers and Tao, to Φ_{dBN} defined before Theorem 17.1.16, already produces a kernel whose transform has only real zeros.

Proposition 11.1.15. *Let G be a real even entire function of order less than 2 with $G(0)\neq 0$. Then G is in the Laguerre Pólya class if and only if $s\mapsto G(0)/G(i\sqrt{s})$ is the Laplace transform of a GGC whose Thorin measure is $\sum_k m_k\delta_{\alpha_k^2}$ with integers $m_k\geq 1$, where $\pm\alpha_k$ are the zeros of G with multiplicities m_k.*

Proof. By Hadamard, $G(z)=G(0)\prod_k(1-z^2/\alpha_k^2)^{m_k}$ with no exponential factor, since G is even and of order less than 2. All α_k are real if and only if $G(0)/G(i\sqrt{s})=\prod_k(1+s/\alpha_k^2)^{-m_k}$ is a product of gamma transforms, which is the stated GGC. If some α_k^2 is not positive, the product has a zero or a pole off $(-\infty,0]$, which a GGC transform cannot have. □

The Laguerre Pólya class consists of the locally uniform limits of real polynomials with only real zeros; a real entire function of order less than 2 belongs to it exactly when its zeros are real. In the language of Proposition 11.1.15, Pólya's multiplier theorem [55, Proposition 1], for even G of order less than 2, reads: if f is even and integrable with $f(t)=O(e^{-|t|^b})$ for some $b>2$, $\int f(t)e^{izt}dt$ has only real zeros, and $G(0)/G(i\sqrt{s})$ is an integer atom GGC, then $\int G(it)f(t)e^{izt}dt$ has only real zeros. RH is the statement that $R=\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ is an integer atom GGC. The Thorin measures in the Pólya theory are always integer valued.

11.1.3.2 Lee and Yang, and Newman

Pólya's Hilfssatz II has a physical reading, to which Kac drew attention in his comments on Pólya's 1926 paper [309, pp. 424 to 426]: it is the one spin case

of the Lee and Yang circle theorem for the Ising ferromagnet [206]. Cardon's exponential sums make this explicit. For $\varepsilon \in \{\pm 1\}^n$ the sum

$$\mathcal{Z}_n(z) = \sum_{\varepsilon} G\Big(i \sum_k a_k \varepsilon_k\Big) \exp\Big(iz \sum_k b_k \varepsilon_k\Big)$$

is a partition function. The spins are weighted by $G(i \sum_k a_k \varepsilon_k)$, a function of a magnetisation, and they feel the purely imaginary external fields izb_k. For the Gaussian member of the Laguerre Pólya class, $G(x) = e^{-\beta x^2/2}$, which has genus 2 and lies outside Cardon's theorem, the weight is $\exp(\beta(\sum_k a_k \varepsilon_k)^2/2)$, the Curie Weiss ferromagnet with couplings $\beta a_j a_k \geq 0$. The zeros of $\mathcal{Z}_n$ in z are then real by the Lee and Yang theorem [206], since off the real axis the fields izb_k have real parts of one sign. A general even G in the class is a limit of products of such Gaussians and of factors $1 + M^2/\alpha^2$ in the magnetisation $M = \sum_k a_k \varepsilon_k$, and Cardon's theorem is the Lee and Yang theorem for this family of ferromagnetic interactions of the magnetisation.

Newman made the connection with the Riemann Hypothesis precise [242]. In a 1974 paper on generalised Ising systems [240] he had characterised the single spin laws with the Lee and Yang property: a symmetric single spin law ρ with $\int e^{bt^2} \rho(dt) < \infty$ for every b has the Lee and Yang property, that ferromagnetic systems of ρ spins have zeros only on the imaginary field axis, exactly when its moment generating function $\int e^{ht} \rho(dt)$ has only purely imaginary zeros, that is when $\int e^{izt} \rho(dt)$ is in the Laguerre Pólya class. In Thorin language this reads as follows.

Proposition 11.1.16. *Let ρ be a symmetric law with Gaussian moments of all orders. Then ρ has the Lee and Yang property if and only if $s \mapsto 1/\int e^{\sqrt{s}t} \rho(dt)$ is the Laplace transform of a GGC with zero drift whose Thorin measure has integer atoms. The atoms are at the squared zeros α_k^2 of $\int e^{izt} \rho(dt)$, with the multiplicities of the zeros.*

Proof. By Newman's theorem [240] the Lee and Yang property holds exactly when $\int e^{izt} \rho(dt)$ is in the Laguerre Pólya class, and then, by Hadamard, $\int e^{ht} \rho(dt) = e^{bh^2} \prod_k (1 + h^2/\alpha_k^2)^{m_k}$ with $b \geq 0$. In fact $b = 0$: for every $\epsilon > 0$, $e^{ht} \leq e^{\epsilon h^2 + t^2/4\epsilon}$ gives $\int e^{ht} \rho(dt) \leq C_\epsilon e^{\epsilon h^2}$ for real h, while the right side is at least e^{bh^2}. With $h = \sqrt{s}$ the reciprocal is $\prod_k (1 + s/\alpha_k^2)^{-m_k}$, a GGC with zero drift and Thorin measure $\sum m_k \delta_{\alpha_k^2}$. Conversely, as in Proposition 11.1.15, a GGC transform with zero drift and integer atoms m_k at α_k^2 is $\prod_k (1 + s/\alpha_k^2)^{-m_k}$, whose reciprocal at $s = -z^2$ is $\prod_k (1 - z^2/\alpha_k^2)^{m_k}$, a function in the Laguerre Pólya class. □

Apply this to the Pólya variable X: since $\mathbb{E}e^{hX} = \xi(\frac{1}{2} + h)/\xi(\frac{1}{2})$, we have $1/\mathbb{E}e^{\sqrt{s}X} = R(s)$, the reciprocal Xi function of Section 10.1.1. Three statements therefore coincide: RH; the law of X is a Lee and Yang single spin law; R is a GGC with no drift whose Thorin measure has an atom at each γ^2, of mass the

multiplicity of the zero. The de Bruijn Newman flow at time λ multiplies $\Phi(t)$ by $e^{\lambda t^2/4}$, which in Ising terms adds the ferromagnetic self interaction $e^{\lambda t^2/4}$ to the Pólya variable viewed as a spin, and Λ is the least λ for which the resulting spin is Lee and Yang. Newman's conjecture $\Lambda \geq 0$, proved by Rodgers and Tao [318], says that no antiferromagnetic self interaction can be tolerated.

11.1.3.3 A Cardon route

Cardon's Fourier theorem [55, Theorem 1] gives a constructive sufficient condition. If

$$\Phi(t)\,dt = \Xi(0)\,G(it)\,\Pr\Big(\sum_k a_k\varepsilon_k \in dt\Big), \tag{11.1.4}$$

where the ε_k are independent fair signs, the $a_k > 0$ are nonincreasing, and G is real, even, entire of order less than 2 with only real zeros, then Ξ has only real zeros. In the Ising reading, (11.1.4) says that the Pólya variable is the magnetisation of a weighted Ising paramagnet with weights a_k, tilted by a ferromagnetic interaction $G(iM)$ of its magnetisation $M = \sum_k a_k\varepsilon_k$. Since $|G(it)| \geq |G(0)|$ for even G in the class, the law of $\sum_k a_k\varepsilon_k$ must carry the doubly exponential decay of Φ, and among Rademacher laws the random harmonic series does.

Proposition 11.1.17. *For $\kappa > -1$ let $H_\kappa = \sum_{k\geq 1} \varepsilon_k/(2(k+\kappa))$, with characteristic function $\hat f_\kappa(v) = \prod_k \cos(v/(2(k+\kappa)))$ and density f_κ.*

(i) As $s \to \infty$, $\log \mathbb{E}e^{sH_\kappa} = \frac{s}{2}\log s + A(\kappa)s + o(s)$ with $A(\kappa) = A(0) - \frac12(\psi(1+\kappa)+\gamma_E)$ (here ψ is the digamma function), $A(0) = -0.1485757\ldots$, so $\log\Pr(H_\kappa > u) = -\frac12 e^{2u-2A(\kappa)-1}(1+o(1))$ as $u\to\infty$. Since $\log\Phi(t) = -\pi e^{2t} + \frac92 t + O(1)$ as $t\to+\infty$, the double exponential terms agree exactly when $A(\kappa) = -\frac12(1+\log 2\pi)$, that is at $\kappa^* = 6.61845\ldots$.

(ii) $\limsup_{|v|\to\infty} |v|^{-1}\log|\hat f_\kappa(v)| = -\pi/4$ for every κ. Hence f_κ is analytic in the strip $|\operatorname{Im} t| < \pi/4$, which is exactly the strip of analyticity of Φ.

Proof. (i) Put $\ell_\kappa(s) = \log\mathbb{E}e^{sH_\kappa} = \sum_{k\geq1}\log\cosh\frac{s}{2(k+\kappa)}$. For $\kappa = 0$ put $\lambda = s/2$ and split the sum at $k = \lambda$, using $\log\cosh y = y - \log 2 + \log(1+e^{-2y})$ for $k \leq \lambda$. Since $\sum_{k\leq\lambda} 1/k = \log\lambda + \gamma_E + O(1/\lambda)$, and since $k \mapsto \log(1+e^{-2\lambda/k})$ and $k\mapsto\log\cosh(\lambda/k)$ are monotone, so that their sums differ from the corresponding integrals by $O(1)$,

$$\ell_0(s) = \lambda\log\lambda + (\gamma_E - \log 2)\lambda + \lambda\int_1^\infty \frac{\log(1+e^{-2y})}{y^2}\,dy + \lambda\int_0^1 \frac{\log\cosh y}{y^2}\,dy + O(1),$$

which is $\frac{s}{2}\log s + A(0)s + O(1)$ with $A(0) = \frac12\big(\gamma_E - 2\log 2 + \int_1^\infty y^{-2}\log(1+e^{-2y})\,dy + \int_0^1 y^{-2}\log\cosh y\,dy\big)$. For general κ the mean value theorem gives $s^{-1}(\ell_\kappa - \ell_0)(s) =$

$\sum_k \tanh(s\xi_k)\big(\frac{1}{2(k+\kappa)} - \frac{1}{2k}\big)$ with ξ_k between $\frac{1}{2(k+\kappa)}$ and $\frac{1}{2k}$, and since $\tanh(s\xi_k) \to 1$, dominated convergence gives the limit $\frac{1}{2}\sum_k(\frac{1}{k+\kappa} - \frac{1}{k}) = -\frac{1}{2}(\psi(1+\kappa)+\gamma_E)$. For the tail put $s_u = e^{2u-2A(\kappa)-1}$. Chebyshev's exponential inequality at $s = s_u$ gives $\log \Pr(H_\kappa > u) \le \ell_\kappa(s_u) - s_u u = -\frac{1}{2}s_u(1+o(1))$. For the lower bound tilt the law of H_κ by $e^{sx-\ell_\kappa(s)}$. By convexity of ℓ_κ the tilted mean is $\ell_\kappa'(s) = \frac{1}{2}\log s + A(\kappa) + \frac{1}{2} + o(1)$, and the tilted variance is $\ell_\kappa''(s) = \sum_k (2(k+\kappa))^{-2}\operatorname{sech}^2 \frac{s}{2(k+\kappa)} = O(1/s)$, by comparison with an integral. Given $\delta > 0$, choose s with $\ell_\kappa'(s) = u + \delta/2$; by Chebyshev's inequality the tilted law gives probability at least $\frac{1}{2}$ to $(u, u+\delta]$ for large u, so $\log \Pr(H_\kappa > u) \ge \ell_\kappa(s) - s(u+\delta) - \log 2 = -\frac{1}{2}s_u e^{\delta}(1+\delta)(1+o(1))$. Letting $\delta \to 0$ gives the tail.

(ii) Since $\hat f_\kappa$ is even we take $v \to +\infty$. First, $\int_0^\infty \log|\cos y|\, y^{-2}dy = -\pi/2$: as $\log|\cos|$ is even and π periodic and $\sum_{n\in\mathbb{Z}}(y+n\pi)^{-2} = \csc^2 y$, the integral equals $\int_0^{\pi/2}\log(\cos y)\csc^2 y\, dy$, and an integration by parts against $-\cot y$ turns this into $-\int_0^{\pi/2}\cot y \tan y\, dy = -\pi/2$. For the upper bound fix $M, Y > 0$ and put $g_M = \max(\log|\cos|, -M)$, which is continuous, of bounded variation on $[0,Y]$ and $O(y^2)$ at 0. The points $y_k = v/(2(k+\kappa))$ have spacing at most $2y_k^2/v$, so $v^{-1}\sum_{y_k \le Y} g_M(y_k) \to \frac{1}{2}\int_0^Y g_M(y) y^{-2}dy$ as a Riemann sum. Since $\log|\cos| \le g_M \le 0$, this gives $\limsup_{v\to\infty} v^{-1}\log|\hat f_\kappa(v)| \le \frac{1}{2}\int_0^Y g_M(y)y^{-2}dy$, and letting $Y \to \infty$ and then $M \to \infty$ gives the bound $-\pi/4$. For the lower bound, the same Riemann sum argument, applied to the bounded continuous function $\Psi(y)/y$ with $\Psi(y) = \int_0^y \log|\cos u|\, du$, gives

$$\begin{aligned}\int_0^V \log|\hat f_\kappa(v)|\, dv &= \sum_k 2(k+\kappa)\Psi\Big(\frac{V}{2(k+\kappa)}\Big)\\ &= \frac{V^2}{2}\int_0^\infty \frac{\Psi(y)}{y^3}\, dy + o(V^2) = -\frac{\pi V^2}{8} + o(V^2),\end{aligned}$$

since an integration by parts gives $\int_0^\infty \Psi(y)y^{-3}dy = \frac{1}{2}\int_0^\infty \log|\cos y|\, y^{-2}dy = -\pi/4$. If $\log|\hat f_\kappa(v)| \le -(\pi/4+\epsilon)v$ for all large v, the left side would be at most $-(\pi/8+\epsilon/2)V^2 + O(1)$. Finally, $|\hat f_\kappa(v)| \le C_\epsilon e^{-(\pi/4-\epsilon)|v|}$ makes the Fourier integral of $\hat f_\kappa$ converge locally uniformly in $|\operatorname{Im} t| < \pi/4 - \epsilon$, so f_κ is analytic there. □

The coincidence of the strips in (ii) has a common source: both densities are Fourier transforms of functions that decay at the exponential rate $\pi/4$ on the real axis. For f_κ this is (ii). For Φ, $|\Xi(\tau)| \le e^{-\pi|\tau|/4+o(|\tau|)}$ by Stirling's formula, since $\zeta(\frac{1}{2}+i\tau)$ grows at most polynomially, and the strip ends at $|\operatorname{Im} t| = \pi/4$ because Φ is a theta series in e^{2t}, which reaches the imaginary axis there.

With both leading orders matched, the question is the ratio

$$r(t) = \frac{\Phi(t)}{\Xi(0) f_{\kappa^*}(t)},$$

which must equal $G(it)$. We computed f_{κ^*} on the real axis by a saddle point contour integral of $\mathbb{E}e^{sH_{\kappa^*}}$, and on the imaginary axis by the Fourier integral $f_{\kappa^*}(ix) = \frac{1}{\pi}\int_0^\infty \hat{f}_{\kappa^*}(v)\cosh(vx)\,dv$. The two methods agree at the origin, $f_{\kappa^*}(0) = 2.104533$, to seven digits.

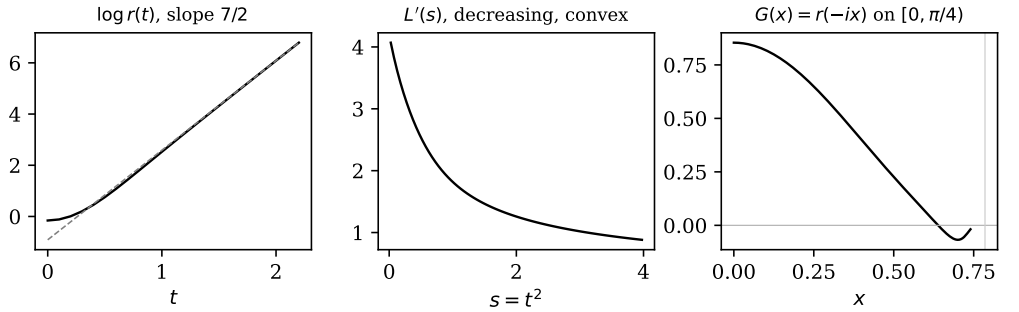


Figure 11.2: The Cardon ratio for the random harmonic law H_{κ^*}. Left: $\log r(t)$ on the real axis, asymptotically linear with slope $\frac{7}{2}$ (dashed). Centre: $L'(s)$ for $L(s) = \log(r(\sqrt{s})/r(0))$, positive, decreasing and convex as a Stieltjes function must be. Right: the continuation $G(x) = r(-ix)$ on $[0, \pi/4)$, with a simple zero at $x = 0.6388$.

Proposition 11.1.18. *The ratio r satisfies the following necessary conditions for* (11.1.4) *(computed).*

(i) Growth. r is even, $r(0) = 0.853936$, and $\log r(t)$ becomes linear with slope approaching $\frac{7}{2}$: the increments over steps of 0.1 fall from 0.3534 to 0.3514 between $t = 1.5$ and $t = 2.2$. The slope is explained exactly: Φ carries the prefactor $e^{9t/2}$, and the saddle point prefactor $\sqrt{s^*/\pi} \approx e^t$ of f_{κ^*} removes one unit. So G would have exponential type $\frac{7}{2}$, compatible with order one.

(ii) Stieltjes signs. With $L(s) = \log r(\sqrt{s}) - \log r(0)$, a Laguerre Pólya G requires $L'(s) = \sum m_k/(s + \alpha_k^2)$, a Stieltjes function. On a grid of 81 points in $s \in [0, 4]$ the difference quotients of L give $L' > 0$, $L'' < 0$ and $L''' > 0$ everywhere, and the first order quotients fall from 4.07 on the first grid interval to 0.94 on the last.

(iii) The imaginary axis. The continuation $G(x) = r(-ix)$, computed independently from $\Phi(ix)$ and $f_{\kappa^*}(ix)$, is real, agrees with r at second order ($\log(G(0.1)/G(0)) = -0.0423$ against $\log(r(0.1)/r(0)) = 0.0417$), decreases from 0.8539, has a simple zero at $\alpha_1 = 0.6388$, and satisfies the Laguerre inequality $G'^2 - GG'' \geq 2.4 > 0$ on $[0, 0.735]$.

Everything visible inside the strip is consistent with G being a Laguerre Pólya function of exponential type $\frac{7}{2}$; the decisive question lies outside it.

PROPOSITION 11.1.19. *If* (11.1.4) *holds with G entire, then*

$$f_{\kappa^*}(t) = \Phi(t)/(\Xi(0)G(it)),$$

so f_{κ^*} continues meromorphically exactly as far as Φ does, and the two share their singular behaviour on the lines $\operatorname{Im} t = \pm\pi/4$, apart from the zeros of G. Since Φ has a natural boundary on these lines, coming from the Jacobi theta function at the edge of its modular domain, f_{κ^*} must have the same natural boundary, with singular behaviour proportional to that of Φ through a factor that continues to an entire function.

The first sentence of Proposition 11.1.19 is immediate. The second uses the classical fact that the theta function $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ has the line $\operatorname{Re} x = 0$ as a natural boundary [313, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], together with $\Phi(t) = 4x^{9/4}\omega''(x) + 6x^{5/4}\omega'(x)$ at $x = e^{2t}$.

So the random harmonic law fits the Pólya kernel inside the strip, and the question is the boundary, where the two functions have different origins: Φ is a modular theta series and f_{κ^*} a Rademacher product whose characteristic function has real zeros at $v = (2j+1)\pi(k+\kappa^*)$, increasingly dense. We see no reason for the Rademacher product to reproduce the modular singularities of Φ, and this suggests that the right Ising system for the Pólya variable is not the harmonic paramagnet but one whose magnetisation law carries theta function structure.

11.1.3.4 The Pólya variable as the logarithm of a GGC

PROPOSITION 11.1.20. *For $0 < q < 1$ the triple product*

$$\sum_{n\in\mathbb{Z}} q^{n^2} e^{inv} = \prod_{m\geq 1} (1-q^{2m})\big(1 + 2q^{2m-1}\cos v + q^{4m-2}\big)$$

exhibits the discrete Gaussian law $\Pr(J=n) \propto q^{n^2}$ as the law of $\sum_m (B_m - B'_m)$ with independent Bernoulli variables of parameter $q^{2m-1}/(1+q^{2m-1})$. Each factor vanishes at $v = \pi \pm i(2m-1)\log(1/q)$, so the characteristic function has no real zeros and the moment generating function has no purely imaginary zeros: the triple product spins are not Lee and Yang spins, and neither is the discrete Gaussian.

Proof. $1 + 2q^{2m-1}\cos v + q^{4m-2} = (1+q^{2m-1}e^{iv})(1+q^{2m-1}e^{-iv})$, and each factor, normalised, is the characteristic function of a Bernoulli variable or its negative. The zeros are those of $1 + q^{2m-1}e^{\pm iv}$. □

The triple product places the theta function in the field variable. For Φ the theta function enters multiplicatively, through the modular variable e^{2t}; the identity of Biane, Pitman and Yor [33], going back to Williams [363] (Section 2.1), makes this exact.

THEOREM 11.1.21. *Let* $S_2 = \sum_{k\ge1} \frac{2}{\pi^2k^2}(e_k + e'_k)$ *with independent standard exponentials* e_k, e'_k, *so that* $\mathbb{E}e^{-sS_2} = \big(\sqrt{2s}/\sinh\sqrt{2s}\big)^2$ *and* S_2 *is a GGC with Thorin measure* $2\sum_k \delta_{\pi^2k^2/2}$. *Then*

$$2\xi(z) = \mathbb{E}\Big[\Big(\frac{\pi S_2}{2}\Big)^{z/2}\Big].$$

Consequently, with $Y = \frac12\log(\pi S_2/2)$, $2\Xi(\tau) = \mathbb{E}[e^{Y/2}e^{i\tau Y}]$, *and the Pólya variable* X *is* Y *tilted by* $e^{Y/2}$: $\Phi(t)\,dt = \frac12 e^{t/2}\Pr(Y \in dt)$.

Proof. This is the identity of [33, Section 2]; the density of S_2 is a theta series by the partial fraction expansion of $(x/\sinh x)^2$ and Poisson summation, and its Mellin transform is 2ξ. □

Section 12.1.1 tilts Y to every real basepoint σ: with no hypothesis, $\xi(\sigma + z)/\xi(\sigma)$ is the moment generating function of $\frac12\log(\frac{\pi}{2}S_2)$ under the law of S_2 size biased by $S_2^{\sigma/2}$.

Theorem 11.1.21 shows where the modular boundary comes from: the quadratic lattice $\pi^2k^2/2$ of the Thorin atoms of S_2, through Jacobi's theta function and Poisson summation, produces the natural boundary at $\operatorname{Im} t = \pm\pi/4$: the density of S_2 is analytic in $\operatorname{Re} x > 0$ with natural boundary on $\operatorname{Re} x = 0$, and $x = \frac{2}{\pi}e^{2t}$. The Thorin structure thus enters twice: R is the reciprocal moment generating function of the Pólya variable in $\sqrt{s}$, and S_2 is a GGC in the usual sense. RH is the statement that every zero of $w \mapsto \mathbb{E}[(\pi S_2/2)^w]$ in the strip $0 < \operatorname{Re} w < \frac12$ lies on the line $\operatorname{Re} w = \frac14$.

This suggests a multiplicative Lee and Yang property: we say that a positive random variable V has *Mellin zeros on a line* if $\mathbb{E}V^w$, continued, vanishes only on one vertical line. The solved models of the Vere-Jones step have exact multiplicative analogues.

PROPOSITION 11.1.22. *Let* $V = \sum_{k\le n} c_k e_k$ *with independent standard exponentials* e_k *and distinct* $c_k > 0$. *Then* $\mathbb{E}V^{u-1} = \Gamma(u)\sum_k A_k c_k^u$ *with* $A_k = c_k^{n-2}/\prod_{j\ne k}(c_k - c_j)$.

(i) For $n = 2$ *the zeros are* $u = 2\pi i m/\log(c_1/c_2)$, $m \ne 0$: *all on the line* $\operatorname{Re} u = 0$. *The two exponential sum is a multiplicative solved model.*

(ii) For $n = 3$ *with* $c = (1, \frac14, \frac19)$, *a truncation of the quadratic lattice with single atoms, the zeros in* $0 < \operatorname{Im} u < 30$ *have real parts* 0.488, −1.268, 0.300, 0.387, −1.318, 0.457, 0.136, −1.129, 0.500 *(computed): they do not lie on a line.*

Proof. The density of V is $\sum_k A_k e^{-x/c_k}$ by partial fractions of $\prod_k(1 + c_k s)^{-1}$, and its Mellin transform is the stated exponential polynomial times $\Gamma(u)$. For $n = 2$ it is proportional to $c_1^u - c_2^u$. For $n = 3$ the zeros are those of $1.5 - 2.4\cdot4^{-u} + 0.9\cdot9^{-u}$,

which has none with $|\operatorname{Re} u| \geq 8$; the argument principle on $[-8,8] \times [0.01,30]$ counts nine, none lies in $0 < \operatorname{Im} u < 0.01$, and Newton's method locates them. □

So the Lee and Yang mechanism survives the passage from additive to multiplicative magnetisations for two components and fails at three.

11.1.4 Harmonic means and scale mixtures of the reciprocal Xi function

The reciprocal Xi function $1/G(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, the Laplace transform of the clock T (Section 2.1), is a harmonic mean of transforms. For $x \neq 0$ put $L_x(s) = 1/\cosh(x\sqrt{s})$. From $\cosh z = \prod_{k \geq 1} \left(1 + z^2/(\pi^2(k - \frac{1}{2})^2)\right)$, L_x is a GGC transform with Thorin atoms $\pi^2(k - \frac{1}{2})^2/x^2$; probabilistically, if C is the exit time of Brownian motion from $(-1,1)$, then $\mathbb{E}e^{-\theta C} = 1/\cosh\sqrt{2\theta}$ and $L_x(s) = \mathbb{E}\exp(-\frac{1}{2}x^2 sC)$.

Proposition 11.1.23. $\mathbb{E}e^{-sT} = 1/\mathbb{E}[1/L_X(s)]$ *with X the Pólya variable: the reciprocal Xi transform is the harmonic mean, over the Pólya density, of Brownian exit time transforms. By Jensen's inequality,* $\mathbb{E}e^{-sT} \leq \mathbb{E}\exp(-\frac{1}{2}sX^2C)$*, with C independent of X.*

Proof. By (2.1.4), $1/\mathbb{E}e^{-sT} = G(s) = \mathbb{E}\cosh(\sqrt{s}X) = \mathbb{E}[1/L_X(s)]$, and $y \mapsto 1/y$ is convex. □

The arithmetic mean on the right is the Laplace transform of the mixture $\frac{1}{2}X^2C$. If harmonic means of GGC transforms were GGC, Proposition 11.1.23 would make the clock a GGC, which is RH (Proposition 2.1.1). That fails: a harmonic mean of GGC transforms need not even be a transform.

Example 11.1.24. The harmonic mean of the GGC transforms $(1+s)^{-2}$ and $(1+2s)^{-2}$ is $1/(1 + 3s + \frac{5}{2}s^2)$, with poles at $-0.6 \pm 0.2i$. Its inverse transform $2e^{-0.6t}\sin(0.2t)$ is negative on $(5\pi, 10\pi)$, so the harmonic mean is not even the Laplace transform of a positive measure.

Example 11.1.25. Every scale mixture $\Gamma_2 Y$ of a Gamma(2) variable Γ_2 by an independent $Y \geq 0$ is infinitely divisible [194], and it is a GGC if Y is, since products of independent GGC variables are GGC [40]. Without a hypothesis on Y it need not be: let Y equal 1 or $\lambda > 0$, $\lambda \neq 1$, with probability $\frac{1}{2}$ each. The transform $\mathbb{E}(1+sY)^{-2}$ has numerator $(1+\lambda s)^2 + (1+s)^2$, which vanishes at non real s. A GGC transform has no zeros off $(-\infty, 0]$, so $\Gamma_2 Y$ is infinitely divisible but not a GGC.

A last negative result excludes Gamma laws as the outer layer of the clock itself, as Theorem 14.2.10 does for $|X|$, X^2 and S_2.

THEOREM 11.1.26. *For no $a > 0$ and no $Y \geq 0$ independent of Γ_a does the clock T have the law of $\Gamma_a Y$.*

Proof. Suppose it did. Then $Y > 0$ almost surely, since $\mathbb{E}e^{-sT} = 1/G(s) \to 0$ as $s \to \infty$, and $\Gamma_a Y$ has density $t^{a-1}g(t)/\Gamma(a)$ with $g(t) = \mathbb{E}[Y^{-a}e^{-t/Y}]$ completely monotone and not identically zero, so $g \geq g(1) > 0$ on $(0, 1]$ and $\mathbb{E}e^{-sT} \geq c_1 s^{-a}$ for large s. But $G(s) \geq \cosh(x_0\sqrt{s}) \Pr(|X| \geq x_0)$ for every $x_0 > 0$, so $\mathbb{E}e^{-sT} \leq c_2 e^{-x_0\sqrt{s}}$ with $c_2 = 2/\Pr(|X| \geq x_0)$, and the two bounds contradict each other as $s \to \infty$. □

Part III

Basepoints and the critical strip

Chapter 12

The basepoint one and hitting times at the basepoints

We now work at the resolution $\varepsilon = \frac{1}{2}$ of the Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 2.1, the basepoint one. The shifted clocks T_α, with $\mathbb{E}e^{-uT_\alpha} = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, carry the Thorin programme. At a basepoint α to the right of every zero, T_α is a GGC (Theorem 2.1.2 and Proposition 2.1.3), and from $\alpha = 1$ on its Thorin density is positive with no hypothesis, because no zero lies on or to the right of the line $\operatorname{Re} s = 1$.

We work the basepoint one out completely, as a test bed for the tools of the book, and ask three questions of T_1: what its Thorin measure and its law are, whether its Thorin density comes from the sine squared lemma, and whether it is a hitting time to which Kent's theorem applies.

For the Thorin measure and the Lévy form of $\log \xi(1+s)$ alike, what is unconditional at the basepoint one extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH.

12.1 The Thorin measure and the Lévy form at the basepoint one

12.1.1 The Mellin side: Bessel hitting times and size biasing

Let J be the first time a three dimensional Bessel process started at 0 reaches the level one. Its Laplace transform is

$$\mathbb{E}e^{-uJ} = \frac{\sqrt{2u}}{\sinh\sqrt{2u}} = \prod_{n\geq 1}\Big(1 + \frac{2u}{\pi^2 n^2}\Big)^{-1}, \qquad \mathbb{E}J = \frac{1}{3},$$

so $J \stackrel{d}{=} \sum_n \frac{2}{\pi^2 n^2} e_n$ with independent standard exponentials e_n: a GGC with unit Thorin atoms at the Dirichlet eigenvalues $\pi^2 n^2/2$ of the Bessel process on $[0,1]$, and the simplest instance of Kent's theorem (Section 5.2.3), since up to scale the Bessel

process is the string diffusion with that spectrum [184]. With J' an independent copy, $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2}\Gamma_n$, with independent Gamma$(2,1)$ variables Γ_n, is Williams' Gamma(2) structure [363] of Section 3.1.3, and by Biane, Pitman and Yor [33] (Section 2.1 and Theorem 11.1.21)

$$2\xi(s) = \mathbb{E}\Big[\Big(\frac{\pi}{2}S_2\Big)^{s/2}\Big], \qquad s \in \mathbb{C}. \tag{12.1.1}$$

Table 12.1 checks (12.1.1) at real s, and Table 5.3 on the critical line.

s	$\mathbb{E}(\pi S_2/2)^{s/2}$, Monte Carlo	standard error	$2\xi(s)$
0.5	0.99415	0.00017	0.99424
1	0.99981	0.00034	1
2	1.04677	0.00074	1.04720
3	1.14713	0.00127	1.14788

Table 12.1: The identity (12.1.1) of Biane, Pitman and Yor, checked with 4×10^5 draws of S_2. The sample mean of S_2 is 0.6664 against $\mathbb{E}S_2 = \frac{2}{3}$.

Put $Y = \frac{1}{2}\log(\frac{\pi}{2}S_2)$, as in Section 2.1. By (12.1.1), $2\xi(\alpha + w) = \mathbb{E}[e^{\alpha Y}e^{wY}]$, so for every real α

$$\frac{\xi(\alpha + w)}{\xi(\alpha)} = \mathbb{E}e^{wV_\alpha}, \qquad P(V_\alpha \in dy) = \frac{e^{\alpha y}P(Y \in dy)}{\mathbb{E}e^{\alpha Y}}.$$

Since the Pólya variable X is Y tilted by $e^{Y/2}$, V_α has the density $e^{(\alpha-\frac{1}{2})y}p(y)\,\xi(\frac{1}{2})/\xi(\alpha)$, the Esscher tilt of the Pólya density p of (2.1.4); so $V_{1/2} = X$, and we write $V = V_1$. Equivalently, V_α is $\frac{1}{2}\log(\frac{\pi}{2}S_2)$ under the law of S_2 size biased by $S_2^{\alpha/2}$: by $S_2^{1/4}$ at the centre and by $S_2^{1/2}$ at the basepoint one, where the weight e^Y needs no normalisation because $\mathbb{E}e^Y = 2\xi(1) = 1$. Every basepoint thus gives a probability law with no hypothesis, and on this side only the weight distinguishes $\alpha = 1$. The law of $-V_\alpha$, with characteristic function $\xi(\alpha - it)/\xi(\alpha)$, is Nakamura's complete Riemann zeta distribution [238]; he proved that RH holds if and only if, for every $\frac{1}{2} < \alpha < 1$, this characteristic function has a Lévy Khintchine form with a signed measure, with no integrability condition on the measure. Infinite divisibility does distinguish $\alpha = 1$: V_α is quasi infinitely divisible, though not infinitely divisible, for every $\alpha > 1$, and not quasi infinitely divisible at $\alpha = 1$ or at the centre (Proposition 14.1.3; for $\alpha = 1$ see also Section 12.1.4).

Remark 12.1.1 (Ostrovsky's Barnes beta laws). The companion $C_2 = \sum_{k\geq 1}\frac{2}{\pi^2(k-\frac{1}{2})^2}(e_k + e'_k)$ of S_2, with independent standard exponentials, has $\mathbb{E}e^{-uC_2} = \operatorname{sech}^2\sqrt{2u}$ [33], and its Mellin transform is again ξ:

$$\mathbb{E}[C_2^q] = \frac{2^{2q+2}-1}{q+1}\Big(\frac{2}{\pi}\Big)^{q+1}\xi(2q+2),$$

checked at $q = -0.3$, $\frac{1}{2}$, 1, 2 to 10^{-12} by quadrature of the Laplace transform. The Lévy density of Proposition 12.1.9 is

$$\frac{1}{4t}\operatorname{sech}^2\frac{t}{4} = \frac{1}{4t}\,\mathbb{E}e^{-t^2C_2/32},$$

a mixture of Gaussians in t whose mixing law has Mellin transform $\xi(2q+2)$: the positivity comes from the gamma side, and the zeros of ξ enter only through the Mellin transform of C_2. More generally the Barnes beta laws $\beta_{M,N}(a,b)$ of [222] have Lévy density $e^{-b_0 t}\prod_{j=1}^{N}(1-e^{-b_j t})/\big(t\prod_{i=1}^{M}(1-e^{-a_i t})\big)$; for $M = N = 2$, $a = (1,1)$ and $b = (b_0, \frac{1}{2}, \frac{1}{2})$ it is the density above times $e^{-(b_0-\frac{1}{2})t}$, again not that of a GGC, since t times it is the Laplace transform of the signed measure $\sum_{j\ge1}(-1)^{j+1} j\,\delta_{b_0-\frac{1}{2}+\frac{j}{2}}$. The case $M = N = 1$, $a_1 = 1$ [152], is the law of $-\log B$ with B a beta variable of parameters b_0 and b_1: t times its Lévy density, $e^{-b_0 t}(1-e^{-b_1 t})/(1-e^{-t})$, is the Laplace transform of $\sum_{j\ge0}(\delta_{b_0+j} - \delta_{b_0+b_1+j})$, so $-\log B$ is a GGC if and only if b_1 is a positive integer, when it is a sum of independent exponential variables of rates $b_0, \dots, b_0+b_1-1$; for other b_1 the negative atoms meet no positive one, and a Laplace transform determines its measure.

12.1.2 The Thorin measure and the law of T_1

For real $\alpha \ge \frac{1}{2}$ we write $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha+\sqrt{u})$, so that $\varphi_{1/2} = 1/G$ is the clock, and T_α for a random variable with $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha(u)$, $u \ge 0$, whenever one exists: Theorem 2.1.2 provides it, as a GGC, for $\alpha \ge 1$, and Theorem 12.2.5 for every $\alpha > \frac{1}{2}$. As in Section 2.1, $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha+i\theta)$.

Proposition 12.1.2. *The function $\varphi_1(u) = \xi(1)/\xi(1+\sqrt{u})$ is the Laplace transform of a GGC T_1 with no drift, $\log\varphi_1(u) = -\int_0^\infty \log(1+u/z)\,U_1(dz)$, and in the variable $\theta = \sqrt{z}$ its Thorin measure has density*

$$\rho_{1/2}(\theta) = \frac{1}{\pi}v_1(\theta) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(1+i\theta) = \frac{1}{\pi}\sum_\rho \frac{1-\beta}{(1-\beta)^2+(\theta-\gamma)^2}, \qquad (12.1.2)$$

the sum running over all nontrivial zeros $\rho = \beta + i\gamma$ in conjugate pairs. Every term is positive, so U_1 is a positive measure with no hypothesis on the zeros.

Proof. This is Theorem 2.1.2 at $\alpha = 1$. The function $-\frac{d}{du}\log\varphi_1(u) = \frac{1}{2\sqrt{u}}\frac{\xi'}{\xi}(1+\sqrt{u})$ is analytic on $\mathbb{C}\setminus(-\infty,0]$, because the zeros of $\xi(1+w)$ lie at $w = \rho - 1$ with $\operatorname{Re} w = \beta - 1 < 0$, never reached by the principal branch $w = \sqrt{u}$; Stieltjes inversion (4.1.2) of its jump across the cut, where $\sqrt{u} = \pm i\theta$, gives the density (2.1.6) in z, which is (12.1.2) in θ since $dz = 2\theta\,d\theta$. By (2.1.2) with conjugate pairs grouped, $\operatorname{Re} 1/(1+i\theta-\rho) = (1-\beta)/((1-\beta)^2+(\theta-\gamma)^2) = \pi P_{1-\beta}(\theta-\gamma)$, as in the proof of Proposition 2.1.3, and every term is positive because $\beta < 1$, the nonvanishing of ζ on $\operatorname{Re} s = 1$. There is no drift because $\log\xi(1+\sqrt{u})$ grows like $\frac{1}{2}\sqrt{u}\log\sqrt{u}$, more slowly than u. □

Under RH every width equals $\frac12$, and (2.1.8) at $\varepsilon=\frac12$ reads

$$\rho_{1/2}(\theta)=\sum_{\gamma>0}\Big(P_{1/2}(\theta-\gamma)+P_{1/2}(\theta+\gamma)\Big),$$

the RH measure blurred by the Cauchy kernel of width one half. The value at the origin is explicit, $\rho_{1/2}(0)=\frac1\pi\sum_\rho\frac1\rho=\frac1\pi\big(1+\frac{\gamma_E}{2}-\frac12\log 4\pi\big)=\frac{0.0230957089661}{\pi}=0.0073515925$, with γ_E Euler's constant, and it is the minimum of $\rho_{1/2}$, attained only at the origin (Proposition 12.1.11). The cumulative mass $M_{1/2}(T)=\int_0^T\rho_{1/2}(\theta)\,d\theta$ is π^{-1} times the argument of ξ along $\operatorname{Re}s=1$, and since no zero lies on or to the right of that line, Proposition 3.3.2 gives the exact unconditional count $N(T)=M_{1/2}(T)+\tau_{1/2}(T)$, where $\tau_{1/2}(T)$, the change of argument of ξ from $1+iT$ to $\frac12+iT$ divided by π, is $O(\log T)$, like $\arg\zeta(\frac12+iT)$ in Section 2.2.1 [348]. Numerically $\int_0^{100}\rho_{1/2}=29.111$ against $N(100)=29$, so $\tau_{1/2}(100)=-0.111$. Table 12.2 lists values of the density and Figure 12.1 shows it with its mass.

θ	$\rho_{1/2}(\theta)$ from ξ'/ξ	from the arithmetic form	from 1700 zeros, $\beta=\frac12$
0	0.0073515925	0.0073515925	0.0071935
5	0.0084053053	0.0084053053	0.0082472
14.134725	0.6473699273	0.6473699273	0.6472119
20	0.1418590769	0.1418590769	0.1417010
30	0.4070956035	0.4070956035	0.4069375
100	0.2308349819	0.2308349819	0.2306766

Table 12.2: The Thorin density $\rho_{1/2}=v_1/\pi$ of T_1 at the basepoint one, computed from ξ'/ξ, from the arithmetic form $v_1(\theta)=\frac{1}{1+\theta^2}-\frac12\log\pi+\frac12\operatorname{Re}\psi(\frac{1+i\theta}{2})+\operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$ with ζ'/ζ evaluated exactly, and from the Poisson sum over the first 1700 zeros with $\beta=\frac12$. The zero sum misses the zeros above height 2198, whose contribution is a nearly constant 1.6×10^{-4}.

Because $\xi'(1)\neq0$, the Laplace transform of T_1 is not differentiable at the origin:

$$\varphi_1(u)=1-\frac{\xi'}{\xi}(1)\sqrt{u}+O(u)=1-\Big(\sum_\rho\frac1\rho\Big)\sqrt{u}+O(u),\qquad u\downarrow0.$$

By the Tauberian theorem for Laplace transforms [86, XIII.5], T_1 has infinite mean and

$$P(T_1>x)\sim\frac{\sum_\rho\rho^{-1}}{\Gamma(\frac12)}\,x^{-1/2}=0.01303\,x^{-1/2},\qquad x\to\infty.\tag{12.1.3}$$

Equivalently, near the origin the Thorin density in z behaves like $\rho_{1/2}(0)/(2\sqrt{z})$, so $\int U_1(dz)/z$ diverges. At the centre, by contrast, $\xi'(\frac12)=0$ and the clock has

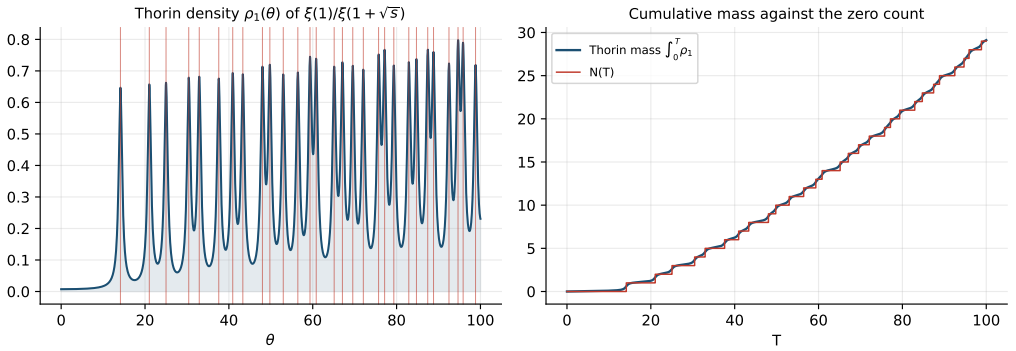


Figure 12.1: Left: the Thorin density $\rho_{1/2}(\theta) = v_1(\theta)/\pi$ of T_1 on $[0, 100]$, with the ordinates of the zeros as thin vertical lines. Right: the cumulative Thorin mass $\int_0^T \rho_{1/2}$ against the zero count $N(T)$. The plot writes ρ_1 for $\rho_{1/2}$ and s for u.

the finite mean $\mathbb{E}T = \sum_k a_k^{-1} = 0.0231050$, which is $\sum_{\gamma>0} \gamma^{-2}$ under RH; at the basepoint one the constant $\sum_\rho \rho^{-1} = 0.0230957$ multiplies $\sqrt{u}$.

12.1.3 The sine squared lemma at the basepoint one

At a basepoint $\alpha > 1$ the sine squared lemma (2.1.9), applied to the canonical measure of $\log \xi$, writes v_α as a drift plus an archimedean, a pole and a prime piece, (1.3.1). The pole piece, the image of $e^{-(\alpha-1)x}dx$, enters with a minus sign; the prime piece is positive. As $\alpha \downarrow 1$ both diverge, because $\sum \Lambda(n)/n$ diverges (Section 1.3.2), so at the basepoint one the lemma applies only to their difference. Throughout Section 12.1, $\psi(y) = \sum_{n \le y} \Lambda(n)$ is Chebyshev's function when its argument is the real variable $y \ge 1$ or the prime cutoff X, and $\psi = \Gamma'/\Gamma$ is the digamma function at every other argument, as in $\psi(\frac{3}{2})$ or $\psi(\frac{1+i\theta}{2})$; from here on X denotes the cutoff, not the Pólya variable of Section 12.1.1.

Proposition 12.1.3. *Let $\nu_1 = \sum_n \Lambda(n) n^{-1} \delta_{\log n} - \mathbf{1}_{(0,\infty)}(x)\, dx$, the prime measure compensated by Lebesgue measure. For every $\theta \neq 0$,*

$$v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+i\theta}{2}\Big) + \gamma_E + \lim_{X\to\infty} \int_0^{\log X} 2\sin^2\Big(\frac{\theta x}{2}\Big)\nu_1(dx), \tag{12.1.4}$$

that is, the limit of $\sum_{n\le X} \Lambda(n) n^{-1} 2\sin^2(\frac{1}{2}\theta \log n) - \big(\log X - \sin(\theta \log X)/\theta\big)$. The limit exists by the prime number theorem, and at $\theta = 0$ the formula reduces to $v_1(0) = \sum_\rho \rho^{-1}$.

Proof. On the line $\operatorname{Re} s = 1$ the pole term $1/(s-1) = 1/(i\theta)$ is purely imaginary, so $v_1(\theta) = \operatorname{Re}\frac{1}{1+i\theta} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$. For $\operatorname{Re} s \ge 1$ with

$s \neq 1$, partial summation gives

$$-\sum_{n\le X}\frac{\Lambda(n)}{n^s}+\int_1^X y^{-s}dy=-\frac{\psi(X)-X}{X^s}-s\int_1^X(\psi(y)-y)\,y^{-s-1}dy-1,$$

and the prime number theorem with the de la Vallée Poussin error term, $\psi(y)-y=O(y\exp(-c\sqrt{\log y}))$, makes the boundary term vanish and the integral converge absolutely as $X\to\infty$, uniformly for $\operatorname{Re}s\ge 1$. For $\operatorname{Re}s>1$ the limit is $\zeta'(s)/\zeta(s)+1/(s-1)$, and by continuity the same holds on $\operatorname{Re}s=1$, $s\neq 1$. Taking real parts at $s=1+i\theta$, where $\operatorname{Re}\int_1^X y^{-1-i\theta}dy=\sin(\theta\log X)/\theta$, gives $\operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)=\lim_X\big[-\sum_{n\le X}\Lambda(n)n^{-1}\cos(\theta\log n)+\sin(\theta\log X)/\theta\big]$. Writing $-\cos=2\sin^2-1$ and using $\sum_{n\le X}\Lambda(n)/n=\log X-\gamma_E+o(1)$, again a consequence of the prime number theorem, gives (12.1.4), since $\int_0^{\log X}2\sin^2(\theta x/2)dx=\log X-\sin(\theta\log X)/\theta$. At $\theta=0$ the bracket vanishes and $1-\frac12\log\pi+\frac12\psi(\frac12)+\gamma_E=1+\frac{\gamma_E}{2}-\frac12\log 4\pi$. □

So the Thorin density comes from the sine squared lemma, but its positivity does not: ν_1 is signed, so its image is not termwise positive. Positivity comes from the zero side, through $\beta<1$ (Proposition 12.1.2), and written arithmetically it is the inequality

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(1+it)<\frac{1}{1+t^2}-\frac12\log\pi+\frac12\operatorname{Re}\psi\Big(\frac{1+it}{2}\Big),$$

whose right side grows like $\frac12\log(t/2\pi)$: the inequality of Hadamard and de la Vallée Poussin behind the prime number theorem [348].

Proposition 12.1.4 (de la Vallée Poussin in Thorin form). *For $\alpha>1$ let $c_\alpha=-\frac{\zeta'}{\zeta}(\alpha)$ and let $Z(\alpha,y)=\sum_{n\ge2}\Lambda(n)n^{-\alpha}(1-\cos(y\log n))$ be the prime part of the Thorin density $g(\alpha,y)=\operatorname{Re}\frac{\xi'}{\xi}(\alpha+iy)$, so that $Z(\alpha,y)=c_\alpha+g(\alpha,y)-\operatorname{Re}\mathcal{A}(\alpha+iy)$ with $\mathcal{A}$ as in Theorem 23.1.1(ii).*

(i) $0\le Z(\alpha,y)\le 2c_\alpha$ *and* $4Z(\alpha,y)+Z(\alpha,2y)\le 8c_\alpha$.

(ii) If $\beta_0+i\gamma_0$ is a zero, then $\dfrac{4}{\alpha-\beta_0}\le 3c_\alpha+4\operatorname{Re}\mathcal{A}(\alpha+i\gamma_0)+\operatorname{Re}\mathcal{A}(\alpha+2i\gamma_0)$. *With $\alpha=1+\frac{1}{4\log\gamma_0}$ this gives $1-\beta_0\ge c/\log\gamma_0$ for $\gamma_0\ge\gamma^*$, with absolute constants $c,\gamma^*>0$.*

Proof. (i) Termwise, $0\le 1-\cos\le 2$, and $3+4\cos\theta+\cos2\theta=2(1+\cos\theta)^2\ge 0$ gives $3c_\alpha+4(c_\alpha-Z(\alpha,y))+(c_\alpha-Z(\alpha,2y))\ge 0$. (ii) For $\alpha>1$ every term $(\alpha-\beta)/|\alpha+iy-\rho|^2$ of g is positive, so $g(\alpha,\gamma_0)\ge 1/(\alpha-\beta_0)$ and $g(\alpha,2\gamma_0)\ge 0$; insert these into (i) through $Z=c_\alpha+g-\operatorname{Re}\mathcal{A}$. Then $c_\alpha=\frac{1}{\alpha-1}+O(1)$ and $\operatorname{Re}\mathcal{A}(\alpha+i\tau)=\frac12\log\tau+O(1)$ for $1<\alpha\le 2$, $\tau\ge 2$, and with $x=(1-\beta_0)\log\gamma_0$ the inequality reads $4/(x+\frac14)\le\frac{29}{2}+o(1)$, so $x\ge\frac{8}{29}-\frac14-o(1)=\frac{3}{116}-o(1)$. □

The sine squared lemma is the lower bound $Z \geq 0$ at one frequency. A zero close to $\operatorname{Re} s = 1$ makes $Z(\alpha, \gamma_0)$ large, and the upper bound in (i), with the two frequencies y and $2y$, excludes it. At $\alpha = \frac{3}{2}$, in floating point over $0 < y \leq 100$, $\max_y Z/(2c_\alpha) = 0.716$ and $\max_y\{4Z(\alpha, y) + Z(\alpha, 2y)\}/(8c_\alpha) = 0.813$.

How far does the lemma alone reach? For $\alpha > 1$ let $D(\alpha) = \frac{\xi'}{\xi}(\alpha) - \frac{1}{\alpha-1} = \frac{1}{2}\psi(1+\frac{\alpha}{2}) - \frac{1}{2}\log\pi + \frac{\zeta'}{\zeta}(\alpha)$, the logarithmic derivative of ξ without the pole.

PROPOSITION 12.1.5. *The function D is increasing and its root is $\alpha_0 = 5.4962215670\ldots$. For every $\alpha \geq \alpha_0$, $u \mapsto \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of a GGC, and the proof uses only the sine squared lemma* (2.1.9)*, with no information on the zeros.*

Proof. With $\xi(s) = (s-1)\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ and $a = \alpha - 1$, write $\xi(\alpha)/\xi(\alpha+s) = \frac{a}{a+s}/G(s)$, where $G(s) = \pi^{-s/2}\Gamma(1+\frac{\alpha+s}{2})\zeta(\alpha+s)/(\Gamma(1+\frac{\alpha}{2})\zeta(\alpha))$. Differentiating $\log\Gamma$ twice, and expanding $\log\zeta$ by the Euler product, gives $\log G(s) = D(\alpha)s + \int_0^\infty (e^{-sx} - 1 + sx)\, e^{-\alpha x}\big(\frac{dx}{e^{2x}-1} + \sum_n \Lambda(n)\delta_{\log n}\big)\frac{1}{x}$, a positive measure. Put $s = \sqrt{u}$. By (2.1.9), the compensated integral makes $\exp(-\cdot)$ the transform of a GGC with positive Thorin density. The pole factor is $\exp(-\log(1+\sqrt{u}/a))$, and $\log(1+\sqrt{u}/a) = \frac{a}{\pi}\int_0^\infty \log(1+u/\theta^2)\frac{d\theta}{a^2+\theta^2}$ (both sides vanish at $u = 0$ and have the derivative $1/(2\sqrt{u}\,(a+\sqrt{u}))$), so it is a GGC factor as well. Finally $e^{-D\sqrt{u}} = \exp(-\frac{D}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta)$ is a GGC transform exactly when $D \geq 0$. The function D increases because ψ increases and $\frac{\zeta'}{\zeta}(\alpha) = -\sum_n \Lambda(n)n^{-\alpha}$ increases, and $D(\alpha) \to \infty$, so $D \geq 0$ exactly for $\alpha \geq \alpha_0$. □

This is weaker than Theorem 2.1.2, which gives the Thorin condition for every $\alpha \geq 1$; its interest is the measure it provides. Below α_0 the factor $e^{|D|\sqrt{u}}$ is not a Laplace transform, and the factorwise use of the lemma stops. Keeping the pole and Gamma factors together reaches lower in floating point, where the condition holds only because the three parts combine.

REMARK 12.1.6. In the variable $y = \sqrt{z}$ the Thorin density of $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ is $\pi^{-1}g(\alpha, y)$ with $g = P + D + A + Z$, where $P = \frac{\alpha-1}{(\alpha-1)^2+y^2}$ comes from the pole, $A = \frac{1}{2}\big(\operatorname{Re}\psi(1+\frac{\alpha+iy}{2}) - \psi(1+\frac{\alpha}{2})\big) \geq 0$ from the Gamma factor, and $Z = \sum_{n\geq 2}\Lambda(n)n^{-\alpha}(1-\cos(y\log n)) \geq 0$ from the primes. Dropping Z leaves $m(\alpha) = \min_y\{P+A\} + D(\alpha)$, which is -0.0133 at $\alpha = 3.5$, 0.0638 at 4, 0.1650 at 5 and 0.2012 at 5.5, and changes sign at $\alpha_1 = 3.5747764\ldots$, with the minimum in y at $y = 3.4506$. Where $m(\alpha) \geq 0$ the Thorin condition holds using only $\Lambda(n) \geq 0$, but this is computed in floating point (20 digits) at sampled points, not proved on $[\alpha_1, \alpha_0)$, where m need not be monotone; a sweep in ball arithmetic is open. No bound on the primes uniform in y goes below α_1, since $\inf_y Z(\alpha, y) = 0$ for every $\alpha > 1$: the $\log p$ are independent over $\mathbb{Q}$, so by Kronecker's theorem, for any finite set of primes, every $\cos(y\log p^k)$ is close to 1 at arbitrarily large y, and the tail of Z is uniformly small. The slice version of this obstruction is in Section 25.2.6.

For every $\alpha > 1$ a march with step $0.9\,g/L$ and a Lipschitz bound $L(\alpha)$ for g on $[0, Y(\alpha)]$, where $A(\alpha, Y) + D(\alpha) = 0$, checks $g > 0$ from values of ζ on $\operatorname{Re} s = \alpha$ alone. Theorem 2.1.2 already gives $\alpha \geq 1$ from the Hadamard product; the march measures the price of avoiding the zeros: $Y(\alpha)$ grows like $e^{2/(\alpha-1)}$, and $\alpha = 1.3$ took 21 931 evaluations (451 s, floating point, no interval arithmetic).

For $F = \mathbb{Q}(\sqrt{-3})$, with $\xi_F(s) = s(s-1)(\sqrt{3}/2\pi)^s\Gamma(s)\zeta_F(s)$ and $\zeta_F = \zeta\, L(\cdot, \chi_{-3})$, the same splitting has the drift $D_F(\alpha) = \psi(1+\alpha) + \log\frac{\sqrt{3}}{2\pi} + \frac{\zeta_F'}{\zeta_F}(\alpha)$, negative on $(1, \alpha_0^F)$ with root $\alpha_0^F = 3.3098807\ldots$, and the merged threshold, in floating point at sampled y as for ζ, is $\alpha_1^F = 2.38426$. The zeros give the Thorin property of $\xi_F(\alpha)/\xi_F(\alpha + \sqrt{u})$ for every $\alpha \geq 1$ (Theorem 16.5.8(a)), and the half plane $\operatorname{Re} s > \frac{7}{8}$ that [257] claims for every Dirichlet L function, which has not been refereed (Section 25.1.2), would give it for every $\alpha > \frac{7}{8}$.

The lemma also acts on the heat trace. Write $p_1 = \int_0^\infty W(t)\,dt = \frac{1}{2}(\log\xi)''(\frac{1}{2}) = 0.0231049931\ldots$, which is $\sum_{\gamma>0}\gamma^{-2}$ under RH, and $F(v) = \xi(\frac{1}{2} + \sqrt{v})/\xi(\frac{1}{2})$. Theorem 8.1.1 gives $\log F(v) = \int_0^\infty (1 - e^{-vt})W(t)\frac{dt}{t}$ with $W > 0$, hence $p_1 v - \log F(v) = \int_0^\infty (e^{-vt} - 1 + vt)W(t)\frac{dt}{t}$.

Proposition 12.1.7. *Unconditionally, for $s \geq 0$,*

$$\Psi(s) := \frac{\xi(\frac{1}{2} + s^{1/4})}{\xi(\frac{1}{2})}\, e^{-p_1\sqrt{s}} = \exp\Big(- \int_0^\infty \log\Big(1 + \frac{s}{z}\Big) U_*(dz)\Big),$$

where

$$U_*(dz) = \frac{dz}{\pi\sqrt{z}} \int_0^\infty \sin^2\Big(\frac{t\sqrt{z}}{2}\Big) W(t)\,dt \geq 0,$$

so Ψ is the Laplace transform of a GGC with a Thorin measure that is absolutely continuous. Under RH, $\Psi(s) = \prod_{\gamma>0}(1 + \sqrt{s}/\gamma^2)\, e^{-\sqrt{s}/\gamma^2}$.

Proof. Apply (2.1.9) with $x = t$ and $s = v$ inside the last integral above, and Tonelli's theorem (the integrand is positive): $p_1 v - \log F(v) = \int_0^\infty \log(1 + v^2/z)\, U_*(dz)$. Put $v = \sqrt{s}$. Under RH $F(v) = \prod_{\gamma>0}(1 + v/\gamma^2)$. □

Remark 12.1.8 (The square root is forced). For every real α, $q \mapsto \xi(\alpha)/\xi(\alpha + q)$ is not $\int_{[0,\infty)} e^{-qt}\mu(dt)$ for a positive measure $\mu \neq 0$ on any half plane. For real $q \to \infty$, $\xi(\alpha + q)$ grows like $\exp(\frac{q}{2}\log q)$, so the function decays faster than e^{-qT} for every T, while $\int e^{-qt}\mu(dt) \geq \mu([0,T])e^{-qT}$; hence $\mu([0,T]) = 0$ for all T. (Landau's theorem [359, Chapter II, Theorem 5b] gives it too: the function has no real singularity, since ξ has no real zeros, so the abscissa would be $-\infty$ and the function entire, but it has poles.) No law on $(0,1]$ has Mellin transform $\xi(\alpha)/\xi(\alpha + q)$, and the clock starts at the variable $u = q^2$.

The construction of Proposition 12.1.7 is explicit for a law of Barnes type. Let $Y_{\mathfrak{b}} \geq 0$ be infinitely divisible with Lévy density $\nu(t) = \operatorname{sech}^2(t/4)/(4t)$, so that

$\mathbb{E}Y_{\mathfrak{b}} = \frac{1}{4}\int_0^\infty \operatorname{sech}^2(t/4)\,dt = 1$, and let $\mathfrak{b} = e^{-Y_{\mathfrak{b}}}$; written as $e^{-t/2}(1-e^{-t/2})^2/(t(1-e^{-t})^2)$, ν is the Lévy density of a Barnes beta law of Ostrovsky [222].

PROPOSITION 12.1.9. *(i) For* $\operatorname{Re} q > -\frac{1}{2}$,

$$\mathbb{E}\mathfrak{b}^q = \frac{\mathsf{G}(q+1)^2\mathsf{G}(\frac{1}{2})\mathsf{G}(\frac{3}{2})}{\mathsf{G}(q+\frac{1}{2})\mathsf{G}(q+\frac{3}{2})} = \frac{2^{2q^2}(2\pi)^{-q}\mathsf{G}(1+q)^4}{\mathsf{G}(1+2q)},$$

with G *the Barnes function;* $\mathbb{E}\mathfrak{b} = 2/\pi$, *and* $\mathbb{E}\mathfrak{b}^q \sim Cq^{-1/4}$. *Since* $t\nu(t) = \sum_{k\ge1}(-1)^{k+1}k\,e^{-kt/2}$, $Y_{\mathfrak{b}}$ *is not a GGC.*

(ii) For $s \ge 0$, $\mathbb{E}\mathfrak{b}^{\sqrt{s}} = e^{-\sqrt{s}}/\Psi_{\mathfrak{b}}(s)$, *where* $\Psi_{\mathfrak{b}}(s) = \exp(-\int_0^\infty \log(1+\frac{s}{z})\,\omega_{\mathfrak{b}}(z)\,dz)$ *is the Laplace transform of a GGC with Thorin density*

$$\omega_{\mathfrak{b}}(z) = \frac{1}{2\pi\sqrt{z}} - \frac{1}{\sinh(2\pi\sqrt{z})} > 0.$$

(iii) For $-\frac{1}{2} < \operatorname{Re} w < \frac{1}{2}$, $\int_0^\infty z^{w-1}\omega_{\mathfrak{b}}(z)\,dz = -4(2\pi)^{-2w}(1-2^{-2w})\Gamma(2w)\zeta(2w)$, *whose zeros in that strip are the points* $\rho/2$ *and* $\pi ik/\log 2$, $k \ne 0$.

Proof. (i) By $\log\Gamma(z) = \int_0^\infty[(z-1)e^{-t} - (e^{-t}-e^{-zt})/(1-e^{-t})]\,dt/t$, both $\log\mathbb{E}\mathfrak{b}^{q+1} - \log\mathbb{E}\mathfrak{b}^q = -\int_0^\infty e^{-qt}(1-e^{-t})\nu(t)\,dt$ and $\log\frac{\Gamma(q+1)^2}{\Gamma(q+\frac{1}{2})\Gamma(q+\frac{3}{2})}$ equal $-\int_0^\infty e^{-qt}\frac{e^{-t/2}(1-e^{-t/2})^2}{t(1-e^{-t})}\,dt$. The first expression for $\mathbb{E}\mathfrak{b}^q$ satisfies the same recursion, by $\mathsf{G}(z+1) = \Gamma(z)\mathsf{G}(z)$, and both sides equal 1 at $q=0$, so their quotient is 1 periodic on $q > -\frac{1}{2}$. Both sides are $q^{-1/4}(C+o(1))$ as $q\to\infty$: the left because $\nu(t) = \frac{1}{4t} + O(1)$ near 0 and decays exponentially, the right by Stirling's formula for G. So the quotient has a limit, is constant, and equals 1; it extends to $\operatorname{Re} q > -\frac{1}{2}$ by analyticity. The second expression follows from the multiplication formula for G at $z = q+\frac{1}{2}$. (ii) The sine squared lemma (2.1.9) with $L(dx) = \nu(x)dx$ gives the Thorin density $\frac{1}{\pi\sqrt{z}}\int_0^\infty \sin^2(\frac{t\sqrt{z}}{2})\frac{\operatorname{sech}^2(t/4)}{4}\,dt$, and $\int_0^\infty \cos(vt)\operatorname{sech}^2(t/4)\,dt = 8\pi v/\sinh(2\pi v)$, from $\int_{\mathbb{R}} e^{i\omega x}\operatorname{sech}^2 x\,dx = \pi\omega/\sinh(\pi\omega/2)$, turns it into $\omega_{\mathfrak{b}}$; positivity is $\sinh x > x$. (iii) With $x = 2\pi\sqrt{z}$ the integral is $2(2\pi)^{-2w}\int_0^\infty x^{2w-1}(\frac{1}{x} - \frac{1}{\sinh x})\,dx$, and $\int_0^\infty x^{s-1}(\frac{1}{\sinh x} - \frac{1}{x})\,dx = 2(1-2^{-s})\Gamma(s)\zeta(s)$ for $-1 < \operatorname{Re} s < 1$, the continuation of $\int_0^\infty x^{s-1}\,dx/\sinh x = 2(1-2^{-s})\Gamma(s)\zeta(s)$ past $s=1$. In the strip $\Gamma(2w)$ has no zeros, its pole at $w=0$ cancels against $1-2^{-2w}$, and $\zeta(2w)$ vanishes only at $2w = \rho$. □

Numerically (25 digits), $\log\mathbb{E}\mathfrak{b} = -0.4515827052894548647$ from the Lévy integral and from both closed forms, and (ii) holds at $s = 10$ and 100 to 10^{-14}. The GGC $\Psi_{\mathfrak{b}}$ is a ratio of Barnes values; ζ enters only through the Mellin transform of its Thorin density, in Riemann's kernel $\frac{1}{x} - \frac{1}{\sinh x}$, which places no constraint

on the zeros. The clock is not of this type: here $t\nu(t)$ is bounded near 0, while $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ (Section 8.1).

The lemma runs outward only: a positive Lévy density in v gives a Thorin measure in $\sqrt{v}$, here $v = \sqrt{s}$ and the argument $s^{1/4}$ of ξ. At the argument $\sqrt{s}$ the Lévy measure of $\log \xi(\frac{1}{2} + v)$ is the signed Weil distribution, the lemma does not apply, and the Thorin form there is equivalent to RH (Theorem 2.1.2). The continuation of Ψ to $\mathbb{C} \setminus (-\infty, 0]$ through the integral is zero free, which says $\xi(\frac{1}{2} + w) \neq 0$ for $|\arg w| < \frac{\pi}{4}$, an empty statement, since every zero has $w = \rho - \frac{1}{2}$ with $|\arg w - \frac{\pi}{2}| \leq \arctan(1/2\gamma_1)$.

12.1.4 The Lévy form of $\xi(1+s)$

Proposition 12.1.10 (Lévy form). *For* $\operatorname{Re} s > 0$,

$$\log \frac{\xi(1+s)}{\xi(1)} = b_1 s + \int_0^\infty \left(e^{-sx} - 1 + sx\right) \Pi_1(dx), \tag{12.1.5}$$

with drift $b_1 = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{3}{2}) + \gamma_E = \sum_\rho \rho^{-1} = 0.0230957089661$ *and the signed Lévy measure*

$$\Pi_1(dx) = \Big(\frac{e^{-3x}}{1 - e^{-2x}} - 1\Big)\frac{dx}{x} + \sum_{n \geq 2} \frac{\Lambda(n)}{n \log n}\, \delta_{\log n}(dx),$$

the integral against the last two pieces being taken as $\lim_{X\to\infty} \int_0^{\log X}$.

Proof. Write $\xi(z) = (z-1)\pi^{-z/2}\Gamma(1 + \frac{z}{2})\zeta(z)$, so that $\xi(1+s) = \pi^{-(1+s)/2}\Gamma(\frac{3}{2} + \frac{s}{2})\, s\zeta(1+s)$ and $\xi(1) = \frac{1}{2}$. The factor $\pi^{-s/2}$ contributes $-\frac{s}{2}\log\pi$. Binet's formula [256, §5.9], $\log\Gamma(a+w) - \log\Gamma(a) = w\psi(a) + \int_0^\infty (e^{-wt} - 1 + wt)\frac{e^{-at}}{t(1-e^{-t})}dt$, with $a = \frac{3}{2}$, $w = \frac{s}{2}$ and $t = 2x$, contributes $\frac{s}{2}\psi(\frac{3}{2})$ and the density $e^{-3x}/(x(1 - e^{-2x}))$. For the last factor, the Euler product gives $\log\zeta(1+s) = \sum_n \frac{\Lambda(n)}{n\log n} e^{-s\log n}$ for $\operatorname{Re} s > 0$, and Frullani's integral gives $\log s = \int_0^\infty (e^{-x} - e^{-sx})\frac{dx}{x}$. Their sum is $\log(s\zeta(1+s))$, which is analytic at $s = 0$ with value 0 and derivative γ_E, and subtracting the linear term puts it in the compensated form $\gamma_E s + \int (e^{-sx} - 1 + sx)\, x^{-1}\nu_1(dx)$, where $x^{-1}\nu_1(dx) = \sum \frac{\Lambda(n)}{n\log n}\delta_{\log n} - \frac{dx}{x}$. The compensated integral converges as $X \to \infty$ because the prime atoms and the measure dx/x cancel at first order, by the prime number theorem in the form $\sum_{n \leq X} \Lambda(n)/n = \log X - \gamma_E + o(1)$, and at zeroth order, by the logarithm of Mertens' product theorem [106, Theorem 429], $\sum_{p^k \leq X} 1/(kp^k) = \log\log X + \gamma_E + o(1)$, which does not need the prime number theorem (the logarithm of the product also contains the terms with $p \leq X < p^k$, whose sum is $O(X^{-1/2})$). The drift is $\frac{d}{ds}\log\xi(1+s)|_{s=0} = \frac{\xi'}{\xi}(1)$. □

The variable $V = V_1$ of Section 12.1.1 has $\mathbb{E}e^{sV} = \xi(1+s)/\xi(1)$, so (12.1.5) has the form of a Lévy Khintchine formula for the moment generating function of V,

with Π_1, reflected by $x \mapsto -x$, as Lévy measure. That measure is signed: beyond $x = 0.28120$ its absolutely continuous part is negative and the only positive mass is the prime atoms, so the uniqueness theorem for Lévy triplets does not apply to Π_1 directly. Nor is Π_1 the signed Lévy measure of a quasi infinitely divisible law in the sense of Lindner, Pan and Sato [213], a law whose characteristic function is the quotient of two infinitely divisible ones: that would require the positive and negative parts of Π_1 to be Lévy measures, of finite mass away from the origin, while on $(1, \infty)$ the positive part is the prime atoms and the negative part exceeds $0.94\,dx/x$, both of infinite mass, cancelling only conditionally, which is why (12.1.5) holds only as a compensated limit. Proposition 14.1.3(3) proves that no other signed measure represents the characteristic function of V, so V is not quasi infinitely divisible, hence not infinitely divisible. The left panel of Figure 12.2 shows the canonical measure $x\Pi_1(dx)$.

In the variable $\theta = \sqrt{z}$ the sine squared lemma (2.1.9) reads

$$e^{-sx} + sx - 1 = \frac{2x}{\pi} \int_0^\infty \log\Big(1 + \frac{s^2}{\theta^2}\Big) \sin^2\Big(\frac{x\theta}{2}\Big) d\theta, \qquad \operatorname{Re} s > 0,\ x > 0,$$

and the drift converts through $\int_0^\infty \log(1 + s^2/\theta^2)\, d\theta = \pi s$. Applying both to (12.1.5) and substituting $s = \sqrt{u}$ gives $\log \varphi_1(u) = -\int_0^\infty \log(1 + u/\theta^2)\rho_{1/2}(\theta)\, d\theta$ with

$$v_1(\theta) = b_1 + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)\, x\Pi_1(dx) = b_1 + A(\theta) + Q(\theta), \tag{12.1.6}$$

where the drift b_1 is the stable part, $A(\theta) = \frac{1}{2}\big[\operatorname{Re}\psi(\frac{3+i\theta}{2}) - \psi(\frac{3}{2})\big]$ is the gamma piece and $Q(\theta) = \lim_{X\to\infty} \int_0^{\log X} 2\sin^2(\frac{x\theta}{2})\, \nu_1(dx)$ is the compensated prime piece. This is Proposition 12.1.3 again: the two constant terms $-\frac{1}{2}\log\pi + \gamma_E$ and $\frac{1}{2}\psi(\frac{3}{2})$ combine into b_1, and $\frac{1}{2}\operatorname{Re}\psi(\frac{3+i\theta}{2}) = \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \frac{1}{1+\theta^2}$ accounts for the term $1/(1+\theta^2)$.

The conversion has three readings, visible in the right panel of Figure 12.2. First, the drift $b_1 = \sum_\rho \rho^{-1}$ becomes the constant Thorin density b_1/π of the positive stable one half factor $e^{-b_1\sqrt{u}}$, so φ_1 is $e^{-b_1\sqrt{u}}$ times the transform of a GGC with Thorin density $(A + Q)/\pi$ (Proposition 12.1.11). The stable factor plays the role of the stable tilt for the floor of Section 10.2.4; at $u = (2k-1)^2$ the factorisation splits each coefficient $1/\zeta(2k)$ of Riesz's function (13.1.5), up to an explicit factor, into a stable value and a GGC value (Section 13.2.3). Second, the gamma factor contributes $A(\theta) \geq 0$, smooth and growing like $\frac{1}{2}\log(\theta/2)$, the average density of zeros. Third, the primes and the pole contribute $Q(\theta)$, which takes both signs, between -1.47 and 1.04 on $[0, 100]$ and carries the peaks at the ordinates.

The first reading needs $A + Q \geq 0$, that is, $v_1 \geq b_1$, at every height.

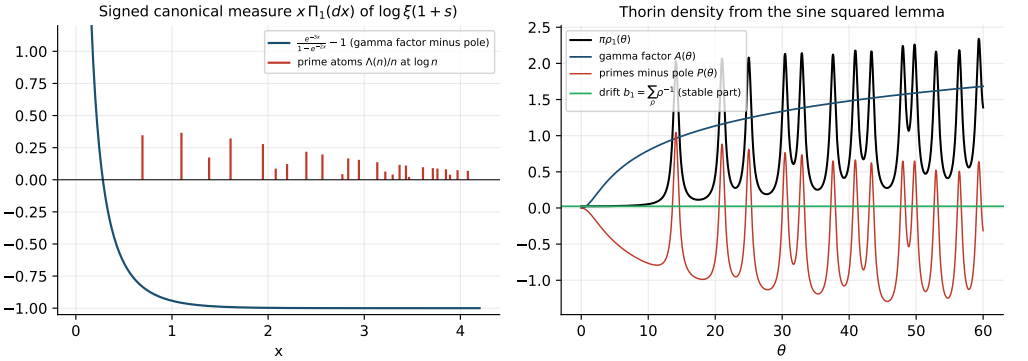


Figure 12.2: Left: the signed canonical measure $x\Pi_1(dx)$ of $\log\xi(1+s)$, made of the gamma density minus the pole, $\big(\frac{e^{-3x}}{1-e^{-2x}}-1\big)dx$, which is negative beyond $x=0.28120$, and the prime atoms $\Lambda(n)/n$ at $\log n$ for $n\le 60$. Right: the bracket $v_1(\theta)=\pi\rho_{1/2}(\theta)$ on $[0,60]$ assembled by the sine squared lemma from the drift b_1, the gamma piece $A(\theta)$ and the compensated prime piece $Q(\theta)$; the plot writes ρ_1 for $\rho_{1/2}$ and $P(\theta)$ for $Q(\theta)$.

Proposition 12.1.11. *With no hypothesis on the zeros, $v_1(\theta)>v_1(0)=b_1$ for every real $\theta\neq 0$; the proof uses a computation of the zeros below height 2000 in ball arithmetic and, above that height, explicit bounds for the number of zeros up to a given height [108, 9]. Consequently $T_1\overset{d}{=}S+T_1^\circ$, where S and T_1° are independent, S is positive stable of index one half with $\mathbb{E}e^{-uS}=e^{-b_1\sqrt{u}}$, and T_1° is a GGC with no drift whose Thorin density in the chart $\theta=\sqrt{z}$ is $(v_1-b_1)/\pi=(A+Q)/\pi$.*

Proof. By (12.1.2), $v_1(\theta)=\pi\sum_\rho P_{1-\beta}(\theta-\gamma)$ is a sum of positive terms, with $\pi P_a(x)=a/(a^2+x^2)$, and v_1 is even, because ξ is real on the real axis. We treat three ranges of $\theta\ge 0$.

On $[0,13.5]$ we use convexity. Every zero has $|\gamma|\ge\gamma_1=14.134725$: the certified zeros of the next step, computed in order from the first, show that none has $0<|\gamma|<\gamma_1$, and none is real, because $\zeta(\sigma)<0$ for $0<\sigma<1$. Since $\pi P_a''(x)=2a(3x^2-a^2)/(a^2+x^2)^3$, the kernel P_a is strictly convex where $|x|>a/\sqrt{3}$, and for $|\theta|\le 13.5$ every zero has $|\theta-\gamma|\ge\gamma_1-13.5>0.63>1/\sqrt{3}>(1-\beta)/\sqrt{3}$. So v_1 is strictly convex on $[-13.5,13.5]$, and, being even, it has there its unique minimum at $\theta=0$ and increases on $[0,13.5]$. By (2.1.2), differentiated twice,

$$v_1''(0)=-\Big(\frac{\xi'}{\xi}\Big)''(1)=\sum_\rho\frac{2(1-\beta)\big(3\gamma^2-(1-\beta)^2\big)}{\big((1-\beta)^2+\gamma^2\big)^3}>0.$$

On $[13.5,2000]$ we use the nearest zero. The first 1800 zeros are computed in ball arithmetic with Arb [169], which isolates them in order by Gram's law and

Rosser's rule (Section 2.2.1), each applied only in the range where it is known to hold [47, Section 3], so that none is missed. They reach height 2304.3, all are simple and on the critical line, and consecutive ordinates differ by at most $\gamma_2 - \gamma_1 < 6.8874$, so every $\theta \in [13.5, 2000]$ lies within 3.4437 of an ordinate, and a zero on the critical line at distance x contributes $\pi P_{1/2}(x) = \frac{1}{2}/(\frac{1}{4} + x^2)$, which exceeds b_1 for $x < 4.6259$. All other terms being positive,

$$v_1(\theta) \geq \frac{1/2}{1/4 + 3.4437^2} > 0.041, \qquad 13.5 \leq \theta \leq 2000.$$

For $\theta \geq 2000$ we count zeros. Write $N(T) = \vartheta(T)/\pi + 1 + S(T)$ (Section 2.2.1). Hasanalizade, Shen and Wong [108, (1.7) and Corollary 1.4] give $|S(T)| \leq 2.5167$ for $0 \leq T \leq X_0 = 30\,610\,046\,000$, from Platt's database of the zeros, and $|S(T)| \leq 0.1095 \log T + 0.2042 \log\log T + 3.0305$ for every $T \geq e$, and the smooth part $\vartheta(T)/\pi + 1$ of $N(T)$ differs from $\frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}$ by at most $\frac{1}{50T}$ for $T \geq \frac{5}{7}$ [108, (2.7)]. So

$$\begin{aligned}\Big|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac{7}{8}\Big| &\leq E(T) \\ &= \begin{cases} 2.5167 + \frac{1}{50T}, & e \leq T \leq X_0, \\ 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T}, & T > X_0, \end{cases}\end{aligned}$$

and $E(T) < 0.3 \log T$ for $T > X_0$, since this holds at X_0 and $0.3 - 0.1095 - 0.2042/\log X_0 > 0$. The bound $0.112\log T + 0.278\log\log T + 2.510 + 0.2/T$ of Trudgian [351], which is often quoted, has an incomplete proof [24, 108]. Put $H = 15$. The window $(\theta - H, \theta + H]$ holds $N(\theta + H) - N(\theta - H)$ zeros, counted with multiplicity, and since the main term has derivative $\frac{1}{2\pi}\log\frac{T}{2\pi}$, their number is at least

$$n(\theta) = \frac{H}{\pi}\log\frac{\theta - H}{2\pi} - E(\theta - H) - E(\theta + H).$$

While $\theta + H \leq X_0$, n increases, because E decreases there, and $n(2000) > 22.44$; for $\theta + H > X_0$, both values of E are below $0.3\log(\theta + H)$, and $n(\theta) > \frac{H}{\pi}\log\frac{\theta - H}{2\pi} - 0.6\log(\theta + H) > 90.$ So for every $\theta \geq 2000$ the window holds at least 23 zeros. Since $\xi(1 - \bar{s}) = \overline{\xi(s)}$, the map $\rho \mapsto 1 - \bar{\rho}$ sends each zero to a zero of the same multiplicity; it keeps the ordinate and exchanges β and $1 - \beta$, so it permutes the zeros of the window, and the sum of $1 - \beta$ over them is half their number. Each of them has $0 < 1 - \beta < 1$ and $|\theta - \gamma| \leq H$, so its term $(1 - \beta)/((1 - \beta)^2 + (\theta - \gamma)^2)$ in $v_1(\theta)$ exceeds $(1 - \beta)/(1 + H^2)$, and all other terms are positive. Hence

$$v_1(\theta) > \frac{23}{2(1 + H^2)} = \frac{23}{452} > 0.05, \qquad \theta \geq 2000,$$

wherever the zeros lie. Backlund's classical constants, valid for $T \geq 200$ [9], give in the same way the increasing bound $E_B(T) = 0.137\log T + 0.443\log\log T + 4.35 +$

$0.2/T$ in place of E, and at least $\frac{H}{\pi}\log\frac{\theta-H}{2\pi} - 2E_B(\theta+H)$ zeros in the window, which increases with θ and exceeds 14.89 at $\theta = 2000$; so the window holds at least 15 zeros, and the bound becomes $15/452 > 0.033$.

It remains to split φ_1. Since $\frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta = \sqrt{u}$, Proposition 12.1.2 gives

$$\varphi_1(u) = e^{-b_1\sqrt{u}} \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\big(v_1(\theta)-b_1\big)\,d\theta\Big\}, \qquad u \ge 0,$$

where the integral is finite because $0 \le v_1 - b_1 \le v_1$ and $\int_0^\infty \log(1+u/\theta^2)\,v_1(\theta)\,d\theta = -\pi\log\varphi_1(u)$ is finite by Theorem 2.1.2 at $\alpha = 1$. The image U_1° of $\pi^{-1}(v_1 - b_1)\,d\theta$ under $\theta \mapsto \theta^2 = z$ is a positive measure with $\int \log(1+u/z)\,U_1^\circ(dz) < \infty$, which is equivalent to the integrability condition in (4.1.1), so the second factor is the Laplace transform of a GGC T_1° with no drift. Hence $T_1 \stackrel{d}{=} S + T_1^\circ$, and $v_1 - b_1 = A + Q$ by (12.1.6). □

The stable summand carries the infinite mean and the whole tail (12.1.3), $P(S > t) \sim b_1/\sqrt{\pi t}$, while T_1° has finite mean and infinite variance. The Laplace exponent of T_1° is $\log(\xi(1+\sqrt{u})/\xi(1)) - b_1\sqrt{u} = \frac{1}{2}(\log\xi)''(1)\,u + O(u^{3/2})$ as $u \downarrow 0$, so that

$$\mathbb{E}T_1^\circ = \frac{1}{2}(\log\xi)''(1) = -\frac{1}{2}\sum_\rho \rho^{-2} = 0.0230772,$$

and its Thorin density $(v_1 - b_1)/\pi$ vanishes at the origin to exactly second order, since v_1 is even and $v_1''(0) > 0$; for instance $v_1(1) - b_1 = 1.12\times10^{-4}$ and $v_1(5) - b_1 = 3.31\times10^{-3}$. Near $z = 0$ the Thorin measure U_1° therefore has a density of exact order $\sqrt{z}$, and the variance $\int z^{-2}\,U_1^\circ(dz)$ is infinite.

For every $\alpha \ge 1$ the same construction gives the Lévy form of $\log(\xi(\alpha+s)/\xi(\alpha))$ with drift $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and

$$x\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)dx + \sum_n \frac{\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx),$$

$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)x\Pi_\alpha(dx),$$

which for $\alpha > 1$ is (1.3.1). For $\alpha > 1$ every piece converges, and the pole contributes the finite negative term $-\theta^2/((\alpha-1)((\alpha-1)^2+\theta^2))$. Expanding $1/(e^{2x}-1) = \sum_{k\ge1} e^{-2kx}$ in the archimedean piece of (1.3.1), and writing $q_c(\theta) = \int_0^\infty (1-\cos\theta x)e^{-cx}\,dx = \theta^2/(c(c^2+\theta^2))$, gives

$$v_\alpha(\theta) - b_\alpha = \sum_{k\ge1} q_{\alpha+2k}(\theta) - q_{\alpha-1}(\theta) + \sum_{n\ge2}\frac{\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big), \qquad \alpha > 1. \tag{12.1.7}$$

Only the pole term is negative. Near the origin $q_c(\theta) \sim \theta^2/c^3$, and the pole term outweighs the gamma terms when $\alpha < 7.789$, where $\frac{1}{8}\zeta(3, \frac{\alpha}{2}+1) = (\alpha-1)^{-3}$; below that basepoint the primes are needed to keep $v_\alpha - b_\alpha$ positive near the origin.

PROPOSITION 12.1.12. *With no hypothesis on the zeros, $v_\alpha(\theta) > v_\alpha(0) = b_\alpha$ for every $\alpha \geq 1$ and every real $\theta \neq 0$; the proof uses the zeros below height* 2000 *of Proposition 12.1.11, the explicit bounds of [108] for the number of zeros above that height, and a computation in ball arithmetic. Consequently, for every $\alpha \geq 1$, $T_\alpha \stackrel{d}{=} S_\alpha + T_\alpha^\circ$, where S_α and T_α° are independent, S_α is positive stable of index one half with $\mathbb{E}e^{-uS_\alpha} = e^{-b_\alpha\sqrt{u}}$, which is the law of the first passage time of Brownian motion to the level $b_\alpha/\sqrt{2}$, and T_α° is a GGC with no drift, with Thorin density $(v_\alpha - b_\alpha)/\pi$ in the chart $\theta = \sqrt{z}$ and mean $\mathbb{E}T_\alpha^\circ = \frac{1}{2}(\log\xi)''(\alpha)$.*

Proof. At $\alpha = 1$ this is Proposition 12.1.11, so let $\alpha > 1$. Since v_α is even, it suffices to take $\theta > 0$. A zero $\rho = \beta + i\gamma$ contributes $a/(a^2 + (\theta-\gamma)^2)$ to $v_\alpha(\theta)$, with $a = \alpha - \beta \in (\alpha-1, \alpha)$, and we put $\theta_0(\alpha) = \gamma_1 - \alpha/\sqrt{3}$. Four steps cover all $\alpha > 1$.

Near the origin, for $\alpha \leq 9$, we use convexity. If $0 < \theta \leq \theta_0(\alpha)$, every zero has $|\theta - \gamma| \geq \gamma_1 - \theta \geq \alpha/\sqrt{3} > a/\sqrt{3}$, so, as in the proof of Proposition 12.1.11, v_α is strictly convex on $[-\theta_0(\alpha), \theta_0(\alpha)]$, and $v_\alpha(\theta) > v_\alpha(0)$ there. Since $\theta_0(9) = 8.94 > 0$, it remains to treat $\theta > \theta_0(\alpha)$ for $\alpha \leq 9$, and every $\theta > 0$ for $\alpha > 9$.

For $1 < \alpha \leq 3$ we use the zeros. Here $\theta_0(\alpha) \geq \theta_0(3) = 12.40$ and $\gamma_1 - 12.40 < 3.4437$, so by the computation in the proof of Proposition 12.1.11 every $\theta \in [\theta_0(\alpha), 2000]$ lies within 3.4437 of the ordinate of a zero on the critical line, whose term exceeds $r(\alpha) = (\alpha - \frac{1}{2})/((\alpha-\frac{1}{2})^2 + 3.4437^2)$. For $\theta \geq 2000$ the window $(\theta - 15, \theta + 15]$ holds at least 20 zeros, the sum of a over them is $\alpha - \frac{1}{2}$ times their number, and each term exceeds $a/(\alpha^2 + 225)$, so that $v_\alpha(\theta) > w(\alpha) = 20(\alpha - \frac{1}{2})/(\alpha^2 + 225)$. The functions r and w increase on $[1, 3]$, and so does b_α, whose derivative $\sum_\rho (\gamma^2 - a^2)/(a^2+\gamma^2)^2$ is positive for $\alpha < \gamma_1$. It therefore suffices that $r(\alpha_k) > b_{\alpha_{k+1}}$ and $w(\alpha_k) > b_{\alpha_{k+1}}$ at the points $\alpha_k = 1 + k/256$, and in ball arithmetic the smallest margins are 0.0180 and 0.0209.

For $3 \leq \alpha \leq 7$ we use the primes. Adding and subtracting $q_\alpha(\theta)$ in (12.1.7), dropping the prime terms and using $q_c(\theta) < 1/c$ gives

$$\begin{aligned} v_\alpha(\theta) - b_\alpha > L(\alpha,\theta) &= \sum_{k\geq 0} q_{\alpha+2k}(\theta) - \frac{1}{\alpha} - \frac{1}{\alpha-1} \\ &= \frac{1}{2}\Big[\operatorname{Re}\psi\Big(\frac{\alpha+i\theta}{2}\Big) - \psi\Big(\frac{\alpha}{2}\Big)\Big] - \frac{1}{\alpha} - \frac{1}{\alpha-1}, \end{aligned}$$

which increases in θ. Since $|\psi'(z)| \leq \psi'(\operatorname{Re} z)$ and $\psi'(x) \leq 1/x + 1/x^2$ for $x > 0$, $|\partial L/\partial\alpha| \leq M(\alpha) = 1/\alpha + 3/\alpha^2 + 1/(\alpha-1)^2$, which decreases in α. It therefore suffices that $L(\alpha_k, \theta_0(\alpha_{k+1})) > M(\alpha_k)/256$ at the points $\alpha_k = 3 + k/256$ of $[3, 7]$;

the smallest margin is 0.0228, at $\alpha = 7$, and the same check succeeds up to $\alpha = 7.707$.

For $\alpha \geq 7$ the gamma terms suffice. Since $q_c(\theta)$ decreases in c, $\sum_{k\geq 1} q_{\alpha+2k}(\theta) \geq \frac{1}{2}\int_{\alpha+2}^{\infty} q_c(\theta)\,dc = \frac{1}{4}\log(1+\theta^2/(\alpha+2)^2)$, so with $A = \alpha+2$, $B = \alpha-1$ and $s = \theta^2$,

$$v_\alpha(\theta) - b_\alpha \geq \Phi_\alpha(s) = \frac{1}{4}\log\Big(1+\frac{s}{A^2}\Big) - \frac{s}{B(B^2+s)}.$$

Here $\Phi_\alpha(0) = 0$, and $\Phi'_\alpha(s) = 1/(4(A^2+s)) - B/(B^2+s)^2$ is nonnegative exactly when $(B^2+s)^2 \geq 4B(A^2+s)$, the two sides differing by a function that increases in s when $B \geq 2$. For $\alpha \geq 9$ they are in this order already at $s = 0$, because $B^3 - 4A^2 = (\alpha-1)^3 - 4(\alpha+2)^2$ is 28 at $\alpha = 9$ and increases beyond, so $\Phi_\alpha(s) > 0$ for every $s > 0$, and no zero enters. For $7 \leq \alpha \leq 9$ and $\theta > \theta_0(\alpha) \geq \theta_0(9)$, we have $s > s_9 = \theta_0(9)^2 = 79.90$, $6 \leq B \leq 8$ and $A \leq 11$, so $(B^2+s_9)^2 \geq (36+s_9)^2 > 32(121+s_9) \geq 4B(A^2+s_9)$; hence Φ_α increases on $[s_9,\infty)$, where $\Phi_\alpha(s_9) \geq \frac{1}{4}\log(1+s_9/121) - s_9/(6(36+s_9)) > 0.0118$.

The checks of the last three steps are carried out in ball arithmetic with Arb [169]. The factorisation follows from the inequality as in the proof of Proposition 12.1.11, Theorem 2.1.2 giving $\int_0^\infty \log(1+u/\theta^2)\,v_\alpha(\theta)\,d\theta < \infty$, and the mean is the coefficient of u in $\log(\xi(\alpha+\sqrt{u})/\xi(\alpha)) - b_\alpha\sqrt{u} = \frac{1}{2}(\log\xi)''(\alpha)\,u + O(u^{3/2})$. □

As at the basepoint one, $v_\alpha - b_\alpha$ starts as $c(\alpha)\theta^2$, with $c(\alpha) = -\frac{1}{2}(\xi'/\xi)''(\alpha) > 0$, so at every basepoint above one the Thorin density of T_α° vanishes to second order at the origin, T_α° has finite mean and infinite variance, and S_α carries the whole tail $P(T_\alpha > t) \sim b_\alpha/\sqrt{\pi t}$.

More precisely [152], for $\alpha \geq 1$ the Lévy density of T_α° is asymptotic to $c(\alpha)\,t^{-5/2}/(4\sqrt{\pi})$ as $t \to \infty$, and $P(T_\alpha^\circ > t) \sim c(\alpha)\,t^{-3/2}/(6\sqrt{\pi})$. Indeed t times the Lévy density is $\pi^{-1}\int_0^\infty e^{-t\theta^2}(v_\alpha(\theta) - b_\alpha)\,d\theta$, where $v_\alpha(\theta) - b_\alpha = c(\alpha)\theta^2 + O(\theta^4)$ near 0 and v_α grows only logarithmically, so Watson's lemma gives $\pi^{-1}c(\alpha)\sqrt{\pi}/(4t^{3/2})$. The Lévy tail is then asymptotic to $c(\alpha)t^{-3/2}/(6\sqrt{\pi})$, a regularly varying and hence subexponential tail, and for an infinitely divisible law with such a Lévy tail the tail of the law is asymptotic to it [83]. At $\alpha = 2$ the ratio of t times the Lévy density to $c(2)/(4\sqrt{\pi}\,t^{3/2})$ is 1.000372, 1.0000929 and 1.0000232 at $t = 25$, 100 and 400, with $c(2) = 3.25029 \times 10^{-4}$ (computed).

For $\alpha < 1$ the Euler product no longer converges near $\operatorname{Re} s = \alpha$, and the prime atoms $\Lambda(n)n^{-\alpha}$ can be compensated only if the primes are well enough distributed; the route through the Lévy form and the sine squared lemma is unconditional on $\alpha \geq 1$.

12.1.5 Below the basepoint one

The construction of Section 12.1.4 can be written down for any basepoint $\alpha = 1-\varepsilon$ with $0 < \varepsilon < \frac{1}{2}$; the question is whether it converges. Formally only the pole term

changes, and it now grows: with $b_\alpha = \frac{\xi'}{\xi}(\alpha)$,

$$x\Pi_\alpha(dx) = \frac{e^{-(\alpha+2)x}}{1-e^{-2x}}\,dx \;+\; \Big(\sum_n \frac{\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx) - e^{\varepsilon x}\,dx\Big),$$

and the converted Thorin density would be π^{-1} times

$$b_\alpha + A_\alpha(\theta) + \lim_{X\to\infty}\Big[\sum_{n\le X}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big) - \int_0^{\log X} 2\sin^2\Big(\frac{\theta x}{2}\Big)e^{\varepsilon x}dx\Big], \quad (12.1.8)$$

with $A_\alpha(\theta) = \frac{1}{2}\big[\operatorname{Re}\psi(\frac{\alpha+2+i\theta}{2}) - \psi(\frac{\alpha+2}{2})\big]$. The prime piece and the pole piece now each grow like $X^\varepsilon/\varepsilon$, and whether their difference converges is a question about how well $\psi(y)$ is approximated by y.

Proposition 12.1.13. *Let $\Theta = \sup\{\operatorname{Re}\rho\}$ over the nontrivial zeros and let $\frac{1}{2} < \alpha < 1$.*

1. *If $\Theta < \alpha$, the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty (e^{-sx} - 1 + sx)\Pi_\alpha(dx)$ holds for $\operatorname{Re} s > 0$, the compensated limit converging, and the sine squared lemma converts it into* (12.1.8)*, which then equals $v_\alpha(\theta)$ and is positive. The error of the compensated sum with prime cutoff X is $O(X^{\Theta-\alpha+\delta})$ for every $\delta > 0$.*

2. *Conversely, if the Lévy form holds for $\operatorname{Re} s > 0$, then $\xi(\alpha+s)$ has no zeros with $\operatorname{Re} s > 0$, so $\Theta \le \alpha$.*

Proof. If $\Theta < \alpha$, the explicit formula gives $\psi(y) - y = O(y^{\Theta+\delta})$ for every $\delta > 0$ [348]. Then the Stieltjes integral $\int_1^\infty y^{-w}\,d(\psi(y)-y)$, which equals $-\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}$ for $\operatorname{Re} w > 1$, converges after one integration by parts for $\operatorname{Re} w > \Theta$ and provides the analytic continuation there, with tail beyond X of order $X^{\Theta-\operatorname{Re} w+\delta}$. Taking $w = \alpha + s$ with $\operatorname{Re} s \ge 0$ and integrating twice in s from $s = 0$ gives the compensated Lévy form, the gamma and polynomial factors being handled exactly as in Proposition 12.1.10. For fixed θ the sine squared lemma may be applied under the limit, because the compensated measure integrated against $2\sin^2(\theta x/2)$ is a combination of the same convergent Stieltjes integrals at $w = \alpha$ and $w = \alpha \pm i\theta$. Positivity follows from the Poisson form of Proposition 2.1.3, as in Proposition 12.1.2, with widths $\alpha - \beta > 0$. Conversely, suppose the Lévy form holds for $\operatorname{Re} s > 0$. Write ν_α for the bracket in $x\Pi_\alpha$, so that the compensated part of the right side is the limit of $\int_0^{\log X}(e^{-sx} - 1 + sx)\,x^{-1}\nu_\alpha(dx)$; its part over $[0,1]$ is an entire function of s. For real $s > 1 - \alpha$ the integral $\int_1^\infty e^{-sx}x^{-1}\nu_\alpha(dx)$ converges absolutely, so the convergence of the compensated limit at two such values of s gives that of $\int_1^{\log X}\nu_\alpha(dx)$ and of $\int_1^{\log X}x^{-1}\nu_\alpha(dx)$. The function $y \mapsto \int_1^y \nu_\alpha(dx)$ is therefore bounded on $[1,\infty)$, and one integration by parts against it shows that $\int_1^{\log X} e^{-sx}x^{-1}\nu_\alpha(dx)$ converges uniformly on compact subsets of $\operatorname{Re} s > 0$. So the compensated limit converges locally uniformly, the right side of the Lévy form is analytic in $\operatorname{Re} s > 0$, hence so is $\log\xi(\alpha+s)$, and $\xi(\alpha+s)$ cannot vanish there. □

So the extension to $\xi(1-\varepsilon+s)$ exists exactly when ζ has no zeros in $\operatorname{Re} s > 1-\varepsilon$, up to the boundary case $\Theta = 1-\varepsilon$. At the basepoint one the convergence of the compensated sum was the prime number theorem; at $1-\varepsilon$ it is $\psi(y) - y = O(y^{1-\varepsilon+\delta})$, the zero free half plane $\operatorname{Re} s > 1-\varepsilon$; and as $\varepsilon \uparrow \frac{1}{2}$ it is $\psi(y) - y = O(y^{1/2+\delta})$, RH. The Lévy form route is an arithmetic restatement of step 7 of the path in Table 25.1, not a way around it.

We evaluate (12.1.8) with primes up to 10^7 at $\theta = \gamma_1$ and compare it with $v_\alpha(\gamma_1)$ computed directly. On the range of the computation the primes behave as RH predicts, $\psi(y) - y = O(y^{1/2}\log^2 y)$, so the errors should decay like $X^{1/2-\alpha}$, and they do (Figure 12.3, left): fast at $\alpha = 0.9$, slowly at $\alpha = 0.6$, not at all at the centre. At $\alpha = \frac{1}{2}$ and $\theta = \gamma_1$ the compensated sum grows like $\log X$ (Figure 12.3, right), the atom δ_{γ_1} of the RH measure forming as the cutoff increases; away from the zeros it oscillates without converging. The exponent in $X^{\Theta-\alpha}$ measures the largest real part of a zero only up to the computable range of X, so it can never certify Θ.

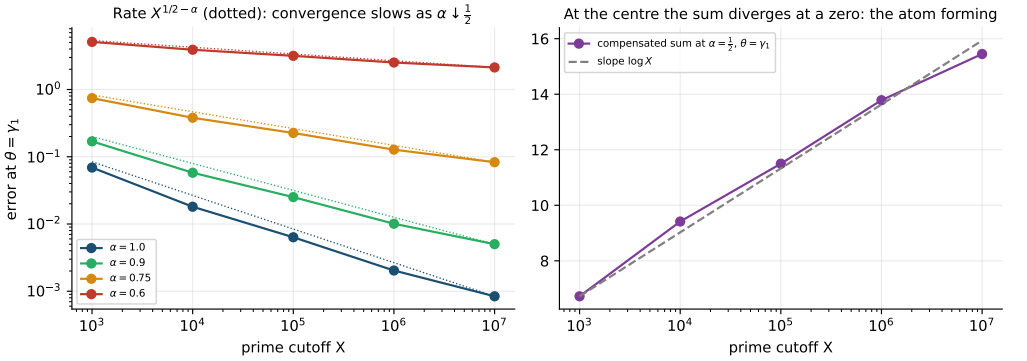


Figure 12.3: Left: absolute error of the compensated formula (12.1.8) at $\theta = \gamma_1$ against the prime cutoff X for $\alpha = 1$, 0.9, 0.75 and 0.6, with the rates $X^{1/2-\alpha}$ (dotted). Right: at the centre, $\alpha = \frac{1}{2}$, the compensated sum at the first ordinate grows like $\log X$ (dashed line), the atom of the RH measure forming.

Writing the tail of the compensated sum beyond X as a Stieltjes integral against $d(\psi(y) - y)$ and inserting the explicit formula $\psi(y) = y - \sum_\rho y^\rho/\rho - \log 2\pi - \frac{1}{2}\log(1-y^{-2})$ [70, Chapter 17] gives, for $\Theta < \alpha$ and X not a prime power,

$$\text{(formula (12.1.8) at cutoff } X) - v_\alpha(\theta) = \sum_\rho \Big[-\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)} \Big] + O(X^{-1-\alpha}), \quad (12.1.9)$$

the sum taken over conjugate pairs. Every term has size $X^{\beta-\alpha}$, which makes the rate $X^{\Theta-\alpha}$ visible term by term, and at $\theta = \gamma$ the third term has denominator

$\beta-\alpha$, so under RH it approaches $X^{1/2-\alpha}/(1-2\alpha)$ and at the centre becomes the $\log X$ growth of Figure 12.3. Table efbp:tab:err, computed, compares the observed error with (12.1.9) truncated to the 1700 zeros below height 2198. The explicit formula accounts for the error to four or five decimal places for $\alpha \geq 0.75$ and to three at $\alpha = 0.6$, and primes up to X plus the first 1700 zeros reduce the error of the primes alone by factors between about 700 and 11000.

α	X	observed error	predicted by (12.1.9)	error after correction
1	10^5	-6.352×10^{-3}	-6.349×10^{-3}	-2.9×10^{-6}
1	10^7	-8.419×10^{-4}	-8.431×10^{-4}	1.2×10^{-6}
0.9	10^7	-5.014×10^{-3}	-5.020×10^{-3}	6.1×10^{-6}
0.75	10^7	-8.297×10^{-2}	-8.304×10^{-2}	6.8×10^{-5}
0.6	10^5	-3.1668	-3.1665	-2.9×10^{-4}
0.6	10^7	-2.1285	-2.1292	7.7×10^{-4}

Table 12.3: The error of the compensated sum (12.1.8) at $\theta=\gamma_1$ with primes up to X, its prediction from the explicit formula (12.1.9) over the first 1700 zeros, and what remains after subtracting the prediction.

This sharpens Proposition 12.1.13: a zero with $\beta>\alpha$ would contribute a term of size $X^{\beta-\alpha}$ growing with the cutoff, so the compensated sum at α would diverge, and the size and phase of that growth would locate the zero. The hybrid formula is a zero detector with the structure of Turing's method (Section 2.2.1): primes up to X and zeros up to height T must balance to within the tail of both.

If instead of compensating one continues the pole contribution analytically, its value at $\alpha=1-\varepsilon$ is $\operatorname{Re}\frac{1}{-\varepsilon+i\theta}+\frac{1}{\varepsilon}=\frac{\theta^2}{\varepsilon(\varepsilon^2+\theta^2)}$, positive, since the pole now lies to the right of the basepoint. The single negative term of (1.3.1) above the basepoint one has become positive: below $\operatorname{Re}s=1$ the negativity that RH must control is carried by the primes, not by the pole.

12.1.6 The minimum at the origin below the basepoint one

Propositions 12.1.11 and 12.1.12 put the minimum of v_α at the origin for every $\alpha\geq1$, with no hypothesis on the zeros. Below the basepoint one the margin at the origin vanishes linearly, $b_\alpha\approx(\alpha-\frac12)(\log\xi)''(\frac12)$, and Table 12.6 found the minimum at the origin on a window by computation. We prove it for every $\alpha\in(\frac12,7.35]$ on the range of heights where the zeros are known to lie on the critical line, up to $\frac{1+\sqrt2}{2}$ below its top, and show that RH is equivalent to it at every height for every $\alpha\in(\frac12,1)$. The tool is a comparison across vertical lines from the first author's note on the Dirichlet beta function [283]. We state it for $f=\xi$, for $f=\xi(\cdot,\chi_4)$ of Section 15.1.7 and for $f=\xi(\cdot,\chi_{12})$ of Section 15.1.1, and write $v_\alpha(\theta)=\operatorname{Re}(f'/f)(\alpha+i\theta)$, $b_\alpha=v_\alpha(0)=(f'/f)(\alpha)$ and $V_f=(\log f)''(\frac12)$.

Each f is entire of order one, real on the real axis, with $f(s)=f(1-s)$ and

no real zeros. Its zeros therefore come in groups: a zero $\frac12+i\gamma$ on the critical line, or a pair $\frac12\pm\delta+i\gamma$ with $0<\delta<\frac12$. At resolution $\varepsilon=\alpha-\frac12$ and offset $u=\theta-\gamma$, a zero on the line contributes $\varepsilon/(\varepsilon^2+u^2)$ to $v_\alpha(\theta)$, and a pair contributes

$$k_\delta(\varepsilon,u)=\frac{\varepsilon-\delta}{(\varepsilon-\delta)^2+u^2}+\frac{\varepsilon+\delta}{(\varepsilon+\delta)^2+u^2}=\frac{2\varepsilon(w+c)}{(w+c)^2+\kappa^2}, \tag{12.1.10}$$

where $w=\varepsilon^2$, $c=u^2-\delta^2$ and $\kappa=2\delta|u|$, since the numerator of the sum is $2\varepsilon(\varepsilon^2+u^2-\delta^2)$ and its denominator is $(\varepsilon^2+\delta^2+u^2)^2-4\varepsilon^2\delta^2=(w+c)^2+\kappa^2$. The sum over the groups converges absolutely, since $\sum_\rho|\rho|^{-2}<\infty$, and $v_\alpha(\theta)/(\alpha-\frac12)$ is the sum of the ratios $1/(w+u^2)$ for the zeros on the line and

$$r_\delta(w,u)=\frac{k_\delta(\varepsilon,u)}{\varepsilon}=2\,\mathrm{Re}\,\frac{1}{w+c+i\kappa}$$

for the pairs, as functions of $w=(\alpha-\frac12)^2$. The ratio of a zero on the line decreases in w; that of a pair need not.

Lemma 12.1.14 (Comparison). *Let $0<\delta<\frac12$ and let u be real. As a function of $w>0$, the ratio $r_\delta(w,u)$*

(a) is nonincreasing on $(0,\infty)$ if $|u|\ge(1+\sqrt2)\delta$;

(b) is nonincreasing on $[2\delta^2,\infty)$ for every u, in particular on $w\ge\frac12$, that is for $\alpha\ge\frac12+\frac{1}{\sqrt2}$, whatever δ;

(c) if $0<|u|<(1+\sqrt2)\delta$, increases exactly on the interval $\max\big(0,2\delta^2-(|u|+\delta)^2\big)<w<2\delta^2-(|u|-\delta)^2$, which begins at 0 if and only if $|u|\ge(\sqrt2-1)\delta$; at $u=0$ it equals $2/(w-\delta^2)$, which has a pole at $w=\delta^2$.

In particular the constant $1+\sqrt2$ in (a) cannot be lowered.

Proof. For $u\ne0$, $\partial_w r_\delta=-2\,\mathrm{Re}(w+c+i\kappa)^{-2}=-2\big((w+c)^2-\kappa^2\big)/\big((w+c)^2+\kappa^2\big)^2$, which is positive exactly when $-\kappa<w+c<\kappa$, that is when $2\delta^2-(|u|+\delta)^2<w<2\delta^2-(|u|-\delta)^2$, since $-\kappa-c=2\delta^2-(|u|+\delta)^2$ and $\kappa-c=2\delta^2-(|u|-\delta)^2$. The right end is at most $2\delta^2$, which gives (b) for $u\ne0$; it is positive exactly when $\big||u|-\delta\big|<\sqrt2\delta$, that is when $|u|<(1+\sqrt2)\delta$, which gives (a) and the first part of (c); and the left end is at most 0 exactly when $|u|+\delta\ge\sqrt2\delta$. At $u=0$, $c=-\delta^2$ and $\kappa=0$, and $2/(w-\delta^2)$ decreases on $w>\delta^2$, which completes (b). Finally $2\delta^2<\frac12$ because $\delta<\frac12$. □

Theorem 12.1.15 (Comparison across vertical lines). *Let $\frac12<\alpha<\alpha'$ and let θ be real. Suppose that every zero $\frac12+\delta+i\gamma$ of f with $\delta>(\alpha-\frac12)/\sqrt2$ satisfies $|\theta-\gamma|\ge(1+\sqrt2)\delta$. Then*

$$\frac{v_\alpha(\theta)}{\alpha-\frac12}\ \ge\ \frac{v_{\alpha'}(\theta)}{\alpha'-\frac12}.$$

The hypothesis holds at every θ under RH for ξ, or GRH for χ_4 or χ_{12}; at every θ when $\alpha \ge \frac12 + \frac{1}{\sqrt2}$, with no hypothesis; and at every θ with $|\theta| \le H_0 - \frac{1+\sqrt2}{2}$, where H_0 is a height below which every zero of f is known to lie on the critical line: $H_0 = 3\times10^{12}$ for ξ [277], and $H_0 = 10^8/q$ for $\xi(\cdot,\chi_q)$, that is 2.5×10^7 for χ_4 and $10^8/12 > 8.33\times10^6$ for χ_{12} [275].

Proof. A zero on the line has the ratio $1/(w+u^2)$, which decreases in w. A pair with $\delta \le (\alpha-\frac12)/\sqrt2$ has $2\delta^2 \le (\alpha-\frac12)^2$, so its ratio is nonincreasing on $[(\alpha-\frac12)^2, (\alpha'-\frac12)^2]$ by Lemma 12.1.14(b), and a pair with $|\theta-\gamma| \ge (1+\sqrt2)\delta$ by Lemma 12.1.14(a). Summing over the groups, which converges absolutely, gives the inequality. Under RH there are no pairs. If $\alpha \ge \frac12 + \frac{1}{\sqrt2}$, every pair has $\delta < \frac12 \le (\alpha-\frac12)/\sqrt2$. If $|\theta| \le H_0 - \frac{1+\sqrt2}{2}$, every zero off the line has $|\gamma| > H_0$, so that $|\theta-\gamma| > \frac{1+\sqrt2}{2} > (1+\sqrt2)\delta$. □

Corollary 12.1.16 (Explicit lower bound). *For $\sigma > 1$ and real θ, $v_\sigma(\theta) \ge A_f^\infty(\sigma,\theta) - S_f(\sigma)$, where*

$$\begin{aligned} A_\xi^\infty(\sigma,\theta) &= -\tfrac12\log\pi + \tfrac12 \operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big),\\ A_{\chi_4}^\infty(\sigma,\theta) &= \tfrac12\log\tfrac{4}{\pi} + \tfrac12 \operatorname{Re}\psi\big(\tfrac{\sigma+1+i\theta}{2}\big),\\ A_{\chi_{12}}^\infty(\sigma,\theta) &= \tfrac12\log\tfrac{12}{\pi} + \tfrac12 \operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big),\\ S_f(\sigma) &= \sum_{(n,q)=1} \frac{\Lambda(n)}{n^\sigma}, \end{aligned}$$

with $q = 1$, 4 and 12 for the three functions, so that $S_\xi = -\zeta'/\zeta$, $S_{\chi_4}(\sigma) = S_\xi(\sigma) - \log 2/(2^\sigma-1)$ and $S_{\chi_{12}}(\sigma) = S_{\chi_4}(\sigma) - \log 3/(3^\sigma-1)$; and $A_f^\infty(\sigma,\theta)$ increases in $|\theta|$. Hence, under the hypothesis of Theorem 12.1.15 with $\alpha' > 1$,

$$\begin{aligned} v_\alpha(\theta) &\ge \frac{\alpha-\frac12}{\alpha'-\frac12}\Big(A_f^\infty(\alpha',\theta) - S_f(\alpha')\Big)\\ &= \frac{\alpha-\frac12}{2\alpha'-1}\log|\theta| + O(1), \qquad |\theta|\to\infty. \end{aligned}$$

Proof. For $\operatorname{Re} s > 1$,

$$\frac{\xi'}{\xi}(s) = \frac1s + \frac{1}{s-1} - \frac12\log\pi + \frac12\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s),$$

where $\operatorname{Re}(\frac1s + \frac{1}{s-1}) \ge 0$ and $\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\theta) = -\sum_n \Lambda(n)n^{-\sigma}\cos(\theta\log n) \ge -S_\xi(\sigma)$; for the two L functions the bound is that of Proposition 15.1.26. The monotonicity in $|\theta|$ follows from the series $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\ge0} y^2/\big((x+k)((x+k)^2+y^2)\big)$, and $\operatorname{Re}\psi(x+iy) = \log|y| + O(1)$ for fixed $x > 0$. Then apply Theorem 12.1.15 between α and α'. □

REMARK 12.1.17 (A bound of order $\log\theta$ outside a sparse set). For fixed $\alpha > \frac{1}{2}$ the hypothesis of Theorem 12.1.15 fails only at heights within $(1+\sqrt{2})\delta < \frac{1+\sqrt{2}}{2}$ of the ordinate of a zero $\frac{1}{2}+\delta+i\gamma$ with $\delta > (\alpha-\frac{1}{2})/\sqrt{2}$. For $f=\xi$ the number of such zeros with $0<\gamma\le T$ is $N(\sigma_0,T)$ with $\sigma_0 = \frac{1}{2}+(\alpha-\frac{1}{2})/\sqrt{2} > \frac{1}{2}$, which is $O(T^{3(1-\sigma_0)/(2-\sigma_0)}\log^5 T) = o(T)$ by Ingham's estimate [163] (Section 14.3.1). So Corollary 12.1.16 with $\alpha' = \alpha+1$ gives $v_\alpha(\theta)/\pi \ge c\log\theta$, for every $c < (\alpha-\frac{1}{2})/(\pi(2\alpha+1))$ and all large θ, outside a set of heights of density zero: the lower bound of the first of the four questions of Section 14.3.7, for v_α/π itself. More directly, by the proof of Corollary 14.3.7, $v_\alpha(\theta)\le 0$ only within $\sqrt{\delta^2-(\alpha-\frac{1}{2})^2} < \frac{1}{2}$ of the ordinate of a zero $\frac{1}{2}+\delta+i\gamma$ with $\frac{1}{2}+\delta>\alpha$, so the set of $\theta\in[0,T]$ with $v_\alpha(\theta)\le 0$ has measure at most $N(\alpha,T+\frac{1}{2}) \le T^{e(\alpha)+o(1)}$, with $e(\alpha)<1$ of Theorem 14.3.2. The exceptional set, however, contains the window of every zero to the right of $\operatorname{Re} s=\alpha$, since such a zero has $\delta > \alpha-\frac{1}{2} > (\alpha-\frac{1}{2})/\sqrt{2}$; so the bound says nothing where Lemma 14.3.9 would need it, and does not by itself confine a failure of positivity to the thin strips of that question.

THEOREM 12.1.18 (Minimum at the origin). *Let V_f, γ_1, H_0 and $\alpha_{\max}$ be as in Table 12.4.*

(i) For every $\alpha>\frac{1}{2}$, $0<b_\alpha\le(\alpha-\frac{1}{2})V_f$; the ratio $b_\alpha/(\alpha-\frac{1}{2})$ decreases in α and tends to V_f as $\alpha\downarrow\frac{1}{2}$.

(ii) For $\frac{1}{2}<\alpha\le\alpha_{\max}$, $v_\alpha(\theta)>b_\alpha$ for $0<|\theta|\le H_0-\frac{1+\sqrt{2}}{2}$.

(iii) For $\frac{1}{2}+\frac{1}{\sqrt{2}}\le\alpha\le\alpha_{\max}$, $v_\alpha(\theta)>b_\alpha$ for every $\theta\ne0$.

(iv) For $f=\xi(\cdot,\chi_4)$ and $f=\xi(\cdot,\chi_{12})$, $v_\alpha(\theta)>b_\alpha$ for every $\alpha\ge\frac{1}{2}+\frac{1}{\sqrt{2}}$ and every $\theta\ne0$.

(v) For $f=\xi(\cdot,\chi_4)$ and $f=\xi(\cdot,\chi_{12})$, $v_\alpha(\theta)>b_\alpha$ for every $\alpha\ge1$ and every $\theta\ne0$.

(vi) RH, respectively GRH for χ_4 or for χ_{12}, holds if and only if $v_\alpha(\theta)>b_\alpha$ for every $\alpha\in(\frac{1}{2},1)$ and every $\theta\ne0$.

Proof. (i) The function $f(\frac{1}{2}+s)=\int_{\mathbb{R}}K(u)e^{su}\,du$ is the two sided Laplace transform of a positive even kernel K: the Pólya kernel Φ for ξ (Section 2.1), the kernel K_4 of Proposition 15.1.22, and the eta kernel of Section 15.1.1. So $\log f(\frac{1}{2}+s)$ is, up to a constant, the cumulant generating function of a nondegenerate law; it is even and strictly convex in real s, and $b_\alpha=(\log f)'(\alpha)=\int_{1/2}^{\alpha}(\log f)''(r)\,dr>0$. At $\theta=0$ the offset of a group with ordinate γ has $|u|=|\gamma|\ge\gamma_1>\frac{1+\sqrt{2}}{2}>(1+\sqrt{2})\delta$, so by Lemma 12.1.14(a) the ratio of every group is nonincreasing in w, strictly for the zeros on the line, and $b_\alpha/(\alpha-\frac{1}{2})$, their sum, decreases in α. As $w\downarrow0$ the ratios increase to γ^{-2} for a zero on the line and to $2\operatorname{Re}(|\gamma|+i\delta)^{-2}$ for a pair,

and by monotone convergence the sum tends to the sum of these limits, which is $-\sum_\rho(\frac12-\rho)^{-2} = (\log f)''(\frac12) = V_f$.

(ii) Put $\alpha' = \alpha_{\max}$ and let t_1 be its value at $\alpha_{\max}$ in Table 12.4, where $A_f^\infty(\alpha', t_1) - S_f(\alpha') > (\alpha'-\frac12)V_f$ is certified in ball arithmetic; since A_f^∞ increases in $|\theta|$, the inequality holds for $|\theta| \ge t_1$. For $t_1 \le |\theta| \le H_0 - \frac{1+\sqrt2}{2}$, Corollary 12.1.16 and (i) give $v_\alpha(\theta) > (\alpha-\frac12)V_f \ge b_\alpha$ (for $\alpha = \alpha'$ the comparison is an identity). For $|\theta| \le t_1$ we show that v_α is strictly convex on $[-t_1, t_1]$; as it is even, it then exceeds $v_\alpha(0) = b_\alpha$ at $0 < |\theta| \le t_1$. The term $\varphi(\theta-\gamma)$, $\varphi(x) = a/(a^2+x^2)$ with $a = \alpha - \frac12$, of a zero on the line is strictly convex where $|\theta-\gamma| > a/\sqrt3$, since $\varphi''(x) = 2a(3x^2-a^2)/(a^2+x^2)^3$. Below H_0 all zeros lie on the line and have $|\gamma| \ge \gamma_1$, and for $|\theta| \le t_1$, $|\theta-\gamma| \ge \gamma_1 - t_1 > (\alpha'-\frac12)/\sqrt3 \ge a/\sqrt3$ by the convexity margin of the table. Every zero above H_0 has $|\theta-\gamma| > H_0 - t_1 > 10^6$. For a pair there, the second derivative in θ of $k_\delta(a, \theta-\gamma)$ is $g(a+\delta, u) + g(a-\delta, u)$ with $g(c,u) = 2c(3u^2-c^2)/(c^2+u^2)^3$, the second derivative of $c/(c^2+u^2)$ in u; since $\partial_c g(c,u) = 6(u^4 - 6c^2u^2 + c^4)/(c^2+u^2)^4 > 0$ for $|c| < (\sqrt2-1)|u|$, and $|a \pm \delta| < \alpha_{\max} < (\sqrt2-1)10^6$, the odd function $g(\cdot, u)$ increases on an interval containing $\pm(a \pm \delta)$, so that $g(a+\delta,u) > g(\delta-a,u) = -g(a-\delta,u)$. The sum of the second derivatives converges uniformly on $[-t_1,t_1]$, so $v_\alpha'' > 0$ there.

(iii) For $\alpha \ge \frac12 + \frac{1}{\sqrt2}$ the hypothesis of Theorem 12.1.15 holds at every θ, so the bound of (ii) holds for every $|\theta| \ge t_1$, and the convexity argument for $|\theta| \le t_1$ is unchanged.

(iv) It remains to treat $\alpha > \alpha_{\max}$, and there we use the primes. For χ_4, $v_\alpha(\theta) - b_\alpha = G_\alpha(\theta) + \varpi_\alpha(\theta)$ by (15.1.9), with $G_\alpha(\theta) = \frac12[\operatorname{Re}\psi(\frac{\alpha+1+i\theta}{2}) - \psi(\frac{\alpha+1}{2})]$; for χ_{12} the same holds with $G_\alpha(\theta) = \frac12[\operatorname{Re}\psi(\frac{\alpha+i\theta}{2}) - \psi(\frac{\alpha}{2})]$ and the weights $\chi_{12}(n)\Lambda(n)$ (Section 15.1.3). Dropping the positive weights and bounding $1-\cos x$ by 2 and by $x^2/2$ gives $\varpi_\alpha(\theta) \ge -R_\alpha(\theta)$ with $R_\alpha(\theta) \le \min\big(2S^-(\alpha), \frac12\theta^2 M_2(\alpha)\big)$, where

$$S^-(\alpha) = \sum_{\chi(n)=-1} \frac{\Lambda(n)}{n^\alpha} = \frac12\Big(S_f(\alpha) + \frac{L'}{L}(\alpha)\Big),$$

$$M_2(\alpha) = \sum_{\chi(n)=-1} \frac{\Lambda(n)\log^2 n}{n^\alpha} = \frac12\Big(S_f''(\alpha) + \Big(\frac{L'}{L}\Big)''(\alpha)\Big).$$

The function $G_\alpha(\theta)$ is a sum of terms $q_c(\theta) = \theta^2/(c(c^2+\theta^2))$, so it increases in $|\theta|$ and decreases in α, and $G_\alpha(\theta)/\theta^2$ decreases in $|\theta|$; and S^- and M_2 decrease in α. So on a cell $s_0 \le \alpha \le s_1$ it suffices that $G_{s_1}(T) > \max\big(2S^-(s_0), \frac12 T^2 M_2(s_0)\big)$ for one $T > 0$: then $G_\alpha(\theta) \ge G_{s_1}(T) > 2S^-(\alpha)$ for $|\theta| \ge T$, and $G_\alpha(\theta) \ge \theta^2 G_{s_1}(T)/T^2 > \frac12\theta^2 M_2(\alpha)$ for $0 < |\theta| \le T$. A computation checks this in ball arithmetic on cells of width $\frac{1}{32}$ covering $[\sigma_2, 20]$, with $T \in \{0.25, 0.5, \dots, 4\}$, where $\sigma_2 = 3.71875$ for χ_4 and 2.8125 for χ_{12}; the cells just below σ_2 fail. For $\alpha \ge 20$ take $T = 2$: since q_c decreases in c, $G_\alpha(2) \ge \frac12\int_{\alpha+1}^\infty q_c(2)\,dc = \frac14\log(1 + 4/(\alpha+1)^2) \ge 1/((\alpha+1)^2+4)$, and since every n with $\chi(n) = -1$ is at least 3, $S^-(\alpha)$ and $M_2(\alpha)$ are at most

$3^{-(\alpha-20)}$ times their values at 20. With $K = \max(2S^-(20), 2M_2(20))$ it therefore suffices that $K((\alpha+1)^2+4)3^{-(\alpha-20)} < 1$ for $\alpha \geq 20$; the left side decreases in α, and at $\alpha = 20$ it is 3.4×10^{-7} for χ_4 and 3.9×10^{-11} for χ_{12}. As $\alpha_{\max} = 4.70 > 3.71875$ for χ_4 and $3.10 > 2.8125$ for χ_{12}, (iii) and this argument cover every $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$.

(v) By (ii) and (iv) it remains to treat $1 \leq \alpha < \frac{1}{2} + \frac{1}{\sqrt{2}}$ and $|\theta| > H_0 - \frac{1+\sqrt{2}}{2}$, and by symmetry $\theta > 0$. Let n be the number of zeros with $|\gamma - \theta| \leq h$, counted with multiplicity, where $h = 15$ for χ_4 and $h = 10$ for χ_{12}. Every zero has $0 < \beta < 1 \leq \alpha$, so every term of $v_\alpha(\theta)$ is positive, and a zero in the window contributes $a/(a^2+(\theta-\gamma)^2) \geq a/(\alpha^2+h^2)$, with $a = \alpha-\beta \in (0,\alpha)$; the window contains both members of each pair $\frac{1}{2}\pm\delta+i\gamma$, whose values of a add up to $2(\alpha-\frac{1}{2})$, so the values of a over the window add up to $n(\alpha-\frac{1}{2})$. Hence $v_\alpha(\theta) > n(\alpha-\frac{1}{2})/(\alpha^2+h^2)$, which is at least $(\alpha-\frac{1}{2})V_f \geq b_\alpha$ as soon as $n \geq V_f(\alpha^2+h^2)$. Since $\alpha^2 < (\frac{1}{2}+\frac{1}{\sqrt{2}})^2 = 1.4571\ldots$, it suffices that $n > 35.335$ for χ_4 and $n > 33.778$ for χ_{12}. By the explicit counting formula of Bennett, Martin, O'Bryant and Rechnitzer [24, Theorem 1.1], for a primitive character χ of conductor $q > 1$, $T \geq \frac{5}{7}$ and $\ell = \log(q(T+2)/(2\pi)) > 1.567$, the number $N(T,\chi)$ of zeros with $|\gamma| \leq T$ satisfies

$$\Big|N(T,\chi) - \frac{T}{\pi}\log\frac{qT}{2\pi e} + \frac{\chi(-1)}{4}\Big| \leq E(T) = 0.22737\,\ell + 2\log(1+\ell) - 0.5.$$

As f has no real zeros and its zeros are symmetric about the real axis, $N(\theta+h,\chi) - N(\theta-h,\chi)$ is at most twice n, whatever the convention for zeros at the ends of the window, and since E increases in T,

$$n \;\geq\; \frac{1}{2\pi}\int_{\theta-h}^{\theta+h}\log\frac{qt}{2\pi}\,dt - E(\theta+h) \;\geq\; \frac{h}{\pi}\log\frac{q(\theta-h)}{2\pi} - E(\theta+h).$$

The right side increases in θ, its derivative being at least $\big(h/\pi - 0.22737 - 2/(1+\ell)\big)/(\theta+h+2)$ with $\ell = \ell(\theta+h) > 16$, which is positive. At $\theta = H_0 - \frac{1+\sqrt{2}}{2}$, where $qH_0 = 10^8$ for both characters, $\log(q(\theta-h)/(2\pi)) = 16.58280\ldots$ and $E(\theta+h) = 9.00428\ldots$, so that $n \geq 79.177 - 9.005 > 70$ for χ_4 and $n \geq 52.785 - 9.005 > 43$ for χ_{12}, well above the 36 and 34 zeros needed. A computation checks these inequalities and the hypotheses of the formula in ball arithmetic.

(vi) Under RH, or GRH, the hypothesis of Theorem 12.1.15 holds at every θ, and the proof of (ii) applies with $H_0 = \infty$ for every $\alpha < 1 < \alpha_{\max}$. Conversely, the condition and (i) give $v_\alpha \geq b_\alpha > 0$ on every line $\operatorname{Re} s = \alpha$ with $\frac{1}{2} < \alpha < 1$, and $v_\alpha > 0$ for $\alpha \geq 1$ (Theorem 2.1.2, Proposition 15.1.25, and Proposition 15.2.1 with Section 15.1.3); so $\operatorname{Re}(f'/f)(s) > 0$ for $\operatorname{Re} s > \frac{1}{2}$, which is RH, or GRH, by Lagarias's criterion [198]. □

For the two L functions, part (v) gives the factorisation of Proposition 12.1.12 at every basepoint $\alpha \geq 1$: for the clock T_α with Laplace transform $f(\alpha)/f(\alpha+\sqrt{u})$, $T_\alpha \stackrel{d}{=} S_\alpha + T_\alpha^\circ$, where S_α and T_α° are independent, S_α is positive stable of index

	ξ	$\xi(\cdot,\chi_4)$	$\xi(\cdot,\chi_{12})$
$V_f=(\log f)''(\frac12)$	0.04620998623	0.1560296499	0.3329236082
first ordinate γ_1	14.1347251	6.0209489	3.8046276
verified height H_0	3×10^{12}	2.5×10^{7}	$10^8/12$
comparison line α' of [283]	4.75	3.15	2.30
t_1 at α'	9.236	2.439	1.685
convexity margin $\gamma_1-t_1-(\alpha'-\frac12)/\sqrt3$	2.4450	2.0520	1.0804
α' minimising t_1, and t_1 there	4.770, 9.2353	3.320, 2.4176	2.395, 1.6717
$\alpha_{\max}$	7.35	4.70	3.10
t_1 at $\alpha_{\max}$	10.124	3.574	2.245
convexity margin at $\alpha_{\max}$	0.0559	0.0221	0.0585
cells of the proof of (iv) pass from $\sigma_2=$		3.71875	2.8125

Table 12.4: Constants of Theorem 12.1.18. For a comparison line $\alpha'>1$, t_1 is the root of $A_f^{\infty}(\alpha',t)=S_f(\alpha')+(\alpha'-\frac12)V_f$, found by bisection and rounded up to three decimals, at which the strict inequality is certified in ball arithmetic; the proof needs the convexity margin to be positive. The first comparison line is that of the first author's note; the minimiser of t_1 is located on a grid of step 0.005; $\alpha_{\max}$ is the largest α' in steps of 0.05 keeping the margin above 0.02. For ξ the range $\alpha>\alpha_{\max}$ is covered by Proposition 12.1.12. Computed; the ordinates of $L(s,\chi_4)$ and $L(s,\chi_{12})$ are certified.

one half with $\mathbb{E}e^{-uS_\alpha}=e^{-b_\alpha\sqrt{u}}$, and T_α° is a GGC with no drift, with Thorin density $(v_\alpha-b_\alpha)/\pi$ in the chart $\theta=\sqrt{z}$ and mean $\frac12(\log f)''(\alpha)$. The proof is that of Proposition 12.1.12: the Thorin representation $\log(f(\alpha+\sqrt{u})/f(\alpha))=\frac1\pi\int_0^\infty\log(1+u/\theta^2)\,v_\alpha(\theta)\,d\theta$ holds at every $\alpha\ge1$ (Proposition 15.1.25 and its proof, which applies to $\xi(\cdot,\chi_{12})$), and $\frac1\pi\int_0^\infty\log(1+u/\theta^2)\,d\theta=\sqrt{u}$.

Part (vi) sharpens Lagarias's criterion, which asks only for $v_\alpha>0$: RH is equivalent to the Thorin density on every line $\operatorname{Re}s=\alpha$, $\frac12<\alpha<1$, being smallest at the origin and only there, and by (ii) this holds with no hypothesis for $|\theta|\le3\times10^{12}-1.21$. Since b_α/π is the density of the positive stable part of the clock, (vi) says that RH holds exactly when, for every $\alpha>\frac12$, T_α splits as $S_\alpha+T_\alpha^{\circ}$ with T_α° a GGC with no drift (under RH every T_α is a GGC by Theorem 2.1.2, and the factorisation is proved as in Proposition 12.1.12). Part (i) shows that the stable part is present at every basepoint: in the chart $z=\theta^2$ the density $v_\alpha(\sqrt{z})/(2\pi\sqrt{z})$ behaves like $b_\alpha/(2\pi\sqrt{z})$ as $z\downarrow0$, with $b_\alpha>0$ for every $\alpha>\frac12$. For ξ, part (iii) gives a second proof of Proposition 12.1.12 for $\frac12+\frac{1}{\sqrt2}\le\alpha\le7.35$ that uses of the zeros only the first ordinate and the verification height; for $1\le\alpha<\frac12+\frac{1}{\sqrt2}$ the comparison reaches only the verified height, and there and for $\alpha>7.35$ Proposition 12.1.12 is still needed. For the two L functions no zero beyond the first is used; the zeros enter only through Platt's verification heights and the counting formula of Bennett, Martin, O'Bryant and Rechnitzer.

Remark 12.1.19 (The margin at the origin). For ξ, parts (i) and (ii) explain the margin of Section 1.3. For every $\varepsilon \in (0, 6.85]$ the minimum of $\rho_\varepsilon = v_{1/2+\varepsilon}/\pi$ over $0 \le \theta \le 3 \times 10^{12} - 1.21$ is $\rho_\varepsilon(0) = b_{1/2+\varepsilon}/\pi$, attained only at the origin, and $\rho_\varepsilon(0) \le \varepsilon V_\xi/\pi$, with $V_\xi/\pi = 2\pi^{-1}\sum_k a_k^{-1} = 0.0147091$, the constant of the computed minimum $0.0147\,\varepsilon$ (Section 1.3 and Table 12.6). By (i), $\rho_\varepsilon(0)/\varepsilon$ increases to V_ξ/π as $\varepsilon \downarrow 0$.

Remark 12.1.20 (The nearest zero under GRH). Under RH, or GRH, a shorter route gives the minimum at the origin on a finite range of heights; the first author's note gives it for χ_4 [283]. Let $0 < a \le \frac{1}{2}$ and assume that every zero of f lies on the critical line. Then $v_{1/2+a}(\theta) = \sum_\gamma a/(a^2+(\theta-\gamma)^2)$ over all ordinates, and $b_{1/2+a} = \sum_\gamma a/(a^2+\gamma^2) \le aV_f$, since $V_f = \sum_\gamma \gamma^{-2}$. If every gap between consecutive positive ordinates up to γ_N is less than $2\sqrt{1/V_f - a^2}$, then $v_{1/2+a}(\theta) > b_{1/2+a}$ for $0 < |\theta| \le \gamma_N$. Indeed, on $|\theta| < \gamma_1 - a/\sqrt{3}$ the sum is even and strictly convex; within $a/\sqrt{3}$ of an ordinate the nearest term alone is at least $3/(4a) > aV_f$, as $a^2 \le \frac{1}{4} < 3/(4V_f)$ for each of the three functions; and otherwise θ lies between two consecutive ordinates at distance at most half their gap g from one of them, whose term is at least $a/(a^2 + g^2/4)$, which exceeds aV_f exactly when $g < 2\sqrt{1/V_f - a^2}$. For χ_4 the threshold is 4.963 at $a = \frac{1}{2}$ and tends to $2/\sqrt{V_{\chi_4}} = 5.063$ as $a \downarrow 0$, and the largest gap below height 100 is the first, 4.2228 (Section 15.1.7.1), so under GRH the conclusion holds for $|\theta| \le 98.75$. Under GRH the gaps tend to zero, since the argument of $L(\frac{1}{2} + iT, \chi_4)$ is $O(\log T/\log\log T)$ by Littlewood's argument for ζ [348, Theorem 14.13], so the gap condition can fail only on a bounded range of heights. Theorem 12.1.18 needs no gaps: its part (ii) gives the conclusion with no hypothesis for $0 < |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, and part (vi) at every height under RH, or GRH.

Remark 12.1.21 (Complete monotonicity in the resolution). Under RH, or GRH, the ratio of Theorem 12.1.15 is $v_{1/2+\sqrt{w}}(\theta)/\sqrt{w} = \sum_\gamma 1/(w + (\theta-\gamma)^2)$, the sum over all ordinates, a Stieltjes function of $w > 0$ and in particular completely monotone. With no hypothesis the ratio is a Stieltjes function at no height, and fails to be completely monotone near every zero off the line. The ratio of a pair is $r_\delta(w,u) = 2\int_0^\infty e^{-wx}e^{-cx}\cos(\kappa x)\,dx$ for $w > -c$, so the representing function of the whole ratio is $\sum e^{-(\theta-\gamma')^2 x}$, over the zeros γ' on the line, plus $2e^{-cx}\cos\kappa x$ for each pair. If one pair has $\kappa \ne 0$ and $c = u^2 - \delta^2$ smaller than $(\theta-\gamma')^2$ for every other zero and than the value of c of every other pair, its term dominates the representing function as $x \to \infty$ and changes sign, so by Bernstein's theorem the ratio is not completely monotone at that height; at $u = 0$ the pair contributes the pole $2/(w - \delta^2)$. So, answering a question of [283], no constant plays for complete monotonicity the role of $1 + \sqrt{2}$ in Lemma 12.1.14: near a zero off the line, complete monotonicity depends on the neighbouring zeros, not only on the distance to that zero in units of δ.

12.1.7 The heat trace and the Chebyshev error

The compensated sums of Sections 12.1.3 and 12.1.5 converge only conditionally, and at $\alpha = 1$ an explicit bound for Chebyshev's error $\psi(y)-y$ bounds the error of a truncation at X only by an amount growing linearly with the height θ. Averaging over the height against a Gaussian changes this. For $\alpha \geq 1$ the heat trace of T_α is the Laplace transform $k_\alpha(t) = \pi^{-1}\int_0^\infty e^{-t\theta^2} v_\alpha(\theta)\,d\theta$ of its Thorin measure in the variable $z = \theta^2$. The average turns the kernel $2\sin^2(x\theta/2)$ of the sine squared lemma (2.1.9) into a positive, increasing, bounded kernel, and one integration by parts then turns an explicit bound for $\psi(y) - y$ into an error bound valid for all t at once.

Lemma 12.1.22. *For $x \geq 0$ and $t > 0$,*

$$\frac{1}{\pi}\int_0^\infty e^{-t\theta^2}\,2\sin^2\Big(\frac{x\theta}{2}\Big)\,d\theta = \frac{g_t(x)}{2\sqrt{\pi t}}, \qquad g_t(x) = 1 - e^{-x^2/4t}.$$

Proof. Write $2\sin^2(x\theta/2) = 1-\cos x\theta$ and use $\int_0^\infty e^{-t\theta^2}\cos x\theta\,d\theta = \frac{1}{2}\sqrt{\pi/t}\,e^{-x^2/4t}$. □

The kernel g_t increases from 0 to 1 on $[0,\infty)$, it satisfies $g_t(x) \leq \min(1, x^2/4t)$, and for fixed x it decreases in t.

Proposition 12.1.23 (Heat kernel form). *For $\alpha \geq 1$ and $t > 0$,*

$$2\sqrt{\pi t}\,k_\alpha(t) = b_\alpha + \bar{A}_\alpha(t) + \int_{[1,\infty)} g_t(\log y)\,y^{-\alpha}\,d\big(\psi(y) - y\big), \tag{12.1.11}$$

where the gamma piece

$$\bar{A}_\alpha(t) = \int_0^\infty g_t(x)\,\frac{e^{-(\alpha+2)x}}{1-e^{-2x}}\,dx = \sum_{k\geq 1}\Big(\frac{1}{c_k} - \sqrt{\pi t}\,e^{c_k^2 t}\operatorname{erfc}\big(c_k\sqrt{t}\big)\Big), \qquad c_k = \alpha+2k,$$

is positive and the Stieltjes integral converges absolutely after one integration by parts. For $\alpha > 1$ that integral is $\sum_n \Lambda(n)n^{-\alpha}g_t(\log n) - L_\alpha(t)$, with the Lebesgue piece

$$L_\alpha(t) = \int_0^\infty g_t(x)\,e^{-(\alpha-1)x}\,dx = \frac{1}{\alpha-1} - \sqrt{\pi t}\,e^{(\alpha-1)^2 t}\operatorname{erfc}\big((\alpha-1)\sqrt{t}\big).$$

Proof. Let $\alpha > 1$ and write $v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2(x\theta/2)\,x\Pi_\alpha(dx)$, as at the end of Section 12.1.4. Near the origin $|x\Pi_\alpha|$ has density about $1/(2x)$, and away from it the gamma and pole densities and the prime atoms have finite mass, so $\int_0^\infty \min(1,x^2)\,|x\Pi_\alpha|(dx) < \infty$. By Lemma 12.1.22 the integral of $e^{-t\theta^2}2\sin^2(x\theta/2)$ against $d\theta\,|x\Pi_\alpha|(dx)$ is $\frac{\sqrt{\pi}}{2\sqrt{t}}\int g_t\,d|x\Pi_\alpha| < \infty$, so the order of integration may be exchanged, and the lemma gives $2\sqrt{\pi t}\,k_\alpha(t) = b_\alpha + \int_0^\infty g_t(x)\,x\Pi_\alpha(dx)$. The gamma

density gives $\bar{A}_\alpha$, whose series follows from $e^{-(\alpha+2)x}/(1-e^{-2x}) = \sum_{k\ge1} e^{-c_k x}$ and $\int_0^\infty g_t(x)e^{-cx}\,dx = \frac{1}{c} - \sqrt{\pi t}\,e^{c^2t}\operatorname{erfc}(c\sqrt{t})$ for $c>0$, the case $c=\alpha-1$ being L_α; the prime atoms and the pole density give the Stieltjes integral, written in the variable $y=e^x$.

At $\alpha=1$ the same computation applies to $v_1 = b_1 + A + Q$ of (12.1.6), in which $Q(\theta) = \lim_{X\to\infty} Q_X(\theta)$ with $Q_X(\theta) = \int_{[1,X]} 2\sin^2(\frac{1}{2}\theta\log y)\,y^{-1}d(\psi(y)-y)$. Integration by parts gives

$$Q_X(\theta) = w(X)\big(\psi(X)-X\big) - \int_1^X \big(\psi(y)-y\big)\,w'(y)\,dy, \qquad w(y) = \frac{2\sin^2(\frac{1}{2}\theta\log y)}{y},$$

and since $|w'(y)| \le (2+|\theta|)\,y^{-2}$, while $\int_1^\infty |\psi(y)-y|\,y^{-2}dy < \infty$ by the prime number theorem with the de la Vallée Poussin error term, $|Q_X(\theta)| \le C(1+|\theta|)$ for all X. By dominated convergence the limit in X may therefore be taken outside the integral over θ, and for each X Lemma 12.1.22, applied to the finitely many atoms and to Lebesgue measure on $[0,\log X]$, gives $\frac{1}{\pi}\int_0^\infty e^{-t\theta^2}Q_X(\theta)\,d\theta = (2\sqrt{\pi t})^{-1}\int_{[1,X]} g_t(\log y)\,y^{-1}d(\psi(y)-y)$; the gamma piece A gives $\bar{A}_1$ as before. For every $\alpha\ge1$ the same integration by parts, now with $|\frac{d}{dy}(g_t(\log y)\,y^{-\alpha})| \le (g_t'(\log y)+\alpha)\,y^{-\alpha-1}$ and g_t' bounded, shows that the Stieltjes integral converges absolutely after one integration by parts, the boundary terms vanishing because $g_t(0)=0$ and $y^{-\alpha}(\psi(y)-y)\to0$. □

For $\alpha>1$ every prime power enters (12.1.11) with the positive weight $\Lambda(n)n^{-\alpha}g_t(\log n)$, and the other terms are explicit, so keeping the primes up to X gives a lower bound with no error term,

$$2\sqrt{\pi t}\,k_\alpha(t) \ge b_\alpha + \bar{A}_\alpha(t) + \sum_{n\le X}\frac{\Lambda(n)}{n^\alpha}\,g_t(\log n) - L_\alpha(t), \qquad \alpha>1, \tag{12.1.12}$$

which uses only $\Lambda(n)\ge0$. By Table 12.5, the right side is positive for every t once $\alpha\ge3.191$ with no prime atoms, the primes then entering only through b_α, and once $\alpha\ge1.533$, 1.367 and 1.284 with the primes up to 10^3, 10^5 and 10^7; the thresholds, to four decimals, are 3.1900, 1.5320, 1.3667 and 1.2840. The same argument pointwise in θ, applied to the sine squared form of v_α with no prime atoms, needs $\alpha\ge3.575$ (threshold 3.5748). At α equal to the four decimal value less 5×10^{-5} the right side is negative at one t, or one θ. From the smaller of the four decimal value plus 5×10^{-5} and the bound in the table up to $\alpha=10$, cells cover every α and every $t>0$, or $\theta\ge0$, and the right side is positive on each cell: $\bar{A}_\alpha(t)$, the prime sum and $L_\alpha(t)$ are monotone in α and in t, so each is bounded at a corner of the cell or, near the minimum over t, by a mean value form in t, and b_α by its mean value form in α. No finite X reaches $\alpha=1$: for fixed t, $L_\alpha(t)$ increases to $\int_0^\infty g_t(x)\,dx=\infty$ as $\alpha\downarrow1$, while the other terms stay bounded.

At the basepoint one the primes and the pole must be kept together. Splitting the Stieltjes integral at X, and writing $\int_0^{\log X} g_t(x)\,dx = \log X - \sqrt{\pi t}\,\mathrm{erf}\big(\log X/(2\sqrt{t})\big)$,

$$2\sqrt{\pi t}\,k_1(t) = b_1 + \bar{A}_1(t) + \sum_{n\le X} \frac{\Lambda(n)}{n}\, g_t(\log n) - \Big(\log X - \sqrt{\pi t}\,\mathrm{erf}\,\frac{\log X}{2\sqrt{t}}\Big) + R_X(t), \tag{12.1.13}$$

with $R_X(t) = \int_{(X,\infty)} g_t(\log y)\, y^{-1} d(\psi(y)-y)$. Suppose that $|\psi(y)-y| \le \omega(\log y)\, y$ for $y \ge X$. Integrating by parts, with $|\frac{d}{dy}(g_t(\log y)/y)| \le (g_t'(\log y) + g_t(\log y))\, y^{-2}$, $g_t \le 1$ and $\int_{\log X}^{\infty} g_t' = 1 - g_t(\log X)$,

$$\begin{aligned} |R_X(t)| &\le g_t(\log X)\,\frac{|\psi(X)-X|}{X} + \big(1 - g_t(\log X)\big) \sup_{x \ge \log X} \omega(x) \\ &\quad + \int_{\log X}^{\infty} \omega(x)\,dx \le \eta_X \end{aligned} \tag{12.1.14}$$

for every $t > 0$, where $\eta_X = \sup_{x\ge \log X} \omega(x) + \int_{\log X}^{\infty} \omega(x)\,dx$, the last step because $|\psi(X) - X| \le \omega(\log X) X$. Explicit forms of the prime number theorem supply ω: $|\psi(y)-y| \le 0.94\sqrt{y}$ for $11 < y \le 10^{19}$ [51]; $|\psi(y)-y| < 1.93378 \times 10^{-8}\, y$ for $y \ge 10^{19}$ and $|\psi(y)-y| < 6.95 \times 10^{-9}\, y$ for $y \ge e^{46}$ [48]; and $|\psi(y)-y| \le 8.86\, y (\log y)^{1.514} e^{-0.8288\sqrt{\log y}}$ for $\log y \ge 3000$ [170]. Since ω must be integrable, the constants alone do not suffice; the decaying bound of Johnston and Yang closes the integral. With $X = 10^7$ they give $\eta_X < 9.2 \times 10^{-4}$.

Proposition 12.1.24. *For every $t > 0$, $2\sqrt{\pi t}\,k_1(t) > 0.021$. The proof uses the primes up to 10^7, the bounds for $\psi(y) - y$ quoted above and a computation in ball arithmetic.*

Proof. By (12.1.13) and (12.1.14) it suffices that the right side of (12.1.13), with $R_X(t)$ replaced by $-\eta_X$, exceed 0.021 for every t. The three terms of it that depend on t, the gamma piece, the prime sum and the bracket, all decrease in t, because g_t does, so on a cell $[t_0, t_1]$ the value obtained by evaluating the first two at t_1 and the bracket at t_0 is a lower bound. A computation evaluates it in ball arithmetic with Arb [169], on 65137 cells that cover $[10^{-10}, 10^8]$ and are refined until each bound exceeds 0.021. It bounds $\bar{A}_1$ from below by the terms of its series with $k \le 1000$ and $c_k\sqrt{t} \le 25$, all terms being positive, and the prime sum from below by grouping the prime powers above 10^4 into blocks of width 5×10^{-4} in $\log n$, each weighted by g_t at its left end. For $t \le 10^{-10}$ the same monotonicity, with the bracket at most $\log X$, gives the lower bound 2.86, and for $t \ge 10^8$ dropping $\bar{A}_1$ and the prime sum gives 0.0221. □

From the zero side, Proposition 12.1.11 gives $v_1 \ge b_1$, hence $2\sqrt{\pi t}\,k_1(t) > b_1 = 0.0230957$ for every t, so Proposition 12.1.24 is weaker than what the zeros give.

Its interest is its input, the primes up to 10^7 and published bounds for $\psi(y) - y$, with no zero computed here. It does not avoid the zeros altogether, since those bounds are proved from them: from the verification of RH to a finite height, now 3×10^{12} [277], and from explicit zero free regions and zero density estimates.

input from the primes	positivity	basepoints
$\Lambda(n) \geq 0$	$v_\alpha(\theta) > 0$ for every θ	$\alpha \geq 3.575$
$\Lambda(n) \geq 0$	$k_\alpha(t) > 0$ for every t	$\alpha \geq 3.191$
primes up to 10^3	$k_\alpha(t) > 0$ for every t	$\alpha \geq 1.533$
primes up to 10^5	$k_\alpha(t) > 0$ for every t	$\alpha \geq 1.367$
primes up to 10^7	$k_\alpha(t) > 0$ for every t	$\alpha \geq 1.284$
primes up to 10^7 and bounds for $\psi(y) - y$	$2\sqrt{\pi t}\, k_1(t) > 0.021$ for every t	$\alpha = 1$

Table 12.5: What the primes show about the Thorin density and the heat trace of T_α, with no information on the zeros beyond what the published bounds for $\psi(y) - y$ contain. The first five rows are proved in ball arithmetic on cells covering every α from the bound shown to 10 and every time, or height; for $\alpha \geq 10$ positivity is immediate: $b_\alpha = \mathbb{E}V_\alpha$ increases with α, its derivative being $\operatorname{Var} V_\alpha$, $b_{10} = 0.391$, and the pole term is at most $\frac{1}{\alpha-1} \leq \frac{1}{9}$ in both forms. The last row is Proposition 12.1.24, also in ball arithmetic. Below $\alpha = 1$ this route needs an explicit quasi Riemann hypothesis as input.

Below the basepoint one this method stops, again by Proposition 12.1.13. For $\frac{1}{2} < \alpha < 1$ the integration by parts bounds the tail by a boundary term plus $\int_{\log X}^\infty \omega(x)\, e^{(1-\alpha)x}(g_t'(x) + \alpha g_t(x))\, dx$, and since $g_t(x) \to 1$ as $x \to \infty$, this is finite only if $\int^\infty \omega(x) e^{(1-\alpha)x}\, dx < \infty$, that is, only if ω decays exponentially. Every explicit unconditional bound decays more slowly, like $e^{-c\sqrt{x}}$ from the zero free region of de la Vallée Poussin or like $e^{-cx^{3/5}(\log x)^{-1/5}}$ from that of Vinogradov and Korobov, and so do the published bounds above beyond their finite ranges. The half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 (Section 25.1.2) would give $\psi(y) - y = O(y^{7/8+\delta})$ for every $\delta > 0$ [348], enough in principle for every $\alpha > \frac{7}{8}$; its constant has not been made explicit, and an explicit one would let this method pass a little below the basepoint one. A bound with $\int^\infty \omega(x) e^{(1-\alpha)x}\, dx < \infty$ is itself a quasi Riemann hypothesis: it makes $\int_1^\infty y^{-w}\, d(\psi(y) - y) = -\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}$ converge for $\operatorname{Re} w \geq \alpha$ after one integration by parts, so it forces $\Theta \leq \alpha$, and it follows from $\Theta < \alpha$. Along this route the step below the basepoint one needs a quasi Riemann hypothesis as input, and every $\alpha > \frac{1}{2}$ needs RH itself. The obstruction belongs to the route: it does not settle whether $k_\alpha > 0$ just below $\alpha = 1$ needs any hypothesis, since $k_\alpha > 0$ is weaker than $v_\alpha \geq 0$ and only zeros with real part above α contribute negatively to v_α.

Remark 12.1.25 (The centre). At $\alpha = \frac{1}{2}$ the prime sum and the Lebesgue piece of (12.1.11) diverge separately, but the identity survives with a finite part, with

no hypothesis [152]. Let $W(t) = \sum_{\operatorname{Re}\gamma>0} e^{-t\gamma^2}$ and

$$\mathcal{F}(\tfrac12) = 2 - \frac{\zeta'}{\zeta}(\tfrac12) = 2 - \tfrac12\log\pi + \tfrac12\frac{\Gamma'}{\Gamma}(\tfrac14) = -0.686091709613,$$

the value at $s=\frac12$ of the continuation of $\int_1^\infty y^{-s}\,d(\psi(y)-y) = -\frac{\zeta'}{\zeta}(s) - \frac{1}{s-1}$. Then for every $t>0$

$$2\sqrt{\pi t}\,W(t) = \bar A_{1/2}(t) + \mathcal{F}(\tfrac12) - \int_{[1,\infty)} e^{-(\log y)^2/4t}\,y^{-1/2}\,d\big(\psi(y)-y\big),$$

with $\bar A_{1/2}$ the gamma piece of Proposition 12.1.23 at $\alpha=\frac12$, and the integral converges absolutely. The kernel is $1-g_t(\log y)$ instead of $g_t(\log y)$ because the divergent constant $\int_{[1,\infty)} y^{-1/2}d(\psi(y)-y)$ has been replaced by $\mathcal{F}(\frac12)$. With a sharp cutoff, $\int_{[1,X]} y^{-1/2}d(\psi(y)-y)$ has no limit as $X\to\infty$, since a limit would give $\psi(X)-X = o(\sqrt{X})$, which the pole of $\frac{\zeta'}{\zeta}$ at $\frac12+i\gamma_1$ excludes; its Gaussian means do converge: as $t\to\infty$, $W(t)\to 0$ exponentially and $\bar A_{1/2}(t) = O(1/t)$, so the integral above tends to $\mathcal{F}(\frac12)$. By Theorem 8.1.1 the left side is positive for every t, so the Gaussian smoothed Chebyshev error at the centre stays below $\bar A_{1/2}(t) + \mathcal{F}(\frac12)$ at every time, unconditionally.

Proof. The arithmetic form (2.1.5) of W, multiplied by $2\sqrt{\pi t}$, reads

$$\begin{aligned} 2\sqrt{\pi t}\,W(t) = 2\sqrt{\pi t}\,e^{t/4} - \tfrac12\log\pi + \sqrt{\tfrac{t}{\pi}}\int_0^\infty e^{-tr^2}\operatorname{Re}\frac{\Gamma'}{\Gamma}\Big(\frac14+\frac{ir}{2}\Big)dr \\ - \sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\,e^{-(\log n)^2/4t}. \end{aligned}$$

Completing the square in $\int_{\mathbb{R}} e^{x/2-x^2/4t}dx$ splits the polar term as

$$\begin{aligned} 2\sqrt{\pi t}\,e^{t/4} = \int_{[1,\infty)} e^{-(\log y)^2/4t}y^{-1/2}dy \\ + \sqrt{\pi t}\,e^{t/4}\operatorname{erfc}\big(\tfrac{\sqrt{t}}{2}\big), \end{aligned}$$

and the first piece compensates the prime sum. By Lemma 12.1.22,

$$\bar A_{1/2}(t) = 2\sqrt{\pi t}\,\pi^{-1}\int_0^\infty e^{-tr^2}\sum_{k\ge1} q_{\frac12+2k}(r)\,dr,$$

and

$$\begin{aligned} \sum_{k\ge1} q_{\frac12+2k}(r) = \frac12\Big[\operatorname{Re}\frac{\Gamma'}{\Gamma}\Big(\frac14+\frac{ir}{2}\Big) - \frac{\Gamma'}{\Gamma}\Big(\frac14\Big)\Big] \\ - 2 + \frac{1/2}{1/4+r^2}. \end{aligned}$$

With $\int_0^\infty e^{-tr^2}\frac{1/2}{1/4+r^2}dr = \frac{\pi}{2}e^{t/4}\operatorname{erfc}(\frac{\sqrt{t}}{2})$ and $\int_0^\infty e^{-tr^2}dr = \frac{1}{2}\sqrt{\pi/t}$, this gives

$$\begin{aligned}\bar{A}_{1/2}(t) + \mathcal{F}(\tfrac{1}{2}) = \sqrt{\tfrac{t}{\pi}}\int_0^\infty e^{-tr^2}\operatorname{Re}\frac{\Gamma'}{\Gamma}(\tfrac{1}{4}+\tfrac{ir}{2})\,dr \\ + \sqrt{\pi t}\,e^{t/4}\operatorname{erfc}(\tfrac{\sqrt{t}}{2}) \\ - \tfrac{1}{2}\log\pi,\end{aligned}$$

because $-\frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{1}{4}) - 2 + \mathcal{F}(\frac{1}{2}) = -\frac{1}{2}\log\pi$, and the identity follows. The two forms of $\mathcal{F}(\frac{1}{2})$ agree because $\frac{\xi'}{\xi}(\frac{1}{2}) = 0$. For the sharp cutoff: if $I(X) = \int_{[1,X]} y^{-1/2}d(\psi(y)-y)$ converged, partial summation would give $\psi(X) - X = o(\sqrt{X})$, and then $\varepsilon\int_1^\infty y^{-s}d(\psi(y)-y) \to 0$ at $s = \frac{1}{2}+\varepsilon+i\gamma_1$ as $\varepsilon \downarrow 0$, whereas $-\frac{\zeta'}{\zeta}(s) - \frac{1}{s-1}$ times ε tends to minus the multiplicity of the zero. Finally $\bar{A}_{1/2}(t) \le \int_0^\infty \min(1, \frac{x^2}{4t})\frac{e^{-5x/2}}{1-e^{-2x}}\,dx = O(1/t)$. □

At $t = 0.01$, 0.025, 0.125, 0.5 and 2 the two sides agree to 3×10^{-15} (computed). For $t \ge 0.5$ the left side is below 10^{-40} and the prime sum is in double precision, so the residual there is the rounding of the prime sum. The compensator $\int_{[1,\infty)} e^{-(\log y)^2/4t}y^{-1/2}dy$ and the prime sum both grow like $2\sqrt{\pi t}\,e^{t/4}$, while their difference stays bounded.

12.2 Hitting times at the basepoints

The third question, whether T_1 is a hitting time, is probabilistic. We keep the notation φ_α, T_α, b_α, v_α and $\rho_{1/2} = v_1/\pi$ of Section 12.1. In addition B is a standard Brownian motion, $H_a^{(c)}$ is the first passage time of $B_t + ct$ to the level $a \ge 0$, and $H_a = H_a^{(0)}$; this H has nothing to do with the de Bruijn Newman flow H_t of Section 4.1.

12.2.1 Kent's theorem and Brownian first passage to a random level

The first candidate is the hitting time of a level by a diffusion. By Kent's theorem [184] and Krein's inverse spectral theory [78], recalled in Section 5.2.3, the first hitting times of levels by diffusions started at a reflecting or entrance end are exactly the GGCs without drift whose Thorin measures consist of unit atoms at distinct points, the Dirichlet eigenvalues of the interval up to the level. The Thorin measure of φ_1 is absolutely continuous, so T_1 is not such a hitting time. The centre clock has the discrete form: under RH its Thorin measure has an atom at each γ^2, of mass the multiplicity of the zero, so it is such a hitting time exactly when the zeros are also simple. A second candidate is Brownian first passage to a random level, which Proposition 3.1.1 excludes for the clock.

PROPOSITION 12.2.1. *For no drift $c \geq 0$ and no random level $L \geq 0$ independent of the Brownian motion is T_1 equal in law to the first passage time $H_L^{(c)}$. (For $c < 0$ the first passage time is infinite with positive probability, so equality is impossible.)*

Proof. Conditionally on L, the Laplace transform of $H_L^{(c)}$ is $\exp\{-L(\sqrt{c^2+2u} - c)\}$, so $\mathbb{E}e^{-uH_L^{(c)}} = \mathbb{E}\exp\{-L(\sqrt{c^2+2u}-c)\}$ extends analytically, and bounded by one, to the region $R_c = \{u : \operatorname{Re}\sqrt{c^2+2u} > c\}$, which contains the right half plane and lies inside $\mathbb{C}\setminus(-\infty,0]$. The function φ_1 is analytic there as well, by the proof of Proposition 12.1.2, so if the two agreed on $\operatorname{Re} u > 0$ they would agree on R_c, and $|\varphi_1| \leq 1$ on R_c and on its boundary, the parabola $u_q = \frac{1}{2}(-q^2 + 2icq)$, $q \in \mathbb{R}$, on which $\sqrt{c^2+2u_q} = c + iq$, as in the proof of Proposition 3.1.1. On that parabola $\operatorname{Re}\sqrt{u_q}$ stays bounded while $\operatorname{Im}\sqrt{u_q} \sim q/\sqrt{2}$, so by Stirling's formula $|\xi(1+\sqrt{u_q})|$ decays exponentially and $|\varphi_1(u_q)| \to \infty$. For $c = 0$ the parabola is the cut itself and $|\varphi_1(-t^2)| = |\xi(1)/\xi(1+it)|$, which equals 13.0, 1.2×10^4, 2.0×10^{10} and 1.5×10^{23} at $t = 10, 20, 40, 80$, as computed. □

A subordinator with a drift as well as a stable one half part escapes this obstruction: on the cut, where $\sqrt{u} = it$, its exponent $u + \sqrt{u} = -t^2 + it$ has negative real part, so mixtures of its transforms can grow.

12.2.2 SUBORDINATION OF THE CENTRE CLOCK

Let T be the clock variable of Section 2.1, with $\mathbb{E}e^{-uT} = \varphi_{1/2}(u) = 1/G(u)$, a positive random variable that exists with no hypothesis (Theorem 8.1.1, whose proof uses the verification of RH below height 1000); it is the clock of [279], the centre clock.

THEOREM 12.2.2. *Let $\widetilde{T}$ have the tilted law $P(\widetilde{T} \in dt) = e^{-t/4}P(T \in dt)/\mathbb{E}e^{-T/4}$, and let $(\sigma_\ell)_{\ell\geq 0}$ be an independent subordinator with Laplace exponent $u + \sqrt{u}$. Then*

$$T_1 \stackrel{d}{=} \sigma_{\widetilde{T}} \stackrel{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}},$$

where H_a is the first passage time of a standard Brownian motion to the level a, independent of $\widetilde{T}$.

Proof. For $\operatorname{Re}\sqrt{u} \geq 0$ the number $\frac{1}{2} + \sqrt{u}$ has positive real part and $(\frac{1}{2}+\sqrt{u})^2 = \frac{1}{4} + u + \sqrt{u}$, so $\xi(1+\sqrt{u}) = \xi\big(\frac{1}{2} + \sqrt{\frac{1}{4}+u+\sqrt{u}}\big)$ with the principal root. Hence

$$\begin{aligned}\varphi_1(u) &= \frac{\xi(1)}{\xi(1+\sqrt{u})} = \frac{\varphi_{1/2}(\frac{1}{4}+u+\sqrt{u})}{\varphi_{1/2}(\frac{1}{4})} \\ &= \frac{\mathbb{E}\big[e^{-T/4}e^{-(u+\sqrt{u})T}\big]}{\mathbb{E}e^{-T/4}} = \mathbb{E}e^{-(u+\sqrt{u})\widetilde{T}} = \mathbb{E}e^{-u\sigma_{\widetilde{T}}}.\end{aligned}$$

Finally $\sigma_\ell \stackrel{d}{=} \ell + H_{\ell/\sqrt{2}}$, because $\mathbb{E}e^{-uH_a} = e^{-a\sqrt{2u}}$. □

The subordinator σ has a classical description: a subordinator with exponent $\kappa_1 u + \kappa_2\sqrt{2u}$ is the inverse local time at zero of a Brownian motion that is sticky at zero, the drift κ_1 measuring the time spent at zero per unit of local time [166, Section 5.7, p. 187]; here $\kappa_1 = 1$ and $\kappa_2 = 1/\sqrt{2}$. So T_1 is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$.

Corollary 12.2.3. *Assume RH and that the zeros are simple. Then $\widetilde{T}$ is Kent's hitting time for a Krein string whose Dirichlet spectrum is $\{\gamma^2 + \frac14\} = \{|\rho|^2\}$, and T_1 is the first hitting time of the level y_* by the process $D(\ell_t)$, where D is that string diffusion started at its left end, y_* is the level whose hitting time by D is $\widetilde{T}$, and ℓ_t is the local time at zero of an independent sticky Brownian motion B. Equivalently, T_1 is the first hitting time of the line $\{y = y_*\}$ by the two dimensional Markov process $(D(\ell_t), B_t)$.*

Proof. Under RH, $\mathbb{E}e^{-u\widetilde{T}} = \varphi_{1/2}(\frac14+u)/\varphi_{1/2}(\frac14) = \prod_{\gamma>0}(\gamma^2+\frac14)/(\gamma^2+\frac14+u)$, the product running over the ordinates $\gamma > 0$ repeated according to multiplicity. As the zeros are simple, this is a GGC without drift with unit atoms at the distinct points $|\rho|^2$, and Kent's theorem and Krein's inverse theory provide the string. The first time $D(\ell_t)$ reaches y_* is the first t with $\ell_t \ge \widetilde{T}$, which is $\sigma_{\widetilde{T}}$. □

The tail (12.1.3) follows from the theorem: $P(H_a > x) \sim a\sqrt{2/(\pi x)}$, so $P(T_1 > x) \sim \mathbb{E}[\widetilde{T}]/\sqrt{\pi x}$, and $\mathbb{E}\widetilde{T} = -\frac{d}{du}\log\varphi_{1/2}(\frac14+u)\big|_{u=0} = \frac{\xi'}{\xi}(1) = \sum_\rho \rho^{-1}$.

12.2.3 The density of T_1 and harmonic measure

Conditioning on $\widetilde{T}$ in Theorem 12.2.2 and using the density of H_a gives the density of T_1 as an expectation over the tilted centre clock,

$$f_{T_1}(x) = \mathbb{E}\Big[\frac{\widetilde{T}}{2\sqrt{\pi}}\,(x-\widetilde{T})^{-3/2}\exp\Big(-\frac{\widetilde{T}^2}{4(x-\widetilde{T})}\Big)\,\mathbf{1}\{\widetilde{T} < x\}\Big], \tag{12.2.1}$$

since H_a has density $a(2\pi h^3)^{-1/2}e^{-a^2/2h}$ and $a = \widetilde{T}/\sqrt{2}$. Figure 12.4 compares (12.2.1), averaged over 4×10^5 draws of $\widetilde{T}$, with Talbot inversion of φ_1 [341]; they agree to about 0.1%, except near the mode $x = 0.024$, where the Monte Carlo average is within 1%. For large x the density behaves like $\frac{1}{2\sqrt{\pi}}\sum_\rho \rho^{-1}\,x^{-3/2} = 0.00652\,x^{-3/2}$, the derivative of (12.1.3). Near the origin the density is super flat, inherited from the centre clock through $\widetilde{T}$.

The Cauchy kernels of Proposition 12.1.2 and the stable one half subordinator of Theorem 12.2.2 are both laws of planar Brownian motion at the moment it first reaches a line. Let $B = (B^1, B^2)$ be a planar Brownian motion started at a zero $\rho = \beta + i\gamma$, identified with the point (β, γ), and let H be the first time B^1 reaches the line $\operatorname{Re} s = 1$, a distance $1-\beta$ away, that is, the first passage time $H_{1-\beta}$ of

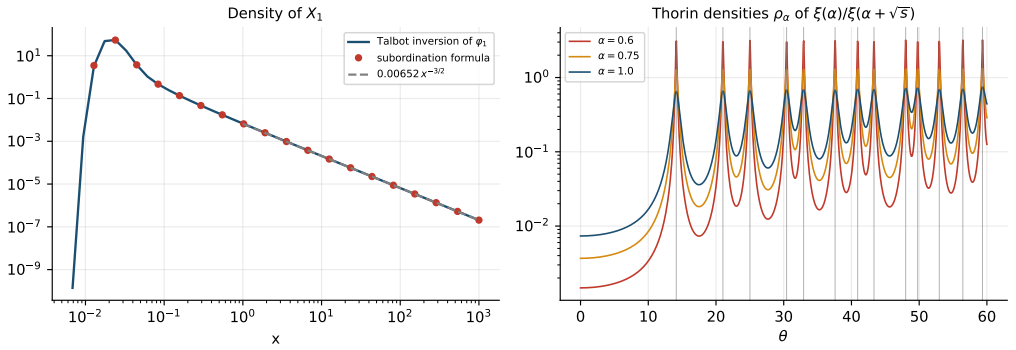


Figure 12.4: Left: the density of T_1 by Talbot inversion of φ_1 and by the subordination formula (12.2.1), with the tail $0.00652\,x^{-3/2}$. Right: the densities v_α/π at the basepoints $\alpha = 0.6$, 0.75 and 1, with the ordinates of the zeros; they sharpen onto the zeros as α decreases, and for $\alpha < 1$ they are Thorin densities of T_α only if no zero lies to the right of the line $\operatorname{Re} s = \alpha$. The plots write X_1 for T_1, ρ_α for v_α/π and s for u.

the one dimensional Brownian motion $B^1 - \beta$. Then B_H^2 has the Cauchy law with centre γ and width $1-\beta$, which is the harmonic measure of the half plane, and H has Laplace transform $e^{-(1-\beta)\sqrt{2u}}$, the stable one half law. So, exactly,

$$\rho_{1/2}(\theta)\,d\theta = \sum_\rho P\big(B_H^2 \in d\theta \mid B_0 = \rho\big).$$

Up to scale, H has the law of the stable part of the subordinator σ of Theorem 12.2.2; the correspondence is of type, not of scale, since there the stable part runs per unit of local time $\widetilde{T}$, not per unit of distance.

12.2.4 The exact Brownian boundary and Wald's identity

The inverse first passage question asks for a boundary b such that the first time $H_b = \inf\{t > 0 : B_t \geq b(t)\}$ at which a standard Brownian motion started at zero crosses b has the law of T_1. A deterministic curved boundary always exists: Anulova proved this for two sided symmetric boundaries [4], and Chen, Cheng, Chadam and Saunders for one sided boundaries when the survival function is continuous and positive [57, Theorem 1]. By the theorems of Ekström and Janson [81, Theorems 2.3, 8.2 and 9.2], every law on $(0,\infty)$ is the law of H_b for a lower semicontinuous boundary b, which is continuous on every interval where the density of the law is bounded below by a positive constant; when that holds on every compact subinterval of $(0,\infty)$, b is the unique continuous solution of the Volterra equation (3.1.1) with $\liminf_{t\downarrow 0} b(t) \geq 0$. The density (12.2.1) of T_1 is continuous and positive on $(0,\infty)$, so the boundary of T_1 is continuous there and

is computed from the law of T_1 alone through (3.1.1), with f the density of T_1, in the complementary form of Section 3.1.2.2, with $P(T_1 > t)$ in place of $P(T > t)$, once the survival probability falls below one half. We computed the distribution function of T_1 by Talbot inversion of $\varphi_1(u)/u$ for $t < 0.03$, where T_1 is super flat, and by the subordination representation of Theorem 12.2.2 beyond, in the form

$$P(T_1 > t) = \mathbb{E}\Big[\mathbf{1}\{\widetilde{T} \geq t\} + \mathbf{1}\{\widetilde{T} < t\}\Big(2\mathcal{N}\Big(\frac{\widetilde{T}}{\sqrt{2(t-\widetilde{T})}}\Big) - 1\Big)\Big],$$

with $\mathcal{N}$ the standard normal distribution function, as in Section 3.1.2.2. This form is smooth and keeps relative accuracy even at $t = 100$, where $\sqrt{t}\,P(T_1 > t) = 0.013037$ against the limit 0.013030. As at the centre, the boundary is computed over the three zones of Section 3.1.2.2.

In the small time zone the law of T_1 is that of the tilted centre clock, and the boundary starts from the reflection level 0.505 at $t = 0.009$. In the bulk, from $t = 0.011$ to $t = 0.05$, the boundary falls steeply, to 0.363 at $t = 0.012$ and 0.111 at $t = 0.02$. Here the tangent approximation in density form equates the first passage density of the tangent line with the density of T_1, whereas the boundary of Roberts and Shortland [317] in Section 3.1.2.1 equates hazards. Its solution, the boundary b with $(b - tb')(2\pi t^3)^{-1/2}e^{-b^2/2t} = f(t)$ computed, agrees with the exact boundary to 10^{-4} at $t = 0.012$ and to 1.5×10^{-3} at $t = 0.02$.

In the right tail the basepoint one differs completely from the centre. At the centre the survival function is exponential and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1$ (Section 3.1.2.2). At the basepoint one the survival function is the stable one half tail $0.01303\,t^{-1/2}$, and a Brownian motion survives that long only below a boundary that stops moving. The exact boundary flattens at the level $b_\infty = -0.19277$: it stays within 10^{-3} of this level from $t = 0.056$ on and within 10^{-4} from $t = 1.2$ on, up to the end of the computation at $t = 100$; that $b_\infty = \lim_{t\to\infty} b(t)$ is observed, not proved. The tangent approximation fails here: it overshoots downward, to -0.332 at $t = 0.05$ against the exact -0.188, and has no solution beyond $t = 0.091$. It treats the crossing density at time t as if the path had just started, so a flat boundary at level a would have to produce the density $a\,(2\pi)^{-1/2}t^{-3/2}$, forcing $a = \sqrt{2\pi} \times 0.00652 = 0.0163 > 0$; but by the time the tail is reached the tangent boundary is already negative, and a negative linear boundary can only produce an exponential tail.

The number 0.0163 does appear in the exact picture, in a different role. Optional stopping for the martingale $B_{t\wedge H_b}$ gives $\mathbb{E}[b(H_b)\mathbf{1}\{H_b \leq t\}] = -\mathbb{E}[B_t\mathbf{1}\{H_b > t\}]$, and once the boundary is flat, a survivor at depth y below it survives a further time r with probability about $y\sqrt{2/(\pi r)}$; the next lemma matches these two facts exactly. For a continuous monotone boundary that is positive at 0 the limit is Theorem 2 of Novikov [249]; the lemma assumes boundedness in place of monotonicity.

LEMMA 12.2.4. *Let b be continuous on $(0,\infty)$ and bounded on $[1,\infty)$, and let $H_b = \inf\{t > 0 : B_t \geq b(t)\}$ be positive almost surely. Then $b(H_b) = B_{H_b}$ is integrable and $\sqrt{t}\,P(H_b > t) \to \sqrt{2/\pi}\,\mathbb{E}\,b(H_b)$ as $t \to \infty$.*

Proof. Since b is bounded above on $[1,\infty)$ and $\limsup_t B_t = \infty$, H_b is finite, and $B_{H_b} = b(H_b)$ by continuity. Choose $u_0 > 0$ with $P(H_b > u_0) > 0$; on that event $b > B$ on $(0,u_0]$, so b, which is also continuous on $[u_0,1]$ and bounded on $[1,\infty)$, is bounded below on $(0,\infty)$, say by $-K$. Optional stopping gives $A(s) = \mathbb{E}[b(H_b)\mathbf{1}\{H_b \leq s\}] = -\mathbb{E}[B_s\mathbf{1}\{H_b > s\}]$. Fix $s \geq 1$, let c_1 and c_2 be the infimum and the supremum of b on $[s,\infty)$, and let $g(y,\tau) = (2\mathcal{N}(y/\sqrt{\tau}) - 1)^+$, the probability that a Brownian motion started at zero stays below the level y up to time τ. By the Markov property at time s, on $\{H_b > s\}$ the conditional probability of $\{H_b > t\}$ given the path up to time s lies between $g(c_1 - B_s, t-s)$ and $g(c_2 - B_s, t-s)$, and $c_2 - B_s > 0$ there. Since $\sqrt{t}\,g(y,t-s) \to \sqrt{2/\pi}\,y^+$ as $t \to \infty$, and $\sqrt{t}\,g(y,t-s) \leq 2y^+$ for $t \geq 2s$, dominated convergence and $y^+ \geq y$ give

$$\liminf_{t\to\infty} \sqrt{t}\,P(H_b > t) \geq \sqrt{\tfrac{2}{\pi}}\big(c_1 P(H_b > s) + A(s)\big),$$
$$\limsup_{t\to\infty} \sqrt{t}\,P(H_b > t) \leq \sqrt{\tfrac{2}{\pi}}\big(c_2 P(H_b > s) + A(s)\big).$$

The upper bound at $s = 1$ and the lower bound at every s bound $A(s)$ uniformly, so, since $b(H_b) \geq -K$, monotone convergence makes $b(H_b)$ integrable with $A(s) \to \mathbb{E}\,b(H_b)$. As $s \to \infty$, c_1 and c_2 stay bounded and $P(H_b > s) \to 0$, so both bounds tend to $\sqrt{2/\pi}\,\mathbb{E}\,b(H_b)$. □

The exact boundary of T_1 is continuous; its boundedness on $[1,\infty)$, which the lemma needs, is observed, not proved. Granting it, the lemma and the tail (12.1.3) give

$$\mathbb{E}\,b(T_1) = \lim_{t\to\infty} \mathbb{E}\big[(b_\infty - B_t)\mathbf{1}\{H_b > t\}\big] = \sqrt{\frac{\pi}{2}} \times \frac{\sum_\rho \rho^{-1}}{\sqrt{\pi}} = \frac{1}{\sqrt{2}} \sum_\rho \frac{1}{\rho} = 0.0163311,$$

the first equality by optional stopping, since $P(H_b > t) \to 0$. So, if the boundary is bounded on $[1,\infty)$, Wald's identity $\mathbb{E}B_{H_b} = 0$, which holds for every crossing time of finite mean, fails here, and the defect is exactly $\sum_\rho \rho^{-1}/\sqrt{2}$. Numerically, integrating the computed boundary against the law of T_1 gives $\mathbb{E}\,b(T_1) = 0.0163370$. The defect is also the level of the stable summand of T_1: since $\mathbb{E}e^{-uH_a} = e^{-a\sqrt{2u}}$, the variable S of Proposition 12.1.11 has the law of the first passage time $H_{b_1/\sqrt{2}}$ (Section 13.2.4).

Simulating 1.5×10^5 Brownian paths against the exact boundary on $[0,1]$, with a Brownian bridge crossing correction, gives a Kolmogorov distance of 0.0019 between the simulated and exact survival functions, at the level of the Monte Carlo noise. Figure 12.5 shows the boundary and the survival check.

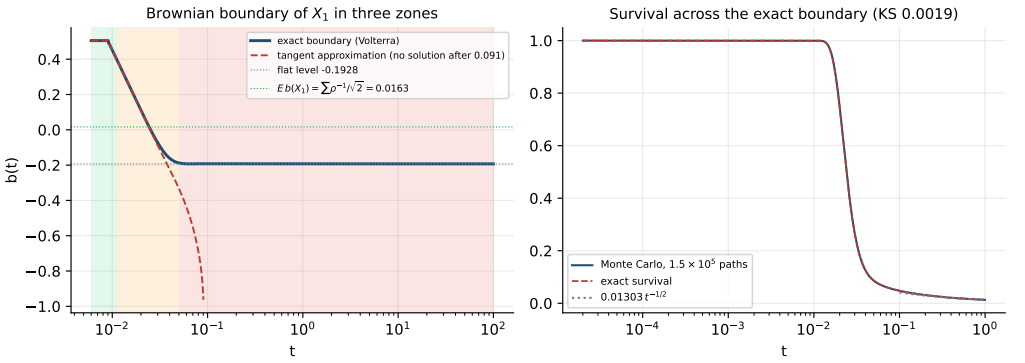


Figure 12.5: Left: the exact Brownian boundary of T_1, written X_1 in the plot, from the Volterra equation over the three zones of Section 3.1.2.2 (shaded: small time, bulk, right tail), the tangent approximation, which has no solution beyond $t = 0.091$, the flat level -0.19277, and the value $\mathbb{E}\,b(T_1) = \sum_\rho \rho^{-1}/\sqrt{2} = 0.0163$. Right: survival of 1.5×10^5 simulated Brownian paths across the exact boundary against the exact survival function of T_1 and the tail $0.01303\,t^{-1/2}$; the Kolmogorov distance is 0.0019.

12.2.5 Every basepoint above the centre

Nothing in Theorem 12.2.2 is special to the basepoint one: since $(\alpha - \frac{1}{2} + \sqrt{u})^2 = (\alpha - \frac{1}{2})^2 + u + 2(\alpha - \frac{1}{2})\sqrt{u}$, the computation applies at every basepoint to the right of the centre.

Theorem 12.2.5. *For every $\alpha > \frac{1}{2}$, with no hypothesis on the zeros (the existence of the centre clock, Theorem 8.1.1, uses the verification of RH below height 1000), $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha}$, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and $\sigma^{(\alpha)}$ is an independent subordinator with exponent $u + 2(\alpha - \frac{1}{2})\sqrt{u}$, that is $T_\alpha \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$. The law has infinite mean and $P(T_\alpha > x) \sim \frac{\xi'}{\xi}(\alpha)\,(\pi x)^{-1/2}$. The variable T_α is a GGC if and only if no zero has $\beta > \alpha$, in particular for every $\alpha \geq 1$, and when every zero has $\beta < \alpha$ its Thorin density in the chart $\theta = \sqrt{z}$ is $\frac{1}{\pi}\sum_\rho (\alpha - \beta)/((\alpha - \beta)^2 + (\theta - \gamma)^2)$.*

Proof. The subordination is the computation of Theorem 12.2.2 with $\frac{1}{4}$ replaced by $(\alpha - \frac{1}{2})^2$ and $\sqrt{u}$ by $2(\alpha - \frac{1}{2})\sqrt{u}$, and H_a has exponent $a\sqrt{2u}$ with $a = \sqrt{2}(\alpha - \frac{1}{2})\ell$. The tail follows as in Section 12.1.2 from $\varphi_\alpha(u) = 1 - \frac{\xi'}{\xi}(\alpha)\sqrt{u} + O(u)$. The Thorin density and the GGC criterion are the threshold theorem, Theorem 2.1.2: when every zero has $\beta < \alpha$ each Cauchy term is positive, and a zero with $\beta > \alpha$ puts a pole of φ_α at the point $(\rho - \alpha)^2$ of $\mathbb{C} \setminus (-\infty, 0]$ (its principal root is $\rho - \alpha$, since $\mathrm{Re}(\rho - \alpha) > 0$), where the Laplace transform of a GGC is holomorphic, whatever

the sign of the density near its ordinate (Remark 2.1.4). In the boundary case, when some zero has $\beta = \alpha$ and none has $\beta > \alpha$, every zero has $\beta < \alpha'$ for $\alpha' > \alpha$, so $T_{\alpha'}$ is a GGC; as $\alpha' \downarrow \alpha$, $\varphi_{\alpha'} \to \varphi_\alpha$ pointwise on $[0,\infty)$, so $T_{\alpha'} \to T_\alpha$ in law, and T_α is a GGC by the closure theorem of Section 9.1.1. There the Cauchy kernel of a zero $\alpha + i\gamma$, $\gamma > 0$, of multiplicity m becomes an atom: with its conjugate it contributes the factor $(1+u/\gamma^2)^{-m}$ to φ_α, a Thorin atom of mass m at $z = \gamma^2$. □

COROLLARY 12.2.6. *Let $\alpha > \frac{1}{2}$ and $a = \alpha - \frac{1}{2}$. Then T_α is infinitely divisible, with no drift and Lévy measure $W_\alpha(x)\,dx/x$, where*

$$\begin{aligned} W_\alpha(x) &= \frac{ax}{\sqrt{\pi}} \int_0^x (x-s)^{-3/2} e^{-a^2 xs/(x-s)}\, W(s)\, ds \\ &= \mathbb{E}\Big[(1+4xV)^{-1/2}\, W\Big(\frac{x}{1+4xV}\Big)\Big], \end{aligned}$$

with $V = a^2/(2Z^2)$ and Z standard normal. Hence $W_\alpha > 0$ and $W_\alpha' < 0$ on $(0,\infty)$, so that T_α is self decomposable; and on every interval $(0,T]$ on which W is completely monotone, so is W_α.

Proof. By Theorem 12.2.5, T_α is the subordinator $\sigma_s^{(\alpha)} = s + H_{\sqrt{2}as}$ evaluated at the independent time $\widetilde{T}_\alpha$, which is infinitely divisible with no drift and Lévy measure $e^{-a^2 s} W(s)\, ds/s$, and $H_{\sqrt{2}as}$ has the density $(as/\sqrt{\pi}) y^{-3/2} e^{-a^2 s^2/y}$. By Bochner's subordination [325, Theorem 30.1], T_α is infinitely divisible with no drift and Lévy measure $\int_0^\infty P(\sigma_s^{(\alpha)} \in dx)\, e^{-a^2 s} W(s)\, ds/s$, which, since $a^2 s^2/(x-s) + a^2 s = a^2 xs/(x-s)$, is $W_\alpha(x)\,dx/x$ with the first form of W_α. The substitution $s = x/(1+4xv)$ turns the integral into an average against the density $\frac{a}{2\sqrt{\pi}} v^{-3/2} e^{-a^2/4v}$ of V, which is the second form. There the derivative in x of $(1+4xv)^{-1/2} W(x/(1+4xv))$ is $-2v(1+4xv)^{-3/2} W + (1+4xv)^{-5/2} W'$, negative because $W > 0$ and $W' < 0$ (Theorem 8.1.1), and differentiation under the average is legitimate, because $W^{(n)}(s) = O(s^{-n-1/2} \log(1/s))$ as $s \to 0$ makes every derivative in x of the integrand $O(\log(1+V))$ on compact subsets of $(0,\infty)$, and $\mathbb{E}\log(1+V) < \infty$. Finally, $x \mapsto x/(1+4xv)$ has a completely monotone derivative and maps $(0,T]$ into itself, and $(1+4xv)^{-1/2}$ is completely monotone, so Faà di Bruno's formula and Leibniz's rule carry complete monotonicity of W on $(0,T]$ to every term of the average. □

With Theorem 8.1.9, W_α is therefore completely monotone on $(0, 1.29]$ for every $\alpha > \frac{1}{2}$. When no zero has real part above α, in particular for every $\alpha \geq 1$, T_α is a GGC and W_α is the Laplace transform k_α of its Thorin measure (Section 14.1.6); when some zero has real part above α, T_α is not a GGC, so W_α is not completely monotone on $(0,\infty)$, although it is on $(0, 1.29]$.

Table 12.6 and the right panel of Figure 12.4 give the constants and densities at three basepoints. As $\alpha \downarrow \frac{1}{2}$ the tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$ tends to zero, because

$\xi'(\frac{1}{2}) = 0$, and the law acquires the finite mean $\sum_k a_k^{-1}$ of the centre clock; at the same time the subordinator tends to pure drift and the Cauchy widths to $\frac{1}{2} - \beta$, which is where RH is decided.

α	$\frac{\xi'}{\xi}(\alpha)$	tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$	$\min_{[0,60]} \rho_{\alpha-1/2}$
0.6	0.00462092	0.002607	0.00147
0.75	0.01155134	0.006517	0.00368
1	0.02309571	0.013030	0.00735

Table 12.6: Basepoints above the centre: the drift $\frac{\xi'}{\xi}(\alpha)$, the tail constant of T_α, and the minimum on $[0, 60]$, attained at the origin, of the density $\rho_{\alpha-1/2} = v_\alpha/\pi$, the Thorin density of T_α when every zero lies to the left of the line $\operatorname{Re} s = \alpha$. At $\alpha = 1$ the minimum at the origin holds at every height (Proposition 12.1.11), as it does at every $\alpha \geq 1$ (Proposition 12.1.12); for $\frac{1}{2} < \alpha < 1$ the minimum at the origin holds on the verified window $|\theta| \leq 3 \times 10^{12} - 1.21$ (Theorem 12.1.18(ii)), and at every height for every such α it is equivalent to RH (part (vi) of that theorem).

12.2.6 Boundaries across the basepoints

We solved the Volterra equation (3.1.1) [270] at $\alpha = 0.6$ and 0.75, with the law of T_α from Theorem 12.2.5. As at the basepoint one, the distribution function comes from Talbot inversion of $\varphi_\alpha(u)/u$ for $t < 0.03$ and from the subordination identity beyond, with 3×10^5 draws of T_α, made under RH as a sum of exponential variables over the first 200 zeros plus a gamma variable with the mean and variance of the remaining terms. The bottom left panel of Figure 14.2 in Section 14.1 draws the four boundaries together.

In the small time zone and the bulk the boundaries are close: they start near the reflection level 0.50, and at $t = 0.02$ they are at 0.072 for the centre and at 0.080, 0.092 and 0.111 for $\alpha = 0.6$, 0.75 and 1. They differ in the right tail. At the centre the survival function is exponential with rate γ_1^2, and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1 = -19.99$ and the logarithmic correction $\frac{3}{2\sqrt{2}\gamma_1} \log t$; the slope of the computed boundary over $[0.2, 0.3]$ is -19.77 (Section 3.1.2.2). Above the centre the exact boundaries flatten: each stays within 0.011 of its level at $t = 100$, the end of the computation, from $t = 0.1$ on, and within 10^{-3} from $t = 1$ on. As $\alpha \downarrow \frac{1}{2}$ the flat level moves down, towards the linear boundary of the centre.

The Wald type identity of Section 12.2.4 extends to every basepoint above the centre. The density of T_α is again continuous and positive on $(0, \infty)$, so its boundary is continuous [81, Theorem 8.2], and as computed it is bounded up to $t = 100$. Granting boundedness at infinity, again observed, not proved, Lemma 12.2.4 and the stable tail $P(T_\alpha > t) \sim b_\alpha (\pi t)^{-1/2}$ of Theorem 12.2.5 give

$\mathbb{E}\, b(T_\alpha) = b_\alpha/\sqrt{2}$, so Wald's identity $\mathbb{E}B_{H_b} = 0$ fails by exactly $b_\alpha/\sqrt{2}$. Table 12.7 compares the identity and the tail constant with the computed values; all six pairs agree to within 6×10^{-4} in relative terms. The tangent approximation fails in the right tail at every basepoint above the centre, for the reason given in Section 12.2.4.

α	flat level	$\mathbb{E}\, b(T_\alpha)$	$b_\alpha/\sqrt{2}$	$\sqrt{100}\, P(T_\alpha > 100)$	$b_\alpha/\sqrt{\pi}$
0.6	-0.33384	0.0032687	0.0032675	0.0026083	0.0026071
0.75	-0.25537	0.0081710	0.0081680	0.0065203	0.0065171
1	-0.19277	0.016337	0.016331	0.013037	0.013030

Table 12.7: Exact Brownian boundaries above the centre: the flat level, the mean boundary value $\mathbb{E}\, b(T_\alpha)$ at the crossing, obtained by integrating the computed boundary against the law of T_α, against the Wald type identity, and the computed $\sqrt{100}\, P(T_\alpha > 100)$ against the tail constant of Theorem 12.2.5. Computed for $\alpha = 0.6$, 0.75 and 1.

Chapter 13

The critical strip, Riesz's criterion and the squarefree series

Chapter 12 read the reciprocal Xi function at a basepoint α as a Laplace transform on the half line, $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ for $u \geq 0$. At the basepoint one its Thorin measure is the Poisson smoothing of the zeros and its Lévy form carries the primes, and by Theorem 12.2.5 it is, for every $\alpha > \frac{1}{2}$ and with no hypothesis, the Laplace transform of a positive random variable T_α obtained by subordinating the centre clock. Two further readings of the same function, as the characteristic function of a normal scale mixture and through the completeness of the translates of the tilted Pólya kernels, run across the whole critical strip and show how much of the unconditional picture at the basepoint one survives inside it. Section 13.1 develops both and ends with Riesz's criterion, which reads the zero free half planes through growth rather than completeness. Section 13.2 writes that criterion in the language of the clocks, where it says that every T_α with $\alpha > \frac{1}{2}$ is a GGC, and shows why none of the resulting representations of its coefficients supplies the growth bound the criterion needs. Section 13.3 turns to the squarefree integers, whose Dirichlet series $\zeta(s)/\zeta(2s)$ carries the same reciprocal $1/\zeta(2s)$: the constant of $\sum_{n \leq x} \mu(n)^2/n$ is a combination of two tilted Pólya means, and smoothing the count by a gamma law gives criteria for RH of Riesz's kind.

As for the Thorin measure and the Lévy form in Chapter 12, what is unconditional at the basepoint one for the scale mixtures and the completeness statements extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH. Chapters 12 to 14 prove no new zero free region.

13.1 Normal scale mixtures and completeness across the strip

The zero free half planes of Chapter 12 have two readings, the second being completeness (Section 13.1.6). The first evaluates φ_α at u^2: since $\varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha + |u|)$, the reciprocal of ξ along the real axis is a characteristic function, that of a normal scale mixture, at every basepoint and with no hypothesis

(Section 13.1.1); what the zeros decide is the class of the law.

We keep the notation of Chapter 12: T is the centre clock and $T_{1/2} = T$, $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac12)^2 T}$, H_a is the first passage time of a standard Brownian motion to the level a, $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$. When every zero lies to the left of the line $\mathrm{Re}\, s = \alpha$, the Thorin density of T_α in the chart $\theta = \sqrt{z}$ is v_α/π, the density $\rho_{\alpha-1/2}$ of (2.1.7) (Theorem 2.1.2); when a zero lies to the right of that line, T_α is not a GGC and v_α/π is not a Thorin density for it (Proposition 14.3.3). Throughout, Z is a standard normal variable independent of everything else.

13.1.1 The reciprocal on the real axis is a characteristic function

If $X \ge 0$ is independent of Z, then $\mathbb{E}e^{iu\sqrt{2X}Z} = \mathbb{E}e^{-u^2X}$ for real u. The centre clock T of [279] is infinitely divisible and self decomposable with no hypothesis (Theorem 8.1.1, whose proof uses the verification of RH below height 1000). For $\alpha > \frac12$, Theorem 12.2.5 writes $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha} \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac12)\widetilde{T}_\alpha}$, where $\sigma^{(\alpha)}$ is an independent subordinator with Laplace exponent $u + 2(\alpha - \frac12)\sqrt{u}$. With $X = T_\alpha$ the evaluation gives the following.

Proposition 13.1.1. *For every $\alpha \ge \frac12$ the function $u \mapsto \varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha+|u|)$ is the characteristic function of $Y_\alpha = \sqrt{2T_\alpha}\,Z$. The law of Y_α is symmetric, unimodal and infinitely divisible with no hypothesis. At the centre it has variance $2\mathbb{E}T = 2\sum_k a_k^{-1} = 0.046210$. For $\alpha > \frac12$ its characteristic function has a corner at the origin, $\xi(\alpha)/\xi(\alpha+|u|) = 1 - b_\alpha|u| + O(u^2)$, so Y_α lies in the domain of attraction of the Cauchy law, with $P(|Y_\alpha| > y) \sim 2b_\alpha/(\pi y)$ as $y \to \infty$.*

Proof. The first statement is the evaluation with $X = T_\alpha$. Symmetry and unimodality hold for every normal scale mixture. The tilt $\widetilde{T}_\alpha$ of the infinitely divisible law of T is infinitely divisible, with Lévy measure $e^{-(\alpha-\frac12)^2 t}W(t)\,dt/t$, and a subordinator run for an independent infinitely divisible time is infinitely divisible, because a Bernstein function composed with a Bernstein function is again one [327, Corollary 3.8]. So T_α is infinitely divisible, and so is Y_α, a Brownian motion run for the independent time $2T_\alpha$. At the centre $\xi'(\frac12) = 0$, so $\xi(\frac12)/\xi(\frac12 + |u|) = 1 - \mathbb{E}T\,u^2 + O(u^4)$, with $\mathbb{E}T = \sum_k a_k^{-1}$ by (2.1.1). For $\alpha > \frac12$ the expansion follows from $\xi(\alpha + |u|) = \xi(\alpha)(1 + b_\alpha|u| + O(u^2))$. The tail comes from that of T_α: by Theorem 12.2.5, $P(T_\alpha > t) \sim b_\alpha(\pi t)^{-1/2}$ as $t \to \infty$, so $P(T_\alpha > t) \le Kt^{-1/2}$ for all $t > 0$ and some constant K. Conditioning on Z, $yP(|Y_\alpha| > y) = \mathbb{E}\big[yP(T_\alpha > y^2/(2Z^2) \mid Z)\big]$, and the integrand is at most $\sqrt{2}K|Z|$ and tends to $\sqrt{2/\pi}\,b_\alpha|Z|$ as $y \to \infty$; by dominated convergence $yP(|Y_\alpha| > y) \to \sqrt{2/\pi}\,b_\alpha\mathbb{E}|Z| = 2b_\alpha/\pi$. □

At the centre $Y_{1/2} = Z\sqrt{2T}$ is the second member of the van Dantzig pair of Section 2.1; it has the same variance as the Pólya variable X, the first member,

as the relation $\mathbb{E}e^{-\theta^2 T} = 1/\mathbb{E}e^{\theta X}$ requires.

With no hypothesis the law of $Y_{1/2}$ is also self decomposable. For $0 < c < 1$ the self decomposability of T (Theorem 8.1.1) gives $T \stackrel{d}{=} cT' + R_c$ with $T' \stackrel{d}{=} T$ and $R_c \geq 0$ independent, and then $Y_{1/2} \stackrel{d}{=} \sqrt{c}\,Y' + Z'\sqrt{2R_c}$, where $Y' = Z\sqrt{2T'}$ has the law of $Y_{1/2}$ and Z' is standard normal and independent of (Z, T', R_c).

Under RH each factor $(1 + u^2/\gamma^2)^{-1}$ of $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ is a characteristic function, that of L/γ for a standard Laplace variable L, and a zero off the line would break this. The quadruple $\frac{1}{2} \pm \delta \pm i\gamma$, with $\gamma > 0$ and $0 < \delta < \frac{1}{2}$, contributes to $\xi(\frac{1}{2})/\xi(\frac{1}{2}+|u|)$, by (2.1.1), the factor $g(u) = |1+u^2/(\gamma-i\delta)^2|^{-2}$, which is positive and integrable with $g(0) = 1$, and

$$\frac{1}{2\pi}\int_{\mathbb{R}} g(u)e^{-iux}\,du = \frac{\gamma^2+\delta^2}{4\gamma\delta}\,e^{-\gamma|x|}\big(\gamma\sin\delta|x| + \delta\cos\delta|x|\big).$$

For $x > 0$ the derivative of the right side is $-\frac{(\gamma^2+\delta^2)^2}{4\gamma\delta}e^{-\gamma x}\sin\delta x$, so its least value, attained at $x = \pi/\delta$, is $-e^{-\pi\gamma/\delta}$ times its value $(\gamma^2+\delta^2)/(4\gamma)$ at $x = 0$. Since g is integrable, it could only be the characteristic function of the law with this density, which is negative there. So g is not a characteristic function, and $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ would still be one, as $W > 0$ ensures, but no longer factor by factor. The defect is tiny: every ordinate is at least γ_1 and $\delta < \frac{1}{2}$, so the least value exceeds -9.5×10^{-39}, the value of the formula at $\gamma = \gamma_1$ and $\delta = \frac{1}{2}$; and a zero off the line would lie above T_0, where $e^{-\pi\gamma/\delta} < e^{-2\pi T_0}$.

Away from the centre neither V_α nor Y_α belongs to a pair. Take $\alpha > \frac{1}{2}$; for V_α the case $\alpha < \frac{1}{2}$ follows from $V_\alpha \stackrel{d}{=} -V_{1-\alpha}$. The tilted Pólya variable V_α of Section 2.1 has characteristic function $\varphi(\theta) = \xi(\alpha + i\theta)/\xi(\alpha)$, and $1/\varphi(i\theta) = \xi(\alpha)/\xi(\alpha - \theta)$ exceeds 1 for $0 < \theta < 2\alpha - 1$, because $|\alpha - \theta - \frac{1}{2}| < \alpha - \frac{1}{2}$ there and $\xi(\frac{1}{2} + s) = \xi(\frac{1}{2})\mathbb{E}\cosh(sX)$ increases with $|s|$; at $\alpha = 0.75$ and $\theta = 0.25$ it is 1.00145. So it is not a characteristic function. By its Cauchy tail Y_α has no exponential moments, so it cannot be a first member. As a second member its partner φ would satisfy $\varphi(i\theta) = \xi(\alpha+|\theta|)/\xi(\alpha)$, and a moment generating function finite on the whole line is analytic, while $\xi(\alpha + |\theta|)/\xi(\alpha) = 1 + b_\alpha|\theta| + O(\theta^2)$ has a corner at $\theta = 0$, since $b_\alpha > 0$ (Proposition 13.1.1).

13.1.2 From the Lévy form in $|u|$ to the Thorin form in u^2

For $\alpha \geq 1$ the function $\log(\xi(\alpha+s)/\xi(\alpha))$ has, for $\operatorname{Re} s > 0$, the Lévy form (12.1.5) with b_α and Π_α in place of b_1 and Π_1, where $x\Pi_\alpha(dx) = \big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\big)dx + \sum_n \Lambda(n)n^{-\alpha}\delta_{\log n}(dx)$ is the signed measure that Section 12.1.4 gives for every basepoint $\alpha \geq 1$. Putting $s = |u|$ gives

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = e^{-b_\alpha|u|}\exp\Big\{-\int_0^\infty \big(e^{-|u|x} - 1 + |u|x\big)\Pi_\alpha(dx)\Big\}, \tag{13.1.1}$$

a representation in $|u|$. The first factor is the characteristic function of a Cauchy law of scale b_α. The second is not in Lévy Khintchine form, because $e^{-|u|x}$ is a function of $|u|$ rather than of u, and the sine squared lemma (2.1.9) converts it: in the chart $\theta = \sqrt{z}$ and at $s = |u|$, as in the derivation of (12.1.6), it writes $e^{-|u|x} + |u|x - 1$ as $\frac{2x}{\pi}\int_0^\infty \log(1 + u^2/\theta^2)\sin^2(x\theta/2)\,d\theta$ for $x > 0$, and the drift converts through $|u| = \frac{1}{\pi}\int_0^\infty \log(1 + u^2/\theta^2)\,d\theta$. Both identities follow from $\int_0^\infty \log(1 + u^2/\theta^2)\cos(x\theta)\,d\theta = \pi(1 - e^{-|u|x})/x$.

By Section 4.1.2, a symmetric law is an extended generalised gamma convolution (EGGC) if and only if it is the law of $Z\sqrt{2T'}$ with T' a GGC independent of Z.

THEOREM 13.1.2. *Let $\alpha \geq 1$, or more generally let $\alpha > \frac{1}{2}$ be such that $\operatorname{Re}\rho < \alpha$ for every zero ρ. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha + |u|)} = \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1 + \frac{u^2}{\theta^2}\Big) v_\alpha(\theta)\,d\theta \Big\}, \tag{13.1.2}$$

where

$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big) x\Pi_\alpha(dx) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + i\theta) > 0,$$

the first expression being available for $\alpha \geq 1$, and as the compensated limit (12.1.8) *(in which $\varepsilon = 1-\alpha$) when $\sup_\rho \operatorname{Re}\rho < \alpha$, so Y_α is a symmetric EGGC with Thorin density v_α/π in the chart θ. With no hypothesis on the zeros, for $\frac{1}{2} < \alpha < 1$ the law of Y_α is an EGGC if and only if ζ has no zeros with real part greater than α, and at the centre if and only if RH holds. Under RH,*

$$\frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + |u|)} = \prod_{\gamma > 0}\Big(1 + \frac{u^2}{\gamma^2}\Big)^{-1}, \qquad Y_{1/2} \stackrel{d}{=} \sum_k \frac{L_k}{\gamma_k},$$

with independent standard Laplace variables L_k, of density $\frac{1}{2}e^{-|y|}$.

Proof. For $\alpha \geq 1$, insert the two identities of the lemma into (13.1.1) and exchange the order of integration, which is justified for $\alpha > 1$ by absolute convergence and at $\alpha = 1$ by the compensation argument of Proposition 12.1.10. This gives the first expression for v_α; the second is its identification, by (12.1.6) (Proposition 12.1.3) at $\alpha = 1$ and (1.3.1) for $\alpha > 1$; for $\frac{1}{2} < \alpha < 1$ with $\sup_\rho \operatorname{Re}\rho < \alpha$ the first expression holds in the compensated form (12.1.8) of Proposition 12.1.13, in which $\varepsilon = 1 - \alpha$. In every case (13.1.2) is the Thorin representation of T_α given by Theorem 2.1.2, $\varphi_\alpha(u) = \exp\{-\int \log(1 + u/z)\,U_\alpha(dz)\}$ with U_α as in (2.1.6), evaluated at u^2 in the chart $z = \theta^2$, and $v_\alpha > 0$ because every Cauchy kernel of Proposition 2.1.3 then has positive width $\alpha - \beta$. Since T_α is a GGC, $Y_\alpha = Z\sqrt{2T_\alpha}$ is an EGGC. Conversely, if $Y_\alpha \stackrel{d}{=} Z\sqrt{2T'}$ with T' a GGC, then $\mathbb{E}e^{-u^2T'} = \mathbb{E}e^{-u^2T_\alpha}$ for real u, so $T' \stackrel{d}{=} T_\alpha$. Hence Y_α is an EGGC exactly when T_α is a GGC, which by Theorem 2.1.2 happens exactly when $\alpha \geq \sup_\rho \operatorname{Re}\rho$. For $\frac{1}{2} < \alpha < 1$ this is the absence of zeros with real

part greater than α, and at the centre, by the symmetry $\rho \mapsto 1-\rho$, it is RH (Proposition 2.1.1). Under RH the Hadamard product (2.1.1) at $s = |u|$ is the product displayed, and each factor $(1+u^2/\gamma^2)^{-1}$ is the characteristic function of L/γ, as in Proposition 10.2.8(iii). □

The hypothesis that every zero lies to the left of the line $\operatorname{Re} s = \alpha$ cannot be dropped from (13.1.2). By Proposition 2.1.3, v_α/π is, off the ordinates of the zeros on the line $\operatorname{Re} s = \alpha$ and whatever the zeros do, the sum over the zeros with $\gamma > 0$ of $P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)$, with $P_{-c} = -P_c$, and by the semigroup property of the Cauchy kernels $\int_0^\infty \frac{2s}{s^2+\theta^2}[P_c(\theta-\gamma) + P_c(\theta+\gamma)]\,d\theta = \frac{2(s+|c|)\operatorname{sgn} c}{(s+|c|)^2+\gamma^2}$ for $s > 0$ and real $c \neq 0$. Integrating in s from 0 to $|u|$ shows that a zero $\rho = \beta + i\gamma$ with $\gamma > 0$ contributes to the right side of (13.1.2) the factor $((\alpha-\beta)^2+\gamma^2)/((|u|+\alpha-\beta)^2+\gamma^2)$ when $\beta < \alpha$, which is its factor in $\xi(\alpha)/\xi(\alpha+|u|)$, but the factor $((|u|+\beta-\alpha)^2+\gamma^2)/((\beta-\alpha)^2+\gamma^2)$ when $\beta > \alpha$. If no zero lies on the line $\operatorname{Re} s = \alpha$, the integral with the signed density therefore converges and equals $\xi(\alpha)/\xi(\alpha+|u|)$ multiplied by

$$\prod_{\beta>\alpha,\ \gamma>0} \frac{\big((|u|+\beta-\alpha)^2+\gamma^2\big)\big((|u|-\beta+\alpha)^2+\gamma^2\big)}{\big((\beta-\alpha)^2+\gamma^2\big)^2},$$

which exceeds one for $u \neq 0$ as soon as there is such a zero, since each factor equals $1+u^2(u^2+2\gamma^2-2(\beta-\alpha)^2)/((\beta-\alpha)^2+\gamma^2)^2$ and $0 < \beta-\alpha < \frac{1}{2} < \gamma$. The series of Proposition 2.1.3 may be integrated term by term, because the absolute value of the term of a zero integrates to $\log(((|\alpha-\beta|+|u|)^2+\gamma^2)/((\alpha-\beta)^2+\gamma^2)) = O(\gamma^{-2})$. A zero on the line $\operatorname{Re} s = \alpha$ contributes instead an atom of mass one at its ordinate, as every zero does at the centre under RH.

The drift b_α becomes, through $|u| = \frac{1}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\,d\theta$, the constant Thorin density b_α/π, which is also the value $v_\alpha(0)/\pi$. At the basepoint one it is the minimum of v_1/π, attained only at $\theta = 0$ (Proposition 12.1.11; confirmed numerically on $[0, 200]$). So Y_1 splits as an independent sum of a Cauchy variable of scale $b_1 = \sum_\rho \rho^{-1} = 0.0230957$ and a symmetric EGGC with Thorin density $(v_1 - b_1)/\pi$ (Figure 13.1, left). This is the image at u^2 of the factorisation $T_1 \overset{d}{=} S + T_1^\circ$ of Proposition 12.1.11 into a positive stable variable of index one half and a GGC with Thorin density $(A+Q)/\pi$: since $\mathbb{E}e^{-u^2 S} = e^{-b_1|u|}$, the Cauchy summand is $Z\sqrt{2S}$, and the EGGC summand is $\sqrt{2T_1^\circ}$ times an independent standard normal variable, with characteristic function $e^{b_1|u|}\xi(1)/\xi(1+|u|)$. Table 13.1 checks (13.1.2); at $\alpha = 0.75$ the formula presupposes that no zero has real part 0.75 or more.

13.1.3 Extended generalised gamma convolutions in the critical strip

Thorin introduced generalised gamma convolutions [344, 345] and extended them to laws on the whole line [346], the extended generalised gamma convolutions of

u	Thorin form, $\alpha=1$	$\xi(1)/\xi(1+\lvert u\rvert)$	Thorin form, $\alpha=0.75$	$\xi(0.75)/\xi(0.75+\lvert u\rvert)$
0.5	0.982809	0.982838	0.988506	0.988520
1	0.954904	0.954930	0.965969	0.965980
2	0.871156	0.871171	0.891317	0.891317
5	0.507984	0.507962	0.536812	0.536769
20	0.0003491	0.0003487	0.0004116	0.0004110

Table 13.1: The Thorin form (13.1.2) against the reciprocal Xi function on the real axis, computed.

Bondesson [39, Section 7.1] (Section 4.1.2). A symmetric law without Gaussian part is an EGGC exactly when its characteristic function is

$$\exp\Big\{-\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big)m(d\theta)\Big\}$$

with a positive measure m on $(0,\infty)$ that, under $z=\theta^2$, is the Thorin measure of the mixing GGC T' in the representation $Z\sqrt{2T'}$ recalled before Theorem 13.1.2. Whether T_α is a GGC is decided by the location of the zeros and not by the sign of v_α on the single line $\operatorname{Re} s=\alpha$: for ξ it is not known whether positivity on one line excludes zeros to its right (Remark 2.1.4).

THEOREM 13.1.3. *Let $\frac12<\alpha<1$ and write $\varepsilon=\alpha-\frac12$.*

1. *With no hypothesis, $\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$ is the characteristic function of the infinitely divisible, symmetric, unimodal law Y_α. If every zero has real part less than α, then* (13.1.2) *holds with the positive density $v_\alpha(\theta)/\pi=\frac1\pi\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(\theta-\gamma)^2}$.*

2. *Y_α is an EGGC if and only if no zero has $\beta>\alpha$, that is, if and only if ζ has no zeros in the half plane $\operatorname{Re} s>\alpha$.*

3. *Let $T_0\ge 2$ be a height up to which all zeros are known to lie on the critical line. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha+\lvert u\rvert)}=\prod_{0<\gamma\le T_0}\frac{\varepsilon^2+\gamma^2}{(\varepsilon+\lvert u\rvert)^2+\gamma^2}\times\mathcal{E}_{\alpha,T_0}(u),$$

where the finite product is the characteristic function of an EGGC whose Thorin density $\sum_{0<\gamma\le T_0}[P_\varepsilon(\theta-\gamma)+P_\varepsilon(\theta+\gamma)]$ is positive, and the remainder satisfies $\lvert\log\mathcal{E}_{\alpha,T_0}(u)\rvert\le C(\lvert u\rvert+u^2)\frac{\log T_0}{T_0}$ for an absolute constant C.

Proof. (1) is Proposition 13.1.1 and Theorem 13.1.2, the sum being that of Proposition 2.1.3. (2) is part of Theorem 13.1.2: Y_α is an EGGC exactly when T_α is a GGC, and a zero with $\beta>\alpha$ puts a pole of φ_α at the point $(\rho-\alpha)^2$

of $\mathbb{C} \setminus (-\infty, 0]$, where the Laplace transform of a GGC is holomorphic (Theorem 2.1.2). (3) Each pair $\frac{1}{2} \pm i\gamma$ with $0 < \gamma \le T_0$ contributes the factor $\frac{\varepsilon^2+\gamma^2}{(\varepsilon+|u|)^2+\gamma^2}$, an EGGC characteristic function whose Thorin density is the pair of Cauchy kernels of width ε at $\pm\gamma$, by the computation after Theorem 13.1.2. Every other zero has $|\gamma| > T_0$, and a pair $\rho, \bar\rho$ with $\gamma > T_0$ contributes $-\log(1+x)$ to $\log \mathcal{E}_{\alpha,T_0}(u)$, with $x = (2(\alpha-\beta)|u| + u^2)/((\alpha-\beta)^2+\gamma^2)$. Since $|\alpha-\beta| < 1$, $x \ge -(\alpha-\beta)^2/((\alpha-\beta)^2+\gamma^2) > -\frac{1}{2}$ and $|x| \le 2(|u|+u^2)/\gamma^2$, so the contribution is at most $4(|u|+u^2)/\gamma^2$ in absolute value. Summing over the zeros with $\gamma > T_0$ gives at most $4(|u|+u^2)\sum_{\gamma>T_0}\gamma^{-2}$, and by partial summation this sum is at most $2\int_{T_0}^{\infty} N(t)t^{-3}\,dt$. For $t \ge 2$ we have $N(t) \le \frac{t}{2\pi}\log t$: no zero has $0 < \gamma < \gamma_1$, and for $t \ge \gamma_1$ the explicit bound of [108], recalled in the proof of Proposition 12.1.11, gives $N(t) \le \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8} + E(t)$, where $\frac{7}{8} + E(t) < \frac{t}{2\pi}\log(2\pi e)$: up to X_0 the left side is at most 3.40 and the right side is at least 6.38, its value at $t = \gamma_1$, and beyond X_0 the left side is less than $\frac{7}{8} + 0.3\log t$ and the right side exceeds $0.45\,t$. So $\sum_{\gamma>T_0}\gamma^{-2} \le \frac{1}{\pi}\int_{T_0}^{\infty} t^{-2}\log t\,dt = \frac{1+\log T_0}{\pi T_0} \le \frac{\log T_0}{T_0}$, the last step because $T_0 \ge 2$, and the bound holds with $C = 4$. □

With the verification height $T_0 = 3 \times 10^{12}$ of Platt and Trudgian [277], which also enters Corollary 9.3.7, the remainder is of relative size at most 8×10^{-11} for $|u| \le 1$, so Y_α is, to that accuracy in characteristic function, an EGGC for every $\alpha > \frac{1}{2}$, whether or not RH holds; Theorem 14.3.6 gives the same statement for the Laplace transform of T_α with explicit constants. Grouping each zero above T_0 with its reflection in the critical line makes the contribution of every off line pair nonnegative below $T_0 - \frac{1}{2}$, so $v_\alpha > 0$ on $[0, T_0 - \frac{1}{2}]$ for every $\alpha > \frac{1}{2}$ (Corollary 14.3.7). A zero above T_0 with $\beta > \alpha$ would stop T_α from being a GGC, and Y_α from being an EGGC, whether or not it made v_α negative near its ordinate (Exercise 1.7), and no finite computation excludes such a zero (Section 1.3.1). What can be said with no hypothesis is in Theorem 14.3.2.

Table 13.2 checks the product over the computed zeros against $\xi(\alpha)/\xi(\alpha+|u|)$ at $\alpha = 0.6$ and 0.75. On $[0, 200]$ the Thorin density is positive at both basepoints, and in both cases its minimum is at $\theta = 0$ and equals the Cauchy floor b_α/π: 0.0014709 at $\alpha = 0.6$ and 0.0036769 at $\alpha = 0.75$ (Figure 13.1, left).

13.1.4 The density, the Lévy measure and the Poisson trace

Because $\xi(\alpha)/\xi(\alpha+|u|)$ is integrable, Y_α has the density

$$f_{Y_\alpha}(y) = \frac{1}{\pi}\int_0^{\infty} \cos(uy)\frac{\xi(\alpha)}{\xi(\alpha+u)}du,$$

which is computed. At the origin $f_{Y_\alpha}(0) = 1.8994,\ 1.8680,\ 1.8225,\ 1.7507$ for $\alpha = \frac{1}{2},\ 0.6,\ 0.75,\ 1$. The corner of the characteristic function at the origin gives the tail $f_{Y_\alpha}(y) \sim b_\alpha/(\pi y^2)$ for $\alpha > \frac{1}{2}$ (Figure 13.1, middle). At the centre there

α	u	Thorin integral (13.1.2)	product over zeros	$\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$
0.6	1	0.9726796	0.9726812	0.9726812
0.6	5	0.5549761	0.5549190	0.5549193
0.6	20	0.0004547	0.0004539	0.0004539
0.75	1	0.9659691	0.9659798	0.9659798
0.75	5	0.5368120	0.5367687	0.5367691
0.75	20	0.0004116	0.0004110	0.0004110

Table 13.2: The EGGC representation in the strip, computed: the Thorin integral over $\theta \le 200$ plus the Weyl tail, and the product of the factors of Theorem 13.1.3(3) over the 1700 computed zeros plus the Weyl tail.

is no corner, and the density decays like $e^{-\gamma_1|y|}$, the tail of L_1/γ_1 under RH; unconditionally, the characteristic function continues to the meromorphic function $\xi(\frac12)/\xi(\frac12+u)$ of u, whose poles nearest the real axis are $\pm i\gamma_1$.

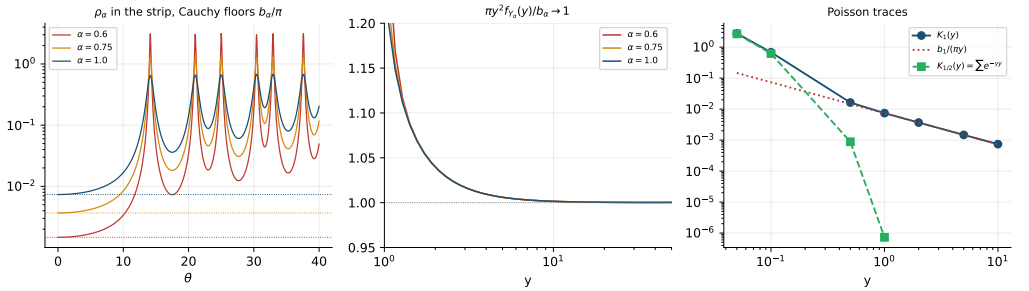


Figure 13.1: Left: the Thorin densities v_α/π at $\alpha = 0.6$, 0.75, 1, written ρ_α in the plot, and their Cauchy floors b_α/π (dotted). Middle: $\pi y^2 f_{Y_\alpha}(y)/b_\alpha$, tending to one. Right: the Poisson traces K_1 and $K_{1/2}$ against $b_1/(\pi y)$.

Frullani's integral gives $\log(1+u^2/\theta^2) = \int_{\mathbb{R}}(1-\cos uy)\,e^{-\theta|y|}\,\frac{dy}{|y|}$, the sine squared lemma read in the other direction, since $1-\cos uy = 2\sin^2(uy/2)$, and the cosine dictionary (10.2.4) of Section 10.2. Inserting it in (13.1.2) puts the characteristic function of Y_α in Lévy Khintchine form in u:

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \exp\Big\{-\int_{\mathbb{R}} 2\sin^2\Big(\frac{uy}{2}\Big)\,\frac{K_\alpha(|y|)}{|y|}\,dy\Big\},$$
$$K_\alpha(y) = \frac{1}{\pi}\int_0^\infty e^{-\theta y} v_\alpha(\theta)\,d\theta.$$

So the Lévy density of Y_α is $K_\alpha(|y|)/|y|$, as Section 4.1.2 gives for a symmetric EGGC; in the normalisation of (10.2.4), $K_\alpha = \frac12 K_{U_\alpha}$. We call K_α the Poisson trace, because at the centre, where under RH the Thorin measure in the chart θ

is $\sum_{\gamma>0}\delta_\gamma$, it is $K_{1/2}(y)=\sum_{\gamma>0}e^{-\gamma y}$, which is $e^{-iy/2}V(iy)$ for Cramér's function $V(z)=\sum_{\gamma>0}e^{\rho z}$ [66] (compare Proposition 10.2.8(iii)): the sum over the zeros of the Poisson kernel in place of the heat kernel of the heat trace $W(t)=\sum_k e^{-a_k t}$, which is $\sum_{\gamma>0}e^{-\gamma^2 t}$ under RH. Since $Y_{1/2}=B_{2T}$ for an independent Brownian motion B, and T has Lévy density $W(t)/t$ (Theorem 8.1.1), the Lévy density of $Y_{1/2}$ is $\int_0^\infty (4\pi t)^{-1/2}e^{-y^2/4t}\,W(t)\,t^{-1}dt$, so

$$K_{1/2}(y)=\int_0^\infty \frac{y}{\sqrt{4\pi t^3}}\,e^{-y^2/4t}\,W(t)\,dt=\mathbb{E}\,W(H_{y/\sqrt{2}}),\qquad y>0.$$

So the Poisson trace is the heat trace subordinated by the stable one half first passage kernel, whose Laplace transform $e^{-y\sqrt{a}}$ is the stable one half average of the heat kernel recalled after Proposition 10.2.9. This defines $K_{1/2}$ unconditionally, as $K_{1/2}(y)=\sum_k e^{-y\sqrt{a_k}}$ with the principal root, and makes it computable from the explicit formula (2.1.5) for W.

Corollary 13.1.4. *RH holds if and only if the Poisson trace $K_{1/2}$ is completely monotone, and if and only if the heat trace W is completely monotone. For $\alpha\geq 1$ the Poisson trace K_α is completely monotone with no hypothesis, and $K_\alpha(y)\sim b_\alpha/(\pi y)$ as $y\to\infty$, so that the Lévy density of Y_α is asymptotic to $b_\alpha/(\pi y^2)$, the Lévy density of the Cauchy component.*

Proof. For $\alpha\geq 1$, K_α is the Laplace transform of the positive density v_α/π, hence completely monotone, and since v_α is continuous with $v_\alpha(0)=b_\alpha$, $yK_\alpha(y)\to b_\alpha/\pi$ as $y\to\infty$. At the centre, W is completely monotone exactly when T is a GGC (Section 2.1), that is exactly when RH holds (Proposition 2.1.1). Under RH, $K_{1/2}(y)=\sum_{\gamma>0}e^{-\gamma y}$ is completely monotone. Conversely, suppose that $K_{1/2}(y)=\int_{[0,\infty)}e^{-y\theta}\nu(d\theta)$ with a positive measure ν. For $w>0$ the unconditional formula $K_{1/2}(y)=\sum_k e^{-y\sqrt{a_k}}$ gives

$$\int_0^\infty e^{-wy}\,y\,K_{1/2}(y)\,dy=\sum_k\frac{1}{(w+\sqrt{a_k})^2}=\int_{[0,\infty)}\frac{\nu(d\theta)}{(w+\theta)^2}.$$

The first equality integrates term by term, which is legitimate because $\sqrt{a_k}=\gamma_k-i(\beta_k-\frac12)$ has real part γ_k when $\rho_k=\beta_k+i\gamma_k$ is taken with $\gamma_k>0$, so the absolute values of the terms integrate to $\sum_k(w+\gamma_k)^{-2}<\infty$. The second is Tonelli's theorem, and it shows that the right side is finite for $w>0$. The middle expression is meromorphic in $w\in\mathbb{C}$, with a double pole at each point $-\sqrt{a_k}$, while the right side is holomorphic on $\mathbb{C}\setminus(-\infty,0]$, since $|w+\theta|$ is bounded below by a positive multiple of $1+\theta$ for w in a compact subset of $\mathbb{C}\setminus(-\infty,0]$. The two sides agree for $w>0$, hence on $\mathbb{C}\setminus(-\infty,0]$ with the poles removed, so no pole $-\sqrt{a_k}$ lies in $\mathbb{C}\setminus(-\infty,0]$. Since $\operatorname{Re}\sqrt{a_k}>0$, every $\sqrt{a_k}$ is therefore real and positive, that is, every $a_k=-(\rho_k-\frac12)^2$ is positive, which is RH. □

The traces K_1 and $K_{1/2}$ are drawn in Figure 13.1, right. Propositions 13.1.5 and 13.1.6 are based on [139].

PROPOSITION 13.1.5 (The positive half of the reciprocal law). *Let A be the centred infinitely divisible law without Gaussian part whose Lévy measure is $K_{1/2}(x)\,dx/x$ on $(0,\infty)$, a positive measure by Proposition 14.4.1, and let A' be an independent copy of A. Then $Y_{1/2} \stackrel{d}{=} A - A'$, and $\kappa_m(A) = \int_0^\infty x^{m-1} K_{1/2}(x)\,dx$ for $m \geq 2$; in particular* $\operatorname{Var} A = \mathbb{E}T$.

(i) Under RH, with γ_k the positive ordinates counted with multiplicity and E_k independent standard exponential variables, A has the law of $\sum_k (E_k - 1)/\gamma_k$, a series that converges almost surely and in L^2; $\mathbb{E}e^{-sA} = 1/\mathcal{P}(s)$ for $s > -\gamma_1$, where $\mathcal{P}(s) = \prod_k (1 + s/\gamma_k) e^{-s/\gamma_k}$ and $\mathcal{P}(s)\mathcal{P}(-s) = \xi(\frac{1}{2} + is)/\xi(\frac{1}{2})$; and $\kappa_m(A) = (m-1)! \sum_k \gamma_k^{-m}$.

(ii) If A has the law of $\sum_k (E_k - 1)/c_k$ for positive numbers c_k with $\sum_k c_k^{-2} < \infty$, then RH holds.

(iii) For positive γ_k with $\sum_k \gamma_k^{-2} < \infty$, the function $u \mapsto \mathcal{P}(\sqrt{u})$ is the Laplace transform of a GGC whose Thorin density in the chart θ is $\frac{1}{\pi} \sum_k \theta^2 / \big(\gamma_k(\gamma_k^2 + \theta^2)\big)$.

For $t \geq \Lambda$, unconditionally for $t \geq 0.2$, the series $A_t = \sum_k (E_k - 1)/\theta_k(t)$ over the zeros $\frac{1}{2} \pm i\theta_k(t)$ of ξ_t satisfies $B_{2T_t} \stackrel{d}{=} A_t - A_t'$, with T_t the tilted clock of Section 4.1.8, and $\mathbb{E}e^{-sA_t} = 1/\mathcal{P}_t(s)$, $\mathcal{P}_t(s)\mathcal{P}_t(-s) = \xi_t(\frac{1}{2} + is)/\xi_t(\frac{1}{2})$, with $\mathcal{P}_t$ the product of (i) over the $\theta_k(t)$.

Proof. By Section 13.1.4, $Y_{1/2}$ is symmetric and infinitely divisible without Gaussian part, with Lévy density $K_{1/2}(|y|)/|y|$. The law A exists because $K_{1/2}(x) = O(x^{-1}\log(1/x))$ at 0 (Proposition 13.1.11) and $K_{1/2}$ decays exponentially at infinity, and $\mathbb{E}e^{iu(A-A')} = \exp\{-\int_0^\infty 4\sin^2(ux/2) K_{1/2}(x)\,dx/x\}$ is the characteristic function of $Y_{1/2}$. The cumulants of A are the moments of its Lévy measure, and $\int_0^\infty x K_{1/2}(x)\,dx = \sum_k a_k^{-1} = \mathbb{E}T$ by $K_{1/2}(x) = \sum_k e^{-x\sqrt{a_k}}$. (i) Under RH $K_{1/2}(x) = \sum_k e^{-\gamma_k x}$, and by Frullani's integral $(E-1)/\gamma$ has Lévy measure $e^{-\gamma x}\,dx/x$ on $(0,\infty)$. The series has independent centred terms with $\sum_k \gamma_k^{-2} < \infty$, and $\mathbb{E}e^{-s(E-1)/\gamma} = e^{s/\gamma}\gamma/(\gamma + s)$ for $s > -\gamma$; the partial products are bounded uniformly on compact subsets of $(-\gamma_1, \infty)$, which gives uniform integrability and $\mathbb{E}e^{-sA} = 1/\mathcal{P}(s)$. The product $\mathcal{P}(s)\mathcal{P}(-s) = \prod_k (1 - s^2/\gamma_k^2)$ is $\xi(\frac{1}{2} + is)/\xi(\frac{1}{2})$ by Lemma 6.2.1. (ii) Uniqueness of the Lévy measure gives $K_{1/2}(x) = \sum_k e^{-c_k x}$, which is completely monotone, and Corollary 13.1.4 gives RH. (iii) For $c > 0$, differentiation in $\sqrt{u}$ gives $\log(1 + \sqrt{u}/c) = \int_0^\infty \log(1 + u/\theta^2)\, \frac{c}{\pi(c^2+\theta^2)}\,d\theta$, and $\sqrt{u}/c = \int_0^\infty \log(1 + u/\theta^2)\, \frac{d\theta}{\pi c}$. At $c = \gamma_k$ the difference has the positive integrand

$\log(1+u/\theta^2)\,\theta^2/\big(\pi\gamma_k(\gamma_k^2+\theta^2)\big)$, and summing over k by Tonelli's theorem,

$$-\log \mathcal{P}(\sqrt{u}) = \int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\frac{1}{\pi}\sum_k \frac{\theta^2}{\gamma_k(\gamma_k^2+\theta^2)}\,d\theta,$$

where the left side is finite because $x-\log(1+x)\le x^2/2$ for $x\ge 0$ and $\sum_k \gamma_k^{-2}<\infty$. For the flow, $\mathbb{E}e^{iuB_{2T_t}} = \mathbb{E}e^{-u^2T_t} = R_t(u^2) = \xi_t(\frac12)/\xi_t(\frac12+u) = \prod_k(1+u^2/\theta_k(t)^2)^{-1}$ by Proposition 4.1.18 and Lemma 6.2.1, and the rest is (i) with $\theta_k(t)$ in place of γ_k. □

Thus $\operatorname{sd} A = (\mathbb{E}T)^{1/2} = 0.152003$, the excess kurtosis of A is $6\operatorname{Var}T/(\mathbb{E}T)^2 = 0.41779$, and under RH its skewness is $2\sum_k\gamma_k^{-3}/(\mathbb{E}T)^{3/2} = 0.4155$ (computed). For the lattice $\gamma_k = k$ the series is the Gumbel law less Euler's constant (Section 19.20), with skewness $12\sqrt{6}\,\zeta(3)/\pi^3 = 1.1395$ and excess kurtosis $\frac{12}{5}$; the ordinates spread the mass over many comparable rates, and A is much closer to normal.

Proposition 13.1.6 (Primes as poles of the Poisson trace). *With Cramér's function V defined above, whose sum runs over the zeros $\rho=\beta+i\gamma$ with $\gamma>0$, $K_{1/2}(x)=e^{-ix/2}V(ix)$, and the series converges for $\operatorname{Re}x>0$. The function $K_{1/2}$ continues to a meromorphic function on $\mathbb{C}\setminus(-\infty,0]$ (the point $x=0$ is a logarithmic branch point), whose poles are simple and lie at $x=\pm i\log n$ for the prime powers n, with residue $-\Lambda(n)/(2\pi\sqrt{n})$.*

Proof. The relation to V is the definition read at $z=ix$, since $\sqrt{a_k}=-i(\rho-\frac12)$ for the zero ρ with $\gamma>0$ that belongs to a_k, and the continuation of V with its simple poles at $z=\pm m\log p$, p prime and $m\ge1$, is due to Cramér [66]. For the residue fix $n\ge2$. By Landau's formula [203], $\sum_{0<\gamma\le T} n^\rho = -\frac{T}{2\pi}\Lambda(n)+O(\log T)$, and since $\rho\mapsto 1-\bar\rho$ permutes the zeros with $\gamma>0$, $S(T)=\sum_{0<\gamma\le T}n^{-\rho} = n^{-1}\overline{\sum_{0<\gamma\le T}n^\rho} = -cT+O(\log T)$ with $c=\Lambda(n)/(2\pi n)$. For $\varepsilon>0$,

$$K_{1/2}(\varepsilon+i\log n) = \sqrt{n}\sum_{\gamma>0} e^{i\varepsilon(\rho-\frac12)}n^{-\rho}, \qquad e^{i\varepsilon(\rho-\frac12)} = e^{-\varepsilon\gamma}e^{i\varepsilon(\beta-\frac12)}.$$

Since $|e^{i\varepsilon(\beta-\frac12)}-1|\le\varepsilon/2$, $|n^{-\rho}|\le1$ and $\sum_{\gamma>0}e^{-\varepsilon\gamma}=O(\varepsilon^{-1}\log(1/\varepsilon))$ by the zero count, replacing $e^{i\varepsilon(\beta-\frac12)}$ by 1 changes $\varepsilon K_{1/2}(\varepsilon+i\log n)$ by $O(\varepsilon\log(1/\varepsilon))$. Abel summation against S gives

$$\varepsilon\sum_{\gamma>0}e^{-\varepsilon\gamma}n^{-\rho} = \varepsilon^2\int_0^\infty e^{-\varepsilon T}S(T)\,dT = -c+O\big(\varepsilon\log(1/\varepsilon)\big),$$

hence $\varepsilon K_{1/2}(\varepsilon+i\log n)\to-\sqrt{n}\,c=-\Lambda(n)/(2\pi\sqrt{n})$ as $\varepsilon\to0^+$. This limit is the residue: near $i\log n$ Cramér's continuation has the form $K_{1/2}(x)=r\,(x-i\log n)^{-1}+h(x)$ with h holomorphic, so along the horizontal ray $x=i\log n+\varepsilon$

from the right, on which the series converges, $\varepsilon K_{1/2}(x) = r + \varepsilon h(x) \to r$ (a pole of higher order would make the limit infinite). Since the zeros ρ and $1-\bar\rho$ have conjugate values of $\sqrt{a_k}$, $K_{1/2}(\bar x) = \overline{K_{1/2}(x)}$, so the residue at $-i\log n$ is the conjugate of the real number $-\Lambda(n)/(2\pi\sqrt{n})$, the same value. □

From the 10378 zeros below 10200, $\varepsilon K_{1/2}(\varepsilon + i\log n)$ at $\varepsilon = 0.05$, 0.02, 0.01, 0.005, extrapolated linearly to $\varepsilon = 0$ from the values at 0.01 and 0.005, agrees with $-\Lambda(n)/(2\pi\sqrt{n})$ to within 7.1×10^{-4} for $n = 2,3,4,5,7,8,9$, and for the control $n = 6$, where $\Lambda(6) = 0$, the extrapolated value is 5.5×10^{-4}, of the size of the errors (computed). The Poisson trace carries the zeros as exponentials on the positive axis and the primes as poles on the imaginary axis.

13.1.5 The complete monotonicity conditions of the Poisson trace

By Corollary 13.1.4, RH is the complete monotonicity of the Poisson trace $K_{1/2}$. This subsection is based on [284] and compares $K_{1/2}$ with the corresponding trace of Pólya's function $P(\theta) = K_{i\theta/2}(2\pi)$ of Section 18.1, which is completely monotone with no hypothesis.

Write the zeros as $\rho_k = \beta_k + i\gamma_k$ with $\gamma_k > 0$, so that $\sqrt{a_k} = \gamma_k + i\delta_k$ with $\delta_k = \frac12 - \beta_k \in (-\frac12, \frac12)$, and put $\varphi_k = \arctan(\delta_k/\gamma_k)$. The series $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ converges absolutely, with all its derivatives, for $y > 0$, and the zeros ρ_k and $1-\bar\rho_k$ have conjugate values of $\sqrt{a_k}$. Hence

$$(-1)^n K_{1/2}^{(n)}(y) = \sum_k \operatorname{Re}\left((\sqrt{a_k})^n e^{-y\sqrt{a_k}}\right) = \sum_k |\sqrt{a_k}|^n\, e^{-\gamma_k y}\cos(n\varphi_k - \delta_k y), \tag{13.1.3}$$

with each zero counted once, with its multiplicity; a zero on the line contributes $\gamma_k^n e^{-\gamma_k y} > 0$. So $K_{1/2}$ is completely monotone, and by Corollary 13.1.4 RH holds, exactly when every one of the quantities (13.1.3) is nonnegative, for $n = 0, 1, 2, \ldots$ and $y > 0$.

Proposition 13.1.7. *The term of a zero with $\delta_k \neq 0$ in* (13.1.3) *is positive whenever*

$$n\arctan\frac{|\delta_k|}{\gamma_k} + |\delta_k|\, y < \frac{\pi}{2}.$$

If $\delta_k > 0$ and $\delta_k y < \frac{\pi}{2}$, the first order n at which the term is negative is the least integer with $n\varphi_k - \delta_k y > \frac{\pi}{2}$, that is the least integer above $(\frac{\pi}{2} + \delta_k y)/\varphi_k$ when this ratio is not an integer. The partner $1-\bar\rho_k$ has the same term.

Proof. $|n\varphi_k - \delta_k y| \le n|\varphi_k| + |\delta_k| y$, and the cosine is positive on $(-\frac{\pi}{2}, \frac{\pi}{2})$. For $\delta_k > 0$ the argument $n\varphi_k - \delta_k y$ increases with n from $-\delta_k y \in (-\frac{\pi}{2}, 0)$, and the cosine is first negative when it passes $\frac{\pi}{2}$. The partner has $-\delta_k$ and $-\varphi_k$, so its argument is the negative of this one. □

A computation finds the first negative order by evaluating the sign factor; Table 13.3 compares it with the formula.

γ	δ	y	first negative order	$(\frac{\pi}{2}+\delta y)/\varphi$
50	0.25	1	365	364.16
50	0.25	0.3	330	329.16
200	0.1	1	3342	3341.59
14.13	0.5	0.5	52	51.48

Table 13.3: The first order at which the term of a hypothetical zero $\frac{1}{2}-\delta+i\gamma$ in (13.1.3) is negative, against Proposition 13.1.7. The last row is a zero at the edge of the strip at the height of the first ordinate. Computed.

THEOREM 13.1.8. *Suppose that every zero with $0<\gamma_k\le H$ lies on the critical line and that every zero has $|\delta_k|\le\Delta$, with $0<\Delta\le\frac{1}{2}$. Then for $0<y<\pi/(2\Delta)$ and every integer n with $0\le n<\pi H/(2\Delta)$, every term of* (13.1.3) *is positive, and $(-1)^nK_{1/2}^{(n)}(y)>0$. Since $0<\beta_k<1$, $\Delta=\frac{1}{2}$ is admissible with no hypothesis, and with $H=T_0=3\times10^{12}$ [277] this covers $0<y<\pi$ and every $n<\pi T_0=9.42\times10^{12}$.*

Proof. A zero with $\gamma_k\le H$ has $\delta_k=0$, and its term $\gamma_k^n e^{-\gamma_k y}$ is positive. A zero with $\gamma_k>H$ has φ_k of the sign of δ_k, $n|\varphi_k|\le n|\delta_k|/\gamma_k\le n\Delta/H<\frac{\pi}{2}$ and $|\delta_k|y\le\Delta y<\frac{\pi}{2}$. So $n\varphi_k-\delta_k y$, a difference of two numbers of the same sign and of absolute value less than $\frac{\pi}{2}$, lies in $(-\frac{\pi}{2},\frac{\pi}{2})$, and the cosine in (13.1.3) is positive. The series converges absolutely, so its sum is positive. □

Every condition that could fail has $y\ge\pi$ or $n\ge\pi H$; there complete monotonicity is a characterisation of RH, not a test that a computation could run to a failure.

REMARK 13.1.9. The rectangle uses that φ_k and δ_k have the same sign, which the sum bound of Proposition 13.1.7 discards; the observation is from the first author's note on Ostrovsky's laws and the quasi Riemann hypothesis. Under $(\mathrm{Q}_{7/8})$ of Section 23.1, which rests on the unrefereed half plane of [257], $\Delta=\frac{3}{8}$ and the theorem holds for $0<y<\frac{4\pi}{3}=4.1888$ and $n<\frac{4\pi}{3}T_0=1.26\times10^{13}$. Liu's claimed half plane $\operatorname{Re}s>0.874957$ [223], also unrefereed, would give $y<4.1893$. The half plane at $\frac{7}{8}$ changes the constants by the factor $\frac{4}{3}$ and nothing else.

The weight of the zero γ_k in the condition (n,y) is about $\gamma_k^n e^{-\gamma_k y}$, whose logarithm $n\log\gamma-\gamma y$ is largest at

$$\gamma_*=\frac{n}{y},$$

with a width of order $\sqrt{n}/y$. So the condition (n, y) probes the zeros near the height n/y. Among the first 300 ordinates the dominant one is, in every case tested, the ordinate nearest to n/y, for instance 98.831 for $(n, y) = (100, 1)$. Figure 13.2 shows two weight profiles and two sign factors. By Proposition 13.1.7 a zero at height γ can make its term negative only when $n \gtrsim \frac{\pi}{2}\gamma/|\delta| \geq \pi\gamma$, that is when $\gamma \lesssim n/\pi$, below the peak at n/y when $y < \pi$; and since $\gamma^n e^{-\gamma y}$ increases up to n/y, its value at $\gamma \leq n/\pi$ is at most $\exp(n(\log \frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its value at the peak, which is exponentially small in n. A proof along these lines would need a bound for the number of zeros off the line near each height.

For the whole sum the first failure lies far above the termwise order. We replaced two adjacent ordinates among the first 500 by a pair $\frac{1}{2} \pm \delta + i\bar{\gamma}$ at their midpoint, and located by bisection over integers the order n^* at which the minimum of $(-1)^n K_{1/2}^{(n)}(y)$ over $y \in [0.85, 1.15]\, n/\bar{\gamma}$ first becomes negative; the bisection assumes that the sign is monotone in n, and at sampled lower orders no negative value occurs for $y \leq 30$ or $y \leq 5n/\bar{\gamma}$. For $\bar{\gamma} = 54.7$, 125.9, 272.5 and $\delta = 0.1$, 0.375 the ratio $n^*\delta/\bar{\gamma}^2$ lies between 0.63 and 1.19, the law $n^* \approx \gamma^2/\delta$ of the heat trace (Table 17.6), against the termwise order $\pi\bar{\gamma}/(2\delta)$ near $y = 0$.

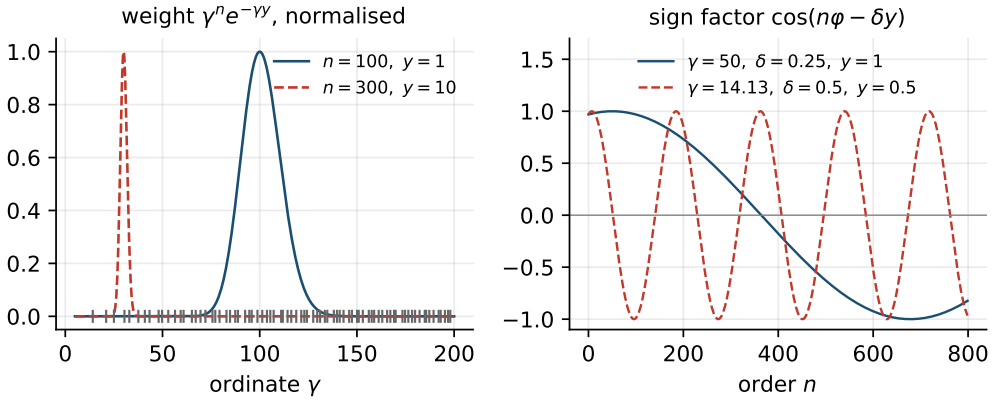


Figure 13.2: Left: the weight $\gamma^n e^{-\gamma y}$ of the condition (n, y) in (13.1.3), normalised to maximum one, for $(n, y) = (100, 1)$ and $(300, 10)$, peaking at $\gamma = n/y$; the marks are the ordinates below 200. Right: the sign factor $\cos(n\varphi - \delta y)$ of a hypothetical zero off the line for $(\gamma, \delta, y) = (50, 0.25, 1)$ and $(14.13, 0.5, 0.5)$, positive below the orders 365 and 52, at which it is first negative.

Proposition 13.1.10. *The following are equivalent:*

(i) RH;

(ii) $K_{1/2}$ is completely monotone on $(0, \infty)$;

(iii) for every $x > 0$ the function $y \mapsto K_{1/2}(x + iy)$ is positive definite on $\mathbb{R}$;

(iv) for some $x > 0$ the function $y \mapsto K_{1/2}(x+iy)$ is bounded on $[0,\infty)$.

Under these conditions $|K_{1/2}(x+iy)| \le K_{1/2}(x)$.

Proof. The series $K_{1/2}(z) = \sum_k e^{-z\sqrt{a_k}}$ converges absolutely and locally uniformly, to an analytic function, for $\operatorname{Re} z > 0$, since $|e^{-z\sqrt{a_k}}| = e^{-\gamma_k \operatorname{Re} z + \delta_k \operatorname{Im} z} \le e^{-\gamma_k \operatorname{Re} z + |\operatorname{Im} z|/2}$. (i) implies (ii) by Corollary 13.1.4. If (ii) holds, Bernstein's theorem gives $K_{1/2}(y) = \int_{[0,\infty)} e^{-y\theta}\nu(d\theta)$ with a positive measure ν, the integral converges for $\operatorname{Re} z > 0$ and agrees there with $K_{1/2}(z)$ by analytic continuation, and $K_{1/2}(x+iy) = \int e^{-iy\theta}e^{-x\theta}\nu(d\theta)$ is the Fourier transform of the finite positive measure $e^{-x\theta}\nu(d\theta)$, which gives (iii) and the bound. A positive definite function f satisfies $|f(y)| \le f(0)$, so (iii) implies (iv). Suppose (iv) holds at x. For $\operatorname{Re} w > \frac12$ the bound above allows term by term integration, and

$$\int_0^\infty e^{-wy} K_{1/2}(x+iy)\,dy = \sum_k \frac{e^{-x\sqrt{a_k}}}{w + i\sqrt{a_k}}.$$

The right side is a meromorphic function of $w \in \mathbb{C}$, since $\sum_k e^{-x\gamma_k} < \infty$, with a pole at each point $-i\sqrt{a_k} = \delta_k - i\gamma_k$, of residue $e^{-x\sqrt{a_k}}$ times the multiplicity, which is not zero. Since $K_{1/2}(x+iy)$ is bounded, the left side is analytic in $\operatorname{Re} w > 0$, so no pole has $\delta_k > 0$. The zeros ρ_k and $1-\bar\rho_k$ have opposite δ_k, so every δ_k is 0, which is RH. □

Written out, (iii) says that for every $x > 0$, all real $y_1, \dots, y_m$ and all complex $c_1, \dots, c_m$,

$$\sum_{j,l} c_j \bar c_l\, K_{1/2}\big(x + i(y_j - y_l)\big) = \sum_k e^{-x\gamma_k} \Big| \sum_j c_j e^{-i\gamma_k y_j} \Big|^2 \ge 0,$$

the equality holding under RH.

One would like to turn each condition into a statement about primes with the explicit formula of Section 2.1 (the display before (2.1.5)). The test function of the condition (n, y) is $h(r) = |r|^n e^{-y|r|}$, with

$$g(x) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r) e^{-irx}\,dr = \frac{n!}{\pi} \operatorname{Re} \frac{1}{(y+ix)^{n+1}},$$

which decays only like a power of $|x|$ and has one sign for large $|x|$. The prime side $-2\sum_{m\ge2} \Lambda(m) m^{-1/2} g(\log m)$ then diverges, since $\sum_p p^{-1/2}(\log p)^{-j} = \infty$ for every j; the formula holds for g decaying faster than $e^{-|x|/2}$, which goes with h analytic in a strip $|\operatorname{Im} r| < \frac12 + \epsilon$, and this h is not analytic at $r = 0$. Smoothing in the variable of the zeros removes the divergence without changing the problem: the Gaussian weight $e^{-r^2 t}$ in place of $e^{-y|r|}$ gives the heat trace $W(t)$, whose explicit formula (2.1.5) converges absolutely, and whose complete monotonicity is again

equivalent to RH (Corollary 13.1.4). The note [284] identifies this smoothing with the de Bruijn Newman flow, but the Gaussian weight acts on the test function, while the flow multiplies the kernel Φ by $e^{tu^2/4}$ and moves the zeros.

By Proposition 18.1.13,

$$\log \frac{K_{\lambda/2}(2\pi)}{K_0(2\pi)} = \int_0^\infty 4\sin^2\Big(\frac{\lambda y}{2}\Big) K^P(y)\,\frac{dy}{y},$$
$$K^P(y) = \sum_n e^{-2\nu_n y} = \operatorname{tr} e^{-2y\sqrt{A_P}},$$

the sum over the zeros $\theta_n = 2\nu_n$ of P, so K^P is the counterpart of $K_{1/2}$ in the formula for $\xi(\frac12)/\xi(\frac12+u)$, and it is completely monotone because A_P is selfadjoint and positive.

PROPOSITION 13.1.11. *Let* $M(y) = \frac{1}{2\pi y}\big(\log\frac{1}{2\pi y} - \gamma_E\big)$ *and* $c = 2\pi - \frac{1}{6\pi} - \frac{1}{48\pi} = 6.2235022035$. *As* $y \to 0+$:

(i) $K_{1/2}(y) = M(y) + \frac78 + O(y\log\frac1y)$, *and under RH* $K_{1/2}(y) = M(y) + \frac78 + \frac{1}{48\pi} y\log\frac1y + o(y\log\frac1y)$;

(ii) $K^P(y) = M(y) - \frac14 + \big(2\pi - \frac{1}{6\pi}\big) y\log\frac1y + O(y)$;

(iii) $D(y) = K_{1/2}(y) - K^P(y) = \frac98 + O(y\log\frac1y)$, *and under RH* $D(y) = \frac98 - c\,y\log\frac1y + o(y\log\frac1y)$.

Proof. (i) The real part of the term $e^{-y\sqrt{a_k}}$ is $e^{-\gamma_k y}\cos(\delta_k y) = e^{-\gamma_k y} + O(y^2 e^{-\gamma_k y})$, and the imaginary parts cancel in pairs, so $K_{1/2}(y) = \sum_k e^{-\gamma_k y} + O(y^2\sum_k e^{-\gamma_k y})$, and the sum will turn out to be $O(y^{-1}\log\frac1y)$. By Abel summation $\sum_k e^{-\gamma_k y} = y\int_0^\infty e^{-Ty} N(T)\,dT$, with $N(T) = \vartheta(T)/\pi + 1 + S(T)$ [348, Section 9.3]. The Riemann Siegel function satisfies $\vartheta(T)/\pi = N_0(T) - \frac18 + \frac{1}{48\pi T} + O(T^{-3})$ with $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$, so $N = N_0 + \frac78 + S + E$ with $E(T) = \frac{1}{48\pi T} + O(T^{-3})$ for $T \ge 1$ and E bounded on $(0,1]$. Now $\int_0^\infty T e^{-Ty}\log T\,dT = y^{-2}(1 - \gamma_E - \log y)$ gives $y\int_0^\infty e^{-Ty} N_0(T)\,dT = M(y)$; the constant gives $\frac78$; $y\int_1^\infty e^{-Ty}\frac{dT}{48\pi T} = \frac{1}{48\pi} y\log\frac1y + O(y)$, and the rest of E gives $O(y)$. With $S_1(T) = \int_0^T S(t)\,dt$, an integration by parts gives $y\int_0^\infty e^{-Ty} S(T)\,dT = y^2\int_0^\infty e^{-Ty} S_1(T)\,dT$, which is $O(y\log\frac1y)$ since $S_1(T) = O(\log T)$ [348, Theorem 9.9(A)]. Under RH there are no terms $O(y^2 e^{-\gamma_k y})$, and $S_1(T) = O(\log T/(\log\log T)^2)$ [348, Theorem 14.13] makes the last integral $o(y\log\frac1y)$. (ii) By Proposition 18.1.8, the number of zeros $\theta_n \le T$ is $\lfloor x(T)\rfloor$, with $x(T) = N_0(T) + \frac14 + (2\pi - \frac{1}{6\pi})T^{-1} + O(T^{-3})$ for $T \ge 1$. Write $\lfloor x\rfloor = x - \frac12 + (\frac12 - \{x\})$. The terms N_0, $\frac14 - \frac12$ and $(2\pi - \frac{1}{6\pi})T^{-1}$ give $M(y)$, $-\frac14$ and $(2\pi - \frac{1}{6\pi}) y\log\frac1y + O(y)$ as in (i). The sawtooth $\frac12 - \{x(T)\}$ has integral over each interval on which x increases by one equal to $O(T^{-1}(\log T)^{-3})$, because such an interval has length about $2\pi/\log T$ and $x'(T) = \frac{1}{2\pi}\log\frac{T}{2\pi} + O(T^{-2})$ varies over

it by $O(T^{-1}(\log T)^{-2})$ relative to its size, so its integral from 0 to T is bounded, and another integration by parts makes its contribution $O(y)$. (iii) is (i) minus (ii). □

A computation tests the proposition. It takes the first 30000 ordinates from Arb, up to 25755.348, cuts at $G = 25754.905$, midway between the last two, and adds the tail $\int_G^\infty e^{-Ty}\vartheta'(T)\,dT/\pi - e^{-Gy}S(G)$, with $S(G) = -0.569$ exact; the neglected term $y\int_G^\infty e^{-Ty}S(T)\,dT$ equals $y^2\int_G^\infty e^{-Ty}\int_G^T S(t)\,dt\,dT$, and Trudgian's bound $|\int_{t_1}^{t_2} S(t)\,dt| \le 2.067 + 0.059\log t_2$ for $t_2 > t_1 > 168\pi$ [350], with Jensen's inequality for the logarithm, bounds it by $ye^{-Gy}(0.059\log(G+1/y) + 2.067)$; the computation records twice this bound, 4.1×10^{-5} and 4.3×10^{-5} at $y = 10^{-4}$ and 10^{-5}, so the entries of Table 13.4 at $y = 10^{-4}$ and 10^{-5} may be off by up to two units in the fifth decimal. The zeros of P come from the phase condition of Proposition 18.1.8, solved by Newton's method for all 30132 zeros to just above the cut, with the tail beyond G added. The factor e^{-Gy} that weights the tails is 0.076 at $y = 10^{-4}$ and 0.77 at $y = 10^{-5}$, so the last row of Table 13.4 rests mostly on the smooth tails. The constant $\frac{9}{8}$ is clear, and D approaches it along $\frac{9}{8} - cy\log\frac{1}{y}$.

y	$K_{1/2}(y)$	$K^P(y)$	$K_{1/2} - M$	$K^P - M$	$D(y)$	$\frac{9}{8} - cy\log\frac{1}{y}$
0.05	2.686015	2.230883	0.837782	0.382650	0.455132	0.192803
0.02	12.772471	12.033884	0.860279	0.121691	0.738587	0.638070
0.01	35.723862	34.836495	0.867696	−0.019670	0.887366	0.838397
0.005	92.647265	91.663043	0.871374	−0.112848	0.984222	0.960130
0.003	180.932794	179.901902	0.872836	−0.158057	1.030892	1.016540
0.001	715.903738	714.816981	0.874286	−0.212471	1.086757	1.082010
10^{-4}	10815.847446	10814.727704	0.874931	−0.244811	1.119742	1.119268
10^{-5}	144797.380092	144796.255761	0.874994	−0.249338	1.124332	1.124283

Table 13.4: The Poisson trace $K_{1/2}$ of the zeros of ζ, Pólya's trace K^P, their distances from the common main term M, which tend to $\frac{7}{8}$ and $-\frac{1}{4}$, and their difference D, which tends to $\frac{9}{8}$, against the prediction of Proposition 13.1.11 under RH. Computed.

The note [284] states $K_{1/2}(y) = M(y) + O(\log\frac{1}{y})$, which is true but loses the constant, and reports $D(y)/\log\frac{1}{y} = 0.195, 0.194, 0.193, 0.193$ at $y = 0.003, 0.005, 0.01, 0.02$, from which it concludes that D grows like $0.19\log\frac{1}{y}$. The accurate values of $D/\log\frac{1}{y}$ in this range, 0.177, 0.186, 0.193, 0.189, are also close to 0.19, but that is a coincidence of the range: by Proposition 13.1.11, D tends to $\frac{9}{8}$, and $D/\log\frac{1}{y}$ is 0.098 at $y = 10^{-5}$.

K^P is completely monotone as the trace of the semigroup $e^{-2y\sqrt{A_P}}$. By Corollary 13.1.4, RH says that $K_{1/2} = K^P + D$ is completely monotone as well, where

D is bounded, tends to $\frac{9}{8}$ while both traces grow like $\frac{1}{2\pi y}\log\frac{1}{y}$, and carries the whole difference between Φ and Pólya's kernel, the arithmetic included. Complete monotonicity constrains derivatives of every order, and no bound on the size of D can decide it.

13.1.6 Wiener's theorem and the basepoint kernels

The zero free half planes have a second reading, as completeness statements, through Wiener's closure theorem [360], for which Kac [173] and Korevaar [193] gave short alternative treatments: the translates $f(\cdot - x)$, $x \in \mathbb{R}$, of a function $f \in L^1(\mathbb{R})$ span a dense subspace of $L^1(\mathbb{R})$ if and only if its Fourier transform $\hat f(\theta) = \int f(u)e^{i\theta u}du$ has no real zeros.

Let Φ be Pólya's kernel (2.1.3), so that $\xi(\frac12 + s) = \int_{\mathbb{R}} \Phi(u)e^{su}du$ for every complex s by (2.1.4), and for $\alpha \ge \frac12$ put

$$\Phi_\alpha(u) = e^{(\alpha-\frac12)u}\,\Phi(u), \qquad \hat\Phi_\alpha(\theta) = \int_{\mathbb{R}} \Phi_\alpha(u)e^{i\theta u}du = \xi(\alpha + i\theta). \tag{13.1.4}$$

Up to the factor $\xi(\alpha)$, Φ_α is the density of the tilted Pólya variable V_α of Section 12.1.1, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$; it is integrable for every α because Φ decays doubly exponentially. The real zeros of $\hat\Phi_\alpha$ are the ordinates of the zeros of ζ on the line $\operatorname{Re} s = \alpha$, so Wiener's theorem gives the following.

Proposition 13.1.12. *1. The translates of Φ_α are dense in $L^1(\mathbb{R})$ if and only if ζ has no zeros on the line $\operatorname{Re} s = \alpha$.*

2. At the centre they are not dense. For every finite combination $h = \sum_k c_k\Phi(\cdot - x_k)$ and every $g \in L^1(\mathbb{R})$,

$$\|g - h\|_1 \;\ge\; \sup_\gamma |\hat g(\gamma)|,$$

the supremum over the ordinates of the zeros on the critical line.

3. At the basepoint one they are dense, with no hypothesis, because $\xi(1+i\theta) \neq 0$.

4. For $\alpha_0 \in [\frac12, 1)$, ζ has no zeros in the half plane $\operatorname{Re} s > \alpha_0$ if and only if the translates of Φ_α are dense in $L^1(\mathbb{R})$ for every $\alpha > \alpha_0$. In particular RH holds if and only if they are dense for every $\alpha > \frac12$.

5. If the translates of Φ_α are dense, approximation near the zeros is expensive: if $h = \sum_k c_k\Phi_\alpha(\cdot - x_k)$ and $\|g-h\|_1 \le \delta$, then $\sum_k |c_k| \;\ge\; (|\hat g(\gamma)| - \delta)/|\xi(\alpha + i\gamma)|$ for every ordinate γ.

Proof. (1), (3) and (4) are Wiener's theorem applied to $\hat\Phi_\alpha(\theta) = \xi(\alpha + i\theta)$, together with the nonvanishing of ζ on the line $\operatorname{Re} s = 1$. For (2) and (5), $\hat h(\theta) = \hat\Phi_\alpha(\theta)\sum_k c_k e^{i\theta x_k}$, so at the centre $\hat h(\gamma) = 0$ at every ordinate of a zero on the line, while $|\hat g(\theta) - \hat h(\theta)| \le \|g - h\|_1$ for every θ; for (5), $|\hat h(\gamma)| \le |\xi(\alpha + i\gamma)|\sum_k |c_k|$. □

Wiener's proof of the prime number theorem runs through statement (3), and the family interpolates between that theorem at $\alpha = 1$ and RH at $\alpha = \frac{1}{2}$. It is a restatement, not a new criterion: by statement (4), completeness at every $\alpha' > \alpha$ is the condition $\alpha \geq \sup_\rho \operatorname{Re}\rho$ under which T_α is a GGC (Theorem 2.1.2) and Y_α an EGGC (Theorem 13.1.2), and, by Lagarias's argument [198], it is also the positivity of $v_{\alpha'}$ for every $\alpha' > \alpha$ (Remark 2.1.4). What the completeness form adds is an explicit, computable obstruction at the centre, statement (2), and an explicit cost of approximation above it, statement (5): near the centre $|\xi(\alpha+i\gamma)| \approx (\alpha-\frac{1}{2})|\xi'(\frac{1}{2}+i\gamma)|$ at the ordinate γ of a simple zero, so the coefficients needed to reproduce $\hat{g}$ at an ordinate grow like $1/(\alpha - \frac{1}{2})$.

We tested the dichotomy numerically, approximating the Gaussian $g(u) = (2\pi\kappa^2)^{-1/2}e^{-u^2/2\kappa^2}$ with $\kappa = 0.15$, for which $\hat{g}(\gamma_1) = e^{-\kappa^2\gamma_1^2/2} = 0.10565$, by combinations of N equally spaced translates of Φ_α on $[-1.5, 1.5]$, minimising the L^1 error on $[-3, 3]$ exactly by linear programming (Table 13.5). At the centre every approximant has $\hat{h}(\gamma_1) = 0$ up to rounding while the coefficient sums grow to 4×10^9, and the best error stalls at 0.1345, above the barrier 0.1056 of statement (2), however many translates are used. Above the centre the error falls below the value at which the centre stalls and keeps decreasing, to below the barrier at $\alpha = 0.75$ and $\alpha = 1$; what remains is limited by conditioning, not completeness. With 80 translates the coefficient sums reach 3×10^7 to 2×10^8 above the centre, as statement (5) and the rapid decay of $\hat{\Phi}_\alpha$ require, and double precision linear programming cannot go much further. The closer α is to $\frac{1}{2}$, the larger the coefficients and the slower the approach: the completeness counterpart of the Cauchy kernels sharpening onto the zeros.

N	$\alpha = \frac{1}{2}$	$\alpha = 0.6$	$\alpha = 0.75$	$\alpha = 1$
10	0.4705	0.4698	0.4664	0.4540
20	0.1345	0.1221	0.1060	0.0852
40	0.1345	0.1152	0.0907	0.0614
80	0.1365	0.1139	0.0878	0.0570
$\lvert\hat{h}(\gamma_1)\rvert$ at $N = 80$	0	0.0167	0.0377	0.0635

Table 13.5: Best L^1 error in approximating a Gaussian of width 0.15 by N translates of Φ_α, computed. At the centre the error cannot fall below $|\hat{g}(\gamma_1)| = 0.1056$, and $\hat{h}(\gamma_1)$ vanishes exactly; the computed value is 1.2×10^{-8}.

13.1.7 Salem's criterion and its Pólya analogue

Wiener's theorem has a dual form: for $k \in L^1(\mathbb{R})$, the convolution equation $\int k(x-y)f(y)\,dy = 0$ for all x has a bounded solution $f \neq 0$ if and only if $\hat{k}$ has a real zero. Salem [322] used it to give one of the classical equivalents of RH: RH holds

if and only if, for every $\frac12<\sigma<1$, the equation

$$\int_{\mathbb{R}} \frac{e^{-\sigma y}f(y)}{e^{e^{x-y}}+1}\,dy=0 \qquad (x\in\mathbb{R})$$

has no bounded solution other than $f=0$. Multiplied by $e^{\sigma x}$, it is the convolution equation with the kernel $k_\sigma(u)=e^{\sigma u}/(e^{e^u}+1)$, the Fermi Dirac function $1/(e^w+1)$ in the variable $w=e^u$, whose Fourier transform is

$$\hat{k}_\sigma(\theta)=\int_{\mathbb{R}}\frac{e^{\sigma u}e^{i\theta u}}{e^{e^u}+1}\,du=\int_0^\infty\frac{w^{\sigma+i\theta-1}}{e^w+1}\,dw=\Gamma(\sigma+i\theta)\,\eta(\sigma+i\theta),$$
$$\eta(s)=(1-2^{1-s})\zeta(s),$$

with η Dirichlet's alternating zeta function (not the Dedekind eta function of Section 15.2). So for $\frac12<\sigma<1$ the real zeros of $\hat{k}_\sigma$ are exactly the ordinates of the zeros of ζ on the line $\operatorname{Re}s=\sigma$, and Salem's criterion is Proposition 13.1.12(4) with the Fermi Dirac kernel in place of the tilted Pólya kernel (13.1.4).

Corollary 13.1.13. *RH holds if and only if, for every $\frac12<\alpha<1$, the equation $\int_{\mathbb{R}}\Phi_\alpha(x-y)f(y)\,dy=0$ for all x has no bounded solution $f\neq0$. At the centre the equation has the explicit bounded solutions $f(y)=\cos(\gamma y)$, one for each ordinate γ of a zero on the critical line, because $\Xi(\gamma)=0$ there; at the basepoint one it has none.*

Proof. By the dual form of Wiener's theorem with $k=\Phi_\alpha$, a bounded solution $f\neq0$ exists exactly when $\hat{\Phi}_\alpha(\theta)=\xi(\alpha+i\theta)$ has a real zero, that is, when ζ has a zero on the line $\operatorname{Re}s=\alpha$. That no such zero exists for any $\alpha\in(\frac12,1)$ means that no zero has $\frac12<\operatorname{Re}\rho<1$, which, since no zero lies on the line $\operatorname{Re}s=1$, is RH by the symmetry $\rho\mapsto1-\rho$. At the centre Φ is even, so $\int_{\mathbb{R}}\Phi(x-y)\cos(\gamma y)\,dy=\Xi(\gamma)\cos(\gamma x)$, which vanishes when $\frac12+i\gamma$ is a zero; at the basepoint one $\xi(1+i\theta)\neq0$ for every real θ. □

The factor $1-2^{1-s}$ in η vanishes at $s=1+2\pi ik/\log2$ with $k\neq0$, so Salem's kernel has real Fourier zeros on the line $\sigma=1$, at $\theta=9.0647,\,18.1294,\,27.1942,\ldots$, and its translates are not complete there, whereas the Pólya family is complete at $\alpha=1$ because $\xi(1+i\theta)\neq0$. The left panel of Figure 13.3 shows $|\eta(\sigma+i\theta)|$: at $\sigma=\frac12$ it dips to zero at every ordinate, and at $\sigma=0.6$ and 0.75 the dips are shallow. The depth of the dips, of order $\sigma-\frac12$ near each ordinate, plays for Salem's kernel the role of $|\xi(\alpha+i\gamma)|$ in the cost of approximation of Proposition 13.1.12(5).

13.1.8 Riesz's criterion and multiplicative completeness

Riesz [314] found the first criterion of this type, in terms of growth rather than completeness. With the Möbius function $\mu(n)$ (unrelated to the Thorin measure

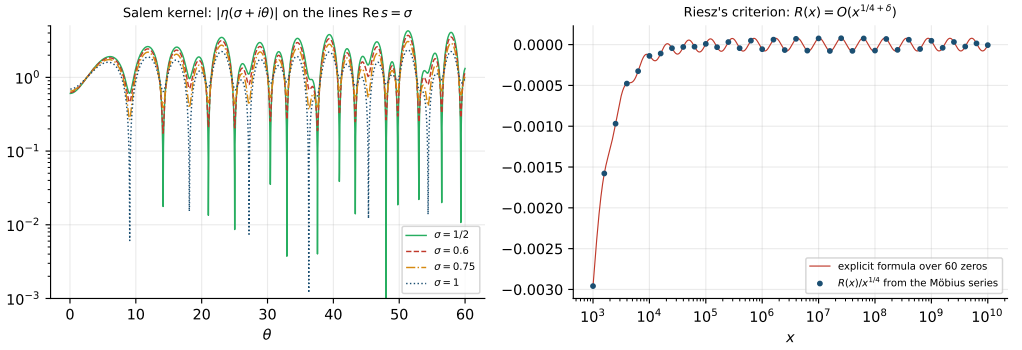


Figure 13.3: Left: $|\eta(\sigma + i\theta)|$, the Fourier transform of Salem's kernel divided by $\Gamma(\sigma + i\theta)$, at $\sigma = \frac{1}{2}$, 0.6, 0.75, 1. Right: the Riesz function $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula (13.1.7) over 60 zeros.

μ_ε of Section 2.1), put

$$R(x) = -\sum_{k\geq 1} \frac{(-x)^k}{(k-1)!\,\zeta(2k)} = x \sum_{n\geq 1} \frac{\mu(n)}{n^2}\, e^{-x/n^2}, \tag{13.1.5}$$

the second form following from $1/\zeta(2k) = \sum_n \mu(n) n^{-2k}$. Riesz proved that RH holds if and only if $R(x) = O(x^{1/4+\delta})$ for every $\delta > 0$. The Mellin transform shows why: for $\frac{1}{2} < \operatorname{Re} s < 1$,

$$\int_0^\infty R(x)\, x^{-s-1} dx = \frac{\Gamma(1-s)}{\zeta(2s)}, \tag{13.1.6}$$

whose poles in the strip $0 < \operatorname{Re} s < \frac{1}{2}$ sit at $s = \rho/2$, and shifting the contour gives, when the zeros are simple, the explicit formula

$$R(x) = \sum_\rho \frac{\Gamma(1-\rho/2)}{2\zeta'(\rho)}\, x^{\rho/2} + \sum_{k\geq 1} \frac{\Gamma(1+k)}{2\zeta'(-2k)}\, x^{-k}, \tag{13.1.7}$$

in which the sum over the zeros is the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence of heights $T_\nu \to \infty$. The same argument gives the basepoint version: for $\frac{1}{2} \leq \alpha < 1$, ζ has no zeros in $\operatorname{Re} s > \alpha$ if and only if $R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. Proposition 13.2.1 gives the proof for a twisted form of R: the zero free half plane bounds $1/\zeta(\sigma + it)$ by a power of $|t|$ on $\sigma \geq \alpha + 2\delta$, by the Borel Carathéodory lemma as in [348, Section 14.2] with the line $\operatorname{Re} s = \alpha$ in place of the critical line.

In the language of this book, $R(x)/x = \sum_n \mu(n) n^{-2} e^{-x/n^2}$ is the Laplace transform of the signed Möbius measure $\sum_n \mu(n) n^{-2} \delta_{1/n^2}$, a heat trace on the

arithmetic side with the Möbius function in place of the zeros, and the Möbius series that underlies the representation of the reciprocal zeta function in Section 10.1 (Theorem 10.1.5). The coefficients $1/\zeta(2k)$ of the power series of R are, up to elementary factors, values of the Laplace transforms of the centre clock T and of T_1 (Section 13.2.2).

We compute $R(x)$ from the Möbius series with $n \le N = 10^7$ as

$$R(x) \approx x\Big[\sum_{n\le N} \mu(n)n^{-2}\big(e^{-x/n^2}-1\big) + \frac{6}{\pi^2}\Big]$$

to avoid the slowly converging tail, for $10 \le x \le 10^{12}$. Beyond $x \approx 10^{10}$ the truncation at $N = 10^7$ limits the accuracy: the omitted terms are about $-x^2\sum_{n>N}\mu(n)n^{-4} \approx 9\times10^{-26}x^2$, that is 3×10^{-8} in $R(x)/x^{1/4}$ at $x = 10^{10}$ and 9×10^{-5} at $x = 10^{12}$. For $10^4 \le x \le 10^{10}$ the ratio $R(x)/x^{1/4}$ oscillates with amplitude about 1.1×10^{-4}, and for $10^5 \le x \le 10^{10}$ with amplitude 7.8×10^{-5}, which is the amplitude $|\Gamma(1-\rho_1/2)/\zeta'(\rho_1)|$ of the term of the first zero. The larger swing near $x = 10^4$ comes from the term $\Gamma(2)x^{-1}/(2\zeta'(-2)) = -16.42\,x^{-1}$ of the first trivial zero, which contributes -1.6×10^{-4} to the ratio at $x = 10^4$. The explicit formula (13.1.7) with the first 60 zeros and the first three trivial terms reproduces the computed values to 3.1×10^{-8} on $[10^3, 10^{10}]$, against a signal of size up to 3×10^{-3} (Figure 13.3, right). The largest difference, at $x = 10^{10}$, is the truncation error just described, and on $[10^3, 10^9]$ the difference is at most 6.2×10^{-10}. Section 13.2.1 twists the Möbius series by $(x/n^2)^{ih}$ so that it responds to the zeros near height $2h$.

Wiener's theorem concerns translates; its multiplicative counterpart concerns dilates, and there completeness is equivalent to RH itself. Nyman and Beurling [251, 31] proved that RH holds if and only if the constant function 1 lies in the $L^2(0,1)$ closure of the span of the functions $r_\lambda(x) = \{\lambda/x\} - \lambda\{1/x\}$, $0 < \lambda \le 1$, where $\{\cdot\}$ is the fractional part, and Báez Duarte [10] showed that $\lambda = 1/n$ suffices. Under $x = e^{-u}$ this is a closure problem for translates in a weighted L^2 space on the half line, and the Mellin transform of r_λ is $-\zeta(s)(\lambda^s - \lambda)/s$, so the zeros of ζ in $\operatorname{Re} s > \frac12$ are what obstructs the closure. The same framework governs the dilates $\operatorname{sech}(nx)$ of the Lai and Vere-Jones identity (Proposition 10.1.3), which lies behind the expansion of Theorem 10.1.5: the Mellin transform of sech is $2\Gamma(s)L(s,\chi_4)$, and completeness of systems of dilates is controlled by where the associated Dirichlet series vanishes, in the sense of Beurling's periodic dilation problem and the work of Hedenmalm, Lindqvist and Seip [113]. How the two questions are related is the Nyman Beurling problem of Section 17.1.4. The Nyman Beurling closure problem is equivalent to RH; for the dilates $\operatorname{sech}(nx)$ the answer is different: completeness in one weighted space $L^2(x^{2c-1}dx)$, $c > 0$, holds unconditionally (Theorem 17.3.2), and only the two weight version is equivalent to the generalised Riemann hypothesis for $L(s,\chi_4)$ (Theorem 17.5.2, Chapter 17). The additive family of Proposition 13.1.12 is the probabilistic one, and like every

construction of Chapters 12 to 14 it reaches RH only as $\alpha \downarrow \frac{1}{2}$.

13.2 Riesz's criterion and the Thorin path

Section 13.1.8 introduced Riesz's criterion [314] for the Riesz function (13.1.5) and its basepoint form, of which RH is the case $\alpha = \frac{1}{2}$. Hardy and Littlewood [105] gave a companion criterion with $\zeta(2k+1)$ in place of $\zeta(2k)$ and the exponent $-\frac{1}{4}$, to which this section applies in the same way. We develop the criterion computationally (Section 13.2.1) and through the clocks of Chapter 12, which read the coefficients $1/\zeta(2k)$; none of this supplies the growth bound (Section 13.2.6).

We keep the notation of Chapter 12 and Section 13.1 for $T = T_{1/2}$, T_α, φ_α, H_a, b_α and v_α; in particular $b_1 = \sum_\rho \rho^{-1} = 0.0230957$. We write $\widetilde{T} = \widetilde{T}_1$ for the centre clock tilted by $e^{-T/4}$, as in Theorem 12.2.2, $\psi(y) = \sum_{n \le y} \Lambda(n)$ for Chebyshev's function, and

$$\kappa_k = \frac{2k(2k-1)(k-1)!}{\pi^k}, \qquad c_k = \frac{1}{\kappa_k\, \zeta(2k)},$$

so that $\xi(2k) = \frac{1}{2}\kappa_k \zeta(2k)$ and $c_k = \xi(1)/\xi(2k)$.

13.2.1 Which zeros the Riesz function sees

On the range computed in Section 13.1.8 the fluctuations of the Riesz function are accounted for by very few known zeros, as RH predicts. In the explicit formula (13.1.7) a zero ρ enters $R(x)$ through the term $r_\rho x^{\rho/2}$ with the weight $r_\rho = \Gamma(1-\rho/2)/(2\zeta'(\rho))$. By Stirling's formula $|\Gamma(\frac{3}{4} - \frac{i}{2}\gamma)|$ decays like $e^{-\pi\gamma/4}$. The weight is 3.9×10^{-5} for the first zero, whose term and that of its conjugate give the amplitude 7.8×10^{-5} found in Section 13.1.8, 2.1×10^{-17} for the tenth, 1.5×10^{-49} for the fiftieth and 2.2×10^{-81} for the hundredth (Figure 13.4, left). An off line zero at height 1000 would enter with a weight near 10^{-340}, invisible on any feasible range of x: as a computational test Riesz's criterion examines only the bottom of the spectrum. The remedy is to change the kernel. For real h put

$$R_h(x) = x \sum_{n \ge 1} \frac{\mu(n)}{n^2} \Big(\frac{x}{n^2}\Big)^{ih} e^{-x/n^2}, \qquad x > 0,$$

so that $R_0 = R$.

Proposition 13.2.1. *For every real h and $\frac{1}{2} < \operatorname{Re} s < 1$,*

$$\int_0^\infty R_h(x)\, x^{-s-1} dx = \frac{\Gamma(1-s+ih)}{\zeta(2s)}.$$

For $\frac{1}{2} \le \alpha < 1$, ζ has no zeros in the half plane $\operatorname{Re} s > \alpha$ if and only if $R_h(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. In particular RH holds if and only if $R_h(x) = O(x^{1/4+\delta})$ for every $\delta > 0$.

Proof. Term by term, $\int_0^\infty x^{1+ih} n^{-2-2ih} e^{-x/n^2} x^{-s-1}\,dx = n^{-2s}\Gamma(1-s+ih)$, and summing against $\mu(n)$ gives $\Gamma(1-s+ih)\sum_n \mu(n)n^{-2s}$; the interchange is justified because, with $\sigma = \operatorname{Re} s$, the integrals of the absolute values sum to $\Gamma(1-\sigma)\zeta(2\sigma) < \infty$. Suppose that ζ has no zeros in $\operatorname{Re} s > \alpha$, so that $1/\zeta(2s)$ is analytic in $\operatorname{Re} s > \frac{\alpha}{2}$. By the argument of [348, Section 14.2] with the line of real part α in place of the critical line, as in Section 13.1.8, $|1/\zeta(2s)| \le |2\operatorname{Im} s|^{C_\delta}$ on $\operatorname{Re} s \ge \frac{\alpha}{2}+\delta$, $|\operatorname{Im} s| \ge 5$, while $|\Gamma(1-s+ih)|$ decays like $e^{-\pi|\operatorname{Im} s - h|/2}$ in vertical strips. So the Mellin inversion integral $R_h(x) = \frac{1}{2\pi i}\int_{(c)} \Gamma(1-s+ih)\zeta(2s)^{-1}x^s\,ds$, with $\frac12 < c < 1$, can be moved to the line $\operatorname{Re} s = \frac{\alpha}{2}+\delta$, which gives $R_h(x) = O(x^{\alpha/2+\delta})$. Conversely, if $R_h(x) = O(x^{\alpha/2+\delta})$ as $x \to \infty$, then, since also $|R_h(x)| \le \zeta(2)\,x$, the Mellin integral converges absolutely for $\frac{\alpha}{2}+\delta < \operatorname{Re} s < 1$ and continues $\Gamma(1-s+ih)/\zeta(2s)$ analytically there. As Γ has no zeros, $\zeta(2s)$ has no zeros with $\alpha + 2\delta < \operatorname{Re} 2s < 2$; since $\delta > 0$ is arbitrary and ζ has no zeros with $\operatorname{Re} s \ge 1$, it has none with $\operatorname{Re} s > \alpha$. □

Shifting the contour further gives, when the zeros are simple, the twisted form of the explicit formula (13.1.7), with the sum over the zeros again the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence $T_\nu \to \infty$:

$$R_h(x) = \sum_\rho \frac{\Gamma(1-\rho/2+ih)}{2\zeta'(\rho)}\,x^{\rho/2} + \sum_{k\ge1} \frac{\Gamma(1+k+ih)}{2\zeta'(-2k)}\,x^{-k}. \tag{13.2.1}$$

The heights T_ν are those of [348, Theorem 9.7], on which $1/\zeta(\sigma+iT_\nu) = O(T_\nu^A)$ for $-1 \le \sigma \le 2$; with the functional equation for $\operatorname{Re} 2s \le -1$ this makes the integrals over the horizontal segments $\operatorname{Im} 2s = \pm T_\nu$ tend to 0. On the line $\operatorname{Re} s = -q - \frac12$ we write $1/\zeta(2s) = \pi^{1/2-2s}\Gamma(s)/(\Gamma(\frac12 - s)\zeta(1-2s))$; since $|\Gamma(1-s+ih)\Gamma(s)| \le e^{\pi|h|/2}\pi/\cosh(\pi\operatorname{Im} s)$, $|\Gamma(\frac12 - s)| \ge q!\,(\pi t/\sinh \pi t)^{1/2}$ with $t = \operatorname{Im} s$, and $|\zeta(1-2s)| \ge 2 - \zeta(4)$, the integral over that line is at most $Ce^{\pi|h|/2}\pi^{2q}x^{-q-1/2}/q!$, which tends to 0 as $q \to \infty$.

The twist changes the weight. Since $|\Gamma(\frac34 + i(h-\gamma/2))|$ is largest when $\gamma \approx 2h$ and decays like $e^{-\pi|h-\gamma/2|/2}$ away from it, R_h sees the zeros in a band of width about ten around height $2h$. At $h = 50$ the largest weight among the first 300 zeros is 0.124, at $\gamma = 101.3$, where the untwisted weight is 2.9×10^{-35} (Figure 13.4, left). We compute $R_{50}(x)$ from the Möbius series with μ sieved to 10^7 at forty values of x in $[10^4, 10^{10}]$; against (13.2.1) over the first 300 zeros and three trivial terms it agrees to 1.2×10^{-8} in $|R_h(x)|/x^{1/4}$, against a signal of size up to 0.27, and the zeros with $|\gamma - 100| < 15$ carry 99.9997% of the modulus of the explicit formula, in the median over the forty values of x (Figure 13.4, right).

An off line zero with real part β near height $2h$ would contribute a term of size $x^{\beta/2}$, growing relative to the background $x^{1/4}$ by the factor $x^{(\beta-1/2)/2}$: a factor 3.2 between $x = 10^{10}$ and $x = 10^{20}$ when $\beta = 0.6$.

REMARK 13.2.2. Twisted Riesz functions are not competitive with direct zero finding as a verification method; their value is independence. They are computed from $\mu(n)$ only, with no evaluation of ζ, so agreement between R_h and the explicit formula over computed zeros is a check of the zeros that shares no code or method with their computation.

The cost of $R(x)$ or $R_h(x)$ is dominated by sieving $\mu(n)$ to $N = K\sqrt{x}$. Terms with $n < \sqrt{x}/12$ are below e^{-144} and can be omitted, and the terms with $n > N$ are summed through $\sum_n \mu(n)n^{-2} = 6/\pi^2$, or $1/\zeta(2+2ih)$ for R_h: we compute $R(x)/x$ as $\sum_{\sqrt{x}/12 \le n \le N} \mu(n)n^{-2}e^{-x/n^2} + \big(6/\pi^2 - \sum_{n\le N} \mu(n)n^{-2}\big)$, and the neglected remainder $\sum_{n>N} \mu(n)n^{-2}(e^{-x/n^2} - 1) \approx -x\sum_{n>N} \mu(n)n^{-4}$ is of order $xN^{-7/2}$ under square root cancellation in the Möbius tail, that is, of order $K^{-7/2}$ in $R(x)/x^{1/4}$. That cancellation is proved in the range used: the Mertens function $\sum_{n\le t} \mu(n)$ is less than $0.571\sqrt{t}$ in absolute value for $33 \le t \le 10^{16}$ [157, Section 6.1], and partial summation then bounds the remainder in $R(x)/x^{1/4}$ by 1.1×10^{-8} for $K = 200$ and $1 \le x \le 10^{12}$. We implement this with a segmented sieve and extended precision; with $K = 200$ it reproduces the explicit formula over 60 zeros and three trivial terms to about 2×10^{-9} in $R(x)/x^{1/4}$ at $x = 10^8$, 10^{10}, 10^{11} and 10^{12}. The twisted version agrees too: at $x = 10^{10}$ with $h = 50$ it gives $R_{50}(x) = 37.441416 - 26.042495\,i$, within 3.1×10^{-9} of (13.2.1) over 300 zeros in $|R_h(x)|/x^{1/4}$.

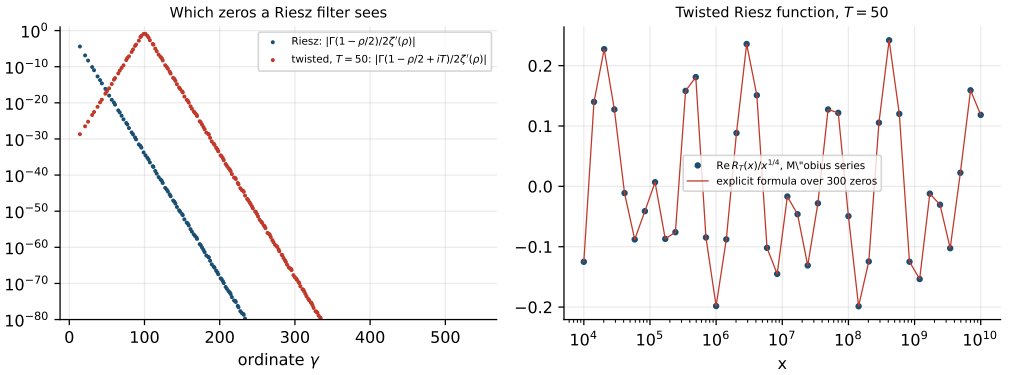


Figure 13.4: Left: the weights $|\Gamma(1-\rho/2)/(2\zeta'(\rho))|$ of the Riesz function and $|\Gamma(1-\rho/2+ih)/(2\zeta'(\rho))|$ of the twisted Riesz function with $h = 50$, for the first 300 zeros. Right: $\operatorname{Re} R_{50}(x)/x^{1/4}$ from the Möbius series against the explicit formula (13.2.1) over 300 zeros. The plot writes T for h.

13.2.2 The Riesz coefficients through the centre clock

The Riesz coefficients are values of two Laplace transforms: that of the centre clock T of [279] and that of the basepoint one clock T_1 (Proposition 12.1.2).

Proposition 13.2.3. *For every integer $k \geq 1$, with no hypothesis,*

$$\frac{1}{\zeta(2k)} = \frac{\kappa_k}{2\xi(\frac{1}{2})}\,\mathbb{E}\,e^{-(2k-\frac{1}{2})^2 T} = \kappa_k\,\mathbb{E}\,e^{-(2k-1)^2 T_1}.$$

Proof. By the definition of ξ, $\xi(2k) = k(2k-1)\pi^{-k}\Gamma(k)\zeta(2k) = \frac{1}{2}\kappa_k\zeta(2k)$. Since $2k-\frac{1}{2} > 0$, the Laplace transform of T at $(2k-\frac{1}{2})^2$ is $\varphi_{1/2}((2k-\frac{1}{2})^2) = \xi(\frac{1}{2})/\xi(2k)$; similarly $\mathbb{E}e^{-(2k-1)^2T_1} = \xi(1)/\xi(2k)$, with $\xi(1) = \frac{1}{2}$. □

Proposition 13.2.4. *For every $x > 0$, with no hypothesis,*

$$R(x) = -\frac{1}{\xi(\frac{1}{2})}\,\mathbb{E}\Big[\sum_{k\geq 1} k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-\frac{1}{2})^2 T}\Big].$$

Proof. Since $\kappa_k/(2(k-1)!) = k(2k-1)\pi^{-k}$, Proposition 13.2.3 turns the term $-(-x)^k/((k-1)!\,\zeta(2k))$ of Riesz's series into $-\xi(\frac{1}{2})^{-1}k(2k-1)(-x/\pi)^k\,\mathbb{E}e^{-(2k-\frac{1}{2})^2T}$. The exchange of sum and expectation is justified by Fubini's theorem, because

$$\sum_{k\geq 1} k(2k-1)\Big(\frac{x}{\pi}\Big)^k \mathbb{E}\,e^{-(2k-\frac{1}{2})^2 T} = \xi(\tfrac{1}{2})\sum_{k\geq 1}\frac{x^k}{(k-1)!\,\zeta(2k)} < \infty.$$

□

The terms averaged in Proposition 13.2.4 decay like e^{-4k^2T} rather than like $1/(k-1)!$. Table 13.6 checks both propositions with 4×10^5 draws of T, generated under RH as $\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ: the first 200 zeros are used exactly, and the remaining terms are replaced by a gamma variable with the same mean and variance, computed from the exact power sums $\sum\gamma^{-2}$ and $\sum\gamma^{-4}$. The coefficients agree within about one standard error, and so do the values of R from the Möbius series and the clock average.

The clock average is far better conditioned than Riesz's power series, which is numerically hopeless for large x: its largest term is about $e^x\sqrt{x/2\pi}$, that is $10^{4344.5}$ at $x = 10^4$, while $R(10^4)$ is of order 10^{-3}. In the clock average the largest term of the inner series, $\max_k k(2k-1)(x/\pi)^k e^{-(2k-\frac{1}{2})^2T}$, is about 10^{81} at $x = 10^4$ for the typical value $T = 0.023$, and its logarithm grows like $(\log x)^2/(16T)$ instead of like x (Figure 13.5, right).

k	$1/\zeta(2k)$	via the clock	standard error
1	0.6079271	0.6079115	0.0000131
2	0.9239384	0.9238153	0.0001046
3	0.9829526	0.9826581	0.0002603
4	0.9959392	0.9954609	0.0004615

x	$R(x)$, Möbius	clock average	standard error
0.5	0.1252199	0.1252287	0.0000074
1	0.0439818	0.0440007	0.0000157
2	−0.2677934	−0.2677918	0.0000026
5	−0.7705597	−0.7706050	0.0000397

Table 13.6: Left: the Riesz coefficients from the clock, Proposition 13.2.3. Right: the Riesz function from the Möbius series, which agrees with the power series to fourteen digits, and from the clock average of Proposition 13.2.4.

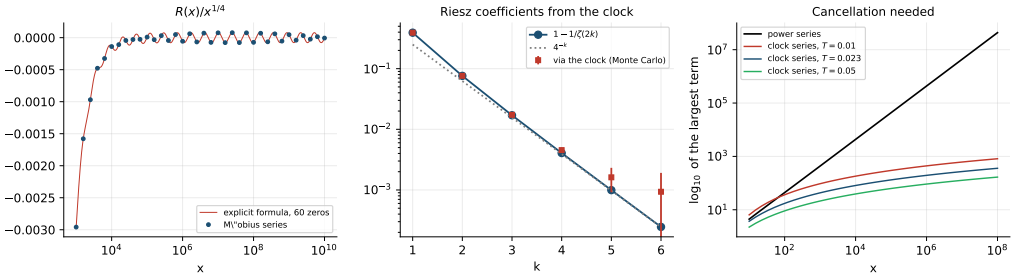


Figure 13.5: Left: $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula over 60 zeros, as in the right panel of Figure 13.3. Middle: $1 - 1/\zeta(2k)$ and its Monte Carlo value from the clock (Proposition 13.2.3), against 4^{-k}. Right: base ten logarithm of the largest term in Riesz's power series and in the clock series of Proposition 13.2.4 for $T = 0.01$, 0.023 and 0.05.

13.2.3 The sine squared lemma and a GGC form of Riesz's criterion

Proposition 13.2.3 extends to every basepoint between the centre and one. For $\alpha > \frac{1}{2}$ the clock T_α exists with no hypothesis (Theorem 12.2.5), and it is a GGC exactly when ζ has no zeros with $\operatorname{Re} s > \alpha$ (Theorem 2.1.2).

Proposition 13.2.5 (Riesz coefficients as GGC values). *For every $\alpha \in [\frac{1}{2}, 1]$ and every integer $k \geq 1$,*

$$\frac{1}{\zeta(2k)} = \frac{\kappa_k}{2\xi(\alpha)}\, \mathbb{E}\, e^{-(2k-\alpha)^2 T_\alpha}.$$

At $\alpha = 1$, with no hypothesis, T_1 is a GGC with Thorin density $\rho_{1/2} = v_1/\pi$ in

the chart $\theta = \sqrt{z}$*, and*

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ - \int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big)\rho_{1/2}(\theta)\, d\theta\Big\}.$$

Proof. Since $2k-\alpha > 0$, the Laplace transform of T_α at $(2k-\alpha)^2$ is $\xi(\alpha)/\xi(2k)$, and $\xi(2k) = \frac{1}{2}\kappa_k\zeta(2k)$. At $\alpha = 1$, $\xi(1) = \frac{1}{2}$, and the Thorin representation $\log\varphi_1(u) = -\int_0^\infty \log(1 + u/\theta^2)\rho_{1/2}(\theta)\, d\theta$ of Proposition 12.1.2, taken at $u = (2k-1)^2$, gives the second formula. □

Remark 13.2.6. All the Riesz coefficients are samples of one GGC Laplace transform: $c_k = \varphi_1((2k-1)^2)$, where $\varphi_1(u) = \mathbb{E}e^{-uT_1}$. As a Laplace transform φ_1 is log convex, by Hölder's inequality, so the slopes $(\log c_{k+1} - \log c_k)/8k$ increase with k. As the transform of a GGC, φ_1 is also hyperbolically completely monotone [39, Theorem 6.1.1]: for each $u > 0$ the function $v \mapsto \varphi_1(uv)\varphi_1(u/v)$ is a completely monotone function of $v + 1/v$. In particular it decreases as v moves away from 1, and differentiating $\log\varphi_1(uv) + \log\varphi_1(u/v)$ in $\log v$ shows that $t \mapsto \log\varphi_1(e^t)$ is concave. At the sample points this gives inequalities in the other direction: the slopes $(\log c_{k+1} - \log c_k)/\log\frac{2k+1}{2k-1}$ decrease with k, and $c_j c_l \le c_k^2$ whenever $(2j-1)(2l-1) = (2k-1)^2$, for instance $c_1 c_5 \le c_2^2$.

The Thorin density v_1/π is built from the primes by the sine squared lemma (2.1.9). Proposition 12.1.10 writes $\log(\xi(1+s)/\xi(1))$ in the Lévy form (12.1.5), with the drift b_1 and a signed Lévy measure Π_1 made of the gamma density, the pole of ζ and the prime atoms, and the lemma converts it term by term into the Thorin form $v_1 = b_1 + A + Q$ of (12.1.6), with the gamma piece A and the compensated prime piece Q. The Lévy form in $s = 2k-1$ is

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ - b_1(2k-1) - \int_0^\infty \big(e^{-(2k-1)x} - 1 + (2k-1)x\big)\,\Pi_1(dx)\Big\},$$

and in the Thorin form in $u = (2k-1)^2$ the drift splits off a positive stable one half factor:

$$\frac{1}{\zeta(2k)} = \kappa_k\, e^{-b_1(2k-1)}\, \exp\Big\{ - \frac{1}{\pi}\int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big)\big(A(\theta) + Q(\theta)\big)\, d\theta\Big\}. \tag{13.2.2}$$

This is the factorisation $T_1 \overset{d}{=} S + T_1^\circ$ of Proposition 12.1.11 evaluated at $u = (2k-1)^2$: the middle factor is $\mathbb{E}e^{-(2k-1)^2 S}$ for the positive stable variable S of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$, and the last factor is $\mathbb{E}e^{-(2k-1)^2 T_1^\circ}$ for the independent GGC T_1° with Thorin density $(A+Q)/\pi = (v_1 - b_1)/\pi$.

Inserting Proposition 13.2.5 at $\alpha = 1$ into the power series gives the Riesz function as a GGC mixture,

$$R(x) = -2\,\mathbb{E}\Big[\sum_{k\ge1} k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-1)^2 T_1}\Big], \tag{13.2.3}$$

where the exchange of sum and expectation is justified as in Proposition 13.2.4, since $\sum_k 2k(2k-1)(x/\pi)^k c_k = \sum_k x^k/((k-1)!\,\zeta(2k)) < \infty$. The variable T_1 has two realisations: it is the tilted centre clock run through the inverse local time of a sticky Brownian motion (Theorem 12.2.2 and the paragraph after it), and, like every GGC without drift [168], it is the Wiener gamma integral (14.1.4) against the inverse of the Poisson smoothed zero count $M_{1/2}(\theta) = \pi^{-1}\int_0^\theta v_1$ (Section 14.1.6).

THEOREM 13.2.7. *For $\frac12 \le \alpha < 1$ the following are equivalent:*

1. *$R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$;*
2. *ζ has no zeros with $\operatorname{Re} s > \alpha$;*
3. *$T_{\alpha'}$ is a GGC for every $\alpha' > \alpha$;*
4. *for every $\alpha' > \alpha$, $v_{\alpha'}(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha' + i\theta) \ge 0$ at every real θ for which $\alpha' + i\theta$ is not a zero of ξ;*
5. *for every $\alpha' > \alpha$ and every real θ the compensated sine squared sum*

$$\lim_{L\to\infty}\Big[\sum_{\log n \le L}\frac{\Lambda(n)}{n^{\alpha'}}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big) - \int_0^L 2\sin^2\Big(\frac{\theta x}{2}\Big)e^{(1-\alpha')x}\,dx\Big]$$

converges.

In particular RH holds if and only if T_α is a GGC for every $\alpha > \frac12$.

Proof. Write $\Theta = \sup\operatorname{Re}\rho$ and $D(y) = \psi(y) - y$. Conditions (1) and (2) are equivalent by the basepoint version of Riesz's criterion (Section 13.1.8, and Proposition 13.2.1 with $h = 0$). Conditions (2) and (3) are equivalent by Theorem 2.1.2: $\varphi_{\alpha'}$ is the Laplace transform of a GGC exactly when $\alpha' \ge \Theta$, and this holds for every $\alpha' > \alpha$ exactly when $\Theta \le \alpha$, which is (2).

(2) implies (4), because if $\Theta \le \alpha < \alpha'$ every Cauchy kernel in $v_{\alpha'}(\theta) = \sum_\rho(\alpha'-\beta)/((\alpha'-\beta)^2+(\theta-\gamma)^2)$ has positive width (Proposition 2.1.3). Conversely, let $\rho_0 = \beta_0 + i\gamma_0$ be a zero of multiplicity m with $\beta_0 > \alpha$. The zeros of ξ are isolated, so only finitely many lie on the segment $\{\sigma + i\gamma_0 : 0 \le \sigma \le 1\}$. Let n_0 be the number of those other than ρ_0, counted with multiplicity, and choose $\eta \in (0, \beta_0 - \alpha)$ with $|\beta - \beta_0| \ge 2\eta$ for each of them. For $\alpha' \in (\beta_0 - \eta, \beta_0)$ the point $\alpha' + i\gamma_0$ is not a zero, and Proposition 2.1.3 gives

$$v_{\alpha'}(\gamma_0) = \frac{m}{\alpha' - \beta_0} + \sum_{\rho\ne\rho_0,\ \gamma=\gamma_0}\frac{1}{\alpha'-\beta} + \sum_{\gamma\ne\gamma_0}\frac{\alpha'-\beta}{(\alpha'-\beta)^2+(\gamma-\gamma_0)^2}.$$

The middle sum is at most n_0/η in absolute value. Since α' and β lie in $(0,1)$, each term of the last sum is at most $(\gamma-\gamma_0)^{-2}$ in absolute value, and $\sum_{\gamma\ne\gamma_0}(\gamma-\gamma_0)^{-2} < \infty$: finitely many zeros have $0 < |\gamma - \gamma_0| < 1$, and for $|\gamma - \gamma_0| \ge 1$ we have

$(\gamma-\gamma_0)^{-2} \le 5/(4+(\gamma-\gamma_0)^2) \le 5(2-\beta)/((2-\beta)^2+(\gamma-\gamma_0)^2)$, a series bounded by $5v_2(\gamma_0) < \infty$. Hence $v_{\alpha'}(\gamma_0) \le m/(\alpha'-\beta_0) + n_0/\eta + \sum_{\gamma\ne\gamma_0}(\gamma-\gamma_0)^{-2} < 0$ for α' close to β_0, and (4) fails at (α',γ_0).

(2) implies (5). This is part 1 of Proposition 12.1.13 at the basepoint α', since $\Theta \le \alpha < \alpha'$; directly, choose σ with $\Theta < \sigma < \alpha'$, so that $D(y) = O(y^\sigma)$ [348]. With $f(y) = y^{-\alpha'}\,2\sin^2(\frac12\theta\log y)$ the bracket in (5) is

$$\int_{(1,e^L]} f\,dD = f(e^L)D(e^L) - \int_1^{e^L} D(y)f'(y)\,dy,$$

since $f(1) = 0$. The boundary term tends to 0, and the integral converges absolutely because $|f'(y)| \le (2\alpha' + |\theta|)y^{-\alpha'-1}$.

(5) implies (2). Suppose (5), and let ρ_0 be a zero of multiplicity m with $\operatorname{Re}\rho_0 > \alpha$. Fix $\alpha' \in (\alpha, \operatorname{Re}\rho_0)$ and $\theta \notin \mathcal{E} = \{\theta : \rho_0 + i\theta \text{ or } \rho_0 - i\theta \text{ is a zero}\}$, a countable set that contains 0, and let $M(t) = \int_{(0,t]}(1-\cos\theta s)\,dD(e^s)$, a function of locally bounded variation. The bracket in (5) is $J(L) = \int_{(0,L]} e^{-\alpha' t}dM(t)$; by hypothesis it converges as $L\to\infty$, so J is bounded. For $\operatorname{Re} w > \alpha'$, integration by parts gives

$$\int_{(0,L]} e^{-wt}\,dM(t) = e^{-(w-\alpha')L}J(L) + (w-\alpha')\int_0^L e^{-(w-\alpha')t}J(t)\,dt,$$

which converges as $L\to\infty$, locally uniformly in w, to an analytic function $F_\theta(w)$ on $\operatorname{Re} w > \alpha'$. For $\operatorname{Re} w > 1$ the integral converges absolutely, and writing $1-\cos\theta t = 1 - \frac12 e^{i\theta t} - \frac12 e^{-i\theta t}$ gives

$$F_\theta(w) = g(w) - \tfrac12 g(w-i\theta) - \tfrac12 g(w+i\theta),$$
$$g(w) = \int_{(1,\infty)} y^{-w}\,dD(y) = -\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}.$$

The right side, call it $\widetilde{F}_\theta(w)$, is meromorphic in the plane with a discrete set $\mathcal{P}$ of poles. On the connected open set $\Omega = \{\operatorname{Re} w > \alpha'\}\setminus\mathcal{P}$ both $\widetilde{F}_\theta$ and F_θ are analytic, and they agree on $\operatorname{Re} w > 1$, so $\widetilde{F}_\theta = F_\theta$ on Ω by the identity theorem. The function g is analytic at 1, and in $\operatorname{Re} w > 0$ its only poles are the zeros of ζ, each simple with residue minus its multiplicity. Since $\theta\notin\mathcal{E}$, $g(w\pm i\theta)$ is analytic at ρ_0, so $\widetilde{F}_\theta$ has a pole at ρ_0 with residue $-m\ne 0$, and $|F_\theta(w)|\to\infty$ as $w\to\rho_0$ in Ω. This contradicts the continuity of F_θ at ρ_0, where $\operatorname{Re}\rho_0 > \alpha'$. Hence no zero has real part greater than α, which is (2). □

13.2.4 The coefficients as killing probabilities

For a positive random variable X and an independent exponential variable E_λ of rate λ, $\mathbb{E}e^{-\lambda X} = P(X < E_\lambda)$, the probability that X occurs before a killing at

rate λ. So by Propositions 13.2.3 and 13.2.5 every Riesz coefficient is a killing probability, in two ways and with no hypothesis:

$$\frac{1}{\zeta(2k)} = \kappa_k \, P\big(T_1 < E_{(2k-1)^2}\big) = \frac{\kappa_k}{2\xi(\frac{1}{2})} \, P\big(T < E_{(2k-\frac{1}{2})^2}\big). \tag{13.2.4}$$

By Theorem 12.2.2, $T_1 \stackrel{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$, and by the paragraph after it T_1 is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$. So $c_k = 1/(\kappa_k \zeta(2k))$ is the probability that this hitting happens before an independent killing at rate $(2k-1)^2$.

The factorisation (13.2.2) reads the same way. Since $\mathbb{E}e^{-\lambda H_a} = e^{-a\sqrt{2\lambda}}$, the stable factor is

$$e^{-b_1(2k-1)} = P\big(H_{b_1/\sqrt{2}} < E_{(2k-1)^2}\big),$$

the probability that Brownian motion reaches the level $b_1/\sqrt{2} = 0.016331$ before it is killed. So $S \stackrel{d}{=} H_{b_1/\sqrt{2}}$ and $T_1 \stackrel{d}{=} H_{b_1/\sqrt{2}} + T_1^\circ$ with T_1° an independent GGC. Thus $c_k = P(H_{b_1/\sqrt{2}} < E_{(2k-1)^2})\, P(T_1^\circ < E'_{(2k-1)^2})$ with independent exponential clocks of the same rate. If the exact Brownian boundary b of T_1 is bounded at infinity, as Section 12.2.4 observes but does not prove, Lemma 12.2.4 and the tail of T_1 make the level $b_1/\sqrt{2} = \sum_\rho \rho^{-1}/\sqrt{2}$ its Wald defect $\mathbb{E}\, b(T_1)$.

Under RH with simple zeros, the centre clock is a GGC with unit atoms at the distinct points γ^2 (Proposition 2.1.1), hence, by Kent's theorem and Krein's inverse spectral theory (Section 5.2.3), the hitting time of a level by the diffusion of a Krein string with Dirichlet spectrum $\{\gamma^2\}$ (Table 9.1). The second form of (13.2.4) then says that $1/\zeta(2k)$ is proportional to the probability that this Riemann string diffusion reaches its level before it is killed at rate $(2k-\frac{1}{2})^2$. At the basepoint one, Corollary 12.2.3 gives the first form the same reading under the same hypotheses.

The Riesz function becomes an alternating sum of killing probabilities,

$$R(x) = -\sum_{k\geq 1} \frac{(-x)^k}{(k-1)!} \, \kappa_k \, P\big(T_1 < E_{(2k-1)^2}\big),$$

and the criterion asks that these alternating sums grow no faster than $x^{1/4+\delta}$. The killing rates $(2k-\alpha)^2$ that the criterion needs, with k in the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$, are complex, and at a complex rate the Laplace transform is no longer a probability; this is the hitting time form of the obstruction of Section 13.2.6.

A simulation checks these identities with 10^6 draws of $T_1 = \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ generated under RH as in Section 12.2.3. In Table 13.7 the killing probabilities match c_k, and the Brownian hitting probabilities match $e^{-b_1(2k-1)}$, within two standard errors. With the averaged killing probabilities for $k \leq 30$ from a second run of 10^6 draws, the alternating sum gives $R(0.5) = 0.125242$, $R(1) = 0.043990$, $R(2) = -0.267895$ and $R(3) = -0.526639$, against the exact 0.125220, 0.043982, -0.267793 and -0.526356.

k	$P(T_1 < E)$, drawn	$\mathbb{E}e^{-(2k-1)^2 T_1}$	$c_k = \frac{1}{\kappa_k \zeta(2k)}$	$P(H_{b_1/\sqrt{2}} < E)$	$e^{-b_1(2k-1)}$
1	0.95481 ± 0.00021	0.954928 ± 0.000112	0.954930	0.97699 ± 0.00015	0.97717
2	0.75965 ± 0.00043	0.759835 ± 0.000161	0.759909	0.93352 ± 0.00025	0.93306
3	0.50738 ± 0.00050	0.507865 ± 0.000156	0.507962	0.89098 ± 0.00031	0.89094
4	0.28834 ± 0.00045	0.288646 ± 0.000122	0.288731	0.85059 ± 0.00036	0.85072
5	0.14179 ± 0.00035	0.141473 ± 0.000080	0.141535	0.81195 ± 0.00039	0.81232
6	0.06045 ± 0.00024	0.060638 ± 0.000045	0.060679	0.77555 ± 0.00042	0.77565

Table 13.7: Riesz coefficients as killing probabilities of the basepoint one clock, and the stable factor as a Brownian hitting probability; E is an independent exponential clock of rate $(2k-1)^2$, and the errors are one standard error of 10^6 draws.

13.2.5 Half Cauchy variables, the sech clock and a duality

Bourgade, Fujita and Yor [43] derived Euler's formulae for $\zeta(2n)$ from products of Cauchy variables, which gives a representation of $\zeta(2n)$ dual to that of $1/\zeta(2n)$. Let C and C' be independent standard Cauchy variables, so that $|C|$ and $|C'|$ are half Cauchy. By Section 9.2.2, $\mathbb{E}|C|^s = \sec(\pi s/2)$ for $|\operatorname{Re} s| < 1$, so $\sec^2(\pi s/2) = \mathbb{E}|CC'|^s$ is the moment generating function of $\log|CC'|$ on $|s| < 1$. Differentiating the series $\tan z = \sum_{n\geq 1} 2^{2n}(2^{2n}-1)|B_{2n}|z^{2n-1}/(2n)!$ and inserting Euler's formula $\zeta(2n) = (2\pi)^{2n}|B_{2n}|/(2(2n)!)$ gives

$$\sec^2 \frac{\pi s}{2} = \sum_{n\geq 1} \frac{8(2n-1)(1-2^{-2n})\,\zeta(2n)}{\pi^2}\, s^{2n-2},$$

and comparing coefficients with the moment generating function,

$$\zeta(2n) = \frac{\pi^2}{8(2n-1)(1-2^{-2n})}\, \frac{\mathbb{E}\big[(\log|CC'|)^{2n-2}\big]}{(2n-2)!}, \qquad n \geq 1. \tag{13.2.5}$$

Let τ be the exit time of a standard Brownian motion from $(-1,1)$, so that $\mathbb{E}e^{-\lambda\tau} = 1/\cosh\sqrt{2\lambda}$ (Section 9.2.2), and let B be a Brownian motion independent of τ. Then $\mathbb{E}e^{i\theta B_\tau} = \mathbb{E}e^{-\theta^2\tau/2} = \operatorname{sech}\theta$, and since $\log|C|$ has characteristic function $\operatorname{sech}(\pi t/2)$ (Section 9.2.2),

$$\log|C| \stackrel{d}{=} \frac{\pi}{2}\, B_\tau.$$

We call τ the sech clock. By the product $\cosh\sqrt{2\lambda} = \prod_{k\geq 1}\big(1 + 8\lambda/((2k-1)^2\pi^2)\big)$ it is a GGC with unit Thorin atoms at $(2k-1)^2\pi^2/8$, the GGC $\sum_k e_k/(2k-1)^2$ of Section 9.2.2 multiplied by $8/\pi^2$, and it is Kent's hitting time for the interval: the hitting time of 1 by reflected Brownian motion started at 0, whose eigenvalues on $[0,1]$, with reflection at 0 and killing at 1, are these atoms. Its spectrum is proportional to the points $(2k-1)^2$ at which the Riesz coefficients sample

the Laplace transform of T_1. For two independent copies, $\log|CC'| \stackrel{d}{=} \frac{\pi}{2}B_{\tau+\tau'}$. Combining (13.2.5) with Proposition 13.2.5 gives a duality between the two clocks,

$$\frac{\mathbb{E}\big[(\log|CC'|)^{2k-2}\big]}{(2k-2)!}\,\mathbb{E}\,e^{-(2k-1)^2T_1} = \frac{8(2k-1)(1-2^{-2k})}{\pi^2\kappa_k}, \tag{13.2.6}$$

in which a moment of Brownian motion stopped at a sum of two independent sech clocks, times a killing probability of the basepoint one clock, is an elementary constant.

The moment formula (13.2.5), checked exactly through the Taylor coefficients of $\sec^2$, reproduces $\zeta(2)$, $\zeta(4)$, $\zeta(6)$ and $\zeta(8)$ to ten digits. With the killing probabilities of the second run of Section 13.2.4, the products in (13.2.6) are 0.95508, 1.87524, 2.06191 and 1.63889 for $k = 1, \dots, 4$, against 0.95493, 1.87500, 2.06167 and 1.63851.

The duality is structural rather than quantitative: $\zeta(2k)$ as a moment lives at the integers, while Riesz's criterion needs $1/\zeta(2s)$ inside the strip. The half Cauchy law also enters the book through Bosch's theorem [41] (Section 9.2), whose dictionary, Table 9.1, pairs the odd integers $2k-1$ with the ordinates γ.

13.2.6 Why the representation cannot supply the growth bound

The representations of Sections 13.2.2 to 13.2.5 are exact, and all except the Kent string readings of Section 13.2.4 hold with no hypothesis. They do not supply the bound $R(x) = O(x^{1/4+\delta})$, for four reasons.

1. The integer values carry almost no information. The growth of R is fixed by the Mellin transform $\Gamma(1-s)/\zeta(2s)$, that is, by the analytic interpolation of $k \mapsto 1/\zeta(2k)$ to complex k with $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$. At the integers $1/\zeta(2k) = 1 - 4^{-k} + O(9^{-k})$, and these values, which Proposition 13.2.3 represents, determine the interpolation only through its analytic continuation.

2. The representation interpolates only on one side of the line $\operatorname{Re} 2k = \alpha$, and as an expectation only on a cone. For complex k the expectation $\mathbb{E}e^{-(2k-\alpha)^2T_\alpha}$ is finite exactly when $\operatorname{Re}(2k-\alpha)^2 \geq 0$, because T_α has a tail of order $t^{-1/2}$ (Theorem 12.2.5) and no exponential moments; there it equals $\varphi_\alpha((2k-\alpha)^2) = \xi(\alpha)/\xi(\alpha + \sqrt{(2k-\alpha)^2})$, and the principal square root equals $2k-\alpha$ only when $\operatorname{Re}(2k-\alpha) > 0$. So the representation through T_α returns $\xi(\alpha)/\xi(2k)$ as an expectation on the cone $|\operatorname{Im} 2k| \leq \operatorname{Re} 2k - \alpha$, and on the rest of the half plane $\operatorname{Re} 2k > \alpha$ only through the analytic continuation of φ_α, which is analytic there exactly when no zero lies to the right of α. It is a GGC representation with no hypothesis only for $\alpha \geq 1$, that is, on the half plane $\operatorname{Re} 2k > 1$ of absolute convergence, where the criterion carries no information. To reach $\frac{1}{2} < \operatorname{Re} 2k < 1$ one needs a basepoint $\alpha < 1$, and T_α is a GGC there exactly when ζ has no zeros in $\operatorname{Re} s > \alpha$.

3. The centre clock's Laplace integral converges only up to its first singularity, which under RH is at $-\gamma_1^2$. In the k plane, with $u = (2k - \frac{1}{2})^2$, the region $\operatorname{Re} u > -\gamma_1^2$ is a neighbourhood of the segment $|\operatorname{Im} 2k| < \gamma_1$ of the critical line $\operatorname{Re} 2k = \frac{1}{2}$, not the strip; beyond it one needs analytic continuation, which is again a statement about the zeros.

4. Positivity does not survive. Each coefficient becomes a positive average, but the series is alternating, and the bound requires, at $x = 10^4$ and $T = 0.023$, the cancellation of terms as large as 10^{81} after averaging. No monotonicity or positivity argument controls that cancellation; it is carried by the oscillating terms $x^{\rho/2}$ of the explicit formula.

What would help is what the basepoint version of the criterion (Proposition 13.2.1 with $h = 0$) requires: an analytic, and ideally positive, representation of $1/\zeta(2s)$ on whole vertical lines of the strip $\frac{1}{4} < \operatorname{Re} s < \frac{1}{2}$. A representation of the Thorin kind that is positive there is a GGC at a basepoint below one, that is, a zero free half plane. The Thorin representation of the Riesz coefficients is therefore an exact reformulation, with better cancellation and a probabilistic reading, that reaches the criterion only through the step every construction of Chapters 12 to 14 requires. At a basepoint $\frac{1}{2} < \alpha < 1$, zero density estimates bound the number of Mellin poles to the right of the line $\operatorname{Re} s = \alpha/2$ up to any given height, as they bound the negative part of v_α/π (Theorem 14.3.2), but a single such pole breaks the bound $R(x) = O(x^{\alpha/2+\delta})$ for small δ.

13.3 The squarefree series and the clock

Riesz's criterion (Sections 13.1.8 and 13.2) reads RH from a function whose Mellin transform is $\Gamma(1-s)/\zeta(2s)$. The squarefree integers carry the same reciprocal $1/\zeta(2s)$. Their Dirichlet series is

$$D(s) = \sum_{n \geq 1} \frac{\mu(n)^2}{n^s} = \prod_p \left(1 + p^{-s}\right) = \frac{\zeta(s)}{\zeta(2s)}, \qquad \operatorname{Re} s > 1, \tag{13.3.1}$$

whose singularities in the half plane $\operatorname{Re} s > 0$ are the pole at $s = 1$ and the poles at the points $\rho/2$ at which $\zeta(\rho/2) \neq 0$ (all of them under RH), and the number $Q(x)$ of squarefree integers up to x oscillates about $6x/\pi^2$ on the scale $x^{1/4}$ that these poles set when they lie on the line $\operatorname{Re} s = \frac{1}{4}$, that is under RH. This section is based on [289], which starts from a Math StackExchange thread [233] on the partial sums of $\mu(n)^2 n^{-\alpha}$ over $\mathbb{Q}$ and over number fields. For the error $\Delta(x) = Q(x) - 6x/\pi^2$ the bound $\Delta(x) = O(x^{1/4+\varepsilon})$ is open even under RH, but it is equivalent to RH once the count is smoothed by an exponential or a Gamma(2) law, and in mean square; the order $x^{1/4}$ is attained with no hypothesis; and the bound for the count smoothed by a Gamma(n) law loses a factor $n^{1/8}$, which still

gives $\Delta(x) = O(x^{2/5+\varepsilon})$ under RH (Sections 13.3.5 and 13.3.6). The Pólya kernels of the Dedekind zeta functions of imaginary quadratic fields, whose positivity can fail, are in Section 15.1.6 of Chapter 18.

We keep the notation of Section 2.1 for the Pólya variable X, with $\mathbb{E}e^{sX} = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, the clock T, $G(u) = \xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2})$, the numbers $a_k = -(\rho_k - \frac{1}{2})^2$, the heat trace $W(t) = \sum_k e^{-a_k t}$ and the measure U_{RH}, and the notation of Section 12.1.1 for the tilted variables V_α, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, and $b_\alpha = \frac{\xi'}{\xi}(\alpha) = \mathbb{E}V_\alpha$. Throughout, μ is the Möbius function, $M(t) = \sum_{d \le t} \mu(d)$ is the Mertens function, $\{y\}$ is the fractional part of y, and

$$Q(x) = \sum_{n \le x} \mu(n)^2, \qquad \Delta(x) = Q(x) - \frac{6}{\pi^2}\, x.$$

13.3.1 The thread: Perron's formula and an elementary method

The thread asks for the behaviour as $x \to \infty$ of

$$S_\alpha(x) = \sum_{N(\mathfrak{a}) \le x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})^\alpha},$$

the sum running over the integral ideals $\mathfrak{a}$ of a number field K, with μ the Möbius function of ideals and N the norm, and, in a comment, for its expansion at $\alpha = 1$. Over $\mathbb{Q}$, $S_\alpha(x) = \sum_{n \le x} \mu(n)^2 n^{-\alpha}$, and another comment points to Abel summation from the classical count

$$Q(x) = \frac{6}{\pi^2}\, x + O(\sqrt{x}). \tag{13.3.2}$$

The accepted answer applies Perron's formula over K, and the other answer carries out the elementary method over $\mathbb{Q}$. Neither evaluates the constant at $\alpha = 1$.

Since $\mu(\mathfrak{p}^k)^2$ is 1 for $k \le 1$ and 0 for $k \ge 2$, the Euler factor of $\sum_{\mathfrak{a}} \mu(\mathfrak{a})^2 N(\mathfrak{a})^{-s}$ at a prime ideal $\mathfrak{p}$ is $1 + N(\mathfrak{p})^{-s} = (1 - N(\mathfrak{p})^{-2s})/(1 - N(\mathfrak{p})^{-s})$, so the series is $\zeta_K(s)/\zeta_K(2s)$, with ζ_K the Dedekind zeta function of K, and

$$S_\alpha(x) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{\zeta_K(\alpha+s)}{\zeta_K(2\alpha+2s)}\, x^s\, \frac{ds}{s}, \qquad \sigma > \max\{1-\alpha, 0\}. \tag{13.3.3}$$

The integrand of (13.3.3) has a simple pole at $s = 1-\alpha$, a pole at $s = 0$, and poles at the points $s = \rho/2 - \alpha$ for the zeros ρ of ζ_K, trivial and nontrivial, at which $\zeta_K(\rho/2) \ne 0$. Let $c_K = \lim_{s \to 1}(s-1)\zeta_K(s) = 2^{r_1}(2\pi)^{r_2} hR/(w\sqrt{|d_K|})$ be the residue of the class number formula.

The case $\alpha \ne 1$. The residue at $s = 1-\alpha$ is $c_K x^{1-\alpha}/((1-\alpha)\zeta_K(2))$, as the accepted answer states, and the residue at $s = 0$, which it does not compute, is

$\zeta_K(\alpha)/\zeta_K(2\alpha)$, the value of the continued series. Over $\mathbb{Q}$, for $\alpha > \frac12$ and $\alpha \neq 1$, the two residues are the main terms of

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\,\frac{x^{1-\alpha}}{1-\alpha} + \frac{\zeta(\alpha)}{\zeta(2\alpha)} + O\big(x^{\frac12-\alpha}\big). \tag{13.3.4}$$

Indeed, Abel summation and (13.3.2) give, for $\alpha > \frac12$, $\alpha \neq 1$,

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\,\frac{x^{1-\alpha}}{1-\alpha} + \Big(\alpha\int_1^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt - \frac{6}{\pi^2}\,\frac{\alpha}{1-\alpha}\Big) + \frac{\Delta(x)}{x^\alpha} - \alpha\int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt. \tag{13.3.5}$$

The last two terms are $O(x^{\frac12-\alpha})$, and the bracket is $D(\alpha)$, because $D(s) = s\int_1^\infty Q(t)t^{-s-1}dt = \frac{6}{\pi^2}\frac{s}{s-1} + s\int_1^\infty \Delta(t)t^{-s-1}dt$ for $\operatorname{Re} s > 1$ and the right side is analytic in $\operatorname{Re} s > \frac12$ apart from $s = 1$. For $\frac12 < \alpha < 1$ the constant is negative (Corollary 13.3.4). Write $\mathcal{E}_\alpha(x)$ for the left side of (13.3.4) minus its two main terms, so that by (13.3.5)

$$\mathcal{E}_\alpha(x) = \frac{\Delta(x)}{x^\alpha} - \alpha\int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt. \tag{13.3.6}$$

The case $\alpha = 1$. The poles at $1-\alpha$ and at 0 merge into a double pole at $s = 0$ of $\zeta_K(1+s)\zeta_K(2+2s)^{-1}e^{s\log x}/s$. The accepted answer computes the coefficient of $\log x$ and calls the constant term much more complicated when $K \neq \mathbb{Q}$; it is explicit. Define the Euler Kronecker constant γ_K of K by $\zeta_K(s) = c_K/(s-1) + c_K\gamma_K + O(s-1)$ [160], so that $\gamma_{\mathbb{Q}} = \gamma_E$. With $1/\zeta_K(2+2s) = \zeta_K(2)^{-1}\big(1 - 2s\,\zeta_K'(2)/\zeta_K(2) + O(s^2)\big)$ the residue at $s = 0$ is

$$\frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big), \qquad C_K = \gamma_K - 2\,\frac{\zeta_K'(2)}{\zeta_K(2)}. \tag{13.3.7}$$

The elementary method. The identity $\mu(n)^2 = \sum_{d^2\mid n}\mu(d)$, the coefficientwise form of $D(s) = \zeta(s)\cdot\zeta(2s)^{-1}$, gives

$$Q(x) = \sum_{d\le\sqrt{x}} \mu(d)\Big\lfloor \frac{x}{d^2}\Big\rfloor = x\sum_{d\le\sqrt{x}}\frac{\mu(d)}{d^2} - \sum_{d\le\sqrt{x}}\mu(d)\Big\{\frac{x}{d^2}\Big\}, \tag{13.3.8}$$

and since $\sum_{d>\sqrt{x}} d^{-2} = O(x^{-1/2})$ and the last sum has at most $\sqrt{x}$ terms, each in $[0,1)$, this is (13.3.2). The second answer then sets

$$B(x) = \sum_{n\le x}\Big(\mu(n)^2 - \frac{6}{\pi^2}\Big) = O(\sqrt{x}),$$

$$D_0(s) = \sum_{n\ge1}\frac{\mu(n)^2 - 6/\pi^2}{n^s} = D(s) - \frac{6}{\pi^2}\,\zeta(s).$$

By Abel summation, $D_0(s) = \lim_{x\to\infty}\big(B(x)x^{-s} + s\int_1^x B(t)t^{-s-1}dt\big)$ converges for $\operatorname{Re} s > \frac12$, so $\sum_{n\le x}\mu(n)^2/n = \frac{6}{\pi^2}(\log x + \gamma_E) + D_0(1) + o(1)$, where the answer stops. Since $D_0(s) = \zeta(s)\big(\zeta(2s)^{-1} - \zeta(2)^{-1}\big)$ and ζ has a simple pole with residue 1 at $s = 1$,

$$D_0(1) = \lim_{s\to1}\zeta(s)\Big(\frac{1}{\zeta(2s)} - \frac{1}{\zeta(2)}\Big) = -\frac{2\zeta'(2)}{\zeta(2)^2} = 0.69298946940360442669,$$

in agreement with (13.3.7) at $K = \mathbb{Q}$, where $c_{\mathbb{Q}} = 1$ and $\gamma_{\mathbb{Q}} = \gamma_E$.

Proposition 13.3.1. (i) *Let K be a number field of degree $n \ge 2$. As $x \to \infty$,*

$$\sum_{N(\mathfrak{a})\le x}\frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} = \frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big) + O\big(x^{-1/n}\log x\big).$$

(ii) *Over $\mathbb{Q}$,*

$$\begin{aligned}\sum_{n\le x}\frac{\mu(n)^2}{n} &= \frac{6}{\pi^2}\big(\log x + C\big) + O\big(x^{-1/2}\big),\\ C &= \gamma_E - 2\,\frac{\zeta'(2)}{\zeta(2)} = 24\log A - \gamma_E - 2\log 2\pi,\end{aligned} \tag{13.3.9}$$

where A is the Glaisher Kinkelin constant; $C = 1.71713765109059847340624$1 and $\frac{6}{\pi^2}C = 1.0438945157119382974$. Moreover $\mathcal{E}_1(x) = \sum_{n\le x}\mu(n)^2/n - \frac{6}{\pi^2}(\log x + C)$ is given by (13.3.6) *with $\alpha = 1$, and*

$$\int_1^\infty \frac{\Delta(t)}{t^2}\,dt = \frac{6}{\pi^2}\,(C-1) = 0.43596741385791166874.$$

Proof. (i) The number of ideals of norm at most y is $c_K y + O(y^{1-1/n})$ [204, Chapter VI], so by Abel summation $L(y) = \sum_{N(\mathfrak{c})\le y} N(\mathfrak{c})^{-1} = c_K(\log y + \gamma_K) + O(y^{-1/n})$ for $y \ge 1$, the constant being the one in the Laurent expansion of ζ_K at $s = 1$. Since $\mu(\mathfrak{a})^2 = \sum_{\mathfrak{b}^2\mid\mathfrak{a}}\mu(\mathfrak{b})$, as one checks on prime powers,

$$\begin{aligned}\sum_{N(\mathfrak{a})\le x}\frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} &= \sum_{N(\mathfrak{b})\le\sqrt{x}}\frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\,L\Big(\frac{x}{N(\mathfrak{b})^2}\Big)\\ &= c_K\sum_{N(\mathfrak{b})\le\sqrt{x}}\frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\big(\log x + \gamma_K - 2\log N(\mathfrak{b})\big)\\ &\quad + O\Big(x^{-1/n}\sum_{N(\mathfrak{b})\le\sqrt{x}} N(\mathfrak{b})^{-2+2/n}\Big).\end{aligned}$$

The last sum is $O(\log x)$, since $-2 + 2/n \le -1$. Completing the sums over $\mathfrak{b}$ costs $O(x^{-1/2}\log x)$, which is $O(x^{-1/n}\log x)$ as $n \ge 2$, since $\sum_{N(\mathfrak{b})>Y} N(\mathfrak{b})^{-2}\log N(\mathfrak{b}) = O(Y^{-1}\log Y)$, and the completed sums are

$\sum_{\mathfrak{b}} \mu(\mathfrak{b}) N(\mathfrak{b})^{-2} = 1/\zeta_K(2)$ and $\sum_{\mathfrak{b}} \mu(\mathfrak{b}) \log N(\mathfrak{b})\, N(\mathfrak{b})^{-2} = \zeta_K'(2)/\zeta_K(2)^2$, which give (13.3.7).

(ii) By Abel summation, $\sum_{n\le x} \mu(n)^2/n = Q(x)/x + \int_1^x Q(t)t^{-2}dt$, and inserting $Q(t) = 6t/\pi^2 + \Delta(t)$ gives

$$\sum_{n\le x} \frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\log x + \frac{6}{\pi^2} + \int_1^\infty \frac{\Delta(t)}{t^2}\,dt + \frac{\Delta(x)}{x} - \int_x^\infty \frac{\Delta(t)}{t^2}\,dt,$$

whose last two terms are $O(x^{-1/2})$ by (13.3.2). The constant is identified through D: for $\operatorname{Re} s > 1$, $\int_1^\infty \Delta(t)t^{-s-1}dt = D(s)/s - \frac{6}{\pi^2}\frac{1}{s-1}$, and $D(s) = \frac{6}{\pi^2}\zeta(s) + D_0(s) = \frac{6}{\pi^2}\big(\frac{1}{s-1} + \gamma_E - 2\frac{\zeta'(2)}{\zeta(2)}\big) + O(s-1)$, so that $\int_1^\infty \Delta(t)t^{-2}dt = \frac{6}{\pi^2}(C-1)$ by continuity at $s=1$. This gives (13.3.9) and the formula for $\mathcal{E}_1$, and the classical evaluation $\zeta'(2)/\zeta(2) = \gamma_E + \log 2\pi - 12\log A$ gives the second form of C. □

The sieve gives $Q(2\times 10^8) = 121585426$. Table 13.8 shows that $\mathcal{E}_1(x)$ is of order $x^{-3/4}$ and $\Delta(x)$ of order $x^{1/4}$, the sizes that the poles at $\rho/2-1$ and $\rho/2$ predict under RH (Section 13.3.4).

x	$Q(x)$	$\Delta(x)/x^{1/4}$	$\mathcal{E}_1(x)$	$\mathcal{E}_1(x)\,x^{3/4}$	$\mathcal{E}_{3/4}(x)\,x^{1/4}$
10^3	608	0.0130	-9.130×10^{-4}	−0.162	−0.0291
10^4	6083	0.3729	3.031×10^{-4}	0.303	0.0296
10^5	60794	0.0725	-2.308×10^{-6}	−0.0130	−0.00031
10^6	607926	−0.0348	-1.665×10^{-6}	−0.0526	−0.00177
10^7	6079291	0.3553	1.729×10^{-6}	0.308	0.00561
10^8	60792694	−0.1619	-1.737×10^{-7}	−0.174	−0.00174

Table 13.8: The sharp counts: the number $Q(x)$ of squarefree integers up to x, the normalised error $\Delta(x)/x^{1/4}$, the error $\mathcal{E}_1(x)$ of (13.3.9), which confirms the constant C, also multiplied by $x^{3/4}$, and the error $\mathcal{E}_{3/4}(x)$ of (13.3.4) at $\alpha = \frac{3}{4}$ multiplied by $x^{1/4}$. Computed; the sums of reciprocals are checked in extended precision.

13.3.2 The Pólya form of the series and the constant

Lemma 13.3.2. *For all s,*

$$D(s) = \Gamma_{\mathrm{sq}}(s)\,\frac{\xi(s)}{\xi(2s)}, \qquad \Gamma_{\mathrm{sq}}(s) = \frac{2s-1}{s-1}\,2^s\pi^{-(s+1)/2}\,\Gamma\Big(\frac{s+1}{2}\Big).$$

Proof. By the definition of ξ, $\zeta(s)/\zeta(2s) = \frac{\xi(s)}{\xi(2s)}\cdot\frac{2(2s-1)}{s-1}\,\pi^{-s/2}\,\frac{\Gamma(s)}{\Gamma(s/2)}$, and the duplication formula $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$ gives the factor Γ_{sq}. □

A computation checks the lemma at $s = 3$, $0.3+2.1i$, $\frac{3}{4}$ and -1.5, with relative errors below 5×10^{-41}.

PROPOSITION 13.3.3. *For every s with $\xi(2s) \neq 0$,*

$$\sum_{n\geq 1} \frac{\mu(n)^2}{n^s} = \Gamma_{\mathrm{sq}}(s)\, \frac{\mathbb{E}e^{(s-\frac12)X}}{\mathbb{E}e^{(2s-\frac12)X}},$$

the left side read as the continuation $D(s)$ where the series diverges.

Proof. Lemma 13.3.2 and (2.1.4), which gives $\xi(s) = \xi(\frac12)\mathbb{E}e^{(s-\frac12)X}$ and $\xi(2s) = \xi(\frac12)\mathbb{E}e^{(2s-\frac12)X}$. □

COROLLARY 13.3.4. *For real $\alpha \notin \{1, \frac12, -1, -3, -5, \dots\}$ the sign of $\zeta(\alpha)/\zeta(2\alpha)$ is the sign of $\frac{2\alpha-1}{\alpha-1}\Gamma(\frac{\alpha+1}{2})$. It is positive for $\alpha > 1$ and for $-1 < \alpha < \frac12$, negative for $\frac12 < \alpha < 1$, and equal to $(-1)^k$ on the interval $(-2k-1, -2k+1)$ for $k \geq 1$.*

Proof. At real α the ratio of moment generating functions in Proposition 13.3.3 is positive, and so is $2^\alpha \pi^{-(\alpha+1)/2}$. The rational factor is positive except on $(\frac12, 1)$, and $\Gamma(\frac{\alpha+1}{2})$ has the sign $(-1)^k$ when $\frac{\alpha+1}{2} \in (-k, -k+1)$, that is when $\alpha \in (-2k-1, -2k+1)$. □

At the even negative integers $\zeta(\alpha)$ and $\zeta(2\alpha)$ both vanish and the quotient keeps its sign. At $\alpha = \frac34$, where the series diverges, $\Gamma_{\mathrm{sq}}(\frac34) = -1.34612167295$, $\mathbb{E}e^{X/4} = 1.00144503253351$ and $\mathbb{E}e^{X} = 1.02335500958202$, the last two by the trapezoid rule on the Pólya kernel (2.1.3); the ratio 0.978590052481 multiplies $\Gamma_{\mathrm{sq}}(\frac34)$ to $-1.31730127858 = \zeta(\frac34)/\zeta(\frac32)$.

THEOREM 13.3.5. *The constant of* (13.3.9) *is*

$$C = 2 + \log 2 - \tfrac12 \log \pi - \tfrac12 \gamma_E + b_1 - 2b_2 = 2 + \log 2 - \tfrac12 \log \pi - \tfrac12 \gamma_E + \mathbb{E}V_1 - 2\,\mathbb{E}V_2.$$

With no hypothesis on the zeros,

$$\begin{aligned} b_1 - 2b_2 &= \frac{G'}{G}\Big(\frac14\Big) - 6\,\frac{G'}{G}\Big(\frac94\Big) = \sum_k \Big(\frac{1}{\frac14 + a_k} - \frac{6}{\frac94 + a_k}\Big) \\ &= \int_0^\infty \big(e^{-t/4} - 6e^{-9t/4}\big)\, W(t)\, dt, \end{aligned} \tag{13.3.10}$$

the sum running over the pairs $\{\rho, 1-\rho\}$ and converging absolutely. Under RH, $a_k = \gamma_k^2$ and $b_1 - 2b_2 = \int \big((z+\frac14)^{-1} - 6(z+\frac94)^{-1}\big)\, U_{\mathrm{RH}}(dz)$, a combination of two values of the Stieltjes transform of the Thorin measure of the clock.

Proof. By the proof of Proposition 13.3.1, $(s-1)D(s) = \frac{6}{\pi^2}\big(1 + C(s-1) + O((s-1)^2)\big)$, so C is the derivative of $\log((s-1)D(s))$ at $s = 1$. By Proposition 13.3.3,

$$\begin{aligned} \log\big((s-1)D(s)\big) = \log(2s-1) + s\log 2 - \tfrac{s+1}{2}\log\pi + \log\Gamma\big(\tfrac{s+1}{2}\big) \\ + \log \mathbb{E}e^{(s-\frac12)X} - \log \mathbb{E}e^{(2s-\frac12)X}. \end{aligned}$$

At $s=1$ the derivatives of the first four terms are 2, $\log 2$, $-\frac12\log\pi$ and $\frac12\psi(1)=-\frac12\gamma_E$, with ψ the digamma function here. The derivative of $\log \mathbb{E}e^{(s-\frac12)X}$ at $s=1$ is $\mathbb{E}Xe^{X/2}/\mathbb{E}e^{X/2}$, the mean of X tilted by $e^{X/2}$, which is $\mathbb{E}V_1=b_1$ (Section 12.1.1); likewise the last term gives $-2\mathbb{E}V_2=-2b_2$. For the zero form, differentiating $\log G(w^2)=\log\xi(\frac12+w)-\log\xi(\frac12)$ gives $\frac{\xi'}{\xi}(\frac12+w)=2w\frac{G'}{G}(w^2)$, so $b_1=\frac{G'}{G}(\frac14)$ and $b_2=3\frac{G'}{G}(\frac94)$. Then $\frac{G'}{G}(u)=\sum_k(u+a_k)^{-1}$ by (2.1.1), and this is $\int_0^\infty e^{-ut}W(t)\,dt$ for $u\ge0$, by termwise integration, since $\operatorname{Re}a_k\ge\gamma_1^2-\frac14>0$ and $\sum_k|a_k|^{-1}<\infty$. Under RH, $\frac{G'}{G}(u)=\int(u+z)^{-1}U_{\mathrm{RH}}(dz)$ by Proposition 2.1.1. □

The elementary part is $2+\log 2-\frac12\log\pi-\frac12\gamma_E=1.83217440518447879 2$, and $b_1=0.023095708966121033814$, which is $1+\frac12\gamma_E-\frac12\log4\pi=\sum_\rho\rho^{-1}$, and $b_2=0.069066231530000676225$, by the closed forms of ξ'/ξ and by numerical differentiation, which agree to every digit printed. So $b_1-2b_2=-0.11503675409388031864$, and the two parts reproduce C to all 25 digits.

The variance identity $\sum_k a_k^{-1}=\frac12\operatorname{Var}X$ of [289] is the formula $\mathbb{E}T=\sum_k a_k^{-1}=\xi''(\frac12)/(2\xi(\frac12))$ of Section 2.1, and the alternation of the even cumulants of X, derived there under RH, holds with no hypothesis (Section 20.1.2).

13.3.3 Number fields

Let K be a number field with r_1 real and r_2 pairs of complex embeddings and discriminant d_K, and put $\Gamma_{\mathbb{R}}(s)=\pi^{-s/2}\Gamma(\frac{s}{2})$, $\Gamma_{\mathbb{C}}(s)=2(2\pi)^{-s}\Gamma(s)$ and

$$\xi_K(s)=s(s-1)\,|d_K|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_K(s).$$

Then ξ_K is entire of order one, $\xi_K(s)=\xi_K(1-s)$ by Hecke's functional equation [204, Chapter XIII], ξ_K is real on the real axis, and its zeros are the nontrivial zeros of ζ_K, all in the strip $0<\operatorname{Re}s<1$; for $K=\mathbb{Q}$, $\xi_{\mathbb{Q}}=2\xi$. For $K=\mathbb{Q}(i)$, where $\zeta_K(s)=\zeta(s)L(s,\chi_{-4})$, the duplication formula gives $\xi_K(s)=\xi(s)\,\xi(s,\chi_{-4})$ exactly, with $\xi(s,\chi_{-4})=(4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_{-4})$ the completed function of the odd character modulo 4 in the normalisation of Section 15.1.1.

Proposition 13.3.6. *Suppose $\zeta_K(\frac12)\neq0$ and let $G_K(u)=\xi_K(\frac12+\sqrt{u})/\xi_K(\frac12)$. Then $1/G_K$ is the Laplace transform of a GGC if and only if every zero of ζ_K in the critical strip lies on the line $\operatorname{Re}s=\frac12$, the Riemann hypothesis for ζ_K. Its drift is then 0 and its Thorin measure is $\sum_{\gamma>0}\delta_{\gamma^2}$, over the ordinates of the zeros counted with multiplicity. A real zero $\beta\in(\frac12,1)$ of ζ_K would be a pole of $1/G_K$ at the point $(\beta-\frac12)^2$ of the positive axis.*

Proof. Since $\xi_K(\frac12+w)$ is even and entire of order one, G_K is entire of order $\frac12$, and Hadamard's theorem gives $G_K(u)=\prod_k(1+u/a_{K,k})$ with $a_{K,k}=-(\rho_k-\frac12)^2$, one factor for each pair $\{\rho,1-\rho\}$ of zeros of ξ_K, and $\sum_k|a_{K,k}|^{-1}<\infty$. If every zero lies on the line, every $a_{K,k}=\gamma_k^2$ is positive and $1/G_K$ is a convergent product of gamma Laplace transforms, a GGC with the stated Thorin measure and no

drift. Conversely, the Laplace transform of a GGC is holomorphic and zero free on $\mathbb{C}\setminus(-\infty,0]$, so G_K has no zero there, and its zeros $(\rho-\frac12)^2$ lie in $(-\infty,0]$; since $\rho\neq\frac12$, this means $\rho-\frac12\in i\mathbb{R}\setminus\{0\}$, that is $\operatorname{Re}\rho=\frac12$. A real zero β gives the zero $(\beta-\frac12)^2>0$ of G_K. □

The hypothesis $\zeta_K(\frac12)\neq0$ fixes the normalisation; Armitage found number fields whose Dedekind zeta function vanishes at the centre [7]; there the zero of $\xi_K(\frac12+w)$ at $w=0$ has even order $2m$, and the proposition applies to $u^{-m}\xi_K(\frac12+\sqrt{u})$ normalised at $u=0$.

THEOREM 13.3.7. *Let*

$$A_K=2+\tfrac12\log|d_K|+r_1\big(\log2-\tfrac12\log\pi-\tfrac12\gamma_E\big)+r_2\big(2-\gamma_E-\log2\pi\big).$$

Then the constant of (13.3.7) *is*

$$C_K=A_K+\frac{\xi_K'}{\xi_K}(1)-2\,\frac{\xi_K'}{\xi_K}(2)=A_K+\sum_k\Big(\frac{1}{\frac14+a_{K,k}}-\frac{6}{\frac94+a_{K,k}}\Big),$$

the sum running over the pairs $\{\rho,1-\rho\}$ of zeros of ξ_K, with multiplicity, a zero of order $2m$ at $s=\frac12$ counting as m pairs with $a_{K,k}=0$. Under the Riemann hypothesis for ζ_K, and $\zeta_K(\frac12)\neq0$, the zero part is $\frac{G_K'}{G_K}(\frac14)-6\frac{G_K'}{G_K}(\frac94)$, the same combination of values of the Stieltjes transform of the Thorin measure of Proposition 13.3.6 as over $\mathbb{Q}$.

Proof. From the definition of ξ_K,

$$(s-1)\frac{\zeta_K(s)}{\zeta_K(2s)}=\frac{\xi_K(s)}{\xi_K(2s)}\cdot2(2s-1)\,|d_K|^{s/2}\,\frac{\Gamma_{\mathbb{R}}(2s)^{r_1}\Gamma_{\mathbb{C}}(2s)^{r_2}}{\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}},$$

and C_K is the logarithmic derivative of the left side at $s=1$, by the Laurent expansions behind (13.3.7). With $\Gamma_{\mathbb{R}}'/\Gamma_{\mathbb{R}}(s)=-\frac12\log\pi+\frac12\psi(\frac s2)$, $\Gamma_{\mathbb{C}}'/\Gamma_{\mathbb{C}}(s)=-\log2\pi+\psi(s)$, $\psi(1)=-\gamma_E$, $\psi(\frac12)=-\gamma_E-2\log2$ and $\psi(2)=1-\gamma_E$, the logarithmic derivative of the elementary factor at $s=1$ is A_K, and that of $\xi_K(s)/\xi_K(2s)$ is $\frac{\xi_K'}{\xi_K}(1)-2\frac{\xi_K'}{\xi_K}(2)$. The Hadamard product of $\xi_K(\frac12+w)$, which is even in w, gives $\frac{\xi_K'}{\xi_K}(\frac12+w)=\sum_k 2w/(w^2+a_{K,k})$, and the values at $w=\frac12$ and $w=\frac32$ give the zero form. □

For the Gaussian field $\mathbb{Q}(i)$, with $r_2=1$ and $d_K=-4$, we find $c_K/\zeta_K(2)=3/(2\pi L(2,\chi_{-4}))=0.521269392989129$, with $L(2,\chi_{-4})$ Catalan's constant, and

$$\gamma_K=\gamma_E+\frac{L'}{L}(1,\chi_{-4})=0.822825249678847033,$$
$$\frac{\zeta_K'}{\zeta_K}(2)=-0.48089570872664776864,$$

and $C_K = 1.784616667132142570284$, where $\frac{L'}{L}(1,\chi_{-4}) = \gamma_E + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac14) = 0.24560958477731417239$. In Theorem 13.3.7, $A_K = 4 - \gamma_E - \log\pi = 2.2780544492490669653$ and the zero part is the sum of the zero part $b_1 - 2b_2$ of ζ and the zero part $\frac{\xi_\chi'}{\xi_\chi}(1) - 2\frac{\xi_\chi'}{\xi_\chi}(2) = -0.378401028023044$ of $\xi_\chi = \xi(\cdot,\chi_{-4})$: the zeros of ζ and of $L(s,\chi_{-4})$ simply add their Thorin atoms, and the two sides of the theorem differ by 4.6×10^{-41}. The 289 zeros of $L(s,\chi_{-4})$ below height 400, located in Arb, give -0.365385920572 for the L part, and their smooth tail -0.013015062 brings the sum within 4.6×10^{-8} of the closed form.

13.3.4 The Thorin atoms in the squarefree count

Put $s = \frac14 + \frac12\sqrt{u}$, so that $2s = \frac12 + \sqrt{u}$. Lemma 13.3.2 and the clock give, for $u \ge 0$ and with no hypothesis,

$$D\Big(\frac14 + \frac{\sqrt{u}}{2}\Big) = \Gamma_{\mathrm{sq}}\Big(\frac14 + \frac{\sqrt{u}}{2}\Big)\,\frac{\xi(\frac14 + \frac12\sqrt{u})}{\xi(\frac12)}\,\mathbb{E}e^{-uT},$$

since $1/G$ is the Laplace transform of the clock T (Section 2.1). The poles of D in the strip $0 < \operatorname{Re} s < \frac12$ are therefore among the points where G vanishes, $u = -a_k$, that is $s = \rho_k/2$, and they are all of these points when $\zeta(\rho_k/2) \neq 0$, as under RH; under RH they are the atoms γ_k^2 of U_{RH}, placed at $u = -\gamma_k^2$, that is at $s = \frac14 \pm \frac{i}{2}\gamma_k$.

At a simple zero ρ, $\zeta(2s) = 2\zeta'(\rho)(s-\rho/2) + O((s-\rho/2)^2)$, so the residue of $D(s)x^s/s$ at $s = \rho/2$ is $R_\rho x^{\rho/2}$, and formally

$$\Delta(x) \approx 1 + \sum_{\rho} R_\rho\, x^{\rho/2} + \sum_{k=1,3,5,\dots} r_k\, x^{-k}, \qquad R_\rho = \frac{\zeta(\rho/2)}{\rho\,\zeta'(\rho)}, \qquad r_k = \frac{\zeta(-k)}{-2k\,\zeta'(-2k)}, \tag{13.3.11}$$

where $1 = D(0)$ is the residue at $s = 0$ and $r_k x^{-k}$ is the residue at the trivial pole $s = -k$. For the first zero $R_{\rho_1} = 0.028803 - 0.0961835i$ and $|R_{\rho_1}| = 0.100404$, with $\zeta(\rho_1/2) = 1.03142 + 0.452556i$, and for the next four zeros $|R_\rho| = 0.069179$, 0.0338528, 0.0260772 and 0.0464698; $r_1 = -1.3684327776202059$, $r_3 = 0.23541450677954054$ and $r_5 = -0.020962860062791976$. Under RH every term of the sum over ρ has size $x^{1/4}$, so $\Delta(x) = O(x^{1/4+\varepsilon})$ is the natural conjecture and $\Delta(x)/x^{1/4}$ stays of order one in Table 13.8; by Corollary 13.3.11 the exponent $\frac14$ cannot be lowered. The series over ρ is not expected to converge absolutely, and the bound $O(x^{1/4+\varepsilon})$ is open even under RH: the known conditional exponents are larger than $\frac14$, among them $\frac{7}{22}$ of Baker and Pintz [12], since improved, and the obstacle is the control of $1/\zeta$ near the critical line.

Figure 13.6 compares $\Delta(x)$ on $[100, 3000]$ with the right side of (13.3.11) truncated after the first K zeros, both divided by $x^{1/4}$ and sampled on the grid $x = 100 + j/20$. The Pearson correlation of the two curves is 0.828, 0.866, 0.915, 0.927 and 0.945 for $K = 10$, 20, 60, 100 and 200, and the root mean square of

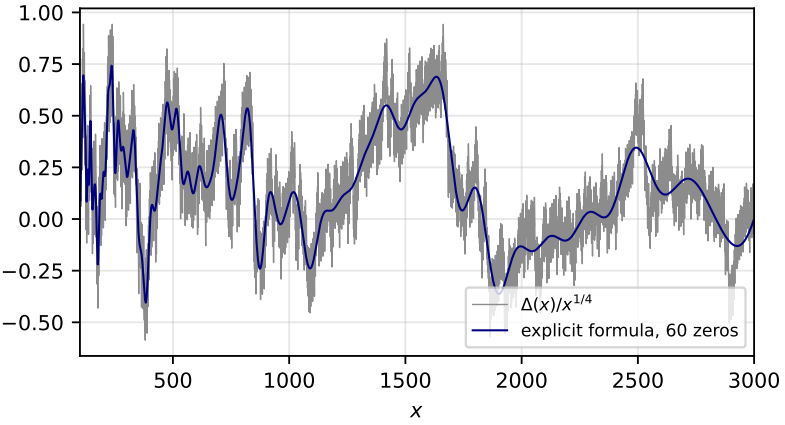


Figure 13.6: The normalised error $\Delta(x)/x^{1/4}$ of the squarefree count on $[100, 3000]$ (grey) and the explicit formula (13.3.11) with the first 60 zeros and the trivial terms at $s = -1$, -3 and -5, divided by $x^{1/4}$ (dark).

their difference falls from 0.149 to 0.0868, against 0.304 for $\Delta(x)/x^{1/4}$ itself; what remains includes the unit jumps of Q, which no finite sum of atoms reproduces.

For the sum $\sum_{n\le x}\mu(n)^2/n$ of Proposition 13.3.1 the same atoms appear at $s = \rho/2 - 1$, with residues $\zeta(\rho/2)x^{\rho/2-1}/((\rho-2)\zeta'(\rho))$ of size $x^{-3/4}$ under RH (of modulus $0.09991\,x^{-3/4}$ for the first zero), inside the error $O(x^{-1/2})$. Likewise the error $\mathcal{E}_\alpha$ of (13.3.4), for $\frac12 < \alpha < 1$, has atoms of size $x^{1/4-\alpha}$. By (13.3.6), a pointwise bound of that order in either case would follow from $\Delta(x) = O(x^{1/4+\varepsilon})$, and is open for the same reason.

13.3.5 Smoothed counts

Smoothing the count removes the obstacle. For $x > 0$ let

$$Q_1(x) = \sum_{n\ge1}\mu(n)^2 e^{-n/x} = \sum_{d\ge1}\frac{\mu(d)}{e^{d^2/x}-1}, \qquad Q_2(x) = \sum_{n\ge1}\mu(n)^2\Big(1+\frac{2n}{x}\Big)e^{-2n/x},$$

the second form of Q_1 by $\mu(n)^2 = \sum_{d^2\mid n}\mu(d)$. If $G_{1,x}$ is exponential and $G_{2,x}$ has the Gamma(2) law, both with mean x, then $P(G_{1,x}\ge n) = e^{-n/x}$ and $P(G_{2,x}\ge n) = (1+2n/x)e^{-2n/x}$, so $Q_1(x) = \mathbb{E}Q(G_{1,x})$ and $Q_2(x) = \mathbb{E}Q(G_{2,x})$. Since $\int_0^\infty e^{-n/x}x^{-s-1}dx = \Gamma(s)n^{-s}$ and $\int_0^\infty(1+2n/x)e^{-2n/x}x^{-s-1}dx = (1+s)\Gamma(s)(2n)^{-s}$, the Mellin transforms are

$$\begin{aligned}\int_0^\infty Q_1(x)\,x^{-s-1}dx &= \Gamma(s)D(s),\\ \int_0^\infty Q_2(x)\,x^{-s-1}dx &= (1+s)\Gamma(s)\,2^{-s}D(s), \qquad \operatorname{Re}s>1.\end{aligned} \tag{13.3.12}$$

The kernels $\Gamma(s)$ and $(1+s)\Gamma(s)2^{-s}$ have no zeros in $\operatorname{Re} s > 0$ and decay exponentially on vertical lines.

LEMMA 13.3.8. *If $D(s) = \zeta(s)/\zeta(2s)$ is analytic in the half plane $\operatorname{Re} s > \frac{1}{4}$ apart from $s = 1$, then RH holds.*

Proof. Let ρ be a zero with $\beta = \operatorname{Re}\rho > \frac{1}{2}$. Then $\operatorname{Re}(\rho/2) > \frac{1}{4}$, so the pole of $1/\zeta(2s)$ at $s = \rho/2$ must be cancelled, that is $\zeta(\rho/2) = 0$. Since $\frac{1}{4} < \operatorname{Re}(\rho/2) < \frac{1}{2}$, the functional equation and the reflection $s \mapsto \bar{s}$ make $\rho' = 1 - \bar{\rho}/2$ a zero, with $\operatorname{Re}\rho' = 1 - \beta/2 > \frac{1}{2}$ and $\operatorname{Im}\rho' = \operatorname{Im}\rho/2$. Iterating gives zeros ρ_j whose real parts tend to $\frac{2}{3}$, the fixed point of $\beta \mapsto 1 - \beta/2$, and whose imaginary parts are $\operatorname{Im}\rho/2^j$. Since $\zeta < 0$ on $(0,1)$, $\operatorname{Im}\rho \neq 0$, so the ρ_j are distinct and accumulate at $\frac{2}{3}$, where ζ is analytic and not identically zero, a contradiction. □

THEOREM 13.3.9. (i) *RH holds if and only if $Q_1(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$, and if and only if $Q_2(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$.*

(ii) *Under RH, if the zeros are simple, then*

$$Q_1(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho A_\rho\, x^{\rho/2} + \frac{c_1 \log x + c_2}{x} + O(x^{-2}),$$

$$Q_2(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho B_\rho\, x^{\rho/2} + \frac{2c_1}{x} + O(x^{-2}),$$

$$A_\rho = \frac{\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)}, \qquad B_\rho = \Big(1 + \frac{\rho}{2}\Big)\frac{2^{-\rho/2}\,\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)},$$

the sums over ρ taken over $|\operatorname{Im}\rho| < 2T_j$ along a sequence $T_j \to \infty$, with

$$c_1 = -\frac{\zeta(-1)}{2\zeta'(-2)} = -1.36843277762021,$$

$$c_2 = c_1\Big(1 - \gamma_E + \frac{\zeta'(-1)}{\zeta(-1)} - \frac{\zeta''(-2)}{\zeta'(-2)}\Big) = -0.339381226129599.$$

Proof. (i) Suppose RH. By Section 14.2 of [348], $1/\zeta(\sigma + it) = O(|t|^\varepsilon)$ uniformly for $\sigma \geq \frac{1}{2} + \varepsilon$, and $\zeta(\sigma + it)$ grows at most polynomially in $|t|$ on vertical strips, while $\Gamma(s)$ decays exponentially. Move the line of integration in $Q_1(x) = \frac{1}{2\pi i}\int_{(2)} \Gamma(s)D(s)x^s\,ds$ to $\operatorname{Re} s = \frac{1}{4} + \varepsilon$. The only pole crossed is $s = 1$, with residue $\Gamma(1)/\zeta(2) \cdot x = 6x/\pi^2$; the point $s = \frac{1}{2}$ is regular because $\zeta(2s)$ has a pole there. On the new line $|\Gamma(s)D(s)|$ is integrable and the integral is $O(x^{1/4+\varepsilon})$. The same argument with the kernel $(1+s)\Gamma(s)2^{-s}$ treats Q_2.

Conversely, suppose $Q_1(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$. For $\operatorname{Re} s > 1$, by (13.3.12),

$$\Gamma(s)D(s) - \frac{6}{\pi^2}\,\frac{1}{s-1} = \int_1^\infty \Big(Q_1(x) - \frac{6}{\pi^2}x\Big)x^{-s-1}dx + \int_0^1 Q_1(x)\,x^{-s-1}dx.$$

The first integral converges and is analytic in $\operatorname{Re} s > \frac{1}{4}$, by the hypothesis, and the second is entire, since $Q_1(x) = O(e^{-1/x})$ as $x \to 0$. As Γ has no zeros, D is analytic in $\operatorname{Re} s > \frac{1}{4}$ apart from $s = 1$, and Lemma 13.3.8 gives RH. The same argument applies to Q_2.

(ii) By Theorem 9.7 of [348] there are heights $T_j \in [j, j+1]$ at which $1/\zeta(\sigma \pm 2iT_j) = T_j^{O(1)}$ uniformly for $-1 \le \sigma \le 2$, and on the left of the line $\operatorname{Re} s = -\frac{1}{2}$ the functional equation bounds $1/\zeta(2s)$ polynomially. Move the line of integration to $\operatorname{Re} s = -\frac{5}{2}$ along the rectangles of height T_j; the exponential decay of Γ makes the horizontal sides vanish as $j \to \infty$, and the integral on the new line is $O(x^{-5/2})$. The poles crossed are $s = 1$; the points $\rho/2$, with the residues $A_\rho x^{\rho/2}$ and $B_\rho x^{\rho/2}$; $s = 0$, where $\Gamma(s)D(s)$ and $(1+s)\Gamma(s)2^{-s}D(s)$ have residue $D(0) = \zeta(0)/\zeta(0) = 1$; $s = -1$; and $s = -2$, a simple pole of both, which gives a multiple of x^{-2}. At $s = -1$, Γ has a simple pole and $1/\zeta(2s)$ another, from the trivial zero -2 of ζ, so $\Gamma(s)D(s) = c_1(s+1)^{-2} + c_2(s+1)^{-1} + O(1)$, with c_1 and c_2 from the expansions $\Gamma(s) = -(s+1)^{-1}(1 + (1-\gamma_E)(s+1) + \cdots)$, $\zeta(s) = \zeta(-1)(1 + \frac{\zeta'(-1)}{\zeta(-1)}(s+1) + \cdots)$ and $\zeta(2s) = 2\zeta'(-2)(s+1)(1 + \frac{\zeta''(-2)}{\zeta'(-2)}(s+1) + \cdots)$, and the residue of $\Gamma(s)D(s)x^s$ is $x^{-1}(c_1 \log x + c_2)$. The factor $1 + s$ makes the pole of $(1+s)\Gamma(s)2^{-s}D(s)$ at $s = -1$ simple, with residue $2c_1$ times x^{-1}. □

At the even negative integers $\Gamma(s)D(s)$ has simple poles, since $\zeta(s)$ vanishes there together with $\zeta(2s)$, and at the odd ones double poles: the Laurent coefficients at $s = -1, \dots, -6$, computed by the trapezoid rule on circles of radius 0.3, reproduce the closed forms of c_1 and c_2 within 5×10^{-46}. The coefficients A_ρ are tiny, $|A_{\rho_1}| = 1.648 \times 10^{-5}$ and 6.29×10^{-82} at the hundredth zero, because $|\Gamma(\rho/2)|$ decays like $e^{-\pi\gamma/4}$. A computation checks part (ii) for Q_1 to the working precision. Table 13.10 in Section 13.3.7 does the same for Q_2.

The Perron kernel x^s/s decays only like $1/|t|$, so the atom series for $Q(x)$ has no exponential damping, and bounding its partial sums would need control of $\sum_{|\gamma| \le T} |\zeta(\rho/2)/(\rho\zeta'(\rho))|$, which is out of reach.

13.3.6 Mean square, and how much smoothing the sharp count needs

Two further results bracket the sharp problem. The first, presumably folklore, settles it on average. For $\operatorname{Re} s > 1$ let

$$F(s) = \int_1^\infty \Delta(x)\, x^{-s-1} dx = \frac{D(s)}{s} - \frac{6}{\pi^2}\,\frac{1}{s-1},$$

the second form because $\int_1^\infty Q(x) x^{-s-1} dx = D(s)/s$.

Theorem 13.3.10. *RH holds if and only if $\int_X^{2X} \Delta(x)^2\, dx = O(X^{3/2+\varepsilon})$ for every $\varepsilon > 0$, that is, if and only if the root mean square of Δ on every range $[X, 2X]$ is $O(X^{1/4+\varepsilon})$.*

Proof. Suppose RH. Then F continues analytically to $\operatorname{Re} s > \frac14$: the poles of D at the points $\rho/2$ lie on the line $\operatorname{Re} s = \frac14$, $s = \frac12$ is regular, and the pole at $s = 1$ cancels. By Section 14.2 of [348], under RH $\zeta(\sigma + it) = O(|t|^{\max(0,\frac12-\sigma)+\varepsilon})$ and $1/\zeta(2\sigma + 2it) = O(|t|^{\varepsilon})$ uniformly for $\sigma \ge \frac14 + \varepsilon$ and $|t| \ge 1$, so $F(\sigma + it) = O(|t|^{-3/4+2\varepsilon})$ uniformly there. For $w > 2$ let $I(w) = \int_1^\infty \Delta(x)^2 x^{-w-1}dx$, the Laplace transform in $v = \log x$ of the nonnegative function $\Delta(e^v)^2$. Plancherel's theorem for the function $v \mapsto \Delta(e^v)e^{-wv/2}$ on $(0,\infty)$, whose Fourier transform is $t \mapsto F(\frac{w}{2} + it)$, and the symmetry $F(\bar s) = \overline{F(s)}$ give

$$I(w) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F\Big(\frac{w}{2} + it\Big)\,F\Big(\frac{w}{2} - it\Big)\,dt, \qquad w > 2.$$

The right side is analytic in w on $\operatorname{Re} w > \frac12$, since the integrand is analytic there and $O(|t|^{-3/2+4\varepsilon})$ locally uniformly. So I, the Laplace transform of a nonnegative function, continues analytically to a neighbourhood of every point of $(\frac12,\infty)$, and Landau's theorem [359, Chapter II, Theorem 5b] puts its abscissa of convergence at or below $\frac12$. Hence $I(w) < \infty$ for every $w > \frac12$, and $\int_X^{2X} \Delta^2\,dx \le (2X)^{w+1}I(w) = O(X^{3/2+\varepsilon})$ with $w = \frac12 + \varepsilon$.

Conversely, by the Cauchy Schwarz inequality the hypothesis gives $\int_X^{2X} |\Delta(x)|x^{-\sigma-1}dx = O(X^{1/4+\varepsilon-\sigma})$, so, summing over $X = 2^j$, the integral defining F converges absolutely for $\operatorname{Re} s > \frac14$. Then $D(s) = s\big(F(s) + \frac{6}{\pi^2}\frac{1}{s-1}\big)$ is analytic in $\operatorname{Re} s > \frac14$ apart from $s = 1$, and Lemma 13.3.8 gives RH. □

Table 13.9 gives the root mean square of Δ on dyadic ranges, computed exactly from the sieve: on $[n, n+1)$ the error decreases linearly from $\Delta_n = Q(n) - 6n/\pi^2$ to $\Delta_n - 6/\pi^2$, so $\int_n^{n+1} \Delta^2 = \Delta_n^2 - \frac{6}{\pi^2}\Delta_n + \frac{12}{\pi^4}$, and the extremes over real x are taken at integers or as left limits there. Over the eighteen ranges $[X, 2X]$ with $X = 10^4 \cdot 2^j$, $j = 0,\dots,13$, and $X = 10^4,\dots,10^8$, the root mean square divided by $X^{1/4}$ lies between 0.2337 and 0.3606 and the largest $|\Delta(x)|/X^{1/4}$ between 0.6851 and 1.0361; on $[10^3, 2\times10^8]$ the largest value of $\Delta(x)/x^{1/4}$ is 0.9807, near $x = 47523$, and the least -0.8685, near $x = 80157$.

X	$\big(\frac1X\int_X^{2X}\Delta^2\big)^{1/2}/X^{1/4}$	$\max_{[X,2X]}\lvert\Delta(x)\rvert/X^{1/4}$	near $x =$
10^4	0.2701	0.8979	18471
10^5	0.2494	0.6851	128599
10^6	0.2779	0.8382	1618611
10^7	0.3541	0.9931	17199381
10^8	0.2812	0.8636	144151023

Table 13.9: Root mean square and largest absolute value of the squarefree error Δ on the ranges $[X, 2X]$, divided by $X^{1/4}$, over real x; the largest value on $[10^7, 2\times10^7]$ is the left limit at 17199381, and over the integers alone it is 0.982. Computed.

COROLLARY 13.3.11. (i) *Under RH, for every $\varepsilon > 0$ the set of $x \in [X, 2X]$ with $|\Delta(x)| > X^{1/4+\varepsilon}$ has measure $O(X^{1-\varepsilon})$.*

(ii) *With no hypothesis,*

$$\limsup_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \geq |R_{\rho_1}|, \qquad \liminf_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \leq -|R_{\rho_1}|,$$

where $R_{\rho_1} = \zeta(\rho_1/2)/(\rho_1\zeta'(\rho_1))$ and $|R_{\rho_1}| = 0.100404$. The same holds for $Q_2(x) - \frac{6}{\pi^2}x - 1$ with $|B_{\rho_1}| = 9.944 \times 10^{-5}$ in place of $|R_{\rho_1}|$.

Proof. (i) is Chebyshev's inequality applied to Theorem 13.3.10. For (ii) suppose that $\Delta(x) \leq c\,x^{1/4}$ for $x \geq x_0$, with $0 < c < |R_{\rho_1}|$. The function $L(s) = \int_{x_0}^{\infty}(cx^{1/4} - \Delta(x))x^{-s-1}dx$ is the Laplace transform in $\log x$ of a nonnegative function, and for $\operatorname{Re} s > 1$

$$L(s) = \frac{c\,x_0^{1/4-s}}{s - \frac{1}{4}} - F(s) + E(s),$$

with E entire. The right side has no singularity on the real half line $(\frac{1}{4}, \infty)$: a real pole of D there would be a real zero of ζ in $(\frac{1}{2}, \infty)$, of which there are none, $s = \frac{1}{2}$ is regular, and the pole at $s = 1$ cancels. By Landau's theorem the integral $L(s)$ converges for $\operatorname{Re} s > \frac{1}{4}$, so F is analytic there (which already gives RH, by Lemma 13.3.8), and $|L(\sigma + it)| \leq L(\sigma)$ for $\sigma > \frac{1}{4}$. As $\sigma \downarrow \frac{1}{4}$, $L(\sigma) = c/(\sigma - \frac{1}{4}) + O(1)$, since F is analytic at $s = \frac{1}{4}$, where $\zeta(2s) = \zeta(\frac{1}{2}) \neq 0$. At $s = \sigma + \frac{i}{2}\gamma_1$ the first and last terms of the display are $O(1)$, while $F(s) = R_{\rho_1}/(\sigma - \frac{1}{4}) + O(1)$, because $D(s)/s$ has a simple pole at $\rho_1/2$ with residue R_{ρ_1} (the zero ρ_1 is simple and $\zeta(\rho_1/2) \neq 0$). Hence $|R_{\rho_1}| \leq c$, a contradiction. The lower bound is the same argument applied to $c\,x^{1/4} + \Delta(x)$, and the smoothed case is the same argument with the kernel $(1+s)\Gamma(s)2^{-s}$, whose residue at $\rho_1/2$ is B_{ρ_1}. □

The second measures the cost of sharpening. Let $G_{n,x}$ have the Gamma(n) law with mean x, so that $G_{n,x}$ concentrates at x in a window of width $x/\sqrt{n}$; $Q_1(x) = \mathbb{E}Q(G_{1,x})$ and $Q_2(x) = \mathbb{E}Q(G_{2,x})$.

PROPOSITION 13.3.12. *Under RH, uniformly in $n \geq 1$,*

$$\mathbb{E}Q(G_{n,x}) = \frac{6}{\pi^2}\,x + O_\varepsilon\big(x^{1/4+\varepsilon}\,n^{1/8+\varepsilon}\big).$$

Proof. With rate $\lambda = n/x$, $P(G_{n,x} \geq m) = \Gamma(n, \lambda m)/\Gamma(n)$, whose Mellin transform in λm is $\Gamma(s+n)/(s\Gamma(n))$, so $\mathbb{E}Q(G_{n,x}) = \frac{1}{2\pi i}\int_{(2)} D(s)\frac{\Gamma(s+n)}{s\Gamma(n)}\big(\frac{x}{n}\big)^s ds$. Move the line to $\operatorname{Re} s = c = \frac{1}{4} + \varepsilon$, collecting $6x/\pi^2$ at $s = 1$. On the new line $|\Gamma(n + c + it)|/\Gamma(n) = O\big(n^c \exp(-\kappa \min(t^2/n, |t|))\big)$ for an absolute $\kappa > 0$, since $\Gamma(n+c)/\Gamma(n) = O(n^c)$ and $|\Gamma(n+c+it)/\Gamma(n+c)|^2 = \prod_{k\geq 0}\big(1 + t^2/(n+c+k)^2\big)^{-1}$;

the factors with $k < n$ give $e^{-\kappa t^2/n}$ when $|t| \le n$, and those with $k \le |t|$ give $e^{-\kappa|t|}$ when $|t| > n$. Under RH, $|D(c+it)| = O(|t|^{1/4+\varepsilon})$ for $|t| \ge 1$ (Section 14.2 of [348] and the functional equation). The remaining integral is therefore $O\big(x^c \int (1+|t|)^{-3/4+\varepsilon} e^{-\kappa \min(t^2/n,|t|)}dt\big) = O(x^c n^{1/8+\varepsilon})$, the integral being dominated by $|t| \lesssim \sqrt{n}$. $\square$

Balancing the window against the loss gives a bound for the sharp count.

Corollary 13.3.13. *Under RH, $\Delta(x) = O(x^{2/5+\varepsilon})$ for every $\varepsilon > 0$.*

Proof. Since Q is nondecreasing and counts integers, $|Q(y) - Q(x)| \le |y-x| + 1$, so $|\mathbb{E}Q(G_{n,x}) - Q(x)| \le \mathbb{E}|G_{n,x} - x| + 1 \le x/\sqrt{n} + 1$. With Proposition 13.3.12, $\Delta(x) = O(x^{1/4+\varepsilon} n^{1/8+\varepsilon} + x/\sqrt{n})$, and $n = \lceil x^{6/5} \rceil$ balances the two terms at $x^{2/5+O(\varepsilon)}$. $\square$

The exponent $\frac{2}{5}$ is weaker than the known conditional exponents. If the loss $n^{1/8}$ were replaced by n^{δ}, the same balance would give $\Delta(x) = O(x^{1-3/(8\delta+4)+\varepsilon})$, which tends to $x^{1/4+\varepsilon}$ as $\delta \to 0$; conversely $\Delta(x) = O(x^{1/4+\varepsilon})$ gives $|\mathbb{E}Q(G_{n,x}) - \frac{6}{\pi^2}x| \le \mathbb{E}|\Delta(G_{n,x})| = O(x^{1/4+\varepsilon})$ uniformly in n, by Jensen's inequality. So the sharp conjecture amounts to removing the loss, which needs cancellation in $\int_{|t|\le T} D(\frac14 + it)x^{it}\,dt/t$ and not only a bound on the size of the integrand.

13.3.7 Gamma(2) smoothing and the Mertens function

The elementary method rests on the terms $\mu(d)(\lfloor x/d^2 \rfloor - x/d^2) = -\mu(d)\{x/d^2\}$ of (13.3.8). Following Steutel and Thiemann, who split a random variable into its integer and fractional parts [339], we randomise x by a gamma law; by Theorem 13.3.18, shape two is the largest for which the average is taken under a positive law.

Let Y have the Gamma(ν) law with rate λ, let $a > 0$ and $y = \lambda a$. Then $\mathbb{E}\lfloor Y/a \rfloor = \sum_{m\ge1} P(Y \ge ma) = \sum_{m\ge1} \Gamma(\nu, my)/\Gamma(\nu)$ and $\mathbb{E}Y/a = \nu/y$, so the expected fractional part is

$$\mathbb{E}\{Y/a\} = g_\nu(y) = \frac{\nu}{y} - \sum_{m\ge1} \frac{\Gamma(\nu, my)}{\Gamma(\nu)}.$$

For $\nu = 1$, Steutel and Thiemann observed that $\lfloor Y/a \rfloor$ is geometric and $\{Y/a\}$ an independent truncated exponential variable, with $\mathbb{E}\lfloor Y/a \rfloor = 1/(e^y - 1)$ and $g_1(y) = 1/y - 1/(e^y - 1)$.

Lemma 13.3.14. *For $\nu = 2$,*

$$\mathbb{E}\lfloor Y/a \rfloor = \frac{1}{e^y - 1} + \frac{y}{4\sinh^2(y/2)}, \qquad g_2(y) = \frac{2}{y} - \frac{1}{e^y - 1} - \frac{y}{4\sinh^2(y/2)}.$$

For $0 < |y| < 2\pi$, with the Bernoulli numbers B_{2k},

$$g_2(y) = \frac{1}{2} + \sum_{k\ge 2} \frac{(2k-2)B_{2k}}{(2k)!}\, y^{2k-1} = \frac{1}{2} - \frac{y^3}{360} + \frac{y^5}{7560} - \cdots,$$

$g_2(y) = 2/y + O(ye^{-y})$ as $y \to \infty$, and for every $\nu > 0$

$$\int_0^\infty g_\nu(y)\, y^{s-1}\, dy = -\frac{\Gamma(s+\nu)\,\zeta(s)}{s\,\Gamma(\nu)}, \qquad 0 < \operatorname{Re} s < 1,$$

which is $-(1+s)\Gamma(s)\zeta(s)$ for $\nu = 2$.

Proof. For $\nu = 2$, $P(Y \ge ma) = (1+my)e^{-my}$, and $\sum_m e^{-my} = 1/(e^y - 1)$, $\sum_m mye^{-my} = y/(4\sinh^2(y/2))$. For the expansion use $1/(e^y-1) = 1/y - \frac{1}{2} + \sum_{k\ge1} B_{2k}y^{2k-1}/(2k)!$ and $y/(4\sinh^2(y/2)) = -y\frac{d}{dy}(e^y-1)^{-1}$. For the Mellin transform write $g_\nu(y) = \mathbb{E}\{Y/y\}$ with Y of law Gamma(ν,1), substitute $u = Y/y$, and use the classical formula $\int_0^\infty \{u\}u^{-s-1}du = -\zeta(s)/s$ for $0 < \operatorname{Re} s < 1$ [348, Section 2.1]: the transform is $\mathbb{E}Y^s \cdot (-\zeta(s)/s)$, with $\mathbb{E}Y^s = \Gamma(s+\nu)/\Gamma(\nu)$. □

The terms of the elementary method now read $\mathbb{E}\big[\mu(d)\big(\lfloor G_{2,x}/d^2\rfloor - G_{2,x}/d^2\big)\big] = -\mu(d)g_2(2d^2/x)$ for $G_{2,x}$ of the Gamma(2) law with mean x, and since $0 \le g_2(y) \le \min(1, 2/y)$ the sum over all d converges absolutely, with no truncation at $d^2 \le x$:

$$Q_2(x) = \sum_{n\ge1} \mu(n)^2\Big(1 + \frac{2n}{x}\Big)e^{-2n/x} = \frac{6}{\pi^2}\,x - \sum_{d\ge1} \mu(d)\, g_2\Big(\frac{2d^2}{x}\Big). \tag{13.3.13}$$

THEOREM 13.3.15. *Let*

$$f_2(y) = -2g_2'(y) = \frac{4}{y^2} - \frac{y\cosh(y/2)}{2\sinh^3(y/2)}, \qquad y > 0.$$

Then $f_2 > 0$, $f_2(y) = y^2/60 + O(y^4)$ as $y \to 0$ and $f_2(y) \sim 4/y^2$ as $y \to \infty$, and f_2 is the density of a probability law $\mathcal{V}_2$ with $P(\mathcal{V}_2 \ge y) = 2g_2(y)$ and

$$\mathbb{E}\mathcal{V}_2^s = -2\,\Gamma(s+2)\,\zeta(s), \qquad -3 < \operatorname{Re} s < 1.$$

For every $x > 0$,

$$Q_2(x) = \frac{6}{\pi^2}\,x - \frac{1}{2}\,\mathbb{E}M\Big(\sqrt{x\mathcal{V}_2/2}\Big).$$

Moreover f_2 is twice the difference between the continuous and the integer mixtures of the Gamma(2) densities $\gamma_{2,\lambda}(y) = \lambda^2 ye^{-\lambda y}$ over their rate:

$$f_2(y) = 2\Big(\int_0^\infty \gamma_{2,\lambda}(y)\,d\lambda - \sum_{m\ge1}\gamma_{2,m}(y)\Big) = -2\int_0^\infty \Big(\{\lambda\} - \frac{1}{2}\Big)\,\partial_\lambda\gamma_{2,\lambda}(y)\,d\lambda.$$

Proof. With $z = y/2$, $f_2(y) > 0$ is $(\sinh z/z)^3 > \cosh z$, Lazarević's inequality [205]. The expansion at 0 follows from Lemma 13.3.14. The mass is $2(g_2(0+) - g_2(\infty)) = 1$ and the tail is $2g_2(y)$. Integrating by parts in the Mellin formula of Lemma 13.3.14 gives $\mathbb{E}\mathcal{V}_2^s = 2s\int_0^\infty y^{s-1}g_2(y)\,dy = -2s(1+s)\Gamma(s)\zeta(s)$ for $0 < \operatorname{Re} s < 1$, and the behaviour at 0 and at infinity extends the integral and the identity to $-3 < \operatorname{Re} s < 1$. For the representation, $P(\sqrt{x\mathcal{V}_2/2} \ge d) = P(\mathcal{V}_2 \ge 2d^2/x) = 2g_2(2d^2/x)$, so $\frac12\mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = \sum_d \mu(d)g_2(2d^2/x)$, the interchange being justified by absolute convergence, and (13.3.13) gives the formula. Finally $\int_0^\infty \gamma_{2,\lambda}(y)\,d\lambda = 2/y^2$ and $\sum_m \gamma_{2,m}(y) = y\sum_m m^2e^{-my} = y\cosh(y/2)/(4\sinh^3(y/2))$, and the last form is the Euler Maclaurin formula $\int_0^\infty h - \sum_{m\ge1} h(m) = -\int_0^\infty(\{\lambda\} - \frac12)h'(\lambda)\,d\lambda$ for $h = \gamma_{2,\cdot}(y)$, which vanishes at $\lambda = 0$. □

The exponential case is the same with

$$f_1(y) = -2g_1'(y) = \frac{2}{y^2} - \frac{1}{2\sinh^2(y/2)} = 2\Big(\int_0^\infty \lambda e^{-\lambda y}d\lambda - \sum_{m\ge1} me^{-my}\Big) > 0,$$

positive because $\sinh z > z$, with $f_1(0+) = \frac16$, $\mathbb{E}\mathcal{V}_1^s = -2\Gamma(s+1)\zeta(s)$ for $-1 < \operatorname{Re} s < 1$, and

$$Q_1(x) = \frac{6}{\pi^2}\,x - \frac12\,\mathbb{E}M\Big(\sqrt{x\mathcal{V}_1}\Big).$$

Corollary 13.3.16. (i) *With no hypothesis,* $Q_\nu(x) = \frac{6}{\pi^2}x + o(x^{1/2})$ *for* $\nu = 1, 2$. (ii) *Under RH,* $|Q_2(x) - \frac{6}{\pi^2}x| \le \frac12\mathbb{E}|M(\sqrt{x\mathcal{V}_2/2})| = O(x^{1/4+\varepsilon})$, *and likewise for* Q_1*; this is a second proof of the forward half of Theorem 13.3.9(i).*

Proof. (i) $|M(t)| \le t$ and $M(t) = o(t)$, and $\mathbb{E}\mathcal{V}_\nu^{1/2} < \infty$, so dominated convergence gives $\mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = o(x^{1/2})$, and likewise for $\mathcal{V}_1$. (ii) RH gives $M(t) = O(t^{1/2+\varepsilon})$ [348, Section 14.25], and $\mathbb{E}\mathcal{V}_\nu^{1/4+\varepsilon} < \infty$. □

Table 13.10 checks (13.3.13) and Theorem 13.3.9(ii) for Q_2, computing $Q_2(x) = \sum_d \mu(d)\,\mathbb{E}\lfloor G_{2,x}/d^2\rfloor = \sum_d \mu(d)\big(1/(e^t-1) + t/(4\sinh^2(t/2))\big)$, $t = 2d^2/x$, whose terms with $t > 140$ are below 10^{-55}, and this agrees with the direct sum over n, in double precision, within 10^{-10} up to $x = 10^6$. The sum (13.3.13) truncated at $d \le 2\times10^5$, as in [289], loses 2.2×10^{-4} at $x = 10^6$, because $g_2(2d^2/x)$ is close to x/d^2 for large d and the tail $x\sum_{d>2\times10^5}\mu(d)/d^2$ is left out.

13.3.8 Gamma shapes: two is the threshold

For every real shape $\nu > 0$, $f_\nu = -2g_\nu'$ is a signed density of total mass $2g_\nu(0+) = 1$, and the argument of Theorem 13.3.15 gives, for Y of the Gamma(ν) law with

x	$Q_2(x)-\frac{6}{\pi^2}x$	$d\le 2\cdot 10^5$	$\frac{Q_2(x)-\frac{6}{\pi^2}x-1}{x^{1/4}}$	100 atoms	trivial terms	residual
10^3	0.997161220898	0.997161002208	-5.048×10^{-4}	-1.0571×10^{-4}	-2.73307×10^{-3}	-1.0×10^{-23}
10^4	1.00094524529	1.00094305839	9.452×10^{-5}	1.21889×10^{-3}	-2.73648×10^{-4}	-1.3×10^{-30}
10^5	0.996620193442	0.99659832449	-1.9006×10^{-4}	-3.35244×10^{-3}	-2.73683×10^{-5}	-1.5×10^{-37}
10^6	1.00610290943	1.00588421991	1.9299×10^{-4}	6.10565×10^{-3}	-2.73686×10^{-6}	-1.8×10^{-41}
10^7	0.992114718377		-1.4022×10^{-4}	-7.88501×10^{-3}	-2.73687×10^{-7}	-3.9×10^{-42}
10^8	1.00440085379		4.4009×10^{-5}	4.40088×10^{-3}	-2.73687×10^{-8}	-2.2×10^{-38}

Table 13.10: The Gamma(2) smoothed count: $Q_2(x)-6x/\pi^2$, the sum (13.3.13) truncated at $d\le 2\times10^5$, the normalised error, the contribution $2\,\mathrm{Re}\sum_{k\le100}B_{\rho_k}x^{\rho_k/2}$ of the first hundred atoms, the residues at $s=-1,\dots,-6$, whose leading part is $2c_1/x$, and the residual. The constant 1 is the residue $D(0)$ at $s=0$. Computed at 45 digits.

rate λ,

$$\mathbb{E}Q(Y)=\frac{6}{\pi^2}\,\mathbb{E}Y-\frac12\int_0^\infty M\Bigl(\sqrt{y/\lambda}\Bigr)f_\nu(y)\,dy,$$
$$\int_0^\infty y^s f_\nu(y)\,dy=-\frac{2\,\Gamma(s+\nu)\,\zeta(s)}{\Gamma(\nu)},\qquad -\nu<\mathrm{Re}\,s<1.$$

Write $\gamma_a(t)=t^{a-1}e^{-t}/\Gamma(a)$ for the Gamma(a) density, and let

$$\mathcal{R}_\nu(y)=y\sum_{m\ge1}\gamma_{\nu+1}(my)=\frac{y^{\nu+1}}{\Gamma(\nu+1)}\sum_{m\ge1}m^\nu e^{-my},$$
$$q_\nu(y)=\frac{\Gamma(\nu+1)\bigl(1-\mathcal{R}_\nu(y)\bigr)}{y^{\nu+1}},$$

so that $\mathcal{R}_\nu(y)$ is the right Riemann sum with step y of the Gamma($\nu+1$) density, whose integral is 1.

Lemma 13.3.17. *For $\nu>0$ and $y>0$,*

$$f_\nu(y)=\frac{2\nu\bigl(1-\mathcal{R}_\nu(y)\bigr)}{y^2}=\frac{2y^{\nu-1}}{\Gamma(\nu)}\,q_\nu(y),$$

so $f_\nu\ge0$ exactly when $\mathcal{R}_\nu\le1$. For $0<y<2\pi$,

$$q_\nu(y)=-\sum_{k\ge0}\zeta(-\nu-k)\,\frac{(-y)^k}{k!},$$

and $\mathcal{R}_1(y)=(y/2)^2/\sinh^2(y/2)$, $\mathcal{R}_2(y)=(y/2)^3\cosh(y/2)/\sinh^3(y/2)$.

Proof. With Y of the Gamma(ν,1) law and $c=1/y$, $g_\nu(y)=\mathbb{E}\{cY\}=c\nu-\sum_{m\ge1}P(Y\ge m/c)$, whose derivative in c is $\nu-\sum_m(m/c^2)p_\nu(m/c)$, with $p_\nu=\gamma_\nu$

the density of Y. Since $tp_\nu(t) = \nu\gamma_{\nu+1}(t)$, the derivative is $\nu(1 - \mathcal{R}_\nu(1/c))$, and returning to y gives $g'_\nu(y) = -\nu(1 - \mathcal{R}_\nu(y))/y^2$. The series is the expansion $\sum_m m^\nu e^{-my} = \Gamma(\nu+1)y^{-\nu-1} + \sum_{k\ge0}\zeta(-\nu-k)(-y)^k/k!$ of the polylogarithm, valid for $0 < y < 2\pi$. □

So $\mathcal{R}_1 \le 1$ is $\sinh z \ge z$ and $\mathcal{R}_2 \le 1$ is Lazarević's inequality. Near the origin f_ν has the sign of $q_\nu(0+) = -\zeta(-\nu)$, and at an even integer $\nu = 2m$, where $\zeta(-2m) = 0$, the sign of $q_{2m}(y) = \zeta(-2m-1)y + O(y^2)$. Since ζ is positive on $(-4,-2)$, $(-8,-6)$, … and $\zeta(-2m-1) = -B_{2m+2}/(2m+2)$, f_ν is negative near the origin exactly when $\nu \in (4j+2, 4j+4]$ for some $j \ge 0$.

THEOREM 13.3.18. $\{\nu > 0 :\ f_\nu \ge 0 \text{ on } (0,\infty)\} = (0,2]$. *Thus two is the largest shape for which the gamma smoothed squarefree error is a positive average of the Mertens function, and the obstruction is the trivial zero of ζ at -2.*

Proof. Let $\nu > 2$ and suppose $f_\nu \ge 0$. Since $f_\nu(y) \sim 2\nu/y^2$ at infinity, $\int_1^\infty y^s f_\nu(y)\,dy$ converges and is analytic for $\operatorname{Re} s < 1$. The function $m(s) = -2\Gamma(s+\nu)\zeta(s)/\Gamma(\nu)$ is analytic in a neighbourhood of $(-\nu, 1)$, so $\int_0^1 y^s f_\nu(y)\,dy = m(s) - \int_1^\infty y^s f_\nu(y)\,dy$, a Laplace transform in $-\log y$ of a nonnegative function, continues analytically along the real interval $(-\nu,1)$. By Landau's theorem [359, Chapter II, Theorem 5b] it converges on all of $(-\nu,1)$, and there $m(s) = \int_0^\infty y^s f_\nu(y)\,dy > 0$. But $\Gamma(s+\nu) > 0$ on $(-\nu,1)$, while $\zeta(s) > 0$ on $(-4,-2)$, which meets $(-\nu,1)$ because $\nu > 2$, so $m(s) < 0$ there, a contradiction.

Let $0 < \nu \le 2$. The cases $\nu = 1$ and $\nu = 2$ are the inequalities above. Otherwise write $\nu = k + \mu$ with $k \in \{0,1\}$ and $0 < \mu < 1$. Bernstein's formula $t^\mu = \frac{\mu}{\Gamma(1-\mu)}\int_0^\infty (1 - e^{-st})s^{-\mu-1}ds$, applied to $t = mh$ in $\sum_m (mh)^{k+\mu}e^{-mh}$ and to t in $\int_0^\infty t^{k+\mu}e^{-t}dt$, together with $h\sum_m (mh)^k e^{-mhc} = k!\,\mathcal{R}_k(hc)/c^{k+1}$ and $\int_0^\infty t^k e^{-ct}dt = k!/c^{k+1}$ for $c = 1+s$, gives

$$\Gamma(\nu+1)\big(\mathcal{R}_\nu(h) - 1\big) = \frac{\mu\,h^{k+1}}{\Gamma(1-\mu)}\int_0^\infty \big(q_k(h(1+s)) - q_k(h)\big)\,s^{-\mu-1}\,ds, \qquad h > 0.$$

So $\mathcal{R}_\nu \le 1$ follows if q_k is nonincreasing. For $k = 0$, $q_0(y) = 1/y - 1/(e^y - 1) = g_1(y)$, which decreases because $f_1 > 0$. For $k = 1$, $q_1(y) = 1/y^2 - 1/(4\sinh^2(y/2)) = f_1(y)/2$, and with $z = y/2$

$$q_1'(y) = \frac{1}{4}\Big(\frac{\cosh z}{\sinh^3 z} - \frac{1}{z^3}\Big) < 0$$

by Lazarević's inequality. □

We evaluate $1 - \mathcal{R}_\nu$ for twenty shapes at 40 digits, by the series of Lemma 13.3.17 for $y \le 2.5$ and by the direct sum beyond, on 400 geometrically spaced points in $[10^{-4}, 10^{0.4}]$ and 799 equally spaced points in $(2.5, 60]$. For

the ten shapes $\nu = 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 1.9, 1.99, 2$ no grid point gives a negative value; on the range $[0.02, 15]$ the least values, from 2.68×10^{-3} at $\nu = 0.25$ to 6.90×10^{-10} at $\nu = 2$, are taken at its left end. Table 13.11 shows the shapes above 2: the negative set starts at the origin for the shapes in $(2, 4]$ and in $(6, 8]$, as the sign rule predicts.

ν	$-\zeta(-\nu)$	least value of $1 - \mathcal{R}_\nu$	at $y =$	grid points with $1 - \mathcal{R}_\nu < 0$
2.01	-3.012×10^{-4}	-7.310×10^{-10}	0.0273	from 10^{-4} to 0.0362
2.1	-2.7295×10^{-3}	-4.941×10^{-6}	0.265	from 10^{-4} to 0.347
2.5	-8.517×10^{-3}	-1.0763×10^{-3}	1.187	from 10^{-4} to 1.512
3	-8.333×10^{-3}	-7.2304×10^{-3}	2.144	from 10^{-4} to 2.716
3.5	-4.441×10^{-3}	-0.019006	2.9745	from 10^{-4} to 3.723
4	0	-0.035249	3.726	from 10^{-4} to 4.659
5	3.968×10^{-3}	-0.076713	5.084	from 1.716 to 6.386
6	0	-0.12519	6.323	from 3.076 to 7.969
7	-4.167×10^{-3}	-0.17715	7.4915	from 10^{-4} to 1.234, from 4.227 to 9.409
8	0	-0.23064	8.614	from 10^{-4} to 2.269, from 5.307 to 10.85

Table 13.11: Gamma(ν) smoothing above the threshold of Theorem 13.3.18: the limit $-\zeta(-\nu)$ of q_ν at 0, the least value of $1-\mathcal{R}_\nu$, which has the sign of f_ν, with the point where it is attained (grid search refined by golden section), and the maximal runs of grid points on which $1 - \mathcal{R}_\nu < 0$. Computed.

13.3.9 The Euler product of the squarefree law

For $\sigma > 1$ let N_σ have the law $P(N_\sigma = n) = \mu(n)^2 n^{-\sigma} \zeta(2\sigma)/\zeta(\sigma)$ on the squarefree integers, the squarefree analogue of the zeta distribution of Section 10.1.6. By the Euler product in (13.3.1), $\log N_\sigma = \sum_p B_p \log p$ with independent Bernoulli variables B_p of success probability $p^{-\sigma}/(1 + p^{-\sigma})$, and

$$\mathbb{E} e^{-it \log N_\sigma} = \prod_p \frac{1 + p^{-\sigma - it}}{1 + p^{-\sigma}}, \qquad \log\left(1 + p^{-s}\right) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k}\, p^{-ks}.$$

So the would be Lévy measure of $\log N_\sigma$ is the signed measure

$$\sum_p \sum_{k \geq 1} \frac{(-1)^{k+1}}{k}\, p^{-k\sigma}\, \delta_{k \log p}$$

of finite total variation, a sum over the primes of alternating lattices like the fair coin lattices of Section 10.1.6, here with the geometric weights $p^{-k\sigma}$ of a biased coin: the law is quasi infinitely divisible, and by the uniqueness of the signed Lévy measure it is not infinitely divisible [213]. Its negative masses, at even k, come from the factor $1/\zeta(2s)$, whose weights $-2/k$ at even k outweigh the weights $1/k$ of $\zeta(s)$.

13.3.10 Questions

Question 13.3.19. Prove $\Delta(x) = O(x^{1/4+\varepsilon})$ under RH. Equivalent forms above: remove the loss $n^{1/8}$ in Proposition 13.3.12; or show that the exceptional set of Corollary 13.3.11(i), with $X^{1/4+\varepsilon}$ replaced by a constant multiple, is empty for large X. By that corollary, the latter would follow from a bound $O(X^{1/4+\varepsilon})$ for the oscillation of Δ on the subintervals of $[X, 2X]$ of length $X^{1-\varepsilon/2}$.

Question 13.3.20. Sharpen the constant $|R_{\rho_1}| = 0.1004$ of Corollary 13.3.11(ii) with a Fejér kernel over many zeros, as Ingham did for $M(x)$ [164], and compare it with the observed $\max|\Delta(x)|/x^{1/4}$, about 1 up to 2×10^8 (Table 13.9). Is $\limsup|\Delta(x)|/x^{1/4} = \infty$, as the linear independence of the ordinates would suggest?

Chapter 14

Below the basepoint one: a map of processes, free probability, almost GGCs and the Lambert law

The four sections of this chapter read the laws of Chapters 12 and 13, the shifted clocks T_α, the normal scale mixtures Y_α and the tilted Pólya variables V_α, in other settings: the general theory of infinitely divisible laws and the processes behind them (Section 14.1), free probability and the Upsilon transform (Section 14.2), zero density estimates (Section 14.3), and the work of Pakes around the Lambert law (Section 14.4). The question throughout is what can be said below the basepoint one without a zero free half plane, which none of the four supplies.

14.1 Infinite divisibility and a map of processes

The general theory is that of Sato's book on Lévy processes and infinitely divisible laws [325] and Bondesson's book on generalised gamma convolutions [39]. None of the results of this section supplies positivity of the Thorin measure uniformly in the height (Section 14.1.5). The laws are also marginals of processes: self decomposability makes the centre clock the stationary law of an Ornstein Uhlenbeck process, and the Thorin measure at the basepoint one makes T_1 the integral of a gamma process against the inverse of the Poisson smoothed zero count (Section 14.1.6). Section 14.1.7 adds Brownian motion run on the clocks and sorts all these processes by what each needs: nothing, a zero free half plane, or RH.

We use the notation of Chapter 12. The centre clock T has Laplace transform $\varphi_{1/2}(u) = 1/G(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$; for $\alpha > \frac{1}{2}$ the shifted clock T_α has Laplace transform $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ and the representation $T_\alpha \stackrel{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$ of Theorem 12.2.5, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and H_a is the first passage time of an independent standard Brownian motion to the level a. For $\alpha \geq 1$, T_α is a GGC with Thorin density v_α/π in the chart $\theta = \sqrt{z}$, $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$ (Theorem 2.1.2 and (2.1.7)); at the basepoint one this is $\rho_{1/2} =$

v_1/π of (12.1.2). We write $b_\alpha = \frac{\xi'}{\xi}(\alpha)$, so that $b_1 = \sum_\rho \rho^{-1} = 0.0230957089661$, and V_α for the tilted Pólya variable of Section 12.1.1, the Esscher tilt of the Pólya density p of (2.1.4) with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$; thus $V_{1/2}$ is the Pólya variable X and V_1 is the variable V of Section 12.1.4.

14.1.1 Self decomposability: the driving process and unimodality

The Laplace exponent of the centre clock is $\log G(u) = \int_0^\infty (1-e^{-ut})\,W(t)\,dt/t$, where $W(t) = \sum_k e^{-a_k t}$, $a_k = -(\rho_k - \frac12)^2$, is the heat trace of the zeros, equal to $\sum_{\gamma>0} e^{-\gamma^2 t}$ under RH and to the arithmetic expression (2.1.5) in any case (Section 2.1). Since $W > 0$ and $W' < 0$ on $(0,\infty)$ with no hypothesis (Theorem 8.1.1, whose proof uses the verification of RH below height 1000), T is infinitely divisible with Lévy measure $W(t)\,dt/t$ and self decomposable. A law on $[0,\infty)$ is self decomposable if and only if it is the law of $\int_0^\infty e^{-t}\,dZ_t$ for a subordinator Z with $\mathbb{E}\log(1+Z_1) < \infty$, its background driving Lévy process [365, 172], [325, Theorem 17.5]. The Lévy measure of a self decomposable law on the half line is $k(x)\,dx/x$ with k nonincreasing [325, Corollary 15.11], and the Lévy measure of Z gives mass $k(x)$ to (x,∞), so it has density $-k'$ when k is differentiable. For the centre clock $k = W$.

Proposition 14.1.1. *With no hypothesis, $T \overset{d}{=} \int_0^\infty e^{-t}\,dZ_t$ for a subordinator Z with Lévy density $h(x) = -W'(x)$, which is $\sum_{\gamma>0}\gamma^2 e^{-\gamma^2 x}$ under RH and is given for every $x > 0$ by the derivative of the explicit formula* (2.1.5) *for W. The condition $\mathbb{E}\log(1+Z_1) < \infty$ holds because h decays exponentially. Moreover RH holds if and only if h is completely monotone.*

Proof. The representation is the theorem of Wolfe and of Jurek and Vervaat; we check the Lévy density directly. If $\Psi_Z(u) = -\log \mathbb{E}e^{-uZ_1}$, the Laplace exponent of $\int_0^\infty e^{-t}\,dZ_t$ is $\int_0^\infty \Psi_Z(ue^{-t})\,dt = \int_0^u \Psi_Z(w)\,dw/w$, so $\Psi_Z(u) = u(\log G)'(u)$. By Section 2.1, $(\log G)'(u) = \int_0^\infty e^{-ut}W(t)\,dt$, and since $W(t) = \int_t^\infty h(x)\,dx$ with $h > 0$, Fubini's theorem gives

$$u(\log G)'(u) = \int_0^\infty h(x)\int_0^x ue^{-ut}\,dt\,dx = \int_0^\infty \bigl(1-e^{-ux}\bigr)h(x)\,dx.$$

Here $\int_1^\infty h = W(1)$ and $\int_0^1 xh(x)\,dx = \int_0^1 (W(y)-W(1))\,dy < \infty$, because $W(y)\,dy/y$ is a Lévy measure, so h is the Lévy density of a subordinator without drift. It is not integrable at the origin, since $\int_t^1 h = W(t) - W(1)$ and $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 8.1), so Z makes infinitely many small jumps. As $x \to \infty$, $h(x) \sim \gamma_1^2 e^{-\gamma_1^2 x}$, which gives $\mathbb{E}\log(1+Z_1) < \infty$ [325, Theorem 25.3]. Finally h is completely monotone exactly when $W = \int_x^\infty h$ is, and complete monotonicity of the heat trace is equivalent to RH (Section 2.1 and Table 4.1). □

Table 14.1 checks the explicit formula and its derivative against the zero sums.

t	$W(t)$, zeros	$W(t)$, explicit formula
0.005	0.537191964091	0.537191964091
0.01	0.149698011254	0.149698011254
0.02	0.0185412917313	0.0185412917313
0.05	$4.587835048574 \times 10^{-5}$	$4.587835048574 \times 10^{-5}$
0.1	$2.104799795822 \times 10^{-9}$	$2.104799798080 \times 10^{-9}$
t	$-W'(t)$, zeros	$-W'(t)$, explicit formula
0.005	165.1022811964	165.1022811964
0.01	33.72924560562	33.72924560562
0.02	3.741068142123	3.741068142123
0.05	$9.166117863113 \times 10^{-3}$	$9.166117863113 \times 10^{-3}$
0.1	$4.205189085540 \times 10^{-7}$	$4.205189085735 \times 10^{-7}$

Table 14.1: The heat trace W and the Lévy density $h = -W'$ of the driving process of the centre clock, from the first 1700 zeros and from the explicit formula (2.1.5) and its derivative with the primes below 2×10^4. Computed.

At $t = 0.1$ the terms of the explicit formula cancel to about nine digits (Table 8.4), and the computation takes the weights $\log p$ in double precision; the cancellation magnifies their rounding into the differences of 2×10^{-18} in W and 2×10^{-17} in h in the last rows. With the weights in multiple precision both explicit columns agree with the zero sums at $t = 0.1$ to every digit shown, as in Table 2.1.

Yamazato's theorem states that every self decomposable law on the line is unimodal [366], [325, Theorem 53.1]. The centre clock and every T_α, $\alpha > \frac{1}{2}$, are self decomposable with no hypothesis (Theorem 8.1.1 and Corollary 12.2.6), so:

Corollary 14.1.2. *The centre clock is unimodal with no hypothesis, and so is T_α for every $\alpha > \frac{1}{2}$.*

The computed densities (Figure 14.1) have modes near 0.020 for the centre clock and 0.024 for T_1, the latter located only to the spacing of the geometric Talbot grid, whose points next to 0.024 are 0.0175 and 0.0328. Yamazato's theorem does not reach the reciprocal zeta law of Section 11.1, which is infinitely divisible but not self decomposable (Theorem 11.1.3); the unimodality of its density remains open (Section 17.1.4). The density of the centre clock is C^∞ by Orey's condition [325, Proposition 28.3], since $\int_0^r tW(t)\,dt \geq cr^{3/2}$ by the expansion of W at 0 (Section 8.1).

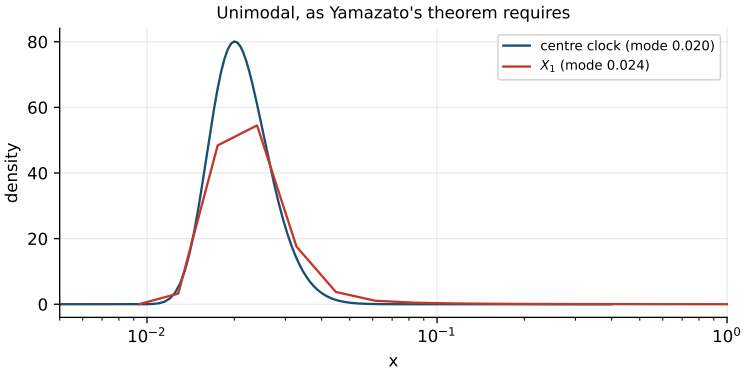


Figure 14.1: The densities of the centre clock T and of the shifted clock T_1, written X_1 in the plot, on a logarithmic axis. The density of T is drawn on a grid of spacing 0.0005 on $[0.004, 0.4]$, from the Talbot inversion and the spectral expansion of Section 5.2; that of T_1 on the 40 point geometric grid of the Talbot inversion, which is why its curve is piecewise linear. Computed.

14.1.2 Moments, the tail and the two ends of the Thorin measure

For a GGC with Thorin measure U the Lévy density is $x^{-1}\int e^{-xz}\,U(dz)$ (Section 4.1.1), so in the chart $\theta=\sqrt{z}$ the Lévy measure of T_1 and its tail are

$$\nu_{T_1}(dx)=\frac{dx}{x}\int_0^\infty e^{-x\theta^2}\rho_{1/2}(\theta)\,d\theta,\qquad \nu_{T_1}((x,\infty))=\int_0^\infty \rho_{1/2}(\theta)\,E_1(x\theta^2)\,d\theta, \tag{14.1.1}$$

with $E_1(y)=\int_y^\infty e^{-w}\,dw/w$. As $x\to\infty$ only θ near zero contributes, and $\int_0^\infty E_1(w^2)\,dw=\sqrt{\pi}$ gives $\nu_{T_1}((x,\infty))\sim\rho_{1/2}(0)\sqrt{\pi/x}=b_1(\pi x)^{-1/2}$, regularly varying of index $-\frac12$, hence subexponential. By the theorem of Embrechts, Goldie and Veraverbeke [83], an infinitely divisible law on $[0,\infty)$ is subexponential if and only if its Lévy measure, restricted to $(1,\infty)$ and normalised, is, and if and only if the tail of the law is asymptotic to that of the Lévy measure. Hence

$$P(T_1>x)\sim\nu_{T_1}((x,\infty))\sim\frac{b_1}{\sqrt{\pi x}}=0.01303\,x^{-1/2},\qquad x\to\infty. \tag{14.1.2}$$

Theorem 12.2.5 gives the infinite mean and the tail of T_α for every $\alpha>\frac12$ by the Tauberian route; since T_α is infinitely divisible for every $\alpha>\frac12$ (Section 14.1.4), the same theorem, read from the law to the Lévy measure, shows that the Lévy tail of T_α is asymptotic to $b_\alpha(\pi x)^{-1/2}$ wherever the zeros lie.

Table 14.2 compares the survival function of T_1 from Section 12.2.4 with the Lévy tail computed from (14.1.1) and the Thorin density on $[0,100]$. In the bulk they differ widely: at $x=0.02$, just below the mode, the survival probability is 0.747 and the Lévy tail 0.101.

x	$P(T_1 > x)$	$\nu_{T_1}((x,\infty))$	ratio	$b_1(\pi x)^{-1/2}$
0.1	4.7639×10^{-2}	4.1558×10^{-2}	1.1463	4.1206×10^{-2}
0.5	1.8907×10^{-2}	1.8458×10^{-2}	1.0243	1.8428×10^{-2}
1	1.3199×10^{-2}	1.3041×10^{-2}	1.0121	1.3030×10^{-2}
10	4.1272×10^{-3}	4.1209×10^{-3}	1.0015	4.1206×10^{-3}
100	1.3037×10^{-3}	1.3030×10^{-3}	1.0005	1.3030×10^{-3}

Table 14.2: The subexponential principle for T_1: the tail of the law is asymptotic to the tail of its Lévy measure. The ratio is that of the second column to the third, and the last column is the common asymptote of (14.1.2). Computed.

For a GGC the Thorin measure near zero governs the right tail, and its total mass governs the density at the origin [39]. The Thorin measure U_1 of T_1 (Proposition 12.1.2) has $U_1((0,z]) \sim (b_1/\pi)\sqrt{z}$ as $z \downarrow 0$, which gives $1-\varphi_1(u) \sim b_1\sqrt{u}$ (Section 12.1). Its total mass is infinite, since $U_1((0,Y^2]) = M_{1/2}(Y)$ differs from the zero count $N(Y)$ by $\tau_{1/2}(Y) = O(\log Y)$ (Section 12.1.2); this is why T_1, like the centre clock (Section 3.1.2.2), is super flat at the origin. Indeed, for every $z_0 > 0$ the Thorin form gives $\varphi_1(u) \le (1+u/z_0)^{-U_1((0,z_0])}$, and $U_1((0,z_0]) \to \infty$, so $\varphi_1(u)$ decays faster than any power of u. Markov's inequality $P(T_1 \le x) \le e\,\varphi_1(1/x)$ carries this to the distribution function, and since the density f_{T_1} is nondecreasing up to the mode (Corollary 14.1.2), $x f_{T_1}(x) \le P(T_1 \le 2x)$ for small x carries it to the density.

14.1.3 Uniqueness and quasi infinite divisibility

Lindner, Pan and Sato [213] call a law quasi infinitely divisible if its characteristic function is the quotient of two infinitely divisible characteristic functions, equivalently if it has a Lévy Khintchine form with a signed measure $\nu^+ - \nu^-$ whose parts are both Lévy measures, so in particular of finite mass away from the origin; the Gaussian coefficient is still nonnegative [213, Lemma 2.7], and the triplet is unique. The signed measures in question are those of the Lévy forms of Section 12.1.4: for $\alpha \ge 1$,

$$\log \frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty \left(e^{-sx} - 1 + sx\right) \Pi_\alpha(dx),$$

which at $\alpha = 1$ is (12.1.5), a compensated limit, and for $\alpha > 1$ converges absolutely when $\operatorname{Re} s > 1-\alpha$. For $\alpha > 1$, read at $s = it$, it is a Lévy Khintchine form for the characteristic function of V_α, with Π_α carried to the negative half line by $x \mapsto -x$. Let $x_* = 0.28120$ be the root of $e^{-3x} = 1 - e^{-2x}$.

Proposition 14.1.3. *1. The Pólya variable $X = V_{1/2}$ is not quasi infinitely divisible, since its characteristic function $\Xi(t)/\Xi(0)$ vanishes at the ordinates.*

2. For every $\alpha > 1$, V_α is quasi infinitely divisible and not infinitely divisible.

Its signed Lévy measure is the reflection to the negative half line of

$$\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)\frac{dx}{x} + \sum_n \frac{\Lambda(n)}{n^\alpha \log n}\,\delta_{\log n}(dx),$$

whose atoms have total mass $\log\zeta(\alpha)$ *and whose absolutely continuous part is positive on* $(0, x_*)$ *and negative beyond. The negative part of* Π_α *is therefore carried by* (x_*, ∞)*, and its mass on* $(1,\infty)$ *is* $E_1(\alpha-1) - \sum_{k\ge 0} E_1(\alpha+2+2k)$*: it is dominated by the pole term* $e^{-(\alpha-1)x}dx/x$*, of mass* $E_1(\alpha-1)$ *on* $(1,\infty)$*, but is not equal to it.*

3. *At* $\alpha = 1$*,* V_1 *is not quasi infinitely divisible. The candidate measure* Π_1 *of Proposition 12.1.10 has the pole term* $-dx/x$ *and the prime atoms* $\Lambda(n)/(n\log n)$ *at* $\log n$*, each of infinite mass on* $(1,\infty)$*; since the gamma density* $e^{-3x}(1-e^{-2x})^{-1}/x$ *is less than* $0.06/x$ *there, the positive and the negative part of* Π_1 *both have infinite mass on* $(1,\infty)$*, and its Lévy form converges only as a compensated limit, which exists by the prime number theorem.*

Items (1) and (2) are due to Nakamura [238]: (2), with the same measure, is his Theorem 1.4, and (1) is the remark after it. The argument of (1) applied to the moment generating function shows that no V_α, $\alpha \in \mathbb{R}$, is infinitely divisible, because $\xi(\alpha+w)/\xi(\alpha)$ is entire with zeros [325, Theorem 25.17].

Proof. (1) A quasi infinitely divisible characteristic function is the exponential of a finite exponent, so it never vanishes, while $\mathbb{E}e^{itX} = \Xi(t)/\Xi(0)$ by (2.1.4) vanishes at $t = 14.134725$ and at every other ordinate of a zero on the critical line.

(2) Let $\alpha > 1$. Every piece of the Lévy form converges absolutely and is a Lévy measure: the positive density $e^{-(\alpha+2)x}(1-e^{-2x})^{-1}dx/x$, the pole term and the atoms. Removing the factor $s(s-1)$ from $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ gives, for $\operatorname{Re} s > 1-\alpha$,

$$\frac{\xi(\alpha+s)}{\xi(\alpha)}\cdot\frac{\alpha}{\alpha+s}\cdot\frac{\alpha-1}{\alpha-1+s} = \pi^{-s/2}\,\frac{\Gamma(\frac{\alpha+s}{2})}{\Gamma(\frac{\alpha}{2})}\,\frac{\zeta(\alpha+s)}{\zeta(\alpha)}.$$

Let e_1, e_2 be standard exponential, $\Gamma_{\alpha/2}$ gamma of shape $\frac{\alpha}{2}$ and unit scale, and $\mathcal{Z}_\alpha$ of zeta law $P(\mathcal{Z}_\alpha = n) = n^{-\alpha}/\zeta(\alpha)$, all independent of each other and of V_α. Since $\mathbb{E}e^{-se_1/\alpha} = \alpha/(\alpha+s)$, $\mathbb{E}\Gamma_{\alpha/2}^{s/2} = \Gamma(\frac{\alpha+s}{2})/\Gamma(\frac{\alpha}{2})$ and $\mathbb{E}\mathcal{Z}_\alpha^{-s} = \zeta(\alpha+s)/\zeta(\alpha)$, the identity says

$$V_\alpha - \frac{e_1}{\alpha} - \frac{e_2}{\alpha-1} \overset{d}{=} \frac12\log\Gamma_{\alpha/2} - \frac12\log\pi - \log\mathcal{Z}_\alpha.$$

The right side is infinitely divisible: by Binet's formula [256, §5.9], as in the proof of Proposition 12.1.10, $\frac12\log\Gamma_{\alpha/2}$ has Lévy measure $e^{-\alpha x}(1-e^{-2x})^{-1}dx/x$

reflected, and by the Euler product $-\log \mathcal{Z}_\alpha$ is compound Poisson with Lévy measure $\sum_n \Lambda(n)(n^\alpha \log n)^{-1}\delta_{-\log n}$, of total mass $\log\zeta(\alpha)$. The variable $-e_1/\alpha - e_2/(\alpha{-}1)$ is infinitely divisible with Lévy measure $(e^{-\alpha x}{+}e^{-(\alpha-1)x})\,dx/x$ reflected. So V_α convolved with an infinitely divisible law is infinitely divisible, V_α is quasi infinitely divisible, and its signed Lévy measure is the difference, which is Π_α reflected since $e^{-\alpha x}(1-e^{-2x})^{-1} - e^{-\alpha x} = e^{-(\alpha+2)x}(1-e^{-2x})^{-1}$. The absolutely continuous part of Π_α has density $e^{-(\alpha-1)x}\big(e^{-3x}(1-e^{-2x})^{-1}-1\big)x^{-1}$, whose sign is that of $e^{-3x}-(1-e^{-2x})$ for every α: positive on $(0,x_*)$ and negative beyond, where it is at least $-e^{-(\alpha-1)x}/x$, of integral $E_1(\alpha-1)$ over $(1,\infty)$; expanding $(1-e^{-2x})^{-1}$ in powers of e^{-2x} and integrating termwise over $(1,\infty)$ gives the mass of the negative part there. If V_α were infinitely divisible, uniqueness of the triplet [325, Theorem 8.1], applied to the identity in law, would make its Lévy measure Π_α reflected, which is negative on the reflection of $(x_*,\log 2)$, an interval with no atom.

(3) Suppose that V_1 is quasi infinitely divisible, so that $\mathbb{E}e^{itV_1} = \exp\Psi_{V_1}(t)$ with

$$\Psi_{V_1}(t) = ict - \frac{A}{2}t^2 + \int_{\mathbb{R}} \big(e^{ity} - 1 - ity\mathbf{1}\{|y|\le 1\}\big)\,\mu(dy),$$

where A, c are real and μ is a signed measure on $\mathbb{R}\setminus\{0\}$ with $\int \min(1,y^2)\,|\mu|(dy) < \infty$. As in (1), the characteristic function never vanishes, so $\xi(1+it)\neq 0$ for real t; and $\xi(1+s)\neq 0$ for $\operatorname{Re} s>0$ by the Euler product. So $\ell(s)=\log(\xi(1+s)/\xi(1))$, $\ell(0)=0$, is a continuous logarithm on $\operatorname{Re} s\ge 0$, holomorphic inside, and $\Psi_{V_1}(t) = \ell(it)$, both being continuous logarithms of $\xi(1+it)/\xi(1)$ vanishing at 0. For $r>0$ put $\Delta_r F(t) = F(t+r)+F(t-r)-2F(t)$. Then

$$\Delta_r\Psi_{V_1}(t) = \int_{\mathbb{R}} e^{ity}\,\lambda_r(dy), \qquad \lambda_r(dy) = -Ar^2\delta_0(dy) + (2\cos ry - 2)\,\mu(dy),$$

a finite signed measure because $|2\cos ry - 2| \le \min(4, r^2y^2)$. Let $\check{\Pi}_1$ be the reflection of Π_1. For $\sigma>0$, $V_{1+\sigma}$ has characteristic function $\exp(\ell(\sigma+it)-\ell(\sigma))$, and by (2) its signed Lévy measure is the reflection of $\Pi_{1+\sigma}(dx) = e^{-\sigma x}\Pi_1(dx)$, that is $e^{\sigma y}\check{\Pi}_1(dy)$. The second difference removes drift and compensator, so

$$\Delta_r\ell(\sigma+i\,\cdot\,)(t) = \int_{\mathbb{R}} e^{ity}\,m_\sigma(dy), \qquad m_\sigma(dy) = e^{\sigma y}(2\cos ry-2)\,\check{\Pi}_1(dy),$$

a finite signed measure, the atoms contributing at most $4\log\zeta(1+\sigma)$ to its total variation. As $\sigma\downarrow 0$, $m_\sigma \to m_0 = (2\cos ry-2)\,\check{\Pi}_1$ as tempered distributions, by dominated convergence, because $|m_0|$ is locally finite and $|m_0|((-R,0)) = O(R)$: the factor $2\cos ry-2$ tames $\check{\Pi}_1$ near the origin, and $\Lambda(n)\le\log n$ bounds the atom at $-\log n$ by $1/n$. So the Fourier transforms converge as tempered distributions. On the other hand $\Delta_r\ell(\sigma+it)\to\Delta_r\Psi_{V_1}(t)$ uniformly on compact sets, ℓ being uniformly continuous on compact subsets of the closed half plane. Hence the Fourier transforms of m_0 and λ_r agree on smooth test functions of

compact support, which are dense in the Schwartz space, and by injectivity $m_0 = \lambda_r$. The mass at the origin gives $A = 0$. On a compact $K \subset \mathbb{R}\setminus\{0\}$, $2\cos ry - 2 < 0$ once $r < 2\pi/\max_K |y|$, so $\mu = \check{\Pi}_1$ on K, hence on $\mathbb{R} \setminus \{0\}$. But the atom of Π_1 at $\log p$ has mass $1/p$, so $|\mu|((-\infty,-1)) \geq \sum_{p\geq 3} 1/p = \infty$, and $|\mu|$ is not a Lévy measure. □

The proof of (3) uses about the primes only the Euler product: the nonvanishing of ζ on $\operatorname{Re} s = 1$ enters only through the hypothesis, which forces it. Section 12.1.4 reads the conclusion of (3) from the Lévy form itself, and the proof supplies the uniqueness on which that reading rests. The same argument, with $\xi(1+s, \chi_{12})$ in place of $\xi(1+s)$, shows that the tilted eta variable of Section 15.2.2 is neither quasi infinitely divisible nor infinitely divisible.

14.1.4 Bondesson's classes and Thorin Bernstein functions

The class of HCM densities is closed under products and quotients of independent variables and under powers X^q, $|q| \geq 1$ [39, Section 5.1]; with the characterisation of GGCs by hyperbolic complete monotonicity [39, Theorem 6.1.1] (Section 4.1.2) these are the tools behind the Bosch threshold of Theorem 9.3.5.

Bondesson's mixtures of exponential laws and their generalised convolutions are the natural class for Kent's hitting times started from an interior point, whose Laplace transform is a ratio of two products with interlacing zeros. The law of T_1 is not a mixture of exponentials, since such a mixture has a completely monotone density while that of T_1 vanishes at the origin and has an interior mode (Figure 14.1); this agrees with Section 12.2, where T_1 is a subordinated rather than a direct Kent hitting time (Section 12.2.1 and Theorem 12.2.2).

In the language of Schilling, Song and Vondraček [327, Definition 9.10], a law on $[0,\infty)$ is a GGC exactly when its Laplace exponent is a Thorin Bernstein function, equivalently when the derivative of the exponent is a Stieltjes function, the function f_H of Section 4.1.1. The subordination identity of Theorem 12.2.5 reads

$$\begin{aligned} -\log \varphi_\alpha(u) &= \log G\big((\alpha-\tfrac12)^2 + f_\alpha(u)\big) - \log G\big((\alpha-\tfrac12)^2\big), \\ f_\alpha(u) &= u + 2(\alpha-\tfrac12)\sqrt{u}, \end{aligned} \tag{14.1.3}$$

with f_α, the Laplace exponent of the subordinator $\sigma^{(\alpha)}$ of that theorem, a complete Bernstein function. The derivative $\frac{d}{du}[-\log\varphi_\alpha(u)] = \frac{1}{2\sqrt{u}}\frac{\xi'}{\xi}(\alpha+\sqrt{u})$ is a Stieltjes function exactly when $\alpha \geq \sup_\rho \operatorname{Re}\rho$, the criterion of Theorem 2.1.2: a zero with $\operatorname{Re}\rho > \alpha$ makes $\xi(\alpha+\sqrt{u})$ vanish at the point $u = (\rho-\alpha)^2$ of the cut plane $\mathbb{C}\setminus(-\infty,0]$, where a Stieltjes function is holomorphic.

Proposition 14.1.4. *For every $\alpha \geq 1$ the composition $\log G((\alpha-\frac12)^2+f_\alpha) - \log G((\alpha-\frac12)^2)$ is a Thorin Bernstein function with no hypothesis, while $\log G$ itself is a Thorin Bernstein function if and only if RH holds. For $\frac12 < \alpha < 1$ the*

composition is a Thorin Bernstein function if and only if ζ has no zeros with real part greater than α.

Proof. By (14.1.3) the composition is the Laplace exponent of T_α, a Thorin Bernstein function exactly when T_α is a GGC, which by Theorem 2.1.2 happens exactly when no zero has real part greater than α; for $\alpha \geq 1$ there is none. The exponent $\log G$ is a Thorin Bernstein function exactly when T is a GGC, which is RH by Proposition 2.1.1. □

For every $\alpha > \frac{1}{2}$ the composition is at least a Bernstein function: $u \mapsto \log G((\alpha-\frac{1}{2})^2+u) - \log G((\alpha-\frac{1}{2})^2)$ is the Laplace exponent of $\widetilde{T}_\alpha$, and a Bernstein function composed with a Bernstein function is one [327, Corollary 3.8]. So T_α is infinitely divisible for every $\alpha > \frac{1}{2}$ with no hypothesis, and the zeros decide only whether it is a GGC; composing with f_α creates the Thorin property from a function only known to be Bernstein.

General composition theorems do not settle this. For a Bernstein function κ and $a \geq 0$ let $\kappa_a(u) = \kappa((\sqrt{u}+a)^2) - \kappa(a^2)$, so that (14.1.3) is κ_a for $\kappa = \log G$, $a = \alpha - \frac{1}{2}$. If κ is Thorin Bernstein then so is every κ_a: the shift turns a drift cu into $c(u + 2a\sqrt{u})$ and each factor $(1+u/t)^{-1}$ of the Thorin form into $(a^2+t)/((\sqrt{u}+a)^2+t)$, the Laplace transform of a GGC whose Thorin density in θ is the pair of Cauchy kernels of width a at $\pm\sqrt{t}$, as in Section 12.2.2. Since $\kappa_{a+a'} = (\kappa_a)_{a'}$, the shifts a for which κ_a is Thorin Bernstein form a half line, which for $\kappa = \log G$ Proposition 14.1.4 identifies as $[a_*, \infty)$, $a_* = \sup_\rho \operatorname{Re}\rho - \frac{1}{2}$; RH is $a_* = 0$. For a fixed shift the implication cannot be reversed. The function $\kappa(u) = 2\log(1+u) + \log(1 + (120u + 16u^2)/289)$ is a Bernstein function, since κ' is the Laplace transform of $2e^{-x} + 2e^{-15x/4}\cos 2x \geq 0$, and not Thorin Bernstein, since κ' has poles at $-\frac{15}{4} \pm 2i$; yet

$$\kappa_{1/2}(u) = \log\left(1+\frac{u}{4}\right) + 2\log\frac{4((\sqrt{u}+\frac{1}{2})^2+1)}{5} + \log\frac{(\sqrt{u}+1)^2+4}{5}$$

is Thorin Bernstein, the exponent of an exponential law plus those of GGCs whose Thorin densities are pairs of Cauchy kernels, of width $\frac{1}{2}$ at ± 1 and of width 1 at ± 2; $u \mapsto \kappa(u/4a^2)$ gives the same example at any shift $a > 0$.

The Dirichlet mean representation of James, Roynette and Yor [168] writes a GGC whose Thorin measure has finite total mass c as $\Gamma_c\mathcal{M}$, with Γ_c gamma of shape c and $\mathcal{M}$ an independent Dirichlet mean. It applies to the truncated lattice variables $\mathcal{S}_n$ of Section 17.1.1, of Thorin mass $2n$, and to the finite sums of exponentials of Proposition 11.1.22, but not to T_α or, under RH, to T, whose Thorin measures have infinite mass.

14.1.5 What these results cannot give

Every GGC criterion in these books is, for the reciprocal Xi function at the centre, equivalent to RH: the HCM property of the Laplace transform and the Stieltjes

property of the derivative of the exponent (Theorem 9.2.4), the Pick function property (Corollary 9.3.6), and the complete monotonicity of the Lévy density of the driving process (Proposition 14.1.1). The consequences of Sections 14.1.1 to 14.1.4, collected in Table 14.3, are mostly unconditional, for one of two reasons. Some use only the infinite divisibility and self decomposability of the centre clock, the inequalities $W > 0$ and $W' < 0$, which hold wherever the zeros above height 1000 lie (Theorem 8.1.1). The others use the region $\operatorname{Re} s \geq 1$, through positivity of the Thorin measure at a basepoint $\alpha \geq 1$, which rests on the nonvanishing of ζ on $\operatorname{Re} s = 1$ (Theorem 2.1.2), or through the Euler product (Proposition 14.1.3). So none supplies positivity of the Thorin measure uniformly in the height at a basepoint $\alpha < 1$, that is the GGC property of T_α in the sense of Theorem 2.1.2, which is the zero free half plane $\operatorname{Re} s > \alpha$ of step 7 of the path in Table 25.1, and still less at the centre, where it is RH. That positivity is the one ingredient still missing on the path to RH; the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 would supply it for every $\alpha > \frac{7}{8}$, by arithmetic from outside the theory (Section 25.1.2).

14.1.6 Wiener gamma integrals and the Riemann Ornstein Uhlenbeck process

A gamma process realises the superposition of gamma laws in a GGC pathwise. Let $(\Gamma_t)_{t\geq 0}$ be the standard gamma process, $\mathbb{E}e^{-u\Gamma_t} = (1+u)^{-t}$; for measurable $f \geq 0$ the Wiener integral $\int_0^\infty f(t)\, d\Gamma_t$ has Laplace exponent $\int_0^\infty \log(1+uf(t))\, dt$. Let U be the Thorin measure of a GGC without drift, $M(\theta) = U((0,\theta^2])$ its cumulative mass in the chart $\theta = \sqrt{z}$, and $M^{-1}(t) = \inf\{\theta : M(\theta) > t\}$. The image of Lebesgue measure on $(0, M(\infty))$ under M^{-1} is dM, so $t = M(\theta)$ turns $\int_0^\infty \log(1 + u/M^{-1}(t)^2)\, dt$ into $\int \log(1+u/z)\, U(dz)$, and the GGC is the Wiener gamma integral [168]

$$\int_0^\infty \frac{d\Gamma_t}{M^{-1}(t)^2}. \tag{14.1.4}$$

At the centre, under RH, $M = N$, $M^{-1}(t) = \gamma_k$ for $k-1 \leq t < k$ (ordinates in increasing order, with multiplicity), and (14.1.4) is $\sum_k (\Gamma_k - \Gamma_{k-1})/\gamma_k^2$, the series $\sum_{\gamma>0} e_\gamma/\gamma^2$ of Proposition 2.1.1. At a basepoint α the count is replaced by the Poisson smoothed count $M_{\alpha-1/2}(\theta) = \pi^{-1}\int_0^\theta v_\alpha$ of Section 3.3, π^{-1} times the argument of ξ along $\operatorname{Re} s = \alpha$. For $\alpha \geq 1$, $v_\alpha > 0$, so with no hypothesis T_α is (14.1.4) with $M = M_{\alpha-1/2}$. For $\frac{1}{2} < \alpha < 1$ the representation holds exactly when T_α is a GGC, that is when no zero has real part greater than α (Theorem 12.2.5), and then M is $M_{\alpha-1/2}$ plus a unit jump at the ordinate of each zero on $\operatorname{Re} s = \alpha$ (Theorem 11.1.14). It needs $v_\alpha \geq 0$ on the line $\operatorname{Re} s = \alpha$, but for ξ it is not known whether positivity on one line excludes zeros to its right (Remark 2.1.4).

Self decomposability gives a second process. If Z is a subordinator with $\mathbb{E}\log(1+Z_1) < \infty$, the Ornstein Uhlenbeck type process $dF_t = -F_t\, dt + dZ_t$,

result	source	consequence here	status
background driving Lévy process	Wolfe; Jurek and Vervaat; Sato, Theorem 17.5	centre clock driven by a subordinator with Lévy density $-W'$; RH if and only if it is completely monotone	proved
Yamazato's theorem	Yamazato; Sato, Theorem 53.1	centre clock and T_α, $\alpha > \frac{1}{2}$, unimodal	proved
Urbanik's classes L_n	Urbanik; Sato 1980; Berg and Forst	centre clock in L_2, not in L_3; its driving process and the driving process of that self decomposable, with unimodal marginals	proved
moment theorem	Sato, Theorem 25.3	$\mathbb{E}T_\alpha = \infty$ for $\alpha \geq 1$	proved
subexponential tails	Embrechts, Goldie, Veraverbeke	$P(T_\alpha > x) \sim b_\alpha(\pi x)^{-1/2}$; ratio 1.0005 at $x = 100$	proved, checked
uniqueness, quasi infinite divisibility	Sato, Theorem 8.1; Lindner, Pan, Sato	V_α quasi infinitely divisible for $\alpha > 1$, not at $\alpha = 1$; Pólya variable not quasi infinitely divisible	proved
HCM characterisation and closure	Bondesson	Bosch threshold for powers	proved
Thorin measure at the two ends	Bondesson	$x^{-1/2}$ tail and super flat origin	proved
Thorin Bernstein composition	Schilling, Song, Vondraček	smoothing by f_α creates the Thorin property for $\alpha \geq 1$	proved
Dirichlet means	James, Roynette, Yor	applies to truncations only	remark

Table 14.3: What the books of Sato and Bondesson, and the papers around them, give the reciprocal Xi function. Theorem numbers, here and in the text, refer to the first edition of Sato's book [325].

$F_t = e^{-t}F_0 + \int_0^t e^{-(t-r)}\,dZ_r$, has a unique stationary law, that of $\int_0^\infty e^{-t}\,dZ_t$ [325, Theorem 17.5 and Corollary 17.9]. With the driver $h = -W'$ of Proposition 14.1.1, which the explicit formula computes from the primes, there is, with no hypothesis, an Ornstein Uhlenbeck process F whose stationary law is the centre clock, which we call the Riemann Ornstein Uhlenbeck process. Under RH the Laplace exponent of Z is $\Psi_Z(u) = u(\log G)'(u) = \sum_{\gamma>0} u/(\gamma^2 + u)$, so the driver superposes independent compound Poisson processes, one for each zero with multiplicity, each of rate one with exponential jumps of mean γ^{-2}.

The simulation is exact for the 1700 computed zeros; the zeros above the 1700th enter as a drift of the driver, which adds the mean $\mathbb{E}T - \sum_{k\leq 1700} \gamma_k^{-2}$ of

their components to the stationary law. The time average is 0.023094 against $\mathbb{E}T = \sum_k a_k^{-1} = 0.023105$ (Section 12.1.2). The empirical Laplace transform of the stationary law is 0.79520 ± 0.00096, 0.32828 ± 0.00180 and 0.016404 ± 0.00030 at $u = 10$, 50 and 200, with standard errors from 100 batch means, against $\varphi_{1/2}(u) = 0.79514$, 0.32827 and 0.016399. The stationary process at time $\tau \geq 0$ is $e^{-\tau}$ times its value at 0 plus an independent variable, and the centre clock has finite variance, so the autocorrelation at lag τ is $e^{-\tau}$ with no hypothesis. The estimates at lags 0.5, 1 and 2 are 0.602, 0.352 and 0.119 against 0.607, 0.368 and 0.135, within 1.6 standard errors, which are about 0.008, 0.010 and 0.013 by batch estimates over 20 blocks. The top right panel of Figure 14.2 shows ten units of time of the path.

For $\alpha \geq 1$, T_α is a GGC, hence self decomposable, with its own driver: by (14.1.1) with v_α/π in place of $\rho_{1/2}$, its Lévy measure is $k_\alpha(x)\,dx/x$ with $k_\alpha(x) = \pi^{-1}\int_0^\infty e^{-x\theta^2} v_\alpha(\theta)\,d\theta$, and its driver has the completely monotone Lévy density $-k'_\alpha(x) = \pi^{-1}\int_0^\infty \theta^2 e^{-x\theta^2} v_\alpha(\theta)\,d\theta$. For $\frac{1}{2} < \alpha < 1$ the process exists as well, T_α being self decomposable with no hypothesis (Corollary 12.2.6); its driver has Lévy density $-W'_\alpha$, completely monotone exactly when T_α is a GGC.

14.1.7 Subordinated Brownian motion and the map of processes

Running an independent standard Brownian motion B on a clock gives a symmetric law with

$$\mathbb{E}e^{iuB_{T_\alpha}} = \mathbb{E}e^{-u^2 T_\alpha/2} = \frac{\xi(\alpha)}{\xi(\alpha + |u|/\sqrt{2})}, \qquad u \in \mathbb{R},$$

so B_{T_α} has the law of $Y_\alpha/\sqrt{2}$, the normal scale mixture of Proposition 13.1.1, and Section 13.1 carries over with a factor $\sqrt{2}$.

Table 14.4 arranges the processes of Chapter 12 and of this section by basepoint, with $a = \alpha - \frac{1}{2}$. The Mellin side is the Bessel picture of Williams [363] and of Biane, Pitman and Yor [33] (Sections 2.1 and 12.1.1). The clock side starts from the centre clock, under RH with simple zeros Kent's hitting time for a string with spectrum $\{\gamma^2\}$, passes through the tilt $e^{-a^2 T}$ and the sticky Brownian subordination of Theorem 12.2.5 to T_α, and continues to the Wiener gamma integral of Section 14.1.6, to Brownian motion run on the clock, and to the Brownian boundaries of Section 12.2.6.

Three conclusions come out of the map. First, the Mellin side and the clock side differ in kind. The Bessel picture and the size biased variables V_α are unconditional at every basepoint, and there the zeros enter only as zeros, of the characteristic function $\Xi(t)/\Xi(0)$ and of $\xi(\alpha+s)$ in the Lévy forms of Section 14.1.3; the positivity statements equivalent to RH, the GGC property of the centre clock and the EGGC property of $Y_{1/2}$, live on the clock side, in the reciprocal.

Second, on the clock side the Ornstein Uhlenbeck process at the centre and, at every basepoint above it, the tilted and subordinated clock, Brownian motion

	$\alpha = \frac{1}{2}$	$\frac{1}{2} < \alpha < 1$	$\alpha = 1$	$\alpha > 1$
Mellin side	Pólya variable, $S_2^{1/4}$ bias	$S_2^{\alpha/2}$ bias	$S_2^{1/2}$ bias	$S_2^{\alpha/2}$ bias
Lévy form of $\log \xi(\alpha + s)$	none: the characteristic function vanishes at the ordinates	exists if $\sup \operatorname{Re} \rho < \alpha$, only if $\sup \operatorname{Re} \rho \leq \alpha$	compensated limit, by the prime number theorem	absolutely convergent; quasi infinitely divisible
clock law	T, infinitely divisible and self decomposable	$\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$	$\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$	$\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$
GGC	if and only if RH	if and only if no zero has $\beta > \alpha$	yes	yes
Thorin measure	$\sum_{\gamma>0} \delta_{\gamma^2}$ (RH)	Cauchy widths $\alpha - \beta$	Cauchy widths $1 - \beta$	Cauchy widths $\alpha - \beta$
Kent string	spectrum γ^2 (RH)	spectrum $\gamma^2 + a^2$ on the sticky clock (RH)	spectrum $\lvert\rho\rvert^2$ on the sticky clock (RH)	spectrum $\gamma^2 + a^2$ on the sticky clock (RH)
tail of the clock	exponential, rate γ_1^2	$b_\alpha(\pi x)^{-1/2}$	$0.01303\, x^{-1/2}$	$b_\alpha(\pi x)^{-1/2}$
Brownian boundary	asymptotically linear, slope $-\sqrt{2}\gamma_1$	flat: -0.3338 ($\alpha = 0.6$), -0.2554 ($\alpha = 0.75$)	flat at -0.19277	flat
Wald defect $\mathbb{E}\, b(T_\alpha)$	0	$b_\alpha/\sqrt{2}$	$\sum_\rho \rho^{-1}/\sqrt{2}$	$b_\alpha/\sqrt{2}$
B_{T_α}	finite variance	Cauchy domain	Cauchy domain, tail $0.0104/y$	Cauchy domain
Ornstein Uhlenbeck driver	jump density $-W'$	jump density $-W'_\alpha$	jump density $-k'_1$	jump density $-k'_\alpha$
symmetric EGGC Y_α	if and only if RH	if and only if no zero has $\beta > \alpha$	yes	yes

Table 14.4: The processes by basepoint, with $a = \alpha - \frac{1}{2}$. (RH) marks statements that hold under RH, and the Kent strings need the zeros to be simple as well. The Mellin row gives the size bias of S_2 that produces V_α (Section 12.1.1); $\widetilde{T}_\alpha$, H and the sticky clock are those of Theorem 12.2.5 and Corollary 12.2.3, with $\widetilde{T} = \widetilde{T}_1$; and the boundaries b and their Wald defects are those of Sections 12.2.4 and 12.2.6. The boundary row reports computations at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 made under RH beyond $t = 0.03$, and its entry for $\alpha > 1$ is extrapolated, not computed; the flatness is observed, not proved, and above the centre the Wald defect $b_\alpha/\sqrt{2}$ requires the boundary to be bounded at infinity (Lemma 12.2.4). All other entries hold as stated.

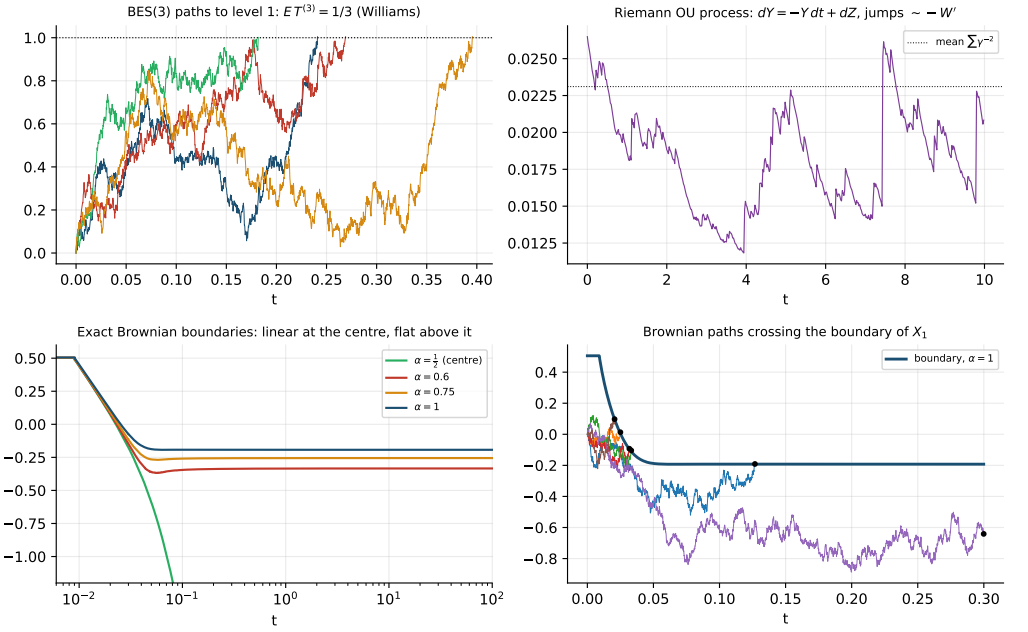


Figure 14.2: Four processes of the map. Top left: paths of a three dimensional Bessel process run until they reach the level one; the hitting time has mean $\frac{1}{3}$, and two independent copies make up the variable S_2 of Section 12.1.1. Top right: ten units of time of the Riemann Ornstein Uhlenbeck process F, written Y in the plot, whose stationary law is the centre clock, with the mean $\mathbb{E}T$, written $\sum\gamma^{-2}$ in the plot. Bottom left: the exact Brownian boundaries of T_α at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 computed under RH beyond $t = 0.03$ (Section 12.2.6), asymptotically linear at the centre and flat above it. Bottom right: Brownian paths crossing the boundary of T_1, written X_1 in the plot (Section 12.2.4).

run on the clock and the Brownian boundary exist with no hypothesis. The Kent strings need RH and simple zeros, because a Kent hitting time is a GGC whose Thorin measure consists of unit atoms at distinct points (Section 12.2.1). For $\alpha > \frac{1}{2}$, T_α is never a Kent hitting time: when it is a GGC, each pair of zeros $\rho, \bar{\rho}$ with $\beta < \alpha$ contributes the pair of Cauchy kernels of width $\alpha - \beta$ at $\pm\gamma$ (Section 14.1.4), so its Thorin measure is not a sum of atoms.

Third, the Bessel process is the template: its hitting time of the level one is Kent's hitting time for a string with spectrum $\{\pi^2 n^2/2\}$, as under RH with simple zeros the centre clock is for the spectrum $\{\gamma^2\}$.

14.2 Free probability and the Upsilon transform

Wigner's semicircle law [362] is the limit law of the free central limit theorem [355, 247]. This section, based on [291], applies the Bercovici Pata bijection [25] and the Upsilon transform [22, 20] to the centre clock T. With no hypothesis, T has an Upsilon image $\Upsilon(T)$, a GGC whose Thorin measure has distribution function $u \mapsto W(1/u)$, and a free counterpart $\Lambda_{\mathrm{BP}}(T)$, freely self decomposable on $[0, \infty)$; RH holds exactly when T is itself the Upsilon image of an infinitely divisible law (Theorem 14.2.3).

14.2.1 The Bercovici Pata bijection and the Upsilon transform

As in Section 4.1.4, $G_\mu(z) = \int \mu(dx)/(z-x)$ is the Cauchy transform of a probability measure μ on $\mathbb{R}$ (the subscript distinguishes it from the function G of the clock) and σ_t the semicircle law of variance t. G_μ maps $\mathbb{C}^+$ into $\mathbb{C}^-$, $1/G_\mu$ has a right inverse on a region $\Gamma_{\alpha,M} = \{z : |\operatorname{Re} z| < \alpha \operatorname{Im} z,\ \operatorname{Im} z > M\}$, and the Voiculescu transform $\varphi_\mu(z) = (1/G_\mu)^{-1}(z) - z$ linearises free additive convolution, $\varphi_{\mu \boxplus \nu} = \varphi_\mu + \varphi_\nu$ [355, 27], where $\mu \boxplus \nu$ is the law of $X+Y$ for free self adjoint X, Y with laws μ, ν [247]. The free cumulant transform $\mathcal{C}_\mu(z) = z\varphi_\mu(1/z)$, for $z \in \mathbb{C}^-$ with $1/z \in \Gamma_{\alpha,M}$, is additive under $\boxplus$; μ is freely infinitely divisible if for every n it is the n fold free convolution of a law with itself, and then φ_μ extends analytically to $\mathbb{C}^+$ [27], so $\mathcal{C}_\mu$ is analytic on $\mathbb{C}^-$.

A law μ on $\mathbb{R}$ is infinitely divisible exactly when its cumulant function has the Lévy Khintchine form

$$C_\mu(u) = i\eta u - \tfrac{1}{2}Au^2 + \int_{\mathbb{R}} \left(e^{iut} - 1 - iut\,\mathbf{1}_{[-1,1]}(t)\right) \Pi(dt), \qquad u \in \mathbb{R}, \quad (14.2.1)$$

with $\eta \in \mathbb{R}$, $A \geq 0$ and a Lévy measure Π; the triplet is unique [325, Theorem 8.1]. A law is freely infinitely divisible exactly when

$$\mathcal{C}_\mu(z) = \eta z + Az^2 + \int_{\mathbb{R}} \Big(\frac{1}{1-tz} - 1 - tz\,\mathbf{1}_{[-1,1]}(t)\Big)\, \Pi(dt), \qquad z \in \mathbb{C}^-, \quad (14.2.2)$$

with a triplet of the same kind, again unique [27, 25], [5, Proposition 6].

Definition 14.2.1. The Bercovici Pata bijection Λ_{BP} maps the infinitely divisible law with triplet (η, A, Π) in (14.2.1) to the freely infinitely divisible law with the same triplet in (14.2.2) [25], [5, Definition 7].

The map is a homomorphism, $\Lambda_{\mathrm{BP}}(\mu * \nu) = \Lambda_{\mathrm{BP}}(\mu) \boxplus \Lambda_{\mathrm{BP}}(\nu)$, and a homeomorphism for weak convergence [25], [5, Section 3.2]. When Π has moments of all orders, comparing the expansions of e^{iut} and $1/(1-tz)$ shows that the free cumulants of $\Lambda_{\mathrm{BP}}(\mu)$ are the classical cumulants of μ, both $A\mathbf{1}_{\{n=2\}} + \int t^n\, \Pi(dt)$ for $n \geq 2$; $N(0,t)$ goes to σ_t [25], [5, Section 3.2 and Remark 8].

Barndorff-Nielsen and Thorbjørnsen [21] proved that Λ_{BP} maps the self decomposable laws onto the freely self decomposable laws, those μ with $\mu = D_c\mu \boxplus \mu_c$ for every $c \in (0,1)$ and some law μ_c, $D_c\mu$ being the law of cX for $X \sim \mu$; Hasebe and Thorbjørnsen [109] proved that freely self decomposable laws are unimodal, the free analogue of Yamazato's theorem (Corollary 14.1.2). By [5, Theorem 9] a freely infinitely divisible μ is the image of an infinitely divisible law on $[0,\infty)$ if and only if $\mu^{\boxplus t}$ is carried by $[0,\infty)$ for every $t > 0$; these are the free regular laws. If μ is infinitely divisible on $[0,\infty)$, with drift $d \geq 0$ and Lévy measure Π on $(0,\infty)$, $\int \min(1,x)\,\Pi(dx) < \infty$, then (14.2.2) reduces to [5, (7)]

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(z) = dz + \int_{(0,\infty)} \Big(\frac{1}{1-zx} - 1\Big)\,\Pi(dx) = dz + \int_{(0,\infty)} \frac{zx}{1-zx}\,\Pi(dx), \quad (14.2.3)$$

whose right side is analytic on $\mathbb{C} \setminus (0,\infty)$ and extends $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}$ to the negative half line.

The Upsilon transform [22, 20] maps an infinitely divisible μ with Lévy process (X_t), $X_1 \sim \mu$, to the law $\Upsilon(\mu)$ of $\int_0^1 \log(1/t)\,dX_t$, which is infinitely divisible with

$$C_{\Upsilon(\mu)}(u) = \int_0^1 C_\mu\big(u\log(1/t)\big)\,dt = \int_0^\infty C_\mu(ux)\,e^{-x}\,dx. \quad (14.2.4)$$

Barndorff-Nielsen, Maejima and Sato [20] proved that Υ maps the infinitely divisible laws onto the Goldie Steutel Bondesson class (Lévy measure with a completely monotone density in $|x|$ on each half line) and the self decomposable laws onto the Thorin class; on $[0,\infty)$ these are the class BO and the GGCs of Section 4.1.2. Barndorff-Nielsen and Thorbjørnsen [22] found the link with the free side:

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(i\zeta) = C_{\Upsilon(\mu)}(\zeta), \qquad \zeta < 0. \quad (14.2.5)$$

14.2.2 The Upsilon transform on the half line

Let μ be infinitely divisible on $[0,\infty)$ with drift $d \geq 0$ and Lévy measure Π on $(0,\infty)$,

$$\Psi_\mu(s) = ds + \int_{(0,\infty)} \big(1 - e^{-sx}\big)\,\Pi(dx), \qquad \int \min(1,x)\,\Pi(dx) < \infty.$$

If μ is self decomposable, $\Pi(dx) = k(x)\,dx/x$ with canonical function k nonincreasing, right continuous, $k(\infty) = 0$; if μ is a GGC with Thorin measure U, $k(x) = \int e^{-tx}\,U(dt)$ (Section 4.1.1). The cumulants, when finite, are $\kappa_1 = d + \int x\,\Pi(dx)$ and $\kappa_n = \int x^n\,\Pi(dx)$, $n \geq 2$.

Proposition 14.2.2. *Let μ be as above.*

(a) $\Upsilon(\mu)$ *is the infinitely divisible law on* $[0,\infty)$ *with drift* d*, Laplace exponent*

$$\Psi_{\Upsilon(\mu)}(s) = \int_0^\infty \Psi_\mu(sx)\, e^{-x}\, dx = ds + \int_{(0,\infty)} \frac{sy}{1+sy}\, \Pi(dy) \tag{14.2.6}$$

and completely monotone Lévy density $\ell(z) = \int y^{-1} e^{-z/y}\, \Pi(dy)$*. Every infinitely divisible law on* $[0,\infty)$ *with a completely monotone Lévy density, that is every law in BO, is* $\Upsilon(\mu)$ *for exactly one* μ*. Moreover* $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(-s) = -\Psi_{\Upsilon(\mu)}(s)$ *for* $s > 0$*, and* $\kappa_n(\Upsilon(\mu)) = n!\, \kappa_n(\mu)$ *whenever* $\int x^n\, \Pi(dx) < \infty$*.*

(b) $\Upsilon(\mu)$ *is a GGC if and only if* μ *is self decomposable. In that case* $\Upsilon(\mu)$ *has drift* d *and the Thorin measure* $\widetilde{U}$ *with* $\widetilde{U}((0,u]) = k(1/u)$*, the image of the measure* $-dk$ *under* $v \mapsto 1/v$*, and* Υ *is a bijection from the self decomposable laws on* $[0,\infty)$ *onto the GGCs, with inverse given by* $k(v) = \widetilde{U}((0,1/v])$*.*

(c) If μ *is a GGC with Thorin measure* U*, then* $\widetilde{U}$ *is the mixture over* $U(dt)$ *of the laws of* t/E*, with* E *standard exponential:*

$$\widetilde{U}(du) = u^{-2}(-k')(1/u)\, du, \qquad \Psi_{\Upsilon(\mu)}(s) = ds + \int e^{t/s} E_1(t/s)\, U(dt), \tag{14.2.7}$$

and $\int u^p\, \widetilde{U}(du) = \Gamma(1-p) \int t^p\, U(dt)$ *for* $p < 1$*.*

Proof. (a) The first equality in (14.2.6) is (14.2.4) for Laplace transforms: for a subordinator, $\mathbb{E}\exp(-s\int_0^1 f\, dX) = \exp(-\int_0^1 \Psi_\mu(sf(t))\, dt)$ for bounded measurable $f \geq 0$, by step functions and independent increments, and $f = \log(1/t)$ follows by monotone convergence. Fubini's theorem and $\int_0^\infty (1-e^{-sxy}) e^{-x} dx = sy/(1+sy) = \int_0^\infty (1-e^{-sz})\, y^{-1} e^{-z/y}\, dz$ give the second equality and $\Psi_{\Upsilon(\mu)}(s) = ds + \int_0^\infty (1-e^{-sz})\, \ell(z)\, dz$. The function ℓ is a mixture of exponentials, hence completely monotone, and a Lévy density: $\int_0^\infty \min(1,z)\, y^{-1} e^{-z/y} dz = y(1-e^{-1/y})$ lies between $(1-e^{-1})\min(1,y)$ and $\min(1,y)$, so $\int \min(1,z)\ell(z)\, dz$ is comparable with $\int \min(1,y)\, \Pi(dy)$. Conversely, let λ be infinitely divisible on $[0,\infty)$ with drift d and completely monotone Lévy density $\ell(z) = \int_{[0,\infty)} e^{-zr}\, \sigma(dr)$ (Bernstein's theorem); integrability on $(1,\infty)$ forces $\sigma(\{0\}) = 0$. With Π the image of $r^{-1}\sigma(dr)$ under $r \mapsto 1/r$, $\int y^{-1} e^{-z/y}\, \Pi(dy) = \ell(z)$, the comparison shows that Π is a Lévy measure with $\int \min(1,y)\, \Pi(dy) < \infty$, and $\lambda = \Upsilon(\mu)$ for the μ with drift d and Lévy measure Π. Uniqueness: ℓ determines σ, hence Π, and the drift is $\lim \Psi(s)/s$. By (14.2.3), $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(-s) = -ds - \int sx(1+sx)^{-1}\, \Pi(dx) = -\Psi_{\Upsilon(\mu)}(s)$. Finally $\int z^n \ell(z)\, dz = n! \int y^n\, \Pi(dy)$ gives $\kappa_n(\Upsilon(\mu)) = n!\, \kappa_n(\mu)$ for $n \geq 2$; for $n = 1$ the common drift is added on both sides.

(b) Let μ be self decomposable with canonical function k. With $r = 1/y$,

$$z\, \ell(z) = \int_0^\infty \frac{z}{y^2}\, e^{-z/y}\, k(y)\, dy = \int_0^\infty z\, e^{-zr}\, k(1/r)\, dr = \int_{(0,\infty)} e^{-zr}\, d_r\big[k(1/r)\big],$$

integrating by parts: $k(1/r) \to 0$ as $r \to 0$, and $e^{-zr}k(1/r) \to 0$ as $r \to \infty$ because $yk(y) \le \int_0^y k \to 0$ as $y \to 0$. Since $r \mapsto k(1/r)$ is nondecreasing, $\widetilde{U} = d_r[k(1/r)]$ is a positive measure with $\widetilde{U}((0,u]) = k(1/u)$, and $z\ell(z) = \int e^{-zr}\,\widetilde{U}(dr)$ says that $\Upsilon(\mu)$ is a GGC with Thorin measure $\widetilde{U}$ (Section 4.1.1). Conversely, let $\Upsilon(\mu)$ be a GGC with drift d and Thorin measure $\widetilde{U}$, and put $k(v) = \widetilde{U}((0,1/v])$, nonincreasing with $k(\infty) = 0$. By Fubini's theorem $\int_0^1 k(y)\,dy = \int \min(1,1/r)\,\widetilde{U}(dr)$ and $\int_1^\infty k(y)\,dy/y = \int_{(0,1)} \log(1/r)\,\widetilde{U}(dr)$, both finite by the integrability conditions of (4.1.1). So the self decomposable law μ' with drift d and canonical function k exists, $\Upsilon(\mu') = \Upsilon(\mu)$ by the first part, and $\mu = \mu'$ by the uniqueness in (a). The same construction from an arbitrary GGC gives surjectivity, and (a) gives injectivity.

(c) Now $\widetilde{U}((0,u]) = k(1/u) = \int e^{-t/u}\,U(dt) = \int \Pr(t/E \le u)\,U(dt)$, and differentiating, $\widetilde{U}(du) = u^{-2}\int te^{-t/u}\,U(dt)\,du = u^{-2}(-k')(1/u)\,du$. Since $\Psi_\mu(s) = ds + \int \log(1+s/t)\,U(dt)$ and, by parts, $\int_0^\infty \log(1+sx/t)\,e^{-x}dx = \int_0^\infty e^{-x}(x + t/s)^{-1}dx = e^{t/s}E_1(t/s)$, the formula for $\Psi_{\Upsilon(\mu)}$ follows from (14.2.6). Finally $\int u^p\,\widetilde{U}(du) = \int \mathbb{E}[(t/E)^p]\,U(dt)$ and $\mathbb{E}E^{-p} = \Gamma(1-p)$. □

Part (b) is the half line reading of $\Upsilon(\mathrm{SD}) =$ Thorin class [20], which [291] pointed out through (14.2.7). For the standard exponential law, $k(x) = e^{-x}$ and $\Upsilon(\mathrm{Exp}(1))$ has Lévy density $\int_0^\infty y^{-2}e^{-z/y-y}dy = 2z^{-1/2}K_1(2\sqrt{z})$.

14.2.3 The Upsilon image and the free counterpart of the clock

The centre clock T has Laplace transform $1/G(u)$, $G(u) = \prod_k(1+u/a_k)$ (2.1.1); it is infinitely divisible with no drift and Lévy measure $W(t)\,dt/t$, and with no hypothesis self decomposable with canonical function W (Theorem 8.1.1); $h = -W'$ is the Lévy density of its driver (Proposition 14.1.1). The Stieltjes moments of Section 8.1 are $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_k a_k^{-n-1}$, and since $-\log G(u) = \sum_{n\ge1}(-1)^n m_{n-1}u^n/n$ near 0, $\kappa_n(T) = (n-1)!\,m_{n-1}$.

Theorem 14.2.3. *(a) With no hypothesis, $\Upsilon(T)$ is a GGC with no drift and Thorin measure $\widetilde{U}(du) = u^{-2}h(1/u)\,du$, so that $\widetilde{U}((0,u]) = W(1/u) > 0$ for every $u > 0$. Its Laplace exponent is*

$$\Psi_{\Upsilon(T)}(s) = \int_0^\infty \log G(sx)\,e^{-x}\,dx = \sum_k e^{a_k/s}E_1(a_k/s), \qquad s > 0, \quad (14.2.8)$$

with the principal branch of E_1; its Lévy density is the completely monotone function $\ell(z) = z^{-1}\int_0^\infty W(z/w)\,e^{-w}\,dw$; its cumulants are $\kappa_n(\Upsilon(T)) = n!\,(n-1)!\,m_{n-1}$; and $\int u^{-n}\,\widetilde{U}(du) = n!\,m_{n-1}$ for $n \ge 1$. Moreover $\mathbb{E}e^{\lambda\Upsilon(T)} = \infty$ for every $\lambda > 0$, while $\mathbb{E}e^{\lambda T} < \infty$ exactly when $\lambda < \gamma_1^2$.

(b) With no hypothesis, $\Lambda_{\mathrm{BP}}(T)$ is free regular, freely self decomposable and unimodal. Its free cumulant transform is

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(z) = \int_0^\infty \frac{z\,W(x)}{1-zx}\,dx, \qquad z \in \mathbb{C}^-, \tag{14.2.9}$$

where the right side continues it analytically to $\mathbb{C}\backslash(0,\infty)$, with $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ for $s>0$; its free cumulants are $(n-1)!\,m_{n-1}$, the cumulants of T. Its Cauchy transform satisfies

$$\Psi_{\Upsilon(T)}\big(-G_{\Lambda_{\mathrm{BP}}(T)}(z)\big) = 1 - z\,G_{\Lambda_{\mathrm{BP}}(T)}(z), \qquad z \in \mathbb{C}^+, \tag{14.2.10}$$

where $\Psi_{\Upsilon(T)}$ is continued analytically to $\mathbb{C}\setminus(-\infty,0)$ by the right side of (14.2.6)*, and its support lies in $[c,\infty)$, where $c = \max_{s>0}(\Psi_{\Upsilon(T)}(s)-1)/s > 0$.*

(c) RH holds if and only if T is the Upsilon image of an infinitely divisible law on $[0,\infty)$. In that case $T = \Upsilon(\mu_0)$, where μ_0 is the self decomposable law with no drift and canonical function $N(v^{-1/2})$, that is with Lévy measure $N(x^{-1/2})\,dx/x$; μ_0 is neither a GGC nor in BO. With no hypothesis, T is not the Upsilon image of a law in BO.

Proof. (a) Proposition 14.2.2(b) applies with $k = W$, $W' = -h$, and the drift is zero. The first form of $\Psi_{\Upsilon(T)}$ is (14.2.6) with $\Psi_T = \log G$. For the second, $\mathrm{Re}(1/a_k) > 0$, so $w = sx/a_k$ has $\mathrm{Re}\,w \ge 0$ and $|\log(1+w)| = |\int_0^1 w\,d\tau/(1+\tau w)| \le |w|$; since $\sum_k |a_k|^{-1} < \infty$, the series $\log G(sx) = \sum_k \log(1+sx/a_k)$ integrates term by term, and each term gives $e^{a_k/s}E_1(a_k/s)$ as in the proof of Proposition 14.2.2(c), valid for $\mathrm{Re}\,a_k > 0$ because $e^z E_1(z) = \int_0^\infty e^{-w}(w+z)^{-1}dw$ for $\mathrm{Re}\,z > 0$. The Lévy density of Proposition 14.2.2(a) is $\int_0^\infty y^{-2}e^{-z/y}W(y)\,dy$, and $y = z/w$ gives the stated form. The cumulants follow from $\kappa_n(\Upsilon(T)) = n!\,\kappa_n(T)$, and $\int u^{-n}\widetilde{U}(du) = \int_0^\infty v^n h(v)\,dv = n\int_0^\infty v^{n-1}W(v)\,dv = n!\,m_{n-1}$, the boundary terms vanishing because $W(v) = O(v^{-1/2}\log(1/v))$ at 0 and decays exponentially. By Sato's moment theorem [325, Theorem 25.17], $\mathbb{E}e^{\lambda X} < \infty$ exactly when the Lévy measure integrates $e^{\lambda x}$ over $(1,\infty)$. For $\Upsilon(T)$, $z\ell(z) = \int e^{-zr}\widetilde{U}(dr) \ge e^{-z\varepsilon}W(1/\varepsilon)$, and with $\varepsilon < \lambda$ the integral diverges. For T, every zero but the first has $\mathrm{Re}\,a_k \ge \gamma_2^2 - \frac14 > \gamma_1^2$, so $W(t) = e^{-\gamma_1^2 t}(1+o(1))$ and $\int_1^\infty e^{\lambda t}W(t)\,dt/t$ converges exactly when $\lambda < \gamma_1^2$.

(b) With $\Pi(dx) = W(x)\,dx/x$, $\int \min(1,x)\,\Pi(dx) < \infty$, $\Lambda_{\mathrm{BP}}(T)$ is free regular by [5, Theorem 9], freely self decomposable by [21] and unimodal by [109]. Formula (14.2.9) is (14.2.3), and $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ is Proposition 14.2.2(a); the right sides of (14.2.9) and (14.2.6) give $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(w) = -\Psi_{\Upsilon(T)}(-w)$ on $\mathbb{C}\setminus(0,\infty)$. Since Π has all moments, the free cumulants are the cumulants of T. Write $G_\Lambda = G_{\Lambda_{\mathrm{BP}}(T)}$. Where $(1/G_\Lambda)^{-1}(1/G_\Lambda(z)) = z$, the definition of φ gives $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(G_\Lambda(z)) = zG_\Lambda(z) - 1$; both sides are analytic on $\mathbb{C}^+$, which G_Λ maps into

$\mathbb{C}^-$, so this is (14.2.10) on $\mathbb{C}^+$. For the support, put $x(s) = (\Psi_{\Upsilon(T)}(s)-1)/s$. Then $s^2x'(s) = f(s) = s\Psi'_{\Upsilon(T)}(s)-\Psi_{\Upsilon(T)}(s)+1$, with $f(0+) = 1$, $f'(s) = s\Psi''_{\Upsilon(T)}(s) < 0$, and $f(s) \to -\infty$, since $s\Psi' - \Psi = -\int(1 - e^{-sz} - sze^{-sz})\ell(z)\,dz$ and ℓ is not integrable at 0. So x increases from $-\infty$ to its maximum $c = x(s^*) = \Psi'_{\Upsilon(T)}(s^*)$ on $(0, s^*]$ and decreases to 0 on $[s^*, \infty)$; in particular $c > 0$. For $x < 0$, outside the support, $G_\Lambda(x) < 0$ and (14.2.10) extends by continuity, so $S = -G_\Lambda(x) > 0$ solves $x(S) = x$, whose only positive solution is $s(x)$, the inverse of $x(\cdot)$ on $(0, s^*)$. This inverse is analytic on a neighbourhood V of $(-\infty, c)$, a union of discs centred on that half line missing $[c, \infty)$, and $-s$ agrees with G_Λ on $(-\infty, 0)$, hence on the connected set $V \setminus [0, c)$. So G_Λ extends analytically across $(-\infty, c)$ with real values, and by Stieltjes inversion $\Lambda_{\mathrm{BP}}(T)$ gives no mass to $(-\infty, c)$.

(c) By Proposition 14.2.2(a) the Upsilon images of the infinitely divisible laws on $[0, \infty)$ form the class BO, and the clock is in BO if and only if RH holds (Section 15.1.4 and Table 4.1). Under RH, $W(z)/z = \sum_{\gamma>0}\int_{\gamma^2}^\infty e^{-zr}\,dr = \int_0^\infty e^{-zr}N(\sqrt{r})\,dr$, so $\sigma(dr) = N(\sqrt{r})\,dr$ in the proof of Proposition 14.2.2(a), and the image of $r^{-1}\sigma(dr)$ under $r \mapsto 1/r$ is $N(x^{-1/2})\,dx/x$. Its canonical function $N(v^{-1/2})$ is nonincreasing, so μ_0 is self decomposable, and by Proposition 14.2.2(b) the Thorin measure of $\Upsilon(\mu_0)$ has distribution function $N(\sqrt{u})$, that is it is $\sum_{\gamma>0}\delta_{\gamma^2}$, as Proposition 2.1.1 requires. The canonical function and Lévy density of μ_0 vanish on (γ_1^{-2}, ∞) without vanishing identically, so neither is completely monotone: μ_0 is not a GGC and not in BO. If $T = \Upsilon(\mu)$ with μ in BO, then T is in BO, RH holds, and $\mu = \mu_0$ by the uniqueness in Proposition 14.2.2(a), which is impossible. □

The preimage μ_0 of Theorem 14.2.3(c) has a reading through the Dickman law D, with $\mathbb{E}e^{-sD} = \exp(-\mathrm{Ein}(s))$ and $\mathrm{Ein}(s) = \int_0^s(1 - e^{-v})\,dv/v$. Integrating $\mathrm{Ein}(ax) = \int_0^1(1 - e^{-axv})\,dv/v$ against e^{-x} gives $\int_0^\infty \mathrm{Ein}(ax)e^{-x}dx = \log(1 + a)$, so $\Upsilon(D/\gamma^2)$ is the exponential law of rate γ^2, and since $\mathrm{Ein}(s/\gamma^2) = \int_0^{\gamma^{-2}}(1 - e^{-sy})\,dy/y$, under RH

$$\mu_0 = \text{law of } \sum_{\gamma>0} D_\gamma/\gamma^2, \qquad D_\gamma \text{ independent Dickman variables,}$$

with Lévy measure $\sum_\gamma \mathbf{1}_{(0,\gamma^{-2})}(y)\,dy/y = N(y^{-1/2})\,dy/y$ and all jumps at most γ_1^{-2}. With no hypothesis $\psi_0(s) = \sum_k \mathrm{Ein}(s/a_k)$ is entire and $\int_0^\infty \psi_0(sx)e^{-x}dx = \log G(s)$ for $s > 0$, termwise because $\mathrm{Re}(sx/a_k) > 0$ and $|\mathrm{Ein}(z)| \le |z|$ for $\mathrm{Re}\,z \ge 0$. Under RH the density of μ_0, computed by Fourier inversion since the Laplace transform grows along the negative axis, has mass 0.99993 on $[0.008, 0.045]$ and mean 0.023104, against $p_1 = 0.0231050$.

Proposition 14.2.4. *Let $\mathcal{C}_\xi(z) = \log\{\xi(\frac12)/\xi(\frac12 + \sqrt{-z})\} = \sum_{n\ge1} m_{n-1}z^n/n$ for $|z| < \gamma_1^2$. RH holds if and only if there is a freely infinitely divisible probability measure ν_ξ whose free cumulant transform is $\mathcal{C}_\xi$. Under RH, $\nu_\xi = \Lambda_{\mathrm{BP}}(\mu_0)$, its*

free cumulants are m_{n-1}/n, its free Lévy measure is $N(y^{-1/2})\,dy/y$, its support is compact and it is freely self decomposable.

Proof. Under RH, (14.2.5) gives $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu_0)}(i\zeta) = C_{\Upsilon(\mu_0)}(\zeta) = C_T(\zeta) = \mathcal{C}_\xi(i\zeta)$, and Λ_{BP} preserves Lévy measures and self decomposability. Conversely let ν_ξ exist. Since $\mathcal{C}_\xi$ is analytic at 0, ν_ξ has compact support and free cumulants m_{n-1}/n; for $n \ge 3$ they are the moments $\int t^n \rho(dt)$ of its free Lévy measure ρ, and $|m_{n-1}|/n \le \gamma_1^{-2(n-1)} \sum_k |a_k|^{-1}$ because $|a_k| = \gamma^2 + \delta^2 \ge \gamma_1^2$, so $t^2\rho$ lives on $[-\gamma_1^{-2}, \gamma_1^{-2}]$. Then $\sum_{n\ge3} m_{n-1} z^n/n = -\sum_k \{\log(1 - z/a_k) + z/a_k + z^2/2a_k^2\} = \int \frac{z^3 t^3}{1-zt}\,\rho(dt)$ is analytic off a subset of the real axis, while the left side has a logarithmic singularity at every a_k. So every a_k is real, and RH holds since ξ has no real zeros. □

The Cauchy transform of ν_ξ solves $\mathcal{C}_\xi(G_{\nu_\xi}(z))/G_{\nu_\xi}(z) + 1/G_{\nu_\xi}(z) = z$, so the density follows from ξ alone. Solved at 30 digits along $z = x + i\eta$ with η down to 10^{-11}, it is supported on an interval close to $[0.0166, 0.0348]$ with steep edges, with mode 0.0193, where it equals 98.6. The trapezoid rule on a grid of spacing 0.00025 gives mass 0.99992, mean 0.0231057 and variance 1.8592×10^{-5}, against p_1 and $m_1/2 = 1.8586 \times 10^{-5}$. This is consistent with RH and no evidence beyond the zeros already verified, since the low cumulants are fixed by the first zeros while the existence in Proposition 14.2.4 concerns all of them.

Under RH, $\widetilde{U}$ spreads the atom at γ^2 into the law of γ^2/E (Proposition 14.2.2(c)), and in the chart $\theta = \sqrt{u}$

$$\widetilde{U}((0,\theta^2]) = W(\theta^{-2}) = \sum_{\gamma>0} e^{-\gamma^2/\theta^2} = \mathbb{E}\, N(\theta\sqrt{E}),$$

a smoothing of the zero count. With no hypothesis, $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 8.1) and the Riemann von Mangoldt formula give $W(\theta^{-2})/N(\theta) \to \sqrt{\pi}/2 = 0.8862$. Under RH the expectation gives the next term: with $N_0(\theta) = (\theta/2\pi)\log(\theta/2\pi e) + \frac{7}{8}$, $\mathbb{E}N_0(\theta\sqrt{E}) = (\theta/4\sqrt{\pi})\big(\log(\theta/2\pi e) + \frac{1}{2}\psi(\frac{3}{2})\big) + \frac{7}{8}$, where $\psi(\frac{3}{2}) = 2 - \gamma_E - 2\log 2 = 0.0365$; so $W(\theta^{-2})/N_0(\theta)$ exceeds $\sqrt{\pi}/2$ by a relative amount of about $\psi(\frac{3}{2})/(2\log(\theta/2\pi e))$.

At the left end the free counterpart differs. The clock and its Upsilon image have no drift and infinite Lévy measure, so their supports are $[0,\infty)$ [325, Section 24] and their densities are super flat at the origin; the free law has a hard edge at $c > 0$. Since $\Psi''_{\Upsilon(T)}(s^*) \ne 0$, the solution of $x(s) = x$ leaves the real axis as a square root at $x = c$, and the density $\operatorname{Im} s/\pi$ of $\Lambda_{\mathrm{BP}}(T)$ vanishes like $(x-c)^{1/2}$. At the other end only the first zero matters in $\ell(z) = z^{-1}\int_0^\infty W(z/w)e^{-w}dw$, the saddle point of $e^{-\gamma_1^2 z/w - w}$ is at $w = \gamma_1\sqrt{z}$, and

$$\ell(z) \sim 2\gamma_1 z^{-1/2} K_1(2\gamma_1\sqrt{z}) \sim \sqrt{\pi}\,\gamma_1^{1/2} z^{-3/4} e^{-2\gamma_1\sqrt{z}};$$

under RH, $\ell(z) = 2z^{-1/2}\sum_{\gamma>0} \gamma K_1(2\gamma\sqrt{z})$ exactly. The computed density of $\Upsilon(T)$ is 2.552 times ℓ at $x = 0.2$ and 1.828 times ℓ at $x = 0.4$: the Lévy density takes over the tail only slowly.

Table 14.5 checks (14.2.8). The integral is computed from ξ alone at 30 digits, and the sums use one of two tails: the tail of order one, $s(\log(\gamma_N/2\pi)+1)/(2\pi\gamma_N) = s\int_{\gamma_N}^{\infty}\theta^{-2}dN_0(\theta)$, the choice of [291], and the full smooth tail $\int_{\gamma_N}^{\infty}\phi(\theta^2/s)\,dN_0(\theta)+\phi(\gamma_N^2/s)(N_0(\gamma_N)-N)$, with $\phi(z)=e^zE_1(z)$ and N the number of zeros used. The note's Table 1 (integral and sums over 300 zeros without and with the tail of order one at $s=1$, 10, 100) is reproduced to every digit shown (0.0230681050, 0.0214685871 and 0.0230714840 at $s=1$). With 1700 zeros and the full tail the difference is $-6.29\times10^{-11}s$ for every s up to 10^3: it is the smooth tail of $\sum\gamma^{-2}$ beyond $\gamma_{1700}=2197.26$, where $\phi(\gamma^2/s)\approx s/\gamma^2$.

		sum over the zeros minus the integral				
s	$\int_0^\infty \log G(sx)\,e^{-x}\,dx$	300 zeros	300, tail 1	300, full tail	1700, tail 1	1700, full tail
1	0.0230681049610841	-0.0016	3.38×10^{-6}	-1.32×10^{-9}	1.2×10^{-7}	-6.29×10^{-11}
10	0.227587470315411	-0.016	3.39×10^{-5}	-1.32×10^{-8}	1.2×10^{-6}	-6.29×10^{-10}
100	2.07560251202393	-0.16	3.54×10^{-4}	-1.32×10^{-7}	1.23×10^{-5}	-6.29×10^{-9}
10^3	15.2403221009594	-1.6	4.96×10^{-3}	-1.31×10^{-6}	1.51×10^{-4}	-6.29×10^{-8}
10^4	87.4852790298693	-15.8	0.187	-1.24×10^{-5}	4.28×10^{-3}	-6.27×10^{-7}

Table 14.5: The Laplace exponent of $\Upsilon(T)$, which is minus the free cumulant transform of $\Lambda_{\mathrm{BP}}(T)$ at $-s$: the integral of (14.2.8) from ξ alone, and the sums over the first 300 and 1700 zeros, with no tail, with the tail of order one and with the full smooth tail, as differences from the integral. Computed.

Table 14.6 and the right panel of Figure 14.3 compare the three laws. The density of T is the fixed Talbot inversion of $1/G$ at 25 digits on the grid $0.002(0.0005)0.15$. That of $\Upsilon(T)$ is the fixed Talbot inversion of $e^{-\Psi_{\Upsilon(T)}}$, with $\Psi_{\Upsilon(T)}$ from the 1700 zeros and the tail $u\tau_1-u^2\tau_2+2u^3\tau_3$, τ_j the smooth tail of $\sum\gamma^{-2j}$ with its boundary term, in double precision from $x=0.0125$ on and at 30 digits below. That of $\Lambda_{\mathrm{BP}}(T)$ is $\operatorname{Im}s/\pi$, with s the solution of $\Psi_{\Upsilon(T)}(s)=1+xs$ in the upper half plane given by (14.2.10), found by Newton's method on 500 points of $[c,0.15]$, spaced quadratically near c, with residuals below 4.6×10^{-15}. Here $c=0.01529224618$ and $s^*=387.9402$.

14.2.4 The semicircle as a local model for the Jacobi parameters

Section 8.1 builds from $m_0,\dots,m_{2N-1}$, that is from the Taylor coefficients of Ξ at the centre alone, the Jacobi matrix J_N of $\nu=\sum_j a_j^{-1}\delta_{1/a_j}$, with diagonal $\alpha_1,\dots,\alpha_N$ and positive off diagonal $\beta_1,\dots,\beta_{N-1}$.

For constant parameters $\alpha_n=\alpha$, $\beta_n=\beta$ the Jacobi matrix is $\alpha+\beta(S+S^*)$ on $\ell^2(\{0,1,2,\dots\})$, S the unilateral shift, and the spectral measure of its first basis vector is the semicircle law on $[\alpha-2\beta,\alpha+2\beta]$. The $N\times N$ truncations have eigenvalues $\alpha+2\beta\cos(\pi k/(N+1))$, $1\le k\le N$, distributed in the limit by the arcsine law $\left(\pi\sqrt{4\beta^2-(x-\alpha)^2}\right)^{-1}$ on the same interval. For slowly varying

	T	$\Upsilon(T)$	$\Lambda_{\mathrm{BP}}(T)$
total mass	1.00000000	1.00000053	1.00000021
mean	0.023104993	0.023104991	0.023104991
variance	3.71726×10^{-5}	7.43455×10^{-5}	3.71726×10^{-5}
third central moment	2.8835×10^{-7}	1.7301×10^{-6}	2.8835×10^{-7}
mode	0.0200	0.0185	0.017711
density at the mode	80.1042	72.6771	91.3878
left end of the support	0	0	0.01529224618

Table 14.6: The clock T, its Upsilon image and its free counterpart, from the densities of Figure 14.3. The exact means are $m_0 = 0.023104993$; the variances $m_1 = 3.71726 \times 10^{-5}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $2m_1 = 7.43452 \times 10^{-5}$ for $\Upsilon(T)$; the third central moments $2m_2 = 2.8835 \times 10^{-7}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $12m_2 = 1.7301 \times 10^{-6}$ for $\Upsilon(T)$. The modes of T and $\Upsilon(T)$ are grid points of spacing 0.0005, that of $\Lambda_{\mathrm{BP}}(T)$ of spacing about 7×10^{-5}. Computed.

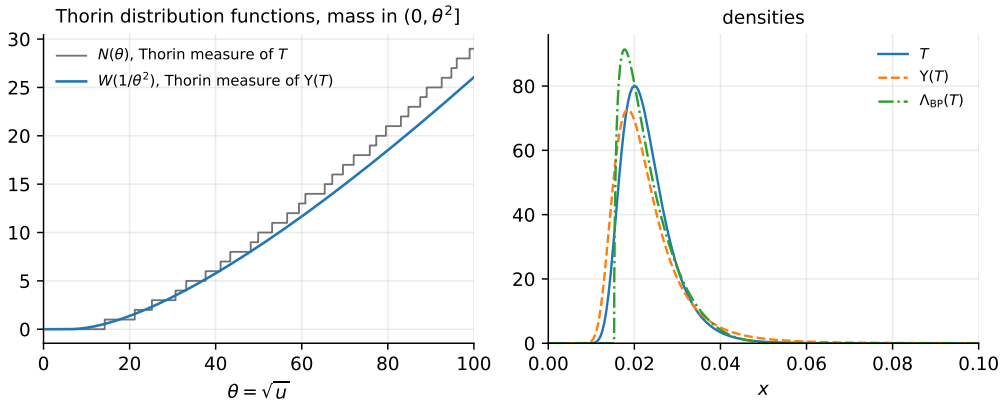


Figure 14.3: Left: the distribution function $W(\theta^{-2})$ of the Thorin measure of $\Upsilon(T)$ in the chart $\theta = \sqrt{u}$, against the zero count $N(\theta)$, which under RH is the distribution function of the Thorin measure of T in the same chart. Right: the densities of T, of $\Upsilon(T)$ and of $\Lambda_{\mathrm{BP}}(T)$ on $[0, 0.1]$; the last vanishes to the left of $c = 0.0153$. Computed.

parameters one integrates local arcsine laws over the rows; for $\alpha_{n,N} \to a(n/N)$, $\beta_{n,N} \to b(n/N)$ this is the theorem of Kuijlaars and Van Assche [195]: the zero distribution of the orthogonal polynomials tends to $\int_0^1 \omega_{[a(t)-2b(t),a(t)+2b(t)]}\,dt$, with $\omega_{[A,B]}$ the arcsine law on $[A,B]$. The parameters of ν decrease to zero with no such scaling, and we need the largest eigenvalues, an edge statement; for ν the local model is a heuristic. Since ν has its atoms in $(0,\gamma_1^{-2}]$ and accumulates only at 0, the local interval should reach down to zero, which suggests $\alpha_n \approx 2\beta_n$. Write $L(n) = \alpha_n + 2\beta_n$ for its right end, taken to decrease in n, with inverse $n(L)$.

The eigenvalues of J_N above $x > 0$ approximate the $N(x^{-1/2})$ atoms $1/\gamma_j^2$ above x. In the local model row n contributes the mass $\frac{2}{\pi}\arccos\sqrt{x/L(n)}_+$ of the arcsine law on $[0,L(n)]$ above x, and nothing when $x \ge L(n)$, so that

$$N\big(x^{-1/2}\big) = \int_0^\infty \frac{2}{\pi}\arccos\sqrt{x/L(n)}_+\,dn = \frac{1}{\pi}\int_x^\infty n(L)\,\frac{\sqrt{x}\,dL}{L\sqrt{L-x}}, \qquad (14.2.11)$$

by parts in L. With $u = 1/L$ and $w = 1/x$ this is Abel's equation

$$\int_0^w \frac{f(u)\,du}{\sqrt{w-u}} = \pi N(\sqrt{w}), \qquad f(u) = u^{-1/2}\,n(1/u). \qquad (14.2.12)$$

Proposition 14.2.5. *Let N be nondecreasing and right continuous on $[0,\infty)$, with $N = 0$ on $[0,\theta_0)$ for some $\theta_0 > 0$. Then*

$$n(L) = \int_{(0,L^{-1/2})} \frac{dN(T)}{\sqrt{1-LT^2}} \qquad (14.2.13)$$

gives a locally integrable solution f of (14.2.12), with $N(\sqrt{w})$ replaced by its left limit at the jumps of N, and it is the only locally integrable solution. If $dN(T) = c\,dT$ on $(0,\infty)$, then $n(L) = \frac{\pi}{2}cL^{-1/2}$, so that $L_n = (\pi c/2n)^2$, which is $\gamma_{2n/\pi}^{-2}$ for the ordinates $\gamma_k = k/c$ of that count.

Proof. For $0 < a < b$ the substitution $u = 1/L$ gives

$$\int_a^b \frac{dL}{L\sqrt{(L-a)(b-L)}} = \int_{1/b}^{1/a} \frac{du}{\sqrt{(1-au)(bu-1)}}$$
$$= \frac{1}{\sqrt{ab}}\int_{1/b}^{1/a} \frac{du}{\sqrt{(1/a-u)(u-1/b)}} = \frac{\pi}{\sqrt{ab}}.$$

Insert (14.2.13) in the right side of (14.2.11), equivalent to (14.2.12), and exchange the integrals, all terms being positive. A point T of the support of dN contributes only when $T < x^{-1/2}$, through $L \in (x, T^{-2})$, and with $b = T^{-2}$ and $\sqrt{1-LT^2} = \sqrt{(b-L)/b}$ its contribution is

$$\frac{\sqrt{x}}{\pi}\int_x^b \frac{\sqrt{b}\,dL}{L\sqrt{(L-x)(b-L)}} = 1.$$

So the right side is N of $(0, x^{-1/2})$, the left limit of N at $x^{-1/2}$. Local integrability: n vanishes for $L > \theta_0^{-2}$, and for $0 < a < b$, $\int_a^b n(L)\,dL \le \int_{(0,a^{-1/2})} dN(T) \int_0^{T^{-2}} dL/\sqrt{1-LT^2} = 2\int_{(0,a^{-1/2})} T^{-2}\,dN(T) < \infty$. If g is the difference of two locally integrable solutions, $\int_0^w g(u)(w-u)^{-1/2}du = 0$ for every w; integrating against $(y-w)^{-1/2}$ over $(0,y)$ and using $\int_u^y dw/\sqrt{(y-w)(w-u)} = \pi$ gives $\int_0^y g = 0$ for every y, so $g = 0$ almost everywhere. Finally $\int_0^{L^{-1/2}} c\,dT/\sqrt{1-LT^2} = cL^{-1/2}\arcsin 1$. $\square$

The local model therefore predicts $\alpha_n \approx 2\beta_n \approx \tfrac12 L_n$ with $n(L_n) = n$ [291]. For the zeros of ξ we use the smooth density $dN_0(T) = (2\pi)^{-1}\log(T/2\pi)\,dT$ on $T > 2\pi$, for which $T = L^{-1/2}\sin\varphi$ gives

$$n(L) = \frac{1}{2\pi\sqrt{L}} \int_{\varphi_0}^{\pi/2} \log\frac{\sin\varphi}{2\pi\sqrt{L}}\,d\varphi, \qquad \sin\varphi_0 = 2\pi\sqrt{L}.$$

As $L \to 0$, $\int_0^{\pi/2}\log\sin\varphi\,d\varphi = -\tfrac{\pi}{2}\log 2$ gives $n(L) = (4\sqrt{L})^{-1}\log(1/(4\pi\sqrt{L})) + O(1)$, while the linear count rule of Proposition 14.2.5 with the ordinates of the smooth count, $N_0(\gamma_r) = r$, gives $\tfrac12 L_n = \tfrac12\gamma_{2n/\pi}^{-2}$, corresponding to $n = (4\sqrt{L})^{-1}\log(1/(2\pi e\sqrt{L}))$. The two differ by a factor that tends to one only logarithmically, which is why the cruder rule $\alpha_n \approx \tfrac12\gamma_{2n/\pi}^{-2}$ is off by a factor near 1.3.

Table 14.7 compares the computed parameters with the two rules. We compute $M_0, \dots, M_{100}$ at 1000 digits by the trapezoid rule of step $1/600$ on $[0, 3.6]$, the moments $m_0, \dots, m_{99}$, and the Cholesky factorisation of $\mathcal{H}_{50}$. The matrix $L^{-1}[m_{i+j+1}]L^{-\top}$ is tridiagonal to 1.3×10^{-836}, and its entries change by at most 2.3×10^{-429} relatively when the factorisation runs at 600 digits. The first parameters are irregular, $\alpha_1, \dots, \alpha_5 = (1.61, 3.50, 1.80, 1.22, 1.07) \times 10^{-3}$ and $\beta_1, \dots, \beta_5 = (1.91, 1.15, 0.697, 0.554, 0.498) \times 10^{-3}$, and the model is meant for larger n. Over $17 \le n \le 49$ the ratio $\alpha_n/\tfrac12 L_n$ has mean 1.033 and standard deviation 0.060 (range 0.940 to 1.178), and $2\beta_n/\tfrac12 L_n$ has mean 1.012 and standard deviation 0.060 (range 0.893 to 1.134); the columns α_n, $2\beta_n$, $\tfrac12 L_n$ and these ratios reproduce Table 6 of [291] to the digits shown there. Over the same range $\alpha_n/(2\beta_n)$ has mean 1.021 and standard deviation 0.047, and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\tfrac14$ in the local model, has mean 0.2502 and standard deviation 0.013 (mean 0.256 and standard deviation 0.059 over $1 \le n \le 49$). The cruder rule falls short: α_n and $2\beta_n$ exceed it by 1.286 and 1.26 on average, with standard deviations 0.081 and 0.078.

Two cautions apply. The rule $n(L_n) = n$ and the midpoint rule $n(L_n') = n - \tfrac12$ differ by a shift of order one, which the asymptotics cannot see; they move the mean ratios by two per cent, to 1.011 and 0.991, within the scatter. And the scatter itself, about six per cent, is unexplained: the note attributes it to the discreteness of the zeros, which the smooth density ignores, and that remains to be tested (Problem 25.3). So the local semicircle model predicts the size of the

Jacobi parameters to a few per cent over $17 \le n \le 49$ with no fitted constant, but this is not a proof of an asymptotic law.

	$10^5\times$			ratio to $\frac12 L_n$		$10^5\times$	ratio to $\frac12 L'_n$		$10^5\times$	
n	α_n	$2\beta_n$	$\frac12 L_n$	α_n	$2\beta_n$	$\frac12 L'_n$	α_n	$2\beta_n$	$\frac12\gamma_{2n/\pi}^{-2}$	$\frac{\beta_n^2}{\alpha_n\alpha_{n+1}}$
5	107.3	99.54	84.84	1.27	1.17	94.11	1.14	1.06	61.8	0.2665
9	56.96	51.12	45.73	1.25	1.12	48.70	1.17	1.05	34.5	0.2548
13	30.20	32.52	30.07	1.00	1.08	31.48	0.959	1.03	23.2	0.2611
17	21.71	21.51	21.83	0.995	0.986	22.64	0.959	0.950	17.0	0.2362
21	18.50	16.55	16.82	1.10	0.983	17.34	1.07	0.954	13.3	0.2479
25	15.90	14.58	13.50	1.18	1.08	13.85	1.15	1.05	10.7	0.2588
29	11.25	11.45	11.16	1.01	1.03	11.41	0.986	1.00	8.93	0.2550
33	9.530	9.336	9.424	1.01	0.991	9.615	0.991	0.971	7.59	0.2647
37	8.704	8.163	8.101	1.07	1.01	8.249	1.06	0.990	6.56	0.2463
41	6.998	7.053	7.062	0.991	0.999	7.179	0.975	0.982	5.74	0.2618
45	6.498	6.150	6.228	1.04	0.987	6.323	1.03	0.973	5.08	0.2481
49	5.957	6.172	5.547	1.07	1.11	5.625	1.06	1.10	4.54	0.2738

Table 14.7: The Jacobi parameters of ν against the local semicircle model: α_n and $2\beta_n$; half the right end $\frac12 L_n$ of the local interval from (14.2.13) with the smooth density and $n(L_n) = n$, and the ratios to it; the same with the midpoint rule $n(L'_n) = n - \frac12$; the cruder rule $\frac12\gamma_{2n/\pi}^{-2}$, with $N_0(\gamma_r) = r$; and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac14$ in the local model. Computed.

14.2.5 Gamma(2), Laguerre polynomials and finite free convolution

Williams' variable [363] $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2}\Gamma_n$, with independent Gamma(2) variables Γ_n, the sum of two independent copies of the hitting time $J = \sum_n \frac{2}{\pi^2 n^2} e_n$ of Section 12.1.1, satisfies $2\xi(s) = \mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$ (12.1.1) [33]; the RH content of this identity lies on its Mellin side (Theorem 11.1.21 and Section 3.1.3). The cumulants of S_2 are $\kappa_m = 2(m-1)!\,(2/\pi^2)^m\zeta(2m)$.

Proposition 14.2.6. *For every complex s,*

$$\mathbb{E}\Big[\Big(\frac{\pi}{2}J\Big)^{s/2}\Big] = \frac{2(1-2^{1-s})\,\xi(s)}{s-1} = s\,\pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)(1-2^{1-s})\,\zeta(s), \qquad (14.2.14)$$

the value at $s = 1$ being $2\xi(1)\log 2 = \log 2$. Equivalently $J \overset{d}{=} V^{-2}S_2$, with V uniform on $[1,2]$ and independent of S_2. The function (14.2.14) *vanishes at the zeros of ξ and at the points $1 + 2\pi i k/\log 2$, $k \neq 0$, of the line $\operatorname{Re} s = 1$, and, unlike the zero set of ξ, its zero set is not symmetric under $s \mapsto 1 - s$.*

Proof. This is the relation of Biane, Pitman and Yor between J, their Σ_1, and S_2, their Σ_2 [33, Table 1 and (43) to (45)]. For $\operatorname{Re} p < 0$, $\int_0^\infty \lambda^{-p-1}e^{-\lambda x}d\lambda = \Gamma(-p)x^p$

and $\mathbb{E}e^{-\lambda J} = \sqrt{2\lambda}/\sinh\sqrt{2\lambda}$ give, with $r = \sqrt{2\lambda}$,

$$\mathbb{E}J^p = \frac{2^{p+1}}{\Gamma(-p)} \int_0^\infty \frac{r^{-2p}}{\sinh r}\, dr = \frac{2^{p+2}(1-2^{2p-1})\,\Gamma(1-2p)\,\zeta(1-2p)}{\Gamma(-p)},$$

by $1/\sinh r = 2\sum_{k\geq 0} e^{-(2k+1)r}$, which gives $\int_0^\infty r^{z-1}/\sinh r\, dr = 2(1-2^{-z})\Gamma(z)\zeta(z)$ for $\operatorname{Re} z > 1$. Put $p = s/2$ with $\operatorname{Re} s < 0$. The functional equation $\zeta(1-s) = \pi^{1/2-s}\Gamma(\frac{s}{2})\zeta(s)/\Gamma(\frac{1-s}{2})$, the duplication formula $\Gamma(1-s) = 2^{-s}\pi^{-1/2}\Gamma(\frac{1-s}{2})\Gamma(1-\frac{s}{2})$ and $\Gamma(1-\frac{s}{2})/\Gamma(-\frac{s}{2}) = -\frac{s}{2}$ turn this into $\mathbb{E}J^{s/2} = -2^{1-s/2}s(1-2^{s-1})\pi^{-s}\Gamma(\frac{s}{2})\zeta(s)$, and multiplying by $(\pi/2)^{s/2}$ gives (14.2.14). Both sides are entire, since J has moments of all real orders and the zero of $1-2^{1-s}$ at $s=1$ cancels the pole of ζ, so the identity holds for every s. Since $\mathbb{E}V^{-s} = \int_1^2 v^{-s}dv = (1-2^{1-s})/(s-1)$, the right side is $\mathbb{E}V^{-s}\,\mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$. The factor $1-2^{1-s}$ vanishes exactly at $1+2\pi ik/\log 2$; on the line $\operatorname{Re} s = 0$, the image of $\operatorname{Re} s = 1$ under $s \mapsto 1-s$, it equals -1 at the corresponding points $2\pi ik/\log 2$ and ξ has no zeros, so the zero set is not symmetric. □

The factor $(1-2^{1-s})/(s-1)$ adds zeros on $\operatorname{Re} s = 1$, the first at height $2\pi/\log 2 = 9.0647$, those of the alternating zeta function of Section 13.1.7.

Proposition 14.2.7. *Let $z_1, z_2, \ldots$ be independent standard complex Gaussian vectors in $\mathbb{C}^2$, with $\mathbb{E}z_n z_n^* = I_2$, let $c_n = 2/(\pi^2 n^2)$ and $W = \sum_n c_n z_n z_n^*$. Then $\operatorname{tr} W \overset{d}{=} S_2$, so that $2\xi(s) = \mathbb{E}(\frac{\pi}{2}\operatorname{tr} W)^{s/2}$, and*

$$\mathbb{E}\det(x-W) = x^2 - 2m_1 x + m_1^2 - m_2, \qquad m_k = \sum_n c_n^k, \quad m_1 = \tfrac{1}{3},\ m_2 = \tfrac{2}{45},$$

with the real roots $m_1 \pm \sqrt{m_2} = 0.122515$ and 0.544152. This is the finite free convolution $\boxplus_2$ of the polynomials $\mathbb{E}\det(x - c_n z_n z_n^*) = x(x-2c_n)$, and the mixed characteristic polynomial of Marcus, Spielman and Srivastava [230] of the matrices $c_n I_2$.

Proof. $|z_n|^2$ is the sum of two independent standard exponentials, so $\operatorname{tr} W \overset{d}{=} S_2$. For 2×2 matrices $\det(x-W) = x^2 - x\operatorname{tr} W + \det W$ and $\det W = \frac{1}{2}((\operatorname{tr} W)^2 - \operatorname{tr} W^2)$. Here $\mathbb{E}\operatorname{tr} W = 2m_1$, $\mathbb{E}(\operatorname{tr} W)^2 = 2m_2 + 4m_1^2$, and $\mathbb{E}\operatorname{tr} W^2 = \sum_{n,k} c_n c_k \mathbb{E}|z_n^* z_k|^2 = 6m_2 + 2(m_1^2 - m_2)$, since $\mathbb{E}|z_n|^4 = 6$ and $\mathbb{E}|z_n^* z_k|^2 = 2$ for $n \neq k$; hence $\mathbb{E}\det W = m_1^2 - m_2$. The terms $c_n z_n z_n^*$ are independent and unitarily invariant in law, so by [231, Theorem 1.2] and bilinearity the expected characteristic polynomial of a finite partial sum is the $\boxplus_2$ convolution of the $x^2 - 2c_n x$; adding $x(x-2c)$ to $x^2 - Ax + B$ gives $x^2 - (A+2c)x + B + Ac$, so the constant term is $2\sum_{n<k} c_n c_k = m_1^2 - m_2$ in the limit. □

Reading the exponent two of $(\sqrt{2\lambda}/\sinh\sqrt{2\lambda})^2$ as the size of the matrix is special to the sinh family of [33].

In the normalisation of [231] write $p(x)=\sum_k(-1)^k\binom{d}{k}e_kx^{d-k}$; the multiplicative convolution $\boxtimes_d$ multiplies the normalised coefficients, $e_k(p\boxtimes_d q)=e_k(p)e_k(q)$ [231, Definition 1.4], and by Szegő's theorem it preserves having only nonnegative roots [231, Theorem 1.6].

LEMMA 14.2.8. *Let $a>0$ and $(a)_k=a(a+1)\cdots(a+k-1)$.*

(i) The reversed Laguerre polynomial $q_a(x)=x^dL_d^{(a-1)}(1/x)$ has d positive simple roots and normalised coefficients $e_k=e_0/(a)_k$, $e_0=\Gamma(d+a)/(d!\,\Gamma(a))$. Hence $p\boxtimes_d(q_a/e_0)$ has the normalised coefficients $e_k(p)/(a)_k$, and it has only real roots whenever the roots of p are real and of one sign.

(ii) $\{1/(a)_k\}_{k\ge0}$ is a multiplier sequence: the function

$$\sum_k\frac{x^k}{k!\,(a)_k}=\Gamma(a)x^{(1-a)/2}I_{a-1}(2\sqrt{x})$$

is entire of order $\frac12$ with only real negative zeros, at $-j_{a-1,m}^2/4$.

(iii) A density f on $(0,\infty)$ is a scale mixture of Gamma(a) densities if and only if $x^{1-a}f(x)$ is completely monotone; for $a=2$, if and only if $f(x)/x$ is.

Proof. (i) By [256, (18.5.12)], $x^dL_d^{(a-1)}(1/x)=\sum_k(-1)^k\binom{d+a-1}{d-k}x^{d-k}/k!$, and $\binom{d+a-1}{d-k}/(k!\binom{d}{k})=\Gamma(d+a)/(d!\,\Gamma(k+a))=e_0/(a)_k$. The zeros of $L_d^{(a-1)}$, orthogonal for $x^{a-1}e^{-x}$ on $(0,\infty)$, are positive and simple [256, Section 18.2(vi)], and the roots of q_a are their reciprocals. The statement on $\boxtimes_d$ is Szegő's theorem; for nonpositive roots apply it to $(-1)^dp(-x)$, with normalised coefficients $(-1)^ke_k(p)$. (ii) The series is that of the modified Bessel function [256, (10.25.2)]; for $x=-y^2/4<0$ it equals $\Gamma(a)(y/2)^{1-a}J_{a-1}(y)$, and for $a-1>-1$ the zeros of $J_{a-1}(y)/y^{a-1}$ are real [256, Section 10.21(i)]. A real entire function of order less than one with only real zeros is in the Laguerre Pólya class, and by Pólya and Schur [310] a sequence whose exponential generating function is in that class with coefficients of one sign is a multiplier sequence. (iii) $f(x)=\int_0^\infty\theta^{-a}x^{a-1}e^{-x/\theta}\,\varpi(d\theta)/\Gamma(a)$ says that $x^{1-a}f(x)$ is the Laplace transform of a positive measure in $1/\theta$; conversely Bernstein's theorem writes a completely monotone $x^{1-a}f(x)$ in this form. □

Let X be the Pólya variable, with density $p=\Phi/\Xi(0)$ for the kernel Φ of (2.1.3) and $\mathbb{E}e^{sX}=G(s^2)$ (2.1.4), and write $G(u)=\sum_k b_ku^k$, so that $b_k=\mathbb{E}X^{2k}/(2k)!=M_k/((2k)!\,M_0)$ with $M_k=\int_{\mathbb{R}}v^{2k}\Phi(v)\,dv$ (Sections 8.1 and 20.4.1).

PROPOSITION 14.2.9. *For $a>0$ let $G_a(u)=\sum_k b_ku^k/(a)_k$. If RH holds, then for every $a>0$ the function G_a is in the Laguerre Pólya class with only real negative zeros, and all its Jensen polynomials $\sum_{i=0}^d\binom{d}{i}(n+i)!\,b_{n+i}x^i/(a)_{n+i}$, $d,n\ge0$, have only real negative zeros. Multiplying the coefficients by $(a)_k$ instead does not preserve real zeros: it turns $(1+u)^2=1+2u+u^2$ into $1+2au+a(a+1)u^2$, whose discriminant is $-4a<0$.*

Proof. Under RH, $G(u) = \prod_j (1 + u/\gamma_j^2)$ with $\sum_j \gamma_j^{-2} < \infty$ is in the Laguerre Pólya class, and by Pólya's theorem its Jensen polynomials $J_{d,n}(x) = \sum_i \binom{d}{i}(n+i)!\, b_{n+i} x^i$ (Section 20.4.1) have only real zeros. Since $(a)_{n+i} = (a)_n (a+n)_i$, the Jensen polynomial of G_a of degree d and shift n is $1/(a)_n$ times the image of $J_{d,n}$ under the multiplier sequence $\{1/(a+n)_i\}_i$ of Lemma 14.2.8(ii), so its zeros are real, and negative because its coefficients are positive. For $n = 0$, $\binom{d}{i} i!\, d^{-i} \to 1$ gives $\sum_i \binom{d}{i} i!\, b_i x^i/((a)_i d^i) \to G_a(x)$ locally uniformly, so by Hurwitz's theorem G_a has only real negative zeros; since $(a)_k \geq a(k-1)!$, G_a has order at most $\frac{1}{3}$, and it is in the Laguerre Pólya class. □

By Lemma 14.2.8(i), dividing the normalised coefficients by $(a)_k$ is the multiplicative convolution with q_a/e_0, so peeling a Gamma(a) layer, which passes from the moments of a Gamma(a) scale mixture to those of its mixing law, keeps zeros real, while the inverse operation can destroy them. The note calls the peeling a free Poisson operation. The example does not show that no argument could pass from real zeros of some G_a to real zeros of G.

Lemma 14.2.8(iii) makes the Gamma(2) mixture route of [282, Section 12.6], recalled as excluded in Section 3.1.3, a statement about complete monotonicity.

THEOREM 14.2.10. *For no $a > 0$ is $|X|$, X^2 or S_2 a scale mixture of Gamma(a) laws.*

Proof. By Lemma 14.2.8(iii) a mixture would make $g(x) = x^{1-a} f(x)$ completely monotone, f the density. A completely monotone $g \not\equiv 0$ is positive and log convex, so $g(x) \geq g(1) \exp\big((x-1) g'(1)/g(1)\big)$ for $x \geq 1$: it decays at most exponentially. For $v \geq 0$, dropping the negative terms of (2.1.3) and using $e^{2v} \geq 1$ gives

$$\Phi(v) \leq 4\pi^2 e^{9v/2} \exp(-\pi e^{2v}) \sum_{n \geq 1} n^4 e^{-\pi(n^2-1)},$$

a constant multiple of $e^{9v/2} \exp(-\pi e^{2v})$. The densities of $|X|$ and X^2 are $2\Phi(v)/\Xi(0)$ and $\Phi(\sqrt{w})/(\Xi(0)\sqrt{w})$, so $v^{1-a}\Phi(v)$ and $w^{1/2-a}\Phi(\sqrt{w})$ would be completely monotone, yet both decay faster than any exponential. For S_2 the obstruction is at the origin. For $\lambda > 0$, $\Pr(S_2 \leq x) \leq e^{\lambda x} \mathbb{E} e^{-\lambda S_2} = e^{\lambda x} \big(\sqrt{2\lambda}/\sinh\sqrt{2\lambda}\big)^2$, and $\lambda = 2/x^2$ gives

$$\Pr(S_2 \leq x) \leq \frac{16\, e^{-2/x}}{x^2 (1 - e^{-4/x})^2} \leq 17\, x^{-2} e^{-2/x}, \qquad 0 < x \leq 1.$$

A mixture would give, g being nonincreasing and $g(1) > 0$, $f(x) \geq g(1) x^{a-1}$ and $\Pr(S_2 \leq x) \geq g(1) x^a/a$ for $0 < x \leq 1$, impossible as $x \downarrow 0$. □

The Hankel test sees the failure at low order. If $Y = \Theta\, \Gamma$ with Γ Gamma(a) independent of $\Theta \geq 0$, then $\mu_k = \mathbb{E} Y^k/(a)_k = \mathbb{E}\Theta^k$ is a Stieltjes moment sequence, and $[\mu_{i+j}]_{i,j<N}$ and $[\mu_{i+j+1}]_{i,j<N}$ are positive semidefinite for every N. Table 14.8

gives the least order N at which each fails to be positive definite, by Cholesky factorisation at 400 digits, with the same orders at 250. The moments of X^2 are M_k/M_0; those of $|X|$ come from a tanh sinh rule, since odd powers of $|v|$ are not smooth at the origin; those of S_2 follow from its cumulants. The table of [291] is the Hamburger row.

a	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	4	6	8	12	20
$\lvert X\rvert$, Stieltjes	2	2	2	2	2	3	3	3	4	4	5
$\lvert X\rvert$, Hamburger	2	2	2	3	3	3	3	4	4	5	6
X^2, Stieltjes	2	3	3	4	4	4	5	5	6	7	9
X^2, Hamburger	2	3	4	4	4	5	5	6	6	7	9
S_2, both	2	2	2	2	2	2	2	3	4	6	12

Table 14.8: The first order N at which the moments $\mu_k = \mathbb{E}Y^k/(a)_k$ fail the Hankel test for a Gamma(a) scale mixture, for $Y = |X|$, X^2 and S_2: the Hamburger test uses $[\mu_{i+j}]_{i,j<N}$ alone, the Stieltjes test also $[\mu_{i+j+1}]_{i,j<N}$. Computed.

Kristiansen proved Steutel's conjecture that scale mixtures of Gamma(2) laws are infinitely divisible [194]; since $x^{\alpha-2}e^{-x}$ is completely monotone for $\alpha \le 2$, Lemma 14.2.8(iii) makes Gamma(α) a scale mixture of Gamma(2) laws, so his theorem covers scale mixtures of Gamma(α) laws for every $\alpha \le 2$. By Theorem 14.2.10 it cannot be applied to $|X|$, X^2 or S_2. The Gamma(2) structure that is true is Williams' convolution, a GGC and hence infinitely divisible directly.

14.2.6 The heat flow and finite free convolution

Remark 4.1.4 identifies the forward heat flow on polynomials with finite free convolution by a Hermite polynomial and, as the degree grows, with free convolution by a semicircle law whose variance is the time. For a real polynomial p of degree $d \ge 2$ write $p_t = e^{-(t/2)D^2}p$, a finite sum, and

$$q_t(x) = e^{-(t/2)D^2}x^d = t^{d/2}H_d\big(x/\sqrt{t}\big), \qquad t > 0,$$

with $q_0 = x^d$ and $H_d = e^{-D^2/2}x^d$ the Hermite polynomial of Remark 4.1.4, whose zeros are real. The time t is twice that of Section 4.1.1, where $\partial_t H_t = -\partial_z^2 H_t$.

Proposition 14.2.11. *Let p be a real polynomial of degree $d \ge 2$ and $s, t \ge 0$.*

(i) $p_t = p \boxplus_d q_t$.

(ii) $q_s \boxplus_d q_t = q_{s+t}$.

(iii) If p has only real roots, then so has p_t, and the largest root of p_t is at most the largest root of p plus $\sqrt{t}\,h_d$, where h_d is the largest zero of H_d.

(iv) While the roots $z_1(t), \dots, z_d(t)$ of p_t are simple, $\dot{z}_k = \sum_{j\ne k}(z_k - z_j)^{-1}$.

Proof. (i) In operator form, $p \boxplus_d q = \hat{p}(D)\hat{q}(D)x^d$ when $p = \hat{p}(D)x^d$ and $q = \hat{q}(D)x^d$ [231, Definition 1.1 and (2)]. For $q = q_t$ the operator $\hat{q}(D)$ acts on polynomials of degree at most d as $e^{-(t/2)D^2}$, and operators in D commute, so $p \boxplus_d q_t = e^{-(t/2)D^2}\hat{p}(D)x^d = p_t$. (ii) is (i) for $p = q_s$, since $e^{-(t/2)D^2}e^{-(s/2)D^2} = e^{-((s+t)/2)D^2}$; it is the identity $a^d H_d(x/a) \boxplus_d b^d H_d(x/b) = c^d H_d(x/c)$, $c^2 = a^2 + b^2$, of [231, Section 1.2]. (iii) The roots of q_t are $\sqrt{t}$ times those of H_d, and by Walsh's theorem [231, Theorem 1.3] the convolution $p \boxplus_d q$ of two polynomials of degree d with only real roots has only real roots, the largest at most the sum of the largest roots of p and q. (iv) Differentiate $p_t(z_k(t)) = 0$ in t and use $\partial_t p_t = -\frac{1}{2}p_t''$: $\dot{z}_k = \frac{1}{2}p_t''(z_k)/p_t'(z_k)$. Writing $p_t = (x - z_k)r$ gives $p_t'(z_k) = r(z_k)$ and $p_t''(z_k) = 2r'(z_k)$, so that $\dot{z}_k = r'(z_k)/r(z_k) = \sum_{j\neq k}(z_k - z_j)^{-1}$. □

For a conjugate pair $x \pm iy$, $y > 0$, the partner contributes $-1/(2y)$ to $\dot{y}$, and each real root r contributes $\operatorname{Im}(x + iy - r)^{-1} = -y/((x - r)^2 + y^2) < 0$: the pair is drawn to the axis by its partner and pulled further by every real root. Alone, $(x - 1)^2 + 0.64$ becomes $(x - 1)^2 + 0.64 - t$, which reaches the axis at $t = 0.64$. The example of the note adds four real roots,

$$p(x) = (x + 3)(x + 1)\big(x - \tfrac{1}{2}\big)(x - 2)\big((x - 1)^2 + 0.64\big).$$

The discriminant of p_t in x is a polynomial of degree 15 in t whose only root in $(0, 2)$ is $t^* = 0.3203291143987806$. Before t^* the roots are distinct, so the pair stays off the axis; since $p_t = e^{-((t-t^*)/2)D^2}p_{t^*}$ for $t > t^*$, part (iii) keeps all roots real afterwards. So the pair meets the axis at t^*, at the point 1.25743, in about half the time it needs alone. The discriminant is computed exactly.

H_d is the expected characteristic polynomial of a symmetric Gaussian matrix [231, Section 1.2], the finite free counterpart of the semicircle law, and the roots of $H_d(\sqrt{d}\,x)$, those of $q_{1/d}$, are close to the quantiles x_k of σ_1 given by $\sigma_1([-2, x_k]) = (k - \frac{1}{2})/d$. The largest deviation is at the edge, and its products with $d^{2/3}$, from 0.3906 to 0.4962 for $d = 20$ to 320, grow slowly, as they should at the edge scale $d^{-2/3}$. As $d \to \infty$ finite free convolution becomes free convolution [6], [231, Section 1.2]: if the root distributions of real rooted p_d tend to μ in moments, those of $p_d \boxplus_d q_{t/d}$ tend to $\mu \boxplus \sigma_t$.

For H_t the identification is a heuristic. The motion of the zeros is the same up to the factor two in time (Section 4.1.1), but H_t has infinitely many zeros with logarithmically growing density, while the limit theorem concerns all d roots after scaling; for H_t it would have to be applied to windows of zeros, the zeros outside acting as an external field, and no theorem of this kind is known. De Bruijn's theorem that real zeros of H_t stay real forward in time [73] is the counterpart of (iii), proved directly. In Thorin terms the forward flow is the path of clocks R_t of Section 4.1.1 (Theorem 17.1.16).

Griffin, Ono, Rolen and Zagier proved that for each degree d the Jensen polynomials $J_{d,n}$ of Section 20.4.1, after an affine change of variable and a normalisation

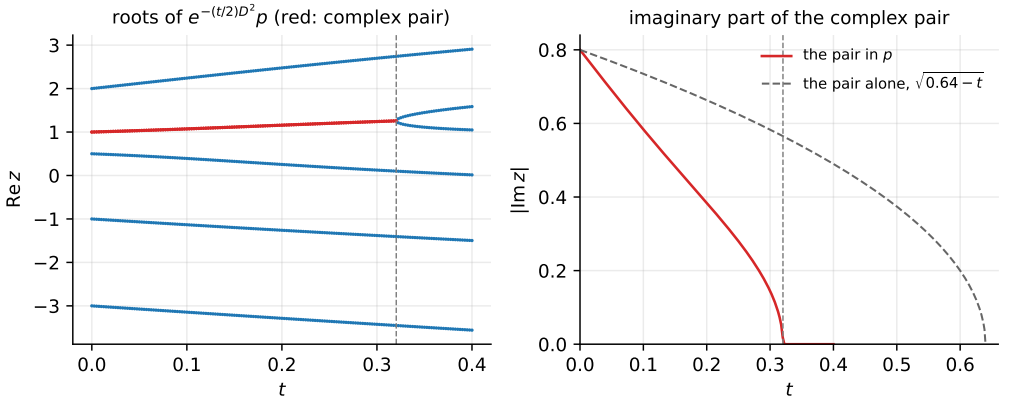


Figure 14.4: The flow $e^{-(t/2)D^2}p$ for p with the roots -3, -1, $\frac{1}{2}$, 2 and $1 \pm 0.8i$. Left: the real parts of the six roots for $0 \le t \le 0.4$; the complex pair has one real part until it meets the axis at $t^* = 0.3203$ (dashed), where its two roots split. Right: the imaginary part of the pair, against $\sqrt{0.64 - t}$ for the pair on its own, which reaches the axis at $t = 0.64$. Computed.

depending on n, tend to the Hermite polynomial of degree d as $n \to \infty$, which gives real zeros for all large shifts [97]: the Jensen polynomials of large shift are asymptotically finite free Gaussian. Problem 25.1 asks for an exact finite free structure at every shift.

These limits see the zeros only through their distribution. Moving $o(d)$ of the d roots changes no limit law, so a single nonreal pair is invisible to free convolution limits, and they cannot decide RH by themselves [291].

14.3 Almost generalised gamma convolutions

Below the basepoint one, as Section 15.2.4 concluded, every positive object built so far from the shifted clock T_α, with Laplace transform $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, is equivalent, up to the boundary case, to the absence of zeros in the half plane $\operatorname{Re} s > \alpha$. These equivalences are restatements, not reductions. Since no zero free half plane is known to the left of the line $\operatorname{Re} s = \frac{7}{8}$, announced in 2026 (Section 25.1.2), the question that remains is quantitative: how negative can the Thorin density be below the basepoint one? Zero density estimates answer it with no hypothesis: with those of Ingham, of Guth and Maynard and of Huxley, every T_α with $\alpha > \frac{1}{2}$ is a GGC up to a defect carried by the zeros to the right of α, whose Cauchy kernels carry a vanishing fraction of the mass of the others, with a power saving; we call this property almost GGC.

With ξ again in place of $\xi(\cdot, \chi_{12})$, we keep the notation of Sections 12.1 and 14.1: $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha + i\theta)$, and the Thorin density at the basepoint α in

the chart $\theta = \sqrt{z}$ is v_α/π, the density $\rho_{\alpha-1/2}$ of (2.1.7); here the basepoint, not the resolution, is the index. $P_w(x) = w/(\pi(w^2+x^2))$ is the Poisson kernel of Section 2.1, $P_{-w} = -P_w$, and P_0 is the unit point mass, as in Proposition 3.2.1. A zero is $\rho = \beta + i\gamma$; $N(T)$ counts the zeros with $0 < \gamma \le T$, and $N(\sigma, T)$ those among them with $\beta \ge \sigma$, so that $N(\sigma,T) \ge \nu(\sigma,T)$ in the notation of Proposition 3.3.2.

14.3.1 Mass bookkeeping and the almost GGC theorem

The Poisson form of Proposition 2.1.3 splits according to the side of the line $\operatorname{Re} s = \alpha$ on which each zero lies, $v_\alpha = v_\alpha^+ - v_\alpha^-$, with

$$\begin{aligned} \frac{v_\alpha^+(\theta)}{\pi} &= \sum_{\beta\le\alpha,\ \gamma>0} \big[P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)\big], \\ \frac{v_\alpha^-(\theta)}{\pi} &= \sum_{\beta>\alpha,\ \gamma>0} \big[P_{\beta-\alpha}(\theta-\gamma) + P_{\beta-\alpha}(\theta+\gamma)\big]. \end{aligned} \tag{14.3.1}$$

Both parts are positive; a zero on the line $\operatorname{Re} s = \alpha$ contributes a unit atom at its ordinate to the positive part, and each pair of kernels has mass one on $(0,\infty)$. A zero with $\beta > \alpha$ has the partner $1 - \bar\rho = 1 - \beta + i\gamma$, with $1 - \beta < \frac12 < \alpha$, in the positive part. Write $m_\alpha^\pm(T) = \pi^{-1}\int_0^T v_\alpha^\pm(\theta)\,d\theta$.

Lemma 14.3.1 (Mass bookkeeping). *Let $\alpha > \frac12$, and suppose that $N(\alpha,t) \le t^{e+o(1)}$ as $t\to\infty$ for some $e < 1$. Then*

$$m_\alpha^-(T) \le N(\alpha,2T) + O\big(T^{e-1+o(1)}\big), \qquad m_\alpha^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} + O(T).$$

Proof. A zero with $\beta > \alpha$ and $\gamma \le 2T$ contributes at most its total mass one. For a zero with $\gamma > 2T$ put $w = \beta - \alpha \in (0, \frac12)$. Since $\int_0^T P_w(\theta+\gamma)\,d\theta = \int_{-T}^0 P_w(\theta-\gamma)\,d\theta$, its pair has mass on $[0,T]$ at most

$$\int_{-\infty}^{T} P_w(\theta-\gamma)\,d\theta \le \frac{w}{\pi(\gamma-T)} \le \frac{1}{\pi\gamma},$$

and $\sum_{\beta>\alpha,\ \gamma>2T}\gamma^{-1} = \int_{2T}^\infty t^{-1}dN(\alpha,t) \le \int_{2T}^\infty N(\alpha,t)t^{-2}dt \ll T^{e-1+o(1)}$. The positive part contains the kernels of all zeros with $\beta \le \alpha$, among them every zero on the critical line and every partner $1-\bar\rho$; its mass on $[0,T]$ is $N(T)$ up to the $N(\alpha,T) = O(T^{e+o(1)})$ zeros to the right of α and up to $O(\log^2 T)$ from the tails of the kernels near height T. Since $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, the second claim follows. □

Theorem 14.3.2 (Almost GGC). *For every $\frac12 < \alpha < 1$, with no hypothesis,*

$$\frac{m_\alpha^-(T)}{m_\alpha^+(T)} \le T^{e(\alpha)-1+o(1)}, \qquad e(\alpha) = \begin{cases} \dfrac{3(1-\alpha)}{2-\alpha}, & \frac12 < \alpha < \frac{7}{10}, \\[2ex] \dfrac{15(1-\alpha)}{3+5\alpha}, & \frac{7}{10} \le \alpha < 1, \end{cases}$$

and $e(\alpha) < 1$ throughout, so the negative part v_α^-/π of (14.3.1), the Cauchy kernels of the zeros to the right of α, carries on $[0,T]$ a vanishing fraction of the mass of the positive part, with a power saving. For $\alpha \geq 1$ no zero lies to the right of α, and $m_\alpha^- = 0$. Under the density hypothesis $e(\alpha) = 2(1-\alpha)$. RH holds if and only if v_α^- vanishes identically for every $\alpha > \frac{1}{2}$.

Proof. Combine Lemma 14.3.1 with three zero density estimates: Ingham's $N(\sigma,T) \ll T^{3(1-\sigma)/(2-\sigma)}\log^5 T$ [163]; Guth and Maynard's $N(\sigma,T) \leq T^{15(1-\sigma)/(3+5\sigma)+o(1)}$ for $\frac{7}{10} \leq \sigma \leq \frac{4}{5}$ [103, Theorem 1.2]; and Huxley's $N(\sigma,T) \leq T^{3(1-\sigma)/(3\sigma-1)+o(1)}$ for $\sigma \geq \frac{3}{4}$ [159], whose exponent is at most $15(1-\sigma)/(3+5\sigma)$ when $\sigma \geq \frac{4}{5}$. Since $3(1-\alpha) < 2-\alpha$ exactly when $\alpha > \frac{1}{2}$, and $15(1-\alpha) < 3+5\alpha$ exactly when $\alpha > \frac{3}{5}$, the exponent is below one for every $\frac{1}{2} < \alpha < 1$. Finally, $v_\alpha^- \equiv 0$ exactly when no zero has $\beta > \alpha$, the GGC criterion of Theorem 2.1.2. □

Even the estimate $N(\sigma,T) = O(T)$ of Bohr and Landau [36], for fixed $\sigma > \frac{1}{2}$, gives a vanishing fraction $O(1/\log T)$; the zero density estimates improve this to a power of T. The theorem bounds the decomposition mass $m_\alpha^-(T)$, which overcounts the negative part $\int_0^T \max(-v_\alpha/\pi, 0)\,d\theta$ of the density itself, since positive kernels of nearby zeros cancel much of each negative kernel. The partner already cancels part of it: with $\varepsilon = \alpha - \frac{1}{2}$ and $\delta = \beta - \frac{1}{2}$, twice the mass in Proposition 3.2.1(iii) gives the negative part of $P_{\alpha+\beta-1} - P_{\beta-\alpha} = P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ the mass $\frac{4}{\pi}\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} - 1 = \frac{2}{\pi}\arcsin\frac{\varepsilon}{\delta} < 1$; Section 14.3.4 shows the effect of the other zeros. On the verified range the negative mass is essentially zero: all zeros with $0 < \gamma \leq T_0$ lie on the critical line [277], so for $T \leq T_0/2$ the negative contributions come from zeros above T_0, of total mass $O(T_0^{e(\alpha)-1+o(1)})$ on $[0,T]$ by the proof of Lemma 14.3.1.

The left panel of Figure 14.6 draws the exponents. The saving $1 - e(\alpha)$ is small near the centre, where Ingham's exponent gives $(2\alpha-1)/(2-\alpha)$. Sharper exponents are known near $\alpha = 1$: the density hypothesis holds for $\alpha > \frac{25}{32}$ [45, p. 133], and Huxley's exponent [159] is 0.176 at $\alpha = 0.9$; those of the theorem suffice here.

14.3.2 The canonical factorisation and explicit unconditional bounds

Proposition 14.3.3 (Canonical factorisation). *For every $\alpha > \frac{1}{2}$ and $u \geq 0$, $\varphi_\alpha(u) = \varphi_\alpha^+(u)\,D_\alpha(u)$, where*

$$\varphi_\alpha^+(u) = \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{u}{\theta^2}\Big)\frac{v_\alpha^+(\theta)}{\pi}\,d\theta\Big\},$$

$$D_\alpha(u) = \prod_{\beta>\alpha,\ \gamma>0} \frac{(\beta-\alpha)^2+\gamma^2}{(\sqrt{u}-\beta+\alpha)^2+\gamma^2};$$

here φ_α^+ is the Laplace transform of a GGC T_α^+ with Thorin density v_α^+/π in the chart $\theta = \sqrt{z}$, and the product defining the defect D_α runs over the zeros to the right of the line $\operatorname{Re} s = \alpha$ and converges absolutely. The defect is identically one if and only if ζ has no zeros with $\operatorname{Re} s > \alpha$, and then $T_\alpha \stackrel{d}{=} T_\alpha^+$. Otherwise $D_\alpha(u) \to 0$ as $u \to \infty$, and T_α^+ is not the sum of T_α and an independent nonnegative variable.

Proof. Grouping the Hadamard product of ξ in conjugate pairs, as in the proof of Proposition 2.1.3, gives for $u \geq 0$

$$\varphi_\alpha(u) = \prod_{\gamma>0} \frac{(\alpha-\beta)^2+\gamma^2}{(\alpha-\beta+\sqrt{u})^2+\gamma^2},$$

over the zeros with $\gamma > 0$, with multiplicity; each factor is $1 + O((u+\sqrt{u})\gamma^{-2})$ and $\sum \gamma^{-2} < \infty$. For $\beta < \alpha$, with $w = \alpha - \beta > 0$, the factor is $\exp\{-\int_0^\infty \log(1 + u/\theta^2)[P_w(\theta-\gamma) + P_w(\theta+\gamma)]\, d\theta\}$, the Laplace transform of a GGC whose Thorin density is the pair of Cauchy kernels of width w at $\pm\gamma$ (the computation after Theorem 13.1.2); for $\beta = \alpha$ it is $\gamma^2/(\gamma^2+u)$, an exponential law with Thorin atom at γ. The partial products over the zeros with $\beta \leq \alpha$ are therefore Laplace transforms of GGCs converging for every $u \geq 0$ to a limit continuous at the origin, and by the closure theorem of Section 9.1.1 the limit is that of a GGC with Thorin density v_α^+/π. The remaining factors form D_α. For a zero to the right of α, $w = \beta - \alpha \in (0, \frac{1}{2})$, its factor tends to zero as $u \to \infty$, and every factor of D_α is at most one for $u \geq 1$, because $(\sqrt{u} - w)^2 \geq w^2$ when $\sqrt{u} \geq 2w$; so $D_\alpha(u) \to 0$. If $T_\alpha^+ = T_\alpha + S$ with $S \geq 0$ independent, $\mathbb{E}e^{-uS} = 1/D_\alpha(u)$ would be unbounded. □

When every zero lies to the left of the line $\operatorname{Re} s = \alpha$, the signed density v_α/π is the Thorin density of T_α (Theorem 2.1.2); a zero on the line adds an atom at its ordinate, the term P_0 of (14.3.1), which v_α does not show. When a zero lies to the right, T_α is not a GGC and v_α/π is not a signed Thorin density for it in any sense: the negative kernel $-P_w$ integrates to the factor $((w+\sqrt{u})^2+\gamma^2)/(w^2+\gamma^2)$, while the zero contributes $(w^2+\gamma^2)/((\sqrt{u}-w)^2+\gamma^2)$ to φ_α. The bookkeeping of Theorem 14.3.2 applies to the other reformulations in Chapters 12 to 14: in the normal scale mixture of Theorem 13.1.3 it bounds the negative part of the candidate EGGC density v_α/π; in Wiener's theorem (Proposition 13.1.12) the proportion of frequencies at which the Fourier transform of the tilted Pólya kernel can vanish; and in Riesz's criterion (Section 13.1.8) the number of Mellin poles to the right of $\alpha/2$.

The pole factors can be written down. Let U_α° be the signed measure on $\theta > 0$ with density v_α/π and a unit atom at γ for every zero $\beta + i\gamma$, $\gamma > 0$, with $\beta = \alpha$, counted with multiplicity.

THEOREM 14.3.4 (Signed Thorin representation). *For every $\alpha \ge \frac12$ and $u \ge 0$,*

$$\varphi_\alpha(u) = \exp\Big\{ -\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big) U_\alpha^\circ(d\theta)\Big\} \prod_{\beta>\alpha,\ \gamma>0} \Big|1-\frac{u}{(\rho-\alpha)^2}\Big|^{-2},$$

the integral and the product converging absolutely. For a zero to the right of α, with $p=(\rho-\alpha)^2=-c+id$, $c=\gamma^2-(\beta-\alpha)^2>0$ and $d=2\gamma(\beta-\alpha)>0$, the factor is

$$\Big|1-\frac{u}{p}\Big|^{-2} = \frac{|p|^2}{(u+c)^2+d^2} = |p|^2\int_0^\infty e^{-ut}\,\frac{e^{-ct}\sin dt}{d}\,dt,$$

the Laplace transform of a damped sinusoid, which changes sign, so the factor is not completely monotone.

Proof. Write $f_w(u)=(w^2+\gamma^2)/((w+\sqrt{u})^2+\gamma^2)$. As in the proof of Proposition 14.3.3, $\varphi_\alpha(u)=\prod_{\gamma>0} f_{\alpha-\beta}(u)$, and for $w\ge 0$, $f_w(u)=\exp\{-\int_0^\infty \log(1+u/\theta^2)[P_w(\theta-\gamma)+P_w(\theta+\gamma)]\,d\theta\}$, with the atom at γ for $w=0$. For $w=-\omega<0$ the kernels carry the sign $P_{-\omega}=-P_\omega$, so the same integral equals f_ω^{-1}, and $f_{-\omega}=f_\omega^{-1}\,|p|^2/|p-u|^2$ with $p=(\omega+i\gamma)^2$, since $|p-u|^2=((\sqrt{u}-\omega)^2+\gamma^2)((\sqrt{u}+\omega)^2+\gamma^2)$. Multiplying over the zeros gives the formula; the integrals may be summed because each is $\pm\log f_{|w|}=O_\alpha((u+\sqrt{u})\gamma^{-2})$, and $\sum\gamma^{-2}<\infty$. Finally $|p-u|^2=(u+c)^2+d^2$, and $c>0$ because $\beta-\alpha<\frac12<\gamma_1$. □

The theorem needs no hypothesis on the zeros, the atoms carrying the zeros on the line $\operatorname{Re} s=\alpha$, which is why $\alpha=\frac12$ is included. Proposition 14.3.3 combines its pole factors with the negative kernels they cancel into D_α. At $\alpha=\frac12$ every term is explicit.

COROLLARY 14.3.5 (The critical line). *Unconditionally, $U_{1/2}^\circ=\sum_{\beta=1/2,\ \gamma>0}\delta_\gamma\ge 0$ and*

$$\varphi_{1/2}(u) = \prod_{\beta=1/2,\ \gamma>0}\frac{\gamma^2}{\gamma^2+u}\ \prod_{\beta>1/2,\ \gamma>0}\Big|1-\frac{u}{(\rho-\frac12)^2}\Big|^{-2}.$$

Proof. A zero with $\beta=\frac12+\delta$, $\delta>0$, and its partner $1-\bar\rho$ have the same ordinate, and their kernels $P_{-\delta}+P_\delta$ cancel, so $v_{1/2}=0$ away from the atoms (the term by term form in the proof of Proposition 2.1.3, valid at $\alpha=\frac12$ away from the ordinates). The atoms come from the zeros with $\beta=\frac12$, each with the factor $f_0=\gamma^2/(\gamma^2+u)$; apply Theorem 14.3.4. □

So on the critical line positivity of $U_{1/2}^\circ$ costs nothing, and RH is the statement that the product of damped sinusoids is empty. For $\frac12<\alpha<\sup_\rho\operatorname{Re}\rho$ the negative kernels are present, and whether they can be outweighed on the single line $\operatorname{Re} s=\alpha$ is open, except near a zero, where they cannot (Remark 2.1.4).

THEOREM 14.3.6 (Explicit bounds from verification). *Let $T_0 = 3 \times 10^{12}$, below which all zeros lie on the critical line [277], and let $\Sigma_0 = \sum_{\gamma > T_0} \gamma^{-2}$, the sum over all zeros above T_0, counted with multiplicity. By the Riemann von Mangoldt formula with the explicit error term of Hasanalizade, Shen and Wong [108], recalled in the proof of Proposition 12.1.11, Σ_0 differs from $\frac{\log(T_0/2\pi)+1}{2\pi T_0} = 1.4797 \times 10^{-12}$ by less than 2×10^{-24}, so $\Sigma_0 < 1.48 \times 10^{-12}$. Then for every $\alpha > \frac{1}{2}$ and every $u \geq 0$:*

1. *$|\log D_\alpha(u)| \leq (u + \sqrt{u})\,\Sigma_0$, so that*
$$e^{-1.5\times 10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u) \leq \mathbb{E}e^{-uT_\alpha} \leq e^{1.5\times 10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u);$$
2. *the negative part of the density satisfies $v_\alpha^-(\theta)/\pi \leq 2\Sigma_0/\pi < 9.43 \times 10^{-13}$ for $0 \leq \theta \leq T_0/2$;*
3. *consequently $v_\alpha(\theta) > 0$ for all $0 \leq \theta \leq T_0/2$ whenever $\alpha - \frac{1}{2} > 6 \times 10^{-10}$.*

Proof. Every zero to the right of α has $\gamma > T_0$ and $\beta > \frac{1}{2}$. The zeros above T_0 off the critical line come in pairs $\beta + i\gamma$, $1 - \beta + i\gamma$ of equal multiplicity, and at most one member of each pair lies to the right of α, so $\sum_{\beta>\alpha,\ \gamma>0} \gamma^{-2} \leq \Sigma_0/2$. (1) For a zero to the right of α put $w = \beta - \alpha \in (0, \frac{1}{2})$ and $x = (u - 2w\sqrt{u})/(w^2 + \gamma^2)$, so that its factor in D_α is $1/(1 + x)$. Since $u - 2w\sqrt{u} \geq -w^2$, $x \geq -w^2/\gamma^2 > -\frac{1}{2}$, and $|x| \leq (u + \sqrt{u})/\gamma^2$; for $x \geq -\frac{1}{2}$, $|\log(1 + x)| \leq 2|x|$. Summing gives $|\log D_\alpha(u)| \leq (u + \sqrt{u})\Sigma_0$, and the inequalities for $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha^+(u)D_\alpha(u)$ follow. (2) For $\theta \leq T_0/2$ and $\gamma > T_0$, $P_w(\theta - \gamma) + P_w(\theta + \gamma) \leq \frac{2w}{\pi(\gamma-\theta)^2} \leq \frac{4}{\pi\gamma^2}$, and summing gives at most $\frac{4}{\pi} \cdot \frac{\Sigma_0}{2}$. (3) Write $a = \alpha - \frac{1}{2}$. If $a \geq \frac{1}{2}$, no zero lies to the right of α and $v_\alpha = v_\alpha^+ > 0$; so let $6 \times 10^{-10} < a < \frac{1}{2}$. The zeros below T_0 lie on the line and contribute the kernels $P_a(\theta - \gamma)$ to v_α^+/π, and all other kernels of v_α^+ are positive, so $v_\alpha^+(\theta)/\pi \geq \frac{a}{\pi(a^2+\gamma_1^2)}$ at every θ within γ_1 of an ordinate below T_0. Every $\theta \in [0, T_0/2]$ is such a point. On $[0, \gamma_1]$ the ordinate γ_1 serves. On $[\gamma_1, 2000]$ one of the first 1800 ordinates serves: they are the certified zeros of the proof of Proposition 12.1.11, they reach height 2304.3, and consecutive ones differ by less than $6.8874 < 2\gamma_1$. For $2000 \leq \theta \leq T_0/2$, the count of that proof, with $H = 14$ in place of 15, shows that $(\theta - 14, \theta + 14]$ holds at least $n(\theta) = \frac{14}{\pi}\log\frac{\theta-14}{2\pi} - E(\theta - 14) - E(\theta + 14)$ zeros; while $\theta + 14 \leq X_0$, n increases and $n(2000) > 20$, and for $\theta + 14 > X_0$, $n(\theta) > \frac{14}{\pi}\log\frac{\theta-14}{2\pi} - 0.6\log(\theta + 14) > 80$, so the window holds a zero, within $14 < \gamma_1$ of θ and, since $\theta + 14 < T_0$, on the line. A computation recomputes the 1800 zeros and checks these inequalities in ball arithmetic. Finally, since $a < \frac{1}{2}$,
$$\frac{a}{\pi(a^2 + \gamma_1^2)} > \frac{a}{\pi(\frac{1}{4} + \gamma_1^2)} > \frac{6 \times 10^{-10}}{\pi(\frac{1}{4} + \gamma_1^2)} > 9.5 \times 10^{-13},$$
more than the bound $2\Sigma_0/\pi < 9.43 \times 10^{-13}$ of (2) on v_α^-/π, so $v_\alpha > 0$ on $[0, T_0/2]$. □

Theorem 14.3.6 restates the verification of RH to height 3×10^{12} in Thorin form. Part (3) of Theorem 13.1.3 is the same statement for the characteristic function of the normal scale mixture, with a remainder of relative size $\log T_0/T_0$, and the paragraph after its proof bounds the negative part of v_α/π for $\theta \le 10^9$ as in part (2).

Part (3) treats every zero above T_0 as if its kernel could be negative wherever the density is evaluated. As the first author's note on the Dirichlet beta function observes [283], grouping each such zero with its reflection in the critical line does better, because the two kernels of an off line pair sum to a negative function only near its ordinate.

Corollary 14.3.7 (Positivity on the verified window). *Let* $T_0 = 3 \times 10^{12}$ *and* $V_\xi = (\log \xi)''(\frac{1}{2}) = 0.04620998623$.

1. *For every* $\alpha > \frac{1}{2}$, $v_\alpha(\theta) > 0$ *for* $|\theta| \le T_0 - \frac{1}{2}$.

2. *For every* $\alpha \in (\frac{1}{2}, 7.35]$, $v_\alpha(\theta) > v_\alpha(0)$ *for* $0 < |\theta| \le T_0 - \frac{1+\sqrt{2}}{2}$, *and* $0 < v_\alpha(0) \le (\alpha - \frac{1}{2})V_\xi$.

Proof. (1) Put $\varepsilon = \alpha - \frac{1}{2}$. By the Hadamard product, as in Proposition 2.1.3, $v_\alpha(\theta)$ is the absolutely convergent sum over the zeros, with multiplicity, of $(\alpha-\beta)/((\alpha-\beta)^2+(\theta-\gamma)^2)$. A zero on the critical line contributes $\pi P_\varepsilon(\theta-\gamma) > 0$. Zeros off the line come in pairs $\frac{1}{2} \pm \delta + i\gamma$, $0 < \delta < \frac{1}{2}$, of equal multiplicity, and a pair contributes $\pi(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})(\theta-\gamma)$, which is nonnegative if $\delta \le \varepsilon$ and, if $\delta > \varepsilon$, negative exactly where $|\theta-\gamma| < \sqrt{\delta^2-\varepsilon^2} < \frac{1}{2}$, by the proof of Proposition 3.2.1(ii). Every zero off the line has $|\gamma| > T_0$ [277], so for $|\theta| \le T_0 - \frac{1}{2}$ every pair contributes a nonnegative term, while $\frac{1}{2} + i\gamma_1$ contributes a positive one.

(2) This is Theorem 12.1.18(i) and (ii) for ξ, with $\alpha_{\max} = 7.35$ and $H_0 = T_0$ (Table 12.4). □

Remark 14.3.8. The corollary does not touch the defect of part (1) of Theorem 14.3.6: a zero above T_0 to the right of the line $\operatorname{Re} s = \alpha$ stops T_α from being a GGC whatever v_α does on the window (Proposition 14.3.3), and near its own ordinate it can make v_α negative (Section 14.3.3).

The proof of part (1) gives, with no hypothesis, $|\log D_\alpha(u)| \le 2(u + \sqrt{u}) \sum_{\beta>\alpha,\ \gamma>0} \gamma^{-2}$, and the sum is at most $\Sigma_0/2$ by the verification height alone. The zero density estimates of Section 14.3.1 [163, 103, 159] improve this to $\sum_{\beta>\alpha} \gamma^{-2} \le \int_{T_0}^{\infty} t^{-2} dN(\alpha, t) \ll T_0^{e(\alpha)-2+o(1)}$, smaller by a further power of T_0.

14.3.3 Where negativity can come from: a cancellation lemma

Pointwise, a negative kernel is cancelled wherever the rest of the density exceeds it, and the kernel of width w has height at most $1/(\pi w)$.

LEMMA 14.3.9 (Cancellation). *Let $\rho = \beta + i\gamma$ be a zero with $\beta > \alpha$, and let $m > 0$ be a lower bound on an interval I for the rest of the density, $v_\alpha/\pi + P_{\beta-\alpha}(\cdot - \gamma)$. If $\beta - \alpha \geq 1/(\pi m)$, the kernel of ρ cannot make v_α negative on I.*

Proof. On I, $v_\alpha/\pi \geq m - \max_x P_{\beta-\alpha}(x) = m - 1/(\pi(\beta-\alpha)) \geq 0$. □

Heuristically, at height γ the density v_α/π averages $\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$, the density of the ordinates, and stays close to it when $\alpha - \frac{1}{2}$ is large compared with the mean spacing $2\pi/\log\gamma$. Taking this average for m, a zero to the right of α can produce negativity only if

$$\beta - \alpha < \frac{2\pi}{\pi\log(\gamma/2\pi)} = \frac{2}{\log(\gamma/2\pi)},$$

so positivity at the basepoint α can be broken at height γ only by zeros in the thin strip $\alpha < \beta < \alpha + 2/\log(\gamma/2\pi)$. The local visibility condition of Section 1.4.2, for small resolutions, has the same scale $1/\log\gamma$. The heuristic proves nothing, because a rigorous lower bound for the background near an arbitrary height would itself control ζ'/ζ in the strip; it suggests combining zero density estimates for thin strips with lower bounds for $\operatorname{Re}\xi'/\xi$ on average.

We tested the lemma at the heights 300 and 10^5 numerically, computing the background $v_{0.6}/\pi$ from ξ'/ξ on a window of width 8 and adding the kernels of one hypothetical pair of zeros at $\beta + i\gamma_0$ and $1 - \beta + i\gamma_0$, γ_0 the centre of the window (Figure 14.5, middle and right). At height 300 the background has mean 0.51 and minimum 0.069; at height 10^5, mean 1.59, close to $\frac{1}{2\pi}\log\frac{10^5}{2\pi} = 1.54$, and minimum 0.34. Table 14.9 shows the result. At both heights zeros with $\beta - \alpha \leq 0.2$ produce negativity and zeros with $\beta - \alpha \geq 0.3$ are absorbed. At height 10^5 the heuristic threshold $2/\log(10^5/2\pi) = 0.21$ is close to the observed one, while the rigorous guarantee of Lemma 14.3.9 with the window minimum, $1/(\pi m) = 0.93$, is conservative. For a fixed width the dip is deeper at the greater height, because the kernel is the same while the local minimum of the background near the hypothetical zero differs.

14.3.4 Hypothetical off line zeros in the computed density

No off line zero is known, so the negative part cannot be observed. To see the mechanism, we insert fifteen hypothetical quadruples of zeros $0.75 \pm i\gamma_j$, $0.25 \pm i\gamma_j$, at the heights $\gamma_j = 20\,j^{5/3}$ below 2000, into the Poisson form of Proposition 2.1.3 together with the first 1700 ordinates, taken on the line; the zeros above height 2198 are omitted. The hypothetical count grows like $T^{0.6}$, as a zero density estimate with $e = 0.6$ would allow. Table 14.10 compares the negative mass $\int_0^T \max(-v_\alpha/\pi, 0)\,d\theta$ with the positive mass and with the counting bound $N_{\text{off}}(2T)$ of Lemma 14.3.1, the number of hypothetical zeros with ordinate at most $2T$.

Three features are visible. The negative mass is well below the counting bound, each negative kernel being partly cancelled by its partner and by the positive

	height 300		height 10^5	
β	min of $v_{0.6}/\pi$	negative mass	min of $v_{0.6}/\pi$	negative mass
0.62	-13.54	0.521	-13.97	0.567
0.65	-4.16	0.299	-4.59	0.365
0.70	-1.24	0.121	-1.62	0.193
0.75	-0.41	0.041	-0.72	0.097
0.80	-0.10	0.008	-0.32	0.041
0.90	0.072	0	0.023	0
0.95	0.075	0	0.100	0

Table 14.9: A single hypothetical pair of zeros at real parts β and $1-\beta$ inserted into the density $v_{0.6}/\pi$, computed from ξ'/ξ on a window of width 8 at the heights 300 and 10^5: the minimum of the density and the negative mass $\int \max(-v_{0.6}/\pi, 0)$ on the window. Computed.

α	T	negative mass	positive mass	fraction	counting bound $N_{\text{off}}(2T)$
0.6	250	0.842	108.8	7.7×10^{-3}	6
0.6	1000	1.655	650.5	2.5×10^{-3}	15
0.6	2000	2.278	1519.2	1.5×10^{-3}	15
0.7	250	2.130	110.1	1.9×10^{-2}	6
0.7	1000	4.815	653.6	7.4×10^{-3}	15
0.7	2000	6.938	1523.8	4.6×10^{-3}	15
0.8	any	0		0	0

Table 14.10: Negative and positive mass of the density v_α/π on $[0, T]$ with fifteen hypothetical quadruples of zeros at real part 0.75. Computed.

kernels of the neighbouring zeros on the line: at $\alpha = 0.6$ the negative kernel has width 0.15 and height $1/(0.15\pi) = 2.1$, and most of it is cancelled, while at $\alpha = 0.7$ the width is 0.05 and the kernel, of height 6.4, appears as a sharp spike that takes the density to -5.39 at the height 292.40 (Figure 14.6; minimum computed). The fraction decreases with T at the rate the counting predicts. And at $\alpha = 0.8$, to the right of the hypothetical zeros, the density is positive everywhere, as Theorem 2.1.2 requires.

A zero is most visible from basepoints just to its left. The left panel of Figure 14.5, computed, shows the negative mass on $[0, 2000]$ as a function of α for the same fifteen heights and $\beta = 0.6$, 0.75 and 0.9. As α increases towards β the negative kernels narrow, cancellation fails, and the negative mass climbs towards the number of hypothetical zeros, reaching 7.55 at $\alpha = 0.85$ for $\beta = 0.9$. At $\alpha = \beta$ the zeros lie on the line of evaluation and the negative mass drops to zero: the Thorin counterpart of the failure of completeness of the translates of the tilted Pólya kernel exactly on the line through the zero (Proposition 13.1.12).

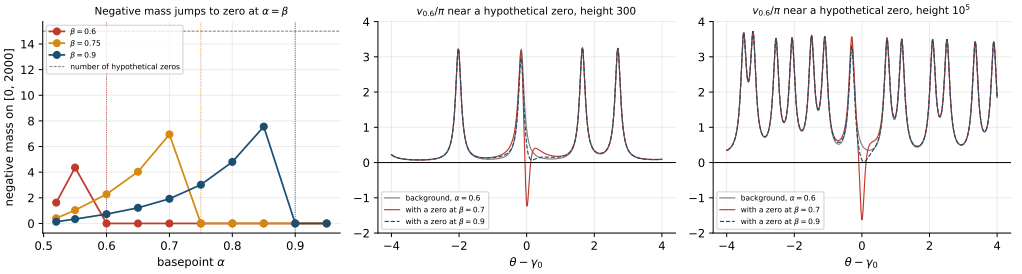


Figure 14.5: Left: negative mass of the density on $[0, 2000]$ against the basepoint α for fifteen hypothetical zeros at real part β. Middle and right: the density $v_{0.6}/\pi$ near the heights 300 and 10^5, alone and with a hypothetical pair of zeros at $\beta = 0.7$ or 0.9 at the centre of the window. Computed.

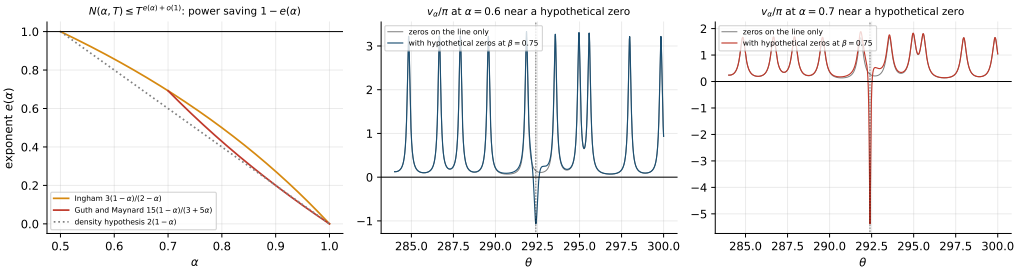


Figure 14.6: Left: the zero density exponents of Theorem 14.3.2 and the region where they give a power saving; the exponent of Guth and Maynard is drawn on $[\frac{7}{10}, 1)$, where it holds by their estimate and, above $\frac{4}{5}$, by Huxley's [159]. Middle and right: the density v_α/π at $\alpha = 0.6$ and 0.7 near the hypothetical off line zero at height 292.4 (dotted), with and without the hypothetical zeros, evaluated as in Table 14.10 with step 0.005.

14.3.5 Positivity margins and the circular unitary ensemble

By Lemma 14.3.9, a zero to the right of α at height γ can make the density negative only if its distance to the line $\operatorname{Re} s = \alpha$ is below $1/(\pi m)$, m the local minimum of the rest of the density. The distribution of these minima, the positivity margins, therefore determines how much room RH has at each height. The minima occur between consecutive ordinates. After unfolding the ordinates to unit mean spacing, $x_n = \vartheta(\gamma_n)/\pi + 1$ with ϑ the Riemann Siegel theta function (Section 8.3.2), the density at the resolution of κ mean spacings is

$$\bar{\rho}_\kappa(x) = \sum_n P_\kappa(x - x_n) = \sum_n \frac{\kappa}{\pi(\kappa^2 + (x - x_n)^2)},$$

of mean one.

We evaluate $\bar\rho_\kappa$ at the 1600 midpoints between consecutive unfolded ordinates below height 2198, leaving out 50 mean spacings at each end, and the same statistic at about 3600 midpoints between the eigenvalues of twelve Haar distributed unitary matrices of size 400, unfolded and extended periodically; this circular unitary ensemble (CUE) models the local statistics of the zeros. Table 14.11 shows the result. The medians agree to 1% at every resolution. In the lower tail the zeros have slightly larger margins than the CUE, a finite height effect in the expected direction, the low zeros being more rigid than the limit of large matrices.

κ	zeta, 0.1%	CUE, 0.1%	zeta, 1%	CUE, 1%	zeta, median	CUE, median
0.1	0.1075	0.0919	0.1177	0.1095	0.3349	0.3343
0.25	0.2617	0.2178	0.2849	0.2659	0.7116	0.7092
0.5	0.4803	0.4047	0.5150	0.4645	0.9739	0.9796
1	0.7477	0.6552	0.7730	0.7313	1.0264	1.0240

Table 14.11: Quantiles of the normalised Thorin density $\bar\rho_\kappa$ at the midpoints between consecutive unfolded ordinates, for the zeros below height 2198 and for CUE eigenvalues. Computed.

The next step is to repeat the comparison with Odlyzko's tables near the 10^{12}th zero [252] and near the 10^{21}st and 10^{22}nd zeros [254, 255], where the finite height corrections are smaller, and to fit the lower tail of the margin distribution. The CUE lower tail is governed by large gaps, whose probability decays like a Gaussian in the gap length, so the smallest margin among the first N midpoints should decrease only like a power of $\sqrt{\log N}$. Through the cancellation lemma this predicts the width, as a function of the height, of the strip to the right of each basepoint in which an off line zero could hide without being cancelled: the quantitative input that question (1) of Section 14.3.7 needs.

14.3.6 Moments of zeta, the Barnes function and zero density

The strength of Theorem 14.3.2 is set by the zero density exponents, which rest on mean value estimates on the critical line, Ingham's on the fourth moment of ζ. Keating and Snaith [180] modelled $\zeta(\frac12+it)$ by the characteristic polynomial $Z_N(\theta)=\det(I-Ae^{-i\theta})$ of a CUE matrix A of size N, with N in the role of $\log(T/2\pi)$. By analogy with the exact formula

$$\mathbb{E}|Z_N(\theta)|^{2k}=\prod_{j=1}^{N}\frac{\Gamma(j)\Gamma(j+2k)}{\Gamma(j+k)^2}\sim\frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)}\,N^{k^2},\qquad N\to\infty,$$

they conjectured that the moments of ζ grow in the same way, with $\log(T/2\pi)$ in place of N and an arithmetic factor a_k in front (Section 14.3.6.1). Here G is the Barnes G function, $\mathsf{G}(1+z)=\Gamma(z)\mathsf{G}(z)$, $\mathsf{G}(1)=1$, in sans serif to keep G

for the clock. Nikeghbali and Yor [248] gave the random matrix factor a meaning in GGC language: the Mellin transform of $|Z_N|$ and the Barnes function are expressed through products and sums of independent gamma, beta and log gamma variables, and $1/\mathsf{G}$ has a Lévy Khintchine type representation. The source is the decomposition of Bourgade, Hughes, Nikeghbali and Yor [44],

$$Z_N \stackrel{d}{=} \prod_{j=1}^{N} \Big(1 + e^{i\omega_j}\sqrt{B_{1,j-1}}\Big),$$

with independent angles ω_j uniform on $[0, 2\pi)$ and independent beta variables $B_{1,j-1}$ with parameters 1 and $j-1$, $B_{1,0} = 1$.

14.3.6.1 The Keating and Snaith moments

Write $I_k(T) = \frac{1}{T}\int_0^T |\zeta(\frac{1}{2} + it)|^{2k} dt$. The conjecture of Keating and Snaith [180], in the form of Conrey, Farmer, Keating, Rubinstein and Snaith [62], is

$$I_k(T) \sim a_k \, \frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)} \Big(\log \frac{T}{2\pi}\Big)^{k^2}, \qquad a_k = \prod_p \Big(1 - \frac{1}{p}\Big)^{k^2} \sum_{m \ge 0} \Big(\frac{\Gamma(m+k)}{m!\,\Gamma(k)}\Big)^2 p^{-m}.$$

It is a theorem for $k = 1, 2$ and open for $k \ge 3$. For $k = 1$ Hardy and Littlewood [105] proved the leading term, and Ingham [162] the mean square with its secondary term, $I_1(T) = \log \frac{T}{2\pi} + 2\gamma_E - 1 + o(1)$; for $k = 2$ Ingham [162] proved the asymptotic with leading coefficient $\frac{1}{2\pi^2}$. The local factor of a_k is $(1 - \frac{1}{p})^{k^2} {}_2F_1(k, k; 1; \frac{1}{p})$; since ${}_2F_1(1, 1; 1; x) = (1-x)^{-1}$ and ${}_2F_1(2, 2; 1; x) = (1+x)(1-x)^{-3}$, it is 1 for $k = 1$ and $1 - p^{-2}$ for $k = 2$, so $a_1 = 1$ and $a_2 = 6/\pi^2$. The Euler product over the primes below 2×10^5 gives $a_3 = 0.0493218$ (computed); since the local factor for $k = 3$ is $1 - 9p^{-2} + O(p^{-3})$, the omitted primes lower it by a relative $9\sum_{p > 2\times 10^5} p^{-2} \approx 4 \times 10^{-6}$, to about 0.0493217. With $\mathsf{G}(3)^2/\mathsf{G}(5) = \frac{1}{12}$ and $\mathsf{G}(4)^2/\mathsf{G}(7) = \frac{1}{8640}$ the leading coefficients are 1, $0.0506606 = \frac{1}{2\pi^2}$, Ingham's constant, and 5.709×10^{-6}, the value $42a_3/9!$ of [62].

Table 14.12, computed, compares the moments with the leading terms. For $k = 1$ the full asymptotic is already accurate at $T = 1000$. For $k = 2$ and 3 the leading term is a small part of the moment at these heights, the lower order terms of the polynomial in $\log T$ predicted in [62] dominating.

14.3.6.2 The moments as exponential moments of the Thorin density

The densities v_σ are the derivatives of $\log|\xi|$ across the basepoints: $\frac{\partial}{\partial\sigma}\log|\xi(\sigma + it)| = v_\sigma(t)$ wherever $\xi(\sigma + it) \ne 0$. Integrating from the critical line to the basepoint one, for every t such that ξ has no zero on the segment from $\frac{1}{2} + it$ to $1 + it$, which excludes only countably many t,

$$\log|\xi(\tfrac{1}{2} + it)| = \log|\xi(1 + it)| - \int_{1/2}^{1} v_a(t)\, da. \tag{14.3.2}$$

T	$I_1(T)$	$\log\frac{T}{2\pi}+2\gamma_E-1$	$I_2(T)$	leading, $k=2$	$I_3(T)$	leading, $k=3$
125	3.062	3.145	28.0	4.05	396	0.11
250	3.869	3.838	53.1	9.33	1136	0.71
500	4.550	4.531	84.8	18.6	2632	3.36
1000	5.211	5.224	127.2	33.5	5309	12.6

Table 14.12: Moments of ζ on the critical line against the asymptotic of Hardy, Littlewood and Ingham for $k=1$ and the leading Keating and Snaith terms for $k=2$ and 3. Computed.

With the gamma factor $h(s)=\frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, $\xi=h\zeta$, this becomes

$$|\zeta(\tfrac12+it)|^{2k}=|\zeta(1+it)|^{2k}\Big|\frac{h(1+it)}{h(\frac12+it)}\Big|^{2k}\exp\Big\{-2k\int_{1/2}^{1}v_a(t)\,da\Big\},$$

and Stirling's formula gives $|h(1+it)/h(\frac12+it)|\sim(t/2\pi)^{1/4}$. So the Keating and Snaith moments are exponential moments, over t uniform on $[0,T]$, of the density $v_a(t)$ integrated across the basepoints from $\frac12$ to 1, with a weight $|\zeta(1+it)|^{2k}|h(1+it)/h(\frac12+it)|^{2k}$ involving no zero of ζ. In the same terms RH, the positivity of v_a for every $a>\frac12$ (Lagarias's criterion, Remark 2.1.4), is the statement that $|\xi(\sigma+it)|$ increases with $\sigma\ge\frac12$ for every t, a reformulation due to Sondow and Dumitrescu [333].

In the language of (14.3.2) the conjecture of Keating and Snaith is a large deviation statement for the integrated Thorin density. By Selberg's central limit theorem [330], for t uniform on $[T,2T]$, $\log|\zeta(\frac12+it)|$ is approximately Gaussian with variance $\frac12\log\log T$; Gaussian exponential moments would give $(\log T)^{k^2}$, and $a_k\mathsf{G}(1+k)^2/\mathsf{G}(1+2k)$ is the correction. On $[500,1000]$ the sample variance of $\log|\zeta(\frac12+it)|$ is 1.53, against $\frac12\log\log 750=0.95$. The difference is a constant term, not slow convergence: the random matrix model predicts the variance $\frac12\log\log\frac{t}{2\pi}+c+o(1)$ with $c=\frac12\big(\gamma_E+1-\sum_p\sum_{m\ge2}(\frac1m-\frac{1}{m^2})p^{-m}\big)=0.70$ [180], a conjecture, which gives 1.48 on $[500,1000]$, close to the sample value.

14.3.6.3 What the moments give

Ingham's exponent $3(1-\sigma)/(2-\sigma)$, which gives Theorem 14.3.2 its saving $(2\alpha-1)/(2-\alpha)$ for $\frac12<\alpha<\frac{7}{10}$, rests on the fourth moment [162]. The Lindelöf hypothesis, $\int_0^T|\zeta(\frac12+it)|^{2k}dt\ll T^{1+\varepsilon}$ for every k, implies the density hypothesis $N(\sigma,T)\ll T^{2(1-\sigma)+\varepsilon}$, which would give the saving $2\alpha-1$, larger than that of Theorem 14.3.2 for $\frac12<\alpha<\frac{9}{10}$; the Keating and Snaith asymptotics would give this and much more. The GGC structure of Nikeghbali and Yor belongs to the random matrix model, not to ζ: it explains the conjecture without proving it, and only the second and fourth moments of ζ are known asymptotically. And

moments, even the Lindelöf hypothesis, bound the number of zeros off the line but do not exclude them.

14.3.7 From almost to exactly

Theorem 14.3.2 cannot be upgraded to exact positivity by improving the exponents or by moment bounds. Even the density hypothesis, or the Lindelöf hypothesis, leaves room for a sparse set of zeros off the line, and a single off line zero $\beta + i\gamma$ stops T_α from being a GGC for every $\alpha \in [\frac{1}{2}, \beta)$ (Theorem 2.1.2) and makes v_α negative near γ for every α close enough to β (Section 14.3.3). The step from "almost" to "exactly" is a step from counting zeros to excluding them.

Four facts say that this step needs input from outside the programme. The smoothing runs the wrong way: when no zero lies to the right of α the density at $\alpha' > \alpha$ is the Poisson smoothing of the density at α by the kernel of width $\alpha' - \alpha$, so positivity propagates to the right; RH asks to propagate it to the left, which is deconvolution, and deconvolution does not preserve positivity. Arithmetic domination requires a zero free region: below the basepoint one the compensated Lévy form, which the sine squared lemma converts into the density, holds if $\psi(x) - x = O(x^{\alpha'})$ for some $\alpha' < \alpha$, and only if $\psi(x) - x = O(x^{\alpha+\delta})$ for every $\delta > 0$, ψ Chebyshev's function (Proposition 12.1.13); either condition is a zero free half plane, none was known for any $\alpha < 1$ before the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026, from outside the programme (Section 25.1.2), and removing the pole does not help: for $L(s, \chi_{12})$ the prime sums need no compensation and the boundary is the same (Section 15.2.2). There is no margin: along the de Bruijn Newman flow the reciprocal is a GGC for $t \geq 0.2$ [277] (Theorem 17.1.16 and Section 4.1), RH is the statement at $t = 0$, and $\Lambda \geq 0$ [318], so a proof must establish a closed condition at the boundary itself. And stronger positivity conditions can be false: Conrey and Li showed that a positivity condition in de Branges' approach fails [61], so any positivity statement stronger than RH should be tested numerically first (Section 25.2.4).

In the Stieltjes form (iv) of Theorem 9.2.4 the difference between counting and excluding is explicit. Split $G'/G = S_{\text{on}} + R_{\text{off}}$, where $S_{\text{on}}(u) = \sum (u + \gamma^2)^{-1}$ runs over the zeros $\frac{1}{2} + i\gamma$ on the line with $\gamma > 0$ and is a Stieltjes function with measure U_{line}, and $R_{\text{off}}(u) = \sum (u - (\rho - \frac{1}{2})^2)^{-1}$ runs over the zeros off the line with $\gamma > 0$. By Theorem 9.3.9 all poles of R_{off} lie in a parabolic sliver $\mathcal{E}_{T_0}$ beside the negative axis, beginning at $-T_0^2 + T_0 + \frac{1}{4}$, and in the upper half plane outside it $\operatorname{Im} R_{\text{off}} \leq 0$ and $\operatorname{Im}(G'/G) < 0$. The size of R_{off} away from the sliver needs no count: if $|u| \leq T_0^2/2$, a disc missing $\mathcal{E}_{T_0}$, every term has $|u - (\rho - \frac{1}{2})^2| \geq |\rho - \frac{1}{2}|^2 - |u| > \gamma^2/2$, since $\gamma > T_0$, so $|R_{\text{off}}(u)| \leq 2\Sigma_0 < 3 \times 10^{-12}$ by Theorem 14.3.6.

The number of terms of R_{off}, on the other hand, is not known to be small. Zero density estimates bound those with $|\beta - \frac{1}{2}| \geq \sigma - \frac{1}{2}$ and $\gamma \leq T$ by $T^{e(\sigma)+o(1)}$, $e(\sigma) < 1$, for each fixed $\sigma > \frac{1}{2}$ (Theorem 14.3.2), which is $o(T)$; but zeros off the line may approach the line as γ grows, and for all of them together only a

proportion is known: at most $(1-\kappa+o(1))N(T)$ of the zeros up to height T lie off the line, with $\kappa > \frac{5}{12}$ in the last published record [312] and $\kappa > 0.6725$, counting only simple zeros on the line, in two preprints of 2026 [2, 202]; that is not $o(T)$, nor even $o(N(T))$. No count touches the Thorin condition, which requires $R_{\text{off}} \equiv 0$, because a single pair puts a pole of G'/G off the cut. Section 14.4.5 gives a test for arguments that claim to exclude such poles: since the de Bruijn Newman flow has nonreal zeros at every $t < 0$ [318], no property that the kernel $e^{tv^2/4}\Phi(v)$ keeps for some $t < 0$ can exclude them, so a proof must use one that fails for every $t < 0$; the Euler product and the explicit formula are the natural candidates.

The forms the new input could take are those of Section 25.1. The first new theorem any of them would give is the GGC property of T_α at a single basepoint $\alpha < \frac{7}{8}$, that is $v_\alpha^- \equiv 0$, a zero free half plane beyond the one announced in 2026 (Section 25.1.2); positivity of v_α on the single line $\operatorname{Re} s = \alpha$ is not known to suffice (Remark 2.1.4).

Sections 14.3.2 and 14.3.3 make the remaining problem concrete. The growth of the defect is not part of it. For $u \geq 1$ every factor of D_α is at most one, and a factor with $\gamma \leq \sqrt{u}$ is at least $\gamma^2/(u+\gamma^2) \geq \gamma^2/(2u) \geq 1/u$, since $\gamma \geq \gamma_1 > \sqrt{2}$. Bounding the factors with $\gamma > \sqrt{u}$ as in the proof of Theorem 14.3.6 gives, for $u \geq 1$,

$$\begin{aligned} |\log D_\alpha(u)| &\leq N(\alpha, \sqrt{u}) \log u + 2(u+\sqrt{u}) \sum_{\beta>\alpha,\ \gamma>\sqrt{u}} \gamma^{-2} \\ &\leq u^{e(\alpha)/2+o(1)}, \qquad u \to \infty, \end{aligned}$$

by the exponents of Theorem 14.3.2. So bounds on the growth of the defect follow from zero density estimates alone, and what remains open is whether it is identically one. Four questions would each be a genuine step.

1. Cancellation on average: show that for fixed $\alpha > \frac{1}{2}$ the background density v_α/π is bounded below by $c \log \gamma$ outside a set of heights of density zero. With Lemma 14.3.9 this would confine any failure of positivity to zeros in the strips $\alpha < \beta < \alpha + O(1/\log \gamma)$. Remark 12.1.17 proves the lower bound for v_α/π itself, but its exceptional set contains a neighbourhood of the ordinate of every zero to the right of α, which is where Lemma 14.3.9 would need it.

2. Thin strip density: bound the number of zeros with $\alpha < \beta < \alpha + 2/\log \gamma$ and $\gamma \leq T$ by $o(T^{1-\delta})$ for some $\delta > 0$, uniformly in α. Combined with (1) this would give a pointwise almost positivity theorem.

3. Explicit defect: bound $\log D_\alpha(u)$ explicitly, as described at the end of Section 14.3.2.

4. Positivity at one basepoint: prove $D_\alpha \equiv 1$, that is $v_\alpha^- \equiv 0$, for a single $\alpha < \frac{7}{8}$, which is a zero free half plane beyond the one announced in 2026 (Section 25.1.2).

What the Thorin framework contributes is a precise target, the GGC property of one explicit clock T_α at one basepoint below $\frac{7}{8}$, together with tools that test candidate arguments against it: certified densities on windows, positivity margins calibrated against random matrix theory, and the explicit defect. Chapter 25 takes stock of the whole path, beginning with Section 25.1.

14.4 Pakes, the Lambert law and the Thorin condition

Anthony Pakes has worked on the objects around the Thorin condition: size biasing and the moment problem, Laplace transforms of positive infinitely divisible laws, the Lambert function, and ladders of self decomposable laws. This section is based on [292], verified and corrected where needed. In Pakes's language the question of Section 4.1 is whether $\log G$ is a Thorin Bernstein function, a Bernstein function whose Lévy measure is $k(t)\,dt/t$ with k completely monotone [327, Chapter 8], equivalently the Laplace exponent of a GGC; by Theorem 9.2.4 this holds if and only if G'/G is a Stieltjes function, and if and only if RH holds.

The section gives a dictionary between Pakes's language and the book's, a few unconditional facts about the clock, the solved Lambert problem as a template whose single failing step for the clock is RH itself, a list of closed routes, and one property, the GGC property of the variable of Biane, Pitman and Yor, that passes the filter of Section 14.4.5 wherever tested but fails for most real Dirichlet L functions tested, so that it does not detect the Euler product.

14.4.1 Two strands of Pakes's work

The first strand is the moment problem. For $X \geq 0$ with $0 < \mathbb{E}X < \infty$, the size biased (Pakes: length biased) variable $\widehat{X}$ has law $x\,P(X \in dx)/\mathbb{E}X$. Pakes and Khattree [268] characterise laws by in law equations linking $\widehat{X}$ and X, such as $X \stackrel{d}{=} \mathsf{U}\widehat{X}$ with U uniform on $(0,1)$ and independent, which holds exactly for the exponential laws; Pakes studies size biasing for laws with the log normal moments [259], and in [260] proves converse Carleman and Krein criteria and introduces the size biasing and stationary excess operations of order r (Berg [28] showed, correcting a remark there, that a Stieltjes determinate law can have indeterminate size biased and stationary excess versions of order one), criteria for moment determinacy [267] and the Stieltjes classes of moment equivalent laws [261].

Size biasing meets infinite divisibility in Steutel's theorem: X is infinitely divisible if and only if $\widehat{X} \stackrel{d}{=} X + A$ for some $A \geq 0$ independent of X, the law of A then being unique [338], [325, Theorem 51.1], [340], [8, Theorem 11.1], [269]. With $\Psi(u) = -\log \mathbb{E}e^{-uX}$, A has Laplace transform $\Psi'(u)/\Psi'(0)$. In [263] Pakes

asks when A or $\widehat{X}$ is again infinitely divisible; he shows that size biasing up to the fourth order preserves infinite divisibility for the Lambert law and for a compound Poisson law, and conjectures it for every order.

For the laws of this book the moment problem is determinate. The moment generating function $\xi(\frac{1}{2}+z)/\xi(\frac{1}{2})$ of the Pólya density is entire (2.1.4). The clock has $\mathbb{E}e^{sT} < \infty$ exactly for $s < \gamma_1^2 = 199.7904548\ldots$: every zero has $|\rho - \frac{1}{2}| \geq |\operatorname{Im}\rho| \geq \gamma_1$, so $1/G$ is analytic in $|u| < \gamma_1^2$ with a pole at $-\gamma_1^2$, and by Landau's theorem [359, Chapter II, Theorem 5b] the abscissa of convergence is $-\gamma_1^2$. Under RH the measure $\nu = \sum_k a_k^{-1}\delta_{1/a_k}$ of the Stieltjes moments m_n is supported in $[0, \gamma_1^{-2}]$, a determinate Hausdorff problem (Section 8.1), while U_{RH} has unbounded support and infinite mass.

The second strand concerns Laplace transforms of positive infinitely divisible laws and the Lambert function $\mathcal{W}$ (Section 14.4.3). Pakes [262] and, independently, Kalugin and Jeffrey [174] proved that $\mathcal{W}$ is a Bernstein function, from a formula for its derivatives involving a sequence of polynomials [266, (8)]; so $e^{-\mathcal{W}(u)} = \mathcal{W}(u)/u$ is the Laplace transform of an infinitely divisible law, the Lambert law. In the same paper, by Pakes's account [266, Section 2], a complex variable argument shows that $\mathcal{W}'$ is a Stieltjes function, so $\mathcal{W}$ is Thorin Bernstein and the Lambert law is a GGC. Kalugin, Jeffrey, Corless and Borwein [176] obtained Stieltjes representations independently and showed that $\mathcal{W}$ is a complete Bernstein function; the Thorin representation of $\mathcal{W}$ is the theorem proved below. Pakes gave two further proofs that $\mathcal{W}$ is Thorin Bernstein: from Nuttall's integral for the coefficients $n^n/n!$ of $\mathcal{W}'$, which yields an explicit quantile function of the Thorin measure [264], and from continuous state branching processes, one of whose martingale limits has cumulant function $\mathcal{W}$ [265], which also gives self decomposability of the Lambert law.

In [266] Pakes identifies the background driving Lévy process of the Lambert law: compound Poisson with unit rate and jump density $\int ze^{-xz}\,U_{\mathcal{W}}(dz)$, $U_{\mathcal{W}}$ the Thorin measure of $\mathcal{W}$. With $T_{\mathcal{W}}$ Lambert distributed and J a jump, this gives $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}} + J)$ and $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$, with $\mathcal{S}$ the stationary excess operator and the factors independent; if both hold for some X and $J \geq 0$ of finite mean $\mu > 0$, then $X \stackrel{d}{=} \mu T_{\mathcal{W}}$ [266, Theorem 1] (Remark 14.4.5). He embeds the Lambert law in a family of GGCs whose Laplace transforms solve Lambert's trinomial equation [266, Theorem 6], another proof that $\mathcal{W}$ is Thorin Bernstein [266, Corollary 2], and introduces BDLP ladders: from an infinitely divisible $X_0 \geq 0$ with $\mathbb{E}(\log(1+X_0))^n < \infty$ for every n, X_{n+1} is the self decomposable law driven by the Lévy process of X_n [266, Section 5]. His X_1 is self decomposable, a unit shift from Urbanik's classes L_n of Section 14.4.4, in which L_0 is the class of self decomposable laws [354, 324, 29]. Along a ladder he finds $\widehat{X}_{n+1} \stackrel{d}{=} X_{n+1} + A_n$, with A_n of density proportional to the tail of the Lévy measure of X_n [266, (53)], the form in which Section 14.4.2 applies Steutel's theorem to the clock.

14.4.2 The Thorin question in Pakes's form

If $\log G$ is Thorin Bernstein (Theorem 9.2.4), its Thorin measure is $U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}$, with no drift (Proposition 2.1.1), and Stieltjes inversion (4.1.2) recovers it from the boundary values of G'/G on the cut, or in integrated form from those of $\log G$ (Section 4.1.1). With no hypothesis, $\log G$ is a Bernstein function with Lévy measure $W(t)\,dt/t$ (Theorem 8.1.1); under RH, $W(t) = \int e^{-tz}\, U_{\mathrm{RH}}(dz)$, the counterpart of the formula that gives the canonical function of the Lambert law as the Laplace transform of its Thorin measure (Corollary 14.4.4; [266, (12)]).

The Bernstein property is weaker but not blind to zeros off the line: $W(t) \geq 0$ would exclude, for instance, a zero $\rho = \frac{1}{2} + \delta + i\gamma$ off the line below all the other zeros, since if $\gamma^2 - \delta^2$ were smaller than every other $\operatorname{Re} a_k$, its four zeros would contribute $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, which dominates as $t \to \infty$ and changes sign. The property holds with no hypothesis (Theorem 8.1.1, using the verification of RH below height 1000), so it cannot decide RH.

Size biasing gives the Thorin question a form with one variable. The size biased clock $\widehat{T}$ has law $t\,P(T \in dt)/\mathbb{E}T$, and $\mathbb{E}T = m_0 = \sum_k a_k^{-1} = 0.0231049931\ldots$.

Proposition 14.4.1. *With no hypothesis, $\widehat{T} \stackrel{d}{=} T + A$, with A independent of T and of density $W(t)/m_0$ on $(0,\infty)$. RH holds if and only if A is a mixture of exponential laws, and then $A \stackrel{d}{=} E/V$, with E standard exponential and independent of V, and V of law $m_0^{-1}\sum_{\gamma>0}\gamma^{-2}\delta_{\gamma^2}$, the ordinates counted with multiplicity.*

Proof. The Laplace transform of $\widehat{T}$ is

$$\frac{\mathbb{E}[Te^{-uT}]}{m_0} = -\frac{1}{m_0}\,\frac{d}{du}\,\frac{1}{G(u)} = \frac{1}{m_0}\,\frac{1}{G(u)}\,\frac{G'}{G}(u).$$

By Section 2.1, $(G'/G)(u) = \int_0^\infty e^{-ut}W(t)\,dt$ for $u > 0$ with no hypothesis, and $W > 0$ (Theorem 8.1.1); letting $u \downarrow 0$, monotone convergence gives $\int_0^\infty W(t)\,dt = (G'/G)(0) = G'(0) = \sum_k a_k^{-1} = m_0$. So W/m_0 is a probability density with Laplace transform $(G'/G)(u)/m_0$, the Laplace transform of $\widehat{T}$ is the product of those of T and of a variable A with this density, and by the uniqueness of Laplace transforms $\widehat{T} \stackrel{d}{=} T + A$ with A independent of T. A variable $A > 0$ is a mixture of exponential laws, $A \stackrel{d}{=} E/V'$ with $V' > 0$ independent of E, exactly when it has a completely monotone density: the density of E/V' is $\mathbb{E}[V'e^{-V't}]$, and conversely, by Bernstein's theorem, a completely monotone probability density is $\int_{[0,\infty)} e^{-vt}\,\sigma(dv)$ with $\int v^{-1}\sigma(dv) = 1$, which forces $\sigma(\{0\}) = 0$, and this is the density of E/V' with $P(V' \in dv) = v^{-1}\sigma(dv)$. The density W/m_0 is completely monotone exactly when RH holds (Section 2.1 and Table 4.1), and then $W(t) = \sum_{\gamma>0} e^{-\gamma^2 t} = m_0\mathbb{E}[Ve^{-Vt}]$ with V as stated. □

This is Steutel's decomposition with $\Psi = \log G$, and it is Pakes's identity along BDLP ladders [266, (53)] for the clock and its driver of Proposition 14.1.1,

with Lévy density $h = -W'$: Pakes's A_n has density the tail $\int_x^\infty \omega_n$ of the Lévy density of the driving law normalised by its mean, and here $\int_t^\infty h = W(t)$ and $\int_0^\infty x h(x)\,dx = m_0$. The variable A has mean $\mathbb{E}A = m_1/m_0 = \operatorname{Var} T/\mathbb{E}T = 1.60885567 \times 10^{-3}$, from the Taylor coefficients of ξ at $\frac12$, with no zero entering. Under RH, A is infinitely divisible as a mixture of exponentials [94, 337], and so is $\widehat{T} = T + A$: both parts of Pakes's question in [263] then have a positive answer for the clock. The weights $P(V = \gamma^2) = \gamma^{-2}/m_0$ of the first six ordinates are 0.21663, 0.09794, 0.06919, 0.04676, 0.03990 and 0.03064, together 0.50105; the first 1700 carry 0.978508 (computed).

The moment problem enters through the Taylor coefficients at the origin: for $|u| < \gamma_1^2$ and $|u| < 1/e$ respectively,

$$(\log G)'(u) = \sum_{n\ge 0} m_n (-u)^n, \qquad \mathcal{W}'(u) = \sum_{n\ge 1} \frac{n^n}{n!}(-u)^{n-1},$$

the second the derivative of the Lagrange series of $\mathcal{W}$ [65, (3.1)]. Under RH, $m_{n-1} = \int z^{-n}\, U_{\mathrm{RH}}(dz)$ for $n \ge 1$, just as $n^n/n! = \int z^{-n}\, U_{\mathcal{W}}(dz)$ (Corollary 14.4.6). For the clock positive definiteness of every $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is equivalent to RH (Section 8.1); the number of negative eigenvalues of $\mathcal{H}_N$ is nondecreasing in N and tends to the number of distinct zeros with $\operatorname{Re}\rho > \frac12$, $\operatorname{Im}\rho > 0$ (Theorem 8.1.7).

Li's criterion is a positivity condition on the same zeros, read on the Thorin functional through the test functions φ_n of Section 23.6.2. It sees a zero off the line more slowly: the term of $\rho = \frac12 - d + i\gamma$, $0 < d < \frac12$, grows like e^{nd/γ^2}, so it can dominate only from n of order $\gamma^2/d > 2\gamma^2$, while in the data of Section 4.1.3 the Hankel test often registers a pair at an order close to $2k+2$, set by its index k, at a precision cost growing linearly with the order (Section 8.1).

14.4.3 The Lambert law, a Thorin problem solved

Let $\mathcal{W}$ be the principal branch of Lambert's function, the solution of $\mathcal{W}(u)e^{\mathcal{W}(u)} = u$ analytic on $\mathbb{C} \setminus (-\infty, -1/e]$, real and increasing on $(-1/e, \infty)$, with $\mathcal{W}(0) = 0$, $\mathcal{W}(-1/e) = -1$ and $|\operatorname{Im}\mathcal{W}| < \pi$. Its Lagrange series $\mathcal{W}(u) = \sum_{n\ge1}(-n)^{n-1}u^n/n!$ has radius $1/e$, and $\mathcal{W}(u) = \log u - \log\log u + o(1)$ as $u \to \infty$ [65, (3.1), (4.19)]. We write $T_{\mathcal{W}}$ for a variable with the Lambert law. The proof that it is a GGC is a template that fails for the clock at exactly one step.

Proposition 14.4.2 ([65, Section 4, (4.4)], [175, (10), (11)]). *For $0 < b < \pi$ put $r(b) = b\,e^{-b\cot b}/\sin b$. Then r increases from $1/e$ to ∞ on $(0,\pi)$, and the boundary values of $\mathcal{W}$ on the cut, from the upper half plane, are $\mathcal{W}(-r(b)+i0) = -b\cot b + ib$.*

Proof. Put $w = \alpha + ib$ with $0 < b < \pi$ in $we^w = -x$, $x > 0$. The imaginary part gives $e^\alpha(\alpha\sin b + b\cos b) = 0$, so $\alpha = -b\cot b$, and then the real part gives $e^\alpha(\alpha\cos b - b\sin b) = -e^{-b\cot b}\, b/\sin b$, so that $x = r(b)$. Next, $(\log r)'(b) =$

$1/b - 2\cot b + b/\sin^2 b$ and $b\sin^2 b\,(\log r)'(b) = (b - \sin b\cos b)^2 + \sin^4 b > 0$, while $r(0+) = 1/e$ and $r(\pi-) = \infty$; so for each $x > 1/e$ there is exactly one point w of the curve $\{-b\cot b + ib : 0 < b < \pi\}$ with $we^w = -x$, namely the point with $r(b) = x$. This curve bounds the range of the principal branch in the upper half plane [65, Section 4], and $\mathcal{W}$ is continuous on the cut from above, with values on it; so $\mathcal{W}(-r(b) + i0) = -b\cot b + ib$. □

Kalugin, Jeffrey and Corless obtain the next theorem from the Stieltjes representation of $\mathcal{W}'$ [175, Theorem 3.3] and the characterisation of Thorin Bernstein functions in [327, Chapter 8].

THEOREM 14.4.3 (Kalugin, Jeffrey and Corless [175, Theorem 6.2], [177]). *Let $U_{\mathcal{W}}$ be the image of the uniform law db/π on $(0,\pi)$ under r, a probability measure on $(1/e,\infty)$. Then for $u \in \mathbb{C}\setminus(-\infty,-1/e]$*

$$\mathcal{W}(u) = \int \log\Big(1 + \frac{u}{z}\Big)\,U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi \log\Big(1 + \frac{u}{r(b)}\Big)\,db.$$

Hence $\mathcal{W}$ is a Thorin Bernstein function, and $T_{\mathcal{W}}$ is a GGC with no drift and Thorin measure $U_{\mathcal{W}}$, of total mass one, whose distribution function is $U_{\mathcal{W}}((1/e,y]) = \pi^{-1}\operatorname{Im}\mathcal{W}(-y+i0)$ for $y > 1/e$.

Proof. Let $\Psi(u)$ be the integral. Since $U_{\mathcal{W}}$ is carried by $(1/e,\infty)$, Ψ is analytic on $\mathbb{C}\setminus(-\infty,-1/e]$, real on $(-1/e,\infty)$, and $\Psi(\bar u) = \overline{\Psi(u)}$; so is $\mathcal{W}$. Let $x > 1/e$ and $r(b(x)) = x$. As $\eta \downarrow 0$, $1 + (-x + i\eta)/z$ tends to a negative number from the upper half plane when $z < x$ and to a positive number when $z > x$, and $U_{\mathcal{W}}$ has no atoms, so $\operatorname{Im}\Psi(-x+i0) = \pi\,U_{\mathcal{W}}((1/e,x)) = b(x)$; by Proposition 14.4.2 also $\operatorname{Im}\mathcal{W}(-x+i0) = b(x)$, which is the formula for the distribution function. Near $1/e$, $r(b) - 1/e \sim b^2/(2e)$, so $U_{\mathcal{W}}$ has a density asymptotic to $(e/2)^{1/2}\pi^{-1}(z - 1/e)^{-1/2}$ there, and a bounded continuous density on $[1/e + c,\infty)$ for every $c > 0$. Hence Ψ and $\mathcal{W}$ extend continuously to the closed upper half plane, and $D = \mathcal{W} - \Psi$ is continuous there and real on the real axis, the imaginary parts of the boundary values agreeing on $(-\infty,-1/e)$. Schwarz reflection continues D across $(-\infty,-1/e)$, consistently with $D(\bar u) = \overline{D(u)}$, and the bounded singularity at $-1/e$ is removable. So D is entire, with $D(0) = 0$.

It remains to bound D, and one sided bounds on real parts suffice. First, $\operatorname{Re}\Psi(u) = \int \log|1 + u/z|\,U_{\mathcal{W}}(dz) \le \log(1 + e|u|)$. Next let $|u| = R \ge e$ and $w = \mathcal{W}(u)$, so that $|w|e^{\operatorname{Re} w} = R$ and $|\operatorname{Im} w| < \pi$. If $|w| > \log R + \pi$, then $|\operatorname{Re} w| > \log R$; this is impossible when $\operatorname{Re} w \ge 0$, since then $\operatorname{Re} w = \log R - \log|w| < \log R$, and when $\operatorname{Re} w < 0$ it gives $|w| - \log|w| < \pi - \log R \le \pi - 1$, which fails for $|w| > 1 + \pi$, since $x - \log x$ increases for $x > 1$ and exceeds $\pi - 1$ at $x = 1 + \pi$. Hence $|w| \le \log R + \pi$ and $\operatorname{Re}\mathcal{W}(u) \ge \log R - \log(\log R + \pi)$. By continuity both bounds hold on the whole circle, so

$$\max_{|u|=R} \operatorname{Re}\big(-D(u)\big) \le \log(1 + eR) - \log R + \log(\log R + \pi), \qquad R \ge e.$$

The Borel Carathéodory lemma for $-D$, which vanishes at 0, gives $\max_{|u|=R/2}|D(u)| \le 2\max_{|u|=R}\mathrm{Re}(-D(u)) = O(\log\log R)$, and by Cauchy's estimates $D \equiv 0$.

□

Corollary 14.4.4 (first part from [175, Theorem 6.5] and [177]). *The canonical function of $T_{\mathcal{W}}$ is*

$$k_{\mathcal{W}}(x) = \int e^{-xz}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi e^{-x\, r(b)}\, db,$$

so that $\mathcal{W}(u) = \int_0^\infty (1-e^{-ux})\, k_{\mathcal{W}}(x)\, dx/x$ for $u \ge 0$. The symmetric law of $Z\sqrt{2T_{\mathcal{W}}}$, with Z standard normal and independent of $T_{\mathcal{W}}$, has characteristic function $e^{-\mathcal{W}(w^2)}$, is an EGGC (Section 4.1.2), and has Lévy density $k^{\mathrm{s}}_{\mathcal{W}}(|x|)/|x|$ with

$$k^{\mathrm{s}}_{\mathcal{W}}(x) = \int e^{-x\sqrt{z}}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi e^{-x\sqrt{r(b)}}\, db,$$

that is, $\mathcal{W}(w^2) = \int_0^\infty 4\sin^2(wx/2)\, k^{\mathrm{s}}_{\mathcal{W}}(x)\, dx/x$ for real w.

Proof. Insert Frullani's integral $\log(1+u/z) = \int_0^\infty (1-e^{-ux})e^{-zx}x^{-1}dx$ in Theorem 14.4.3 and use Tonelli's theorem. For the symmetric law, $\mathbb{E}e^{iwZ\sqrt{2T_{\mathcal{W}}}} = \mathbb{E}e^{-w^2 T_{\mathcal{W}}}$, and the cosine dictionary (10.2.4) with $U = U_{\mathcal{W}}$, for which $K_U = 2k^{\mathrm{s}}_{\mathcal{W}}$, gives the Lévy density $K_U(|x|)/(2|x|)$. □

Remark 14.4.5 (Self decomposability, after Pakes). Since $k_{\mathcal{W}}$ is nonincreasing, $T_{\mathcal{W}}$ is self decomposable, and as in Proposition 14.1.1 its driving subordinator Z has Lévy density $-k'_{\mathcal{W}}(x) = \int z e^{-xz}\, U_{\mathcal{W}}(dz)$, of total mass $k_{\mathcal{W}}(0+) = 1$ and mean $\int_0^\infty k_{\mathcal{W}} = \int z^{-1}\, U_{\mathcal{W}}(dz) = 1$ (Corollary 14.4.6). So the driver is compound Poisson with unit rate and jumps J of mean one; its Laplace exponent is $u\mathcal{W}'(u) = \mathcal{W}(u)/(1+\mathcal{W}(u))$, and $\mathbb{E}e^{-uJ} = 1/(1+\mathcal{W}(u))$ [266, (5), (6), (18)]. Stopping $T_{\mathcal{W}} = \int_0^\infty e^{-t}\, d\mathsf{Z}_t$ at the first jump, at an exponential time τ, gives the perpetuity $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}} + J)$, with $\mathsf{U} = e^{-\tau}$ uniform on $(0,1)$ and the three variables independent. The stationary excess $\mathcal{S}J$ has density $P(J > x)$ and Laplace transform $(1 - \mathbb{E}e^{-uJ})/u = \mathcal{W}'(u)$; since $\mathbb{E}e^{-u\mathsf{U}Y} = u^{-1}\int_0^u \mathbb{E}e^{-sY}ds$, also $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$, and $\mathcal{S}J \stackrel{d}{=} J + T_{\mathcal{W}}$.

Corollary 14.4.6 ([175, Corollary 3.4], [176]; Nuttall's integral [250, 46]). *For $n \ge 0$,*

$$\int z^{-n}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi \Big(\frac{\sin b\; e^{b\cot b}}{b}\Big)^n db = \frac{n^n}{n!} \qquad (0^0 = 1),$$

so $(n^n/n!)_{n\ge 0}$ is the moment sequence of the image of $U_{\mathcal{W}}$ under $z \mapsto 1/z$, a probability measure on $(0,e)$. Nuttall conjectured, and Bouwkamp proved, the identity with $\nu^\nu/\Gamma(\nu+1)$ for every real $\nu \ge 0$.

Proof. Differentiating Theorem 14.4.3 gives $\mathcal{W}'(u) = \int (u+z)^{-1} U_{\mathcal{W}}(dz) = \sum_{n\geq 1} (-u)^{n-1} \int z^{-n}\, U_{\mathcal{W}}(dz)$ for $|u| < 1/e$, uniformly in $z > 1/e$; compare with the derivative of the Lagrange series, whose coefficients are $(-n)^{n-1}/(n-1)! = (-1)^{n-1} n^n / n!$. For $n = 0$ the integral is the total mass. □

The section was checked numerically.

What transfers to the clock. The proof of Theorem 14.4.3 has four steps: the boundary values on the cut, the positivity of the boundary measure, the identification by reflection and a growth bound, and, for the symmetric law, the cosine dictionary (10.2.4). For the clock the first step is the argument count: continued along the upper side of the negative axis, past the zeros of G by small half circles in the upper half plane, $\operatorname{Im} \log G(-x+i0) = \pi N_{\text{line}}(\sqrt{x})$, where $N_{\text{line}}(t)$ counts with multiplicity the zeros on the critical line with ordinate in $(0,t)$, because the four zeros of a zero off the line contribute the factor $|1+u/a|^2 > 0$ at real u. So the distribution function $N_{\text{line}}(\sqrt{x})$ of U_{line} (Proposition 2.1.3) is read from the boundary values of $\log G$ with no hypothesis, and it is $N(\sqrt{x})$ under RH. The second step is then automatic, a counting function being nondecreasing. The fourth gives the Poisson trace $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$, which is $\sum_{\gamma>0} e^{-\gamma y}$ under RH, as the Lévy side of $Y_{1/2} = Z\sqrt{2T}$ (Section 13.1.4), while W is the canonical function of T (Section 4.1.1); the two traces are linked with no hypothesis by subordination. The third step is the obstruction: $\mathcal{W}$ is analytic off $(-\infty, -1/e]$ by construction, whereas $\log G$ is analytic off $(-\infty, 0]$ if and only if RH holds (Proposition 2.1.1 and the argument after it).

14.4.4 Multiply self decomposable laws and branching

The centre clock is self decomposable with no hypothesis (Theorem 8.1.1), so it is $\int_0^\infty e^{-t}\, dZ_t$ for its driving subordinator Z (Proposition 14.1.1). The law of Z_1 may be self decomposable in turn, with a driver of its own, and so on; read upwards such a chain is Pakes's BDLP ladder [266, Section 5], and its length places a law in one of Urbanik's classes L_n. For the centre clock the chain has exactly three steps, with no hypothesis on the zeros (Theorem 14.4.8), and the clock is not the martingale limit of a supercritical continuous state branching process, the construction behind one of Pakes's proofs for the Lambert law (Theorem 14.4.11). Neither says anything about RH.

14.4.4.1 Urbanik's classes and the ladder functions

A law μ on $\mathbb{R}$ is self decomposable if for every $c \in (0,1)$ there is a law μ_c, its c component, with $\widehat{\mu}(z) = \widehat{\mu}(cz)\, \widehat{\mu}_c(z)$. Urbanik [354] let L_0 be the class of self decomposable laws, L_n, $n \geq 1$, the laws in L_{n-1} all of whose components lie in L_{n-1}, and $L_\infty = \bigcap_n L_n$; Sato proved that L_∞ is the smallest class containing the

stable laws and closed under convolution and weak limits [324, Theorem 3.5]. A law lies in L_n, $n \geq 1$, exactly when it is self decomposable and the law at time one of its driving process lies in L_{n-1} [161]. For laws on $[0,\infty)$ with Laplace exponent f and $(Sf)(u) = uf'(u)$, Berg and Forst proved that $\mu \in L_n$ if and only if $Sf, S^2f, \ldots, S^{n+1}f$ are Bernstein functions [29, Theorem 1.3].

For the centre clock $f = \log G$, and S acts on the canonical function W through the ladder functions

$$W^{[0]} = W, \qquad W^{[j+1]}(t) = -t\,\frac{d}{dt}W^{[j]}(t), \qquad \text{so that} \quad W^{[j]} = \Big(-t\frac{d}{dt}\Big)^j W.$$

For one exponential, $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)\,e^{-at}$, where $\beta_0 = 1$ and $\beta_{j+1}(u) = u\beta_j(u) - u\beta_j'(u)$:

$$\beta_1(u) = u, \quad \beta_2(u) = u^2 - u, \quad \beta_3(u) = u^3 - 3u^2 + u, \quad \beta_4(u) = u^4 - 6u^3 + 7u^2 - u.$$

In general $\beta_j(u) = \sum_{i=1}^{j} (-1)^{j-i} S(j,i)\, u^i = (-1)^j B_j(-u)$, with $S(j,i)$ the Stirling numbers of the second kind and B_j the Touchard polynomials, and since $-u\frac{d}{du}$ multiplies Mellin transforms by s,

$$\int_0^\infty \beta_j(u)\, e^{-u} u^{s-1}\, du = s^j\, \Gamma(s) \qquad (s > 0). \tag{14.4.1}$$

The positive roots of β_3 are $(3 \mp \sqrt{5})/2 = 0.381966$ and 2.618034, and those of β_4 are 0.165757, 1.343380 and 4.490864, so $\beta_4 < 0$ on $(0, 0.165757) \cup (1.343380, 4.490864)$. Since $W(t) = \sum_k e^{-a_k t}$ with $\operatorname{Re} a_k \geq \gamma_k^2 - \frac{1}{4} > 0$ for every zero (Section 14.1.1),

$$W^{[j]}(t) = \sum_k \beta_j(a_k t)\, e^{-a_k t}, \tag{14.4.2}$$

absolutely and locally uniformly on $(0,\infty)$.

Lemma 14.4.7. *For every $j \geq 0$ and $u \geq 0$,*

$$\Big(u\frac{d}{du}\Big)^j \log G(u) = \int_0^\infty \big(1 - e^{-ut}\big)\, W^{[j]}(t)\, \frac{dt}{t}. \tag{14.4.3}$$

For $n \geq 0$ the centre clock lies in L_n if and only if $W^{[n+1]} \geq 0$ on $(0,\infty)$, and then $W^{[1]}, \ldots, W^{[n]}$ are nonnegative too. In that case the driving processes $Z^{[1]} = Z$, $Z^{[2]}$, …, $Z^{[n+1]}$ exist, each the driving process of the law at time one of the one before; $Z^{[j]}$ has Laplace exponent $(u\frac{d}{du})^j \log G(u)$ and Lévy density $W^{[j]}(t)/t$; and for $1 \leq j \leq n$ the law of $Z_s^{[j]}$ is self decomposable for every $s > 0$.

Proof. By (14.4.2), $W^{[j]}(t) = O(t^j e^{-\gamma_1^2 t})$ as $t \to \infty$, and by (14.4.4) below, $W^{[j]}(t) = O(t^{-1/2}\log(1/t))$ as $t \downarrow 0$. The case $j = 0$ of (14.4.3) is the Lévy

Khintchine form of $\log G$. If it holds for j, then, since $ue^{-ut} = \frac{d}{dt}(1 - e^{-ut})$,

$$u\frac{d}{du}\int_0^\infty \bigl(1 - e^{-ut}\bigr)W^{[j]}(t)\,\frac{dt}{t} = \int_0^\infty ue^{-ut}\,W^{[j]}(t)\,dt$$
$$= \int_0^\infty \bigl(1 - e^{-ut}\bigr)\,W^{[j+1]}(t)\,\frac{dt}{t},$$

the boundary terms vanishing by the two bounds. A function of the form (14.4.3) is Bernstein exactly when $W^{[j]} \geq 0$: if so it is a Lévy Khintchine form, and conversely its derivative $\int_0^\infty e^{-ut}W^{[j]}(t)\,dt$ is then completely monotone, the Laplace transform of a nonnegative measure, which by uniqueness is $W^{[j]}(t)\,dt$. So by Berg and Forst the clock lies in L_n exactly when $W^{[1]}, \ldots, W^{[n+1]} \geq 0$. If $W^{[n+1]} = -t\,(W^{[n]})' \geq 0$, then $W^{[n]}$ is nonincreasing with limit 0 at infinity, hence nonnegative; descending, so are $W^{[n-1]}, \ldots, W^{[1]}$. For the drivers, the Laplace exponent of $\int_0^\infty e^{-t}\,dY_t$ is $\int_0^u \Psi_Y(w)\,dw/w$ (proof of Proposition 14.1.1), so a self decomposable law with exponent f has a driver with exponent $uf'(u)$. Starting from $\log G$ and using (14.4.3), $Z^{[j]}$ has exponent $(u\frac{d}{du})^j \log G$ and Lévy density $W^{[j]}(t)/t$ while the laws at time one are self decomposable, and the law of $Z_s^{[j]}$, with Lévy density $sW^{[j]}(t)/t$, is self decomposable exactly when $W^{[j]}$ is nonincreasing [325, Corollary 15.11], that is when $W^{[j+1]} \geq 0$. □

14.4.4.2 The ladder index of the centre clock

THEOREM 14.4.8. *Unconditionally, that is, using only the verification of RH below height* 2197.3 *[277], $W^{[3]}(t) > 0$ for every $t > 0$, and $W^{[4]}(e^{-4.15}) < 0$. Consequently W, $W^{[1]}$ and $W^{[2]}$ are positive on $(0, \infty)$; the centre clock lies in L_2 and not in L_3; its driving process $Z = Z^{[1]}$ and the driving process $Z^{[2]}$ of Z are self decomposable at every time, and the driving process $Z^{[3]}$ of $Z^{[2]}$ is self decomposable at no time.*

Proof. By Lemma 14.4.7 it suffices to prove the two inequalities; the lower ladder functions are then positive, since $W^{[j]}$ strictly decreases to 0 when $W^{[j+1]} > 0$.

Large t. Let $\tau_1 = (3 + \sqrt{5})/(2\gamma_1^2) = 0.0131039$, so that $\gamma_1^2\tau_1 = 2.618034$ is the largest root of β_3, beyond which β_3 is positive and increasing. The zeros with ordinates below $\Gamma = 2197.3$ lie on the line [277]; there are $N(\Gamma) = 1700$ of them, the last at 2197.2587, and each is enclosed in ball arithmetic. For $t \geq \tau_1$ each contributes $\beta_3(\gamma^2 t)e^{-\gamma^2 t} \geq 0$ to $W^{[3]}(t)$, and the second contributes at least $\beta_3(\gamma_2^2\tau_1)e^{-\gamma_2^2 t} > 99.38\,e^{-\gamma_2^2 t}$, since $\gamma_2^2\tau_1 = 5.7910$. A zero with ordinate $\gamma > \Gamma$, wherever it lies, has $\operatorname{Re} a \geq \gamma^2 - \frac{1}{4}$ and $|a| \leq \gamma^2 + \frac{1}{4}$, so its term has modulus at most $f(\gamma) = B_3((\gamma^2 + \frac{1}{4})t)\,e^{-(\gamma^2 - \frac{1}{4})t}$, where the Touchard polynomial B_j has the moduli of the coefficients of β_j. Since $vB_j'(v) \leq jB_j(v)$, $\log(B_j(v)e^{-v})$ decreases at rate at least $\frac{1}{2}$ for $v \geq 2j$, so for $(\Gamma^2 + \frac{1}{4})t \geq 6$, $f(\gamma) \leq f(\Gamma)e^{-(\gamma^2 - \Gamma^2)t/2}$. Let

$$N^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + 0.1038\log T + 0.2573\log\log T + 9.3675,$$

an upper bound for $N(T)$ for $T \ge e$ [108, Corollary 1.2], with derivative at most $\frac{1}{2\pi}\log\frac{T}{2\pi} + \frac{0.4}{T}$ for $T \ge \Gamma$. Partial summation against $N \le N^+$, with $\gamma^2 - \Gamma^2 \ge 2\Gamma(\gamma-\Gamma)$ and $\log\frac{\gamma}{2\pi} \le \log\frac{\Gamma}{2\pi} + \frac{\gamma-\Gamma}{\Gamma}$, bounds the sum of these moduli by

$$\varepsilon(t) = f(\Gamma)\Big(N^+(\Gamma) - 1700 + \frac{L}{\Gamma t} + \frac{1}{2\pi\Gamma^3 t^2}\Big), \qquad L = \frac{1}{2\pi}\log\frac{\Gamma}{2\pi} + \frac{0.4}{\Gamma}.$$

The logarithmic derivative of $e^{\gamma_2^2 t}\varepsilon(t)$ is at most $3/t - (\Gamma^2 - \frac14 - \gamma_2^2) < 0$ for $t \ge \tau_1$, and $e^{\gamma_2^2\tau_1}\varepsilon(\tau_1) < 10^{-27458}$. So $W^{[3]}(t) \ge e^{-\gamma_2^2 t}\big(99.38 - e^{\gamma_2^2 t}\varepsilon(t)\big) > 0$ for $t \ge \tau_1$.

Small t. Applying $(-t\frac{d}{dt})^j$ term by term to (2.1.5), which the Gaussian factors in r and in $\log n$ allow, gives with no hypothesis

$$W^{[j]}(t) = \beta_j\big(-\tfrac{t}{4}\big)e^{t/4} + A_j(t) - P_j(t), \tag{14.4.4}$$

where

$$A_j(t) = \frac{1}{2\pi}\int_0^\infty \beta_j(r^2 t)\, e^{-r^2 t}\, m(r)\, dr,$$
$$P_j(t) = \frac{1}{2\sqrt{\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\, e^{-v_n} q_j(v_n), \qquad v_n = \frac{(\log n)^2}{4t},$$

and $q_0 = 1$, $q_{j+1}(v) = (\frac12 - v)q_j(v) + vq_j'(v)$, because $-t\frac{d}{dt}$ maps $t^{-1/2}$ to $\frac12 t^{-1/2}$ and v_n to v_n; thus

$$q_3(v) = -v^3 + \tfrac92 v^2 - \tfrac{13}{4}v + \tfrac18, \qquad q_4(v) = v^4 - 8v^3 + \tfrac{29}{2}v^2 - 5v + \tfrac{1}{16}.$$

In the notation of the proof of Theorem 8.1.1, $\frac{1}{2\pi}\int_0^\infty e^{-r^2 t}\log\frac{r}{2\pi}\,dr = s_0(t)(\ell+\alpha_0)$, and since $-t\frac{d}{dt}$ multiplies s_0 by $\frac12$ and maps ℓ to $\frac12$, replacing $m(r)$ by $\log\frac{r}{2\pi}$ in A_j gives

$$\bar A_j(t) = 2^{-j} s_0(t)\,(\ell + \alpha_0 + j),$$

which also follows from (14.4.1) and its derivative in s at $s = \frac12$. For the difference we split the integral at $r = 5$: for $r \le 5$ we use $|\beta_j(u)|e^{-u} \le u$, valid for $j = 2,3,4$, and the bound $\int_0^5 r^2|m(r) - \log\frac{r}{2\pi}|\,dr < 0.404$ stated after the proof of Theorem 8.1.1; for $r \ge 5$ the bound $r^2|m(r) - \log\frac{r}{2\pi}| < 0.05$ stated there and $u = r^2 t$. This gives

$$|A_j(t) - \bar A_j(t)| \le E_j(t) = \frac{1}{2\pi}\Big(0.404\, t + 0.025\sqrt{t}\, J_j\Big),$$

where $J_j = \int_0^\infty |\beta_j(u)|\, e^{-u} u^{-3/2}\, du$ satisfies $J_3 < 1.3313$ and $J_4 < 1.8295$. Let $\tau_2 = 1/75$. For $t \le \tau_2$ the main term $\bar A_3(t)$ decreases in t, because $-t\frac{d}{dt}\bar A_3 = \bar A_4$ and $\ell + \alpha_0 + 4 > 0$; the modulus $|\beta_3(-\frac{t}{4})|e^{t/4}$ of the polar term and $E_3(t)$ increase; and so does the modulus of each term of $P_3(t)$, because $v_n \ge v_2 > 9$, $q_3 < 0 < q_4$ on $[9,\infty)$, and $-t\frac{d}{dt}$ maps the nth term of P_3 to that of P_4. At

$t = \tau_2$ the four quantities are $\bar{A}_3(\tau_2) = 0.357155$, 0.003378, $E_3(\tau_2) < 0.001469$, and at most 0.057882 for the sum of the moduli of the terms of P_3, the terms with $n > 200$ bounded through $e^{-(\log n)^2/8t} \le n^{-\sigma}$, $\sigma = (\log 200)/8t$, as in the proof of Theorem 8.1.1. Hence $W^{[3]}(t) > 0.2944$ for $t \le \tau_2$, and since $\tau_2 > \tau_1$, $W^{[3]} > 0$ on $(0, \infty)$.

Rung four. Let $t_0 = e^{-4.15}$. Then $\gamma_1^2 t_0 = 3.1496$ lies in $(1.343380, 4.490864)$, where $\beta_4 < 0$. The first five zeros contribute -0.97604, 0.62189, 0.22811, 0.01295 and 0.00216 to $W^{[4]}(t_0)$, and all zeros together, with the tail bound above, give $W^{[4]}(t_0) = -0.11089729616255 \pm 10^{-23}$ in ball arithmetic. The explicit formula confirms the sign without the zeros: at t_0 the polar term is 0.004066, $\bar{A}_4(t_0) = 0.228563$, $P_4(t_0) = 0.344026$ and $E_4(t_0) < 0.001928$, so that $W^{[4]}(t_0) < 0.004066 + 0.228563 + 0.001928 - 0.344026 < -0.109$. □

The ladder breaks where the first zero dominates. At t_0 its term -0.97604 is only partly cancelled: the second zero, with $\gamma_2^2 t_0$ already beyond the largest root 4.490864 of β_4, contributes 0.62189, and the zeros above it are too far out. On the arithmetic side the failure is the prime 2: its term in $P_4(t_0)$, 0.343448, outweighs the archimedean main term $\bar{A}_4(t_0) = 0.228563$. On $-\log t \in [0, 10]$ a scan in double precision of step 0.001, with the sign at the ends of every interval certified from the zeros, finds no negative value of W, $W^{[1]}$, $W^{[2]}$ or $W^{[3]}$, and finds $W^{[4]} < 0$ exactly for $-\log t \in (3.946, 4.335)$, $W^{[5]} < 0$ on $(3.443, 4.126)$ and $W^{[6]} < 0$ on $(3.146, 3.773) \cup (4.403, 5.051)$. Towards the origin $W^{[j]}(t) = \bar{A}_j(t) + O(1)$ by (14.4.4), so $W^{[2]}/W \to \frac{1}{4}$ and $W^{[3]}/W \to \frac{1}{8}$; the two ratios fall from 5.73 and 5.95 at $-\log t = 4.30$ to 0.614 and 0.404 at 8 and to 0.470 and 0.291 at 10, decreasing throughout $[4.30, 10]$. The least values of $W^{[2]}$ and $W^{[3]}$ on $4.30 \le -\log t \le 8$ are 0.396547, at 4.30, and 0.410976, at 4.335.

Corollary 14.4.9. *With no hypothesis, for every $s > 0$ the laws of Z_s and of $Z_s^{[2]}$ are self decomposable, hence unimodal, and the law of Z_s lies in L_1.*

Proof. The first statement is Theorem 14.4.8 with Lemma 14.4.7, and Yamazato's theorem [366], [325, Theorem 53.1] gives unimodality. The Laplace exponent of Z_s is $s\,u(\log G)'(u)$, so by (14.4.3) its images under S and S^2 correspond to $sW^{[2]}$ and $sW^{[3]}$, which are positive, and Berg and Forst place Z_s in L_1. □

Remark 14.4.10. The argument of Remark 14.4.5 applies to any compound Poisson driver: if the driver Y of a self decomposable law is compound Poisson with rate c and jumps distributed as J, stopping $X = \int_0^\infty e^{-t}\,dY_t$ at the first jump gives $X \stackrel{d}{=} \mathsf{U}^{1/c}(X + J)$, with U uniform on $(0, 1)$ and the three variables independent. The centre clock satisfies no relation of this form, for any $c > 0$ and $J \ge 0$. Such a J would have $\mathbb{E}\log(1 + J) < \infty$, since $\log(1 + \mathsf{U}^{1/c}J) \ge \log(1 + J) + c^{-1}\log \mathsf{U}$ and $\mathbb{E}\log(1 + \mathsf{U}^{1/c}J) \le \mathbb{E}\log(1 + T) \le \mathbb{E}T = m_0$, and iterating would write T as $\int_0^\infty e^{-t}\,dY_t$ with Y compound Poisson; but the driver of a self decomposable law

is unique in law, its Laplace exponent being $u(\log G)'(u)$, and that of the clock has the infinite Lévy measure $-W'(x)\,dx$ (Proposition 14.1.1).

This ladder is not the derivative ladder of Theorems 8.1.1, 8.1.6, 8.1.9 and 8.1.13 and Table 4.1, which climbs the signs $c_k = (-1)^k W^{(k)} > 0$; its rungs are alternating combinations of the same c_k with Stirling numbers as coefficients:

$$W^{[j]}(t) = \sum_{i=1}^{j} (-1)^{j-i} S(j,i)\, t^i c_i(t),$$

$$W^{[3]} = t^3 c_3 - 3t^2 c_2 + t c_1, \qquad W^{[4]} = t^4 c_4 - 6t^3 c_3 + 7t^2 c_2 - t c_1.$$

Of the BDLP ladder RH gives only the first rung, since a GGC is self decomposable and a gamma law, a GGC, lies outside L_1. Conversely, membership of the clock in L_∞ would imply RH [29, Section 4], but no finite rung does: for every n the canonical function $(\log \frac{1}{t})_+^n$, which vanishes on $[1,\infty)$ and so is not completely monotone, defines a law in L_n that is not a GGC, its ladder functions being $n!/(n-j)!\,(\log \frac{1}{t})_+^{n-j}$ for $j \le n$ and the measure $n!\,\delta_1(dt)$ for $j = n+1$.

14.4.4.3 The ladder along the de Bruijn Newman flow

By Theorem 17.1.16, for every $t \ge \Lambda$, and so for every $t \ge 0.2$ with no hypothesis, $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ (Section 4.1.1) is the Laplace transform of a GGC with Thorin atoms $a_k(t) = (z_k(t)/2)^2$, $0 < z_1(t) < z_2(t) < \cdots$ the zeros of H_t, and $R_0 = 1/G$. Writing x for the variable of the canonical function $W(x;t) = \sum_k e^{-a_k(t)x}$ of R_t, the ladder functions are

$$W^{[j]}(x;t) = \Big(-x\frac{\partial}{\partial x}\Big)^j W(x;t) = \sum_k \beta_j\big(a_k(t)x\big)\, e^{-a_k(t)x}.$$

Lemma 14.4.7 holds for these laws with the same proof, the bound for small x coming from the number of zeros of H_t, so R_t lies in L_n exactly when $W^{[n+1]}(\cdot\,;t) \ge 0$.

We compute H_t by Gauss Legendre quadrature in ball arithmetic, its zeros on $(0,600]$ by a sign scan of step 0.2 and bisection, and $W^{[j]}(x;t)$, $j \le 5$, from these zeros on the window $0 \le -\log x \le 8$, grid step 0.001, for nine values of t (Table 14.13). Nothing needs computing for $-\log x \le 0$: the largest roots of $\beta_2, \ldots, \beta_5$ are 1, 2.618034, 4.490864 and 6.510134, and $a_1(t)x \ge a_1(t) \ge 12.47$ in every row. The computation is numerical for $t > 0$: the quadrature agrees with a rule of 1024 nodes to relative differences below 10^{-46} at $z = 100$, 300 and 590, but its error is not bounded, and the zeros above $z = 600$ are left out; an estimate from the density $\frac{1}{4\pi}\log\frac{z}{4\pi}$ of the zeros of H_0 puts their share of $W^{[3]}$ at $-\log x = 8$ below 2×10^{-9}, and of $W^{[5]}$ below 10^{-5}. At $t = 0$ the computed zeros are the numbers $2\gamma_k$ to double precision, $N(300) = 138$ of them, as Arb's count of the zeros of ζ gives, and H_0 agrees with $\Xi(z/2)/8$ to 10^{-46}.

t	$z_1(t)$	a_2/a_1	$W^{[2]}/W$	$W^{[3]}/W$	$W^{[3]}<0$	$W^{[4]}<0$	n
0	28.2695	2.2119	0.6144	0.4038	none	$(3.946, 4.335)$	2
0.2	28.1536	2.2142	0.6131	0.4029	none	$(3.935, 4.333)$	2
1	27.6920	2.2239	0.6078	0.3993	none	$(3.893, 4.321)$	2
4	25.9874	2.2678	0.5887	0.3864	none	$(3.737, 4.263)$	2
10	22.7127	2.3915	0.5545	0.3630	none	$(3.426, 4.090)$	2
20	17.7111	2.7188	0.5069	0.3301	none	$(2.884, 3.675)$	2
30	13.3727	3.2530	0.4687	0.3030	$(2.983, 3.343)$	$(2.303, 3.153)$	1
40	9.7966	4.0960	0.4374	0.2807	$(2.236, 3.014)$	$(1.676, 2.593)$	1
50	7.0629	5.3383	0.4115	0.2620	$(1.563, 2.539)$	$(1.022, 2.035)$	1

Table 14.13: The ladder functions of R_t along the de Bruijn Newman flow, on the window $0 \le -\log x \le 8$: the first zero $z_1(t)$ of H_t, the ratio a_2/a_1 of the first two Thorin atoms, the ratios $W^{[2]}/W$ and $W^{[3]}/W$ at $-\log x = 8$, the sets of $-\log x$ on which $W^{[3]}(x;t) < 0$ and $W^{[4]}(x;t) < 0$, to the grid step 0.001, and the largest n with $W^{[1]}, \dots, W^{[n+1]} \ge 0$ on the window: R_t lies in no class above L_n, and in L_n if the signs persist for $x < e^{-8}$. On the window W, $W^{[1]}$ and $W^{[2]}$ are positive and $W^{[5]}$ takes negative values for every t. Numerical for $t > 0$; the row $t = 0$ is Theorem 14.4.8. Computed.

Along the flow the zeros move towards the origin and apart, and the number in $(0, 600]$ rises from 138 to 149. The set where $W^{[4]} < 0$ follows the first atom to smaller $-\log x$: it lies where $a_1(t)x \in (1.343380, 4.490864)$, less where the other atoms compensate, and at $t = 50$, where it covers most of that interval, the first atom alone almost decides the sign. Up to $t = 20$, $W^{[3]} \ge 0$ on the window and R_t keeps the index two of the clock, as far as the window shows. Between $t = 27.8$ and 27.9 it turns negative: the program finds $W^{[3]} \ge 0$ on the window at $t = 21, \dots, 27$ and $t = 27.1, \dots, 27.8$, hence $W^{[2]}, W^{[1]} \ge 0$ there by the descent in the proof of Lemma 14.4.7 on $[e^{-8}, \infty)$; at $t = 27.9$ it finds $W^{[3]} < 0$ for $3.230 < -\log x < 3.307$, with least value -0.0015, and at $t = 30$, 40, 50 on the intervals of the table. So, by this computation, for these t the law R_t, a GGC with no hypothesis, is not in L_2, and its index on the window drops to one. The mechanism is that of Theorem 14.4.8, one rung lower: the term of the first atom in $W^{[3]}$ is negative where $a_1(t)x \in (0.381966, 2.618034)$, and once the first two atoms are far enough apart the second no longer compensates; at the threshold a_2/a_1 is between 3.113 and 3.119, against 2.212 for the clock. At $t = 50$ the ratio $W^{[2]}/W$, which for $t \le 40$ decreases across the window, has an interior minimum, 0.276 at $-\log x = 2.682$: the same mechanism has begun on the second rung, which a single atom already violates.

Below the window $W^{[2]}/W$ and $W^{[3]}/W$ tend to $\frac{1}{4}$ and $\frac{1}{8}$, at $t = 0$ by (14.4.4) and for every $t \ge 0.2$ by the count of the zeros of H_t. For fixed $t > 0$ the number $N_t(T)$ of zeros of H_t in $(0, T]$ is $\Psi(T) + \frac{t}{16}\log T + O(1)$, with $\Psi(T) = \frac{T}{4\pi}\log\frac{T}{4\pi} - \frac{T}{4\pi}$: Ki, Kim and Lee proved a formula of this kind for every $t > 0$ [189, Theorem 1.4],

as recalled in [318, Section 3], and the Polymath project proved it in this form for $0 < t \le \frac{1}{2}$, with error $O(1)$ for $T \ge e^{C/t}$ and $O(\log T)$ for $T \ge 2$, the constants absolute but not numerical [311, Theorem 1.5]. An error $O(\log T)$ would do; a bound with numerical constants is Problem 6.6(b). Partial summation against $d\Psi$, with $u = z^2x/4$ and (14.4.1) at $s = \frac{1}{2}$, gives

$$W^{[j]}(x;t) = \frac{\log(1/x)}{2^{j+3}\sqrt{\pi x}}\big(1 + o(1)\big) \qquad (x \downarrow 0),$$

at $t = 0$ the leading part of $\bar{A}_j$. The approach is slow, so for $t > 0$ the signs below $x = e^{-8}$ are plausible but not proved. Either way the index is set by the positions and spacing of the first zeros: it changes along the flow where every law is a GGC with no hypothesis, and carries no information about RH.

14.4.4.4 No branching process realisation

One of Pakes's proofs obtains the Lambert law as the limit of a branching process [265]. Let Y be a spectrally positive Lévy process with Laplace exponent $\Psi_Y(u) = -\log \mathbb{E}e^{-uY_1}$, sign as in Proposition 14.1.1, and finite Malthusian parameter $\Psi_Y'(0) \in (0, \infty)$. Lamperti's time change turns Y into a continuous state branching process $(\mathsf{B}_t)_{t\ge 0}$; from $\mathsf{B}_0 = z > 0$ it has $\mathbb{E}\mathsf{B}_t = ze^{\Psi_Y'(0)t}$, and the martingale $e^{-\Psi_Y'(0)t}\mathsf{B}_t$ converges almost surely to M. By the branching property $\mathbb{E}e^{-u\mathsf{M}} = e^{-z\omega(u)}$ with ω independent of z. Conditioning on B_s gives $\omega(u) = v_s\big(\omega(ue^{-\Psi_Y'(0)s})\big)$, where $e^{-zv_s(\theta)} = \mathbb{E}e^{-\theta \mathsf{B}_s}$ and $\partial_s v_s = \Psi_Y(v_s)$, and differentiating at $s = 0$,

$$\Psi_Y'(0)\, u\, \omega'(u) = \Psi_Y\big(\omega(u)\big), \tag{14.4.5}$$

which is [266, (68)]. For the Lambert law Y is compound Poisson with unit rate and standard exponential jumps, $\Psi_Y(u) = u/(1+u)$, $\Psi_Y'(0) = 1$, and $\omega = \mathcal{W}$ solves (14.4.5), since $u\mathcal{W}' = \mathcal{W}/(1+\mathcal{W})$ [266, Section 7].

For λT, $\lambda > 0$, to be such a limit we need $z\omega(u) = \log G(\lambda u)$. Then (14.4.5) determines Ψ_Y on the range $[0, \infty)$ of ω:

$$\Psi_Y(w) = \frac{\Psi_Y'(0)}{z}\, g(zw), \qquad g = \Psi_Z \circ H = \frac{H}{H'},$$

where H is the inverse function of $\log G$ and $\Psi_Z(u) = u(\log G)'(u)$ is the Laplace exponent of the driver of Proposition 14.1.1. The function g is analytic at 0, $g(w) = \sum_{k\ge 1} g_k w^k$, $g_1 = 1$. For every spectrally positive Y, $-\Psi_Y''(u) = \sigma^2 + \int_0^\infty x^2 e^{-ux}\,\Pi(dx)$ is completely monotone, and since $\Psi_Y^{(k)}(0) = \Psi_Y'(0)\, z^{k-1} k!\, g_k$, this forces

$$(-1)^{k-1} g_k \ge 0 \qquad (k \ge 2). \tag{14.4.6}$$

The coefficients g_k need no hypothesis: the Taylor series of ξ at $\frac{1}{2}$ in ball arithmetic gives that of $\log G(u) = \sum_{n\ge 1}(-1)^{n+1} m_{n-1} u^n/n$; reverting gives H, and

composing gives g. The computation does this at 3200 bits for $k \le 60$, checks it against $g = H/H'$, which agrees for $k \le 59$, and checks $m_1, \dots, m_{11}$ against the sums over the first 1700 zeros. Every sign for $k \le 60$ is certified.

THEOREM 14.4.11. *With no hypothesis,* $(-1)^{15} g_{16} < 0$*: in ball arithmetic* $g_{16} = 1.49524474945112 \times 10^{-15} \pm 2 \times 10^{-30}$*. Consequently no positive multiple of the centre clock is the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter, from any initial state.*

Proof. Such a limit would satisfy (14.4.6), which the sign of g_{16} violates. □

The signs keep failing for a structural reason, Pringsheim's theorem, the power series form of Landau's theorem [359, Chapter II, Theorem 5b]: if $(-1)^{k-1} g_k \ge 0$ for all large k and $\sum g_k w^k$ has finite radius R, its sum is singular at $w = -R$. But g is analytic near the negative axis. With no hypothesis, $(\log G)'(u) = \sum_k 1/(u + a_k) > 0$ for $-\gamma_1^2 < u \le 0$: a zero on the line contributes $1/(u + \gamma^2) > 0$, and the conjugate atoms of an off line zero contribute $2\,\mathrm{Re}(u+a)/|u+a|^2$, with $\mathrm{Re}(u+a) \ge \gamma^2 - \frac{1}{4} - \gamma_1^2 > 0$. So $\log G$ increases from $-\infty$ to 0 on $(-\gamma_1^2, 0]$, H is analytic near $(-\infty, 0]$, and so is $g = H/H'$; hence (14.4.6) fails for infinitely many k unless g is entire.

Numerically the radius is finite, $R = 7.1565$, fixed by singularities off the axis. Ξ has a critical point $t^* = 15.5857085898$ between γ_1 and γ_2, certified by bisection on Ξ', and at $u^* = -t^{*2} = -242.914312$, $G(u) = \Xi(\sqrt{-u})/\Xi(0)$ has $G'(u^*) = 0$ and $G(u^*) = -0.00161241048$. Continued from 0 around the zero $-\gamma_1^2$ through the upper half plane, $\log G$ reaches u^* with the value

$$\lambda^* = \log |G(u^*)| + i\pi = -6.43002502663 + i\pi,$$

of modulus $|\lambda^*| = 7.156453$ and argument $\arg \lambda^* = 2.687121$; since $(\log G)'(u^*) = 0$, H has a square root branch point at λ^*, and by symmetry at $\bar{\lambda}^*$. Continuing H numerically along the segment from 0 to λ^* in 3000 steps keeps $|G| = e^{\mathrm{Re}\,\lambda} \ge 0.0016$ on the path, away from the zeros of G, and brings the preimage to within 2.86 of u^*, at a distance proportional to the square root of the distance from λ^* over the last steps, as at a simple critical point; so λ^* and $\bar{\lambda}^*$ are singularities of the principal branch of g (a numerical continuation, not a certificate). The root test agrees: $|g_k|^{1/k}$ is 0.1269, 0.1274, 0.1296, 0.1288 and 0.1317 at $k = 20$, 30, 40, 50 and 60, approaching $1/|\lambda^*| = 0.13973$ slowly, as the factor $k^{-3/2}$ of a square root singularity predicts. The conjugate pair makes $(-1)^{k-1} g_k$ oscillate in k with period $2\pi/(\pi - \arg \lambda^*) = 13.8253$.

The next critical points of Ξ do not matter. The second, $t^{**} = 22.0979772804$ between γ_2 and γ_3, has $G(-t^{**2}) = 1.56051 \times 10^{-5} > 0$ and $\log G(-t^{**2}) = -11.0679$, and the third, 26.2722473569 between γ_3 and γ_4, has $G = -1.34670 \times 10^{-6}$. So the branching obstruction, like the end of the ladder, comes from the bottom of the spectrum: from the critical point of Ξ between γ_1 and γ_2.

14.4.5 The criticality filter

Along the de Bruijn Newman flow, R_t of Theorem 17.1.16 is the Laplace transform of a GGC exactly when $t \geq \Lambda$ (Corollary 9.2.5), and $\Lambda \geq 0$ [318]. So the Pólya kernel sits at the edge of a family of kernels $e^{tv^2/4}\Phi(v)$, $t < 0$, as close to it as one wishes, whose transforms have nonreal zeros and for which the Thorin condition fails. This subsection turns that fact into a test for properties of the kernel that might imply RH, and applies it to two mixture representations of the Pólya law and to the GGC property of the variable of Biane, Pitman and Yor, which passes the test at every $t < 0$ tested but does not detect the Euler product. Every inequality below called certified is checked in ball arithmetic [169].

14.4.5.1 The filter

For an even, positive, integrable g on $\mathbb{R}$ put

$$\widehat{g}(\theta) = \int_{\mathbb{R}} g(v)\cos(\theta v)\,dv, \qquad R_g(s) = \frac{\widehat{g}(0)}{\widehat{g}(i\sqrt{s})} = \frac{\int_{\mathbb{R}} g(v)\,dv}{\int_{\mathbb{R}} g(v)\cosh(\sqrt{s}\,v)\,dv}, \quad (14.4.7)$$

for the $s \geq 0$ at which the last integral converges. For the damped Pólya kernel $\Phi_t(v) = e^{tv^2/4}\Phi(v)$, Φ as in (2.1.3), $\widehat{\Phi_t}$ is Ξ_t of Section 18.1.6, and R_{Φ_t} is R_t of Theorem 17.1.16, written φ_t in Corollary 9.2.5; at $t = 0$ it is the clock $1/G$.

Proposition 14.4.12. *Let $\mathcal{P}$ be a property of even, positive, integrable functions on $\mathbb{R}$ that Φ_t has for some $t < 0$. Then $\mathcal{P}$ implies neither that $\widehat{g}$ has only real zeros nor that R_g is the Laplace transform of a GGC, for the functions g with $\mathcal{P}$. At every $t \leq 0$ the kernel Φ_t*

(i) is positive and even, and strictly decreasing on $(0, \infty)$;

(ii) is log concave;

(iii) satisfies $\log \Phi_t(v) = -\pi e^{2|v|} + O(v^2)$ as $|v| \to \infty$;

(iv) extends analytically to the strip $|\operatorname{Im} v| < \pi/4$.

So no combination of (i) to (iv) implies that $\widehat{g}$ has only real zeros, or that R_g is the Laplace transform of a GGC.

Proof. Let $t < 0$ be such that Φ_t has $\mathcal{P}$. The zeros of H_t, and so of $\Xi_t(\theta) = 8H_t(2\theta)$, are all real if and only if $t \geq \Lambda$ (de Bruijn [73], Newman [242]), and $\Lambda \geq 0$ [318], so Ξ_t has a zero $z \notin \mathbb{R}$, not on the imaginary axis either, since $\Xi_t(iy) = \int \Phi_t(v)\cosh(yv)\,dv > 0$. Hence $s_0 = -z^2$ has $\operatorname{Im} s_0 = -2\operatorname{Re} z \operatorname{Im} z \neq 0$. As Ξ_t is even and entire, $s \mapsto \Xi_t(i\sqrt{s})$ is entire and vanishes at s_0, so R_t is meromorphic with a pole at $s_0 \notin (-\infty, 0]$. By (F7) of Section 9.2.1 an HCM function extends analytically to $\mathbb{C} \setminus (-\infty, 0]$, and the extension would agree with

R_t off its poles; so R_t is not HCM, and by (F6) not the Laplace transform of a GGC.

For (i), $\Phi_t'(v) = e^{tv^2/4}\big(\Phi'(v) + \frac{t}{2}v\Phi(v)\big) < 0$ for $v > 0$, since $\Phi' < 0$ on $(0,\infty)$ [69, Theorem A] and $tv\Phi(v) \le 0$. For (ii), $(\log\Phi_t)'' = (\log\Phi)'' + \frac{t}{2} \le (\log\Phi)'' \le 0$, Φ being log concave [59]. Properties (iii) and (iv) hold for Φ by (2.1.3), and the factor $e^{tv^2/4}$ is entire and adds $tv^2/4$ to the logarithm. □

A property that is to imply RH through the kernel alone must fail for Φ_t at every $t < 0$; we then say that it *passes the filter*. Passing is necessary and far from sufficient: it says only that the theorem of Rodgers and Tao does not rule the property out. For (i), (ii) and (iv) nothing so deep is needed: the convolution of $(1-|u|)_+^2$ with $e^{-\cosh u}$ in Section 15.1.5 is even, positive, entire, log concave and doubly exponentially decaying, and its transform has nonreal zeros. The filter adds a counterexample with every property of the list that differs from Φ only by the factor $e^{tv^2/4}$, with t as close to 0 as one wishes.

The filter is the kernel counterpart of the perturbation test of Section 25.2.6, where F of Theorem 1.4.1 keeps the zeros of ξ below a height R, the order, the type and the functional equation, and has four zeros off the line above R; Φ_t keeps every kernel property that a Gaussian factor with $t < 0$ preserves, and has nonreal zeros, though Rodgers and Tao do not say where. For $t > 0$ the zeros of H_t obey $N_t(T) = \Psi(T) + \frac{t}{16}\log T + O(1)$ (Section 14.4.4.3); for $t < 0$, the case the filter needs, we know of no count in the literature: Rodgers and Tao prove one for $\Lambda < t \le 0$, with error $O(\log^2 T)$ [318, Section 3], as a step in showing that this range is empty. A proof of RH must therefore use, at some step, a property that Φ_t lacks for every $t < 0$. The Euler product and the explicit formula are the natural candidates, as for F, but this is a reading and not a theorem: no Euler product for H_t with $t \ne 0$ is known (see the discussion after Theorem 17.1.16), and none is known not to exist. The Lambert proof of Section 14.4.3 is untouched by the filter: it concerns a single function, whose analyticity off the cut is forced by $\mathcal{W}e^{\mathcal{W}} = u$, with no family of comparison functions in sight.

14.4.5.2 Mixtures of arcsine laws

For $t \le 0$ let X_t have density $p_t = \Phi_t/\Xi_t(0)$, so $X_0 = X$. Since p_t is even and decreasing on $(0,\infty)$ by Proposition 14.4.12(i), Khinchine's theorem writes $X_t = UQ_t$ with U uniform on $[-1,1]$ and $Q_t \ge 0$ independent, of density $-2rp_t'(r)$; at $t = 0$ this is the uniform mixture of Section 20.4.2.

Proposition 14.4.13. *For every $t \le 0$ the function*

$$h_t(\rho) = -2\rho\int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2-\rho^2}}, \qquad \rho > 0,$$

is a positive probability density, and

$$p_t(v) = \frac{1}{\pi}\int_{|v|}^{\infty} \frac{h_t(\rho)\,d\rho}{\sqrt{\rho^2 - v^2}}, \qquad v \neq 0.$$

Equivalently, $X_t \stackrel{d}{=} \varrho_t \cos\Theta$ *with* Θ *uniform on* $[0, 2\pi)$ *and* ϱ_t *of density* h_t *independent of* Θ*:* X_t *is a mixture of arcsine laws, and it is the first coordinate of the rotation invariant vector* $\varrho_t(\cos\Theta, \sin\Theta)$ *in* $\mathbb{R}^2$*, as well as of a rotation invariant vector in* $\mathbb{R}^3$*.*

Proof. By Archimedes' theorem the coordinates of a uniform point of the unit sphere in $\mathbb{R}^3$ are uniform on $[-1, 1]$. Writing the point as $(B\cos\Theta, B\sin\Theta, V)$, V uniform on $[-1, 1]$, $B = \sqrt{1 - V^2}$ and Θ uniform and independent, $U \stackrel{d}{=} B\cos\Theta$, where B has density $b/\sqrt{1 - b^2}$ on $(0, 1)$. Hence $X_t \stackrel{d}{=} (BQ_t)\cos\Theta$, and Q_t times the uniform point is rotation invariant in $\mathbb{R}^3$ with first coordinate X_t. The density of $\varrho_t = BQ_t$ is

$$\int_\rho^\infty \big(-2rp_t'(r)\big)\,\frac{\rho/r}{\sqrt{1 - \rho^2/r^2}}\,\frac{dr}{r} = -2\rho\int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2 - \rho^2}} = h_t(\rho),$$

positive since $p_t' < 0$. Given $\varrho_t = \rho$, $\rho\cos\Theta$ has the arcsine density $1/(\pi\sqrt{\rho^2 - v^2})$ on $(-\rho, \rho)$, which gives the formula for p_t. □

At $t = 0$ this is the arcsine route of [282, Chapter 12] recalled in Section 1.4, which writes X as $2\sqrt{I}$ times the cosine of an independent angle; since $\mathbb{E}|\cos\Theta|^s \neq 0$ for $s > -1$, the Mellin transform of $|X|$ determines the law of the radius, so $\varrho_0 \stackrel{d}{=} 2\sqrt{I}$. Both mixtures hold at every $t \leq 0$, so by Proposition 14.4.12 neither can decide RH or the Thorin condition by itself: whatever is to be proved about Q_0 or ϱ_0 must use something that Q_t or ϱ_t lacks for $t < 0$. The Gamma scale mixtures, the other mixture route of [282, Section 12.6], are excluded already at $t = 0$, for every shape (Theorem 14.2.10).

14.4.5.3 The variable of Biane, Pitman and Yor under damping

Let $\mathcal{B} = \frac{\pi}{2}S_2$, $S_2 = J + J'$ the sum of two independent hitting times of 1 by a three dimensional Bessel process from 0, as in (12.1.1) and Theorem 11.1.21. Then

$$\mathbb{E}e^{-\lambda\mathcal{B}} = \Big(\frac{\sqrt{\pi\lambda}}{\sinh\sqrt{\pi\lambda}}\Big)^2 = \prod_{m\geq 1}\Big(1 + \frac{\lambda}{\pi m^2}\Big)^{-2}, \qquad \mathbb{E}\mathcal{B}^{s/2} = 2\xi(s), \tag{14.4.8}$$

so $\mathcal{B}$ is a GGC with Thorin measure $2\sum_m \delta_{\pi m^2}$, and its density is

$$f(b) = \sum_{m\geq 1}\big(4\pi^2 m^4 b - 6\pi m^2\big)e^{-\pi m^2 b} = b^{-5/4}\,\Phi\big(\tfrac{1}{2}\log b\big), \qquad b > 0.$$

For an even positive kernel g with $\int g(v)e^{|v|/2}\,dv < \infty$ let $\mathcal{B}_g$ have density proportional to $b^{-5/4}g(\frac{1}{2}\log b)$; the Pólya variable of g, of density $g/\widehat{g}(0)$, is $\frac{1}{2}\log \mathcal{B}_g$ tilted by $\mathcal{B}_g^{1/4}$, and $\mathcal{B}_\Phi = \mathcal{B}$ (Theorem 11.1.21). For the damped kernels $\mathcal{B}_t = \mathcal{B}_{\Phi_t}$ has density proportional to $e^{t(\log b)^2/16}f(b)$, finite for every real t because $f(b)$ decays like $e^{-\pi b}$ as $b \to \infty$ and, up to a power of b, like $e^{-\pi/b}$ as $b \downarrow 0$, and $\mathbb{E}\mathcal{B}_t^{s/2} = \Xi_t(i(\frac{1}{2}-s))/\Xi_t(\frac{i}{2})$. Damping the kernel tilts $\mathcal{B}$ by a lognormal factor, and "$\mathcal{B}_g$ is a GGC" is a property of g that Φ has. The filter asks whether Φ_t lacks it for every $t < 0$.

Two tests are available for a positive variable $\mathcal{V}$ with $\ell(\lambda) = \mathbb{E}e^{-\lambda\mathcal{V}}$. Let $\kappa_k(\lambda) = (-1)^k \frac{d^k}{d\lambda^k}\log \ell(\lambda)$, $k \geq 1$, the cumulants of the tilted law $e^{-\lambda b}\,P(\mathcal{V} \in db)/\ell(\lambda)$. The variable is infinitely divisible if and only if $-\log \ell$ is Bernstein, that is $\kappa_k(\lambda) \geq 0$ for all $k \geq 1$, $\lambda > 0$ [327]. If it is a GGC, $-\log \ell(\lambda) = d\lambda + \int \log(1+\lambda/z)\,U(dz)$, then for $k \geq 0$

$$c_k(\lambda) = \frac{\kappa_{k+1}(\lambda)}{k!} = d\,\delta_{k0} + \int \frac{U(dz)}{(\lambda+z)^{k+1}} = \int x^k\,\sigma_\lambda(dx), \tag{14.4.9}$$

where σ_λ is $d\,\delta_0$ plus the image of $(\lambda+z)^{-1}U(dz)$ under $z \mapsto (\lambda+z)^{-1}$. Hence $\mathcal{K}_n(\lambda) = [c_{i+j}(\lambda)]_{i,j=0}^{n-1}$ and $\mathcal{K}_n'(\lambda) = [c_{i+j+1}(\lambda)]_{i,j=0}^{n-1}$ are positive semidefinite for every n and $\lambda \geq 0$, since $\sum_{i,j} y_i y_j c_{i+j+r}(\lambda) = \int x^r \big(\sum_i y_i x^i\big)^2 \sigma_\lambda(dx)$ for $r = 0, 1$, and a negative eigenvalue of either proves that $\mathcal{V}$ is not a GGC. The smallest case is already useful.

Lemma 14.4.14. *If $\mathcal{V}$ is a GGC with a finite fourth moment, its cumulants $\kappa_k = \kappa_k(0)$ satisfy $2\kappa_2\kappa_4 \geq 3\kappa_3^2$, with equality for the gamma laws.*

Proof. With $c_k = \kappa_{k+1}/k!$ the inequality reads $c_1c_3 \geq c_2^2$, which is $\det \mathcal{K}_2'(0) \geq 0$. For a gamma law U is a single atom $\alpha\delta_z$ and $c_k = \alpha z^{-k-1}$, so equality holds. □

We compute the moments

$$\int_{\mathbb{R}} e^{2kv}\,e^{-\lambda e^{2v}}\,e^{tv^2/4}\,e^{-v/2}\,\Phi(v)\,dv,$$

those of the tilted law of $\mathcal{B}_t$ up to normalisation, by the trapezoid rule on $[-5,5]$, forms $\kappa_1, \ldots, \kappa_{26}$ at $\lambda = 0$, $\frac{1}{2}$, 2, 10, 20, 50 and 200, and the matrices $\mathcal{K}_n(\lambda)$, $\mathcal{K}_n'(\lambda)$ of orders $n \leq 12$. Dividing c_{i+j+r} by c^{i+j+r}, $c = c_1/c_0$, is a congruence that keeps the inertia, and the eigenvalues quoted are those of the scaled matrices. The certificate encloses the moments in balls (Arb, 256 bits, each tail bounded by 2^{-3000}), computes the cumulants and the scaled matrix M in ball arithmetic, and encloses $y^{\mathsf{T}}My$ for a rational y close to the eigenvector of the least eigenvalue; an enclosure inside $(-\infty, 0)$ proves that M is not positive semidefinite. In every certificate below the cumulants used are certified positive and $y^{\mathsf{T}}My$ agrees with the computed least eigenvalue to all digits quoted.

Infinite divisibility shows nothing. In all 301 cases computed (the tilts $\mathcal{B}_t$ for the twenty values of t below, the reciprocals $1/\mathcal{B}_t$ for $t = -1$, 0, $\frac{1}{5}$, and the ten functions other than ξ of Table 14.15, on both sides, each at the seven values of λ), none of $\kappa_1, \dots, \kappa_{26}$ is negative.

The GGC test separates $t = 0$ from every other value. It was run for $t = -20$, -10, -5, -2, -1, $-\frac{1}{2}$, $-\frac{1}{5}$, $-\frac{1}{10}$, $-\frac{1}{20}$, $-\frac{1}{50}$, $-\frac{1}{100}$, 0, $\frac{1}{100}$, $\frac{1}{50}$, $\frac{1}{20}$, $\frac{1}{10}$, $\frac{1}{5}$, $\frac{1}{2}$, 1 and 5. At $t = 0$ no matrix of order at most 12 has a negative eigenvalue, as the theorem requires, though the matrices are nearly singular. For every $t \neq 0$ some $\mathcal{K}'_n(\lambda)$ with $n \leq 12$ has a negative eigenvalue at each $\lambda \leq 50$. Table 14.14 shows eleven of the twenty values; the other nine give the same picture.

t	$\lambda = 0$	$\frac{1}{2}$	2	10	20	50	eigenvalue at $\lambda = 2$
-20	8	9	8	8	7	9	-1.582×10^{-7}
-5	8	8	7	7	8	10	-8.095×10^{-8}
-1	8	8	7	7	8	11	-1.533×10^{-8}
$-\frac{1}{2}$	8	8	7	7	9	11	-4.725×10^{-9}
$-\frac{1}{10}$	8	8	8	8	9	12	-1.800×10^{-9}
$-\frac{1}{100}$	9	9	8	8	10	12	-4.119×10^{-11}
0	none	none	none	none	none	none	1.247×10^{-13}
$\frac{1}{100}$	4	4	5	6	7	10	-2.357×10^{-5}
$\frac{1}{5}$	4	4	4	5	7	9	-1.101×10^{-4}
1	3	3	4	5	6	8	-2.102×10^{-3}
5	3	3	3	5	6	8	-2.141×10^{-3}

Table 14.14: The GGC test for the tilted variables $\mathcal{B}_t$ of Section 14.4.5.3: the least order $n \leq 12$ at which $\mathcal{K}'_n(\lambda)$ has a negative eigenvalue, and the least eigenvalue of the scaled matrix $\mathcal{K}'_n(2)$ at that order; for $t = 0$, where no order fails, the last column gives $\mathcal{K}'_7(2)$. At $\lambda = 200$ no matrix of order at most 12 fails, for any t. Every negative entry of the last column is certified, and so is the entry for $\mathcal{K}'_9(0)$ at $t = -\frac{1}{100}$, -2.899×10^{-12}. Computed.

Proposition 14.4.15. *(i) $\mathcal{B}_t$ is not a GGC for $t = -20$, -5, -1, $-\frac{1}{2}$, $-\frac{1}{10}$, $-\frac{1}{100}$, $\frac{1}{100}$, $\frac{1}{5}$, 1 and 5.*

(ii) There is $t_0 > 0$ such that $\mathcal{B}_t$ is not a GGC for $0 < |t| < t_0$.

Proof. (i) For each of these t the matrix $\mathcal{K}'_n(2)$ at the order n of Table 14.14 is certified not to be positive semidefinite.

(ii) Put $\ell_t(\lambda) = \mathbb{E}e^{-\lambda \mathcal{B}_t}$ and split the numerator $\int_0^\infty e^{-\lambda b} e^{t(\log b)^2/16} f(b)\, db$ at $b = 1$. The part on $(0, 1]$ is entire in $(t, \lambda) \in \mathbb{C}^2$, since $|e^{t(\log b)^2/16}| = e^{O((\log b)^2)}$ is negligible against $f(b) = O(b^{-7/2} e^{-\pi/b})$ as $b \downarrow 0$, and the denominator, the value at $\lambda = 0$, is analytic in t and positive for real t. On $[1, \infty)$ write f as the sum of its terms. Fix $M \geq 1$ and x with $\pi M^2 < x < \pi(M+1)^2$. The terms with $m > M$ give a function analytic in $t \in \mathbb{C}$ and $\operatorname{Re} \lambda > -\pi(M+1)^2$. For $m \leq M$ turn the path $[1, \infty)$ into the ray $b = 1 + re^{-3\pi i/4}$, $r \geq 0$, which avoids $(-\infty, 0]$ and along

which $(\log b)^2$ grows like $(\log r)^2$. Writing $\lambda+\pi m^2 = u+iv$, the integrand decays along every ray $b = 1+re^{i\alpha}$ with $-\frac{3\pi}{4} \le \alpha \le 0$ when $v > u > 0$, so there the rotated integral equals the original one; and it converges, analytically in (t,λ), on the half plane $v > u$, which contains $\lambda = -x$. So ℓ_t continues analytically, jointly in (t,λ) near $(0,-x)$, from the upper half plane across the negative axis at $-x$.

Suppose that $\mathcal{B}_t$ is a GGC, with Thorin measure U_t. Then $-\log \ell_t(\lambda) = d_t\lambda + \int \log(1+\lambda/z)\, U_t(dz)$ on $\mathbb{C}\setminus(-\infty,0]$, and as $\varepsilon \downarrow 0$ its imaginary part at $\lambda = -y+i\varepsilon$ tends to $\pi\big(U_t((0,y)) + \frac{1}{2}U_t(\{y\})\big)$. For t near 0, ℓ_t has no zero near $-x$, since $\ell_0(-x) = \pi x/\sin^2\sqrt{\pi x} > 0$ by (14.4.8). So the limit is real analytic in y near x, U_t has no atom there, and $U_t((0,y)) = -\frac{1}{\pi}\arg \ell_t(-y+i0)$, the argument continued from 0 on $(0,\infty)$ through the upper half plane, is nondecreasing in y; at $t=0$ it equals $2M$. Write $\ell_t = \ell_0 + t\big(\dot\ell - \dot\ell(0)\,\ell_0\big) + O(t^2)$, with $\dot\ell(\lambda) = \frac{1}{16}\int_0^\infty e^{-\lambda b}(\log b)^2 f(b)\,db$ continued in the same way; by joint analyticity the remainder is $O(t^2)$ with its λ derivative, uniformly near $-x$. Since ℓ_0 and $\dot\ell(0)$ are real there,

$$U_t((0,y)) = 2M - \frac{t}{\pi}\mathcal{J}(y) + O(t^2), \qquad \mathcal{J}(y) = \frac{\operatorname{Im}\dot\ell(-y+i0)}{\ell_0(-y)},$$

with the same expansion for the derivatives in y. In $\dot\ell(-y+i0)$ the part on $(0,1]$ and the terms with $m > M$ are real, and for $m \le M$ the rotated integral differs by a real number from $\int_0^\infty e^{-\sigma b}(\log b)^2(4\pi^2m^4b - 6\pi m^2)\,db$ with $\sigma = -s_m + i0$, $s_m = y-\pi m^2$, a combination of $\partial_a^2\big[\Gamma(a)\sigma^{-a}\big]$ at $a=2$ and $a=1$, with $\log\sigma = \log s_m + i\pi$. This gives

$$\operatorname{Im}\dot\ell(-y+i0) = \frac{\pi}{8}\sum_{m\le M}\Big(\frac{4\pi^2m^4(\log s_m - 1 + \gamma_E)}{s_m^2} + \frac{6\pi m^2(\log s_m + \gamma_E)}{s_m}\Big).$$

At $y = 1.015\pi$, where $M=1$, the certificate program gives $\mathcal{J}'(y) = 8.52630079173 \pm 3.4\times10^{-12}$, and at $y = 2.47\pi$ it gives $\mathcal{J}'(y) = -0.0671813667518 \pm 10^{-14}$. So for small $t > 0$ the derivative of $U_t((0,y))$ at $y = 1.015\pi$ is $-\frac{t}{\pi}\mathcal{J}'(1.015\pi) + O(t^2) < 0$, and for small $t < 0$ the derivative at $y = 2.47\pi$ is negative; both contradict monotonicity. □

We scan $\mathcal{J}'$ at 399 points in each of the first four gaps $(\pi M^2, \pi(M+1)^2)$: in each, $\mathcal{J}'$ is positive on a left part and negative on a right part, for y/π in $[1.8925, 3.9925]$, $[5.5125, 8.9875]$, $[11.1, 15.9825]$ and $[18.6775, 24.9775]$, so every one of these gaps excludes both signs of t on its own.

Remark 14.4.16. The variable $\mathcal{B}$ is a GGC, but its density f is not HCM: f is analytic in $\operatorname{Re} b > 0$ with the natural boundary $\operatorname{Re} b = 0$ (the discussion after Theorem 11.1.21), while an HCM function extends analytically to $\mathbb{C}\setminus(-\infty,0]$ by (F7).

The GGC property of $\mathcal{B}_g$ passes the filter wherever tested, the only property of this subsection that does; whether $\mathcal{B}_t$ fails to be a GGC for every $t < 0$ is open (Problem 14.2). It is not a property of the zeros alone: for $t \geq 0.2$ the zeros of Ξ_t are all real [277] and R_t is a GGC transform (Theorem 17.1.16), yet $\mathcal{B}_t$ is not a GGC at $t = \frac{1}{5}$, 1 and 5.

14.4.5.4 Theta kernels, characters and the function of Davenport and Heilbronn

The variable $\mathcal{B}_g$ can be formed for other Dirichlet series with a functional equation. Let $q \geq 3$, $p \in \{0, 1\}$, and let $a = (a_n)_{n\geq 1}$ be real of period q with $a_1 = 1$ and $a_{q-n} = (-1)^p a_n$, whose theta series

$$\vartheta_a(x) = \sum_{n\geq 1} n^p a_n e^{-\pi n^2 x/q}, \qquad x > 0,$$

satisfies $\vartheta_a(1/x) = x^{p+1/2}\vartheta_a(x)$. Then

$$\xi(s, a) = \int_0^\infty \vartheta_a(x)\, x^{(s+p)/2}\, \frac{dx}{x} = \Big(\frac{q}{\pi}\Big)^{(s+p)/2} \Gamma\Big(\frac{s+p}{2}\Big) \sum_{n\geq 1} \frac{a_n}{n^s}$$

is entire with $\xi(s, a) = \xi(1 - s, a)$, and $K_a(v) = 2e^{(p+1/2)v}\vartheta_a(e^{2v})$ is an even kernel with $\xi(\frac{1}{2} + s, a) = \int_{\mathbb{R}} K_a(v)e^{sv}\, dv$. The real primitive characters χ_d, d a fundamental discriminant, are of this kind, with $q = |d|$ and $p = 1$ for $d < 0$; for χ_{-4}, written χ_4 in Section 15.1.7, $\vartheta_a(x) = \eta(ix)^3$ and K_a is the kernel K_4 of Proposition 15.1.22. When $\vartheta_a > 0$, $\mathcal{B}_a = \mathcal{B}_{K_a}$ has density $b^{p/2-1}\vartheta_a(b)/\xi(0, a)$ and $\mathbb{E}\mathcal{B}_a^{s/2} = \xi(s, a)/\xi(0, a)$, and by the functional equation its reciprocal has density $x^{(p-1)/2}\vartheta_a(x)/\xi(0, a)$. We test both. For χ_{-4}, $1/\mathcal{B}_a$ is a GGC: it is $\frac{\pi}{2}$ times the exit time of Remark 15.1.23. Whether $\mathcal{B}_{\chi_{-4}}$ itself is a GGC is open, and HCM cannot decide it: the density of $1/\mathcal{B}_{\chi_{-4}}$ is not HCM (Theorem 9.3.20), so by (F2) neither is that of $\mathcal{B}_{\chi_{-4}}$.

Lemma 14.4.17. *Let $S_a = \sum_{n\geq 2} n^p |a_n| e^{-\pi(n^2-1)/q}$. If $S_a < 1$, then $\vartheta_a > 0$ on $(0, \infty)$.*

Proof. For $x \geq 1$, $\vartheta_a(x) = e^{-\pi x/q}\big(1 + \sum_{n\geq 2} n^p a_n e^{-\pi(n^2-1)x/q}\big) \geq e^{-\pi x/q}(1 - S_a) > 0$, and for $0 < x < 1$, $\vartheta_a(x) = x^{-p-1/2}\vartheta_a(1/x) > 0$. □

The certificate encloses S_a: it is at most 0.659 for the nine characters and the function of Davenport and Heilbronn below, so these kernels are positive. Positivity can fail: for $d = -19$ and $d = -40$, a computation finds $e^{\pi/q}\vartheta_a(1) = -0.567$ and -0.421. For $d = -19 \equiv 5 \pmod{24}$ this is a case of Theorem 15.1.19, and kernels of this kind go back to Pólya's examples recalled in [33, Section 6.1].

The function of Davenport and Heilbronn [72] is $f_{\mathrm{DH}}(s) = \sum_n a_n n^{-s}$ with a of period 5 and $(a_1, \dots, a_5) = (1, \varkappa, -\varkappa, -1, 0)$, where

$$\varkappa = \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} = 0.2840790438404122960283.$$

With χ the character modulo 5 with $\chi(2) = i$, $f_{\mathrm{DH}} = \frac{1-i\varkappa}{2} L(s, \chi) + \frac{1+i\varkappa}{2} L(s, \bar\chi)$, and $\varkappa$ makes $(1 + i\varkappa)/(1 - i\varkappa)$ the root number

$$\varepsilon(\chi) = 0.85065080835203993218 + 0.52573111211913360603\, i;$$

then $p = 1$ and $\vartheta_a(1/x) = x^{3/2}\vartheta_a(x)$. The coefficients are not multiplicative ($a_6 = 1$, $a_2 a_3 = -\varkappa^2$), so f_{DH} has no Euler product, and it has infinitely many zeros in $\operatorname{Re} s > 1$ [348, Chapter 10]. Zeros off the critical line inside the strip have been computed by Spira [335] and by Balanzario and Sánchez-Ortiz [16]; the program confirms two of them,

$$0.808517182456637 + 85.6993484853776\, i,$$
$$0.650830080609737 + 114.163342730757\, i,$$

where $|f_{\mathrm{DH}}| < 10^{-25}$. Modulo 5 there is no other function of the class built from χ and $\bar\chi$: a combination $\alpha L(s, \chi) + \beta L(s, \bar\chi)$ with $a_1 = 1$ whose completed function is w times its reflection must have $w = \pm 1$ and $\beta = w\,\varepsilon(\chi)\alpha$, and $w = -1$ gives the coefficients $(1, -1/\varkappa, 1/\varkappa, -1, 0)$, whose theta series satisfies $\vartheta(1/x) = -x^{3/2}\vartheta(x)$, so it vanishes at $x = 1$ and takes both signs.

Table 14.15 gives the test for ξ, nine real characters and f_{DH}. For the characters and f_{DH} the moments are computed as for $\mathcal{B}_t$, from the kernels K_a, and at $\lambda = 0$ also from L values alone: $\mathbb{E}\mathcal{B}_a^k = \xi(2k, a)/\xi(0, a)$ and $\mathbb{E}\mathcal{B}_a^{-k} = \xi(2k + 1, a)/\xi(0, a)$, with $\sum_n a_n n^{-s} = q^{-s} \sum_{r=1}^{q} a_r \zeta(s, r/q)$ through the Hurwitz zeta function. The two routes agree, and the failures marked are certified, by the first route for $\lambda > 0$ and by the second at $\lambda = 0$.

Proposition 14.4.18. *Neither $\mathcal{B}_{\chi_{-8}}$ nor $1/\mathcal{B}_{\chi_{-8}}$ is a GGC.*

Proof. From the values of $L(s, \chi_{-8})$ at $s = 0, 2, 4, 6, 8$ and at $s = 3, 5, 7, 9$, the certificate program gives

$$3\kappa_3^2 - 2\kappa_2\kappa_4 = 7.55465911093 \pm 4 \times 10^{-12} \qquad \text{and} \qquad 78.8854168657 \pm 2 \times 10^{-11}$$

for $\mathcal{B}_{\chi_{-8}}$, where $(\kappa_2, \kappa_3, \kappa_4) = (3.0767, 15.997, 123.54)$, and for its reciprocal. Both are positive, against Lemma 14.4.14. □

The same four numbers settle $1/\mathcal{B}_{\chi_{-7}}$ (27.8926758865) and $1/\mathcal{B}_a$ for f_{DH} (0.738104955084). So $L(s, \chi_{-8})$, which has an Euler product and whose zeros lie on the critical line to height $10^8/8$ at least [275], fails the test already at order two, on both sides.

function	S_a	$\mathcal{K}_7'(0)$	$\mathcal{K}_7'(2)$	$\mathcal{B}_a$	$1/\mathcal{B}_a$
ξ		6.2×10^{-16}	1.2×10^{-13}	GGC	fails at $\mathcal{K}_3'(0)$
χ_{-3}	0.086	2.4×10^{-8}	4.9×10^{-8}	passes	GGC
χ_{-4}	0.006	1.2×10^{-8}	4.6×10^{-8}	passes	GGC
χ_5	0.158	7.8×10^{-8}	7.1×10^{-8}	passes	fails at $\mathcal{K}_{10}'(0)$
χ_8	0.043	4.8×10^{-8}	3.8×10^{-8}	passes	fails at $\mathcal{K}_7'(2)$
χ_{12}	0.002	2.1×10^{-8}	4.2×10^{-8}	passes	fails at $\mathcal{K}_9'(10)$
χ_{24}	0.045	7.5×10^{-9}	2.0×10^{-8}	fails at $\mathcal{K}_{11}'(2)$	fails at $\mathcal{K}_6'(0)$
χ_{-7}	0.608	-0.149	-0.221	fails at $\mathcal{K}_3'(0)$	fails, zero
χ_{-8}	0.130	-0.057	-0.477	fails at $\mathcal{K}_2'(0)$	ID, fails, zero
χ_{13}	0.659	-0.0023	-0.256	fails at $\mathcal{K}_3'(0)$	fails at $\mathcal{K}_3'(0)$
f_{DH}	0.092	-1.2×10^{-5}	-1.4×10^{-4}	fails at $\mathcal{K}_6'(0)$	fails, zero

Table 14.15: The GGC test for the variables $\mathcal{B}_a$ of Section 14.4.5.4 and their reciprocals: the sum S_a of Lemma 14.4.17, the least eigenvalues of the scaled matrices $\mathcal{K}_7'(0)$ and $\mathcal{K}_7'(2)$ of $\mathcal{B}_a$, and the verdicts. "Passes": no matrix $\mathcal{K}_n(\lambda)$ or $\mathcal{K}_n'(\lambda)$ with $n \le 12$ has a negative eigenvalue at $\lambda = 0$, $\frac12$, 2, 10, 20, 50 and 200. "Fails at $\mathcal{K}_n'(\lambda)$": the least failing order, certified at that λ. "Zero": the Laplace transform vanishes on $(-\infty, 0)$ (Proposition 14.4.19). GGC for ξ by (14.4.8), for $1/\mathcal{B}_a$ with χ_{-3} and χ_{-4} by Proposition 14.4.19. All the functions but f_{DH} have an Euler product. Computed.

Proposition 14.4.19. *Let $p = 1$ and $\vartheta_a > 0$, and put $w = \sqrt{\pi q\lambda}$ and $\Sigma_a(w) = \sum_{1\le n<q/2} a_n \sinh\frac{(q-2n)w}{q}$. Then*

$$\int_0^\infty e^{-\lambda x}\vartheta_a(x)\,dx = \frac{\sqrt{q}\,\Sigma_a(w)}{\sinh w}, \qquad \mathbb{E}e^{-\lambda/\mathcal{B}_a} = \frac{\sqrt{q}\,\Sigma_a(w)}{\xi(0,a)\sinh w}, \tag{14.4.10}$$

and $\xi(0,a) = q^{-1/2}\sum_{1\le n<q/2}(q-2n)a_n$. In particular:

(i) for χ_{-4}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a} = \operatorname{sech}\sqrt{\pi\lambda}$, and $1/\mathcal{B}_a$, which is $\frac{\pi}{2}$ times the exit time of Brownian motion from $(-1,1)$, is a GGC;

(ii) for χ_{-3}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a} = 3\sinh(w/3)/\sinh w = \prod_{3\nmid m}\left(1+3\lambda/\pi m^2\right)^{-1}$, so $1/\mathcal{B}_a$ is a GGC with Thorin measure $\sum_{3\nmid m}\delta_{\pi m^2/3}$, and it is $\frac{3\pi}{2}$ times the hitting time of 1 by a three dimensional Bessel process started at $\frac13$;

(iii) for χ_{-8}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a} = \cosh(w/4)/\cosh(w/2) = \prod_{k\ge1}(1+\lambda/4\alpha_k)/(1+\lambda/\alpha_k)$ with $\alpha_k = \frac{\pi}{8}(2k-1)^2$, so $1/\mathcal{B}_a$ is infinitely divisible, with Lévy density $\sum_k(e^{-\alpha_k x} - e^{-4\alpha_k x})/x > 0$, and it is π times the hitting time of 1 by a reflected Brownian motion started at $\frac12$; it is not a GGC, because its Laplace transform vanishes at $\lambda = -\pi/2$;

(iv) if $a = \chi_d$ with $d < 0$ and $\chi_d(2) = 1$, that is $d \equiv 1 \pmod 8$, the Laplace transform of $1/\mathcal{B}_a$ vanishes at a point of $(-4\pi/q, -\pi/q)$, so $1/\mathcal{B}_a$ is not a

GGC; for χ_{-7} the point is $\lambda = -1.033610573$. The same holds for f_{DH}, at $\lambda = -2.046575671$.

Proof. By the functional equation $\vartheta_a(x) = x^{-3/2}\sum_n n a_n e^{-\pi n^2/qx}$, and

$$\int_0^\infty e^{-\lambda x - c/x} x^{-3/2}\,dx = \sqrt{\pi/c}\,e^{-2\sqrt{c\lambda}}$$

with $c = \pi n^2/q$ gives $\int_0^\infty e^{-\lambda x}\vartheta_a(x)\,dx = \sqrt{q}\sum_{n\ge1} a_n e^{-2nw/q}$ for $\lambda > 0$. Summing over a period and pairing n with $q-n$, $a_{q-n} = -a_n$, gives the first formula of (14.4.10), which is [33, (96)] in our normalisation; at $\lambda = 0$ it gives $\int_0^\infty \vartheta_a = \xi(1,a) = \xi(0,a)$, and the second formula follows since $1/\mathcal{B}_a$ has density $\vartheta_a/\xi(0,a)$. (i) and (ii) follow from the product formulas for cosh and sinh, and [33, (92)] identifies the transforms in (ii) and (iii) with those of the hitting times. In (iii) Frullani's integral gives the Lévy density. A GGC transform ℓ cannot vanish on $(-\infty,0)$ where it extends meromorphically: for $0 < \varepsilon \le 1$, $|\ell(-x+i\varepsilon)| \ge e^{dx-C}$ with $C = \int \log(1+(x+1)/z)\,U(dz) < \infty$, because $|1+(-x+i\varepsilon)/z| \le 1+(x+1)/z$. In (iii) the zero is that of $\cosh(w/4)$ at $w = 2\pi i$. In (iv) put $w = iy$; then $\Sigma_a(iy)/\sinh(iy) = \nu(y)/\sin y$ with $\nu(y) = \sum_{1\le n<q/2} a_n \sin\frac{(q-2n)y}{q}$, and since $a_{q-n} = -a_n$, $\nu(\pi) = \frac12\sum_{n=1}^{q-1} a_n \sin\frac{2\pi n}{q}$ and $\nu(2\pi) = -\frac12\sum_{n=1}^{q-1} a_n \sin\frac{4\pi n}{q}$. For an odd real primitive character the Gauss sum is $i\sqrt{q}$, so $\nu(\pi) = \frac{\sqrt{q}}{2}$ and $\nu(2\pi) = -\chi_d(2)\frac{\sqrt{q}}{2}$. If $\chi_d(2) = 1$, ν vanishes at some $y_0 \in (\pi, 2\pi)$, where $\sin y_0 \ne 0$, and the transform vanishes at $\lambda = -y_0^2/\pi q$. For f_{DH}, $\nu(\pi) = \frac{\sqrt{5}}{2}$ and $\nu(2\pi) = -0.317610026505$. □

A survey extends the table. For each of the 26 fundamental discriminants with $3 \le |d| \le 40$, we test the positivity of ϑ_{χ_d} and, where it holds, run the test at 80 digits to order 10 at $\lambda = 0$, $\frac12$, 2, 10 and 20, on both sides. Of the 24 positive kernels, $\mathcal{B}_{\chi_d}$ passes only for $d = -4$, -3, 5, 8, 12 and 24 (the last fails at order 11, beyond the survey), and $1/\mathcal{B}_{\chi_d}$ only for $d = -4$ and -3. The 18 failures of $\mathcal{B}_{\chi_d}$ occur at order 2 or 3 for $d = -39$, -35, -31, -24, -23, -20, -15, -11, -8, -7, 13, 21, 29 and 37, and at orders 6 to 8 for $d = 17$, 28, 33 and 40; the failures of $1/\mathcal{B}_{\chi_d}$ for $d = -7$, -15, -23, -31 and -39 are cases of Proposition 14.4.19(iv).

Combinations of L functions do not change the picture. Modulo 7 the odd characters are χ_{-7} and a complex pair χ, $\bar\chi$, with $\chi(3) = e^{\pi i/3}$, and for real r

$$f_r = \big(1 - 2r\cos\tfrac{\phi}{2}\big)L(s,\chi_{-7}) + re^{-i\phi/2}L(s,\chi) + re^{i\phi/2}L(s,\bar\chi), \qquad e^{i\phi} = \varepsilon(\chi),$$

has real coefficients, $a_1 = 1$ and $\vartheta(1/x) = x^{3/2}\vartheta(x)$. For 17 values of r between -3 and 3 the kernel is positive for $-3 \le r \le 2$ and not for $r = 3$, and for each of the 16 positive kernels both $\mathcal{B}$ and $1/\mathcal{B}$ fail at $\lambda = 0$, at order 2 or 3. None of the passing characters has a complex character of the same conductor and parity to combine with, so the smallest family through a passing character would need a larger modulus.

The pattern is not that of the Euler product. All nine L functions of Table 14.15 have one, with GRH verified to height at least $10^8/q$ [275]; yet $\mathcal{B}_{\chi_d}$ fails for χ_{24}, χ_{-7}, χ_{-8} and χ_{13}, and for 18 of the 24 positive kernels of the survey. It fails for f_{DH}, which has zeros off the line, but in the same way for χ_{-7} and χ_{13}, whose zeros are on the line as far as computed. At $\lambda = 0$ the test of order n reads only the moments $\xi(2k, a)/\xi(0, a)$, $1 \le k \le 2n$, whose numerators are values of Dirichlet series where they converge absolutely, and for $\lambda > 0$ it reads Laplace transforms of the theta series: it sees the shape of the theta transform, not the arithmetic of the coefficients. Where the GGC property is proved it comes from a product formula for that transform, $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$ for ξ and those of Proposition 14.4.19(i) and (ii) for the reciprocals of χ_{-4} and χ_{-3}. The computations find no function with $\mathcal{B}_a$ a GGC and a zero off the line.

14.4.6 What survives, and open problems

Apart from the Lambert template, the results of this section and those it uses fall into three classes. The first consists of equivalences with RH: the Thorin Bernstein property of $\log G$ (Theorem 9.2.4), the exponential mixture in the size biased clock (Proposition 14.4.1), the vanishing of the Hankel inertia indices (Theorem 8.1.7), and the GGC property of R_t at $t = 0$ (Theorem 17.1.16 and Proposition 2.1.1). The second consists of facts with no hypothesis: the clock is infinitely divisible and self decomposable (Theorem 8.1.1), lies in L_2 and not in L_3 (Theorem 14.4.8), and its driver has self decomposable, hence unimodal, laws (Corollary 14.4.9); the points of the upper half plane where $\mathrm{Im}(G'/G)$ could fail to be negative lie in a thin parabola beyond about $-T_0^2$ (Theorem 9.3.9); the clock is not the martingale limit of a supercritical continuous state branching process (Theorem 14.4.11); R_t is a GGC for $t \ge 0.2$ [277] and for no $t < 0$ (Corollary 9.2.5); and $\mathcal{B}_t$ is not a GGC for t near 0, $t \ne 0$ (Proposition 14.4.15). None of these implies RH, not for being unconditional (a proof of RH would be too) but because none distinguishes ξ from functions with zeros off the line: the statements about the flow at $t \ne 0$ are removed by the filter, infinite divisibility and self decomposability hold also for the comparison function F of Section 25.2.6, the ladder index and the branching obstruction are set by the first zeros (Section 14.4.4), and the localisation uses only the zeros below the verification height. The third class consists of closed routes. The Hankel route is closed for practical purposes, on the heuristic of Table 17.6. Properties of the kernel and its mixture representations are removed by the filter (Propositions 14.4.12 and 14.4.13), and the GGC property of the variable of Biane, Pitman and Yor does not detect the Euler product (Section 14.4.5.4).

14.4.6.1 Open problems

Problem 14.1. Find a natural class of functions $\xi(s, a)$ with positive kernels, containing ξ, in which the GGC property of $\mathcal{B}_a$ implies that every zero of $\xi(s, a)$

lies on the critical line, or a function in such a class whose variable $\mathcal{B}_a$ is a GGC and which has a zero off the line. Since $\mathcal{B}$ is a GGC, the first answer for a class containing ξ proves RH. No candidate for the second is known: the function of Davenport and Heilbronn, the family f_r modulo 7 at the 16 tested values of r with a positive kernel and the tilts $\mathcal{B}_t$ at the eleven tested values of $t < 0$ all fail the test. The question is related to Question 18.1.20 and to the structural reason for R to be a GGC that the discussion of Table 17.6 asks for.

PROBLEM 14.2. Is $\mathcal{B}_t$ a GGC for no $t \neq 0$? For $t < 0$ this would be a property of the kernel that passes the filter at every t. Proposition 14.4.15 proves it for $0 < |t| < t_0$, through the first order change of the Thorin measure in the gaps between its atoms, and at the ten values of part (i).

PROBLEM 14.3. Which of the variables $\mathcal{B}_{\chi_d}$ for $d = -4$, -3, 5, 8 and 12, which pass the test to order 12, are GGCs? (For $d = 24$ the test fails at order 11.) On the reciprocal side, $1/\mathcal{B}_{\chi_d}$ is a GGC for $d = -3$ and -4 (Proposition 14.4.19); for $d = -4$ it is the exit time variable of Remark 15.1.23, whose density is not HCM (Theorem 9.3.20), so the question for $\mathcal{B}_{\chi_{-4}}$ cannot be settled through HCM. For even characters the reciprocal transform is the series $\frac{2}{\xi(0,a)} \sum_n a_n K_0\big(2n\sqrt{\pi\lambda/q}\big)$ of Bessel functions, and no product formula for it is known to us.

PROBLEM 14.4. With no hypothesis, the Li coefficients of Section 23.6.2 are

$$\lambda_n = \int_0^\infty k_n(t) W(t)\, dt = m_0\, \mathbb{E} k_n(A),$$

where W is the heat trace, A is the variable of Proposition 14.4.1, and k_n is the function with $\int_0^\infty e^{-zt} k_n(t)\, dt = \varphi_n(z)$. Since φ_n is a polynomial of degree n in $(z+\frac{1}{4})^{-1}$ with no constant term, $k_n(t) = e^{-t/4} p_n(t)$ with p_n a polynomial of degree $n-1$; for instance $k_1(t) = e^{-t/4}$ and $k_2(t) = (4-t)e^{-t/4}$. Which unconditional properties of W, or of A, force $\mathbb{E} k_n(A) \geq 0$ for every n, which is RH by Li's criterion [208, 37]? Positivity of W (Theorem 8.1.1) gives $n = 1$; complete monotonicity of W, which is RH, gives every n, since $\varphi_n \geq 0$ on $(0, \infty)$.

PROBLEM 14.5. Describe the ladder index of R_t along the de Bruijn Newman flow. The computation of Section 14.4.4.3 finds the index two of the clock up to $t = 27.8$ and one from $t = 27.9$ on, on its window. Is the index nonincreasing in t, and is it one for all large t? The answer carries no information about RH, but it would show how the spacing of the first zeros sets the index.

Part IV
Beyond zeta

Chapter 15

The Dedekind eta function, $L(s, \chi_{12})$ and Ramanujan's Δ

15.1 Beyond zeta: the Dedekind eta kernel, $L(s, \chi_{12})$ and Ramanujan's Δ

The Pólya route uses little that is special to ζ. The zero density identity of Section 3.3 needs a completed function that is positive on the real axis and zero free on $\operatorname{Re} s = 1$, and the sine squared decomposition (1.3.1) needs an Euler product. The completed L functions of the real character χ_{12} modulo 12 and of Ramanujan's cusp form Δ have both, and their kernels, built from the Dedekind eta function, are positive by the product formula. From the eta kernel alone, the zero density identity and a count of sign changes place every zero of $L(s, \chi_{12})$ below height 100 on the critical line: the generalised Riemann hypothesis (GRH) for χ_{12} on that window, a question raised in [285, Theorem 12]. Section 15.2 returns to $L(s, \chi_{12})$ at the basepoint one.

15.1.1 The eta kernel and GRH for $L(s, \chi_{12})$ below height 100

The character χ_{12} is the even primitive character modulo 12: $\chi_{12}(n) = 1$ for $n \equiv \pm 1$, -1 for $n \equiv \pm 5 \pmod{12}$, and 0 otherwise. Euler's pentagonal number theorem gives

$$\eta(ix) = e^{-\pi x/12} \prod_{n \geq 1} \bigl(1 - e^{-2\pi n x}\bigr) = \sum_{n \geq 1} \chi_{12}(n)\, e^{-\pi n^2 x/12}, \qquad x > 0,$$

so the theta series of χ_{12} is the Dedekind eta function on the imaginary axis. The completed L function

$$\xi(s, \chi_{12}) = \Bigl(\frac{12}{\pi}\Bigr)^{s/2} \Gamma\Bigl(\frac{s}{2}\Bigr) L(s, \chi_{12}) = \int_0^\infty \eta(ix)\, x^{s/2} \frac{dx}{x}$$

satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$ because $\eta(i/x) = \sqrt{x}\,\eta(ix)$, the case $z = ix$ of $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$. With $x = e^{2u}$,

$$\xi\Big(\frac12 + s, \chi_{12}\Big) = \int_{\mathbb{R}} K(u)\, e^{su}\, du, \qquad K(u) = 2e^{u/2}\eta\big(ie^{2u}\big),$$

and the modular transformation makes K even. The product formula makes $\eta(ix) > 0$, so $K/\xi(\frac12,\chi_{12})$ is a probability density, the analogue of the Pólya density: if Y has this density, $\mathbb{E}e^{sY} = \xi(\frac12+s,\chi_{12})/\xi(\frac12,\chi_{12})$ for every complex s. Like Φ, K is analytic in $|\operatorname{Im} u| < \pi/4$ and decays doubly exponentially, so the trapezoid analysis of Section 2.1 applies unchanged.

The construction extends to every real primitive character χ modulo m, whose root number is one. For even χ the kernel is $2e^{u/2}\sum_{n\ge1}\chi(n)e^{-\pi n^2e^{2u}/m}$; for odd χ, whose completed function is $(m/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi)$, it is $2e^{3u/2}\sum_{n\ge1} n\chi(n)e^{-\pi n^2e^{2u}/m}$. Both kernels are even and analytic in $|\operatorname{Im} u| < \pi/4$, but their positivity has to be checked case by case.

The two polynomial evaluator of Section 2.1, run with the eta kernel, agrees with `mpmath`'s Dirichlet L function [237] to sixteen digits. Sign changes of $\xi(\frac12 + i\theta,\chi_{12})$ on a grid of step 0.02, refined by Newton's method, give 67 zeros below height 100, the first of them at

$$3.8046276,\ 6.6922233,\ 8.8905930,\ 11.1883927,\ 12.9661788,\ 15.1814809,\ \ldots.$$

The Thorin density at resolution ε, $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac12+\varepsilon+i\theta,\chi_{12})$ as in (2.1.7), is again a sum of Poisson kernels over the zeros (Proposition 2.1.3).

The zero density identity, Proposition 3.3.2, holds for $\xi(s,\chi_{12})$ with the same hypotheses and proof, since $\xi(s,\chi_{12})$ is entire, symmetric under $s\mapsto 1-s$, positive on the real axis because K is, and zero free on $\operatorname{Re} s \ge 1$. Here $N_{\chi_{12}}(T)$ counts the zeros with $0 < \operatorname{Im}\rho \le T$, and $\nu(\sigma,T)$ those among them with $\operatorname{Re}\rho > \sigma$; $M_\varepsilon(T) = \int_0^T \rho_\varepsilon(\theta)\,d\theta$ is the Thorin mass up to height T, and $\tau_\varepsilon(T)$ is the change of argument of $\xi(s,\chi_{12})$ from $\frac12+\varepsilon+iT$ to $\frac12+iT$, divided by π. Table 15.1 evaluates the identity at $T = 100$.

ε	$M_\varepsilon(100)$	$\tau_\varepsilon(100)$	$M_\varepsilon + \tau_\varepsilon = N - 2\nu$
0.5	67.604974186	−0.604974186	67.00000000000003
0.25	67.476432836	−0.476432836	67.00000000010355
0.1	67.308171973	−0.308171973	67.00000000014663
0.05	67.185437421	−0.185437421	67.00000000009639
0.02	67.080023774	−0.080023774	67.00000000005207

Table 15.1: The zero density identity for $L(s,\chi_{12})$ at height 100, computed from the eta kernel. The last column is $N_{\chi_{12}}(100) - 2\nu(\frac12+\varepsilon,100)$; the row $\varepsilon = \frac12$ gives $N_{\chi_{12}}(100)$.

The row $\varepsilon = \frac{1}{2}$ gives $N_{\chi_{12}}(100) = 67$, and with the 67 sign changes every zero below height 100 is simple and lies on the critical line. The other rows give $\nu(\frac{1}{2}+\varepsilon, 100) = 0$ without using the sign changes. This is GRH for χ_{12} below height 100, computed, not certified: the argument is followed on a grid as in Section 3.3, and its increments are summed in double precision. The deviations from 67, at most 1.5×10^{-10}, are far below the one half that fixes the integer. GRH for small moduli has long been verified to far greater heights by other methods, by Rumely [321] and later by Platt [275].

15.1.2 Ramanujan's Δ and the cost of the three evaluators

The construction works for Ramanujan's cusp form $\Delta(z) = \eta(z)^{24} = \sum_{n\geq 1} \tau(n) q^n$, $q = e^{2\pi i z}$, of weight 12 and level 1. Its L function, normalised so that the critical line is $\operatorname{Re} s = \frac{1}{2}$, is $L(s,\Delta) = \sum_{n\geq 1} \tau(n) n^{-s-11/2}$, and the completed function $\xi(s,\Delta) = (2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ satisfies $\xi(s,\Delta) = \xi(1-s,\Delta)$. The Mellin transform of $\Delta(iy)$ gives the Pólya type representation

$$\xi\Big(\frac{1}{2}+s,\Delta\Big) = \int_{\mathbb{R}} K_\Delta(u)\, e^{su}\, du,$$
$$K_\Delta(u) = e^{6u}\,\Delta(ie^u) = e^{6u}e^{-2\pi e^u} \prod_{n\geq 1} \big(1-e^{-2\pi n e^u}\big)^{24}.$$

The kernel is even because $\Delta(i/y) = y^{12}\Delta(iy)$, positive by the product formula, and analytic in the strip $|\operatorname{Im} u| < \pi/2$.

We computed $\tau(n)$ for $n \leq 3000$, the range needed for the prime weights of Section 15.1.3, and evaluated the kernel by its q expansion. At 60 digits the trapezoid evaluator finds the first zeros on the critical line at

$$9.222379,\ 13.907550,\ 17.442777,\ 19.656513,\ 22.336104,\ \ldots,$$

in agreement with the known values [220, modular form 1.12.a.a], and 20 sign changes below height 50, the number of zeros there.

15.1.3 The sine squared lemma without a pole

The eta based L functions are entire, so two consequences of the pole of ζ disappear: the negative pole piece of the sine squared decomposition, and the factor that Chapters 10 and 11 need to remove the pole, which by Proposition 11.1.13 alone prevents the normalised reciprocal zeta function from being a GGC. A floor in the sense of Theorem 10.2.15 is still needed for χ_{12}: its tilt, the factor $e^{-c\sqrt{u}}$ in the clock variable u (written $e^{-c\sqrt{s}}$ in Section 10.2), removes the negative part of the Thorin density of the gamma factor, and the gamma factor $(12/\pi)^{s/2}\Gamma(s/2)$ is that of ξ times $12^{s/2}$, so Theorem 10.2.12 holds with $c = 2.6861$ replaced by

$c - \frac{1}{2}\log 12 = 1.4436$. The positivity of such a floor would need an analogue of Theorem 11.1.1.

For $f = \chi_{12}$ or Δ write $\xi(s,f) = L_\infty(s,f)L(s,f)$, with L_∞ the gamma factor, and $-(L'/L)(s,f) = \sum_{n\ge2}\Lambda_f(n)n^{-s}$ for $\operatorname{Re} s > 1$. At a basepoint $\alpha > 1$ the sine squared lemma gives, as (1.3.1) does for ζ,

$$v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta,f) = \underbrace{\frac{\xi'}{\xi}(\alpha,f)}_{\text{drift}} + \underbrace{\operatorname{Re}\frac{L'_\infty}{L_\infty}(\alpha+i\theta,f) - \frac{L'_\infty}{L_\infty}(\alpha,f)}_{\text{archimedean } \ge 0} + \underbrace{\sum_{n\ge2}\Lambda_f(n)\,n^{-\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big)}_{\text{primes}},$$

where $v_\alpha(\theta)$ is π times the Thorin density at resolution $\alpha - \frac{1}{2}$, and the archimedean piece is nonnegative because the digamma function $\psi = \Gamma'/\Gamma$ satisfies $\operatorname{Re}\psi(x+iy) \ge \psi(x)$ for $x > 0$. For ζ, whose gamma factor $\pi^{-s/2}\Gamma(1+\frac{s}{2})$ absorbs the factor s of ξ, the remaining factor $s-1$ adds the pole piece $\operatorname{Re}\frac{1}{s-1} - \frac{1}{\alpha-1}$, which is $-\theta^2/\big((\alpha-1)((\alpha-1)^2+\theta^2)\big)$ at $s = \alpha+i\theta$, and the identity becomes (1.3.1).

The prime weights are $\Lambda(n) = \log p$ at $n = p^k$ for ζ, all positive; $\Lambda_{\chi_{12}}(p^k) = \chi_{12}(p)^k\log p$ for χ_{12}, of both signs and zero at $p = 2$ and 3; and $\Lambda_\Delta(p^k) = (\alpha_p^k + \bar\alpha_p^k)\log p$ for Δ, where the Satake parameters α_p and $\bar\alpha_p$ are the roots of $X^2 - \tau(p)p^{-11/2}X + 1$. Deligne's theorem $|\tau(p)| \le 2p^{11/2}$ [75] makes them complex conjugates of modulus one, so the weights of Δ lie in $[-2\log p, 2\log p]$ and take both signs.

We computed the three decompositions at $\alpha = 2$ for $0 \le \theta \le 40$, with prime powers up to 3000, and compared their sum with the left side computed from the kernel. They agree to 5×10^{-4} for ζ and 9×10^{-6} for Δ, the difference being the truncation of the prime sum. For χ_{12} the largest difference, 2.5×10^{-5} at $\theta = 40$, is an error of the kernel evaluator, whose truncation at $|u| \le 2.5$ is too short at that height. Of the 466 prime power weights up to 3000, none is negative for ζ, 222 are negative for χ_{12} (whose 18 weights at powers of 2 and 3 vanish) and 230 for Δ.

On $0 \le \theta \le 40$ the prime piece of $L(s,\chi_{12})$ ranges over $[-0.218, 0.053]$, and that of $L(s,\Delta)$ over $[-0.261, 0.222]$ (Figure 15.1, left). The left sides are positive, with minima at $\theta = 0$ equal to the drifts, 0.466 and 0.167: the positivity comes from the drift and the archimedean piece, not from the primes.

For each of the three functions its Riemann hypothesis (RH for ζ, GRH for the other two) is the statement that v_α stays positive for every $\alpha > \frac{1}{2}$: Lagarias's criterion [198], whose proof from the Hadamard product applies unchanged to $\xi(s,\chi_{12})$ and $\xi(s,\Delta)$. Below $\alpha = 1$ the prime piece no longer converges absolutely; for $L(s,\chi_{12})$, Section 15.2.2 shows that the extension of the decomposition down to the centre is GRH for χ_{12}. At resolution 0.1 the Thorin densities of $L(s,\chi_{12})$

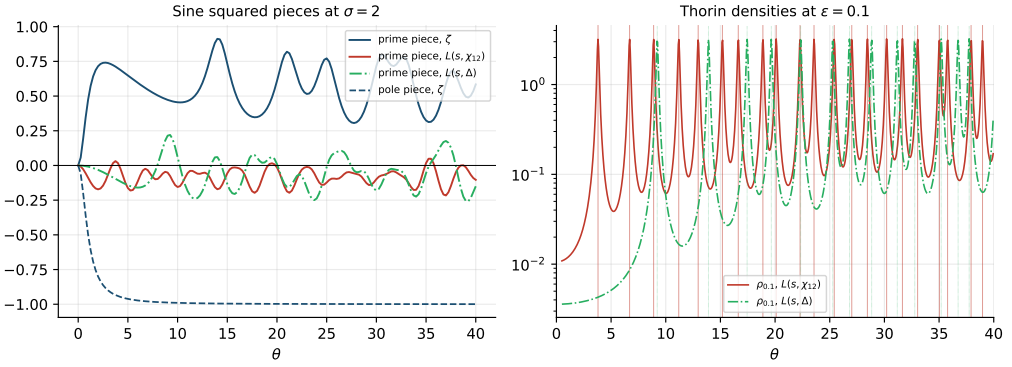


Figure 15.1: Left: the prime pieces of the sine squared decomposition at the basepoint $\alpha = 2$, written σ in the plot, for ζ, $L(s, \chi_{12})$ and $L(s, \Delta)$, and the pole piece of ζ. For ζ the prime piece is positive and the pole piece negative; for the two eta based functions there is no pole piece and the prime piece takes both signs. Right: the Thorin densities $\rho_{0.1}$ of $L(s, \chi_{12})$ and $L(s, \Delta)$ on a logarithmic scale, peaking at their zeros (vertical lines).

and $L(s, \Delta)$ have computed minima 0.011 and 0.004 below height 40 (Figure 15.1, right), positive as GRH predicts.

The distances of Section 3.1, with the zeros of $L(s, \chi_{12})$ or $L(s, \Delta)$ as the reference atoms, Theorem 9.3.5 and the flow of Section 4.1 carry over unchanged, the last two because they use only the Hadamard product, positivity on the real line and the decay of the kernel; we have not computed the first two for these functions, and Section 15.1.5 computes the flow for Δ. The bound $|\arg z| < \pi/4$ of Lemma 9.3.4, which Theorem 9.3.5 needs for the zeros z of $\xi(\frac{1}{2} + iz, f)$ with $\operatorname{Re} z > 0$, holds too: $|\operatorname{Im} z| < \frac{1}{2}$, while the lowest ordinates, 3.80 and 9.22, exceed $\frac{1}{2}$. The arithmetic side, however, offers no termwise positivity even far to the right of the critical line, so the domination of the pole by positive primes that Sections 1.3 and 5.1 formulate for ζ has no direct analogue. Far enough to the right the absolute values suffice: the drift and the archimedean piece outweigh the prime piece bounded by the absolute values of its weights for $\alpha > 1.8349$ for $L(s, \chi_{12})$ and for $\alpha > 2.2823$ for the Dirichlet beta function $L(s, \chi_4)$, and by Kronecker's theorem no better bound from the sizes of the weights holds at all large heights (Propositions 15.1.26 and 15.1.27). So neither eta based function is theoretically easier than ζ, although $L(s, \chi_{12})$ is the cheapest of the three per zero.

15.1.4 Maxima of exponentials

The product formula has a probabilistic reading [286]. The factor $1-e^{-2\pi ny}$ is the distribution function of an exponential variable with rate $2\pi n$, so

$$F(y)=\prod_{n\ge1}\bigl(1-e^{-2\pi ny}\bigr)^{24}=e^{2\pi y}\Delta(iy),\qquad F_1(y)=\prod_{n\ge1}\bigl(1-e^{-2\pi ny}\bigr)=e^{\pi y/12}\eta(iy)$$

are the distribution functions of the maxima M^* and M_1 of independent exponential variables.

15.1.4.1 The max representation

Lemma 15.1.1. *Let $E_{n,j}$, $n\ge1$, $1\le j\le24$, be independent, with $E_{n,j}$ exponential of rate $2\pi n$, and put $M^*=\sup_{n,j}E_{n,j}$ and $M^{(j)}=\sup_n E_{n,j}$. Almost surely these suprema are attained and finite, $\Pr(M^*\le y)=F(y)$ for $y>0$, and each $M^{(j)}$ has the distribution function F_1. So $M^*=\max_{j\le24}M^{(j)}$ is the maximum of 24 independent copies of M_1.*

Proof. For $y>0$, $\sum_{n,j}\Pr(E_{n,j}>y)=24\sum_n e^{-2\pi ny}<\infty$, so by the Borel Cantelli lemma only finitely many $E_{n,j}$ exceed y, almost surely. Taking y rational and below $E_{1,1}$, the supremum is the largest of finitely many values. By independence $\Pr(M^*\le y)=\lim_N\prod_{n\le N}(1-e^{-2\pi ny})^{24}=F(y)$, and in the same way each $M^{(j)}$ has the distribution function F_1; the $M^{(j)}$ are independent because they are built from disjoint sets of the $E_{n,j}$. □

The modular relations $\Delta(i/y)=y^{12}\Delta(iy)$ and $\eta(i/y)=\sqrt{y}\,\eta(iy)$ become self dualities of the two laws,

$$F(y)=y^{-12}e^{2\pi y-2\pi/y}F(1/y),\qquad F_1(y)=y^{-1/2}e^{\pi y/12-\pi/(12y)}F_1(1/y),\quad(15.1.1)$$

so the lower tails are thin: $\Pr(M^*\le y)\sim y^{-12}e^{-2\pi/y}$ and $\Pr(M_1\le y)\sim y^{-1/2}e^{-\pi/(12y)}$ as $y\downarrow0$. Since $F(1/y)\le1$, also $F(y)\le y^{-12}e^{2\pi y-2\pi/y}$ for every $y>0$, a bound increasing in y, since its logarithmic derivative is $(2\pi y^2-12y+2\pi)/y^2$ and $144<16\pi^2$.

The density of M^* is Ramanujan's logarithmic derivative: expanding $1/(1-e^{-2\pi ny})$ gives $(\log F)'(y)=24\sum_n 2\pi n\,e^{-2\pi ny}/(1-e^{-2\pi ny})=48\pi\sum_{m\ge1}\sigma(m)e^{-2\pi my}$, with $\sigma(m)$ the sum of the divisors of m, so that, with Eisenstein's series $E_2(z)=1-24\sum_{m\ge1}\sigma(m)e^{2\pi imz}$,

$$f^*(y)=F(y)\,(\log F)'(y)=2\pi F(y)\bigl(1-E_2(iy)\bigr).\qquad(15.1.2)$$

Differentiating (15.1.1) gives the functional equation of the density,

$$f^*(1/y)=y^{12}e^{2\pi/y-2\pi y}\Bigl(\bigl(2\pi y^2-12y+2\pi\bigr)F(y)-y^2f^*(y)\Bigr),$$

which in terms of E_2 is its quasi modularity, $E_2(i/y)=-y^2E_2(iy)+6y/\pi$.

THEOREM 15.1.2. *For every complex s,*

$$\xi(s,\Delta) = (2\pi)^{-s-\frac{11}{2}}\,\mathbb{E}\,\Gamma\big(s+\tfrac{11}{2},2\pi M^*\big), \qquad L(s,\Delta) = \mathbb{E}\,Q\big(s+\tfrac{11}{2},2\pi M^*\big), \tag{15.1.3}$$

with $\Gamma(a,x) = \int_x^\infty e^{-v}v^{a-1}dv$ and $Q(a,x) = \Gamma(a,x)/\Gamma(a)$, and both expectations converge absolutely. In particular $L(s,\Delta) = 0$ at the points $s = -\frac{11}{2} - k$, $k = 0,1,2,\ldots$, where $Q(-k,x) = 0$.

Proof. Put $a = s + \frac{11}{2}$, so that $\xi(s,\Delta) = \int_0^\infty \Delta(iy)y^{a-1}dy$. Inserting $\Delta(iy) = e^{-2\pi y}\mathbb{E}\,\mathbf{1}\{M^* \le y\}$ and integrating in y first gives $\mathbb{E}\int_{M^*}^\infty e^{-2\pi y}y^{a-1}dy = (2\pi)^{-a}\mathbb{E}\,\Gamma(a,2\pi M^*)$. Fubini's theorem applies: with $\sigma = \operatorname{Re} a$, Tonelli's theorem gives $\mathbb{E}\int_{M^*}^\infty e^{-2\pi y}y^{\sigma-1}dy = \int_0^\infty \Delta(iy)y^{\sigma-1}dy$, which is finite for every real σ, since $\Delta(iy)$ decays exponentially as $y \to \infty$ and like $y^{-12}e^{-2\pi/y}$ as $y \downarrow 0$. The same computation gives $\mathbb{E}|\Gamma(a,2\pi M^*)| \le \mathbb{E}\,\Gamma(\sigma,2\pi M^*) < \infty$. Dividing by $(2\pi)^{-a}\Gamma(a)$, with $1/\Gamma$ entire, gives the second identity, in which $Q(a,x)$, for fixed $x > 0$, is the entire function $\Gamma(a,x)/\Gamma(a)$ of a. At $a = -k$ it vanishes, since $\Gamma(-k,x)$ is finite and $1/\Gamma(-k) = 0$. □

COROLLARY 15.1.3 (Gamma mixture and mixed Poisson forms). *Let G_a be a gamma variable of shape $a > 0$ and scale one, and N a Poisson process of unit rate, both independent of M^*. For real $s > -\frac{11}{2}$, with $a = s + \frac{11}{2}$,*

$$L(s,\Delta) = \Pr\big(G_a > 2\pi M^*\big) = \mathbb{E}F\Big(\frac{G_a}{2\pi}\Big) = \mathbb{E}\prod_{n\ge1}\big(1-e^{-nG_a}\big)^{24},$$

and for every integer $k \ge 1$

$$L\big(k-\tfrac{11}{2},\Delta\big) = \Pr\big(N(2\pi M^*) \le k-1\big) = \mathbb{E}\Big[e^{-2\pi M^*}\sum_{j<k}\frac{(2\pi M^*)^j}{j!}\Big].$$

Proof. For real $a > 0$, $Q(a,x) = \Pr(G_a > x)$, which gives the first form; conditioning on G_a instead gives $\Pr(M^* < G_a/2\pi) = \mathbb{E}F(G_a/2\pi)$, since the law of M^* has no atoms. For an integer k the kth point of N has the law of G_k, so $\Pr(G_k > x) = \Pr(N(x) \le k-1)$. □

So $L(-\frac{9}{2},\Delta) = \mathbb{E}e^{-2\pi M^*} = 0.03744128126851554174$ is a value of the Laplace transform of M^*, and $L(\frac{1}{2},\Delta) = 0.79212283864603056936$ is the probability that a Poisson clock of rate 2π rings at most five times before the time M^*. Write $P_k = \Pr(N(2\pi M^*) \le k-1) = L(k-\frac{11}{2},\Delta)$. The functional equation $\xi(s,\Delta) = \xi(1-s,\Delta)$ at $s = k - \frac{11}{2}$ becomes the family of identities

$$(2\pi)^{-k}(k-1)!\,P_k = (2\pi)^{k-12}(11-k)!\,P_{12-k}, \qquad 1 \le k \le 11, \tag{15.1.4}$$

both sides being $\xi(k-\frac{11}{2},\Delta)$.

Remark 15.1.4. The identities (15.1.4) follow from the self duality (15.1.1), since $(2\pi)^{-k}(k-1)!\,P_k = \int_0^\infty e^{-2\pi y}F(y)y^{k-1}dy$ and the substitution $y \mapsto 1/y$ turns this integral into the one with $12-k$ in place of k. They are five relations, since k and $12-k$ give the same one and $k=6$ gives none, and they do not imply the self duality. Let g be smooth and not identically zero, with compact support in $(0,1)$, such that $g(\log y)$ is orthogonal on $(1,e)$ to the five functions $e^{-2\pi y}F(y)(y^{k-1} - y^{11-k})$, $k \le 5$. For small ε, $\tilde F(y) = F(y)\big(1+\varepsilon g(\log y)\big)$ is again a distribution function, because F' has a positive lower bound and F is below one on $[1,e]$, and it satisfies (15.1.4), because the perturbation changes the difference of the two sides of the kth identity by ε times the integral of $g(\log y)$ against the kth function. It is not self dual: on $(1,e)$ self duality would give $\tilde F(y) = y^{-12}e^{2\pi y - 2\pi/y}\tilde F(1/y) = F(y)$, since $g(\log(1/y)) = 0$. With g even and supported in $(-1,1)$ instead, the same construction gives a self dual distribution function other than F, so self duality does not determine the law of M^* either. Self duality together with the form of F as a power series in $e^{-2\pi y}$ does. If $F(y) = \sum_{n\ge0} c_n e^{-2\pi n y}$ for $y>0$, with $c_0 = 1$, and F satisfies (15.1.1), then $f(z) = \sum_{n\ge0} c_n e^{2\pi i(n+1)z}$ is holomorphic on the upper half plane, satisfies $f(z+1) = f(z)$, and satisfies $f(-1/z) = z^{12}f(z)$ on the imaginary axis, hence everywhere by analytic continuation. So f is a cusp form of weight 12 for $\mathrm{SL}_2(\mathbb{Z})$, a multiple of Δ because these cusp forms make up a space of dimension one [167, Chapter 14], and $c_0 = 1$ gives $f = \Delta$. This argument is the simplest case of Hecke's converse theorem [112]. The distribution function of a maximum of independent exponential variables whose rates are integer multiples of 2π is a power series of this kind, so M^* is the only such maximum with the self duality (15.1.1).

Cutting the integral at $y=1$ and folding $(0,1)$ onto $(1,\infty)$ by modularity gives the symmetric form

$$\xi(s,\Delta) = A(s) + A(1-s), \qquad A(s) = \int_1^\infty \Delta(iy)\, y^{s+\frac{9}{2}}dy, \tag{15.1.5}$$

$$A(s) = (2\pi)^{-s-\frac{11}{2}}\,\mathbb{E}\,\Gamma\big(s+\tfrac{11}{2}, 2\pi\max(M^*,1)\big) = \sum_{n\ge1}\tau(n)\,\frac{\Gamma(s+\frac{11}{2},2\pi n)}{(2\pi n)^{s+11/2}},$$

where the last form, Hecke's, comes from expanding $\Delta(iy) = \sum_n \tau(n)e^{-2\pi n y}$ instead. It is the analogue for Δ of Riemann's second proof of the functional equation [80, Chapter 1], which cuts the Mellin integral of the theta function $\omega(x) = \sum_{n\ge1} e^{-\pi n^2 x}$ of Section 2.1 at $x=1$.

Proposition 15.1.5 (Riemann). *For every complex $s \ne 0,1$,*

$$\pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big)\zeta(s) = -\frac{1}{s} - \frac{1}{1-s} + \sum_{n\ge1}\Big[\frac{\Gamma(\frac{s}{2},\pi n^2)}{(\pi n^2)^{s/2}} + \frac{\Gamma(\frac{1-s}{2},\pi n^2)}{(\pi n^2)^{(1-s)/2}}\Big].$$

Proof. Split the Mellin integral of ω at $x=1$, apply the theta transformation on $(0,1)$, and integrate term by term on $(1,\infty)$. □

The three formulas are incomplete gamma continuations of Mellin integrals that randomise different things. In Riemann's and Hecke's the threshold is the fixed point $x = 1$ or $y = 1$, and the rates πn^2 or $2\pi n$ carry weights: one each for ζ, at the price of the pole terms, and the signed coefficients $\tau(n)$ for Δ. In (15.1.3) there is a single rate 2π, coming from the factor $q = e^{-2\pi y}$ of Δ, and the threshold M^* is random with a positive law; the symmetric form (15.1.5) is Riemann's formula with the sum over n replaced by an expectation over $\max(M^*, 1)$, and with no pole terms.

At $s = \frac{1}{2}$, $\frac{1}{2} + 5i$ and 3 the max representation, the symmetric form and Hecke's formula agree to 25 digits. On the critical line the symmetric form gives $\xi(\frac{1}{2} + i\theta, \Delta) = 2\,\mathrm{Re}\,A(\frac{1}{2} + i\theta)$. It also suggests a Hermite Biehler test, the route that Lagarias studied for ξ [199]: if $|A(s)| > |A(1-s)|$ held throughout $\mathrm{Re}\,s > \frac{1}{2}$, there could be no zeros there. The test fails. On $\mathrm{Re}\,s = 0.6$ the two halves have equal modulus at $\theta = 10.26$, 14.67, 18.04, 20.36 and 22.99, and the dominant half alternates between these ordinates, which separate the zeros 9.22, 13.91, 17.44, 19.66 and 22.34, so the cut at $y = 1$ gives no Hermite Biehler function.

The cut can be moved: for every $c > 0$ the same folding gives $\xi(s, \Delta) = A_c(s) + A_{1/c}(1-s)$ with $A_c(s) = \int_c^\infty \Delta(iy) y^{s+9/2} dy$, and c may be chosen separately on each horizontal ray. The freedom is illusory. As $c \to 0$, $A_{1/c}(1-s) \to 0$ and $A_c(s) \to \xi(s, \Delta)$ uniformly on compact sets, so domination for small c only says that $\xi(s, \Delta)$ does not vanish there, and the certificate turns into an evaluation of ξ; and at a zero $\frac{1}{2} + i\gamma$ the two halves have equal modulus for every c, since their sum vanishes. A Hermite Biehler function that could prove GRH for Δ would have to come from the structure of the law of M^*, not from tuning a cut. The representation is exact and has no signs, but like the kernel it restates the problem without locating the zeros.

15.1.4.2 Other theta products

The construction needs a modular form whose expansion on the imaginary axis is a power of q times a product of factors $1 - q^a$, $a > 0$, with nonnegative exponents, so that the product is a distribution function; eta products with nonnegative exponents qualify.

For χ_{12} the Mellin integral of Section 15.1.1, with $\eta(ix) = e^{-\pi x/12} F_1(x)$, gives

$$L(s, \chi_{12}) = \mathbb{E}\, Q\Big(\frac{s}{2}, \frac{\pi}{12} M_1\Big), \qquad \xi(s, \chi_{12}) = A_{12}(s) + A_{12}(1-s),$$

with $A_{12}(s) = \int_1^\infty \eta(ix) x^{s/2-1} dx = (\frac{\pi}{12})^{-s/2} \mathbb{E}\,\Gamma(\frac{s}{2}, \frac{\pi}{12}\max(M_1, 1))$; the second exponent comes from $\eta(i/x) = \sqrt{x}\,\eta(ix)$. At $s = 1$, where $Q(\frac{1}{2}, x) = \mathrm{erfc}\,\sqrt{x}$, the class number formula reads $\mathbb{E}\,\mathrm{erfc}\,\sqrt{\pi M_1/12} = L(1, \chi_{12}) = \log(2 + \sqrt{3})/\sqrt{3}$; at $s = 2$, where $Q(1, x) = e^{-x}$, $L(2, \chi_{12}) = \mathbb{E} e^{-\pi M_1/12} = \pi^2/(6\sqrt{3})$ is a value of the Laplace transform of M_1, which Section 15.1.4.4 computes in closed form; and at the centre $L(\frac{1}{2}, \chi_{12}) = \Pr(12 G_{1/4} > \pi M_1) = 0.498557002457815436157$.

At $s = 0.6 + 5i$ the max representation, both forms of the symmetric form and `mpmath`'s Dirichlet L function agree to 25 digits, as do the three values above with their closed forms.

For the elliptic curve of conductor 11 the newform is $f(z) = \eta(z)^2\eta(11z)^2 = q\prod_{n\ge1}(1-q^n)^2(1-q^{11n})^2 = \sum_{n\ge1} a_n q^n$ [220, modular form 11.2.a.a], so $f(iy) = e^{-2\pi y}F_{11}(y)$, where F_{11} is the distribution function of the maximum M_{11} of independent exponential variables, two of each rate $2\pi n$ and two of each rate $22\pi n$. The Fricke relation $f(i/(11y)) = 11y^2 f(iy)$ makes $f(iy)$ decay faster than any power at both ends, and the proof of Theorem 15.1.2 gives, in the arithmetic normalisation $L(s,E) = \sum_n a_n n^{-s}$ with centre $s = 1$,

$$L(s,E) = \mathbb{E}\,Q(s, 2\pi M_{11}) \quad \text{for every complex } s,$$
$$L(1,E) = \mathbb{E}e^{-2\pi M_{11}} = 0.2538418608559106843378,$$

equal in all digits to the rapidly convergent series $2\sum_n a_n n^{-1} e^{-2\pi n/\sqrt{11}}$ of root number one.

Jacobi's triple product gives $\vartheta_4(x) = \sum_{n\in\mathbb{Z}}(-1)^n e^{-\pi n^2 x} = \prod_{m\ge1}(1 - e^{-2\pi m x})(1 - e^{-\pi(2m-1)x})^2$, a distribution function with no prefactor: that of the maximum M_4 of independent exponential variables, one of each rate $2\pi m$ and two of each rate $\pi(2m-1)$. The partial sums of the alternating series lie between -1 and 1, so the series can be integrated term by term against $\theta e^{-\theta y}$, and the partial fraction expansion of $z/\sinh z$ gives

$$\mathbb{E}e^{-\theta M_4} = \sum_{n\in\mathbb{Z}} (-1)^n \frac{\theta}{\theta + \pi n^2} = \frac{\sqrt{\pi\theta}}{\sinh\sqrt{\pi\theta}} = \prod_{n\ge1}\Big(1 + \frac{\theta}{\pi n^2}\Big)^{-1},$$

so $M_4 \overset{d}{=} \sum_{n\ge1} E_n/(\pi n^2)$ with independent standard exponential variables E_n: a GGC with Thorin measure $\sum_n \delta_{\pi n^2}$. Since $\Pr(K \le k) = \sum_n (-1)^n e^{-2n^2k^2} = \vartheta_4(2k^2/\pi)$, $\pi M_4 = 2K^2$ has the Kolmogorov law of Table 4.3. With no prefactor the Mellin transform gives moments instead of incomplete gamma functions: $\mathbb{E}M_4^{s/2} = \frac{s}{2}\int_0^\infty y^{s/2-1}(1 - \vartheta_4(y))\,dy$ for $\operatorname{Re} s > 0$, and integrating $1 - \vartheta_4(y) = 2\sum_{n\ge1}(-1)^{n+1}e^{-\pi n^2 y}$ term by term, for $\operatorname{Re} s > 1$ and then by analytic continuation, gives

$$(1 - 2^{1-s})\,\zeta(s) = \frac{\mathbb{E}\big[(2K^2)^{s/2}\big]}{2\,\Gamma(1 + \frac{s}{2})}$$

for every complex s, both sides being entire, since $\vartheta_4(y) = y^{-1/2}\sum_n e^{-\pi(n+\frac12)^2/y}$ makes every moment of M_4 finite. This is a case of the dictionary of Biane, Pitman and Yor [33]. At $s = 1$ it gives $\mathbb{E}K = \sqrt{\pi/2}\,\log 2 = 0.868731160636159141830$, which quadrature of $1 - \vartheta_4$ reproduces to 2×10^{-23}; at $s = 3$ and $\frac12 + 5i$ the two sides agree to 25 digits.

For ζ the construction stops. The theta function $\vartheta_3(x) = 1 + 2\omega(x) = \prod_{m\ge1}(1-q^{2m})(1+q^{2m-1})^2$, $q = e^{-\pi x}$, has factors larger than one and is not a distribution function, and the transformation $\vartheta_4(1/x) = \sqrt{x}\,\vartheta_2(x)$ leads to $\vartheta_2(x) = \sum_n e^{-\pi(n+\frac12)^2x}$, which is not one either. So the max structure reaches ζ only through the factor $1-2^{1-s}$, in moment form, and not through $\zeta_{\mathbb{Q}(\sqrt3)}(s) = \zeta(s)L(s,\chi_{12})$: the eta law represents the factor $L(s,\chi_{12})$ only, and no product of this kind can represent a Dedekind zeta function, since an eta product with non-negative exponents decays faster than any power of y at both ends of the imaginary axis, so its Mellin transform is entire, as in the proof of Theorem 15.1.2, while $\zeta_{\mathbb{Q}(\sqrt3)}$ has a pole at $s=1$.

15.1.4.3 Gamma shapes and infinite divisibility at real points

The kernels of ζ and Δ differ in their gamma shapes. On the logarithmic scale the function x^ae^{-x} of $x = ce^{ku}$ is proportional to the density of $(\log G_a - \log c)/k$, a log gamma density of shape a, and its Fourier transform $\int_{\mathbb{R}} x^ae^{-x}e^{i\theta u}du = k^{-1}c^{-i\theta/k}\Gamma(a+i\theta/k)$ has no zeros. With $x = 2\pi e^u$, $K_\Delta(u) = e^{6u}\Delta(ie^u) = (2\pi)^{-6}x^6e^{-x}F(e^u)$ is a single gamma kernel of shape 6 times the distribution function of $\log M^*$, and the gamma kernel's own transform is $(2\pi)^{-6-i\theta}\Gamma(6+i\theta)$, so every zero of $\xi(\frac12+i\theta,\Delta)$ is produced by the multiplier $F(e^u)$. The Pólya kernel (2.1.3) is, with $x_n = \pi n^2e^{2u}$ and the gamma densities $g_k(x) = x^{k-1}e^{-x}/\Gamma(k)$,

$$\Phi(u) = 2e^{u/2}\sum_{n\ge1}\big(2x_n^2 - 3x_n\big)e^{-x_n} = 2e^{u/2}\sum_{n\ge1}\big(4g_3(x_n) - 3g_2(x_n)\big),$$

a signed mixture of Gamma(2) and Gamma(3) densities over the scales πn^2. As a function of $x = e^{2u}$ the sum is, up to the factor $x/2$, the density $2\sum_n(2\pi^2n^4x - 3\pi n^2)e^{-\pi n^2x}$ of $\pi S_2/2$ in the representation $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Williams and of Biane, Pitman and Yor (Section 3.1.3). On the logarithmic scale, with the weight $e^{u/2} = (x_n/\pi n^2)^{1/4}$, the two terms have shapes $\frac94$ and $\frac54$, while the theta function itself contributes $x_n^{1/4}e^{-x_n}$, of shape $\frac14$.

Remark 15.1.6. Let $D = -x\,d/dx$. Integrating by parts, $\int_0^\infty (Df)(x)x^{s/2-1}dx = \frac{s}{2}\int_0^\infty f(x)x^{s/2-1}dx$ whenever the boundary terms vanish, so the operator $2D(D-\frac12)$ corresponds to the factor $\frac12 s(s-1)$ of $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$. It annihilates $-\frac12$ and $\frac12x^{-1/2}$, the two terms by which $\omega(x)$ and $x^{-1/2}\omega(1/x)$ differ, and it maps e^{-x} to $(2x^2-3x)e^{-x}$. Hence $[2D(D-\frac12)\omega](x)$, which is also $2D(D-\frac12)$ applied to $x^{-1/2}\omega(1/x)$, decays faster than any power at both ends,

$$\xi(s) = \int_0^\infty \big[2D(D-\tfrac12)\omega\big](x)\,x^{s/2-1}dx \quad \text{for every } s,$$
$$\Phi(u) = 2e^{u/2}\big[2D(D-\tfrac12)\omega\big](e^{2u}),$$

the first for $\operatorname{Re} s > 1$ by parts and then by analytic continuation, the second by $x = e^{2u}$. So the gamma shapes of the terms $e^{-\pi n^2 x}$ of ω, one in x and $\frac{1}{4}$ on the logarithmic scale, are raised by one and by two, to the Gamma(2) and Gamma(3) of the Pólya kernel, by the operator that removes the poles of ζ. For Δ, whose L function is entire, no such operator is needed and the kernel keeps the single shape 6; the cancellation between the Gamma(2) and Gamma(3) terms of Φ is replaced by the positive multiplier $F(e^u)$.

At real points the Gamma(2) shape enters through Kristiansen's theorem [194]: every scale mixture of Gamma(2) laws is infinitely divisible. Since $G_a \stackrel{d}{=} G_2 B$ for $0 < a < 2$, with B a beta variable of parameters a and $2 - a$ independent of G_2, the scale mixtures of Gamma(a) laws with $a \le 2$ are scale mixtures of Gamma(2) laws.

Proposition 15.1.7. *For real $a > 0$ let $V_a = G_a/(2\pi M^*)$, with G_a independent of M^*. Then V_a is a scale mixture of Gamma(a) laws, the mixing law being that of $1/(2\pi M^*)$, and $L(a - \frac{11}{2}, \Delta) = \Pr(V_a > 1)$. For $0 < a \le 2$, that is $-\frac{11}{2} < s \le -\frac{7}{2}$, V_a is infinitely divisible. In the same way $L(s, \chi_{12}) = \Pr\left(12G_{s/2}/(\pi M_1) > 1\right)$ for real $s > 0$, where $12G_{s/2}/(\pi M_1)$ is infinitely divisible for $s \le 4$, and is a mixture of exponential laws for $s \le 2$, in particular at the centre $s = \frac{1}{2}$.*

Proof. The identities are Corollary 15.1.3 and its analogue for M_1. For $a \le 2$, $V_a = G_2 \cdot B/(2\pi M^*)$ is a scale mixture of Gamma(2) laws, infinitely divisible by Kristiansen's theorem. For $s \le 2$, $G_{s/2} \stackrel{d}{=} G_1 B'$ with B' a beta variable of parameters $\frac{s}{2}$ and $1 - \frac{s}{2}$ (or $B' = 1$ at $s = 2$), so $12G_{s/2}/(\pi M_1)$ is a scale mixture of exponential laws, which is infinitely divisible by the theorem of Goldie and Steutel [94, 337]. □

The proposition says nothing about zeros: $\Pr(V_a > 1)$ is defined for real a only, since the shape $6 + i\theta$ on the critical line is not that of a probability law; for Δ the centre $a = 6$ lies outside Kristiansen's range, while for χ_{12} the centre shape $\frac{1}{4}$ lies inside it. The two Gamma(2) structures differ in kind: in the Pólya kernel it is a kernel shape forced by the poles of ζ, in Proposition 15.1.7 a mixing law.

Remark 15.1.8 (Gamma(k) scale mixtures, the Kolmogorov law and a Kristiansen law for ζ). Kristiansen's class also gives a reformulation of RH, but not a route to the GGC property [152]. Let $W(t) = \sum_{\operatorname{Re}\gamma > 0} e^{-t\gamma^2}$ be the canonical function of the clock T, whose Lévy density is $W(t)/t$. For every integer $k \ge 1$, RH holds if and only if $t^{k-1}W(t)$ is proportional to a scale mixture of Gamma(k) densities. Indeed RH is equivalent to the complete monotonicity of W (Section 2.1), that is to $W(t) = \int e^{-tz} U(dz)$ with $U \ge 0$ by Bernstein's theorem, and then $t^{k-1}W(t) = \int \Gamma(k)^{-1} z^k t^{k-1} e^{-tz}\, \Gamma(k) z^{-k} U(dz)$; conversely a Gamma($k$) scale mixture divided by t^{k-1} is completely monotone. Under RH the mixing law is discrete: the law with density proportional to $t^{k-1}W(t)$ is that of G_k/γ_J^2, with $\Pr(J = \gamma)$ proportional

to γ^{-2k}. For $k=1$ it is a mixture of exponential laws, infinitely divisible by the theorem of Goldie and Steutel [94, 337]; for $k=2$ it is the Kolmogorov law of T, with density $tW(t)/\operatorname{Var} T$, infinitely divisible by Kristiansen's theorem. Neither is a GGC, by the following fact, and Kristiansen's theorem can be applied only once the Gamma(2) mixture is known, which is RH itself.

A discrete mixture of Gamma(k) laws, with k a positive integer and at least two distinct rates $\lambda_1 < \lambda_2 < \cdots$, finitely many or tending to infinity, is not a GGC. Its Laplace transform $\varphi(z) = \sum_n w_n(1+z/\lambda_n)^{-k}$ is meromorphic on $\mathbb{C}$, with a pole of order k at each $-\lambda_n$. If it were the transform of a GGC, then $-\varphi'/\varphi = a + \int U(dt)/(z+t)$ with $a \ge 0$ and $U \ge 0$, a function holomorphic off $(-\infty, 0]$ whose poles have positive residues. So φ has no zero off the negative axis, and none on it, where $-\varphi'/\varphi$ would have the residue $-m$ at a zero of order m; by Stieltjes inversion $U = k\sum_n \delta_{\lambda_n}$, and $\varphi(z) = e^{-az}\prod_n(1+z/\lambda_n)^{-k} \le (1+z/\lambda_1)^{-k}(1+z/\lambda_2)^{-k}$ for $z>0$, which contradicts $\varphi(z) \ge w_1(1+z/\lambda_1)^{-k}$ as $z \to \infty$.

Kristiansen's class carries the completed zeta function itself, with no hypothesis. Let N have the zeta law $\Pr(N=n) = n^{-4}/\zeta(4)$, independent of G_2, and put $Q = G_2/(\pi N^2)$. Then Q has the density $\frac{\pi^2}{\zeta(4)}\, x\,\omega(x)$, with $\omega(x) = \sum_{n\ge1} e^{-\pi n^2 x}$, and $\mathbb{E}Q^q = \Gamma(q+2)\pi^{-q}\zeta(2q+4)/\zeta(4)$, so that

$$\pi^{-s/2}\Gamma\tfrac{s}{2}\zeta(s) = \frac{\pi^2}{90}\,\mathbb{E}\big[Q^{s/2-2}\big], \qquad \xi(s) = \frac{\pi^2}{180}\,s(s-1)\,\mathbb{E}\big[Q^{s/2-2}\big], \qquad \operatorname{Re} s > 1,$$

a normalisation of Riemann's integral $\int_0^\infty x^{s/2-1}\omega(x)\,dx$; beyond $\operatorname{Re} s > 1$ the identities hold after continuation, not as expectations. So Q is infinitely divisible by Kristiansen's theorem and, by the fact above with the rates πn^2, not a GGC. Williams' variable $\pi^2 S_2/2 = \sum_n \Gamma_n/n^2$ is a convolution of Gamma(2) laws of rates n^2 and carries the functional equation (Section 3.1.3); Q is a mixture of Gamma(2) laws of rates πn^2 and carries the Dirichlet series.

15.1.4.4 The law of M_1

Put $b_n = \pi(n^2-1)/12$; the pentagonal number theorem of Section 15.1.1 reads $F_1(y) = \sum_{n\ge1}\chi_{12}(n)e^{-b_n y}$, so the rates in this expansion are $b_{6k\pm1} = \pi k(3k\pm1)$, 2π times the generalised pentagonal numbers $k(3k\pm1)/2$.

Proposition 15.1.9. *For $\theta > 0$ let $w = \sqrt{1-12\theta/\pi}$, positive or purely imaginary. Then*

$$\mathbb{E}e^{-\theta M_1} = \frac{\pi(1-w^2)\sin(\pi w/3)}{2\sqrt{3}\,w\cos(\pi w/2)} = \prod_{k\ge1}\frac{1+\theta/b_{6k}}{(1+\theta/b_{6k-1})(1+\theta/b_{6k+1})}.$$

Proof. Since $\sum_n \theta/(\theta+b_n) < \infty$, $\theta e^{-\theta y}F_1(y)$ can be integrated term by term, and $\theta + b_n = \pi(n^2-w^2)/12$ gives $\mathbb{E}e^{-\theta M_1} = \sum_{n\ge1}\chi_{12}(n)\theta/(\theta+b_n) = (1-$

$w^2)\sum_{n\ge1}\chi_{12}(n)/(n^2-w^2)$. The function $g(w)=\pi\sin(\pi w/3)/(2\sqrt{3}\,w\cos(\pi w/2))$ is even and meromorphic, its only poles are simple and at the odd integers prime to 3, and its residue at $w=n$ is $-\chi_{12}(n)/(2n)$, as for the series, so their difference is entire. On the circles $|w|=2k$, $|\cos(\pi w/2)|$ is bounded below, so $g=O(1/k)$ there, and the series is $O(k^{-1}\log k)$; by the maximum modulus principle the difference vanishes. For the product, insert $\sin(\pi w/3)=\frac{\pi w}{3}\prod_{m\ge1}(1-\frac{w^2}{9m^2})$ and $\cos(\pi w/2)=\prod_{n\text{ odd}}(1-\frac{w^2}{n^2})$. The factors with $n=3m$, m odd, cancel, $1-w^2$ cancels the factor $n=1$, and what remains is a constant times the factors $1-w^2/n^2$ with $n=6k$ above and $n=6k\pm1$ below. The value 1 at $\theta=0$ fixes the constant, and $(1-w^2/n^2)/(1-1/n^2)=1+\theta/b_n$. □

At $\theta=\pi/12$, where $w=0$, the closed form gives $\mathbb{E}e^{-\pi M_1/12}=\pi^2/(6\sqrt{3})$, which is $L(2,\chi_{12})$ by Section 15.1.4.2.

Corollary 15.1.10 (A Rényi type identity). *$M_1\overset{d}{=}\sum_{k\ge1}Y_k$ with independent Y_k, where Y_k is exponential of rate b_{6k-1} with probability p_k and of rate b_{6k+1} with probability $1-p_k$, and*

$$p_k=\Big(1-\frac{b_{6k-1}}{b_{6k}}\Big)\frac{b_{6k+1}}{b_{6k+1}-b_{6k-1}}=\frac{(12k-1)(3k+1)}{2(36k^2-1)},$$

so $p_1=\frac{22}{35}$, $p_2=\frac{161}{286}$, and $p_k\downarrow\frac12$.

Proof. For $\alpha<\gamma<\beta$ the partial fractions $\frac{1+\theta/\gamma}{(1+\theta/\alpha)(1+\theta/\beta)}=p\frac{\alpha}{\alpha+\theta}+(1-p)\frac{\beta}{\beta+\theta}$ hold with $p=(1-\alpha/\gamma)\beta/(\beta-\alpha)$ and $1-p=\alpha(\beta-\gamma)/(\gamma(\beta-\alpha))$, both in $(0,1)$; take $\alpha=b_{6k-1}=\pi k(3k-1)$, $\gamma=b_{6k}=\pi(36k^2-1)/12$ and $\beta=b_{6k+1}=\pi k(3k+1)$ in each factor of Proposition 15.1.9, and use the continuity theorem for Laplace transforms. □

By Frullani's integral, as in the first reading of Section 4.1.1, the product gives

$$-\log\mathbb{E}e^{-\theta M_1}=\int_0^\infty\big(1-e^{-\theta t}\big)\frac{k_1(t)}{t}\,dt,\quad k_1(t)=\sum_{k\ge1}\big(e^{-b_{6k-1}t}+e^{-b_{6k+1}t}-e^{-b_{6k}t}\big),$$

so M_1 is infinitely divisible, with no drift and Lévy density $k_1(t)/t$; this also follows from Corollary 15.1.10, since mixtures of exponential laws are infinitely divisible [94, 337]. The Lévy density is completely monotone, since $(e^{-b_{6k-1}t}-e^{-b_{6k}t})/t=\int_{b_{6k-1}}^{b_{6k}}e^{-zt}dz$, so M_1 lies in the Bondesson class BO of Section 4.1.2. It is not a GGC: its Thorin measure would have to be $U_1=\sum_k(\delta_{b_{6k-1}}+\delta_{b_{6k+1}}-\delta_{b_{6k}})$, negative at the points b_{6k} where χ_{12} vanishes between $6k-1$ and $6k+1$. Indeed the transform vanishes at $\theta=-b_{6k}$, where $w=6k$, so $-(\log\mathbb{E}e^{-\theta M_1})'$ has residue -1 there, and if M_1 were a GGC the Stieltjes inversion (4.1.2) would give $U(\{b_{6k}\})=-1$. The poles of the transform are at $\theta=-b_n$, $n\equiv\pm1\pmod6$, $n\ge5$, and its zeros at $-b_{6k}$: each zero lies between its own two poles $-b_{6k\pm1}$, but zeros and poles

do not alternate, since $-b_7$ and $-b_{11}$, for instance, are consecutive poles. The next lemma reads Lemma 10.2.5 and the cosine dictionary (10.2.4) for a signed Thorin measure.

LEMMA 15.1.11. *Let $X \ge 0$ have Laplace exponent $-\log \mathbb{E}e^{-\theta X} = \int_{(0,\infty)} \log(1+\theta/z)\, U(dz)$ for a signed measure U with $\int \log(1+1/z)\, |U|(dz) < \infty$, and put $N(z) = U((0,z])$, with $N(z)\log(1+u^2/z^2) \to 0$ as $z \to \infty$ for real u. Then $-\log \mathbb{E}e^{-\theta X} = \int_0^\infty (1-e^{-\theta t})\nu(t)\, dt$ for $\theta \ge 0$, with the signed function $\nu(t) = t^{-1}\int e^{-zt} U(dz) = \int_0^\infty e^{-zt} N(z)\, dz$, and for real u*

$$\begin{aligned} -\log \left|\mathbb{E}e^{iuX}\right|^2 &= \int_0^\infty 4\sin^2\Big(\frac{ut}{2}\Big)\nu(t)\, dt \\ &= \int \log\Big(1+\frac{u^2}{z^2}\Big) U(dz) = \int_0^\infty N(z)\, \frac{2u^2}{z(z^2+u^2)}\, dz. \end{aligned} \tag{15.1.6}$$

The law of X is a GGC if and only if $U \ge 0$, and it lies in BO, that is, it is infinitely divisible with a completely monotone Lévy density, if and only if $N \ge 0$.

Proof. Frullani's integral $\log(1+\theta/z) = \int_0^\infty (1-e^{-\theta t})e^{-zt}t^{-1}dt$ gives the Lévy form, the exchange of integrals being justified by $\log(1+\theta/z) \le C_\theta \log(1+1/z)$, and integrating e^{-zt} against dN by parts gives the second expression of ν. The first equality of (15.1.6) is twice the real part of the Lévy form at $\theta = -iu$, since $1-\cos ut = 2\sin^2(ut/2)$; the second is the Frullani identity $\int_0^\infty 4\sin^2(ut/2)e^{-zt}t^{-1}dt = \log(1+u^2/z^2)$; the third is an integration by parts, with $\frac{d}{dz}\log(1+u^2/z^2) = -2u^2/(z(z^2+u^2))$, whose boundary terms vanish by the assumptions. The signed function ν is determined by the law of X, since $t\nu(t)$ is the inverse Laplace transform of the derivative of $-\log \mathbb{E}e^{-\theta X}$, and so is the signed measure U, by the Stieltjes inversion of $\int U(dz)/(\theta+z)$. If $U \ge 0$ the law is a GGC, and if it is a GGC, its Laplace exponent grows more slowly than θ, so it has no drift, and its Thorin measure is U. If $N \ge 0$, ν is completely monotone and the law is in BO; if the law is in BO, its Lévy density is $\int e^{-zt}\mu(dz)$ with $\mu \ge 0$, it equals ν, and the uniqueness of Laplace transforms gives $\mu(dz) = N(z)\, dz$, so $N \ge 0$, N being right continuous. □

For M_1 the Thorin measure is $U_1 = \sum_{n\ge 2} a(n)\delta_{b_n}$ with weights of sine squared form,

$$a(n) = \sin^2\frac{\pi n}{2} + \frac{4}{3}\sin^2\frac{\pi n}{3} - 1 = \mathbf{1}\{2 \nmid n\} - \mathbf{1}\{3 \mid n\} = \frac{1}{6} - \frac{1}{2}\cos \pi n - \frac{2}{3}\cos\frac{2\pi n}{3},$$

equal to 1 for $n \equiv \pm 1 \pmod 6$, to -1 for $6 \mid n$, and to 0 otherwise.

PROPOSITION 15.1.12. *The canonical function $k_1(t) = t\nu_1(t)$ of M_1 has the theta form*

$$k_1(t) = \tfrac{1}{2}e^{\pi t/12}\Big(1 + \frac{\vartheta_0 - 3\vartheta_{1/2} - 4\vartheta_{1/3}}{\sqrt{3t}}\Big) - 1, \qquad \vartheta_\alpha = \sum_{m\in\mathbb{Z}} e^{-12\pi(m+\alpha)^2/t},$$

so $\nu_1(t) = (2\sqrt{3})^{-1}t^{-3/2} - (2t)^{-1} + O(t^{-1/2})$ *as* $t \downarrow 0$ *and* $\nu_1(t) \sim e^{-2\pi t}/t$ *as* $t \to \infty$. *Its Bondesson function* $N_1(z) = U_1((0,z])$ *is nonnegative, and* $N_1(z) \sim \frac{1}{3}\sqrt{3z/\pi}$ *as* $z \to \infty$.

Proof. Since a is even, $a(0) = -1$ and $a(1) = 1$,

$$k_1(t) = \sum_{n\geq 2} a(n)e^{-b_n t} = \tfrac{1}{2}e^{\pi t/12}\Big(1 + \sum_{n\in\mathbb{Z}} a(n)e^{-\pi n^2 t/12}\Big) - 1.$$

Poisson summation, $\sum_{n\in\mathbb{Z}} \cos(2\pi n\alpha)e^{-\pi n^2 v} = v^{-1/2}\sum_{m\in\mathbb{Z}} e^{-\pi(m+\alpha)^2/v}$ with $v = t/12$, applied to the three terms of the cosine form of $a(n)$, gives the theta form. As $t \downarrow 0$, $\vartheta_0 = 1 + O(e^{-12\pi/t})$, while $\vartheta_{1/2}$ and $\vartheta_{1/3}$ are $O(e^{-4\pi/(3t)})$, so $k_1(t) = (2\sqrt{3t})^{-1} - \frac{1}{2} + O(\sqrt{t})$; as $t \to \infty$ the first term $e^{-b_5 t} = e^{-2\pi t}$ of the series dominates. Since $b_{6k-1} < b_{6k} < b_{6k+1}$, every negative atom of U_1 is preceded by its own positive atom, so $N_1 \geq 0$. Up to z there are about $X/3$ integers $n \equiv \pm 1 \pmod 6$ and $X/6$ multiples of 6 with $b_n \leq z$, where $X = \sqrt{12z/\pi + 1}$, so $N_1(z) = X/6 + O(1) \sim \frac{1}{3}\sqrt{3z/\pi}$. □

The Lévy density has the index $\frac{1}{2}$ of a Brownian first passage time, matching $-\log \mathbb{E}e^{-\theta M_1} \sim \sqrt{\pi\theta/3}$, which the lower tail $F_1(y) \sim y^{-1/2}e^{-\pi/(12y)}$ gives; the index comes from modularity. By (15.1.6),

$$\left|\mathbb{E}e^{iuM_1}\right|^{-2} = \prod_{k\geq 1} \frac{(1 + u^2/b_{6k-1}^2)(1 + u^2/b_{6k+1}^2)}{1 + u^2/b_{6k}^2} = \exp\int_0^\infty N_1(z)\,\frac{2u^2\,dz}{z(z^2+u^2)},$$

the exponential of a positive kernel integrated against a nonnegative function. Figure 15.2 (left) shows N_1 against its asymptote.

The sine squared form of $a(n)$ splits U_1 into two positive measures: $\sin^2(\pi n/2)$ selects the odd n, and $1 - \frac{4}{3}\sin^2(\pi n/3)$ the multiples of 3.

THEOREM 15.1.13. *With independent standard exponential variables* E_n *and* E'_m, *let* $S = \sum_{n\ odd,\, n\geq 3} E_n/b_n$ *and* $Z = \sum_{m\geq 1} E'_m/b_{3m}$, *with* Z *independent of* M_1. *Then* S *and* Z *are GGCs,*

$$\mathbb{E}e^{-\theta S} = \frac{\pi}{4}\,\frac{1 - w^2}{\cos(\pi w/2)}, \qquad \mathbb{E}e^{-\theta Z} = \frac{\sqrt{3}\,w/2}{\sin(\pi w/3)},$$

and $M_1 + Z \stackrel{d}{=} S$. *Equivalently* $U_1 = U_S - U_Z$, *with* $U_S = \sum_{n\ odd,\, n\geq 3} \delta_{b_n}$ *and* $U_Z = \sum_{m\geq 1} \delta_{b_{3m}}$, *the atoms at* b_{3m} *with* m *odd cancelling.*

Proof. With $1 + \theta/b_n = (n^2 - w^2)/(n^2 - 1)$, the product $\cos(\pi w/2) = \prod_{n\ \mathrm{odd}}(1 - w^2/n^2)$ gives $\prod_{n\ \mathrm{odd},\, n\geq 3}(1 + \theta/b_n) = \frac{4}{\pi}\cos(\pi w/2)/(1 - w^2)$, the constant $\frac{\pi}{4} = \prod_{n\ \mathrm{odd},\, n\geq 3}(1 - n^{-2})$ being the limit at $w = 1$; and $\sin(\pi w/3)/(\pi w/3) = \prod_m(1 - w^2/9m^2)$ gives $\prod_m(1 + \theta/b_{3m}) = \frac{2}{\sqrt{3}}\sin(\pi w/3)/w$. The quotient of the two transforms is the closed form of Proposition 15.1.9, so $\mathbb{E}e^{-\theta(M_1+Z)} = \mathbb{E}e^{-\theta S}$. □

The rates are shifted eigenvalues of Brownian hitting times [33]. With $\lambda_j = \pi^2(2j-1)^2/8$, the rates of the exit time C of Brownian motion from $(-1,1)$, whose transform is $1/\cosh\sqrt{2\theta} = \prod_j (1+\theta/\lambda_j)^{-1}$, $b_{2j-1} = \frac{2}{3\pi}(\lambda_j - \lambda_1)$, so $S \stackrel{d}{=} \frac{3\pi}{2}\sum_{j\ge 2} E_j/(\lambda_j - \lambda_1)$: the variable $C = \sum_j E_j/\lambda_j$ with its ground mode $j=1$ removed and the others shifted by λ_1. With $\mu_m = \pi^2 m^2/2$, the rates of the first passage T of the three dimensional Bessel process from 0 to 1, whose transform is $\sqrt{2\theta}/\sinh\sqrt{2\theta}$, $b_{3m} = \frac{3}{2\pi}(\mu_m - \frac{\pi^2}{18})$, so $Z \stackrel{d}{=} \frac{2\pi}{3} T^{(\pi^2/18)}$, where $T^{(c)}$ has the law of T tilted by e^{cT}, a proper law since $\frac{\pi^2}{18} < \mu_1$. By Kent's theorem with Krein's inverse spectral theory (Section 5.2.3), S and Z, GGCs without drift with unit Thorin atoms at distinct points, are hitting times of levels by generalised diffusions started at a reflecting or entrance end, so

$$M_1 + \big(\text{a tilted Bessel first passage time}\big) \stackrel{d}{=} \big(\text{the reduced Brownian exit time } S\big),$$

and M_1 itself is not a hitting time from such an end, since those are GGCs. Every first hitting time of a one dimensional diffusion lies in BO [185], as M_1 does, and from an interior point the transform is a quotient of the kind discussed in Section 14.1.4; whether M_1 is the hitting time of some level by some diffusion started at an interior point we do not know.

15.1.4.5 The law of M^*

The law of M^* has no such product, but its transform $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*} = \theta\int_0^\infty e^{-\theta y}F(y)\,dy$ is meromorphic: since $F(y) = \sum_{n\ge1}\tau(n)e^{-2\pi(n-1)y}$, for any $c>0$

$$\psi^*(\theta) = \theta\int_0^c e^{-\theta y}F(y)\,dy + \theta\sum_{n\ge1}\tau(n)\frac{e^{-(\theta+2\pi(n-1))c}}{\theta+2\pi(n-1)},$$

with at most simple poles at $\theta = -2\pi m$, $m\ge1$, of residue $-2\pi m\,\tau(m+1)$. Just to the right of a pole ψ^* has the sign of the residue and just to the left the opposite sign, so it has a real zero between $-2\pi(m+1)$ and $-2\pi m$ whenever $\tau(m+1)\tau(m+2)>0$. The first such pair is $\tau(6)=-6048$, $\tau(7)=-16744$, and a computation with $c=0.3$ at 40 digits finds the zero at $\theta = -34.48400851340497048 52 = -2\pi\times 5.48830041253$. It is the only sign change of ψ^* on $(-15\pi,0)$, and the argument principle on the rectangle $|\operatorname{Re}\theta|\le 15\pi$, $|\operatorname{Im}\theta|\le 60$, which contains the seven poles $-2\pi,\dots,-14\pi$, gives winding number -6, so it is the only zero there. As for M_1, the zero shows that M^* is not a GGC.

The zeros are certified in ball arithmetic, with $c=\frac{3}{100}$, the series to $n=1500$, its tail bounded by Deligne's bound $|\tau(n)|\le d(n)n^{11/2}\le 2n^6$, and the integral bounded by $F(y)\le y^{-12}e^{2\pi y-2\pi/y}$, which is increasing.

PROPOSITION 15.1.14. *1. For each of the 25 values $m\le 60$ with $\tau(m+1)\tau(m+2)>0$, ψ^* changes sign in an interval of length 2×10^{-20} inside the gap*

$(-2\pi(m+1), -2\pi m)$; *the zeros begin* $\theta/2\pi = -5.48830041$, -8.66373325, -11.4203774, -13.2533839, -14.7973058, *and the last is* -60.7670708.

2. ψ^* *has exactly one zero in the disc of radius* 10^{-7} *about each of the ten points*

$$\begin{aligned}
&-155.5940822775511402804 \pm 2.7631188861212609656\,i,\\
&-215.6211277570694988944 \pm 2.9647292148515237348\,i,\\
&-307.4008549057072052743 \pm 3.2283287024583906394\,i,\\
&-322.6135369899940182007 \pm 3.0449254921259967464\,i,\\
&-347.8816009468099010890 \pm 2.7171007701068929577\,i,
\end{aligned}$$

and it lies within 10^{-19} *of that point.*

Proof. For (1), the values of ψ^* at the two ends of each interval are balls of opposite signs. For (2), let z_0 be the approximation to the zero computed by Newton's method at 230 bits; the point shown is z_0 rounded to nineteen decimals, at distance at most 6.1×10^{-20} from it. With $A \geq |\psi^*(z_0)|$, $B \leq |\psi^{*\prime}(z_0)|$ and $C \geq |\psi^{*\prime\prime}|$ on the square of half side r about z_0, each computed as a ball, $A + 2Cr^2 < Br/2$ holds for $r = 10^{-6}$ and for $r = 10^{-29}$. On the circle $|\theta - z_0| = r$, inside the square, $|\psi^*(\theta) - \psi^{*\prime}(z_0)(\theta - z_0)| \leq A + Cr^2/2 < Br$, so by Rouché's theorem ψ^* has as many zeros in the disc $|\theta - z_0| < r$ as $\psi^{*\prime}(z_0)(\theta - z_0)$, namely one. The disc of radius 10^{-7} about the point shown lies in the disc of radius 10^{-6} about z_0, and the disc of radius 10^{-29} about z_0 lies within 10^{-19} of the point shown. The zeros come in conjugate pairs because ψ^* is real on the real axis. □

The argument principle on the rectangles $|\operatorname{Re}\theta| \leq 2\pi(k+\frac{1}{2})$, with $|\operatorname{Im}\theta| \leq 30$ for $k = 15$, 25, 40 and 60 and $|\operatorname{Im}\theta| \leq 100$ for $k = 60$, and on a grid of cells, 2π wide and at most 5 high, covering $-2\pi \times 60.5 \leq \operatorname{Re}\theta \leq 2\pi \times 5.5$, $\frac{1}{2} \leq \operatorname{Im}\theta \leq 30$, finds exactly these zeros: the winding number on the largest rectangle, which contains the 60 poles $-2\pi, \ldots, -120\pi$, is -26, so it contains 34 zeros, the 24 real zeros with $m \leq 59$ and the ten nonreal zeros of the proposition. This count, unlike the zeros themselves, is computed and not certified: the change of argument is followed along the contours with steps halved until it is below 0.3, in ball arithmetic whose radii stay below a relative 10^{-46}. Every real zero lies in a gap where the residues force it, every gap with $\tau(m+1)\tau(m+2) < 0$ is free of real zeros, and the five nonreal pairs have their real parts in such gaps, those with $m = 24$, 34, 48, 51 and 55 ($\operatorname{Re}\theta/2\pi = -24.764$, -34.317, -48.924, -51.346, -55.367), at heights between 2.72 and 3.23 (Figure 15.2, right). In a gap where the residues have opposite signs ψ^* has the same sign at both ends, so its real zeros there come in pairs; the computation finds none, and the five nonreal pairs sit in such gaps, as if a pair of real zeros had collided and left the axis.

Corollary 15.1.15. M^* *is not in the Bondesson class BO; in particular it is not a GGC.*

Proof. If M^* were in BO, its Lévy density would be $\int e^{-zt}\mu(dz)$ with $\mu \geq 0$, so $-(\log\psi^*)'(\theta) = d + \int \mu(dz)(\theta+z)^{-2}$, with $d \geq 0$ the drift, for $\theta > 0$. The right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, since $\int \mu(dz)(1+z)^{-2} < \infty$ and $|\theta + z|$ is bounded below by a multiple of $1+z$ for θ in a compact subset of $\mathbb{C} \setminus (-\infty, 0]$. Then the meromorphic function $\psi^{*\prime}/\psi^*$ would have no poles in $\mathbb{C} \setminus (-\infty, 0]$, and ψ^* no zeros there, against Proposition 15.1.14(2). □

Nor does M^* have a signed Thorin measure in the sense of Lemma 15.1.11: for a signed measure U on $(0,\infty)$, $-(\log\psi^*)'(\theta) = \int U(dz)(\theta+z)^{-1}$ would again be holomorphic on $\mathbb{C}\setminus(-\infty,0]$. Unit atoms at the points $2\pi m$ with $\tau(m+1) \neq 0$, less unit atoms at the real zeros, account for the poles and the real zeros of ψ^*, but no measure on the positive axis accounts for the nonreal zeros. Whether M^* is infinitely divisible remains open. If it is, with Lévy measure ν^*, then $-(\log\psi^*)'(\theta) = d + \int t\, e^{-\theta t}\nu^*(dt)$ is completely monotone on $(0,\infty)$: the cumulants $\kappa_j(\theta) = (-1)^j(\log\psi^*)^{(j)}(\theta)$ of M^* tilted by $e^{-\theta M^*}$ are all positive. At $\theta = 0.5$, 2, 8 and 30 the first eight are positive [286]. As $\theta \to \infty$ the Laplace method, at the saddle point $y_0 = \sqrt{2\pi/\theta}$ of $e^{-\theta y - 2\pi/y}$ and with the lower tail $F(y) \sim y^{-12}e^{-2\pi/y}$, gives

$$\psi^*(\theta) \sim 2^{-1/2}(2\pi)^{-21/4}\,\theta^{25/4}\,e^{-2\sqrt{2\pi\theta}},$$

so $-\log\psi^*(\theta) = 2\sqrt{2\pi\theta} - \frac{25}{4}\log\theta + \frac{21}{4}\log 2\pi + \frac{1}{2}\log 2 + o(1)$. If M^* is infinitely divisible, it has no drift, and Karamata's Tauberian theorem turns the leading term into $\nu^*((t,\infty)) \sim 2\sqrt{2}\,t^{-1/2}$ as $t \downarrow 0$.

The first author conjectures that every maximum of independent exponential variables, such as M_1, M_4, M^* and M_{11}, is infinitely divisible [286]. For finitely many rates $\lambda_1, \ldots, \lambda_n$ the transform is rational: with $\lambda_S = \sum_{i\in S}\lambda_i$,

$$\mathbb{E}e^{-\theta M} = \theta\int_0^\infty e^{-\theta y}\prod_{i\leq n}(1-e^{-\lambda_i y})\,dy = \sum_{S\neq\emptyset}(-1)^{|S|+1}\frac{\lambda_S}{\theta+\lambda_S},$$

with simple poles at the values $-\lambda_S$ whose grouped coefficients do not vanish, generically $2^n - 1$ of them, and, since $\mathbb{E}e^{-\theta M} \sim n!\,\lambda_1\cdots\lambda_n\theta^{-n}$, with n fewer zeros. Writing the poles as $-p$ and the zeros as $-z$, Frullani's integral gives $-\log\mathbb{E}e^{-\theta M} = \int_0^\infty(1-e^{-\theta t})k(t)t^{-1}dt$ with $k(t) = \sum_p e^{-pt} - \sum_z e^{-zt}$, as long as every zero has $\operatorname{Re} z > 0$, and by the uniqueness of the Lévy measure the law is infinitely divisible if and only if $k \geq 0$. In particular infinite divisibility forces every zero to have $\operatorname{Re} z$ at least the smallest pole $p_{\min}$, since otherwise k would change sign at large t. For two rates $a < b$ the transform is $ab(a+b+2\theta)/((a+\theta)(b+\theta)(a+b+\theta))$ and the canonical function is $e^{-at} + e^{-bt} + e^{-(a+b)t} - e^{-(a+b)t/2}$, positive by the inequality of arithmetic and geometric means; the signed Thorin measure $\delta_a + \delta_b + \delta_{a+b} - \delta_{(a+b)/2}$ has a nonnegative distribution function, so this maximum is in BO and is not a GGC, and $e^{at}k(t) = 1 - x + x^2 + x^{2b/(b-a)} > \frac{3}{4}$ with $x = e^{-(b-a)t/2}$, a bound approached as $b/a \downarrow 1$. Numerically, k is evaluated at 181

points of $p_{\min}t \in [10^{-3}, 40]$ for 300 families of random rational rates in $[0.1, 5]$ for each $n = 2, 3, 4, 5$ and 40 families with $n = 6$. In every family k is positive at every point, $e^{p_{\min}t}k(t) \geq 0.53$, and every zero has $\operatorname{Re} z > p_{\min}$. Nonreal zeros occur in 140, 222, 266 and 40 of the families with 3, 4, 5 and 6 rates, up to 2, 8, 22 and 52 of them; so nonreal zeros, which exclude BO, do not seem to exclude infinite divisibility. The truncations of M_1, with the rates $2\pi n$, $n \leq N$, have only real zeros for $N = 4, \dots, 10$, and those of M^*, with each rate $2\pi n$, $n \leq N$, taken 24 times, have 24 real zeros for $N = 2$ and 72 zeros, 52 of them nonreal, for $N = 3$; in each case k is positive on the grid. None of this is a proof.

Question 15.1.16. Is every maximum of independent exponential variables infinitely divisible? In particular, is M^*?

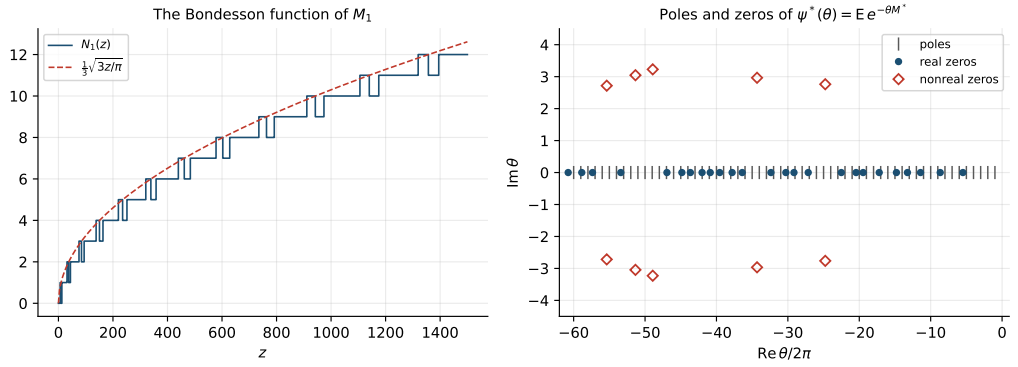


Figure 15.2: Left: the Bondesson function N_1 of M_1, nonnegative, with its asymptote $\frac{1}{3}\sqrt{3z/\pi}$; each drop is a zero of the transform at $-b_{6k}$. Right: the poles $-2\pi m$, $m \leq 60$, and the zeros of $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*}$: the 25 real zeros above $-2\pi \times 61$, one in each gap with $\tau(m+1)\tau(m+2) > 0$, and the five pairs of nonreal zeros, the only ones with $|\operatorname{Re}\theta| \leq 2\pi \times 60.5$ and $|\operatorname{Im}\theta| \leq 100$.

For the clock, relaxing GGC to BO gains nothing. If its Lévy density $W(t)/t$ were completely monotone, equal to $\int e^{-zt}\mu(dz)$ with $\mu \geq 0$, then the formula $(G'/G)(u) = \int_0^\infty e^{-ut}W(t)\,dt$ of Section 2.1 would give $(G'/G)(u) = \int \mu(dz)(u+z)^{-2}$, holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. But G'/G has a pole at every zero $-a_k = (\rho_k - \frac{1}{2})^2$ of G, and this point lies in $(-\infty, 0]$ only when ρ_k is on the critical line. So the clock is in BO if and only if RH holds (Table 4.1). The signed Thorin measures of the max laws come from zeros of their Laplace transforms, and the transform $1/G$ of the clock has none. The two laws show the two ways out of the Thorin class: M_1 leaves it through real zeros, each preceded by its own pole, and stays in BO, while M^* leaves BO too, through zeros off the real axis, the way a zero of ξ off the critical line would take the clock out of both classes.

15.1.5 The de Bruijn Newman flow for Δ

The flow of Section 4.1 needs only an even kernel that decays faster than any Gaussian, so it applies to K_Δ:

$$H_t^\Delta(\theta) = \int_0^\infty e^{tu^2} K_\Delta(u)\cos(\theta u)\,du, \qquad 2H_0^\Delta(\theta) = \xi\big(\tfrac12 + i\theta, \Delta\big).$$

In the variable θ the times of Section 4.1, whose variable is $z = 2\theta$, are divided by four, and the isolated pair estimate for two zeros at distance D is $-D^2/8$. The zeros of H_0^Δ lie in the strip $|\operatorname{Im}\theta| < \frac12$, so by de Bruijn's theorem [73] every zero of H_t^Δ is real for $t \ge \frac18$. Dobner's theorem for the extended Selberg class [77] gives the other side: Δ has a de Bruijn Newman constant Λ_Δ, and H_t^Δ has nonreal zeros for every $t < 0$. So $0 \le \Lambda_\Delta \le \frac18$, and GRH for Δ is the statement $\Lambda_\Delta = 0$.

As for the eta kernel in Section 15.1.1, if V has the density $K_\Delta(v)/\xi(\frac12,\Delta)$ on $\mathbb{R}$, then $\mathbb{E}e^{i\theta V} = \xi(\frac12 + i\theta,\Delta)/\xi(\frac12,\Delta)$, and GRH for Δ is the statement that this characteristic function has only real zeros. In the language of Section 15.1.4, e^V has the density proportional to $y^5\Delta(iy) = y^5 e^{-2\pi y}F(y)$, so V is the logarithm of $G_6/2\pi$ conditioned on $G_6/2\pi \ge M^*$, with G_6 a gamma variable of shape 6 independent of M^*; its law is symmetric by the self duality (15.1.1), with no symmetrisation. The variable V is not even quasi infinitely divisible, because its characteristic function vanishes at $\theta = 9.2223793999$ (the argument of Proposition 14.1.3(1)). Since $1 - \cos x = 2\sin^2(x/2)$ and K_Δ is even,

$$\xi\big(\tfrac12,\Delta\big) - \xi\big(\tfrac12 + i\theta,\Delta\big) = 4\int_0^\infty K_\Delta(v)\sin^2\Big(\frac{\theta v}{2}\Big)dv,$$

so the zeros on the critical line are the solutions of $\mathbb{E}[2\sin^2(\theta V/2)] = 1$. At the level of logarithms, the factorisation $\xi(s,\Delta) = (2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ splits $\log|\xi(\frac12+i\theta,\Delta)/\xi(\frac12,\Delta)|^2$ into $\log|\Gamma(6+i\theta)/\Gamma(6)|^2$ and $\log|L(\frac12+i\theta,\Delta)/L(\frac12,\Delta)|^2$. The first has the positive sine squared form given by Binet's formula,

$$\log\Big|\frac{\Gamma(6+i\theta)}{\Gamma(6)}\Big|^2 = -\int_0^\infty 4\sin^2\Big(\frac{\theta u}{2}\Big)\frac{e^{-6u}}{u(1-e^{-u})}\,du,$$

the Lévy Khintchine form of the infinitely divisible variable $\log G_6$, and the second carries every zero. To the right of the critical strip the second has a sine squared form too: for $\sigma > 1$, expanding $\log L(s,f) = \sum_{n\ge2}\Lambda_f(n)n^{-s}/\log n$ gives

$$\log\Big|\frac{L(\sigma+i\theta,f)}{L(\sigma,f)}\Big|^2 = -\sum_{n\ge2}\frac{4\Lambda_f(n)}{n^\sigma\log n}\sin^2\Big(\frac{\theta\log n}{2}\Big),$$

for $f = \chi_{12}$ and Δ, and for ζ with Λ in place of Λ_f. Since $\int_\sigma^\infty n^{-\alpha}d\alpha = n^{-\sigma}/\log n$, the right side is minus twice the integral of the prime piece of the decomposition of Section 15.1.3 over the basepoints $\alpha > \sigma$. For ζ the weights are positive, which

is the infinite divisibility of log of the zeta distribution (Section 10.1.6). For Δ they are $8\cos(k\vartheta_p)/(kp^{k\sigma})$ at $n = p^k$, where $\alpha_p = e^{i\vartheta_p}$, $0 \le \vartheta_p \le \pi$, and they take both signs. The Sato Tate law for Δ, proved by Barnet-Lamb, Geraghty, Harris and Taylor [23], makes the angles ϑ_p equidistributed for the measure $\frac{2}{\pi}\sin^2\vartheta\, d\vartheta$ on $[0,\pi]$, under which $\cos k\vartheta < 0$ with probability $\frac12$ for $k \ne 2$ and $\frac12 + \frac{1}{\pi}$ for $k = 2$. So even where the Euler product converges absolutely the sine squared form of $\log|L(\sigma+i\theta,\Delta)|$ has a signed measure, and on the critical line the series diverges: positivity lives in the kernel and the gamma factor, not in the arithmetic factor.

The kernel factors as $K_\Delta(u) = e^{-2\pi\cosh u}H(u)$, with $H(u) = \sqrt{F(e^u)F(e^{-u})}$ because $K_\Delta(u)^2 = K_\Delta(u)K_\Delta(-u)$, and $\log H(u) = -12\sum_m \frac{\sigma(m)}{m}(e^{-2\pi m e^u} + e^{-2\pi m e^{-u}})$. The expansion comes from $-\log F(y) = 24\sum_{m\ge1}\frac{\sigma(m)}{m}e^{-2\pi m y}$, so $-\log F$ is completely monotone, the Laplace transform of the positive measure with mass $24\sigma(m)/m$ at each point $2\pi m$; for η the masses are $\sigma(m)/m$, and every maximum of independent exponential variables has this property, since $-\log(1-e^{-\lambda y}) = \sum_k e^{-k\lambda y}/k$. The transform $K_{i\theta}(2\pi)$ of Pólya's factor has only real zeros [302], the first at $\theta = 9.768770083509977867$, and the multiplier H moves it to 9.2223793999. Pólya's theorem on universal factors [73, p. 197], which covers the Gaussian multiplier e^{tu^2} of the flow for $t > 0$, does not apply to H, which is bounded by one and supplies half of the doubly exponential decay of K_Δ. The positivity of the measure of $-\log F$ is automatic and does not by itself locate the zeros; any answer to the following question must account for Δ, χ_{12} and the conductor 11 curve, whose kernels have this form.

Question 15.1.17. For which $a > 0$, $c_m \ge 0$ and $\beta_m > 0$ does the Fourier transform of $e^{-a\cosh u}\exp\big(-\sum_m c_m(e^{-\beta_m e^u} + e^{-\beta_m e^{-u}})\big)$ have only real zeros?

The kernel K_Δ is strictly log concave [290]. For $c > 0$, $\log(1-e^{-ce^u})$ is the logarithm of the distribution function of $\log E$, with E exponential of rate c, and it is strictly concave in u: with $x = ce^u$ its second derivative is $x(e^x - 1 - xe^x)/(e^x-1)^2 < 0$. So $\log F(e^u)$, a sum of such terms, is strictly concave (in the language of Section 15.1.4, $\log M^*$ has a log concave distribution function), and since $\log K_\Delta(u) = 6u - 2\pi e^u + \log F(e^u)$ and K_Δ is even,

$$-\big(\log K_\Delta\big)''(u) > 2\pi e^{|u|}, \qquad u \in \mathbb{R},$$

where the ratio of the two sides tends to 1 as $|u| \to \infty$. The multiplier H is log concave as well, since $\log H(u) = \frac12\big(\log F(e^u) + \log F(e^{-u})\big)$. In the same way $\log K(u) = \log 2 + \frac{u}{2} - \frac{\pi}{12}e^{2u} + \log F_1(e^{2u})$ gives $-(\log K)''(u) > \frac{\pi}{3}e^{2|u|}$ for the kernel of Section 15.1.1. The Pólya kernel Φ is log concave too [59], though not a product of this kind. Log concavity alone does not put the zeros of a transform on a line, or in any strip. The function $(1-|u|)_+^2$ is log concave, and its transform $4(\theta-\sin\theta)/\theta^3$ vanishes at the nonzero roots of $\sin\theta = \theta$, which are nonreal, the first at $7.4977 + 2.7687i$, with imaginary parts that grow like $\log|\theta|$. Its convolution with $e^{-\cosh u}$ is even, positive, entire and log concave [326], and decays doubly

exponentially, like K_Δ; its transform $8(\theta-\sin\theta)K_{i\theta}(1)/\theta^3$ has the same nonreal zeros.

In the language of Section 15.1.4 the flow is a family of Gaussian tilts: for every real t, $e^{tv^2}K_\Delta(v)$ is integrable, because K_Δ decays doubly exponentially, so after normalisation it is the density of the law of V tilted by e^{tV^2}, and $H_t^\Delta(\theta)/H_t^\Delta(0)$ is its characteristic function. The question $\Lambda_\Delta=0$ asks whether the smallest tilt whose characteristic function has only real zeros is $t=0$.

With Gauss Legendre quadrature at 40 digits the 21 zeros below height 52, from 9.2224 to 51.1566, stay real for $t=0.05,0.1,\dots,0.5$ and all move down. Backward in time each pair of neighbours is followed through the critical point of H_t^Δ between them until the critical value changes sign, and each event is tested by sign counts: a collision at (t_*,θ_*) leaves two sign changes on $[\theta_*-1.2,\theta_*+1.2]$ at $t_*+0.005$ and none at $t_*-0.005$, and the colliding pair is identified by the number of sign changes below $\theta_*-1.2$, since real zeros keep their order. Table 15.2 lists the seven collisions. The numbers of sign changes on $(0.5,52]$ at eleven values of t from 0 to -1.5 fall by two at each collision and once by one, between $t=-0.2$ and -0.3, where the highest zero, followed back from 51.1566, passes 52 at $t\approx-0.29$.

pair at $t=0$	$-D^2/8$	t_*	θ_*
48.3591, 49.2761	−0.1051	−0.12622	49.18
36.7415, 37.7539	−0.1281	−0.15864	37.60
45.0401, 46.1973	−0.1674	−0.24305	46.19
25.2746, 26.8044	−0.2925	−0.40713	27.26
40.2190, 41.7305	−0.2856	−0.48271	42.49
31.1782, 32.7749	−0.3187	−0.48637	33.30
17.4428, 19.6565	−0.6126	−0.98116	21.09

Table 15.2: The collisions of zeros of H_t^Δ below height 52, in the order in which they occur as t decreases: the pair at $t=0$, the isolated pair estimate with D the gap of the pair, and the time t_* and height θ_* of the collision, computed.

The table is recomputed independently, at 70 digits with a different tracking, and finds the same seven collisions to all digits of the table, with the highest zero passing 52 between $t=-0.29$ and -0.30. Figure 15.3 shows the trajectories.

The low zeros of Δ are robust: below height 33 the first pair to leave the line, $(25.27,26.80)$, does so only at $t=-0.407$. Collisions do not come in order of height: the pairs near 37.6 and 49.2 collide at -0.159 and -0.126, and under GRH for Δ Dobner's theorem forces collisions in every interval $(t,0)$, higher up. So no window of heights decides whether $\Lambda_\Delta=0$; a computation can show how fast the collision times approach 0 with height. The neighbours delay every collision beyond the isolated pair estimate, by factors from 1.20 to 1.69. In the variable θ the closest pair of ζ below height 200 collides at -0.072, and Lehmer's pair near height 7005 at -1.8×10^{-4} (Section 4.1.2).

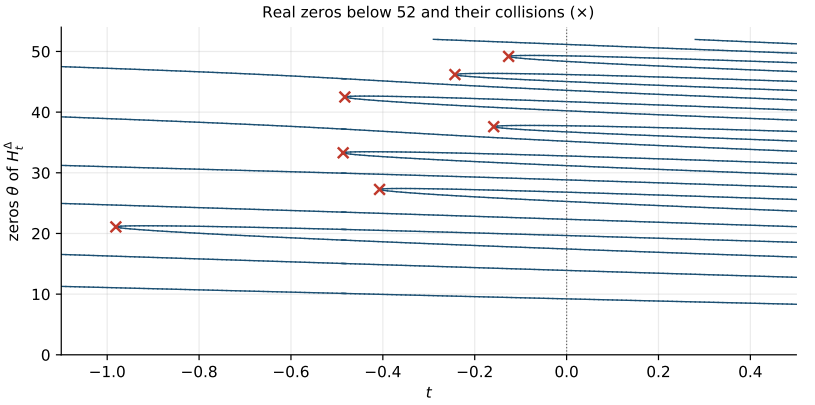


Figure 15.3: The real zeros of H_t^Δ below height 52 for $-1.1 \le t \le 0.5$, and the seven collisions of Table 15.2; all zeros move down as t grows.

15.1.6 A Pólya kernel for imaginary quadratic fields

This subsection is based on [289]. Let $d < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{d})$, χ_d the Kronecker symbol of d, a real primitive odd character modulo $|d|$, h the class number of K and w the number of its roots of unity. Then $\zeta_K(s) = \zeta(s)L(s,\chi_d) = \sum_{n\ge1} a_n n^{-s}$, where $a_n = \sum_{m\mid n}\chi_d(m) \ge 0$ is the number of ideals of norm n, and we put

$$\Lambda_K(s) = \Big(\frac{\sqrt{|d|}}{2\pi}\Big)^s \Gamma(s)\,\zeta_K(s), \qquad \xi_K(s) = 2s(s-1)\Lambda_K(s),$$

the function ξ_K of Section 13.3.3 with $r_1 = 0$ and $r_2 = 1$. The duplication formula gives $\xi_K(s) = \frac{2}{\sqrt{|d|}}\,\xi(s)\,\xi(s,\chi_d)$, with $\xi(s,\chi_d) = (|d|/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_d)$ the completed function of Section 15.1.1, so $\xi_K(s) = \xi_K(1-s)$ because the root number of χ_d is one. Hence $\Lambda_K(s) = \Lambda_K(1-s)$, and Λ_K is meromorphic with simple poles at $s = 1$ and $s = 0$ only, with residues h/w and $-h/w$, by the class number formula $L(1,\chi_d) = 2\pi h/(w\sqrt{|d|})$ and the symmetry. Let

$$\vartheta_K(x) = \sum_{n\ge1} a_n\, e^{-2\pi n x/\sqrt{|d|}}, \qquad \Psi(y) = y(y-2)\,e^{-y}, \qquad H_K(x) = \big(x^2\vartheta_K'(x)\big)',$$

and $\Phi_K(u) = 2e^{u/2}H_K(e^u)$.

Theorem 15.1.18. (i) $\vartheta_K(1/x) = x\,\vartheta_K(x) + \frac{h}{w}\,(x-1)$ *for* $x > 0$.

(ii) $H_K(x) = \sum_{n\ge1} a_n \Psi(2\pi n x/\sqrt{|d|})$ *and* $H_K(1/x) = x\,H_K(x)$. *The kernel* Φ_K *is even, analytic in the strip* $|\operatorname{Im} u| < \frac{\pi}{2}$, *and decays doubly exponentially.*

(iii) $\xi_K(\frac12 + s) = \int_{\mathbb{R}} \Phi_K(u)\,e^{su}\,du$ *for every* $s \in \mathbb{C}$; *in particular* $\xi_K(\frac12 + i\theta) = \int_{\mathbb{R}} \Phi_K(u)\cos(\theta u)\,du$.

(iv) $\Phi_K > 0$ *on* $\mathbb{R}$ *if and only if* $H_K > 0$ *on* $[1, \sqrt{|d|}/\pi]$. *In particular* $\Phi_K > 0$ *when* $|d| < \pi^2$, *that is for* $d = -3, -4, -7, -8$.

(v) *If* $\Phi_K > 0$, *then* $\zeta_K(\frac{1}{2}) < 0$ *and* $L(\frac{1}{2}, \chi_d) > 0$. *In particular* $L(\frac{1}{2}, \chi_d) > 0$ *for* $d = -3, -4, -7, -8$.

Proof. (i) For $c > 1$, Mellin inversion of $\int_0^\infty e^{-2\pi n x/\sqrt{|d|}} x^{s-1} dx = \Gamma(s)\big(\sqrt{|d|}/2\pi n\big)^s$ gives $\vartheta_K(x) = \frac{1}{2\pi i}\int_{(c)} \Lambda_K(s) x^{-s} ds$. Move the line of integration to $\operatorname{Re} s = 1 - c$, which the exponential decay of Γ and the polynomial growth of ζ_K on vertical strips allow; the residues at $s = 1$ and $s = 0$ are $\frac{h}{w}x^{-1}$ and $-\frac{h}{w}$. On the new line substitute $s \mapsto 1 - s$ and use $\Lambda_K(1 - s) = \Lambda_K(s)$: the integral becomes $x^{-1}\vartheta_K(1/x)$. So $\vartheta_K(x) = \frac{h}{w}(x^{-1} - 1) + x^{-1}\vartheta_K(1/x)$, which is (i).

(ii) Termwise, $\big(x^2 \frac{d}{dx} e^{-\kappa x}\big)' = (\kappa^2 x^2 - 2\kappa x) e^{-\kappa x} = \Psi(\kappa x)$. Differentiating (i) gives $\vartheta_K'(1/x) = -x^2\big(\vartheta_K(x) + x\vartheta_K'(x) + \frac{h}{w}\big)$, so $g(x) = x^2\vartheta_K'(x)$ satisfies $g(1/x) = -\big(\vartheta_K(x) + x\vartheta_K'(x) + \frac{h}{w}\big)$, and differentiating again gives $g'(1/x) = x^2\big(2\vartheta_K'(x) + x\vartheta_K''(x)\big) = x\, g'(x)$, that is $H_K(1/x) = x H_K(x)$. Then $\Phi_K(-u) = 2e^{-u/2} H_K(e^{-u}) = 2e^{u/2} H_K(e^u) = \Phi_K(u)$. Since $a_n \le n$, the series for $H_K(e^u)$ converges locally uniformly where $\operatorname{Re} e^u > 0$, that is for $|\operatorname{Im} u| < \frac{\pi}{2}$; and $\Phi_K(u) = O\big(e^{5u/2}\exp(-2\pi e^u/\sqrt{|d|})\big)$ as $u \to \infty$, and likewise as $u \to -\infty$ by evenness.

(iii) By (i), $x\vartheta_K(x) \to \frac{h}{w}$ and $x^2\vartheta_K'(x) \to -\frac{h}{w}$ as $x \to 0$, while ϑ_K and ϑ_K' decay exponentially at infinity. For $\operatorname{Re} z > 1$ two integrations by parts, whose boundary terms vanish, give $\int_0^\infty H_K(x) x^{z-1} dx = z(z-1)\int_0^\infty \vartheta_K(x) x^{z-1} dx = z(z-1)\Lambda_K(z)$. With $x = e^u$ and $z = \frac{1}{2} + s$, $\int_{\mathbb{R}} \Phi_K(u) e^{su} du = 2\int_0^\infty H_K(x) x^{z-1} dx = \xi_K(z)$. Both sides are entire in s, the left by (ii), so the identity holds for every s.

(iv) By (ii), $\Phi_K > 0$ exactly when $H_K > 0$ on $(0, \infty)$, and by the symmetry exactly when $H_K > 0$ on $[1, \infty)$. For $x \ge \sqrt{|d|}/\pi$ every argument $2\pi n x/\sqrt{|d|}$ is at least $2n \ge 2$, where $\Psi \ge 0$, and the term $n = 4$ is positive, since $a_4 = 1 + \chi_d(2) + \chi_d(2)^2 \ge 1$; so $H_K > 0$ there. If $|d| < \pi^2$ the interval $[1, \sqrt{|d|}/\pi]$ is empty.

(v) By (iii), $\xi_K(\frac{1}{2}) = \int_{\mathbb{R}} \Phi_K > 0$, while

$$\xi_K\big(\tfrac{1}{2}\big) = -\tfrac{1}{2}\Lambda_K\big(\tfrac{1}{2}\big) = -\tfrac{1}{2}\big(\sqrt{|d|}/2\pi\big)^{1/2}\sqrt{\pi}\,\zeta_K\big(\tfrac{1}{2}\big).$$

So $\zeta_K(\frac{1}{2}) < 0$, and $L(\frac{1}{2}, \chi_d) = \zeta_K(\frac{1}{2})/\zeta(\frac{1}{2}) > 0$ because $\zeta(\frac{1}{2}) < 0$. □

When $\Phi_K > 0$, the normalised kernel $\Phi_K/\xi_K(\frac{1}{2})$ is the density of a symmetric variable X_K with $\mathbb{E}e^{sX_K} = \xi_K(\frac{1}{2} + s)/\xi_K(\frac{1}{2})$, the Pólya variable of K, and its tilts $V_{K,\alpha}$, with $\mathbb{E}e^{sV_{K,\alpha}} = \xi_K(\alpha + s)/\xi_K(\alpha)$, have the means $\frac{\xi_K'}{\xi_K}(\alpha)$. The zero part of the constant C_K of Theorem 13.3.7 is then $\mathbb{E}V_{K,1} - 2\,\mathbb{E}V_{K,2}$, as over $\mathbb{Q}$ in Theorem 13.3.5; by (iv) this applies to the Gaussian field $d = -4$ of Section 13.3.3. If moreover the Riemann hypothesis holds for ζ_K, then $1/\mathbb{E}\cosh(\sqrt{u}\,X_K) = \xi_K(\frac{1}{2})/\xi_K(\frac{1}{2} + \sqrt{u})$ is the Laplace transform of a GGC, the clock of K (Proposition 13.3.6).

The kernel Φ_K is tied to the kernel of χ_d in Section 15.1.1,

$$K_{\chi_d}(u) = 2e^{3u/2} \sum_{n\ge1} n\chi_d(n)\, e^{-\pi n^2 e^{2u}/|d|}, \qquad \xi\big(\tfrac12 + s, \chi_d\big) = \int_{\mathbb{R}} K_{\chi_d}(u)\, e^{su}\, du.$$

The two sided Laplace transform turns convolution into multiplication, so $\xi_K = \frac{2}{\sqrt{|d|}}\xi \cdot \xi(\cdot, \chi_d)$ and (iii) give $\Phi_K = \frac{2}{\sqrt{|d|}}\, \Phi * K_{\chi_d}$, with Φ the Pólya kernel (2.1.3). As $\Phi > 0$, the kernel Φ_K is positive whenever K_{χ_d} is, and if Φ_K takes a negative value, so does K_{χ_d}. For $d = -4$ Jacobi's identity $\sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8} = q^{1/8}\prod_{n\ge1}(1-q^n)^3$ gives $K_{\chi_{-4}}(u) = 2e^{3u/2}\eta(ie^{2u})^3 > 0$, positive by the product formula as the kernel of χ_{12} is.

The theorem is checked at 30 digits: for $d = -4$, where $h/w = \frac14$, the trapezoid rule with step $1/200$ gives $\int \Phi_K(u)\cos(\theta u)\, du = 0.48753311500024448536$ at $\theta = 0$ and 0.25376667901239784985 at $\theta = 2.5$, equal to $\xi_K(\frac12 + i\theta)$ in every digit shown. The values $\zeta_K(\frac12) = -0.70223711$, -0.97506623, -1.6744215 and -1.6070054 for $d = -3, -4, -7, -8$ are negative, as (v) requires.

The sign of H_K on $[1, \sqrt{|d|}/\pi]$ depends strongly on the first few weights: Ψ is negative on $(0, 2)$ and positive beyond, and the weights $a_2 = 1 + \chi_d(2)$ and $a_3 = 1 + \chi_d(3)$ vanish when 2 and 3 are inert in K.

THEOREM 15.1.19. *Let $d < 0$ be a fundamental discriminant with $\chi_d(2) = \chi_d(3) = -1$, that is $d \equiv 5 \pmod{24}$. Then $H_K(x) < 0$ at $x = 1.42\sqrt{|d|}/2\pi$, so Φ_K takes negative values, and so does the kernel K_{χ_d}. There are infinitely many such d, for instance $d = -p$ for every prime $p \equiv 19 \pmod{24}$.*

Proof. For a fundamental discriminant $d < 0$, $\chi_d(2) = -1$ exactly when $d \equiv 5 \pmod 8$, and $\chi_d(3) = -1$ exactly when $d \equiv 2 \pmod 3$; together they give $d \equiv 5 \pmod{24}$. The function $n \mapsto a_n$ is multiplicative, with $a_{p^k} = \frac12(1 + (-1)^k)$ when $\chi_d(p) = -1$ and $a_{p^k} \le k + 1$ for every p. Hence $a_1 = 1$, $a_2 = a_3 = 0$ and $a_n \le a_n^*$, where $a_n^* = 0$ unless 2 and 3 divide n to even powers, and otherwise a_n^* is the number of divisors of n with its factors 2 and 3 removed. At $x = 1.42\sqrt{|d|}/2\pi$ the nth argument of Ψ in Theorem 15.1.18(ii) is $1.42n$, which exceeds 2 for $n \ge 2$, where $\Psi > 0$. So

$$\begin{aligned} H_K(x) &\le \Psi(1.42) + \sum_{n\ge4} a_n^*\, \Psi(1.42n) \\ &= -0.19907566 + 0.13917904 = -0.05989662 < 0, \end{aligned}$$

the sum taken to $n = 2000$ and the rest bounded, with $a_n^* \le 2\sqrt{n}$, by 10^{-1225}. Then $\Phi_K(\log x) < 0$, and K_{χ_d} takes negative values by the convolution identity above. If $p \equiv 19 \pmod{24}$ is prime, then $p \equiv 3 \pmod 4$, so $-p$ is a fundamental discriminant, and $-p \equiv 5 \pmod{24}$; by Dirichlet's theorem there are infinitely many such primes. □

Table 15.3 shows the discriminants discussed in [289]. The four of class number one with 2 and 3 inert, $d = -19, -43, -67, -163$, take the least value of H_K on $[1, \sqrt{|d|}/\pi]$ at the centre $x = 1$, the two composite ones with $d \equiv 5 \pmod{24}$ inside the interval, and most of the positive kernels at its right end $\sqrt{|d|}/\pi$, where the term $n = 1$ vanishes. All twenty values of $\zeta_K(\frac{1}{2})$ computed are negative, so $L(\frac{1}{2}, \chi_d) > 0$ for all of them whether or not the kernel is positive: Theorem 15.1.18(v) is a sufficient condition only.

d	$\chi_d(2)$	$\chi_d(3)$	$\Phi_K > 0$	least H_K on $[1, \sqrt{\lvert d\rvert}/\pi]$	at $x =$	$\zeta_K(\frac{1}{2})$
-11	-1	1	yes	0.14236	1.0557	-1.4480539
-15	1	0	yes	0.40606	1.2328	-2.6973176
-20	0	1	yes	0.287033	1.4235	-2.4529153
-23	1	1	yes	0.4634	1.5266	-3.5856997
-24	0	0	yes	0.225215	1.2409	-2.2955534
-31	1	-1	yes	0.349008	1.7723	-3.2253855
-39	1	0	yes	0.409694	1.9878	-4.0547932
-47	1	1	yes	0.46368	2.1822	-4.8243519
-71	1	1	yes	0.470667	2.6821	-5.9925928
-191	1	1	yes	0.470667	4.3991	-8.1483722
-19	-1	-1	no	-0.0596605	1	-1.1747378
-43	-1	-1	no	-0.214707	1	-0.76635571
-67	-1	-1	no	-0.251274	1	-0.54460945
-163	-1	-1	no	-0.315818	1	-0.10008066
-427	-1	-1	no	-0.181949	3.1887	-0.20993694
-1003	-1	-1	no	-0.228309	3.9479	-0.64248774

Table 15.3: The Pólya kernel Φ_K of $K = \mathbb{Q}(\sqrt{d})$ for the discriminants named in [289]: the characters at 2 and 3, the sign of the kernel, the least value of H_K on $[1, \sqrt{|d|}/\pi]$, which decides it by Theorem 15.1.18(iv), with the point where it is attained, and $\zeta_K(\frac{1}{2}) = \zeta(\frac{1}{2})L(\frac{1}{2}, \chi_d)$.

A computation also scans all 6079 fundamental discriminants $-20000 \le d < 0$. It evaluates H_K in double precision on 3000 geometrically spaced points of $[1, \sqrt{|d|}/\pi]$, refines the least value by golden section search, and would recompute at 30 digits any least value within 10^{-8} of zero; there is none. The kernel is positive for 2738 of the discriminants. Table 15.4 sorts them by the splitting of 2 and 3: the proportion of positive kernels falls as either prime goes from split to ramified to inert. By blocks of 2000 in $|d|$ the proportion falls from 337/611 for $|d| \le 2000$ to between 0.41 and 0.45 in each of the five blocks above 10000.

Question 15.1.20. Does the proportion of fundamental discriminants $-X \le d < 0$ with $\Phi_K > 0$ tend to a limit as $X \to \infty$, and is the limit positive? Is Φ_K positive for infinitely many d, for instance among the d in which 2 and 3 split, where the proportion up to 20000 is 676/767?

	$\chi_d(3)=1$	$\chi_d(3)=0$	$\chi_d(3)=-1$
$\chi_d(2)=1$	676/767	334/507	412/757
$\chi_d(2)=0$	489/753	178/514	131/757
$\chi_d(2)=-1$	397/761	121/500	0/763

Table 15.4: Positive Pólya kernels Φ_K among the 6079 fundamental discriminants $-20000 \le d < 0$, by the splitting of 2 and 3 in $K=\mathbb{Q}(\sqrt{d})$ ($\chi_d(p)=1$ split, 0 ramified, -1 inert): the number of positive kernels over the number of discriminants in each class, 2738 positive in all. The empty corner is Theorem 15.1.19.

Question 15.1.21. For real quadratic fields and for fields of higher degree the gamma factor $\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}$ replaces the exponentials in ϑ_K by other kernels, for $\Gamma_{\mathbb{R}}(s)^2$ a Bessel function K_0. Is there a criterion for the positivity of the Pólya kernel of ξ_K as simple as Theorem 15.1.18(iv)?

15.1.7 The Dirichlet beta function and the eta cube kernel

The L function of the odd character χ_4 modulo 4,

$$L(s,\chi_4)=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)^s}=\prod_p\bigl(1-\chi_4(p)p^{-s}\bigr)^{-1},\qquad \operatorname{Re}s>1,$$

is the Dirichlet beta function, with $L(1,\chi_4)=\frac{\pi}{4}$ and $L(2,\chi_4)$ Catalan's constant. It is the character χ_{-4} of Section 15.1.6, whose kernel is the cube of the Dedekind eta function, positive by the product formula. This subsection carries the Thorin framework over to it, following [283]; the comparison across vertical lines, which applies to ξ and $L(s,\chi_{12})$ as well, is in Section 12.1.6.

15.1.7.1 The kernel, the exit time and the zeros

Jacobi's identity $\eta(\tau)^3=q^{1/8}\prod_{m\ge1}(1-q^m)^3=\sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8}$, with $q=e^{2\pi i\tau}$ (Section 15.1.6), gives at $\tau=ix$ the theta series of χ_4,

$$\eta(ix)^3=\sum_{n\ge1}\chi_4(n)\,n\,e^{-\pi n^2x/4}=e^{-\pi x/4}\prod_{m\ge1}\bigl(1-e^{-2\pi mx}\bigr)^3,\qquad x>0,$$

and the transformation $\eta(i/x)=\sqrt{x}\,\eta(ix)$ gives $\eta(i/x)^3=x^{3/2}\eta(ix)^3$. In the normalisation of Section 15.1.1 for odd characters the completed function is

$$\xi(s,\chi_4)=\Bigl(\frac{4}{\pi}\Bigr)^{(s+1)/2}\Gamma\Bigl(\frac{s+1}{2}\Bigr)L(s,\chi_4)=\int_0^\infty\eta(ix)^3\,x^{(s+1)/2}\,\frac{dx}{x},$$

the integral by termwise integration for $\operatorname{Re}s>1$, since $\int_0^\infty e^{-\pi n^2x/4}x^{(s+1)/2-1}dx=\Gamma(\frac{s+1}{2})(4/\pi n^2)^{(s+1)/2}$, and for every s by analytic continuation, since $\eta(ix)^3$ decays

like $e^{-\pi x/4}$ as $x\to\infty$ and, by the transformation, like $x^{-3/2}e^{-\pi/(4x)}$ as $x\to 0$. The transformation gives $\xi(s,\chi_4)=\xi(1-s,\chi_4)$, and $\xi(s,\chi_4)$ is entire of order one, with no factor $s(s-1)$ to remove. It is $2/\sqrt{\pi}$ times the function Λ of [283], and $\xi(1,\chi_4)=\frac{4}{\pi}L(1,\chi_4)=1$.

PROPOSITION 15.1.22. *The kernel $K_4(u)=2e^{3u/2}\eta(ie^{2u})^3$ satisfies $\xi(\frac12+s,\chi_4)=\int_{\mathbb R}K_4(u)\,e^{su}\,du$ for every complex s. It is even, positive and analytic in the strip $|\operatorname{Im}u|<\pi/4$, it has the symmetric product form*

$$K_4(u)=2\exp\Big(-\frac{\pi}{4}\cosh 2u\Big)\prod_{m\ge1}\big(1-e^{-2\pi me^{2u}}\big)^{3/2}\big(1-e^{-2\pi me^{-2u}}\big)^{3/2},$$

so that $K_4(u)=2\exp\big(\frac32|u|-\frac{\pi}{4}e^{2|u|}\big)\big(1+O(e^{-2\pi e^{2|u|}})\big)$, and $K_4(0)=2\eta(i)^3=\Gamma(\frac14)^3/(4\pi^{9/4})=0.906767655167731$.

Proof. With $x=e^{2u}$, $x^{(s+1)/2}dx/x=2e^{(s+1)u}du$, so the Mellin form with $\frac12+s$ in place of s is the stated integral. The product formula makes every factor of $K_4(u)=2e^{3u/2}e^{-\frac{\pi}{4}e^{2u}}\prod_m(1-e^{-2\pi me^{2u}})^3$ positive. The transformation gives $K_4(-u)=2e^{-3u/2}\eta(ie^{-2u})^3=2e^{-3u/2}e^{3u}\eta(ie^{2u})^3=K_4(u)$. Hence $K_4(u)=\sqrt{K_4(u)K_4(-u)}$, and in the product $K_4(u)K_4(-u)$ the factors $e^{\pm3u/2}$ cancel while $e^{-\frac{\pi}{4}(e^{2u}+e^{-2u})}=e^{-\frac{\pi}{2}\cosh 2u}$, which gives the symmetric form. For $u\to+\infty$ the product for $K_4(u)$ itself gives the asymptotic, and evenness gives it for $u\to-\infty$. The function $\eta(ie^{2u})$ is analytic where $\operatorname{Re}e^{2u}>0$, that is for $|\operatorname{Im}u|<\pi/4$. Finally $\eta(i)=\Gamma(\frac14)/(2\pi^{3/4})$. □

We compute $\xi(\frac12+s,\chi_4)$ from K_4 by the trapezoid rule with step $1/64$ on $|u|\le 3$ and agrees with the Dirichlet L function of Arb [169] in all fourteen digits printed at $s=0$, $\frac12$, $0.1+5i$ and $0.1+20i$; for instance $\xi(\frac12,\chi_4)=0.98071361405771$, $\xi(0.6+5i,\chi_4)=0.080871220951695+0.011736751015145i$ and $\xi(0.6+20i,\chi_4)=(-2.3183027026127-0.20613003173368i)\times10^{-6}$. Also $L(\frac12,\chi_4)=0.667691457189609$.

REMARK 15.1.23 (The exit time). By Theorem 9.3.20, $\frac{4}{\pi}\eta(4it/\pi)^3$ is the density of $X_C=\sum_k e_k/(2k-1)^2$, which is $\frac{\pi^2}{8}$ times the exit time τ of Brownian motion from $(-1,1)$ (Section 9.2.2). With $x=4t/\pi$, $\eta(ix)^3$ is therefore the density on $(0,\infty)$ of $Z=\frac{4}{\pi}X_C=\frac{\pi}{2}\tau$ [290, Section 3], and

$$\xi(s,\chi_4)=\mathbb E\big[Z^{(s-1)/2}\big],\qquad \mathbb Ee^{-uZ}=\operatorname{sech}\sqrt{\pi u}=\prod_{k\ge1}\Big(1+\frac{4u}{\pi(2k-1)^2}\Big)^{-1}.$$

So Z is a GGC with Thorin measure $\sum_k\delta_{\pi(2k-1)^2/4}$; its total mass $\xi(1,\chi_4)=1$ is the value $L(1,\chi_4)=\frac{\pi}{4}$, and its mean $\xi(3,\chi_4)=\frac{4}{\pi}\sum_k(2k-1)^{-2}=\frac{\pi}{2}$ is the value $L(3,\chi_4)=\pi^3/32$. The transformation $\eta(i/x)^3=x^{3/2}\eta(ix)^3$ says that $1/Z$ has the density $x^{-2}\eta(i/x)^3=x^{-1/2}\eta(ix)^3$, that is $\mathbb Eg(1/Z)=\mathbb E[Z^{-1/2}g(Z)]$ for $g\ge0$,

and with $g(z) = z^{(1-s)/2}$ this is the functional equation. The primal variable Y_4, with density $K_4/\xi(\frac12,\chi_4)$ and $\mathbb{E}e^{sY_4} = \xi(\frac12+s,\chi_4)/\xi(\frac12,\chi_4)$, is $\frac12\log Z$ under the size biasing by $Z^{-1/4}$, since $\mathbb{E}[Z^{-1/4}e^{(s/2)\log Z}]/\mathbb{E}[Z^{-1/4}] = \mathbb{E}[Z^{s/2-1/4}]/\mathbb{E}[Z^{-1/4}]$. So the analogue of the Pólya variable is half the logarithm of a Brownian exit time biased by $Z^{-1/4}$, as for ζ it is half the logarithm of the variable $\pi S_2/2$ of Williams and of Biane, Pitman and Yor biased by $(\pi S_2/2)^{1/4}$ (Section 3.1.3). In the notation of Section 14.4.5.4, $Z = 1/\mathcal{B}_{\chi_{-4}}$ (Proposition 14.4.19).

The basepoint one has an explicit drift. By the functional equation $b_1 = \frac{\xi'}{\xi}(1,\chi_4) = -\frac{\xi'}{\xi}(0,\chi_4)$. With $L(s,\chi_4) = 4^{-s}\big(\zeta(s,\frac14) - \zeta(s,\frac34)\big)$, $\zeta(0,a) = \frac12 - a$ and Lerch's formula $\partial_s\zeta(0,a) = \log\Gamma(a) - \frac12\log 2\pi$ [256, §25.11], one has $L(0,\chi_4) = \frac12$ and $L'(0,\chi_4) = \log\big(\Gamma(\frac14)/2\Gamma(\frac34)\big)$, and with $\psi(\frac12) = -\gamma_E - 2\log 2$ and $\Gamma(\frac14)\Gamma(\frac34) = \pi\sqrt{2}$,

$$b_1 = 3\log 2 + \frac{5}{2}\log\pi + \frac{\gamma_E}{2} - 4\log\Gamma\Big(\frac14\Big) = 0.0777839899618, \tag{15.1.7}$$

and so, since $b_1 = \frac12\log\frac{4}{\pi} - \frac{\gamma_E}{2} + \frac{4}{\pi}L'(1,\chi_4)$, $L'(1,\chi_4) = \frac{\pi}{4}\big(\gamma_E + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac14)\big) = 0.192901316797$. At the centre $\xi'(\frac12,\chi_4) = 0$, and the variance of the primal variable is $V_4 = \operatorname{Var} Y_4 = (\log\xi)''(\frac12,\chi_4) = 0.156029649890$.

Since $K_4 > 0$, $\xi(\frac12+s,\chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}du > 0$ for every real s, so $\xi(s,\chi_4)$ has no real zeros. We compute the zeros of $L(s,\chi_4)$ below height 100 in ball arithmetic with Arb. The function $\xi(\frac12 + i\theta,\chi_4)$ is real; its sign is certified on a grid of step $1/32$, and each sign change is refined by bisection to 10^{-12}. The argument principle along $2 \to 2+100i \to \frac12+100i$, on pieces halved until the enclosure of $\xi(s,\chi_4)$ over each piece lies in a disc excluding zero (1633 pieces), gives $N(100) = 50$ for the number of zeros with $0 < \gamma \le 100$, counted with multiplicity, the remaining sides of the rectangle $[-1,2]\times[0,100]$ following by the functional equation and $\xi(\bar s,\chi_4) = \overline{\xi(s,\chi_4)}$. This equals the number of sign changes, so all fifty zeros are simple and on the critical line. The zero counting function satisfies

$$N(T) = \frac{T}{2\pi}\log\frac{2T}{\pi e} + \frac18 + \frac{1}{\pi}\arg L\big(\tfrac12 + iT,\chi_4\big) + \frac{1}{48\pi T} + O(T^{-3}), \tag{15.1.8}$$

with the argument continued from $L(2,\chi_4) > 0$ along $2 \to 2+iT \to \frac12+iT$, for T not an ordinate. Indeed, by the argument principle on $[-1,2]\times[0,T]$, the functional equation, the reflection and the positivity of $\xi(s,\chi_4)$ on the real axis, $\pi N(T)$ is the change of $\arg\xi(s,\chi_4)$ along $2 \to 2+iT \to \frac12+iT$, and Stirling's series [256, §5.11] for the gamma factor, as in the proof of Proposition 15.1.31 at $\alpha = \frac12$, gives (15.1.8); compare [70, Chapter 16], where the main term is derived with an error $O(\log T)$. Under GRH for χ_4, $V_4 = \sum_\gamma \gamma^{-2}$ over all the ordinates of both signs; twice the sum over the fifty computed ones is 0.1395621, and the main term $\frac{1}{\pi}(\log\frac{2T}{\pi}+1)/T$ of the rest, from $dN(t) \approx \frac{1}{2\pi}\log\frac{2t}{\pi}\,dt$, is 0.0164 at $T = 100$, together 0.1560.

15.1.7.2 The Lévy form and the sine squared decomposition

The Lévy form of Proposition 15.2.2 for χ_{12} has an analogue for χ_4 at every basepoint $\alpha \geq 1$.

PROPOSITION 15.1.24. *For $\alpha \geq 1$,*

$$\log \frac{\xi(\alpha+s,\chi_4)}{\xi(\alpha,\chi_4)} = b_\alpha s + \int_0^\infty \left(e^{-sx} - 1 + sx\right) \Pi_\alpha(dx), \qquad b_\alpha = \frac{\xi'}{\xi}(\alpha,\chi_4),$$

with the signed Lévy measure

$$\Pi_\alpha(dx) = \frac{e^{-\alpha x}}{2x\sinh x}\,dx + \sum_{n\geq 2} \frac{\chi_4(n)\Lambda(n)}{n^\alpha \log n}\,\delta_{\log n}(dx),$$

whose canonical measure $x\Pi_\alpha(dx)$ has the positive density $e^{-\alpha x}/(2\sinh x)$ and the atoms $\chi_4(n)\Lambda(n)n^{-\alpha}$ at $\log n$. For $\alpha > 1$ the form holds for $\operatorname{Re} s > 1-\alpha$, with absolute convergence. At $\alpha = 1$ it holds for $\operatorname{Re} s > 0$, the prime sum converging conditionally, as a limit over $n \leq X$.

Proof. The factor $(4/\pi)^{(s+1)/2}$ is linear in log and is absorbed by the drift. Binet's formula [256, §5.9], in the integrated form

$$\log \frac{\Gamma(z+v)}{\Gamma(z)} - v\psi(z) = \int_0^\infty (e^{-vx} - 1 + vx)\frac{e^{-zx}}{x(1-e^{-x})}\,dx$$

for $\operatorname{Re} z > 0$ and $\operatorname{Re}(z+v) > 0$, which one checks by differentiating twice in v, gives at $z = \frac{\alpha+1}{2}$, $v = \frac{s}{2}$ and after $x \mapsto 2x$ the density $\frac{e^{-(\alpha+1)x}}{x(1-e^{-2x})} = \frac{e^{-\alpha x}}{2x\sinh x}$, for $\operatorname{Re} s > -1-\alpha$. For $\operatorname{Re}(\alpha+s) > 1$ the Euler product gives $\log L(\alpha+s,\chi_4) = \sum_n \chi_4(n)\Lambda(n)n^{-\alpha-s}/\log n$, and subtracting the value and the derivative at $s=0$ gives the atoms; for $\alpha > 1$ the sum converges absolutely for $\operatorname{Re} s > 1-\alpha$, because $|e^{-sx} - 1 + sx| \leq e^{-x\operatorname{Re} s} + 1 + |s|x$ and $\sum_n \Lambda(n)n^{-\alpha}\left(n^{-\operatorname{Re} s} + 1 + |s|\log n\right)/\log n < \infty$ there. At $\alpha = 1$ the linear coefficient is $-\sum_n \chi_4(n)\Lambda(n)/n = \frac{L'}{L}(1,\chi_4)$ and the constant term is $\sum_n \chi_4(n)\Lambda(n)/(n\log n) = \log L(1,\chi_4)$; both series converge by the prime number theorem for arithmetic progressions [70, Chapter 20], as in the proof of Proposition 15.2.2. □

The archimedean density is that of ξ shifted by one basepoint: at $\alpha+1$ the gamma density $e^{-(\alpha+2)x}/(1-e^{-2x})$ of $x\Pi_{\alpha+1}$ is the gamma density of ξ at the basepoint α (Section 12.1.4), because the gamma factor $\Gamma(\frac{s+1}{2})$ at $\alpha+1$ and the gamma factor $\Gamma(1+\frac{s}{2})$ of ξ, which absorbs the factor s, at α have the same argument $\frac{\alpha+2+s}{2}$. There is no pole term, and the atoms carry the sign of $\chi_4(n)$. As for χ_{12} in Section 15.2.2, the tilted variable $V_\alpha^{(4)}$ with moment generating function $\xi(\alpha+s,\chi_4)/\xi(\alpha,\chi_4)$ is quasi infinitely divisible for $\alpha > 1$, the Euler product writing its characteristic function as that of an infinitely divisible gamma piece

times a quotient of two compound Poisson characteristic functions, and it is not infinitely divisible: by the uniqueness of the Lévy measure of a quasi infinitely divisible law [213], the atom $-3^{-\alpha}$ of Π_α at $\log 3$, reflected to $-\log 3$, would have to be positive. At $\alpha = 1$, $V_1^{(4)}$ is not quasi infinitely divisible, since the atoms of each sign have infinite mass, $\sum_{p\equiv 1} 1/p$ and $\sum_{p\equiv 3\,(4)} 1/p$ both diverging, and cancel only conditionally; the argument of Proposition 14.1.3(3) applies.

The sine squared lemma (2.1.9) turns the Lévy form into the Thorin density, the bracket being $\mathrm{Re}(\xi'/\xi)(\alpha+i\theta,\chi_4)$ [279, Theorem 8.2]. With $v_\alpha(\theta) = \mathrm{Re}\,\frac{\xi'}{\xi}(\alpha+i\theta,\chi_4)$ and $q_c(\theta) = \theta^2/(c(c^2+\theta^2))$ as in (12.1.7),

$$v_\alpha(\theta) = b_\alpha + G_\alpha(\theta) + \varpi_\alpha(\theta), \qquad \alpha \ge 1, \tag{15.1.9}$$

where the gamma piece and the prime piece are

$$\begin{aligned} G_\alpha(\theta) &= \int_0^\infty \frac{1-\cos\theta x}{2\sinh x}\, e^{-\alpha x} dx = \frac{1}{2}\Big[\,\mathrm{Re}\,\psi\Big(\frac{\alpha+1+i\theta}{2}\Big) - \psi\Big(\frac{\alpha+1}{2}\Big)\Big] \\ &= \sum_{k\ge 0} q_{\alpha+1+2k}(\theta) \ge 0, \\ \varpi_\alpha(\theta) &= \sum_{n\ge 2} \frac{\chi_4(n)\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big), \end{aligned}$$

the series for G_α from $1/(2\sinh x) = \sum_{k\ge 0} e^{-(2k+1)x}$, and ϖ_α converging conditionally at $\alpha = 1$. So $G_\alpha = \sum_{k\ge 1} q_{(\alpha-1)+2k}$ is the gamma piece of ξ at the basepoint $\alpha - 1$ in (12.1.7), without its pole term $-q_{\alpha-2}$. For $\alpha \ge 1$ the Thorin representation is

$$\log\frac{\xi(\alpha+\sqrt{u},\chi_4)}{\xi(\alpha,\chi_4)} = \frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\, v_\alpha(\theta)\, d\theta, \qquad u \ge 0, \tag{15.1.10}$$

by Proposition 15.1.25 below.

The first zero of $\xi(\frac{1}{2}+i\theta,\chi_4)$ is $\gamma_1 = 6.0209489$.

Figure 15.4 shows the density and the decomposition. On the left, at $\alpha = 1$, 0.75 and 0.55, the minimum of v_α is b_α, at the origin (0.0777840, 0.0389784 and 0.0078013), as Theorem 12.1.18 proves for every $\alpha \in (\frac{1}{2}, 4.70]$ on the verified range and for every $\alpha \ge 1$ at every height. Near each ordinate v_α rises to about $1/(\alpha-\frac{1}{2})$, the height of the Cauchy kernel of a zero on the line, plus the contribution of the other zeros. As $\alpha \downarrow \frac{1}{2}$ the margin at the origin shrinks linearly, $b_\alpha \approx (\alpha-\frac{1}{2})V_4$. On the right, at $\alpha = \frac{3}{2}$, the prime piece ϖ_α is negative on 71% of the grid of step 0.01 on $[0,30]$, with maximum 0.33562 at $\theta = 6.0$, next to γ_1; the minimum of v_α on the grid is $b_{3/2} = 0.15421515$, at the origin. For ξ the same figure would show a positive prime piece and the negative pole piece (Figure 15.1), while for χ_4 the negative input is spread over the primes $p \equiv 3 \pmod 4$.

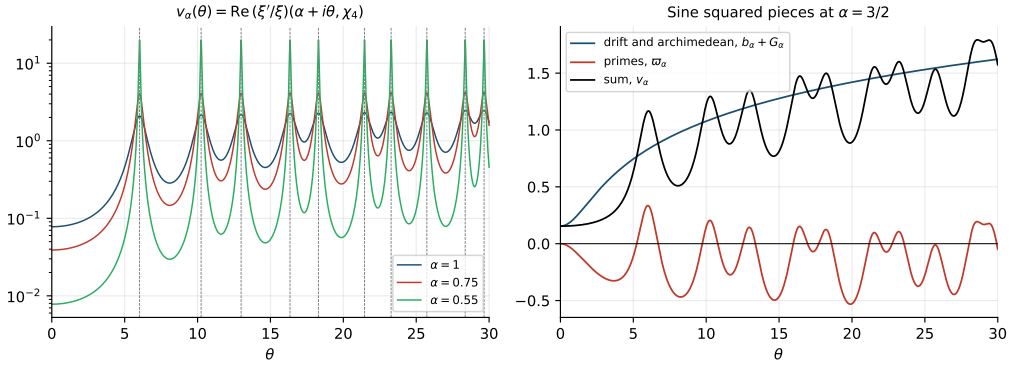


Figure 15.4: Left: $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta, \chi_4) = \pi\rho_{\alpha-1/2}(\theta)$ for $\alpha = 1$, 0.75, 0.55 on a logarithmic scale, with the ordinates of the zeros of $L(s,\chi_4)$ (dashed). Right: at $\alpha = \frac{3}{2}$, the drift and gamma piece $b_\alpha + G_\alpha$, the prime piece ϖ_α, of both signs, and their sum v_α.

PROPOSITION 15.1.25. *For every $\alpha \geq 1$, $u \mapsto \xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$ is the Laplace transform of a GGC with Thorin density $v_\alpha(\theta)/\pi > 0$ in the chart $\theta = \sqrt{z}$, and* (15.1.10) *holds. For $\alpha > \frac{1}{2}$ it is the Laplace transform of a GGC if and only if no zero of $L(s,\chi_4)$ has real part greater than α. GRH for χ_4 holds if and only if $v_\alpha \geq 0$ for every $\alpha > \frac{1}{2}$, and the Thorin measures at resolution $\varepsilon = \alpha - \frac{1}{2}$ converge vaguely, as $\varepsilon \downarrow 0$, to the sum of the point masses δ_{γ^2} over the zeros on the critical line, which is the Thorin measure of $\xi(\frac{1}{2},\chi_4)/\xi(\frac{1}{2}+\sqrt{u},\chi_4)$ under GRH.*

Proof. The proofs of Theorem 2.1.2, of Lagarias's criterion [198] and of the vague limit stated after Remark 2.1.4 use only the Hadamard product of an entire function of order one whose zeros are symmetric under $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1-\rho$, with $\sum_\rho |\rho|^{-2} < \infty$; they apply to $\xi(s,\chi_4)$ [70, Chapter 12], as Section 15.1.3 and Proposition 15.2.1 apply them to $\xi(s,\chi_{12})$. Positivity at $\alpha \geq 1$ needs every zero to have real part less than one, which is the absence of zeros of $L(s,\chi_4)$ on $\mathrm{Re}\,s = 1$ [70, Chapter 14] with the Euler product and the functional equation. □

At the basepoint one the clock $T_1^{(4)}$, with Laplace transform $1/\xi(1+\sqrt{u},\chi_4)$, therefore has the expansion $1 - b_1\sqrt{u} + O(u)$ at the origin, and as for ξ in (12.1.3) the Tauberian theorem [86, XIII.5] gives the tail $P(T_1^{(4)} > x) \sim (b_1/\sqrt{\pi})\, x^{-1/2} = 0.0438849\, x^{-1/2}$, with b_1 from (15.1.7).

Positivity at $\alpha \geq 1$ uses the zeros, not the Lévy measure: bounding the prime weights by their absolute values gives positivity only far to the right, and no such bound does better.

PROPOSITION 15.1.26. *For $\alpha > 1$ and real θ,*

$$v_\alpha(\theta) \ge A_4^\infty(\alpha,\theta) - S_4(\alpha),$$
$$A_4^\infty(\alpha,\theta) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{\alpha+1+i\theta}{2}\Big),$$
$$S_4(\alpha) = \sum_{n\ \text{odd}} \frac{\Lambda(n)}{n^\alpha} = -\frac{\zeta'}{\zeta}(\alpha) - \frac{\log 2}{2^\alpha - 1},$$

and $A_4^\infty(\alpha,\theta)$ increases in $|\theta|$. The bound is positive at every θ exactly when $\alpha > \alpha_c(\chi_4) = 2.2823106368$, the root of $A_4^\infty(\alpha,0) = S_4(\alpha)$. For χ_{12} the same bound, with $A_{12}^\infty(\alpha,\theta) = \frac{1}{2}\log\frac{12}{\pi} + \frac{1}{2}\operatorname{Re}\psi(\frac{\alpha+i\theta}{2})$ and $S_{12}(\alpha) = S_4(\alpha) - \log 3/(3^\alpha - 1)$, is positive at every θ exactly when $\alpha > \alpha_c(\chi_{12}) = 1.8348743295$.

Proof. For $\operatorname{Re} s > 1$,

$$\frac{\xi'}{\xi}(s,\chi_4) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\psi\Big(\frac{s+1}{2}\Big) + \frac{L'}{L}(s,\chi_4),$$

and $|\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)| \le \sum_n |\chi_4(n)|\Lambda(n)n^{-\alpha} = S_4(\alpha)$. The series $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\ge0} y^2/\big((x+k)((x+k)^2+y^2)\big)$ shows that A_4^∞ increases in $|\theta|$. Since $A_4^\infty(\alpha,0)$ increases in α and $S_4(\alpha)$ decreases, the root is unique, and bisection with certified signs locates it with certified signs. For χ_{12} the primes 2 and 3 drop out of the sum. □

PROPOSITION 15.1.27. *For $\alpha > 1$ and every $T > 0$,*

$$\inf_{\theta\ge T}\Big(v_\alpha(\theta) - A_4^\infty(\alpha,\theta)\Big) = \inf_{\theta\ge T}\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4) = -S_4(\alpha),$$

so no lower bound for $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)$ better than $-S_4(\alpha)$ holds at all large heights. The bound of Proposition 15.1.26 certifies $v_\alpha(\theta) > 0$ exactly for $|\theta| > t_0(\alpha)$, where $A_4^\infty(\alpha,t_0) = S_4(\alpha)$, for $1 < \alpha \le \alpha_c$ (so that $t_0(\alpha_c) = 0$), and at every real θ for $\alpha > \alpha_c$, and

$$\log t_0(\alpha) = \frac{2}{\alpha-1} - 2\gamma_E - \log\frac{8}{\pi} + O(\alpha-1), \qquad \alpha \downarrow 1,$$

with $-2\gamma_E - \log\frac{8}{\pi} = -2.0891$.

Proof. From the Euler product, $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4) = -\sum_p \log p\ \operatorname{Re}\frac{w_p}{1-w_p}$ with $w_p = \chi_4(p)p^{-\alpha-i\theta}$, and on the circle $|w| = r < 1$ the maximum of $\operatorname{Re}\frac{w}{1-w} = \operatorname{Re}\frac{1}{1-w} - 1$ is $\frac{r}{1-r}$, at $w = r$; this gives the lower bound $-S_4(\alpha)$. Since the numbers $\log p$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem [106, Chapter 23] gives, for every P and $\eta > 0$, a height $\theta \ge T$ at which $|\chi_4(p)p^{-i\theta} - 1| < \eta$ for all odd $p \le P$, and the primes beyond P contribute at most $\sum_{p>P}\log p\, p^{-\alpha}/(1-p^{-\alpha})$, which

is small for P large. For the asymptotic, $S_4(\alpha) = \frac{1}{\alpha-1} - \gamma_E - \log 2 + O(\alpha-1)$ and $A_4^\infty(\alpha,t) = \frac{1}{2}\log\frac{2t}{\pi} + O(t^{-2})$ as $t\to\infty$, uniformly for $1 \le \alpha \le 2$, from $\psi(z) = \log z - \frac{1}{2z} + O(|z|^{-2})$; solving $A_4^\infty(\alpha,t_0) = S_4(\alpha)$ gives the formula, since $t_0(\alpha)\to\infty$. □

α	$S_4(\alpha)$	$t_0(\alpha)$	$\log t_0(\alpha) - \frac{2}{\alpha-1}$
2	0.33891193	2.36723	−1.1383
1.75	0.57950941	4.71677	−1.1155
1.5	1.1261406	14.8728	−1.3005
1.25	2.9637961	589.44	−1.6208
1.1	8.8348984	7.41336×10^{7}	−1.8786
1.05	18.784554	3.25219×10^{16}	−1.9793

Table 15.5: The height $t_0(\alpha)$ above which the bound of Proposition 15.1.26 certifies $v_\alpha > 0$, for basepoints below $\alpha_c(\chi_4) = 2.2823$; the last column tends to $-2\gamma_E - \log\frac{8}{\pi} = -2.0891$ (Proposition 15.1.27).

So for $1 < \alpha < \alpha_c$ the absolute values of the weights reduce the positivity of v_α to a computation on $[0, t_0(\alpha)]$, but t_0 grows like $e^{2/(\alpha-1)}$ (Table 15.5). At $\alpha = 1$ the prime piece converges only conditionally, by the prime number theorem for progressions, whose analytic input is the absence of zeros of $L(s,\chi_4)$ on $\operatorname{Re} s = 1$; the positivity of v_1, which Proposition 15.1.25 draws from that input, cannot come from the sizes of the weights.

15.1.7.3 The Gaussian field

The pairing of the primes $p\equiv 1$ and $p\equiv 3 \pmod 4$ is their splitting in $K = \mathbb{Q}(i)$. Accordingly $\zeta_K(s) = \zeta(s)L(s,\chi_4)$, and in the normalisation of Section 13.3.3,

$$\xi_K(s) = 2s(s-1)\pi^{-s}\Gamma(s)\zeta_K(s) = \xi(s)\,\xi(s,\chi_4),$$

by the duplication formula $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$. Write v_α^K and v_α^ξ for the brackets $\operatorname{Re}(\xi_K'/\xi_K)(\alpha+i\theta)$ and $\operatorname{Re}(\xi'/\xi)(\alpha+i\theta)$, and v_α as before for χ_4.

Proposition 15.1.28. *On every vertical line, $v_\alpha = v_\alpha^K - v_\alpha^\xi$. For $\alpha > 1$ the canonical measure of ξ_K at the basepoint α is*

$$x\Pi_\alpha^K(dx) = \Big(\frac{e^{-(\alpha+1)x}}{1-e^{-x}} - e^{-(\alpha-1)x}\Big)dx + \sum_{n\ge 2}\frac{(1+\chi_4(n))\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx),$$

and the canonical measure of Proposition 15.1.24 is $x\Pi_\alpha = x\Pi_\alpha^K - x\Pi_\alpha^\xi$, with $x\Pi_\alpha^\xi$ that of Section 12.1.4: the pole densities $-e^{-(\alpha-1)x}$ cancel, the gamma densities differ by $e^{-\alpha x}/(2\sinh x)$, and the atoms by $\chi_4(n)\Lambda(n)n^{-\alpha}$.

Proof. Take logarithmic derivatives in $\xi_K = \xi \cdot \xi(\cdot, \chi_4)$. For the measures write $\xi_K(s) = (s-1) \cdot 2\pi^{-s}\Gamma(s+1)\zeta_K(s)$: Binet's formula for $\Gamma(s+1)$ at α gives the density $e^{-(\alpha+1)x}/(x(1-e^{-x}))$, the factor $s-1$ gives $-e^{-(\alpha-1)x}/x$ as for ξ, and $-\frac{\zeta_K'}{\zeta_K} = -\frac{\zeta'}{\zeta} - \frac{L'}{L}$ has the coefficients $(1+\chi_4(n))\Lambda(n) \geq 0$. Finally $\frac{e^{-(\alpha+1)x}}{1-e^{-x}} - \frac{e^{-(\alpha+2)x}}{1-e^{-2x}} = e^{-\alpha x}\Big(\frac{1}{e^x-1} - \frac{1}{e^{2x}-1}\Big) = \frac{e^{-\alpha x}e^x}{e^{2x}-1} = \frac{e^{-\alpha x}}{2\sinh x}$. □

Corollary 15.1.29. (i) *For every $\alpha \geq 1$ there are independent GGC variables H_α^ξ and $H_\alpha^{(4)}$, with Laplace transforms $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ and $\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$, whose sum $H_\alpha^K = H_\alpha^\xi + H_\alpha^{(4)}$ is the GGC with Laplace transform $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ and Thorin density v_α^K/π.*

(ii) *GRH for χ_4 holds if and only if $v_\alpha^K \geq v_\alpha^\xi$ on every line $\operatorname{Re} s = \alpha > \frac{1}{2}$: the Thorin density of ξ_K dominates that of ξ at every basepoint.*

(iii) *Under RH, GRH for χ_4 holds if and only if, for every $\alpha > \frac{1}{2}$, $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ is the Laplace transform of $H_\alpha^\xi + H$ for some GGC H independent of H_α^ξ.*

Proof. (i) By Theorem 2.1.2 and Proposition 15.1.25 both factors are GGC transforms for $\alpha \geq 1$, and transforms of independent sums multiply. (ii) By Proposition 15.1.28, $v_\alpha^K \geq v_\alpha^\xi$ is $v_\alpha \geq 0$, and Proposition 15.1.25 applies. (iii) Under RH, H_α^ξ is a GGC for every $\alpha > \frac{1}{2}$ (Theorem 2.1.2). If GRH holds for χ_4, take $H = H_\alpha^{(4)}$. Conversely, the transform of H is then $\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$, and if it is a GGC transform for every $\alpha > \frac{1}{2}$, Proposition 15.1.25 gives GRH. □

Neither v_α^K nor v_α^ξ is positive term by term: each has the pole piece of (1.3.1). The factorisation is the right language for the pairing of the primes, but it does not by itself lower α_c: by Proposition 15.1.27 no argument using only the sizes of the weights reaches further. Table 15.6 shows the three brackets at $\alpha = \frac{3}{2}$.

θ	0	2	5	10	20
$v_{3/2}^K(\theta)$	0.200351	0.225243	0.685189	1.32766	1.49863
$v_{3/2}^\xi(\theta)$	0.0461359	0.0470396	0.0526739	0.102889	0.606114
$v_{3/2}(\theta)$	0.154215	0.178204	0.632515	1.22477	0.892517

Table 15.6: The brackets of $\xi_{\mathbb{Q}(i)}$, ξ and $\xi(\cdot,\chi_4)$ at the basepoint $\frac{3}{2}$; the last row is the difference of the first two (Proposition 15.1.28).

The same holds for every real primitive character χ_D, with D a fundamental discriminant and $K = \mathbb{Q}(\sqrt{D})$: ξ_K is a constant multiple of $\xi \cdot \xi(\cdot,\chi_D)$, the constant being $2/\sqrt{|D|}$ for $D < 0$ (Section 15.1.6) and 2 for $D > 0$, since then $\xi_K(s) = s(s-1)D^{s/2}\Gamma_{\mathbb{R}}(s)^2\zeta_K(s)$ and $\xi(s,\chi_D) = D^{s/2}\Gamma_{\mathbb{R}}(s)L(s,\chi_D)$. So $v_\alpha^{(\chi_D)} = v_\alpha^K - v_\alpha^\xi$; the GGC property at every $\alpha \geq 1$, the equivalence of GRH for χ_D with the domination of the Thorin density of ξ by that of ξ_K, and the localisation of Corollary 15.1.32

hold for $\xi(s,\chi_D)$ with the same proofs. For χ_{12}, with $K=\mathbb{Q}(\sqrt{3})$: $\alpha_c(\chi_{12})=1.8348743295$ (Proposition 15.1.26); $V_{12}=(\log\xi)''(\frac12,\chi_{12})=0.332923608212$; and at $\alpha=1$, 0.75 and 0.6 the drift b_α is 0.1650833, 0.0830566 and 0.0332812, the minimum of v_α on a grid of step 0.01 on $[0,60]$, attained at the origin (computed; Theorem 12.1.18). A computation certifies the first five ordinates of $L(s,\chi_{12})$, 3.8046276331, 6.6922233205, 8.8905929587, 11.1883927451 and 12.9661788080, all simple and on the critical line, in agreement with Section 15.1.1; their largest gap, the first, 2.8876, lies below $2/\sqrt{V_{12}}=3.4662$, and below the value 3.3189 at $a=\frac12$ of the threshold of Remark 12.1.20 for χ_{12}.

Question 15.1.30. Is there a construction of $H_\alpha^K=H_\alpha^\xi+H_\alpha^{(4)}$ from the Gaussian primes themselves, two of norm p for each $p\equiv1$ and one of norm p^2 for each $p\equiv3 \pmod 4$, that persists below the basepoint one? By Proposition 15.1.27 such a construction must see the phases $p^{-i\theta}$, not only the sizes of the weights; below $\alpha=1$ its existence is equivalent to GRH for χ_4 under RH, by Corollary 15.1.29(iii). In the language of Section 15.1.7.4, the decomposition is the additivity $\arg\xi_K=\arg\xi+\arg\xi(\cdot,\chi_4)$ on vertical lines, and the domination of Corollary 15.1.29(ii) says that $\arg\xi_K-\arg\xi$ is nondecreasing on every line $\operatorname{Re}s=\alpha>\frac12$.

15.1.7.4 The Thorin measure as an argument

Proposition 15.1.31. *Let $\alpha>\frac12$ be such that no zero of $\xi(s,\chi_4)$ lies on the line $\operatorname{Re}s=\alpha$; this holds for every $\alpha\ge1$. Then for $\theta>0$*

$$\frac{1}{\pi}\int_0^\theta v_\alpha(t)\,dt=\frac{1}{\pi}\arg\xi(\alpha+i\theta,\chi_4),$$

with the argument continuous along the segment from α to $\alpha+i\theta$ and zero at α; for $\alpha\ge1$ the left side is the mass of $(0,\theta^2]$ under the Thorin measure of Proposition 15.1.25. As $\theta\to\infty$, with $a=\frac{\alpha+1}{2}$,

$$\frac{1}{\pi}\arg\xi(\alpha+i\theta,\chi_4)=\frac{\theta}{2\pi}\log\frac{2\theta}{\pi e}+\frac{\alpha}{4}+\frac{1}{\pi}\arg L(\alpha+i\theta,\chi_4)-\frac{a(a-1)+\frac16}{\pi\theta}+O(\theta^{-3}),$$

with $\arg L$ continuous along the same segment. As $\alpha\downarrow\frac12$, for θ not an ordinate, the left side tends to the number of zeros on the critical line with $0<\gamma<\theta$, which is $N(\theta)$ under GRH; at $\alpha=\frac12$ the expansion is the counting formula (15.1.8).

Proof. Since $\frac{d}{dt}\log\xi(\alpha+it,\chi_4)=i\frac{\xi'}{\xi}(\alpha+it,\chi_4)$, the derivative of $\arg\xi(\alpha+it,\chi_4)=\operatorname{Im}\log\xi(\alpha+it,\chi_4)$ is $v_\alpha(t)$, and $\xi(\alpha,\chi_4)>0$. The expansion is $\frac{\theta}{2}\log\frac{4}{\pi}$ plus Stirling's series $\log\Gamma(z)=(z-\frac12)\log z-z+\frac12\log2\pi+\frac{1}{12z}+O(|z|^{-3})$ [256, §5.11] at $z=a+\frac{i\theta}{2}$: with $y=\frac{\theta}{2}$,

$$\operatorname{Im}\log\Gamma(a+iy)=y\log y-y+\frac{\pi}{2}\Big(a-\frac12\Big)-\Big(\frac{a(a-1)}{2}+\frac{1}{12}\Big)\frac1y+O(y^{-3}),$$

from $\arg z = \frac{\pi}{2} - \frac{a}{y} + O(y^{-3})$ and $\log|z| = \log y + \frac{a^2}{2y^2} + O(y^{-4})$; and $\frac{\theta}{2}\log\frac{4}{\pi} + \frac{\theta}{2}\log\frac{\theta}{2} - \frac{\theta}{2} = \frac{\theta}{2}\log\frac{2\theta}{\pi e}$, $\frac{1}{2}(a-\frac{1}{2}) = \frac{\alpha}{4}$. For the limit, write v_α by the Hadamard product, as in Proposition 2.1.3, as the absolutely convergent sum over the zeros $\rho = \beta + i\gamma$ of $(\alpha-\beta)/((\alpha-\beta)^2 + (t-\gamma)^2)$; each term integrates over $[0,\theta]$ to $\arctan\frac{\theta-\gamma}{\alpha-\beta} + \arctan\frac{\gamma}{\alpha-\beta}$, which for a zero on the line tends to π if $0<\gamma<\theta$ and to 0 if $\gamma>\theta$ or $\gamma<0$, while the two terms of an off line pair $\frac{1}{2}\pm\delta+i\gamma$ have the limits $\mp\big(\arctan\frac{\theta-\gamma}{\delta} + \arctan\frac{\gamma}{\delta}\big)$, which cancel. The terms are dominated by π and by $C\theta/\gamma^2$ for $|\gamma| > 2\theta+1$, uniformly for $\frac{1}{2} < \alpha \le 1$, so dominated convergence applies. At $\alpha = \frac{1}{2}$, $a = \frac{3}{4}$ and $a(a-1) + \frac{1}{6} = -\frac{1}{48}$. □

The Thorin measure at resolution $\varepsilon = \alpha - \frac{1}{2}$ is therefore the zero counting function smoothed by the Cauchy kernel of width ε, the analogue for χ_4 of the cumulative Thorin mass of ξ (Section 3.3.1). At $\alpha = 0.75$ the difference between the left side and the first three terms of the expansion is $-0.0182/\theta$ at $\theta = 50$, 100 and 200, against $-(a(a-1)+\frac{1}{6})/\pi = -11/(192\pi) = -0.018237$ at $a = \frac{7}{8}$.

Corollary 15.1.32. *GRH for χ_4 holds if and only if, for every $\alpha > \frac{1}{2}$, $\theta \mapsto \arg\xi(\alpha+i\theta,\chi_4)$ is nondecreasing. With no hypothesis, the heights $\theta>0$ at which it decreases lie in the union of the windows $|\theta-\gamma| < \sqrt{\delta^2-(\alpha-\frac{1}{2})^2}$, each of length less than 1, over the zeros $\frac{1}{2}+\delta+i\gamma$ with $\delta > \alpha - \frac{1}{2}$, and there are none at heights below $2.5\times10^7 - \frac{1}{2}$.*

Proof. The derivative is $v_\alpha(\theta)$, and Proposition 15.1.25 gives the equivalence. By the Hadamard product, v_α is a sum of positive Cauchy kernels, one for each zero on the line and two for each off line pair $\frac{1}{2}\pm\delta+i\gamma$ with $\delta \le \alpha - \frac{1}{2}$, plus, for each pair with $\delta > \alpha - \frac{1}{2}$, the difference $\pi\big(P_{\varepsilon+\delta} - P_{\delta-\varepsilon}\big)(\theta\mp\gamma)$ with $\varepsilon = \alpha - \frac{1}{2}$, which is negative exactly where $|\theta\mp\gamma| < \sqrt{\delta^2-\varepsilon^2}$ (proof of Proposition 3.2.1(ii)); for $\theta>0$ and $\gamma>0$ the reflected term at $\theta+\gamma$ is positive, since $\theta+\gamma > \gamma > \delta$. A window has length $2\sqrt{\delta^2-\varepsilon^2} < 2\delta < 1$. Platt's verification of GRH for the modulus 4 to height $10^8/4$ [275] puts every zero off the line above 2.5×10^7, and so every window above $2.5\times10^7 - \frac{1}{2}$. □

A window holds a decrease only if the negative term of its pair outweighs all the positive terms there, and the decrease is bounded: the negative part of the term $\pi(P_{\varepsilon+\delta} - P_{\delta-\varepsilon})(\theta-\gamma)$ of a pair has integral $4\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} - \pi < \pi$, by Proposition 3.2.1(iii), so over any interval the argument falls by less than π times the number of pairs whose windows meet it, whatever the other zeros, and the fall caused by a pair tends to zero as $(\alpha-\frac{1}{2})/\delta \to 0$. This answers the quantitative half of a question of [283]. No lower bound for the positive terms over a window, such as a mean value estimate for the contribution of the zeros on the line, can exclude this for every pair: at the centre $\theta=\gamma$ of its window the pair contributes $\frac{1}{\varepsilon+\delta} - \frac{1}{\delta-\varepsilon} = -\frac{2\varepsilon}{\delta^2-\varepsilon^2}$, which is unbounded below as $\delta \downarrow \varepsilon$, that is as the zero approaches the line $\operatorname{Re} s = \alpha$ from the right, while by Lemma 14.3.9 a zero far

enough to the right of the line cannot produce a decrease against a given positive floor. Outside the strips $|\theta - \gamma| < (1 + \sqrt{2})\delta$ around the ordinates of the zeros off the line, Theorem 12.1.15 gives the argument a rate of increase that grows like $\log \theta$ (Corollary 12.1.16). Figure 15.5 shows the smoothing.

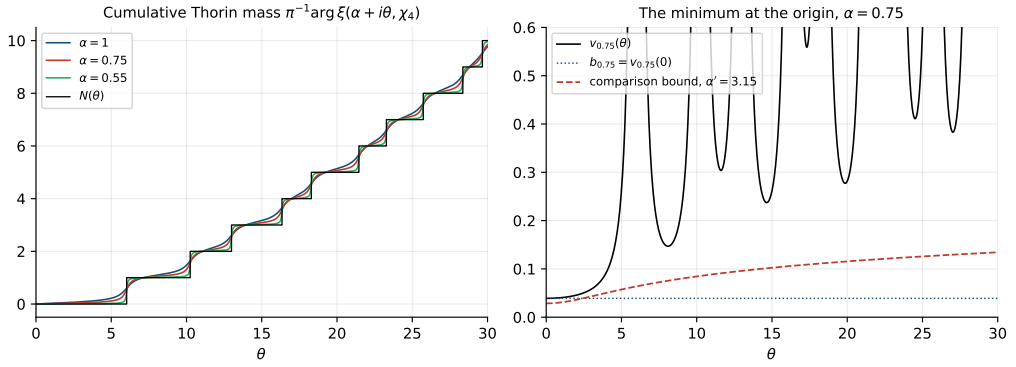


Figure 15.5: Left: the cumulative Thorin mass $\frac{1}{\pi} \arg \xi(\alpha + i\theta, \chi_4) = \frac{1}{\pi} \int_0^\theta v_\alpha$ for $\alpha = 1$, 0.75 and 0.55, against the zero count $N(\theta)$ of $L(s, \chi_4)$. Right: $v_{0.75}(\theta)$, its minimum $b_{0.75} = 0.0389784$ at the origin, and the lower bound of Corollary 12.1.16 with $\alpha' = 3.15$, which exceeds $b_{0.75}$ for $\theta > 2.434$.

15.1.7.5 The centre

For ζ, and likewise for χ_4, the positive structure at the centre is the Thorin measure of the clock, not a structure of the Pólya variable itself (Proposition 14.1.3(1)). The characteristic function $\xi(\frac{1}{2} + i\theta, \chi_4)/\xi(\frac{1}{2}, \chi_4)$ of Y_4 is real and changes sign at $\gamma_1 = 6.0209489$ and at each of the other ordinates below height 100; Hardy's argument for ζ [348, Chapter 10], applied to the theta series $\eta(ix)^3$ of χ_4, gives infinitely many such sign changes. So Y_4 is not quasi infinitely divisible, since such a characteristic function never vanishes. Nor is $\xi(\frac{1}{2} + w, \chi_4)/\xi(\frac{1}{2}, \chi_4)$ a positive mixture $\int M(d\lambda)/(w+\lambda)^2$ of Gamma(2) transforms, for a more elementary reason: such a mixture decreases on $w > 0$, while $\xi(\frac{1}{2} + w, \chi_4) = \int_{\mathbb{R}} K_4(u) \cosh(wu)\, du$ increases there. A one sided object remains, the half kernel $K_4(u)\mathbf{1}\{u > 0\}$, and it is not a positive gamma mixture either.

Proposition 15.1.33. *For no $a > 0$ is $u^{1-a}K_4(u)$ completely monotone on $u > 0$; in particular the half kernel is not a positive mixture of Gamma(a) densities $u^{a-1}e^{-\lambda u}$, for any shape a. The theta function $\eta(ix)^3$ is not completely monotone on $x > 0$.*

Proof. If $g(u) = \int_{[0,\infty)} e^{-\lambda u} M(d\lambda)$ with $M \geq 0$ and $M \neq 0$, choose L with $M([0, L]) > 0$; then $g(u) \geq M([0, L])e^{-Lu}$, so g decays at most exponentially,

while $u^{1-a}K_4(u)$ decays like $\exp(-\frac{\pi}{4}e^{2u})$ (Proposition 15.1.22). The Jacobi series $\eta(ix)^3 = \sum_{n\ge0}(-1)^n(2n+1)e^{-\pi(2n+1)^2x/4}$ is the Laplace transform of the signed measure with the weights $(-1)^n(2n+1)$ at the points $\pi(2n+1)^2/4$, which by the uniqueness of the Laplace transform is its only representing measure, and it is not positive, so Bernstein's theorem excludes complete monotonicity. □

The decay argument applies to the kernel of every completed L function of degree one, all of which decay like $\exp(-ce^{2|u|})$; for ξ it shows that no positive mixture of Gamma(a) densities represents the Pólya kernel on the half line, for any shape a, as [279, Section 13.6] observes (see also [282, Section 12.6]).

The positive structure is in the reciprocal of the product. With $q = e^{-2\pi x}$,

$$\frac{e^{-\pi x/4}}{\eta(ix)^3} = \prod_{m\ge1}\left(1-q^m\right)^{-3} = \exp\Big(3\sum_{N\ge1}\sigma_{-1}(N)\,e^{-2\pi Nx}\Big), \qquad \sigma_{-1}(N) = \sum_{d|N}\frac{1}{d}, \tag{15.1.11}$$

from $-\log(1-q^m) = \sum_k q^{mk}/k$; so the logarithm of the left side is completely monotone, and hence so is the left side. In the language of Section 15.1.4, $\prod_m(1-e^{-2\pi mx})^3$ is the distribution function of the maximum M_3 of independent exponential variables, three of each rate $2\pi m$, and the max representation carries over: since $\eta(ix)^3 = e^{-\pi x/4}\Pr(M_3\le x)$ and $x^{c-1}e^{-\pi x/4}(\pi/4)^c/\Gamma(c)$ is the density of $\frac{4}{\pi}G_c$, with G_c a gamma variable of shape c independent of M_3, the Mellin form gives $\xi(s,\chi_4) = (\frac{4}{\pi})^c\Gamma(c)\Pr(\frac{4}{\pi}G_c \ge M_3)$ with $c = \frac{s+1}{2}$, that is

$$L(s,\chi_4) = \Pr\big(4G_{(s+1)/2} \ge \pi M_3\big), \qquad s > -1,$$

which at $s=1$ is $\mathbb{E}e^{-\pi M_3/4} = \frac{\pi}{4}\int_0^\infty \eta(ix)^3dx = \frac{\pi}{4}$.

The half kernel has the Laplace transform

$$\int_0^\infty e^{-wu}K_4(u)\,du = A_4\big(\tfrac12 - w\big), \tag{15.1.12}$$

where

$$A_4(s) = \int_1^\infty \eta(ix)^3x^{(s+1)/2}\frac{dx}{x} = \sum_{n\ge1}\chi_4(n)\,n\,\frac{\Gamma(\frac{s+1}{2},\frac{\pi n^2}{4})}{(\pi n^2/4)^{(s+1)/2}}$$

is an incomplete Mellin transform of η^3, and the transformation $\eta(i/x)^3 = x^{3/2}\eta(ix)^3$, applied to the integral over $(0,1)$, gives $\xi(s,\chi_4) = A_4(s) + A_4(1-s)$ for every s. This is the analogue of Riemann's formula (Proposition 15.1.5) and of the symmetric form (15.1.5) for Δ, with no pole terms, since $\eta(ix)^3$ has no constant term.

Question 15.1.34. Can the complete monotonicity of (15.1.11), and of its analogue $\prod_m(1-q^m)^{-1}$ for the kernel of χ_{12}, be turned into a positive representation on the reciprocal side, in the spirit of Thorin's construction for the reciprocal of a GGC transform?

15.2 The Dedekind eta analogue at the basepoint one

The completed L function of χ_{12} (Section 15.1) is entire, a Mellin transform of the Dedekind eta function with a positive kernel. We repeat for it the calculations that Chapter 12 made for ζ at the basepoint one.

15.2.1 The completed L function and its Thorin measure

We use the completed function $\xi(s,\chi_{12}) = (12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12})$, its kernel K and the variable Y with density $K/\xi(\frac12,\chi_{12})$ of Section 15.1.1. Sign changes of $\xi(\frac12+it,\chi_{12})e^{\pi t/4}$, computed, give 297 zeros on the critical line below height 340. In this section $b_\alpha = (\xi'/\xi)(\alpha,\chi_{12})$ and $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha+i\theta,\chi_{12})$, and φ_α, T_1, $\widetilde{T}$, Π, A, A_α, Q and $a_k = -(\rho_k - \frac12)^2$ are formed with $\xi(s,\chi_{12})$ and its zeros $\rho = \beta + i\gamma$ in place of ξ. So is the tilted eta variable V_α, the Esscher tilt of Y by $e^{(\alpha-\frac12)Y}$, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s,\chi_{12})/\xi(\alpha,\chi_{12})$. The constants we need, computed, are

$$L(1,\chi_{12}) = \frac{\log(2+\sqrt3)}{\sqrt3} = 0.760346, \qquad L'(1,\chi_{12}) = 0.362495,$$

$$b_1 = \frac{\xi'}{\xi}(1,\chi_{12}) = \frac12\log\frac{12}{\pi} + \frac12\psi\Big(\frac12\Big) + \frac{L'}{L}(1,\chi_{12}) = \sum_\rho \frac{1}{\rho} = 0.1650833123,$$

where ψ is the digamma function. The Taylor coefficients of $\xi(\frac12+s,\chi_{12})/\xi(\frac12,\chi_{12})$ give the power sums $p_1 = \sum_k a_k^{-1} = 0.1664618$ and $p_2 = \sum_k a_k^{-2} = 0.0055985$, which are $\sum_{\gamma>0}\gamma^{-2}$ and $\sum_{\gamma>0}\gamma^{-4}$ under GRH; p_1 is the mean of the centre clock.

Proposition 15.2.1. *The function $\varphi_1(u) = \xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$ is the Laplace transform of a GGC with no drift, with Thorin density in $\theta = \sqrt{z}$*

$$\frac{1}{\pi}v_1(\theta) = \frac{1}{\pi}\,\mathrm{Re}\,\frac{\xi'}{\xi}(1+i\theta,\chi_{12}) = \frac{1}{\pi}\sum_\rho \frac{1-\beta}{(1-\beta)^2+(\theta-\gamma)^2} > 0,$$

the sum over the nontrivial zeros $\rho = \beta+i\gamma$ of $L(s,\chi_{12})$ in conjugate pairs. Positivity is unconditional because $L(1+it,\chi_{12}) \neq 0$.

Proof. As for Proposition 12.1.2, with the Hadamard product of the entire function $\xi(s,\chi_{12})$ in place of that of ξ. □

The minimum on $[0,60]$ is at the origin, $v_1(0)/\pi = b_1/\pi = 0.0525477$, and the cumulative mass tracks the zero count: $\int_0^{60} v_1/\pi = 35.81$ against $N_{\chi_{12}}(60) = 36$, where $N_{\chi_{12}}(T)$ counts the zeros with $0 < \gamma \le T$ as in Section 15.1.1. Table 15.7, computed, compares direct values with the Poisson sum over the 297 computed zeros; the nearly constant difference of 1.11×10^{-3} is the contribution of the zeros above height 340. Figure 15.6 compares the density with that of ζ: on this window the zeros of $L(s,\chi_{12})$ are more than twice as dense, and the density

correspondingly higher. As for ζ, $\varphi_1(u) = 1 - b_1\sqrt{u} + O(u)$, so T_1 has infinite mean and $P(T_1 > x) \sim b_1(\pi x)^{-1/2} = 0.09314\, x^{-1/2}$.

θ	$v_1(\theta)/\pi$ from $\xi'/\xi(\cdot, \chi_{12})$	from 297 zeros, $\beta = \frac{1}{2}$
0	0.0525477	0.0514359
3.8046276	0.6889076	0.6877958
5	0.1852757	0.1841637
10	0.2694467	0.2683341
30	0.7445558	0.7434362
60	0.7298854	0.7287413

Table 15.7: The Thorin density v_1/π of $\xi(1, \chi_{12})/\xi(1 + \sqrt{u}, \chi_{12})$, computed from $\xi'/\xi(\cdot, \chi_{12})$ and from the Poisson sum over the 297 zeros of $L(s, \chi_{12})$ below height 340 with $\beta = \frac{1}{2}$.

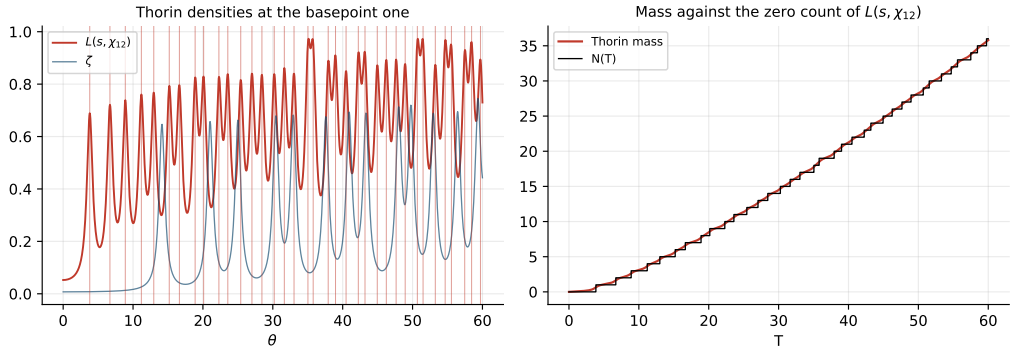


Figure 15.6: Left: the Thorin densities at the basepoint one for $L(s, \chi_{12})$ and for ζ on $[0, 60]$, with the ordinates of the zeros of $L(s, \chi_{12})$ (vertical lines). Right: the cumulative Thorin mass for $L(s, \chi_{12})$ against its zero count $N_{\chi_{12}}(T)$, written $N(T)$ in the plot.

15.2.2 The Lévy form without a pole, at and below the basepoint one

Proposition 15.2.2. *For* $\operatorname{Re} s > 0$,

$$\log \frac{\xi(1+s, \chi_{12})}{\xi(1, \chi_{12})} = b_1 s + \int_0^\infty (e^{-sx} - 1 + sx)\, \Pi(dx),$$

where, with $\Lambda_{\chi_{12}}(n) = \chi_{12}(n)\Lambda(n)$,

$$\Pi(dx) = \frac{e^{-x}}{x(1 - e^{-2x})}\, dx + \sum_{n \geq 2} \frac{\Lambda_{\chi_{12}}(n)}{n \log n}\, \delta_{\log n}(dx),$$

the prime sum converging conditionally, as a limit over $n \le X$.

Proof. Binet's formula [256, §5.9] for $\log\Gamma(\frac{1}{2}+\frac{s}{2})$, in the integrated form of the proof of Proposition 12.1.10 with $a = \frac{1}{2}$, gives the gamma density; the factor $(12/\pi)^{s/2}$ and the linear terms combine into b_1; and for $\operatorname{Re} s > 0$ the Euler product gives

$$\log L(1+s,\chi_{12}) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n\log n}\, e^{-s\log n}.$$

The linear coefficient is $-\sum_n \Lambda_{\chi_{12}}(n)/n = \frac{L'}{L}(1,\chi_{12})$, which converges by the prime number theorem for arithmetic progressions [70, Chapter 20], in the form $\psi(y,\chi_{12}) = O(y\,e^{-c\sqrt{\log y}})$, where $\psi(y,\chi_{12}) = \sum_{n\le y}\Lambda_{\chi_{12}}(n)$ is the Chebyshev function of χ_{12}. □

The form differs from that of ζ in two ways. There is no pole, so no compensating measure: the prime atoms converge on their own at the basepoint one. And the measure is signed for another reason. For ζ the gamma density minus the pole term dx/x is negative beyond $x = 0.281$ and the prime atoms are positive; here the gamma density $e^{-x}/(x(1-e^{-2x}))$ is positive everywhere and the atoms carry the sign of $\chi_{12}(n)$ (Figure 15.7, left). Table 15.8, computed, checks the form.

The tilted eta variable V_1 is not quasi infinitely divisible. Its positive and its negative atoms each have infinite total mass, since $\sum_{\chi_{12}(p)=1} 1/p$ and $\sum_{\chi_{12}(p)=-1} 1/p$ both diverge by Dirichlet's theorem, and they cancel only conditionally, so the argument of Proposition 14.1.3(3) applies with $\xi(1+s,\chi_{12})$ in place of $\xi(1+s)$, as the discussion after its proof explains. At every basepoint $\alpha > 1$, by contrast, V_α is quasi infinitely divisible: the atoms of each sign have finite total mass, and the Euler product writes its characteristic function as that of an infinitely divisible gamma piece times a quotient of two compound Poisson characteristic functions.

s	X	$b_1 s$	gamma piece	primes	total	exact
0.5	10^5	0.08254	0.121808	−0.082565	0.121785	0.122329
0.5	10^7	0.08254	0.121808	−0.081931	0.122419	0.122329
1	10^7	0.16508	0.409141	−0.254214	0.320011	0.320099
2	10^7	0.33017	1.270363	−0.689198	0.911332	0.911001

Table 15.8: The Lévy form of Proposition 15.2.2 against $\log(\xi(1+s,\chi_{12})/\xi(1,\chi_{12}))$, with primes up to X.

The sine squared lemma in the form of Section 12.1.4, with $\int_0^\infty \log(1+s^2/\theta^2)\,d\theta = \pi s$, converts the Lévy form, with $s = \sqrt{u}$, into

$$v_1(\theta) = b_1 + A(\theta) + Q(\theta),$$

with

$$A(\theta) = \tfrac{1}{2}\big[\operatorname{Re}\psi(\tfrac{1+i\theta}{2}) - \psi(\tfrac{1}{2})\big], \qquad Q(\theta) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n}\, 2\sin^2\Big(\frac{\theta \log n}{2}\Big).$$

The drift becomes the stable one half factor $e^{-b_1\sqrt{u}}$, the gamma factor the smooth positive growth A, and the primes the oscillating Q, with no divergent partner. With primes to 10^7 the formula gives 0.84633 at $\theta = 10$ against the exact 0.84649. Figure 15.7 (right) shows the three pieces: the prime piece takes both signs, and the positivity of the sum comes from the drift and the gamma factor, as Section 15.1.3 found at the basepoint two.

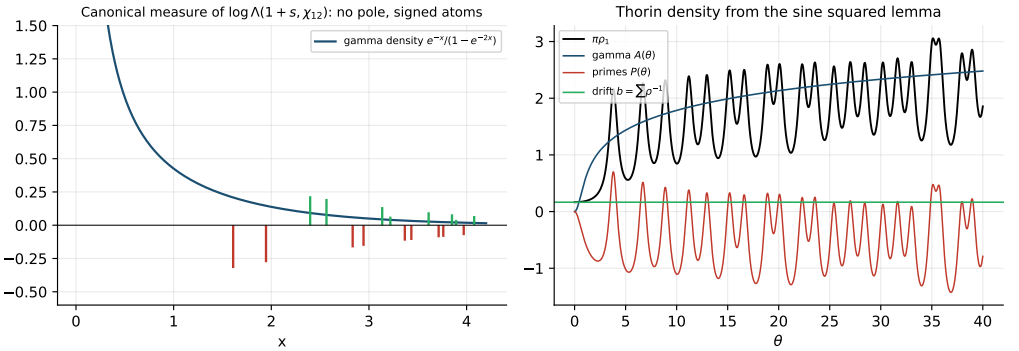


Figure 15.7: Left: the canonical measure $x\Pi(dx)$ of $\log \xi(1+s, \chi_{12})$, written with $\Lambda(1+s, \chi_{12})$ in the plot: the positive gamma density $e^{-x}/(1-e^{-2x})$ and the prime atoms $\Lambda_{\chi_{12}}(n)/n$ at $\log n$, of both signs, the positive ones drawn upward and the negative ones downward; there is no pole. Right: $v_1(\theta)$ on $[0, 40]$, written $\pi\rho_1$ in the plot, as the sum of the drift b_1, written b, the gamma piece $A(\theta)$ and the prime piece $Q(\theta)$, written $P(\theta)$ in the plot.

Below the basepoint one, at $\alpha = 1 - \varepsilon$, the converted Thorin density is π^{-1} times

$$b_\alpha + A_\alpha(\theta) + \lim_{X\to\infty} \sum_{n\le X} \frac{\Lambda_{\chi_{12}}(n)}{n^\alpha}\, 2\sin^2\Big(\frac{\theta \log n}{2}\Big),$$

with $A_\alpha(\theta) = \frac{1}{2}[\operatorname{Re}\psi(\frac{\alpha+i\theta}{2}) - \psi(\frac{\alpha}{2})]$. As for ζ in Proposition 12.1.13, the limit exists and gives the positive Thorin density v_α/π when $L(s, \chi_{12})$ has no zeros with $\operatorname{Re} s > \alpha'$ for some $\alpha' < \alpha$, and conversely its existence for $\operatorname{Re} s > 0$ in the Lévy form forces the absence of zeros with $\operatorname{Re} s > \alpha$. The error at cutoff X is given by the same explicit formula,

$$\sum_\rho \Big[-\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)}\Big] + O(X^{-\alpha}\log X),$$

because $\psi(y, \chi_{12}) = -\sum_\rho y^\rho/\rho + O(\log y)$ has no main term [70, Chapter 19].

For ζ the prime piece and the pole piece each grow like $X^{1-\alpha}/(1-\alpha)$ and must cancel; for $L(s, \chi_{12})$ the prime piece stays bounded. Table 15.9, computed, shows the errors, which decay like $X^{1/2-\alpha}$, and what remains after subtracting the explicit formula over the 297 computed zeros. The correction removes most of the error at $\theta = \gamma_1$; at $\theta = 10$, between zeros, it helps less, because 297 zeros are fewer than the 1700 used for ζ and the missing zeros above height 340 dominate what remains. At the centre and the first ordinate the sum grows like $\log X$: the atom δ_{γ_1} forming (Figure 15.8).

α	θ	X	prime piece	error	error after zero correction
1	γ_1	10^5	0.6964	-5.55×10^{-3}	4.2×10^{-6}
1	γ_1	10^7	0.7016	-4.15×10^{-4}	2.5×10^{-5}
0.75	γ_1	10^5	2.0933	-2.13×10^{-1}	1.4×10^{-4}
0.75	γ_1	10^7	2.2471	-5.92×10^{-2}	1.4×10^{-3}
0.6	γ_1	10^5	4.8292	-3.10	1.1×10^{-3}
0.6	γ_1	10^7	6.0670	-1.86	1.6×10^{-2}
0.6	10	10^7	-2.4525	-5.68×10^{-2}	2.6×10^{-2}

Table 15.9: The sine squared formula below the basepoint one for $L(s, \chi_{12})$, with primes up to X: the prime piece, the error against $v_\alpha(\theta)$, and the error left after subtracting the explicit formula over the 297 computed zeros; $\gamma_1 = 3.8046276$.

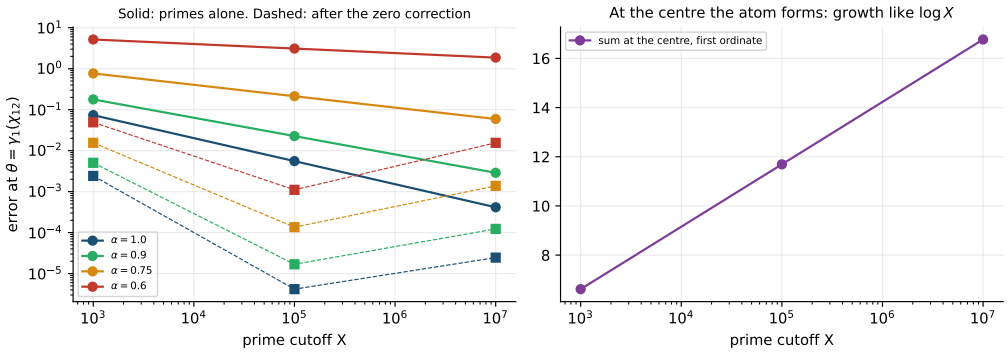


Figure 15.8: Left: absolute error at $\theta = \gamma_1 = 3.8046$ for $L(s, \chi_{12})$ against the prime cutoff X, for $\alpha = 1$, 0.9, 0.75 and 0.6, from primes alone (solid) and after the zero correction over 297 zeros (dashed). Right: at the centre and the first ordinate the sum grows like $\log X$, the atom of the GRH measure forming.

15.2.3 The hitting time and its boundary

We computed the Brownian boundary $b(t)$ whose crossing time has the law of T_1 from the Volterra equation (3.1.1), with the distribution function from Talbot inversion for $t < 0.12$ and, under GRH, from the subordination representation of Theorem 12.2.2 beyond; its time scale is about seven times that of ζ, the ratio of the means of the two centre clocks. In the right tail the exact boundary flattens at -0.2747, the Brownian form of the stable one half tail.

15.2.4 The two examples compared

Table 15.10 sets the two basepoint one examples side by side. Structurally the eta case is the cleaner: below the basepoint one its prime sums stay bounded, so the computation is far better conditioned. Nothing essential becomes easier, however. The negativity that ζ concentrates in the pole piece is carried by the signs of $\chi_{12}(p)$, and under GRH for χ_{12} the probabilistic picture is the same, every constant scaled by the denser zeros.

	ζ	$L(s, \chi_{12})$
kernel	theta series	$\eta(ix)$, a positive product
pole	at $s = 1$	none
first zero	14.134725	3.804628
drift $b_1 = \sum_\rho \rho^{-1}$	0.0230957	0.1650833
tail constant $b_1/\sqrt{\pi}$	0.01303	0.09314
mean of the centre clock $\sum_k a_k^{-1}$	0.0231050	0.1664618
Lévy measure, continuous part	gamma minus pole, negative beyond 0.281	gamma only, positive
Lévy measure, prime atoms	all positive	signs of $\chi_{12}(n)$
compensation at $\alpha = 1$	primes against dx/x (prime number theorem)	none needed
prime piece at $\alpha = 0.6$, $X = 10^7$, $\theta = \gamma_1$	about 1537, cancelled by the pole	6.07
flat level of the exact boundary	-0.19277	-0.27471
$\mathbb{E}\, b(T_1) = b_1/\sqrt{2}$	0.016331	0.116732
tangent approximation fails at	$t = 0.091$	$t = 1.83$

Table 15.10: The two basepoint one examples side by side: the reciprocal Xi function of Chapter 12 and the reciprocal completed L function of χ_{12} of this section.

The comparison confirms two conclusions that fix the shape of the path to RH. First, at the basepoint one the constructions for the reciprocal Xi function are, with two exceptions, unconditional and explicit. The exceptions are the Krein

string representation of Corollary 12.2.3, which assumes RH and simple zeros, and the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$: Lemma 12.2.4 gives it for a boundary bounded at infinity, and the boundedness of the computed Brownian boundary is observed, not proved. Second, below the basepoint one each positive statement for ζ holds, up to the boundary case, exactly when no zero lies to the right of the line $\operatorname{Re} s = \alpha$ (Propositions 12.1.13 and 13.1.12, Theorems 12.2.5 and 13.1.3). The eta example shows that the pole changes how the prime sums converge, not where they stop.

Chapter 16

Dirichlet L functions and the Linnik constant

For a number field F, the reciprocal of the completed Dedekind zeta function, tilted by $u_0 \geq 0$ and so read at $\frac{1}{2} + \sqrt{u_0}$, is a generalised gamma convolution (GGC) exactly when every nontrivial zero lies on the critical line or is real and at distance less than $\sqrt{u_0}$ from the centre, with no zero at the centre when $u_0 = 0$; its Thorin measure is then the zero counting measure shifted by u_0 (Theorem 16.2.3). For cyclotomic fields the Thorin property at the centre, for every modulus, is equivalent to GRH for all Dirichlet L functions together with $L(\frac{1}{2}, \chi) \neq 0$ for every primitive χ, so it implies the GRH value $K = 7$ of the Linnik constant (Corollary 16.7.1). This is weaker than the known conditional result, since the central nonvanishing is an extra hypothesis, and with the present methods it gives nothing beyond GRH (Section 16.1.2). Nothing here lowers the Linnik constant; statements resting on RH, GRH or the unrefereed half plane $\operatorname{Re} s > \frac{7}{8}$ say so.

16.1 Linnik's problem and where zeros enter

16.1.1 The theorem and the records

Linnik [214, 215] proved that there is an absolute constant K such that every sufficiently large even integer N can be written

$$N = p_1 + p_2 + 2^{k_1} + \cdots + 2^{k_K}, \qquad p_1, p_2 \text{ prime}, \quad k_i \in \mathbb{Z}_{\geq 0}. \tag{16.1.1}$$

The 1951 paper assumes GRH, the 1953 paper does not, and neither gives a value of K. Gallagher [90] recast the argument so that explicit constants became possible, and K was then lowered in a series of papers [218, 219, 209, 356, 210], from 54000 (Liu, Liu and Wang) to 1906 (Li), and under GRH to 200 (Liu, Liu and Wang), until Heath-Brown and Puchta [111] reached the first line of the table below.

Authors	K unconditionally	K under GRH
Heath-Brown and Puchta [111]	13	7
Pintz and Ruzsa [271]		7
Pintz and Ruzsa [272]	8	

Platt and Trudgian [276] sharpened the numerical constants in the method of [111].

16.1.2 Where K is determined

Fix a small constant $\varpi > 0$, write $e(x) = \exp(2\pi i x)$ and $L = \lfloor \log_2(N/2K) \rfloor$, and put

$$\mathcal{S}(\alpha) = \sum_{\varpi N < p \le N} e(p\alpha), \qquad \mathcal{D}(\alpha) = \sum_{1 \le \nu \le L} e(2^\nu \alpha).$$

The number of representations (16.1.1) with both primes in $(\varpi N, N]$ and the exponents restricted to $1 \le k_i \le L$, so that the powers of two add up to at most $N/2$, is

$$R(N) = \int_0^1 \mathcal{S}(\alpha)^2 \, \mathcal{D}(\alpha)^K \, e(-N\alpha) \, d\alpha. \tag{16.1.2}$$

Let $\mathfrak{M} \subset [0,1]$ be the set of major arcs, the neighbourhoods of rationals with small denominator, and $\mathfrak{m} = [0,1] \setminus \mathfrak{M}$ the minor arcs.

We follow the five steps of Heath-Brown and Puchta [111], quoting their constants with pinpoints.

(1) *Major arcs.* Gallagher's method, with the major arcs $q \le P$, $|\alpha - a/q| \le P/(qN)$ and $P \le N^{45/154 - 4\varpi}$, gives on $\mathfrak{M}$ a main term of at least

$$2.7895\,(1 - 3\varpi)\, C_0\, N (\log 2)^{-2} L^{K-2}$$

(Lemma 7 of [111], stated there for $K \ge 9$), where $C_0 = \prod_{p>2} \big(1 - (p-1)^{-2}\big) = 0.6601\ldots$ is the twin prime constant, and 2.7895 is $2\big(1 + \frac{1}{3}(1 - 2^{-6}) + \frac{1}{15}(1 - 4^{-6})\big) = 2.78955\ldots$ rounded down, the contribution of the divisors 1, 3 and 5 to the singular series of Goldbach's problem averaged over the powers of two (equation (25) there). No hypothesis on zeros and no explicit numerical zero free region is used. The error terms, character sums over primes, are controlled by the zero density estimates of Huxley and Jutila, $\sum_{r \le R} N(\sigma, r, T) \ll (R^2 T)^{\kappa(\sigma)(1-\sigma)}$ with $\kappa(\sigma) = \frac{3}{2-\sigma} + \varpi$ for $\frac{1}{2} \le \sigma \le \frac{3}{4}$ and $\frac{12}{5} + \varpi$ for $\frac{3}{4} \le \sigma \le 1$, which also fix the restriction on P, and, for the few characters with a zero near 1 and for the principal character, by the zero free regions of Iwaniec (for nonreal characters, with an implied constant that is not explicit) and of Vinogradov. Zeros close to $\sigma = 1$ play no role.

(2) *Minor arc mean square.* $\int_0^1 |\mathcal{S}(\alpha)\mathcal{D}(\alpha)|^2 d\alpha$ counts the solutions of $p_1 - p_2 = 2^\mu - 2^\nu$ with $\mu, \nu \le L$. Chen's sieve bound for the number of prime differences, $r(n) \le C_0 C_1 h(n) N (\log N)^{-2}$ with $C_1 = 7.8342$ (Theorem 3 of Chen's 1978 paper on Goldbach's problem and the sieve methods, as used in [111]), bounds the integral over $[0,1]$; subtracting a lower bound for the part over $\mathfrak{M}$ gives $\int_{\mathfrak{m}} |\mathcal{S}\mathcal{D}|^2 d\alpha \le (13.968 + o(1))\, C_0 N (\log 2)^{-2}$ (Lemmas 8 to 10 of [111]).

(3) *Minor arc sup bound for the primes.* $\mathcal{S}(\alpha) \ll N^{\theta + o(1)}$ on $\mathfrak{m}$, with $\theta = 263/308$ unconditionally, from Theorem 3.1 of Vaughan's book on the Hardy Littlewood method with $P = N^{45/154 - 4\varpi}$, and $\theta = 3/4$ under GRH, from Lemma 12 of Baker and Harman (Section 6 of [111]).

(4) *Large deviations for the powers of two.* The set $\mathcal{A}_\lambda = \{\alpha : |\mathcal{D}(\alpha)| \ge \lambda L\}$ has measure $\ll N^{-E(\lambda)}$, with $E(0.863665) > \frac{109}{154}$ and $E(0.722428) > \frac{1}{2}$ (Lemma 1 of [111], proved there by a martingale argument suggested by Keith Ball). The contribution of $\mathfrak{m} \cap \mathcal{A}_\lambda$ is $\ll N^{-E(\lambda) + 2\theta + o(1)} L^K$, negligible once $E(\lambda) > 2\theta - 1$, that is $\frac{109}{154}$ unconditionally and $\frac{1}{2}$ under GRH.

(5) *Comparison.* Off $\mathcal{A}_\lambda$, $|\mathcal{D}| < \lambda L$, so the contribution of $\mathfrak{m} \setminus \mathcal{A}_\lambda$ is at most $(\lambda L)^{K-2} \cdot 13.968\, C_0 N (\log 2)^{-2}$, and the count is positive, for N large and ϖ small, when $13.968\, \lambda^{K-2} < 2.7895$ (end of Section 6 of [111]). This needs $K - 2 > 10.99$ at $\lambda = 0.863665$ and $K - 2 > 4.95$ at $\lambda = 0.722428$, which gives $K = 13$ unconditionally and $K = 7$ under GRH.

A hypothesis on zeros enters only at step (3): GRH lowers θ to $\frac{3}{4}$ and hence λ. The remaining gains in [111] come from the constants of steps (1), (2) and (4), which use no hypothesis on zeros; Elsholtz is reported there to have obtained $K = 12$ unconditionally, in unpublished work, by improving the constants of the first two steps. The refinements of Pintz and Ruzsa [271, 272] differ from this argument in several respects, as [111] remark, and are not described. In the scheme of [111] an input delivering GRH is worth the passage from $K = 13$ to $K = 7$; against the records it is worth $K = 8$ to $K = 7$.

The zero free half plane of October 2026. OpenAI released in October 2026 a proof, produced with one of its AI models, that $\zeta(s)$ and every Dirichlet L function have no zeros in $\operatorname{Re} s > \frac{7}{8}$ [257], and a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$ [258]. The claims have not been refereed (Section 25.1.2). A zero free region can act in step (3), through the error term of the exponential sum, and in step (1), through the zero density estimates. In step (3) the exponent $\frac{3}{4}$ under GRH comes from the lemma of Baker and Harman used in [111]: for $\alpha = a/q + \beta$ with $|\beta| \le N^{-1/2}$, GRH gives $\mathcal{S}(\alpha) \ll (\log N)\{N/q + N^{1/2} q^{1/2} + N(q|\beta|)^{1/2}\}$. By Dirichlet's theorem every α has such an approximation with $q \le N^{1/2}$ and $|\beta| \le 1/(qN^{1/2})$, which makes each of the three terms at most $N^{3/4}$ once $q > N^{1/4}$,

and $q \le N^{1/4}$ does not occur on the minor arcs because $P = N^{45/154-4\varpi} \ge N^{1/4}$. In the second term GRH enters as the factor $N^{1/2}$, the size of the error term in the explicit formula for $\psi(x,\chi)$. Replacing GRH by a zero free half plane $\operatorname{Re} s > \theta_0$ for all Dirichlet L functions would replace that factor by N^{θ_0}, and the same choice of approximation, with $q \le N^{1-\theta_0}$, would give $\theta = \frac{1}{2}(1+\theta_0)$. This count is heuristic: the lemma is proved under GRH, and its half plane analogue is not proved here. It beats the unconditional exponent $\frac{263}{308}$ only if $\theta_0 < \frac{109}{154} = 0.7078$. The half plane $\operatorname{Re} s > \frac{7}{8}$ gives $\frac{15}{16}$, worse than the unconditional exponent, so used this way it does not lower K. In step (1) the constant 2.7895 is arithmetic and zeros close to $\sigma = 1$ play no role, so a half plane adds nothing by excluding exceptional zeros. Zeros enter the major arcs through the zero density estimates, which fix the restriction $P \le N^{45/154-4\varpi}$ by way of the bound $(\frac{45}{77}\kappa(\sigma) - 1)(1-\sigma) \le \frac{31}{77}(1-\sigma) \le \frac{31}{154}$ for the exponent of the estimate (Section 3 there), taken at the left end $\sigma = \frac{1}{2}$; a half plane $\operatorname{Re} s > \theta_0$ removes only the zeros to the right of θ_0, so it does not relax this restriction. Whether a finer combination of the half plane with Vaughan's identity does better, on the major or the minor arcs, is Problem 16.9. In the Thorin language of this chapter, the half plane would give the Thorin property of the shifted clock at every basepoint $\alpha > \frac{7}{8}$ for ξ_F with F abelian over $\mathbb{Q}$ (quadratic and cyclotomic fields in particular, by (16.5.4)) and for every real character (Theorem 16.5.8; for ξ this is the remark after Theorem 2.1.2). Liu [224] claims the half plane $\operatorname{Re} s > B_\flat = 0.87489$ for every Hecke L function of finite order over every number field (Section 23.1.3); if that unrefereed and unformalised claim stands, the same holds at every basepoint $\alpha > B_\flat$ for ξ_F with F any number field, abelian or not.

16.2 The Thorin property at every tilt

16.2.1 Conventions

The centre clock is the book's T (Section 2.1), the variables e_γ are independent standard exponentials as in the book, and $Y = \sqrt{2T}\,Z$, with Z a standard normal variable independent of T, is the variable $Y_{1/2}$ of Section 13.1. The letter T also denotes a height, in $N(T)$ and $S(T)$ and as the cut height of the anchored tail of Section 16.3.3. The transfer function from the real axis to the critical line is written $\mathfrak{T}(t)$ (Proposition 16.3.6), and $\mathfrak{B}(z)$ is its Blaschke product in $z = t^2$, so that $\mathfrak{T}(t) = \mathfrak{B}(t^2)$ (Proposition 16.3.8). Completed Dirichlet L functions are written $\xi(s,\chi)$ and Λ is the von Mangoldt function. The tilt u_0 and the basepoint α are different parameters: the clock tilted by u_0 reads ξ at $\frac{1}{2} + \sqrt{u_0+u}$, the shifted clock at basepoint α reads it at $\alpha + \sqrt{u}$, and the two agree only at the centre. In Section 16.1.2 only, α is also the circle method variable, and N, L, λ, θ, c and ϖ have their circle method meanings there.

16.2.2 Generalised gamma convolutions

A probability law on $[0,\infty)$ is a generalised gamma convolution (GGC) if its Laplace transform has the form

$$\varphi(s) = \exp\Big(- bs - \int_{(0,\infty)} \log\Big(1 + \frac{s}{t}\Big)\, U(dt)\Big), \qquad s \ge 0, \tag{16.2.1}$$

with $b \ge 0$ and a nonnegative Thorin measure U satisfying $\int_{(0,1]} |\log t|\, U(dt) < \infty$ and $\int_{(1,\infty)} t^{-1} U(dt) < \infty$; see Bondesson [39] and (4.1.1), where the drift is d and the Thorin variable z. Three standard facts are used.

(G1) The right side of (16.2.1) extends to an analytic, zero free function on $\mathbb{C} \setminus (-\infty, -t_0]$, where $t_0 = \inf \operatorname{supp} U$.

(G2) If $t_j > 0$ and $\sum_j 1/t_j < \infty$, then $\prod_j t_j/(t_j + s)$ is the Laplace transform of $\sum_j e_j/t_j$ with e_j independent unit exponentials, a GGC with $b = 0$ and $U = \sum_j \delta_{t_j}$.

(G3) If $\inf \operatorname{supp} U > c$ with $c > 0$, then $\varphi(-c) < \infty$ and $\varphi(s-c)/\varphi(-c)$ is again of the form (16.2.1), with Thorin measure $U(\cdot + c)$ (closure under exponential tilting).

The class is also closed under independent sums, positive scaling and weak limits (equation (4.1.1)), and, by a theorem of Bondesson [40], under products of independent GGC variables. It is not closed under products with arbitrary independent positive variables (an exponential times an independent two point variable is a mixture of two exponentials with rates $a \ne b$, whose Thorin measure would be the signed measure $\delta_a + \delta_b - \delta_c$ for a rate c between a and b), under harmonic means (Proposition 11.1.7), or under mixtures (Proposition 16.6.3).

16.2.3 Completed Dedekind zeta functions

Let F be a number field of degree $n = r_1 + 2r_2$ and discriminant d_F. Put $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ and

$$\xi_F(s) = s(s-1)\, |d_F|^{s/2}\, \Gamma_{\mathbb{R}}(s)^{r_1}\, \Gamma_{\mathbb{C}}(s)^{r_2}\, \zeta_F(s). \tag{16.2.2}$$

Then ξ_F is entire of order one, real on the real axis, and $\xi_F(s) = \xi_F(1-s)$. For $F = \mathbb{Q}$ it is $2\xi(s)$, with Riemann's $\xi(s) = \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta(s)$. Its zeros are the nontrivial zeros of ζ_F, written $\rho = \frac{1}{2} + w_\rho$ with $|\operatorname{Re} w_\rho| < \frac{1}{2}$.

Lemma 16.2.1 (Evenness). *$E_F(w) := \xi_F(\frac{1}{2} + w)$ is even and entire, so $E_F(w) = G_F(w^2)$ with G_F entire. The zeros of G_F are the points $z_\rho = w_\rho^2$, one for each pair $\{\rho, 1-\rho\}$.*

Proof. The functional equation gives $E_F(-w) = \xi_F(\frac{1}{2}-w) = \xi_F(\frac{1}{2}+w) = E_F(w)$, and an even entire function is an entire function of w^2. □

Lemma 16.2.2 (Product formula). *Let Z_F be the multiset of the values $z_\rho \neq 0$, one for each pair of zeros, and let $2m_F$ be the order of vanishing of ξ_F at $\frac{1}{2}$. Then, with absolute convergence since $\sum_{z\in Z_F} |z|^{-1} < \infty$,*

$$G_F(u) = c_F\, u^{m_F} \prod_{z\in Z_F} \Big(1 - \frac{u}{z}\Big), \qquad c_F \neq 0. \tag{16.2.3}$$

Proof. G_F is entire of order $\frac{1}{2}$ because E_F has order one, so Hadamard's theorem gives the product with no exponential factor. Convergence of $\sum |z_\rho|^{-1} = \sum |\rho - \frac{1}{2}|^{-2}$ follows from the Riemann von Mangoldt formula $N_F(T) \asymp T\log T$. □

Unlike G of Section 2.1, G_F is not normalised: $G_{\mathbb{Q}} = 2\xi(\frac{1}{2})\, G$. Under GRH every $z \in Z_F$ equals $-\gamma^2$ for an ordinate $\gamma > 0$, while a real zero $\beta \in (\frac{1}{2}, 1)$, which GRH excludes, would give $z = (\beta - \frac{1}{2})^2 \in (0, \frac{1}{4})$.

16.2.4 The equivalence theorem

For $u_0 \geq 0$ with $\xi_F(\frac{1}{2} + \sqrt{u_0}) \neq 0$, tilt the centre clock by u_0:

$$\varphi_{F,u_0}(s) := \frac{\xi_F(\frac{1}{2} + \sqrt{u_0})}{\xi_F(\frac{1}{2} + \sqrt{u_0 + s})} = \frac{G_F(u_0)}{G_F(u_0 + s)}, \qquad s \geq 0. \tag{16.2.4}$$

By Lemma 16.2.1 the branch of the square root does not matter. For $F = \mathbb{Q}$ this is the function $\widetilde{\varphi}_{u_0}$ of Section 8.1.1.

Theorem 16.2.3 (Thorin property of the tilted clock). *Let $u_0 \geq 0$. The following are equivalent.*

(i) $\xi_F(\frac{1}{2} + \sqrt{u_0}) \neq 0$ and φ_{F,u_0} is the Laplace transform of a GGC.

(ii) Every z_ρ is real with $z_\rho < u_0$, and $m_F = 0$ if $u_0 = 0$. Equivalently, every nontrivial zero of ζ_F lies on the critical line or is a real zero β with $|\beta - \frac{1}{2}| < \sqrt{u_0}$, and $\zeta_F(\frac{1}{2}) \neq 0$ when $u_0 = 0$.

When they hold, $b = 0$ and the Thorin measure is

$$U_{F,u_0} = \sum_{\gamma>0} \delta_{u_0+\gamma^2} + m_F\, \delta_{u_0} + \sum_{\beta} \delta_{u_0-(\beta-\frac{1}{2})^2}, \tag{16.2.5}$$

where $\frac{1}{2}+i\gamma$ runs over critical line zeros with $\gamma > 0$ and β over real zeros in $(\frac{1}{2}, 1)$, all with multiplicity.

Proof. (i) implies (ii). By (G1) the Laplace transform extends to an analytic zero free function on $\mathbb{C} \setminus (-\infty, 0]$, and it agrees with $G_F(u_0)/G_F(u_0+s)$, which is meromorphic, on $s > 0$. Hence $G_F(u_0+s)$ has no zeros off $(-\infty, 0]$, so every zero z of G_F satisfies $z - u_0 \le 0$, and $z \ne u_0$ because $G_F(u_0) \ne 0$. Imaginary w_ρ is a critical line zero, real w_ρ a real zero with $(\beta - \frac{1}{2})^2 < u_0$. At $u_0 = 0$ a central zero would make $G_F(0) = 0$.

(ii) implies (i). Under (ii), Lemma 16.2.2 gives

$$\varphi_{F,u_0}(s) = \Big(\frac{u_0}{u_0+s}\Big)^{m_F} \prod_{z \in Z_F} \frac{u_0 - z}{u_0 - z + s},$$

where every rate $u_0 - z$ is positive, equal to $u_0 + \gamma^2$ or to $u_0 - (\beta - \frac{1}{2})^2$, and the reciprocals are summable. Fact (G2) gives the GGC with Thorin measure (16.2.5). □

At $u_0 = 0$ the theorem is Proposition 13.3.6, which assumes $\zeta_F(\frac{1}{2}) \ne 0$. That is not automatic, since Armitage found fields with $\zeta_F(\frac{1}{2}) = 0$ [7]; here a central zero appears as the atom $m_F \delta_{u_0}$.

Corollary 16.2.4 (The tilts). *(a) At $u_0 = 0$: $\xi_F(\frac{1}{2})/\xi_F(\frac{1}{2} + \sqrt{s})$ is a GGC Laplace transform if and only if all nontrivial zeros of ζ_F lie on the critical line and $\zeta_F(\frac{1}{2}) \ne 0$.*

(b) For every $u_0 \ge \frac{1}{4}$, in particular at $u_0 = \frac{1}{4}$ and $u_0 = 1$, where ξ_F is read from 1 and from $\frac{3}{2}$: φ_{F,u_0} is a GGC Laplace transform if and only if every nontrivial zero lies on the critical line or on the real axis.

(c) Define the Thorin threshold $u_F^* = \max\{(\beta - \frac{1}{2})^2\}$ over real zeros $\beta \in [\frac{1}{2}, 1)$, with $u_F^* = 0$ if there are none. If all nonreal zeros are on the critical line, the Thorin property holds at tilt u_0 exactly for $u_0 > u_F^*$, and also at $u_0 = 0$ when there are no real zeros at all.

(d) When the property holds at two tilts $u_1 < u_0$, the measures are shifts of one another, $U_{F,u_0} = U_{F,u_1}(\cdot - (u_0 - u_1))$, and $\varphi_{F,u_1}(s) = \varphi_{F,u_0}(s - (u_0 - u_1))/\varphi_{F,u_0}(-(u_0 - u_1))$, the tilt (G3).

Proof. (a) and (b) are Theorem 16.2.3 at $u_0 = 0$ and for $u_0 \ge \frac{1}{4}$, since every real zero in $(0,1)$ has $(\beta - \frac{1}{2})^2 < \frac{1}{4}$. (c) restates the condition $z_\rho < u_0$ for the real values z_ρ. (d) follows from (16.2.5) and (16.2.4); for ξ it is the forward and backward tilt of Theorem 8.1.2(a). □

Remark 16.2.5 (The tilt and Landau's theorem). Suppose the Thorin property holds at the tilt 1. Descending from tilt 1 to tilt 0 is the tilt (G3) with $c = 1$, admissible exactly when $\varphi_{F,1}(-1) < \infty$. As $\varphi_{F,1}$ is the Laplace transform of a positive measure, Landau's theorem [359, Ch. II, Theorem 5b] puts its abscissa

of convergence at a real singularity. So $\varphi_{F,1}(-1) < \infty$ if and only if G_F has no zero in $[0,1]$, that is, ξ_F has no real zero in $[\frac{1}{2}, \frac{3}{2}]$. For ξ there is no real zero, so under RH the descent is possible; it is Theorem 8.1.2(a) (Section 8.1.1). Specific to number fields is that a central zero or a real zero in $(\frac{1}{2}, 1)$ blocks the descent.

Remark 16.2.6 (The Thorin measure is the zero counting measure). For $F = \mathbb{Q}$ there are no central or real zeros, since $\zeta(\sigma) < 0$ on $(0,1)$. By Corollary 16.2.4 the Thorin property of ξ at any tilt u_0 is equivalent to RH, with Thorin measure $\sum_{\gamma>0} \delta_{u_0+\gamma^2}$; at tilt $u_0 = 1$ the smallest atom is $1 + \gamma_1^2 = 200.7905$.

16.3 The centre clock: moments, transfer function and negative moments

16.3.1 The zeta function at $\sigma > 1$

Lemma 16.3.1. *For $\sigma > 1$ and $\mathrm{Re}(s) \geq 0$,*

$$\log \frac{\zeta(\sigma + s)}{\zeta(\sigma)} = \int_0^\infty (e^{-sx} - 1)\, \mu^\sigma(dx), \tag{16.3.1}$$

where $\mu^\sigma = \sum_{p,\, r \geq 1} r^{-1} p^{-r\sigma}\, \delta_{r \log p}$, the sum over primes p and integers $r \geq 1$, has total mass $\log \zeta(\sigma)$.

Proof. Take logarithms of the Euler product, expand $\log(1-z)^{-1} = \sum_{r \geq 1} z^r / r$, subtract $\log \zeta(\sigma)$ and write $p^{-rs} = e^{-rs \log p}$. □

Lemma 16.3.2. *Let N follow the Riemann zeta distribution $\mathrm{RZ}(\sigma)$, $\Pr(\mathsf{N} = n) = n^{-\sigma}/\zeta(\sigma)$, and $J^\sigma := \log \mathsf{N}$. Then J^σ is compound Poisson with Lévy measure μ^σ and $\mathbb{E}[e^{itJ^\sigma}] = \zeta(\sigma - it)/\zeta(\sigma)$.*

Proof. $\mathbb{E}[e^{itJ^\sigma}] = \zeta(\sigma)^{-1} \sum_n n^{-\sigma} n^{it}$, and (16.3.1) at $s = -it$ identifies the exponent. □

The Riemann zeta distribution and the infinite divisibility of $\log \mathsf{N}$ go back to Khinchin [187]; see Lin and Hu [212], Theorem 19.3.1 and Section 10.1.6.

Theorem 16.3.3 (Wiener spectral identity). *For $\sigma > 1$ and integer $k \geq 1$,*

$$\lim_{X \to \infty} \frac{1}{2X} \int_{-X}^{X} \left|\zeta(\sigma + it)\right|^{2k} dt = \sum_{n \geq 1} \frac{d_k(n)^2}{n^{2\sigma}}, \tag{16.3.2}$$

where $d_k(n)$ counts ordered factorisations $n = n_1 \cdots n_k$.

Proof. By Lemma 16.3.2, $|\zeta(\sigma+it)|^{2k} = \zeta(\sigma)^{2k} \mathbb{E}[e^{itV}]$ with $V = \sum_{i \leq k} J_i - \sum_{i \leq k} J_i'$ for independent copies of J^σ. Wiener's theorem on the atom of a measure at 0 [361] gives the limit $\Pr(V = 0) = \Pr(\mathsf{N}_1 \cdots \mathsf{N}_k = \mathsf{N}_1' \cdots \mathsf{N}_k')$. The product $\mathsf{N}_1 \cdots \mathsf{N}_k$ has law $d_k(m) m^{-\sigma}/\zeta(\sigma)^k$, so the collision probability is $\zeta(\sigma)^{-2k} \sum_m d_k(m)^2 m^{-2\sigma}$. □

Corollary 16.3.4. *The right side of* (16.3.2) *is* $\zeta(2\sigma)$ *for* $k=1$, $\zeta(2\sigma)^4/\zeta(4\sigma)$ *for* $k=2$ *(through Ramanujan's identity), and in general the Euler product*

$$\prod_p \sum_{r\ge 0} \binom{r+k-1}{k-1}^2 p^{-2r\sigma}.$$

These are the classical mean value theorems of Carlson ($k=1$) and Ingham ($k=2$), which hold for $\sigma > \frac{1}{2}$ (see [348, Chapter VII]); for $k \ge 3$ they are known only on a smaller range of σ, and on the whole range $\sigma > \frac{1}{2}$ they follow from the Lindelöf hypothesis [348, Chapter XIII].

16.3.2 The Riemann case in detail

Take $F=\mathbb{Q}$ and $u_0=0$, and assume RH. By Corollary 16.2.4,

$$\frac{\xi(\frac12)}{\xi(\frac12+\sqrt{u})} = \mathbb{E}\big[e^{-uT}\big], \qquad T \overset{d}{=} \sum_{\gamma>0} \frac{e_\gamma}{\gamma^2}, \qquad u \ge 0, \tag{16.3.3}$$

with independent standard exponential variables e_γ. The measure $\sum_{\gamma>0}\delta_{\gamma^2}$ is the measure U_{RH} of Proposition 2.1.1, which Chapters 3 to 8 compute at finite resolution. Without RH the clock still exists, as a self decomposable variable (Theorem 8.1.1), with the same Laplace transform, and the moment identities, the Mellin transform and the interpolation below hold for it with γ^2 replaced by $a_k = -(\rho_k - \frac12)^2$, the numbers of the product $G(u) = \prod_k (1+u/a_k)$ of equation (2.1.1).

16.3.2.1 The normal scale mixture and the moments

Define $Y := \sqrt{2T}\,Z$ with $Z \sim N(0,1)$ independent of T. Conditioning on T,

$$\mathbb{E}\big[e^{itY}\big] = \mathbb{E}\big[e^{-t^2T}\big] = \frac{\xi(\frac12)}{\xi(\frac12+|t|)}, \tag{16.3.4}$$

which evaluates ξ on the real axis, where it is positive, even and super exponentially growing. This is the variable $Z\sqrt{2T}$ of Section 2.1 and $X = Z\sqrt{2Y}$ of Proposition 22.5.1, where Chapter 22 writes Y for the clock. The moments of T are the Taylor coefficients of (16.3.3): with $p_j := \sum_{\gamma>0}\gamma^{-2j}$,

$$\mathbb{E}[T^k] = k!\,h_k\big(\{\gamma^{-2}\}\big), \qquad \mathbb{E}[Y^{2k}] = (2k-1)!!\,2^k\,\mathbb{E}[T^k], \tag{16.3.5}$$

where h_k is the complete homogeneous symmetric polynomial. In particular $\mathbb{E}[T]=p_1$ and $\mathrm{Var}(T)=p_2$.

16.3.2.2 The Hadamard product and the transfer function

The same factorisation gives the critical line value

$$\frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2})} = \prod_{\gamma>0}\Big(1-\frac{t^2}{\gamma^2}\Big), \tag{16.3.6}$$

which alternates in sign; read in $u = t^2$ it is the Laplace transform of no measure, while in t it is the characteristic function of the Pólya density (Section 2.1). Multiplying (16.3.4) by (16.3.6) gives

$$\mathbb{E}[e^{itY}]\cdot\frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2})} = \frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2}+|t|)} = \prod_{\gamma>0}\frac{\gamma^2-t^2}{\gamma^2+t^2}, \tag{16.3.7}$$

with zeros at $\pm\gamma$ and, continued in t, poles at $\pm i\gamma$, where the Laplace transform of T stops being analytic. The left side is the characteristic function of Y plus an independent Pólya variable, so the middle term is a characteristic function with no hypothesis; the product on the right is its form under RH. The critical line moments $\int_0^X |\zeta(\frac{1}{2}+it)|^{2k}dt$ are governed by (16.3.6), not by Y; the transfer function links the two and has a closed form (Proposition 16.3.6).

16.3.3 Closed forms and their verification

16.3.3.1 The moment identity

$\mathbb{E}[T^k]$ is computed two ways. The zero side uses (16.3.5), with p_j summed over the first 800 ordinates plus the Riemann von Mangoldt tail $\int_{\gamma_{800}}^{\infty} t^{-2j}\,\frac{1}{2\pi}\log\frac{t}{2\pi}\,dt$, started at the ordinate $\gamma_{800} = 1183.7128$. The function side uses the Taylor expansion of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+w)$ in powers of w^2 at $w=0$, from the Taylor coefficients of $\xi(\frac{1}{2}+s)$ at $s=0$ (Table 16.1). The residual is mostly the error of starting the

Table 16.1: The moments $\mathbb{E}[T^k]$ from the zeros (800 ordinates and the smooth tail from γ_{800}) and from the function (100 digits).

k	zero side	function side	relative error
1	0.0231054	0.0231050	1.9×10^{-5}
2	5.71033×10^{-4}	5.71013×10^{-4}	3.5×10^{-5}
3	1.52001×10^{-5}	1.51994×10^{-5}	4.9×10^{-5}
4	4.38851×10^{-7}	4.38824×10^{-7}	6.0×10^{-5}

smooth tail at an ordinate: with the first 800 ordinates and the tail started at the midpoint of γ_{800} and γ_{801} the relative errors for $k=1,\dots,4$ are 3.1×10^{-7}, 5.8×10^{-7}, 8.1×10^{-7} and 1.0×10^{-6}, and with the anchored tail below they are 8.8×10^{-10}, 1.6×10^{-9}, 2.3×10^{-9} and 2.8×10^{-9}. The function side computes

$\sum_k a_k^{-j}$ with no hypothesis (the cumulants of T from $G(u) = \prod_k (1 + u/a_k)$, Section 14.2.3), so the agreement checks the zero list and the tail. One missing zero at γ_{800} would change $\mathbb{E}[T]$ by 3.1×10^{-5} in relative terms, the order of the residuals in Table 16.1, so the table alone cannot exclude a missing zero; the anchored residual is smaller than that change by a factor of 3.5×10^4, and shows that no zero below the cut is missing. It does not test that the zeros lie on the line: moving a pair of zeros at height γ_{800} off the line by 0.25 changes p_1 by 8.3×10^{-12} in relative terms, below every residual. Nor does it check the Thorin property, which no finite computation can establish (Theorem 1.4.1).

Anchoring the tail. Most of the residual comes from the fluctuation $S(T)$ of the zero count at the cut. Write $N(t) = N_0(t) + 1 + S(t)$ with $N_0 = \vartheta/\pi$. Integration by parts gives, exactly,

$$\sum_{\gamma>T} \gamma^{-2j} = \int_T^\infty t^{-2j}\, \frac{\vartheta'(t)}{\pi}\, dt + \big(N_0(T) + 1 - N(T)\big) T^{-2j} + 2j \int_T^\infty S(t)\, t^{-2j-1}\, dt. \tag{16.3.8}$$

The smooth tail keeps only the first term, with the error $-S(T)T^{-2j}$ of the second, of order $T^{-2j}\log T$ at most. Since $\int_T^X S(t)\,dt = O(\log X)$ [348, Section 9.9], the third term is $O(jT^{-2j-1}\log T)$, so the anchored tail, which keeps the first two terms, typically gains a factor of the order of the cut height. Table 16.2 gives the relative errors of p_1, p_2, p_3 against the values of the function side, for the first N ordinates and the cut at the midpoint of γ_N and γ_{N+1}, so that $N(T) = N$. The

Table 16.2: Relative errors of p_1, p_2, p_3 for the smooth and the anchored tail with the first N ordinates; the cuts are $T = 1183.1475$ ($S(T) = 0.083$) for $N = 799$, $T = 397.1503$ ($S(T) = 0.244$) for $N = 200$ and $T = 680.8186$ ($S(T) = -0.211$) for $N = 400$ (100 digits).

N	tail	p_1	p_2	p_3
799	smooth	2.6×10^{-6}	1.1×10^{-9}	2.1×10^{-13}
	anchored	6.3×10^{-9}	5.6×10^{-12}	1.6×10^{-15}
200	smooth	6.7×10^{-5}	2.6×10^{-7}	4.3×10^{-10}
	anchored	6.8×10^{-8}	5.1×10^{-10}	1.2×10^{-12}
400	smooth	2.0×10^{-5}	2.6×10^{-8}	1.4×10^{-11}
	anchored	1.4×10^{-7}	3.7×10^{-10}	3.1×10^{-13}

gain is typical, not guaranteed: the smooth error is $|S(T)|T^{-2j}$ to 0.1%, so the gain is large only when $|S(T)|$ is not small. Over the 1650 cuts $N = 50$ to 1699 the ratio of the smooth to the anchored error of p_1 has median 354, about a third of the height of the cut, and deciles 42 and 2700, and it is below 10 at 3.2% of the cuts. This is the first half of Problem 16.3.

16.3.3.2 The tilt by one quarter and a classical constant

The classical evaluation

$$\sum_{\rho} \frac{1}{\rho} = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi = 0.0230957089661\ldots,$$

with γ_E Euler's constant, which under RH is $\sum_{\gamma>0}(\gamma^2+\frac14)^{-1}$, differs from $\mathbb{E}[T] = 0.0231049931\ldots$ by about 9.3×10^{-6}. The shift by $\frac14$ is the tilt $u_0 = \frac14$. The constant is the mean of this tilted clock (Section 12.2, after Corollary 12.2.3) and π times the Thorin density at the origin of the shifted clock at basepoint one (Proposition 12.1.2); new here is its expansion (16.3.11) in the cumulants.

Proposition 16.3.5 (Esscher tilt). *Assume* (16.3.3) *and let* $u_0 > 0$. *The tilted law* $\Pr_{u_0}(dh) = e^{-u_0 h}\Pr(T \in dh)/\mathbb{E}[e^{-u_0 T}]$ *is the GGC with Thorin measure* $U_{\mathbb{Q},u_0} = \sum_{\gamma>0}\delta_{u_0+\gamma^2}$ *of Theorem 16.2.3, which is the forward tilt of Theorem 8.1.2(a), and*

$$\mathbb{E}\big[e^{-u_0 T}\big] = \frac{\xi(\frac12)}{\xi(\frac12+\sqrt{u_0})}, \tag{16.3.9}$$

$$\mathbb{E}_{u_0}[T] = \sum_{\gamma>0}\frac{1}{\gamma^2+u_0} = \frac{1}{2\sqrt{u_0}}\,\frac{\xi'}{\xi}\big(\tfrac12+\sqrt{u_0}\big). \tag{16.3.10}$$

At $u_0 = \frac14$, *since* $\xi(1) = \frac12$,

$$\mathbb{E}\big[e^{-T/4}\big] = 2\,\xi(\tfrac12) = 0.99424155637\ldots,$$
$$\mathbb{E}_{1/4}[T] = \frac{\xi'(1)}{\xi(1)} = 1 + \frac{\gamma_E}{2} - \frac12\log 4\pi.$$

Proof. $\mathbb{E}_{u_0}[e^{-vT}] = \mathbb{E}[e^{-(u_0+v)T}]/\mathbb{E}[e^{-u_0T}] = \prod_{\gamma>0}(\gamma^2+u_0)/(\gamma^2+u_0+v)$, which is $\varphi_{\mathbb{Q},u_0}(v)$. Equation (16.3.9) is (16.3.3) at $u = u_0$, and $\mathbb{E}_{u_0}[T] = -\frac{d}{du}\log\mathbb{E}[e^{-uT}]$ at u_0, computed either from the product or from $\log\xi(\frac12+\sqrt{u})$. At $u_0 = \frac14$ the evaluation point is $\frac12+\sqrt{u_0} = 1$, where $\xi(1) = \xi(0) = \frac12$ and $\xi'(1)/\xi(1) = \sum_\rho \rho^{-1}$. □

From the first 800 zeros with the anchored tail, $\prod_{\gamma>0}\gamma^2/(\gamma^2+\frac14) = 0.99424155637$ and $\sum_{\gamma>0}(\gamma^2+\frac14)^{-1} = 0.02309570899$, within 5×10^{-12} and 2×10^{-11} of $2\xi(\frac12)$ and of the constant. Expanding $(\gamma^2+\frac14)^{-1}$ in powers of γ^{-2}, valid since $\gamma_1 > \frac12$,

$$1 + \frac{\gamma_E}{2} - \frac12\log 4\pi \;=\; \sum_{j\ge1}(-1)^{j+1}4^{1-j}\,p_j, \tag{16.3.11}$$

ties it to the whole cumulant sequence of T; with the p_j of the function side it agrees with the constant to a relative 2×10^{-30} after ten terms, the size of the first omitted term.

16.3.3.3 The transfer function

PROPOSITION 16.3.6. *Assume RH. For all real t,*

$$\mathfrak{T}(t) \;:=\; \frac{\xi(\frac12+it)}{\xi(\frac12+|t|)} \;=\; \prod_{\gamma>0}\frac{\gamma^2-t^2}{\gamma^2+t^2}, \tag{16.3.12}$$

and $\mathfrak{T}$ is real, even, bounded by 1 in modulus, and vanishes exactly at $t=\pm\gamma$. For $|t|<\gamma_1$,

$$\log\mathfrak{T}(t) \;=\; -2\sum_{\gamma>0}\operatorname{artanh}\Big(\frac{t^2}{\gamma^2}\Big) \;=\; -2\sum_{m\ge1}\frac{p_{2m-1}}{2m-1}\,t^{4m-2}, \tag{16.3.13}$$

which involves only the odd index power sums $p_1,p_3,p_5,\dots$.

Proof. Lemma 16.2.2 gives $\xi(\frac12+s)/\xi(\frac12)=\prod_{\gamma>0}(1+s^2/\gamma^2)$ for all $s\in\mathbb{C}$. Dividing the values at $s=it$ and $s=|t|$ gives (16.3.12), each factor real with modulus at most 1. For (16.3.13), $\log\frac{\gamma^2-t^2}{\gamma^2+t^2}=-2\operatorname{artanh}(t^2/\gamma^2)$; expand and interchange the absolutely convergent double sum. □

PROPOSITION 16.3.7. *Assume RH. The odd index power sums determine the zeros: the series $\sum_{m\ge1}p_{2m-1}z^{2m-1}$ converges for $|z|<\gamma_1^2$ and continues to the meromorphic function*

$$\sum_{m\ge1}p_{2m-1}z^{2m-1}=\frac12\sum_{\gamma>0}\Big(\frac{1}{1-z/\gamma^2}-\frac{1}{1+z/\gamma^2}\Big),$$

whose poles are the points $\pm\gamma^2$, each with residue $-\frac12\gamma^2$ times the multiplicity of γ. Hence $\{p_{2m-1}\}$ determines $\{\gamma\}$ with multiplicities, and with it every p_j.

Proof. For $|z|<\gamma_1^2$, $\sum_m(z/\gamma^2)^{2m-1}=(z/\gamma^2)/(1-z^2/\gamma^4)$, which is half the bracket; summing over γ and interchanging is justified by $\sum\gamma^{-2}<\infty$. The sum over γ converges locally uniformly off $\{\pm\gamma^2\}$, since the bracket is $O(|z|\gamma^{-2})$ for $\gamma^2>2|z|$, so it defines a meromorphic function with the stated poles. A meromorphic function is determined by its Taylor coefficients at 0, and its poles and residues give back the multiset $\{\gamma^2\}$. □

Numerically, $\mathfrak{T}(2)=0.8312$, $\mathfrak{T}(5)=0.3145$ and $\mathfrak{T}(13)=1.7242\times10^{-4}$, the last close to the first zero $\gamma_1\approx14.13$. The Hadamard product over the first 1699 ordinates with the anchored tail reproduces these three values, computed from ξ at the two points $\frac12+it$ and $\frac12+t$, to relative errors 3.8×10^{-10}, 2.4×10^{-9} and 1.6×10^{-8}.

PROPOSITION 16.3.8. *Let $\mathfrak{B}(z)=\prod_{\gamma>0}(\gamma^2-z)/(\gamma^2+z)$, so $\mathfrak{T}(t)=\mathfrak{B}(t^2)$. Under RH, $\mathfrak{B}$ is a Blaschke product for the right half plane: it converges there, $|\mathfrak{B}|<1$ for $\operatorname{Re}(z)>0$, $|\mathfrak{B}|=1$ for $\operatorname{Re}(z)=0$, its zeros are the points γ^2, and it has no singular inner factor.*

Proof. Each factor $b_\gamma(z) = (\gamma^2 - z)/(\gamma^2 + z)$ is a Cayley map taking $\mathrm{Re}(z) > 0$ into the unit disc with $|b_\gamma| = 1$ on $\mathrm{Re}(z) = 0$. Since $1 - b_\gamma(z) = 2z/(\gamma^2 + z)$ and $\sum \gamma^{-2} < \infty$, the product converges locally uniformly away from $\{-\gamma^2\}$; this is the half plane Blaschke condition. A convergent product of Blaschke factors has no singular inner factor, and since the product of Lemma 16.2.2 has no exponential factor, $\mathfrak{B}(z) = \xi(\frac{1}{2} + i\sqrt{z})/\xi(\frac{1}{2} + \sqrt{z})$. □

Remark 16.3.9 (Without RH). Let $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2}) = \prod_k (1 + u/a_k)$ be the function of Section 2.1 (equation (2.1.1)), with $a_k = -(\rho_k - \frac{1}{2})^2$ and one factor for each pair $\{\rho_k, 1 - \rho_k\}$ of zeros. Then $\mathfrak{T}(t) = G(-t^2)/G(t^2)$ for real t with no hypothesis, and $\mathfrak{B}(z) := G(-z)/G(z) = \prod_k (a_k - z)/(a_k + z)$ is still a Blaschke product for the right half plane, with zeros at the points $a_k = -(\rho_k - \frac{1}{2})^2$. They lie in the open right half plane, since $\mathrm{Re}\, a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > 0$ for $\rho_k = \beta_k + i\gamma_k$ ($|\beta_k - \frac{1}{2}| < \frac{1}{2}$ and $|\gamma_k| \geq \gamma_1$). The product converges because $1 - (a_k - z)/(a_k + z) = 2z/(a_k + z)$ and $\sum_k |a_k|^{-1} < \infty$ (Lemma 16.2.2). RH is the statement that these zeros are real, hence equal to the γ^2, not that $\mathfrak{B}$ is inner. In particular $|\mathfrak{B}(iy)| = 1$ is an identity for every real entire G, and the reality of $\mathfrak{T}(t)$ for real t is the functional equation; a numerical check of either tests rounding only.

16.3.3.4 Negative moments as a Mellin transform

Proposition 16.3.10. *For* $\mathrm{Re}(s) > 0$*, with no hypothesis on the zeros,*

$$\begin{aligned} \mathbb{E}\big[T^{-s}\big] &= \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \, \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + \sqrt{u})} \, du \\ &= \frac{2\,\xi(\frac{1}{2})}{\Gamma(s)} \int_0^\infty \frac{w^{2s-1}}{\xi(\frac{1}{2} + w)} \, dw. \end{aligned} \tag{16.3.14}$$

Proof. Since $T > 0$ almost surely, $T^{-s} = \Gamma(s)^{-1} \int_0^\infty u^{s-1} e^{-uT} \, du$. Take expectations (Fubini by positivity), substitute the Laplace transform (16.3.3), which holds without RH, and put $u = w^2$. The integral converges at 0 for $\mathrm{Re}(s) > 0$ and at ∞ for every s. □

In a computation at 30 digits the two forms agree to 10^{-18} at $s = \frac{1}{2}$ and to 1.4×10^{-27} at $s = \frac{3}{4}$; at $s = \frac{1}{4}$ the integrand $u^{-3/4}$ of the first form is singular at the origin, and the value below is that of the second form. The values are

$$\mathbb{E}[T^{-1/2}] = 6.7332723127566\ldots,$$
$$\mathbb{E}[T^{-1/4}] = 2.590049154\ldots, \qquad \mathbb{E}[T^{-3/4}] = 17.567694990\ldots.$$

The value $\mathbb{E}[T^{-1/2}]$ is a spectral average weighted toward the low lying zeros; Problem 16.4 compares it with random matrix theory.

PROPOSITION 16.3.11. *With no hypothesis on the zeros, $s \mapsto \mathbb{E}[T^{-s}]$ is an entire function, equal for $\operatorname{Re} s > 0$ to the Mellin integral* (16.3.14)*, and at the nonpositive integers it takes the values of the positive moments:*

$$\mathbb{E}[T^{-s}]\big|_{s=-n} \;=\; \mathbb{E}[T^n], \qquad n = 0, 1, 2, \dots \tag{16.3.15}$$

Proof. Since $T > 0$ almost surely and T has all positive and all negative moments, $\mathbb{E}|T^{-s}| = \mathbb{E}[T^{-\operatorname{Re} s}] < \infty$ for every s, so $\mathbb{E}[T^{-s}]$ is entire and (16.3.15) holds by definition. The residues show that the continuation of the Mellin formula is this function. Write $M(s) = \int_0^\infty u^{s-1}\varphi(u)\,du$ with $\varphi(u) = \xi(\tfrac12)/\xi(\tfrac12 + \sqrt{u}) = \sum_{n\ge 0} c_n u^n$ near 0, $c_n = (-1)^n\mathbb{E}[T^n]/n!$. Subtracting Taylor polynomials continues M to a meromorphic function with simple poles at $s = -n$ and $\operatorname{Res}_{s=-n} M = c_n$. Since $1/\Gamma$ is entire with simple zeros exactly there, M/Γ is entire and

$$\mathbb{E}[T^{-s}]\big|_{s=-n} = \frac{\operatorname{Res}_{s=-n} M(s)}{\operatorname{Res}_{s=-n}\Gamma(s)} = \frac{(-1)^n\mathbb{E}[T^n]/n!}{(-1)^n/n!} = \mathbb{E}[T^n].$$

□

The entire function $\mathbb{E}[T^{-s}]$ is not of exponential type, because the radius of convergence of $\sum_n \mathbb{E}[T^n]\lambda^n/n!$ is γ_1^2 (Theorem 14.2.3(a)), so $\mathbb{E}[T^n]$ is not bounded by any geometric sequence, whereas the values of a function of exponential type along the integers grow at most geometrically. On every vertical line $|\mathbb{E}[T^{-s}]| \le \mathbb{E}[T^{-\operatorname{Re} s}]$. No uniqueness theorem of Carlson type is claimed.

16.3.3.5 Total positivity of the density of Y

The density of $Y = \sqrt{2T}\,Z$ is

$$f(y) = \frac{1}{\pi}\int_0^\infty \cos(ty)\,\frac{\xi(\frac12)}{\xi(\frac12 + t)}\,dt,$$

computed from ξ on the real axis alone. By Proposition 22.5.1, f is unconditionally a smooth symmetric density, and RH holds if and only if f is a Pólya frequency function of infinite order. One direction is Theorem 20.1.4(iv); the equivalence, in the language of the Laguerre Pólya class, is in [192, Theorem 2.3 and Corollary 2.4].

The low orders are decided by the low zeros (Chapter 22). Order two holds without RH, that is, f is log concave (Theorem 22.11.5, a floating point certificate), and, given the verification of RH to height 3×10^{12} [277], f lies within 1.6×10^{-10} of a Pólya frequency function of every order (Theorem 22.11.6), so that every minor of order 3 and 4 is at least -1.1×10^{-8} and -1.2×10^{-7}. We test f at orders two to eight on other grids, at 70 digits, with the trapezoid rule in t; $f(2)$ agrees to 3×10^{-62} with the residue series $\sum_k r_k e^{-\gamma_k y}$, $r_k = -\xi(\frac12)/\Xi'(\gamma_k)$, over the first 60 zeros.

At the origin $f(0) = 1.8994210510239\ldots$, which equals $\mathbb{E}[T^{-1/2}]/(2\sqrt{\pi})$ with $\mathbb{E}[T^{-1/2}] = 6.7332723127566\ldots$ (Proposition 16.3.10 at $s = \frac{1}{2}$). On $[0, 2]$ with step 0.05, $ff'' - f'^2 < 0$ at all 41 grid points; the largest value, -1.37×10^{-22}, is at $y = 2$, where $f(2) = 1.89857001286 \times 10^{-10}$. The 121 minors $\det[f(x_r - y_c)]_{r,c<3}$ of order three with $x_r = (i+r)/10$ and $y_c = (j+c)/10$, $0 \le i, j \le 10$, depend only on $i - j$ and are all positive; the smallest, 6.8275×10^{-16} at $|i - j| = 10$, is far smaller than the 1.1×10^{-8} by which Theorem 22.11.6 allows a minor of order three to be negative. The Toeplitz minors $\det[f((r - c)h)]_{r,c<n}$ are positive for $n = 3$ to 8 at $h = 0.05$, 0.2 and 0.4 (Table 16.3), the smallest being 1.4971×10^{-21}. These are

Table 16.3: Toeplitz minors $\det[f((r - c)h)]_{r,c<n}$ of the density of Y (70 digits).

n	$h = 0.05$	$h = 0.2$	$h = 0.4$
3	2.7009×10^{-3}	2.1714	6.4789
4	6.4860×10^{-6}	2.0489	11.961
5	3.8720×10^{-9}	1.9020	22.083
6	7.2088×10^{-13}	1.7560	40.770
7	5.0043×10^{-17}	1.6184	75.271
8	1.4971×10^{-21}	1.4906	138.97

finite order tests: a negative minor would refute RH, and positive minors cannot prove it (Problem 16.10).

16.4 The master theorem route

By Theorem 16.2.3 and Remark 16.2.6, the Thorin property of the centre clock is equivalent to RH, so a proof of it that avoids the zeros (Problem 16.1) cannot take RH as an input. The route of [279] expands $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ term by term and passes to an integral representation of the GGC through the Ramanujan master theorem. We isolate what the extension of Chaudhry and Qadir [56] settles in that step and what it leaves open.

16.4.1 The growth hypothesis

Chaudhry and Qadir [56, §4] enlarge Hardy's class to $Q_\infty(\kappa, \delta)$ (their ε is called κ here, because $\varepsilon = \alpha - \frac{1}{2}$ in Section 16.5): meromorphic $F(s; 0)$, analytic for $\mathrm{Re}(s) < \delta$, not identically zero at the nonpositive integers, with

$$|\Gamma(s)F(s;0)| \le C\,|\tau|^{\kappa(\sigma-\delta)}, \qquad s = \sigma + i\tau,\ |\tau| \to \infty, \tag{16.4.1}$$

uniformly for $\sigma_1 \le \sigma \le \sigma_2 < \delta$. Their Theorem 4.1 grants the master theorem to the class, and their Examples 5.5 and 5.6 place $\zeta(s)/[\Gamma(1 - s)\zeta(1 - s)]$ and $\zeta(1 - s)/[\Gamma(s)\zeta(s)]$ in $Q_\infty(1, \frac{1}{2})$. The function to which the route applies the theorem is $F(s; 0) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + s)$.

LEMMA 16.4.1. *Assume RH, and let* $0 < \sigma_1 \le \sigma_2 < \frac{1}{2}$. *Then* $F(s;0) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+s)$ *satisfies*

$$|\Gamma(s)\,\xi(\tfrac{1}{2})/\xi(\tfrac{1}{2}+s)| \le C_\eta\, e^{-\pi|\tau|/4}\,|\tau|^{\sigma/2-9/4+\eta}, \qquad |\tau| \ge 1,$$

uniformly for $\sigma_1 \le \sigma \le \sigma_2$, *for every* $\eta > 0$. *So the growth condition* (16.4.1) *of* $Q_\infty(1, \frac{1}{2})$ *holds on such strips with exponential room. By the functional equation the same argument gives, for* $-\sigma_2 \le \sigma \le -\sigma_1$, *the exponent* $\frac{3}{2}\sigma - \frac{9}{4} + \eta$, *which is smaller.*

Proof. Write $\xi(\frac{1}{2}+s) = \frac{1}{2}w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w)$ with $w = \frac{1}{2}+s$, so $\frac{1}{2} < \operatorname{Re} w < 1$. Stirling's formula $|\Gamma(x+iy)| \asymp |y|^{x-1/2}e^{-\pi|y|/2}$ gives $|\Gamma(w/2)| \asymp |\tau|^{\sigma/2-1/4}e^{-\pi|\tau|/4}$, and $|w(w-1)| \asymp \tau^2$. Under RH, $1/\zeta(w) = O(|\tau|^\eta)$ uniformly for $\operatorname{Re} w \ge \frac{1}{2} + \sigma_1$ [348, Section 14.2]. So $|1/\xi(\frac{1}{2}+s)| \ll e^{\pi|\tau|/4}|\tau|^{-\sigma/2-7/4+\eta}$, and multiplying by $|\Gamma(s)| \asymp |\tau|^{\sigma-1/2}e^{-\pi|\tau|/2}$ gives the bound. For $-\sigma_2 \le \sigma \le -\sigma_1$ use $F(s;0) = F(-s;0)$, which is the functional equation $\xi(\frac{1}{2}+s) = \xi(\frac{1}{2}-s)$, and $|\Gamma(s)/\Gamma(-s)| \asymp |\tau|^{2\sigma}$: the bound at $-s$ has the exponent $-\sigma/2 - \frac{9}{4} + \eta$, and the factor $|\tau|^{2\sigma}$ turns it into $\frac{3}{2}\sigma - \frac{9}{4} + \eta$. □

The strip must avoid $\sigma = 0$, where $F(s;0)$ has its poles $s = i\gamma$. Without RH no polynomial bound for $1/\zeta$ is known on any fixed vertical strip $\sigma_0 < \operatorname{Re} w < 1$ of the critical strip, so unconditionally the growth condition is only known where $\operatorname{Re}(\frac{1}{2}+s) \ge 1$, outside the range of the class.

16.4.2 The absolute convergence hypothesis

With g the density of T (which exists, T being self decomposable and nondegenerate), interchanging sum and integral requires

$$\int_0^\infty g(y)\,\mathfrak{R}(z/y)\,dy < \infty, \qquad \mathfrak{R}(w) := \sum_{k\ge1} b_k w^k, \quad b_k := \frac{1}{\Gamma(k)}\,\frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2}+k)}. \tag{16.4.2}$$

LEMMA 16.4.2. *With no hypothesis on the zeros,* $\mathfrak{R}$ *is entire of order* $\frac{2}{3}$, *and* (16.4.2) *holds for every* $z > 0$.

Proof. Stirling applied to $\Gamma(k/2+1/4)$ gives $\log\xi(\frac{1}{2}+k) = \frac{k}{2}\log k + O(k)$, so the coefficients satisfy $\log(1/b_k) = \frac{3k}{2}\log k + O(k)$ and $\mathfrak{R}$ has order $\frac{2}{3}$. In particular $\mathfrak{R}(w) \le e^{w^{3/4}}$ for $w \ge w_0$, so $\mathfrak{R}(z/y)$ grows at most like $e^{(z/y)^{3/4}}$ as $y \to 0^+$, and convergence is a condition on the left tail of g. Stirling's formula for $\xi(\frac{1}{2}+x)$ gives $\log\mathbb{E}[e^{-uT}] = -\log\big(\xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2})\big) \sim -\frac{1}{4}\sqrt{u}\log u$ as $u \to \infty$, and the exponential Tauberian theorem of Kasahara [178] converts this into $-\log\Pr(T \le y) \sim \frac{1}{16}\,y^{-1}\log^2(1/y)$ as $y \to 0^+$. The clock is self decomposable (Theorem 8.1.1) with the infinite Lévy measure $W(x)\,dx/x$, so it has no

atom, and it is unimodal [366] (Section 14.1.1), with a mode $m > 0$ and a density g that is nondecreasing on $(0, m)$. Hence, for $2y \le m$, $g(y) \le y^{-1} \Pr(T \le 2y) \le y^{-1} \exp\{-\frac{1}{32}\, y^{-1} \log^2(1/y)\,(1+o(1))\}$, which beats $e^{(z/y)^{3/4}}$ since $1 > \frac{3}{4}$. For $y \ge y_0$ the argument z/y stays in $[0, z/y_0]$, where $\mathfrak{R}$ is bounded, and $\int g = 1$. $\square$

16.4.3 What remains open

(i) *The poles at the zeros.* The class requires analyticity on $\operatorname{Re} s < \frac{1}{2}$, and the residue argument of Chaudhry and Qadir assumes that the only poles are those of $\Gamma(s)$. But $\xi(\frac{1}{2})/\xi(\frac{1}{2}+s)$ has further poles at $s = \rho - \frac{1}{2}$, which lie on $\operatorname{Re} s = 0$ under RH and in the strip $|\operatorname{Re} s| < \frac{1}{2}$ in general, so it is not in $Q_\infty(1, \frac{1}{2})$ with or without RH. A contour argument that handles them needs their location, that is, RH.

(ii) *The $\sqrt{u}$ distortion.* The theorem of Chaudhry and Qadir is stated on vertical strips in the variable s, while the Laplace variable of the clock is $u = s^2$, and the squaring map carries the strip $|\operatorname{Re} s| < 1$ onto the parabolic region $\{|\operatorname{Im} u| < 2\sqrt{1 - \operatorname{Re} u}\}$. Whether the master theorem survives this change of variable needs a separate argument; in the shifted variable $w = u - 1$ it holds and is empty (Proposition 16.4.3).

16.4.4 The master theorem in other variables

Hardy's class consists of the φ analytic on $\operatorname{Re} w \ge -\delta$ $(0 < \delta < 1)$ with $|\varphi(v+i\tau)| \le Ce^{Pv + A|\tau|}$, $A < \pi$; for them $\int_0^\infty x^{w-1} \sum_{k\ge 0} (-x)^k \varphi(k)/k!\, dx = \Gamma(w)\varphi(-w)$ on $0 < \operatorname{Re} w < \delta$.

Proposition 16.4.3. *The function $\varphi(w) = \xi(\frac{3}{2})/\xi(\frac{1}{2}+\sqrt{1+w})$, with the principal root, is analytic and bounded on $\operatorname{Re} w \ge -\delta$, hence in Hardy's class with $P = 0$ and any A, with no hypothesis on the zeros. The image of $\operatorname{Re} w \ge -\delta$ under $w \mapsto \frac{1}{2} + \sqrt{1+w}$ lies in $\{\sigma + i\tau : \sigma - \frac{1}{2} \ge \sqrt{1 - \delta + \tau^2}\}$, which contains no zero, so the identity carries no information on them.*

Proof. Let $z = x + iy = \sqrt{1+w}$. Then $x^2 - y^2 = \operatorname{Re}(1+w) \ge 1 - \delta > 0$, which gives the region $(\sigma - \frac{1}{2} = x,\ \tau = y)$ and shows that $|\arg z| < \frac{\pi}{4}$ and $|z|^2 \ge 1 - \delta$. A zero $\rho = \beta + i\gamma$ with $\frac{1}{2} + z = \rho$ would need $|\gamma| < \beta - \frac{1}{2} < \frac{1}{2}$, but $|\gamma| > 14$, so φ is analytic. For large $|z|$, $\operatorname{Re}(\frac{1}{2} + z) \ge \frac{1}{2} + |z|/\sqrt{2} > 1$ and $\zeta(\frac{1}{2} + z) \to 1$, so Stirling gives $\log|\xi(\frac{1}{2} + z)| = \frac{1}{2}|z|(\cos\theta \log|z| - \theta \sin\theta)(1 + o(1))$ with $\theta = \arg z$, and $\cos\theta \ge 1/\sqrt{2}$. Thus $\xi(\frac{1}{2}+z)$ grows faster than any $e^{c|z|}$ and φ is bounded. $\square$

The Thorin property, by contrast, is a statement on the cut $w \in (-\infty, -1]$, which maps onto the critical line; Lemma 16.4.1 concerns the variable s and needs RH. The other two variables fail for exact reasons.

The variable α. Fix $s > 0$ and $\delta < \min(1, \sqrt{s})$. The function $a \mapsto \xi(1 + a)/\xi(1 + a + \sqrt{s})$ is analytic on $\operatorname{Re} a \ge -\delta$, since $\operatorname{Re}(1 + a + \sqrt{s}) > 1$ there, and of

polynomial growth by Stirling and the convexity bound for ζ. The master theorem evaluates $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ at $\alpha = 1-w \in (1-\delta, 1)$, but only while $\alpha+\sqrt{s} > 1$, in the region of the Euler product. Letting $s \to 0$, which the Thorin property at α requires, needs $\delta \to 0$ unless there is a zero free strip $\beta \le 1-\delta$, that is $\Theta \le 1-\delta$ for $\Theta = \sup \beta$, which is not known for any $\delta > 0$. So the route is exactly as strong as the hypothesis on Θ it needs.

The Thorin variable. Unconditionally $\xi(\frac12)/\xi(\frac12+\sqrt{u}) = \exp\{-\int_0^\infty (1-e^{-ut})W(t)\,dt/t\}$ with $W(t) = \sum_{\operatorname{Im}\rho>0} e^{-\lambda_\rho t}$, $\lambda_\rho = -(\rho-\frac12)^2$, as in Theorem 8.1.1. The master theorem needs W to be given by a Taylor series at 0. But $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ (Section 8.1), so W has none: its Mellin transform is $\Gamma(w)Z(w)$ with $Z(w) = \sum_{\operatorname{Im}\rho>0} \lambda_\rho^{-w}$, convergent for $\operatorname{Re} w > \frac12$, and the singular term gives the double pole $\int_0^1 t^{w-3/2}\log(1/t)\,dt = (w-\frac12)^{-2}$, whereas a Taylor series would give only simple poles at $w = 0, -1, \dots$.

16.4.5 Why the square root stays

Let L be the class of the functions $s \mapsto \mathbb{E}e^{-sX}$, X a real random variable, finite on an interval.

Theorem 16.4.4. *(a) For every real α, $s \mapsto \xi(\alpha)/\xi(\alpha+s)$ is not in L, whether or not RH holds.*

(b) For every real α, $t \mapsto \xi(\alpha+it)/\xi(\alpha)$ is, unconditionally, the characteristic function of the density $\Phi(u)e^{(\alpha-\frac12)u}/\xi(\alpha)$, with Φ of (2.1.3), but it is never infinitely divisible, hence never an EGGC. At $\alpha = \frac12$ it is the member $\Xi(t)/\Xi(0)$ of the van Dantzig pair of Theorem 8.1.1.

(c) For every real α, $t \mapsto \xi(\alpha)/\xi(\alpha+it)$ is unbounded, so it is not a characteristic function.

Proof. (a) By Stirling, $\log\{\xi(\alpha)/\xi(\alpha+s)\} = -\frac12 s\log s\,(1+o(1))$, while, if $\mathbb{E}e^{-sX}$ agreed with it on an interval, it would be finite and agree with it on all of $\mathbb{R}$, since the transform of a positive measure is singular at the real end points of its interval of convergence [359] and $\xi(\alpha)/\xi(\alpha+s)$ is analytic near the real axis; then $\mathbb{E}e^{-sX} \ge \Pr(X \le M)e^{-sM}$ decays at most exponentially as $s \to \infty$. (b) Φ is even and positive, since $2\pi n^2 e^{2u} > 3$ for $u \ge 0$, and $\xi(\frac12+iz) = \int_{\mathbb{R}} \Phi(u)e^{izu}du$ continues to $z = t - i(\alpha-\frac12)$, which gives the density, integrable because $\Phi(u) = O(e^{\frac92|u|-\pi e^{2|u|}})$. The function is entire in t and vanishes at $t = \gamma - i(\beta-\alpha)$ for every zero ρ, whereas an infinitely divisible law with $\mathbb{E}e^{cX} < \infty$ for every real c has $\mathbb{E}e^{zX} = e^{\psi(z)}$ for every complex z, with ψ its Lévy Khintchine exponent [325], so its characteristic function has no zero in $\mathbb{C}$. (c) $|\xi(\alpha+it)| \to 0$ faster than any power. □

The EGGC reading of the reciprocal that survives is the symmetric one, $\xi(\alpha)/\xi(\alpha+|t|) = \mathbb{E}e^{-t^2 H}$, with H a GGC when $\alpha \geq \sup_\rho \operatorname{Re}\rho$ (Theorem 2.1.2, Section 4.1.2), which contains the square root. Without it, the structure that exists lives on the ξ side and stops at $\alpha = 1$. For $\alpha > 1$, with $G \sim \text{Gamma}(1+\frac{\alpha}{2}, 1)$ and N independent, $\Pr(N = n) = n^{-\alpha}/\zeta(\alpha)$, the identity $\xi(s) = (s-1)\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ gives $\xi(\alpha+s)/\xi(\alpha) = (1+\frac{s}{\alpha-1})\mathbb{E}e^{-sL}$ with $L = \frac{1}{2}\log\pi - \frac{1}{2}\log G + \log N$ infinitely divisible, and $\xi(\alpha)/\xi(\alpha+s)$ is the quotient of a gamma transform by an infinitely divisible one, whose Lévy measure enters with its sign reversed, $1/\mathbb{E}N^{-s} = \zeta(\alpha)\sum_n \mu(n)n^{-\alpha-s}$ being Möbius signed.

16.5 Dirichlet L functions and number fields

We carry Section 16.2 from ξ to Dirichlet L functions and Dedekind zeta functions; the numbers are computed.

Let χ be a primitive character modulo $q > 1$ of parity $a \in \{0, 1\}$, $\chi(-1) = (-1)^a$, and

$$\xi(s,\chi) = \Big(\frac{q}{\pi}\Big)^{(s+a)/2}\Gamma\Big(\frac{s+a}{2}\Big)L(s,\chi),$$
$$\xi(s,\chi) = W_\chi\,\xi(1-s,\bar\chi), \qquad |W_\chi| = 1,$$

an entire function of order one [70].

16.5.1 The positivity obstacle

For $\operatorname{Re}(s) > 1$,

$$\log L(s,\chi) = \sum_{n\geq 2} \frac{\chi(n)\Lambda(n)}{\log n}\, n^{-s}, \tag{16.5.1}$$

so the analogue of μ^σ (Lemma 16.3.1) carries the weights $\chi(p)^r$: complex for complex χ, signed for real nonprincipal χ. Then $L(\sigma+it,\chi)/L(\sigma,\chi)$ is the characteristic function of no probability law: by uniqueness its only candidate has the weights $\chi(n)n^{-\sigma}/L(\sigma,\chi)$ at $-\log n$, and these are not all nonnegative.

On the critical line the Thorin construction needs no Euler product, only evenness and the Hadamard product of the completed function.

16.5.2 Real characters

For a real primitive character, $\bar\chi = \chi$ and $W_\chi = 1$, so $\xi(s,\chi) = \xi(1-s,\chi)$ and $\xi(\cdot,\chi)$ is real on the real axis.

Proposition 16.5.1. *Let χ be a real primitive character, and assume GRH for $L(s,\chi)$ and $L(\frac{1}{2},\chi) \neq 0$. Write the zeros as $\frac{1}{2} \pm i\gamma$ with $\gamma > 0$, counted with*

multiplicity. Then $T_\chi \stackrel{d}{=} \sum_{\gamma>0} e_\gamma/\gamma^2$ *is a GGC with*

$$\frac{\xi(\frac12,\chi)}{\xi(\frac12+\sqrt{u},\chi)} = \mathbb{E}\big[e^{-uT_\chi}\big], \qquad u \ge 0, \tag{16.5.2}$$

with cumulants $(j-1)!\,p_j(\chi)$, $p_j(\chi) = \sum_{\gamma>0}\gamma^{-2j}$. *Propositions 16.3.5 to 16.3.11 hold with* ξ *replaced by* $\xi(\cdot,\chi)$, *with GRH for* $L(s,\chi)$ *and* $L(\frac12,\chi)\neq 0$ *in the place of RH and* $\xi(\frac12)\neq 0$, *and with the values at* $u_0 = \frac14$, *which use* $\xi(1) = \frac12$, *replaced by those below; in particular*

$$\sum_{\gamma>0}\frac{1}{\gamma^2+\frac14} = \frac{\xi'}{\xi}(1,\chi) = \frac12\log\frac{q}{\pi} + \frac12\,\psi\Big(\frac{1+a}{2}\Big) + \frac{L'}{L}(1,\chi), \tag{16.5.3}$$

$$\mathbb{E}\big[e^{-T_\chi/4}\big] = \frac{\xi(\frac12,\chi)}{\xi(1,\chi)}.$$

Proof. $\xi(\frac12+s,\chi)$ is even, entire of order one and nonzero at $s=0$, so the argument of Lemmas 16.2.1 and 16.2.2 gives $\xi(\frac12+s,\chi)/\xi(\frac12,\chi) = \prod_\gamma(1+s^2/\gamma^2)$, and (G2) gives (16.5.2). The later propositions use only this product, evenness and growth on the real axis. For (16.5.3), take the logarithmic derivative of the product at $s=\frac12$ and of the definition of $\xi(\cdot,\chi)$ at $s=1$. □

The hypothesis $L(\frac12,\chi)\neq 0$ is Chowla's conjecture for real characters, still open; a central zero would put an atom at u_0 in the tilted Thorin measure (Theorem 16.2.3) and block the untilted clock (Corollary 16.2.4(a)). Lemma 16.4.1 holds for $\xi(\cdot,\chi)$ under GRH, with the same decay $e^{-\pi|\tau|/4}$ and the power $|\tau|^{\sigma/2-a/2-1/4+\eta}$ in place of $|\tau|^{\sigma/2-9/4+\eta}$, so the master theorem analysis of Section 16.4 carries over with the same two obstructions.

16.5.3 Complex characters

For complex χ the functional equation links $\xi(s,\chi)$ to the different function $\xi(1-s,\bar\chi)$, so $\xi(\frac12+s,\chi)$ is not even, its ordinates are not symmetric under $\gamma\mapsto-\gamma$, and the pairing behind (16.5.2) fails. Rotating by the root number makes the function real on the critical line, but not even, and it stays complex on the real axis, where a Laplace transform must be positive.

Proposition 16.5.2. *Let* χ *be a complex primitive character and* $\xi_{\chi\bar\chi}(s) := \xi(s,\chi)\xi(s,\bar\chi)$. *Then* $\xi_{\chi\bar\chi}(s) = \xi_{\chi\bar\chi}(1-s)$ *and* $\xi_{\chi\bar\chi}(\sigma) = |\xi(\sigma,\chi)|^2$ *for real* σ. *If GRH holds for* $L(s,\chi)$ *and* $L(\frac12,\chi)\neq 0$, *then*

$$\frac{\xi_{\chi\bar\chi}(\frac12)}{\xi_{\chi\bar\chi}(\frac12+\sqrt{u})} = \mathbb{E}\big[e^{-uT_{\chi\bar\chi}}\big], \qquad T_{\chi\bar\chi} \stackrel{d}{=} \sum_{\gamma\in\Gamma_\chi}\frac{e_\gamma}{\gamma^2},$$

where Γ_χ *is the multiset of all ordinates of zeros of* $L(s,\chi)$, *of both signs.*

Proof. The root numbers multiply to $W_\chi W_{\bar\chi} = |W_\chi|^2 = 1$, and $\xi(\sigma,\bar\chi) = \overline{\xi(\sigma,\chi)}$ for real σ. The zeros of $L(s,\bar\chi)$ are the conjugates $\frac12 - i\gamma$, so $\xi_{\chi\bar\chi}$ has zeros at $\frac12 \pm i\gamma$ for every $\gamma \in \Gamma_\chi$, and the even product argument applies: each ordinate of $L(s,\chi)$ gives one factor $1 + u/\gamma^2$, its partner $\frac12 - i\gamma$ being a zero of $L(s,\bar\chi)$, so no ordinate is counted twice. □

16.5.4 The Dedekind route

Let χ be a real primitive character of conductor q and $F = \mathbb{Q}(\sqrt{D})$, with $D = \chi(-1)q$ a fundamental discriminant. Then $\zeta_F(s) = \zeta(s)L(s,\chi)$, and F is real quadratic, $(r_1,r_2) = (2,0)$, for even χ and imaginary quadratic, $(r_1,r_2) = (0,1)$, for odd χ. More generally, for $q \geq 3$ the cyclotomic field $F_q = \mathbb{Q}(\zeta_q)$ is totally complex of degree $\varphi(q)$ and

$$\zeta_{F_q}(s) = \prod_{\chi \bmod q} L(s,\chi^*), \tag{16.5.4}$$

with χ^* the primitive character inducing χ. Every primitive character of conductor dividing q occurs, complex ones with their conjugates. Dedekind zeta functions have nonnegative coefficients,

$$\log\zeta_F(s) = \sum_{n\geq 2} \frac{\Lambda_F(n)}{\log n}\, n^{-s}, \qquad \Lambda_F(n) = \sum_{N\mathfrak{p}^r = n} \log N\mathfrak{p} \;\geq 0,$$

which is why one passes to fields.

Proposition 16.5.3. *For a quadratic field, $\xi_F(s) = 2\,\xi(s)\xi(s,\chi)$ when χ is even and $\xi_F(s) = 2q^{-1/2}\xi(s)\xi(s,\chi)$ when χ is odd. Consequently, at any tilt $u_0 \geq 0$ at which ξ and $\xi(\cdot,\chi)$ both have the Thorin property in the sense of Theorem 16.2.3, so does ξ_F, its tilted clock is the independent sum of the tilted clocks of ξ and of $\xi(\cdot,\chi)$, and*

$$U_{F,u_0} = U_{\mathbb{Q},u_0} + U_{\chi,u_0},$$

where U_{χ,u_0} is the Thorin measure of the tilted clock of $\xi(\cdot,\chi)$, defined as U_{F,u_0} is in Theorem 16.2.3; at $u_0 = 0$ this reads $T_F \overset{d}{=} T_\zeta + T_\chi$, with $T_\zeta = T$ the centre clock of ξ. For a cyclotomic field, under the hypotheses of Propositions 16.5.1 and 16.5.2, T_{F_q} is the independent sum of T_ζ, of T_χ over the real primitive characters of conductor $f \mid q$ with $f > 1$, and of $T_{\chi\bar\chi}$ over the conjugate pairs of complex primitive characters of conductor dividing q. In particular the cumulants add: $p_j(F) = p_j(\zeta) + p_j(\chi)$ for a quadratic field.

For $\mathbb{Q}(i)$ at basepoints $\alpha \geq 1$, where no hypothesis is needed, this is Proposition 15.1.28 and Corollary 15.1.29, with the constant $2 \cdot 4^{-1/2} = 1$.

Proof. For even χ the gamma factors agree, $\Gamma_{\mathbb{R}}(s)^2$ on both sides, and $|d_F| = q$. For odd χ the duplication formula gives $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$, while $\xi(s,\chi) =$

$q^{(s+1)/2}\Gamma_{\mathbb{R}}(s+1)L(s,\chi)$ carries an extra $q^{1/2}$. The constants cancel in the ratios (16.2.4), so the Laplace transforms multiply and the variables add. The cyclotomic case follows from (16.5.4) in the same way: the factors there are primitive, half of the characters modulo $q \geq 3$ are odd, so the gamma factors multiply to $\Gamma_{\mathbb{C}}(s)^{\varphi(q)/2}$, and the discriminant is the product of the conductors. □

16.5.5 The Lévy decomposition of ξ_F, with signs

For $\alpha > 1$ and $s \geq 0$ put $h_s(x) = e^{-sx} - 1 + sx \geq 0$. Binet's formula gives

$$\log \frac{\Gamma_{\mathbb{R}}(\alpha+s)}{\Gamma_{\mathbb{R}}(\alpha)} - s\frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}(\alpha) = \int_0^\infty h_s(x)\frac{e^{-\alpha x}}{1-e^{-2x}}\frac{dx}{x},$$
$$\log \frac{\Gamma_{\mathbb{C}}(\alpha+s)}{\Gamma_{\mathbb{C}}(\alpha)} - s\frac{\Gamma'_{\mathbb{C}}}{\Gamma_{\mathbb{C}}}(\alpha) = \int_0^\infty h_s(x)\frac{e^{-\alpha x}}{1-e^{-x}}\frac{dx}{x},$$

a linear factor gives $\log\frac{z+s}{z} - \frac{s}{z} = -\int_0^\infty h_s(x)e^{-zx}\frac{dx}{x}$ for $z > 0$, and

$$\log \frac{\zeta_F(\alpha+s)}{\zeta_F(\alpha)} - s\frac{\zeta'_F}{\zeta_F}(\alpha) = \sum_{n\geq 2}\frac{\Lambda_F(n)}{\log n}\, n^{-\alpha} h_s(\log n).$$

Proposition 16.5.4. *For $\alpha > 1$ and $s \geq 0$,*

$$\log \frac{\xi_F(\alpha+s)}{\xi_F(\alpha)} - s\,\frac{\xi'_F}{\xi_F}(\alpha) = \int_0^\infty h_s(x)\, e^{-\alpha x}\,\frac{\nu_F(dx)}{x} \tag{16.5.5}$$

with the signed measure

$$\nu_F(dx) = \Big(\frac{r_1}{1-e^{-2x}} + \frac{r_2}{1-e^{-x}} - 1 - e^x\Big)dx + \sum_{n\geq 2}\Lambda_F(n)\,\delta_{\log n}(dx). \tag{16.5.6}$$

Proof. Add the decompositions above. The density $-(1+e^x)$ comes from the factors $\alpha+s$ and $\alpha-1+s$ of $s(s-1)$, both with positive z since $\alpha > 1$, and $|d_F|^{s/2}$ is linear in s and only changes the drift. □

For $F = \mathbb{Q}$, where $\xi_{\mathbb{Q}} = 2\xi$, this is Proposition 12.1.10; for χ_4 and χ_{12} it is Propositions 15.1.24 and 15.2.2.

Remark 16.5.5 (Where positivity fails, and why it is the same question). The polynomial factor contributes the density $-(1+e^x)$, which dominates the positive continuous part for large x. By the prime ideal theorem Λ_F has mean value one, so on average the atoms $\sum\Lambda_F(n)\delta_{\log n}$ cancel $-e^x\,dx$, and by the explicit formula what remains is a sum over the zeros of ζ_F. Positivity of any smoothed ν_F is again a statement about zeros: the Dedekind route does not give the Thorin property for free.

16.5.6 The sine squared lemma: from primes to the Thorin measure

Proposition 16.5.4 writes $\log \xi_F$ against a signed Lévy measure. The sine squared lemma [279, Lemma 8.1], first in [280, Lemma 1], converts any such decomposition into a Thorin representation of the reciprocal [279, Theorem 8.2]; in the book it is equation (2.1.9) of Section 2.1, applied to ξ in Proposition 12.1.3 and to $\xi(\cdot,\chi_4)$ after the Lévy form of Proposition 15.1.24 (Section 15.1.3). In the chart $z = b^2$ the kernel identity (2.1.9) reads, for $w, x > 0$,

$$e^{-wx} - 1 + wx = \frac{x}{\pi} \int_0^\infty \log\Big(1 + \frac{w^2}{b^2}\Big)\, 2\sin^2\Big(\frac{bx}{2}\Big)\, db, \tag{16.5.7}$$

proved there. For a function f write $v_\alpha(b) = \operatorname{Re} \frac{f'}{f}(\alpha + ib)$. For $f = \xi$ this is the function v_α of Section 2.1, v_α/π is the density $\rho_{\alpha-1/2}$ of (2.1.7), and the measure $U_\alpha(dt) = v_\alpha(\sqrt{t})\, dt/(2\pi\sqrt{t})$ is that of (2.1.6).

Proposition 16.5.6 (The lemma on the prime side). *Let f be ξ_F, or $\xi(\cdot,\chi)$ for a real primitive χ, and let $\alpha > 1$. Then*

$$\frac{f(\alpha)}{f(\alpha + \sqrt{u})} = \exp\Big(- \int_0^\infty \log\Big(1 + \frac{u}{b^2}\Big)\, \frac{v_\alpha(b)}{\pi}\, db\Big), \qquad u \ge 0, \tag{16.5.8}$$

where

$$v_\alpha(b) = \frac{f'}{f}(\alpha) + \int_0^\infty 2\sin^2\Big(\frac{bx}{2}\Big) e^{-\alpha x}\, \nu(dx) = \operatorname{Re} \frac{f'}{f}(\alpha + ib), \tag{16.5.9}$$

and ν is the Lévy measure of f: ν_F of (16.5.6) *for ξ_F, and for $\xi(\cdot,\chi)$*

$$\nu_\chi(dx) = \sum_{n \ge 2} \chi(n)\Lambda(n)\, \delta_{\log n}(dx) + \frac{e^{(1-a)x}}{2\sinh x}\, dx,$$

that is $(2\sinh x)^{-1}dx$ for odd χ and $(1 - e^{-2x})^{-1}dx$ for even χ, from the gamma factor $\Gamma_{\mathbb{R}}(s + a)$.

Proof. For ξ_F the decomposition

$$\log \frac{f(\alpha + w)}{f(\alpha)} - w\, \frac{f'}{f}(\alpha) = \int_0^\infty h_w(x)\, e^{-\alpha x}\, \frac{\nu(dx)}{x}$$

is Proposition 16.5.4, and for $\xi(\cdot,\chi)$ it follows in the same way from Binet's formula for $\Gamma_{\mathbb{R}}(s + a)$ and (16.5.1). Insert (16.5.7) and interchange the integrals, which is justified for $\alpha > 1$ because the kernels are nonnegative and $\int \min(1, x)\, e^{-\alpha x} |\nu|(dx) < \infty$ (near 0 the density of $|\nu|$ is of order $1/x$, and at infinity $e^{-\alpha x}|\nu|(dx)$ is dominated by $e^{-(\alpha-1)x}dx$ and by the prime sum $\sum \Lambda_F(n) n^{-\alpha}$). The linear term is absorbed using $\int_0^\infty \log(1 + w^2/b^2)\, db = \pi w$, which gives the

first form of v_α with $u = w^2$. For the second, differentiate the decomposition in w: $\int(1-e^{-wx})e^{-\alpha x}\nu(dx) = (f'/f)(\alpha+w) - (f'/f)(\alpha)$. The left side is analytic for $\operatorname{Re} w > 1-\alpha$ by the same bound, the right side is meromorphic with poles at the zeros, and they agree for $w>0$, so both continue to $w = ib$; taking real parts gives $\int 2\sin^2(bx/2)e^{-\alpha x}\nu(dx) = \operatorname{Re}(f'/f)(\alpha+ib) - (f'/f)(\alpha)$. □

For a real character the arithmetic part of $v_\alpha(b)$ is

$$-\sum_{n\ge 2}\chi(n)\Lambda(n)\,n^{-\alpha}\cos(b\log n),$$

absolutely convergent for $\alpha>1$. At $\alpha=1$ it converges conditionally for nonprincipal χ, and for ξ_F the prime sum diverges and converges only together with the density $-e^x$ of (16.5.6), as in Proposition 12.1.3.

LEMMA 16.5.7. *For $c\in\mathbb{C}$ with $\operatorname{Re} c>0$ and $w>0$,*

$$\frac{1}{\pi}\int_{-\infty}^{\infty}\log\Big(1+\frac{w^2}{b^2}\Big)\operatorname{Re}\frac{1}{c+ib}\,db = 2\log\Big|1+\frac{w}{c}\Big|.$$

For $\operatorname{Re} c<0$ the right side is replaced by $-2\log|1-w/c|$.

Proof. Write $\log(1+w^2/b^2) = \int_0^w 2w'(b^2+w'^2)^{-1}dw'$. For $w'>0$ the function $b\mapsto (c+ib)^{-1}(b^2+w'^2)^{-1}$ is rational and $O(|b|^{-3})$, with poles at ic, iw' and $-iw'$. If $\operatorname{Re} c>0$ the pole ic lies in the upper half plane, and closing the contour in the lower half plane picks up only $-iw'$, so that $\int_{-\infty}^{\infty} db/((c+ib)(b^2+w'^2)) = -2\pi i/((c+w')(-2iw')) = \pi/(w'(c+w'))$. Since $|\operatorname{Re}(c+ib)^{-1}|\le|\operatorname{Re} c|^{-1}$ and $\int_{-\infty}^{\infty}2w'(b^2+w'^2)^{-1}db = 2\pi$, Fubini applies, and taking real parts,

$$\frac{1}{\pi}\int_{-\infty}^{\infty}\log\Big(1+\frac{w^2}{b^2}\Big)\operatorname{Re}\frac{1}{c+ib}\,db = \operatorname{Re}\int_0^w\frac{2\,dw'}{c+w'} = 2\log\Big|\frac{c+w}{c}\Big|.$$

If $\operatorname{Re} c<0$ the same computation, closing in the upper half plane, gives $-2\log|(c-w)/c|$. □

Zeros $\rho=\beta+i\gamma$ are counted with multiplicity. Say that f has the *Thorin property at the basepoint* $\alpha>\frac12$ if $f(\alpha)\neq 0$ and $u\mapsto f(\alpha)/f(\alpha+\sqrt{u})$ is the Laplace transform of a GGC.

THEOREM 16.5.8 (The basepoint family of the lemma). *Let f be as in Proposition 16.5.6, with zeros $\rho=\beta+i\gamma$.*

(a) If every zero has $\beta<\alpha$, then (16.5.8) *holds with*

$$\frac{v_\alpha(b)}{\pi} = \frac{1}{\pi}\sum_\rho \operatorname{Re}\frac{1}{\alpha+ib-\rho} = \frac{1}{\pi}\sum_\rho\frac{\alpha-\beta}{(\alpha-\beta)^2+(b-\gamma)^2} > 0,$$

so f has the Thorin property at the basepoint α, with no drift and Thorin measure $U_\alpha(dt) = v_\alpha(\sqrt{t})\,dt/(2\pi\sqrt{t})$. Since every zero has $\beta < 1$, this holds unconditionally for every $\alpha \geq 1$.

(b) If some zero has $\beta > \alpha$ and $f(\alpha) \neq 0$, then f does not have the Thorin property at the basepoint α. Hence f has it at every basepoint $\alpha > \frac{1}{2}$ if and only if every zero lies on the critical line.

(c) Under GRH, and if $f(\frac{1}{2}) \neq 0$, with $\varepsilon = \alpha - \frac{1}{2}$,

$$\frac{v_\alpha(b)}{\pi} = \sum_{\gamma>0} \big(P_\varepsilon(b-\gamma) + P_\varepsilon(b+\gamma)\big), \qquad P_\varepsilon(x) = \frac{\varepsilon}{\pi(\varepsilon^2+x^2)},$$

the Poisson smoothing at height ε of the zero counting measure. As $\alpha \downarrow \frac{1}{2}$, $(v_\alpha(b)/\pi)\,db \to \sum_{\gamma>0}\delta_\gamma$ vaguely on $[0,\infty)$, and U_α tends to the untilted Thorin measure $\sum_\gamma \delta_{\gamma^2}$ of Theorem 16.2.3.

Proof. (a) By Hadamard's factorisation $f(s) = e^{A+Bs}\prod_\rho (1-s/\rho)e^{s/\rho}$ and the functional equation, $\operatorname{Re}(f'/f)(s) = \sum_\rho \operatorname{Re}(s-\rho)^{-1}$, absolutely convergent, for all s off the zeros, and $\operatorname{Re} B = -\sum_\rho \operatorname{Re}\rho^{-1}$ [70, Ch. 12]. Each term of the sum is positive when $\operatorname{Re} s > \beta$, which gives the formula for v_α and its positivity. For a zero ρ put $\omega_\rho = \alpha - \rho$, so that $\operatorname{Re}\omega_\rho = \alpha - \beta > 0$ and the term of ρ in $v_\alpha(b)$ is $\operatorname{Re}(\omega_\rho + ib)^{-1}$, and for $w > 0$ put

$$I_\rho = \frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{w^2}{b^2}\Big)\operatorname{Re}\frac{1}{\omega_\rho+ib}\,db \;\geq 0.$$

Since $\operatorname{Re}(\omega_{\bar\rho}+ib)^{-1} = \operatorname{Re}(\omega_\rho - ib)^{-1}$, the substitution $b \mapsto -b$ and Lemma 16.5.7 give $I_\rho + I_{\bar\rho} = 2\log|1+w/\omega_\rho|$, and $|1+w/\omega_{\bar\rho}| = |1+w/\omega_\rho|$. As f is real on the real axis, $\rho \mapsto \bar\rho$ permutes the zeros, and the terms being nonnegative we may sum:

$$\int_0^\infty \log\Big(1+\frac{w^2}{b^2}\Big)\frac{v_\alpha(b)}{\pi}\,db = \sum_\rho I_\rho = \sum_\rho \log\Big|1+\frac{w}{\omega_\rho}\Big|.$$

On the other side the Hadamard product gives $f(\alpha+w)/f(\alpha) = e^{Bw}\prod_\rho(1+w/\omega_\rho)e^{w/\rho}$, an absolutely convergent product, and the left side is positive because f is real and has no zero on $[\alpha,\infty)$. Taking logarithms and using $\operatorname{Re} B = -\sum_\rho \operatorname{Re}\rho^{-1}$ gives $\log f(\alpha+w)/f(\alpha) = \sum_\rho \log|1+w/\omega_\rho|$, the series converging absolutely. With $u = w^2$ this is (16.5.8) with density v_α/π. In the variable $t = b^2$ the exponent is $\int_0^\infty \log(1+u/t)\,U_\alpha(dt)$ and it is finite; this is equivalent to the integrability conditions on a Thorin measure stated with (16.2.1), so U_α is a Thorin measure with no drift.

(b) Let ρ be a zero with $\beta > \alpha$ and put $u_\rho = (\rho-\alpha)^2$. Since $\operatorname{Re}(\rho-\alpha) = \beta-\alpha > 0$, the principal square root of u_ρ is $\rho-\alpha$, and $u_\rho \notin (-\infty, 0]$, because otherwise $\rho-\alpha$

would be purely imaginary. The function $f(\alpha+\sqrt{u})$ is analytic on $\mathbb{C}\setminus(-\infty,0]$ and vanishes at u_ρ, so $f(\alpha)/f(\alpha+\sqrt{u})$ has a pole there. If f had the Thorin property at α, this function would coincide on $u>0$, hence on $\mathbb{C}\setminus(-\infty,0]$, with a GGC transform, which is analytic there by (G1): a contradiction. If every zero lies on the critical line, then $\beta=\frac12<\alpha$ for every $\alpha>\frac12$ and (a) gives the Thorin property at every basepoint $\alpha>\frac12$. If some zero is off the line, the functional equation gives one with $\beta>\frac12$, and the property fails at every $\alpha\in(\frac12,\beta)$, by the argument above where $f(\alpha)\neq0$ and by definition where $f(\alpha)=0$.

(c) Under GRH every zero is $\frac12\pm i\gamma$ with $\gamma>0$, since $f(\frac12)\neq0$, and (a) applies for every $\alpha>\frac12$; with $\alpha-\beta=\varepsilon$ the two zeros $\frac12\pm i\gamma$ contribute $P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)$. Let h be continuous with compact support in $[0,R]$ and let $0<\varepsilon\le1$. For $\gamma\le R+1$ the integral of h against $P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)$ over $[0,\infty)$ tends to $h(\gamma)$, because $\gamma>0$ is a point of continuity of h extended by zero and $\int_0^\infty P_\varepsilon(b+\gamma)\,db\le\varepsilon/(\pi\gamma)$. For $\gamma>R+1$ and $b\in[0,R]$ one has $|b\mp\gamma|\ge\gamma-R\ge\gamma/(R+1)$, so the integral is at most $2R\|h\|_\infty(R+1)^2\varepsilon/(\pi\gamma^2)$. The zeros with $\gamma>R+1$ therefore contribute at most a constant times $\varepsilon\sum_\gamma\gamma^{-2}$, which tends to 0, and the finitely many others give $\sum_{\gamma>0}h(\gamma)$ in the limit, so $\int h\,(v_\alpha/\pi)\,db\to\sum_{\gamma>0}h(\gamma)$. Under $b\mapsto b^2$ this is the convergence of U_α to $\sum_\gamma\delta_{\gamma^2}$. □

If the half plane of [257], which has not been refereed (Section 25.1.2), stands, then every zero of every Dirichlet L function has $\beta\le\frac78$, and part (a) gives the Thorin property of $\xi(\cdot,\chi)$ for every real primitive character χ, and of ξ_F for every quadratic field F, at every basepoint $\alpha>\frac78$, with Thorin density v_α/π a sum of Lorentzians of widths at least $\alpha-\frac78$. This is Theorem 23.1.1(i) carried from ζ to the family; it lowers the unconditional end $\alpha\ge1$ of part (a) to $\alpha>\frac78$ under that hypothesis. At $\alpha=\frac78$ part (a) does not apply, since the half plane does not exclude a zero on the line $\operatorname{Re}s=\frac78$.

For ξ: part (a) is the first sentence of Theorem 2.1.2; parts (a) and (b) together, with the borderline case in which a zero on the line $\operatorname{Re}s=\alpha$ contributes an atom, are Theorem 11.1.14, and Theorem 13.2.7 treats the basepoints $\frac12\le\alpha<1$; and the Poisson form of part (c) is Proposition 2.1.3. For χ_4 the corresponding result is Proposition 15.1.25, and for χ_{12} at the basepoint one Proposition 15.2.1. In part (c) the vague convergence holds without RH, as shown after Remark 2.1.4 and in [279, Proposition 9.5]: the limit is then the counting measure of the ordinates of the zeros on the line, the pairs off the line contributing signed mass whose total variation tends to zero. Without $f(\frac12)\neq0$ a central zero of order $2m$ adds the term $2m\,P_\varepsilon(b)$ to v_α/π, of mass m on $(0,\infty)$, which concentrates at $b=0$ as $\varepsilon\downarrow0$; the limit is then not a Thorin measure (Corollary 16.2.4(a)).

The tilt family and the basepoint family meet at the centre. Theorem 16.2.3 tilts by u_0, reads f at $\frac12+\sqrt{u_0+u}$, and shifts each atom from γ^2 to $u_0+\gamma^2$. The lemma's family reads f at $\alpha+\sqrt{u}$ and smooths each atom by a Poisson kernel of width $\alpha-\frac12$. One consequence differs: at the centre $f(\frac12+w)$ is even in w and $v_{1/2}(0)=0$; for $\alpha>\frac12$ the function $f(\alpha+w)$ is not even, and under the hypothesis

of part (a) $v_\alpha(0) = (f'/f)(\alpha) > 0$ is a sum of positive terms. The Thorin measure then has a component like that of a stable law of index $\frac{1}{2}$, its density behaving like $v_\alpha(0)/(2\pi\sqrt{t})$ as $t \downarrow 0$, and the shifted clock T_α has $\mathbb{E}[T_\alpha] = \infty$. At every basepoint $\alpha \geq 1$ the book proves the exact split into a positive stable variable of index $\frac{1}{2}$ and a GGC, for ξ in Propositions 12.1.11 and 12.1.12 and for χ_4 and χ_{12} in Theorem 12.1.18(v) with the paragraph after it. The moment dictionary of Section 16.3.3 lives only at the centre.

Remark 16.5.9 (What the lemma buys, and what it cannot). Reaching the centre means running the Poisson smoothing backwards from α to $\frac{1}{2}$. In the variable $x = \log n$ the smoothing multiplies the weight of n by $n^{-(\alpha-1/2)}$, so undoing it replaces $n^{-\alpha}$ by the divergent $n^{-1/2}$: the backward step is the explicit formula, and it is ill posed. Chapter 8 measures its cost. By Theorem 8.2.2 the Thorin density at a basepoint beyond the zeros is the Poisson smoothing of a positive measure down to the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon_1$ exactly when no zero has $\beta - \frac{1}{2} > \varepsilon_1$. Under GRH, by Proposition 8.2.3 and Table 8.1 (for χ_4) the number of primes needed for accuracy δ is of order $\delta^{-1/(\alpha-1/2)}$, unbounded as $\alpha \downarrow \frac{1}{2}$. Section 9.1.1 of Chapter 9 shows that the deconvolved object is Weil's distribution of the explicit formula. Positivity on a single line is not known to exclude zeros to its right (Remark 2.1.4 and Section 25.1.2). For ξ, Proposition 12.1.12 proves that the minimum of v_α on the line is at $b = 0$ for every $\alpha \geq 1$, and for χ_4 and χ_{12} Theorem 12.1.18(v) does; the theorem also certifies it for every $\alpha \in (\frac{1}{2}, \alpha_{\max}]$ up to the height to which GRH is verified (Table 12.4), and shows that the statement on the whole lines for all $\alpha \in (\frac{1}{2}, 1)$ together is equivalent to GRH. Our computations for χ_4 below agree. A real zero $\beta > \alpha$ adds the term $-(\beta-\alpha)/((\beta-\alpha)^2 + b^2)$ to $v_\alpha(b)$, whose modulus is largest at $b = 0$, so the origin is where a real exceptional zero acts first: a Siegel zero is a failure of the lemma's positivity at the origin. The classical zero free region arguments live on the Lévy side of the lemma, which restates their conclusions as positivity of a Thorin density. That is a translation, and no new estimate follows from it here, for Linnik's constant K or for the least prime in a progression.

Numerical check for χ_4. A computation checks the lemma for χ_4 at 30 digits. At $\alpha = 2$, with the prime powers up to N and the integral over b done term by term (each prime term is the kernel identity), the integral in (16.5.8) differs from $\log \xi(\alpha + w, \chi_4)/\xi(\alpha, \chi_4)$ by -4.4×10^{-8}, -1.2×10^{-9} and 6.2×10^{-11} for $N = 10^4$, 10^5 and 10^6 at $w = 1$ and $w = 3$; this difference is the tail of the prime sum beyond N. The prime side of (16.5.9), over the prime powers up to 2×10^6, agrees with $v_\alpha(b)/\pi$ computed directly from ξ'/ξ to 9.4×10^{-11} at $\alpha = 2$ and to 1.9×10^{-7} at $\alpha = \frac{3}{2}$. The zero side, the sum of part (a) over the 203 zeros of $L(s, \chi_4)$ below height 300 found numerically, with the tail beyond height 300 taken from the smooth zero density, differs from the direct value by 1.1×10^{-6} to 1.7×10^{-6} at $\alpha = \frac{3}{2}$ and 2 for $0 \leq b \leq 50$; once the tail is anchored to the exact

zero count at the cut, as in (16.3.8), the difference falls to between 9.7×10^{-9} and 1.7×10^{-8}, far inside the enclosure of 1.6×10^{-5} to 2.7×10^{-5} that the explicit bound for the zero count used in the proof of Theorem 12.1.18(v) guarantees.

On the lines $\alpha = 1, \frac{3}{4}, \frac{3}{5}, \frac{11}{20}$ the minimum of $v_\alpha(b)$ over $0 \le b \le 150$, on grids of step 0.1 and 0.005, is at $b = 0$ (Table 16.4), where $v_\alpha(0) = b_\alpha = (\xi'/\xi)(\alpha, \chi_4)$.

α	$v_\alpha(0) = b_\alpha$	$2(\alpha - \frac{1}{2})\, p_1(\chi_4)$	ratio
1	0.0777839900	0.0780148250	0.99704
$\frac{3}{4}$	0.0389784293	0.0390074125	0.99926
$\frac{3}{5}$	0.0156011077	0.0156029650	0.99988
$\frac{11}{20}$	0.0078012503	0.0078014825	0.99997

Table 16.4: The minimum of $v_\alpha(b)$ over $0 \le b \le 150$ for χ_4, always at $b = 0$, against the bound $2(\alpha - \frac{1}{2})\, p_1(\chi_4) = (\alpha - \frac{1}{2})V_f$ of Theorem 12.1.18(i), with $V_f = 2p_1(\chi_4) = 0.1560296499$. Computed.

For $\alpha = 2$ and $\frac{3}{2}$ every zero of $L(s, \chi_4)$ has real part below $1 < \alpha$, so Proposition 16.5.6 and part (a) of Theorem 16.5.8 hold with no hypothesis, and the comparisons above are identities up to truncation: they test the quadrature, the truncation of the prime sums, the zero list and its tail, and the code, not the theorem and not GRH. The grid of the table is a finite test of the function v_α, consistent with GRH; part (ii) of Theorem 12.1.18 turns it into a theorem for $0 < |b| \le 2.5 \times 10^7 - 1.21$, using the verification of GRH for χ_4 to height 2.5×10^7 [275], and for $\alpha < 1$ the function v_α/π is the Thorin density of $\xi(\alpha, \chi_4)/\xi(\alpha + \sqrt{u}, \chi_4)$ only if no zero lies to the right of α, which is known below that height and open above it. The lemma's output is not computable from the primes alone in practice for $\alpha \le 1$: the plain prime sum at $b = 0$ over the prime powers up to 2×10^6 differs from $v_\alpha(0)$ by 0.17, 12, 260 and 1100 percent of the value at $\alpha = 1, \frac{3}{4}, \frac{3}{5}, \frac{11}{20}$, so the minima of the table come from the zeros and the direct values of ξ'/ξ, not from the primes.

Finally, with the smoothing run backwards by a smooth cutoff, the peaks of $-\sum_{n \le N} \chi_4(n)\Lambda(n)n^{-1/2} \cos(b \log n)\, \cos^2\left(\frac{\pi}{2}\frac{\log n}{\log N}\right)$, $N = 2 \times 10^6$, lie within 0.003 of all fourteen zeros of $L(s, \chi_4)$ below height 40. Up to height 45 the seventeen zeros are matched by seventeen peaks and there is no other peak above half the largest value. This tests the explicit formula and locates the ordinates from the primes; it does not test the lemma.

16.6 GRH from the Thorin property, and families of characters

Theorem 16.6.1. *Let χ be a real primitive character with quadratic field F.*

(a) If $\varphi_{F,1}$ is the Laplace transform of a GGC, then RH holds and every nontrivial zero of $L(s, \chi)$ lies on the critical line or is real.

(b) *If $\varphi_{F,0}$ is the Laplace transform of a GGC, equivalently $\varphi_{F,1}$ is one with Thorin measure in $(1,\infty)$, then $L(s,\chi)$ satisfies GRH and $L(\frac{1}{2},\chi)\neq 0$.*

If $\varphi_{F_q,0}$ is a GGC Laplace transform for every $q\geq 3$, with $F_q=\mathbb{Q}(\zeta_q)$, then GRH holds for all Dirichlet L functions, and $L(\frac{1}{2},\chi)\neq 0$ for every primitive character χ.

Proof. Apply Corollary 16.2.4 to $\zeta_F=\zeta\,L(s,\chi)$, part (b) of the corollary for the tilt $u_0=1$ and part (a) for $u_0=0$; for ζ itself real zeros in $(0,1)$ are excluded because $\zeta(\sigma)<0$ there, and $\zeta(\frac{1}{2})\neq 0$. The last statement uses equation (16.5.4): the zeros of ζ_{F_q} are those of the functions $L(s,\chi)$ for the primitive characters χ of conductor dividing q, and every primitive character of conductor $f\geq 3$ occurs for $q=f$. Part (a) of the corollary then puts all of them on the critical line and excludes $L(\frac{1}{2},\chi)=0$. □

Remark 16.6.2 (The Siegel zero is a threshold crossing). A real zero β contributes the atom $u_0-(\beta-\frac{1}{2})^2$, positive while $u_0>(\beta-\frac{1}{2})^2$ and crossing zero as the tilt descends to $(\beta-\frac{1}{2})^2$. A Siegel zero near 1 pushes the threshold u_F^* of Corollary 16.2.4(c) towards $\frac{1}{4}$, leaving only tilts above a value near $\frac{1}{4}$.

16.6.1 Numerical verification for the characters modulo 4 and 5

We take χ_4 (odd, so $a=1$) and the real character χ_5 modulo 5, the Legendre symbol (even, so $a=0$). The numbers are computed in ball arithmetic where a count or a sign is certified and otherwise in mpmath, except those of the last paragraph. Write $N(T,\chi)$ for the number of zeros of $\xi(s,\chi)$ with $0<\operatorname{Im}s<T$, $\theta_\chi(T)/\pi=\frac{T}{2\pi}\log\frac{qT}{2\pi e}+\frac{a}{4}-\frac{1}{8}$ for the smooth count and $S(T,\chi)=N(T,\chi)-\theta_\chi(T)/\pi$ for its fluctuation.

Zeros. Sign changes of the real function $\xi(\frac{1}{2}+it,\chi)$ on $0<t<300$ give the first ordinates 6.0209489 and 6.6484533. The counts, 203 and 214, are certified: the argument principle in ball arithmetic gives $N(300,\chi)=203$ and 214, equal to the numbers of sign changes, so no zero below 300 was missed and all are simple and on the line. The smooth counts $\theta_\chi(T)/\pi$ at $T=300$ are 203.15 and 213.56, so $S(300,\chi_4)=-0.153$ and $S(300,\chi_5)=+0.443$.

No real or central zeros. On a grid of step 0.005 in $[\frac{1}{2},1)$ the smallest values of $L(\sigma,\chi)$ are the central ones, $L(\frac{1}{2},\chi_4)=0.66769$ and $L(\frac{1}{2},\chi_5)=0.23175$, and ball arithmetic certifies $L(\sigma,\chi)>0$ on $[\frac{1}{2},2]$, with 32 pieces and smallest lower bound 0.033 for χ_4 and 101 pieces and 0.0031 for χ_5. Since $\xi(\sigma,\chi)=\xi(1-\sigma,\chi)$ and the gamma factor is positive, $\xi(\sigma,\chi)>0$ on $[0,1]$, so neither function has a real zero in the critical strip, and neither has a central zero. Both Thorin thresholds are therefore 0, and the untilted clock is available under GRH. For χ_4 the book also certifies the absence of real zeros with a positive kernel (Section 15.1.7.1).

Power sums. Let $p_j(\chi) = \sum_{\gamma>0} \gamma^{-2j}$ be the power sums of the ordinates. The zero side is

$$\sum_{0<\gamma<X} \gamma^{-2j} + \int_X^\infty t^{-2j}\, \frac{1}{2\pi} \log \frac{qt}{2\pi}\, dt, \qquad X = 300,$$

the sum below X plus the tail started at X. The function side is read from the Taylor coefficients of $\log \xi(\frac{1}{2}+s,\chi)$, in which s^{2j} has the coefficient $(-1)^{j+1}p_j/j$; it uses no zeros (Table 16.5).

Table 16.5: The power sums $p_j(\chi)$ of the ordinates for the characters modulo 4 and 5: the zero side, the function side, the relative error, and the difference of the two sides times 300^{2j}.

q	j	zero side	function side	relative error	difference $\times 300^{2j}$
4	1	0.0780131	0.0780148	2.2×10^{-5}	-0.1512
	2	9.288458×10^{-4}	9.288458×10^{-4}	2.0×10^{-8}	-0.1498
	3	2.216962×10^{-5}	2.216962×10^{-5}	9.2×10^{-12}	-0.1484
5	1	0.0784620	0.0784571	6.3×10^{-5}	0.4423
	2	7.178001×10^{-4}	7.178001×10^{-4}	7.6×10^{-8}	0.4414
	3	1.316152×10^{-5}	1.316152×10^{-5}	4.6×10^{-11}	0.4405

With $N(t,\chi) = \theta_\chi(t)/\pi + S(t,\chi)$, integration by parts gives the error of a tail started at the height X as $S(X,\chi)X^{-2j} - 2j\int_X^\infty S(t,\chi)\, t^{-2j-1}\, dt$, up to the difference between the density of the tail and θ'_χ/π, which is of order X^{-2j-1}. So the error falls like X^{-2j}, not like the size $X^{1-2j}\log X$ of the tail itself. In Table 16.5 the last column is within 3 percent of $S(300,\chi_4) = -0.1526$ and $S(300,\chi_5) = +0.4431$ for every j.

Transfer function and products. The transfer function $\mathfrak{T}_\chi(t)$, the ratio $\xi(\frac{1}{2}+it,\chi)/\xi(\frac{1}{2}+t,\chi)$, has the values $\mathfrak{T}_{\chi_4}(2) = 0.53522$ and $\mathfrak{T}_{\chi_4}(5) = 0.014573$. The Hadamard product $\prod_{\gamma>0}(1+s^2/\gamma^2)$ at $s=3$, over the 203 and 214 ordinates with the tail started at 300, is 1.9524380 and 1.9736682, against $\xi(\frac{7}{2},\chi)/\xi(\frac{1}{2},\chi) =$ 1.9524675 and 1.9735809 for χ_4 and χ_5. The relative differences, -1.5×10^{-5} and $+4.4\times10^{-5}$, are the size $S(300,\chi)\, s^2/300^2$ of the fluctuation term of the tail. Such a match does not show that the product has no exponential factor; that is Lemma 16.2.2, whose argument covers $\xi(s,\chi)$ for a real character, because $\xi(\frac{1}{2}+s,\chi)$ is even in s (Proposition 16.5.1).

Negative moments and the tilt. The negative moments, computed as the Mellin integral of equation (16.3.14) for $\xi(s,\chi)$, are

$$\mathbb{E}\big[T_{\chi_4}^{-1/2}\big] = 3.7574677\ldots, \qquad \mathbb{E}\big[T_{\chi_5}^{-1/2}\big] = 3.7097072\ldots,$$

smaller than 6.7333 for ζ because the first zeros are lower. In equation (16.5.3), $(\xi'/\xi)(1,\chi_4) = 0.0777840$ and $(\xi'/\xi)(1,\chi_5) = 0.0782785$. The difference from $p_1(\chi)$

is 2.30835×10^{-4} for χ_4 and 1.78632×10^{-4} for χ_5; the series of equation (16.3.11) through $j = 12$ agrees to 3×10^{-28}. Since $\xi(1, \chi_4) = \frac{4}{\pi} \cdot \frac{\pi}{4} = 1$, $\mathbb{E}[e^{-T\chi_4/4}] = \xi(\frac{1}{2}, \chi_4) = 0.9807136$, and the product over the zeros with the tail gives 0.9807140, a relative difference of 4.2×10^{-7}.

The tilt $u_0 = 1$. For ζ, the product $\prod_\gamma (1+\gamma^2)/(1+\gamma^2+s)$ over the first 400 ordinates with the Riemann von Mangoldt tail started at γ_{400} matches $\xi(\frac{3}{2})/\xi(\frac{1}{2} + \sqrt{1+s})$ to the tail error: the zero side is 0.9332865 against 0.9332901 at $s = 3$, a relative difference of 3.8×10^{-6}, and 0.3287603 against 0.3287814 at $s = 50$, a relative difference of 6.4×10^{-5}. With the anchored tail of equation (16.3.8) the differences are 9.5×10^{-9} and 1.6×10^{-7}.

16.6.2 The family: sum and mixture

The cyclotomic field settles one family question: by Proposition 16.5.3 the independent sum of the character variables of conductor dividing q is the Thorin variable of $\mathbb{Q}(\zeta_q)$, a GGC whenever the Thorin property holds there. Averages are another matter. Let $\mathcal{F}_q$ be the real primitive characters modulo q together with one representative of each conjugate pair of complex ones, and let $T^{(q)}$ be the variable of a uniformly chosen member, under GRH for the members and with their central values nonzero, so that their clocks exist (Propositions 16.5.1 and 16.5.2). Its moments are the averages

$$\mathbb{E}\big[(T^{(q)})^k\big] = \frac{1}{|\mathcal{F}_q|} \sum_{\mathcal{F}_q} \mathbb{E}[T_\chi^k],$$

but the next proposition shows that $T^{(q)}$ is in general not a GGC. Here the Thorin measures of the members are sums of unit point masses, repeated according to multiplicity, so their Laplace transforms $\varphi_i(s) = \prod_t t/(t+s)$ are meromorphic on $\mathbb{C}$ with poles at the points $s = -t$.

Proposition 16.6.3 (Mixtures of clocks). *Let φ_1, φ_2 be Laplace transforms of GGCs with zero drift and Thorin measures U_1, U_2 that are sums of unit point masses, repeated according to multiplicity, let $0 < \lambda < 1$, and put $\varphi = \lambda\varphi_1 + (1-\lambda)\varphi_2$.*

(a) If $\varphi(-x) = 0$ for some $x > 0$ that is not an atom of U_1 or U_2, then φ is not the Laplace transform of a GGC.

(b) Let $t_1 < t_2$ be consecutive points of $\operatorname{supp} U_1 \cup \operatorname{supp} U_2$ with t_1 an atom of only one of the measures and t_2 an atom of only the other. If $U_1 + U_2$ has an odd number of atoms in $(0, t_1]$, counted with multiplicity, then φ has a zero in $(-t_2, -t_1)$, so φ is not the Laplace transform of a GGC, for any λ.

Proof. (a) Suppose $\varphi = g$ on $(0, \infty)$ with $g(s) = \exp(-ds - \int \log(1 + s/t)\, U(dt))$. Both are analytic on $\mathbb{C} \setminus (-\infty, 0]$, so they agree there. For $|s| \le x + 1$ off the

negative axis, $|1+s/t| \le 1+(x+1)/t$, so $|g(s)| \ge \exp(-d(x+1) - \int \log(1+(x+1)/t)\,U(dt)) > 0$, the integral being finite by the conditions on a Thorin measure. Letting $s \to -x$ from the upper half plane, $\varphi(s) \to \varphi(-x) = 0$ because $-x$ is not a pole, a contradiction. (b) For $y > 0$ off the atoms, $\varphi_i(-y) = \prod_t t/(t-y)$ has the sign $(-1)^{N_i(y)}$, with $N_i(y)$ the number of atoms of U_i in $(0,y)$. Say t_1 is an atom of U_1 and t_2 of U_2. As $y \downarrow t_1$, $|\varphi_1(-y)| \to \infty$ while $\varphi_2(-y)$ stays bounded, so $\varphi(-y)$ has the sign $(-1)^{n_1}$, where n_i is the number of atoms of U_i in $(0,t_1]$. As $y \uparrow t_2$, φ_2 dominates and the sign is $(-1)^{n_2}$, since U_2 has no atom in (t_1,t_2). These signs differ when n_1+n_2 is odd, and the intermediate value theorem gives the zero; (a) applies. □

Part (b) concerns two clocks. If no atom is common to the two measures, its hypothesis fails only when every maximal run of consecutive atoms of one measure, except the last run, has an even number of atoms (counted with multiplicity), because the numbers of atoms at the changes from one measure to the other are the partial sums of the run lengths; the odd configuration of (b) is the case in which some such partial sum is odd.

Modulo 5 the family has two members, χ_5 and the conjugate pair $\{\psi, \bar\psi\}$ with $\psi(2) = i$, so that $\lambda = \frac12$ is the uniform average. Under GRH and with $L(\frac12,\chi_5)L(\frac12,\psi) \ne 0$ the clocks of the two members are the Thorin variables of Propositions 16.5.1 and 16.5.2, the second for the product $\xi_{\psi\bar\psi}(s) = \xi(s,\psi)\,\xi(s,\bar\psi)$; numerically $L(\frac12,\chi_5) = 0.23175$ and $L(\frac12,\psi) = 0.76375 + 0.21696\,i$, with enclosures that exclude 0 (at 128 bits). In the variable $y = \sqrt{-s}$ the atoms are the ordinates: for the pair 4.1329037, 6.1835782, 8.4572292, 9.4429311 (the absolute values of the ordinates of $L(s,\psi)$), and for χ_5 6.6484533, 9.8314444. These are all the atoms below 9.84: the argument principle certifies exactly 3 zeros of $\xi(\frac12+it,\chi_5)$ and 5 zeros of $L(s,\psi)$, two with positive and three with negative ordinate, below height 12. Three atoms lie in $(0, 6.6484533]$, and the next atom belongs to the pair, so by (b) the mixture vanishes between $y = 6.6484533$ and $y = 8.4572292$ for every λ; the same happens between 9.4429311 and 9.8314444, with five atoms below.

Numerically, on the negative axis the mixture is $M_\lambda(y) = \lambda A(y) + (1-\lambda)B(y)$, with $A(y) = \xi(\frac12,\chi_5)/\xi(\frac12+iy,\chi_5)$ and $B(y) = \xi_{\psi\bar\psi}(\frac12)/\xi_{\psi\bar\psi}(\frac12+iy)$. It is computed in ball arithmetic at 128 bits for $\lambda = 0.01$, 0.1, 0.5, 0.9 and 0.99. In all ten cases the signs at the two ends of the interval are opposite, and each interval contains exactly one zero (the argument principle on a thin rectangle about the interval gives the count); Table 16.6 lists the zeros.

Proposition 16.6.4 (The family modulo 5). *Under GRH for $L(s,\chi_5)$ and $L(s,\psi)$, for every $\lambda \in (0,1)$ the function $\lambda\,\xi(\frac12,\chi_5)/\xi(\frac12+\sqrt{u},\chi_5) + (1-\lambda)\,\xi_{\psi\bar\psi}(\frac12)/\xi_{\psi\bar\psi}(\frac12+\sqrt{u})$ of $u \ge 0$ is not the Laplace transform of a GGC. In particular the variable $T^{(5)}$, the case $\lambda = \frac12$, is not a GGC.*

Proof. The central values are nonzero, as computed, so under GRH the clocks of the two members are the Thorin variables of Propositions 16.5.1 and 16.5.2,

Table 16.6: The zero y of the mixture $M_\lambda = \lambda A + (1-\lambda)B$ modulo 5 in each of the two intervals between a pole of A and a pole of B, for five weights; each interval contains exactly one zero.

λ	zero in (6.6484533, 8.4572292)	zero in (9.4429311, 9.8314444)
0.01	6.64874689	9.83141547
0.1	6.65169528	9.83112598
0.5	6.67865792	9.82859297
0.9	6.98112355	9.80692785
0.99	8.03540132	9.65957988

whose Thorin measures are sums of unit point masses at the squared ordinates. The certified atoms above put the two members in the odd configuration of Proposition 16.6.3(b). The location of the zeros uses no hypothesis; the hypothesis enters only through the identification of the Thorin measures of the two members. □

16.6.3 Landau Siegel zeros and the twisted heat trace

Let χ be a primitive real character of conductor $q \geq 3$. Siegel's theorem gives $1-\beta \gg_\varepsilon q^{-\varepsilon}$ for a real zero β of $L(s,\chi)$, with an ineffective constant, and the classical effective bound is only $1-\beta \gg q^{-1/2}\log^{-2} q$ [70].

Theorem 16.6.5 (Banks, Kerr; unrefereed). *There are absolute constants q_0 and $\delta_0 > 0$ such that for every primitive real character χ of conductor $q \geq q_0$, every real zero $\beta \in (0,1)$ of $L(s,\chi)$ satisfies $(1-\beta)\log q \geq \delta_0$. Banks gives $q_0 = 10^{10}$ and $\delta_0 = 3.0 \times 10^{-3}$.*

The preprints [14, 181] are dated October 2026 and have not been refereed (Section 16.1). Banks describes the proof as an arithmetic comparison of the archimedean and p adic sizes of an interpolation determinant in $\mathbb{Q}(\sqrt{2}, \sqrt{\chi(-1)q})$, and Kerr gives a simplification; we have not checked either. The half plane $\operatorname{Re} s > \frac{7}{8}$ for every Dirichlet L function reported in Section 16.1 [257], if it stands, would give $1-\beta \geq \frac{1}{8}$ for every real zero and so imply Theorem 16.6.5 for every $q \geq 3$. Banks refers to a proof by OpenAI; whether that is the release of [257] we cannot say.

The twisted heat trace. Let $z_\rho = (\rho - \frac{1}{2})^2$, the variable of this chapter (the atom of a zero is $u_0 - z_\rho$, Theorem 16.2.3), so that $z_\rho = -\gamma^2$ for $\rho = \frac{1}{2} + i\gamma$ and $z_\rho = (\beta - \frac{1}{2})^2 > 0$ for a real zero β. As in Section 8.1, with the zeros counted with multiplicity and $t > 0$ the heat time, the twisted heat trace is

$$W(t;\chi) = \sum_{\operatorname{Im}\rho > 0} e^{t z_\rho} + \tfrac{1}{2} \sum_{\rho \text{ real}} e^{t z_\rho}.$$

It is not the root number W_χ of Section 16.5, which equals 1 for real χ. If $L(\frac12,\chi)\neq 0$ and $|\gamma|>|\beta-\frac12|$ for every zero $\beta+i\gamma$ (so that $\mathrm{Re}(-z_\rho)>0$ and the integral converges), then for $u\geq 0$

$$\log\frac{\xi(\frac12+\sqrt{u},\chi)}{\xi(\frac12,\chi)}=\int_0^\infty(1-e^{-ut})\,\frac{W(t;\chi)}{t}\,dt,$$

and the ratio $\xi(\frac12,\chi)/\xi(\frac12+\sqrt{u},\chi)$ is the Laplace transform of a GGC if and only if $W(\cdot\,;\chi)$ is completely monotone, which as in Section 8.1 is the ladder $(-1)^kW^{(k)}(t;\chi)>0$. The Gaussian explicit formula, with $a\in\{0,1\}$ the parity of χ and ψ the digamma function, reads

$$\begin{aligned}W(t;\chi)=\frac{\log(q/\pi)}{4\sqrt{\pi t}}+\frac{1}{4\pi}\int_{-\infty}^{\infty}e^{-tr^2}\,\mathrm{Re}\,\psi\Big(\frac{1+2a}{4}+\frac{ir}{2}\Big)dr\\ -\frac{1}{2\sqrt{\pi t}}\sum_{n\geq 2}\frac{\Lambda(n)\chi(n)}{\sqrt{n}}\,e^{-(\log n)^2/4t}.\end{aligned}\tag{16.6.1}$$

Compared with the formula for ξ in equation (2.1.5) the polar term $e^{t/4}$ is absent, because $L(s,\chi)$ is entire, $-\log\pi$ in $m(r)$ becomes $-\log(\pi/q)$, and the prime sum carries the factor $\chi(n)$.

A real zero as a failure of rung one. A real zero $\beta\in(\frac12,1)$ and its mirror $1-\beta$ contribute together the single term e^{ct} with $c=(\beta-\frac12)^2=\frac14-\beta(1-\beta)$, which grows with t. This is the defect that a complex off line pair cannot model in Theorem 8.1.14, whose defect needs a nonzero imaginary part.

Proposition 16.6.6. *Let $\beta\in(\frac12,1)$ be the largest real zero of $L(s,\chi)$, suppose every nonreal zero of $L(s,\chi)$ lies on the critical line, and put $c=(\beta-\frac12)^2$. Then $W(t;\chi)\to\infty$ and $-W'(t;\chi)<0$ for all $t>t_1=\max\{1,c^{-1}\log(C'\log q/c)\}$, where C' is an absolute constant. If moreover $q\geq q_0$, put $\eta=\delta_0/\log q$; then $c\leq\frac14-\eta(1-\eta)$, so by Theorem 16.6.5 the growth rate c of such a term is at least $\eta(1-\eta)$ below the rate $\frac14$ of the pole term $e^{t/4}$ of ζ.*

Proof. Under the hypotheses,

$$W(t;\chi)=\sum_{\gamma>0}e^{-t\gamma^2}+me^{ct}+(\text{terms }e^{c't}\text{ from the other real zeros, }c'\leq c),$$

and $-W'(t;\chi)\leq\sum_{\gamma>0}\gamma^2e^{-t\gamma^2}-ce^{ct}$. By the bound $N(T,\chi)\ll(T+1)\log(q(T+2))$ for the zero counting function, the sum is at most $C'\log q$ at $t=1$ and decreases in t, so $-W'<0$ as soon as $ce^{ct}>C'\log q$, which holds for $t\geq t_1$; and $W\geq e^{ct}\to\infty$. For the last statement, the identity $c=\frac14-\beta(1-\beta)$ and $\beta\leq 1-\eta$ (Theorem 16.6.5), with $\beta(1-\beta)$ decreasing on $(\frac12,1)$, give $\beta(1-\beta)\geq\eta(1-\eta)$. $\square$

Thus a real zero with $\beta > 1 - \eta$, which Theorem 16.6.5 excludes for $q \geq q_0$, would, under the hypotheses of Proposition 16.6.6, be seen on rung one at the scale t_1. The difficulty is uniformity in q: for a given χ of small conductor the verification of GRH by Platt [275] covers a real zero, and for general q equation (16.6.1) is the only handle. For complex pairs the book's detection of a pair at height a is Proposition 8.1.10, and nothing in Theorem 16.6.5 touches complex zeros.

The pole lock, a heuristic. The prime sum of equation (16.6.1) suggests why a zero near 1 is tied to the primes with $\chi(p) = -1$. Suppose, as a model and not as a claim that such a character exists, that $\chi(p) = -1$ for every prime p. The prime term of equation (16.6.1) is then

$$\frac{1}{2\sqrt{\pi t}} \sum_{p} \frac{\log p}{\sqrt{p}}\, e^{-(\log p)^2/4t} + O(1) \;\approx\; \frac{1}{2\sqrt{\pi t}} \int_0^\infty e^{u/2 - u^2/4t}\, du = e^{t/4}\, \Phi\big(\sqrt{t/2}\big),$$

where Φ is the standard normal distribution function and the $O(1)$, which comes from the prime powers (mainly the squares, where $\chi(p^2) = 1$) and tends to $-\frac{1}{4}$ as $t \to \infty$, is not negligible for $t \leq 3$. As $t \to \infty$ this is $e^{t/4}(1 + o(1))$: on the prime side, such a character produces the growth that the pole of ζ produces in equation (2.1.5). A real zero β is the same picture with the rate reduced from $\frac{1}{4}$ to $(\beta - \frac{1}{2})^2$, and the reduction $\beta(1 - \beta)$ is what Theorem 16.6.5 bounds below. Numerically (primes up to 6×10^6) the ratio of the prime sum to $e^{t/4}$ is 0.402, 0.510, 0.645, 0.735 at $t = 0.5$, 1, 2, 3. This is a heuristic about a hypothetical character.

What does not transfer. The proof of Theorem 16.6.5 as Banks describes it is not a positivity statement and not a complete monotonicity statement, and it concerns the single point $s = 1$ through the pole of $\zeta_K(s) = \zeta(s)L(s, \chi)$ of the quadratic field $K = \mathbb{Q}(\sqrt{\chi(-1)q})$ of χ. The Thorin measure concerns the whole zero set in the variable z_ρ, and we see no way to recast the determinant argument as a rung of the ladder.

16.7 Consequences for the Linnik constant and for RH

16.7.1 The Linnik constant

Corollary 16.7.1. *Suppose that for every $q \geq 3$ the function $\varphi_{\mathbb{Q}(\zeta_q),0}$ is the Laplace transform of a GGC. Then every sufficiently large even integer is a sum of two primes and at most seven powers of two.*

Proof. Theorem 16.6.1 gives GRH for all Dirichlet L functions, and under GRH every sufficiently large even integer is a sum of two primes and at most seven powers of two, by the theorem of Heath-Brown and Puchta [111] (see also Pintz and Ruzsa [271]). □

The corollary is weaker than the known conditional result: by Corollary 16.2.4(a) and equation (16.5.4), its hypothesis is GRH for every Dirichlet L function together with $L(\frac{1}{2},\chi)\neq 0$ for every primitive character χ. The central nonvanishing is not known to follow from GRH (for real characters it is Chowla's conjecture), whereas $K=7$ follows from GRH alone. A direct proof of the Thorin property for all cyclotomic fields would prove GRH, and with it $K=7$ unconditionally, one better than the record $K=8$ of Pintz and Ruzsa [272].

For K itself the Thorin property gives what GRH gives. Nothing in the proof of Corollary 16.7.1 uses the nonvanishing at the centre, and the inputs that decide K below seven, the large deviation estimate for the sum $\mathcal{D}(\alpha)$ over the powers of two (step (4) of Section 16.1.2), the sieve constant for prime differences (step (2)) and the refinements of Pintz and Ruzsa, are additive questions about the powers of two and not statements about zeros. With the present methods the framework does not reach below seven.

A weaker hypothesis may suffice for seven. The Thorin property at tilt $u_0=1$ for cyclotomic fields (Corollary 16.2.4(b) with equation (16.5.4)) is GRH with possible real zeros. To see where a real zero could hurt, write $\mathcal{S}(a/q+\eta)$ as a combination of twisted prime sums over the characters modulo q. A real zero β of $L(s,\chi^*)$, with χ^* of conductor $q^*\mid q$, contributes a term of size about $q^{1/2}\varphi(q)^{-1}N^{\beta}(1+|\eta|N)^{-1}$, which exceeds $N^{3/4}$ only if $q<N^{2\beta-3/2}$ and $|\eta|<N^{\beta-7/4}$. Those arcs lie close to the major arcs, and moving them there, where a real zero changes the main term by a factor of the shape $1-\chi^*(\cdot)N^{\beta-1}$ that Gallagher's method [90] already handles for one exceptional zero, would reduce the question to major arc estimates with real zeros (Problem 16.2). If it works, real zeros, and with them Siegel zeros, become irrelevant to $K=7$.

16.7.2 What would help towards RH

The three smoothings of the centre clock are compared in Chapter 8. A proof has to establish a closed positivity condition at the boundary, from input that sees the arithmetic beyond positivity. Zero free regions of the kind announced in October 2026 [257] would be such input, if confirmed, and each one would move the unconditional end of the basepoint family inwards (Section 16.1.2); averages over families of L functions, where the low lying zeros are conjectured to obey the symmetry laws of Katz and Sarnak [179], are another (Problem 16.4). Section 8.4 says what would get past each of the three obstructions of Chapter 8.

16.8 Open problems

Problem 16.1 (The Thorin property without the zeros). Establish condition (i) of Theorem 16.2.3 for $\mathbb{Q}$, or for cyclotomic fields, without first knowing (ii). In the master theorem route, settle the parabolic distortion of Section 16.4.3, uniformly in the field, and find a substitute for the residue bookkeeping at the zeros. *Status.*

Open. For $\mathbb{Q}$ at $u_0 = 0$ the statement to be proved is RH, so a growth bound that is itself proved under RH, such as Lemma 16.4.1, cannot be an input to its proof.

PROBLEM 16.2 (Seven with real zeros). Show that $K = 7$ follows from the Thorin property at tilt $u_0 = 1$ for cyclotomic fields, that is from GRH with possible real zeros. Section 16.7 locates the arcs where a real zero can exceed the bound $N^{3/4}$; what remains is to move them into the major arcs and to redo the main term with several real zeros, checking that the constant 2.7895 survives. In the paper of Heath-Brown and Puchta [111] the major arc estimate that gives this constant (their Lemma 7) is stated for $K \geq 9$, with zero density estimates in place of GRH, and the paper does not say how it is used for $K = 7$; the computation of the constant itself holds for $K \geq 7$. *Status.* Open; only the location of the arcs is done, and it is heuristic.

PROBLEM 16.3 (The moment identity at scale). Anchoring the tail to the exact zero count, equation (16.3.8), already brings the check of p_1 from 2.6×10^{-6} to 6.3×10^{-9} with 799 zeros. Carry it to large height with Odlyzko's tables [255], where the remaining term of (16.3.8), the integral of $S(t)\, t^{-2j-1}$ over the tail, sets the limit and calls for an explicit formula for $\int S$, and to $L(s, \chi)$ with the zeros of Platt [275]. *Status.* Partly done. A computation reproduces the anchoring with the first 1700 ordinates at the cuts $N = 200$ and $N = 799$ and at every cut $N = 50, \dots, 1699$, at 100 digits. For p_1 the anchored tail typically gains about a third of the cut height (deciles 42 and 2700), and little where $|S|$ is small at the cut: the gain is below 10 at 3.2% of the cuts. Large heights and the characters are open.

PROBLEM 16.4 (The negative moment as a family statistic). The value $\mathbb{E}[T^{-1/2}] = 6.7333$ (and 3.7575, 3.7097 for T_{χ_4}, T_{χ_5}) is computed from ξ alone: $\mathbb{E}[T^{-1/2}] = \Gamma(\frac{1}{2})^{-1} \int_0^\infty u^{-1/2} \prod_\gamma (1 + u/\gamma^2)^{-1}\, du$, the product being over the ordinates $\gamma > 0$. It is mostly a smooth statistic. Let E_m be this integral with the first m ordinates exact and the factors of the others replaced by the smooth model $\exp\big(-\frac{1}{\pi}\int_{t_m}^\infty \log(1+u/t^2)\, \vartheta'(t)\, dt - a_m \log(1+u/t_m^2)\big)$, where ϑ is the Riemann Siegel theta function, t_m is the midpoint of γ_m and γ_{m+1}, and $a_m = N_0(t_m) + 1 - m$, with $N_0 = \vartheta/\pi$, is the mass of the atom at t_m that the anchored tail (16.3.8) adds. For $m = 0$ the cut is $t_0 = 9.6669$, the point where $N_0 + 1 = 0$, and there is no atom. The relative error $E_m/\mathbb{E}[T^{-1/2}] - 1$ is -7.0×10^{-3} for $m = 0$, -2.5×10^{-3} for $m = 1$, 1.0×10^{-4} for $m = 2$, -3.9×10^{-4} for $m = 5$ and 1.4×10^{-5} for $m = 20$. The atom matters: cutting at the midpoint without it gives -5.5×10^{-3}, -1.2×10^{-3}, -1.4×10^{-3} and -6.5×10^{-4} for $m = 1$, 2, 5, 20, and cutting at the point where $N_0 + 1 = m$ without it gives -2.5×10^{-3}, 9.1×10^{-5}, -4.0×10^{-4} and 1.0×10^{-5}. The elasticity $(\gamma_k/E)\, \partial E/\partial \gamma_k$ of $E = \mathbb{E}[T^{-1/2}]$ with respect to one ordinate, the others fixed in the integral above, is 0.1799 for γ_1 and falls to 0.0319 for γ_6 (over all ordinates the elasticities sum to 1, by scaling). A single function says little; the natural comparison with random matrices is in families.

Compute the average of $\mathbb{E}[T_\chi^{-1/2}]$ over real primitive characters of conductor in $[Q, 2Q]$ and compare it with the prediction of the symplectic ensemble. Since $\mathbb{E}[T^{-1/2}]$ is not a linear statistic of the zeros, the one level density of Katz and Sarnak [179] alone does not give this prediction, and all the correlation functions enter. The circular unitary comparisons of Section 14.3.5 and the moments of Keating and Snaith [180] (Section 14.3.6.1) are the models for a single function. *Status.* Partly done: the single function is computed, with the model above, ; the family average is open.

Problem 16.5 (The family mixture). Let $T^{(q)}$ be the variable of a uniformly chosen member of $\mathcal{F}_q$ (Section 16.6.2, under GRH for the members and with their central values nonzero). Is $T^{(q)}$ infinitely divisible, and does any q with two or more members avoid the odd configuration of Proposition 16.6.3(b), which is stated for two clocks, for every pair of its members? *Status.* Open. That $T^{(q)}$ is not a GGC is settled for $q = 5$ and every weight, under GRH for the two members (their values at $\frac{1}{2}$ are nonzero numerically; Proposition 16.6.4), and holds, under the same hypotheses, for every q whose family has exactly two members, when these are in the odd configuration (Proposition 16.6.3(b)). It does not decide whether $T^{(5)}$ is infinitely divisible.

Problem 16.6 (Sign definite explicit formulae). Under the hypothesis of Corollary 16.7.1 the explicit formula for $\psi(x; q, 1)$, which up to the factor $\varphi(q)$ counts the prime ideals of $\mathbb{Q}(\zeta_q)$, becomes an integral against the nonnegative measure $U_{F_q,0}$, and Theorem 16.5.8 supplies the positive family $\pi^{-1}\operatorname{Re}(\xi'_{F_q}/\xi_{F_q})(\alpha + ib)$ joining it to the primes. For $a \not\equiv 1 \pmod q$ the characters enter the formula for $\psi(x; q, a)$ with the weights $\bar\chi(a)$, which are complex or negative, and no sign is available. These are zero counting measures and their Poisson smoothings, so any gain must come from structure beyond counting, such as monotonicity or log concavity of the associated GGC densities [39, Ch. 4], transported to bounds on L'/L near the line. Is there a sign definite explicit formula for $\psi(x; q, a)$ with $a \not\equiv 1 \pmod q$, in terms of such measures? *Status.* Open.

Problem 16.7 (Certified backward continuation). The Thorin density $\pi^{-1}\operatorname{Re}(f'/f)(\alpha + ib)$ of Theorem 16.5.8, which is $\rho_{\alpha-1/2}(b)$ for $f = \xi$, is computable from the primes, with rigorous error bounds, for $\alpha > 1$, and its positivity on every line $\operatorname{Re} s = \alpha'$ with $\alpha' > \alpha$ is a zero free half plane $\operatorname{Re} s > \alpha$. The book certifies the minimum at the origin for ξ, χ_4 and χ_{12} for $\alpha < 1$ up to a large height, from the zeros (Theorem 12.1.18). How far into the strip can positivity be certified from the primes alone, using smoothed prime sums of the kind that located the zeros of $L(s, \chi_4)$ in Section 16.5.6, before the ill posedness of the backward step takes over? Chapter 8 measures the cost of the step: by Theorem 8.2.2 it is a deconvolution, which is unbounded for the basepoint family, and Proposition 8.2.3 counts the primes it needs, up to about $\eta^{-1/(\alpha-\sigma_{\max})}$ for accuracy η; for the heat flow the corresponding step is stable but cannot carry

real zeros (Section 8.4). Section 9.1.1 of Chapter 9 reads the same step as a deconvolution of the explicit formula. *Status.* Partly done. For χ_4 the prime side (16.5.9) of π times the density at $b = 0$, summed over the prime powers up to 2×10^6 without smoothing, differs from $(\xi'/\xi)(\alpha, \chi_4)$ by 1.3×10^{-4}, 4.8×10^{-3}, 4.1×10^{-2} and 8.3×10^{-2} at $\alpha = 1$, $\frac{3}{4}$, $\frac{3}{5}$ and $\frac{11}{20}$, that is by 0.2%, 12%, 260% and 1100% of the value, so that for $\alpha \le 1$ the minima computed in Section 16.5.6 come from the zeros and not from the primes. What smoothed prime sums can certify below $\alpha = 1$ is open.

Problem 16.8 (The arithmetic Lévy measure and the minor arcs). For $F = \mathbb{Q}$ the atoms of ν_F in (16.5.6) at the primes p have mass $\log p$, the weights that enter $\mathcal{S}(\alpha)$; the prime powers of (16.5.6) do not enter $\mathcal{S}(\alpha)$. Can $\mathcal{S}(\alpha)$ on the minor arcs be written as an expectation against the GGC laws above? Such a representation would be a different route to the sup bound in step (3); nothing in the chapter shows that it exists. *Status.* Open.

Problem 16.9 (Zero free half planes in the circle method). Assume $L(s, \chi) \neq 0$ for $\operatorname{Re} s > \theta_0$ and every Dirichlet character. What is the best minor arc exponent $\theta(\theta_0)$ obtainable by combining the explicit formula with Vaughan's identity? The one line balance of Section 16.1.2 gives $\frac{1}{2}(1+\theta_0)$, which is better than the unconditional exponent $\frac{263}{308}$ of step (3) exactly when $\theta_0 < \frac{109}{154}$, so it does nothing at $\theta_0 = \frac{7}{8}$; it is not shown to be optimal. Does a hybrid bound, using the half plane only for the characters of small conductor, improve λ in step (4)? And does the half plane, which excludes exceptional zeros of large conductor, raise the constant 2.7895 of step (1)? The paper of Heath-Brown and Puchta [111] suggests that it does not by itself: they state that zeros close to $\sigma = 1$ play no role and that no explicit numerical zero free region is needed, the constant is a sum over the divisors $d \le 5$ (the improvement of Elsholtz to 2.96169 that they report takes $d \le 21$ and a better sieve constant), and the restriction $q \le P = N^{45/154-4\varpi}$ of their major arcs comes from zero density estimates, whose critical zeros have $\sigma \le \frac{3}{4}$ and are not removed by a half plane at $\frac{7}{8}$. That restriction is not sharp for those estimates: the exponent $(\frac{45}{77}\kappa(\sigma) - 1)(1 - \sigma)$ of their Section 3 has its maximum 0.1050 at $\sigma = 0.676$, against the bound $\frac{31}{154} = 0.2013$ that they use. *Status.* Open.

Problem 16.10 (Finite order total positivity). RH holds if and only if the density of $Y = \sqrt{2T}\, Z$ is a Pólya frequency function of infinite order (Proposition 22.5.1, where the variable is called X; Section 16.7.2). For this one density the question whether a finite order implies RH is trivial: the orders decrease, so if RH holds every such implication is true, and if RH fails the property fails at some finite order, where the implication is vacuous. The question has content for a class of functions. Let $\mathcal{E}$ be the class of even entire functions F with $F(0) = 1$ and the order, type and reflection symmetry of $\xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$ for which $1/F$ is the characteristic function of a probability density f_F; the density of Y is f_F for $F = \xi(\frac{1}{2} + \cdot)/\xi(\frac{1}{2})$ (Proposition 22.5.1). Is there a finite k such that, for every

$F \in \mathcal{E}$, the property that f_F is a Pólya frequency function of order k implies that every zero of F lies on the imaginary axis? *Status.* Open. Chapter 22 shows that the low orders are decided by the low zeros: order two holds without RH (Theorem 22.11.5), and the density lies within 1.6×10^{-10} of a Pólya frequency function of every order (Theorem 22.11.6), so that, given the verification of RH to height 3×10^{12} [277], every minor of order 3 and 4 is at least -1.1×10^{-8} and -1.2×10^{-7}. Problem 22.1 asks for proofs of the low orders, not for the converse asked here. The computations of this chapter are in Section 16.3.3.5. The perturbation of Theorem 1.4.1, made at a height that is large compared with k, is the natural candidate for a negative answer in the class $\mathcal{E}$; whether it keeps the order k has not been computed here.

Notes

The Thorin representation and the van Dantzig pair are from [279, 281]; the GGC background is Bondesson's monograph [39]; the master theorem class is from Chaudhry and Qadir [56], and the exponential Tauberian theorem from Kasahara [178]. Completed Dirichlet L functions and Dedekind zeta functions are in Davenport [70], mean values of ζ in Titchmarsh [348]. The account of the circle method in Section 16.1.2 is a summary of the paper of Heath-Brown and Puchta [111].

Chapter 17

Completeness of the secant dilates and the Möbius problem

Chapter 10 writes the exponential as a Möbius weighted sum of hyperbolic secants (Proposition 10.1.3) and records the question of Lai and Vere-Jones [200] on the completeness of the dilates $\operatorname{sech}(\lambda x)$ as open (Problem 17.10 asks for its relation to the Nyman Beurling criterion). We answer it in weighted L^2 spaces on the half line and say exactly where arithmetic enters. In one weight the integer dilates $\operatorname{sech}(nx)$ are complete unconditionally, by Carleman's formula, whatever the zeros of $L(s,\chi_4)$ do. In two weights at once completeness is equivalent to a zero free half plane of $L(s,\chi_4)$, with the generalised Riemann hypothesis at the threshold. This is a Nyman Beurling type criterion for the Dirichlet beta function, with an analogue for ζ.

The chapter is based on two notes of the first author, one on the secant dilates, Lerch's theorem and the Möbius function (Sections 17.1 to 17.8) and one on its four open problems (Sections 17.10 to 17.16). Every number printed is recomputed for this book. Nothing is attributed to [200] beyond the question and the identity of Proposition 10.1.3; the paper itself was not consulted.

Notation. For $c > 0$ the weighted space is $\mathcal{W}_c = L^2((0,\infty), x^{2c-1}dx)$; $\beta(s) = L(s,\chi_4) = 1 - 3^{-s} + 5^{-s} - \cdots$ is the Dirichlet beta function, a symbol local to this chapter (the real part of a zero is always written $\operatorname{Re}\rho$); μ is the Möbius function, $M_\chi(x) = \sum_{n\le x}\mu(n)\chi_4(n)$ and $M(x) = \sum_{n\le x}\mu(n)$. The Dirichlet polynomial $\mathcal{A}_N(s) = \sum_{n\le N}\mu(n)\chi_4(n)n^{-s}$ is written with a script letter because A_N is the coefficient of Theorem 10.1.5. The completed beta function is $\xi(s,\chi_4)$ of Section 15.1.7. The probability measures of one pair of zeros (Section 17.14) are written $Q_{a,\gamma}$. Results that rest on the quasi Riemann hypothesis [257, 258, 223] are conditional on it, and on one further point: the preprint is announced for Dirichlet L functions, and whether its theorem covers the character modulo 4 has not been checked for this book.

The results:

1. The integer dilates $\{\operatorname{sech}(nx)\}$ are complete in L^1 with the weight of

Lerch's theorem, and in every $\mathcal{W}_c$, $c > 0$, unconditionally; more generally $\{\mathrm{sech}(\lambda x)\}_{\lambda\in\Lambda}$, $\Lambda \subset [1,\infty)$, is complete in $\mathcal{W}_c$ whenever $\sum_\Lambda 1/\lambda = \infty$ (Theorems 17.1.4 and 17.3.2, proved). The proof of the second statement replaces Lerch by Carleman's formula and uses no arithmetic.

2. Completeness in two weights $\mathcal{W}_a \cap \mathcal{W}_b$ for all $\theta < a < b < 1$ is equivalent to $L(s,\chi_4) \neq 0$ for $\mathrm{Re}\, s > \theta$, and to convergence of the Möbius secant series in every $\mathcal{W}_c$, $\theta < c < 1$ (Theorem 17.5.2, proved); completeness in two weights straddling a strip forces that strip to be zero free (Proposition 17.5.1, proved). The zeta kernel $1/(e^x-1) - 1/x$ has the same two statements, with ζ and RH (Theorem 17.6.1, proved). The consequences of the quasi Riemann hypothesis are in Section 17.7 (conditional).

3. Under a polynomial bound for $1/\beta$ on the line $\mathrm{Re}\, s = c$, which holds for $c \geq 1$, conditionally on a zero free half plane for c to its right, and for $c > 0$, $c \neq \frac{1}{2}$ under GRH, the condition $\sum_\Lambda 1/\lambda = \infty$ is also necessary (Theorem 17.10.2, proved); the line $c = \frac{1}{2}$ stays open. The L^1 criterion of Salem for the integer dilates, and its beta analogue, are Theorem 17.11.1 (proved).

4. The optimal distance of e^{-x} from the span of $\mathrm{sech}(nx)$, $n \leq N$, is given by a Gram system (computed, Table 17.2); the optimal coefficients are huge, and the explanation by the frequencies $\log n$ is heuristic (Section 17.12). The residual identity of Proposition 17.12.1, with its factor 2, is proved.

5. For the Meixner family of order t the Mellin transform is M_t of Theorem 17.13.1, single weight completeness holds throughout, and for $t = 2$ the Möbius route is unconditional (Theorem 17.13.3, proved).

6. For $\xi(s,\chi_4)$ the equivalence of the zero free half plane with a generalised gamma convolution form of $\xi(\theta,\chi_4)/\xi(\theta+\sqrt{s},\chi_4)$ and the explicit Thorin measure are Theorem 17.14.2 (proved; for ξ this is Theorem 2.1.2); the link with the two weight criterion is new (Section 17.8).

The open problems close the chapter.

17.1 The secant family and the identity

On the whole line no symmetric scale family can be complete, because odd statistics have mean zero, so the dilates are folded to $(0,\infty)$. The symmetric Meixner density of order $t > 0$ is proportional to $|\Gamma(\frac{t}{2} + \frac{ix}{2})|^2$ (Section 10.2.3), and $t = 1$ gives the hyperbolic secant density because $|\Gamma(\frac{1}{2} + iy)|^2 = \pi/\cosh(\pi y)$. Folded to $x > 0$ it is $p_\lambda(x) = \frac{2\lambda}{\pi}\,\mathrm{sech}(\lambda x)$, and a family $\{p_\lambda : \lambda \in \Lambda\}$ is complete when

$\mathbb{E}_\lambda g(X_\lambda) = 0$ for all $\lambda \in \Lambda$ forces $g = 0$ almost everywhere. For $\Lambda = \mathbb{N}$, completeness says that the dilates $\operatorname{sech}(nx)$ separate functions; this is the integer case of the question of Lai and Vere-Jones [200].

LEMMA 17.1.1. *For $y > 0$, $\operatorname{sech} y = 2\sum_{m\ge1} \chi_4(m) e^{-my}$, and $\sum_{n\ge1} \operatorname{sech}(ny) \le 2/(e^y - 1)$.*

Proof. The identity is the geometric series of Proposition 10.1.3; the bound follows from $\operatorname{sech}(ny) \le 2e^{-ny}$. □

LEMMA 17.1.2. *For $\operatorname{Re} s > 0$ and $\lambda > 0$, $\int_0^\infty x^{s-1} \operatorname{sech}(\lambda x)\, dx = 2\lambda^{-s}\Gamma(s)\beta(s)$.*

Proof. For $\lambda = 1$ integrate Lemma 17.1.1 termwise for $\operatorname{Re} s > 1$, where $\sum_m \int_0^\infty x^{\sigma-1} e^{-mx} dx = \Gamma(\sigma)\zeta(\sigma) < \infty$, and continue analytically; the integral is holomorphic for $\operatorname{Re} s > 0$ and β is entire. The substitution $y = \lambda x$ gives general λ. □

By Proposition 10.1.3, $e^{-x} = \frac{1}{2}\sum_{n\ge1} \mu(n)\chi_4(n) \operatorname{sech}(nx)$ with absolute convergence for $x > 0$, and replacing x by kx,

$$e^{-kx} = \frac{1}{2} \sum_{n\ge1} \mu(n)\chi_4(n) \operatorname{sech}(nkx), \qquad k \in \mathbb{N}, \tag{17.1.1}$$

and every function on the right is again one of the dilates. In Mellin language the identity is the factorisation $\Gamma(s) = \frac{1}{2} \cdot 2\Gamma(s)\beta(s) \cdot \beta(s)^{-1}$ with $\beta(s)^{-1} = \sum \mu(n)\chi_4(n) n^{-s}$.

THEOREM 17.1.3 (Lerch). *Let $\int_0^\infty |h(x)| e^{-x} dx < \infty$. If $\int_0^\infty e^{-kx} h(x)\, dx = 0$ for all $k \in \mathbb{N}$, then $h = 0$ almost everywhere.*

Proof. With $u = e^{-x}$ and $q(u) = h(-\log u)$ we have $q \in L^1(0,1)$ and $\int_0^1 u^j q(u)\, du = 0$ for $j \ge 0$. The measure $q\, du$ annihilates the polynomials, hence $C[0,1]$ by Weierstrass, hence $q = 0$. □

THEOREM 17.1.4 (completeness in L^1). *Let g satisfy*

$$\int_0^1 \frac{|g(x)|}{x}\, dx + \int_1^\infty |g(x)| e^{-x} dx < \infty. \tag{17.1.2}$$

If $\int_0^\infty g(x) \operatorname{sech}(nx)\, dx = 0$ for all $n \in \mathbb{N}$, then $g = 0$ almost everywhere. The same holds with $\mathbb{N}$ replaced by a set $\Lambda \subset (0,\infty)$ containing an interval, provided the integrals $\int g \operatorname{sech}(\lambda x)\, dx$, $\lambda \in \Lambda$, converge absolutely and vanish.

Proof. By Lemma 17.1.1, $\sum_n |\operatorname{sech}(nkx)| \le 2/(e^{kx} - 1)$, which is $O(1/x)$ at the origin and $O(e^{-x})$ at infinity, so (17.1.2) justifies integrating (17.1.1) termwise against g. Every term vanishes, hence $\int g e^{-kx} dx = 0$ for all k, and Theorem 17.1.3 applies. For the interval clause, let $I \subset \Lambda$ be an interval and λ_0 a point of its interior.

Absolute convergence at λ_0 gives $\int_1^\infty |g|e^{-\lambda_0 x}dx < \infty$, because $\operatorname{sech}(\lambda_0 x) \ge e^{-\lambda_0 x}$. The function $F(z) = \int_0^\infty g(x)\operatorname{sech}(zx)\,dx$ is then holomorphic on $\operatorname{Re} z > \lambda_0$ (on compact subsets the large x part is dominated by $C|g(x)|e^{-(\lambda_0+\delta)x}$ and the small x part by $C|g(x)|$), and it vanishes on a subinterval of I to the right of λ_0, hence on the half plane. Take an integer $K > \max(1, \lambda_0)$. The integrals $\int g(x)\operatorname{sech}(nkx)\,dx$ vanish for all integers $n \ge 1$ and $k \ge K$, so (17.1.1) (again integrated termwise, the majorant being $|g|\,2/(e^{kx}-1)$ with $k \ge K$) gives $\int g(x)e^{-kx}dx = 0$ for $k \ge K$. Apply Theorem 17.1.3 to $h(x) = g(x)e^{-(K-1)x}$, whose moments of orders $1, 2, \dots$ are these integrals. □

The proof of the first part uses only $\sum_{d|N}\mu(d) = [N = 1]$, the complete multiplicativity of χ_4 and $|\mu| \le 1$. The condition $\int_0^1 |g|/x < \infty$ is the L^1 shadow of the half plane $\operatorname{Re} s > 1$ of absolute convergence of the Möbius series. The theorem is not claimed to be the form in which the question was posed in [200].

17.2 Weighted spaces and the Mellin line

Let $\mathcal{W}_c = L^2((0,\infty), x^{2c-1}dx)$. The change of variable $x = e^u$ and Plancherel give

$$\langle f, g\rangle_c = \int_0^\infty f\bar{g}\,x^{2c-1}dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(c+it)\overline{G(c+it)}\,dt, \qquad F(s) = \int_0^\infty f(x)x^{s-1}dx. \tag{17.2.1}$$

So $\mathcal{W}_c$ is the space of functions whose Mellin transform lives on the line $\operatorname{Re} s = c$. For $c > 0$ both e^{-x} and every $\operatorname{sech}(\lambda x)$ lie in $\mathcal{W}_c$, and dilation acts by $\|f(k\,\cdot)\|_c = k^{-c}\|f\|_c$.

Lemma 17.2.1 (reduction). *For $c > 0$, or for $\mathcal{W}_a \cap \mathcal{W}_b$ with $0 < a < b$ and norm $\|f\|_a + \|f\|_b$, the exponentials $\{e^{-kx}\}_{k\in\mathbb{N}}$ are complete. Hence a system whose linear span is invariant under the dilations $f \mapsto f(k\,\cdot)$, $k \in \mathbb{N}$, is complete as soon as its closed span contains e^{-x}.*

Proof. A functional on $\mathcal{W}_c$ is $f \mapsto \int fh\,dx$ with $h = \bar{g}x^{2c-1}$, $g \in \mathcal{W}_c$, and $\int |h|e^{-x} \le \|g\|_c\|e^{-x}\|_c$; apply Theorem 17.1.3. On the intersection the dual is the sum of the two duals (Hahn Banach) and the same estimate applies to each piece. If $f_j \to e^{-x}$ with f_j in the span, then $f_j(k\,\cdot) \to e^{-k\,\cdot}$ in the same space and $f_j(k\,\cdot)$ is again in the span. □

Write $S_N(x) = \frac{1}{2}\sum_{n\le N}\mu(n)\chi_4(n)\operatorname{sech}(nx)$. By Lemma 17.1.2 and (17.2.1),

$$\|S_N - e^{-x}\|_c^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\Gamma(c+it)|^2\,\bigl|\beta(c+it)\mathcal{A}_N(c+it) - 1\bigr|^2\,dt, \tag{17.2.2}$$

so convergence of the Möbius secant series in $\mathcal{W}_c$ is the statement that $\mathcal{A}_N \to 1/\beta$ on the line $\operatorname{Re} s = c$ in a norm weighted by $|\Gamma(c+it)|^2 \asymp |t|^{2c-1}e^{-\pi|t|}$.

THEOREM 17.2.2 (the classical range). *For $c \geq 1$, $S_N \to e^{-x}$ in $\mathcal{W}_c$.*

Proof. For $c > 1$ use $|\mathcal{A}_N| \leq \zeta(c)$, $|\beta| \leq \zeta(c)$ and dominated convergence in (17.2.2). For $c = 1$ the prime number theorem for $\mu\chi_4$ in the form $M_\chi(x) \ll x\exp(-a\sqrt{\log x})$ and partial summation,

$$\mathcal{A}_N(s) = \frac{M_\chi(N)}{N^s} + s\int_1^N \frac{M_\chi(u)}{u^{s+1}}du, \tag{17.2.3}$$

give $|\mathcal{A}_N(1+it) - 1/\beta(1+it)| \leq C(1+|t|)\varepsilon_N$ with $\varepsilon_N \to 0$ (the integral converges on $\operatorname{Re} s = 1$ by the same bound for M_χ); with $|\beta(1+it)| \ll \log(2+|t|)$, (17.2.2) tends to zero. □

17.3 Completeness in one weight is unconditional

The Möbius series is one approximation of e^{-x}; completeness asks only that some approximation exists, and in a single weight this needs no arithmetic. The tool is Carleman's formula for a half plane (Titchmarsh [226]; Boas [15]): if f is analytic in a neighbourhood of $\operatorname{Re} w \geq 0$, $f \not\equiv 0$, with zeros $r_k e^{i\theta_k}$, then for $R > 1$

$$\sum_{1<r_k<R}\Big(\frac{1}{r_k} - \frac{r_k}{R^2}\Big)\cos\theta_k = \frac{1}{\pi R}\int_{-\pi/2}^{\pi/2}\log|f(Re^{i\theta})|\cos\theta\,d\theta + \frac{1}{2\pi}\int_1^R\Big(\frac{1}{y^2} - \frac{1}{R^2}\Big)\log|f(iy)f(-iy)|\,dy + O_f(1). \tag{17.3.1}$$

The form of the formula was checked against the standard statement; the section numbers of the two books are not claimed.

LEMMA 17.3.1 (uniqueness on a divergent set). *Let Φ be analytic on $\operatorname{Re} w > 0$ with $|\Phi(w)| \leq C(|w|/\operatorname{Re} w)^B$ for some $B \geq 0$. If $\Phi(\lambda) = 0$ for every λ in a set $\Lambda \subset [1,\infty)$ with $\sum_{\lambda\in\Lambda} 1/\lambda = \infty$, then $\Phi \equiv 0$.*

Proof. If Λ has a finite accumulation point, $\Phi \equiv 0$ by the identity theorem, so let the points of Λ be isolated. Fix $\varepsilon \in (0,1)$ and apply (17.3.1) to $f(w) = \Phi(w+\varepsilon)$, which is analytic on $\operatorname{Re} w > -\varepsilon$, assuming $f \not\equiv 0$. Its zeros include $\lambda - \varepsilon$, $\lambda \in \Lambda$, on the positive axis (the finitely many with $\lambda - \varepsilon \leq 1$ lie outside the range of the sum and cost $O(1)$). The other zeros of f contribute nonnegative terms ($\cos\theta_k \geq 0$, $r_k < R$). Since $\frac{1}{r} - \frac{r}{R^2} \geq \frac{3}{4}\cdot\frac{1}{r}$ for $r < R/2$, the left side is at least $\frac{3}{4}\sum_{\lambda<R/2} 1/\lambda - O(1) \to \infty$. On the right, for $w = Re^{i\theta}$ we have $|w+\varepsilon|/\operatorname{Re}(w+\varepsilon) \leq 2/\cos\theta$, so $\log|f| \leq C' + B\log(1/\cos\theta)$ and the first integral is at most $O(1/R)$, because $\int \log(1/\cos\theta)\cos\theta\,d\theta < \infty$. On the imaginary axis $|f(\pm iy)| \leq C(1+|y|/\varepsilon)^B$, so the second integral is at most $C_\varepsilon\int_1^\infty \log(2+y)\,y^{-2}dy < \infty$, the weight $\frac{1}{y^2} - \frac{1}{R^2}$ being nonnegative. The right side is bounded above, a contradiction. Hence $f \equiv 0$. □

THEOREM 17.3.2 (completeness in one weight). *Let $c > 0$ and let $\Lambda \subset [1,\infty)$ satisfy $\sum_{\lambda\in\Lambda} 1/\lambda = \infty$. Then $\{\operatorname{sech}(\lambda x)\}_{\lambda\in\Lambda}$ is complete in $\mathcal{W}_c$. In particular the integer dilates are complete in every $\mathcal{W}_c$, and so are the prime dilates $\{\operatorname{sech}(px)\}_{p \text{ prime}}$.*

Proof. Let $g \in \mathcal{W}_c$ be orthogonal to every $\operatorname{sech}(\lambda x)$, and let G be its Mellin transform on $\operatorname{Re} s = c$. By Lemma 17.1.2 and (17.2.1),

$$\int_{-\infty}^{\infty} H(t)\lambda^{-it}dt = 0 \quad (\lambda \in \Lambda), \qquad H(t) = \Gamma(c+it)\beta(c+it)\overline{G(c+it)}.$$

Stirling and the convexity bound give $|\Gamma(c+it)\beta(c+it)| \le C(1+|t|)^A e^{-\pi|t|/2}$, and $G \in L^2$, so $H \in L^1$. For $w = |w|e^{i\varphi}$ with $|\varphi| < \pi/2$ define

$$\Phi(w) = \int_{-\infty}^{\infty} H(t)\, w^{-it}\, dt, \qquad |w^{-it}| = e^{t\varphi}.$$

By Cauchy Schwarz,

$$|\Phi(w)| \le C\|G\|_2 \Big(\int (1+|t|)^{2A} e^{-(\pi - 2|\varphi|)|t|} dt \Big)^{1/2} \le C' \Big(\frac{\pi}{2} - |\varphi| \Big)^{-A-1/2},$$

and since $\frac{\pi}{2} - |\varphi| \ge \cos\varphi = \operatorname{Re} w/|w|$, Φ satisfies the growth bound of Lemma 17.3.1 with $B = A + \frac{1}{2}$. It is analytic and vanishes on Λ, so $\Phi \equiv 0$. On the positive axis $\Phi(e^{\xi}) = \int H(t)e^{-i\xi t}dt$ is the Fourier transform of $H \in L^1$, so $H = 0$ almost everywhere. The zeros of β on the line $\operatorname{Re} s = c$ form a discrete set, hence $G = 0$ almost everywhere and $g = 0$. □

REMARK 17.3.3 (the threshold, heuristic). If the Mellin transform of a kernel is bounded by $e^{-\alpha|t|}$ times a power, the function Φ above is analytic at least in the sector $|\arg w| < \alpha$. The map $w \mapsto w^{\pi/(2\alpha)}$ sends the sector to a half plane and λ to $\lambda^{\pi/(2\alpha)}$, so the Carleman sum becomes $\sum \lambda^{-\pi/(2\alpha)}$; for $\Lambda = \mathbb{N}$ this diverges if and only if $\alpha \ge \pi/2$. The gamma factor of sech decays at precisely $\alpha = \pi/2$. This is the threshold of the method: failure of completeness for $\alpha < \pi/2$ is not shown, and the sharpness question is Open Problem 17.1.

REMARK 17.3.4 (Müntz and Szász). For the kernel e^{-x} itself ($\beta \equiv 1$) the proof of Theorem 17.3.2 gives the sufficiency half of the Müntz and Szász theorem: under $u = e^{-x}$, completeness of $\{e^{-\lambda x}\}$ in $\mathcal{W}_{1/2}$ is completeness of $\{u^{\lambda - 1/2}\}$ in $L^2(0,1)$, and divergence of $\sum 1/\lambda$ is the classical sufficient condition. Lerch's theorem is the case $\Lambda = \mathbb{N}$.

Three consequences. First, taking $c = \frac{1}{2}$, the integer dilates of sech are complete in $L^2(0,\infty)$, even though β has infinitely many zeros on the very line $\operatorname{Re} s = \frac{1}{2}$ on which the Mellin transform lives. Second, no property of μ is needed for completeness in one weight. Third, the same proof applies to any kernel whose Mellin

transform on the line is $\Gamma(s)$ times a polynomially bounded function vanishing only on a null set of the line, in particular to the zeta kernel of Section 17.6. Theorem 17.3.2 settles the one weight form of the completeness question unconditionally; only the two weight form is equivalent to a zero free half plane (Theorem 17.5.2).

17.4 The Möbius route in the strip $\frac{1}{2} < c < 1$

Arithmetic enters in the explicit approximation by the Möbius series.

Lemma 17.4.1 (partial summation). *Suppose $M_\chi(x) = O_\varepsilon(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$, with $\frac{1}{2} \le \theta < 1$. Then for $\sigma = \operatorname{Re} s > \theta$, $t = \operatorname{Im} s$ and $0 < \varepsilon < \sigma - \theta$, with constants depending on σ and ε,*

$$|\mathcal{A}_N(s)| \le C(1+|t|), \qquad |\mathcal{A}_N(s) - 1/\beta(s)| \le C(1+|t|)N^{\theta+\varepsilon-\sigma}.$$

Proof. In (17.2.3) the integral converges absolutely for $\sigma > \theta$ to an analytic function equal to $1/(s\beta(s))$ for $\sigma > 1$, hence throughout; the boundary term and the tail $|s| \int_N^\infty u^{\theta+\varepsilon-\sigma-1} du$ give the rate. □

Theorem 17.4.2 (the strip). *Under the hypothesis of Lemma 17.4.1, for every $c \in (\theta, 1)$,*

$$\|S_N - e^{-x}\|_c = O_\varepsilon(N^{\theta-c+\varepsilon}).$$

Under the generalised Riemann hypothesis for β this holds for every $c > \frac{1}{2}$.

Proof. Insert Lemma 17.4.1 in (17.2.2). The integral $\int |\Gamma\beta|^2(1+|t|)^2 dt$ on the line $\operatorname{Re} s = c$ is finite by Stirling and convexity. □

Proposition 17.4.3 (classical equivalences). *For $\frac{1}{2} \le \theta < 1$ the following are equivalent: (a) $M_\chi(x) = O(x^{\theta+\varepsilon})$ for all $\varepsilon > 0$; (b) $\sum \mu(n)\chi_4(n)n^{-s}$ converges for $\operatorname{Re} s > \theta$; (c) $\beta(s) \ne 0$ for $\operatorname{Re} s > \theta$.*

Proof. (a) implies (b) by partial summation. (b) implies (c): the sum is analytic on $\operatorname{Re} s > \theta$ and equals $1/\beta$ for $\operatorname{Re} s > 1$, so $1/\beta$ continues analytically there and β has no zeros. (c) implies (a) is Littlewood's theorem for $L(s, \chi_4)$: a zero free half plane gives $1/\beta(\sigma + it) \ll |t|^\varepsilon$ for $\sigma > \theta$ by the Borel and Carathéodory argument, and Perron's formula then bounds M_χ; compare Titchmarsh [348, Section 14.25] for ζ, whose proof carries over to the character. □

17.5 Two weights: where the zeros are felt

The one weight proof fails for two weights for a structural reason. If $f \in \mathcal{W}_a \cap \mathcal{W}_b$ and $a < \sigma < b$, splitting at $x = 1$ and using Cauchy Schwarz with weight a on

$(0,1)$ and weight b on $(1,\infty)$ gives

$$\Big|\int_0^\infty f(x)x^{s-1}dx\Big| \le \frac{\|f\|_a}{\sqrt{2(\sigma-a)}} + \frac{\|f\|_b}{\sqrt{2(b-\sigma)}}. \tag{17.5.1}$$

So evaluation of the Mellin transform at any point of the open strip is continuous, and zeros of β in the strip are inherited by limits. In a single weight there is no strip and evaluation off the line is not continuous.

PROPOSITION 17.5.1 (necessity). *Let $0 < a < b$. If e^{-x} is a limit in $\mathcal{W}_a \cap \mathcal{W}_b$ of finite combinations of the* $\operatorname{sech}(nx)$*, then $\beta(s) \neq 0$ for $a < \operatorname{Re} s < b$.*

Proof. Each approximant has Mellin transform $2\Gamma(s)\beta(s)D(s)$ with D a Dirichlet polynomial. By (17.5.1) these converge pointwise in the strip to $\Gamma(s)$, which is nonzero; a zero ρ of β in the strip would force $\Gamma(\rho) = 0$. □

THEOREM 17.5.2 (two weight criterion). *For $\frac{1}{2} \le \theta < 1$ the following are equivalent:*

(i) $L(s,\chi_4) \neq 0$ *for* $\operatorname{Re} s > \theta$*;*

(ii) $\{\operatorname{sech}(nx)\}_{n\in\mathbb{N}}$ *is complete in $\mathcal{W}_a \cap \mathcal{W}_b$ for all $\theta < a < b < 1$;*

(iii) $S_N \to e^{-x}$ *in $\mathcal{W}_c$ for every $c \in (\theta, 1)$.*

With $\theta = \frac{1}{2}$, (i) is the generalised Riemann hypothesis for $L(s,\chi_4)$.

Proof. (i) gives (a) of Proposition 17.4.3, then Theorem 17.4.2 gives (iii). Convergence in each $\mathcal{W}_c$, $c \in [a,b]$, gives convergence in $\mathcal{W}_a \cap \mathcal{W}_b$, and Lemma 17.2.1 gives (ii). If (ii) holds and $\beta(\rho) = 0$ with $\theta < \operatorname{Re}\rho < 1$, choose $a < \operatorname{Re}\rho < b$ in $(\theta, 1)$ and apply Proposition 17.5.1; there are no zeros with $\operatorname{Re}\rho \ge 1$. Thus (ii) implies (i). □

In summary, completeness in one weight is free (Theorem 17.3.2). Completeness in two weights straddling a strip forces that strip to be zero free (Proposition 17.5.1), and completeness for all pairs $\theta < a < b < 1$ is equivalent to a zero free half plane (Theorem 17.5.2); whether a zero free strip $a < \operatorname{Re} s < b$ alone gives completeness in $\mathcal{W}_a \cap \mathcal{W}_b$ is not shown. At a single endpoint $c < 1$, convergence of S_N in $\mathcal{W}_c$ does imply zero freeness strictly to the right of c, by combining it with the convergence at any $b > 1$ and applying Proposition 17.5.1 in (c,b), but a bare zero free half plane $\operatorname{Re} s > c$ does not by this proof give convergence on the line c itself. The properties of μ that matter form a ladder.

(P0) Nothing: completeness in one weight, every $c > 0$.

(P1) Inversion $\sum_{d|N}\mu(d) = [N = 1]$: the pointwise identity of Proposition 10.1.3.

(P2) $|\mu| \le 1$: the L^1 theorem and convergence in $\mathcal{W}_c$, $c > 1$.

(P3) The prime number theorem for $\mu\chi_4$: convergence at $c = 1$.

(P4) Power cancellation $M_\chi(x) = O(x^{\theta+\varepsilon})$: convergence and two weight completeness for $c \in (\theta, 1)$, equivalent to a zero free half plane for β. Square root cancellation is GRH for β.

The arithmetic function is the twisted $\mu\chi_4$, not μ: the alternating signs of the secant produce β, not ζ. The barrier $c = \frac{1}{2}$ is genuine: β vanishes at $\frac{1}{2} + 6.0209489\ldots i$ (computed, 6.0209489046976), so two weights straddling $\frac{1}{2}$ can never work, and $M_\chi(x) \neq o(x^{1/2})$, since $M_\chi(x) = o(x^{1/2})$ would give $(\sigma-\frac{1}{2})/|\beta(\sigma+i\gamma)| \to 0$ as $\sigma \downarrow \frac{1}{2}$ at a zero, which is impossible.

17.6 The zeta version

To involve μ itself, use the Bose kernel with its pole removed,

$$\varphi(x) = \frac{1}{e^x - 1} - \frac{1}{x}, \qquad \int_0^\infty x^{s-1}\varphi(x)\,dx = \Gamma(s)\zeta(s) \quad (0 < \operatorname{Re} s < 1). \tag{17.6.1}$$

Since $\varphi(x) \to -\frac{1}{2}$ at the origin and $\varphi(x) \sim -1/x$ at infinity, $\varphi \in \mathcal{W}_c$ exactly for $0 < c < 1$. The cosecant $\operatorname{csch} x$, with transform $2\Gamma(s)(1 - 2^{-s})\zeta(s)$, is unsuitable because the pole of ζ removes it from $\mathcal{W}_c$ for $c \leq 1$. Write $T_N(x) = \sum_{n\leq N} \mu(n)\varphi(nx)$.

Theorem 17.6.1. *(1) For every $c \in (0,1)$ the dilates $\{\varphi(nx)\}_{n\in\mathbb{N}}$ are complete in $\mathcal{W}_c$, unconditionally. (2) Let $\frac{1}{2} \leq \theta < 1$ and suppose $M(x) = O_\varepsilon(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$. Then $T_N \to e^{-x}$ in $\mathcal{W}_c$ with $\|T_N - e^{-x}\|_c = O_\varepsilon(N^{\theta-c+\varepsilon})$ for $\theta < c < 1$ and $0 < \varepsilon < c - \theta$. (3) For $\frac{1}{2} \leq \theta < 1$, completeness of $\{\varphi(nx)\}$ in $\mathcal{W}_a \cap \mathcal{W}_b$ for all $\theta < a < b < 1$ is equivalent to $\zeta(s) \neq 0$ for $\operatorname{Re} s > \theta$ (the pole at 1 being allowed); with $\theta = \frac{1}{2}$ it is equivalent to the Riemann hypothesis.*

Proof. The proofs of Theorems 17.3.2, 17.4.2, 17.5.2 carry over with $(\beta, \mu\chi_4, \operatorname{sech})$ replaced by (ζ, μ, φ), with the following changes. The Mellin transform of $\varphi(n\,\cdot)$ is $n^{-s}\Gamma(s)\zeta(s)$ on $0 < \operatorname{Re} s < 1$, with no factor 2, and $|\zeta(c+it)| \ll (1+|t|)^{1/2}$ there, so $|\Gamma\zeta| \leq C(1+|t|)^A e^{-\pi|t|/2}$ and the proof of part (1) is that of Theorem 17.3.2; ζ has discrete zeros on the line. For part (2) the Mellin transform of $T_N - e^{-x}$ on the line $\operatorname{Re} s = c$ is $\Gamma(s)\big(\zeta(s)\mathcal{A}_N(s) - 1\big)$ with $\mathcal{A}_N = \sum_{n\leq N} \mu(n)n^{-s}$, and Lemma 17.4.1 holds for $\mathcal{A}_N$ and $1/\zeta$ with M in place of M_χ ($1/\zeta$ is analytic on $\operatorname{Re} s > \theta$ because ζ has no zeros there, and ζ is analytic on $0 < \operatorname{Re} s < 1$). For part (3) Proposition 17.5.1 holds with ζ because the approximants have Mellin transform $\Gamma\zeta D$ and converge pointwise in the strip $a < \operatorname{Re} s < b \subset (0,1)$ to Γ; zeros with $\operatorname{Re}\rho \geq 1$ do not exist; and the classical equivalence of Proposition 17.4.3 for ζ is Littlewood's theorem. □

The contrast with the secant is that φ has no unconditional Möbius range at all: its Mellin line must lie inside $0 < c < 1$. Part (3) is a Nyman and Beurling type criterion on the half line, and part (1) explains why it needs two weights: in the classical Nyman and Beurling problem the depth comes from restricting to $(0, 1)$, which makes Mellin transforms analytic in a half plane, while here the second weight plays that role.

17.7 What the quasi Riemann hypothesis gives

OpenAI released in October 2026 a preprint [257] (dated 30 September) announcing a zero free half plane $\operatorname{Re} s > \frac{7}{8}$ for Dirichlet L functions, with a formalisation in Lean, and a companion [258] with the weaker half plane $\operatorname{Re} s > \frac{11}{12}$; Liu [223] claims a small improvement of the exponent $\frac{7}{8}$. A preprint attributed to Thorner, extending a zero free half plane $\operatorname{Re} s > \frac{31}{32}$ to the Dedekind zeta function of every number field, is cited here from the source as unchecked [225]. None has been refereed or checked for this book. Every statement below that uses them is conditional on them, and those about β also on the assumption, a question for the authors, that the $\frac{7}{8}$ theorem covers $L(s, \chi_4)$. If these results stand, the functional equation puts the nontrivial zeros of ζ and of β in $\frac{1}{8} \le \operatorname{Re} \rho \le \frac{7}{8}$.

input	secant, $\mu\chi_4$	zeta kernel, μ
none	one weight, all $c > 0$	one weight, $0 < c < 1$
PNT	Möbius series, $c \ge 1$ (rate subexponential in $\sqrt{\log N}$)	nothing new
Dedekind QRH ($\frac{31}{32}$), unchecked	$c > \frac{31}{32}$	$\frac{31}{32} < c < 1$
QRH for Dirichlet L ($\frac{7}{8}$), conditional	$\frac{7}{8} < c$	$\frac{7}{8} < c < 1$
GRH	$c > \frac{1}{2}$	$\frac{1}{2} < c < 1$

For a zero free half plane $\operatorname{Re} s > \theta$ the entries mean: the Möbius series converges in $\mathcal{W}_c$ at the rate $N^{\theta - c + \varepsilon}$ for every $c > \theta$ (secant) or $\theta < c < 1$ (zeta kernel), and two weight completeness holds on that range. For the secant kernel and $c > 1$ the elementary absolute convergence argument gives $O_c(N^{1-c})$, and at $c = 1$ the prime number theorem gives $O((1 + \sqrt{\log N})e^{-a\sqrt{\log N}})$; neither is a power law with $\theta = 1$, so the PNT row is not an instance of the rate $N^{\theta - c + \varepsilon}$. The input $\frac{7}{8}$ pushes the Möbius route below $c = 1$, from $c \ge 1$ down to $c > \frac{7}{8}$, for both kernels at once if the theorem covers β as well as ζ. The stated half plane does not supply convergence in the remaining range $\frac{1}{2} < c \le \frac{7}{8}$; GRH is sufficient there, whereas a weaker zero free bound already gives convergence above its own boundary. For the one weight question QRH is irrelevant, that question being settled unconditionally.

17.8 Relation to the Thorin programme

Three remarks connect this part to the rest of the book; the second and third are analogies (heuristic), not consequences of the theorems above.

First, the secant is a Thorin object, in the form of Lemma 10.1.4: $\operatorname{sech}\sqrt{2s} = \prod_{k\geq 1}(1+s/\lambda_k)^{-1}$ with $\lambda_k = (2k-1)^2\pi^2/8$ is the Laplace transform of the exit time of Brownian motion from $(-1,1)$, a generalised gamma convolution with discrete Thorin measure at the λ_k (Table 4.3). Dilation of sech corresponds to a time scaling of that law after the substitution $x = \sqrt{2s}$.

Second, the book's Thorin type expansion of $1/\zeta(\frac{1}{2}+\sqrt{s})$ (Theorem 10.1.5) applies Proposition 10.1.3 pointwise at $x = \sqrt{s}\log n$ to the Dirichlet series of $1/\zeta$. Theorems 17.5.2 and 17.6.1 are statements in L^2 in the variable x about approximating e^{-x} in weighted spaces; they are not statements about that series in s, and the kernel φ is not used in it. The parallel is that the convergence of the construction needs the same zero free region as the Möbius series, and no new zero free region comes out of it.

Third, the dichotomy between one and two weights is a pattern for the programme. Statements made on a single Mellin line (completeness, L^2 approximation) are cheap, because point evaluations off the line are not controlled, while statements uniform on a strip carry analyticity into the strip and force zero freeness there (Proposition 17.5.1). A GGC Laplace transform is analytic and zero free in a half plane, so a Thorin representation of $1/\xi$ is of the second kind; Section 17.14 makes this precise for $\xi(s,\chi_4)$.

17.9 Numerical check

The weighted errors $\|S_N - e^{-x}\|_c$ and $\|T_N - e^{-x}\|_c$ were computed by quadrature on logarithmic grids; the secant table was computed on two grid steps and agrees to 10^{-15} relative, the zeta kernel table to $5\cdot 10^{-8}$, the part of the zeta integrals beyond $x = 60$ being added exactly ($T_N - e^{-x} = -m_N/x$ there, $m_N = \sum_{n\leq N}\mu(n)/n$). The pointwise identity of Proposition 10.1.3 is proved and is not checked numerically here.

For $c \geq 1$ Theorem 17.2.2 gives convergence but no rate, and at $c = 1$ the computed errors rise from $N = 10$ to $N = 100$. Inside the strip the GRH rates $N^{1/2-c}$ are slow ($N^{-0.25}$ at $c = 0.75$, $N^{-0.1}$ at $c = 0.6$) and are masked at this range by the oscillation of $M_\chi(N)$, whose values at $N = 10, 10^2, 10^3, 10^4$ are $2, 3, 6, -24$; the error is concentrated at $x \lesssim 1/N$, where $S_N(x) \approx \frac{1}{2}M_\chi(N)$ (heuristic, a Taylor approximation). Finite computations prove no asymptotic rate or zero free half plane. The largest value of $|M_\chi(n)|/\sqrt{n}$ for $n \leq 10^4$ is $2/\sqrt{3} = 1.1547$, attained at $n = 3$, and says nothing about large n. The zeta kernel series converges only conditionally pointwise, since $\varphi(nx) \approx -1/(nx)$ and $\sum \mu(n)/n = 0$ is the prime number theorem; its weighted errors nevertheless decrease steadily.

secant, $\|S_N - e^{-x}\|_c$				
N	$c = 0.60$	$c = 0.75$	$c = 1.00$	$c = 1.25$
10	7.3×10^{-3}	5.1×10^{-3}	2.9×10^{-3}	1.7×10^{-3}
10^2	3.9×10^{-2}	1.9×10^{-2}	6.3×10^{-3}	2.2×10^{-3}
10^3	3.5×10^{-2}	1.2×10^{-2}	2.2×10^{-3}	4.2×10^{-4}
10^4	5.3×10^{-2}	1.3×10^{-2}	1.2×10^{-3}	1.3×10^{-4}
zeta kernel, $\|T_N - e^{-x}\|_c$				
N	$c = 0.75$	$c = 0.875$	$c = 0.95$	
10	2.1×10^{-1}	2.3×10^{-1}	3.1×10^{-1}	
10^2	1.2×10^{-1}	1.0×10^{-1}	1.2×10^{-1}	
10^3	3.0×10^{-2}	1.9×10^{-2}	1.9×10^{-2}	
10^4	2.6×10^{-2}	1.2×10^{-2}	1.0×10^{-2}	

Table 17.1: Weighted errors of the Möbius approximants (computed).

17.10 Necessity of the divergence condition

The sufficiency proof of Theorem 17.3.2 used nothing about β; the necessity proof divides by β on the line, so it needs a hypothesis. Write $K_c(t) = \Gamma(c+it)\beta(c+it)$, which is half the Mellin transform of sech on the line $\operatorname{Re} s = c$, and let $c > 0$ (the spaces $\mathcal{W}_c$ contain the dilates only for $c > 0$).

(N_c) $\quad 1/\beta(c+it) \ll (1+|t|)^A$ for some $A \geq 0$.

Lemma 17.10.1. *Let $c > 0$. Then (N_c) holds unconditionally for $c \geq 1$. If β has no zeros in $\operatorname{Re} s > \theta$, it holds for every $c > \theta$ and, by the functional equation, for every $c \in (0, 1-\theta)$. In particular it holds for all $c > 0$ outside $[\frac{1}{8}, \frac{7}{8}]$ given a zero free half plane $\operatorname{Re} s > \frac{7}{8}$ for $L(s, \chi_4)$ (conditional on the quasi Riemann hypothesis of Section 17.7), and for all $c > 0$, $c \neq \frac{1}{2}$, under GRH for β.*

Proof. For $c > 1$, $|1/\beta(c+it)| \leq \zeta(c)$; for $c = 1$, $1/\beta(1+it) \ll \log(2+|t|)$ as for ζ. Given a zero free half plane $\operatorname{Re} s > \theta$, the Borel and Carathéodory argument of Titchmarsh [348, Theorem 14.2], which uses only the zero free region, gives $\log \beta(\sigma+it) \ll (\log|t|)^B$ with $B \leq 1$ for $\sigma > \theta$, hence (N_c). For $c < 1-\theta$ put $s = 1-c-it$, so that $\operatorname{Re} s > \theta$ and the functional equation $\beta(1-s) = (2/\pi)^s \sin(\pi s/2)\Gamma(s)\beta(s)$ gives $\beta(c+it) = (2/\pi)^s \sin(\pi s/2)\Gamma(s)\beta(s)$. Since $|\sin(\pi s/2)\Gamma(s)| \asymp |t|^{1/2-c}$ for large $|t|$ on the line $\operatorname{Re} s = 1-c \in (\theta, 1)$, which is not an even integer, $1/\beta(c+it)$ is bounded by $|t|^{c-1/2}$ times a polynomial bound for $1/\beta(s)$. □

Theorem 17.10.2 (necessity). *Let $c > 0$ and assume (N_c). If $\Lambda \subset [1, \infty)$ and $\sum_\Lambda 1/\lambda < \infty$, then $\{\operatorname{sech}(\lambda x)\}_{\lambda \in \Lambda}$ is not complete in $\mathcal{W}_c$. So under (N_c) the dilates are complete in $\mathcal{W}_c$ if and only if $\sum_\Lambda 1/\lambda = \infty$.*

Proof. Step 1, a Blaschke product. Put $B(w)=\prod_{\lambda\in\Lambda}\frac{1-w/\lambda}{1+w/\lambda}$. Since $\log\frac{1-w/\lambda}{1+w/\lambda}=-2w/\lambda+O(|w|^2/\lambda^2)$ and $\sum 1/\lambda<\infty$, the product converges uniformly on compact subsets of $\mathbb{C}\setminus(-\Lambda)$ to an analytic function with zeros exactly at Λ and poles at $-\Lambda$. Each factor has modulus at most 1 on $\operatorname{Re} w\ge 0$ and modulus 1 on the imaginary axis, so $|B|\le 1$ on $\operatorname{Re} w\ge 0$. On the imaginary axis

$$\Big|\frac{d^j}{dw^j}\log B(iy)\Big|=(j-1)!\Big|\sum_\lambda\Big[\frac{1}{(\lambda-iy)^j}+\frac{(-1)^{j-1}}{(\lambda+iy)^j}\Big]\Big|\le 2(j-1)!\sum_\lambda\frac{1}{\lambda},$$

because $|\lambda\pm iy|\ge\lambda\ge 1$. Hence every derivative $B^{(j)}$ is bounded on the imaginary axis.

Step 2, the strip. Let $S=\{z:|\operatorname{Im} z|<\pi/2\}$ and $w=e^z$, which maps S onto the right half plane and ∂S onto the imaginary axis. Put $\widetilde{\Phi}(z)=B(e^z)e^{-z^2}$. It is analytic on $|\operatorname{Im} z|<\pi$ (the poles of $B(e^z)$ are at $z=\log\lambda+(2j+1)i\pi$), vanishes at $z=\log\lambda$, $\lambda\in\Lambda$, and $|\widetilde{\Phi}(\xi+i\eta)|\le e^{\pi^2/4-\xi^2}$ on $\overline{S}$. On the boundary lines $z=\xi\pm i\pi/2$, $\frac{d^k}{d\xi^k}[B(e^z)]=\sum_{j=1}^k S(k,j)e^{jz}B^{(j)}(e^z)$ with Stirling numbers $S(k,j)$, which by Step 1 is $O_k(\sum_{j\le k}e^{j\xi})=O_k(e^{k\max(\xi,0)})$, while the derivatives of e^{-z^2} are polynomials times e^{-z^2}. Every ξ derivative of $\widetilde{\Phi}$ on either boundary line is therefore in $L^2(d\xi)$.

Step 3, the inverse transform. Put $H(t)=\frac{1}{2\pi}\int\widetilde{\Phi}(\xi)e^{it\xi}d\xi$. Shifting the contour to $\operatorname{Im} z=\eta$, allowed by the Gaussian decay, gives $H(t)e^{t\eta}=\frac{1}{2\pi}\int\widetilde{\Phi}(\xi+i\eta)e^{it\xi}d\xi$ for $|\eta|\le\pi/2$, so $H(t)e^{\pm\pi t/2}$ is the inverse Fourier transform of the boundary values. Since these have all derivatives in L^2, Plancherel gives $H(t)e^{\pi|t|/2}(1+|t|)^{A'}\in L^2(\mathbb{R})$ for every $A'\ge 0$, in particular $H(t)e^{\pi|t|/2}(1+|t|)^{A'}\in L^1$, so inversion holds on the closed strip: $\int H(t)e^{-itz}dt=\widetilde{\Phi}(z)$. Hence $\int H(t)\lambda^{-it}dt=\widetilde{\Phi}(\log\lambda)=0$ for $\lambda\in\Lambda$, and $H\not\equiv 0$ because $\widetilde{\Phi}\not\equiv 0$.

Step 4, division by the kernel. Define G on $\operatorname{Re} s=c$ by $\overline{G(c+it)}=H(t)/K_c(t)$. By Stirling, $|1/\Gamma(c+it)|\asymp e^{\pi|t|/2}|t|^{1/2-c}$ for large $|t|$, and by (N_c), $|G(c+it)|\le C|H(t)|e^{\pi|t|/2}(1+|t|)^{A+1/2-c}\in L^2$. Let $g\in\mathcal{W}_c$ have Mellin transform G; $g\not\equiv 0$. By (17.2.1) and Lemma 17.1.2,

$$\langle\operatorname{sech}(\lambda\,\cdot),g\rangle_c=\frac{\lambda^{-c}}{\pi}\int\lambda^{-it}K_c(t)\overline{G(c+it)}\,dt=\frac{\lambda^{-c}}{\pi}\int H(t)\lambda^{-it}dt=0$$

for every $\lambda\in\Lambda$. □

Corollary 17.10.3. *For $c\ge 1$, for $c>\frac{7}{8}$ given the zero free half plane $\operatorname{Re} s>\frac{7}{8}$ for $L(s,\chi_4)$ (conditional), and for every $c>0$, $c\ne\frac12$, under GRH for β, $\{\operatorname{sech}(\lambda x)\}_{\lambda\in\Lambda}$, $\Lambda\subset[1,\infty)$, is complete in $\mathcal{W}_c$ if and only if $\sum_\Lambda 1/\lambda=\infty$.*

Remark 17.10.4 (the barrier $c=\frac12$). At $c=\frac12$ the line carries the zeros $\frac12+i\gamma$ of β and $(\mathrm{N}_{1/2})$ fails. The construction survives if H can be made to vanish at each γ to the order of the zero, with H/β still square integrable. Vanishing of

H at a point γ is the linear condition $\int \widetilde{\Phi}(\xi) e^{i\gamma\xi} d\xi = 0$, a moment condition and not a zero of $\widetilde{\Phi}$, so the Blaschke product does not produce it; finitely many such conditions could be imposed, but this does not handle all the zeros or the square integrability after division. Whether the zeros of β on the critical line can rescue completeness for a set with $\sum 1/\lambda < \infty$ is open (Problem 17.1). Numerically, for $\Lambda = \{n^2\}$ the distance from e^{-x} to the span decreases only slowly at $c = \frac{1}{2}$, as it does at $c = \frac{3}{2}$ where Theorem 17.10.2 applies, and at $N = 24$ it is more than an order of magnitude above the distance for the integers (Table 17.4). This is consistent with a positive limit, and so with the critical line not being special here; it is not decisive, since incompleteness of a span does not put the particular target e^{-x} outside it.

Remark 17.10.5. The argument is the half plane Blaschke construction of Müntz and Szász, transported to the strip. What the strip adds is the Gaussian factor e^{-z^2}, which supplies the Sobolev regularity of the boundary values needed to absorb the polynomial part of $1/K_c$, and what the kernel adds is the division by β, which is where (N_c) enters. The same proof gives necessity for the zeta kernel φ of (17.6.1), $0 < c < 1$, under the corresponding hypothesis $1/\zeta(c+it) \ll (1+|t|)^A$.

17.11 Wiener, Kac and Salem: the L^1 picture

The completeness problems of this chapter have an older L^1 form, treated in Section 13.1.6 and Section 13.1.7: Wiener's theorem [360] in its homogeneous form [173], and Salem's criterion [322] for the Fermi kernel $1/(e^x + 1)$, whose Mellin transform is $\Gamma(s)\eta(s)$ (Proposition 13.1.12, Corollary 13.1.13). In dilation form, for k on $(0,\infty)$ with $\int_0^\infty |k(x)| x^{\sigma-1} dx < \infty$, the span of $\{x^{\sigma-1}k(\lambda x) : \lambda > 0\}$ is dense in $L^1(0,\infty)$ if and only if the Mellin transform of k has no zero on the line $\operatorname{Re} s = \sigma$, and the bounded annihilators that exist when it vanishes at $\sigma + i\gamma$ are the characters $x^{i\gamma}$ (under $x = e^u$ translates become dilates). The beta analogue for the secant kernel is

$$\overline{\operatorname{span}}\{x^{\sigma-1}\operatorname{sech}(\lambda x) : \lambda > 0\} = L^1(0,\infty) \iff \beta(\sigma+it) \neq 0 \ \text{ for all } t \in \mathbb{R}, \tag{17.11.1}$$

for $\sigma > 0$. This holds for $\sigma \geq 1$, fails for $\sigma = \frac{1}{2}$ (the zero $\frac{1}{2} + 6.02\ldots i$), and for $\frac{1}{2} < \sigma < 1$ it holds for every such σ if and only if GRH holds for β; the mirror lines $0 < \sigma < \frac{1}{2}$ carry the same zero free statements by the functional equation. At $\sigma = 1$ Salem's own kernel is incomplete, because $1 - 2^{1-s}$ vanishes at $1 + 2\pi i k/\log 2$ for nonzero integers k (the zero at $k = 0$ cancels the pole of ζ, leaving $\eta(1) = \log 2 \neq 0$); the secant kernel has no such defect.

Salem used every positive dilation; the Carleman argument removes that.

Theorem 17.11.1 (Salem's criterion on a divergent set). *Let $\sigma > 0$, let k be $1/(e^x+1)$ or $\operatorname{sech} x$, and let $\Lambda \subset [1,\infty)$ satisfy $\sum_{\lambda\in\Lambda} 1/\lambda = \infty$. Then $\{x^{\sigma-1}k(\lambda x)\}_{\lambda\in\Lambda}$ is complete in $L^1(0,\infty)$ if and only if the Mellin transform of k*

has no zero on the line $\operatorname{Re} s = \sigma$*. In particular, for* $\frac{1}{2} < \sigma < 1$*, the integer dilates of the Fermi kernel are complete in* L^1 *if and only if* ζ *has no zero on* $\operatorname{Re} s = \sigma$*, and the integer dilates of* sech *if and only if* β *has none.*

Proof. If the transform vanishes at $\sigma + i\gamma$ then $x^{i\gamma}$ is a bounded annihilator of every dilate. Conversely let $\varphi \in L^\infty$ annihilate $x^{\sigma-1}k(\lambda x)$ for $\lambda \in \Lambda$ and put $\Phi(w) = \int_0^\infty x^{\sigma-1}\varphi(x)k(wx)\,dx$, $\operatorname{Re} w > 0$. Both kernels extend analytically to $\operatorname{Re} w > 0$, since $e^{wx} + 1$ and $\cosh(wx)$ vanish only for $wx \in i\mathbb{R}$. For $w = re^{i\psi}$, $|\psi| < \pi/2$, split the integral at $x = 1/r$. On $x \le 1/r$ the kernel is bounded by an absolute constant C_0, since $|wx| \le 1$ keeps wx away from the zeros. On $x > 1/r$, $|e^{wx} + 1| \ge e^{rx\cos\psi} - 1$ and $|\cosh(wx)| \ge \sinh(rx\cos\psi) \ge \frac{1}{2}(e^{rx\cos\psi} - 1)$, so after $y = rx\cos\psi$ and with a further absolute constant,

$$|\Phi(w)| \le \|\varphi\|_\infty \Big[\frac{C_0}{\sigma} r^{-\sigma} + C_1 (r\cos\psi)^{-\sigma} \int_{\cos\psi}^\infty \frac{y^{\sigma-1}dy}{e^y - 1}\Big]$$
$$\le C_\sigma \|\varphi\|_\infty\, r^{-\sigma} (\cos\psi)^{-\sigma-1} \log\frac{2}{\cos\psi}$$

for $r \ge 1$. So $\Phi(w+1)$ is analytic in $\operatorname{Re} w > 0$, and since $|w+1|/\operatorname{Re}(w+1) \le |w|/\operatorname{Re} w$ it is bounded by $C(|w|/\operatorname{Re} w)^{\sigma+2}$. If $\Lambda \cap [1,2)$ is infinite, Φ vanishes on a set with a limit point in $[1,2]$ and $\Phi \equiv 0$ by the identity theorem; otherwise $\Phi(w+1)$ vanishes at $w = \lambda - 1$, $\lambda \in \Lambda$, $\lambda \ge 2$, a set in $[1,\infty)$ with divergent reciprocal sum. Lemma 17.3.1 gives $\Phi \equiv 0$. In particular φ annihilates $x^{\sigma-1}k(\lambda x)$ for every $\lambda > 0$, and the dilation form of Wiener's theorem gives $\varphi = 0$. □

Three completeness statements for the same dilates are each governed by which point evaluations of the Mellin transform are continuous on the space.

space	obstruction	annihilators
L^1, Salem	zeros on $\operatorname{Re} s = \sigma$	$x^{i\gamma}$
$\mathcal{W}_c$	none	none
$\mathcal{W}_a \cap \mathcal{W}_b$	zeros in $a < \operatorname{Re} s < b$	$f \mapsto \widehat{f}(\rho)$

In L^1 the Mellin transform is continuous on the line and the dual contains the characters, so a zero on the line is seen. In L^2 with one weight the transform is only defined almost everywhere on the line, the characters are not square integrable, and nothing off the line is visible: Theorem 17.3.2 holds even at $c = \frac{1}{2}$, where Salem's family is unconditionally incomplete. With two weights the bound (17.5.1) makes evaluation continuous throughout the strip, and the annihilator attached to a zero ρ is the functional $f \mapsto \widehat{f}(\rho)$. Salem's criterion and Theorem 17.5.2 are the boundary and interior versions of one statement, detecting zeros on a line and in a strip. The quasi Riemann hypothesis is a statement of the second kind: if it covers $L(s, \chi_4)$, it makes the two weight criterion hold on $(\frac{7}{8}, 1)$ (conditional) but, with only this half plane, does not resolve zero freeness on individual lines in $\frac{1}{8} \le \operatorname{Re} s \le \frac{7}{8}$.

Remark 17.11.2. Nyman and Beurling's criterion is a fourth regime: the dilates are restricted to $(0,1)$, which makes the Mellin transforms of the space analytic in a half plane, so every zero in $\operatorname{Re} s > \frac12$ is seen in a single L^2 norm, by the bound $|\int_0^1 f(x)x^{s-1}dx| \le \|f\|_2/(2\operatorname{Re} s-1)^{1/2}$ for $\operatorname{Re} s > \frac12$. The space $\mathcal{W}_a \cap \mathcal{W}_b$ is the half line substitute for that restriction. Together with Section 17.5 this answers the first half of Problem 17.10, in the variable x on $\mathcal{W}_c$; the L^2 version in s on the boundary $\operatorname{Re}\sqrt{s} = \frac12$ is not addressed.

17.12 Explicit approximants in one weight

17.12.1 The Gram system

Let $d_N(c)$ be the distance in $\mathcal{W}_c$ from e^{-x} to the span of $\operatorname{sech}(nx)$, $n \le N$. With $G_{nm} = \langle \operatorname{sech}(n\,\cdot), \operatorname{sech}(m\,\cdot)\rangle_c$ and $b_n = \langle e^{-x}, \operatorname{sech}(n\,\cdot)\rangle_c$,

$$d_N(c)^2 = \|e^{-x}\|_c^2 - b^{\mathsf T} G^{-1} b, \qquad \|e^{-x}\|_c^2 = \frac{\Gamma(2c)}{2^{2c}}, \tag{17.12.1}$$

and the optimal coefficients are $a = G^{-1}b$ (the matrix G is positive definite, since a nonzero finite combination of distinct dilates cannot vanish). The data have closed forms. Expanding $\operatorname{sech}(nx) = 2\sum_m \chi_4(m)e^{-mnx}$,

$$b_n = 2\Gamma(2c)\sum_{m\ge1}\frac{\chi_4(m)}{(1+nm)^{2c}}, \qquad G_{nm} = \frac{2(nm)^{-c}}{\pi}\int_{-\infty}^{\infty}|K_c(t)|^2\cos\bigl(t\log\tfrac{m}{n}\bigr)dt,$$

the first an alternating series convergent for every $c > 0$, the second a rapidly convergent integral because $|K_c(t)|^2 \asymp |t|^{2c-1}e^{-\pi|t|}|\beta(c+it)|^2$.

17.12.2 What the numbers say

Table 17.2 gives $d_N(c)$, the largest optimal coefficient, and the error of the Möbius approximant S_N for comparison, computed in multiple precision (two precisions and step sizes agree to 3×10^{-28} relative in every d_N and $\max|a_n|$; 40 digits are not enough at $N = 32$).

The optimal distance is about one order of magnitude below the Möbius error at $N = 4$ and 8 (ratios 0.10 to 0.17), and two to three orders below it at $N = 16$ and 32 (ratios 1.5×10^{-3} to 2.6×10^{-3}). The optimal coefficients explode, between 10^{16} and 10^{18} at $N = 32$: the problem is ill conditioned, and beyond $N \approx 16$ the optimal approximant is numerically useless in ordinary precision. The Möbius approximant, whose coefficients are $0, \pm\frac12$, is stable but converges slowly and not monotonically, its error tracking the oscillation of $M_\chi(N)$. This is the same trade off as in the Nyman and Beurling problem, where Báez Duarte [10] observed that the natural Möbius approximants are far from optimal.

c	N	$d_N(c)$	$\max_n \lvert a_n\rvert$	$\lVert S_N - e^{-x}\rVert_c$
0.60	4	8.1×10^{-3}	0.55	4.8×10^{-2}
	8	9.7×10^{-4}	72	7.3×10^{-3}
	16	1.6×10^{-4}	3.3×10^{6}	7.8×10^{-2}
	32	9.5×10^{-5}	7.3×10^{17}	6.2×10^{-2}
0.75	4	5.8×10^{-3}	0.48	4.0×10^{-2}
	8	5.7×10^{-4}	54	5.1×10^{-3}
	16	1.0×10^{-4}	1.2×10^{6}	4.7×10^{-2}
	32	6.5×10^{-5}	7.2×10^{16}	3.5×10^{-2}
1.00	4	3.7×10^{-3}	0.49	3.0×10^{-2}
	8	2.8×10^{-4}	36	2.9×10^{-3}
	16	5.5×10^{-5}	4.0×10^{5}	2.1×10^{-2}
	32	2.8×10^{-5}	5.4×10^{17}	1.4×10^{-2}

Table 17.2: Optimal distance, size of the optimal coefficients and the Möbius error (computed).

17.12.3 Why the coefficients explode

The explanation is heuristic. On the line $\operatorname{Re} s = c$ the approximation problem is to make $2\beta D_N - 1$ small in $L^2(|\Gamma(c+it)|^2 dt)$, where $D_N(c+it) = \sum_{n\le N} a_n n^{-c-it}$. The weight $e^{-\pi|t|}$ confines the problem to a segment $|t| \le T$. On a segment the exponentials $e^{-it\log n}$, $n \ge 1$, are complete in $L^2(-T,T)$ for every fixed T: if h annihilates them, $E(z) = \int_{-T}^{T} h(t)e^{-itz}dt$ is entire of exponential type at most T, so has $O(r)$ zeros in $|z| \le r$, while E vanishes at the e^r points $\log n \le r$ with $n \le e^r$, so $E = 0$. Near linear dependence is quantitative: $\|e^{-it\log(n+1)} - e^{-it\log n}\|^2_{L^2(-T,T)} \le \frac{2T^3}{3}\log(1+\frac{1}{n})^2$, so the finite Gram matrices have no uniform lower spectral bound. This proves ill conditioning of the restricted systems but does not determine the size of the optimal coefficient vector for this target, and the effective cutoff of the segment depends on the coefficient growth through $\log\big((1+\|a\|_{N,c})/\varepsilon\big)$, $\|a\|_{N,c} = \sum_{n\le N} |a_n| n^{-c}$. The qualitative picture, completeness together with increasingly close frequencies, is what Table 17.2 shows.

Proposition 17.12.1 (what the weight sees). *Let $c > 0$, $T \ge 1$ and a_n be arbitrary coefficients, with $D_N(s) = \sum_{n\le N} a_n n^{-s}$ and $\|a\|_{N,c} = \sum_{n\le N} |a_n| n^{-c}$. Then $\|e^{-x} - \sum_{n\le N} a_n \operatorname{sech}(nx)\|_c^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\Gamma(c+it)|^2 |1 - 2\beta(c+it)D_N(c+it)|^2 dt$, and*

$$\Big\|e^{-x} - \sum_{n\le N} a_n \operatorname{sech}(nx)\Big\|_c^2 \le \frac{1}{2\pi}\int_{|t|\le T} |\Gamma(c+it)|^2 \big|1 - 2\beta D_N\big|^2 dt + C_c\, e^{-\pi T} T^{2c} (1 + \|a\|_{N,c})^2.$$

Where $\beta \ne 0$ the integrand of the first term is $|K_c(t)|^2 |2D_N - 1/\beta|^2$.

Proof. The identity is (17.2.1) with Lemma 17.1.2; the Mellin transform of

$\operatorname{sech}(nx)$ carries the factor 2, which is why $2D_N - 1/\beta$ and not $D_N - 1/\beta$ appears (for $a_n = \mu(n)\chi_4(n)/2$ the first term is the error of S_N, which tends to 0; the form with $D_N - 1/\beta$ gives the error of an approximant with twice the coefficients). For the tail $|t| > T$ use $|D_N| \le \|a\|_{N,c}$, $|\Gamma(c+it)|^2 \ll_c |t|^{2c-1}e^{-\pi|t|}$ and $|\beta(c+it)| \ll_c (1+|t|)^{1/2}$ (functional equation, absolute convergence to the right of 1 and interpolation), so the tail is at most $C(1+\|a\|_{N,c})^2 \int_T^\infty t^{2c}e^{-\pi t}dt \ll_c e^{-\pi T}T^{2c}(1+\|a\|_{N,c})^2$ for $T \ge 1$. The first form holds for every $T > 0$ if T^{2c} is replaced by $(1+T)^{2c}$ (take all $a_n = 0$ and $T \downarrow 0$ to see that some such change is needed). □

A computed check at $N = 8$, $c = 0.75$, $a_n = \mu(n)\chi_4(n)/2$: the x space error squared is $2.5846230428 \times 10^{-5}$ and so is $\frac{1}{2\pi}\int |K_c|^2|2D_N - 1/\beta|^2 dt$, while the form with $D_N - 1/\beta$ gives 7.8×10^{-2}.

So an upper bound for $d_N(c)$ follows from approximating $1/\beta$ on a growing segment $|t| \le T_N$ by Dirichlet polynomials with coefficients of size $\|a\|_{N,c} = e^{o(T_N)}$ and a vanishing segment residual (a sufficient route, not an equivalence); the obstruction to an explicit rate is that the coefficient growth is not controlled.

Remark 17.12.2 (what is known conditionally). If β has no zeros in $\operatorname{Re} s > \theta$, then $d_N(c) \le \|S_N - e^{-x}\|_c = O_\varepsilon(N^{\theta-c+\varepsilon})$ for $c \in (\theta, 1)$ by Theorem 17.4.2, so $d_N(c)$ decays polynomially; given the quasi Riemann hypothesis for $L(s, \chi_4)$ (conditional) this is a polynomial rate for $c > \frac{7}{8}$. The question of a lower bound is open. In the Nyman and Beurling setting Burnol [26] proves a zero dependent asymptotic lower bound of order $1/\log N$ for the squared distance δ_N^2, of the form $\liminf \delta_N^2 \log N \ge \sum_{\operatorname{Re}\rho = 1/2} m_\rho^2/|\rho|^2$; it comes from the continuity of point evaluation at the zeros, which the single weight $\mathcal{W}_c$ lacks, and it does not transfer to this Hilbert space. So either there is no polynomial lower bound in one weight or it has a different source; this is the refined form of Problem 17.2.

17.13 The Meixner family

For $t > 0$ let $f_t(x) = |\Gamma(\frac{t}{2} + \frac{ix}{2})|^2$, $x > 0$, the symmetric Meixner density of order t up to normalisation (Section 10.2.3); $f_1(x) = \pi \operatorname{sech}(\pi x/2)$ and $f_2(x) = (\pi x/2)/\sinh(\pi x/2)$.

Theorem 17.13.1 (Mellin transform of the Meixner density). *For* $0 < \operatorname{Re} s < 1$,

$$M_t(s) = \int_0^\infty x^{s-1} f_t(x)\, dx = \frac{2^s \pi\, \Gamma(t)}{\sin(\pi s/2)}\, Z_t(1-s), \tag{17.13.1}$$

where Z_t is entire, given by

$$\begin{aligned} Z_t(w) &= \frac{1}{\Gamma(w)} \int_0^\infty u^{w-1} (2\cosh(u/2))^{-t}\, du \\ &= \sum_{k\ge 0} (-1)^k \binom{t+k-1}{k} \Big(k + \frac{t}{2}\Big)^{-w} \qquad (\operatorname{Re} w > t-1). \end{aligned}$$

For $t = 1, 2, 3$,

$$Z_1(w) = 2^w \beta(w), \qquad Z_2(w) = \eta(w-1) = (1 - 2^{2-w})\zeta(w-1),$$
$$Z_3(w) = 2^{w-3}\big(\beta(w) - \beta(w-2)\big).$$

For integer t, Z_t is a linear combination of $\eta(w-j)$, j odd, when t is even, and of $2^{w-2i}\beta(w-2i)$ when t is odd.

Proof. The beta integral with $r = e^u$ gives the Fourier representation

$$|\Gamma(a+iy)|^2 = \Gamma(2a) \int_{\mathbb{R}} e^{iyu} (2\cosh(u/2))^{-2a} du,$$

hence with $a = \frac{t}{2}$, $y = \frac{x}{2}$,

$$f_t(x) = 2\Gamma(t) \int_0^\infty \cos(xu/2)(2\cosh(u/2))^{-t} du.$$

Insert an Abel regulator $e^{-\epsilon x}$ in the Mellin integral, so that Fubini applies: the inner integral is

$$\tfrac{1}{2}\Gamma(s)[(\epsilon - iu/2)^{-s} + (\epsilon + iu/2)^{-s}],$$

bounded by $C_s u^{-\operatorname{Re} s}$ uniformly in $\epsilon > 0$ for $0 < \operatorname{Re} s < 1$, which is integrable against $(2\cosh(u/2))^{-t}$; dominated convergence as $\epsilon \downarrow 0$ gives

$$\begin{aligned} M_t(s) &= 2\Gamma(t)\, 2^s \Gamma(s) \cos(\pi s/2) \int_0^\infty u^{-s} (2\cosh(u/2))^{-t} du \\ &= 2\Gamma(t) 2^s \Gamma(s) \cos(\pi s/2) \Gamma(1-s) Z_t(1-s), \end{aligned}$$

and

$$\Gamma(s)\Gamma(1-s)\cos(\pi s/2) = \pi/(2\sin(\pi s/2))$$

gives (17.13.1). At $u = 0$, $(2\cosh(u/2))^{-t}$ has an even analytic expansion $\sum b_j u^{2j}$; subtracting finitely many terms on $(0,1)$ continues $\int_0^\infty u^{w-1}(\cdots)du$ meromorphically with simple poles at $w = -2j$ only, and division by $\Gamma(w)$ cancels them, so Z_t is entire (and vanishes at $-1, -3, \dots$). Absolute termwise integration of $(2\cosh(u/2))^{-t} = \sum_k \binom{-t}{k} e^{-(k+t/2)u}$ gives the series for $\operatorname{Re} w > t$, and analytic continuation gives it for $\operatorname{Re} w > t-1$, where the alternating series converges. For

$t=1$, $(k+\frac12)^{-w}=2^w(2k+1)^{-w}$. For $t=2$, $\binom{k+1}{k}(k+1)^{-w}=(k+1)^{1-w}$. For $t=3$,

$$\binom{k+2}{2}=\tfrac12\big((k+\tfrac32)^2-\tfrac14\big)\quad\text{and}\quad\sum_k(-1)^k(k+\tfrac32)^{-u}=2^u(1-\beta(u)),$$

the constant terms cancelling. For integer t, $\binom{k+t-1}{t-1}$ is a polynomial in $m=k+\frac t2$, odd of degree $t-1$ for even t and even for odd t, vanishing at

$$m=1,\dots,\tfrac t2-1,\quad\text{respectively}\quad m=\tfrac12,\dots,\tfrac{t-2}{2},$$

so the sums may be started at $m=1$, respectively $m=\frac12$, and become shifted η or β series. □

The formulas for $t=2,3$ are elementary and were not checked numerically; the statement for general t is proved above and the explicit combinations are stated for integer t only.

Corollary 17.13.2 (one weight completeness throughout the family). *For every $t>0$ and every $c>0$, $\{f_t(\lambda x)\}_{\lambda\in\Lambda}$ with $\Lambda\subset[1,\infty)$ is complete in $\mathcal W_c$ whenever $\sum_{\lambda\in\Lambda}1/\lambda=\infty$; in particular the integer dilates are complete.*

Proof. f_t is bounded at 0 and $f_t(x)\sim 2\pi(x/2)^{t-1}e^{-\pi x/2}$, so $f_t\in\mathcal W_c$ for all $c>0$. For $c\ge1$ the bounds on M_t below refer to its continuation from $0<\operatorname{Re}s<1$, and at even c the pole of $1/\sin(\pi s/2)$ cancels a zero of Z_t. On $\operatorname{Re}s=c$, $|1/\sin(\pi s/2)|\asymp e^{-\pi|\tau|/2}$, and $Z_t(1-c-i\tau)$ is polynomially bounded: with $a_w(k)=\frac{\Gamma(k+t)}{\Gamma(t)\Gamma(k+1)}(k+\frac t2)^{-w}$ and $\Delta a(k)=a(k)-a(k+1)$, alternating summation gives $Z_t(w)=\sum_{j<r}2^{-j-1}\Delta^j a_w(0)+2^{-r}\sum_{k\ge0}(-1)^k\Delta^r a_w(k)$, and on a fixed line $|\Delta^r a_w(k)|\le C(1+|\operatorname{Im}w|)^r(1+k)^{t-1-\operatorname{Re}w-r}$ by Stirling and Cauchy's estimate, so for $r>t-\operatorname{Re}w$ the last series converges absolutely with a polynomial bound in $|\operatorname{Im}w|$. The function Z_t is not identically zero ($M_t(c)>0$), so $M_t(c+i\tau)\ne0$ for almost every τ, and $|M_t(c+i\tau)|\le C(1+|\tau|)^Ae^{-\pi|\tau|/2}$. These are the only properties of the kernel used in the proof of Theorem 17.3.2. □

The hypothesis $\Lambda\subset[1,\infty)$ cannot be dropped: for $\Lambda=\{2^{-n}:n\ge1\}$, whose reciprocal sum diverges, take a smooth $h\ne0$ supported in a short interval avoiding the zeros of $M_t(c+i\tau)$ together with its translate by $P=2\pi/\log2$, put $H(\tau)=h(\tau)-h(\tau-P)$ and $\overline G=H/M_t$; then $\int H(\tau)\lambda^{-i\tau}d\tau=0$ for every $\lambda=2^{-n}$, and G is a nonzero orthogonal vector. The threshold of Remark 17.3.3 is universal in the Meixner family: f_t is analytic in the open right half plane, with poles on the imaginary axis at $\pm i(t+2k)$, and the rate $\pi/2$ is the half opening of that sector. What changes with t is the arithmetic.

THEOREM 17.13.3 (order two is free). *Let $c_n = \mu(n)$ for odd n and $c_n = 0$ for even n. Then for every $x > 0$, with absolute convergence,*

$$e^{-x} = \frac{1}{2}\sum_{n\geq 1} c_n \operatorname{csch}(nx), \qquad xe^{-x} = \frac{1}{2}\sum_{n\geq 1} \frac{c_n}{n}\,(nx)\operatorname{csch}(nx), \tag{17.13.2}$$

and $\sum_n (c_n/n)n^{-s} = 1/\big((1-2^{-s-1})\zeta(s+1)\big)$ converges absolutely for $\operatorname{Re} s > 0$. Consequently the series for xe^{-x} converges absolutely in every $\mathcal{W}_c$, $c > 0$, and $\{(nx)\operatorname{csch}(nx)\}_{n\in\mathbb{N}}$ is complete in every $\mathcal{W}_c$ and in every $\mathcal{W}_a \cap \mathcal{W}_b$, $0 < a < b$, unconditionally.

Proof. (Equivalently $c_n = \sum_{2^j|n} \mu(n/2^j)$: for $n = 2^k m$ with m odd and $k \geq 1$ the two nonzero terms $\mu(m)$ and $\mu(2m) = -\mu(m)$ cancel.) Since $\operatorname{csch}(nx) = 2\sum_{m \text{ odd}} e^{-mnx}$, absolute convergence at fixed $x > 0$ permits grouping by $mn = N$: $\frac{1}{2}\sum_n c_n \operatorname{csch}(nx) = \sum_{N \text{ odd}} e^{-Nx}\sum_{n|N}\mu(n) = e^{-x}$. Multiply by x for the second identity. The Mellin transform of $(nx)\operatorname{csch}(nx)$ is $2n^{-s}\Gamma(s+1)(1-2^{-s-1})\zeta(s+1)$ for $\operatorname{Re} s > 0$. Put $h(x) = x\operatorname{csch} x \in \mathcal{W}_c$; then $\sum |c_n/n|\,\|h(n\cdot)\|_c \leq \|h\|_c \sum n^{-c-1} < \infty$, so the series converges absolutely in each $\mathcal{W}_c$, and simultaneously in $\mathcal{W}_a \cap \mathcal{W}_b$, to its pointwise value xe^{-x}. The span is invariant under integer dilations, so its closure contains every nxe^{-nx}. These are complete: if g is orthogonal to them in $\mathcal{W}_a \cap \mathcal{W}_b$ (or $\mathcal{W}_c$) and $w(x) = x^{2a-1} + x^{2b-1}$ (respectively x^{2c-1}), the finite complex measure $d\nu = xe^{-x}\bar{g}\,w\,dx$ has $\int e^{-(n-1)x}d\nu = 0$ for $n \geq 1$; pushing forward by $y = e^{-x}$ and using the density of polynomials in $C[0,1]$ gives $\nu = 0$, hence $g = 0$. □

The function $M_2(s) = 2(2/\pi)^s\Gamma(s+1)(1-2^{-s-1})\zeta(s+1)$ has no zeros in $\operatorname{Re} s > 0$; the zeros of $Z_2(1-s) = \eta(-s)$ at the positive even integers cancel the apparent poles of $1/\sin(\pi s/2)$, and the nontrivial zeros of ζ mapped by $s \mapsto -s$ lie to the left of the admissible half plane (the critical line of $\eta(-s)$ is $\operatorname{Re} s = -\frac{1}{2}$). There is nothing for the two weight criterion to detect, and Theorem 17.13.3 is the Meixner analogue of the classical range $c \geq 1$ for the secant, now covering every weight. For integer $t \geq 3$ the zero set of $Z_t(1-s)$ in the strip is that of a combination such as $\beta(1-s) - \beta(-1-s)$ or $\eta(-s) - \eta(-2-s)$; it is not governed by a single L function, and for noninteger t the series Z_t remains the object to study.

17.14 The Thorin form of the two weight criterion

Let Ξ denote ξ or $\xi(\cdot, \chi_4)$, with $\xi(s,\chi_4) = (4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})\beta(s)$ as in Section 15.1.7. Both are entire of order one, real and positive on the real axis, satisfy $\Xi(s) = \Xi(1-s)$, and have the Hadamard product $\Xi(s) = \Xi(0)\prod_\rho(1-s/\rho)$ over the nontrivial zeros, with ρ and $\bar{\rho}$ paired (Davenport [70]); grouping the zeros into reflected quartets makes the product invariant under $s \mapsto 1-s$, so the functional

equation forces the exponential factor of Hadamard's theorem to be trivial. For $\theta \geq \frac{1}{2}$ put

$$F_\theta(s) = \frac{\Xi(\theta)}{\Xi(\theta + \sqrt{s})}, \qquad s \in \mathbb{C} \setminus (-\infty, 0], \tag{17.14.1}$$

with the principal square root. A probability law on $[0, \infty)$ is a generalised gamma convolution (GGC) if its Laplace transform is $\exp(-as - \int_0^\infty \log(1 + s/u)U(du))$ with $a \geq 0$ and a Thorin measure U, a positive Radon measure on $(0, \infty)$ with $\int_{(0,1)} |\log u| U(du) < \infty$ and $\int_1^\infty u^{-1} U(du) < \infty$ (local finiteness is part of the definition: the two integrals alone allow infinite mass accumulating at 1); the Thorin measure is unique [39].

For $\Xi = \xi$ and $\theta \geq 1$, and more generally when $\theta > \operatorname{Re}\rho$ for every zero, the GGC statement and the Thorin density are Theorem 2.1.2 of Chapter 2 (with the density (2.1.6) and its Poisson structure, Proposition 2.1.3); the strip form is Theorem 13.1.2 and Theorem 13.1.3, and Chapter 12 computes the Thorin measure at the basepoint one. New here are the case of $\xi(s, \chi_4)$, the description of one conjugate pair of zeros as the law of $|\gamma + aC|$, and the link with the two weight criterion of Theorem 17.5.2.

LEMMA 17.14.1 (one pair of zeros). *Let $a \geq 0$ and $\gamma > 0$. For $s \in \mathbb{C} \setminus (-\infty, 0]$,*

$$\log \frac{(a + \sqrt{s})^2 + \gamma^2}{a^2 + \gamma^2} = \int_0^\infty \log\Big(1 + \frac{s}{u}\Big) Q_{a,\gamma}(du),$$

where $Q_{0,\gamma} = \delta_{\gamma^2}$ and, for $a > 0$, $Q_{a,\gamma}$ is the probability measure on $(0, \infty)$ with density $q_{a,\gamma}(u) = \frac{1}{2\pi\sqrt{u}}\big[\frac{a}{a^2 + (\sqrt{u} - \gamma)^2} + \frac{a}{a^2 + (\sqrt{u} + \gamma)^2}\big]$. In the variable $v = \sqrt{u}$, $Q_{a,\gamma}$ is the law of $|\gamma + aC|$ with C standard Cauchy.

Proof. Both sides vanish at $s = 0$, so it suffices to compare s derivatives for $s > 0$ and continue analytically (the positivity of the measure and the bounds below justify differentiation under the integral). The left side has derivative $(a + \sqrt{s})/(\sqrt{s}((a + \sqrt{s})^2 + \gamma^2))$. With $u = v^2$ the right side has derivative $\frac{1}{\pi}\int_{\mathbb{R}} \frac{1}{v^2 + s} \frac{a}{a^2 + (v - \gamma)^2} dv$, the Poisson integral of $v \mapsto 1/(v^2 + s)$ at the point $\gamma + ia$ of the upper half plane. Since $1/(v^2 + s) = \operatorname{Re}[1/(\sqrt{s}(\sqrt{s} - iv))]$ and $z \mapsto 1/(\sqrt{s}(\sqrt{s} - iz))$ is bounded and analytic in $\operatorname{Im} z > 0$, the Poisson integral equals $\operatorname{Re}[1/(\sqrt{s}(\sqrt{s} + a - i\gamma))]$, which is the left derivative. The mass is 1 because the two Cauchy densities integrate over $v > 0$ to the integral of one Cauchy density over $\mathbb{R}$; the case $a = 0$ is the limit. The condition $a \geq 0$ is indispensable. □

THEOREM 17.14.2 (Thorin form). *Let $\theta \geq \frac{1}{2}$. The following are equivalent.*

(i) Ξ has no zeros in $\operatorname{Re} s > \theta$.

(ii) F_θ is the Laplace transform of a generalised gamma convolution.

(iii) F_θ is analytic in $\mathbb{C} \setminus (-\infty, 0]$.

When they hold, the Thorin measure of F_θ is

$$U_\theta = \sum_{\rho,\ \mathrm{Im}\,\rho>0} m_\rho\, Q_{\theta-\mathrm{Re}\,\rho,\ \mathrm{Im}\,\rho}, \tag{17.14.2}$$

over the distinct zeros in the upper half plane with multiplicity m_ρ, and $F_\theta(s) = \exp(-\int_0^\infty \log(1+s/u)U_\theta(du))$ with no linear term. For $\Xi = \xi(\cdot,\chi_4)$, (i) is equivalent to two weight completeness of the secant dilates on $(\theta,1)$ (Theorem 17.5.2, for $\theta<1$); for $\Xi=\xi$, to two weight completeness of the dilates of φ (Theorem 17.6.1).

Proof. (ii) implies (iii) because a GGC Laplace transform is analytic in $\mathbb{C}\backslash(-\infty,0]$. (iii) implies (i): a zero ρ with $\mathrm{Re}\,\rho>\theta$ makes F_θ have a pole at $s=(\rho-\theta)^2$, which is not on $(-\infty,0]$ since $\mathrm{Re}(\rho-\theta)>0$ (and $\Xi(\theta)>0$, the zero is nonreal and the square root map is locally invertible there). (i) implies (ii): by the Hadamard product and the pairing, for $s>0$,

$$\log\frac{\Xi(\theta+\sqrt{s})}{\Xi(\theta)} = \sum_{\mathrm{Im}\,\rho>0} m_\rho \log\frac{(a_\rho+\sqrt{s})^2+\gamma_\rho^2}{a_\rho^2+\gamma_\rho^2},$$
$$a_\rho = \theta-\mathrm{Re}\,\rho\in[0,\theta],\ \ \gamma_\rho=\mathrm{Im}\,\rho,$$

with $a_\rho\ge 0$ by (i); the summands are nonnegative and $O_{\theta,s}(\gamma_\rho^{-2})$, so the series converges normally. Lemma 17.14.1 turns each term into $\int\log(1+s/u)Q_{a_\rho,\gamma_\rho}(du)$, and by Tonelli the sum of the positive measures is a Thorin measure: for large γ, $0\le a\le\theta$ and $0<u<1$, $q_{a,\gamma}(u)\le C_\theta\gamma^{-2}u^{-1/2}$, which makes the small u integral summable (finitely many low zeros cost nothing), and $\int_1^\infty u^{-1}Q_{a,\gamma}(du)$ is the integral of v^{-2} against the folded Cauchy law over $v\ge1$, which is $\ll_\theta\gamma^{-2}$ (on $|v-\gamma|<\gamma/2$ use $v^{-2}\le 4\gamma^{-2}$ and mass at most one, elsewhere the density is at most $C_\theta\gamma^{-2}$ and $\int_1^\infty v^{-2}dv<\infty$; for $a=0$ the atom at γ^2 gives the same). Hence U_θ is a locally finite Thorin measure, $F_\theta(s)=\exp(-\int\log(1+s/u)U_\theta(du))$ for $s>0$, and both sides continue to $\mathbb{C}\setminus(-\infty,0]$. The absence of a linear term is the absence of an exponential factor in the Hadamard product. The completeness statements are Theorems 17.5.2 and 17.6.1. □

Corollary 17.14.3 (Thorin representations above the zero free abscissa). *For every $\theta\ge1$, $\xi(\theta)/\xi(\theta+\sqrt{s})$ and $\xi(\theta,\chi_4)/\xi(\theta+\sqrt{s},\chi_4)$ are Laplace transforms of GGCs with the absolutely continuous Thorin measures* (17.14.2)*, since the zeros have $\mathrm{Re}\,\rho<1$. Conditionally on the zero free half plane $\mathrm{Re}\,s>\frac78$ of the quasi Riemann hypothesis (for β, on the theorem covering $L(s,\chi_4)$), the GGC representation holds for every $\theta\ge\frac78$, and the Thorin measure is absolutely continuous for $\theta>\frac78$, and at $\theta=\frac78$ if no zero lies on $\mathrm{Re}\,s=\frac78$; a zero on that line contributes an atom $m_\rho\delta_{\gamma_\rho^2}$. For $\theta=\frac12$ the statement for ξ is the Riemann hypothesis and the Thorin measure is then $\sum_{\gamma>0}m_\gamma\delta_{\gamma^2}$.*

The last sentence is the reciprocal Xi representation of the book; Theorem 17.14.2 embeds it in a one parameter family and describes the family above the zero free abscissa.

THEOREM 17.14.4 (the Cauchy semigroup of Thorin measures; for ξ the Cauchy flow of Proposition 2.1.3 and [282, Chapter 8]). *Suppose Ξ has no zeros in $\operatorname{Re} s > \theta_0$. For $\theta \geq \theta_0$ let $\widetilde{U}_\theta = \frac{1}{2}\sum_\rho m_\rho \operatorname{Cauchy}(\operatorname{Im}\rho, \theta - \operatorname{Re}\rho)$ be the symmetric measure on $\mathbb{R}$ (in the variable v), summed over the distinct zeros of both half planes, with $\operatorname{Cauchy}(\gamma, 0) = \delta_\gamma$. Then U_θ is the image of $\widetilde{U}_\theta$ under $v \mapsto v^2$, and for $\theta_0 \leq \theta < \theta'$,*

$$\widetilde{U}_{\theta'} = \widetilde{U}_\theta * \operatorname{Cauchy}(0, \theta' - \theta).$$

In particular, if all zeros lie on $\operatorname{Re} s = \theta_0$ (which for the completed functions forces $\theta_0 = \frac{1}{2}$), then $\widetilde{U}_\theta$ is the Poisson smoothing at width $\theta - \theta_0$ of $\frac{1}{2}\sum_\rho m_\rho \delta_{\operatorname{Im}\rho}$, and $\widetilde{U}_\theta \to \frac{1}{2}\sum_\rho m_\rho \delta_{\operatorname{Im}\rho}$ vaguely as $\theta \downarrow \theta_0$.

Proof. Each of $\operatorname{Cauchy}(\gamma, a)$ and $\operatorname{Cauchy}(-\gamma, a)$ is carried by $v \mapsto v^2$ to $Q_{a,\gamma}$ (Lemma 17.14.1), so the factor $\frac{1}{2}$ makes the image equal to (17.14.2); without it the image of the sum over all zeros is $2U_\theta$. The Cauchy laws form a convolution semigroup in the scale, $\operatorname{Cauchy}(\gamma, a) * \operatorname{Cauchy}(0, b) = \operatorname{Cauchy}(\gamma, a + b)$, and $\theta' - \operatorname{Re}\rho = (\theta - \operatorname{Re}\rho) + (\theta' - \theta)$; Tonelli permits summing the positive measures, which are locally finite by the tail estimate of Theorem 17.14.2. For vague convergence, a compactly supported continuous test function sees each Cauchy term converge to its atom, and the contribution of large $|\gamma|$ is bounded by $C\sum \gamma^{-2}$ uniformly for widths in a bounded interval. The measures have infinite total mass, so weak convergence is not the notion. □

Three remarks on what this gives the programme. First, a checkable statement. For $\theta \geq 1$ the Thorin measure of $\xi(\theta)/\xi(\theta + \sqrt{s})$ is the Cauchy smoothing at width $\theta - \frac{1}{2}$ of the zero measure, plus a correction from any zeros off the critical line. For $\theta = 1$ and the first 1000 zeros, taken on the critical line, the sum $\sum_\gamma \int \log(1 + s/u) Q_{1/2,\gamma}(du)$ plus a smooth tail from the zero density reproduces $\log \xi(1 + \sqrt{s})/\xi(1)$ (Table 17.3, computed). The tail is a model and not a bound, so the totals are computed, not certified. A Thorin measure obtained by any other computational route must agree with this one; the comparison at level $\theta \geq 1$ is a test of the computational machinery with no hypothesis attached.

s	$\log \xi(1 + \sqrt{s})/\xi(1)$	zero sum	tail (model)	total
1	0.0461176	0.044678	0.00144	0.046118
4	0.137917	0.133599	0.00432	0.137917
16	0.454853	0.440461	0.0144	0.454854

Table 17.3: The Thorin check at $\theta = 1$ (computed). The first column is the closed forms $\log(\pi/3)$, $\log(3\zeta(3)/\pi)$, $\log(15\zeta(5)/\pi^2)$ evaluated.

Second, the right metric. Above the zero free abscissa the Thorin measure is absolutely continuous and the RH measure $\sum \delta_{\gamma^2}$ is atomic, and both have infinite mass, so their total variation distance is infinite and not 2; the meaningful comparison is vague convergence (Theorem 17.14.4), or a metric after a weight with finite integral. The idea that a zero to the right of θ shows up as a negative bump of mass -1 at γ, by putting $a < 0$ in $Q_{a,\gamma}$, is false as a characterisation and is not used: formula (17.14.2) was proved only for $a \geq 0$. For $a = -h < 0$ one has $L_{-h}(s) = -L_h(s) + \log \frac{(s+h^2+\gamma^2)^2 - 4h^2 s}{(h^2+\gamma^2)^2}$, where $L_a(s) = \log \frac{(a+\sqrt{s})^2+\gamma^2}{a^2+\gamma^2}$; a negative Cauchy density accounts for $-L_h$ only, and the last term has its zeros at $(h \pm i\gamma)^2$, off the cut, which are precisely the singularities that obstruct the GGC representation. At $\theta = \frac{1}{2}$ an off critical reflected pair with widths $-h$ and $+h$ has formally cancelling densities, while $F_{1/2}$ still has off cut poles; so positivity of the formal signed density does not characterise RH (compare Remark 2.1.4, where a positive density with a zero to its right is exhibited). What is proved is Theorem 17.14.2: RH is equivalent to analyticity of $F_{1/2}$ on $\mathbb{C} \setminus (-\infty, 0]$, or to its GGC property, and above a zero free boundary the positive Cauchy mixture is the genuine Thorin measure. No theorem here identifies the first failure of the GGC property with the appearance of a negative part.

Third, the flow. Above the zero free abscissa $\theta \mapsto \widetilde{U}_\theta$ is the Poisson extension of the already smoothed boundary measure $\widetilde{U}_{\theta_0}$, a Cauchy semigroup running forward, and it is the Poisson extension of an atomic zero measure on one line only when all real parts agree. The two weight criterion of Theorem 17.5.2 is, through Theorem 17.14.2, the statement that F_θ is a GGC at every level above θ. Forward Cauchy convolution is a positive operation; the inverse is not automatically one, so running the semigroup downward from a computed $\widetilde{U}_1$ is an inversion problem, not a theorem (Problem 17.5).

17.15 Numerical checks of the sharpness experiment

The sharpness experiment compares $\Lambda = \{n^2\}$ with $\Lambda = \{n\}$ at $c = \frac{3}{2}$ and at $c = \frac{1}{2}$ (computed with the Gram system of Section 17.12).

	$c = 3/2$		$c = 1/2$	
N	squares	integers	squares	integers
4	9.9×10^{-3}	1.9×10^{-3}	1.2×10^{-2}	1.0×10^{-2}
8	6.5×10^{-3}	1.3×10^{-4}	7.2×10^{-3}	1.4×10^{-3}
16	5.2×10^{-3}	1.6×10^{-5}	4.9×10^{-3}	2.4×10^{-4}
24	4.8×10^{-3}	9.3×10^{-6}	4.2×10^{-3}	1.2×10^{-4}

Table 17.4: Distance from e^{-x} to the span of the first N dilates, squares against integers (computed).

For the squares the distance flattens, consistent with Theorem 17.10.2 at $c = \frac{3}{2}$

(the theorem gives a nonzero orthogonal vector, not that e^{-x} lies outside the closed span), and it behaves similarly at $c = \frac{1}{2}$; for the integers it keeps falling. The ratio of squares to integers is 5, 49, 322, 519 at $c = \frac{3}{2}$ and 1.1, 5.0, 20, 35 at $c = \frac{1}{2}$, so at $c = \frac{1}{2}$ it exceeds one order of magnitude from $N = 16$ on and does not reach two.

17.16 Open problems

Problem 17.1 (sharpness at $c = \frac{1}{2}$). Does every $\Lambda \subset [1, \infty)$ with $\sum 1/\lambda < \infty$ admit a nonzero annihilator of $\{\operatorname{sech}(\lambda x)\}$ in $\mathcal{W}_{1/2}$, that is, can the zeros of β on the critical line be used to build, from a convergent set, functions that the Blaschke construction cannot see? A separate target question is whether e^{-x} has positive distance from the closed span of $\{\operatorname{sech}(n^2 x)\}$ in $\mathcal{W}_{1/2}$. The numerical evidence for the squares (Table 17.4) is consistent with a positive limit and proves nothing.

Problem 17.2 (rate and lower bound for $d_N(c)$). Find an unconditional rate for $d_N(c)$, $\frac{1}{2} < c < 1$, or a lower bound. By Proposition 17.12.1 a sufficient route to an upper bound is approximating $1/\beta$ on the segment $|t| \le T_N$ by Dirichlet polynomials with coefficients at most $e^{o(T_N)}$.

Problem 17.3 (the strip converse). Does a zero free strip $a < \operatorname{Re} s < b$ alone give completeness of $\{\operatorname{sech}(nx)\}$ in $\mathcal{W}_a \cap \mathcal{W}_b$? Proposition 17.5.1 is the converse direction, and Theorem 17.5.2 needs the whole half plane.

Problem 17.4 (Meixner order $t \ge 3$). Describe the zero sets of $Z_t(1 - s)$ in the strip for integer $t \ge 3$, beginning with $\beta(1 - s) - \beta(-1 - s)$. Are there infinitely many zeros in $0 < \operatorname{Re} s < 1$, and is there a two weight criterion with an arithmetic meaning?

Problem 17.5 (below the zero free abscissa). Is there a direct characterisation, not passing through the zeros, of which positive measures $\widetilde{U}_1$ arise at level $\theta = 1$ from an L function? Given $\widetilde{U}_1$, does the level one Thorin measure together with an explicit analytic reconstruction determine the first abscissa at which F_θ develops an off cut pole or loses the GGC property? The level one function $\Xi(1 + \sqrt{s})$ is available from the absolutely convergent Euler product, but recovering $\widetilde{U}_1$ from it is an ill posed inversion.

Part V

Probability, hitting times and statistical mechanics

Chapter 18

Hitting times, Krein strings and the Pólya Bessel clock

This chapter studies Pólya's function $P(\theta) = K_{i\theta/2}(2\pi)$, the building block of his approximation to Ξ, as a model in which every statement of the Thorin programme is a theorem with an explicit witness: Biane's Brownian hitting times, a Liouville operator whose Dirichlet spectrum gives the zeros, a GGC with the squared zeros as Thorin atoms whose clock is the hitting time of an explicit diffusion, and a Klein Gordon criterion. The de Bruijn Newman flow shows where the analogy with Ξ stops. The exercises at the end also cover sections now placed elsewhere: the law of the clock of Ξ (Section 5.2), its Brownian boundaries (Section 3.1), Li's criterion (Section 23.6), the de Bruijn Newman flow (Section 4.1) and the Dirichlet beta function (Section 15.1.7).

18.1 Pólya's Bessel function: hitting times, a clock and the flow

Pólya replaced the kernel Φ of (2.1.3) by its asymptote Φ^* and proved that the transform Ξ^* has only real zeros [302] (Section 19.1.1). The building block of his proof is the Macdonald function

$$P(\theta) = K_{i\theta/2}(2\pi) = \int_{\mathbb{R}} e^{-2\pi \cosh 2u}\, e^{i\theta u}\, du, \tag{18.1.1}$$

which has only real zeros and whose zero counting function has the main term of the Riemann von Mangoldt formula. Biane [32] showed that Ξ and P are Mellin transforms of two Brownian hitting times whose Thorin measures are spectral measures of the radial Brownian motions on $SU(2)$ and on $SL_2(\mathbb{C})/SU(2)$. This section is based on [284]. For P every statement of the Thorin programme is a theorem with an explicit witness: the zeros are the Dirichlet spectrum of a Liouville operator, $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ is a GGC with the squared zeros as Thorin atoms, its clock is the hitting time of an explicit diffusion, and its Poisson trace is completely monotone as the trace of a semigroup. The analogy stops at the

flow: by a theorem of Newman, Pólya's function has a finite de Bruijn Newman constant, although its zeros show no sign of colliding in the range we compute.

18.1.1 Two hitting times and Biane's dictionary

As $u \to \pm\infty$ only the term $n = 1$ of (2.1.3) matters, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2} \exp(-\pi e^{2|u|})$. Pólya replaced Φ by the even function with the same asymptotics on both sides,

$$\Phi^*(u) = 8\pi^2 \cosh \tfrac{9u}{2}\, e^{-2\pi \cosh 2u},$$
$$\Xi^*(\theta) = \int_{\mathbb{R}} \Phi^*(u) e^{i\theta u}\, du = 4\pi^2 \Big[P\big(\theta - \tfrac{9i}{2}\big) + P\big(\theta + \tfrac{9i}{2}\big) \Big],$$

since the factor $e^{\pm 9u/2}$ shifts the argument by $\mp \frac{9i}{2}$. This is the form $G(z+ia) + G(z-ia)$ of Pólya's Hilfssatz II (Section 11.1.3) with $G = P$ and $a = \frac{9}{2}$: Pólya proved that P, entire of order one, has only real zeros, and the Hilfssatz transfers this to Ξ^*. In the Bessel index the shift is $\frac{9}{4}$, since $P(\theta \mp \frac{9i}{2}) = K_{i\theta/2 \pm 9/4}(2\pi)$, the form of Ξ^* in Section 19.1.1. Pointwise Φ^* is poor near the origin, $\Phi^*(0) = 8\pi^2 e^{-2\pi} = 0.147447$ against $\Phi(0) = 0.893394$, and the kernel $e^{-2\pi\cosh 2u}$ of P is narrower still (Figure 18.1); the three share the doubly exponential tail, which governs the zero counts (Table 18.1).

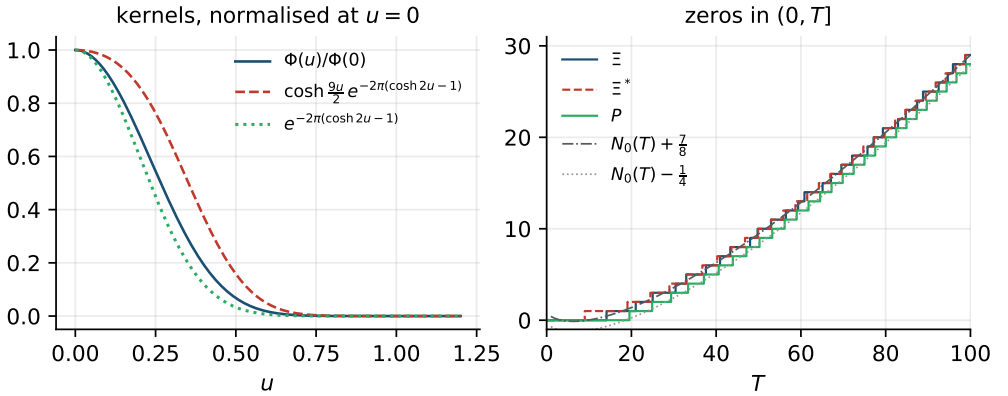


Figure 18.1: Left: the kernel Φ of (2.1.3), Pólya's asymptote Φ^* and the kernel $e^{-2\pi\cosh 2u}$ of P, each divided by its value at $u = 0$. Right: the zero counting functions of Ξ, Ξ^* and P on $(0, 100]$, against $N_0(T) + \frac{7}{8}$ and $N_0(T) - \frac{1}{4}$, the mean values of the Riemann von Mangoldt formula and of Proposition 18.1.8, with $N_0(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e}$.

The argument 2π is forced. Pólya's asymptotics for the zeros of $y \mapsto K_{iy}(2a)$, which Biane derives by stationary phase [32, Section 3.5], give $\frac{T}{\pi} \log \frac{T}{a} - \frac{T}{\pi} + O(1)$

n	Ξ	Ξ^*	P
1	14.135	8.993	19.538
2	21.022	19.065	24.897
3	25.011	24.475	29.370
4	30.425	28.975	33.383
5	32.935	33.007	37.099
6	37.586	36.737	40.600
7	40.919	40.249	43.936
8	43.327	43.595	47.140
9	48.005	46.807	50.234
10	49.774	49.908	53.236
11	52.970	52.916	56.158
12	56.446	55.843	59.009
13	59.347	58.700	

T	Ξ	Ξ^*	P	$N_0(T)$
60	13	13	12	11.998
100	29	29	28	28.127
240	102	102	101	100.946
1000	649	649	647	647.741

Table 18.1: Left: the positive zeros below 60 of Ξ, of Pólya's Ξ^* and of $P(\theta) = K_{i\theta/2}(2\pi)$. Right: the numbers of zeros in $(0,T]$ against $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Computed.

zeros with $0 < y \le T$; for P the zeros are at $\theta = 2y$ with $a = \pi$, and the count in $(0,T]$ becomes $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(1) = N_0(T) + O(1)$. The constant $a = \pi$ comes from the factor $e^{-\pi e^{2u}}$ of the term $n = 1$.

Let $H_a^{(c)}$ be the first passage time of $B_t + ct$ to the level $a > 0$, for standard Brownian motion B and drift $c \ge 0$ (Appendix A). Its density is $a(2\pi\tau^3)^{-1/2}e^{-(a-c\tau)^2/2\tau}$, the inverse Gaussian law of Table 4.3, and its Laplace transform is $e^{a(c-\sqrt{c^2+2u})}$.

Proposition 18.1.1. *Let $H = H_a^{(c)}$ with $a = c = \sqrt{2\pi}$.*

(i) H has the density $\tau^{-3/2}e^{2\pi}e^{-\pi(\tau+1/\tau)}$ on $(0,\infty)$, and $1/H$ has the law of H size biased by H.

(ii) $\mathbb{E}H^s = 2e^{2\pi}K_{s-\frac12}(2\pi)$ for every $s \in \mathbb{C}$, so that $\mathbb{E}H^s = \mathbb{E}H^{1-s}$.

(iii) $P(\theta) = \frac12 e^{-2\pi}\,\mathbb{E}H^{\frac12+\frac{i\theta}{2}}$ for every $\theta \in \mathbb{C}$.

(iv) $U = \frac12\log H$ has the density $2e^{2\pi}e^{-u}e^{-2\pi\cosh 2u}$, and the law of U size biased by e^U has the density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, whose characteristic function is $P(\theta)/P(0)$.

Proof. (i) With $a = c = \sqrt{2\pi}$, $(a-c\tau)^2/2\tau = \pi(\tau+1/\tau) - 2\pi$ and $a/\sqrt{2\pi} = 1$. The density of $1/H$ at σ is σ^{-2} times that of H at $1/\sigma$, which is $\sigma^{-1/2}e^{2\pi}e^{-\pi(\sigma+1/\sigma)}$, that is σ times the density of H at σ; in particular $\mathbb{E}H = 1$. (ii) The substitution $\tau = e^w$ and $K_\nu(x) = \int_0^\infty e^{-x\cosh w}\cosh(\nu w)\,dw$ [256, (10.32.9)] give $\int_0^\infty \tau^{\nu-1}e^{-\pi(\tau+1/\tau)}\,d\tau = 2K_\nu(2\pi)$, here with $\nu = s - \frac12$; the integral converges for

every complex ν, and $K_\nu = K_{-\nu}$ gives the symmetry. (iii) is (ii) at $s = \frac12 + \frac{i\theta}{2}$. (iv) The substitution $\tau = e^{2u}$ turns the density of (i) into $2e^{2\pi}e^{-3u}e^{-2\pi\cosh 2u}e^{2u}$; multiplying by e^u and dividing by $\mathbb{E}H^{1/2} = 2e^{2\pi}K_0(2\pi)$ gives the last density, and its characteristic function is $\int e^{-2\pi\cosh 2u}e^{i\theta u}du/K_0(2\pi) = P(\theta)/P(0)$ by (18.1.1). □

The parallel with Ξ is exact. By the identity (12.1.1) of Biane, Pitman and Yor [33], with $V = \frac{\pi}{2}S_2$, $2\xi(s) = \mathbb{E}V^{s/2}$, so $\Xi(\theta) = \frac12\mathbb{E}V^{\frac14+\frac{i\theta}{2}}$, and the kernel is $\Phi(u) = \frac12 e^{u/2}f_Y(u)$ with $Y = \frac12\log V$: the Pólya law of (2.1.4) is the law of Y size biased by $e^{Y/2}$, as the law of (iv) is that of U size biased by e^U. In both cases the evenness of the kernel is a reflection symmetry of the hitting time: $\tau \mapsto 1/\tau$ for H, which is (i), and the functional equation $\mathbb{E}V^{s/2} = \mathbb{E}V^{(1-s)/2}$ for V, which comes from Jacobi's theta transformation.

Both hitting times are GGCs (Table 4.3). For $H_a^{(c)}$ the Thorin measure is $a\{\pi\sqrt{2z-c^2}\}^{-1}\mathbf{1}_{z>c^2/2}\,dz$, and the Lévy density, t^{-1} times its Laplace transform, is $a\,e^{-c^2t/2}(2\pi t^3)^{-1/2}$. For the Bessel hitting time, $\mathbb{E}e^{-uJ_a} = a\sqrt{2u}/\sinh(a\sqrt{2u})$ with $J_a \stackrel{d}{=} a^2J$, so $J_a + J_a' = a^2S_2$ has Thorin measure $2\sum_{n\ge1}\delta_{\pi^2n^2/2a^2}$ and Lévy density $2t^{-1}\sum_n e^{-\pi^2n^2t/2a^2}$. The Thorin measure can also be read from the continuation of $\Psi(u) = -\log\mathbb{E}e^{-uX}$ across the cut: integrating the Stieltjes inversion (4.1.2),

$$U\big((0,z)\big) = \frac{1}{\pi}\operatorname{Im}\Psi(-z+i0) \qquad (z>0 \text{ not an atom}),$$

because each factor $\log(1+u/w)$ with $w < z$ contributes $i\pi$ at $u = -z+i0$. For $H_a^{(c)}$, $\Psi(-z+i0) = a\big(i\sqrt{2z-c^2}-c\big)$ for $z > \frac{c^2}{2}$, so $U((0,z)) = \frac{a}{\pi}\sqrt{2z-c^2}$; for $J_a + J_a'$, $\Psi(-\frac{\lambda^2}{2}+i0) = 2\log\frac{\sin\lambda a}{\lambda a}$, which gains $2\pi i$ each time λa passes a multiple of π. Biane's dictionary identifies these Thorin measures with spectral measures. In his normalisation [32, (2.1) to (2.3)] the generator $L_1 = \frac12\frac{d^2}{dx^2}$ on $[0,\pi]$ with Dirichlet conditions has the eigenfunctions $\sin(kx)$, $k = 1,2,\dots,$ with eigenvalues $-\frac{k^2}{2}$, and the spectral measure $\sigma_1 = \frac{1}{\pi}\sum_{n\ge1}\delta_{n^2/2}$; the generator $L_2 = \frac12\frac{d^2}{dx^2} + \frac{d}{dx}$ on $[0,\infty)$ with a Dirichlet condition at 0, selfadjoint in $L^2(e^{2x}dx)$, has the generalised eigenfunctions $e^{-x}\sin(kx)$, $k>0$, with eigenvalues $-\frac{k^2+1}{2}$, and the spectral measure $\sigma_2(d\lambda) = d\lambda/(\pi\sqrt{2\lambda-1})$ on $\lambda > \frac12$.

THEOREM 18.1.2 (Biane's dictionary, with corrected constants). *(i) The Thorin measure of $J_\pi + J_\pi' = \pi^2S_2$ is $2\pi\sigma_1$.*

(ii) The Thorin measure of $H_1^{(1)}$ is σ_2.

(iii) The radial generator $\frac12\frac{d^2}{dx^2} + \cot x\frac{d}{dx}$ of Brownian motion on the unit sphere $S^3 \cong SU(2)$ is the Doob transform of L_1 by its ground state $\sin x$, $\sin^{-1}(L_1 + \frac12)\sin$, and the radial generator $\frac12\frac{d^2}{dx^2} + \coth x\frac{d}{dx}$ of Brownian motion on $\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$ is the Doob transform of L_2 by the harmonic function $1-e^{-2x}$.

Proof. (i) With $a = \pi$ the atoms $\pi^2 n^2/2a^2$ are at $n^2/2$, with mass $2 = 2\pi \cdot \frac{1}{\pi}$. (ii) With $a = c = 1$ the density $a\{\pi\sqrt{2z - c^2}\}^{-1}$ on $z > \frac{c^2}{2}$ is that of σ_2. (iii) For f smooth, $\frac{1}{2}(f \sin)'' = \sin \cdot(\frac{1}{2}f'' + \cot x\, f' - \frac{1}{2}f)$, so $\sin^{-1}(L_1 + \frac{1}{2})(f \sin) = \frac{1}{2}f'' + \cot x\, f'$; and $L_2 h = 0$ for $h = 1 - e^{-2x}$, while $h^{-1}L_2(hf) = \frac{1}{2}f'' + (1 + h'/h)f'$ with $1 + h'/h = 1 + \frac{2e^{-2x}}{1-e^{-2x}} = \coth x$. The eigenfunctions are checked the same way: $L_2(e^{-x}s) = e^{-x}(\frac{1}{2}s'' - \frac{1}{2}s)$. □

For P itself $a = c = \sqrt{2\pi}$ rather than 1. Brownian scaling gives $H_a^{(c)} \overset{d}{=} \frac{a}{c} H_{\sqrt{ac}}^{(\sqrt{ac})}$, so level and drift can always be made equal, which is the reciprocal symmetry of Proposition 18.1.1(i), but their common value $\sqrt{ac}$ is $\sqrt{2\pi}$ for P and 1 in the dictionary. The Doob transforms of (iii) go in the direction Biane states [32, Section 2.4]; the note [284] reverses them.

Remark 18.1.3 (Biane's constants). Biane's $K_\mu(x) = \int_0^\infty t^{\mu-1} e^{-\frac{x}{2}(t+t^{-1})} dt$ is twice the standard Macdonald function, and his kernel (3.1) is the Φ of (2.1.3); with that kernel $\Xi(z) = 2\int_0^\infty \Phi(u)\cos(zu)\, du$ holds for the standard $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, so the factor $\frac{1}{2}$ is missing from his displayed definition of ξ, and his Mellin formula $\mathbb{E}W_a^s = 2(2a^2/\pi)^s \xi(2s)$ also uses the standard ξ (both sides are $\frac{2}{3}$ at $s = 1$, $a = 1$). In his Section 4.3 the Lévy measure of $W_a = J_a + J_a'$ should be $2t^{-1}\sum_n e^{-\pi^2 n^2 t/2a^2} dt$, with the factor t^{-1} and the $\frac{1}{2}$ in the exponent, and that of the drifted first passage time should carry the level x as a factor. His Thorin measure (4.1) should have its atoms at $\pi^2 n^2/2a^2$, not at n^2/a^2; the density (4.2) should be $x\, dc/(\pi\sqrt{2c - a^2})$, which for $x = 1$ is his density divided by $\sqrt{2}$, and the identity offered for it should read $(\pi t)^{-1/2} e^{-a^2 t/2} = \int_{a^2/2}^\infty e^{-ct}\, dc/(\pi\sqrt{c - a^2/2})$, with t in place of t^3. The measure (2.3) lives on $(\frac{1}{2}, \infty)$, not on $[1, \infty)$. In Propositions 2.3 and 2.4 the drift and the harmonic function are oriented for the cone opposite to the one stated: with the cone $\{y_1 \geq \cdots \geq y_n\}$ the drift is $\rho' = (n-1, n-3, \ldots, 1-n)$ and the positive harmonic function is $\prod_{i<j}(1 - e^{-2(y_i - y_j)}) = e^{-\langle \rho', y\rangle} \prod_{i<j} 2\sinh(y_i - y_j)$; the Vandermonde function $\prod_{i>j}(x_i - x_j)$ of his Proposition 2.1 is reversed in the same way, and is positive on the cone as $\prod_{i<j}(x_i - x_j)$. The eigenvalue of Proposition 2.6 is discussed in Remark 18.1.5. With these corrections the dictionary of his Section 4.5 is the exact statement of Theorem 18.1.2.

18.1.2 The Wick rotation and the curvature family

Biane closes his paper by observing that ξ and Pólya's function are related, through $SU(2)$ and $SL_2(\mathbb{C})/SU(2)$, "in this non obvious way" [32, Section 4.5]. The relation is a Wick rotation, the substitution of iy for a real variable y. Wick's device [358] rotates time, $t = -i\tau$, turning the unitary group e^{-itH} into the heat semigroup $e^{-\tau H}$; it keeps the spectrum of H and exchanges oscillation and positivity: probability laws, Laplace transforms and completely monotone functions live on

the Euclidean side, Fourier transforms and their zeros on the real time side, and a Thorin measure is spectral data readable on either side. Through Cartan's duality between a symmetric space G/K of noncompact type and its compact dual U/K, $y \mapsto iy$ reverses the sign of the curvature. Biane's paper contains the rotation in four places: in space (the curvature), in the Laplace variable, in time (heat kernels) and in the Mellin index.

Let $H_n = \{y \in \mathbb{R}^n : \sum_i y_i = 0\}$ and $\mathcal{C} = \{y \in H_n : y_1 \geq \cdots \geq y_n\}$. Biane describes three matrix valued Brownian motions of type A [32, Section 2]. For traceless Hermitian Brownian motion the ordered eigenvalues perform Brownian motion in $\mathcal{C}$ conditioned in Doob's sense never to reach the boundary: the semigroup of Brownian motion killed at $\partial\mathcal{C}$ is the alternating sum $p^0_t(x,y) = \sum_{w\in S_n} \varepsilon(w) p_t(x, wy)$ (the reflection principle), the Vandermonde product $\prod_{i<j}(y_i - y_j)$ is its positive harmonic function vanishing on $\partial\mathcal{C}$, and the Doob transform is Dyson's Brownian motion [79]. For Brownian motion g_t on $SL_n(\mathbb{C})$, the logarithms of the singular values, the radial part of Brownian motion on $SL_n(\mathbb{C})/SU(n)$, perform Brownian motion in $\mathcal{C}$ with generator $\frac{1}{2}\Delta + \langle \nabla \log \tilde{h}, \nabla\rangle$, $\tilde{h}(y) = \prod_{i<j} \sinh(y_i - y_j)$ [171], which is Brownian motion with drift $\rho' = (n-1, n-3, \ldots, 1-n)$, killed at $\partial\mathcal{C}$, transformed by the harmonic function of Remark 18.1.3. For Brownian motion on $SU(n)$ the eigenvalues $e^{i\theta_j}$, with $\sum_j \theta_j = 0$, live in the alcove $\theta_1 \geq \cdots \geq \theta_n$, $\theta_1 - \theta_n \leq 2\pi$, a fundamental domain of the affine Weyl group $S_n \ltimes ((2\pi\mathbb{Z})^n \cap H_n)$; the killed semigroup is the alternating sum over that group, and the eigenvalue process is the Doob transform by the Dirichlet ground state $\prod_{j>k}(e^{i\theta_j} - e^{i\theta_k})$, which is a constant multiple of $\prod_{j<k} \sin\frac{\theta_j - \theta_k}{2}$ because $\sum_j \theta_j = 0$.

The three Doob functions are one function at three values of a curvature parameter. For real κ put $s_\kappa(z) = \sin(\sqrt{\kappa}z)/\sqrt{\kappa}$, so that $s_{-1} = \sinh$, $s_0(z) = z$ and $s_1 = \sin$, and

$$h_\kappa(y) = \prod_{i<j} s_\kappa(y_i - y_j), \qquad |\rho|^2 = \sum_{j=1}^{n} (n+1-2j)^2 = \frac{n(n^2-1)}{3}.$$

Proposition 18.1.4. *For every real κ and every $n \geq 2$, $\Delta h_\kappa = -\kappa|\rho|^2 h_\kappa$ on H_n.*

Proof. For $\kappa = -1$, write $\prod_{i<j} 2\sinh(y_i - y_j) = \prod_{i<j} e^{-(y_i+y_j)}(e^{2y_i} - e^{2y_j})$. The Vandermonde determinant gives $\prod_{i<j}(e^{2y_i} - e^{2y_j}) = \det(e^{2(n-j)y_i})$, and on H_n the factor $\prod_{i<j} e^{-(y_i+y_j)} = e^{-(n-1)\sum_i y_i}$ is one, so the exponents $2(n-j)$ may be replaced by $2(n-j)-(n-1) = n+1-2j$: $\prod_{i<j} 2\sinh(y_i - y_j) = \sum_{w\in S_n} \varepsilon(w) e^{\langle w\rho', y\rangle}$, the Weyl denominator formula. Each $w\rho'$ lies in H_n and has $|w\rho'|^2 = |\rho|^2$, so each exponential is an eigenfunction of the Laplacian of H_n with eigenvalue $|\rho|^2$, and $\Delta h_{-1} = |\rho|^2 h_{-1}$. This identity between entire functions of $y \in H_n \otimes \mathbb{C}$ continues analytically. For $\kappa < 0$, $h_\kappa(y)$ is a constant times $h_{-1}(\sqrt{-\kappa}\, y)$, and the scaling multiplies the Laplacian by $-\kappa$; for $\kappa > 0$, $s_\kappa(z) = \sinh(i\sqrt{\kappa}z)/(i\sqrt{\kappa})$, so $h_\kappa(y)$

is a constant times $h_{-1}(i\sqrt{\kappa}\,y)$, and the substitution multiplies the Laplacian by $(i\sqrt{\kappa})^2 = -\kappa$. At $\kappa = 0$, h_0 is the Vandermonde product, the limit of h_κ as $\kappa \to 0$, since $s_\kappa(z) = z - \kappa z^3/6 + O(\kappa^2)$, and it is harmonic. □

So Biane's three semigroups are one formula,

$$q_t^{(\kappa)}(x,y) = \frac{h_\kappa(y)}{h_\kappa(x)}\, e^{\kappa|\rho|^2 t/2}\, p_t^0(x,y), \qquad \kappa \in \{-1,0,1\},$$

with generator $\frac{1}{2}\Delta + \langle \nabla \log h_\kappa, \nabla\rangle$, where p^0 is the killed kernel of the chamber $\mathcal{C}$ for $\kappa \le 0$ and of the alcove $\{y_1 \ge \cdots \ge y_n,\ y_1 - y_n \le \pi\}$ for $\kappa = 1$; for $\kappa = 1$ the coordinates are the half angles $y = \theta/2$, in which h_1 is the positive Dirichlet ground state of the alcove. The rotation $y \mapsto iy$ sends h_κ to a constant times $h_{-\kappa}$, since $s_\kappa(iz) = is_{-\kappa}(z)$: it exchanges the noncompact and the compact Doob functions and reverses the sign of the ground energy, $\frac{1}{2}\Delta h_\kappa = -\frac{\kappa}{2}|\rho|^2 h_\kappa$, which is $+\frac{1}{2}|\rho|^2$ for $SL_n(\mathbb{C})/SU(n)$ and $-\frac{1}{2}|\rho|^2 = \frac{n-n^3}{6}$ for $SU(n)$ in the half angles.

Remark 18.1.5 (The eigenvalue of Biane's Proposition 2.6). Biane endows the Lie algebra of $SU(n)$ with the form $\langle A, B\rangle = -\operatorname{Tr}(AB)$, in which the angles θ_j perform standard Brownian motion in H_n, and states that the ground state $\prod_{j<k} \sin\frac{\theta_j - \theta_k}{2}$ has the eigenvalue $\frac{n-n^3}{6}$ for $\frac{1}{2}\Delta$ [32, Proposition 2.6]. By Proposition 18.1.4 in the half angles, the eigenvalue for $\frac{1}{2}\Delta_\theta$ is $\frac{1}{4}\cdot\frac{n-n^3}{6} = \frac{n-n^3}{24}$; more generally, for the form $-c\operatorname{Tr}(AB)$ it is $\frac{n-n^3}{24c}$. Biane's value holds for $c = \frac{1}{4}$, that is when the half angles are standard Brownian motion. His rank one discussion uses a third normalisation, $c = \frac{1}{2}$, in which $SU(2)$ is the unit sphere S^3, the angle $x \in [0,\pi]$ has the generator $\frac{1}{2}\frac{d^2}{dx^2} + \cot x \frac{d}{dx}$ of Theorem 18.1.2(iii), and the ground energy is $\frac{2-8}{12} = -\frac{1}{2}$. A computation confirms $\frac{1}{2}\Delta_\theta h/h = -\frac{1}{4}, -1, -\frac{5}{2}$ for $n = 2,3,4$.

In rank one the curvature family is the radial part of Brownian motion on the three dimensional space form of curvature κ, with generator $\frac{1}{2}\frac{d^2}{dx^2} + \sqrt{\kappa}\cot(\sqrt{\kappa}x)\frac{d}{dx}$: the sphere of radius $\kappa^{-1/2}$ for $\kappa > 0$, the Bessel process of dimension three for $\kappa = 0$, and the hyperbolic space of curvature κ for $\kappa < 0$. Let $J_a^{(\kappa)}$ be its hitting time of the radius a from the centre, with $a < \pi/\sqrt{\kappa}$ when $\kappa > 0$; then $J_a^{(0)} = J_a$, whose Laplace transform is classical [58]. For $\kappa < 0$ the generator is that of the Bessel process of dimension three with drift $\sqrt{-\kappa}$ [323, (2.5)], the norm of a three dimensional Brownian motion with drift of that length.

Proposition 18.1.6. *For $u \ge 0$,*

$$\mathbb{E}e^{-uJ_a^{(\kappa)}} = \frac{\sqrt{2u-\kappa}\; s_\kappa(a)}{\sinh\left(a\sqrt{2u-\kappa}\right)},$$

the Laplace transform of a GGC with Thorin measure $\sum_{n\ge1}\delta_{\pi^2n^2/2a^2-\kappa/2}$. Equivalently, the law of $J_a^{(\kappa)}$ is the exponential tilt $e^{\kappa t/2}\Pr(J_a \in dt)/\mathbb{E}e^{\kappa J_a/2}$ of the law of J_a.

Proof. The generator is the Doob transform of $\frac{1}{2}\frac{d^2}{dx^2}$ by $s = s_\kappa$, since $s'' = -\kappa s$ gives $s^{-1}(\frac{1}{2}\frac{d^2}{dx^2} + \frac{\kappa}{2})(sf) = \frac{1}{2}f'' + \frac{s'}{s}f'$ and $s'/s = \sqrt{\kappa}\cot(\sqrt{\kappa}x)$. Hence $f(x) = \sinh(x\sqrt{2u-\kappa})/s_\kappa(x)$ satisfies $\frac{1}{2}f'' + \frac{s'}{s}f' = s^{-1}(\frac{1}{2}g'' + \frac{\kappa}{2}g) = uf$ with $g = \sinh(x\sqrt{2u-\kappa})$, and f is, up to a constant factor, the positive increasing solution that stays bounded at 0, with $f(0+) = \sqrt{2u-\kappa}$. Hence $\mathbb{E}_x e^{-uJ_a^{(\kappa)}} = f(x)/f(a)$ for $0 < x < a$, and letting $x \downarrow 0$ gives $\mathbb{E}e^{-uJ_a^{(\kappa)}} = f(0+)/f(a)$, which is the formula; for $\kappa > 2u$ the square roots are imaginary and the formula is read through $\sinh(ix)/i = \sin x$. As a function of u the denominator divided by $\sqrt{2u-\kappa}$ is entire of order $\frac{1}{2}$, with zeros where $a\sqrt{2u-\kappa} = i\pi n$, that is at $u = -(\pi^2 n^2/2a^2 - \kappa/2)$, all negative under the hypothesis on a; Hadamard's product in genus zero gives the Thorin measure. The tilt is the formula at $u - \frac{\kappa}{2}$ for $\kappa = 0$, divided by its value at $u = 0$, since $\sqrt{-\kappa}/\sinh(a\sqrt{-\kappa}) = 1/s_\kappa(a)$. □

Three consequences clarify the dictionary. First, Riemann's clock $J_a + J_a'$ is the flat member $\kappa = 0$, and its link with $SU(2)$ in Theorem 18.1.2 is spectral rather than geometric: the atoms $\pi^2 n^2/2a^2$ are the Dirichlet spectrum of $\frac{1}{2}\frac{d^2}{dx^2}$ on $[0, a]$, which for $a = \pi$ is the spectrum of the radial generator of $SU(2)$ shifted by its ground energy $\frac{1}{2}$. Second, the hyperbolic member $\kappa = -1$ has the discrete atoms $\pi^2 n^2/2a^2 + \frac{1}{2}$; Pólya's continuous measure comes instead from a first passage over a level on the half line, a Dirichlet problem on $[0, \infty)$ with continuous spectrum. Third, the tilt destroys the reciprocal symmetry of $J_a + J_a'$ that encodes the functional equation. With $W = J_1^{(\kappa)} + J_1'^{(\kappa)}$, the function $(\frac{\pi}{2})^s \mathbb{E}W^s$ equals $2\xi(2s)$ for $\kappa = 0$ and is symmetric under $s \mapsto \frac{1}{2} - s$: it takes the value 0.9963110325 at both $s = 0.1$ and $s = 0.4$. For $\kappa = 1$ the two values are 1.002917197 and 1.023703077. A curvature interpolation between the two clocks must therefore be sought in a class that keeps the symmetry (Question 18.1.20).

Replacing u by a negative value in $\mathbb{E}e^{-uX}$ is the rotation of the Laplace variable used in the inversion formula above. For the curvature family the reciprocal of the transform becomes, for $u < \frac{\kappa}{2}$, the damped sine $\sin(a\sqrt{\kappa - 2u})/(\sqrt{\kappa - 2u}\, s_\kappa(a))$, whose zeros sit at minus the Thorin atoms and slide with κ; for the drifted clock the reciprocal $e^{a(\sqrt{c^2+2u}-c)}$ becomes $e^{-ac}e^{ia\sqrt{2|u|-c^2}}$ beyond the branch point $u = -\frac{c^2}{2}$, which has constant modulus and no zeros.

The rotation of time relates the heat kernels of the two dual spaces. The radial heat kernels of $\frac{1}{2}\Delta$, relative to the Riemannian volume, are

$$p_t^{\mathbb{H}^3}(r) = \frac{e^{-t/2}}{(2\pi t)^{3/2}} \frac{r}{\sinh r} e^{-r^2/2t},$$

$$p_t^{S^3}(\omega) = \frac{e^{t/2}}{(2\pi t)^{3/2}} \sum_{k\in\mathbb{Z}} \frac{\omega + 2\pi k}{\sin\omega} e^{-(\omega+2\pi k)^2/2t},$$

and the term $k = 0$ of the second is, up to the constant factor $(-1)^{-3/2}$, the first

after $r \mapsto i\omega$, $t \mapsto -t$, a genuine Wick rotation of time; the sum over k is the method of images over the affine Weyl group, which makes the kernel periodic.

The fourth rotation acts on the Mellin index. The substitution $s = \frac{1}{2} + i\theta$ turns real Mellin moments into Fourier transforms of a tilted law, and Biane's contour shift $w \mapsto w + \frac{i\pi}{2}$ [32, Section 3.5], with $\cosh(w + \frac{i\pi}{2}) = i \sinh w$, turns the damped integrand $e^{zw - 2a\cosh w}$ into an oscillating one on which stationary phase locates the zeros. For P this is the map $\mu \mapsto i\mu$ on the index of K_μ, and Section 18.1.5 shows that the index is the eigenvalue of a boost, which the rotation turns into an angular momentum. Atoms of a Thorin measure on the Euclidean side become poles after the rotation, and poles require discrete spectrum, which for these radial motions without potential means a compact space. That is why ξ, whose time side Thorin measure is atomic, is attached to $SU(2)$, while P, whose time side measure has a density, is attached to $\mathbb{H}^3$.

18.1.3 The zeros of P as a Dirichlet spectrum

Put $\psi_\mu(x) = K_\mu(e^x)$. Since $\frac{d}{dx} = r\frac{d}{dr}$ for $r = e^x$, the modified Bessel equation $r^2K'' + rK' - (r^2 + \mu^2)K = 0$ becomes

$$-\psi_\mu'' + e^{2x}\psi_\mu = -\mu^2\psi_\mu,$$

a Schrödinger equation with the Liouville potential e^{2x}, the index entering only through the energy $-\mu^2$. Let $x_0 = \log 2\pi$ and let $A_P = -\frac{d^2}{dx^2} + e^{2x}$ act in $L^2([x_0, \infty))$ with a Dirichlet condition at x_0. Biane recalls this spectral reading of the zeros of the Macdonald function as well known (Titchmarsh [347]) and notes that Pólya does not mention it [32, Section 3.3]. We write $\nu_1 < \nu_2 < \cdots$ for the positive zeros of $\nu \mapsto K_{i\nu}(2\pi)$, so that P vanishes at $\theta = \pm 2\nu_n$.

Proposition 18.1.7. *(i) A_P is selfadjoint, has compact resolvent, and $A_P \geq 4\pi^2$.*

(ii) A number λ is an eigenvalue of A_P if and only if $\lambda = \nu^2$ with $\nu > 0$ and $K_{i\nu}(2\pi) = 0$, and the eigenfunction is then $K_{i\nu}(e^x)$. In particular every zero of P is real, of absolute value at least 4π, and the zeros of P are the numbers $\pm 2\nu_n$ with ν_n^2 running over the spectrum of A_P.

(iii) Every eigenvalue of A_P is simple, and every zero of $\mu \mapsto K_\mu(2\pi)$ and of P is simple. More precisely, for $\nu > 0$ and every $y \in \mathbb{R}$,

$$\psi_{i\nu}'(y)\,\partial_\nu\psi_{i\nu}(y) - \psi_{i\nu}(y)\,\partial_\nu\psi_{i\nu}'(y) = -2\nu\int_y^\infty \psi_{i\nu}(x)^2\,dx < 0. \qquad (18.1.2)$$

Proof. (i) The endpoint x_0 is regular, and the potential is bounded below, so the equation is in the limit point case at $+\infty$ and the Dirichlet realisation is selfadjoint; since $e^{2x} \to \infty$, the resolvent is compact. For f in the form domain,

$\langle A_P f, f\rangle = \int_{x_0}^{\infty}(|f'|^2 + e^{2x}|f|^2)\,dx \geq e^{2x_0}\|f\|^2 = 4\pi^2\|f\|^2$. (ii) For every complex μ, $K_\mu(r) = \sqrt{\pi/2r}\,e^{-r}(1+O(1/r))$ as $r\to\infty$ [256, (10.40.2)], so ψ_μ is a nonzero solution in L^2 near $+\infty$, and in the limit point case it is the only one up to a constant factor. Hence $-\mu^2$ is an eigenvalue exactly when $\psi_\mu(x_0) = K_\mu(2\pi) = 0$. By (i) such an eigenvalue is real and at least $4\pi^2$, so $\mu = \pm i\nu$ with $\nu \geq 2\pi$; since $P(\theta) = K_{i\theta/2}(2\pi)$, the zeros of P are $\theta = \pm 2\nu$. (iii) Two eigenfunctions with the same eigenvalue both vanish at x_0, so they are proportional. For (18.1.2), put $\phi = \partial_\nu \psi_{i\nu}$; differentiating $\psi_{i\nu}'' = (e^{2x}-\nu^2)\psi_{i\nu}$ in ν gives $\phi'' = (e^{2x}-\nu^2)\phi - 2\nu\psi_{i\nu}$, so that $(\psi_{i\nu}'\phi - \psi_{i\nu}\phi')' = 2\nu\psi_{i\nu}^2$. Both functions and their derivatives decay like $\exp(-e^x)$, and integrating from y to ∞ gives the identity. The function $\psi_{i\nu}$ is real valued and not identically zero, so the right side is negative. At $y = x_0$ and $\nu = \nu_n$ the second term on the left vanishes, so $\partial_\nu K_{i\nu}(2\pi) \neq 0$ at ν_n, which is the simplicity of the zero. □

The proposition is Pólya's theorem, obtained from selfadjointness: A_P is an explicit Hilbert Pólya operator for P. Biane also writes the equation in Dirac form [32, Section 3.4]. With $D = \frac{d}{dx}$ and $\mathcal{A} = D + \frac12 + e^x$, whose formal adjoint is $\mathcal{A}^\dagger = -D + \frac12 + e^x$, the system $\mathcal{A}f = \gamma g$, $\mathcal{A}^\dagger g = \gamma f$ gives $\mathcal{A}^\dagger\mathcal{A} f = \gamma^2 f$, and a direct computation gives $\mathcal{A}^\dagger\mathcal{A} = A_P + \frac14$. With the Dirichlet condition on f the Dirac eigenvalues are therefore $\gamma = \pm(\nu_n^2 + \frac14)^{1/2}$, not $\pm\nu_n$ (Biane writes the shift as $\gamma^2 - \frac14$). In the variable $u = e^x$ the system reads $(u\frac{d}{du} + \frac12 + u)f = \gamma g$, $(-u\frac{d}{du} + \frac12 + u)g = \gamma f$, and since $u\frac{d}{du} + \frac12$ is i times the symmetrised dilation operator $\frac12(up + pu)$, $p = -i\frac{d}{du}$, it is the Hamiltonian xp of Berry and Keating [30] perturbed by the potential u, as Biane remarks.

The zeros of P obey an exact quantisation rule. Let I_μ be the modified Bessel function of the first kind, and for real ν put

$$\vartheta_P(\nu) = \operatorname{Im}\log\Gamma(1+i\nu) - \nu\log\pi - \arg {}_0F_1(;1+i\nu;\pi^2), \tag{18.1.3}$$

with the branches continuous in ν and equal to zero at $\nu = 0$, where ${}_0F_1(;1;\pi^2) = I_0(2\pi) > 0$. Write $N_P(T)$ for the number of zeros of P in $(0,T]$ and

$$x(T) = \frac{1}{\pi}\vartheta_P\big(\tfrac{T}{2}\big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}.$$

Proposition 18.1.8. *(i) For $\nu > 0$, $I_{i\nu}(2\pi) \neq 0$ and*

$$K_{i\nu}(2\pi) = \frac{\pi\,|I_{i\nu}(2\pi)|\,\sin\vartheta_P(\nu)}{\sinh\pi\nu}.$$

(ii) $\vartheta_P(\nu) > 0$ for $\nu > 0$, $\vartheta_P(\nu_n) = n\pi$ for every n, and the number of zeros of $\nu \mapsto K_{i\nu}(2\pi)$ in $(0,\nu)$ is $\lceil\vartheta_P(\nu)/\pi\rceil - 1$. Equivalently, $N_P(T) = \lfloor x(T)\rfloor$ for every $T > 0$ that is not a zero of P.

(iii) As $\nu \to \infty$,

$$\vartheta_P(\nu) = \nu \log \frac{\nu}{\pi} - \nu + \frac{\pi}{4} + \frac{\pi^2 - \frac{1}{12}}{\nu} + O(\nu^{-3}),$$

so that $x(T) = N_0(T) + \frac{1}{4} + \big(2\pi - \frac{1}{6\pi}\big)\frac{1}{T} + O(T^{-3})$ *as* $T \to \infty$.

Proof. (i) For real $r > 0$, $I_{-i\nu}(r) = \overline{I_{i\nu}(r)}$, and $K_\mu = \pi(I_{-\mu} - I_\mu)/(2\sin\mu\pi)$ gives $K_{i\nu}(r) = -\pi \operatorname{Im} I_{i\nu}(r)/\sinh\pi\nu$. The series $I_{i\nu}(r) = (r/2)^{i\nu}\,{}_0F_1(;1 + i\nu; r^2/4)/\Gamma(1+i\nu)$ shows that, for $r > 0$,

$$\vartheta(\nu, r) = \operatorname{Im}\log\Gamma(1+i\nu) - \nu\log\frac{r}{2} - \arg {}_0F_1\big(;1+i\nu;\tfrac{r^2}{4}\big)$$

satisfies $I_{i\nu}(r) = |I_{i\nu}(r)|e^{-i\vartheta(\nu,r)}$, and $\vartheta_P(\nu) = \vartheta(\nu, 2\pi)$; this gives the formula. If $I_{i\nu}(r)$ vanished, so would $I_{-i\nu}(r)$, against the Wronskian $I_\mu I'_{-\mu} - I'_\mu I_{-\mu} = -2\sin(\mu\pi)/(\pi r)$ [256, (10.28.1)], which is not zero for $\mu = i\nu$. So ${}_0F_1$ does not vanish on $[0,\infty)\times(0,\infty)$, and ϑ is defined there as a continuous function with $\vartheta(0,r) = 0$. (ii) The same Wronskian, written for $I = I_{i\nu}$ and $\bar I = I_{-i\nu}$, gives $\operatorname{Im}(I'/I) = \sinh(\pi\nu)/(\pi r|I|^2)$, hence $\partial_r\vartheta(\nu,r) = -\sinh(\pi\nu)/(\pi r|I_{i\nu}(r)|^2) < 0$. As $r \to \infty$, $I_{i\nu}(r) \sim e^r/\sqrt{2\pi r}$ and $K_{i\nu}(r) \sim \sqrt{\pi/2r}\,e^{-r}$ uniformly for ν in compact sets, so $\sin\vartheta(\nu,r) = \sinh(\pi\nu)K_{i\nu}(r)/(\pi|I_{i\nu}(r)|) \to 0$ and $\cos\vartheta(\nu,r) > 0$ for large r; by continuity in ν from $\nu = 0$, $\vartheta(\nu,r) \to 0$. So for fixed $\nu > 0$ the function $r \mapsto \vartheta(\nu,r)$ decreases from $\vartheta_P(\nu)$ at $r = 2\pi$ to 0 at $r = \infty$; in particular $\vartheta_P(\nu) > 0$, and $r \mapsto K_{i\nu}(r)$ has exactly $\lceil\vartheta_P(\nu)/\pi\rceil - 1$ zeros in $(2\pi,\infty)$, one at each crossing of a multiple of π. On the other hand, by (18.1.2) at a zero y of $\psi_{i\nu}$, $\psi'_{i\nu}(y)\,\partial_\nu\psi_{i\nu}(y) < 0$, so every zero of $x \mapsto K_{i\nu}(e^x)$ moves to the right as ν increases; no zero comes in from $+\infty$, where $K_{i\nu}(r) > 0$ uniformly for ν in compact sets, and zeros cannot appear in pairs, since they are simple. So the number of zeros in (x_0,∞) is zero for small ν and increases by one exactly when ν passes some ν_n, where a zero enters through x_0: it equals the number of $\nu_n < \nu$. Comparing the two counts gives the statement, and $\vartheta_P(\nu_n) \in \pi\mathbb{Z}$ with the count passing from $n-1$ to n gives $\vartheta_P(\nu_n) = n\pi$. (iii) Stirling's series gives $\operatorname{Im}\log\Gamma(1+i\nu) = \nu\log\nu - \nu + \frac{\pi}{4} - \frac{1}{12\nu} + O(\nu^{-3})$, and with $b = 1+i\nu$, $\log {}_0F_1(;b;z) = z/b - z^2/(2b^2(b+1)) + O(b^{-3})$, so $\arg {}_0F_1(;1+i\nu;\pi^2) = \operatorname{Im}\frac{\pi^2}{1+i\nu} + O(\nu^{-3}) = -\frac{\pi^2}{\nu} + O(\nu^{-3})$. With $\nu = T/2$, $\frac{1}{\pi}(\nu\log\frac{\nu}{\pi} - \nu) = N_0(T)$. □

Writing $s_P(T) = \{x(T)\} - \frac{1}{2}$ for the sawtooth,

$$N_P(T) = N_0(T) - \frac{1}{4} - s_P(T) + \Big(2\pi - \frac{1}{6\pi}\Big)\frac{1}{T} + O(T^{-3}),$$

against the Riemann von Mangoldt formula $N(T) = \vartheta(T)/\pi + 1 + S(T)$ of Section 2.2, which with $\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \frac{1}{48T} + O(T^{-3})$ reads

$$N(T) = N_0(T) + \frac{7}{8} + S(T) + \frac{1}{48\pi T} + O(T^{-3}).$$

The two counts share the main term N_0, their mean constant terms differ by $\frac{7}{8}+\frac{1}{4}=\frac{9}{8}$, and the fluctuation of N_P is the bounded sawtooth of a quantisation rule where that of N is $S(T)$; these two facts govern the comparison of the Poisson traces in Section 13.1.5. In Table 18.1, $N(1000)-N_P(1000)=2$. The semiclassical count of A_P, which the note [284] states is exactly the main term of the Riemann von Mangoldt formula, is

$$N_{\mathrm{WKB}}(\nu)=\frac{1}{\pi}\int_{x_0}^{\log\nu}\sqrt{\nu^2-e^{2x}}\,dx=\frac{1}{\pi}\Big(\nu\operatorname{arccosh}\frac{\nu}{2\pi}-\sqrt{\nu^2-4\pi^2}\Big),$$

and $N_{\mathrm{WKB}}(T/2)=N_0(T)+\frac{2\pi}{T}+O(T^{-3})$: it agrees with the main term up to $2\pi/T$, not exactly. Its phase $\varphi(\nu)=\pi N_{\mathrm{WKB}}(\nu)+\frac{\pi}{4}$ is the phase of Debye's expansion of $K_{i\nu}(2\pi)$ for large ν, the term $\frac{\pi}{4}$ being the Maslov correction at the turning point, and by (iii) $\vartheta_P(\nu)-\varphi(\nu)=-\frac{1}{12\nu}+O(\nu^{-3})$.

The 1516 zeros $\nu_n\le 1000$ are computed from (18.1.3) with mpmath [237].

18.1.4 The index side: a GGC with the squared zeros as Thorin atoms, and its clock

Biane's Thorin measures in Theorem 18.1.2 live on the time side: they describe the hitting time in the energy variable of its Laplace transform. The Thorin programme works on the index side, with the reciprocal $1/\mathbb{E}\cosh(\sqrt{u}\,X)$ of the moment generating function (2.1.4) of the Pólya law, whose Thorin atoms would be the squared ordinates. The same hitting time carries both measures, through the Mellin transforms of Proposition 18.1.1 and (12.1.1).

Theorem 18.1.9. *For $u\ge 0$,*

$$\frac{K_0(2\pi)}{K_{\sqrt{u}}(2\pi)}=\frac{\mathbb{E}H^{1/2}}{\mathbb{E}H^{1/2+\sqrt{u}}}=\prod_{n\ge1}\Big(1+\frac{u}{\nu_n^2}\Big)^{-1},$$

with $H=H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$ as in Proposition 18.1.1. This is the Laplace transform of the GGC $T_P=\sum_{n\ge1}e_n/\nu_n^2$, with independent standard exponentials e_n, no drift and Thorin measure $\sum_n\delta_{\nu_n^2}$.

Proof. $K_\mu(2\pi)=\int_0^\infty e^{-2\pi\cosh t}\cosh(\mu t)\,dt$ is entire and even in μ, so $w\mapsto K_{\sqrt{w}}(2\pi)$ is entire. Since $|\cosh(\mu t)|\le\cosh(t\operatorname{Re}\mu)$, $|K_\mu(x)|\le K_{\operatorname{Re}\mu}(x)$, and for $\nu>0$, $K_\nu(x)=\frac12\int_0^\infty s^{\nu-1}e^{-\frac{x}{2}(s+1/s)}ds\le\frac12\Gamma(\nu)(2/x)^\nu$. Hence $\log|K_{\sqrt{w}}(2\pi)|=O(|w|^{1/2}\log|w|)$, the function has order at most $\frac12$ in w, and Hadamard's theorem gives a product of genus zero over its zeros. By Proposition 18.1.7 these are the points $w=-\nu_n^2$, each simple. So $K_{\sqrt{u}}(2\pi)=K_0(2\pi)\prod_n(1+u/\nu_n^2)$, and Proposition 18.1.1(ii) with $s=\frac12+\sqrt{u}$ gives the middle expression. Each factor $(1+u/\nu_n^2)^{-1}$ is the Laplace transform of e_n/ν_n^2, and $\sum_n\nu_n^{-2}<\infty$. □

The theorem is an instance of two results of the book. In the language of Section 20.1.3, the law U of Proposition 18.1.1(iv), with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, has the moment generating function $L_U(s) = K_{s/2}(2\pi)/K_0(2\pi)$, which by Proposition 18.1.7 vanishes only at $s = \pm 2i\nu_n$; so U is ferromagnetic, with $c_k = 2\nu_k$ and $b = 0$, and Theorem 20.1.4 gives the clock $T_U = \sum_k e_k/(4\nu_k^2) = T_P/4$, the same variable in the normalisation of the ordinate θ. And Proposition 11.1.15, applied to the even function P of order one, which is in the Laguerre Pólya class by Proposition 18.1.7, gives the GGC $P(0)/P(i\sqrt{s}) = K_0(2\pi)/K_{\sqrt{s}/2}(2\pi)$ with atoms at the squared zeros $4\nu_n^2$. The mean and variance of T_P follow from $K_\mu(x) = k_0(x) + \frac{\mu^2}{2}k_2(x) + \frac{\mu^4}{24}k_4(x) + \cdots$, with $k_{2j}(x) = \int_0^\infty t^{2j}e^{-x\cosh t}dt$:

$$\mathbb{E}T_P = \sum_n \frac{1}{\nu_n^2} = \frac{k_2(2\pi)}{2k_0(2\pi)} = 0.0741239453481,$$

$$\operatorname{Var} T_P = \sum_n \frac{1}{\nu_n^4} = 2.25103144196 \times 10^{-4},$$

and the sums over the 1516 zeros $\nu_n \le 1000$, with the tail described next, agree.

Table 18.2 compares the closed forms with the products over the zeros, for P, for Pólya's Ξ^* and for Ξ. The tail beyond the last zero V is $\int_V^\infty \log(1 + u/\nu^2)\, dN(\nu) - \frac{1}{2}\log(1 + u/V^2)$, with dN the smooth density of zeros, $\pi^{-1}\vartheta'_P$ from Proposition 18.1.8(iii) for P and $\frac{1}{2\pi}\log\frac{T}{2\pi}$ for Ξ^* and Ξ; the subtracted half term accounts for the last zero at the lower end of the integral. The truncation of the note [284], at a fixed height with the plain density from the last zero, agrees to three or four digits (last column); that is a limit of the tail, not of the identities. For Ξ the agreement is a check, not a test, since RH is verified far beyond this range.

Pólya's Ξ^* belongs to the same class. The law with density proportional to Φ^* is ferromagnetic (Section 19.1.1), so by Theorem 20.1.4, or Proposition 11.1.15, $\Xi^*(0)/\Xi^*(i\sqrt{u}) = \prod_n (1 + u/\gamma_n^{*2})^{-1}$ is the Laplace transform of a GGC with the positive Thorin measure $\sum_n \delta_{\gamma_n^{*2}}$ over the positive zeros γ_n^* of Ξ^*. In the language of Hilfssatz II this is closure under the cosine tilt: the kernel of $G(\theta+ia)+G(\theta-ia)$ is $2\cosh(au)$ times that of G, and Pólya's passage from P to Ξ^* is this closure with $a = \frac{9}{2}$ in θ, that is $\frac{9}{4}$ in the Bessel index (Section 11.1.3).

T_P can be realised pathwise. The Bessel index enters as a centrifugal term $\mu^2/2r^2$, so a Feynman Kac weight $\exp(-\frac{u}{2}\int dt/R_t^2)$ along a radial path, the clock of the Hartman Watson law and of Brownian winding, shifts the index from 0 to $\sqrt{u}$.

THEOREM 18.1.10. *Let R be the diffusion on $(0,\infty)$ with generator*

$$\mathcal{G} = \frac{1}{2}\frac{d^2}{dr^2} + \Big(\frac{1}{2r} - \frac{K_1(r)}{K_0(r)}\Big)\frac{d}{dr},$$

the radial part of planar Brownian motion killed at rate $\frac{1}{2}$ and Doob transformed

function	u	closed form	product	difference	truncated
P	0.5	0.0370339026997	$\nu_n \le 1000$	-4.6×10^{-11}	1.1×10^{-5}
	2	0.147801988307		-1.8×10^{-10}	4.5×10^{-5}
	5	0.367871890035		-4.6×10^{-10}	1.1×10^{-4}
	100	6.62276702182		-9.1×10^{-9}	2.3×10^{-3}
Ξ^*	10	0.3044209914	$\gamma_n^* \le 1000$	1.6×10^{-8}	8.8×10^{-5}
	60	1.661131318		9.4×10^{-8}	5.3×10^{-4}
	200	4.771881099		3.1×10^{-7}	1.8×10^{-3}
Ξ	10	0.2292377633	1700 zeros	1.6×10^{-7}	3.4×10^{-5}
	60	1.328022599		9.8×10^{-7}	2.0×10^{-4}
	200	4.11056511		3.3×10^{-6}	6.8×10^{-4}

Table 18.2: Index side products. The closed forms are $\log(K_{\sqrt{u}}(2\pi)/K_0(2\pi))$, $\log(\Xi^*(i\sqrt{u})/\Xi^*(0))$ and $\log(\xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2}))$; the next column says which zeros enter the product $\sum \log(1+u/z^2)$, whose difference from the closed form, after the tail of the text is added, is the fifth column. The last column is the difference for the truncation of the note: the 137 zeros $\nu_n < 150$, the 102 zeros of Ξ^* below 240 and 300 zeros of Ξ, each with the plain density from the last zero. Computed.

by K_0. For $r > b > 0$ let T_b be the hitting time of b and $\tau_r = \frac{1}{2}\int_0^{T_b} dt/R_t^2$ under $\Pr_r$. Then $T_b < \infty$ almost surely and, for $u \ge 0$,

$$\mathbb{E}e^{-u\tau_r} = \frac{K_{\sqrt{u}}(r)}{K_0(r)} \cdot \frac{K_0(b)}{K_{\sqrt{u}}(b)}.$$

As $r \to \infty$, τ_r converges in law to τ_∞ with $\mathbb{E}e^{-u\tau_\infty} = K_0(b)/K_{\sqrt{u}}(b)$. For $b = 2\pi$, $\tau_\infty \stackrel{d}{=} T_P$.

Proof. Let $\mathcal{A} = \frac{1}{2}(\frac{d^2}{dr^2} + \frac{1}{r}\frac{d}{dr} - 1)$, the generator of the radial part of planar Brownian motion killed at rate $\frac{1}{2}$. The Bessel equation gives $\mathcal{A}K_\mu = \frac{\mu^2}{2r^2}K_\mu$, in particular $\mathcal{A}K_0 = 0$, and since $K_0' = -K_1$, $\mathcal{G}f = K_0^{-1}\mathcal{A}(K_0 f)$. Hence $v = K_{\sqrt{u}}/K_0$ satisfies $\mathcal{G}v = \frac{u}{2r^2}v$. For real order $K_{\sqrt{u}} \ge K_0$, so $v \ge 1$, and $v(r) \to 1$ as $r \to \infty$ because $K_\mu(r) = \sqrt{\pi/2r}\, e^{-r}(1 + O(1/r))$ uniformly for bounded μ; so v is bounded on $[b, \infty)$. The scale density of $\mathcal{G}$ is $s'(r) = 1/(rK_0(r)^2)$, which is not integrable at ∞, so from $r > b$ the diffusion reaches b before any level $M > r$ with probability $(s(M) - s(r))/(s(M) - s(b)) \to 1$, and $T_b < \infty$ almost surely; the drift is $-1 + \frac{1}{8r^2} + O(r^{-3})$ at infinity. By Itô's formula $v(R_{t\wedge T_b}) \exp(-\frac{u}{2}\int_0^{t\wedge T_b} ds/R_s^2)$ is a bounded martingale, and optional stopping gives $v(b)\mathbb{E}_r e^{-u\tau_r} = v(r)$. The limit follows from $v(r) \to 1$ and the continuity theorem for Laplace transforms, and the last statement from Theorem 18.1.9. □

In the logarithmic coordinate $X = \log R$, run on the clock $\tau = \frac{1}{2}\int dt/R_t^2$, Itô's

formula gives

$$dX_\tau = -2e^{X_\tau}\frac{K_1(e^{X_\tau})}{K_0(e^{X_\tau})}\,d\tau + \sqrt{2}\,dW_\tau,$$

a diffusion with generator $\frac{d^2}{dx^2} + 2\frac{\psi'}{\psi}\frac{d}{dx} = -\psi^{-1}A_P\,\psi$, where $\psi(x) = K_0(e^x)$ is the ground solution of $-\psi'' + e^{2x}\psi = 0$. Its drift is about $-2e^x$, so $+\infty$ is an entrance boundary, and τ_∞ is its hitting time of $x_0 = \log 2\pi$ from $+\infty$. This is the Kent form of Section 5.2.3: by Kent's theorem [184] the hitting time from an entrance boundary is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues of the generator, here the ν_n^2 of A_P. For Pólya's function Theorem 18.1.10 thus solves the inverse problem that Sections 12.2.1 and 5.2.3 pose for ξ, exhibiting the diffusion whose hitting time has its Thorin atoms at the squared zeros, without Krein's inverse spectral theory. For Ξ one wants a half line operator with Dirichlet spectrum $\{\gamma^2/4\}$, the counterpart of $\{\nu_n^2\} = \{\theta_n^2/4\}$ over the zeros $\theta_n = 2\nu_n$ of P; the potential e^{2x} on $[\log 2\pi, \infty)$ has the right Weyl law (Proposition 18.1.8), so it is the natural reference to perturb, and the perturbation must account for the difference between Φ and Pólya's asymptote seen in Figure 18.1 (Question 18.1.17).

Corollary 18.1.11. *For real λ, with τ_∞ the clock of Theorem 18.1.10 for $b = 2\pi$ and a standard Brownian motion B independent of it,*

$$\frac{K_0(2\pi)}{K_\lambda(2\pi)} = \mathbb{E}e^{i\lambda B_{2\tau_\infty}}.$$

Proof. $\mathbb{E}e^{i\lambda B_{2\tau}} = \mathbb{E}e^{-\lambda^2\tau}$ for an independent random time τ; apply Theorem 18.1.10 with $u = \lambda^2$. □

For Ξ the corresponding identity $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\lambda) = \mathbb{E}e^{i\lambda B_{2T}}$ holds with no hypothesis, with the clock T of Theorem 8.1.1, and the variable B_{2T} is the variable $Y_{1/2}$ of Section 13.1.4; what RH adds is that T is the GGC $\sum_\gamma e_\gamma/\gamma^2$ (Proposition 2.1.1). The note [284] states the identity under RH and claims the random time exists only under RH; in fact the time exists unconditionally, and only its GGC form is equivalent to RH.

Remark 18.1.12 (The Hartman Watson law). Theorem 18.1.10 is the K counterpart of a formula of Yor [368] for the Hartman Watson law [107]: for planar Brownian motion started at distance r from the origin and conditioned on $|B_t| = \rho$, $\mathbb{E}[\exp(-\frac{u}{2}\int_0^t ds/|B_s|^2)] = I_{\sqrt{u}}(r\rho/t)/I_0(r\rho/t)$. There the endpoint is fixed and the function is I; here the path is killed at rate $\frac{1}{2}$, transformed by K_0 and stopped at a level, and the function is K. Pitman and Yor develop the general theory of such Bessel laws [273].

We simulate the clock. With $R_0 = 2\pi e^{2.5} = 76.544867$, so that X starts at $\log 2\pi + 2.5$ as in the note, the exact law of τ_{R_0} has, from the expansion of $\log v$ in u, mean 0.06763390577 and standard deviation 0.01499747823, against

0.07412394535 and 0.01500343775 for τ_∞. The simulation runs 100000 Euler paths of X with the step $\Delta = 4\times10^{-6}$ of the note and records two hitting times per path: the plain one, the first step ending below $\ell = \log 2\pi$, and one with the Brownian bridge correction, under which a step from x to x', both above ℓ, crosses with probability $\exp(-(x-\ell)(x'-\ell)/\Delta)$, the step variance being 2Δ. The plain scheme overshoots and is late by about $0.5826\sqrt{2\Delta}/|\beta|$, with $\beta = -13.5317$ the drift at the barrier and $0.5826 = -\zeta(\frac12)/\sqrt{2\pi}$ the constant of Broadie, Glasserman and Kou [49], that is by 1.218×10^{-4}. In Table 18.3 the plain mean is 2.9 standard errors late, by 1.39×10^{-4}, and the corrected one agrees with the exact mean (Figure 18.2). The exact Laplace transform is computed with Arb and the exact densities by the fixed Talbot contour [341, 1]. The note's run, 40000 plain paths with mean 0.067695 ± 0.000075, is consistent with the exact mean and the predicted lateness.

	mean	standard deviation	z of the mean	quantiles 0.01, 0.5, 0.99
exact	0.0676339	0.0149975		
plain Euler	0.0677726	0.0150584	2.91	0.04190, 0.06559, 0.11391
bridge corrected	0.0676538	0.0150221	0.42	0.04185, 0.06548, 0.11364
u	25	100	200	400
exact $\mathbb{E}e^{-u\tau_{R_0}}$	0.196345	2.54238×10^{-3}	1.65093×10^{-5}	2.91684×10^{-9}
z, bridge corrected	-0.25	0.36	0.72	0.12
z, plain Euler	-2.81	-1.84	-0.76	-0.48

Table 18.3: The clock from $R_0 = 2\pi e^{2.5}$: exact moments against 100000 Euler paths with step 4×10^{-6}, with and without the bridge correction, and the Laplace transform, with the z scores (Monte Carlo − exact)/standard error. The standard error of the mean is 4.75×10^{-5}; over $0 < u \le 400$ the largest $|z|$ of the corrected transform is 0.72. Computed.

The cosine dictionary (10.2.4) turns each Thorin measure of this section into a Lévy measure of the reciprocal transform, in the variable conjugate to the square root of the energy.

Proposition 18.1.13. *For real λ, $a > 0$, $c > 0$ and $\ell > 0$,*

$$2\log\frac{\sinh\lambda a}{\lambda a} = \int_0^\infty (1-\cos\lambda v)\,\frac{4\,dv}{v\,(e^{\pi v/a}-1)},$$

$$\ell\big(\sqrt{c^2+\lambda^2}-c\big) = \int_0^\infty (1-\cos\lambda v)\,\frac{2\ell c}{\pi}K_1(cv)\,\frac{dv}{v},$$

$$\log\frac{K_\lambda(2\pi)}{K_0(2\pi)} = \int_0^\infty 4\sin^2\Big(\frac{\lambda v}{2}\Big)\,\Theta_P(v)\,\frac{dv}{v}, \qquad \Theta_P(v) = \sum_{n\ge1} e^{-\nu_n v} = \operatorname{tr} e^{-v\sqrt{A_P}}.$$

The left sides are $-\log\mathbb{E}e^{-\lambda^2 W/2}$ for $W = J_a + J_a'$ and $W = H_\ell^{(c)}$, and $-\log\mathbb{E}e^{-\lambda^2 T_P}$.

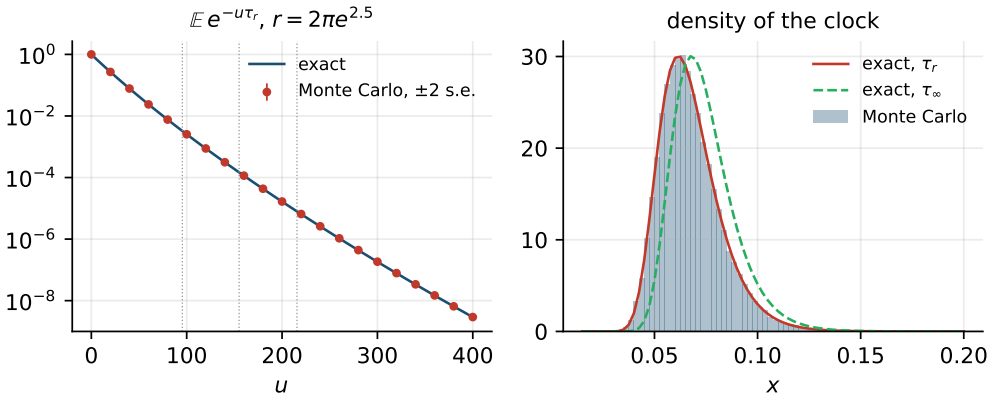


Figure 18.2: The clock from $R_0 = 2\pi e^{2.5}$. Left: $\mathbb{E}e^{-u\tau_{R_0}}$, exact and from the bridge corrected Monte Carlo with two standard errors, on a logarithmic scale; the dotted lines mark the first three Thorin atoms $\nu_1^2 = 95.4289$, $\nu_2^2 = 154.965$, $\nu_3^2 = 215.648$ of τ_∞. Right: the histogram of the corrected hitting times against the exact densities of τ_{R_0} and of τ_∞, by Talbot inversion.

Proof. By Frullani's integral, $\log(1+\lambda^2/\beta^2) = \int_0^\infty (1-\cos\lambda v)\, 2e^{-\beta v} dv/v$. For $W = J_a + J_a'$ the Thorin measure $2\sum_n \delta_{\pi^2 n^2/2a^2}$ of Section 18.1.1 gives, at $u = \lambda^2/2$, the rates $\beta = \pi n/a$ with weight 2, and $\sum_n e^{-\pi n v/a} = (e^{\pi v/a}-1)^{-1}$. For $H_\ell^{(c)}$ the density $\ell\{\pi\sqrt{2z-c^2}\}^{-1}$ on $z > \frac{c^2}{2}$ becomes, under $\beta = \sqrt{2z}$, the measure $\ell\beta\, d\beta/(\pi\sqrt{\beta^2-c^2})$ on (c,∞), and $\int_c^\infty e^{-\beta v}\beta(\beta^2-c^2)^{-1/2} d\beta = cK_1(cv)$. For T_P take Theorem 18.1.9 at $u = \lambda^2$, with rates ν_n, and use $1-\cos = 2\sin^2(\cdot/2)$. □

The Bessel clock has the Planck measure, a geometric superposition of the frequencies $\pi n/a$, and the drifted clock the Macdonald measure, absolutely continuous because its spectrum is. The dictionary acts on Thorin measures, not on Biane's Lévy measures in the time variable, which are heat traces. For Pólya's function the third identity is the exact analogue of the Lévy Khintchine form of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ in Section 13.1.4, with the Poisson trace $K_{1/2}$ replaced by the wave trace Θ_P of the selfadjoint operator A_P, which is completely monotone with no hypothesis. In the variable θ, where the zeros of P are $2\nu_n$, the trace is $K^P(y) = \Theta_P(2y) = \sum_n e^{-2\nu_n y}$, and $\log(K_{\lambda/2}(2\pi)/K_0(2\pi)) = \int_0^\infty 4\sin^2(\lambda y/2) K^P(y)\, dy/y$ has the form of the formula for ξ with K^P in place of $K_{1/2}$; Section 13.1.5 compares the two traces.

18.1.5 Rindler modes and a Klein Gordon criterion

Undoing the Wick rotation of the angle in the plane of Theorem 18.1.10 gives a Lorentzian picture in which Pólya's function is a standard object of quantum field

theory. Put

$$F(x,w)=\exp(-e^x\cosh w),\qquad \text{so that}\qquad K_\mu(e^x)=\frac{1}{2}\int_{\mathbb{R}}F(x,w)\,e^{\mu w}\,dw.$$

Since $\partial_x^2F=(e^{2x}\cosh^2 w-e^x\cosh w)F$ and $\partial_w^2F=(e^{2x}\sinh^2 w-e^x\cosh w)F$,

$$\Big(\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial w^2}-e^{2x}\Big)F=0.$$

In the coordinates $X_1=e^x\cosh w$, $X_0=e^x\sinh w$ of the Rindler wedge $X_1>|X_0|$, in which $\rho=e^x$ is the proper distance from the edge of the wedge and w the rapidity, $\partial_{X_1}^2-\partial_{X_0}^2=e^{-2x}(\partial_x^2-\partial_w^2)$, so the equation is $(\partial_{X_1}^2-\partial_{X_0}^2-1)F=0$ with $F=e^{-X_1}$: the Klein Gordon equation of unit mass in $1+1$ dimensional Minkowski space, solved by a plane wave. The index μ is the eigenvalue of the boost generator ∂_w, and $K_{i\omega}(\rho)$ is the Rindler mode of boost frequency ω [89]. The zeros of $P(\theta)=K_{i\theta/2}(2\pi)$ are therefore, with $\theta=2\omega$, the boost frequencies of the normal modes of a unit mass scalar field in the Rindler wedge with a Dirichlet mirror on the hyperbola $\rho=2\pi$, the configuration of 't Hooft's brick wall model of a black hole horizon [156]. Biane's Dirac system of Section 18.1.3 is the corresponding Dirac field, close to the Rindler space models of Sierra for the Hamiltonian xp [331].

The Wick rotation $w=i\phi$ turns the wedge into the Euclidean plane in polar coordinates (ρ,ϕ), the boost into a rotation, the Klein Gordon operator into $\Delta-1$, and the plane wave e^{-X_1} into $e^{-\rho\cos\phi}$, which is harmonic for $\frac{1}{2}(\Delta-1)$, the generator of planar Brownian motion killed at rate $\frac{1}{2}$ of Theorem 18.1.10. The boost frequency becomes the angular momentum μ, and the radial operator $\frac{1}{2}(\partial_\rho^2+\rho^{-1}\partial_\rho-\mu^2/\rho^2-1)$ carries the index as the centrifugal term that the Feynman Kac weight of the clock produces.

Theorem 18.1.14 (A Klein Gordon criterion). *Let g be even, positive and continuous with $\int g(w)e^{c|w|}dw<\infty$ for every c, and $G(t)=\int g(w)e^{itw}dw$. Suppose that there are $x_0\in\mathbb{R}$, a real continuous function V on $[x_0,\infty)$ bounded below, and $F\in C^2([x_0,\infty)\times\mathbb{R})$ such that*

(a) $(\partial_x^2-\partial_w^2-V(x))F=0$ and $F(x_0,\cdot)=g$;

(b) F and its partial derivatives of order at most two are $O(e^{-c|w|})$ for every $c>0$, uniformly for x in compact subsets of $[x_0,\infty)$, and $\Psi_\mu(x)=\int F(x,w)e^{\mu w}dw$ lies in $L^2((x_0,\infty))$ for every $\mu\in\mathbb{C}$;

(c) Ψ_μ is not identically zero for any $\mu\in\mathbb{C}$.

Then every zero of G is real, and its square is an eigenvalue of $-\frac{d^2}{dx^2}+V$ on $[x_0,\infty)$ with a Dirichlet condition at x_0.

Proof. By (b), Ψ_μ is twice differentiable under the integral sign, and two integrations by parts in w, whose boundary terms vanish, give $\Psi_\mu'' = \int(\partial_w^2 F + VF)e^{\mu w}dw = (\mu^2 + V)\Psi_\mu$. Since V is bounded below, the operator $-\frac{d^2}{dx^2} + V$ is in the limit point case at ∞, and its Dirichlet realisation at x_0 is selfadjoint. If $G(t) = 0$, then $\Psi_{it}(x_0) = G(t) = 0$ by (a), and by (b) and (c) Ψ_{it} is an eigenfunction with eigenvalue $-(it)^2 = t^2$, which is therefore real. So t is real or purely imaginary, and $t = -i\mu$ with real μ is excluded because $G(-i\mu) = \int g(w)e^{\mu w}dw > 0$. □

For Pólya's function the hypotheses hold in the rapidity variable $w = 2u$: with $g(w) = e^{-2\pi\cosh w}$, $G(t) = 2K_{it}(2\pi) = 2P(2t)$, $V(x) = e^{2x}$, $x_0 = \log 2\pi$, $F = \exp(-e^x \cosh w)$ and $\Psi_\mu(x) = 2K_\mu(e^x)$, and the theorem gives Proposition 18.1.7(ii) again (the equation for F is checked numerically). So the zeros of an even Fourier transform are real once its kernel is the boundary value of a field, decaying fast enough in the rapidity, that satisfies a Klein Gordon equation with a real, time independent potential.

For Ξ a Klein Gordon extension of Φ would prove RH (Question 18.1.18). The scaling matters. For P the eigenvalues of A_P are $\nu_n^2 = \theta_n^2/4$, and the function whose zeros they are, as a function of the spectral parameter μ at the boundary x_0, is the Jost function $\mu \mapsto \Psi_\mu(x_0) = 2K_\mu(2\pi) = 2P(-2i\mu)$. A potential asymptotic to e^{2x} at $+\infty$ has the Weyl law of Proposition 18.1.8, which matches the zeros of Ξ only in the same scaling, $g(w) = \Phi(w/2)$ and $G(t) = 2\Xi(2t)$, and then the Jost function is $\mu \mapsto 2\Xi(-2i\mu) = 2\xi(\frac{1}{2} + 2\mu)$. The choice $\mu \mapsto \xi(\frac{1}{2} + \mu)$ of the note [284] corresponds to the eigenvalues γ^2, of half the density of the Weyl law of e^{2x} (a potential growing like e^{4x} would match them). Inverse spectral theory reconstructs an operator from its eigenvalues and norming constants, a potential in the forms of Gelfand and Levitan or of Marchenko and a string in Krein's form, and under RH and the simplicity of the zeros Krein's theory realises the eigenvalues $\gamma^2/4$ (Section 5.2.3); the content of the question is that the operator be of the form $-\frac{d^2}{dx^2} + V$ and that V admit a Klein Gordon solution whose boundary value is $\Phi(w/2)$, which is a statement about the whole function ξ and not only about its zeros.

18.1.6 The de Bruijn Newman flow of P against that of Ξ

For real t put

$$P_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4} e^{-2\pi\cosh 2u}\, e^{i\theta u}\, du,$$

$$\Xi_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4}\, \Phi(u)\cos(\theta u)\, du = 8H_t(2\theta),$$

so that $P_0 = P$, $\Xi_0 = \Xi$, and t is the time of the de Bruijn Newman flow in the normalisation of Section 4.1.1. Decreasing t is the direction in which real zeros

can be lost: RH is the statement that Ξ_t keeps only real zeros down to $t = 0$, and by the theorem of Rodgers and Tao [318] it has nonreal zeros for every $t < 0$.

PROPOSITION 18.1.15. *There is a constant Λ_P with $-\infty < \Lambda_P \le 0$ such that P_t has only real zeros if and only if $t \ge \Lambda_P$.*

Proof. Let S be the set of t for which P_t has only real zeros. By Proposition 18.1.7, $0 \in S$. If $t \in S$ and $t' > t$, then $t' \in S$ by de Bruijn's theorem on the Gaussian factor $e^{\lambda u^2}$, $\lambda > 0$ [73, Theorem 13], which applies to the kernel $e^{tu^2/4}e^{-2\pi\cosh 2u}$ because it is $O(e^{-|u|^b})$ for every $b > 2$. If $t_k \in S$ decrease to t, then $P_{t_k} \to P_t$ locally uniformly on $\mathbb{C}$ by dominated convergence, with $|e^{t_k u^2/4}e^{i\theta u}| \le e^{t_1 u^2/4}e^{|\operatorname{Im}\theta||u|}$, and $P_t(0) > 0$, so by Hurwitz's theorem every zero of P_t is a limit of real zeros and $t \in S$. Hence $S = [\Lambda_P, \infty)$ with $\Lambda_P \in [-\infty, 0]$. Suppose $\Lambda_P = -\infty$. Then, for the even probability measure ρ with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, the Fourier transform of $e^{\lambda u^2}\rho(du)$ has only real zeros for every real λ, and by Newman's theorem [242], [244, Theorem 10], ρ is either $\frac{1}{2}(\delta_{u_0} + \delta_{-u_0})$ or has a density $Ku^{2m}\exp(-\alpha u^4 - \beta u^2)\prod_j(1 + u^2/a_j^2)e^{-u^2/a_j^2}$, with $K > 0$, $m \ge 0$ an integer, $a_j > 0$, $\sum_j a_j^{-4} < \infty$, and $\alpha > 0$, or $\alpha = 0$ and $\beta + \sum_j a_j^{-2} > 0$. The inequality $(1 + y)e^{-y} \ge e^{-y^2/2}$ for $y \ge 0$ bounds such a density below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$, while $e^{-2\pi\cosh 2u}$ decays faster than e^{-Cu^4} for every C. This contradiction gives $\Lambda_P > -\infty$. □

The proof is Newman's proof that $\Lambda > -\infty$ [242], [244, Theorem 9], which uses only decay faster than e^{-Cu^4} for every C; it applies verbatim to Φ^* and to every even positive kernel with that decay whose Fourier transform has only real zeros. It refutes the conjecture $\Lambda_P = -\infty$ of the note [284], which rested on the computation below.

We compute both transforms by the trapezoid rule with step $1/250$ on $[0, 2]$ at 90 digits. The kernels are analytic in $|\operatorname{Im} u| < \pi/4$ (that of P_t is entire) and below 10^{-68} beyond $u = 2$, and at $t = 0$ the sums reproduce Arb's $K_{i\theta/2}(2\pi)$ and mpmath's Ξ up to $\theta = 90$ with relative errors below 1.4×10^{-45}. The real zeros in $(0, 90)$ are located by sign changes on a grid of step 0.05 and refined, for $t = 0, -0.5, \dots, -20$ and $t = -21, \dots, -60$. The real zeros of Ξ_t in the window fall from 25 at $t = 0$ to 2 at $t = -40$, and they are lost in pairs, neighbouring zeros colliding and leaving the axis: the first collisions in the window, as t decreases, are those of the census of Section 4.1.3, beginning with $(\gamma_{24}, \gamma_{25})$ at $t = -1.144$ and height 88.47. For P_t nothing of the kind occurs down to $t = -60$: between consecutive grid times every real zero increases, and the count falls only as zeros leave through $\theta = 90$ (Figure 18.3).

Since sign changes see only real zeros, we also count all zeros in the box $0.5 \le \operatorname{Re}\theta \le 60.3$, $|\operatorname{Im}\theta| \le 3$ by the argument principle, on a contour refined until the argument changes by less than 0.5 between neighbours. For P_t the box holds 12,

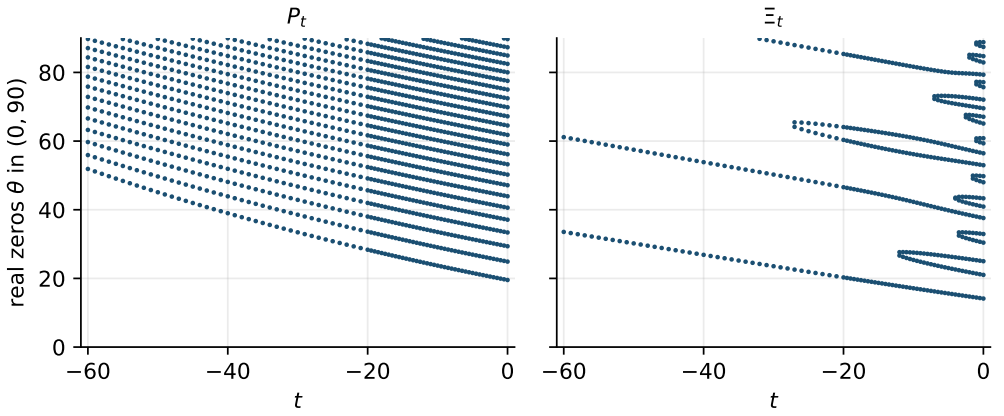


Figure 18.3: The real zeros in $(0, 90)$ of P_t (left) and of Ξ_t (right) for $-60 \le t \le 0$, on a grid (step 0.5 above $t = -20$, step 1 below). The zeros of P_t drift upward and leave the window one at a time; those of Ξ_t collide in pairs.

11, 8 and 3 zeros at $t = 0, -10, -30$ and -60, all of them real. For Ξ_t it holds 13, 12 and 12 zeros at $t = 0, -2$ and -4, of which 13, 10 and 8 are real. The three collisions below height 60 listed in the note [284] are located by Newton's method for the double zero $\Xi_t = \partial_\theta \Xi_t = 0$ in (θ, t):

$$\begin{aligned} (\gamma_9, \gamma_{10}) &: \ t_c = -1.877909073, \ \theta_c = 49.66423465; \\ (\gamma_4, \gamma_5) &: \ t_c = -3.954792524, \ \theta_c = 33.09873381; \\ (\gamma_2, \gamma_3) &: \ t_c = -12.49451313, \ \theta_c = 27.3212277. \end{aligned}$$

The first agrees with the census of Section 4.1.3 (-1.87791 at height 49.664); the other two lie below its range $[-2.05, 0]$. They are not the only collisions below height 60: on the grid of step 0.5 in t a fourth occurs before $t = -12.5$: the zeros γ_7 and γ_8 are at 43.263 and 43.522 at $t = -4$ and have left the axis at $t = -4.5$, and the isolated pair estimate of Section 4.1.1 puts their collision at about $t = -4.03$. In the variable θ a simple zero moves by

$$\frac{d\theta_k}{dt} = \frac{\partial_\theta^2 F_t(\theta_k)}{4\, \partial_\theta F_t(\theta_k)} = \frac{1}{2} \sum_{j \neq k} \frac{1}{\theta_k - \theta_j},$$

for $F_t = P_t$ and $F_t = \Xi_t$, the equation of motion of Section 4.1.1 with $z = 2\theta$. As t decreases zeros attract; a pair collides when its mutual attraction overcomes the difference of the pulls of the other zeros on its two members.

The mean motion can be predicted. For $\theta > 4\pi$ the integral $P(\theta)$ is dominated by the conjugate saddle points $u_\pm = \pm \frac{x}{2} + \frac{i\pi}{4}$ of $i\theta u - 2\pi \cosh 2u$, with $x = \operatorname{arccosh}(\theta/4\pi)$, where the exponent is $-\frac{\pi\theta}{4} \pm i(\frac{1}{2}\theta x - 2\pi \sinh x)$. So $P(\theta)$ is $e^{-\pi\theta/4}$

times a slowly varying amplitude times a cosine of phase $\varphi(\theta) = \frac{1}{2}\theta x - 2\pi \sinh x$ plus a constant; since $\varphi'(\theta) = \frac{x}{2}$, the zeros have density $\frac{x}{2\pi} = \frac{1}{2\pi}\log\frac{\theta}{2\pi} + O(\theta^{-2})$. To first order in t the factor $e^{tu^2/4}$ adds to the phase the imaginary part $\frac{\pi t}{16}x$ of $\frac{t}{4}u_+^2$, so the number of zeros below a fixed height changes at the rate $\frac{x}{16}$ per unit of t, and each zero moves with

$$\frac{d\theta}{dt} \approx -\frac{x/16}{x/2\pi} = -\frac{\pi}{8},$$

whatever its height. The equation of motion agrees: with the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$ on both half lines, $\frac{1}{2}\sum_{j\neq k}(\theta_k - \theta_j)^{-1}$ becomes $\frac{1}{4\pi}$ times the principal value integral of $\log|s|/(\theta_k - s)$, which is $-\frac{\pi^2}{2}$ for $\theta_k > 0$. Over $-0.5 \le t \le 0$ the 24 real zeros of P in the window rise at mean speeds decreasing with height from 0.3990 (the first) to 0.3949 (the last), towards $\pi/8 = 0.3927$, while the 25 zeros of Ξ rise at speeds between 0.0040, for γ_{25}, held back by γ_{24} closing on it, and 0.7350, for γ_{19}, closing on γ_{20}, with mean 0.3647. The zeros of P move almost as a rigid block; those of Ξ fluctuate about the same mean motion, and the fast and slow members of close pairs are the ones that collide.

In the language of Lee and Yang [206] (Section 11.1.3, Proposition 11.1.16), the single spin law with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, the ferromagnetic law U of Section 18.1.4, has the Lee and Yang property, since its moment generating function $K_{h/2}(2\pi)/K_0(2\pi)$ vanishes only at imaginary h. Newman's theorem says that this property survives every Gaussian factor e^{-bu^2}, $b > 0$, only for two point laws and for the densities of the proof of Proposition 18.1.15, which include the φ^4 laws $e^{-au^4-bu^2}$ that Simon and Griffiths realised as limits of Ising ferromagnets [332]. The law with density proportional to $e^{-2\pi\cosh 2u-bu^2}$ has the property exactly when its Fourier transform, a multiple of P_{-4b}, has only real zeros, that is when $b \le -\Lambda_P/4$. So the Simon Griffiths type construction for $e^{-2\pi\cosh 2u-bu^2}$ at every real b proposed in the note cannot exist.

18.1.7 Pólya's function and Ξ side by side

Table 18.4 collects the comparison; the Wick rotation of Section 18.1.2 relates the two entries of each of the second, third and fourth rows. Against the note [284] three rows are corrected: the index side clock of Ξ exists with no hypothesis, Λ_P is finite, and the flow computation covers a window, not the whole flow.

The statements about Ξ in the right column and their relatives, Proposition 2.1.1, Corollary 13.1.4, Proposition 11.1.15 and the GGC form of the clock T discussed after Corollary 18.1.11, are proved by Hadamard factorisation, Bernstein's theorem and integral identities. No property of the primes enters, and none could, since each is an equivalence between two statements about the zeros of one entire function: they reorganise RH (Section 25.2) and do not approach it. The

	Pólya's $P(\theta) = K_{i\theta/2}(2\pi)$	Riemann's $\Xi(\theta)$
kernel	$e^{-2\pi\cosh 2u}$, (18.1.1)	$\Phi(u)$, (2.1.3)
hitting time	$H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$, Brownian motion with drift (Proposition 18.1.1)	$\frac{\pi}{2}S_2$, three dimensional Bessel process, (12.1.1)
symmetric space	$\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$	$SU(2) \cong S^3$
Thorin measure of the hitting time (energy)	continuous: σ_2 for $H_1^{(1)}$	discrete: $2\pi\sigma_1$ for $\pi^2 S_2$ (Theorem 18.1.2)
index side Thorin measure (angular momentum)	$\sum_n \delta_{\nu_n^2}$, a theorem (Theorem 18.1.9)	$\sum_\gamma \delta_{\gamma^2}$ if and only if RH (Proposition 2.1.1)
index side clock	τ_∞, a hitting time of an explicit diffusion (Theorem 18.1.10)	a random time T (Theorem 8.1.1); no diffusion is known (Question 18.1.17)
trace of the sine squared form	$\Theta_P = \operatorname{tr} e^{-v\sqrt{A_P}}$, completely monotone (Proposition 18.1.13)	$K_{1/2}$, completely monotone if and only if RH (Corollary 13.1.4)
operator for the zeros	$-\frac{d^2}{dx^2} + e^{2x}$ on $[\log 2\pi, \infty)$ (Proposition 18.1.7)	not known
Klein Gordon extension	$\exp(-e^x \cosh w)$ (Theorem 18.1.14)	not known (Question 18.1.18)
de Bruijn Newman constant	$-\infty < \Lambda_P \le 0$ (Proposition 18.1.15)	$\Lambda \ge 0$ [318]; RH if and only if $\Lambda = 0$
real zeros in $(0, 90)$ for $-60 \le t \le 0$	no collision; mean speeds 0.395 to 0.399 on $[-0.5, 0]$	collisions in pairs from $t = -1.144$; mean speeds 0.004 to 0.735 on $[-0.5, 0]$

Table 18.4: Pólya's function and Ξ. Apart from the computations of the last row, the left column is a list of theorems; in the right column the entries that concern the zeros are equivalent to RH or open.

entries on the left that concern the zeros all go through the selfadjoint operator A_P, exactly what the right column lacks.

Part (a) of the following is in [284]; part (b) is the strengthening that Newman's theorem gives.

PROPOSITION 18.1.16. *Let $\mathcal{P}$ be a property of even, positive, integrable kernels such that*

(i) if $\mathcal{P}(g)$ holds, then $\int g(u)e^{i\theta u}du$ has only real zeros, and

(ii) there is $\varepsilon > 0$ such that $\mathcal{P}(g)$ implies $\mathcal{P}(e^{-\varepsilon u^2/4}g)$.

(a) If (ii) holds for $g = \Phi$, then $\mathcal{P}(\Phi)$ is false.

(b) If (ii) holds for all g with the same ε, then $\mathcal{P}(g)$ is false for every kernel g that is $O(e^{-|u|^b})$ for some $b > 2$ and satisfies $g(u)e^{Cu^4} \to 0$ as $|u| \to \infty$ for every C; in particular $\mathcal{P}$ fails for Φ, for Φ^* and for $e^{-2\pi\cosh 2u}$.

Proof. (a) If $\mathcal{P}(\Phi)$ held, then by (ii) and (i) the transform of $e^{-\varepsilon u^2/4}\Phi$, which is $\Xi_{-\varepsilon}$ of Section 18.1.6, would have only real zeros, so $\Lambda \le -\varepsilon < 0$, against the theorem of Rodgers and Tao [318]. (b) If $\mathcal{P}(g)$ held, then by induction $\mathcal{P}(e^{-k\varepsilon u^2/4}g)$ would hold for $k = 0, 1, 2, \dots$, so the transform of $e^{-k\varepsilon u^2/4}g$ would have only real zeros for each k. This kernel is again $O(e^{-|u|^b})$, so by de Bruijn's theorem [73, Theorem 13] the transform of $e^{tu^2/4}g$ would have only real zeros for every $t \ge -k\varepsilon$, and hence for every real t. By Newman's theorem, as in the proof of Proposition 18.1.15, $g/\int g$ would be a density of the form given there, which is bounded below by $Ku^{2m}\exp(-C'u^4 - \beta u^2)$ for some C', against the decay of g. □

Part (b), which uses only de Bruijn's theorem and Newman's theorem of 1976, shows that the obstruction has nothing to do with Φ lying on the boundary of the class of kernels with real zeros: a property that survives a fixed damping for all kernels fails for P too, although the zeros of P are real for a structural reason, the selfadjointness of A_P. So the reading of the note, that P tolerates every Gaussian damping and has slack where Ξ has none, becomes quantitative: P has the finite slack $|\Lambda_P|$, of unknown size, and Ξ has none if RH holds. The Lee and Yang property of a single spin law is preserved by the ferromagnetic factor e^{bu^2}, $b > 0$, which runs the flow forward, but by (b) not by a fixed damping outside Newman's class. A Lee and Yang representation of the law with density proportional to Φ, which is equivalent to RH (Proposition 11.1.16), or a Simon Griffiths type representation of it, is therefore not excluded by the proposition, contrary to the list of excluded arguments in the note.

Three questions raised by the note are not equivalences and can be studied without assuming what is to be proved. The first two ask for an object, a diffusion or a potential, and part (b) does not apply to them, because the existence of such an object is not preserved by a fixed Gaussian damping of every kernel: for P it is lost at every $t < \Lambda_P$. The third concerns an explicit kernel with no arithmetic in it and measures the slack of Pólya's function. A fourth concerns the interpolation between the two clocks of Section 18.1.2.

Question 18.1.17. Is there a diffusion on a half line, with an entrance boundary, whose hitting time of a level from the entrance boundary has the Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + 2\sqrt{u})$, the transform of the clock T of Theorem 8.1.1 scaled by 4, as τ_∞ has the transform $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ in Theorem 18.1.10? By Kent's theorem [184] such a hitting time is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues $\lambda_n > 0$ of the generator, so a positive answer implies RH; under RH and the simplicity of the zeros, Krein's inverse spectral theory gives a generalised diffusion with the eigenvalues $\gamma^2/4$ (Section 5.2.3). The question asks for one constructed without

RH, for example as a perturbation of the diffusion X of Section 18.1.4, whose potential e^{2x} has the right Weyl law.

Question 18.1.18. Does $g(w) = \Phi(w/2)$ admit a Klein Gordon extension in the sense of Theorem 18.1.14, with a real potential V asymptotic to e^{2x} at $+\infty$? By the theorem a positive answer implies RH. Under RH the question asks whether $\mu \mapsto 2\xi(\frac{1}{2} + 2\mu)$ is, up to normalisation, the Jost function of such a potential whose Klein Gordon solution has the boundary value g. A first numerical step is to reconstruct a potential from the first few thousand ordinates and a choice of norming constants by the method of Gelfand and Levitan or of Marchenko, and to test whether it is stable under truncation and close to e^{2x}.

Question 18.1.19. What is the de Bruijn Newman constant Λ_P of Pólya's function, and that of Ξ^*, which is also finite and at most 0 by the proof of Proposition 18.1.15? At what height do the first nonreal zeros of P_t appear as t decreases through Λ_P, and do they appear, as for Ξ_t, through the collision of two real zeros?

Question 18.1.20. Is there a one parameter family of laws joining the drifted first passage time and the sum of two Bessel hitting times, each with a reciprocal symmetry that encodes a functional equation, as $\tau \mapsto 1/\tau$ does for H in Proposition 18.1.1(i) and Jacobi's transformation does for S_2, along which the index side Thorin measure can be followed? The curvature family of Proposition 18.1.6 is an Esscher tilt and breaks the symmetry. Positivity of the index side Thorin measure at the Bessel end is RH.

18.2 Exercises

On Section 5.2.

18.1. Assuming that the zeros are on the critical line and simple, show that the residues of the clock transform at $u = -\gamma_k^2$ are $R_k = 2i\gamma_k \xi(\frac{1}{2})/\xi'(\frac{1}{2} + i\gamma_k)$ and that they alternate in sign. (They are the coefficients c_n of Theorem 9.3.22.)

18.2. Explain why the spectral expansion of the density diverges at $t = 0$ while the Talbot inversion does not, and compare the abscissa of convergence in Theorem 9.3.22.

18.3. Show that the hazard of a finite sum of exponentials with rates $\lambda_1 < \lambda_2 < \cdots$ tends to λ_1 and that $h(t) - \lambda_1$ decays like $e^{-(\lambda_2 - \lambda_1)t}$.

18.4. Sample Williams' variable with 10^5 draws and estimate $\Xi(5)$; compare the standard error with Table 5.3.

18.5. Assume RH, let Θ_N be the cutoff of Section 5.2.3, and assume that $S(\Theta_N) = 0$, as for the values of N used there. Show that $\log G(w) - \log G_N^{\mathrm{W}}(w)$ tends

to $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\,d\gamma$ as $w \to \infty$. Then, using Stirling's formula for $\log \xi(\frac{1}{2}+x)$ as $x \to +\infty$ and $\int_0^\infty \log(1+x^2/t^2)\log\frac{t}{2\pi}\,dt = \pi x \log\frac{x}{2\pi} - \pi x$, show that

$$\ell_N = 2\sum_{k\le N} \log\gamma_k - \log\xi(\tfrac{1}{2}) + \tfrac{1}{4}\log\tfrac{\pi}{2} - \frac{1}{\pi}\int_0^{\Theta_N} \log t\,\log\frac{t}{2\pi}\,dt + \frac{2}{\pi}\int_{\Theta_N}^{\infty}\Big(\vartheta'(t) - \frac{1}{2}\log\frac{t}{2\pi}\Big)\log t\,dt,$$

the terms in $\log x$ cancelling because $\vartheta(\Theta_N) = (N-1)\pi$. So ℓ_N is determined by $\gamma_1,\dots,\gamma_N$ and $\xi(\frac{1}{2})$.

On Section 3.1.

18.6. Show that on the parabola $u = \frac{1}{2}(-q^2 + 2icq)$ one has $\operatorname{Re}\sqrt{c^2+2u} = c$, and hence that every mixture of inverse Gaussian laws with drift c has transform bounded by one there.

18.7. Derive the small q expansion $|G(u)|^2 = 1 - q^2(\mathbb{E}T - c^2\operatorname{Var}T) + O(q^3)$ on the parabola.

18.8. Show that the hazard of the first passage time of Brownian motion across the line $\alpha+\beta t$, with $\alpha > 0$ and $\beta < 0$, which has the inverse Gaussian law with drift $|\beta|$, tends to $\beta^2/2$ with correction $3/(2t)$.

18.9. Implement the tangent boundary ODE with a different starting time and check that the asymptotic slope is unchanged.

18.10. Show that the density f of the clock is continuous and positive on $(0,\infty)$. (The Laplace transform of T equals $1/G$ on $\operatorname{Re} u \ge 0$, by analytic continuation from $(0,\infty)$ and continuity, so the characteristic function is $1/G(-iy)$; Stirling's formula and the bound $|\zeta(s)| \ge 1/\zeta(\operatorname{Re} s)$ for $\operatorname{Re} s > 1$, applied on the rays $\frac{1}{2} + re^{\pm i\pi/4}$, show that it decays faster than any power of $|y|$. Next, $\log G(u) = o(u)$ as $u \to \infty$, so $P(T < t) > 0$ for every $t > 0$; $1/G$ has poles, so the law of T has unbounded support; and the law is unimodal by Corollary 14.1.2, so f itself, which is the derivative of the distribution function at every point, is nondecreasing up to a mode and nonincreasing after it.)

On Section 23.6.

18.11. Derive formula (23.6.1) from the derivative representation of λ_n.

18.12. Show that $\lambda_1 = \sum_\rho \rho^{-1} = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, and that under RH $\lambda_1 = \sum_{\gamma>0}(\gamma^2+\frac{1}{4})^{-1}$.

18.13. Show that under RH $\lambda_n = \sum_{\gamma>0} 4\sin^2(n \arctan \frac{1}{2\gamma})$, and that the terms are of order n^2/γ^2 for $\gamma \gg n$.

18.14. For a hypothetical zero $\rho = \frac{1}{2} + d + i\gamma$ with $d > 0$, show that $|1 - 1/(1 - \bar{\rho})| > 1$, and estimate the index n at which its contribution would make λ_n negative.

On Section 4.1.

18.15. Show that H_t satisfies $\partial_t H_t = -\partial_z^2 H_t$ and derive the equation of motion of a simple zero.

18.16. For an isolated pair at distance D_0, solve $dD/dt = 4/D$ and find the collision time.

18.17. Modify the Pólya evaluator to compute H_t and reproduce Table 4.1.

18.18. Explain why the collision time of a pair of zeros gives numerical evidence for, but not a proof of, a lower bound on Λ.

On Section 15.1.7.

18.19. Show that $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$ is even, using $\eta(i/x) = \sqrt{x}\,\eta(ix)$, and deduce the functional equation $\xi(s, \chi_4) = \xi(1 - s, \chi_4)$ from $\xi(\frac{1}{2} + s, \chi_4) = \int_{\mathbb{R}} K_4(u) e^{su}\, du$.

18.20. Show that for real $\sigma > 0$ the series $\sum_{n\geq 0}(-1)^n(2n+1)^{-\sigma}$ lies between $1 - 3^{-\sigma}$ and 1, and deduce that $L(s, \chi_4)$ has no zeros on $(0, 1)$ without using the kernel.

18.21. For $0 < \varepsilon < \delta$ show that $\pi(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$ integrates over the interval $|x| < \sqrt{\delta^2 - \varepsilon^2}$, where it is negative, to $\pi - 4\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)}$, which lies in $(-\pi, 0)$ and tends to 0 as $\varepsilon/\delta \to 0$; this is the bound on the fall of the argument after Corollary 15.1.32.

18.22. Show that $g(c, u) = 2c(3u^2 - c^2)/(c^2 + u^2)^3$, the second derivative of $c/(c^2 + u^2)$ in u, has $\partial_c g(c, u) = 6(u^4 - 6c^2u^2 + c^4)/(c^2 + u^2)^4$, which is positive exactly when $|c| < (\sqrt{2} - 1)|u|$ or $|c| > (\sqrt{2} + 1)|u|$, and deduce that $g(a + \delta, u) + g(a - \delta, u) > 0$ for $a, \delta > 0$ with $a + \delta < (\sqrt{2} - 1)|u|$, the step for the zeros off the line in the proof of Theorem 12.1.18(ii).

On Section 18.1.

18.23. Show that $1/H$ has the law of H size biased by H, for $H = H^{\frac{(\sqrt{2\pi})}{\sqrt{2\pi}}}$ with the density of Proposition 18.1.1(i), and deduce $\mathbb{E}H^s = \mathbb{E}H^{1-s}$ without computing the Mellin transform.

18.24. Show that $F(x,w) = \exp(-e^x \cosh w)$ satisfies $(\partial_x^2 - \partial_w^2 - e^{2x})F = 0$, and that in the coordinates $X_1 = e^x \cosh w$, $X_0 = e^x \sinh w$ one has $\partial_x^2 - \partial_w^2 = e^{2x}(\partial_{X_1}^2 - \partial_{X_0}^2)$, so that $F = e^{-X_1}$ solves the Klein Gordon equation of unit mass.

18.25. Show that $(1+y)e^{-y} \geq e^{-y^2/2}$ for $y \geq 0$, and deduce that each density of Newman's theorem in the proof of Proposition 18.1.15 is bounded below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$.

18.26. Show that the principal value integral of $\log|s|/(1-s)$ over $\mathbb{R}$, taken with symmetric limits at $\pm\infty$, is $-\frac{\pi^2}{2}$, and deduce the mean speed $-\frac{\pi}{8}$ of the zeros under the flow in Section 18.1.6 from the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$.

18.27. For $0 < y < \pi$ show, as claimed in Subsection 13.1.5, that $\gamma^n e^{-\gamma y}$, for $\gamma \leq n/\pi$, is at most $\exp(n(\log\frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its maximum over $\gamma > 0$, and evaluate the exponent for $n = 10^6$ at $y = 1$ and at $y = 3$.

Chapter 19

The Pólya density as a maximum and as an extreme value law

This chapter is based on [287]. It starts from one observation: the law behind the Pólya kernel of Ξ is built from the maximum M_4 of independent exponential variables of Section 15.1.4.2, in the sense that e^{2Y} is a sum of two independent copies of M_4 (Proposition 19.2.1), and it asks which parts of that observation survive the passage to the Thorin measure of the clock, with Laplace transform $1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{\lambda})$. The unconditional statements about the clock one level below the GGC property are those of Section 8.1. The last sections read the Pólya density as an extreme value law: each term of its theta series is a combination of two Gumbel kernels, which gives explicit approximations, a Hankel test from kernel moments and a reading of the clock as the hitting time of a Krein string.

Table 19.1 sets the trigonometric side against the Riemann side.

19.1 Notation

The Laplace variable is λ, the u of Section 2.1, and the transform $1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{\lambda})$ of the clock is written $\varphi_{1/2}$ when the basepoint family is in view.

19.2 The Pólya density as a maximum order statistic

By Section 15.1.4.2, $\vartheta_4(t) = \sum_{n\in\mathbb{Z}}(-1)^n e^{-\pi n^2 t}$ is the distribution function of the maximum M_4 of independent exponential variables, one of each rate πn, $n \geq 1$, and one of each rate $\pi(2m-1)$, $m \geq 1$: by independence and Jacobi's triple product, $\Pr(M_4 \leq t) = \prod_{n\geq 1}(1-e^{-\pi n t})\prod_{m\geq 1}(1-e^{-\pi(2m-1)t}) = \vartheta_4(t)$. Since the Kolmogorov distribution function is $k \mapsto \vartheta_4(2k^2/\pi)$, πM_4 is twice the square of a variable with the Kolmogorov law. Also $\mathbb{E}e^{-\lambda M_4} = \sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda} = \prod_{n\geq 1}(1+\lambda/(\pi n^2))^{-1}$, so that $M_4 \stackrel{d}{=} \sum_{n\geq 1} E_n/(\pi n^2)$ with independent standard exponential variables E_n, a GGC with Thorin measure $\sum_n \delta_{\pi n^2}$; and $\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\,\xi(s)/(s-1)$ for every complex s, with the right side continued at $s=1$

	Trigonometric side	Riemann side
entire function	$\sin(\sqrt{\pi}\theta)/(\sqrt{\pi}\theta)$	$\Xi(\theta)/\Xi(0)$
real zeros	$\theta = \pm\sqrt{\pi}\, n$	$\theta = \pm\gamma_k$ (all real iff RH)
space component	$\sqrt{\pi/2}\, v$, with v uniform on $(-1,1)$	$X/\sqrt{2}$, with X the Pólya variable
time component	M_4	T, the clock
Laplace transform	$\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda}$	$1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{\lambda})$
exists as a law	yes	yes (Theorem 8.1.1)
ID and SD	yes	yes (Theorem 8.1.1)
x times the Lévy density	$\sum_{n\geq 1} e^{-\pi n^2 x}$	$W(x) = \sum_k e^{-a_k x}$
GGC, Thorin measure	$\sum_{n\geq 1} \delta_{\pi n^2}$	$\sum_k \delta_{\gamma_k^2}$ if and only if RH (Proposition 2.1.1)
maximum of exponentials	yes, with rates πn and $\pi(2m-1)$	no (Proposition 19.9.1)
Gnedenko norming	$a_n = 1/\pi$	$a_n = 1/\gamma_1^2$

Table 19.1: The trigonometric side, where everything is classical, against the Riemann side. The variable θ is that of Section 2.1; ID and SD abbreviate infinitely divisible and self decomposable, and the Wald couple of the third and fourth rows is that of Theorem 19.4.2 and Remark 19.5.1.

by $\log 2$, a case of the dictionary of Biane, Pitman and Yor [33]. Inside the open critical strip the zeros of $s \mapsto \mathbb{E}M_4^{s/2}$ are therefore exactly the nontrivial zeros of ζ, since $1 - 2^{1-s}$ vanishes only on $\operatorname{Re} s = 1$. The same law is a maximum of exponentials with linear rates πn and a sum of exponentials with quadratic rates πn^2; this coincidence is the Jacobi triple product, and it has no counterpart for the clock (Section 19.9).

Up to a constant factor, e^Y is the maximum of a normalised Brownian excursion [33].

Proposition 19.2.1. *Let M_4 and M_4' be independent, and let Y be the variable of Section 2.1, $\mathbb{E}e^{sY} = 2\xi(s)$. Then:*

(i) $e^{2Y} = \frac{\pi}{2} S_2 \stackrel{d}{=} M_4 + M_4'$.

(ii) If f is the density of $e^Y = \sqrt{M_4 + M_4'}$, then the Pólya kernel (2.1.3) *is* $\Phi(u) = \frac{1}{2}\, e^{3u/2} f(e^u)$, *and*

$$f(y) = 4 \sum_{n\geq 1} \big(2\pi^2 n^4 y^3 - 3\pi n^2 y\big) e^{-\pi n^2 y^2}.$$

Proof. (i) By definition $e^{2Y} = \frac{\pi}{2} S_2 = \sum_{k\geq 1} (e_k + e_k')/(\pi k^2)$, and since $M_4 \stackrel{d}{=} \sum_{n\geq 1} E_n/(\pi n^2)$, a sum of two independent copies of M_4 has the same law.

(ii) Let μ be the law of Y. By Section 2.1, $2\Xi(\theta) = 2\xi(\frac{1}{2} + i\theta) = \mathbb{E}e^{(\frac{1}{2}+i\theta)Y} = \int_{\mathbb{R}} e^{u/2}e^{i\theta u}\mu(du)$, and the measure $e^{u/2}\mu(du)$ is finite because $\mathbb{E}e^{Y/2} = 2\xi(\frac{1}{2})$. Also $\Xi(\theta) = \int_{\mathbb{R}} \Phi(u)e^{i\theta u}\,du$ by (2.1.3), Φ being even, so the finite measures $\frac{1}{2}e^{u/2}\mu(du)$ and $\Phi(u)\,du$ are equal: $\mu(du) = 2e^{-u/2}\Phi(u)\,du$. The density of e^Y is $f(y) = y^{-1} \cdot 2y^{-1/2}\Phi(\log y)$, which is the first claim, and substituting (2.1.3) gives the series. □

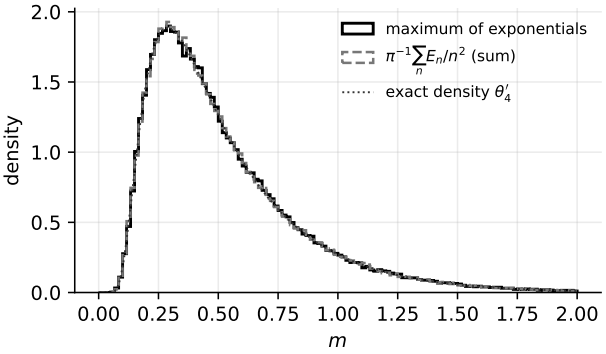


Figure 19.1: Histograms of M_4 simulated as a maximum of exponentials and as a weighted sum of exponentials.

19.3 Khinchin: the zeta distribution inside the tilted maximum

Khinchin [187] proved that for $\sigma > 1$, $\zeta(\sigma + it)/\zeta(\sigma)$ is an infinitely divisible characteristic function; Gnedenko and Kolmogorov [93] reproduce it as an example, and Lin and Hu [212] call the law the Riemann zeta distribution. If $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, the zeta distribution of Sections 10.1.6 and 11.1.2.3, then $\mathbb{E}N_\sigma^{-it} = \zeta(\sigma + it)/\zeta(\sigma)$, and the Euler product gives $N_\sigma = \prod_p p^{G_p}$ with independent geometric variables G_p, $\Pr(G_p = k) = (1 - p^{-\sigma})p^{-k\sigma}$. The work continuing their line stays at $\sigma > 1$, away from the zeros [212, 102, 239]. The completed ratio $\xi(\sigma + it)/\xi(\sigma)$ is the characteristic function of the variable V_σ of Section 12.1.1, whose reflection $-V_\sigma$ is Nakamura's complete Riemann zeta distribution [238], and V_σ is quasi infinitely divisible but not infinitely divisible for $\sigma > 1$ (Proposition 14.1.3). The Pólya density, on the other hand, is not infinitely divisible, because its characteristic function $\Xi(\theta)/\Xi(0)$ has real zeros.

Theorem 19.3.1. *Fix $\sigma > 1$ and let $\widetilde{M}_4$ have the tilted law $\Pr(\widetilde{M}_4 \in dm) \propto m^{\sigma/2}\Pr(M_4 \in dm)$. Let K, N_σ and $\Gamma_{1+\sigma/2}$ be independent of each other and of $\widetilde{M}_4$, with $\Pr(K = k) = (1 - a)a^k$ for $k \geq 0$ and $a = 2^{1-\sigma}$, with $N_\sigma = \prod_p p^{G_p}$ zeta distributed with parameter σ, and with $\Gamma_{1+\sigma/2}$ gamma distributed with shape*

$1+\sigma/2$ *and scale* 1. *Then*

$$\widetilde{M}_4\, 4^{-K} \stackrel{d}{=} \frac{\Gamma_{1+\sigma/2}}{\pi N_\sigma^2} \stackrel{d}{=} \frac{\Gamma_{1+\sigma/2}}{\pi} \prod_p p^{-2G_p}.$$

Proof. By Section 15.1.4.2, $\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\,\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ for every complex s, because $2\xi(s)/(s-1) = 2\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$. With $s=\sigma+it$, $\mathbb{E}\widetilde{M}_4^{it/2}$ is the ratio of this at s and at σ, that is

$$\mathbb{E}\widetilde{M}_4^{it/2} = \frac{1-ae^{-it\log 2}}{1-a}\cdot \pi^{-it/2} \cdot \frac{\Gamma(1+\frac{\sigma+it}{2})}{\Gamma(1+\frac{\sigma}{2})}\cdot\frac{\zeta(\sigma+it)}{\zeta(\sigma)}.$$

The first factor is the reciprocal of $\mathbb{E}e^{-itK\log 2} = \mathbb{E}(4^{-K})^{it/2}$, and the others are $\pi^{-it/2}$, $\mathbb{E}\Gamma_{1+\sigma/2}^{it/2}$ and $\mathbb{E}N_\sigma^{-it}$. Hence $\mathbb{E}(\widetilde{M}_4\,4^{-K})^{it/2} = \mathbb{E}\big(\Gamma_{1+\sigma/2}/(\pi N_\sigma^2)\big)^{it/2}$ for every real t, which is the first identity in law; the second is the Euler product. □

19.4 The Wald couple of the maximum

Definition 19.4.1. A pair (τ,ς) of independent variables with $\tau\ge 0$ is a Wald couple if $\mathbb{E}e^{s\varsigma}<\infty$ and $\mathbb{E}e^{s\varsigma}\,\mathbb{E}e^{-s^2\tau/2}=1$ for all real s. Equivalently, with Z standard normal and independent of τ, the characteristic functions of ς and of $\sqrt{\tau}Z$ form a van Dantzig pair in the sense of Section 2.1: $\mathbb{E}e^{-\theta^2\tau/2} = 1/\mathbb{E}e^{-\theta\varsigma}$ for real θ.

The name comes from Wald's identity $\mathbb{E}\exp(sB_\tau - s^2\tau/2)=1$: if a stopping time τ of Brownian motion B has B_τ independent of τ and the stopped martingales $\exp(sB_{t\wedge\tau}-s^2(t\wedge\tau)/2)$ are uniformly integrable, then (τ,B_τ) is a Wald couple.

Theorem 19.4.2. *Let $\upsilon,\upsilon_1,\upsilon_2$ be uniform on $(-1,1)$, with υ_1 and υ_2 independent, let M_4' be an independent copy of M_4, and let $c=\sqrt{\pi/2}$. Then $(M_4, c\upsilon)$ and $(M_4+M_4', c(\upsilon_1+\upsilon_2))$ are Wald couples. The associated van Dantzig pairs are $(\sin ct/(ct),\ ct/\sinh ct)$ and their squares, and $\sqrt{M_4}\,Z$ has the logistic law with characteristic function $ct/\sinh ct$.*

Proof. $\mathbb{E}e^{-s^2M_4/2} = cs/\sinh(cs)$ by Section 15.1.4.2, with $\lambda=s^2/2$, and $\mathbb{E}e^{sc\upsilon} = \sinh(cs)/(cs)$. Squaring gives the second couple. □

The pair $(\sin\theta/\theta, \theta/\sinh\theta)$ is the classical van Dantzig pair of Section 2.1 [228], at $\theta=ct$, and the language of Wald couples is that of the first author's [281]. The logistic statement is the representation of the logistic law as a normal scale mixture with Kolmogorov mixing of Andrews and Mallows [3] and Stefanski [336], read through the maximum.

19.5 The clock exists and is infinitely divisible

Theorem 8.1.1 proves unconditionally, from the verification of RH below height 1000, that $W > 0$ and $W' < 0$ on $(0, \infty)$, so that the clock is the Laplace transform of an infinitely divisible, self decomposable law with Lévy density $W(x)/x$, unimodal by Yamazato's theorem [366] (Section 14.1), with mean $\mathbb{E}T = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2})) = 0.0231050$ (Section 2.1). The ladder of signs $(-1)^k W^{(k)} > 0$ is Theorem 8.1.6 for $0 \le k \le 10^{17}$, Theorem 8.1.9 for every order on $(0, 1.29]$ and Theorem 8.1.13 for every $t > 0$ and every $k \le 1.16 \times 10^{25}$.

Remark 19.5.1. $(T, X/\sqrt{2})$ is a Wald couple. Indeed T exists by Theorem 8.1.1, $\mathbb{E}e^{sX/\sqrt{2}} = \xi(\frac{1}{2} + s/\sqrt{2})/\xi(\frac{1}{2})$ and $\mathbb{E}e^{-s^2T/2} = \xi(\frac{1}{2})/\xi(\frac{1}{2} + |s|/\sqrt{2})$, and $\xi(\frac{1}{2} + w)$ is even in w. This is the van Dantzig pair $(\Xi(\theta)/\Xi(0),\ \xi(\frac{1}{2})/\xi(\frac{1}{2}+\theta))$ of Theorem 8.1.1 read at $\theta = s/\sqrt{2}$.

19.6 A Stieltjes criterion

Section 4.1 reads the GGC property through the Stieltjes transform of the Thorin measure. For the clock, $G'/G = -\frac{d}{d\lambda}\log(1/G)$ takes the place of the function f_H there, whether or not RH holds, and by Section 2.1 $(G'/G)(\lambda) = \sum_k (\lambda + a_k)^{-1} = \int_0^\infty e^{-\lambda x} W(x)\, dx$. The quantity of this section is

$$r_\eta(x) = \frac{1}{\pi}\ \mathrm{Im}\,\frac{G'}{G}(-x - i\eta), \qquad x \in \mathbb{R},\ \eta > 0,$$

defined wherever $-x - i\eta$ is not a pole of G'/G. If the clock is a GGC with Thorin measure U, then r_η is the Poisson smoothing of U at distance η below the cut, as in (4.1.2).

Theorem 19.6.1. *The following are equivalent.*

(a) The Riemann Hypothesis.

(b) T is a GGC; then $T \overset{d}{=} \sum_{\gamma>0} e_\gamma/\gamma^2$ and its Thorin measure is $U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}$, the ordinates counted with multiplicity.

(c) G'/G is a Stieltjes function.

(d) $r_\eta(x) \ge 0$ for all $x \in \mathbb{R}$ and $\eta > 0$ for which $-x - i\eta$ is not a pole of G'/G.

(e) W is completely monotone.

In that case r_η is the Poisson smoothing of U_{RH} at scale η, and $U_{\mathrm{RH}}((0, x]) = N(\sqrt{x})$.

Items (a), (b), (c) and (e) are known: (a) if and only if (b) is Proposition 2.1.1, (b) implies (c) by the Stieltjes form of the GGC property in Section 4.1, and (b) if and only if (e) is the characterisation of a GGC by the complete monotonicity of k, x times its Lévy density [39], with $k = W$ for the clock (Sections 2.1 and 4.1).

Proof. It remains to prove that (c) implies (d) and that (d) implies (a). (c) implies (d): a Stieltjes function is analytic on $\mathbb{C} \setminus (-\infty, 0]$ and maps the upper half plane into the closed lower half plane, and $(G'/G)(\bar\lambda) = \overline{(G'/G)(\lambda)}$, so $\operatorname{Im}(G'/G)(-x - i\eta) \geq 0$ for $\eta > 0$. (d) implies (a): if RH fails, then some zero has $\beta \neq \frac12$, and since the zeros are symmetric under $\rho \mapsto \bar\rho$ and $\rho \mapsto 1 - \rho$ there is a zero $\rho = \beta + i\gamma$ with $\gamma > 0$ and $\beta > \frac12$; Lemma 19.6.2 gives $r_\eta(x_\rho) \to -\infty$ as $\eta \uparrow b$, so $r_\eta(x_\rho) < 0$ for $\eta < b$ close to b and (d) fails. The last statement is (4.1.2) and the counting formula of Section 4.1 for U_{RH}. □

Lemma 19.6.2 (Detection). *Let $\rho = \beta + i\gamma$ be a zero of multiplicity m with $\beta > \frac12$ and $\gamma > 0$, put $\delta = \beta - \frac12$, and write $a_\rho = x_\rho - ib$ with $x_\rho = \gamma^2 - \delta^2$ and $b = 2\delta\gamma$, so that $\lambda_\rho = -a_\rho = -x_\rho + ib$. Then $r_\eta(x_\rho) \to -\infty$ as $\eta \uparrow b$. Moreover, for $0 < \eta < b$ such that $-x_\rho - i\eta$ is not a pole of G'/G,*

$$\pi r_\eta(x_\rho) = \eta\Big(A_\eta - \frac{2m}{b^2 - \eta^2}\Big),$$

where

$$A_\eta = \frac{1}{\eta} \sum_{\lambda_{\rho'} \neq \lambda_\rho, \bar\lambda_\rho} \operatorname{Im}\big(-x_\rho - i\eta - \lambda_{\rho'}\big)^{-1},$$

the sum running over the zeros ρ' with $\operatorname{Im}\rho' > 0$, counted with multiplicity, whose $\lambda_{\rho'}$ is neither λ_ρ nor $\bar\lambda_\rho$. Hence $r_\eta(x_\rho) < 0$ at every $\eta \in (0, b)$ with $A_\eta < 2m/b^2$. If all the other zeros lie on the line, $A_\eta \to A_0 = \sum_{\gamma'} (\gamma'^2 - x_\rho)^{-2}$ as $\eta \downarrow 0$.

Proof. At $x = x_\rho$ we have $-x_\rho - i\eta - \lambda_\rho = -i(\eta + b)$ and $-x_\rho - i\eta - \bar\lambda_\rho = -i(\eta - b)$. Since $\operatorname{Im}(-iy)^{-1} = 1/y$, each of the m copies of the pair $\lambda_\rho, \bar\lambda_\rho$ contributes $1/(\eta + b) + 1/(\eta - b) = -2\eta/(b^2 - \eta^2)$, which gives the display. This tends to $-\infty$ as $\eta \uparrow b$, while all other terms stay bounded, because the other poles of G'/G lie at a positive distance from $-x_\rho - ib$. If $A_\eta < 2m/b^2$ then $A_\eta < 2m/(b^2 - \eta^2)$ for every $\eta \in (0, b)$. For a zero ρ' on the line, $\lambda_{\rho'} = -\gamma'^2$ and $\operatorname{Im}(-x_\rho - i\eta + \gamma'^2)^{-1} = \eta/((\gamma'^2 - x_\rho)^2 + \eta^2)$, which gives the limit A_0. □

For $x \neq x_\rho$ the same computation shows that the m copies of the pair contribute $m\,[P_{b+\eta}(x - x_\rho) - P_{b-\eta}(x - x_\rho)]$ to $r_\eta(x)$ for $0 < \eta < b$, with P_h the Poisson kernel of width h: a positive kernel of width $b + \eta$ and a negative kernel of width $b - \eta$, which becomes an atom of mass $-m$ at x_ρ as $\eta \uparrow b$. A zero with $b < \sqrt{x_\rho}$ is therefore seen by a search over the window $\{0 < \eta < \sqrt{x},\ x \leq X\}$ as soon as $x_\rho \leq X$. The condition $b < \sqrt{x_\rho}$ is $\delta < \gamma/\sqrt{4\gamma^2 + 1}$, that is $\beta < 1 - 1/(16\gamma^2)$ up to terms of higher order, and it follows from the zero free region $\beta < 1 - c/\log\gamma$

for γ large [348, Theorem 3.8]; it does not follow from $b < \gamma$. A proof that $r_\eta \geq 0$ on the whole window certifies that no such zero has $x_\rho \leq X$; a search on a grid is evidence only. Figure 19.2 tests the detection on a model in which G'/G is built from the first 60 zeros: with all zeros on the line, r_η is positive on the grid of the figure; after moving γ_5 to $\beta = 0.6$ (so $b = 6.59$) a negative region appears at $x \approx 1084.7$ for η up to 6.5, and $r_\eta(x_\rho) \to -\infty$ as $\eta \uparrow b$, as the lemma predicts.

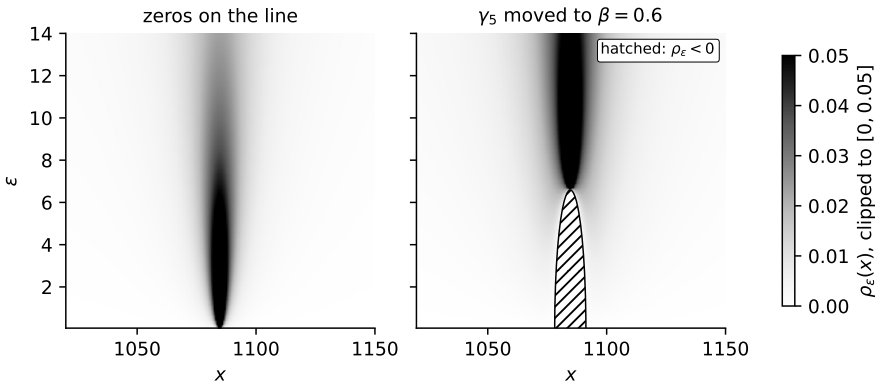


Figure 19.2: The smoothed Thorin density $r_\eta(x)$ in the (x, η) plane near $\gamma_5^2 = 1084.7$. Left: zeros on the line. Right: γ_5 moved off the line; the off line pair is an atom of negative mass at (x_ρ, b).

19.7 Is the clock a weak limit of GGC laws?

The class of GGC laws is closed under weak limits [39, Theorem 3.1.5] (Section 9.1.1), and with Proposition 2.1.1 this gives a sharp answer: T is a weak limit of GGC laws if and only if RH holds (Corollary 9.1.2).

The de Bruijn Newman flow of Section 4.1 gives a second family, and Theorem 17.1.16 places RH at the end of a path of GGC laws. Let $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$ be the kernel in the rescaled variable of Rodgers and Tao [318], as in Theorem 17.1.16, let $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be the de Bruijn Newman deformation of $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$, with Λ the de Bruijn Newman constant in this normalisation, and let V_t be the variable with density proportional to $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$, the Gaussian tilt of the Pólya variable $V_0 = X/2$, so that $\mathbb{E}e^{sV_t} = H_t(is)/H_t(0)$ for real s. Whenever a law T_t makes (T_t, V_t) a Wald couple (Definition 19.4.1), $\mathbb{E}e^{-\lambda T_t} = H_t(0)/H_t(i\sqrt{2\lambda}) = R_t(\lambda/2)$, where $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ is the transform of the flow clock of Section 4.1 and $R_0 = 1/G$. Thus T_t is half the flow clock, $T_0 = T/2$, and $(T/2, X/2)$ is the couple of Remark 19.5.1 rescaled by $\sqrt{2}$.

By Theorem 17.1.16, for every $t \geq \Lambda$ the transform R_t is that of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t) > 0$ are the positive zeros of the even function H_t (each pair $\pm z$ gives one atom). Hence for $t \geq \Lambda$ the law T_t exists

and is a GGC with Thorin measure $\sum_k \delta_{z_k(t)^2/2}$, and since $\Lambda \le 0.2$ [277, 311] this holds unconditionally for $t \ge 0.2$. As $t \downarrow 0$, $H_t \to H_0$ locally uniformly, so $R_t(\lambda/2) \to 1/G(\lambda/2)$ pointwise, which is the transform of $T/2$, and $T_t \to T/2$ in law along any sequence $t_n \downarrow 0$ for which T_{t_n} exists. With the bound $\Lambda \ge 0$ of Rodgers and Tao [318], Theorem 17.1.16 gives that RH holds if and only if T_t is a GGC for every $t > 0$, if and only if $\Lambda = 0$.

Two cautions. Unconditionally T_t is known to exist only for $t \ge 0.2$ and at $t = 0$ (Theorem 8.1.1), so the convergence $T_t \to T/2$ concerns the range where T_t exists: every $t \ge 0$ under RH, while without RH existence for $0 < t < 0.2$ is open (Problem 19.3). And the Thorin atoms $z_k(t)^2/2$ move with the zeros of H_t only for $t \ge \Lambda$, where T_t is a GGC; Table 4.2 gives the zeros of $H_t(2\theta)$, $\theta = z/2$.

Definition 19.7.1 (Laplace distance). For laws P and Q on $[0,\infty)$ with Laplace transforms L_P and L_Q, the Laplace distance is $d_L(P,Q) = \sup_{\lambda \ge 0} |L_P(\lambda) - L_Q(\lambda)|$.

Proposition 19.7.2. *Let $H = 3 \times 10^{12}$, let $N = N(H)$ be the number of zeros ρ_k with $0 < \gamma_k \le H$, and let $T^{[N]} = \sum_{k \le N} e_k/\gamma_k^2$ with independent standard exponential variables e_k, the GGC with transform $1/G_N$ of Section 9.1.1. With a_k as in Section 2.1, put $\varepsilon_H = \sum_{k>N} 1/|a_k|$ and, for $K \le N$, $S_K = \sup_{\lambda \ge 0} \lambda/G_K(\lambda)$. Then*

$$d_L\big(T, T^{[N]}\big) \le S_K\, \varepsilon_H, \qquad \varepsilon_H < 1.4798 \times 10^{-12}, \qquad S_1 = \gamma_1^2, \qquad S_{1700} < 16.85, \tag{19.7.1}$$

so that $d_L(T, T^{[N]}) < 2.5 \times 10^{-11}$, while the first factor alone gives $\gamma_1^2 \varepsilon_H < 2.96 \times 10^{-10}$. The proof uses only the verification of RH up to height H [277], and makes no assumption on the zeros above H.

Proof. By (2.1.1), $1/G = \prod_k f_k$ with $f_k(\lambda) = (1 + \lambda/a_k)^{-1}$, and $\operatorname{Re} a_k = \gamma_k^2 - (\beta_k - \frac12)^2 > 0$ for every zero, so for $\lambda \ge 0$ we have $|1 + \lambda/a_k| \ge \operatorname{Re}(1 + \lambda/a_k) \ge 1$, hence $|f_k| \le 1$ and $|f_k - 1| = |\lambda/a_k|\,|f_k| \le \lambda/|a_k|$. For complex numbers of modulus at most one, $|\prod_j f_j - 1| \le \sum_j |f_j - 1|$. The zeros with $k \le N$ lie on the line, so $\prod_{k \le N} f_k = 1/G_N$ is the transform of $T^{[N]}$, and its factors lie in $(0,1]$; leaving out all but the first K of them can only increase the product, so $1/G_N \le 1/G_K$. Therefore

$$\Big|\frac{1}{G(\lambda)} - \frac{1}{G_N(\lambda)}\Big| = \frac{1}{G_N(\lambda)}\,\Big|\prod_{k>N} f_k(\lambda) - 1\Big| \le \frac{\lambda}{G_K(\lambda)}\,\varepsilon_H \le S_K\,\varepsilon_H.$$

For $K = 1$, $\lambda/G_1(\lambda) = \lambda/(1 + \lambda/\gamma_1^2)$ increases to γ_1^2, a supremum that is not attained. For $K \ge 2$, $\frac{d}{d\lambda}\log(\lambda/G_K(\lambda)) = -\phi_K(\lambda)/\lambda$ with $\phi_K(\lambda) = \sum_{k \le K} \lambda/(\lambda + \gamma_k^2) - 1$, which increases strictly from -1 to $K - 1$, so λ/G_K increases up to the root λ^* of ϕ_K and decreases after it. For $K = 1700$ the root is $\lambda^* = 47.362$, and evaluating ϕ_{1700} and λ/G_{1700} in ball arithmetic with the first 1700 zeros, on both sides of λ^*, gives $S_{1700} < 16.8497$ (code/maxwald_weak.py).

For ε_H, $|a_k| = |\rho_k - \frac{1}{2}|^2 \geq \gamma_k^2$, and partial summation gives $\varepsilon_H \leq \sum_{\gamma_k > H} \gamma_k^{-2} = 2\int_H^\infty (N(t) - N(H))\, t^{-3}\, dt$. Write $N(t) = N_0(t) + E(t)$ with $N_0(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e}$; by [108, Corollary 1.2], $|E(t)| \leq 0.1038\log t + 0.2573\log\log t + 9.3675$ for $t \geq e$. The main term integrates to exactly $(\log\frac{H}{2\pi} + 1)/(2\pi H) = 1.47970 \times 10^{-12}$, and the bound for E, used at t and at H, adds less than 10^{-23}. □

19.8 The sine squared lemma and the basepoint family in the λ chart

The sine squared lemma (2.1.9) [279, Lemma 8.1] turns the Lévy measure of $\log\xi$ at a basepoint α into the Thorin measure of the reciprocal there. For $\alpha > 1$ its input is the signed canonical measure of [282, Theorem 8.1]; for $\frac{1}{2} < \alpha \leq 1$ there is no such Lévy form, and the family is defined by the Hadamard product. The family is that of the shifted clocks T_α with transform $\varphi_\alpha(\lambda) = \xi(\alpha)/\xi(\alpha + \sqrt{\lambda})$, $T_{1/2} = T$. For every $\alpha > \frac{1}{2}$ put $u_\alpha(\lambda) = \mathrm{Re}(\xi'/\xi)(\alpha + i\sqrt{\lambda})/(2\pi\sqrt{\lambda})$, the Thorin density of T_α when every zero has $\beta < \alpha$ (Theorem 2.1.2) and in every case the density of U_α of (2.1.6) in the chart $\lambda = \theta^2$, a sum over the zeros of Poisson kernels (Proposition 2.1.3), and for $\alpha > 1$ it has the arithmetic form (1.3.1) with $\theta = \sqrt{\lambda}$ (drift, archimedean, pole and prime pieces), which Table 2.2 compares with the Pólya route. Since $\varphi_\alpha \to 1/G$ pointwise as $\alpha \downarrow \frac{1}{2}$ and T_α exists for every $\alpha > \frac{1}{2}$, $T_\alpha \to T$ in law, with no hypothesis (Section 9.1.1).

In the chart θ a zero $\rho = \beta + i\gamma$ with $\gamma > 0$, paired with $\bar\rho$, contributes $P_\delta(\theta - \gamma) + P_\delta(\theta + \gamma)$ with $\delta = \alpha - \beta$ (Proposition 2.1.3; $P_{-a} = -P_a$ as there). The map $\theta \mapsto \theta^2 = \lambda$ folds the kernel at $-\gamma$ onto the one at γ and brings in the Jacobian $1/(2\sqrt{\lambda})$.

Proposition 19.8.1. *Let $\alpha > \frac{1}{2}$, let $\rho = \beta + i\gamma$ be a zero with $\gamma > 0$, and put $\delta = \alpha - \beta \neq 0$ and $c = \alpha - \rho = \delta - i\gamma$. The term of ρ in $u_\alpha = \sum_{\gamma>0} u_\rho$, where the zeros ρ and $\bar\rho$ are paired, is*

$$u_\rho(\lambda) = \frac{P_\delta(\sqrt{\lambda} - \gamma) + P_\delta(\sqrt{\lambda} + \gamma)}{2\sqrt{\lambda}}, \qquad \lambda > 0, \tag{19.8.1}$$

and it satisfies the following.

(a) $u_\rho(\lambda) = \kappa_\rho(\lambda)\, C_\rho(\lambda) = \delta\,(\lambda + |c|^2)/(\pi\sqrt{\lambda}\,|c^2 + \lambda|^2)$, *where*

$$\kappa_\rho(\lambda) = \frac{\lambda + \gamma^2 + \delta^2}{2\gamma\sqrt{\lambda}},$$
$$C_\rho(\lambda) = \frac{1}{\pi}\,\frac{2\delta\gamma}{(\lambda - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2}.$$

For $\delta > 0$, C_ρ is the Cauchy density centred at $\gamma^2 - \delta^2$ with half width $2\delta\gamma$.

(b) $\int_0^\infty u_\rho(\lambda)\, d\lambda = \operatorname{sign}\delta$, *so for $\delta > 0$ the term has unit mass.*

(c) $\kappa_\rho(\lambda) \ge \sqrt{1+\delta^2/\gamma^2} > 1$ *for every* $\lambda > 0$, $\kappa_\rho(\gamma^2-\delta^2) = (1-\delta^2/\gamma^2)^{-1/2}$ *when* $|\delta| < \gamma$, *and for* $K \ge 1$ *and* $|\delta| \le \gamma/(2K)$,

$$|\kappa_\rho(\lambda) - 1| \le \Big(\frac{K^2}{2} + 1\Big)\frac{\delta^2}{\gamma^2} \qquad \text{for } |\lambda - \gamma^2| \le K|\delta|\gamma.$$

Hence near its peak the term is a Lorentzian of full width $4|\delta|\gamma$ *at half height, with the sign of* δ, *up to a relative error* $O(\delta^2/\gamma^2)$; *away from the peak it is not:* $u_\rho(\lambda) \sim \delta/(\pi(\gamma^2+\delta^2)\sqrt{\lambda})$ *as* $\lambda \downarrow 0$ *and* $u_\rho(\lambda) \sim \delta/(\pi\lambda^{3/2})$ *as* $\lambda \to \infty$, *while* C_ρ *is bounded at* 0 *and* $O(\lambda^{-2})$ *at infinity.*

Proof. Write $\theta = \sqrt{\lambda}$ and $A = \lambda + \gamma^2 + \delta^2$. Since $(\theta \mp \gamma)^2 + \delta^2 = A \mp 2\gamma\theta$,

$$P_\delta(\theta-\gamma) + P_\delta(\theta+\gamma) = \frac{\delta}{\pi}\Big[\frac{1}{A-2\gamma\theta} + \frac{1}{A+2\gamma\theta}\Big] = \frac{\delta}{\pi}\,\frac{2A}{A^2-4\gamma^2\lambda},$$

and $A^2 - 4\gamma^2\lambda = (\lambda - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2$, since both sides equal $\lambda^2 + 2(\delta^2 - \gamma^2)\lambda + (\gamma^2+\delta^2)^2$. For $c = \delta - i\gamma$ the right side is also $|c^2+\lambda|^2$, because $c^2 + \lambda = (\lambda+\delta^2-\gamma^2) - 2i\delta\gamma$, and $|c|^2 = \gamma^2+\delta^2$. Dividing by 2θ gives $u_\rho = \delta A/(\pi\sqrt{\lambda}\,|c^2+\lambda|^2)$, which is the second form in (a), and writing $\delta A/\sqrt{\lambda} = \frac{A}{2\gamma\sqrt{\lambda}} \cdot 2\delta\gamma$ gives $\kappa_\rho C_\rho$. For (b) substitute $\lambda = \theta^2$: $\int_0^\infty u_\rho\,d\lambda = \int_0^\infty [P_\delta(\theta-\gamma) + P_\delta(\theta+\gamma)]\,d\theta = \int_{-\gamma}^\infty P_\delta + \int_\gamma^\infty P_\delta$, which is 1 for $\delta > 0$ because P_δ is an even probability density, and -1 for $\delta < 0$ because $P_{-a} = -P_a$. For (c), $\lambda + (\gamma^2+\delta^2) \ge 2\sqrt{\lambda}\sqrt{\gamma^2+\delta^2}$ gives the lower bound, and $A = 2\gamma^2$ at $\lambda = \gamma^2 - \delta^2$ gives the value there. For the window put $\lambda = \gamma^2(1+\varepsilon)$, so that $|\varepsilon| \le K|\delta|/\gamma \le \frac12$ and

$$\kappa_\rho(\lambda) = \frac{1+\varepsilon/2}{\sqrt{1+\varepsilon}} + \frac{\delta^2}{2\gamma^2}\,\frac{1}{\sqrt{1+\varepsilon}}.$$

The first term is 1 at $\varepsilon = 0$ and has derivative $\varepsilon/(4(1+\varepsilon)^{3/2})$, so it differs from 1 by at most $\varepsilon^2/2$ for $|\varepsilon| \le \frac12$, and the second term is at most δ^2/γ^2. Take $\delta > 0$, the case $\delta < 0$ being the same with a change of sign. The half height points of C_ρ are at $\gamma^2 - \delta^2 \pm 2\delta\gamma$, inside the window with $K = 3$ (for $\delta \le \gamma/6$), where $|(\log C_\rho)'| = 1/(2\delta\gamma)$; a relative perturbation η of the density therefore moves them by $O(\eta\,\delta\gamma)$, which gives the width $4\delta\gamma\,(1 + O(\delta^2/\gamma^2))$. The behaviour at 0 and at infinity is read off (a). □

Remark 19.8.2. The leading correction to the width is the factor $1+\eta$ with $\eta = \delta^2/(2\gamma^2)$: with $x = (\lambda - \gamma^2 + \delta^2)/(2\delta\gamma)$ the expansion above gives $\kappa_\rho = 1 + \eta(1+x^2) + O(\delta^3/\gamma^3)$, so near the peak $u_\rho \approx C_\rho + \eta/(2\pi\delta\gamma)$, with a relative error $O(\delta^3/\gamma^3)$, is a Lorentzian on a flat background of relative height η, whose half height points are at $x = \pm(1+\eta)$ to first order.

Numerically the three expressions for u_α agree up to their truncations: the line form $\mathrm{Re}(\xi'/\xi)(\alpha + i\sqrt{\lambda})/(2\pi\sqrt{\lambda})$ computed through the Pólya density, the zero form $\sum_\rho u_\rho$ of Proposition 19.8.1, and the prime form (1.3.1). At $\alpha = 2$ the three forms agree to 4 to 5 significant digits at $\lambda = 100, 250, 500, 1000$; at $\alpha = 1.5$ the prime form with $n \le N = 2 \times 10^5$ agrees to 1 to 2 digits, the gap being the truncated prime tail $\sum_{n>N} \Lambda(n) n^{-\alpha} \approx N^{1-\alpha}/(\alpha - 1)$ as for Table 2.2. The minimum of u_α on $[1, 2500]$ is positive for $\alpha = 0.55, 0.6, 0.75, 1, 1.5$ and lies near $\lambda = 59$ for each; on a window where every zero is known to lie on the line this checks the computation and does not test RH.

19.9 The maximum representation does not transfer

The distribution function of T can be computed by residues, as ϑ_4 is for M_4 (Section 15.1.4). Along a vertical line $\mathrm{Re}\,\lambda = -L$ that contains no pole of $1/G$, the point $\frac{1}{2} + \sqrt{\lambda}$ runs off to the right as $|\mathrm{Im}\,\lambda| \to \infty$, where $|\xi|$ grows, so $1/(\lambda G(\lambda))$ is integrable on the line and the Bromwich contour for $F_T(t) = \Pr(T \le t)$, whose transform is $1/(\lambda G(\lambda))$, can be shifted from $\mathrm{Re}\,\lambda > 0$ to $\mathrm{Re}\,\lambda = -L$ past the finitely many poles in between, because $\mathrm{Re}(-a_k) \le \frac{1}{4} - \gamma_k^2$. Write R_k for the residue of $1/G$ at $-a_k$, which for a simple zero $\rho_k = \frac{1}{2} + i\gamma_k$ is $R_k = -2\gamma_k \Xi(0)/\Xi'(\gamma_k)$ (Section 5.2.1). When the poles with $\mathrm{Re}\,a_k < L$ are simple,

$$\Pr(T > t) = \sum_{\mathrm{Re}\,a_k < L} \frac{R_k}{a_k}\, e^{-a_k t} + O(e^{-Lt}), \qquad t \to \infty. \tag{19.9.1}$$

Only this finite shift is used below: the full residue series needs contours between the poles on which $1/|\xi|$ is controlled, and under RH with simple zeros it is the spectral expansion of Section 5.2.1.

The coefficients R_k/a_k alternate in sign, because Ξ' changes sign at consecutive simple zeros, and grow like $e^{\pi\gamma_k/4}$ up to a power of γ_k and the factor $1/|\zeta'(\rho_k)|$ (Section 5.2.1); the factor $e^{-\gamma_k^2 t}$ wins for every $t > 0$.

A law F on $(0, \infty)$ that is the distribution function of a maximum of independent exponential variables E_i/r_i with deterministic rates r_i, where $\sum_i e^{-r_i t} < \infty$ for $t > 0$, has $-\log F(t) = \sum_i \sum_{m \ge 1} e^{-m r_i t}/m$, a sum of exponentials with positive coefficients. The same holds, with $-\log F(t) = \int e^{-xt}\, \nu(dx)$, for the maximum of a Poisson cloud of exponential clocks with a rate measure ν, that is, of the clocks E_i/x_i attached to the points x_i of a Poisson process of intensity ν, since the number of clocks above t is Poisson with mean $\int e^{-xt} \nu(dx)$.

Proposition 19.9.1. *$-\log F_T$ is not completely monotone. Hence T is not distributed as a maximum of independent exponential variables, for any choice of rates, nor as the maximum of a Poisson cloud of exponential clocks.*

Proof. Put $q_k = R_k/a_k$. For $k \le 3$ the zero ρ_k lies on the line, so the $a_k = -(\rho_k - \frac{1}{2})^2$ of Section 2.1 equals γ_k^2. The zeros with $\mathrm{Re}\,a < 3a_1 = 599.37$ are ρ_1

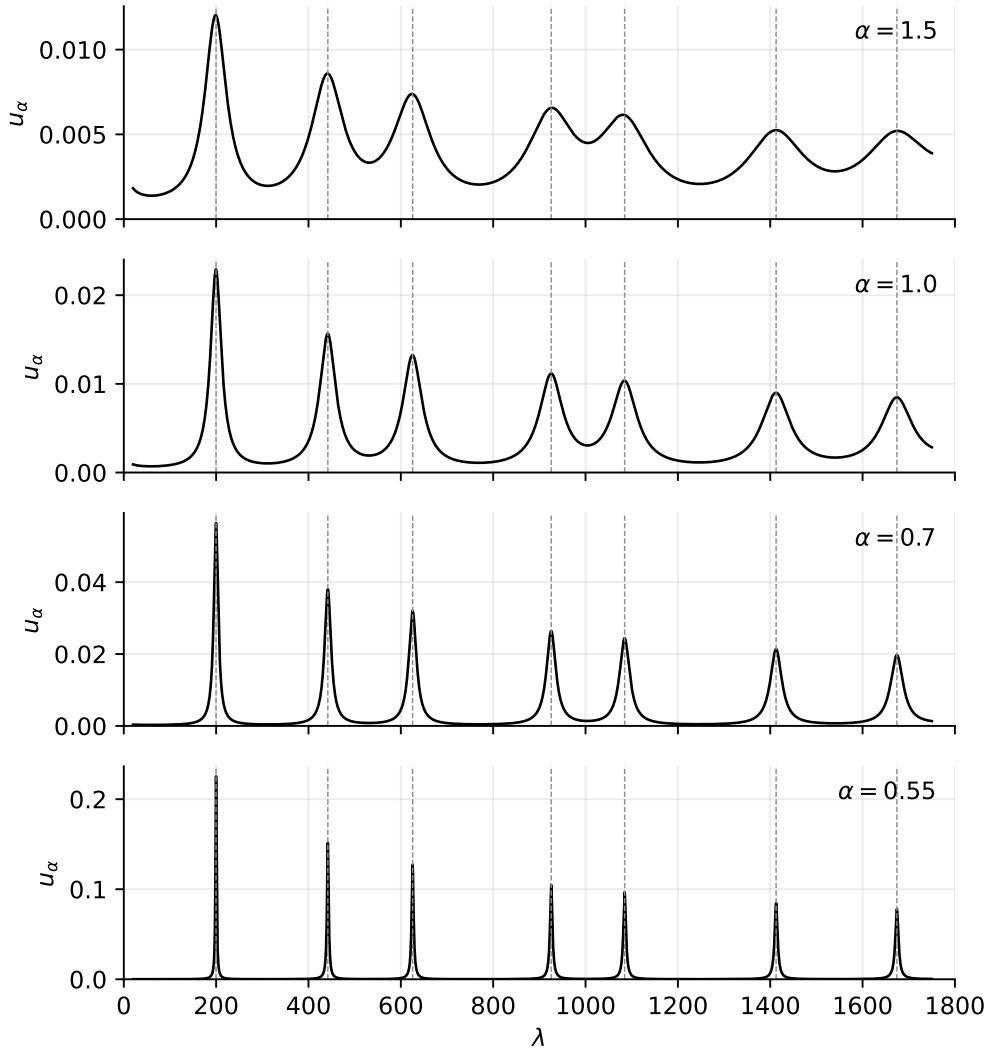


Figure 19.3: The density u_α of (2.1.6), computed from $\operatorname{Re} \xi'/\xi$ on the line $\operatorname{Re} s = \alpha$ with no zero as input, for $\alpha = 1.5, 1, 0.7, 0.55$; for $\alpha = 0.7$ and 0.55 it is the Thorin density of T_α only if no zero lies to the right of the line, as under RH. Dashed lines mark γ_k^2. The term of a zero on the line has unit mass and is close to a Lorentzian of width $4(\alpha - \frac{1}{2})\gamma_k$ near its peak (Proposition 19.8.1); as $\alpha \downarrow \frac{1}{2}$ these terms converge to unit atoms at γ_k^2.

and ρ_2 only: $a_1 = 199.79$, $a_2 = 441.93$ and $a_3 = 625.54$, the zeros below height $H = 3\times10^{12}$ lie on the line [277], and a zero above H has $\operatorname{Re} a \ge \gamma^2 - \frac14 > H^2 - \frac14$ wherever it lies. The zeros below height 200 are simple (Section 1.3.1). Equation (19.9.1) with $L = 3a_1$ gives $\Pr(T > t) = q_1 e^{-a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t})$, and since $-\log(1-y) = y + \frac12 y^2 + O(y^3)$ and $2a_1 < a_2 < 3a_1 < a_1 + a_2$,

$$-\log F_T(t) = q_1 e^{-a_1 t} + \tfrac12 q_1^2 e^{-2a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t}).$$

Suppose that $-\log F_T(t) = \int_{[0,\infty)} e^{-xt}\nu(dx)$ for $t > 0$ with $\nu \ge 0$ (Bernstein's theorem). First $\nu([0,a_1)) = 0$: for $x_0 < a_1$, $\nu([0,x_0])e^{-x_0 t} \le -\log F_T(t) = O(e^{-a_1 t})$. Then $e^{a_1 t}(-\log F_T(t)) = \nu(\{a_1\}) + \int_{(a_1,\infty)} e^{-(x-a_1)t}\nu(dx)$, whose last term tends to 0 by dominated convergence, so $\nu(\{a_1\}) = q_1$, and $f_1(t) = \int_{(a_1,\infty)} e^{-xt}\nu(dx) = \frac12 q_1^2 e^{-2a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t})$. The same two steps give $\nu((a_1, 2a_1)) = 0$ and $\nu(\{2a_1\}) = \frac12 q_1^2$, and then $f_2(t) = \int_{(2a_1,\infty)} e^{-xt}\nu(dx) \ge 0$ satisfies $e^{a_2 t} f_2(t) \to q_2$ because $a_2 < 3a_1$. So $q_2 \ge 0$. But $R_2 = -2\gamma_2 \Xi(0)/\Xi'(\gamma_2)$ and $\Xi'(\gamma_2) > 0$, since Ξ is positive on $[0,\gamma_1)$, negative on (γ_1,γ_2) and changes sign at the simple zero γ_2; so $q_2 = R_2/a_2 < 0$, a contradiction. □

For M_4 the coefficient at $a_2 = 4\pi$ also has the wrong sign: $\Pr(M_4 > t) = \sum_{n\ge1} 2(-1)^{n+1} e^{-\pi n^2 t}$ has $q_1 = 2$ and $q_2 = -2$. But $a_2 = 4\pi = 4a_1$ coincides with a term of the expansion in q_1, and the coefficient of $e^{-4\pi t}$ in $-\log\vartheta_4$ becomes $q_2 + \frac14 q_1^4 = -2 + 4 = 2 > 0$. Commensurability of the exponents πn^2 rescues complete monotonicity (the Jacobi triple product writes $-\log\vartheta_4$ as a sum of exponentials with positive coefficients), while $\gamma_2^2/\gamma_1^2 = 2.2119$ is not an integer, so nothing rescues it for T.

19.10 Gnedenko: the first zero as a norming constant

Gnedenko's theorem [92] classifies the limits of normalised maxima of independent identically distributed variables; a law with tail $\Pr(X > t) \sim Ce^{-at}$ lies in the Gumbel domain with norming constant $1/a$ and centring $\log(Cn)/a$. For T the norming constant is given by the first Thorin atom γ_1^2, and C by the first residue.

Proposition 19.10.1. *(i)* $\Pr(M_4 > t) \sim 2e^{-\pi t}$, *so for independent copies* $M_4^{(j)}$, $j \le n$,

$$\Pr\big(\pi \max_{j\le n} M_4^{(j)} - \log(2n) \le x\big) \to \exp(-e^{-x}).$$

(ii) $\Pr(T > t) \sim (R_1/\gamma_1^2)\, e^{-\gamma_1^2 t}$, *so for independent copies* $T^{(j)}$, $j \le n$, *of* T,

$$\Pr\Big(\gamma_1^2 \max_{j\le n} T^{(j)} - \log\frac{R_1 n}{\gamma_1^2} \le x\Big) \to \exp(-e^{-x}),$$

$$\max_{j\le n} T^{(j)} \Big/ \log n \to \frac{1}{\gamma_1^2} \quad \textit{in probability.}$$

Proof. (i) is $1-\vartheta_4(t) = 2e^{-\pi t} + O(e^{-4\pi t})$. For (ii), T exists by Theorem 8.1.1, and (19.9.1) with $a_1 < L < a_2$ gives $\Pr(T > t) = (R_1/a_1)e^{-a_1 t} + O(e^{-Lt})$, because the simple pole at $-\gamma_1^2$ is the rightmost singularity of $1/(\lambda G(\lambda))$ after 0, every other pole having $\operatorname{Re}\lambda \le -\gamma_2^2$. In both cases, if $\Pr(X > t) \sim Ce^{-at}$ and $y_n = (\log(Cn) + x)/a$, then $n\Pr(X > y_n) \to e^{-x}$ and $\Pr(\max_{j\le n} X^{(j)} \le y_n) = (1 - \Pr(X > y_n))^n \to \exp(-e^{-x})$. The last statement follows because $a\max_{j\le n} X^{(j)} - \log n$ is then tight. □

So γ_1 can be found with no zero as input: invert $(1 - 1/G(\lambda))/\lambda$ numerically and read off the exponential rate of the tail, as in the hazard experiment of Section 5.2.1. The local logarithmic slopes of $\Pr(T > t)$ on $[0.04, 0.05]$, $[0.05, 0.06]$, $[0.06, 0.07]$ and $[0.07, 0.08]$ are 199.496, 199.764, 199.788 and 199.790, converging to $\gamma_1^2 = 199.7905$ at the rate $e^{-(\gamma_2^2-\gamma_1^2)t}$ predicted by the second term of (19.9.1), and the intercept $e^{\gamma_1^2 t}\Pr(T > t)$ at $t = 0.08$ gives $R_1/\gamma_1^2 = 50.87103$ against the residue value 50.87104.

19.11 Finding the zeros from the Thorin density in the z chart

Theorem 19.6.1 gives a procedure in the chart $z = \lambda$: compute r_η on a grid from ξ'/ξ, with no zero used, locate its local maxima, and take square roots. Under RH $r_\eta = P_\eta * U_{\mathrm{RH}}$ is the Poisson smoothing of the Thorin measure $U_{\mathrm{RH}} = \sum_k \delta_{\gamma_k^2}$ at distance η below the cut (Theorem 19.6.1 and Section 4.1), so each atom appears as a spike of unit mass and height near $1/(\pi\eta)$ at $x \approx \gamma_k^2$. The same recovery in the chart $\theta = \sqrt{z}$ is in Sections 1.3.1 and 10.8; Table 19.2 is a cross check in the chart z, at $\eta = 3$.

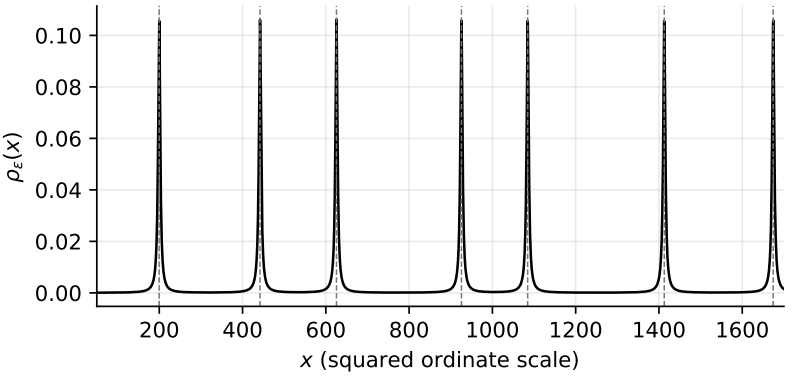


Figure 19.4: The smoothed Thorin density r_η of T, $\eta = 3$, computed from ξ'/ξ with no zero as input. Dashed lines mark γ_k^2. Each spike has height near $1/(\pi\eta)$ and unit mass.

k	peak of r_η	$\sqrt{\text{peak}}$	γ_k
1	199.790462	14.1347254	14.1347251
2	441.926159	21.0220398	21.0220396
3	625.542987	25.0108574	25.0108576
4	925.673105	30.4248764	30.4248761
5	1084.718263	32.9350613	32.9350616
6	1412.720791	37.5861782	37.5861782
7	1674.341571	40.9187191	40.9187190

Table 19.2: Zeros recovered as atoms of the smoothed Thorin measure, $\eta = 3$. The integral of r_η over $[50, 1700]$ is 6.95, against 7 zeros.

The peak of r_η near γ_k^2 is displaced from it by the pull of the neighbouring spikes, to leading order by $\eta^4 \sum_{j\neq k}(\gamma_j^2-\gamma_k^2)^{-3}$, which is of modulus below 2×10^{-5} in the table. On the window $[50, 1700]$ with $\eta = 3$ the minimum of r_η is $5.6\times10^{-5} > 0$, attained at the endpoint $x = 50$. This is a consistency check of the computation with Theorem 19.6.1(d) and Lemma 19.6.2, not a test of RH: the window contains only the squares of the zeros below height $\sqrt{1700} = 41.2$, which are known to lie on the line.

19.12 Finite maxima and a limit argument

The maximum of Section 19.2 suggests a direct strategy: truncate it at N terms, show that the finite object has no zeros off the critical line, and pass to the limit. For the sum representation the analogue is the truncation of the Biane, Pitman and Yor lattice (Subsection 17.1.1).

19.12.1 Finite maxima

For $N \geq 1$ let

$$M_4^{(N)} = \max\Big\{ \frac{E_n}{\pi n} \ (n \leq N), \quad \frac{E'_m}{\pi(2m-1)} \ (m \leq N) \Big\},$$

$$\mathcal{P}_N(q) = \prod_{n\leq N}(1-q^n) \prod_{m\leq N}(1-q^{2m-1}) = \sum_{m\geq 0} p_m^{(N)} q^m,$$

where $(E_n)_{n\geq1}$ and $(E'_m)_{m\geq1}$ are independent standard exponential variables, and $M_4 = \lim_N M_4^{(N)}$ is the maximum of Subsection 15.1.4, written with the rates πn and $\pi(2m-1)$. Then $\Pr(M_4^{(N)} \leq t) = \mathcal{P}_N(e^{-\pi t})$ exactly, a polynomial in $e^{-\pi t}$ with integer coefficients. Expanding the product and integrating term by term, for $\operatorname{Re} s > 0$,

$$\mathbb{E}\big(M_4^{(N)}\big)^{s/2} = -\Gamma(1+\tfrac{s}{2})\,\pi^{-s/2} D_N(s), \qquad D_N(s) = \sum_{m\geq1} p_m^{(N)}\, m^{-s/2},$$

a Dirichlet polynomial with integer coefficients. Since the factors left out of $\mathcal{P}_N$ are $1-q^a$ with $a>N$, $p_m^{(N)}$ for $m\le N$ is the coefficient of ϑ_4, $2(-1)^n$ at $m=n^2$ and 0 otherwise. So the first block of D_N is exactly minus the partial sum $2\sum_{n^2\le N}(-1)^{n+1}n^{-s}$ of $2(1-2^{1-s})\zeta(s)$, followed by a tail of integer coefficients at the frequencies $\frac12\log m$ with $N<m\le N(3N+1)/2$, the degree of $\mathcal{P}_N$.

By Subsection 15.1.4, $(1-2^{1-s})\zeta(s)=\mathbb{E}[(2K^2)^{s/2}]/(2\Gamma(1+\frac{s}{2}))$ for every complex s, with $\pi M_4=2K^2$ and K the Kolmogorov variable, and since $2\xi(s)/(s-1)=2\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ this reads

$$\mathbb{E}M_4^{s/2}=\frac{2(1-2^{1-s})\,\xi(s)}{s-1}\qquad\text{for every complex } s.$$

Define the normalised finite stage and its symmetrisation

$$\xi_N(s)=\frac{(s-1)\,\mathbb{E}(M_4^{(N)})^{s/2}}{2(1-2^{1-s})},\qquad \xi_N^{\mathrm{sym}}(s)=\tfrac12\big(\xi_N(s)+\xi_N(1-s)\big),$$

so that the same expression with M_4 in place of $M_4^{(N)}$ is ξ. In the open strip $0<\operatorname{Re}s<1$ the factor $1-2^{1-s}$ does not vanish and $\mathbb{E}(M_4^{(N)})^{s/2}$ is holomorphic (it is holomorphic for $\operatorname{Re}s>-4N$, since $\Pr(M_4^{(N)}\le t)\sim Ct^{2N}$ as $t\to0$), so ξ_N and ξ_N^{sym} are holomorphic there. The function ξ_N^{sym} is invariant under $s\mapsto1-s$ and, because ξ_N is real on the real axis, $\xi_N^{\mathrm{sym}}(\frac12+it)=\operatorname{Re}\,\xi_N(\frac12+it)$ is real for real t.

THEOREM 19.12.1. *(i)* $M_4^{(N)}\uparrow M_4$ *almost surely, and* $\Pr(M_4^{(N)}\ne M_4)\le C\sqrt{N}\,e^{-\pi\sqrt{N}/2}$.

(ii) On every compact K *in the open strip* $0<\operatorname{Re}s<1$,

$$\sup_K|\xi_N-\xi|\le C_K\,N^{1/4}e^{-\pi\sqrt{N}/4},\qquad \sup_K|\xi_N^{\mathrm{sym}}-\xi|\le C_K\,N^{1/4}e^{-\pi\sqrt{N}/4}.$$

Proof. (i) The variables of the stages are the same for all N, so adding variables can only raise the maximum, and $M_4^{(N)}\ne M_4$ requires a left out variable to exceed $M_4^{(N)}$. For any $\tau>0$,

$$\Pr(M_4^{(N)}\ne M_4)\le\Pr(M_4^{(N)}\le\tau)+\sum_{n>N}e^{-\pi n\tau}+\sum_{m>N}e^{-\pi(2m-1)\tau}.$$

The last two sums are at most $2e^{-\pi N\tau}/(1-e^{-\pi\tau})$. For the first, $\Pr(M_4^{(N)}\le\tau)=\vartheta_4(\tau)/\Pi_N(\tau)$, where $\Pi_N(\tau)=\prod_{n>N}(1-e^{-\pi n\tau})\prod_{m>N}(1-e^{-\pi(2m-1)\tau})$ is the product of the left out factors. Since $1-x\ge e^{-x/(1-x)}$, $\Pi_N(\tau)\ge\exp\big(-2e^{-\pi N\tau}/((1-e^{-\pi\tau})(1-e^{-\pi N\tau}))\big)$, while Jacobi's transformation $\vartheta_4(\tau)=\tau^{-1/2}\sum_n e^{-\pi(n+1/2)^2/\tau}$ gives $\vartheta_4(\tau)\le3\tau^{-1/2}e^{-\pi/(4\tau)}$ for $\tau\le1$. The choice $\tau=1/(2\sqrt{N})$ balances $\pi/(4\tau)$ against $\pi N\tau$, both equal to $\pi\sqrt{N}/2$, and gives the bound.

(ii) For $\sigma = \operatorname{Re} s > 0$, $M_4^{(N)} \le M_4$ gives $|\mathbb{E}M_4^{s/2} - \mathbb{E}(M_4^{(N)})^{s/2}| \le 2\mathbb{E}[M_4^{\sigma/2}\mathbf{1}\{M_4 \ne M_4^{(N)}\}] \le 2(\mathbb{E}M_4^{\sigma})^{1/2}\Pr(M_4^{(N)} \ne M_4)^{1/2}$ by Cauchy and Schwarz. The prefactor $(s-1)/(2(1-2^{1-s}))$ is bounded on K because $1-2^{1-s}$ vanishes only on $\operatorname{Re} s = 1$. The symmetrised bound follows since K and $1-K$ both lie in the strip and $\xi(1-s) = \xi(s)$. □

The exponent in (ii) is not sharp: a heuristic count of the event $M_4^{(N)} \ne M_4$, in which the jth left out variable wins with probability $\mathbb{E}e^{-\pi j M_4} = \pi\sqrt{j}/\sinh(\pi\sqrt{j})$, of order $\sqrt{j}\,e^{-\pi\sqrt{j}}$, gives a rate of order $Ne^{-\pi\sqrt{N}}$.

Corollary 19.12.2 (Hurwitz). *(a) Every zero of ξ in the open strip is a limit of zeros of ξ_N, and of zeros of ξ_N^{sym}, with multiplicity; every limit of zeros of ξ_N inside the open strip is a zero of ξ.*

(b) If $\rho = \frac{1}{2} + i\gamma$ is a simple zero of ξ, then for all large N, ξ_N^{sym} has exactly one zero in a small disc about ρ, and that zero lies exactly on the critical line.

(c) Suppose that there are $N_j \to \infty$ and heights $h_j \to \infty$ such that $\xi_{N_j}^{\mathrm{sym}}$ has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < h_j\}$ for every j. Then the Riemann Hypothesis holds. In particular this is so if there are $c > 0$ and N_0 such that for every $N \ge N_0$, ξ_N^{sym} has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < c\sqrt{N}\}$.

Proof. (a) is Hurwitz's theorem with Theorem 19.12.1(ii). For (b), Hurwitz gives exactly one zero z_N of ξ_N^{sym} in a disc D centred at ρ for large N. The symmetries $s \mapsto 1-s$ and $s \mapsto \bar{s}$ of ξ_N^{sym} make its zero set invariant under $s \mapsto 1-\bar{s}$, which maps D to itself; uniqueness forces $1-\bar{z}_N = z_N$, that is $\operatorname{Re} z_N = \frac{1}{2}$. For (c), fix Γ and let $U = \{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < \Gamma\}$. For j so large that $h_j > \Gamma$ the functions $\xi_{N_j}^{\mathrm{sym}}$ have no zeros in U, and they converge to $\xi \not\equiv 0$ uniformly on compact subsets of U, so by Hurwitz ξ has no zeros in U. Letting $\Gamma \to \infty$ and using $\xi(s) = \xi(1-s)$ gives RH. □

Part (b) is pointwise in ρ and gives no height up to which the stages reproduce all the zeros of ξ: that would need an error bound uniform up to the height and lower bounds for $|\xi|$ on circles around every zero below it, that is bounds for $1/\zeta'(\rho)$, and none is proved here.

19.12.2 What the finite stages look like

The unsymmetrised stage ξ_N has zeros off the line at every N tested from 4 to 128, so the naive strategy fails. Its zeros near ρ_k approach ρ_k as N grows, and more slowly the larger γ_k is (Table 19.3), and it carries spurious zeros elsewhere, such as the zero of ξ_{16} at $1.055 + 17.984i$, just to the right of the strip.

From $N = 32$ to $N = 64$ the error at γ_1 falls by a factor of about 250, against $e^{\pi(8-\sqrt{32})} = 1574$ for the exponential factor of the heuristic rate (about 800 with its prefactor N) and $e^{\pi(8-\sqrt{32})/4} = 6.3$ for that of the proven one; this is a rate of

N	zero near ρ_1	zero near ρ_2	zero near ρ_5
16	$0.51577 + 14.09915i$	$0.95703 + 20.91901i$	$1.56809 + 34.68128i$
32	$0.49948 + 14.13483i$	$0.48159 + 21.02790i$	$1.02797 + 32.22350i$
64	$0.50000052 + 14.1347256i$	$0.50003418 + 21.0220191i$	$0.51254 + 32.92764i$

Table 19.3: Zeros of the unsymmetrised finite stage ξ_N. For $N = 16$ and $N = 32$ the zero listed near ρ_5 does not track ρ_5, since ρ_5 lies above the height $4\sqrt{N}$ discussed below.

N	zeros	$\lvert t_1 - \gamma_1\rvert$	$\lvert t_2 - \gamma_2\rvert$	$\lvert t_3 - \gamma_3\rvert$	$\lvert t_5 - \gamma_5\rvert$
16	7	3.6×10^{-2}	9.9×10^{-2}	1.7×10^{-1}	1.3×10^{-3}
32	10	1.0×10^{-4}	5.9×10^{-3}	6.8×10^{-2}	1.1
64	8	4.1×10^{-7}	2.1×10^{-5}	3.8×10^{-4}	7.5×10^{-3}

Table 19.4: Real zeros t_k of the symmetrised stage $\operatorname{Re}\xi_N(\frac{1}{2} + it)$: the column "zeros" is the number of sign changes on $[10, 44]$, where ξ has 8 zeros ($10 \le \gamma \le 44$). At $N = 16$ the agreement at γ_5 is accidental.

about $e^{-0.75\pi\sqrt{N}}$ (about $e^{-0.60\pi\sqrt{N}}$ between $N = 8$ and $N = 16$). The supremum of $|\xi_N - \xi|$ over the grid $\{0.3, 0.5, 0.7\} + i\{0, 2, \dots, 10\}$ decays like $e^{-c\pi\sqrt{N}}$ with $c \approx 0.93$ between $N = 32$ and $N = 64$ and $c \approx 0.80$ between $N = 8$ and $N = 16$, and that of $|\xi_N^{\mathrm{sym}} - \xi|$ with $c \approx 0.81$ and $c \approx 0.80$, all between the exponent $\frac{1}{4}$ of the proof and the exponent 1 of the heuristic. Between consecutive stages the exponents for ξ_N^{sym} and for the error at γ_1 oscillate, between 0.73 and 1.10 and between 0.49 and 1.37, so the exponent 1 of the heuristic is not a bound. Since $|\xi(\frac{1}{2} + i\gamma)|$ decays like $e^{-\pi\gamma/4}$, the heuristic predicts that zeros up to height γ are captured once $\pi\sqrt{N}$ exceeds $\pi\gamma/4$ by a margin, that is $N \gtrsim \gamma^2/16$. The tables agree: the symmetrised stage $N = 64$ captures every zero up to height 33 to within 10^{-2} (and ξ_{64} itself to within 1.5×10^{-2}), and the accuracy degrades beyond 37. This suggests that the stage of index N tracks the zeros of ξ up to a height of about $4\sqrt{N}$; this tracking height is not the zero free height of Corollary 19.12.2(c), which may grow with N in any way.

19.12.3 The symmetric stages F_N^{sym} and how far they stay on the line

The normalisation ξ_N divides by $1 - 2^{1-s}$, which has zeros on $\operatorname{Re} s = 1$ and, after symmetrisation, on $\operatorname{Re} s = 0$; for counting zeros we keep the factor. Put

$$F_N(s) = \frac{s-1}{2}\,\mathbb{E}\big(M_4^{(N)}\big)^{s/2} = -\frac{s-1}{2}\,\Gamma\big(1 + \tfrac{s}{2}\big)\pi^{-s/2}D_N(s),$$

$$F_N^{\mathrm{sym}}(s) = \tfrac{1}{2}\big(F_N(s) + F_N(1-s)\big),$$

so that $F_N = (1 - 2^{1-s})\xi_N$. These functions are holomorphic on $-4N < \operatorname{Re} s < 4N + 1$, but they are not entire: F_N has a pole at $s = -4N$ and F_N^{sym} also at

$s = 4N + 1$, far from every region used below. As $N \to \infty$, $F_N \to (1 - 2^{1-s})\xi(s)$ and $F_N^{\mathrm{sym}} \to \mathfrak{q}(s)\xi(s)$, with $\mathfrak{q}(s) = 1 - 2^{-s} - 2^{s-1}$. The zeros of $\mathfrak{q}$ are simple and are the solutions of $2^s = 1 \pm i$, all on $\operatorname{Re} s = \frac{1}{2}$, at $t = (2\pi k \pm \pi/4)/\log 2$.

Theorem 19.12.1 and Corollary 19.12.2 hold for F_N^{sym} with $\mathfrak{q}\xi$ in place of ξ, on compact subsets of the open strip, since $1 - 2^{1-s}$ is bounded there. Part (c) becomes: if there are $N_j \to \infty$ and $h_j \to \infty$ such that $F_{N_j}^{\mathrm{sym}}$ has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < h_j\}$, then RH holds, by the same argument, since a zero of ξ in that region is a zero of $\mathfrak{q}\xi$. This is the F_N^{sym} form of Corollary 19.12.2(c), a different statement from (c) for ξ_N^{sym}, since F_N^{sym} is not ξ_N^{sym} times a fixed factor.

Zeros of F_N^{sym} in a rectangle $[-1, 2] \times [t_1, t_2]$ can be counted by the argument principle, and zeros on the line are sign changes of the real function $F_N^{\mathrm{sym}}(\frac{1}{2} + it)$. When the two counts agree, every zero in the rectangle is on the line. Write $T^*(N)$ for the height below which the counts agree in every window of height 10 and above which off line zeros appear in the rectangle. Since the rectangle contains the strip, F_N^{sym} has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < T^*(N)\}$, the hypothesis of the F_N^{sym} form of Corollary 19.12.2(c) at height $T^*(N)$. Table 19.5 reports the counts.

N	terms	height	on the line	first off line window	$T^*(N)$
8	72	400	70 of 192	$(110, 120]$	≈ 110
16	350	1000	146 of 510	$(250, 260]$	≈ 250
32	1478	800	312 of 360	$(530, 540]$	≈ 530
48	3372	1000	494 of 496	$(870, 880]$	≈ 870
64	6034	1500	791 of 797	$(1060, 1070]$	≈ 1060
96	13670	2500	1463 of 1465	$(1590, 1600]$	≈ 1590
128	24354	130	64 of 66	$(60, 70]$	≈ 60

Table 19.5: Zeros of the symmetric stage F_N^{sym} in $[-1, 2] \times [0, T]$ by the argument principle, against sign changes on the critical line: "terms" is the number of nonzero coefficients $p_m^{(N)}$, $m \geq 1$, "height" the height T searched, and "on the line" the number of sign changes out of the number of zeros counted. The windows have height 10. Above $T^*(N)$ the off line zeros arrive in bulk for $N \leq 32$: for $N = 16$ the two windows of $(300, 320]$ have 10 of their 13 zeros off the line. The sum D_N, whose terms are much larger than the sum itself, is evaluated in double precision, with a bound on its rounding error on the critical line in every window, and in multiprecision in the windows below 100 for $N = 96$ and below 130 for $N = 128$; above 130 the double precision margin of $N = 128$ is lost, so that stage is reported only up to 130. Off line zeros come in pairs $\sigma + it$, $1 - \sigma + it$, so an off line pair adds two to the count of its window, and a difference of one between the two counts is an artefact, such as a zero at the window boundary.

The first off line pairs of F_N^{sym} are $1.5643 + 111.7906i$ for $N = 8$, $1.1950 + 254.3196i$ for $N = 16$, $1.2079 + 537.2287i$ for $N = 32$, $1.0434 + 870.3807i$ for $N = 48$, $1.1025 + 1063.9243i$ for $N = 64$, $1.1177 + 1593.1234i$ for $N = 96$ and

$1.5311 + 60.7314i$ for $N = 128$, each with its mirror image under $s \mapsto 1 - \bar{s}$. The first off line zero inside the strip appears for $N = 8$ in $(120, 130]$, for $N = 16$ in $(330, 340]$ and for $N = 32$ in $(550, 560]$, and none was found there for $N = 48$, 64, 96 and 128 up to the heights 1000, 1500, 2500 and 130 searched, so the height up to which F_N^{sym} has no off line zeros in the strip is at least $T^*(N)$ and may be larger. The counts depend on the rectangle: for $N = 16$ the rectangle $[-8, 9] \times [240, 250]$ contains 7 zeros where $[-1, 2] \times [240, 250]$ contains 3, all on the line, so zeros of F_N^{sym} far from the strip exist below $T^*(N)$, and $T^*(N)$ refers to $[-1, 2]$.

Three things follow. First, the symmetric stages are not real rooted: off line zeros exist in $[-1, 2] \times [0, T]$ for every N tested, but only above $T^*(N)$, and the first ones found lie outside the critical strip. Second, up to $N = 96$ the height $T^*(N)$ grows roughly linearly, while the height up to which the stage approximates $\mathfrak{q}\xi$ well enough to capture its zeros is only about $4\sqrt{N}$ (Table 19.4 is for ξ_N^{sym}; the same height is expected for F_N^{sym}, whose limit $\mathfrak{q}\xi$ has the same zeros besides those of $\mathfrak{q}$). At $N = 128$ the growth breaks: $T^*(128)/128 = 0.47$, although the strip has no off line zero of that stage below 130, the height to which it was counted. Below $4\sqrt{N}$ the zeros of F_N^{sym} are those of $\mathfrak{q}\xi$ moved along the line; between $4\sqrt{N}$ and $T^*(N)$ the stage has its own zeros, fewer than $\mathfrak{q}\xi$, still all on the line; above $T^*(N)$ it has zeros off the line. The linear growth is what a heuristic based on the approximate functional equation suggests for a symmetrised Dirichlet polynomial whose longest frequency is $\frac{1}{2}\log m_{\max}$ with $m_{\max} = N(3N + 1)/2$: the two halves of F_N^{sym} can balance off the line only when $(t/2\pi)^{\sigma - 1/2}$ is matched by a ratio of the Dirichlet polynomial at σ and $1 - \sigma$, which first becomes possible at t of order N; the pair of $N = 128$, far from the strip, falls outside this heuristic. Third, the hypothesis of the F_N^{sym} form of Corollary 19.12.2(c) holds up to height $T^*(N)$ for the N tested, but finitely many N cannot verify an asymptotic hypothesis. Since (c) needs only $T^*(N_j) \to \infty$ along some sequence $N_j \to \infty$, the conjecture $T^*(N) \geq \kappa N$ for some $\kappa > 0$ and all N would imply RH, as would the far weaker $T^*(N) \geq c\sqrt{N}$.

We have no mechanism for $T^*(N)$. The classical argument that puts zeros of a symmetrised Dirichlet polynomial on the line, dominance of the first term of D_N, that is $|D_N(s) - p_1| < |p_1|$ on $\operatorname{Re} s \geq \frac{1}{2}$ (with $|p_1| = 2$), fails here because $\sum_m |p_m^{(N)}| m^{-1/4}$ is large: about 50 for $N = 8$ and 2.4×10^9 for $N = 64$. The observed $T^*(N)$ is nevertheless what the balance heuristic predicts, so the right statement to prove may be about the ratio $|D_N(\sigma + it)/D_N(1 - \sigma + it)|$ rather than about D_N alone.

19.12.4 Comparison with the symmetrised lattice truncations

Subsection 17.1.1 carries out the same strategy for the symmetrised lattice truncations K_n, with Proposition 17.1.6 and Theorem 17.1.7 as the analogues of Corollary 19.12.2(b) and (c); the off line zeros of F_N^{sym} above $T^*(N)$ do not bear on

Conjecture S of that subsection, which concerns K_n.

19.13 Bondesson classes containing GGC

The classes BO, GGC and SD (Subsection 4.1.2) have finite order versions. For the clock T finite verification gives them up to order 1.16×10^{25} on the Lévy side and 9×10^{12} on the transform side, and every infinite order version is equivalent to RH.

Definition 19.13.1. A function f on an interval is k monotone if $(-1)^j f^{(j)} \geq 0$ for $0 \leq j \leq k$, and completely monotone if it is k monotone for every k. For $k \geq 1$ let GGC_k be the laws on $[0,\infty)$ with a Lévy density ℓ such that $x\ell(x)$ is k monotone, and BO_k those with ℓ itself k monotone. Thus $\mathrm{GGC}_1 = \mathrm{SD}$ is the self decomposable class, $\bigcap_k \mathrm{GGC}_k = \mathrm{GGC}$ and $\bigcap_k \mathrm{BO}_k = \mathrm{BO}$. A positive function φ on $(0,\infty)$ is HCM of order k if for every $u > 0$ the function $w \mapsto \varphi(uv)\varphi(u/v)$, $w = v + 1/v \geq 2$, is k monotone on $[2,\infty)$; HCM means HCM of every order (Subsection 9.1.2).

These finite order versions are not Urbanik's nested classes, and other finite order notions (Williamson's k monotone functions, Bondesson's classes of finite order for densities) use other conventions. By the Leibniz identity $(-1)^j(fg)^{(j)} = \sum_i \binom{j}{i}[(-1)^i f^{(i)}][(-1)^{j-i} g^{(j-i)}]$ a product of two nonnegative k monotone functions is k monotone; as x^{-1} is completely monotone and $\ell(x) = x\ell(x) \cdot x^{-1}$, this gives $\mathrm{GGC}_k \subset \mathrm{BO}_k$ for every k.

Theorem 19.13.2. *Let T be the clock, with Lévy density $W(x)/x$ and Laplace transform $\varphi_{1/2}(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{\lambda}) = 1/G(\lambda)$, and let $H = 3 \times 10^{12}$. Statements (i) and (ii) are unconditional: they use the verification of RH to height H [277].*

(i) $T \in \mathrm{GGC}_k$, and hence $T \in \mathrm{BO}_k$, for every $k \leq 1.16 \times 10^{25}$.

(ii) $\varphi_{1/2}$ is HCM of order k for every $k \leq 9 \times 10^{12}$.

(iii) The following are equivalent: RH; T is a GGC; $T \in \mathrm{BO}$; $\varphi_{1/2}$ is HCM; $\varphi_{1/2}$ is HCM of every order.

Proof. (i) By Theorem 8.1.1 the clock T exists and is infinitely divisible with Lévy density $\ell(x) = W(x)/x$, and Theorem 8.1.13 gives $(-1)^j W^{(j)}(x) > 0$ for every $x > 0$ and every integer $0 \leq j \leq 1.29(H-23)^2$, which exceeds 1.16×10^{25}. Since $x\ell(x) = W(x)$, this says $T \in \mathrm{GGC}_k$ for every $k \leq 1.16 \times 10^{25}$, and the Leibniz step above gives $T \in \mathrm{BO}_k$.

(ii) Zeros are counted with multiplicity, and $\rho = \frac{1}{2} + b + i\gamma$ with $\gamma > 0$ runs over the zeros of positive ordinate. Put $a_\rho = -(\rho - \frac{1}{2})^2 = \gamma^2 - b^2 - 2ib\gamma$, as in Section 2.1, so that $\mathrm{Re}\, a_\rho > 0$ and $\sum_\rho |a_\rho|^{-1} < \infty$, and by the Hadamard product (2.1.1), $\varphi_{1/2}(\lambda) = \prod_\rho (1 + \lambda/a_\rho)^{-1}$, absolutely convergent. Since $(1 + uv/a)(1 + u/(va)) =$

$1 + uw/a + u^2/a^2$ with $w = v + 1/v$,

$$\varphi_{1/2}(uv)\,\varphi_{1/2}(u/v) = \prod_{b=0,\ \gamma>0} \Big(1 + \frac{uw}{a_\rho} + \frac{u^2}{a_\rho^2}\Big)^{-1} \times \prod_{b>0,\ \gamma>0} \Big|1 + \frac{uw}{a_\rho} + \frac{u^2}{a_\rho^2}\Big|^{-2}, \qquad w = v + \frac{1}{v},$$

where in the second product ρ stands for the pair $\{\rho, 1-\bar\rho\}$ of zeros with $\gamma > 0$ and $b \neq 0$, whose values a_ρ and $a_{1-\bar\rho} = \bar a_\rho$ are conjugate, so that their two factors multiply to a squared modulus. Put $c_\rho(u) = a_\rho/u + u/a_\rho$, so that $1 + uw/a + u^2/a^2 = (u/a)(w + c)$. For a zero on the line $a_\rho = \gamma^2$ and $c_\rho(u) \geq 2$ is real, and the factor $(a_\rho/u)/(w + c_\rho(u))$ is completely monotone in w on $[2, \infty)$. For a pair the factor is $(|a_\rho|^2/u^2)/((w + c)(w + \bar c))$ with $c = c_\rho(u)$ complex. Write $a_\rho = re^{-i\phi}$ with $0 < \phi < \pi/2$. Then $\tan\phi = 2b\gamma/(\gamma^2 - b^2) \leq \gamma/(\gamma^2 - \frac14) < 1.01/\gamma$ for $\gamma \geq 10$, because $0 < b < \frac12$. Moreover $\operatorname{Re} c = (r/u + u/r)\cos\phi$ and $|\operatorname{Im} c| = |r/u - u/r|\sin\phi \leq (r/u + u/r)\sin\phi$, so for every $w \geq 0$ the angle $\vartheta = |\arg(w + c)|$ satisfies $\tan\vartheta = |\operatorname{Im} c|/(w + \operatorname{Re} c) \leq \tan\phi$, that is $\vartheta \leq \phi$. Partial fractions give $1/((w + c)(w + \bar c)) = -\operatorname{Im}[(w + c)^{-1}]/\operatorname{Im} c$, whence

$$(-1)^j \frac{d^j}{dw^j} \frac{1}{(w + c)(w + \bar c)} = \frac{j!\,\sin((j + 1)\vartheta)}{|\operatorname{Im} c|\,|w + c|^{j+1}},$$

which is positive as long as $(j+1)\vartheta < \pi$, hence for all $j < \pi/\phi - 1$ (when $\operatorname{Im} c = 0$, that is $u = r$, the factor is $(w + c)^{-2}$, whose signed derivatives $(j + 1)!\,(w + c)^{-j-2}$ are positive). By the verification of RH to height H [277] every zero with $\gamma \leq H$ is on the line, so every pair has $\gamma > H$ and $\phi < 1.01/H$, and its factor is k monotone for every $k \leq \pi H/1.01 - 1$, which is 9.33×10^{12} and exceeds 9×10^{12}. Products of nonnegative k monotone functions are k monotone by the Leibniz identity, and the partial products converge, with all their w derivatives, locally uniformly on $\operatorname{Re} w > 0$: the factors have no zeros there, since the poles $w = -c$ lie in $\operatorname{Re} w < 0$, and $\sum_\rho |uw/a_\rho + u^2/a_\rho^2| < \infty$ locally uniformly. So the limit is k monotone.

(iii) RH holds if and only if T is a GGC by Proposition 2.1.1, and if and only if $\varphi_{1/2}$ is HCM by Corollary 9.1.3(ii), that is by Bondesson's theorem [39, Theorem 6.1.1] (a function φ with $\varphi(0+) = 1$ is the Laplace transform of a GGC if and only if it is HCM; see also [168]). A GGC is in BO by the Leibniz step. Conversely suppose that $\ell(x) = W(x)/x$ is completely monotone, $\ell(x) = \int_{[0,\infty)} e^{-xy} m(dy)$ with $m \geq 0$. Taking Laplace transforms of $W(x) = x\ell(x)$ for $\operatorname{Re}\lambda > 0$,

$$\int_0^\infty e^{-\lambda x} W(x)\,dx = \int_{[0,\infty)} \frac{m(dy)}{(\lambda + y)^2} = \sum_\rho \frac{1}{\lambda + a_\rho} = \frac{G'}{G}(\lambda).$$

The middle expression is analytic on $\mathbb{C} \setminus (-\infty, 0]$, while G'/G has a pole at every $-a_\rho$, and $-a_\rho \notin (-\infty, 0]$ exactly when $a_\rho \notin (0, \infty)$, that is when ρ is off the line. So BO forces RH. □

For a model pair factor with a zero at $\beta = 0.6$, $\gamma = 100$, the first order at which the closed form fails is 1571 at $u = 1$, $w = 2$, and 1570 as $u \to 0$, against the bound $\pi\gamma/(2(\beta - \frac{1}{2})) = 1570.8$.

19.14 Is there a new path to the Riemann Hypothesis?

No.

19.14.1 Three reformulations and what each costs

The derivative ladder: RH holds if and only if W is completely monotone (Theorem 19.6.1(e)), and order k and height H trade quadratically (Subsection 8.1.6), so climbing the ladder to infinity is verifying RH to infinite height; the ladder is a reformulation, not a path. The basepoint family: the basepoint clock T_α (Section 19.8) is GGC for every $\alpha > \frac{1}{2}$ if and only if RH (Theorem 13.2.7), and this is Lagarias's criterion [198] (Remark 2.1.4), so the Thorin route at basepoints and the Lagarias route are one route. The finite maxima: Corollary 19.12.2(c) reduces RH to zero freeness of the symmetrised stages ξ_N^{sym} off the line up to heights $h_j \to \infty$, for infinitely many N, a hypothesis that is not weaker than RH (whether RH implies it is open). Unsymmetrised sections of ζ are known to have zeros off the line and even to the right of $\mathrm{Re}\, s = 1$ [349, 236], which is why the symmetrisation is essential; for the symmetrised sections there are results of Spira and of Gonek and Montgomery [334, 96] (the paragraph after Table 17.2); we know of no result of this kind for the symmetrised stages of the maximum.

19.14.2 The arithmetic form below $\alpha = 1$, and how far it sees

The sine squared prime form of Section 2.1 (the paragraph after (2.1.9), with Table 2.2) writes $v_\alpha(\theta) = \mathrm{Re}\, \xi'/\xi(\alpha + i\theta)$ for $\alpha > 1$ through the signed canonical measure with atoms $\Lambda(n)n^{-\alpha}$ at $\log n$, an archimedean density and a pole density: the prime and gamma terms are nonnegative and the pole terms negative, so the positivity of v_α for $\alpha > 1$, which is Lagarias positivity there, comes from the absence of zeros with $\beta \geq 1$ and not from the terms one by one. Theorem 13.2.7 already continues it below $\alpha = 1$ by compensation: its last condition, that the compensated sine squared sum converges for every $\alpha' > \alpha$, is equivalent to zero freeness above α. A smoothed form of the continuation, with explicit pole terms, is the following.

Proposition 19.14.1. *Let $\alpha = \frac{1}{2} + \delta$ with $\delta > 0$, $t \in \mathbb{R}$ and $X > 0$. Put $g_X(x) = e^{-\delta|x|}\cos(tx)\,(1 - |x|/X)_+$ and $h_X(r) = \int_{\mathbb{R}} g_X(x)e^{irx}dx$, so that $h_X \geq 0$ on $\mathbb{R}$: it is $(2\pi)^{-1}$ times the convolution of the two sided Poisson kernel $h_\infty(r) = \delta/(\delta^2 + (t - r)^2) + \delta/(\delta^2 + (t + r)^2)$ with the Fejér kernel $4\sin^2(Xr/2)/(Xr^2)$.*

Define

$$A_X(\delta,t)=\frac12\Big[h_X(\tfrac i2)+h_X(-\tfrac i2)-\log\pi+\frac{1}{2\pi}\int_{\mathbb R}h_X(r)\,\operatorname{Re}\psi\big(\tfrac14+\tfrac{ir}{2}\big)\,dr \\ -2\sum_{n<e^X}\frac{\Lambda(n)}{n^\alpha}\cos(t\log n)\Big(1-\frac{\log n}{X}\Big)\Big],$$

where ψ is the digamma function and $h_X(\frac i2)+h_X(-\frac i2)=2\operatorname{Re}[J(-\delta+\frac12+it)+J(-\delta-\frac12+it)]$ with $J(c)=-1/c+(e^{cX}-1)/(Xc^2)$. Then:

(a) $A_X(\delta,t)=\frac12\sum_\rho h_X(\gamma_\rho)$, with $\gamma_\rho=(\rho-\frac12)/i$ and the sum over all nontrivial zeros, unconditionally.

(b) Under RH, $A_X(\delta,t)\ge0$ for every X, δ and t. In particular a single triple (δ,t,X) with $A_X(\delta,t)<0$ would disprove RH.

(c) If every zero has $|\beta-\frac12|<\delta$ (under RH, or unconditionally when $\delta\ge\frac12$), then $A_X(\delta,t)\to\operatorname{Re}\,\xi'/\xi(\alpha+it)$ as $X\to\infty$.

Proof. (a) This is Weil's explicit formula $\sum_\rho h(\gamma_\rho)=h(\frac i2)+h(-\frac i2)-g(0)\log\pi+\frac{1}{2\pi}\int_{\mathbb R}h(r)\operatorname{Re}\psi(\frac14+\frac{ir}{2})\,dr-2\sum_n\Lambda(n)n^{-1/2}g(\log n)$, in the form of [167], which evaluates the distribution D of Proposition 17.1.14 on a test function g with Fourier transform h, applied to the even test function g_X, which is continuous, compactly supported and piecewise smooth, with Fourier transform h_X, so that $g_X(0)=1$ and $n^{-1/2}g_X(\log n)=n^{-\alpha}\cos(t\log n)(1-\log n/X)_+$; the sum over zeros converges absolutely because $h_X(r)=O(r^{-2})$ uniformly in $|\operatorname{Im}r|\le\frac12$. Here $J(c)=\int_0^X e^{cx}(1-x/X)\,dx$, and $h_X(r)=2\int_0^X e^{-\delta x}\cos(tx)\cos(rx)(1-x/X)\,dx=\frac12\sum_{s_1,s_2=\pm1}J(-\delta+i(s_1t+s_2r))$. At $r=\pm\frac i2$ two of the four terms are the complex conjugates of the other two, so $h_X(\pm\frac i2)=\operatorname{Re}[J(-\delta+\frac12+it)+J(-\delta-\frac12+it)]$ and the two pole terms add up to twice that, which is the stated value.

(b) Under RH every γ_ρ is real and $h_X\ge0$ on $\mathbb R$, so every term of (a) is nonnegative.

(c) For a zero $\rho=\frac12+b+i\gamma$ the four exponentials of $h_X(\gamma_\rho)=h_X(\gamma-ib)$ have $\operatorname{Re}c=-\delta\pm b$, which is negative when $|b|<\delta$. Then $J(c)\to-1/c$ as $X\to\infty$, and $|J(c)+1/c|\le2/(X|c|^2)$, so $h_X(\gamma_\rho)\to h_\infty(\gamma_\rho)=\delta/(\delta^2+(t-\gamma_\rho)^2)+\delta/(\delta^2+(t+\gamma_\rho)^2)$, with $|h_X(\gamma_\rho)|\le C(1+|\gamma_\rho|)^{-2}$ uniformly in $X\ge1$, where C depends on δ, t and the finitely many zeros with $||\gamma_\rho|-t|\le1$, because the four terms $-1/c$ combine in pairs to $O(\gamma^{-2})$ and $|c|\ge1$ for the other zeros. By dominated convergence $A_X\to\frac12\sum_\rho h_\infty(\gamma_\rho)$. Finally $h_\infty(\gamma_\rho)$ is the sum of the four terms $\frac12/(\alpha\pm it-\rho)$ and $\frac12/(\rho-(1-\alpha)\pm it)$, and since the zeros are symmetric under $\rho\mapsto1-\rho$ and $\rho\mapsto\bar\rho$, $\frac12\sum_\rho h_\infty(\gamma_\rho)=\operatorname{Re}\sum_\rho(\alpha+it-\rho)^{-1}=\operatorname{Re}\,\xi'/\xi(\alpha+it)$ by the symmetric Hadamard product. □

Remark 19.14.2. The hypothesis of (c) is needed. If a zero has $b = \beta - \frac{1}{2} > \delta$, the terms of $h_X(\gamma - ib)$ with $\operatorname{Re} c = b - \delta > 0$ contain $e^{(b-\delta)X}/(Xc^2)$, which grows with X, and $h_\infty(\gamma - ib)$ is only the analytic continuation of a divergent integral. At $t = \gamma$ the four zeros $\frac{1}{2} \pm b \pm i\gamma$ contribute $e^{(b-\delta)X}/(X(b-\delta)^2)\,(1+O((b-\delta)^2/\gamma^2))+O(1)$ to $A_X(\delta, \gamma)$, which is positive and unbounded in X. No converse is claimed: whether an off line zero forces $A_X(\delta, t) < 0$ somewhere for large X is not shown here.

The proposition is Lagarias's criterion restricted to a family, and a form of Weil's positivity criterion (compare Proposition 17.1.14) for the positive definite test functions g_X, which have corners. Table 19.6 gives the minimum over $t \in [0, 60]$, positive at every X and every α tested.

X	$\alpha = 0.75$	$\alpha = 0.6$	$\alpha = 0.55$
2	0.03275	0.02495	0.02230
4	0.02208	0.01432	0.01160
8	0.01707	0.00954	0.00681
13	0.01510	0.00812	0.00576

Table 19.6: $\min_{0 \le t \le 60} A_X(\alpha - \frac{1}{2}, t)$ from primes below e^X (up to 4.4×10^5 at $X = 13$). The values decrease towards $\xi'/\xi(\alpha) = 0.011551$, 0.0046209, 0.0023105 for the three α.

The positivity in the table is not evidence of anything new. The contribution of zeros above height H to A_X is at most about $(e^{X/2}/X + \delta) \log H/H$, because $|h_X(\gamma_\rho)| \lesssim e^{|\operatorname{Im}\gamma_\rho| X}/(X\gamma^2)$ and $|\operatorname{Im}\gamma_\rho| \le \frac{1}{2}$. With $H = 3 \times 10^{12}$ and $X = 13$ that is below 10^{-8}, about 4.9×10^{-10}, far under the minima in the table. So the verified zeros force $A_X \ge 0$ for t in the range of the table as long as this bound stays below the minima, that is up to X of about 46 to 50, where it reaches the values from 0.0058 down to the limit 0.0023; the bound itself stays small up to about $2 \log H \approx 57$. The arithmetic test at length e^X sees zeros up to a height of order $e^{X/2}$, the usual duality between primes and zeros, and to see past the present verification one would need primes up to about 10^{25}. As a computation the arithmetic form is strictly weaker than zero verification. As a target for proof it is sharper than before: RH needs $A_X \ge 0$ uniformly in X (whether this suffices is open, Remark 19.14.2), with the prime sum, the gamma integral and the pole terms each explicit, and the whole difficulty is that the pole terms, of size $e^{(1/2-\delta)X}/X$, are cancelled by the prime sum only through the prime number theorem with an error term, which is to say through the zeros.

19.15 Open problems

Problem 19.1. Prove the hypothesis of the F_N^{sym} form of Corollary 19.12.2(c): find heights $h_N \to \infty$ such that for infinitely many N the symmetric stage F_N^{sym} has

no zeros with $\frac{1}{2} < \operatorname{Re} s < 1$ and $|\operatorname{Im} s| < h_N$, for instance $h_N = c\sqrt{N}$ with some $c > 0$. For the N tested, Subsection 19.12.3 finds $T^*(N)$ of order N for $N \le 96$, far more than needed (at $N = 128$ a pair outside the strip comes down to height 60.7, while the strip has none below 130); the sharper conjecture is $T^*(N) \ge \kappa N$. Either statement concerns explicit functions built from the integer Dirichlet polynomials D_N, and either implies RH. A mechanism for $T^*(N)$, presumably through the ratio $|D_N(\sigma+it)/D_N(1-\sigma+it)|$, is the first thing to find. The corresponding statement for the symmetrised lattice truncations K_n is Conjecture S of Subsection 17.1.1 (Theorem 17.1.7).

Problem 19.2. Theorem 13.2.7 continues the sine squared prime form below $\alpha = 1$ by compensation, and shows that the compensated sum converges for every $\alpha' > \alpha$ exactly when ζ has no zero with $\operatorname{Re} s > \alpha$; Proposition 19.14.1 gives a smoothed form of the continuation with explicit pole terms, which is not positive term by term. Is there a continuation of the prime form below $\alpha = 1$ that is positive term by term? The prime sum diverges there, but smoothed versions of it, with $n^{-\alpha}$ replaced by a weight that cuts off at $n \approx e^{1/\varepsilon}$, remain positive termwise, and the question is whether the loss from the cutoff can be controlled by $(\xi'/\xi)(\alpha)$, the gamma sum and the pole terms. A positive answer on $\alpha \in (\alpha_0, 1]$ would prove the zero free region $\beta \le \alpha_0$.

Problem 19.3. For $0 < t < 0.2$, does the law T_t of Section 19.7, half of the flow clock of Section 4.1, with transform $R_t(\lambda/2)$, exist as a probability law, and is it infinitely divisible? Unconditionally it is known to be a GGC only for $t \ge 0.2$, and to exist only for $t \ge 0.2$ and at $t = 0$, where $T_0 = T/2$; under RH it exists and is a GGC for every $t \ge 0$. This is the analogue along the flow of the existence of T (Theorem 8.1.1), and would need an explicit formula for the heat trace of the flow, which the Gaussian tilt destroys.

Remark 19.15.1. No published result gives an explicit height below which the zeros of H_t are real for $0 < t < 0.2$: [311] and [277] give $\Lambda \le 0.2$ through a barrier at $|\operatorname{Im} z| \ge y_0 > 0$, and [189] show that all but finitely many zeros are real, with no height. Real zeros do not collide forward in time (a multiple real zero splits into real zeros, as in Lemma 4.1.24, after the statement of [73] cited in Section 4.1), so the verification of RH to 3×10^{12} passes to H_t except for zeros entering from above, which a box barrier touching the real axis would exclude; that route has not been carried out (the barrier hypothesis and its consequence are Proposition 4.1.26). With a certified height H', an explicit count of the zeros of H_t and the low zeros of H_t located, the certificate of Section 8.1.8 would give about $0.76H'$ rungs, and rungs 0 and 1 (infinite divisibility and self decomposability of T_t for every $t \in [0, 0.2]$) would need only an explicit zero density bound for H_t and a small computation. Proposition 6.2.7 reduces an explicit count of the zeros of H_t to two inequalities for the heat average of ζ, which are not proved here (Problem 6.6).

Problem 19.4. Is there an analogue of Theorem 19.3.1 at $\sigma = \frac{1}{2}$, that is, a factorisation of the tilted law of T or of the Pólya variable through a law built from the primes, with the explicit formula replacing the Euler product?

19.16 Numerical checks

The numerical results of Sections 19.2 to 19.14 are recomputed. The zero counts of Table 19.5 are the most delicate; the other checks use multiprecision arithmetic, except the grids of Figure 19.2, the grid values and the gamma factors of the finite stages, and the sums and integrals of Table 19.6, which are in double precision; those of Table 19.6 are checked against the sum over the zeros, to 5×10^{-7}. Table 19.7 lists the checks.

result	test points	agreement
kernel identity of Proposition 19.2.1	six values of u	24 digits
$\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\xi(s)/(s-1)$	$s = 0.5,\ 2,\ 3+2i$	8 to 30 digits
Theorem 19.3.1, Mellin form	$\sigma = 1.5,\ 2,\ 3.5,\ t = 1,\ 5,\ 20$	30 digits
low coefficients of $\mathcal{P}_N$ equal those of ϑ_4	$N = 4, 8, 16$	exact
zero of ξ_{64} near ρ_1	$N = 64$	$\lvert \operatorname{Re} - \frac{1}{2}\rvert = 5.2 \times 10^{-7}$
zero counts of F_N^{sym} in $[-1,2] \times [0,T]$ against sign changes on the line (Table 19.5)	$N = 8,\ 16,\ 32,\ 48,\ 64,\ 96,\ 128$	windows of height 10
unit mass of the term of one zero, Proposition 19.8.1(b)	$\alpha = 0.55$	10 digits
width at half height of the term of ρ_1, Proposition 19.8.1	$\alpha = 0.55$	6 digits, up to the factor $1 + \delta^2/(2\gamma_1^2)$
γ_1^2 by Laplace inversion, Proposition 19.10.1	$t \in [0.04, 0.08]$	199.790

Table 19.7: Numerical checks of results not covered by the other tables.

19.17 The Pólya density as an extreme value law

Write

$$\Phi(u) = \sum_{n \geq 1} \phi_n(u), \qquad \phi_n(u) = \left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right) e^{-\pi n^2 e^{4u}},$$

so that $H(z) = \int_0^\infty \Phi(u)\cos(zu)\,du = \frac{1}{8}\Xi(z/2)$ (checked numerically to 10^{-88} at $z = 0$ and to 10^{-60} at $z = 150 + 150i$). In Sections 19.17 to 19.22 the kernel Φ is this one, the kernel Φ_{dBN} of Section 4.1 in the normalisation of Rodgers and

Tao, not that of (2.1.3); the zeros of H are at $z = \pm 2\gamma_k$, and $x = z/2$ is the variable of the zeros. Let $g_k(x) = e^{-kx}\exp(-e^{-x})/\Gamma(k)$, the density of $-\log G_k$ for a gamma variable G_k of shape k: for integer k the law of the kth largest point of the Gumbel limit process with intensity $e^{-x}dx$, for real k a fractional order statistic in the sense of Stigler [155]. Proposition 19.2.1 and Section 21.2 treat the maximum and the Gumbel law; here the individual terms of the theta series are decomposed.

THEOREM 19.17.1 (Gumbel decomposition). *With $x = -4u - \log(\pi n^2)$, as densities in x,*

$$\phi_n(u)\,du = \pi^{-1/4}n^{-1/2}\Gamma(\tfrac{5}{4})\left[\tfrac{5}{8}g_{9/4}(x) - \tfrac{3}{4}g_{5/4}(x)\right]dx.$$

The signed masses of the terms are $-\frac{1}{8}\pi^{-1/4}\Gamma(\frac{5}{4})n^{-1/2}$, and the formal sum of the masses, regularised by ζ, is

$$-\tfrac{1}{8}\pi^{-1/4}\Gamma(\tfrac{5}{4})\zeta(\tfrac{1}{2}) = \tfrac{1}{4}\xi(\tfrac{1}{2}) = \int_{\mathbb{R}}\Phi.$$

With $s = (1+iz)/2$,

$$\int_{\mathbb{R}}\phi_n(u)e^{izu}du = \tfrac{1}{4}n^{-s}\xi(s)/\zeta(s).$$

Proof. Put $y = e^{-x} = \pi n^2 e^{4u}$, so that $e^{9u} = (y/\pi n^2)^{9/4}$, $e^{5u} = (y/\pi n^2)^{5/4}$ and $du = -dx/4$. The coefficients become $2\pi^2 n^4(\pi n^2)^{-9/4} = 2\pi^{-1/4}n^{-1/2}$ and $3\pi n^2(\pi n^2)^{-5/4} = 3\pi^{-1/4}n^{-1/2}$, and $\Gamma(\frac{9}{4}) = \frac{5}{4}\Gamma(\frac{5}{4})$ turns $\frac{1}{4}\cdot 2\Gamma(\frac{9}{4})$ into $\frac{5}{8}\Gamma(\frac{5}{4})$. The masses are $\frac{5}{8} - \frac{3}{4} = -\frac{1}{8}$ times the common factor. The ζ regularised sum is a formal identity (the series $\sum n^{-1/2}$ diverges), and it agrees with $\int_{\mathbb{R}}\Phi = \xi(\frac{1}{2})/4$ because $\Gamma(\frac{5}{4}) = \frac{1}{4}\Gamma(\frac{1}{4})$ and $\xi(\frac{1}{2}) = -\frac{1}{8}\pi^{-1/4}\Gamma(\frac{1}{4})\zeta(\frac{1}{2})$. For the transform, the same substitution gives $(\pi n^2)^{-iz/4}$ times a combination of $\Gamma(w)$ at $w = (9+iz)/4$ and $(5+iz)/4$, which equals $n^{-s}\pi^{-s/2}s(s-1)\Gamma(s/2)/8$. □

So the Dirichlet series of ζ is a sum over translates of one Gumbel kernel along the lattice $\frac{1}{2}\log n$. The tail of the normalised law $p = 4\Phi/\xi(\frac{1}{2})$ is $S(u) \sim (2\pi/\xi(\frac{1}{2}))e^{5u}\exp(-\pi e^{4u})$ (proved, one integration), so p is in the Gumbel domain of attraction of the theorem of Fisher, Tippett and Gnedenko (Proposition 19.10.1). That the maximum of N independent copies converges at the slow rate $1/\log N$ is a heuristic by analogy with the normal law, not used. Three families of approximations follow.

- *Folded Gumbel.* $\Phi_N^F(u) = \sum_{n\le N}\phi_n(|u|)$. The sharp cutoff of the theta series at N is the omitted set $\{n > N\}$ of Chapter 18 (Section 18.2), and the cosine transform of Φ_N^F is

$$H_N^F(z) = \tfrac{1}{2}\left[F_N(z) + F_N(-z)\right],$$
$$F_N(z) = \sum_{n\le N}\left[\tfrac{1}{2}a_n^{2-w_9}\Gamma(w_9, a_n) - \tfrac{3}{4}a_n^{1-w_5}\Gamma(w_5, a_n)\right],$$

where $a_n = \pi n^2$, $w_j = (j + iz)/4$ and $\Gamma(w, a)$ is the upper incomplete gamma function. For real z this is $\operatorname{Re} F_N(z)$, but off the real axis only the symmetrised form is the transform of Φ_N^F.

- *Two sided Gumbel.* Pólya's kernel of Section 18.1 (equation (18.1.1)),

$$\Phi^P(u) = (4\pi^2 \cosh 9u - 6\pi \cosh 5u) e^{-2\pi \cosh 4u}.$$

Here $e^{-2\pi \cosh 4u} = \mathbb{P}(V \le u < W)$ for independent variables V and W with $\mathbb{P}(V \le u) = \exp(-\pi e^{-4u})$ and $\mathbb{P}(W > u) = \exp(-\pi e^{4u})$, a Gumbel maximum and a Gumbel minimum, and for real z

$$H^P(z) = \operatorname{Re}\left[\pi^2 K_{(9+iz)/4}(2\pi) - \tfrac{3\pi}{2} K_{(5+iz)/4}(2\pi)\right].$$

- *Dyck chain.* With H_n the height of a uniform Dyck path of length $2n$ and

$$Y_n = \sqrt{2/\pi}\, H_n/\sqrt{2n},$$

the Dirichlet polynomial $F_n(s) = \mathbb{E}[Y_n^s]$ tends to $2\xi(s)$ on $\operatorname{Re} s > 0$, because $\mathbb{E}[Y^s] = 2\xi(s)$ for $Y = \sqrt{2/\pi}$ times the maximum of a Brownian excursion (Biane, Pitman and Yor [33]; this is e^Y of Proposition 19.2.1) and the Dyck height scales to the excursion maximum (the height law of Dyck paths is in de Bruijn, Knuth and Rice [71]).

By [33] the normalised law p is the law of $\frac{1}{2}\log Y$ under the measure with density proportional to $Y^{1/2}$: indeed $\mathbb{E}[Y^{1/2}Y^{iz/2}]/\mathbb{E}[Y^{1/2}] = \xi(s)/\xi(\frac{1}{2})$. The Dyck chain is a Markov chain whose extremes arrive in clusters rather than as independent exceedances, as in the stationary chains of Smith, Tawn and Coles [153] (the analogy is ours). Table 19.8 lists the zeros of the approximations. The zero counts are sign changes of $H_N^F(2x)$, with signs certified on a grid of mesh 0.05.

The folded family is accurate up to a height $T_N \approx 4(N+1)^2$ in x and then fails, because of a kink at the origin. Write $R_N = \Phi - \Phi_N^F = \sum_{n>N} \phi_n$ on $u \ge 0$ and $\kappa_N = |R_N'(0)|$, the one sided slope. Its Fourier tail is of order κ_N/z^2, which overtakes $H(z) \approx |z|^{7/4} e^{-\pi z/8}$ when $x \approx 4(N+1)^2$ (heuristic: a balance of leading orders). The slope is $\kappa_N = 8.27 \cdot 10^{-8}$, $1.39 \cdot 10^{-16}$ and $2.87 \cdot 10^{-28}$ for $N = 2, 3, 4$ (computed); the leading order formula $8\pi^3(N+1)^6 e^{-\pi(N+1)^2}$ overstates it by 15%, 8% and 5%. The predicted heights are 36, 64 and 100. Measured by the first error above 10^{-2} in a tracked zero the breakdown is at $\gamma_5 = 32.9$, $\gamma_{15} = 65.1$ and $\gamma_{30} = 101.3$, and measured by the last real zero tracked it is at $x = 39.5$, 65.0 and 103.7 (computed); with error thresholds from 10^{-3} to 10^{-1} the heights lie in 30 to 38, 59 to 65 and 96 to 104, so the rule is a statement about the order of magnitude and the leading constant, not a sharp threshold.

For the census of complex zeros below, the exact law of the height H_n of the Dyck chain is used: by the reflection principle the number of Dyck paths of length

approximation	real zeros	accuracy
folded, $N = 2$	7 below $x = 70$	$3.9 \cdot 10^{-5}$ to γ_3
folded, $N = 3$	15 below $x = 100$	$9.6 \cdot 10^{-6}$ to γ_{11}
folded, $N = 4$	31 below $x = 140$	$5.8 \cdot 10^{-6}$ to γ_{25}
two sided (Pólya)	all real	first zero $x = 7.604$ against $\gamma_1 = 14.135$
Dyck chain, $n = 12800$	zero near γ_1 off the line	$0.4367 + 14.0457i$ against $0.5 + 14.1347i$
Dyck chain, Richardson in $n^{-1/2}$	off the line, to 10^{-3}	$0.50003 + 14.13473i$ (two steps)

Table 19.8: Zeros of the extreme value approximations (computed). For the Dyck chain the real parts of the zero near γ_1 for $n = 800, 1600, 3200, 6400, 12800, 25600$ are $0.2438, 0.3196, 0.3729, 0.4103, 0.4367, 0.4553$, and the extrapolation uses $n = 3200, 6400, 12800$.

$2n$ and height at most h is $\sum_{k\in\mathbb{Z}} \left[\binom{2n}{n+k(h+2)} - \binom{2n}{n-1+k(h+2)}\right]$, out of $\binom{2n}{n}/(n+1)$. At $n = 25600$, with these probabilities as balls at 256 bits, the argument principle on the boundary of $-4 < \operatorname{Re} s < 6$, $0 < \operatorname{Im} s < 60$ counts 13 zeros of the Dirichlet polynomial F_n of the Dyck chain, in the bands $0 < \operatorname{Im} s < 20$, $20 < \operatorname{Im} s < 40$ and $40 < \operatorname{Im} s < 60$ the numbers 1, 5 and 7 that the ordinates γ_k have (the number of γ_k below 60 is 13). There is no zero with $\frac{1}{2} < \operatorname{Re} s < 6$ and $0 < \operatorname{Im} s < 60$, and the real parts of the 13 zeros run from 0.0907 (the highest, at height 59.126) to 0.4553 (the lowest, at height 14.072). Followed in n from 25600 through 12800, 6400 and 3200 down to 1600, with Newton's method started at the zero found for the next larger n, these 13 zeros stay distinct and to the left of the line, and the zero of lowest height lies at distance 0.3082, 0.1544 and 0.0773 from $\frac{1}{2} + i\gamma_1$ for $n = 1600$, 6400 and 25600 (computed). The count is for $n = 25600$ only: at $n = 1600$ the same contour holds 14 zeros, and Newton's method started at $\frac{1}{2} + i\gamma_k$ does not stay near it for $k = 4, \dots, 12$ (the starts $k = 7$ and 8 converge to one zero), so the zeros are identified by following them in n. This paragraph is based on [136].

The side from which these zeros approach is set by the lattice. By de Bruijn, Knuth and Rice [71], $\mathbb{E}H_n = \sqrt{\pi n} - \frac{3}{2} + o(1)$, while $\sqrt{2n}$ times the mean of the excursion maximum $\sqrt{\pi/2}\,Y$ is $\sqrt{\pi n}$; so to first order $H_n/\sqrt{2n}$ is that maximum moved down by

$$\delta_n = \frac{3}{2\sqrt{2n}}.$$

Heuristically, the reflection formula makes $\mathbb{P}(H_n \le h)$ close to $\mathbb{P}(\sqrt{\pi n}\,Y \le h + 2)$, so that H_n behaves like $\sqrt{\pi n}\,Y - 2$ rounded up, which adds $\frac{1}{2}$ on average. For the moments this reads as follows; since $\mathbb{E}Y^{s-1} = 2\xi(s-1)$, it says that $F_n(s)$ is $\mathbb{E}(Y - \sqrt{2/\pi}\,\delta_n)^s$ to first order.

Conjecture 19.17.2. *Uniformly on every compact subset of* $\operatorname{Re} s > 0$,

$$F_n(s) = 2\xi(s) - \frac{3s}{\sqrt{\pi n}}\,\xi(s-1) + O\Big(\frac{1}{n}\Big).$$

At $s = 1$ the conjecture says $\mathbb{E}H_n = \sqrt{\pi n} - \frac{3}{2} + O(n^{-1/2})$, which agrees with de Bruijn, Knuth and Rice, and it implies that $F_n \to 2\xi$ locally uniformly on $\operatorname{Re} s > 0$. Status: open; the zeros below agree with its consequence, Proposition 19.17.3 (computed).

Proposition 19.17.3. *Assume Conjecture 19.17.2. Let ρ be a simple zero of ξ and put*

$$c_{\mathrm{D}}(\rho) = \sqrt{\frac{2}{\pi}}\,\frac{\rho\,\xi(\rho-1)}{\xi'(\rho)} = \sqrt{\frac{2}{\pi}}\,\frac{\rho\,\xi(2-\rho)}{\xi'(\rho)}.$$

There is a constant C such that for all large n, F_n has exactly one zero $z_n(\rho)$ in the disc $|s - \rho - \delta_n c_{\mathrm{D}}(\rho)| < C/n$; thus $z_n(\rho) = \rho + \delta_n c_{\mathrm{D}}(\rho) + O(1/n)$. Status: conditional.

Proof. Put $g(s) = s\,\xi(s-1)$. On a closed disc about ρ in $\operatorname{Re} s > 0$, the conjecture and Taylor's formula give $F_n(s) = 2\xi'(\rho)(s-\rho) - \frac{3}{\sqrt{\pi n}}\,g(\rho) + O\big(|s-\rho|^2 + n^{-1/2}|s-\rho| + n^{-1}\big)$. Since $\frac{3g(\rho)}{2\sqrt{\pi n}\,\xi'(\rho)} = \delta_n c_{\mathrm{D}}(\rho)$, on the circle $|s-\rho-\delta_n c_{\mathrm{D}}(\rho)| = C/n$, where $|s-\rho| = O(n^{-1/2})$, this is $2\xi'(\rho)\big(s - \rho - \delta_n c_{\mathrm{D}}(\rho)\big) + O(1/n)$. The linear term has modulus $2|\xi'(\rho)|C/n$, which exceeds the error once C is large, and Rouché's theorem gives exactly one zero inside, as in Proposition 17.1.2. The second form of c_{D} is the functional equation. □

Proposition 19.17.4. *Let $\rho = \frac{1}{2} + i\gamma$, $\gamma > 0$, be a simple zero on the critical line, with θ and Z as in Theorem 17.1.3. Then*

$$c_{\mathrm{D}}(\rho) = -\frac{C_{\mathrm{D}}(\gamma)\,e^{-i\Theta(\gamma)}}{Z'(\gamma)}, \qquad C_{\mathrm{D}}(\gamma) = \frac{\sqrt{2}}{\pi}\Big|\frac{3}{2} - i\gamma\Big|\,\Big|\frac{\Gamma(\frac{3}{4} + \frac{i\gamma}{2})}{\Gamma(\frac{1}{4} + \frac{i\gamma}{2})}\Big|\,\big|\zeta(\tfrac{3}{2} + i\gamma)\big| > 0,$$

$$\Theta(\gamma) = \theta(\gamma) + \arctan\tanh\frac{\pi\gamma}{2} + \arg\zeta(\tfrac{3}{2} + i\gamma) - \arctan\frac{3}{2\gamma},$$

so that $\operatorname{Re} c_{\mathrm{D}}(\rho)$ has the sign of $-\cos\Theta(\gamma)/Z'(\gamma)$. Status: proved.

Proof. Since $\Xi(t) = \xi(\frac{1}{2} + it) = -\frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|Z(t)$ and $Z(\gamma) = 0$, $\xi'(\rho) = -i\,\Xi'(\gamma)$ with $\Xi'(\gamma) = -\frac{1}{2}(\gamma^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|Z'(\gamma)$, and $\xi(2 - \rho) = \xi(\frac{3}{2} - i\gamma) = \overline{\xi(\frac{3}{2} + i\gamma)}$. By the definition of θ, $\pi^{-i\gamma/2}\Gamma(\frac{3}{4} + \frac{i\gamma}{2}) = |\Gamma(\frac{3}{4} + \frac{i\gamma}{2})|e^{i(\theta(\gamma)+\Delta(\gamma))}$ with $\Delta(\gamma) = \arg\Gamma(\frac{3}{4} + \frac{i\gamma}{2}) - \arg\Gamma(\frac{1}{4} + \frac{i\gamma}{2})$, and the reflection formula $\Gamma(\frac{1}{4} + it)\Gamma(\frac{3}{4} - it) = \pi/\sin(\frac{\pi}{4} + i\pi t)$ gives $\Delta(\gamma) = \arg\sin(\frac{\pi}{4} + \frac{i\pi\gamma}{2}) = \arctan\tanh\frac{\pi\gamma}{2}$. Substituting, $\rho(\frac{1}{2} - i\gamma) = \gamma^2 + \frac{1}{4}$ cancels the factor of Ξ', and

$$c_{\mathrm{D}}(\rho) = -i\,\frac{\sqrt{2}}{\pi}\Big(\frac{3}{2} - i\gamma\Big)\Big|\frac{\Gamma(\frac{3}{4} + \frac{i\gamma}{2})}{\Gamma(\frac{1}{4} + \frac{i\gamma}{2})}\Big|\,e^{-i(\theta+\Delta)}\,\frac{\overline{\zeta(\frac{3}{2} + i\gamma)}}{Z'(\gamma)};$$

writing $\frac{3}{2} - i\gamma = |\frac{3}{2} - i\gamma|\, e^{-i(\pi/2 - \arctan(3/2\gamma))}$ gives the statement. □

With S_{mid} as after Theorem 17.1.3, $\theta(\gamma_n) = \pi(n - \frac{3}{2} - S_{\mathrm{mid}})$ and $Z'(\gamma_n)$ has the sign $(-1)^{n+1}$, so $\operatorname{Re} c_{\mathrm{D}}(\rho_n) < 0$ exactly when $\sin\big(\Theta(\gamma_n) - \theta(\gamma_n) - \pi S_{\mathrm{mid}}\big) > 0$. Here $\Theta - \theta = \frac{\pi}{4} + \pi a_{\mathrm{D}}(\gamma)$, and since $|\arctan\tanh\frac{\pi\gamma}{2} - \frac{\pi}{4}| \le 2e^{-\pi\gamma}$ and, for the branch given by the Euler product, $|\arg\zeta(\frac{3}{2} + i\gamma)| \le \log\zeta(\frac{3}{2}) = 0.9603$, $|a_{\mathrm{D}}(\gamma)| \le \big(0.9603 + \frac{3}{2\gamma} + 2e^{-\pi\gamma}\big)/\pi$, below 0.35 for $\gamma \ge 14$. So $\operatorname{Re} c_{\mathrm{D}}(\rho_n) < 0$ when $-\frac{3}{4} + a_{\mathrm{D}} < S_{\mathrm{mid}} < \frac{1}{4} + a_{\mathrm{D}}$ and $\operatorname{Re} c_{\mathrm{D}}(\rho_n) > 0$ when $\frac{1}{4} + a_{\mathrm{D}} < S_{\mathrm{mid}} < \frac{5}{4} + a_{\mathrm{D}}$, modulo 2 (proved), and under Conjecture 19.17.2 the zero is then approached from the left, respectively from the right (Proposition 19.17.3): a Gram parity, as for the lattice of Section 17.1.1, with the window $(-\frac{3}{4}, \frac{1}{4})$ in place of $(-\frac{1}{2}, \frac{1}{2})$ and an arithmetic correction about three times as large. The first zero with $\operatorname{Re} c_{\mathrm{D}}(\rho_k) > 0$ is ρ_{64}, against ρ_{213} there (computed in ball arithmetic, below).

In ball arithmetic, from the first 200 zeros of the stored list, $\xi'(\rho_k) \ne 0$ for every $k \le 200$, $\operatorname{Re} c_{\mathrm{D}}(\rho_k) < 0$ for every $k \le 63$, $c_{\mathrm{D}}(\rho_1) = -6.72018 - 9.54303i$, $c_{\mathrm{D}}(\rho_{64}) = 53.08757 - 266.82638i$ (radius below 10^{-55}), and among $k \le 200$, $\operatorname{Re} c_{\mathrm{D}}(\rho_k) > 0$ exactly for $k = 64$, 92, 115, 126, 136, 160, 165, 183 and 187 (computed). At $n = 6400$ and 25600, Newton's method started at $\rho_k + \delta_n c_{\mathrm{D}}(\rho_k)$, $k \le 13$, converges to the zeros of the census above, each alone in a square of half side 0.25 by the argument principle, and the distance $|z_n(\rho_k) - \rho_k - \delta_n c_{\mathrm{D}}(\rho_k)|$ falls by a factor between 3.77 and 4.64 from $n = 6400$ to $n = 25600$, for 11 of the 13 zeros between 3.94 and 4.14, as a remainder $O(1/n)$ predicts; for ρ_1 at $n = 25600$ the law gives the real part $\frac{1}{2} - 0.04455$ against $\frac{1}{2} - 0.04470$. The zero of F_n near ρ_{64}, followed from $n = 10^8$ down to 10^5, lies to the right of the line at every computed n: $\operatorname{Re} z_n(\rho_{64}) - \frac{1}{2}$ is 0.1615, 0.1348, 0.0837, 0.0456, 0.0223, 0.0118 and 0.0061 at $n = 10^5$, $3 \cdot 10^5$, 10^6, $3 \cdot 10^6$, 10^7, $3 \cdot 10^7$ and 10^8, against the first order values $\delta_n \operatorname{Re} c_{\mathrm{D}}(\rho_{64})$ from 0.1781 down to 0.00563, and for $10^6 \le n \le 10^8$ the product $n\,|z_n(\rho_{64}) - \rho_{64} - \delta_n c_{\mathrm{D}}(\rho_{64})|$ lies between $6.33 \cdot 10^4$ and $6.61 \cdot 10^4$. The first order regime needs $\delta_n |c_{\mathrm{D}}(\rho_{64})|$ small: $|c_{\mathrm{D}}(\rho_{64})| = 272.1$, so this is 0.91 at $n = 10^5$ and 0.029 at $n = 10^8$. The neighbours ρ_{63} and ρ_{65} stay to the left, with real parts $\frac{1}{2} - 0.0283$ and $\frac{1}{2} - 0.0227$ at $n = 10^8$ (computed; Newton's method on ball enclosures of the exact law, with the terms below 10^{-110} dropped and not bounded, checked against a second law with another cut; uniqueness by the argument principle, sampled). This part is based on [142].

19.18 Smith and Tawn in pictures

The viewpoint of Smith and Tawn enters in three places. Figure 19.5 shows the second and third from stored data; the first has no stored data, and the estimates quoted for it are the source's.

The first is the model of Smith, Tawn and Coles [153]: a first order Markov chain in Gumbel margins whose transitions come from the bivariate logistic ex-

treme value law with $\alpha = \frac{1}{2}$. It is simulated by inverting the conditional distribution

$$F(y \mid x) = \exp\bigl(1/x - (x^{-1/\alpha} + y^{-1/\alpha})^{\alpha}\bigr)\bigl(1 + (x/y)^{1/\alpha}\bigr)^{\alpha-1}$$

in unit Fréchet margins (rederived from the logistic law). Exceedances of a high threshold arrive in clusters. The source reports a runs estimate of the extremal index from the plotted series of about 0.38 and a value of about 0.33 from simulating the tail chain from a very high level, so that the maximum of N steps behaves like $F^{0.33N}$.

The second is the Markov chain behind the Pólya law, a rescaled uniform Dyck path; the left panel follows the zero of its Dirichlet polynomial F_n near $\frac{1}{2} + i\gamma_1$ for $n = 100$ to 12800 (Table 19.8). The Pólya variable is the law of $\frac{1}{2}\log Y$ under the $Y^{1/2}$ biased measure (above), and the visits within two lattice steps of the maximum form one cluster. The extremes are not a limit over a long series: the whole law is a cluster functional, which is why the Pólya density is a theta function and not a generalised extreme value law.

The third is the threshold choice problem of Smith and Tawn transferred to the Thorin variable. The right panel shows its simplest form, the second atom estimated after deflating the first against the threshold x, which approaches γ_2 as the threshold rises (Section 19.19). Prony fits of $K = 6, 8, 10$ exponentials to the tail $\mathbb{P}(T > x)$ on windows starting at a threshold x_0 treat the same problem jointly. Raising the threshold removes the bias of the omitted atoms, raising K allows a lower threshold, and together they give γ_1 to 10^{-39} (Section 19.4).

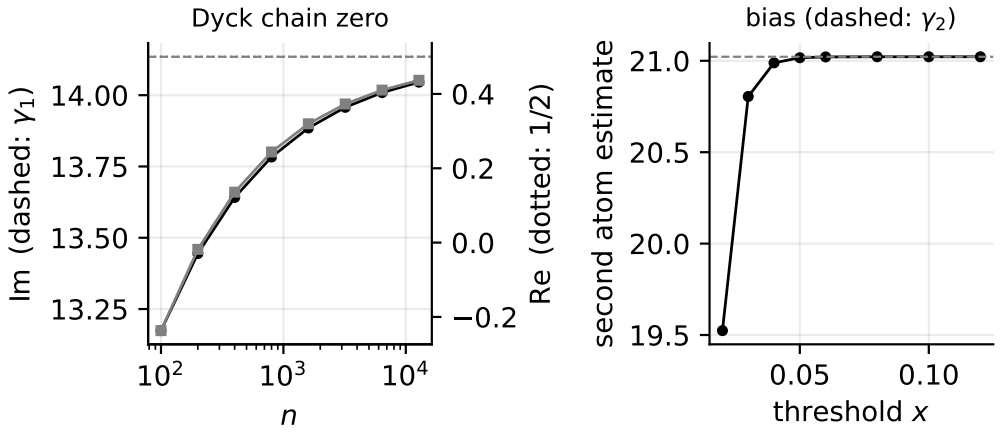


Figure 19.5: Left: real and imaginary part of the zero of the Dirichlet polynomial of the Dyck chain near γ_1 for $n = 100$ to 12800 (dashed: γ_1, dotted: the real part $\frac{1}{2}$). Right: estimate of the second atom of the hitting time law after deflating the first, against the threshold x (dashed: γ_2).

19.19 Hitting times and the Thorin variable

Kent's theorem and Krein's converse (Proposition 5.4.1, Section 5.2.3; Kent [184], Dym and McKean [78], and for generalised diffusions Kotani and Watanabe [129] and Knight [128]) are used in the following form. A string with a reflecting end and a Dirichlet end has simple spectrum, so the converse holds only for strictly increasing sequences $0 < \lambda_1 < \lambda_2 < \cdots$ with $\sum_k \lambda_k^{-1} < \infty$. And the string is not unique: the hitting time law fixes the eigenvalues and not the norming constants (a string with two point masses has three parameters modulo the natural scale and two eigenvalues, so a one parameter family has the same hitting time law). The spectral measure, eigenvalues together with weights, is what determines the string by Krein's algorithm.

Let $\widehat{H}$ be an even real entire function of order less than two with $\widehat{H}(0) > 0$, and put

$$\widehat{L}(s) = \frac{\widehat{H}(0)}{\widehat{H}(2i\sqrt{s})},$$

which for $\widehat{H} = H$ is $L(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s}) = \prod_k (1 + s/\gamma_k^2)^{-1}$. Because the order is below two, $\widehat{H}(2i\sqrt{s})$ has order below one in s and is the canonical product of genus zero over its zeros. Here $\pm\gamma_k$ are the zeros of Ξ, real exactly when the zeros of ζ lie on the critical line, in which case $0 < \gamma_1 \le \gamma_2 \le \cdots$.

Corollary 19.19.1. *Write $\pm\gamma_k$ for the zeros of Ξ, ordered by real part, so that $\frac{1}{2} + i\gamma_k$ are the nontrivial zeros of ζ (the γ_k are ordinates, real under RH). The following are equivalent.*

1. *All zeros of Ξ are real (RH).*
2. *L is the Laplace transform of a generalised gamma convolution whose Thorin measure $\sum_k \delta_{\gamma_k^2}$ is positive (Polson [279], Proposition 2.1.1).*

Moreover, all zeros of Ξ are real and simple if and only if there is a Krein string with positive mass whose hitting time T satisfies $\mathbb{E}e^{-sT} = L(s)$. The same statements hold with $\widehat{H}$ and $\widehat{L}$ in place of H and L.

Proof. The equivalence of (1) and (2) is Proposition 2.1.1 and Theorem 19.6.1. If the zeros are real and simple, the γ_k^2 are strictly increasing and Krein's converse gives a string. If a string exists, its spectrum is real, positive and simple, and by Kent's theorem $\mathbb{E}e^{-sT} = \prod_k (1 + s/\lambda_k)^{-1}$; the zeros of $\widehat{H}(2i\sqrt{s})$ in s are then the $-\lambda_k$, so the zeros of $\widehat{H}$ are real and simple. The same argument covers $\widehat{H}$. □

So the zeros are real and simple exactly when the string exists (Proposition 5.4.1 also assumes simple zeros; simplicity of the zeros of Ξ is not known). Two features of the string are visible without exhibiting it.

Proposition 19.19.2 (Principal eigenvalue from the tail). *If RH holds, the zeros are simple and* $|\xi'(\rho)| \geq e^{-C|\gamma|}$ *for some constant* C *and all zeros* $\rho = \frac{1}{2} + i\gamma$ *(the hypothesis of Theorem 5.3.2), then* $\mathbb{P}(T > x) = \sum_k c_k e^{-\gamma_k^2 x}$ *with* $c_k = \prod_{j \neq k}(1 - \gamma_k^2/\gamma_j^2)^{-1} = 2i\xi(\frac{1}{2})/(\gamma_k \xi'(\frac{1}{2} + i\gamma_k))$*. In particular the excess of* T *over a high threshold is asymptotically exponential with rate* $\lambda_1 = \gamma_1^2$*, the bottom of the spectrum of the string; this last clause needs only the two term tail of Proposition 5.3.3, which has no hypothesis on* ξ'*.*

This is Proposition 5.3.3 and the hazard rate of Section 5.2.1 in the present notation. The exponential excess follows from the expansion and is analogous to the peaks over threshold theorem of Pickands and of Balkema and de Haan [149, 13], which is not used. The estimator is the hazard rate. At a threshold x the ratio of the density to the tail, $f(x)/\mathbb{P}(T > x)$, tends to γ_1^2, and the Laplace inversion of L (the method of Talbot, at 100 and 130 nodes, agreeing to 20 digits) uses no zero. On the window $x \in [0.18, 0.20]$ the square root of this ratio is $14.1347251417346937\ldots$ at all three of $x = 0.18, 0.19, 0.20$ (the values are 14.13472514173469373756, $\ldots 3785760$ and $\ldots 3790040$ against $\gamma_1 = 14.13472514173469379045\ldots$), so γ_1 is recovered to eighteen significant digits (computed). The threshold bias for the later atoms is the point of Smith and Tawn: deflating by the first atom, the second atom is estimated as 21.0167, 21.0212 and 21.02204 from the thresholds $x = 0.05$, 0.06 and 0.1, against $\gamma_2 = 21.02204$ (computed). Figure 19.6 shows the error of these estimates of γ_1.

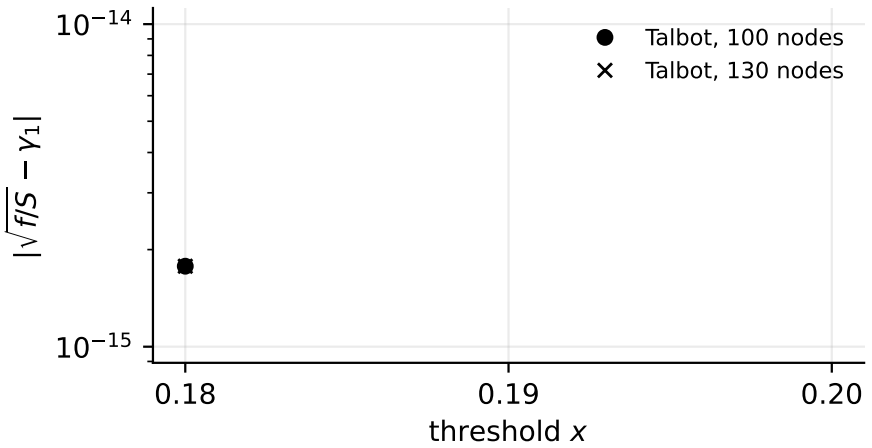


Figure 19.6: Error of the estimate $\sqrt{f(x)/\mathbb{P}(T > x)}$ of γ_1 from the Laplace inversion of L (Talbot, 100 and 130 nodes) at $x = 0.18, 0.19, 0.20$.

19.20 B splines, Hadamard products and Thorin

Is there a B spline view of the Gumbel law that leads to a Hadamard product and from there to a Thorin representation? The first two steps are known. By Rényi's representation the maximum of n independent exponentials is $M_n = \sum_{k\le n} E_k/k$, a generalised gamma convolution with Thorin measure $\sum_{k\le n}\delta_k$ and the passage time of a pure death chain; $M_n - \log n$ tends to the Gumbel law G, with $\mathbb{E}e^{-sG} = \Gamma(1+s) = \exp\big(-\gamma_E s - \sum_{k\ge1}[\log(1+s/k) - s/k]\big)$, a compensated Thorin representation with atoms at the integers (the compensation is needed because $\sum 1/k$ diverges and is what the centring $\log n$ removes). The density of M_n is, up to the factor y^{n-1} and a Laplace transform, a B spline with the knots $1,\dots,n$, and the same holds for the truncations of the reciprocal ξ law with knots $\gamma_1^2,\dots,\gamma_K^2$ (Proposition 21.2.1 and Section 21.2, which also give the Dirichlet mean form; the density of a sum of exponentials is the divided difference of $e^{-\lambda x}$ in λ, which is the B spline only after this transform). If RH holds, the truncated laws shifted by the tail mean $r_K = \sum_{k>K}\gamma_k^{-2}$ converge to the reciprocal ξ law, since the difference has mean zero and variance $\sum_{k>K}\gamma_k^{-4} \to 0$; no compensation is needed because $\sum_k \gamma_k^{-2} < \infty$ (the shift is the tail shift of the Kent approximants of Section 5.2.3).

Figure 19.7 shows the supremum distances against K. For the reciprocal Ξ law the supremum distance between the shifted truncation and the density obtained by Laplace inversion is 128 and 42 for $K = 1, 2$ (attained at the knot r_K, where the truncated density jumps or ends), and 8.1, 0.80 and 0.23 for $K = 5, 20, 40$, against a peak density of 80 (computed; on a uniform grid of mesh $5\cdot10^{-4}$, which misses the knot, they are 110 and 36, and the source printed 114 and 37).

For the Pólya density the B spline view gives the following.

1. Each Gumbel piece of Theorem 19.17.1 has its own exact chain. The piece g_k is the law of $-\log G_k$, so $\mathbb{E}e^{-sX} = \mathbb{E}G_k^s = \Gamma(k+s)/\Gamma(k)$, and the Weierstrass product gives a compensated Thorin form with atoms at $k, k+1, k+2, \dots$. For the Pólya pieces these are $\frac{9}{4}+j$ and $\frac{5}{4}+j$ (proved).

2. In $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ this Gumbel structure is the archimedean factor $\Gamma(s/2)$. Its poles are at $s = 0, -2, -4, \dots$; the pole at 0 is cancelled by the factor s and the others by the trivial zeros $-2, -4, \dots$ of ζ, so the knots of the Gumbel view correspond to the trivial zeros (proved).

3. The atoms at γ_k^2 do not come from any product of Gumbel pieces. They come from the theta superposition $\Phi = \sum_n \phi_n$, which translates one Gumbel kernel along the lattice $\frac12\log n$ and turns the Gamma factor into $\Gamma(s/2)\zeta(s)$ (Theorem 19.17.1). A superposition is a sum, not a product, and a sum does not preserve Hadamard products or Thorin form (a heuristic remark: it explains why the argument stops, and proves nothing).

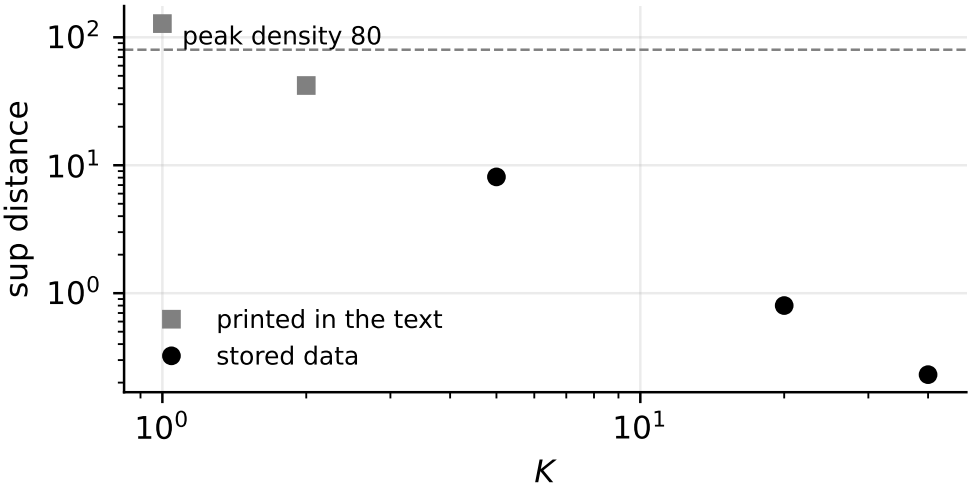


Figure 19.7: Supremum distance between the mean corrected truncation with knots $\gamma_1^2, \ldots, \gamma_K^2$ and the density of the reciprocal Ξ law, against K (dashed: the peak density 80). The values for $K = 5, 20, 40$ are stored data, those for $K = 1, 2$ are the printed values above.

So the B spline view leads to a Hadamard product and a Thorin representation for every Gumbel piece unconditionally and for the reciprocal Ξ law given the zeros. The step that would make it a route to RH, that the superposition of Gumbel pieces still has a positive Thorin measure, is the open question of this chapter in another form.

19.21 An explicit string: the Liouville operator

Pólya proved that his two sided kernels have only real zeros [296, 302]; the spectral reading is Titchmarsh's [347] and the hitting time form Biane's [32] (Section 18.1). For $a > 0$ the function $g(\tau) = K_\nu(ae^\tau)$ solves $g'' - a^2 e^{2\tau} g = \nu^2 g$, the real zeros z_k of $z \mapsto K_{iz}(a)$ are the square roots of the Dirichlet eigenvalues of $-g'' + a^2 e^{2\tau} g$ on $(0, \infty)$ with $g(0) = 0$, and the diffusion obtained by the Doob transform of this operator by its positive solution $K_0(ae^\tau)$, started at the entrance boundary at $+\infty$, has hitting time T_0 of 0 with

$$\mathbb{E} e^{-sT_0} = \frac{K_0(a)}{K_{\sqrt{s}}(a)} = \prod_k \left(1 + \frac{s}{z_k^2}\right)^{-1}$$

(Proposition 18.1.7 and Theorem 18.1.10, proved, with simple spectrum). The product form uses that $K_{\sqrt{s}}(a)$ has order one half in s, because

$|K_\nu(a)| \le K_{|\operatorname{Re}\nu|}(a) \le C\,\Gamma(|\nu|+1)(2/a)^{|\nu|}$, so Hadamard's factorisation has genus zero with no exponential factor.

A second order finite difference discretisation of the operator on $(0,6)$ with $a = 2\pi$ and about 64000 intervals (the size at which the digits of the source are reproduced) gives $\sqrt{\lambda_k} = 9.768770,\ 12.448488,\ 14.684960,\ 16.691578,\ 18.549374$, and the zeros of $K_{iz}(2\pi)$ are $9.768770,\ 12.448488,\ 14.684960,\ 16.691579,\ 18.549375$ (computed; with certified signs of the Bessel values, bisection gives $9.76877008,\ 12.44848789,\ 14.68495978,\ 16.69157894,\ 18.54937520$).

Pólya's kernel involves $K_{(9\pm iz)/4}(2\pi)$, not $K_{iz}(2\pi)$; he passes between them with the shift lemma: if G is real entire of genus at most one with only real zeros, so is $\frac{1}{2}[G(z+ic)+G(z-ic)]$, which is the transform of $\cosh(cu)$ times the original kernel. The lemma deals with one Bessel term, and the combination of the two terms of Φ^P is handled in the refinement of Section 22.3 (which also records that the kernel with the second exponential term kept is real rooted). The lemma preserves real zeros, hence by Corollary 19.19.1 the existence of a string when the zeros are simple, but does not say what the new string is; writing it explicitly is Problem 2 of Section 19.3.

19.22 The string test from kernel moments

For a general approximation the string is not known, but whether the zeros are real, the first condition for it to exist, can be tested from the kernel alone. Expand

$$\widehat{H}(2i\sqrt{s}) = \int_0^\infty \widehat{\Phi}(u)\cosh(2\sqrt{s}\,u)\,du = \sum_{m\ge 0} \frac{4^m M_{2m}}{(2m)!}s^m,$$

$$M_{2m} = \int_0^\infty u^{2m}\widehat{\Phi}(u)\,du.$$

Taking logarithms gives $\log[\widehat{H}(2i\sqrt{s})/\widehat{H}(0)] = \sum_j (-1)^{j+1}\mu_j s^j/j$ with

$$\mu_j = \sum_k \widehat{\gamma}_k^{-2j},$$

the power sums of the inverse Thorin atoms, equivalently $\kappa_j(T)/(j-1)!$ for the cumulants $\kappa_j(T)$ of the candidate hitting time. Put $w_k = \widehat{\gamma}_k^{-2}$ and $\nu = \sum_k w_k\delta_{w_k}$, so that $\mu_{j+1} = \int y^j \nu(dy)$ whenever the w_k are positive.

Proposition 19.22.1 (Hankel test). *Let $\widehat{H}$ be as in Section 19.19, with infinitely many zeros and $\sum_k |w_k| < \infty$. The zeros of $\widehat{H}$ are all real (and nonzero) if and only if all the Hankel determinants*

$$D_n = \det[\mu_{i+j+1}]_{i,j=0}^{n-1}, \qquad D_n' = \det[\mu_{i+j+2}]_{i,j=0}^{n-1}, \qquad n \ge 1,$$

are positive.

Proof. If the zeros are real, ν is a positive measure on $(0,\infty)$ and the determinants are those of the Stieltjes moment problem, hence positive. Conversely, if all are positive, the sequence (μ_{j+1}) is a Stieltjes moment sequence. It grows at most like ρ^j with $\rho = \max_k |w_k|$, so the representing measure ν' is unique and supported in $[0,\rho]$, and its Stieltjes transform $\int \nu'(dy)/(1-zy)$ is analytic off $[1/\rho,\infty)$ and equals $\sum_{j\ge 0} \mu_{j+1} z^j$ near 0. That series is also $\sum_k w_k/(1-zw_k)$, a meromorphic function (the sum converges locally uniformly off the poles because $\sum |w_k| < \infty$) with poles exactly at the points $1/w_k$ and residues that do not cancel, since terms with equal w_k add with the same sign and distinct w_k give distinct poles. By analytic continuation the poles lie in $[1/\rho,\infty)$, so every w_k is real and positive. □

The continued fraction coefficients built from D_n and D'_n are the masses and lengths of a discrete string, a birth and death chain whose hitting time matches the first $2n$ cumulants (classical, Stieltjes and Krein; the wording of the matching was not checked). So the test is a hitting time calculation: build the chain and check that its rates are positive. It is a finite form of the Stieltjes criterion of Theorem 19.6.1, and the moment matching along the Kent approximants is in Section 5.2.4.

The moments M_{2m} were computed by quadrature for the exact kernel, the two term folded kernel and Pólya's two term kernel, then μ_j, then D_n and D'_n. At 180 digits all of $D_1,\dots,D_{11}$ and $D'_1,\dots,D'_{11}$ are positive for all three kernels, with the sign of each determinant certified by ball arithmetic from the computed moments (the quadrature error of the moments is not certified; the relative change between two node counts is below 10^{-89}; computed). For example $D_{11} = 8.99\cdot 10^{-368}$ for the exact kernel, $9.52\cdot 10^{-368}$ for the folded kernel and $1.14\cdot 10^{-350}$ for Pólya's, and $D_6 = 8.1\cdot 10^{-100}$ for the exact kernel. The first moment is $\mu_1 = \sum_k \gamma_k^{-2} = 0.0231049931154190$ for the exact kernel, which equals $\xi''(\frac12)/(2\xi(\frac12))$ computed independently by a Cauchy integral; for the two term folded kernel it is 0.0231049931181204, and for Pólya's kernel 0.0363173599127934.

The positive result for the folded kernel, which has nonreal zeros above height about $x = 40$ (Table 19.8), shows the limitation of the test. A Hankel matrix of order n sees roughly the first n atoms, and the weights of the atoms decay like $\widehat{\gamma}^{-2j}$, so an off line pair at height x needs an order beyond the number of real atoms below x, and precision of the order of $4n\log_{10}(x/\gamma_1)$ digits, before it can make a determinant negative (heuristic; the masking of the first pair is measured in Section 19.4). Finite order positivity certifies a positive string for a truncated moment problem, which is the analogue of verifying RH to a finite height, and it mirrors Li's criterion [208], where finitely many positive coefficients prove nothing about RH.

Chapter 20

Ferromagnetic random variables

A symmetric random variable is ferromagnetic if its moment generating function vanishes only on the imaginary axis. By (2.1.4) the moment generating function of the Pólya variable X is $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, so RH is the statement that one explicit law, the Pólya law with density $p = \Phi/\Xi(0)$, is ferromagnetic.

With no hypothesis, the Pólya law lies within total variation distance 6.1×10^{-6} of a law that is ferromagnetic by an explicit construction, and within 2×10^{-6} of one whose positivity, and hence ferromagnetism, rests on a computer assisted proof in ball arithmetic; these ferromagnetic approximations are in Volume II.

We use the units of (2.1.3) throughout, in which the zeros of the characteristic function of X are the ordinates γ and $\operatorname{Var} X = 2\mathbb{E}T = 0.0462099862$. The normalisation of Rodgers and Tao in Section 4.1 doubles the variable of the kernel; quoted statements have been converted. As in Section 2.1, Z is a standard normal variable, the e_k are standard exponential variables, the L_k are standard Laplace variables, ε and the ε_k are fair random signs, U is uniform on $[-1,1]$, and all of them are independent of each other and of everything else.

20.1 Ferromagnetic laws and their clocks

20.1.1 Definition and background

Definition 20.1.1. A real random variable X is *ferromagnetic* if it is symmetric, $\mathbb{E}e^{b_0X^2} < \infty$ for some $b_0 > 0$, and its moment generating function $L_X(z) = \mathbb{E}e^{zX}$, which is then entire, vanishes only on the imaginary axis. Equivalently, every zero of the characteristic function $\varphi_X(\theta) = \mathbb{E}e^{i\theta X} = L_X(i\theta)$ is real.

These are Newman's laws of type $\mathcal{L}$ [241]. The name comes from statistical mechanics. Lee and Yang proved that the partition function of an Ising ferromagnet with spins ±1 and a common complex field vanishes only when the field is purely imaginary [206], and Newman showed that a symmetric single spin law with Gaussian moments of all orders has this property in every ferromagnetic system exactly when its own moment generating function has only imaginary zeros [240];

Lieb and Sokal gave another proof and extended the theorem to spins with several components [211]. Section 11.1.3 recalls these results, and Proposition 11.1.16 restates Newman's theorem in terms of Thorin measures. Two smaller classes carry the correlation inequalities of the Ising model. The Griffiths Simon class consists of the weak limits of suitably scaled sums of ferromagnetically coupled Ising spins; it contains the laws with density proportional to $e^{-ax^4-bx^2}$, $a>0$, and its laws inherit the Lee and Yang property and the inequalities of Griffiths, Kelly and Sherman and of Griffiths, Hurst and Sherman [332, 98]. The class of Ellis, Monroe and Newman consists of the laws with density $e^{-V(x)}$, V even and V' convex on $[0,\infty)$, and their weak limits; the Griffiths, Hurst and Sherman inequality holds for ferromagnets with such single spin laws [82].

20.1.2 The Hadamard form and the cumulants

Theorem 20.1.2 (Newman [241]). *Let X be symmetric with $\mathbb{E}e^{b_0X^2}<\infty$ for some $b_0>0$. Then X is ferromagnetic if and only if*

$$L_X(s)=e^{bs^2}\prod_k\Big(1+\frac{s^2}{c_k^2}\Big),\qquad b\ge 0,\quad c_k>0,\quad \sum_k c_k^{-2}<\infty, \tag{20.1.1}$$

with each c_k repeated according to the multiplicity of the zero ic_k. If $\mathbb{E}e^{bX^2}<\infty$ for every $b>0$, then $b=0$.

Proof. Since $|L_X(z)|\le \mathbb{E}e^{|\operatorname{Re} z||X|}\le Ce^{|z|^2/4b_0}$, the function φ_X is entire of order at most 2, even, real on $\mathbb{R}$, with $\varphi_X(0)=1$. Suppose its zeros are real, at $\pm c_k$ with $c_k>0$. Hadamard's theorem, with the primary factors of genus 2 paired over $\pm c_k$, gives $\varphi_X(\theta)=e^{-b'\theta^2}\prod_k(1-\theta^2/c_k^2)e^{\theta^2/c_k^2}$ with b' real and $\sum_k c_k^{-4}<\infty$, so that, for real s,

$$\frac{\log L_X(s)}{s^2}=b'+\sum_k\frac{1}{c_k^2}\,\frac{\log(1+x_k)-x_k}{x_k},\qquad x_k=\frac{s^2}{c_k^2}.$$

The function $(\log(1+x)-x)/x$ decreases from 0 to -1 on $(0,\infty)$, so by monotone convergence the sum tends to $-\sum_k c_k^{-2}$ as $s\to\infty$. If this sum diverged, $L_X(s)$ would tend to 0, which is impossible because $L_X(s)=\mathbb{E}\cosh(sX)\ge 1$. Hence $\sum_k c_k^{-2}<\infty$, the factors e^{θ^2/c_k^2} can be collected into the exponential, and (20.1.1) holds with $b=b'-\sum_k c_k^{-2}$. Then $\log L_X(s)=bs^2+o(s^2)$, since $\log(1+x)/x$ is bounded by 1 and tends to 0, and $L_X\ge 1$ forces $b\ge 0$. The converse is clear, since the right side of (20.1.1) vanishes only at $\pm ic_k$. If $\mathbb{E}e^{bX^2}<\infty$ for every b, then $L_X(s)\le C_\epsilon e^{\epsilon s^2}$ for every $\epsilon>0$, so $b=0$, as in the proof of Proposition 11.1.16. □

Corollary 20.1.3 (Newman [241]). *If X is ferromagnetic, with (20.1.1), then* $\operatorname{Var}X=2b+2\sum_k c_k^{-2}$, *and for $m\ge 2$*

$$\frac{\kappa_{2m}(X)}{(2m)!}=\frac{(-1)^{m+1}}{m}\sum_k c_k^{-2m}. \tag{20.1.2}$$

So the even cumulants alternate in sign: $\kappa_4 \le 0$, $\kappa_6 \ge 0$, and so on. Moreover $\mathbb{E}e^{sX} \le e^{s^2 \operatorname{Var} X/2}$ for real s, and $\mathbb{E}X^4 \le 3(\operatorname{Var} X)^2$.

Proof. The first two statements are the expansion of $\log L_X(s) = bs^2 + \sum_k \log(1 + s^2/c_k^2)$. The bound on $\mathbb{E}e^{sX}$ follows from $\log(1+x) \le x$, and the bound on $\mathbb{E}X^4 = \kappa_4 + 3\kappa_2^2$ from $\kappa_4 \le 0$. □

For the Pólya law the alternation holds with no hypothesis: by the formula for $\log G$ in Section 8.1, $(-1)^{n-1}\kappa_{2n}(X) = \frac{(2n)!}{n!}\int_0^\infty t^{n-1}W(t)\,dt > 0$ for every n (Section 4.1), and under RH this is (20.1.2) with $c_k = \gamma_k$ and $b = 0$.

20.1.3 The clock of a ferromagnetic law

Theorem 20.1.4. *Let X be ferromagnetic, with* (20.1.1)*, and put*

$$T_X = b + \sum_k \frac{e_k}{c_k^2}, \qquad Y_X = Z\sqrt{2T_X},$$

the series converging almost surely and in mean.

(i) $\mathbb{E}e^{-uT_X} = 1/L_X(\sqrt{u})$ *for $u \ge 0$, and T_X is a GGC with drift b and Thorin measure $\sum_k \delta_{c_k^2}$, one unit atom for each pair of zeros $\pm ic_k$.*

(ii) $\mathbb{E}e^{i\theta Y_X} = 1/L_X(\theta)$ *for real θ, so that $(\varphi_X, \varphi_{Y_X})$ is a van Dantzig pair, and $Y_X \overset{d}{=} \sqrt{2b}\,Z + \sum_k L_k/c_k$.*

(iii) The moment generating function of Y_X is $1/\varphi_X(t)$ for $|t| < c_1$ and infinite for $|t| \ge c_1$, unless X is Gaussian.

(iv) Unless X is degenerate, Y_X has a Pólya frequency density of infinite order.

(v) $\kappa_m(T_X) = (-1)^{m+1}m!\,\kappa_{2m}(X)/(2m)!$ *for $m \ge 1$; in particular* $\mathbb{E}T_X = \frac{1}{2}\operatorname{Var} X$ *and* $\operatorname{Var} T_X = -\kappa_4(X)/12$.

Proof. (i) $\mathbb{E}e^{-ue_k/c_k^2} = (1 + u/c_k^2)^{-1}$; multiply over k and compare with (20.1.1) at $s = \sqrt{u}$. (ii) Conditionally on T_X, Y_X is centred normal with variance $2T_X$, so $\mathbb{E}e^{i\theta Y_X} = \mathbb{E}e^{-\theta^2 T_X} = 1/L_X(\theta)$, which is also the characteristic function of $\sqrt{2b}\,Z + \sum_k L_k/c_k$, since $\mathbb{E}e^{i\theta L} = (1+\theta^2)^{-1}$. (iii) $\mathbb{E}e^{tY_X} = \mathbb{E}e^{t^2T_X} = e^{bt^2}\prod_k(1 - t^2/c_k^2)^{-1}$ while $t^2 < c_1^2$, and $\mathbb{E}e^{t^2 e_1/c_1^2} = \infty$ for $t^2 \ge c_1^2$. (iv) The Laplace density $\frac{1}{2}e^{-|x|}$ and the normal densities are Pólya frequency functions of infinite order, and the class is closed under convolution and under the limits involved here. Equivalently, the bilateral Laplace transform of Y_X is the reciprocal of $\varphi_X(s) = e^{-bs^2}\prod_k(1 - s^2/c_k^2)$, which has Schoenberg's form [328]. (v) For u near 0, $\log \mathbb{E}e^{-uT_X} = -\log L_X(\sqrt{u}) = -\sum_m \kappa_{2m}(X)u^m/(2m)!$; compare coefficients with $\sum_m \kappa_m(T_X)(-u)^m/m!$. □

Part (i) is Proposition 11.1.16 with the Gaussian factor allowed; the series of exponential variables with its atomic Thorin measure also appears in [281]. For the Pólya law under RH, $b = 0$, $c_k = \gamma_k$, T_X is the clock T of Proposition 2.1.1, and $(\varphi_X, \varphi_{Y_X})$ is the van Dantzig pair of Section 2.1, with $Y_X = Z\sqrt{2T}$ the variable $Y_{1/2}$ of Proposition 13.1.1. Without RH the clock T and the pair still exist (Theorem 8.1.1); what fails is the product form of the clock. Konstantopoulos, Patie and Sarkar study such van Dantzig pairs through the Laguerre Pólya class and Pólya frequency functions, and connect them with the Lee and Yang property [192]. The correspondence runs one way only: every ferromagnetic law gives a Pólya frequency law, but $\varphi(s) = 1 - s^2$, the reciprocal of the bilateral Laplace transform of the Laplace law, is not a characteristic function.

Part (v) holds for the Pólya law with no hypothesis, with the clock T of Theorem 8.1.1 in place of T_X, since the proof uses only $\log \mathbb{E}e^{-uT} = -\log G(u) = -\log \mathbb{E}e^{\sqrt{u}X}$. With the cumulants of Table 20.3 it gives $\mathbb{E}T = 0.0231049931$ and $\operatorname{Var} T = -\kappa_4/12 = 3.71725993 \times 10^{-5}$, the constants of Section 2.1.

20.1.4 Examples

Table 20.1 lists the basic ferromagnetic laws with their clocks. The variables are those of Biane, Pitman and Yor [33]: C is the exit time of Brownian motion from $(-1, 1)$, with $\mathbb{E}e^{-uC} = 1/\cosh\sqrt{2u}$, and J is the hitting time of 1 by a Bessel process of dimension 3 started at 0, with $\mathbb{E}e^{-uJ} = \sqrt{2u}/\sinh\sqrt{2u}$, as in Section 12.1.1. The first two rows are the classical van Dantzig pairs of Section 2.1.

law of X	$L_X(s)$	c_k	clock T_X	Y_X
fair sign ε	$\cosh s$	$(k - \frac{1}{2})\pi$	$C/2$	hyperbolic secant
uniform on $[-1, 1]$	$\sinh s/s$	$k\pi$	$J/2$	logistic, scale $1/\pi$
$c_\nu(1 - x^2)^{\nu - 1/2}$, $\nu > -\frac{1}{2}$	$\Gamma(\nu + 1)(2/s)^\nu I_\nu(s)$	$j_{\nu,k}$	$\frac{1}{2}\tau_\nu$	Bessel type
$N(0, \sigma^2)$	$e^{\sigma^2 s^2/2}$	none	$\sigma^2/2$	$N(0, \sigma^2)$
$e^{-2\pi\cosh 2x}/K_0(2\pi)$	$K_{s/2}(2\pi)/K_0(2\pi)$	$2\nu_k$	$T_P/4$	$Z\sqrt{T_P/2}$
Pólya, under RH	$\xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$	γ_k	T	$Z\sqrt{2T}$

Table 20.1: Ferromagnetic laws and their clocks. Here $j_{\nu,k}$ are the positive zeros of the Bessel function J_ν, and τ_ν is the hitting time of 1 by a Bessel process of dimension $2\nu + 2$ started at 0. In the fifth row ν_k are the zeros of $\nu \mapsto K_{i\nu}(2\pi)$ and T_P is the clock of Theorem 18.1.9.

The third row rests on Kent's theorem [183]: for the Bessel process of dimension $2\nu + 2$ started at 0, the hitting time of 1 satisfies $\mathbb{E}e^{-u\tau_\nu} = (\sqrt{2u})^\nu/(2^\nu\Gamma(\nu + 1)I_\nu(\sqrt{2u}))$, so $\mathbb{E}e^{-u\tau_\nu/2} = 1/L_X(\sqrt{u})$ and $T_X = \frac{1}{2}\tau_\nu = \sum_k e_k/j_{\nu,k}^2$. The zeros $j_{\nu,k}$ are real for $\nu > -1$, so every symmetric beta law $c_\nu(1 - x^2)^{\nu - 1/2}$ with $\nu > -\frac{1}{2}$ is ferromagnetic. The family contains the arcsine law ($\nu = 0$), the uniform law ($\nu = \frac{1}{2}$) and the semicircle law ($\nu = 1$), and tends to the law of ε as $\nu \downarrow -\frac{1}{2}$, where

$\tau_{-1/2} = C$.

20.1.5 Closure

Products of functions of the form (20.1.1) have the same form, so sums of independent ferromagnetic variables are ferromagnetic. The class is closed under convergence in law, which is why Definition 20.1.1 allows the Gaussian factor: the laws of $n^{-1/2}\sum_{k\le n}\varepsilon_k$ have Gaussian moments of all orders, and their limit $N(0,1)$ does not.

Theorem 20.1.5. *Let X_n be ferromagnetic and suppose that $X_n \to X$ in law. Then X is ferromagnetic,* $\operatorname{Var} X_n \to \operatorname{Var} X$*, and $L_{X_n} \to L_X$ locally uniformly on* $\mathbb{C}$.

Proof. Let $\sigma_n^2 = \operatorname{Var} X_n$. By Corollary 20.1.3, $\mathbb{E}X_n^4 \le 3\sigma_n^4$, so the Paley Zygmund inequality applied to X_n^2 gives $P(X_n^2 > \sigma_n^2/4) \ge (\frac{3}{4})^2\sigma_n^4/\mathbb{E}X_n^4 \ge \frac{3}{16}$. A sequence that converges in law is tight, so the σ_n are bounded, say by σ. By Corollary 20.1.3 again, $|L_{X_n}(z)| \le L_{X_n}(\operatorname{Re} z) \le e^{\sigma^2(\operatorname{Re} z)^2/2}$, so the L_{X_n} form a locally bounded, hence normal, family of entire functions, and for each real s the variables e^{sX_n} are uniformly integrable, since their second moments are bounded. Hence $\mathbb{E}e^{sX_n} \to \mathbb{E}e^{sX}$ for real s, which gives $\mathbb{E}e^{sX} \le e^{\sigma^2 s^2/2}$ and, averaging over $s = \sqrt{2b}Z$, $\mathbb{E}e^{bX^2} < \infty$ for $b < 1/2\sigma^2$. So L_X is entire, every locally uniform limit point of the L_{X_n} agrees with it on $\mathbb{R}$, and by Vitali's theorem $L_{X_n} \to L_X$ locally uniformly. Each L_{X_n} has no zero in $\operatorname{Re} z > 0$ and $L_X(0) = 1$, so by Hurwitz's theorem L_X has no zero there either; by symmetry it has none in $\operatorname{Re} z < 0$. The fourth moments are bounded, so the X_n^2 are uniformly integrable and the variances converge. □

Newman and Wu proved this closure [245].

Proposition 20.1.6. *Let X be ferromagnetic with law ρ, and let $\mathbb{E}e^{b_0X^2} < \infty$.*

(i) For every $a > 0$ the law $\cosh(ax)\,\rho(dx)/\mathbb{E}\cosh(aX)$ is ferromagnetic.

(ii) For every $0 < \lambda < b_0$ the law $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$ is ferromagnetic.

Proof. (i) The moment generating function of the tilted law is proportional to $L_X(z+a)+L_X(z-a)$. For $\operatorname{Re} w > 0$ and $a > 0$ we have $|w+a| > |w-a|$. Applied to $w = z \pm ic_k$ this gives $|c_k^2 + (z+a)^2| > |c_k^2 + (z-a)^2|$ when $\operatorname{Re} z > 0$, and $|e^{b(z+a)^2}| = e^{4ab\operatorname{Re} z}|e^{b(z-a)^2}|$. So for $\operatorname{Re} z > 0$ every factor of (20.1.1) has larger modulus at $z+a$ than at $z-a$, and $L_X(z+a) + L_X(z-a) \ne 0$ unless L_X is constant, in which case the sum is 2. The sum is even in z, so it has no zeros with $\operatorname{Re} z < 0$ either, and the tilted law satisfies the moment condition because $\cosh(ax) \le e^{a|x|}$. (ii) By (i) applied n times, the law with density proportional to $\cosh(x\sqrt{2\lambda/n})^n$ with respect to ρ is ferromagnetic. Since $\cosh y \le e^{y^2/2}$, these densities are dominated by the ρ integrable function $e^{\lambda x^2}$ and converge to it,

so the laws converge in total variation to $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$, and Theorem 20.1.5 applies. □

Part (i) is the one spin case of the Lee and Yang theorem, and for the Pólya law, whose transform has genus 1, it is Pólya's Hilfssatz II of Section 11.1.3. For the Pólya law, part (ii) is the forward half of de Bruijn's theorem [73]: multiplying p by $e^{\lambda x^2/4}$ is the de Bruijn Newman flow at time λ (Section 4.1).

Remark 20.1.7. Even polynomial factors with imaginary zeros also preserve the class. If X is ferromagnetic with law ρ and $r > 0$, then $(x^2+r^2)\rho(dx)/\mathbb{E}(X^2+r^2)$ is ferromagnetic. It is symmetric with a Gaussian moment, and its characteristic function is proportional to $r^2\varphi_X - \varphi_X'' = (r-D)(r+D)\varphi_X$, $D = d/d\theta$; by (20.1.1) φ_X is in the Laguerre Pólya class [207], and each factor $r \pm D$ preserves that class by Laguerre's theorem, as in the proof of Proposition 19.1.17. Products of such factors are the polynomial case of the multipliers $\psi(it)$ of Proposition 19.1.8. This remark is based on [138].

Corollary 20.1.8. *RH holds if and only if the Pólya law is a limit in law of ferromagnetic laws.*

Proof. If RH holds, the Pólya law is ferromagnetic by (2.1.4). Conversely, a limit in law of ferromagnetic laws is ferromagnetic by Theorem 20.1.5, and then every zero of Ξ is real. □

Let δ be the infimum of the total variation distances $\sup_A |P(X \in A) - P(Y \in A)|$ from the Pólya variable X to ferromagnetic variables Y; when Y has a density f the distance is $\mathrm{TV}(f,p) = \frac{1}{2}\int |f-p|$. If RH holds, $\delta = 0$; conversely, ferromagnetic laws that converge to the Pólya law in total variation converge in law, so $\delta = 0$ implies RH.

20.2 Laws on an interval

The Pólya law is symmetric and unimodal, hence a mixture of uniform laws. For laws on an interval we ask which mixtures are ferromagnetic. Mixing towards the centre can destroy the property, as an atom at the origin shows; a concave shoulder guarantees it, a convex one rules it out unless the density is a scaled triangle, and mixing towards the edges is always safe. The Pólya law falls between the two shoulder cases.

20.2.1 Mixtures of uniform laws

By Khinchine's theorem [86], a random variable X is symmetric and unimodal about 0 if and only if $X \stackrel{d}{=} UR$ with $R \geq 0$ independent of U. Then $L_X(s) = \mathbb{E}[\sinh(sR)/(sR)]$, and if X has a density p that is absolutely continuous on $(0,\infty)$ and tends to 0 at infinity, R has the density $-2rp'(r)$. Williamson's theorem refines

the representation [364]: a density on $(0,\infty)$ is k monotone exactly when it is a scale mixture of $k(1-x)_+^{k-1}$, so the symmetric k monotone densities are the scale mixtures of

$$f_k(x) = \tfrac{k}{2}(1-|x|)_+^{k-1},$$

of which f_1 is the uniform density and f_2 the triangle. The uniform law itself is ferromagnetic, with $L_U(s) = \sinh s/s$, and Viète's product

$$\frac{\sinh s}{s} = \prod_{k\ge 1} \cosh\frac{s}{2^k}, \qquad U \stackrel{d}{=} \sum_{k\ge 1} 2^{-k}\varepsilon_k,$$

shows why: the uniform variable is a sum of independent fair signs. So is every sum $\sum_k c_k U_k$ of independent scaled uniform variables. Mixtures behave differently.

THEOREM 20.2.1. *For every $0<q<1$ the law $(1-q)\,U + q\,\delta_0$ is symmetric and unimodal but not ferromagnetic. Consequently, for every $0<q<1$ and every small enough $a>0$, the mixture of the uniform laws on $[-1,1]$ and on $[-a,a]$ with weights $1-q$ and q is not ferromagnetic.*

Proof. The moment generating function $L(s) = (1-q)\sinh s/s + q$ vanishes exactly where $\sinh s/s = -c$, with $c = q/(1-q) > 0$. There is no real zero, since $\sinh s/s \ge 1$ on $\mathbb{R}$, and only finitely many imaginary ones, since $s = iy$ needs $\sin y/y = -c$ and $|\sin y/y| \le 1/|y|$. But L has infinitely many zeros. It is entire of order 1, so with finitely many zeros it would be $e^{\alpha s}P(s)$ with a polynomial P; evenness forces $\alpha = 0$, and L is not a polynomial, since it grows like e^s/s on the positive axis. So all but finitely many zeros lie off the imaginary axis. The second statement follows from Theorem 20.1.5, since the mixtures converge in law to $(1-q)U + q\delta_0$ as $a \downarrow 0$. □

Both components of the mixture are ferromagnetic, so the class is not closed under mixing, even of unimodal laws with a common mode. For $c > c_*$, where $c_* = -\min_y \sin y/y = 0.2172336\ldots$, the minimum being at the first positive root $y = 4.4934095\ldots$ of $\tan y = y$, there are no imaginary zeros at all.

PROPOSITION 20.2.2. *For $0 \le q \le 1$ the law $q\,\delta_0 + \frac{1-q}{2}(\delta_{-1}+\delta_1)$ is ferromagnetic if and only if $q \le \frac12$ or $q = 1$.*

Proof. For $q = 1$ the law is δ_0, and $L = 1$ has no zeros. For $q < 1$ the zeros of $L(s) = (1-q)\cosh s + q$ solve $\cosh s = -c$ with $c = q/(1-q)$. If $c \le 1$, write $-c = \cos\theta_0$ with $\theta_0 \in [\frac{\pi}{2}, \pi]$; the solutions $s = \pm i\theta_0 + 2\pi i k$ are imaginary. If $c > 1$ the solutions are $s = \pm \operatorname{arccosh} c + i\pi(2k+1)$, which are not; for $q = 0.6$ the first is $0.9624237 + i\pi$. □

For $k = 2$ the representation is explicit and is a statement about convexity. If p is a symmetric density, twice continuously differentiable on $(0,\infty)$, with $r^2 p'(r) \to$

0 as $r \to \infty$ and $\int_0^\infty (r + r^2)|p''(r)|\,dr < \infty$, two integrations by parts give, for $x \neq 0$ (and at $x = 0$ with $p(0) = \lim_{x \downarrow 0} p(x)$),

$$p(x) = \int_0^\infty \frac{1}{r}\Big(1 - \frac{|x|}{r}\Big)_+ r^2 p''(r)\,dr, \qquad \mathbb{E}e^{i\theta X} = \frac{4}{\theta^2}\int_0^\infty p''(r)\sin^2\frac{\theta r}{2}\,dr \quad (\theta \neq 0),$$

so X is a scale mixture of f_2 for the signed measure $r^2 p''(r)\,dr$, which has mass one and is the only finite signed measure on $(0,\infty)$ with this property (the second distributional derivative of $r^{-1}(1 - x/r)_+$ is $r^{-2}\delta_r$ on $(0,\infty)$); it is a probability law exactly when p is convex on $(0,\infty)$. This is Williamson's representation for $k = 2$ [364], in the form used by Polson, Scott and Windle [278, Section 2.2].

PROPOSITION 20.2.3. *(i) If X is a scale mixture of f_2 for a probability law ν on $(0,\infty)$, then $\mathbb{E}e^{i\theta X} \geq 0$ for real θ, and for $\theta \neq 0$ it vanishes exactly when ν is carried by $\{2\pi j/|\theta| : j \geq 1\}$. In particular, if ν has a nonzero absolutely continuous part, $\mathbb{E}e^{i\theta X}$ has no real zeros.*

(ii) The Pólya law is not a scale mixture of f_2 for any probability law: with $y_n = \pi n^2 e^{2u}$ and $R(y) = 32y^4 - 224y^3 + 330y^2 - 75y$, the kernel (2.1.3) satisfies $\Phi''(u) = \frac{1}{2}e^{u/2}\sum_{n\geq 1} R(y_n)e^{-y_n}$, $\Phi''(0) < 0$, and $\Phi''(u) > 0$ for $u \geq \frac{1}{2}\log(7/\pi)$.

(iii) The kernel Φ has exactly one positive inflection point u^*: $\Phi'' < 0$ on $[0, u^*)$, $\Phi'' > 0$ on (u^*, ∞), and $\Phi'''(u^*) > 0$.

Proof. (i) $\mathbb{E}e^{i\theta X} = \int_0^\infty \hat{f}_2(r\theta)\,\nu(dr)$ with $\hat{f}_2(\theta) = 4\sin^2(\theta/2)/\theta^2$, which is nonnegative and, for $\theta \neq 0$, vanishes exactly on $2\pi\mathbb{Z} \setminus \{0\}$. A nonnegative integral vanishes exactly when its integrand vanishes ν almost everywhere, and a countable set is null for an absolutely continuous measure. (ii) Each $(1 - x/r)_+$ is convex on $(0,\infty)$, so a mixture of them with a probability law is convex there, and it suffices to show $\Phi''(0) < 0$. The formula for Φ'' is termwise differentiation of (2.1.3), written in y_n. At $u = 0$ the term $n = 1$ of the sum is below -34, since $R(\pi) < -800$ and $e^{-\pi} > 0.043$, while $0 < R(\pi n^2) < 32\pi^4 n^8$ for $n \geq 2$ and $32\pi^4\sum_{n\geq 2} n^8 e^{-\pi n^2} < 3$ (the three inequalities are certified in ball arithmetic). For $y \geq 7$, $R(y) = y\{32y^2(y-7) + 330y - 75\} > 0$, so every term is positive when $u \geq \frac{1}{2}\log(7/\pi)$.

(iii) Put $y = \pi e^{2u}$ and $D(y) = \sum_{n\geq 2} R(n^2 y)e^{-(n^2-1)y}$, so that $\Phi''(u) = \frac{1}{2}e^{u/2-y}F(y)$ with $F = R + D$; it suffices to show that F has exactly one zero on $[\pi,\infty)$ and that it is simple. For $y \geq \pi$ and $n \geq 2$, $0 < R(n^2 y) < 32n^8 y^4$ and $y^4 e^{-(n^2-1)y}$ decreases, so $0 < D(y) < 32\pi^4 e^{\pi}\sum_{n\geq 2} n^8 e^{-\pi n^2} < 3e^{\pi} < 72$. Since $R''(y) = 384y^2 - 1344y + 660 > 0$ for $y \geq 3$, R is convex on $[\pi, \frac{9}{2}]$ and lies below the larger of its endpoint values, $R(\pi) < -800$ and $R(\frac{9}{2}) = -945$, so $F < -728$ there. For $y \geq \frac{9}{2}$, $R'(y) \geq R'(\frac{9}{2}) = 951$, and termwise differentiation gives $|D'(y)| \leq 64y^4\sum_{n\geq 2} n^{10}e^{-(n^2-1)y} < 40$ (the sum is at its largest at

$y = \frac{9}{2}$, where its first term dominates). Hence $F' > 911$ on $[\frac{9}{2}, \infty)$, $F(\frac{9}{2}) < 0$ and $F(7) > 0$, so F has exactly one zero, which lies in $(\frac{9}{2}, 7)$ and is simple. The prefactor is positive, which gives every sign claimed in (iii). □

The location of u^* is certified in ball arithmetic, $u^* = 0.23726528718742524911$, so the Pólya density is concave on $|u| \le u^*$ and convex outside. Its mixing measure $r^2 p''(r)\, dr$ has mass -0.05106 on $(0, u^*)$ and 1.05106 on (u^*, ∞) (computed, [145]). Part (i) gives a second reason why this measure is signed: Ξ vanishes at γ_1, and a probability mixing law with a nonzero absolutely continuous part would leave no real zero. At every real zero of Ξ the contributions of $(0, u^*)$ and (u^*, ∞) to the integral above cancel, but this is only the Fourier identity $\int_0^\infty p''(r) \sin^2(\theta r/2)\, dr = \theta^2 \hat{p}(\theta)/4$ at a zero of $\hat{p}$. The heat trace behaves the other way: by Theorem 8.1.13 it is 2 monotone unconditionally, so $W(t) = \int (1 - tz/2)_+ \, U_2(dz)$ with $U_2 \ge 0$ (Section 8.1.4), a positive mixture of triangles, and RH is the same positivity at every order.

20.2.2 The shoulder theorem

Let p be a symmetric density on $[-1, 1]$ that is absolutely continuous and nonincreasing on $[0, 1]$, with $p(1) = 0$, and put $g = -p' \ge 0$ on $(0, 1)$. Integrating $\varphi_X(z) = 2\int_0^1 p(y) \cos(zy)\, dy$ by parts gives

$$z\, \varphi_X(z) = 2 \int_0^1 g(y) \sin(zy)\, dy. \tag{20.2.1}$$

Pólya proved that if f is positive, integrable and nondecreasing on $(0, 1)$, then $\int_0^1 f(t) \cos(zt)\, dt$ and $\int_0^1 f(t) \sin(zt)\, dt$ have only real zeros [297].

Theorem 20.2.4. *Let p be a symmetric density on $[-1, 1]$, absolutely continuous and nonincreasing on $[0, 1]$, with $p(1) = 0$, and let $[-c, c]$, $0 < c \le 1$, be its support.*

(i) If p is concave on $[0, 1]$, then X is ferromagnetic.

(ii) If p is convex on $[0, 1]$ and not linear on $[0, c]$, then X is not ferromagnetic. Moreover $\varphi_X \ge 0$ on $\mathbb{R}$, and its real zeros, if any, are double and lie in a lattice $(2\pi/s_0)\mathbb{Z}$ with $0 < s_0 < c$.

Proof. (i) If p is concave, then $c = 1$ and g is nondecreasing on $(0, 1)$. If $g > 0$ on $(0, 1)$, Pólya's theorem in its sine form and (20.2.1) show that φ_X has only real zeros. Otherwise p is constant near 0; then $(p + \epsilon(1 - |x|))/(1 + \epsilon)$ is concave with a positive slope, so it is ferromagnetic, and Theorem 20.1.5 applies as $\epsilon \downarrow 0$.

(ii) If p is convex, g is nonincreasing on $(0, c)$, so $g(y) = g(c-) + \mu((y, c))$ with a positive measure μ on $(0, c)$, which is not zero because p is not linear. Since $\int_0^s \sin(zy)\, dy = 2\sin^2(zs/2)/z$, Fubini's theorem in (20.2.1) gives, for real $z \ne 0$,

$$\frac{z^2}{4}\, \varphi_X(z) = g(c-) \sin^2 \frac{zc}{2} + \int_{(0,c)} \sin^2 \frac{zs}{2}\, \mu(ds) \ \ge 0.$$

A real zero z_0 needs $z_0 s \in 2\pi\mathbb{Z}$ for every s in the support of μ, so with any such $s_0 < c$ the real zeros lie in $(2\pi/s_0)\mathbb{Z}$. At a real zero the right side has vanishing first derivative and second derivative $\frac{1}{2}g(c-)c^2 + \frac{1}{2}\int s^2\mu(ds) > 0$, because every cosine $\cos(z_0 s)$ that enters equals 1; so the zero is double. Suppose now that every zero of φ_X were real. Since φ_X is even and entire of exponential type, Hadamard's theorem would give $\varphi_X(z) = \prod_{m\in M}(1 - z^2/z_m^2)^2$ for a set M of positive integers, with $z_m = 2\pi m/s_0$, and hence

$$\varphi_X(iy) \le \prod_{m\ge 1}\Big(1 + \frac{y^2 s_0^2}{4\pi^2 m^2}\Big)^2 = \Big(\frac{\sinh(ys_0/2)}{ys_0/2}\Big)^2 \le e^{s_0|y|}.$$

But $p > 0$ on $[0, c)$, so for $0 < \epsilon < c - s_0$ and $y > 0$, $\varphi_X(iy) = L_X(y) \ge \int_{c-\epsilon}^{c} e^{yx}p(x)\,dx \ge C_\epsilon e^{(c-\epsilon)y}$ with $C_\epsilon > 0$, which contradicts the bound as $y \to \infty$. □

Corollary 20.2.5. *(i) The law with density proportional to $1-|x|^b$ on $[-1,1]$, $b > 0$, is ferromagnetic if and only if $b \ge 1$. In Khinchine's form, UR with R distributed as* Beta$(a, 1)$ *is ferromagnetic if and only if $a \ge 2$.*

(ii) The symmetric k monotone kernel f_k is ferromagnetic if and only if $k \le 2$.

(iii) The laws with density proportional to $(1 - x^2)^m$, $0 < m \le 1$, which include the semicircle law, are ferromagnetic.

(iv) The product U_1U_2 of two independent uniform variables, with density $\frac{1}{2}\log(1/|x|)$, is not ferromagnetic.

Proof. (i) $1 - x^b$ is concave on $[0, 1]$ for $b \ge 1$ and convex and not linear for $b < 1$; the mixing variable has density $-2rp'(r) \propto r^b$, that of Beta$(b + 1, 1)$. This settles $a = b + 1 > 1$. For $0 < a \le 1$ the density of UR is unbounded at 0 and the theorem does not apply, but Proposition 20.2.7 below excludes these laws: with $\mu_j = a/(a + 2j)$, its case $j = 1$ requires $a \ge \sqrt{10} - 2 > 1$. (ii) f_1 is uniform and f_2 the triangle; for $k \ge 3$, $(1 - x)^{k-1}$ is convex and not linear. (iii) $\frac{d^2}{dx^2}(1 - x^2)^m = -2m(1 - x^2)^{m-2}(1 - (2m - 1)x^2) \le 0$ on $[0, 1]$ when $m \le 1$. (iv) The density is unbounded at the origin, so Theorem 20.2.4 does not apply, but directly $\varphi(z) = \int_0^1 \sin(zu)/(zu)\,du = \mathrm{Si}(z)/z$, and $\mathrm{Si}(z) > 0$ for $z > 0$. So φ has no real zero, while it has infinitely many zeros, by the argument in the proof of Theorem 20.2.1. □

20.2.3 Mixing towards the edges

Theorem 20.2.6. *Every symmetric density on $[-1, 1]$ that is nondecreasing in $|x|$ is ferromagnetic. Equivalently, every mixture of the uniform laws on the shells $\{a \le |x| \le 1\}$, $0 \le a < 1$, is ferromagnetic.*

Proof. Here $\varphi_X(z) = 2\int_0^1 p(t)\cos(zt)\,dt$ with p nondecreasing on $(0,1)$. If $p>0$ on $(0,1)$, Pólya's theorem in its cosine form [297] applies; otherwise add ϵ times the uniform density, normalise, and let $\epsilon \downarrow 0$ (Theorem 20.1.5). A density nondecreasing in $|x|$ is a mixture of the shell densities, and conversely. □

20.2.4 A moment test

Ferromagnetic laws satisfy inequalities between consecutive moments. If X is ferromagnetic, $L_X(\sqrt{x}) = \sum_j b_j x^j$, with $b_j = \mathbb{E}X^{2j}/(2j)!$, equals $e^{bx}\prod_k(1 + x/c_k^2)$, an entire function of x in the Laguerre Pólya class with only negative zeros. Its coefficients satisfy the Turán inequalities $(j!\,b_j)^2 \geq (j-1)!\,b_{j-1}(j+1)!\,b_{j+1}$, the limit as $n, K \to \infty$ of Newton's inequalities for the polynomials $(1 + bx/n)^n \prod_{k\leq K}(1 + x/c_k^2)$, whose zeros are real; that is

$$b_j^2 \geq \Big(1 + \frac{1}{j}\Big) b_{j-1} b_{j+1}, \qquad j \geq 1. \tag{20.2.2}$$

For a mixture $X = UR$ the moments factor as $\mathbb{E}X^{2j} = \mu_j/(2j+1)$ with $\mu_j = \mathbb{E}R^{2j}$.

Proposition 20.2.7. *If UR is ferromagnetic and R is not 0 almost surely, then for every $j \geq 1$*

$$\frac{2j+1}{2j+3} \leq \frac{\mu_j^2}{\mu_{j-1}\mu_{j+1}} \leq 1.$$

Proof. The moments μ_j are finite, since $\mathbb{E}(UR)^{2j} = \mu_j/(2j+1)$ and L_{UR} is entire, and positive, since $P(R>0) > 0$. Substitute $b_j = \mu_j/(2j+1)!$ in (20.2.2): the factor is $(1+\frac{1}{j})\frac{(2j+1)!^2}{(2j-1)!(2j+3)!} = \frac{2j+1}{2j+3}$. The upper bound is the Cauchy Schwarz inequality and holds for every R. □

The test is necessary but weak. For R distributed as $\mathrm{Beta}(a,1)$, $\mu_j = a/(a+2j)$; at $j=1$ the condition is $a^2+4a-6 \geq 0$, that is $a \geq \sqrt{10}-2 = 1.1622777$, and for $j \geq 2$ it holds for every $a>0$. Corollary 20.2.5(i) gives the true threshold $a \geq 2$.

20.2.5 Summary

Table 20.2 collects the answers. The Pólya law has unbounded support, so none of them applies to it.

20.3 The Mellin transform of a clock

This section is based on [147, 144]. Let X be ferromagnetic with $b=0$ in (20.1.1) and infinitely many zeros, so that $L_X(s) = \prod_k(1+s^2/c_k^2)$, and let $T_X = \sum_k e_k/c_k^2$ be its clock (Theorem 20.1.4). For every m, T_X is at least c_m^{-2} times a sum of m

law on $[-1,1]$	ferromagnetic
uniform, and sums $\sum_k c_k U_k$	yes
symmetric beta $c_\nu(1-x^2)^{\nu-1/2}$, $\nu > -\frac{1}{2}$	yes
concave shoulder	yes
nondecreasing in $\lvert x\rvert$ (shell mixtures)	yes
convex shoulder, not linear on the support	no
$1-\lvert x\rvert^b$	if and only if $b \geq 1$
k monotone kernel f_k	if and only if $k \leq 2$
uniform with an atom at 0, weight in $(0,1)$	no
product of two uniform variables	no
spin one, $q\delta_0 + \frac{1-q}{2}(\delta_{-1}+\delta_1)$	if and only if $q \leq \frac{1}{2}$ or $q = 1$

Table 20.2: Which laws on $[-1,1]$ are ferromagnetic. Every entry is proved in this section or in Section 20.1.

independent standard exponential variables, and $\mathbb{E}e^{tT_X} < \infty$ for $t < c_1^2$, so T_X has moments of every real order and $\mathbb{E}T_X^{-w}$ is entire. Put

$$\mathcal{D}_X(w) = \Gamma(w)\,\mathbb{E}T_X^{-w} = \int_0^\infty u^{w-1}\Big(\frac{1}{L_X(\sqrt{u})} - 1\Big)\,du, \qquad -1 < \operatorname{Re} w < 0,$$

the Mellin transform of the Laplace transform $1/L_X(\sqrt{u})$ of T_X, a meromorphic function whose poles are among those of Γ. For the uniform pair $\pi U + \pi U'$, with U' an independent copy of U, the moment generating function is $(\sinh \pi s/\pi s)^2$, the zeros are $c_k = k$, each counted twice, and the clock is $\sum_k (e_k + e_k')/k^2$, a GGC with Thorin measure $2\sum_k \delta_{k^2}$. The Gamma(2) variables of Williams' structure (Section 3.1.3) are the sums $e_k + e_k'$ over the two spins, and the clock is $\frac{\pi^2}{2}S_2$ in the notation of (12.1.1). By (12.1.1) with $s = -2w$, and the functional equation,

$$\mathcal{D}_{\pi U + \pi U'}(w) = 2\pi^{-w}\,\Gamma(w)\,\xi(1+2w). \tag{20.3.1}$$

So the nonreal zeros of $\mathcal{D}_{\pi U+\pi U'}$ are the points $\frac{\rho-1}{2}$, and RH is the statement that they lie on the line $\operatorname{Re} w = -\frac{1}{4}$. We write $z = 2w+1$, the variable in which they are the zeros of ξ.

Proposition 20.3.1. *Let X be as above, with simple zeros $\pm ic_k$, $0 < c_1 < c_2 < \cdots$, and put*

$$A_k = -\frac{2}{c_k \varphi_X'(c_k)}, \qquad B_k = \frac{2\big(c_k\varphi_X''(c_k) - \varphi_X'(c_k)\big)}{c_k^2\,\varphi_X'(c_k)^3}.$$

(i) Near $u = -c_k^2$, $1/L_X(\sqrt{u}) = A_k(1+u/c_k^2)^{-1} + O(1)$ and $1/L_X(\sqrt{u})^2 = A_k^2(1+u/c_k^2)^{-2} + B_k(1+u/c_k^2)^{-1} + O(1)$.

(ii) If $1/L_X(\sqrt{u}) = 1 + \sum_k A_k\big((1+u/c_k^2)^{-1} - 1\big)$ *for* $u \ge 0$ *and* $\sum_k |A_k| c_k^{2\sigma} < \infty$ *for some* $\sigma \in (-1, 0)$*, then*

$$\mathcal{D}_X(w) = \frac{\pi}{\sin \pi w} \sum_k A_k c_k^{2w}, \qquad -1 < \operatorname{Re} w \le \sigma.$$

(iii) If $1/L_X(\sqrt{u})^2 = 1 + \sum_k \big[A_k^2\big((1+u/c_k^2)^{-2} - 1\big) + B_k\big((1+u/c_k^2)^{-1} - 1\big)\big]$ *for* $u \ge 0$ *and* $\sum_k (A_k^2 + |B_k|) c_k^{2\sigma} < \infty$ *for some* $\sigma \in (-1, 0)$*, then, with* X' *an independent copy of* X*,*

$$\mathcal{D}_{X+X'}(w) = \frac{\pi}{\sin \pi w} \sum_k \big[(1-w)A_k^2 + B_k\big] c_k^{2w}, \qquad -1 < \operatorname{Re} w \le \sigma.$$

Proof. (i) Put $\theta = \sqrt{-u}$, so that $1/L_X(\sqrt{u}) = 1/\varphi_X(\theta)$ and $1 + u/c_k^2 = (c_k - \theta)(c_k + \theta)/c_k^2$, and expand φ_X to second order at its simple zero c_k. (ii) and (iii). For $c > 0$ and $-1 < \operatorname{Re} w < 0$, the beta integral gives $\int_0^\infty u^{w-1}\big((1+u/c)^{-1} - 1\big)\, du = c^w \pi / \sin \pi w$ and $\int_0^\infty u^{w-1}\big((1+u/c)^{-2} - 1\big)\, du = c^w \Gamma(w)\Gamma(2-w) = c^w (1-w)\pi / \sin \pi w$. Since $|(1+u/c)^{-j} - 1| \le \min(2u/c, 1)$ for $u \ge 0$ and $j = 1, 2$, and only finitely many c_k are below 1, the hypothesis makes the series of the integrals of the absolute values converge on every line $\operatorname{Re} w = \sigma' \le \sigma$, so sum and integral can be exchanged. □

The expansion in (ii) has the form of the partial fraction expansion that defines Krein's class, recalled before Proposition 5.4.2, for the function $u \mapsto L_X(\sqrt{u})$, whose zeros are the points $-c_k^2$, with the coefficients $A_k c_k^2$. For $X = \pi U$, $\varphi_X(\theta) = \sin \pi\theta / (\pi\theta)$, so $A_k = -2(-1)^k$ and $B_k = -6$ for every k, and the two expansions are the classical partial fractions

$$\frac{\pi\sqrt{u}}{\sinh \pi\sqrt{u}} = 1 + 2u \sum_{k \ge 1} \frac{(-1)^k}{u + k^2}, \qquad \Big(\frac{\pi\sqrt{u}}{\sinh \pi\sqrt{u}}\Big)^2 = 1 + \sum_{k \ge 1} \frac{2u(u - k^2)}{(u + k^2)^2},$$

the second from $\pi^2 / \sinh^2 \pi x = \sum_{k \in \mathbb{Z}} (x - ik)^{-2}$. The conditions on the weights hold for every $\sigma < -\frac{1}{2}$. So $\mathcal{D}_{\pi U}(w) = (2\pi / \sin \pi w)\, \eta(-2w)$, with $\eta(s) = \sum_k (-1)^{k-1} k^{-s}$, and the weights of the pair, $(1-w)A_k^2 + B_k = -(4w+2)$, do not depend on k:

$$\mathcal{D}_{\pi U + \pi U'}(w) = -\frac{2\pi(2w+1)}{\sin \pi w}\, \zeta(-2w), \qquad -1 < \operatorname{Re} w < -\tfrac{1}{2}. \tag{20.3.2}$$

This is proved. By the functional equation of ζ it is (20.3.1), and read the other way it proves that functional equation: for $\operatorname{Re} w > 0$, $\mathcal{D}_{\pi U + \pi U'}(w) = 2\int_0^\infty x^{2w-1} (\pi x / \sinh \pi x)^2\, dx$, and the expansion $\sinh^{-2} y = 4 \sum_{m \ge 0} (m+1) e^{-(2m+2)y}$ gives $2(2\pi)^{-2w} \Gamma(2w+2) \zeta(2w+1)$, so the two expressions are one meromorphic function. This is the argument of Riemann's first proof of the

functional equation [80, Chapter 1], a sum over residues: one integration by parts turns the integral into $4\pi(2w+1)\int_0^\infty x^{2w}/(e^{2\pi x}-1)\,dx$, and the poles $\pm ik$, $k\ge 1$, of $(\pi x/\sinh \pi x)^2$, over which (20.3.2) sums, are the zeros of the moment generating function of the pair, its Lee and Yang zeros.

The hypotheses of (ii) fail for smooth densities. If X has a density q for which $tq(t)$ is smooth with all derivatives integrable, then $\varphi_X'(\theta)=O(\theta^{-N})$ for every N, so $|A_k|$ grows faster than every power of c_k and $\sum_k A_k c_k^{2w}$ diverges for every w. This covers the laws of Theorem 19.1.4; for $m\equiv 1$, $|A_k|$ grows like $e^{\pi c_k/2\ell}$ up to powers of c_k and of $\log c_k$, by the asymptotic formula for $K_{i\nu}(a)$ as $\nu\to\infty$, as the residues of the Pólya law grow like $\gamma^{-3/4}e^{\pi\gamma/4}/|\zeta'(\rho)|$ in the proof of Proposition 5.4.2. The zeros of $\mathcal{D}$ do not need the expansion: let W have a law of Theorem 19.1.4, with an independent copy W', and put $X_\delta=\pi U+\delta W$, a ferromagnetic law with a smooth density of this kind. Since $0\le 1-1/L_W(\delta\sqrt{u})^2\le \min(1,2\delta^2 u\,\mathbb{E}T_W)$, dominated convergence gives $\mathcal{D}_{X_\delta+X_\delta'}\to\mathcal{D}_{\pi U+\pi U'}$ uniformly on compact subsets of $-1<\operatorname{Re} w<0$ as $\delta\to 0$, and by Hurwitz's theorem, for each k with ρ_k simple, $\mathcal{D}_{X_\delta+X_\delta'}$ has a zero near $\frac{\rho_k-1}{2}$ once δ is small enough (proved).

PROPOSITION 20.3.2. *Let $X_k=\pi(U_1+\cdots+U_k)$ with independent copies U_i of U. For $\operatorname{Re} w>0$,*

$$\mathcal{D}_{X_k}(w)=2(2\pi)^{-2w}\,\Gamma(2w+k)\sum_{m\ge 0}\binom{m+k-1}{k-1}\Big(m+\frac{k}{2}\Big)^{-2w-k}.$$

For $k=2r$ the series is $\sum_{i=0}^{r-1}a_{r,i}\,\zeta(2w+1+2i)$, where $a_{r,i}$ is the coefficient of $j^{2r-1-2i}$ in $j\prod_{l=1}^{r-1}(j^2-l^2)/(2r-1)!$, and for $k=2r+1$ it is $\sum_{i=0}^{r}b_{r,i}\,(2^{2w+1+2i}-1)\,\zeta(2w+1+2i)$, where $b_{r,i}$ is the coefficient of h^{2r-2i} in $\prod_{l=1}^{r}\big(h^2-(l-\frac12)^2\big)/(2r)!$. The first and the last of these coefficients are not 0, so for $k\ge 3$ the series has at least two terms. In particular

$$\mathcal{D}_{X_1}(w)=4\pi^{-2w}\big(1-2^{-2w-1}\big)\Gamma(2w+1)\,\zeta(2w+1),$$
$$\mathcal{D}_{X_4}(w)=\tfrac13(2\pi)^{-2w}\,\Gamma(2w+4)\big[\zeta(2w+1)-\zeta(2w+3)\big],$$

and $\mathcal{D}_{X_6}(w)=\frac{1}{60}(2\pi)^{-2w}\,\Gamma(2w+6)\big[\zeta(2w+1)-5\zeta(2w+3)+4\zeta(2w+5)\big]$, while $\mathcal{D}_{X_2}$ is (20.3.1).

Proof. For $\operatorname{Re} w>0$,

$$\mathcal{D}_{X_k}(w)=\int_0^\infty u^{w-1}L_{X_k}(\sqrt{u})^{-1}\,du=2\int_0^\infty x^{2w-1}\Big(\frac{\pi x}{\sinh \pi x}\Big)^k\,dx.$$

Expand $\sinh^{-k}y=2^k\sum_{m\ge0}\binom{m+k-1}{k-1}e^{-(2m+k)y}$ and integrate term by term. The binomial coefficient is $\prod_{l=1}^{k-1}(m+l)/(k-1)!$, which is the stated polynomial in $j=m+r$ for $k=2r$ and in $h=m+r+\frac12$ for $k=2r+1$. It vanishes at $j=1,\dots,r-1$ and at $h=\frac12,\dots,r-\frac12$, so the sums run over all positive integers j and all positive half integers h, and $\sum_{m\ge0}(m+\frac12)^{-v}=(2^v-1)\zeta(v)$. □

For $k = 1$ the nonreal zeros of $\mathcal{D}_{X_1}$ are those of $\zeta(z)$ and the points $z = 2\pi i m/\log 2$, $m \neq 0$, so they lie on the two lines $\operatorname{Re} z = \frac{1}{2}$ (under RH) and $\operatorname{Re} z = 0$; for $k = 2$ they are the zeros of $\xi(z)$. For $k = 4$ and $k = 6$ they do not lie on one vertical line: $\zeta(z) - \zeta(z+2)$ vanishes at $z = 1.76891972665 + 23.3078297413i$ and $z = 0.650238826497 + 31.7124838244i$, and $\zeta(z) - 5\zeta(z+2) + 4\zeta(z+4)$ at $z = 3.18634599215 + 31.979112615i$ and $z = 1.51363208762 + 41.215007487i$ (computed in ball arithmetic, each zero alone in a square of half side $7.5\cdot 10^{-9}$ by the argument principle).

Laws whose characteristic function is that of πU divided by a polynomial behave in the same way. If $\varphi_X(\theta) = \varphi_{\pi U}(\theta)/Q(\theta^2)$ for a polynomial Q with $Q(0) = 1$, then $1/L_X(\sqrt{u})^2 = Q(-u)^2/L_{\pi U}(\sqrt{u})^2$, and integrating term by term for $\operatorname{Re} w > 0$ gives $\mathcal{D}_{X+X'}(w) = \sum_j q_j\, \mathcal{D}_{\pi U+\pi U'}(w+j)$, where $Q(-u)^2 = \sum_j q_j u^j$, an identity of meromorphic functions (proved).

For the density $(1 + \cos t)/2\pi$ on $[-\pi, \pi]$, which is C^1 and not C^2 at $\pm\pi$, $Q(x) = 1 - x$, the zeros are $c_k = k + 1$, $k \geq 1$, the law is ferromagnetic, and

$$\mathcal{D}_{X+X'}(w) = \mathcal{D}_{\pi U+\pi U'}(w) + 2\,\mathcal{D}_{\pi U+\pi U'}(w+1) + \mathcal{D}_{\pi U+\pi U'}(w+2),$$

a combination of $\zeta(-2w)$, $\zeta(-2w-2)$ and $\zeta(-2w-4)$ by (20.3.2). Its weights $A_k = 2(-1)^{k+1}(k^2 + 2k)$ grow like k^2, so (ii) and (iii) do not apply. In the box $-1 < \operatorname{Re} w < 1$, $\frac{1}{2} < \operatorname{Im} w < 30$, which holds the 13 zeros $\frac{\rho_k - 1}{2}$, $\gamma_k < 60$, of $\mathcal{D}_{\pi U+\pi U'}$, it has no zero, and in $-3 < \operatorname{Re} w < -2$, $\frac{1}{2} < \operatorname{Im} w < 30$ it has 6, at $-2.443554 + 15.837821i$, $-2.976747 + 19.238797i$, $-2.479242 + 21.119434i$, $-2.273308 + 24.426843i$, $-2.862245 + 26.206760i$ and $-2.756588 + 28.553832i$, so not on one vertical line (computed in ball arithmetic).

The line is not stable inside the laws of Theorem 20.2.6, scaled by π. For $\varepsilon \geq 0$ let X_ε have the density $(1 + \varepsilon t^2)/(2\pi + \frac{2}{3}\pi^3\varepsilon)$ on $[-\pi, \pi]$, which is nondecreasing in $|t|$, so that X_ε and the pair $X_\varepsilon + X'_\varepsilon$ are ferromagnetic. At $\varepsilon = 0$ the zero $w_k(0) = \frac{\rho_k - 1}{2}$ of $\mathcal{D}_{\pi U+\pi U'}$ is simple whenever ρ_k is, as for $k = 1, 2$, and then by the implicit function theorem it moves with

$$\frac{dw_k}{d\varepsilon}(0) = -\frac{\dot{\mathcal{D}}(w_k(0))}{\mathcal{D}'_{\pi U+\pi U'}(w_k(0))}, \qquad \dot{\mathcal{D}}(w) = -2\int_0^\infty u^{w-1}\,\frac{\dot{L}(\sqrt{u})}{L_{\pi U}(\sqrt{u})^3}\,du,$$

where $\dot{L}(s) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\big(t^2 - \frac{\pi^2}{3}\big)e^{st}\,dt$ is the derivative of $L_{X_\varepsilon}(s)$ at $\varepsilon = 0$ and $\dot{\mathcal{D}}$ that of $\mathcal{D}_{X_\varepsilon + X'_\varepsilon}$.

The computed values are $\frac{dw_1}{d\varepsilon}(0) = 5.89145 - 13.09321i$ and $\frac{dw_2}{d\varepsilon}(0) = 1.56442 - 16.25014i$, with real parts different from 0 and from each other (computed by quadrature with three rules that agree). So for small $\varepsilon > 0$ the first two zeros leave the line $\operatorname{Re} w = -\frac{1}{4}$ at different rates, and the nonreal zeros of $\mathcal{D}_{X_\varepsilon + X'_\varepsilon}$ do not lie on one vertical line although every X_ε is ferromagnetic. By this computation no theorem whose hypotheses hold for all symmetric densities on an interval nondecreasing in $|t|$ can put the zeros of $\mathcal{D}$ of the pair on one line.

20.4 The Pólya law

The Pólya variable X has the density $p = \Phi/\Xi(0)$, with $\Xi(0) = \xi(\frac{1}{2}) = 0.4971207782$, and $L_X(s) = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$ is an entire function of order 1. Its density decays like $\exp(-\pi e^{2|t|})$, so X has Gaussian moments of all orders, and RH is the statement that X is ferromagnetic; by Theorem 20.1.2, with $b = 0$, it is the product formula $L_X(s) = \prod_k (1 + s^2/\gamma_k^2)$ over the positive ordinates. The law is symmetric and unimodal, since $\Phi'(t) < 0$ for $t > 0$ [69, Theorem A].

20.4.1 Necessary conditions

If X is ferromagnetic, Corollary 20.1.3 with $c_k = \gamma_k$ gives $\kappa_2 = 2\sum_k \gamma_k^{-2}$, $\kappa_4 = -12\sum_k \gamma_k^{-4}$ and $\kappa_6 = 240\sum_k \gamma_k^{-6}$. Table 20.3 computes the cumulants. The agreement of the columns is not evidence for RH: the Hadamard product $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2}) = \prod_k (1+s^2/\gamma_k^2)$, with one factor for each pair of zeros $\frac{1}{2} \pm i\gamma_k$, holds with no hypothesis if γ_k is allowed to be complex, and it converges absolutely because $\sum_k |\gamma_k|^{-2} < \infty$; it gives the same formulas for the cumulants.

	moments of Φ	Taylor coefficients of $\log \xi$	zero sums
κ_2	0.0462099862308	0.0462099862308	0.046209986105
κ_4	$-4.46071191423 \times 10^{-4}$	$-4.46071191423 \times 10^{-4}$	$-4.46071191423 \times 10^{-4}$
κ_6	$3.46017435362 \times 10^{-5}$	$3.46017435362 \times 10^{-5}$	$3.46017435362 \times 10^{-5}$

Table 20.3: Even cumulants of the Pólya law, by quadrature of the moments of Φ, from the Taylor coefficients of $\log \xi(\frac{1}{2}+s)$, and from the sums $(-1)^{m+1}\frac{(2m)!}{m}\sum_k \gamma_k^{-2m}$ over the first 1700 ordinates with the smooth count beyond them. The difference 1.3×10^{-10} in κ_2 is the error of the smooth count, which ignores the oscillation of $S(t)$ above height 2197. Computed.

The moment test of Section 20.2.4 is sharper. For the Pólya law the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ are the Taylor coefficients of Ξ, up to sign and the factor $\Xi(0)$:

$$\frac{\Xi(t)}{\Xi(0)} = \sum_{j \ge 0} (-1)^j b_j t^{2j}.$$

In this form (20.2.2), $b_j^2 \ge (1+\frac{1}{j}) b_{j-1} b_{j+1}$, is the system of Turán inequalities for the moments $\int_0^\infty t^{2j}\Phi(t)\,dt$, which Csordas, Norfolk and Varga proved for every $j \ge 1$, answering a question of Pólya [69]. Table 20.4 shows the margins. They decrease slowly; the Gaussian law, for which $b_j = (\sigma^2/2)^j/j!$, satisfies every inequality with equality.

The Turán inequalities do not characterise the class. The law with mass $\frac{3}{5}$ at 0 and $\frac{1}{5}$ at $\pm\sqrt{5/2}$ has $b_0 = 1$ and $b_j = \frac{2}{5}(\frac{5}{2})^j/(2j)!$ for $j \ge 1$, so $b_1^2 = \frac{1}{4} > \frac{5}{24} = 2b_0 b_2$, and $b_j^2/(b_{j-1}b_{j+1}) = \frac{(j+1)(2j+1)}{j(2j-1)} > 1 + \frac{1}{j}$ for $j \ge 2$: every inequality (20.2.2) holds.

Yet its characteristic function $\frac{3}{5} + \frac{2}{5}\cos(\sqrt{5/2}\,\theta)$ has no real zero. The cumulants detect this: by (20.1.1) a ferromagnetic law has $(-1)^{n+1}\kappa_{2n} = \frac{(2n)!}{n}\sum_k c_k^{-2n} \geq 0$ for $n \geq 2$, while here $\kappa_6 = -\frac{5}{4}$. In the numbers $p_n = (-1)^{n+1} n\kappa_{2n}/(2n)!$, which for a ferromagnetic law are $\sum_k c_k^{-2n}$ when $n \geq 2$ and $b + \sum_k c_k^{-2}$ when $n = 1$, the Hankel determinant $p_1 p_3 - p_2^2$ of p_1, p_2, p_3 is $-\frac{5}{1152}$, whereas for a ferromagnetic law it is $bp_3 + \sum_{j<k} c_j^{-2} c_k^{-2} (c_j^{-2} - c_k^{-2})^2 \geq 0$.

j	b_j	$b_j^2/(b_{j-1}b_{j+1})$	$1 + 1/j$
1	2.3104993×10^{-2}	2.1496879	2
2	2.4833405×10^{-4}	1.5941155	1.5
3	1.6743526×10^{-6}	1.4057378	1.3333333
4	8.0306974×10^{-9}	1.3100631	1.25
5	$2.9401358 \times 10^{-11}$	1.2518550	1.2
10	$1.1310848 \times 10^{-24}$	1.1322525	1.1
15	$1.7510030 \times 10^{-39}$	1.0907154	1.0666667
20	$2.9919818 \times 10^{-55}$	1.0693835	1.05
24	$2.1176410 \times 10^{-68}$	1.0585214	1.0416667

Table 20.4: The moment test (20.2.2) for the Pólya law: the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ and the Turán ratios, against the bound $1 + \frac{1}{j}$ that every ferromagnetic law satisfies. All 24 computed ratios exceed the bound. Computed.

The Turán inequalities are the first of a hierarchy. For $d, n \geq 0$ let

$$J_{d,n}(x) = \sum_{i=0}^{d} \binom{d}{i} (n+i)!\, b_{n+i}\, x^i,$$

the Jensen polynomial of degree d and shift n of the sequence $j!\, b_j$. Pólya proved that RH holds if and only if every $J_{d,n}$ has only real zeros, and for $d = 2$ this is (20.2.2) with $j = n + 1$. Griffin, Ono, Rolen and Zagier proved that for each d all but finitely many of the $J_{d,n}$ have only real zeros, and that all of them do when $d \leq 8$ [97]. Each degree is a necessary condition for the Pólya law to be ferromagnetic, and none is known to fail. With $F(x) = \sum_{j \geq 0} b_j x^j$, so that $F(-t^2) = \varphi_X(t)$, the polynomials $J_{d,n}$ with n fixed are the Jensen polynomials of the derivative $F^{(n)}$, and all of them have only real zeros exactly when $F^{(n)}$ is in the Laguerre Pólya class. For $n \geq 1$ this says nothing about the zeros of φ_X: for the spin one law of Table 20.2 with $q > \frac{1}{2}$, $F(x) = q + (1-q)\cosh\sqrt{x}$ and $F'(x) = (1-q)\sinh\sqrt{x}/(2\sqrt{x})$, whose zeros $-k^2\pi^2$ are real, so every $J_{d,n}$ with $n \geq 1$ has only real zeros, while $\varphi_X(t) = q + (1-q)\cos t$ has no real zero at all. In the other direction, if ξ had only finitely many zeros off the critical line, then F, which has order $\frac{1}{2}$, would have only finitely many nonreal zeros, and $F^{(n)}$ would be in the Laguerre Pólya class for every large n, by the theorem of Craven, Csordas and Smith on real entire functions of order less than two [68].

A necessary condition of a different kind comes from the correlation inequalities of Section 20.1.1. Write $\Phi = e^{-F}$. Newman proved that F' is convex on $[0,\infty)$ [243]; in his normalisation $\Xi(x/2) = 4\int e^{ixt - V(t)}\,dt$, and $V(t) = F(2t) + \log 2$. So the Pólya law belongs to the class of Ellis, Monroe and Newman, and every ferromagnet whose single spins are independent copies of X satisfies the Griffiths, Hurst and Sherman inequality [82].

20.4.2 A mixture of uniform laws

Since p is symmetric and unimodal, Khinchine's theorem of Section 20.2.1 writes $X \stackrel{d}{=} UR$, with R independent of U and with the density $h(r) = -2rp'(r) = -2r\Phi'(r)/\Xi(0)$ on $(0,\infty)$. Since $\mathbb{E}|X|^m = \mathbb{E}R^m/(m+1)$, the mixing variable has $\mathbb{E}R = 2\mathbb{E}|X| = 0.3464543599$ and $\mathbb{E}R^2 = 3\operatorname{Var} X = 0.1386299587$. Integrating by parts as in (20.2.1), now on the half line,

$$z\,\varphi_X(z) = 2\int_0^\infty \big(-p'(u)\big)\sin(zu)\,du,$$

so RH says that the sine transform of the positive function $-\Phi'$ on $(0,\infty)$ has only real zeros. This slope fits neither part of Theorem 20.2.4. Since $p'' = p\,(F'^2 - F'')$, the density is concave near the origin, where $p''(0) = -F''(0)p(0) = -33.65$, and convex for large u, where $F'^2/F'' \to \infty$. On a bounded support, a slope that rises gives only real zeros, by Pólya's theorem, and one that falls, and is not constant on the support, gives zeros off the real line, by Theorem 20.2.4(ii); the slope of the Pólya law does both. The damped densities, proportional to $e^{tu^2/4}\Phi(u)$ with $t < 0$, are symmetric and unimodal too, hence mixtures of uniform laws, and they are mixtures of arcsine laws as well (Proposition 14.4.13), while their transforms have nonreal zeros, so neither mixture can decide RH by itself (Section 14.4.5).

The mixture does not discretise well. Replacing R by J equally weighted atoms at its quantiles r_j of order $(j - \frac12)/J$ gives the characteristic function $\varphi_J(z) = J^{-1}\sum_j \sin(r_j z)/(r_j z)$. For $J = 5$ it has 17 zeros in the rectangle $(0.25, 100) \times (-30, 30)$, of which only 7 are real, and for $J = 20$ only 3 of 19 are real, while φ_X has 29 zeros in $(0, 100)$, all real (computed). As Theorem 20.2.1 suggests, mixing in R destroys the property easily.

20.4.3 No component with finitely many zeros

Sums of independent ferromagnetic variables are ferromagnetic (Section 20.1.5), so one might hope to reach or approximate the Pólya law as such a sum of simple pieces. The growth of L_X rules out most pieces.

Theorem 20.4.1. *Let X be a symmetric random variable with $\mathbb{E}e^{sX} < \infty$ for every real s and $\log \mathbb{E}e^{sX} = o(s^2)$ as $s \to \infty$, and let $X \stackrel{d}{=} D + R$ with D and R*

independent. Then $\log \mathbb{E}e^{sD} = o(s^2)$ *as* $|s| \to \infty$. *In particular, if* D *is ferromagnetic and not* 0, *then* $b = 0$ *in its Hadamard form* (20.1.1), *and its characteristic function has infinitely many zeros. The Pólya law satisfies the hypothesis, so it has no Gaussian component, and it is not a finite or convergent infinite sum of independent ferromagnetic variables of which one is not* 0 *and has finitely many zeros.*

Proof. By independence $\mathbb{E}e^{sX} = \mathbb{E}e^{sD}\,\mathbb{E}e^{sR}$, and both factors are finite because the product is. Jensen's inequality gives $\mathbb{E}e^{sR} \geq e^{s\mathbb{E}R}$, and $\mathbb{E}R = -\mathbb{E}D$ because $\mathbb{E}X = 0$. So $\log \mathbb{E}e^{sD} \leq \log \mathbb{E}e^{sX} + s\mathbb{E}D$, and the right side is $o(s^2)$ as $s \to \pm\infty$, by the symmetry of X. If D is ferromagnetic, $\mathbb{E}e^{sD} = e^{bs^2} \prod_k (1 + s^2/c_k^2) \geq e^{bs^2}$, so $b = 0$. If moreover there are finitely many zeros, then $\varphi_D(t) = \prod_{k \leq K} (1 - t^2/c_k^2)$ is a polynomial bounded on the real line, hence constant, so $K = 0$ and $D = 0$. For the Pólya law, $\log \mathbb{E}e^{sX} = \log \xi(\frac{1}{2} + s) - \log \xi(\frac{1}{2}) = O(s \log s)$, since ξ has order 1. A summand of an almost surely convergent sum of independent variables is an independent component of the sum. □

The laws of Section 19.1 that keep the first K zeros of φ_X exactly carry a Gaussian factor, so none of them is a component of the Pólya law. If RH holds, any decomposition of X into independent ferromagnetic pieces splits the ordinates into infinite sets, each the zero set of a characteristic function $\prod_k (1 - t^2/c_k^2)$ with no Gaussian factor, like those of the uniform law, the fair sign and the symmetric beta laws in Table 20.1.

The simplest such splitting would be a Viète product $L_X(s) = \prod_j \cosh(a_j s)$, which would write $X = \sum_j a_j \varepsilon_j$ as a sum of independent scaled fair signs, a member of the Griffiths Simon class with uncoupled spins. The ordinates would then split into arithmetic progressions, and three consecutive terms of a progression satisfy $\gamma - 2\gamma' + \gamma'' = 0$. The hypothesis that the positive ordinates are linearly independent over the rationals, which Ingham used in his work on the Mertens conjecture [164], forbids such a relation. Under that hypothesis the Pólya law is not a sum of independent scaled fair signs, and if it belongs to the Griffiths Simon class at all, the spins must be coupled. See Section 20.7.

20.5 The exponential of the Pólya variable

The positive variable e^{2X} lives on the Mellin side of Section 12.1.1: it is the variable $\frac{\pi}{2} S_2$ of Biane, Pitman and Yor, size biased by its power $\frac{1}{4}$. Along the family $e^{\beta X}$ the picture is complete except on one interval: $e^{\beta X}$ is not infinitely divisible for $0 < |\beta| < 2$, it is self decomposable and not a GGC at $|\beta| = 2$, and it has a hyperbolically completely monotone density, so it is a GGC, for $|\beta| \geq 5.5791422$, a bound certified in ball arithmetic. None of this assumes anything about the zeros. GGC, HCM and self decomposable laws are as in Section 4.1.2.

20.5.1 Size biased laws and their phase

Write $B_0 = \frac{\pi}{2} S_2 = \sum_{m\ge1} \Gamma_m/(\pi m^2)$, with independent Gamma$(2,1)$ variables Γ_m as in Section 12.1.1. It is a GGC with Thorin atoms of mass 2 at the points πm^2 and Laplace transform $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$, and by (12.1.1) $\mathbb{E}B_0^\sigma = 2\xi(2\sigma)$ for every complex σ. With Riemann's function $\omega(x) = \sum_{n\ge1} e^{-\pi n^2 x}$ of Section 2.1, its density is

$$f_0(y) = \sum_{m\ge1} \big(4\pi^2 m^4 y - 6\pi m^2\big) e^{-\pi m^2 y} = 4y\,\omega''(y) + 6\,\omega'(y)\,:$$

two integrations by parts give $\int_0^\infty y^\sigma (4y\omega'' + 6\omega')\,dy = 2\sigma(2\sigma-1)\pi^{-\sigma}\Gamma(\sigma)\zeta(2\sigma) = 2\xi(2\sigma)$. For real q let B_q be B_0 size biased by B_0^q, with density $y^q f_0(y)/(2\xi(2q))$, so that

$$\mathbb{E}B_q^\sigma = \frac{\xi(2\sigma+2q)}{\xi(2q)}, \qquad e^{2X} \stackrel{d}{=} B_{1/4},$$

the second because $\mathbb{E}e^{2\sigma X} = \xi(\frac12 + 2\sigma)/\xi(\frac12)$. In the notation of Section 12.1.1, where V_α is the Esscher tilt of the Pólya density at the basepoint α, $B_q = e^{2V_{2q}}$; so the B_q with $0 < q < 1$ correspond to the basepoints $0 < \alpha < 2$, and $e^{2X} = e^{2V_{1/2}}$ to the centre. Since X is symmetric, $e^{-\beta X} \stackrel{d}{=} e^{\beta X}$, and for $\beta > 0$, $e^{\beta X} = B_{1/4}^{\beta/2}$.

Proposition 20.5.1. *For $0 < |\beta| < 2$ the variable $e^{\beta X}$ is not infinitely divisible.*

Proof. Take $\beta > 0$. The first term of Φ dominates for large t, so $\log P(X > x) = -\pi e^{2x} + O(x)$ and, with $Y = e^{\beta X}$, $\log P(Y > y) = -\pi y^{2/\beta} + O(\log y)$. Since $2/\beta > 1$, $\mathbb{E}\exp(cY\log Y) < \infty$ for every $c > 0$. By Sato's theorem on the tails of infinitely divisible laws [325, Theorem 26.1], an infinitely divisible law with this property has no Lévy measure, so it is Gaussian or degenerate, and Y is neither. □

The other results come from the continuation of the Laplace transform

$$\Lambda_q(\lambda) = \mathbb{E}e^{-\lambda B_q} = \frac{1}{2\xi(2q)} \int_0^\infty y^q f_0(y)\, e^{-\lambda y}\, dy,$$

which is finite for $\lambda > -\pi$. Let $0 < q < 1$, the range of the results below, fix $M \ge 1$ and let R_M be the tail $\sum_{m>M}$ of the series for f_0. It is bounded near 0, because $f_0(y) = y^{-5/2} f_0(1/y)$ by the evenness of Φ, so that $f_0(y) \to 0$ as $y \downarrow 0$, and it is $O(y\, e^{-\pi(M+1)^2 y})$ at infinity. Integrating the first M terms gives, for $\operatorname{Re}\lambda > -\pi$,

$$2\xi(2q)\,\Lambda_q(\lambda) = \sum_{m\le M} \Big[\frac{4\pi^2 m^4 \Gamma(q+2)}{(\pi m^2+\lambda)^{q+2}} - \frac{6\pi m^2 \Gamma(q+1)}{(\pi m^2+\lambda)^{q+1}}\Big] + \int_0^\infty y^q R_M(y)\, e^{-\lambda y}\, dy, \tag{20.5.1}$$

with principal powers. The last integral is analytic for $\operatorname{Re}\lambda > -\pi(M+1)^2$ and real on the real axis. So (20.5.1) continues Λ_q analytically to the plane cut along

$(-\infty,-\pi]$, and Λ_q has boundary values $\Lambda_q(-r+i0)$ that are continuous in $r\neq\pi m^2$, while near each $-\pi m^2$ it behaves like a multiple of $(\pi m^2+\lambda)^{-q-2}$.

The boundary values carry the Thorin measure: if G is a GGC with drift b and Thorin measure μ, its Laplace transform $H(\lambda)=\exp\{-b\lambda-\int\log(1+\lambda/t)\,\mu(dt)\}$ continues analytically to $\mathbb{C}\setminus(-\infty,0]$, and at a point r that is not an atom of μ and where $H(-r+i0)$ is finite and not zero, the argument of $H(-r+i0)$ is $-\pi\mu((0,r))$, modulo 2π, as the proof of the lemma below shows. We call a continuous determination of $-\frac{1}{\pi}\arg H(-r+i0)$ on an interval a phase of H there. Fourati used the argument of such boundary values to show that certain powers of positive stable variables are not GGC [88].

LEMMA 20.5.2. *Let G be a GGC with drift b, Thorin measure μ and Laplace transform H, and let $I\subset(0,\infty)$ be an open interval such that for each $r\in I$, $H(-r+i\epsilon)$ tends as $\epsilon\downarrow 0$ to a nonzero limit $H(-r+i0)$, continuous in r. Then μ has no atoms in I, and $\mu((0,r))-\phi(r)$ is constant on I for every phase ϕ of H on I.*

Proof. Let $L(\lambda)=-b\lambda-\int\log(1+\lambda/t)\,\mu(dt)$, so that $H=e^L$. For $r\in I$ and $0<\epsilon\le 1$, $\operatorname{Im}\log(1+(-r+i\epsilon)/t)=\arg(t-r+i\epsilon)$ lies in $(0,\pi)$, tends to π, $\frac{\pi}{2}$ or 0 according as $t<r$, $t=r$ or $t>r$, and is at most $2\epsilon/t$ for $t\ge 2r$. Since $\mu((0,2r))<\infty$ and $\int_{[2r,\infty)}\mu(dt)/t<\infty$, dominated convergence gives $\operatorname{Im}L(-r+i\epsilon)\to-\pi\big(\mu((0,r))+\frac{1}{2}\mu(\{r\})\big)$. If $\mu(\{r\})=a>0$ then, since $|t-r+i\epsilon|\le t+r+1$,

$$\operatorname{Re}L(-r+i\epsilon)\;\ge\;br-\int_{t\neq r}\log\Big(1+\frac{r+1}{t}\Big)\,\mu(dt)-a\log\frac{\epsilon}{r}\;\longrightarrow\;\infty,$$

so $|H(-r+i\epsilon)|\to\infty$, which is excluded. Hence μ has no atoms in I, and $e^{i\operatorname{Im}L(-r+i\epsilon)}$, the argument factor of $H(-r+i\epsilon)$, tends to that of $H(-r+i0)$. So $\mu((0,r))-\phi(r)$ is an even integer for each $r\in I$, and being continuous on I it is constant. □

The same lower bound for $\operatorname{Re}L$ shows that the boundary values of a GGC transform never vanish.

THEOREM 20.5.3. *For $0<q<1$ the variable B_q is not a GGC. In particular e^{2X} is not a GGC.*

Proof. Let $r>\pi$, $r\neq\pi m^2$, and take M with $\pi(M+1)^2>r$ in (20.5.1). At $\lambda=-r+i0$ the terms with $\pi m^2>r$ and the last integral are real, and $(\pi m^2-r+i0)^{-p}=(r-\pi m^2)^{-p}e^{-i\pi p}$ for $\pi m^2<r$, so

$$2\xi(2q)\,\operatorname{Im}\Lambda_q(-r+i0)=-\sin(\pi q)\sum_{\pi m^2<r}\Big[\frac{4\pi^2m^4\Gamma(q+2)}{(r-\pi m^2)^{q+2}}+\frac{6\pi m^2\Gamma(q+1)}{(r-\pi m^2)^{q+1}}\Big]<0. \tag{20.5.2}$$

Suppose that B_q is a GGC, and apply Lemma 20.5.2 on $I = (\pi, 4\pi)$. By (20.5.2) a phase ϕ of Λ_q on I takes its values in $(2k, 2k+1)$ for a single integer k. As $r \downarrow \pi$ the first term of (20.5.1) dominates, so that $\Lambda_q(-r + i0) \sim c\,(r-\pi)^{-q-2} e^{-i\pi(q+2)}$ with $c > 0$, and $\phi(r) \to 2k + q$. As $r \uparrow 4\pi$ the term with $m = 2$ dominates and is positive, and $\phi(r) \to 2k$. So ϕ falls by q across I, while $\mu((0,r)) = \phi(r) + \text{const}$ is nondecreasing, a contradiction. □

Size biasing a GGC by a negative power gives a GGC: Fourati's Lemma 6.3 [88] states this for every real exponent, and its proof, which multiplies the variable by one plus an independent gamma variable of shape $\beta > 0$ and lets the rate of the gamma variable tend to 0, covers the size bias by $x^{-\beta}$, once slips in the printed normalising constants are corrected. For positive exponents it fails: for every exponent in $(0,1)$ by Theorem 20.5.3, and at 1 as well, since the size bias of order one of the GGC $e_1 + \frac{1}{2}e_2$ has Laplace transform $\frac{4}{3}(2\lambda+3)(1+\lambda)^{-2}(2+\lambda)^{-2}$, which vanishes at the point $-\frac{3}{2}$ of the cut. With the negative orders, B_q is a GGC for every $q \le 0$ and for no $q \in (0,1)$.

20.5.2 The size bias principle

A gamma variable of shape t size biased by y^q is a gamma variable of shape $t+q$ with the same rate. For a sum of gamma variables with different rates the same happens at every rate, up to an independent GGC that lives between the rates. The proof uses divided differences: For a polynomial Q with real zeros $s_1 \le \cdots \le s_n$, listed with multiplicity, and a function h analytic near $[s_1, s_n]$, put

$$h[Q] = \frac{1}{2\pi i} \oint \frac{h(\zeta)}{Q(\zeta)}\, d\zeta = \sum_{\zeta_0} \mathrm{Res}_{\zeta=\zeta_0} \frac{h(\zeta)}{Q(\zeta)},$$

the contour enclosing the zeros of Q inside the domain of h and the sum running over the distinct zeros ζ_0 of Q, so that $h[Q]$ is the divided difference of h at $s_1, \dots, s_n$ when Q is monic. By the formula of Hermite and Genocchi, $h[Q]$ is the integral of $h^{(n-1)}(\sum_j w_j s_j)$ over the simplex of weights $w_j \ge 0$, $\sum_j w_j = 1$, with its uniform measure of total mass $1/(n-1)!$; so $h[Q]$ has the sign of $h^{(n-1)}$ when that sign is constant on $[s_1, s_n]$.

Lemma 20.5.4. *Let $Q(\zeta) = \prod_{n=1}^{K} (\zeta - \tau_n)^{t_n}$ with $\tau_1 < \cdots < \tau_K$ and integers $t_n \ge 1$, let $1 \le k < K$ and $\tau_k < x < \tau_{k+1}$, and write $Q = Q_- Q_+$ with $Q_-(\zeta) = \prod_{n \le k} (\zeta - \tau_n)^{t_n}$, of degree N_-, and Q_+ of degree N_+. For $p > 0$ put*

$$D_+(x) = h_+[Q_+], \qquad h_+(\zeta) = \frac{(\zeta - x)^{-p}}{Q_-(\zeta)},$$

$$D_-(x) = h_-[Q_-], \qquad h_-(\zeta) = \frac{(x - \zeta)^{-p}}{Q_+(\zeta)}.$$

Then $(-1)^{N_+-1}D_+(x) > 0$ *and* $(-1)^{N_+}D_-(x) > 0$, *and as functions of* $x \in (\tau_k, \tau_{k+1})$, $|D_+|$ *increases and* $|D_-|$ *decreases.*

Proof. On (x, ∞) the function h_+ is a product of the completely monotone functions $(\zeta - x)^{-p}$ and $(\zeta - \tau_n)^{-1}$, $n \le k$, so $(-1)^j h_+^{(j)} > 0$ there for every j, and the formula of Hermite and Genocchi with $j = N_+ - 1$ gives the first sign. On $(-\infty, x)$ the function $(-1)^{N_+}h_-(\zeta) = (x-\zeta)^{-p}\prod_{n>k}(\tau_n - \zeta)^{-t_n}$ is a product of functions all of whose derivatives are positive, which gives the second. Finally $\partial_x(\zeta - x)^{-p} = p(\zeta - x)^{-p-1}$ and $\partial_x(x-\zeta)^{-p} = -p(x-\zeta)^{-p-1}$, so D'_+ and D'_- are p and $-p$ times the same expressions with $p+1$ in place of p: D'_+ has the sign of D_+, and D'_- the sign opposite to that of D_-. □

THEOREM 20.5.5 (size bias principle). *Let* $Y = \sum_{n=1}^K \Gamma_n/\tau_n$ *with* $0 < \tau_1 < \cdots < \tau_K$ *and independent gamma variables* Γ_n *with integer shapes* $t_n \ge 1$ *and unit scale, let* $0 < q < 1$, *and let* Y_q *be* Y *size biased by* Y^q. *Then*

$$Y_q + G \stackrel{d}{=} \sum_{n=1}^K \frac{\Gamma'_n}{\tau_n},$$

where the Γ'_n *are independent gamma variables with shapes* $t_n + q$ *and unit scale, and* G, *independent of* Y_q, *is a GGC without drift whose Thorin measure has a continuous distribution function* θ_G *that vanishes on* $(0, \tau_1]$, *increases by exactly* q *on each interval* $[\tau_k, \tau_{k+1}]$ *and is constant on* $[\tau_K, \infty)$.

Proof. Let $N = \sum_n t_n$ and $Q(\zeta) = \prod_n(\zeta - \tau_n)^{t_n}$. The density of Y is $c_0(-1)^{N-1}(e^{-\zeta y})[Q]$ with $c_0 = \prod_n \tau_n^{t_n}$, as taking Laplace transforms under the contour integral shows: $(1/(\zeta+\lambda))[Q] = -1/Q(-\lambda)$, by the residue at $\zeta = -\lambda$. Multiplying by y^q before integrating gives, for $\lambda \ge 0$,

$$\Lambda(\lambda) = \mathbb{E}e^{-\lambda Y_q} = c\,(-1)^{N-1}g_\lambda[Q], \qquad g_\lambda(\zeta) = (\zeta+\lambda)^{-1-q},$$

with a constant $c > 0$, and since $g_\lambda^{(N-1)} = (-1)^{N-1}(1+q)\cdots(N-1+q)\,(\zeta+\lambda)^{-N-q}$, the formula of Hermite and Genocchi gives

$$\Lambda(\lambda) = \frac{\mathbb{E}(\bar{s}+\lambda)^{-N-q}}{\mathbb{E}\bar{s}^{\,-N-q}}, \qquad \bar{s} = \sum_{j=1}^N w_j s_j,$$

where $s_1, \ldots, s_N$ are the rates listed with multiplicity and (w_j) is uniform on the simplex. This continues Λ analytically to $\mathbb{C} \setminus (-\infty, -\tau_1]$, where the residue form $c(-1)^{N-1}g_\lambda[Q]$ remains valid, and shows that $|\Lambda(\lambda)|$ is of exact order $|\lambda|^{-N-q}$ as $|\lambda| \to \infty$, uniformly up to the cut.

Boundary values. Let $\tau_k < r < \tau_{k+1}$. At points $\zeta < r$ the derivatives of $(\zeta - r + i0)^{-1-q}$ are $e^{-i\pi(1+q)}$ times those of $(r-\zeta)^{-1-q}$. Splitting the residues

of g_λ/Q into those at τ_n with $n > k$ and those with $n \le k$ gives, with $D_\pm$ of Lemma 20.5.4 at $p = 1+q$ and the signs found there,

$$\Lambda(-r+i0) = c\,(-1)^{N-1}\big(D_+(r) + e^{-i\pi(1+q)}D_-(r)\big) = c\,(-1)^{N_-}z(r),$$
$$z(r) = |D_+(r)| + e^{-i\pi q}|D_-(r)|.$$

The point $z(r)$ is the sum of two nonzero vectors at the angle $\pi q < \pi$, so its argument lies in $(-\pi q, 0)$. The ratio $|D_-|/|D_+|$ decreases, from ∞ as $r \downarrow \tau_k$, where D_- has the singular term $(r-\tau_k)^{-t_k-q}$, to 0 as $r \uparrow \tau_{k+1}$; so $\arg z(r)$ increases from $-\pi q$ to 0. For $r > \tau_K$ the same computation with $Q_+ = 1$ gives $\Lambda(-r+i0) = c(-1)^{N-1}e^{-i\pi(1+q)}D_-(r)$ with $D_- > 0$.

The factor G. Put $\theta_G = 0$ on $(0,\tau_1]$, $\theta_G(r) = kq + \frac{1}{\pi}\arg z(r)$ for $\tau_k < r < \tau_{k+1}$, and $\theta_G = (K-1)q$ on $[\tau_K, \infty)$. This function is continuous and nondecreasing. Near τ_k, $|D_+/D_-|$ is $(r-\tau_k)^{t_k+q}$ times a function that is continuously differentiable up to τ_k, and similarly near τ_{k+1}; since $t_k+q > 1$, θ_G is continuously differentiable, with derivative 0 at the points τ_n. Let G be the GGC without drift and with Thorin measure $d\theta_G$, with Laplace transform $H_G(\lambda) = \exp\{-\int \log(1+\lambda/t)\,d\theta_G(t)\}$, and let $\Lambda_+(\lambda) = \prod_n (1+\lambda/\tau_n)^{-t_n-q}$, the Laplace transform of the right side. The function $J = \Lambda H_G/\Lambda_+$ is analytic on $\mathbb{C}\setminus(-\infty,-\tau_1]$ and positive on $(-\tau_1,\infty)$. For $\tau_k < r < \tau_{k+1}$ the arguments of Λ, H_G and $1/\Lambda_+$ at $-r+i0$ are, modulo 2π,

$$\pi N_- + \arg z(r), \qquad -\pi kq - \arg z(r), \qquad \pi(N_- + kq),$$

and for $r > \tau_K$ they are $\pi N - \pi q$, $-\pi(K-1)q$ and $\pi(N+Kq)$. In both cases they add up to a multiple of 2π, so J has real boundary values from above, by symmetry the same boundary values from below, and it continues analytically across the cut between the points $-\tau_n$. Near $-\tau_n$, $\Lambda = O(|\lambda+\tau_n|^{-t_n-q})$ by the residue form, and $1/\Lambda_+$ cancels this, while H_G is bounded because θ_G has a bounded density; so these singularities are removable, and J is entire. As $|\lambda| \to \infty$, $|\Lambda|$, $|1/\Lambda_+|$ and $|H_G|$ are of the exact orders $|\lambda|^{-N-q}$, $|\lambda|^{N+Kq}$ and $|\lambda|^{-(K-1)q}$, so J is bounded. By Liouville's theorem $J \equiv J(0) = 1$, that is $\Lambda H_G = \Lambda_+$, which is the identity in law. □

THEOREM 20.5.6. *For $0 < q < 1$,*

$$B_q + G_q \;\stackrel{d}{=}\; \frac{\pi}{2}S_{2+q},$$

where G_q is independent of B_q and is a GGC without drift whose Thorin measure has a continuous distribution function θ_q with $\theta_q = 0$ on $(0,\pi]$ and $\theta_q(\pi m^2) = (m-1)q$ for every $m \ge 1$. In particular $e^{2X} + G_{1/4} \stackrel{d}{=} \frac{\pi}{2}S_{9/4}$.

Proof. Let $B^{(M)} = \sum_{m\le M}\Gamma_m/(\pi m^2)$ and let $B_q^{(M)}$ be its size bias of order q. By Theorem 20.5.5, $B_q^{(M)} + G_M \stackrel{d}{=} \sum_{m\le M}\Gamma'_m/(\pi m^2)$, with shapes $2+q$ on the

right. As $M \to \infty$, $\mathbb{E}[(B^{(M)})^q e^{-\lambda B^{(M)}}] \to \mathbb{E}[B_0^q e^{-\lambda B_0}]$ for $\lambda \geq 0$ by dominated convergence, since $B^{(M)} \leq B_0$; so $B_q^{(M)} \to B_q$ in law, and the right sides converge in law to $\frac{\pi}{2} S_{2+q}$, whose Laplace transform is $\Lambda_+(\lambda) = (\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^{2+q}$. Hence the Laplace transforms of the G_M converge to $H = \Lambda_+/\Lambda_q$, which is continuous at 0. So G_M converges in law to a variable G_q with Laplace transform H, G_q is a GGC by the closure theorem of Section 9.1.1, and $B_q + G_q \stackrel{d}{=} \frac{\pi}{2} S_{2+q}$. A GGC with drift b is at least b, while $\frac{\pi}{2} S_{2+q}$ is smaller than any given positive number with positive probability, so G_q has no drift.

It remains to find the Thorin measure μ of G_q. By (20.5.1), near each $-\pi m^2$ both Λ_q and Λ_+ are $(\pi m^2 + \lambda)^{-2-q}$ times functions with continuous nonzero boundary values, and between these points $\Lambda_q(-r + i0) \neq 0$ by (20.5.2). So H has continuous nonzero boundary values on all of $(0, \infty)$, and by Lemma 20.5.2 μ has no atoms and $\mu((0, r)) = \phi(r)$, where ϕ is the phase of H that vanishes on $(0, \pi)$, where $H > 0$. On $(\pi m^2, \pi(m+1)^2)$ the argument of $\Lambda_+(-r + i0)$ is $-\pi(2+q)m$, and $\Lambda_q(-r+i0)$ lies in the lower half plane by (20.5.2), with argument tending to $-\pi(2+q)$ modulo 2π at the left end and to 0 at the right end, as in the proof of Theorem 20.5.3. So, modulo 2, ϕ takes its values in $(mq - 1, mq)$ on this interval and tends to $(m-1)q$ at its left end and to mq at its right end. Since ϕ is continuous and $\phi(\pi) = 0$, induction on m gives $\phi(\pi m^2) = (m-1)q$. □

The theorem gives B_q a signed Thorin measure, $(2+q)\sum_m \delta_{\pi m^2} - d\theta_q$, with a negative continuous part of mass q on each gap, which is what Theorem 20.5.3 detects.

20.5.3 Self decomposability and the theta Lévy density

Theorem 20.5.7. *For $0 < q < 1$ the variable B_q is self decomposable, with no drift and Lévy density $k_q(x)/x$, where*

$$k_q(x) = (2+q)\,\omega(x) - \int_0^\infty e^{-tx}\, d\theta_q(t), \qquad 2\,\omega(x) \leq k_q(x) \leq (2+q)\,\omega(x).$$

In particular e^{2X} is self decomposable, hence infinitely divisible and unimodal, its Lévy density lies between $2\omega(x)/x$ and $\frac{9}{4}\omega(x)/x$, and it is not a GGC.

Proof. By Theorem 20.5.6, $-\log\Lambda_q = -\log\Lambda_+ + \log H$, and Frullani's integral $\log(1 + \lambda/t) = \int_0^\infty (1 - e^{-\lambda x}) e^{-tx}\, dx/x$, applied to the atoms of $\frac{\pi}{2} S_{2+q}$ and to the Thorin measure of G_q, gives

$$-\log\Lambda_q(\lambda) = \int_0^\infty (1 - e^{-\lambda x})\, k_q(x)\, \frac{dx}{x}.$$

On $[\pi m^2, \pi(m+1)^2]$ the measure $d\theta_q$ has mass q and $e^{-tx} \leq e^{-\pi m^2 x}$, so $0 \leq \int e^{-tx}\, d\theta_q(t) \leq q\,\omega(x)$, which gives the bounds. In particular $k_q > 0$, so B_q is

infinitely divisible with Lévy density $k_q(x)/x$, and it is self decomposable if k_q is nonincreasing (Section 4.1.2), that is, if

$$\int_0^\infty te^{-tx}\,d\theta_q(t) \;\le\; (2+q)\sum_{m\ge1}\pi m^2e^{-\pi m^2x}, \qquad x>0.$$

Bound the contribution of $[\pi m^2,\pi(m{+}1)^2]$ to the left side by $q\,s_m(x)$, with $s_m(x)=\max\{te^{-tx}:\pi m^2\le t\le\pi(m+1)^2\}$. For $m\ge2$, $s_m(x)\le\pi(m+1)^2e^{-\pi m^2x}\le\frac94\pi m^2e^{-\pi m^2x}$, and $\frac94q<2+q$, which leaves $(2-\frac54q)\pi m^2e^{-\pi m^2x}$ of the mth term on the right. It remains to show $q\,s_1(x)\le(2+q)\pi e^{-\pi x}+(2-\frac54q)4\pi e^{-4\pi x}$. If $x\ge1/\pi$, then $s_1(x)=\pi e^{-\pi x}$. If $x\le1/(4\pi)$, then $s_1(x)=4\pi e^{-4\pi x}$, and $4q\le4(2-\frac54q)+(2+q)$ because $8q\le10$. Otherwise $s_1(x)=1/(ex)$, and $(2+q)\pi x\,e^{1-\pi x}\ge(2+q)\frac14e^{3/4}>1>q$, because ue^{1-u} increases on $(\frac14,1)$. Self decomposable laws are unimodal [366], and $e^{2X}=B_{1/4}$ is not a GGC by Theorem 20.5.3. □

Near 0, $\omega(x)\sim1/(2\sqrt{x})$, and $0\le q\sqrt{t/\pi}-\theta_q(t)\le2q$, since $\theta_q=0$ on $(0,\pi]$ and, on $[\pi m^2,\pi(m+1)^2]$, the first lies between mq and $(m+1)q$ and the second between $(m-1)q$ and mq; so the subtracted fraction $1-k_q/((2+q)\omega)$ tends to $q/(2+q)$ as $x\to0$.

We check the three theorems numerically: we continue Λ_q by (20.5.1) with $M=5$, at $30+r/4$ digits, on 1800 points of $r\in[0.05,90]$, for $q=0.1$, 0.25, 0.5, 0.75 and 0.9. In every case the phase of Λ_q decreases along the grid on each of the four gaps $(\pi m^2,\pi(m+1)^2)$, $m\le4$, so that θ_q increases, and at the last grid point below $\pi(m+1)^2$, θ_q is within 10^{-7} of mq, as Theorem 20.5.6 requires. Extending θ_q beyond the scan by $q(\sqrt{t/\pi}-1)$, the value it takes at the points πm^2, the function k_q is positive and decreasing on 300 points of $[10^{-3},3]$, and the subtracted fraction at $x=10^{-3}$ is 0.046, 0.108, 0.194, 0.264 and 0.300, against the limits 0.048, 0.111, 0.200, 0.273 and 0.310.

The factorisation can also be tested away from the cut, through the characteristic function. For a positive infinitely divisible variable B with no drift and Lévy density $k(x)/x$, $-\log|\mathbb{E}e^{iyB}|^2=\int_0^\infty4\sin^2(\frac{yx}{2})\,k(x)\,dx/x$, and Frullani's integral $\int_0^\infty4\sin^2(\frac{yx}{2})e^{-tx}\,dx/x=\log(1+y^2/t^2)$ turns this into the Thorin form $\int\log(1+y^2/t^2)$ against the Thorin measure, here the signed one. For B_q the atoms sum in closed form, by $\prod_m(1+u^4/m^4)=(\cosh\sqrt2\pi u-\cos\sqrt2\pi u)/(2\pi^2u^2)$:

$$-\log\left|\mathbb{E}e^{iyB_q}\right|^2=(2+q)\log\frac{\cosh\sqrt{2\pi y}-\cos\sqrt{2\pi y}}{2\pi y}-\int_0^\infty\log\left(1+\frac{y^2}{t^2}\right)d\theta_q(t).$$

Table 20.5 compares, for e^{2X}, the direct Fourier integral of the density with the Thorin form and the theta Lévy form, both built from the computed $\theta_{1/4}$. The two forms agree with each other to a relative 2×10^{-7}, and with the direct integral to a relative 1.2×10^{-5}. As a last check, $\mathbb{E}G_{1/4}=\mathbb{E}\frac\pi2S_{9/4}-\mathbb{E}e^{2X}=\frac{3\pi}{8}-\xi(\frac52)/\xi(\frac12)=0.0815947$, and $\int\theta_{1/4}(t)\,t^{-2}dt=0.0815351$ from the computed Thorin measure;

y	direct	Thorin form	theta Lévy form	atoms alone
0.5	0.0588838	0.0588840	0.0588840	0.0609722
1	0.227615	0.227616	0.227616	0.235869
2	0.808286	0.808290	0.808290	0.839868
4	2.347416	2.347431	2.347431	2.457076
8	5.261493	5.261552	5.261553	5.575941

Table 20.5: $-\log|\mathbb{E}e^{iye^{2X}}|^2$ computed four ways: by a direct Fourier integral of the density $y^{1/4}f_0(y)/(2\xi(\frac{1}{2}))$; in the Thorin form with the atoms $\frac{9}{4}$ at the points πm^2 summed in closed form and the continuous part from the computed $\theta_{1/4}$; in the theta Lévy form $\int 4\sin^2(\frac{yx}{2})k_{1/4}(x)\,dx/x$; and from the atoms alone, which is $-\log|\mathbb{E}e^{iy\frac{\pi}{2}S_{9/4}}|^2$. Computed.

the extension beyond $t = 90$ differs from $\theta_{1/4}$ by at most q, which can move the integral by at most $q/90 < 0.003$.

On the dual side the roles are reversed. Section 13.1.4 puts the characteristic function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ of $Y_{1/2}$ in the same form, with no hypothesis: $\log(\xi(\frac{1}{2}+u)/\xi(\frac{1}{2})) = 2\int_0^\infty(1-\cos ux)\,K_{1/2}(x)\,dx/x$ for real u. So the logarithm of the moment generating function of X has the form of $-\log|\mathbb{E}e^{iyB_q}|^2$, with the Poisson trace $K_{1/2}$, a sum over the zeros, in place of the theta kernel k_q. The Poisson trace is completely monotone exactly when RH holds (Corollary 13.1.4), while $k_{1/4}$ is positive and decreasing with no hypothesis.

20.5.4 Hyperbolically completely monotone powers

The kernel Φ is analytic in the strip $|\operatorname{Im} t| < \pi/4$, where $\operatorname{Re} e^{2t} > 0$, and since it is real and even, $F(v) = \Phi(iv)$ is real for real v. We have $F(0) = 0.8933938$; F has a local maximum $F(v_m) = 7.0315837$ at $v_m = 0.5630962646$, and it vanishes at $v = 0.6388301$, which is the zero α_1 of the continuation in Proposition 11.1.18(iii).

Proposition 20.5.8. *The variable $e^{\beta X}$ has an HCM density, and so is a GGC, for $|\beta| \geq 5.5791422$, and it does not have an HCM density for $|\beta| \leq 5.5791395$. The proof of the first statement is computer assisted.*

Proof. Let $\beta > 0$. The density $p(\beta^{-1}\log y)/(\beta y)$ of $e^{\beta X}$ has elasticity $1-\beta^{-1}(\Phi'/\Phi)(\beta^{-1}\log y)$, and the upper half plane in y corresponds to the strip $0 < \operatorname{Im} t < \pi/\beta$. By Theorem 9.3.2, as in Proposition 9.3.26, $e^{\beta X}$ has an HCM density if and only if Φ continues to a zero free analytic function on $|\operatorname{Im} t| < \pi/\beta$ and

$$h(x,v) = \operatorname{Im}\Big(-\frac{\Phi'}{\Phi}\Big)(x+iv) \geq 0 \qquad \text{for } 0 < v < \pi/\beta,$$

and HCM laws are GGC by (F5) of Section 9.2.1. On the imaginary axis $h(0,v) = F'(v)/F(v)$. Ball arithmetic [169] encloses v_m in an interval of width 2×10^{-17}

with certified signs $F' > 0$ at its left end and $F > 0 > F'$ at its right end v_b. If $\pi/\beta > v_b$, the strip contains iv_b, where Φ is analytic and $h < 0$, so $e^{\beta X}$ is not HCM. Since $\pi/v_b > 5.5791395$, this proves the second statement.

For the first, let $v_* = 0.563096$, so that $\pi/v_* < 5.5791422$, and let φ_1 be the term $n = 1$ of Φ. Three estimates, in ball arithmetic with rigorous bounds for the tail of the series, show that $h \geq 0$ on $0 < v \leq v_*$. (Z) For $\operatorname{Re} t \geq 0$ and $|\operatorname{Im} t| \leq 0.5634$, $|\Phi/\varphi_1 - 1| \leq 0.6015$, so $\Phi \neq 0$ there, and also for $\operatorname{Re} t \leq 0$, because $\Phi(-\bar{t}) = \overline{\Phi(t)}$. (T) On the segment $v = v_*$, $0 \leq x \leq 2$, adaptive bisection into 2896 intervals encloses h in positive balls. The smallest lower bound, 1.8×10^{-8}, occurs next to $x = 0$, where $h(0, v)$ tends to 0 as $v \uparrow v_m$. Since $h(-x, v) = h(x, v)$, $h > 0$ on the whole segment $|x| \leq 2$. (S) For $x \geq 2$ and $0 < v \leq v_*$, the first term gives $\operatorname{Im}(-\varphi_1'/\varphi_1) \geq 2\pi e^{2x} \sin 2v \geq 8e^4 v$, and Cauchy's estimate on discs of radius 0.1 bounds the imaginary part of the derivative of $\log(\Phi/\varphi_1)$ by $10^{-40}v$; so $h > 0$ there, and by symmetry also for $x \leq -2$. By (Z), h is harmonic on the strip $|\operatorname{Im} t| < 0.5634$. It vanishes on the real axis and, by (T) and (S), is positive on the top and the sides of the rectangle $[-2, 2] \times [0, v_*]$, so $h \geq 0$ inside by the minimum principle, and with (S), $h \geq 0$ on $0 < v \leq v_*$. So $e^{\beta X}$ is HCM for $\beta \geq \pi/v_*$. □

The certificate stops 3×10^{-6} short of $\pi/v_m = 5.5791396$, because h vanishes at iv_m and the strip of height v_m cannot be covered by balls with positive lower bounds; presumably the threshold is exactly π/v_m. The computation works in the variable $t/2$ of Rodgers and Tao [318], where the heights are halved and the thresholds doubled.

The HCM threshold, the least $|\beta|$ for which $e^{\beta X}$ has an HCM density (these powers form a closed half line, by (F2) and (F4) of Section 9.2.1), lies between 5.5791395 and π/v_*, and for $2 < |\beta|$ below it infinite divisibility and the GGC property are open (Section 20.7). The bracket comes from the kernel, not from the zeros, and needs no hypothesis. It is the Pólya side counterpart of Theorem 9.3.27, where, under RH and two further hypotheses on the zeros, the HCM powers of the Mellin variable W are limited by the first gap $\gamma_2 - \gamma_1$. For the clock, whose Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ is the moment generating function of X inverted at $\sqrt{u}$, the GGC property is RH itself (Proposition 2.1.1).

20.6 Two arithmetic spin chains

In this section $\beta = \sigma + it$ is a complex inverse temperature, $s(n)$ is Stern's diatomic sequence, given by $s(1) = 1$, $s(2n) = s(n)$ and $s(2n + 1) = s(n) + s(n + 1)$ [154], and μ and φ are the Möbius and Euler functions (φ is Euler's function here, not the characteristic function φ_X of Definition 20.1.1). Knauf's number theoretical spin chain [126, 127] has, at length N, 2^N configurations, whose energies are the

numbers $\log s(k)$, $2^N \le k < 2^{N+1}$. Its partition function is

$$Z_N(\beta) = \sum_{k=2^N}^{2^{N+1}-1} s(k)^{-\beta} = \sum_{m\ge 1} c_N(m)\, m^{-\beta}, \tag{20.6.1}$$

where $c_N(m)$ is the number of times m occurs in the row $s(2^N), \dots, s(2^{N+1}-1)$. Knauf proved that the chain is ferromagnetic, in the sense that minus its energy, expanded in products of the spins, has nonnegative coefficients apart from the constant term, and that its interactions are many body [126]; Contucci and Knauf proved that it has a phase transition at $\beta = 2$ [64]. Section 20.6.3 treats a second chain, of Kleban and Özlük, in a magnetic field. This section is based on [137].

20.6.1 Knauf's chain and the Stern row

Proposition 20.6.1. *(i) $c_N(1) = 1$, and for $m \ge 2$, $c_N(m)$ is the number of coprime pairs (p, q) of positive integers with $p + q = m$ for which p/q has depth at most $N-1$ in the Calkin Wilf tree. Hence $c_N(m)$ does not decrease with N, $c_N(m) = \varphi(m)$ for $m \le N+1$, $c_N(m) = 0$ for $m > F_{N+2}$, where F_n are the Fibonacci numbers, $F_1 = F_2 = 1$, and $c_N(m) \to \varphi(m)$ as $N \to \infty$.*

(ii) $Z_N(\beta) \to \zeta(\beta-1)/\zeta(\beta)$ as $N \to \infty$ for $\operatorname{Re}\beta > 2$, increasingly for real $\beta > 2$.

(iii) $Z_N(-1) = \sum_m m\, c_N(m) = 3^N$.

Part (ii) is due to Knauf [126]; the proof below uses the Calkin Wilf enumeration of the positive rationals [52].

Proof. The Calkin Wilf tree has the root $1/1$, and the children of p/q are $p/(p+q)$ and $(p+q)/q$. Read level by level it lists the fractions $s(k)/s(k+1)$, $k \ge 1$, those at depth j being those with $2^j \le k < 2^{j+1}$, and every positive rational occurs in it exactly once, in lowest terms [52]. Since $s(2k) = s(k)$ and $s(2k+1) = s(k) + s(k+1)$, row $N+1$ consists of row N, at its even places, and of the sums $p+q$ over the fractions p/q at depth N, at its odd places. Row 0 is $\{1\}$, so by induction row N consists of 1 and of one entry $p+q$ for each fraction p/q at depth at most $N-1$. The coprime pairs with $p+q = m \ge 2$ are the pairs $(p, m-p)$ with p prime to m; there are $\varphi(m)$ of them, each at one depth. Passing from p/q to a child adds p or q to $p+q$, so a fraction at depth j has $j+2 \le p+q \le F_{j+3}$, the upper bound by induction together with $\max(p,q) \le F_{j+2}$. Hence every pair with $p+q = m \le N+1$ has depth at most $m-2 \le N-1$, and no fraction at depth at most $N-1$ has $p+q > F_{N+2}$. This proves (i). Part (ii) follows from (i) by monotone convergence for real β and by dominated convergence for complex β, since $\sum_m \varphi(m) m^{-\beta} = \zeta(\beta-1)/\zeta(\beta)$ converges absolutely for $\operatorname{Re}\beta > 2$. For (iii), let S_N be the sum of row N. Then $S_{N+1} = S_N + \sum_{k=2^N}^{2^{N+1}-1} \big(s(k)+s(k+1)\big) = 3S_N$, because $s(2^N) = s(2^{N+1}) = 1$, and $S_0 = 1$. □

The finite statements of (i) and (iii) were also checked with exact integers for every $N \le 34$, and the depth count of (i) against Euclid's algorithm for $m \le 800$ (computed). The continuation of $\zeta(\beta-1)/\zeta(\beta)$ vanishes at the points $1+\rho$, for the nontrivial zeros ρ of ζ, at $\beta = 1$ and at $\beta = 1-2k$, $k \ge 1$. The series $\sum_m \varphi(m) m^{-\beta}$ diverges on the strip $1 < \operatorname{Re}\beta < 2$, which contains the points $1+\rho$, but the finite sums Z_N have zeros close to them (Section 20.6.2). As in statistical mechanics, zeros of a partition function in the complex inverse temperature are called Fisher zeros.

20.6.2 The Perron identity

Let $w_N(x) = c_N(m)/\varphi(m)$ for $m \le x < m+1$, $m \ge 1$, and $w_N(x) = 1$ for $0 < x < 1$. By Proposition 20.6.1(i) this depth profile equals 1 on $(0, N+2)$ and 0 on $[F_{N+2}+1, \infty)$, so its Mellin transform $W_N(z) = \int_0^\infty w_N(x) x^{z-1}\,dx$, defined for $\operatorname{Re} z > 0$, is $1/z$ plus an entire function.

Proposition 20.6.2. *For* $\operatorname{Re}\beta > 1$, $\beta \ne 2$,

$$Z_N(\beta) = \frac{\zeta(\beta-1)}{\zeta(\beta)} + \frac{W_N(2-\beta)}{\zeta(2)} + R_N(\beta), \qquad R_N(\beta) = \sum_{d\ge1} \mu(d) d^{-\beta} r_d(\beta),$$

where $r_d(\beta) = \sum_{n\ge1} n^{1-\beta} w_N(dn) - \zeta(\beta-1) - d^{\beta-2} W_N(2-\beta)$, *and the series converges absolutely.* R_N *extends analytically to* $\beta = 2$, *and* $C_N(\beta) = Z_N(\beta) - W_N(2-\beta)/\zeta(2)$ *is meromorphic on the plane with a single pole, at* $\beta = 2$, *which is simple with residue* $1/\zeta(2)$.

Proof. Since $\varphi(m) = \sum_{d\mid m} \mu(d)\, m/d$ and $c_N(m) = \varphi(m) w_N(m)$ vanishes for $m > F_{N+2}$,

$$Z_N(\beta) = \sum_m \varphi(m) w_N(m) m^{-\beta} = \sum_{d\ge1} \mu(d) d^{-\beta} \sum_{n\ge1} n^{1-\beta} w_N(dn),$$

a finite sum. For $\operatorname{Re}\beta > 1$ the series $\sum_d \mu(d) d^{-\beta} = 1/\zeta(\beta)$ and $\sum_d \mu(d) d^{-2} = 1/\zeta(2)$ converge absolutely, and $r_d(\beta) = -\zeta(\beta-1) - d^{\beta-2} W_N(2-\beta)$ for $d > F_{N+2}$, so the series for R_N converges absolutely, and inserting the definition of r_d gives the identity. Z_N is entire, and $W_N(2-\beta)$ is analytic except for a simple pole at $\beta = 2$ with residue -1, which gives the statement on C_N. The main term $\zeta(\beta-1)/\zeta(\beta)$ has a simple pole at $\beta = 2$ with residue $1/\zeta(2)$, so the poles of the two main terms cancel and R_N extends analytically to $\beta = 2$. □

The proposition holds for every N; its content is the form of R_N. For $1 < \operatorname{Re}\beta < 2$, $d^{\beta-2} W_N(2-\beta) = \int_0^\infty w_N(dx) x^{1-\beta}\,dx$, so $r_d(\beta)$ is the error made in replacing the sum $\sum_n n^{1-\beta} w_N(dn)$ by the corresponding integral, less the value $\zeta(\beta-1)$ of the same difference for $w_N \equiv 1$ (the limit of the truncated differences): an Euler Maclaurin remainder. Heuristically R_N is small when the depth profile

t	$N = 16$	$N = 24$	$N = 32$	$\lvert\zeta(\beta-1)/\zeta(\beta)\rvert$
0.3	15.0197 / 14.9781	43.7060 / 43.7165	126.5819 / 126.5812	0.5617
1	4.7475 / 4.7440	10.6377 / 10.6446	23.8043 / 23.7874	0.5873
2	1.0789 / 1.0509	1.4826 / 1.5005	2.0082 / 2.0524	0.6557
5	0.0938 / 0.1117	0.0446 / 0.0652	0.0376 / 0.0503	0.9003
10	0.0660 / 0.0812	0.0762 / 0.0805	0.0415 / 0.0462	1.2084

Table 20.6: Knauf's chain on the line $\operatorname{Re}\beta = \frac{3}{2}$, at $\beta = \frac{3}{2} + it$. Each entry is $|Z_N(\beta) - \zeta(\beta-1)/\zeta(\beta)|$ followed, after a slash, by the profile term $|W_N(2-\beta)|/\zeta(2)$; computed.

is regular on the scale of the integers. The two main terms are the residues that a smooth cutoff produces (Remark 20.6.4(i)).

On the line $\operatorname{Re}\beta = \frac{3}{2}$ the profile term $W_N(2-\beta)/\zeta(2)$ grows with N at small heights and accounts there for almost all of $Z_N - \zeta(\beta-1)/\zeta(\beta)$ (Table 20.6): at the tabulated points with $t \le 2$ the two moduli agree to within 3 percent. The profile term falls below $|\zeta(\beta-1)/\zeta(\beta)|$ at a height t_N between 2.1 and 2.5 for $10 \le N \le 24$ (first crossing on a grid of step 0.02). At the first twenty points $1+\rho_n$ the mean of $|R_N|$ is 0.081 at $N = 12$ and 0.010 at $N = 34$, with maxima 0.22 and 0.026. A power fit over $12 \le N \le 34$ gives the exponent -1.82, but the data fix no law: the exponent is -1.69 over $12 \le N \le 24$ and -2.23 over $24 \le N \le 34$. These values are computed in double precision from the exact counts $c_N(m)$; the double precision sums agree with ball arithmetic at 128 bits to 2×10^{-11} (seven values of N up to 26).

For every even N from 10 to 34 and every $n \le 20$, the disc of radius 0.3 about $1 + \rho_n$ contains exactly one zero of Z_N and exactly one zero of the corrected sum C_N of Proposition 20.6.2, found by Newton's method from $1 + \rho_n$ (the winding number on the circle, sampled; computed, not a ball proof); the largest distance of any of these zeros from $1 + \rho_n$ is 0.243. Table 20.7 gives the mean distance of these zeros from $1 + \rho_n$. Removing the profile term does not bring the zeros consistently closer: the mean distance is smaller for C_N only at $N = 14$ and at every computed N from 24 on, and larger at the other $N \le 22$. At $N = 34$ the correction brings 12 of the twenty zeros closer, 14 zeros of C_N lie within 0.005 of $1 + \rho_n$ against 9 of Z_N, and the largest distances are 0.011 for C_N and 0.021 for Z_N. At the points $1 + \rho_n$, where $\zeta(\beta-1)/\zeta(\beta)$ vanishes, the means of $|Z_N|$ and of the profile term at $N = 34$ are 0.016 and 0.021, against the mean 0.010 of $|R_N|$ given above: there the profile term and R_N are of the same size and partly cancel, and removing the profile term lowers the mean error from 0.016 to 0.010. So the profile term is the systematic error of Z_N at small heights, where it dominates, and at $N = 34$ it is part of the error at the points $1 + \rho_n$ too (computed, and likewise the means at $N = 34$ and the windings for $N = 30$ to 34).

N	10	14	18	22	24	26	30	34
Z_N	0.0676	0.0304	0.0148	0.0107	0.0114	0.0094	0.0076	0.0081
C_N	0.0693	0.0280	0.0186	0.0121	0.0103	0.0092	0.0065	0.0047

Table 20.7: Mean distance from $1+\rho_n$, over $n \le 20$, of the zero of Knauf's partition function Z_N and of the zero of the corrected sum C_N in the disc of radius 0.3 about $1+\rho_n$; computed.

Conjecture 20.6.3. *There is an open strip $\sigma_1 < \operatorname{Re}\beta < \sigma_2$, with $1 \le \sigma_1 < \frac{3}{2} < \sigma_2 \le 2$, on which $R_N \to 0$ locally uniformly as $N \to \infty$.*

If the conjecture holds, then $C_N = \zeta(\beta-1)/\zeta(\beta) + R_N$ converges to $\zeta(\beta-1)/\zeta(\beta)$ locally uniformly on the strip, and by Hurwitz's theorem every point $1+\rho$ in the strip is the limit of exactly as many zeros of C_N, counted with multiplicity, as its multiplicity as a zero of $\zeta(\beta-1)$, and every limit point of zeros of C_N in the strip is such a point (conditional). The computations support the conjecture only at the twenty points $1+\rho_n$ of the line $\operatorname{Re}\beta = \frac{3}{2}$; at $\beta = \frac{3}{2}+2i$ the value of $|R_N|$ lies between 0.021 and 0.051 for every N from 10 to 34, with no decrease (computed).

Remark 20.6.4. (i) For a smooth cutoff the analogue of Conjecture 20.6.3 is a theorem, and its main terms do not depend on the zeros of ζ. Let $\Sigma_Y(\beta) = \sum_m \varphi(m) e^{-m^2/Y^2} m^{-\beta}$, whose cutoff e^{-x^2} has the Mellin transform $W(z) = \frac{1}{2}\Gamma(\frac{z}{2})$. For every $\delta > 0$, as $Y \to \infty$,

$$\Sigma_Y(\beta) - \frac{Y^{2-\beta} W(2-\beta)}{\zeta(2)} = \frac{\zeta(\beta-1)}{\zeta(\beta)} + O\big(Y^{1+\delta-\operatorname{Re}\beta}\big),$$

uniformly on compact subsets of the strip $1+\delta < \operatorname{Re}\beta < 3$ without the point 2. Indeed, $\Sigma_Y(\beta) = \frac{1}{2\pi i}\int_{(c)} \frac{\zeta(\beta+z-1)}{\zeta(\beta+z)} Y^z W(z)\,dz$ for $c > \max(0, 2-\operatorname{Re}\beta)$. Moving the line of integration to $\operatorname{Re}(\beta+z) = 1+\delta$ crosses the poles at $z=0$ and $z=2-\beta$, with residues $\zeta(\beta-1)/\zeta(\beta)$ and $Y^{2-\beta}W(2-\beta)/\zeta(2)$ (no other pole of W is crossed for $\operatorname{Re}\beta < 3$); on the new line $|1/\zeta(\beta+z)| \le \zeta(1+\delta)$, $\zeta(\beta+z-1)$ grows at most polynomially in $|\operatorname{Im} z|$, and $W(z)$ decays exponentially.

(ii) Hence the consequence of Conjecture 20.6.3 does not decide RH: if it holds, the zeros of C_N in the strip converge to the points $1+\rho$ in it whether or not $\operatorname{Re}\rho = \frac{1}{2}$, and by (i) the two terms removed from Z_N are those that a regular cutoff produces whatever the zeros. The conjecture is a statement about the regularity of the depth profile.

(iii) A statement that would imply RH concerns the zeros of $\zeta(\beta)$, which are poles of $\zeta(\beta-1)/\zeta(\beta)$: if C_N converges locally uniformly on the half plane $\operatorname{Re}\beta > \frac{1}{2}$ with the point 2 removed, then RH holds. The limit is analytic there, by Weierstrass's theorem, and it equals $\zeta(\beta-1)/\zeta(\beta)$ on $\operatorname{Re}\beta > 2$, by Proposition 20.6.1(ii) and because $W_N(2-\beta) \to 0$ there: for $\operatorname{Re} z < 0$, $W_N(z) =$

$1/z + \int_1^\infty w_N(x) x^{z-1}\,dx \to 1/z - 1/z$ by dominated convergence, as $w_N \uparrow 1$. So $\zeta(\beta-1)/\zeta(\beta)$ would continue analytically to the half plane without the point 2, while a zero ρ of ζ with $\operatorname{Re}\rho > \frac12$ is a pole of it, because $\zeta(\rho-1) \neq 0$, as $-\frac12 < \operatorname{Re}(\rho-1) < 0$. The hypothesis of this implication, the convergence of C_N on the half plane, is open. For the smooth cutoff of (i), whose term $Y^{2-\beta}W(2-\beta)$ has further poles at $\beta = 4, 6, \dots$, the corresponding convergence on the strip $\frac12 < \operatorname{Re}\beta < \frac52$ without the point 2 holds if and only if RH holds: under RH the line in (i) can be moved to $\operatorname{Re}(\beta+z) = \frac12 + \delta$, where $1/\zeta(\beta+z) = O(|\operatorname{Im} z|^\delta)$ [348, Section 14.2]. That version therefore restates RH, as the convergence of $\sum_n \mu(n) n^{-s}$ on $\operatorname{Re} s > \frac12$ does [348, Section 14.25].

20.6.3 The Farey chain in a field

Kleban and Özlük introduced a second chain [125]. Its configurations are the words $\varpi_1 \cdots \varpi_N$ in the letters $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $B = A^{\mathsf T}$, and the energy of a word is $\log \operatorname{tr}(\varpi_1 \cdots \varpi_N)$. Its free energy equals that of Knauf's chain [87], so it has the same transition at $\beta = 2$, and it is fully magnetised for $\beta > 2$ [63]. Conjugation by $\Pi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ exchanges A and B, so the energy does not change when the two letters are exchanged, which is the flip of all spins; Knauf's chain has a single ground state, $s(k) = 1$ only at $k = 2^N$, and no such symmetry. In a field $x = e^h$ the partition function of the Farey chain is

$$Z_N^{\mathrm F}(\beta, x) = \sum_{\varpi_1, \dots, \varpi_N \in \{A,B\}} \bigl(\operatorname{tr} \varpi_1 \cdots \varpi_N\bigr)^{-\beta} x^{n_B},$$

where n_B is the number of letters B in the word. It is a polynomial of degree N in x with positive coefficients, palindromic by the symmetry. Under the Gibbs law at zero field the centred magnetisation $n_B - \frac{N}{2}$ is bounded and symmetric, with moment generating function $e^{-Nz/2} Z_N^{\mathrm F}(\beta, e^z)/Z_N^{\mathrm F}(\beta, 1)$, so it is ferromagnetic in the sense of Definition 20.1.1 exactly when every zero of $x \mapsto Z_N^{\mathrm F}(\beta, x)$ lies on the unit circle. We then say that the chain has the Lee and Yang property at (β, N).

Lemma 20.6.5. *Let $P(x) = 1 + x^N + Q(x)$ with $Q(x) = \sum_{k=1}^{N-1} a_k x^k$ and $a_k = a_{N-k} \geq 0$. If $Q(1) < 2$, the zeros of P are simple and lie on the unit circle.*

Proof. For real θ, $g(\theta) = e^{-iN\theta/2} P(e^{i\theta}) = 2\cos\frac{N\theta}{2} + \sum_k a_k \cos\bigl((k - \frac{N}{2})\theta\bigr)$ is real, and the sum has modulus at most $Q(1) < 2$. So $g(2\pi j/N)$ has the sign of $(-1)^j$ for $j = 0, 1, \dots, N$, and g has at least N zeros in $(0, 2\pi)$. They give N distinct zeros $e^{i\theta}$ of P, which has degree N. □

For the Farey chain the words A^N and B^N have trace 2. Every other word contains both letters, so it is a matrix with positive integer entries and determinant 1, whose diagonal entries a and d satisfy $ad = 1 + bc \geq 2$, and its trace is at least 3.

N	10	14	18	22	26	30	32
$\beta_{\mathrm{LY}}(N)$	1.2424	1.4885	1.6201	1.7005	1.7543	1.7926	1.8079
θ_N	0.9952	0.6802	0.5130	0.4105	0.3414	0.2918	0.2719
$N\theta_N$	9.952	9.522	9.235	9.030	8.876	8.754	8.702
$\beta^*(N)$	2.1056	2.0924	2.0809	2.0719	2.0648	2.0592	2.0567

Table 20.8: The Farey chain in a field: the computed Lee and Yang threshold $\beta_{\mathrm{LY}}(N)$, the angle θ_N of the double zeros at the threshold, and the root $\beta^*(N)$ of $2^\beta Z_N^{\mathrm{F}}(\beta,1)=4$, above which Lemma 20.6.5 proves the property; computed.

Hence $Z_N^{\mathrm{F}}(\beta,x)=2^{-\beta}\big(1+x^N+Q_{N,\beta}(x)\big)$, where $Q_{N,\beta}(1)=\sum(2/\operatorname{tr}\varpi_1\cdots\varpi_N)^\beta$ over the other words decreases strictly in β, and Lemma 20.6.5 gives the Lee and Yang property at (β,N) for every $\beta>\beta^*(N)$, where $\beta^*(N)$ is the root of $2^\beta Z_N^{\mathrm{F}}(\beta,1)=4$.

The computation finds, for the even N from 10 to 36, a threshold $\beta_{\mathrm{LY}}(N)$ such that the Lee and Yang property holds at every computed $\beta\geq\beta_{\mathrm{LY}}(N)$ and fails at every computed β with $0<\beta<\beta_{\mathrm{LY}}(N)$ (grid of step 0.1 on $0.2\leq\beta\leq4$, one transition at every N, then bisection; for $N\leq32$ the bracket of width 3×10^{-11} is certified by a Sturm count in ball arithmetic at 400 bits, for $N=34,36$ it is a double precision bisection); at $\beta=0$ it holds trivially, since $Z_N^{\mathrm{F}}(0,x)=(1+x)^N$. At the threshold four zeros meet on the circle as two double zeros $e^{\pm i\theta_N}$, and just below it they leave the circle as a quartet $e^{\pm u\pm i\theta}$. The threshold rises with N, from 1.2424 at $N=10$ to 1.8329 at $N=36$ (computed), and stays below $\beta^*(N)$, which falls with N (Table 20.8, which stops at $N=32$; computed).

Conjecture 20.6.6. *For every $\beta>2$ the Farey chain has the Lee and Yang property at (β,N) for all large N, and for every β with $0<\beta<2$ it fails to have it for all large N.*

The value 2 comes from the transition [64, 87], not from the data: free fits of $\beta_{\mathrm{LY}}(N)$ in powers of $1/N$ over $10\leq N\leq22$ extrapolate to 2.086 (linear) and 2.046 (quadratic), and over $20\leq N\leq36$ to 2.044 and 2.017 (computed), all above 2. For every N let $\beta_{\mathrm{LY}}(N)$ be the infimum of the $\beta_0>0$ such that the property holds at (β,N) for all $\beta>\beta_0$; it is the computed threshold for the even N above. By Lemma 20.6.5, $\beta_{\mathrm{LY}}(N)\leq\beta^*(N)$, so a limit of $\beta_{\mathrm{LY}}(N)$ is at most $\liminf_N\beta^*(N)$.

The theorems of Lee and Yang, of Newman and of Lieb and Sokal recalled in Section 20.1.1 concern pair interactions, and the interactions of both chains are many body [126, 125], so these theorems do not apply; for the Farey chain, Lemma 20.6.5 gives the Lee and Yang property above $\beta^*(N)$, and the computation finds it down to $\beta_{\mathrm{LY}}(N)$, below the transition, and not for $0<\beta<\beta_{\mathrm{LY}}(N)$. The zeros of this section are of two kinds: the Fisher zeros of Section 20.6.2 are zeros of Knauf's chain in the inverse temperature, and those of this subsection are zeros of the Farey chain in the field, at real β.

20.7 Open problems

Problems 20.7 and 20.8 concern the variable e^{2X} of Section 20.5, where no hypothesis on the zeros enters, and the last two concern the spin chains of Section 20.6.

Problem 20.1. Mixed shoulders. Theorem 20.2.4 decides the symmetric laws on $[-1,1]$ whose density is concave on $[0,1]$ and those whose density is convex there and not linear on its support: by (20.2.1) the question is whether the sine transform of the slope $g=-p'$ has only real zeros, and the theorem answers it when g is monotone. Decide the slopes that first rise and then fall. One class is settled: if g is symmetric about $\frac12$, then $p(0)=\int_0^1 g=1$, because $1=2\int_0^1 p=2\int_0^1 xg(x)\,dx=\int_0^1 g$ by the symmetry of g, so $u\mapsto g(\frac12+u)$ is a probability density on $[-\frac12,\frac12]$; and $X\stackrel{d}{=}\frac12U+B$, with B independent of U and distributed with that density, since the density of $\frac12U+B$ has slope g on $(0,1)$. As $\sin(z/2)/(z/2)$ has only real zeros, X is then ferromagnetic exactly when B is. For the beta slopes $g(y)\propto y^a(1-y)^b$, $a,b\geq 0$, this decides the diagonal $a=b$, where B is a symmetric beta law and X is ferromagnetic (Section 20.1.4), and Theorem 20.2.4 decides the edges: X is ferromagnetic for $b=0$, where p is concave, and not for $a=0<b$, where p is convex and not linear. The rest of the quadrant is open. On the half line the slope of the Pólya law rises and then falls (Section 20.4.2), and RH is the statement that its sine transform has only real zeros.

Problem 20.2. Lines of zeros of the Mellin transform of a clock. For which ferromagnetic laws X does $\mathcal{D}_X$ of Section 20.3 have nonreal zeros, all of them on one vertical line? The uniform pair $\pi U+\pi U'$ has this property if and only if RH holds, by (20.3.1). The sums of one, four and six independent copies of πU do not have it (Proposition 20.3.2; for four and six computed, Section 20.3), and neither do the pairs $X_\varepsilon+X'_\varepsilon$ of Section 20.3 for small $\varepsilon>0$ (computed), so no theorem whose hypotheses hold for every symmetric density on an interval that is nondecreasing in $|t|$ can give it. Is the uniform pair, up to scale, the only pair $X+X'$ with X in the class of Theorem 20.2.6 that has it? A first step is the first order motion of the zeros along the densities $\frac{1}{2\pi}+\varepsilon g$ on $[-\pi,\pi]$, which is linear in g: is there an even $g\neq 0$ with $\int g=0$ along which every zero $\frac{\rho_k-1}{2}$ keeps its real part to first order? This problem is based on [147, 144].

Problem 20.3. The Griffiths Simon class. Newman's theorem places the Pólya law in the class of Ellis, Monroe and Newman (Section 20.4.1). Does it belong to the Griffiths Simon class of Section 20.1.1? The laws of that class are ferromagnetic, so a positive answer would imply RH, and under Ingham's hypothesis the spins of any approximating system would have to be coupled (Section 20.4.3).

Problem 20.4. Splitting the zeros. Assume RH, so that $\varphi_X(t)=\prod_\gamma(1-t^2/\gamma^2)$ over the positive ordinates, with no Gaussian factor by Theorem 20.4.1. Can the ordinates be split into two nonempty sets A and A' such that $\prod_{\gamma\in A}(1-t^2/\gamma^2)$

and $\prod_{\gamma\in A'}(1-t^2/\gamma^2)$ are both characteristic functions? This is the same as writing $X = D + R$ with D and R independent, ferromagnetic and not 0, and by Theorem 20.4.1 both sets are then infinite. Under Ingham's hypothesis no three ordinates are in arithmetic progression, so no piece is a scaled fair sign or a scaled uniform variable, whose zeros form arithmetic progressions.

Problem 20.5. Positivity of the zero completed laws. The laws f_K of Section 19.1.4 are proved to be ferromagnetic probability densities for $K \le 16$ by the Maxwell construction, and for every $K \le 200$ and for $K = 300$, 400, 600 and 800 by a computer assisted proof in ball arithmetic. Compute the threshold $v^*(K)$ near $K = 5000$, where the linear and the power law fits of the ratio $v^*(K)/2\beta_K$ part; the cancellation there calls for a working precision of tens of thousands of bits. Is $v^*(K) < 2\beta_K$ for every K? If so, every f_K is a ferromagnetic probability density by Theorem 19.1.12(ii), and with no hypothesis their limit in law, with characteristic function $e^{-\beta_\infty\theta^2}\prod_k(1-\theta^2/\gamma_k^2)$ over the zeros on the critical line, is a ferromagnetic law by Theorem 20.1.5; by Theorem 19.1.12(iii) it is the Pólya law exactly when RH holds.

Problem 20.6. The cosine constant. By Section 19.1.2, $\Omega \le \frac{1}{2}$ with no hypothesis, $\Omega \le \frac{3}{8}$ if the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 is free of zeros (Section 25.1.2), and RH holds if and only if $\Omega = 0$. Is the set of $a \ge 0$ for which $\cosh(at)\,p$ is ferromagnetic an interval? Improve these bounds by computation, as computations with the zeros have improved de Bruijn's bound $\Lambda \le \frac{1}{2}$ on the de Bruijn Newman constant, to $\Lambda \le 0.2$ [277]. A first case is $a = \sqrt{0.1}$, below both bounds: the law $\cosh(\sqrt{0.1}\,t)\,p$ is as close to p as the flow at $\lambda = 0.2$, which is ferromagnetic, and deciding it is a computation with the zeros of $\frac{1}{2}(\xi(s+a)+\xi(s-a))$, of the kind that gave the bound 0.2 for the flow.

Problem 20.7. The size bias principle for real shapes. Theorem 20.5.5 is proved for integer shapes. Does it hold for all real shapes $t_n > 0$? For $K = 1$ it is the gamma identity, with $G = 0$. Integer shapes enter the proof twice, through the divided difference formula for the density of Y and through Lemma 20.5.4, and together they show that on each gap (τ_k, τ_{k+1}) the phase of the Laplace transform of Y_q is nonincreasing and falls by exactly q. Is this still true for real shapes? With it the rest of the proof goes through, provided the estimates of the transform near the points $-\tau_n$ and at infinity also hold.

Problem 20.8. The powers of e^X. For $2 < |\beta|$ below the HCM threshold, which lies between 5.5791395 and $\pi/0.563096 = 5.57914219\ldots$ (Proposition 20.5.8), is $e^{\beta X}$ infinitely divisible, self decomposable or a GGC? Below the threshold it has no HCM density, which leaves the GGC property open, and Theorem 20.5.7 gives self decomposability only at $|\beta| = 2$. Is the HCM threshold exactly $\pi/v_m = 5.5791396$? That requires $h \ge 0$ on the open strip $0 < v < v_m$, where $h(0, v) \to 0$ as $v \uparrow v_m$, so a certificate must replace the balls with positive lower bounds near iv_m by a

local argument.

Problem 20.9. The depth profile of Knauf's chain. Prove Conjecture 20.6.3, or find the strips on which it holds. By Proposition 20.6.2, R_N measures, heuristically, how far the profile $w_N = c_N/\varphi$ of Section 20.6.2 is from being regular on the scale of the integers. The fraction a/m is a child of $a/(m-a)$ in the Calkin Wilf tree, and each step down the tree adds one to the sum of the partial quotients, the root $1/1 = [1]$ having sum 1; so by Proposition 20.6.1(i), for $m \geq 2$, $c_N(m)$ is the number of a prime to m, $1 \leq a < m$, for which $a/m = [0; b_1, \dots, b_k]$ has $b_1 + \cdots + b_k \leq N + 1$. Yao and Knuth proved that the mean of $b_1 + \cdots + b_k$ over $1 \leq a \leq m$ is asymptotic to $\frac{6}{\pi^2} \log^2 m$ [367]; over the a prime to m, which are those counted here, a quadratic fit in $\log m$ to the computed means at $m = 10^5, \dots, 10^8$ has leading coefficient 0.6073 (computed). The consequence of the conjecture does not decide RH (Remark 20.6.4), and the convergence in Remark 20.6.4(iii) would imply it.

Problem 20.10. The Lee and Yang threshold of the Farey chain. Prove Conjecture 20.6.6. By Lemma 20.6.5 its first half follows if, for every inverse temperature $\beta > 2$, the words other than A^N and B^N carry less weight than these two for all large N, that is $\sum (2/\operatorname{tr} \varpi_1 \cdots \varpi_N)^\beta < 2$ over the other words. The computed $\beta^*(N)$ falls from 2.1056 to 2.0525 for $10 \leq N \leq 36$, and for $2 < \beta \leq \beta^*(N)$ the lemma does not decide. The second half asks that for every β with $0 < \beta < 2$ some zeros leave the circle once N is large; at the computed thresholds the zeros that leave it first are at the angle θ_N, with $N\theta_N$ between 8.61 and 9.95 (computed; Table 20.8 for $N \leq 32$).

Chapter 21

Cauchy mixtures, the reverse sine squared lemma and a two spin Lee and Yang form of RH

This chapter reads the Thorin density of the reciprocal Ξ function as a mixture of Cauchy densities and asks what that structure says about the ferromagnetism of the Pólya law of Chapter 20. The partial fraction expansion of ξ'/ξ already gives the Cauchy form (Proposition 2.1.3); the sine squared lemma (2.1.9) gives the same density from the primes. We add the reverse direction, from the Cauchy mixture back to the canonical measure, and five results that follow from it.

The results, with their status, are these. For every symmetric law of Xi type the Thorin density at a level is a signed Cauchy mixture, one bump for each zero (Theorem 21.1.2, proved); it is a positive mixture at every level exactly for the ferromagnetic laws, and then the bumps have one common width and the mixing measure is the zero counting measure (Theorem 21.2.1, proved). The reverse sine squared lemma turns each Cauchy bump of width a into a damped cosine $e^{-a|x|}\cos\gamma x$ on the canonical side, unconditionally at every basepoint $\alpha \geq 1$ (Theorem 21.3.2, proved), and undamping to the critical line gives Weil's functional (Theorem 21.3.4, proved). The cosine tilted laws $\cosh(wx)\,p$ are ferromagnetic for every $w \geq \frac{1}{2}$ with no hypothesis (Theorem 21.4.1, proved). Coupling the Pólya spin to one Ising spin gives a two spin ferromagnet that is of Lee and Yang type for every coupling $w > w_0$ exactly when ζ has no zero with real part above $\frac{1}{2} + w_0$ (Theorem 21.5.1 and Corollary 21.5.2, proved); it holds for $w_0 = \frac{1}{2}$ unconditionally, and RH is $w_0 = 0$. Finite Euler products sit exactly at $w = \frac{1}{2}$ (Proposition 21.5.3, proved). Finally, the one variable test, ferromagnetism of $\cosh(wx)\,p$ alone, is weaker than the zero free half plane, proved by an exact two spin law (Proposition 21.6.1) and, for Ξ, conditionally on the positivity of a planted density and on the zeros of ζ outside a computed box: for Ξ with a planted pair of zeros at distance 0.3 from the line it holds from $w \approx 0.2951$ on (Proposition 21.6.2, computed), and a phase criterion explains why (Proposition 21.6.3, proved). With several clusters off the line each is seen through its own field, which includes the other clusters,

and a double pair is missed by the fixed fraction $1-\cos\frac{\pi}{4}$ of its distance from the line rather than by $O(\eta^3)$ (Theorem 21.8.1, proved, and Table 21.2, computed).

Against the path of Section 1.4 the chapter is an exact dictionary and not a new route. The Cauchy mixture of one width, the ferromagnetism of the Pólya law, the positivity of the undamped prime measure and the Lee and Yang property of the two spin system at every coupling are four forms of RH; each is proved at the level one, or the coupling $\frac{1}{2}$, and each needs a zero free half plane to go lower. The descent in coupling is the descent in level, and the first step below $w=\frac{1}{2}$ is the quasi Riemann hypothesis.

21.1 Laws of Xi type and their Thorin densities

Definition 21.1.1. A law of Xi type is the law of a symmetric random variable X with $\mathbb{E}e^{aX^2}<\infty$ for some $a>0$, whose moment generating function has the Hadamard form

$$L(s)=\mathbb{E}e^{sX}=e^{bs^2}\prod_r\Big(1-\frac{s^2}{r^2}\Big),\qquad b\ge 0,\qquad \sum_r|r|^{-2}<\infty,$$

the product over one zero of each pair $\pm r$, with multiplicity. For $y>0$ and real v with $L(y+iv)\neq 0$ its Thorin density at level y is $u_y(v)=\pi^{-1}\operatorname{Re}(L'/L)(y+iv)$.

The Pólya law is of Xi type, with $b=0$ and zeros $r=\rho-\frac{1}{2}$; its Thorin density at level y is $v_{1/2+y}/\pi$ in the notation of Section 1.3, the Thorin density of the shifted clock at the basepoint $\frac{1}{2}+y$ whenever no zero lies to its right (Theorem 2.1.2). Every ferromagnetic law is of Xi type by Newman's theorem [240], since its characteristic function is in the Laguerre Pólya class. We write $P_a(x)=a/(\pi(a^2+x^2))$.

Theorem 21.1.2. *Let X be of Xi type and write its zeros as $r=\delta+i\gamma$, both members of each pair $\pm r$ included. For $y>0$ with $y\neq\delta$ for every zero,*

$$u_y(v)=\frac{2b}{\pi}\,y+\sum_r\operatorname{sgn}(y-\delta)\,P_{|y-\delta|}(v-\gamma),$$

the sum taken over the pairs $\{r,-\bar r\}$, absolutely and locally uniformly convergent in v.

Proof. $L'/L(s)=2bs+\sum_r\big(\frac{1}{s-r}+\frac{1}{s+r}\big)$, each term $O(|r|^{-2})$ on compact sets away from the zeros, and $\operatorname{Re}\frac{1}{s-r}=\pi\operatorname{sgn}(y-\delta)P_{|y-\delta|}(v-\gamma)$ at $s=y+iv$. Since L is even and real, its zero set is invariant under $r\mapsto-\bar r$, and the terms can be regrouped in these pairs. □

Each zero contributes one bump at its ordinate, of width equal to its horizontal distance from the level, with the sign of the side on which it lies; a Gaussian factor contributes the drift $2by/\pi$. For the Pólya law this is Proposition 2.1.3.

21.2 Positive Cauchy mixtures and ferromagnetism

Theorem 21.2.1. *For X of Xi type the following are equivalent.*

(i) $u_y(v) \geq 0$ *for every* $y > 0$ *and every* v *at which it is defined.*

(ii) There are $\eta \geq 0$ *and a positive measure* ν *with* $\int (1+x^2)^{-1}\nu(dx) < \infty$ *such that* $u_y = \eta y + P_y * \nu$ *for every* $y > 0$.

(iii) X *is ferromagnetic in the sense of Definition 20.1.1.*

(iv) $u \mapsto 1/L(\sqrt{u})$ *is the Laplace transform of a GGC with drift* b *and Thorin measure* $\sum_k m_k \delta_{c_k^2}$, *the* c_k *being the zeros of the characteristic function, with multiplicities* m_k.

Then $\eta = 2b/\pi$ *and* $\nu = \sum_k m_k(\delta_{c_k} + \delta_{-c_k})$, *both unique, and* $b = 0$ *if* X *has Gaussian moments of all orders.*

Proof. (iii) implies (ii) by Theorem 21.1.2 with every $\delta = 0$; (ii) implies (i) trivially. For (i) implies (iii), suppose L has a zero r of multiplicity m with $\delta = \operatorname{Re} r > 0$. Near r, $L'/L(s) = m/(s-r) + O(1)$, so at $s = r - \varepsilon$ with small $\varepsilon > 0$ we get $\operatorname{Re} L'/L = -m/\varepsilon + O(1) < 0$ at a point with real part $\delta - \varepsilon > 0$, against (i); by symmetry no zero has $\delta < 0$. (iii) is equivalent to (iv) by Proposition 11.1.16, which also gives $b = 0$ under Gaussian moments of all orders. Uniqueness is that of the Herglotz and Riesz representation (Theorem 7.2.3 and Proposition 7.1.1 of Volume II). □

Remark 21.2.2. Condition (i) says that u is a positive harmonic function on the right half plane, and (ii) is then forced by the Herglotz and Riesz representation. What is special to laws of Xi type is that ν must have integer atoms: a GGC with diffuse Thorin measure has a Laplace transform whose reciprocal has a branch cut along a half line $(-\infty, -c]$ of the negative axis, so it is not $L(\sqrt{u})$ for any law. The diffuse Cauchy mixtures belong to clocks, not to laws.

Definition 21.2.3. For $\delta \geq 0$ let $\mathcal{C}_\delta$ be the set of laws of Xi type whose characteristic function has all its zeros in $|\operatorname{Im}\theta| \leq \delta$. The Cauchy defect of X is the least such δ.

Proposition 21.2.4. *For X of Xi type, $u_y \geq 0$ for every $y > \delta$ if and only if $X \in \mathcal{C}_\delta$; then for $y > \delta$, u_y is a positive Cauchy mixture with widths in $[y-\delta, y+\delta]$, plus the drift. The classes are closed under independent sums (the Thorin densities add) and under the tilts $m(x) = \mathbb{E}e^{xY}$, $Y = \sqrt{2\mu}Z + \sum_j \lambda_j \varepsilon_j$, which map $\mathcal{C}_\delta$ into $\mathcal{C}_{\delta'}$ with $\delta' = (\max(\delta^2 - \operatorname{Var} Y, 0))^{1/2}$ when the density decays faster than $e^{-|x|^\kappa}$ for some $\kappa > 2$. They are not closed under mixtures.*

Proof. The first statement is the proof of Theorem 21.2.1 with $y > \delta$ in place of $y > 0$. Sums multiply moment generating functions. The tilts are de Bruijn's Theorems 12 and 13 [73], as in Proposition 19.1.2 of Volume II. For mixtures, U and $3U$ are ferromagnetic for U uniform on $[-1,1]$, while the equal mixture has characteristic function $\frac{\sin\theta}{\theta}(1-\frac{2}{3}\sin^2\theta)$, which vanishes at $\theta = \frac{\pi}{2} + k\pi \pm i\,\mathrm{arccosh}\,\sqrt{3/2}$, nonreal. □

The Pólya law is in $\mathcal{C}_{1/2}$ with every zero strictly inside the strip, in $\mathcal{C}_{3/8}$ if the half plane $\mathrm{Re}\, s > \frac{7}{8}$ announced in October 2026 and not refereed is confirmed [257], and in $\mathcal{C}_0$ exactly under RH. The class of Ellis, Monroe and Newman, to which the Pólya law belongs by Newman's theorem (Section 20.4.1), is not inside the ferromagnetic class.

Corollary 21.2.5. *Given $\Lambda \geq 0$ [318], the class of Ellis, Monroe and Newman is not contained in the class of ferromagnetic laws.*

Proof. If e^{-V} has V even and V' convex on $[0,\infty)$, so does $e^{-V-\varepsilon x^2}$. If the class were ferromagnetic, $e^{-\varepsilon x^2}p$ would be ferromagnetic for every $\varepsilon > 0$, which is the de Bruijn Newman flow at time -4ε in the normalisation of Section 20.1.1, so $\Lambda < 0$. This is the argument of Proposition 23.2.3. □

21.3 The reverse sine squared lemma

The sine squared lemma (2.1.9) turns a Lévy representation $\log \xi(\alpha + s)/\xi(\alpha) = cs + \int (e^{-sx} + sx - 1) L_\alpha(dx)$ into the Thorin bracket $v_\alpha(\theta) = c + \int 2\sin^2(x\theta/2)\, x L_\alpha(dx)$, and at $\alpha > 1$ the canonical measure xL_α has atoms $\Lambda(n) n^{-\alpha}$ at $\log n$ (Section 2.1). The reverse goes from the zeros to the canonical measure.

Lemma 21.3.1. *For $a > 0$ and real γ, θ, $\int_{\mathbb{R}} e^{-a|x|}\cos(\gamma x)\cos(\theta x)\, dx = \pi\big[P_a(\theta - \gamma) + P_a(\theta+\gamma)\big]$. For a zero $\rho = \beta + i\gamma$ with $\beta < \alpha$, $\log(1 + s/(\alpha - \rho)) = \int_0^\infty (1 - e^{-sx}) e^{-(\alpha-\rho)x}\, dx/x$, and the canonical density of ρ and $\bar\rho$ together is $2e^{-(\alpha-\beta)x}\cos\gamma x$.*

Proof. $\int e^{-a|x|}\cos kx\, dx = 2a/(a^2+k^2)$ and $2\cos\gamma x\cos\theta x = \cos(\theta-\gamma)x + \cos(\theta+\gamma)x$; the second statement is Frullani's integral. □

Theorem 21.3.2. *Let $\alpha \geq 1$, or more generally let $\alpha > \frac{1}{2}$ with every zero having $\beta < \alpha$. The series $K_\alpha(x) = \sum_{\gamma>0} e^{-(\alpha-\beta)|x|}\cos\gamma x$ converges in $\mathcal{S}'(\mathbb{R})$; its Fourier cosine transform is v_α; on $(0,\infty)$ it equals $-\frac{1}{2}xL_\alpha$, with $c = b_\alpha = \sum_\rho (\alpha-\rho)^{-1}$; and it is positive definite, with $v_\alpha > 0$.*

Proof. Each term is bounded by 1, and two integrations by parts against a Schwartz function give $O(\gamma^{-2})$, the only boundary term coming from the kink

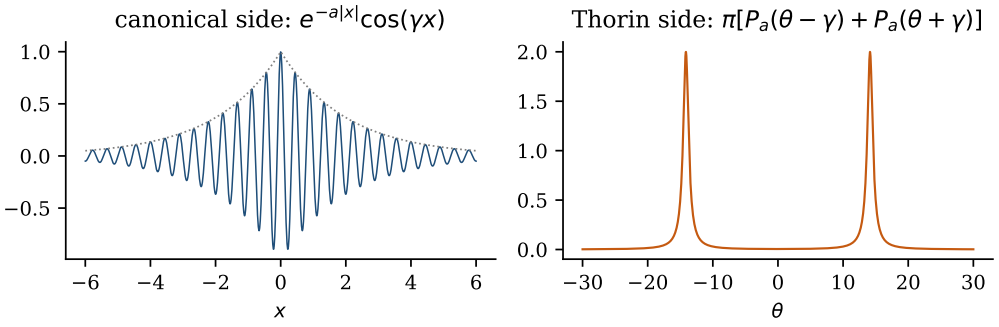


Figure 21.1: The reverse kernel for one zero, $a = \frac{1}{2}$, $\gamma = \gamma_1$: a damped cosine on the canonical side and a Cauchy pair on the Thorin side (Lemma 21.3.1). The damping rate is the width.

at 0; since $\sum \gamma^{-2} < \infty$ the series converges in $\mathcal{S}'$, and the transform follows term by term from Lemma 21.3.1 and Proposition 2.1.3. The Hadamard product and Lemma 21.3.1 give $\log \xi(\alpha + s)/\xi(\alpha) = \int_0^\infty (1 - e^{-sx}) 2K_\alpha(x)\, dx/x$, and $1 - e^{-sx} = sx - (e^{-sx} + sx - 1)$ identifies $L_\alpha = -2K_\alpha\, dx/x$ and c. Every width $\alpha - \beta$ is positive, so $v_\alpha > 0$, and a tempered distribution with positive Fourier transform is positive definite. □

Read with the forward lemma, the canonical measure at α has two expressions, the primes with the archimedean and pole densities, and $-2\sum e^{-(\alpha-\beta)x}\cos\gamma x$ over the zeros. Their equality is the explicit formula.

THEOREM 21.3.3. *Let $\alpha \geq 1$. If $v_\alpha/\pi = P_{\alpha-1/2} * \nu$ for a positive measure ν, so that the Thorin density of $1/\xi$ at the single basepoint α is a positive Cauchy mixture of the single width $\alpha - \frac{1}{2}$, then RH holds, the Pólya law is ferromagnetic, and $\nu = \sum_\gamma \delta_\gamma$. The converse holds at every $\alpha > \frac{1}{2}$.*

Proof. This is Theorem 8.2.2(b) of Volume II. Put $H(t,\theta) = (P_{t-1/2} * \nu)(\theta) \geq 0$, harmonic on $t > \frac{1}{2}$. For $t > \alpha$, $v_t/\pi = P_{t-\alpha} * (v_\alpha/\pi) = H(t,\cdot)$ by the semigroup law. Both sides are harmonic on the connected set $\{t > \frac{1}{2}\}$ minus the zeros, so they agree there, and a zero with $\beta > \frac{1}{2}$ would make $v_t(\gamma) \to -\infty$ as $t \uparrow \beta$, against $H \geq 0$. □

THEOREM 21.3.4. *Let h be even and real with $e^{|x|/2}(|h| + |h''|)$ integrable, and $\alpha \geq 1$, $w = \alpha - \frac{1}{2}$. Group the zeros in pairs $\{\rho, 1-\bar\rho\}$ with real parts $\frac{1}{2} \pm \delta$, a zero on the line ($\delta = 0$) forming a pair by itself, with the single term $e^{(\frac{1}{2}-\alpha)|x|}$ in the sum below. Then*

$$\big\langle e^{w|x|}K_\alpha, h\big\rangle := \sum_{\text{pairs}} \int \big(e^{(\frac{1}{2}+\delta-\alpha)|x|} + e^{(\frac{1}{2}-\delta-\alpha)|x|}\big) e^{w|x|}\cos(\gamma x) h(x)\, dx = \frac{1}{2}\sum_\rho \widehat{h}(t_\rho),$$

with $t_\rho = (\rho - \frac12)/i$ *and* $\widehat{h}(t) = \int h(x)e^{-itx}dx$*; the series converges absolutely and does not depend on* α*. It is Weil's functional [357]; under RH it equals* $\frac12 \sum_\gamma |\widehat{g}(\gamma)|^2 \ge 0$ *for* $h = g * \tilde g$*, and Weil's criterion gives the converse [357, 38].*

Proof. Within a pair $e^{(\frac12 \pm \delta - \alpha)|x|}e^{(\alpha-\frac12)|x|} = e^{\pm\delta|x|}$, whose sum $2\cosh\delta x$ is smooth, so the kink of $e^{w|x|}$ disappears pair by pair, and $\int 2\cosh(\delta x)\cos(\gamma x)h = \widehat{h}(\gamma - i\delta) + \widehat{h}(\gamma + i\delta)$ for even h. The hypothesis on h makes $\widehat{h}$ analytic on $|\operatorname{Im} t| \le \frac12$ and $O(|\operatorname{Re} t|^{-2})$ there, and $|\delta| < \frac12$. □

So the single width hypothesis, read on the canonical side, is Weil's criterion, and the place where the prime measure $\Lambda(n)n^{-\alpha}$ stops being tempered, $\alpha = 1$, is removed by pairing each zero with its reflection: the kernel $\cosh\delta x$ grows at most like $e^{|x|/2}$. The windows of Weil positivity certified in Chapters 15 and following of Volume II are certificates for this functional on test functions of bounded support.

21.4 The cosine tilted laws

For $w \ge 0$ let $p_w = \cosh(wx)\,p/\mathbb{E}\cosh(wX)$. Its characteristic function is $\operatorname{Re}\xi(\frac12 + w + it)/\xi(\frac12 + w)$.

Theorem 21.4.1. *For* X *of Xi type and* $w > 0$*, the law* $\cosh(wx)\,P(dx)/\mathbb{E}\cosh(wX)$ *is ferromagnetic whenever every zero of* L *has* $|\operatorname{Re} r| < w$*. In particular* p_w *is ferromagnetic for every* $w \ge \frac12$*, with no hypothesis.*

Proof. The moment generating function is proportional to $L(s+w) + L(s-w)$, and $|L(s-w)| = |L(w-y+iv)|$ at $s = y + iv$, since L is even and real. A zero $r = \delta + i\gamma$ contributes $|w+y+iv-r|/|w-y+iv-r|$ to $|L(s+w)|/|L(s-w)|$, and $(w+y-\delta)^2 - (w-y-\delta)^2 = 4y(w-\delta) > 0$ for $y > 0$; the partner $-r$ also has real part below w, and e^{bs^2} contributes $e^{4bwy} \ge 1$. So $|L(s+w)| > |L(s-w)|$ for $y > 0$, and the sum has no zeros off the imaginary axis. For the Pólya law $|\operatorname{Re} r| = |\beta - \frac12| < \frac12$. □

In Thorin terms the comparison is $\log|L(w+y+iv)| - \log|L(w-y+iv)| = \pi\int_{w-y}^{w+y} u_t(v)\,dt$: the density is integrated across a horizontal segment centred at the level w, and each Cauchy bump contributes a positive amount as soon as its zero lies within w of the axis. This is de Bruijn's lemma [73, Lemma 1], the simplest case of Proposition 19.1.2 of Volume II and Theorem 22.6.1(ii) of Volume I; Lagarias proved that the zeros of $\xi(s+w)+\xi(s-w)$ are simple for $w \ge \frac12$ [199]. With Theorem 20.1.5, RH holds if and only if p_w is ferromagnetic for every $w > 0$.

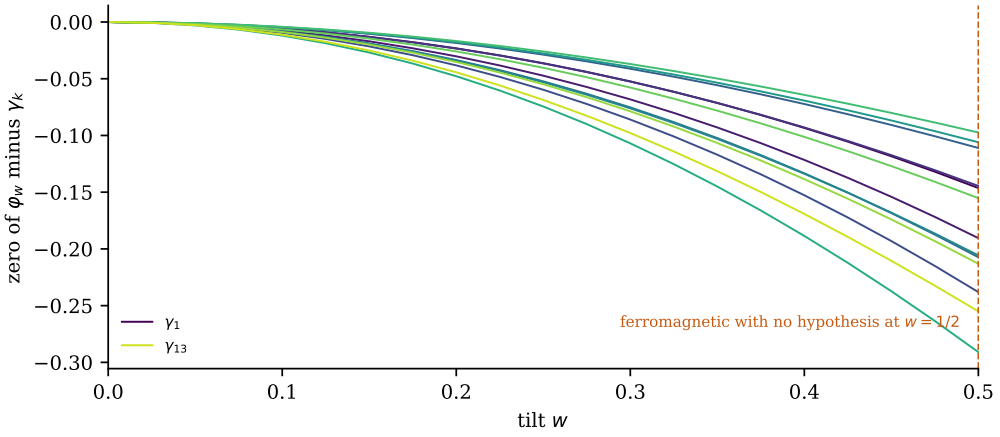


Figure 21.2: Zeros of the characteristic function of p_w on $(0, 60)$ minus γ_k, against w. They move continuously and stay real; at $w = \frac{1}{2}$ the law is ferromagnetic with no hypothesis. The displacement is quadratic in w for small w, as the first order rule of Remark 19.1.3 of Volume II predicts (computed).

21.5 A two spin Lee and Yang form of RH

Let ε be a fair sign independent of X and couple the two spins with strength $w > 0$. With fields h_1 on X and h_2 on ε the partition function is

$$Z_w(h_1, h_2) = \mathbb{E} \sum_{\varepsilon=\pm 1} e^{wX\varepsilon + h_1 X + h_2 \varepsilon} = e^{h_2} L(h_1 + w) + e^{-h_2} L(h_1 - w).$$

The marginal law of X is the cosine tilt, and the pair has the multivariate Lee and Yang property if $Z_w \neq 0$ whenever $\operatorname{Re} h_1 > 0$ and $\operatorname{Re} h_2 > 0$.

Theorem 21.5.1. *Let X be of Xi type and $w > 0$.*

(a) If no zero of L has $\operatorname{Re} r > w$, the pair has the Lee and Yang property.

(b) If the pair has the Lee and Yang property, no zero of L has $\operatorname{Re} r > w$. So the pair has the Lee and Yang property at the coupling w exactly when $X \in \mathcal{C}_w$.

(c) For $w_0 \geq 0$, the pair has the Lee and Yang property for every $w > w_0$ if and only if no zero has $\operatorname{Re} r > w_0$.

Proof. A zero of Z_w solves $e^{2h_2} = g(h_1)$ with $g = -L(h_1 - w)/L(h_1 + w)$, and e^{2h_2} ranges over $\{|z| > 1\}$ when $\operatorname{Re} h_2 > 0$; so the property is $|g| \leq 1$ on $\operatorname{Re} h_1 > 0$, poles excluded. (a) is the factor comparison of Theorem 21.4.1, in which a zero with $\operatorname{Re} r = w$ contributes the ratio 1, so $|g| \leq 1$. (b) Let r be a zero with $\operatorname{Re} r > w$ and put $h_1 = r - w$, so $\operatorname{Re} h_1 > 0$. If $L(r - 2w) = 0$, then $Z_w(h_1, h_2) = 0$ for every

h_2; otherwise g has a pole at h_1 and takes every large modulus near it. Either way the property fails. (c) follows from (a) and (b). □

Corollary 21.5.2. *The Pólya spin coupled to one Ising spin with coupling w is a Lee and Yang ferromagnet for every $w \geq \frac{1}{2}$, with no hypothesis. For $w_0 \in [0, \frac{1}{2})$ it is one for every $w > w_0$ if and only if ζ has no zero with $\operatorname{Re} s > \frac{1}{2} + w_0$. RH holds if and only if it is one for every $w > 0$; under the announced half plane at $\frac{7}{8}$, for every $w > \frac{3}{8}$.*

Proposition 21.5.3. *For $N \geq 2$ let $E_N(s) = \prod_{p \leq N}(1 - p^{-1/2-s})(1 - p^{-1/2+s})$, the truncated Euler product of $\zeta(\frac{1}{2}+s)^{-1}\zeta(\frac{1}{2}-s)^{-1}$. Its zeros are $\pm\frac{1}{2} + 2\pi i k/\log p$, and the two spin system built from E_N in place of L has the Lee and Yang property exactly for $w \geq \frac{1}{2}$.*

Proof. $(1-p^{-1/2-s})(1-p^{-1/2+s}) = 1+p^{-1}-2p^{-1/2}\cosh(s\log p)$ vanishes exactly when $\cosh(s\log p) = \cosh(\frac{1}{2}\log p)$. E_N is even, real and of exponential type, with Hadamard product $E_N(0)\prod(1 - s^2/r^2)$ and $\sum |r|^{-2} < \infty$, so Theorem 21.5.1 applies to it as an entire function. For $w > \frac{1}{2}$ part (a) applies; at $w = \frac{1}{2}$ the zeros with $\operatorname{Re} r = \frac{1}{2}$ contribute factor ratios equal to 1 and their partners ratios above 1, so $|g| < 1$ still. For $w < \frac{1}{2}$ the zeros with $\operatorname{Re} r = \frac{1}{2} > w$ violate part (b). □

So finite Euler products sit exactly at the coupling $\frac{1}{2}$, the line $\operatorname{Re} s = 1$, which is where the Pólya law reaches unconditionally; the symmetrised infinite product converges absolutely nowhere. Coupling below $\frac{1}{2}$ needs the cancellation of the infinite product, which is the zero free region problem.

21.6 The one variable test is weaker

Theorem 21.4.1 gives ferromagnetism of the cosine tilt from the strip. The converse fails for the one variable tilt (Proposition 21.6.1), although it holds for the pair by Theorem 21.5.1.

In two exact models the threshold is the strip. For $X \in \{-1, 0, 1\}$ with $P(X = \pm 1) = q < \frac{1}{4}$, $L(s) = (1-2q) + 2q\cosh s$ has zeros at real parts $\pm a$, $\cosh a = \frac{1-2q}{2q}$, and the tilt $2(1-2q)+4q\cosh s\cosh w$ is ferromagnetic exactly for $w \geq a$ (proved). For $qU_1 + (1-q)U_3$ with $q > \frac{1}{4}$, U_c uniform on $[-c, c]$, the nonreal zeros sit at $\pm i\delta$ from the real axis with $\delta = \operatorname{arccosh}\sqrt{3/(4(1-q))}$; counting by the argument principle in $|\operatorname{Re}| \leq 40\pi + 0.37$, $|\operatorname{Im}| \leq 4$, all 160 nonreal zeros are gone from $w = \delta + 0.002$ on, and a few become real just before δ (152 of 160 remain at $w = \delta - 0.002$ for $q = \frac{1}{2}$; computed).

The first model has a sum of two independent copies that is a sharper example, because there the threshold lies strictly inside the strip.

Proposition 21.6.1 (the sum of two three point spins). *Let $0 < q < \frac{1}{4}$, $\cosh a = \frac{1-2q}{2q}$ (so $a > 0$), and $X = Y_1 + Y_2$ with Y_1, Y_2 independent and $P(Y_i = \pm 1) = q$,*

$P(Y_i = 0) = 1 - 2q$. Then X is of Xi type, and its cosine tilt is ferromagnetic exactly for $w \geq w^*$, where $\cosh 2w^* = \cosh^2 a$. Moreover $\sinh w^* = \sinh a/\sqrt{2}$, $a/\sqrt{2} < w^* < a$, and $w^*/a \to \cos\frac{\pi}{4}$ as $q \uparrow \frac{1}{4}$. Since L has its zeros at real parts $\pm a$, X lies in $\mathcal{C}_a$ and not in $\mathcal{C}_w$ for $w < a$. So for every $w \in [w^*, a)$ the cosine tilt of X at w is ferromagnetic while $X \notin \mathcal{C}_w$. For $w > 0$ and $w < w^*$ the tilt has no zero on the imaginary axis at all. (Proved.)

Proof. Put $A = \cosh a$. Then $L(s) = \mathbb{E}e^{sX} = (1 - 2q + 2q\cosh s)^2 = 4q^2(A + \cosh s)^2$, an even entire function of order one with zeros at $\cosh s = -A$, that is $s = \pm a + i\pi(2k+1)$, double, so X is of Xi type and $X \in \mathcal{C}_a \setminus \mathcal{C}_w$ for $w < a$. The moment generating function of the tilted law is a multiple of $M_w(s) = L(s+w)+L(s-w)$, so ferromagnetism at w means that every zero of M_w has $\operatorname{Re} s = 0$. Write $c = \cosh s$, $C = \cosh w$, $S = \sinh w$. As $\cosh(s \pm w) = cC \pm S\sinh s$ and $\sinh^2 s = c^2 - 1$,

$$M_w(s) = 8q^2 Q_w(\cosh s), \qquad Q_w(c) = \cosh(2w)\, c^2 + 2AC\, c + A^2 - S^2.$$

Since cosh maps $\mathbb{C}$ onto $\mathbb{C}$, and $\cosh s \in [-1, 1]$ exactly when $\operatorname{Re} s = 0$ (for $s = x + iy$, $\cosh s = \cosh x \cos y + i \sinh x \sin y$, and $|\cosh s| = \cosh x > 1$ if $x \neq 0$ and $\cosh s$ is real), the zeros of M_w lie on the axis exactly when both roots $c_\pm$ of Q_w lie in $[-1, 1]$, and none lies on the axis when neither does. The discriminant of Q_w is

$$(AC)^2 - (C^2 + S^2)(A^2 - S^2) = \sinh^2 w\,(\cosh 2w - \cosh^2 a),$$

which for $w > 0$ is negative, zero or positive according as $w < w^*$, $w = w^*$ or $w > w^*$.

If $0 < w < w^*$ the roots are $c_\pm = (-AC \pm iS\sqrt{\cosh^2 a - \cosh 2w})/\cosh 2w$, not real because $S > 0$, so M_w has zeros, and none on the axis.

If $w > w^*$ the roots are real, $c_- \leq c_+$. Here $Q_w(1) = (C + A)^2 > 0$ and $Q_w(-1) = (C - A)^2 \geq 0$, and the vertex $c_0 = -AC/\cosh 2w$ satisfies $-1 \leq c_0 < 0$: from $\cosh 2w \geq A^2$ we get $A \leq \sqrt{2C^2 - 1}$, and $C\sqrt{2C^2 - 1} \leq 2C^2 - 1$ because $C \geq 1$, so $0 < AC \leq \cosh 2w$. If $c_- < -1$, then $Q_w(-1) \geq 0$ would force $c_+ \leq -1$ as well, so $c_0 < -1$, a contradiction; so $c_- \geq -1$. As $Q_w(1) > 0$, both roots lie on the same side of 1, and both above 1 would put $c_0 > 1$; so $c_+ < 1$. Hence $c_\pm \in [-1, 1)$ and every zero is on the axis.

If $w = w^*$ the root is double, $c_0 = -C/A$. Since $C^2 = (1 + A^2)/2 < A^2$ for $A > 1$, we have $c_0 \in (-1, 0)$ and all zeros are on the axis (each of multiplicity two).

Finally $\cosh 2w^* = \cosh^2 a$ reads $\sinh w^* = \sinh a/\sqrt{2}$, so $w^* < a$; and $\sinh x/x$ is increasing on $(0, \infty)$, so $\sinh(a/\sqrt{2}) < \sinh a/\sqrt{2} = \sinh w^*$, which gives $w^* > a/\sqrt{2}$. As $q \uparrow \frac{1}{4}$, $A \downarrow 1$ and $a \downarrow 0$, and $w^*/a = \frac{\operatorname{arsinh} u}{u} \cdot \frac{\sinh a}{a} \cdot \frac{1}{\sqrt{2}} \to \frac{1}{\sqrt{2}}$ with $u = \sinh a/\sqrt{2} \to 0$. □

For $q = 0.2$ ($a = 0.9624$) the threshold is $w^* = 0.7253$. The ratio w^*/a is 0.7536, 0.7165, 0.7080 and 0.70720 at $q = 0.2$, 0.24, 0.249 and 0.2499 (computed).

Plant a pair off the line in Ξ: with $c = \frac{1}{2}(\gamma_k + \gamma_{k+1}) + i\eta$ let

$$F(z) = \Xi(z)\,\frac{(z^2 - c^2)(z^2 - \bar{c}^2)}{(z^2 - \gamma_k^2)(z^2 - \gamma_{k+1}^2)},$$

an even entire function, real on the real axis, with strip half width η.

PROPOSITION 21.6.2. *For $k = 1$ and $\eta = 0.3$, $F(z + iw) + F(z - iw)$ has only real zeros in $[1, 45.3] \times [-2.2, 2.2]$ at $w = 0.299 < \eta$. The nonreal zero born from c reaches the real axis at $w^* \approx 0.2951$, its imaginary part behaving like $(w^* - w)^{1/2}$. The inverse Fourier transform of F is positive at $x = 0, 0.25, \ldots, 1.25$, with ratio to the Pólya density between 0.989 and 1.111 (computed, floating point).*

The exact law of Proposition 21.6.1 proves unconditionally that ferromagnetism of the cosine tilt at w does not force a law of Xi type into $\mathcal{C}_w$, for every $w \in [w^*, a)$. The planted computation illustrates the same fact for Ξ: if the planted density is positive on $\mathbb{R}$ and the tilt has no nonreal zero outside the box (which needs the zeros of ζ above height 45.3 to lie close to the line, for instance RH), ferromagnetism of the cosine tilt at w does not force Ξ with the planted pair into $\mathcal{C}_w$; the two variable property of Theorem 21.5.1 is needed. Table 21.1 repeats the computation for planted pairs at other heights and widths.

k	height m	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$	$\eta = 0.4$
1	17.58	0.0998	0.1985	0.2951	0.3888
3	27.72	0.0997	0.1980	0.2935	0.3854
5	35.26	0.0997	0.1983	0.2944	0.3873
8	45.67	0.0997	0.1978	0.2928	0.3839
11	54.71	0.0997	0.1978	0.2929	0.3842
15	66.10	0.0996	0.1971	0.2907	0.3796

Table 21.1: Slice thresholds w^* for Ξ with the ordinates γ_k, γ_{k+1} replaced by the pair $m \pm i\eta$, $m = \frac{1}{2}(\gamma_k + \gamma_{k+1})$: the least w at which the nonreal zero born from $m + i\eta$ reaches the real axis, found by continuation in w (computed, mpmath at 15 digits, floating point). Every threshold lies below η, as Theorem 21.4.1 requires, and the gap $\eta - w^*$ is close to $C\eta^3$ with C between 0.17 and 0.38, larger at the later ordinates of the table (the run to height 10^4 below shows that $C = \frac{1}{2}\phi^2$ then saturates and fluctuates).

In every case the gap is of third order in the width (computed; Theorem 21.7.1 proves the law $\eta - w^* \sim C\eta^3$ as $\eta \to 0$, with $C = \frac{1}{2}\phi^2$). So the one variable test measures the strip up to a small third order error, and it is not exact: a law can pass it at a coupling below its Cauchy defect (Proposition 21.6.1, proved, and for Ξ with a planted pair Proposition 21.6.2, conditionally).

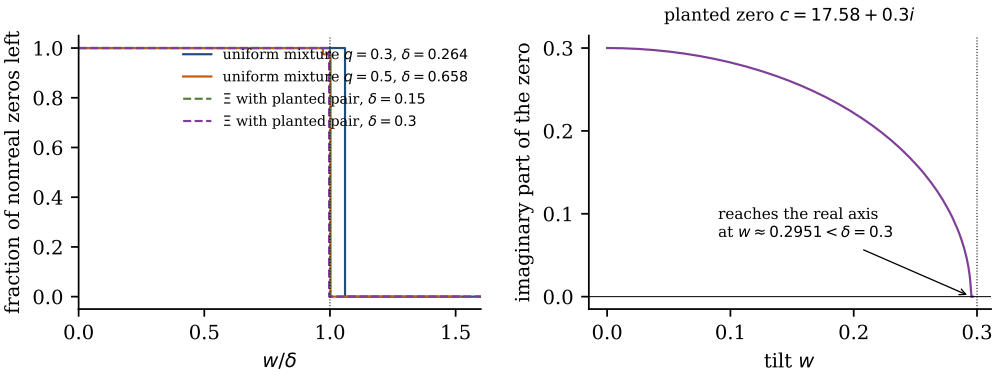


Figure 21.3: Left: fraction of nonreal zeros remaining after the cosine tilt, against w/δ, for two uniform mixtures and for Ξ with a planted pair. Right: imaginary part of the zero born from the planted zero $17.58 + 0.3i$, which reaches the real axis at $w \approx 0.2951 < \delta$ (computed).

The mechanism is a phase criterion. For real x, $F(x - iw) = \overline{F(x + iw)}$, so the tilt is $2 \operatorname{Re} F(x + iw)$ on the real axis.

PROPOSITION 21.6.3. *Let $\Theta_w(x)$ be a continuous argument of $F(x + iw)$ on an interval I where $F(x + iw) \neq 0$. The real zeros of the tilt in I are the points where $\Theta_w \in \frac{\pi}{2} + \pi\mathbb{Z}$. A real zero a of F contributes* $\arg(x - a + iw)$ *to Θ_w, which decreases by π across a; a zero $c = m + i\eta$ contributes* $\arg(x - m + i(w - \eta))$*, which increases by π across m when $w < \eta$ and decreases when $w > \eta$, while $\bar{c}$ contributes a decrease. So for $w < \eta$ the planted pair produces a local rise of Θ_w near m, of height tending to π as $w \uparrow \eta$, and two nonreal zeros of the tilt become real exactly when that rise first reaches a level $\frac{\pi}{2} + k\pi$.*

Proof. On the real axis the tilt is $2|F(x + iw)| \cos \Theta_w(x)$. The contributions are those of the factors of the Hadamard product of F. The total number of zeros of the tilt in a rectangle around I is constant in w as long as none crosses its boundary, so a pair of real zeros appears exactly when a pair of nonreal ones disappears; and a new pair of real zeros appears exactly when a local maximum of Θ_w reaches a level $\frac{\pi}{2} + k\pi$. □

Whether the rise reaches the level before $w = \eta$ depends on where the background phase of the other zeros puts the level, so the slice threshold can lie below the strip by an amount that depends on the configuration, and it can never lie above it (Theorem 21.4.1). Figure 21.4 shows the rise touching the level $\frac{3}{2}\pi$ at $w \approx 0.2951$.

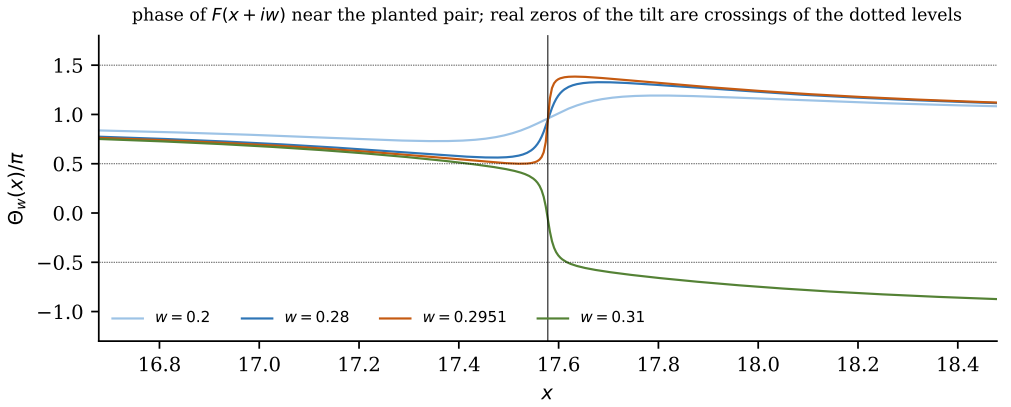


Figure 21.4: The phase Θ_w near the planted pair of Proposition 21.6.2. For $w < 0.2951$ the local rise stays below the level $\frac{3}{2}$; at $w \approx 0.2951$ it touches it, and two real zeros appear; for $w > \eta = 0.3$ the phase drops by about 2π (computed).

21.7 The size of the gap

The phase criterion has an exact form. Write $G = F/((z-c)(z-\bar{c}))$ for F without the planted pair $c = m + i\eta$, $\bar{c}$; it keeps the mirror pair $-c, -\bar{c}$ and the real zeros, it is real on the real axis, and we assume $G(m) \neq 0$. For $0 < w < \eta$ and real $x = m + u$, the two factors of the pair combine:

$$\arg\big((u+iw-i\eta)(u+iw+i\eta)\big) = \arg\big(u^2+\eta^2-w^2+2iwu\big) = \arctan\frac{2wu}{u^2+\eta^2-w^2},$$

the principal value, since the real part is positive. Put $\psi_w(x) = \arg G(x+iw)$, a continuous branch near m.

Theorem 21.7.1. *Let $r > 0$ be such that G has no zero in the disc $|z-m| \le 2r$. For η small enough the tilt $F(z+iw)+F(z-iw)$ has exactly two zeros in $|z-m| < r$ for every $w \in (0,\eta)$, and they are real if and only if*

$$\Phi_w(u) = \arctan\frac{2wu}{u^2+\eta^2-w^2} + \psi_w(m+u)$$

takes a value in $\frac{\pi}{2} + \pi\mathbb{Z}$ for some $u \in (-r, r)$. If $\phi = (G'/G)(m) \neq 0$, the slice threshold satisfies

$$\eta - w^* = \tfrac{1}{2}\,\phi^2\,\eta^3\big(1+O(\eta)\big) \qquad (\eta \to 0),$$

and if $\phi = 0$, then $\eta - w^* = O(\eta^7)$.

Proof. As $\eta \to 0$, uniformly in $w \in [0,\eta]$, the tilt converges on $|z-m| \le r$ to $2(z-m)^2 G(z)$, with G read as its limit as $\eta \to 0$, which has a double zero at m

and no other zero there; by Hurwitz's theorem the tilt has exactly two zeros in $|z-m|<r$ for η small. Nonreal zeros come in conjugate pairs, so the two are both real or both nonreal. On the real axis $F(x-iw)=\overline{F(x+iw)}$, so the tilt is $2|F(x+iw)|\cos\arg F(x+iw)$, and $\arg F(x+iw)=\Phi_w(x-m)$ by the display above; real zeros are the points where $\Phi_w\in\frac{\pi}{2}+\pi\mathbb{Z}$.

For the asymptotics write $w=\eta-\varepsilon$ and $a=(\eta^2-w^2)^{1/2}$. The first term of Φ_w is odd in u, with extreme values $\pm\arcsin(w/\eta)$ at $u=\pm a$, and it is $O(w/|u|)$ for $|u|\gg a$. Since G is real on the real axis, $\psi_w(m)=w\phi+O(w^3)$ modulo π and $\psi_w'(m+u)=\operatorname{Im}(G'/G)(m+u+iw)=O(w)$. Suppose $\phi>0$; the case $\phi<0$ is the mirror image with the level $-\frac{\pi}{2}$. Then Φ_w stays above $-\frac{\pi}{2}$, and its maximum is $\psi_w(m)+\arcsin(w/\eta)+O(\eta a)$, attained near $u=a$. It reaches $\frac{\pi}{2}$ exactly when $\arccos(w/\eta)\le\psi_w(m)+O(\eta a)$. Here $\arccos(w/\eta)=(2\varepsilon/\eta)^{1/2}(1+O(\varepsilon/\eta))$ and $a=(2\eta\varepsilon)^{1/2}(1+O(\varepsilon/\eta))$. At the threshold ε is of order η^3, so $\eta a=O(\eta^3)$, while $\psi_w(m)=\eta\phi(1+O(\eta))$; hence $(2\varepsilon^*/\eta)^{1/2}=\eta\phi(1+O(\eta))$, which is the stated law. If $\phi=0$, then $\psi_w(m)=O(\eta^3)$, the same comparison gives $(2\varepsilon^*/\eta)^{1/2}=O(\eta^3)$, and $\varepsilon^*=O(\eta^7)$. □

The coefficient is the field that all the other zeros exert at the planted pair,

$$\phi(m)=\frac{G'}{G}(m)=\sum_a\frac{1}{m-a}+\frac{4m}{4m^2+\eta^2},$$

the sum over the real zeros of F in symmetric pairs and the last term from the mirror pair; it grows over the first ordinates, as the constants of Table 21.1 do ($\phi=-0.62$ at $k=1$, -0.88 at $k=15$), and then saturates and fluctuates (see the run to height 10^4 below). The exact criterion reproduces the thresholds found by continuation to every printed digit: for $k=1,3,8,15$ and $\eta=0.1,0.3$ it gives 0.099810, 0.295090, 0.099746, 0.293544, 0.099711, 0.292783, 0.099619 and 0.290699, against 0.09981, 0.29509, 0.09975, 0.29354, 0.09971, 0.29278, 0.09962 and 0.29069 (computed).

Remark 21.7.2 (heuristic). Solved as a fixed point with the cubic and quadratic terms kept, the criterion gives

$$\varepsilon^*=\frac{\operatorname{dist}\big(\arg G(m+iw^*),\pi\mathbb{Z}\big)^2}{4\big(b+\frac{1}{2\eta-\varepsilon^*}\big)},\qquad b=-\operatorname{Im}\frac{G'}{G}(m+iw^*),$$

to the order of Theorem 21.7.1.

Two further computations test it.

First, solved as a fixed point, this formula reproduces all twenty four thresholds of Table 21.1 to within about 1.5%, which is the resolution of the table, at every height and width (Figure 21.6, left; computed). An earlier version that left out the mirror pair was off by up to 18% at the first pair; the term $4m/(4m^2+\eta^2)\approx 1/m$ is what matters at low height.

Second, the formula predicts that a pair planted where ϕ vanishes has a gap of much higher order. For $k = 3$ and $k = 5$ we located the zero of ϕ in the gap left by the removed ordinates, at $m = 22.961$ and $m = 32.156$, and planted the pair there. The gaps fell from 2.5×10^{-4} and 2.2×10^{-4} at the midpoint to 1.2×10^{-9} at $\eta = 0.1$, and from 1.5×10^{-2} and 1.3×10^{-2} to 1.5×10^{-6} and 2.8×10^{-6} at $\eta = 0.4$, a reduction by four to five orders of magnitude. Between $\eta = 0.2$ and $\eta = 0.4$ they grow by factors of 100 and 110, close to the factor $2^7 = 128$ of an η^7 law (Figure 21.6, centre; computed, thresholds resolved to about 10^{-9}).

So the one variable test misses the strip by an amount set by a single local quantity, the field of the other zeros at the pair: by $\frac{1}{2}\phi^2\eta^3$, and by almost nothing where the field vanishes.

The field and the thresholds to height 10^4 *(computed, with ball arithmetic where stated).* The field was evaluated for the midpoint pair ($\eta = 0.1$) at each of the first 10^4 ordinates, with the ordinates γ_k as the balls of the certified zero enumeration of `acb.zeta_zeros` and every value a ball of radius below 3×10^{-31} (128 bit). The field is negative for 93.0% of the pairs, with mean -0.785 and standard deviation 0.531; it does not keep growing. By decade of height the mean is -0.736, -0.781, -0.784, -0.785, -0.786 and the median of $|\phi|$ is 0.78 at every height, while only the spread grows (0.12 below height 100, 0.58 above 6000). Values at later ordinates are -1.017 at $k = 100$, -0.396 at $k = 1000$, -0.354 at $k = 3000$ and -0.692 at $k = 9999$. Small values are rare: $|\phi| < 10^{-2}$ for 43 of the 10^4 pairs (0.43%, none below height 1000) and $|\phi| < 0.1$ for 4.8%. For every k the field as a function of the position of the pair has exactly one sign change in the window $(\gamma_{k-1}, \gamma_{k+2})$ (grid and bisection, computed), and about 0.3% of the window lies within 10^{-2} of it.

The slice thresholds were recomputed by the exact criterion of Theorem 21.7.1 (flint, 160 bit balls for the criterion, double precision for the argument) for k up to 9999 at $\eta = 0.1$ and 0.3. The eight values above are reproduced to six digits. The ratio of the measured gap to $\frac{1}{2}\phi^2\eta^3$ falls from 0.995 at $k = 1$ to 0.951 at $k = 9999$ for $\eta = 0.1$, and from 0.954 to 0.688 for $\eta = 0.3$, the $O(\eta)$ correction being larger where the zeros are denser. The fixed point of Remark 21.7.2 reproduces the gap to 0.2% at $\eta = 0.1$ and 1.2% at $\eta = 0.3$ for every k run. At the zero of the field ($|\phi| < 3 \times 10^{-12}$) the gap at $\eta = 0.1$ is 1.9×10^{-10} to 1.6×10^{-9}, which is 1.7×10^4 to 2.3×10^6 times smaller than at the midpoint, and from $\eta = 0.1$ to 0.3 it grows by 1984, 1905, 1540, 1666 and 1499 for $k = 8$, 100, 1000, 3000 and 9999, against $3^7 = 2187$ for an η^7 law.

The two spin test was run as a zero free rectangle. In the three windows $[10.0, 109.1]$, $[1000.6, 1100.0]$ and $[9792.9, 9878.2]$ (33, 81 and 100 zeros), the rectangle $w \le \operatorname{Im} z \le 1.5$ over the window contains no zero of Ξ for each $w = 0.5, 0.4, 0.3, 0.2, 0.1, 0.05$: the winding number of Ξ around the boundary in 96 bit ball arithmetic is 0 in all 18 cases (certified, apart from the Euler product bound

at $\sigma = 2$, which was not computed). This is the zero free region of the window for the pair, not the global property of Corollary 21.5.2. The one variable test on the same windows (every zero of $\Xi(z+iw)+\Xi(z-iw)$ in $|\operatorname{Im} z| \le 1$ real, by the same winding routine and the sign changes on a grid of step 0.04) passes in all 18 cases and costs 13 to 18 thousand evaluations of Ξ, nearly independent of w. The cost of the rectangle grows like $1/w$ times the width of the window: relative to the one variable test, 0.33, 0.62 and 1.0 at $w = 0.5$ for the three windows, and 0.72, 2.9 and 9.9 at $w = 0.05$ (165 against 18 seconds at height 10^4). It is no dearer than the one variable test down to w about 0.3.

Where the two tests differ the run shows it. For pairs planted at $k = 8$ and 300 ($\eta = 0.3$) and $k = 1000$ and 9999 ($\eta = 0.1$), with w just above the threshold of the one variable test and below η ($w = 0.296391$, 0.297784, 0.099962 and 0.099886), the one variable test passes (all 6, 13, 14 and 16 zeros real) while the rectangle contains the planted zero (count 1); below the threshold the one variable test fails, and for $w > \eta$ both are clean. At $k = 1000$ the rectangle detects the pair at $w = \eta - 3.8 \times 10^{-5}$, so the two spin form loses nothing to the gap. This is a test on windows and on planted pairs, and it proves nothing about the zeros of ζ outside them.

21.8 Several pairs and multiple zeros

The proof of Theorem 21.7.1 is local, and it extends to configurations with several pairs off the line and to pairs of higher multiplicity. The answer has two parts. Each cluster of zeros off the line is seen by the one variable test separately, through its own field, which now includes the other clusters. And a multiple zero changes the order of the gap: a double pair is seen only at a fixed fraction of its distance from the line.

Fix distinct heights $m_1, \dots, m_K > 0$, multiplicities $n_j \ge 1$ and ratios $t_j \in (0, 1]$, and let

$$F = G_0 \prod_{j=1}^{K} \big((z^2 - c_j^2)(z^2 - \bar{c}_j^2)\big)^{n_j}, \qquad c_j = m_j + i\eta_j, \quad \eta_j = t_j \eta,$$

with G_0 real entire, real on the real axis, and $G_0(m_j) \ne 0$. Let $G_j = F/((z - c_j)(z - \bar{c}_j))^{n_j}$; as $\eta \to 0$ it converges near m_j to the function G_j^0 in which every other cluster has become a real zero of multiplicity $2n_l$ at m_l, so its field

$$\phi_j = \frac{(G_j^0)'}{G_j^0}(m_j) = \frac{G_0'}{G_0}(m_j) + \sum_{l \ne j} \frac{2n_l}{m_j - m_l} + \sum_{l} \frac{2n_l}{m_j + m_l}$$

contains the other clusters and all the mirror images. The local slice threshold w_j^* is the least w for which all $2n_j$ zeros of the tilt born from c_j and $\bar{c}_j$ are real.

THEOREM 21.8.1. *Let $r > 0$ be such that G_j^0 has no zero in $|z - m_j| \le 2r$ for every j. For η small enough the tilt $F(z+iw) + F(z-iw)$ has exactly $2n_j$ zeros in $|z - m_j| < r$ for every $w \in [0, \eta]$; they are all real for $w \ge \eta_j$, and*

1. *for n_j odd, $\eta_j - w_j^* = \frac{\phi_j^2}{2n_j^2}\,\eta_j^3\big(1+O(\eta)\big)$ when $\phi_j \neq 0$, and $O(\eta^7)$ when $\phi_j = 0$;*
2. *for n_j even, $w_j^* = \eta_j \cos\frac{\pi}{2n_j} + O(\eta^2)$, and more precisely $w_j^* = \eta_j \sin\big(\frac{(n_j-1)\pi}{2n_j} + \frac{|\psi_j|}{n_j}\big) + O(\eta^3)$ with $\psi_j = \operatorname{dist}(\arg G_j(m_j + iw_j^*), \pi\mathbb{Z}) = O(\eta)$.*

The configuration is seen by the one variable test, in the sense that every zero born from a planted zero is real, exactly for $w \ge \max_j w_j^*$.

Proof. Fix j and drop the index. As $\eta \to 0$, uniformly in $w \in [0, \eta]$, the tilt converges on $|z - m| \le r$ to $2(z-m)^{2n}G^0(z)$, so by Hurwitz's theorem it has exactly $2n$ zeros there. On the real axis the tilt is $2|F(x+iw)|\cos\arg F(x+iw)$ and, with $x = m + u$ and $a^2 = \eta^2 - w^2$,

$$\arg F(x+iw) = \Phi(u) = n\,A(u) + \psi_w(m+u), \qquad A(u) = \arg\big(u^2 + a^2 + 2iwu\big),$$

where $\psi_w = \arg G(\cdot + iw)$ varies by $O(\eta r)$ on $(-r, r)$, with derivative $O(\eta)$.

For $w < \eta$ the function A is odd, decreases from $O(\eta/r)$ at $u = -r$ to $-\alpha$ at $u = -a$, increases to α at $u = a$ and decreases to $O(\eta/r)$ at $u = r$, with $\alpha = \arcsin(w/\eta)$. Put $\psi_0 = \psi_w(m)$, taken in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\psi_0 = O(\eta)$. Up to the error terms, Φ therefore passes once through every level of $\frac{\pi}{2} + \pi\mathbb{Z}$ in $(\psi_0 - n\alpha, \psi_0)$ on $(-r, -a)$, once through every level in $(\psi_0 - n\alpha, \psi_0 + n\alpha)$ on $(-a, a)$, and once through every level in $(\psi_0, \psi_0 + n\alpha)$ on (a, r). So the number of real zeros is twice the number of levels in $(\psi_0 - n\alpha, \psi_0 + n\alpha)$, an interval of length $2n\alpha < n\pi$, which contains at most n levels. All $2n$ zeros are real exactly when it contains n of them. For n even the levels needed are $\pm\frac{\pi}{2}, \dots, \pm\frac{(n-1)\pi}{2}$, and the condition is $n\alpha \ge \frac{(n-1)\pi}{2} + |\psi_0|$, which is the formula of (ii). For n odd one side needs the level $\frac{n\pi}{2}$, and the condition is $n\alpha + |\psi_0| \ge \frac{n\pi}{2}$, that is $n \arccos(w/\eta) \le |\psi_0|$; with $\psi_0 = \eta\phi(1 + O(\eta))$ and $\arccos(w/\eta) = (2\varepsilon/\eta)^{1/2}(1 + O(\varepsilon/\eta))$, $\varepsilon = \eta - w$, this is (i), exactly as in the proof of Theorem 21.7.1. The error terms are those of that proof: near the extrema ψ_w moves by $O(\eta a)$, which is $O(\eta^3)$ at the threshold for n odd and $O(\eta^2)$ for n even, giving the stated orders.

For $w \ge \eta$ the real part of $u^2 + a^2 + 2iwu$ is negative near $u = 0$ and its imaginary part changes sign there, so A decreases continuously by $2\pi - O(\eta/r)$ across $(-r, r)$ and Φ decreases by $2n\pi - O(\eta)$. It passes through at least $2n - 1$ levels, hence, since nonreal zeros come in conjugate pairs and there are $2n$ zeros in all, through $2n$, and all the zeros are real. The last statement collects the clusters. □

Three consequences are worth recording.

First, the other clusters enter only through the field. A neighbour at distance d adds $\pm 2/d$ to ϕ_j; it can enlarge the gap, and it can also cancel the field of the real zeros and hide the pair almost completely. Table 21.2 shows both, for two pairs of equal width planted in the gap left by γ_8 and γ_9, at $M \mp \frac{d}{2}$ with $M = \frac{1}{2}(\gamma_8 + \gamma_9)$. At $d = 2$ the field at the upper pair is 0.037 and its gap 6.5×10^{-7}, about 450 times below that of a single pair at the same width. The fixed point of Remark 21.7.2, with G_j in place of G, predicts every gap of two pairs of equal width with $d \geq 0.3$ to within 7%, and to within 3% for $d \geq 1.2$, at widths 0.1 and 0.2, including separations of three times the width, where the theorem does not apply.

d	η	pair	ϕ_j	gap $\eta - w_j^*$	fixed point
3.0	0.1	lower	-1.096	5.88×10^{-4}	5.86×10^{-4}
3.0	0.1	upper	-0.425	8.92×10^{-5}	8.92×10^{-5}
2.0	0.1	lower	-1.536	1.13×10^{-3}	1.13×10^{-3}
2.0	0.1	upper	$+0.037$	6.47×10^{-7}	6.47×10^{-7}
0.8	0.1	lower	-3.123	4.18×10^{-3}	4.12×10^{-3}
0.8	0.1	upper	$+1.636$	1.21×10^{-3}	1.21×10^{-3}
0.3	0.1	lower	-6.712	1.31×10^{-2}	1.25×10^{-2}
0.3	0.1	upper	$+5.227$	8.62×10^{-3}	8.30×10^{-3}
1.2	0.2	lower	-2.242	1.53×10^{-2}	1.49×10^{-2}
1.2	0.2	upper	$+0.753$	1.93×10^{-3}	1.92×10^{-3}

Table 21.2: Two pairs of width η planted at $M \mp \frac{d}{2}$ in place of γ_8, γ_9: field, gap found by continuation, and the fixed point of Remark 21.7.2 with G_j keeping the other pair (computed, mpmath at 20 digits, floating point). When the second pair has half the width, the gaps of the wider pair change by less than 5% for $d \geq 0.8$ and by less than 0.5% at $d = 3$.

Second, among simple pairs the pair nearest to the line does not matter. A simple pair with $\eta_j < \eta$ has $w_j^* < \eta_j$, so a configuration of simple pairs is decided by the widest pairs alone, and this is borne out by the table: halving the width of one pair changes the gap of the other by a few per cent at most.

Third, multiplicity changes the order of the gap. A simple pair is missed by $O(\eta^3)$; a double pair is missed by a fixed fraction, $1 - \cos\frac{\pi}{4} \approx 0.29$, of its distance from the line. Planted in place of γ_8, γ_9, a double pair of width 0.1 has all four zeros real from $w = 0.07263$, close to the value given by (ii), and two of them already from 0.0689; a quadruple pair has all eight real from 0.0930, again close to (ii). At width 0.2 the double pair gives 0.1493 against 0.1490. Two simple pairs that coalesce interpolate between the two laws. At width 0.2 the gap of the configuration, the smaller of the two, rises from 7.0×10^{-4} at $d = 3$ to 5.03×10^{-2} at $d = 0.03$, against 0.0507 for the double pair, and the larger rises from 4.4×10^{-3} to 6.5×10^{-2}, against 0.0655 for the first real pair of zeros of the double pair: the two simple pairs become the two stages of the double one (Figure 21.5; computed).

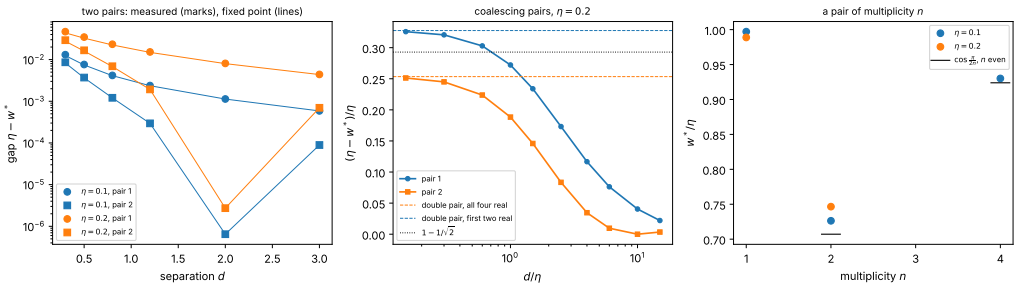


Figure 21.5: Left: gaps of two planted pairs of equal width against their separation, by continuation (marks) and by the fixed point of Remark 21.7.2 (lines). Centre: the gap over the width as two pairs of width 0.2 coalesce, approaching the two stages of the double pair at this width (dashed), near the limit $1 - 1/\sqrt{2}$ (dotted). Right: the slice threshold of a pair of multiplicity n over its width; for even n it is close to $\cos\frac{\pi}{2n}$ (computed).

So the quantitative converse that one might hope for, ferromagnetism of $\cosh(wx)\,p$ implying no zero with $\operatorname{Re} s > \frac{1}{2} + w(1+o(1))$, fails by a constant factor already for double zeros: under Theorem 21.8.1(ii) a double pair at distance δ from the line is invisible to the one variable test for every w above about $\delta/\sqrt{2}$. For isolated clusters this is the worst case, since $\cos\frac{\pi}{2n} \geq \cos\frac{\pi}{4}$ for even n and odd multiplicities have gaps of order η^3; whether some configuration of simple zeros does worse is part of Problem 21.1. The two spin system of Theorem 21.5.1 has no such loss, which is one more reason to state the target in that form.

21.9 A strip Lee and Yang theorem

The Lee and Yang theorem says that a ferromagnetic coupling of spins in $\mathcal{C}_0$ gives a partition function with zeros only on the imaginary axis. For the strip classes the question is whether a coupling can widen the strip. It cannot, for symmetric spins whose Laplace transforms are of exponential type, by the general theorem of Lieb and Sokal.

Theorem 21.9.1. *Let $\chi_1, \dots, \chi_N$ be symmetric probability measures on $\mathbb{R}$ whose Laplace transforms $\varphi_i(x) = \int e^{x\sigma}\chi_i(d\sigma)$ are entire of exponential type, for instance measures of bounded support, and suppose $\varphi_i(x) \neq 0$ whenever $\operatorname{Re} x > \delta_i$. Let $J_{ij} \geq 0$. Then the partition function*

$$Z(h_1, \dots, h_N) = \int \exp\Big(\tfrac{1}{2}\sum_{i,j} J_{ij}\sigma_i\sigma_j + \sum_i h_i\sigma_i\Big) \prod_i \chi_i(d\sigma_i)$$

does not vanish when $\operatorname{Re} h_i > \delta_i$ for every i. It does not vanish when $\operatorname{Re} h_i < -\delta_i$ for every i either, and in a common field all its zeros lie in $|\operatorname{Re} h| \leq \max_i \delta_i$.

Proof. $Z = \exp\left(\frac{1}{2}\sum J_{ij} D_i D_j\right) \prod_i \varphi_i(x_i)$ at $x_i = h_i$, with $D_i = \partial/\partial x_i$. The function $G(x) = \prod_i \varphi_i(x_i + \delta_i)$ is entire of exponential type and does not vanish when every $\operatorname{Re} x_i > 0$. Since φ_i is even, $\varphi_i(x) = \prod_k (1 - x^2/r_{ik}^2)$ with $|\operatorname{Re} r_{ik}| \le \delta_i$, so G is the locally uniform limit of the polynomials $\prod_i \prod_{k \le n} (1 - (x_i + \delta_i)^2/r_{ik}^2)$, which do not vanish when every $\operatorname{Re} x_i > 0$. By Proposition 2.2 of Lieb and Sokal [211] and the domination argument of the proof of Theorem 21.9.2, $\exp(\frac{1}{2}\sum J_{ij} D_i D_j) G$ does not vanish when every $\operatorname{Re} x_i > 0$; since translation commutes with the differential operator, that function is $Z(x_1 + \delta_1, \dots, x_N + \delta_N)$. For $\operatorname{Re} h_i < -\delta_i$ apply the same argument to the reflected measures. □

We have read Proposition 2.2 of [211] in the restatement of [377], not in the original. The theorem settles Problem 21.3 as it was first posed for bounded spins, and it explains the computations of Figure 21.6, right: for spin one laws on $\{-1, 0, 1\}$ with $P(\pm 1) = q < \frac{1}{4}$, whose strip half width a satisfies $\cosh a = \frac{1-2q}{2q}$, the strip of Curie Weiss systems and open chains of 2, 4 and 8 spins never widens, shrinks monotonically with the coupling, and closes to $\mathcal{C}_0$ at a finite coupling. A random search over 1164 systems with single spin laws on $\{-3, \dots, 3\}$ found no widening, and 20000 random tests of the multivariable form, a spin one law coupled to up to five Ising spins in independent complex fields, found no zero with $\operatorname{Re} h_0 > a$ when every Ising field has positive real part (computed). The shrinking is not part of the theorem.

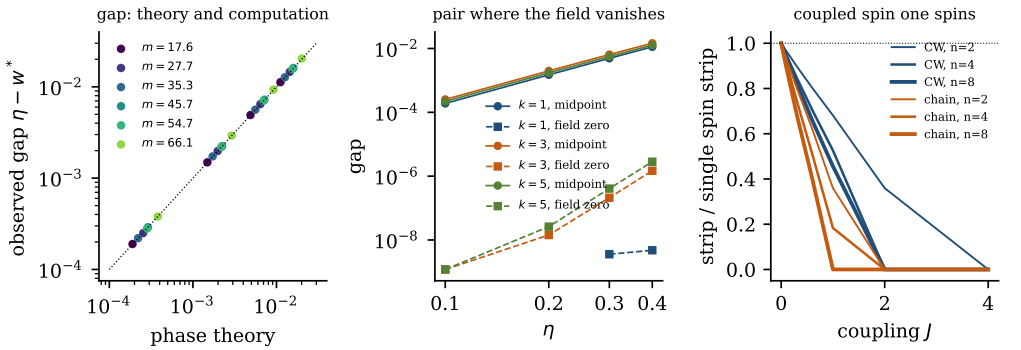


Figure 21.6: Left: the gap $\eta - w^*$ of Table 21.1 against the phase theory of Remark 21.7.2. Centre: gaps for pairs planted at the midpoint and at the zero of the field ϕ; at the zero of the field the gap is smaller by four to five orders of magnitude. Right: strip half width of coupled spin one spins, $q = 0.2$, relative to one spin, against the coupling (computed).

The Pólya law is not of exponential type: its Laplace transform $\xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$ has order one and infinite type. The theorem still extends to it, for every ferromagnetic coupling, by approximating each transform with its truncated Hadamard products.

THEOREM 21.9.2. *Let $\chi_1, \dots, \chi_N$ be laws of Xi type with Gaussian moments of all orders, whose Laplace transforms satisfy $\varphi_i(x) \neq 0$ for $\operatorname{Re} x > \delta_i$, and let $J_{ij} \geq 0$. Then $Z(h_1, \dots, h_N) \neq 0$ whenever $\operatorname{Re} h_i > \delta_i$ for every i. In particular N Pólya spins in any ferromagnetic coupling have all the zeros of their partition function in a common field in the strip $|\operatorname{Re} h| \leq \sup_\rho |\beta - \frac{1}{2}|$, with no hypothesis.*

Proof. With Gaussian moments of all orders $b = 0$ in Definition 21.1.1, so $\varphi_i(x) = \prod_k (1 - x^2/r_{ik}^2)$. Expanding $e^{\frac{1}{2}\sigma \cdot J\sigma}$ in powers and integrating term by term, which Gaussian moments of all orders allow, gives $Z = \exp(\frac{1}{2}\sum J_{ij} D_i D_j) \prod_i \varphi_i$ as a convergent series of derivatives. Let $\varphi_i^{(n)}(x) = \prod_{k \leq n}(1 - x^2/r_{ik}^2)$. These polynomials have their zeros among those of φ_i, so $\varphi_i^{(n)}(x) \neq 0$ for $\operatorname{Re} x > \delta_i$, and they converge to φ_i locally uniformly. Their Taylor coefficients are bounded in modulus by those of $M_i(x) = \prod_k (1 + x^2/|r_{ik}|^2)$, an entire function with nonnegative coefficients and $\log M_i(R) = o(R^2)$, because $\sum_k |r_{ik}|^{-2} < \infty$. Hence every term of the series for $Z^{(n)} = \exp(\frac{1}{2}\sum J_{ij} D_i D_j) \prod_i \varphi_i^{(n)}$ at h is bounded by the corresponding term of $\exp(\frac{1}{2}\sum J_{ij} D_i D_j) \prod_i M_i$ at $(|h_1|, \dots, |h_N|)$. All terms of the latter are nonnegative and increase with the entries of J, so it is at most the same series for $J + cI$, which is positive semidefinite for c large and is at most the Gaussian average $\mathbb{E} \prod_i M_i(|h_i| + |((J + cI)^{1/2}\zeta)_i|)$, finite because $\log M_i(R) = o(R^2)$. By Tannery's theorem $Z^{(n)} \to Z$ locally uniformly on $\mathbb{C}^N$. By Proposition 2.2 of [211] applied to the polynomial $\prod_i \varphi_i^{(n)}(x_i + \delta_i)$, each $Z^{(n)}$ is nonzero when $\operatorname{Re} h_i > \delta_i$ for every i. By Hurwitz's theorem in several variables Z is nowhere zero on that connected domain or identically zero there, and $Z > 0$ for real h. □

This settles the extension asked for in the first version of Problem 21.3, for every nonnegative coupling and every law of Xi type with Gaussian moments of all orders, modulo the reading of Proposition 2.2 of [211] noted above. As a check, two copies of the planted law of Proposition 21.6.2, with strip 0.3, in the Curie Weiss coupling $a \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ have their zeros near the planted pair at real parts 0.300, 0.293, 0.265, 0.229 and 0.149 for $a = 0$, 0.002, 0.01, 0.02 and 0.04, and on the axis from $a = 0.07$ on (computed, Gauss Hermite quadrature with 40 nodes): inside the strip, and moving towards the line. The bare pair coupling of the Pólya spin with an Ising spin is the case below, where a direct proof is available.

PROPOSITION 21.9.3. *Let X be of Xi type in $\mathcal{C}_\delta$ and couple it to a fair Ising spin with strength $w \geq 0$. In a common field h the partition function $e^h L(h + w) + e^{-h} L(h - w)$ has no zero with $|\operatorname{Re} h| > \delta$, and none off the imaginary axis when every zero of L has $|\operatorname{Re} r| < w$.*

Proof. Let $h = y + iv$ with $y > \delta$. A zero r of L contributes $|h + w - r|/|h - w - r|$ to $|L(h+w)|/|L(h-w)|$, and $(y + w - \operatorname{Re} r)^2 - (y - w - \operatorname{Re} r)^2 = 4w(y - \operatorname{Re} r) \geq 0$, with the partner $-r$ likewise; so $|L(h+w)| \geq |L(h-w)|$, while $|e^h| > |e^{-h}|$. Hence

the two terms cannot cancel; $y < -\delta$ follows by symmetry. The second statement is Theorem 21.5.1(a) with $h_1 = h_2 = h$. □

For the descent the theorem is a monotonicity statement of the right kind: adding ferromagnetic interactions never moves zeros away from the line. It does not help with RH directly, because RH asks for an interaction to be removed.

21.10 Approximation of the Pólya density in Cauchy terms

The ferromagnetic approximations of Section 19.1 of Volume II, whose best distance to the Pólya law is TV $< 1.932 \times 10^{-6}$ with no hypothesis (Proposition 19.1.15 of Volume II), have a reading in this chapter. The zero completed law f_K of Theorem 19.1.12 of Volume II has, by Theorem 21.1.2, the Thorin density

$$u_y^{(K)}(v) = \frac{2\beta_K}{\pi} y + \sum_{k \le K} \big(P_y(v - \gamma_k) + P_y(v + \gamma_k) \big), \qquad \beta_K = \tfrac{1}{2} \operatorname{Var} X - \sum_{k \le K} \gamma_k^{-2}.$$

It keeps the first K atoms of the mixing measure and replaces the rest by the drift. If the zeros above γ_K are on the line, $0 \le \frac{2\beta_K}{\pi} y - \sum_{k>K} 2P_y(\gamma_k) \le \frac{2y^3}{\pi} \sum_{k>K} \gamma_k^{-4}$, so the drift is the low frequency limit of the far bumps (proved, conditional on the zeros above γ_K only). At $y = \frac{1}{2}$ the two Thorin densities agree at $v = 0$ to within 4×10^{-8} for $K = 16$ and 5×10^{-9} for $K = 150$, and on $[0, \gamma_K/2]$ to within 6.3×10^{-4} and 2.2×10^{-4} (computed; Figure 21.7). Inside the ferromagnetic class the Thorin side is automatic once the atoms are real, and what must be proved is the positivity of the density, the dual of the deconvolution form of RH in Theorem 8.2.2 of Volume II.

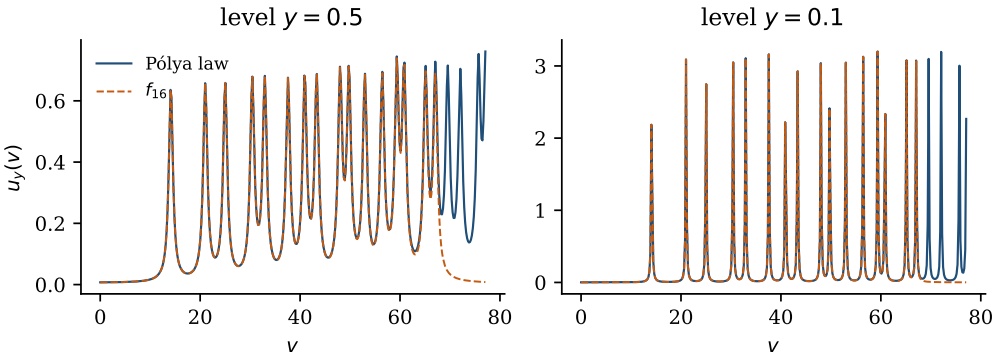


Figure 21.7: Thorin density of the Pólya law (solid) and of the ferromagnetic law f_{16} (dashed) at the levels $y = 0.5$ and $y = 0.1$; the approximant keeps sixteen bumps and replaces the rest by a constant (computed).

21.11 The target as monotonicity, and the function field model

Theorem 21.11.1. *The Pólya spin coupled to one Ising spin is a Lee and Yang ferromagnet for every coupling $w > 0$ if and only if $\sigma \mapsto |\xi(\sigma+it)|$ is nondecreasing on $[\frac{1}{2}, \infty)$ for every t, if and only if RH holds.*

Proof. By the proof of Theorem 21.5.1 the Lee and Yang property at coupling w implies $|L(h_1 - w)| \le |L(h_1 + w)|$ for $\operatorname{Re} h_1 > 0$ (and is equivalent to it), and since $|L(-x+iv)| = |L(x+iv)|$ this reads $|L(|y-w|+iv)| \le |L(y+w+iv)|$ with $h_1 = y + iv$. Taking $y = \frac{x_1+x_2}{2}$ and $w = \frac{x_2-x_1}{2}$ for $0 \le x_1 < x_2$ gives monotonicity of $|L(x+iv)|$ on $[0,\infty)$, and conversely. If $|L(\cdot+iv)|$ is nondecreasing and vanishes at $x_0 > 0$, it vanishes on $[0, x_0]$ and $L \equiv 0$; so every zero is imaginary. Under RH each factor $|1 - (x+iv)^2/r^2|$ with r imaginary is nondecreasing in $x \ge 0$. □

With strict increase this is the criterion of Sondow and Dumitrescu [333], and its infinitesimal form is Lagarias's [198]. Lowering the coupling is shortening the horizontal segments across which $|\xi|$ must increase; at $w \ge \frac{1}{2}$ they reach the line $\operatorname{Re} s = 1$, where the Euler product supplies the increase.

The function field case, where RH is a theorem, shows what a proof along this route needs. For an elliptic curve over $\mathbb{F}_q$ with $\#E(\mathbb{F}_q) = q + 1 - a$, the numerator of the zeta function gives, in $x = s - \frac{1}{2}$,

$$L_E(x) = q^x\big(1 - aq^{-1/2-x} + q^{-2x}\big) = 2\cosh(x \log q) - \frac{a}{\sqrt{q}}.$$

Proposition 21.11.2. *For $a \le 0$, L_E is the moment generating function of the three point measure with masses 1, $|a|/\sqrt{q}$, 1 at $-\log q$, 0, $\log q$; for $a > 0$ the imaginary field shift $x \mapsto x + i\pi/\log q$ turns it into minus such a function and changes no real part. Its zeros are imaginary if and only if $|a| \le 2\sqrt{q}$, and otherwise lie at real parts $\pm\delta$, $\cosh(\delta \log q) = |a|/(2\sqrt{q})$. Hence the Hasse bound holds exactly when this Frobenius spin is ferromagnetic, when its Thorin density is a single width Cauchy mixture at every level, and when the pair formed with an Ising spin is a Lee and Yang ferromagnet at every coupling.*

Proof. $2\cosh z = e^z + e^{-z}$ and $\cosh(z + i\pi) = -\cosh z$; $\cosh(x \log q) = a/(2\sqrt{q})$ has only imaginary solutions exactly when $|a| \le 2\sqrt{q}$. The equivalences are Theorems 21.2.1 and 21.5.1, which use only the zeros. □

For a curve of genus g the completed function is, when $|\alpha_j| = \sqrt{q}$, a product of g factors $2\cosh(x \log q) - 2\operatorname{Re}(\alpha_j)/\sqrt{q}$, one Frobenius spin for each pair of eigenvalues. Hasse proved $|a| \le 2\sqrt{q}$ from $\deg(m - n\pi) = m^2 - amn + qn^2 \ge 0$, the nonnegativity of degrees of isogenies. In our language it is Weil positivity on two point test functions: for $g = m\delta_0 - n\sqrt{q}\,\delta_{\log q}$ on the lattice $\log q\,\mathbb{Z}$, the sum of $|\widehat{g}|^2$ over the two zeros of one period is $2(m^2 - amn + qn^2)$. The positivity comes

from the geometry of the curve, not from the zeta function. For $y^2 = x^3 + x + 1$ and the 165 primes of good reduction among $5 \le p < 1000$ ($p = 31$ is bad), $\max |a_p|/(2\sqrt{p}) = 0.979$, so every Frobenius spin is ferromagnetic; a fake factor with $|a|/(2\sqrt{q}) = 1.3$ at $q = 101$ has strip $\delta = 0.164$ and a Thorin density that is negative below it (Figure 21.8; computed).

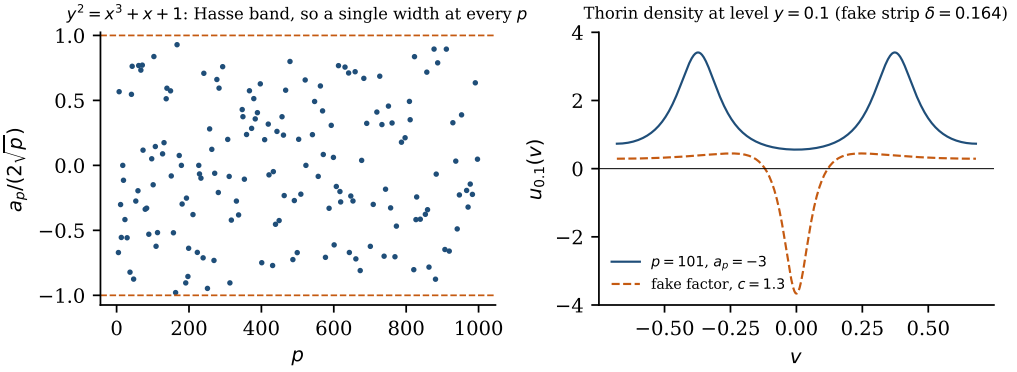


Figure 21.8: Left: normalised Frobenius traces $a_p/(2\sqrt{p})$ of $y^2 = x^3 + x + 1$ for $p < 1000$, all inside the Hasse band. Right: Thorin density at level $y = 0.1$ of the genuine factor at $p = 101$ and of a fake factor beyond the Hasse bound (computed).

The comparison locates the gap exactly. A curve has finitely many Frobenius spins, each of exponential type, so for the Frobenius spins that are laws (a spin with $a > 0$ is a signed measure) the Lee and Yang theory of Theorem 21.9.1 applies, and a finite computation decides each curve; the positivity that makes the spins ferromagnetic comes from intersection theory. The Pólya spin is one spin of infinite type, the Maxwell decomposition of Theorem 19.1.14 of Volume II into infinitely many ferromagnetic spins is RH itself, and no source of positivity outside ζ is known. A proof along this route needs a positive form, built from the primes, whose diagonal values are the Weil functional; that is the Hilbert and Pólya problem in the language of ferromagnets.

21.12 Is this a path to RH?

No. The four forms of RH in this chapter, the single width Cauchy mixture, the ferromagnetism of the Pólya law, Weil's functional and the Lee and Yang property of the two spin system at every coupling, are each proved at the level one or the coupling $\frac{1}{2}$, and each step below is a zero free half plane (Corollary 21.5.2). The descent in coupling is the descent in level of Section 1.4, and the three constraints apply. A proof must be infinite, since finite Euler products sit at $w = \frac{1}{2}$ (Proposition 21.5.3) and finite computations cannot exclude zeros above them

(Theorem 1.4.1 of Volume II). It must use the Euler product, since the Davenport and Heilbronn function has the same Cauchy structure. And it must not rest on a property preserved by Gaussian damping (Corollary 21.2.5). In statistical mechanical terms the Lee and Yang property is preserved when ferromagnetic interactions are added and RH asks for one to be removed, the coupling to the Ising spin; no general theorem goes in that direction, and $\Lambda \geq 0$ says that the self interaction of the de Bruijn Newman flow cannot be removed either. What the chapter adds is the exact form of the target and the fact that the natural one variable test is strictly weaker than it.

21.13 Open problems

Problem 21.1. For the one variable tilt, is the set of w for which $\cosh(wx)\,P(dx)$ is ferromagnetic always an interval? Theorems 21.7.1 and 21.8.1 treat clusters that are small compared with their distance from the other zeros. Extend the gap law to clusters whose width is comparable to the spacing of the neighbouring zeros, where the fixed point of Remark 21.7.2 is still accurate (Table 21.2) but no theorem applies. Is there a constant $\kappa > 0$ such that ferromagnetism of $\cosh(wx)\,P(dx)$ for every $w > w_0$ implies that no zero has $|\operatorname{Re} r| > w_0/\kappa$? Proposition 21.6.1 gives $\kappa \leq \cos\frac{\pi}{4}$: its law is ferromagnetic for every $w > w_0$ exactly when $w_0 \geq w^*$ and has a zero with $|\operatorname{Re} r| = a$, so $\kappa \leq w^*/a$, which tends to $\cos\frac{\pi}{4}$ as $q \uparrow \frac{1}{4}$.

Problem 21.2. Find a coupled system of Lee and Yang spins built from the Euler product, not from ξ, with coupling below $\frac{1}{2}$ and a partition function that contains $L(h_1 \pm w)$; by Corollary 21.5.2 its Lee and Yang property would be a zero free half plane.

Problem 21.3. Theorem 21.9.2 shows that ferromagnetic couplings never widen the strip. Do they shrink it strictly, by at least an amount comparable to the coupling, as the computations of Section 21.9 suggest? A quantitative shrinking theorem for a coupling that can be removed again would be new information about a single spin.

Chapter 22

Pólya and the zeros of entire functions

This chapter reads Pólya's work through the reciprocal Xi law $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$, its Thorin measure, the basepoint laws $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ and the $2\sin^2$ lemma. The statement that carries the chapter is Proposition 22.6.3: the atoms of the even part laws produced by Pólya's universal factor $\cosh(au)$ sit exactly where the argument clock of the basepoint law crosses half integers.

The consequences of a quasi Riemann hypothesis at 7/8, conditional on a half plane that has not been refereed, are in Volume II. A number marked as computed is a floating point value at the precision stated, and a certificate that says so is a floating point certificate with explicit analytic bounds, not an interval arithmetic proof.

22.1 Pólya's question

Pólya spent a decade, from 1918 to 1927, on a single question: when does the Fourier transform of a positive even kernel have only real zeros? The motivation was Riemann's formula

$$\begin{aligned} \Xi(t) &= \xi\big(\tfrac{1}{2}+it\big) = 2\int_0^\infty \Phi(u)\cos(tu)\,du, \\ \Phi(u) &= 2\sum_{n\geq 1}\Big(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\Big)\,e^{-\pi n^2 e^{2u}}, \end{aligned} \tag{22.1.1}$$

in which the Riemann Hypothesis is the statement that Ξ has only real zeros. The Thorin programme of this book reads the same function from the other side, through the reciprocal $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$, which is unconditionally the Laplace transform of a self decomposable law μ on $[0,\infty)$ (Theorem 8.1.1), and is the Laplace transform of a generalised gamma convolution (GGC) exactly when RH holds (Proposition 2.1.1). Pólya works on the kernel Φ, multiplying it by factors that preserve real zeros; the Thorin programme works on the logarithm of the transform, asking for positivity of a measure. We ask which of Pólya's results

are not yet in the Thorin programme, which would help to show RH, and whether the $2\sin^2$ lemma bears on them.

Pólya's papers on zeros are collected in Volume II of his Collected Papers, edited by Boas [309]. We did not have the full text of every paper discussed below. The Acta paper of 1926 [302] was read in part; what is said of it comes from its first page and from the accounts of Newman and Wu [244], Lagarias [199], Gasper [91] and Haglund [104], and every formula used below for Pólya's Ξ^* was rederived and checked numerically. Whether a result is already part of the programme is judged against the earlier chapters, [279] (version 24 of arXiv 1708.02653) and [282].

22.2 Notation and the Thorin dictionary

We use the Titchmarsh normalisation (22.1.1). Since Φ is even and positive,

$$\xi\big(\tfrac12+w\big)=2\int_0^\infty \Phi(u)\cosh(wu)\,du=\sum_{k\ge0}c_k w^{2k},\qquad c_k=\frac{2}{(2k)!}\int_0^\infty \Phi(u)u^{2k}\,du>0. \tag{22.2.1}$$

Put $F(s)=\xi(1/2+\sqrt{s})/\xi(1/2)=\sum_{k\ge0}a_k s^k$ with $a_k=c_k/c_0>0$. Then F is entire of order one half with $F(s)=\prod_\rho(1-s/s_\rho)$, $s_\rho=(\rho-1/2)^2$, the product taken over zeros in the upper half plane (one factor for each pair $\rho,1-\rho$), and $\varphi=1/F$ is the Laplace transform of the reciprocal Xi law μ; we write $Y\sim\mu$. Its exponent $\psi=\log F$ satisfies $\psi'(s)=\int_0^\infty e^{-sx}W(x)\,dx$ with the heat trace $W(x)=\sum_{\gamma>0}e^{(\rho-1/2)^2x}$, and the dictionary of Section 8.1 reads: $W\ge0$ (infinite divisibility, unconditional), $-W'\ge0$ (self decomposability, unconditional), W completely monotone (GGC, equivalent to RH). Under RH the Thorin measure is $U=\sum_{\gamma>0}\delta_{\gamma^2}$ and $Y\overset{d}{=}\sum_\gamma E_\gamma/\gamma^2$.

At a basepoint α the law $\mu^{(\alpha)}$ has Laplace transform $\xi(\alpha)/\xi(\alpha+\sqrt{s})$; it is a GGC if and only if $\alpha\ge\sigma_{\max}=\sup\operatorname{Re}\rho$, unconditionally for $\alpha\ge1$ (Theorem 2.1.2), and its Thorin distribution in the ordinate variable is the clock

$$U_\alpha(\tau)=\frac{1}{\pi}\,\Delta_0^\tau\arg\xi(\alpha+i\tau),\qquad \frac{d}{d\tau}U_\alpha(\tau)=\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau). \tag{22.2.2}$$

With τ_α the inverse of the clock and g a standard gamma subordinator, $\mu^{(\alpha)}$ is the law of $\int_0^\infty dg_v/\tau_\alpha(v)^2$.

The $2\sin^2$ lemma (Lemma 10.2.4) rests on the identity, for $x,w>0$ and

$u = w^2/2$,

$$e^{-wx} + wx - 1 = \int_0^\infty \log\Big(1 + \frac{u}{z}\Big)\, \varpi_x(z)\, dz,$$
$$\varpi_x(z) = \frac{x}{\pi\sqrt{2z}}\, 2\sin^2\Big(x\sqrt{z/2}\Big) \geq 0. \tag{22.2.3}$$

So if $\log f(\alpha + w)/f(\alpha) - wf'(\alpha)/f(\alpha) = \int (e^{-wx} - 1 + wx)e^{-\alpha x}\mu_f(dx)/x$ with $\mu_f \geq 0$, the compensated reciprocal $e^{wf'(\alpha)/f(\alpha)} f(\alpha)/f(\alpha + w)$, $w = \sqrt{2u}$, is a GGC in u with Thorin density $\int \varpi_x(z)e^{-\alpha x}\mu_f(dx)/x$. Differentiating twice in w, the hypothesis $\mu_f \geq 0$ is the statement that $(\log f)''$ is completely monotone on (α, ∞).

Notation. The function F is the function G of Section 2.1, and $Y \sim \mu$ is the clock variable T, so that $\varphi = 1/F$ is its Laplace transform. The coefficients a_k are the scaled moments b_k of Section 20.2.4 (Table 20.4), not the exponents a_j of Section 8.1. The word clock in this chapter means the argument clock U_α, the cumulative Thorin mass $M_\varepsilon(\theta) = \pi^{-1} \arg \xi(\frac{1}{2} + \varepsilon + i\theta)$ of Section 3.3 with $\varepsilon = \alpha - \frac{1}{2}$ and $\theta = \tau$, and not the Laplace transform $1/G$ of the earlier chapters. We write $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ for the drift of Section 12.1. Finally $X = Z\sqrt{2Y}$ is the variable $Y_{1/2}$ of Section 13.1, with density f_X, and not the Pólya variable X of Section 2.1, whose density is $\Phi/\Xi(0)$.

22.3 Pólya's papers on zeros, one by one

P1. Approximation by polynomials with real zeros (1913)

Über Annäherung durch Polynome mit lauter reellen Wurzeln, Rend. Circ. Mat. Palermo 36 [295]. Completing Laguerre, Pólya characterised the entire functions that are locally uniform limits of real polynomials with only real zeros: they are exactly

$$f(z) = cz^m e^{-\lambda z^2 + \kappa z} \prod_k \left(1 - \frac{z}{b_k}\right) e^{z/b_k},$$
$$m \in \{0, 1, 2, \dots\},\ \lambda \geq 0,\ c, \kappa \in \mathbb{R},\ b_k \in \mathbb{R} \setminus \{0\},\ \sum_k b_k^{-2} < \infty, \tag{22.3.1}$$

the Laguerre Pólya class $\mathcal{LP}$. The functions of type I, $cz^m e^{\sigma z} \prod_k (1 + z/b_k)$ with $c \geq 0$, $\sigma \geq 0$, $b_k > 0$ and $\sum_k b_k^{-1} < \infty$, are exactly the locally uniform limits of polynomials with only negative zeros and positive coefficients. RH is the statement $\Xi \in \mathcal{LP}$.

P2. Pólya and Schur, two kinds of factor sequences (1914)

Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen, J. reine angew. Math. 144 [310]. A sequence (g_k) is a multiplier sequence if $\sum p_k x^k$ with only real zeros implies $\sum g_k p_k x^k$ with only real zeros. Pólya and Schur proved that a positive sequence is a multiplier sequence if and only if $\sum g_k z^k/k!$ is in $\mathcal{LP}$ of type I, if and only if every Jensen polynomial $\sum_{k=0}^{d} \binom{d}{k} g_k x^k$ has only real zeros.

P3. Genus zero and one (1915)

Algebraische Untersuchungen über ganze Funktionen vom Geschlechte Null und Eins, J. reine angew. Math. 145 [296]. Operations on entire functions of genus at most one that preserve reality of zeros. The statement used later is the shift lemma, quoted as Hilfssatz II of the 1926 paper P8 in Section 11.1.3: if G is real, of genus 0 or 1, with only real zeros, at least one of them, then $G(z+ia)+G(z-ia)$ has only real zeros for real a.

P4. Zeros of certain entire functions (1918)

Über die Nullstellen gewisser ganzer Funktionen, Math. Z. 2 [297]. The continuous Eneström Kakeya theorem: if f is positive and nondecreasing on $(0,1)$, then $\int_0^1 f(t)\cos(zt)\,dt$ and $\int_0^1 f(t)\sin(zt)\,dt$ have only real zeros. The Riemann kernel is positive but decreasing on $(0,\infty)$, so the theorem does not apply to it.

P5. Number theoretic remarks (1919)

Verschiedene Bemerkungen zur Zahlentheorie, Jahresber. DMV 28 [298]. Among other things, the Pólya conjecture that the summatory Liouville function $L(x) = \sum_{n\le x} \lambda(n)$ satisfies $L(x) \le 0$ for $x \ge 2$. It implies RH and is false: Haselgrove (1958) [110] disproved it and Tanaka (1980) [342] found the first sign change at $x = 906{,}150{,}257$.

P6. Zeros of successive derivatives (1922)

Über die Nullstellen sukzessiver Derivierten, Math. Z. 12 [299]. Pólya introduced the final set of a function, the set of limit points of the zeros of $f^{(n)}$ as $n \to \infty$, and determined it in several classes; the statements we use are recorded under P13.

P7. A Fourier integral (1923)

On the zeros of an integral function represented by Fourier's integral, Messenger of Math. 52 [301]. A short note on the zeros of cosine transforms of positive kernels, a precursor of the 1926 and 1927 papers.

P8. The integral representation of the Riemann xi function (1926)

Bemerkung über die Integraldarstellung der Riemannschen ξ Funktion, Acta Math. 48 [302]. As $u \to +\infty$ the kernel is dominated by its $n = 1$ term, and Pólya replaced it by the even function

$$\Phi^*(u) = 4\pi^2 \left(e^{9u/2} + e^{-9u/2}\right) e^{-\pi(e^{2u}+e^{-2u})} = 8\pi^2 \cosh\left(\tfrac{9u}{2}\right) e^{-2\pi \cosh 2u}, \quad (22.3.2)$$

and proved that $\Xi^*(t) = 2\int_0^\infty \Phi^*(u)\cos(tu)\,du$ has only real zeros, with a zero count asymptotic to that of Ξ. The proof combines the reality of the zeros of $z \mapsto K_{iz}(a)$, $a > 0$, with the shift lemma P3. A refinement keeping the second exponential term is also real rooted.

P9. Zeros of certain trigonometric integrals (1926)

J. London Math. Soc. 1 [303]. Following Hardy's proof of infinitely many zeros on the critical line, Pólya explained why infinitely many real zeros is much easier than no complex zeros, and gave a general criterion for infinitely many real zeros. In Thorin language the infinitude of real zeros is the statement that the on line part $\varphi_{\text{on}} = \prod_{\text{on line}}(1 + s/\gamma^2)^{-1}$, a GGC whatever the zero set, has a Thorin measure with infinitely many atoms.

P10. Trigonometric integrals with only real zeros (1927)

Über trigonometrische Integrale mit nur reellen Nullstellen, J. reine angew. Math. 158 [304]. The central paper. A real analytic ϕ is a universal factor if $\int \mathcal{F}(t)e^{izt}dt$ with only real zeros implies the same for $\int \phi(t)\mathcal{F}(t)e^{izt}dt$, for every admissible kernel $\mathcal{F}$, that is, every real kernel that decays fast enough for these integrals to define entire functions of z. Pólya proved that ϕ is a universal factor if and only if $z \mapsto \phi(iz)$ belongs to $\mathcal{LP}$. Hence $e^{\lambda t^2}$ ($\lambda > 0$) and $\cosh(bt)$ ($b \in \mathbb{R}$) are universal factors. He also proved a Mellin criterion: if $\mathcal{F}$ is admissible, holomorphic near the origin, and $H(z) = \int_0^\infty \mathcal{F}(t)t^{z-1}dt$ has no zeros in $\mathbb{C} \setminus (-\infty, 0]$, then $\int \mathcal{F}(t^{2q})e^{izt}dt$ has only real zeros for every positive integer q. Consequences: $\int e^{-t^{2q}}e^{izt}dt$ and $\int \cosh(bt)e^{-a\cosh t}e^{izt}dt$ with $a > 0$ and $b \in \mathbb{R}$ have only real zeros. The factor $e^{\lambda t^2}$ is the starting point of de Bruijn (1950) and Newman (1976), hence of the de Bruijn Newman constant Λ (Table 4.8).

P11. Jensen's investigations (1927)

Über die algebraisch funktionentheoretischen Untersuchungen von J. L. W. V. Jensen, Kgl. Danske Vid. Selsk. Math. Fys. Medd. 7 [305]. Pólya edited and proved Jensen's criteria: a real entire function of genus at most one is in $\mathcal{LP}$ if and only if all its Jensen polynomials have only real zeros, together with Jensen's

integral criteria. This is the source of the Jensen polynomial approach to RH (Section 20.4.1); Farmer (2022) [84] argued that this direction cannot reach RH.

P12. Fourier's work on transcendental equations (1930)

Quart. J. Math. Oxford 1 [307]. The Fourier Pólya conjectures on the number of nonreal zeros of successive derivatives.

P13. Derivatives and analytic character (1943)

Bull. AMS 49 [308]. The Pólya Wiman conjecture: if f is real entire of order below two with finitely many nonreal zeros, then $f^{(n)}$ has only real zeros for all large n; proved by Craven, Csordas and Smith (1987) [68] and Kim (1990) [190], with the Fourier Pólya conjecture settled by Ki and Kim (2000) [188]. For Ξ, whose zeros lie in a strip, the theorem of Ki and Kim gives the local statement that for every R the zeros of $\Xi^{(n)}$ in $|z| < R$ are real for all large n; Conrey showed that the proportion of real zeros of $\Xi^{(n)}$ tends to one [60]. Global real rootedness of $\Xi^{(n)}$ for some n is not known.

P14. The Hilbert Pólya suggestion

In a 1982 letter to Odlyzko, Pólya recalled suggesting to Landau, around 1914, that the zeros might be the eigenvalues of a self adjoint operator. The Krein string route of Corollary 12.2.3, which assumes RH and simple zeros, is a probabilistic form of this.

22.4 The Laguerre Pólya class in the Thorin variable

Proposition 22.4.1. *The following are equivalent.*

(i) RH.

(ii) $F(s) = \xi(1/2 + \sqrt{s})/\xi(1/2)$ *belongs to the Laguerre Pólya class of type I, that is, it is a locally uniform limit of polynomials with only negative zeros.*

(iii) The sequence $g_k = k!\, a_k$*, where* $F(s) = \sum a_k s^k$*, is a Pólya Schur multiplier sequence.*

(iv) For every d *the Jensen polynomial* $J_d(x) = \sum_{k=0}^{d} \binom{d}{k} g_k x^k$ *has only real, negative zeros.*

(v) The reciprocal Xi law is a GGC.

Proof. Under RH every $s_\rho = -\gamma^2$ is negative and $F(s) = \prod_\gamma (1 + s/\gamma^2)$ with $\sum \gamma^{-2} < \infty$, which is of type I; conversely a function of type I has only real

nonpositive zeros, so every $s_\rho \le 0$, and $s_\rho = (\rho - 1/2)^2 \le 0$ forces $\rho - 1/2 \in i\mathbb{R}$. The equivalence of (ii), (iii) and (iv) is the theorem of Pólya and Schur, since $\sum g_k s^k/k! = F(s)$. The equivalence with (v) is Proposition 2.1.1. □

The ladder (iv) is the coefficient counterpart of the monotonicity ladder $T_k = (-1)^k W^{(k)} \ge 0$ of Theorem 8.1.6. Its first rung, for F and for every derivative of F, is the Turán inequality $g_k^2 \ge g_{k-1} g_{k+1}$ for $k \ge 1$, the condition that the Jensen polynomials $J_{2,n}$, $n \ge 0$, are real rooted; for F alone it is the case $k = 1$. With $m_k = (2k)!\, c_k = 2\int_0^\infty \Phi(u) u^{2k}\, du$ it reads $m_k^2 \ge \frac{2k-1}{2k+1} m_{k-1} m_{k+1}$, which is the Turán inequality proved unconditionally by Csordas, Norfolk and Varga (1986) [69]; in the scaled moments a_k, which are the b_j of Section 20.2.4, it is (20.2.2), whose margins are in Table 20.4.

Numerics. Computing the c_k from (22.2.1) by quadrature on the kernel reproduces $\Xi(0) = \xi(1/2) = 0.497120778188314$ to fifteen digits and gives $a_1 = 0.0231049931$ and the values a_2, a_3, a_4 of Table 20.4. The Turán ratios $g_k^2/(g_{k-1}g_{k+1})$ decrease slowly from 1.0748 at $k = 1$ and stay above 1.0109 for all $k \le 40$. The Jensen polynomials J_d of every degree $2 \le d \le 30$, computed with 120 digits, have only real negative zeros.

Remark 22.4.2. At fixed degree d the ladder (iv) is a finite set of polynomial inequalities in $a_1, \dots, a_d$, checkable to any degree; but for each fixed d it is implied by information about finitely many zeros, so it cannot prove RH by itself.

22.5 Pólya frequency functions: the two sided criterion

Schoenberg (1951) [328] proved that an integrable function f on $\mathbb{R}$ is a Pólya frequency function of every order (PF_∞) if and only if its bilateral Laplace transform is $1/\Psi(s)$ on a strip around the origin, with Ψ a function of the form (22.3.1) with $m = 0$ and $\Psi(0) > 0$ that is not of the form $Ce^{\kappa s}$. The theorem is the two sided counterpart of the Thorin theory.

Proposition 22.5.1. *Let $Y \sim \mu$ be the reciprocal Xi law and Z a standard normal variable independent of Y. Unconditionally, $X = Z\sqrt{2Y}$ has a smooth symmetric density f_X with characteristic function and bilateral Laplace transform*

$$\mathbb{E}e^{itX} = \frac{\xi(1/2)}{\xi(1/2 + t)}, \qquad \mathbb{E}e^{zX} = \frac{\Xi(0)}{\Xi(z)} \quad (|\operatorname{Re} z| < \gamma_1). \tag{22.5.1}$$

Moreover RH holds if and only if f_X is PF_∞.

This is the equivalence of Section 2.1, between RH and the law of $Z\sqrt{2Y}$ as a sum of Laplace variables, read through Schoenberg's theorem; the same equivalence is stated for the Laguerre Pólya class in [192, Theorem 2.3 and Corollary 2.4], and its forward half is item (iv) of Theorem 20.1.4.

Proof. $\mathbb{E}e^{itX} = \mathbb{E}e^{-t^2Y} = \varphi(t^2) = \xi(1/2)/\xi(1/2+t)$, which is integrable in t since $\xi(1/2+t)$ grows like $\exp(\frac{t}{2}\log t)$, so f_X exists and is smooth. For real z, $\mathbb{E}e^{zX} = \mathbb{E}e^{z^2Y} = \varphi(-z^2)$, and $\varphi(-z^2) = \xi(1/2)/\xi(1/2+iz) = \Xi(0)/\Xi(z)$ by evenness; the Laplace transform of the positive measure μ converges up to its first singularity on the negative real axis (Landau's theorem), which is at $-\gamma_1^2$, because a zero off the line gives a nonreal s_ρ and the zero of lowest ordinate is on the line. Under RH, $\Xi(z)/\Xi(0) = \prod_\gamma(1 - z^2/\gamma^2) = \prod_\gamma(1 - z/\gamma)e^{z/\gamma}(1 + z/\gamma)e^{-z/\gamma}$ is of Schoenberg's form, so f_X is PF_∞. Conversely if f_X is PF_∞ then $\Xi(z)/\Xi(0)$ agrees on a strip with an entire function having only real zeros, hence everywhere, and RH follows. □

The Thorin atoms γ^2 become Schoenberg's $\pm 1/\gamma$. Under RH, $X \overset{d}{=} \sum_\gamma L_\gamma/\gamma$ with independent standard Laplace variables L_γ, because a Laplace variable is $Z\sqrt{2E}$. The PF ladder $\mathrm{PF}_2 \supset \mathrm{PF}_3 \supset \cdots \supset \mathrm{PF}_\infty$ is a third ladder, after T_k and the Jensen degrees; PF_2 is log concavity of f_X.

Numerics. Fourier inversion of (22.5.1) gives f_X. On $[0, 1.6]$ with step 0.02 every second difference of $\log f_X$ is negative (the largest is -2.6×10^{-5}), so PF_2 holds on the grid; all 3000 random pairs of ordered triples $x_1 < x_2 < x_3$, $y_1 < y_2 < y_3$ in $[-0.8, 0.8]$ have a positive normalised determinant $\det[f_X(x_i - y_j)]_{i,j\le 3} / \prod_i f_X(x_i - y_i)$, the smallest being 6.7×10^{-9}.

Problem 22.1. Prove unconditionally, in exact or interval arithmetic, that f_X is log concave, that is PF_2 (Theorem 22.11.5 gives a floating point certificate), and then that it is PF_n for $n \ge 3$ (Section 22.11). The case of PF_2 would be a two sided analogue of the unconditional self decomposability of μ, one rung up the Schoenberg ladder.

22.6 The universal factor $\cosh(au)$ and the clock

Multiplying the kernel by e^{au} is an Esscher tilt of the Pólya density and moves the basepoint: $\int_{-\infty}^{\infty} \Phi(u)e^{au}\cosh(wu)\,du = \frac{1}{2}\left(\xi(\frac{1}{2}+a+w) + \xi(\frac{1}{2}+a-w)\right)$. The tilt destroys evenness; Pólya's universal factor $\cosh(au)$ is the even average of the two tilts. Define, for $a \ge 0$ and $\alpha = 1/2 + a$,

$$A_\alpha(w) = 2\int_0^\infty \Phi(u)\cosh(au)\cosh(wu)\,du = \tfrac{1}{2}\left(\xi(\alpha+w) + \xi(\alpha-w)\right), \quad (22.6.1)$$

the even part in w of the basepoint function $w \mapsto \xi(\alpha + w)$. For real t, $A_\alpha(it) = \mathrm{Re}\,\xi(\alpha + it)$, and in the variable $s = 1/2 + w$ this is Lagarias's function $A_h(s) = \frac{1}{2}(\xi(s+h) + \xi(s-h))$ with $h = a$.

Theorem 22.6.1 (Pólya, de Bruijn, Lagarias). *(i) Under RH, A_α has only purely imaginary zeros for every $\alpha > 1/2$.*

(ii) If every zero of Ξ *satisfies* $|\operatorname{Im} t| \le \Delta$*, then every zero of* $t \mapsto A_\alpha(it)$ *satisfies* $|\operatorname{Im} t| \le \max(\Delta^2 - a^2, 0)^{1/2}$*. In particular, unconditionally,* A_α *has only purely imaginary zeros for* $\alpha \ge 1$*, and they are simple.*

(iii) If A_{α_n} *has only imaginary zeros along a sequence* $\alpha_n \downarrow 1/2$*, then RH holds.*

Proof. (i) $\cosh(au)$ is a universal factor since $\cos(az) \in \mathcal{LP}$. (ii) is de Bruijn's strip theorem for the factor cosh, and $\Delta \le 1/2$ because ζ has no zeros with $\operatorname{Re} s > 1$; simplicity is Lagarias's theorem [199]. (iii) $A_\alpha \to \xi(1/2 + w)$ locally uniformly as $\alpha \downarrow 1/2$, and Hurwitz's theorem applies. □

Corollary 22.6.2 (Even part basepoint laws). *Whenever* A_α *has only imaginary zeros* $\pm i\tau_k$*, the function* $\varphi_\alpha^P(s) = \xi(\alpha)/A_\alpha(\sqrt{s})$ *is the Laplace transform of a GGC law* μ_α^P *with Thorin measure* $\sum_k \delta_{\tau_k^2}$*, so* μ_α^P *is the law of* $\sum_k E_k/\tau_k^2$*. This holds unconditionally for* $\alpha \ge 1$*, for all* $\alpha > 1/2$ *under RH, and RH follows if it holds along a sequence* $\alpha_n \downarrow 1/2$*.*

Proof. $A_\alpha(\sqrt{s})$ is entire of order one half in s with positive Taylor coefficients by (22.6.1), value $\xi(\alpha) > 0$ at $s = 0$, and zeros $s = -\tau_k^2$; the genus zero product gives $\varphi_\alpha^P(s) = \prod_k (1 + s/\tau_k^2)^{-1}$. □

Proposition 22.6.3 (Half integer crossings). *Let* α *be such that* ξ *has no zeros in* $\operatorname{Re} s \ge \alpha$ *(this holds for* $\alpha > \sigma_{\max}$*, and unconditionally for every* $\alpha \ge 1$*, since* ζ *has no zeros on* $\operatorname{Re} s = 1$*). Then the clock* U_α *is continuous and strictly increasing from* $U_\alpha(0) = 0$*, and the positive zeros of* $t \mapsto \operatorname{Re} \xi(\alpha + it)$ *are exactly*

$$\tau_k = \tau_\alpha\big(k - \tfrac{1}{2}\big), \qquad k = 1, 2, \dots, \tag{22.6.2}$$

where τ_α *is the inverse of the clock. Consequently the Thorin distribution of the even part law is the rounding of the Thorin distribution of the basepoint law,*

$$U_\alpha^P\big((0, T^2]\big) = \big\lfloor U_\alpha(T) + \tfrac{1}{2} \big\rfloor, \tag{22.6.3}$$

and in law $\mu_\alpha^P = \mathcal{L}\big(\sum_k E_k/\tau_\alpha(k - \frac{1}{2})^2\big)$ *while* $\mu^{(\alpha)} = \mathcal{L}\big(\int_0^\infty dg_v/\tau_\alpha(v)^2\big)$*: the even part law is the midpoint rule for the gamma clock integral.*

Proof. Since no zero has $\beta_\rho \ge \alpha$, the Hadamard sum $\operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau) = \sum_\rho \frac{\alpha - \beta_\rho}{|\alpha + i\tau - \rho|^2}$ has every term positive, so the clock is strictly increasing, and $\xi(\alpha + i\tau) \ne 0$. Write $\xi(\alpha + i\tau) = |\xi(\alpha + i\tau)| e^{i\pi U_\alpha(\tau)}$, which is legitimate since $\xi(\alpha) > 0$. Then $\operatorname{Re} \xi(\alpha + i\tau) = |\xi(\alpha + i\tau)| \cos(\pi U_\alpha(\tau))$ vanishes exactly when $U_\alpha(\tau) \in \mathbb{Z} + \frac{1}{2}$, and a strictly increasing clock starting at 0 and tending to infinity (by Stirling, $U_\alpha(\tau) \sim \frac{\tau}{2\pi} \log \tau$) crosses each level $k - \frac{1}{2}$ exactly once, transversally. This gives (22.6.2) and simple zeros. Every zero ρ has $|\beta_\rho - \frac{1}{2}| < \alpha - \frac{1}{2}$, by the hypothesis and the functional equation, so Theorem 22.6.1(ii) with $\Delta = \alpha - \frac{1}{2}$ shows that A_α has no zeros off the imaginary axis, and the corollary on even part laws applies;

counting levels below $U_\alpha(T)$ then gives (22.6.3). The two laws follow from the gamma process representation: the increments of g over $[k-1,k]$ are independent standard exponentials, and replacing $\tau_\alpha(v)$ on $[k-1,k]$ by its midpoint value gives the sum. □

Under the hypothesis of Proposition 22.6.3, which holds unconditionally for $h \geq \frac{1}{2}$ and under RH for every $h > 0$, the proposition explains why the zeros of A_h are asymptotically equally spaced, as in Lagarias's theorem [199]: they are level crossings of a smooth increasing clock whose derivative $\pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + i\tau)$ is a Cauchy smoothing of the zero density, with fluctuations of lower order than its mean $\frac{1}{2\pi} \log \frac{\tau}{2\pi}$.

Numerics. Table 22.1 lists, for four basepoints, the number of zeros of $\operatorname{Re} \xi(\alpha+it)$ on $(0, 100]$, the clock at 100, its rounding, and the largest distance of the clock from a half integer at the computed zeros (on a grid of step 0.01 with linear interpolation). As $\alpha \downarrow 1/2$ the atoms converge to the ordinates.

α	zeros on $(0, 100]$	$U_\alpha(100)$	$\lfloor U_\alpha(100) + \frac{1}{2} \rfloor$	max distance to $\mathbb{Z} + \frac{1}{2}$
1.00	29	29.1112	29	2.7×10^{-5}
0.75	29	29.0548	29	5.6×10^{-5}
0.60	29	29.0218	29	1.7×10^{-4}
0.52	29	29.0044	29	1.6×10^{-3}

Table 22.1: Atoms of the even part law against the rounded clock of the basepoint law.

Remark 22.6.4 (Is the Pólya family easier?). Real rootedness of A_α is a counting condition: it can be certified to a given height by sign changes of $\operatorname{Re} \xi(\alpha + it)$ together with an argument principle count, exactly as in Turing's method. So the Pólya family is not easier in principle, but converts the analytic positivity of $\operatorname{Re} \xi'/\xi$ into a discrete, verifiable statement about atoms. The zero trajectories $\alpha \mapsto \tau_k(\alpha)$ satisfy $d\tau_k/d\alpha = \operatorname{Re} \xi'(s)/\operatorname{Im} \xi'(s)$ at $s = \alpha + i\tau_k$. Under RH no two trajectories collide for $\alpha > \frac{1}{2}$, because the zeros are simple (Proposition 22.6.3); conversely a nonimaginary zero of A_α can appear only through a collision of two imaginary zeros, provided no zeros enter from infinity as α decreases, which needs uniform control of the zero counts in α. In this heuristic sense RH is the statement that no two trajectories collide before $\alpha = \frac{1}{2}$, a discrete analogue of the de Bruijn Newman dynamics.

22.7 Pólya's Ξ* as the archimedean model

Proposition 22.7.1. *With* (22.3.2),

$$\Xi^*(t) = 4\pi^2\left[K_{\frac{9}{4}+\frac{it}{2}}(2\pi) + K_{\frac{9}{4}-\frac{it}{2}}(2\pi)\right] = 8\pi^2 \operatorname{Re} K_{\frac{9}{4}+\frac{it}{2}}(2\pi). \tag{22.7.1}$$

Proof. Substitute $v = 2u$ in $2\int_0^\infty \Phi^*(u)\cos(tu)\,du$, write $\cosh(9v/4)\cos(tv/2)$ as the average of $\cos((t/2 \mp 9i/4)v)$, and use $\int_0^\infty e^{-a\cosh v}\cos(\omega v)\,dv = K_{i\omega}(a)$ and the evenness of K_ν in ν. □

This is the identity of Section 18.1, where $P(\theta) = K_{i\theta/2}(2\pi)$ and $\Xi^*(\theta) = 4\pi^2[P(\theta - \frac{9i}{2}) + P(\theta + \frac{9i}{2})]$.

So Ξ^* is exactly the construction (22.6.1) applied to the Bessel function: it is the even part, at basepoint $9/4$ in the order variable, of $\nu \mapsto K_\nu(2\pi)$. Pólya's theorem that $K_{iz}(a)$ has only real zeros is the Riemann Hypothesis for this function: its zeros in ν lie on $\operatorname{Re}\nu = 0$. The basepoint $9/4$ lies strictly to the right of that line, so by the argument of Theorem 22.6.1 and Proposition 22.6.3 the even part has only real zeros (the argument needs $\operatorname{Re}\frac{d}{d\nu}\log K_\nu(2\pi) > 0$ on $\operatorname{Re}\nu > 0$, and this holds because $\nu \mapsto K_\nu(2\pi)$ is even, of order one, with only the zeros $\pm iz_n$, $z_n > 0$, so that $K_\nu(2\pi) = K_0(2\pi)\prod_n(1+\nu^2/z_n^2)$ and each term $\frac{d}{d\nu}\log(1+\nu^2/z_n^2) = (\nu+iz_n)^{-1} + (\nu - iz_n)^{-1}$ has positive real part there), located where the Bessel clock $\pi^{-1}\arg K_{9/4+it/2}(2\pi)$ crosses half integers.

Proposition 22.7.2 (Half Gram points). *Let t_k^* be the positive zeros of Ξ^* in increasing order and ϑ the Riemann Siegel theta function. Then*

$$\frac{\vartheta(t_k^*)}{\pi} = k - \frac{3}{2} - \frac{2\pi^2 - 3}{\pi\, t_k^*} + O\big((t_k^*)^{-2}\big). \tag{22.7.2}$$

Hence, with $N^*(T) = \#\{k : t_k^* \le T\}$, $N^*(T) = \lfloor \vartheta(T)/\pi + \frac{3}{2}\rfloor$ for all large T outside the intervals $[t_k^*, t_k^* + O(1/(t_k^*\log t_k^*)))$, on which $N^*(T)$ is one larger; in particular $N^*(T) - \vartheta(T)/\pi$ is bounded, whereas $N(T) = \vartheta(T)/\pi + 1 + S(T)$.

Sketch. For $\nu = \frac{9}{4} + \frac{it}{2}$, the ascending series of $I_{-\nu}$ and the reflection formula give $K_\nu(2\pi) = \frac{1}{2}\Gamma(\nu)\pi^{-\nu}\big(1 + \frac{\pi^2}{1-\nu} + O(t^{-2})\big)$, the companion term $\frac{1}{2}\Gamma(-\nu)\pi^{\nu}$ being smaller by a factor of order $t^{-9/2}$. With $s = \frac{1}{2} + it$ we have $\Gamma(\nu) = (\frac{s}{2}+1)\frac{s}{2}\,\Gamma(\frac{s}{2})$ and $\Gamma(\frac{s}{2})\pi^{-it/2} = |\cdot|e^{i\vartheta(t)}$, so the phase of $K_\nu(2\pi)$ is $\vartheta(t) + \arg s + \arg(s+2) + \operatorname{Im}\frac{\pi^2}{1-\nu} + O(t^{-2})$. Here $\arg s + \arg(s+2) = \pi - \frac{3}{t} + O(t^{-3})$ and $\operatorname{Im}\frac{\pi^2}{1-\nu} = \frac{2\pi^2 t}{t^2+25/4} = \frac{2\pi^2}{t} + O(t^{-3})$, so the phase is $\vartheta(t) + \pi + \frac{2\pi^2-3}{t} + O(t^{-2})$. The zeros of the real part are where this phase lies in $\frac{\pi}{2} + \pi\mathbb{Z}$, which is (22.7.2); the indexing is fixed by the first zero. The integer can also be fixed without computation, by letting the argument 2π of K decrease to 0 as in the proof of Proposition 22.7.3(ii); at basepoint $\frac{9}{4}$ the ratio of $K_\nu(a)$ to $\frac{1}{2}\Gamma(\nu)(a/2)^{-\nu}$ is then $1 + O(a^2)$ uniformly in t, the companion term contributing $O(a^{9/2})$. □

Compare the Thorin form of the zero count, $U((0,t]) = \vartheta(\sqrt{t})/\pi + 1 + S(\sqrt{t})$ under RH. Pólya's Ξ^* replaces the arithmetic fluctuation S by a bounded rounding error: by Proposition 22.7.2 its Thorin distribution $N^*(\sqrt{t})$ is the rounding of the smooth archimedean distribution $\vartheta(\sqrt{t})/\pi + 1$. This is the distribution that the $2\sin^2$ lemma computes for the gamma factor in Theorem 10.2.12, up to an affine function of $\sqrt{t}$: that theorem gives $(\vartheta(\sqrt{t}) - \vartheta'(0)\sqrt{t})/\pi$ for the compensated factor $e^{-c\sqrt{s}}/G(s)$, with $c = -\vartheta'(0)$. In this sense Ξ^* is the zeta function with ζ replaced by its first Dirichlet term and the result symmetrised, and its reciprocal $\Xi^*(0)/\Xi^*(\sqrt{-s})$ is, by Pólya's theorem, an unconditional GGC with Thorin measure $\sum_k \delta_{(t_k^*)^2}$.

Numerics. Ξ^* has 29 zeros in $(0, 100]$, as does Ξ, but they are displaced: the first eight are 8.9928, 19.0654, 24.4748, 28.9750, 33.0069, 36.7367, 40.2491, 43.5947. There are 138 zeros up to height 300; at the last, $t^*_{138} = 299.627$, the deviation from $k - \frac{3}{2}$ times t_k^* is -5.329, against the constant $-(2\pi^2 - 3)/\pi = -5.328$ of (22.7.2) (Figure 22.1, right).

Basepoint $\frac{1}{4}$. At basepoint $\frac{1}{4}$ the zeros of the even part sit at half Gram points with no shift of index, and their displacement from the zeros of Ξ is governed by S [138]. Let

$$\Xi^\circ(t) = \operatorname{Re} K_{\frac{1}{4}+\frac{it}{2}}(2\pi), \qquad \psi(t) = \arg K_{\frac{1}{4}+\frac{it}{2}}(2\pi), \quad \psi(0) = 0,$$

so that Ξ° is, up to a constant, the transform of the law proportional to $\cosh(\frac{x}{2})e^{-2\pi\cosh 2x}$, the member of Theorem 19.1.4 with $a = 2\pi$, $\ell = 2$ and $\mu = 0$ and one field $\lambda_1 = \frac{1}{2}$.

Proposition 22.7.3. *(i) The zeros of Ξ° are real and simple, ψ is strictly increasing on $[0,\infty)$, and the positive zeros $t_1^\circ < t_2^\circ < \cdots$ are the solutions of $\psi(t_n^\circ) = (n - \frac{1}{2})\pi$. So $N^\circ(T) = \#\{n : t_n^\circ \le T\} = \lfloor \psi(T)/\pi + \frac{1}{2} \rfloor$.*

(ii) $\psi(t) = \vartheta(t) + O(t^{-1/2})$. Hence $\vartheta(t_n^\circ)/\pi = n - \frac{1}{2} + O((t_n^\circ)^{-1/2})$, and $t_n^\circ - g_n = O\big((t_n^\circ)^{-1/2}/\log t_n^\circ\big)$ for the half Gram points $\vartheta(g_n) = (n - \frac{1}{2})\pi$.

(iii) Let $0 < \gamma_1 \le \gamma_2 \le \cdots$ be the ordinates of all zeros of ξ in the upper half plane, with multiplicity and whatever their real parts. Suppose that exactly one zero of ξ, counted with multiplicity, has ordinate γ_{n+1}, and let $S(\gamma_{n+1})$ be the mean of the one sided limits of S there. Then $\vartheta(\gamma_{n+1}) = (n - \frac{1}{2})\pi - \pi S(\gamma_{n+1})$, and

$$t_n^\circ - \gamma_{n+1} = \frac{\pi S(\gamma_{n+1})}{\vartheta'(\gamma_{n+1})} + O\Big(\frac{\gamma_{n+1}^{-1/2}}{\log \gamma_{n+1}}\Big)$$

with an absolute implied constant.

Sketch. (i) The zeros are real by Theorem 19.1.4. As in the paragraph before Proposition 22.7.2, $K_\nu(2\pi) = K_0(2\pi)\prod_n(1+\nu^2/z_n^2)$, and at $\nu = \frac14 + \frac{it}{2}$ the factor of index n contributes

$$\frac{1/8}{\frac{1}{16}+(\frac{t}{2}+z_n)^2} + \frac{1/8}{\frac{1}{16}+(\frac{t}{2}-z_n)^2} > 0$$

to $\psi'(t)$. Since $\Xi^\circ(t) = |K_{\frac14+\frac{it}{2}}(2\pi)|\cos\psi(t)$ and the modulus does not vanish for real t, the zeros on $(0,\infty)$ are where ψ crosses $(\frac12+\mathbb{Z})\pi$, each once and simply. (ii) For $\nu = \frac14 + \frac{it}{2}$ the ascending series and the reflection formula give, as in the sketch of Proposition 22.7.2, $K_\nu(2\pi) = \frac12\Gamma(\nu)\pi^{-\nu}R(t)$ with $R(t) = 1 + O(t^{-1/2})$: the companion term $\frac12\Gamma(-\nu)\pi^{\nu}$ is now smaller only by a factor of order $t^{-1/2}$. As $\arg\Gamma(\frac14+\frac{it}{2}) - \frac{t}{2}\log\pi = \vartheta(t)$, $\psi - \vartheta$ is a continuous argument of R, zero at $t = 0$, and it tends to a multiple of 2π. The multiple is 0. Replace 2π by $a \in (0, 2\pi]$ in the argument of K and in $(a/2)^{-\nu}$: the ratio $R_a(t)$ is continuous and nonzero, positive at $t = 0$, equal to $1 + O(t^{-1/2})$ uniformly for a in compact sets, and within $\frac12$ of 1 for every t once a is small, so the limit of its argument is a multiple of 2π that depends continuously on a and vanishes for small a. The statement on g_n follows with $\vartheta'(t) = \frac12\log\frac{t}{2\pi} + O(t^{-2})$. (iii) Away from ordinates $N(T) = \vartheta(T)/\pi + 1 + S(T)$, and N jumps from n to $n+1$ at γ_{n+1}; averaging the one sided limits gives the identity, which needs no hypothesis on the other zeros. Subtracting it from (ii) gives $\vartheta(t_n^\circ) - \vartheta(\gamma_{n+1}) = \pi S(\gamma_{n+1}) + O(\gamma_{n+1}^{-1/2})$, and the mean value theorem with $\vartheta''(t) = O(1/t)$ and $S(T) = O(\log T)$ gives the displacement. □

Numerically, Ξ° has 28 zeros in $(0, 100]$ (the 29th is $t_{29}^\circ = 101.0946$), the first at $t_1^\circ = 19.332507$, and $\vartheta(t_n^\circ)/\pi - (n - \frac12)$ increases from -0.2434 at $n = 1$ to -0.0155 at $n = 28$ (computed; the existence of the 28 zeros is certified by sign changes in ball arithmetic). For $n \le 28$ the displacement $t_n^\circ - \gamma_{n+1}$ has mean absolute value 0.513 and correlation 0.889 with $\pi S(\gamma_{n+1})/\vartheta'(\gamma_{n+1})$ (computed). Each t_n° is paired with γ_{n+1}, and γ_1 has no partner.

22.8 Pólya's Mellin criterion applied to the Riemann kernel

Take $q = 1$ and $\mathcal{F}(v) = \Phi(\sqrt{v})$ in P10, so that $\mathcal{F}(t^2) = \Phi(t)$ and the conclusion would be RH. The criterion asks that

$$M(z) = \int_0^\infty \Phi(u)\,u^{2z-1}\,du = \tfrac12\int_0^\infty \mathcal{F}(v)v^{z-1}dv \tag{22.8.1}$$

have no zeros in $\mathbb{C}\setminus(-\infty, 0]$. The integral converges for $\operatorname{Re} z > 0$, and $M(k+\frac12) = \frac12(2k)!\,c_k > 0$ and $2M(\frac12) = \xi(\frac12)$. Near the origin $\Phi(u) = \sum_{j\ge0} p_j u^{2j}$, with radius of convergence $\pi/4$, $p_0 = \Phi(0) = 0.893394$ and $p_1 = -8.36525$; integrating termwise over $(0,\delta)$, $\delta < \pi/4$, gives $\int_0^\delta \Phi(u)u^{2z-1}du = \sum_j p_j\delta^{2z+2j}/(2z+$

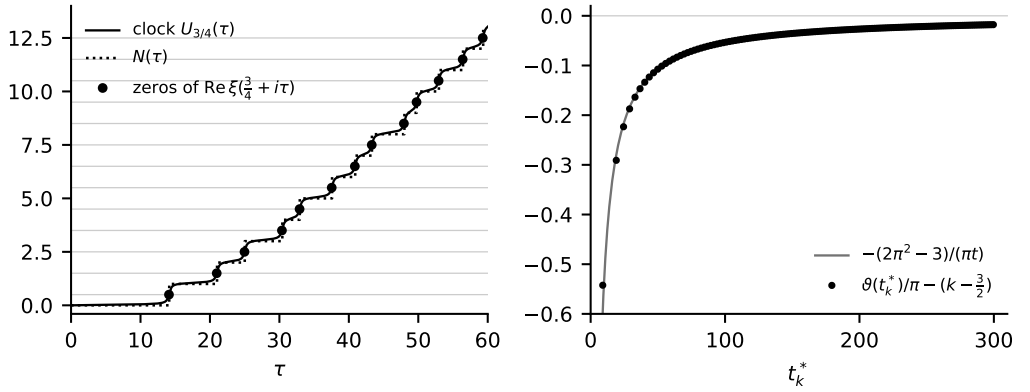


Figure 22.1: Left: the clock $U_{3/4}(\tau)$ of the basepoint law at $\alpha = 3/4$ for $0 \le \tau \le 60$, the zero count $N(\tau)$ (dotted), and the zeros of $\mathrm{Re}\,\xi(3/4 + i\tau)$, which sit exactly on the half integer levels (grey lines). Right: $\vartheta(t_k^*)/\pi - (k - \frac{3}{2})$ for the 138 zeros of Pólya's Ξ^* below height 300, with the asymptote $-(2\pi^2 - 3)/(\pi t)$ of Proposition 22.7.2.

$2j$), which continues M to a meromorphic function with simple poles at $z = 0, -1, -2, \dots$, and this continuation is the function that the criterion refers to.

The criterion does not apply to Φ: if X_Φ has the Pólya density $\Phi/\Xi(0)$ (the variable X of Theorem 1.4.3, and not the X of this chapter), then $\mathbb{E}|X_\Phi|^{2s} = 2M(s+\frac{1}{2})/\Xi(0)$ with the M of (22.8.1), so that theorem says that M has a simple zero at $z_1 = s_1 + \frac{1}{2} = -7.99418226 + 2.93776707\,i$, which lies in $\mathbb{C} \setminus (-\infty, 0]$. We checked this independently (at 60 digits, with two quadrature layouts): $|M| = 8.0 \times 10^{-16}$ at the point $s_1 + \frac{1}{2}$ printed in that theorem, and on the circle $|z - z_1| = \frac{1}{2}$ we have $|M| \ge 0.224$ and the winding number of M is 1, so that z_1 is the only zero in the disc. So the criterion fails for $q = 1$. For $q \ge 2$ the function $\mathcal{F}(t^{2q})$ has no t^2 term whenever $\mathcal{F}$ is holomorphic near the origin, whereas $\Phi(t) = 0.893394 - 8.36525\,t^2 + \cdots$, so there is no factorisation $\Phi(t) = \mathcal{F}(t^{2q})$ with $q \ge 2$. The Mellin route of the 1927 paper is therefore closed for Φ itself, by the zero that closes the Bernstein Pick route in Theorem 1.4.3.

The zero z_1 is the only obstruction we found: an argument principle scan of the continuation (floating point) finds no zero with $0.05 \le |\mathrm{Im}\, z| \le 14.05$ in $-3.95 \le \mathrm{Re}\, z \le 8.05$. A plain Gauss Legendre rule for (22.8.1) produces spurious zeros with $\mathrm{Re}\, z < \frac{1}{2}$, because the integrand has an integrable singularity $u^{2\,\mathrm{Re}\, z - 1}$ at the origin that oscillates like $\cos(2\,\mathrm{Im}\, z \log u)$; the termwise integration on $(0, \frac{1}{4})$ removes the difficulty.

22.9 Is the $2\sin^2$ lemma useful with regard to Pólya's work?

22.9.1 Where it helps

The lemma is applied at a basepoint α through the weight $e^{-\alpha x}$ in its hypothesis, which on the kernel side is the tilt e^{au}, the non even half of Pólya's universal factor $\cosh(au)$; it computes the continuous Thorin densities of which Pólya's approximations are the roundings, up to the affine term of its compensation (Theorem 10.2.12, Proposition 22.7.2); and positivity of the Thorin density at a basepoint makes the clock increasing along $\operatorname{Re} s = \alpha$ and places the real zeros of $\operatorname{Re}\xi(\alpha + it)$ at its half integer crossings (Proposition 22.6.3).

22.9.2 Where it cannot help

The Bessel test. The Bessel model $f(\nu) = K_\nu(2\pi)$ is even in ν, of order one, with zeros $\pm iz_n$ on $\operatorname{Re}\nu = 0$ (Section 22.7) and $\sum z_n^{-2} < \infty$, so the product formula and the basepoint theorem (Theorem 2.1.2) apply verbatim, as in Theorem 18.1.10, which is the case $\alpha = 0$: $K_\alpha(2\pi)/K_{\alpha+\sqrt{s}}(2\pi)$ is a GGC for every $\alpha \geq 0$, with Lorentzian Thorin density when $\alpha > 0$. The hypothesis of the lemma at basepoint α is that $(\log f)''$ is completely monotone on (α, ∞). Table 22.2 shows the signs of $(-1)^j d^{2+j} \log K_\nu(2\pi)/d\nu^{2+j}$: the third entry is negative at every $\nu \leq 5$ and a later entry fails at $\nu = 8$. So the primal measure of the lemma is signed at every basepoint tested, although the reciprocal law is a GGC at every basepoint. The lemma's success for ξ at $\alpha > 1$ is a cancellation, as Chapter 12 already observes.

ν	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
0.25	0.1482	6.7×10^{-4}	-2.7×10^{-3}	-9.8×10^{-5}	3.9×10^{-4}	3.6×10^{-5}
1	0.1469	2.6×10^{-3}	-2.5×10^{-3}	-3.7×10^{-4}	3.2×10^{-4}	1.3×10^{-4}
3	0.1373	6.6×10^{-3}	-1.3×10^{-3}	-6.8×10^{-4}	-1.4×10^{-5}	1.5×10^{-4}
5	0.1223	8.0×10^{-3}	-1.5×10^{-4}	-4.6×10^{-4}	-1.6×10^{-4}	6.9×10^{-6}
8	0.0991	7.1×10^{-3}	5.6×10^{-4}	-6.9×10^{-5}	-7.7×10^{-5}	-3.5×10^{-5}

Table 22.2: $(-1)^j \frac{d^{2+j}}{d\nu^{2+j}} \log K_\nu(2\pi)$.

Kernel multipliers. Pólya's universal factors $e^{\lambda u^2}$ and $\cosh(au)$ multiply the kernel, but the lemma acts on $\log f$ and has no composition rule that carries a lemma statement at one value of λ or a to another, which is why the heat flow row of Table 6.1 stands apart from the basepoint row.

22.9.3 Verdict

The lemma is not a mechanism for producing real zeros: in the Bessel model, where Pólya's theorem gives real zeros and every basepoint law is a GGC, the positivity the lemma needs is false. Any route to RH through the lemma must control a

cancellation between signed primal measures, and Pólya's own tools (universal factors, difference equations for K_ν, Hermite Biehler) are the natural candidates for that control.

22.10 What would help show RH

None of Pólya's theorems crosses into the critical strip; each stops where the basepoint programme stops.

1. The cosh factor and de Bruijn's strip theorem give real zeros of the even part for $\alpha \geq 1/2 + \Delta$, the same threshold $\sigma_{\max}$ as the basepoint laws. Proposition 22.6.3 identifies the two families, so there is no gain in the strip.

2. The heat flow factor $e^{\lambda u^2}$ reaches $\Lambda \leq 0.2$ (Platt and Trudgian, sharpening the bound 0.22 of Polymath 15), inside the strip in the Gaussian sense, and $\Lambda \geq 0$ by Rodgers and Tao; the remaining gap is the whole problem.

3. The Pólya Schur and Schoenberg ladders are equivalent to RH; their first rungs hold: the Turán inequality for the first is a theorem, and log concavity for the second is settled by a floating point certificate in Section 22.11.

4. The Mellin criterion does not apply to Φ (Section 22.8): for $q = 1$ the Mellin transform has the zero of Theorem 1.4.3, and for $q \geq 2$ there is no factorisation $\Phi(t) = \mathcal{F}(t^{2q})$. A different criterion, applied to a different factorisation of the Pólya density, such as its signed mixture of Gamma(2) and Gamma(3) densities (Section 15.1.4.3) or its representation through maxima (Chapter 19), would be needed.

5. The Bessel model is a laboratory: its analogue of RH is a theorem of Pólya, so any proposed mechanism can be tested on it before ξ; the $2\sin^2$ positivity mechanism already fails this test.

Problem 22.2. Find a sign condition on the primal measure of the lemma that holds for the Bessel model at every basepoint $\alpha > 0$, is preserved by the Euler product, that is, by adding the primal measures of the Euler factors, and fails for the Davenport Heilbronn function.

Problem 22.3. Prove real rootedness of $A_\alpha = \frac{1}{2}(\xi(\alpha + w) + \xi(\alpha - w))$ for one fixed $\alpha \in (\frac{1}{2}, 1)$ without proving the quasi Riemann hypothesis at α, by a Pólya type argument that uses the Euler product (the odd part B_α need not be real rooted).

Problem 22.4. Determine whether the derivative laws $\xi^{(2n)}(\frac{1}{2})/\xi^{(2n)}(\frac{1}{2} + \sqrt{s})$, or their counterparts $F^{(n)}(0)/F^{(n)}(s)$ with derivatives taken in the Thorin variable s, are GGCs for some n. For the first family, Ki and Kim give a local statement and

Conrey the proportion of zeros on the line, for large n; for the second, Griffin, Ono, Rolen and Zagier give the first d rungs of the Pólya Schur ladder for $n \geq N(d)$ (Section 22.11). For neither family is a global statement known for any n, and one for a single n would be the first unconditional GGC in its family.

Problem 22.5. Under RH, which is what makes the midpoint rule of Proposition 22.6.3 available for every $\alpha > \frac{1}{2}$, prove that it converges, $\mu_\alpha^P - \mu^{(\alpha)} \to 0$ weakly as $\alpha \downarrow \frac{1}{2}$, with an explicit rate, so that the atomic and continuous descents can be run together.

22.11 Progress on the open problems

22.11.1 Problem 22.5: the even part and basepoint laws differ by an explicit factor

Theorem 22.11.1. *For every $\alpha > 1/2$ and real $s \geq 0$ put $R_\alpha(s) = \xi(\alpha - \sqrt{s})/\xi(\alpha + \sqrt{s})$. Then, unconditionally, $R_\alpha(0) = 1$, $0 < R_\alpha(s) < 1$ for $s > 0$, and*

$$\begin{aligned} \frac{\varphi_\alpha^P(s)}{\varphi^{(\alpha)}(s)} &= \frac{2}{1 + R_\alpha(s)}, \\ \psi_\alpha^P(s) - \psi_\alpha(s) &= \log \frac{1 + R_\alpha(s)}{2}, \end{aligned} \tag{22.11.1}$$

where $\psi = -\log \varphi$ denotes the Laplace exponent, and for $s > 0$ the difference of the exponents lies in $(-\log 2,\, 0)$. Hence $\varphi^{(\alpha)} \leq \varphi_\alpha^P \leq 2\varphi^{(\alpha)}$ on $[0, \infty)$, and as $\alpha = \frac{1}{2} + a \downarrow \frac{1}{2}$,

$$\psi_\alpha^P(s) - \psi_\alpha(s) = -a\, \frac{\xi'}{\xi}\big(\tfrac{1}{2} + \sqrt{s}\big) + O(a^2) \tag{22.11.2}$$

locally uniformly in s. In particular $\varphi_\alpha^P \to \varphi$ and $\varphi^{(\alpha)} \to \varphi$ pointwise, so whenever the two families are laws they converge weakly to the reciprocal Xi law.

Proof. By definition $A_\alpha(\sqrt{s}) = \frac{1}{2}\xi(\alpha + \sqrt{s})\,(1 + R_\alpha(s))$, which gives (22.11.1). For real σ, $\xi(\sigma) > 0$, ξ is symmetric about $\frac{1}{2}$ and increasing on $[\frac{1}{2}, \infty)$, because $\xi(\frac{1}{2} + w) = \sum c_k w^{2k}$ with $c_k > 0$. Since $\xi(\alpha - \sqrt{s}) = \xi(1 - \alpha + \sqrt{s})$ and $|\alpha + \sqrt{s} - \frac{1}{2}| > |\frac{1}{2} - \alpha + \sqrt{s}|$ when $\alpha > \frac{1}{2}$ and $s > 0$, we get $0 < R_\alpha < 1$ for $s > 0$, while $R_\alpha(0) = 1$. For (22.11.2), $\log R_\alpha(s) = \log \xi(\frac{1}{2} + \sqrt{s} - a) - \log \xi(\frac{1}{2} + \sqrt{s} + a) = -2a \frac{\xi'}{\xi}(\frac{1}{2} + \sqrt{s}) + O(a^3)$, and $\log \frac{1+R}{2} = \frac{1}{2}\log R + O((\log R)^2)$. □

With Proposition 22.6.3, Theorem 22.11.1 gives the error of the midpoint rule exactly: for every α as in Proposition 22.6.3, that is for $\alpha > \sigma_{\max}$ and unconditionally for $\alpha \geq 1$, and every $s \geq 0$,

$$\sum_{k \geq 1} \log\!\left(1 + \frac{s}{\tau_\alpha (k - \frac{1}{2})^2}\right) = \int_0^\infty \log\!\left(1 + \frac{s}{\tau_\alpha(v)^2}\right) dv + \log \frac{1 + R_\alpha(s)}{2}. \tag{22.11.3}$$

The defect of the midpoint sum is a ratio of two values of ξ on the real axis, and no modulus of continuity of the clock is needed. Numerically, for $\alpha = 1, 0.75, 0.6, 0.52$ the defect divided by $a = \alpha - \frac{1}{2}$ varies by 1.2% at $s = 1$ and by 10% at $s = 100$, and approaches $-\frac{\xi'}{\xi}(\frac{1}{2} + \sqrt{s}) = -0.0461, -0.408$, as (22.11.2) predicts.

22.11.2 Problem 22.1: f_X is log concave

Write $N = f_X f_X'' - (f_X')^2$, so that PF_2 is $N < 0$, and recall $\varphi(t^2) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+t) > 0$ for real t.

Lemma 22.11.2 (Origin). *$f_X'(0) = 0$ and $f_X''(0) = -\frac{1}{\pi}\int_0^\infty t^2\varphi(t^2)\,dt < 0$, so f_X is log concave on a neighbourhood of 0, unconditionally.*

Lemma 22.11.3 (Residue expansion). *Let $H = 3 \times 10^{12}$ be the height up to which the zeros of ξ are verified to be simple and on the critical line [277], let $0 < T < H$ be different from every ordinate, and put $\eta_k = -\Xi(0)/\Xi'(\gamma_k)$. Then, unconditionally, for $x > 0$,*

$$f_X(x) = \sum_{\gamma_k < T} \eta_k\, e^{-\gamma_k x} + E_T(x), \qquad \bigl|E_T^{(j)}(x)\bigr| \le C_j(T)\, e^{-Tx},$$
$$C_j(T) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{|u - iT|^j\, \xi(\frac{1}{2})}{|\xi(\frac{1}{2} + u - iT)|}\, du. \tag{22.11.4}$$

The residues alternate in sign, $\eta_1 > 0 > \eta_2$, and so on.

Proof. $f_X(x) = \frac{1}{2\pi}\int \varphi(t^2)e^{-itx}dt$. Move the line of integration to $\operatorname{Im} t = -T$. The poles crossed are the zeros of $\xi(\frac{1}{2} + t)$ with $-T < \operatorname{Im} t < 0$, which below H are $t = -i\gamma_k$, simple; the vertical sides vanish because $|\xi(\sigma + i\tau)|$ grows like a gamma function as $|\sigma| \to \infty$, uniformly for bounded τ. The residue at $-i\gamma_k$ is computed from $\xi'(\frac{1}{2} - i\gamma) = i\,\Xi'(\gamma)$. The signs alternate because Ξ changes sign at each simple real zero, starting from $\Xi(0) > 0$. □

Numerically $\log|\eta_k| - \pi\gamma_k/4$ equals $-5.2, -8.0, -8.8, -9.7, -11.1$ for $k = 1, 10, 20, 40, 60$, so $|\eta_k|$ grows like $e^{\pi\gamma_k/4}$ up to a power of γ_k, and the expansion is useful only for $x > \pi/4$.

Lemma 22.11.4 (Two exponentials). *If $f = Ae^{-ax} + Be^{-bx}$, then $ff'' - (f')^2 = AB\,(b-a)^2 e^{-(a+b)x}$. With $A = \eta_1 > 0 > B = \eta_2$ this is negative.*

With $T = 23$, between γ_2 and γ_3, Lemmas 22.11.3 and 22.11.4 give $Ne^{(\gamma_1+\gamma_2)x} = \eta_1\eta_2(\gamma_2 - \gamma_1)^2 + \varepsilon(x)$, where $|\varepsilon(x)|$ is bounded by a sum of terms each decreasing in x, built from C_0, C_1, C_2 and $|\eta_1|, |\eta_2|$. Numerically $\eta_1 = 359.524$, $\eta_2 = -28007.2$, $\eta_1\eta_2(\gamma_2 - \gamma_1)^2 = -4.776 \times 10^8$, and $C_0, C_1, C_2 = 5.99 \times 10^4,\ 1.38 \times 10^6,\ 3.19 \times 10^7$.

THEOREM 22.11.5 (PF_2, floating point certificate). *The density f_X of $X = Z\sqrt{2Y}$ is log concave on $\mathbb{R}$. The certificate has three parts, for $x \geq 0$ (the density is even).*

(A) On $[0, 0.95]$, direct quadrature of $f^{(j)}(x) = \frac{1}{\pi}\int_0^\infty \varphi(t^2)\frac{d^j}{dx^j}\cos(tx)\,dt$ at the left end x_i of each of 770 cells, with $\sup|N'|$ on a cell of length h bounded by $B(h)$ through $|f^{(j)}(y)| \leq |f^{(j)}(x_i)| + h\,m_{j+1}$ and $m_j = \frac{1}{\pi}\int_0^\infty t^j\varphi(t^2)dt$ $(m_0, \ldots, m_4 = 1.899,\ 7.321,\ 44.94,\ 355.8,\ 3361)$, proves $N < 0$. The cells are chosen from the left, each as long as possible with $h\,B(h) \leq 0.99\,|N(x_i)|$, so that $N \leq N(x_i) + h\,B(h) < 0$ on the cell; any factor in $(0, 1)$ in place of 0.99 works (814 cells at 0.9, 1165 at 0.5).

(B) On $[0.95, 2.1]$, the expansion (22.11.4) with the first twelve zeros and $T = \frac{1}{2}(\gamma_{12} + \gamma_{13}) = 57.90$ gives $Ne^{(\gamma_1+\gamma_2)x}$ as an explicit exponential sum plus a remainder with decreasing majorant; 26 equal cells certify negativity (22 equal cells are the least that do, and two adaptive cells suffice). At $x = 0.95$ the sum is -2.18×10^8 and the remainder at most 7.03×10^6.

(C) For $x \geq 2.1$ the two term form with $T = 23$ has $|\varepsilon(x)| \leq 4.687 \times 10^8 < 4.776 \times 10^8$: the majorant decreases and equals 4.687×10^8 at $x = 2.1$.

Figure 22.2 (left) shows $-(\log f_X)''$. It is 23.66 at the origin and then decays like $\frac{|\eta_2|}{\eta_1}(\gamma_2 - \gamma_1)^2 e^{-(\gamma_2-\gamma_1)x}$, which is 4.0×10^{-9} at $x = 4$: the log concavity of the tail is exponentially thin, and it is held up by the sign of η_2 alone.

Numerics. The certificate is reproduced in mpmath at 80 digits. The values $f_X^{(j)}$ on $[0, 1.6]$ come from composite Gauss Legendre quadrature and beyond 1.6 from the residue expansion; the two agree to a relative 2×10^{-20} at $x = 1.6$ with fourteen zeros and to 8×10^{-42} with forty (for the derivatives of orders one to three, to 2×10^{-18} and 6×10^{-39}). The quadrature errors are many orders of magnitude below the margins of Parts (A) and (B). Part (C) is thin: the majorant falls below 4.776×10^8 from $x = 2.0904$ on, so that at $x = 2.1$ the margin is 1.9% of the bound, and a relative error of 2% in C_0, C_1, C_2 would break it; the constants are stable to many digits under two quadrature layouts. The only arithmetic input is that the first thirteen zeros are simple and on the line.

The mechanism of Lemma 22.11.4 extends, as a sketch, to every order n with $\gamma_n < H$ in the far tail. If $x_1 < \cdots < x_n$, $y_1 < \cdots < y_n$ and every $x_i - y_j$ is large, the Cauchy Binet expansion of $\det[f_X(x_i - y_j)]$ is dominated by the term from $\gamma_1, \ldots, \gamma_n$, whose sign is $\operatorname{sgn}(\eta_1 \cdots \eta_n)\,(-1)^{n(n-1)/2} = (-1)^{\lfloor n/2\rfloor + n(n-1)/2} = +1$. So the PF_n minors with $\gamma_n < H$ are positive far in the tail, unconditionally, provided the remainder E_T of (22.11.4), which is not a sum of exponentials, can be dominated inside the determinant; Proposition 22.11.15 does this for coalescing points.

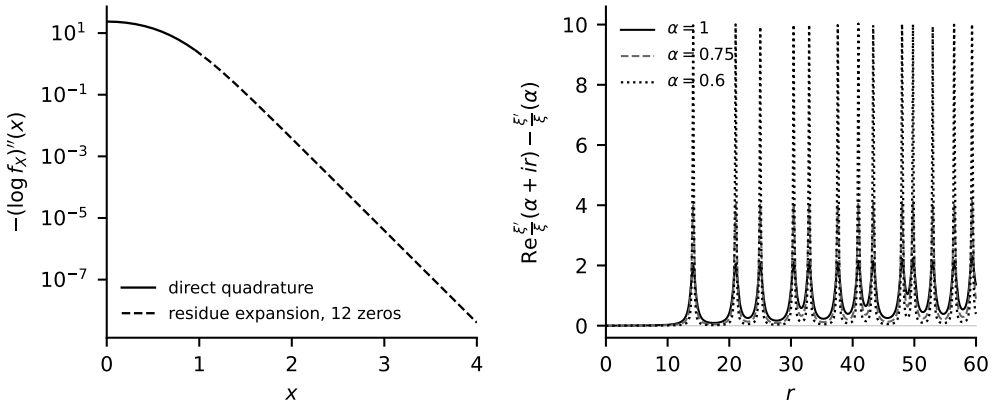


Figure 22.2: Left: $-(\log f_X)''(x)$ on a log scale, from direct quadrature on $[0, 0.95]$ (solid) and from the residue expansion with twelve zeros on $[0.95, 4]$ (dashed); it is positive throughout, which is Theorem 22.11.5. Right: the compensated clock condition $D_\alpha(r) = \mathrm{Re}\,\frac{\xi'}{\xi}(\alpha+ir) - \frac{\xi'}{\xi}(\alpha) \geq 0$ of Theorem 22.11.7 for $\alpha = 1$ (solid), 0.75 (dashed) and 0.6 (dotted), computed from the Hadamard sum over the first 1700 zeros at step 0.02; D_α vanishes at $r = 0$ and is positive after, with a peak of height about $1/(\alpha - \frac{1}{2})$ at each ordinate.

THEOREM 22.11.6 (f_X is uniformly close to a PF_∞ density). *Let $X_{\mathrm{on}} = \sum L_\gamma/\gamma$, the sum over the zeros on the critical line with independent standard Laplace variables. Unconditionally X_{on} has a PF_∞ density f_{on}, and for $j = 0, 1, 2$*

$$\sup_x \left| f_X^{(j)}(x) - f_{\mathrm{on}}^{(j)}(x) \right| \leq \varepsilon_j := \frac{1}{\pi} \int_0^\infty t^j \, \Pi_K(t) \left(2S\, t^2 + S^2 t^4 \right) dt, \qquad (22.11.5)$$

where $\Pi_K(t) = \prod_{k \leq K} (1 + t^2/\gamma_k^2)^{-1}$ for any K with $\gamma_K < H$, and $S = \sum_{\gamma > H} \gamma^{-2} \leq (\log \frac{H}{2\pi} + 2)/(2\pi H) = 1.533 \times 10^{-12}$ for $H = 3 \times 10^{12}$. With $K = 300$, $\varepsilon_0 \leq 1.57 \times 10^{-10}$, $\varepsilon_1 \leq 1.3 \times 10^{-9}$ and $\varepsilon_2 \leq 1.3 \times 10^{-8}$. Consequently every $n \times n$ minor $\det[f_X(x_i - y_j)]$ with ordered x and y is at least $-n \cdot n!\,(2\varepsilon_0 + f_{\mathrm{on}}(0))^{n-1} \varepsilon_0$, which is -1.1×10^{-8} for $n = 3$ and -1.2×10^{-7} for $n = 4$, since $f_{\mathrm{on}}(0) \leq \frac{1}{\pi} \int_0^\infty \Pi_K = 1.976$.

Proof. The characteristic function of X_{on} is $\varphi_{\mathrm{on}}(t^2) = \prod_{\text{on line}} (1 + t^2/\gamma^2)^{-1}$, the reciprocal of a function of type (22.3.1), so f_{on} is PF_∞ by Schoenberg's theorem whatever the zeros off the line do. Write $\varphi(t^2) = \varphi_{\mathrm{on}}(t^2) R(t^2)$, where R collects the off line zeros. Each conjugate pair $s_\rho, \bar{s}_\rho$ with $s_\rho = -a^2$, $a = \gamma - i\delta$, contributes $1/P_\rho(t^2)$ with $P_\rho(v) = 1 + 2v\,\mathrm{Re}(a^{-2}) + v^2 |a|^{-4}$. Since $\gamma > H > \frac{1}{2} > |\delta|$, $\mathrm{Re}\, a^{-2} = (\gamma^2 - \delta^2)/|a|^4 > 0$, so $P_\rho \geq 1$ on $[0, \infty)$, $0 < R \leq 1$ and $1 - R \leq \sum_\rho (P_\rho - 1) \leq 2St^2 + S^2 t^4$. Fourier inversion gives $|f_X^{(j)} - f_{\mathrm{on}}^{(j)}| \leq \frac{1}{\pi} \int_0^\infty t^j \varphi_{\mathrm{on}}(t^2)(1 - R(t^2))\, dt$, and $\varphi_{\mathrm{on}} \leq \Pi_K$ because the first K zeros are on the line. The tail sum S follows

by partial summation from the zero counting formula with an explicit error term [108]; Proposition 19.7.2 gives the sharper value 1.4798×10^{-12} for the same sum, and the constants of the theorem were computed with the larger one. For the minors, expand the determinant multilinearly in the differences $f_X - f_{\mathrm{on}}$ and use $\det[f_{\mathrm{on}}(x_i - y_j)] \geq 0$ and $|f_{\mathrm{on}}| \leq f_{\mathrm{on}}(0)$. $\square$

RH is the statement that the uniform distance of f_X to a PF_∞ density is zero. The bound on the minors grows quickly with n (it exceeds 1 at $n = 10$), and a failure of PF_3 would be invisible at the level 10^{-8}. Numerically there is none: over 6050 Toeplitz type minors for each of $n = 3$ and $n = 4$, with spacings $0.02d$ and $0.02(d+1)$, $1 \leq d \leq 25$, and offsets up to 1.2, all are positive and the smallest normalised minor $\det[f_X(x_i - y_j)] / \prod_i f_X(x_i - y_i)$ is 1.6×10^{-10} for $n = 3$ and 3.5×10^{-19} for $n = 4$. Both occur at the smallest spacing; most of the smallness comes from nearly coinciding points, since for $n = 3$ the smallest value over the offsets grows with d from 1.6×10^{-10} at $d = 1$ to 6.0×10^{-2} at $d = 25$.

22.11.3 Problem 22.2: the exact form of the $2\sin^2$ lemma

Stieltjes inversion gives the Thorin density with no sign assumption on the primal measure.

Theorem 22.11.7. *Let f be a real entire function with no zeros in $\operatorname{Re} z > \alpha - \delta$ for some $\delta > 0$ and with $\frac{f'}{f}(z) = O(\log|z|)$ uniformly on $\operatorname{Re} z \geq \alpha$. This holds for ξ with $\alpha > \sigma_{\max}$, with $\delta = \alpha - \sigma_{\max}$, the bound following from the Hadamard product because $O(\log T)$ zeros have ordinate in $[T, T+1]$, and for $\nu \mapsto K_\nu(2\pi)$ with $\alpha > 0$, with $\delta = \alpha$, the zeros being purely imaginary with spacing about $\pi / \log y$ at height y. Let $\psi(u) = \log f(\alpha + w) - \log f(\alpha) - w \frac{f'}{f}(\alpha)$ with $w = \sqrt{2u}$ the principal root. Then ψ' is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, $\psi'(u) = O(|u|^{-1/2} \log|u|)$, and*

$$\begin{aligned} \psi'(u) &= \int_0^\infty \frac{\omega_\alpha(z)}{z+u}\, dz, \\ \omega_\alpha(z) &= \frac{1}{\pi\sqrt{2z}} \left[\operatorname{Re} \frac{f'}{f}\Big(\alpha + i\sqrt{2z}\Big) - \frac{f'}{f}(\alpha) \right]. \end{aligned} \tag{22.11.6}$$

Hence the compensated reciprocal $e^{wf'(\alpha)/f(\alpha)} f(\alpha)/f(\alpha+w)$ is the Laplace transform in u of a GGC if and only if

$$\operatorname{Re} \frac{f'}{f}(\alpha + ir) \geq \frac{f'}{f}(\alpha) \qquad \text{for all } r > 0, \tag{22.11.7}$$

that is, if and only if the clock at basepoint α runs at least as fast at every height as at height zero.

Proof. For $u \notin (-\infty, 0]$ the root w has positive real part, so $\alpha + w$ stays in the zero free half plane and $\psi'(u) = \big[\frac{f'}{f}(\alpha + w) - \frac{f'}{f}(\alpha)\big]/w$ is holomorphic there. By

the hypothesis on $\frac{f'}{f}$ the numerator of ψ' is $O(\log|w|)$, whence the bound on ψ'; the zero free strip of width δ makes ψ' continuous up to the cut. Cauchy's formula on a keyhole contour around the cut, with the bound to discard the large circle and the integrability of ω_α at 0 (where $\omega_\alpha(z) = O(\sqrt{z})$ by the evenness of $r \mapsto \operatorname{Re}\frac{f'}{f}(\alpha+ir)$), gives the Stieltjes representation with density $-\frac{1}{\pi}\operatorname{Im}\psi'(-z+i0)$. At $u = -z+i0$ the root is $w = ir$, $r = \sqrt{2z}$, and $-\frac{1}{\pi}\operatorname{Im}\psi' = \frac{1}{\pi r}\operatorname{Re}\big[\frac{f'}{f}(\alpha+ir) - \frac{f'}{f}(\alpha)\big]$, which is ω_α. A Stieltjes function with zero constant term is the derivative of a Thorin Bernstein function exactly when its representing density is nonnegative, which is (22.11.7). Finally, if $\log f$ has the representation of the lemma with $\nu(dx) = e^{-\alpha x}\mu_f(dx)$, then $\frac{f'}{f}(\alpha+w) - \frac{f'}{f}(\alpha) = \int(1-e^{-wx})\nu(dx)$ and $\omega_\alpha(z) = \frac{1}{\pi r}\int 2\sin^2(rx/2)\,\nu(dx)$, which is the $2\sin^2$ lemma. □

The right sign condition on the primal measure is not $\nu \geq 0$ but

$$\int \big(1-\cos(rx)\big)\,\nu(dx) \geq 0 \qquad \text{for all } r, \tag{22.11.8}$$

that the cosine transform of ν is maximal at the origin, read in this way because ν has infinite mass for ξ and for the Bessel model. Its local form at $r = 0$ is $-(\log f)'''(\alpha) \geq 0$, the $j = 1$ column of Table 22.2, which is positive at every basepoint; the failures in that table are in higher columns, which (22.11.8) does not require. Numerically, (22.11.7) holds for the Bessel model at $\alpha = 0.25, 1, 3, 8$ on $0 < r \leq 60$ (grid of step 0.02), and for ξ at $\alpha = 1, 0.75, 0.6, 0.55$ on $0 < r \leq 100$ (grid of step 0.01). That the condition holds for the Bessel model at every $\alpha > 0$ rests on these numerics, and that it fails for the Davenport Heilbronn function (Problem 22.2) is not discussed here. For ξ, condition (22.11.7) reads $\operatorname{Re}\frac{\xi'}{\xi}(\alpha + ir) \geq b_\alpha = \frac{\xi'}{\xi}(\alpha)$, the compensated companion of Lagarias's criterion $\operatorname{Re}\frac{\xi'}{\xi} > 0$; in the variable $u = w^2/2$ the Thorin densities of the basepoint law and of the compensated reciprocal differ by the stable term $b_\alpha/(\pi\sqrt{2z})$ of Proposition 12.1.12, where the stable factor is $e^{-b_\alpha\sqrt{u}}$ in the variable u of that proposition. To the right of 1 the primes enter exactly as the lemma predicts; write $\frac{\xi'}{\xi} = \mathcal{A} + \frac{\zeta'}{\zeta}$ with the archimedean part $\mathcal{A}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2})$.

Proposition 22.11.8 (Primes and the compensated clock). *For $\alpha > 1$ and real r, write $D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) - \frac{\xi'}{\xi}(\alpha)$, which is $v_\alpha(r) - b_\alpha$ in the notation of Section 12.1. Then*

$$D_\alpha(r) = \big[\operatorname{Re}\mathcal{A}(\alpha+ir) - \mathcal{A}(\alpha)\big] + \sum_{n\geq 2}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{r\log n}{2}\Big). \tag{22.11.9}$$

The prime sum is nonnegative; it is the $2\sin^2$ transform of the positive measure $\sum\Lambda(n)n^{-\alpha}\delta_{\log n}$. The archimedean bracket equals $\sum_{k\geq 1} q_{\alpha+2k}(r) - q_{\alpha-1}(r)$ with

$q_c(r) = r^2/(c(c^2+r^2))$*, so that only the pole term is negative, and it is nonnegative for every* r *if and only if* $\zeta(3, 1+\frac{\alpha}{2}) \geq 8(\alpha-1)^{-3}$*, that is, if and only if* $\alpha \geq \alpha_1 = 7.78918\ldots$*; the compensated condition then holds from the identity alone. For* $1 < \alpha < \alpha_1$ *the bracket dips below zero (its minimum is* -4.83 *at* $\alpha = 1.2$*,* $r = 1.57$*;* -0.74 *at* $\alpha = 2$*;* -0.011 *at* $\alpha = 6$*), and the primes must make up the deficit. They do, for every* $\alpha \geq 1$*, by Proposition 12.1.12, whose proof combines the zeros below height* 2000*, the bounds of [108] for the zeros above, and a computation in ball arithmetic;* (22.11.9) *is* (12.1.7)*.*

Proof. For $\alpha > 1$, $\operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+ir) - \frac{\zeta'}{\zeta}(\alpha) = \sum \Lambda(n)n^{-\alpha}(1-\cos(r\log n))$ by the Dirichlet series, and $1-\cos x = 2\sin^2(x/2)$. For the bracket, $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\geq 0} y^2/((x+k)((x+k)^2+y^2))$ for the digamma function ψ, so with $x = \alpha/2$, $y = r/2$ the term $\frac{1}{2}\Gamma'/\Gamma(s/2)$ contributes $\sum_{k\geq 0} q_{\alpha+2k}(r)$; the term $k=0$ cancels the contribution $-q_\alpha(r)$ of $1/s$, and $1/(s-1)$ contributes $-q_{\alpha-1}(r)$. Each ratio $((\alpha-1)^2+r^2)/((\alpha+2k)^2+r^2)$ increases with r, so the bracket $r^2\big[\sum_{k\geq 1} \frac{1}{(\alpha+2k)((\alpha+2k)^2+r^2)} - \frac{1}{(\alpha-1)((\alpha-1)^2+r^2)}\big]$ is nonnegative for every r if and only if it is nonnegative as $r \to 0$, that is, if and only if $\sum_{k\geq 1}(\alpha+2k)^{-3} \geq (\alpha-1)^{-3}$, which is $\zeta(3, 1+\frac{\alpha}{2}) \geq 8(\alpha-1)^{-3}$. The quantity $(\alpha-1)^3\sum_{k\geq 1}(\alpha+2k)^{-3} = \sum_{k\geq 1}\big(\frac{\alpha-1}{\alpha+2k}\big)^3$ increases with α, so the condition holds exactly for $\alpha \geq \alpha_1$, where this quantity equals 1. □

As $\alpha \downarrow 1$ the deficit of the bracket comes from the pole term, $\operatorname{Re}\frac{1}{\alpha-1+ir} - \frac{1}{\alpha-1} \sim -\frac{1}{\alpha-1}$ for fixed $r \neq 0$, and the prime sum compensates it because $-\frac{\zeta'}{\zeta}(\alpha) \sim \frac{1}{\alpha-1}$. Grouping $\frac{1}{s-1}$ with $\frac{\zeta'}{\zeta}(s)$, which is holomorphic on $\operatorname{Re} s \geq 1$ by the prime number theorem, both pieces stay finite at $\alpha = 1$. Below 1 the Dirichlet series is gone and (22.11.9) has no direct analogue; this is the wall that Proposition 12.1.12 meets at $\alpha = 1$.

Remark 22.11.9 (The pole needs the gamma factor). The gamma terms of the archimedean bracket of (22.11.9) cannot be dropped while its pole term is kept [152]. Let $f(s) = (s-1)\zeta(s)$, a real entire function with no zeros in $\operatorname{Re} s \geq 1$. For every $\alpha > 1$ the compensated reciprocal $e^{w\frac{f'}{f}(\alpha)}f(\alpha)/f(\alpha+w)$ is not the Laplace transform in $u = w^2/2$ of a GGC. By Theorem 22.11.7, whose hypotheses f satisfies, and the uniqueness of the Stieltjes representation, it would be one only if

$$D_\alpha^f(r) = \operatorname{Re}\frac{f'}{f}(\alpha+ir) - \frac{f'}{f}(\alpha) = \sum_{n\geq 2}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{r\log n}{2}\Big) - q_{\alpha-1}(r)$$

were nonnegative for every r. But the mean of D_α^f over $[0,R]$ tends to $\sum_n \Lambda(n)n^{-\alpha} - \frac{1}{\alpha-1} = -\frac{f'}{f}(\alpha)$ as $R \to \infty$, because the prime series converges absolutely, the mean of $\cos(r\log n)$ over $[0,R]$ is at most $1/(R\log 2)$ in absolute value, and $q_{\alpha-1}(r) \to \frac{1}{\alpha-1}$; and $\frac{f'}{f}(\alpha) > 0$. Indeed $\Gamma(\alpha)\zeta(\alpha) = \int_0^\infty x^{\alpha-1}(e^x-1)^{-1}dx$

gives $f(\alpha) = \mathbb{E}h(G_{\alpha-1})$, with $h(x) = x/(1-e^{-x}) = x + x/(e^x-1)$ and $G_{\alpha-1}$ a Gamma variable of shape $\alpha-1$; since $h' \geq \frac{1}{2}$ and $G_{\alpha-1+\varepsilon}$ is $G_{\alpha-1}$ plus an independent Gamma variable of mean ε, $f'(\alpha) \geq \frac{1}{2}$. So D^f_α is negative somewhere. The primes are bounded on average while the pole term tends to $-\frac{1}{\alpha-1}$; in ξ the gamma terms $\sum_{k\geq 1} q_{\alpha+2k}(r)$, which grow like $\frac{1}{2}\log r$, pay for it, and one prime at a time is harmless (Remark 21.3.2).

Proposition 22.11.10. *If* $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) \geq b_\alpha$ *for all real* r *and every* $\alpha > \alpha_0 \geq \frac{1}{2}$, *then no zero of* ζ *has real part greater than* α_0. *For* $\alpha_0 = \frac{1}{2}$ *it implies RH.*

Proof. $b_\alpha = \frac{\xi'}{\xi}(\alpha) > 0$ for $\alpha > \frac{1}{2}$, because ξ is increasing on $(\frac{1}{2}, \infty)$. So the hypothesis gives $\operatorname{Re}\frac{\xi'}{\xi} > 0$ on $\operatorname{Re} s > \alpha_0$. Near a zero ρ with $\operatorname{Re}\rho > \alpha_0$ and multiplicity $m \geq 1$ the function $\frac{\xi'}{\xi}(s) - m/(s-\rho)$ is holomorphic, hence bounded, so $\operatorname{Re}\frac{\xi'}{\xi}(\rho-\varepsilon) = -m/\varepsilon + O(1)$ as $\varepsilon \downarrow 0$, while $b_{\operatorname{Re}\rho-\varepsilon}$ stays bounded; the hypothesis at $\alpha = \operatorname{Re}\rho - \varepsilon > \alpha_0$ and $r = \operatorname{Im}\rho$ fails for small $\varepsilon > 0$, so there is no such zero; this is condition (iv) of Theorem 13.2.7. □

Under RH the compensated condition holds at every $\alpha > \frac{1}{2}$, and it holds unconditionally for $\alpha \geq 1$ (Theorem 12.1.18 and Proposition 12.1.12), so Proposition 22.11.10 has a converse for $\alpha_0 = \frac{1}{2}$. The delicate range is below the first zero: there the numerics show $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)$ rising from its value at $r=0$ like r^2 (Figure 22.2, right). The refined problem is to prove $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) \geq b_\alpha$ at one $\alpha < 1$ for all heights. By Proposition 22.11.10, doing so for all α above some $\alpha_0 < 1$ implies a quasi Riemann hypothesis at α_0, and by Corollary 23.1.10 it is equivalent to one, whereas on a single line it is not known to imply anything about the zeros to its right (compare Remark 2.1.4).

22.11.4 Problem 22.3: local reality and certificates to height 300

Proposition 22.11.11 (Local reality, unconditional). *Let* K *be such that the first* K *zeros of* ξ *in the upper half plane, ordered by ordinate, are simple and on the line, with ordinates* $\gamma_1 < \cdots < \gamma_K$, *and fix* $\Gamma \in (\gamma_K, \gamma_{K+1})$. *There is* $\alpha_K > \frac{1}{2}$ *such that for* $\frac{1}{2} < \alpha < \alpha_K$ *the zeros of* A_α *in* $|w| < \Gamma$ *are exactly* $\pm i\tau_k(\alpha)$, $k \leq K$, *simple and purely imaginary, with* $\tau_k(\alpha) \to \gamma_k$.

Proof. $A_\alpha \to \xi(\frac{1}{2}+w)$ uniformly on $|w| \leq \Gamma$, so by Hurwitz's theorem A_α has exactly one zero in a small disc around each $\pm i\gamma_k$ and none elsewhere in $|w| < \Gamma$. A_α is even and real on the real axis, so its zero set is invariant under $w \mapsto -\bar{w}$; a unique zero in a disc centred on the imaginary axis is therefore on that axis. □

This is the even part analogue of the local theorem of Ki and Kim for the derivatives of ξ (item P13 of Section 22.3). For $\alpha = 0.6$ and 0.75 the argument principle on the rectangle $|\operatorname{Re} w| \leq 1$, $0 \leq \operatorname{Im} w \leq 100$, which contains every zero

of A_α up to that height by de Bruijn's strip theorem, counts 29.000 zeros, and Table 22.1 lists 29 sign changes of $\operatorname{Re}\xi(\alpha+it)$. So every zero of $A_{0.6}$ and $A_{0.75}$ up to height 100 is purely imaginary. At $\alpha=0.55$ the same count to height 300 gives 138.000 zeros against 138 sign changes, so every zero of $A_{0.55}$ up to height 300 is purely imaginary as well. Together with Proposition 22.6.3 this is a Turing type method for Thorin atoms in the strip: it verifies the even part laws to a given height without any statement about $\operatorname{Re}\xi'/\xi$.

22.11.5 Problem 22.4: derivatives in the Thorin variable are the shifted Jensen polynomials

Since $\xi(\frac12+z)=\frac18(4z^2-1)\widehat{\zeta}(\frac12+z)$ with $\widehat{\zeta}(s)=\pi^{-s/2}\Gamma(s/2)\zeta(s)$, the sequence of Griffin, Ono, Rolen and Zagier [97], defined by $(4z^2-1)\widehat{\zeta}(\frac12+z)=\sum_n \gamma_{\mathrm{GORZ}}(n)z^{2n}/n!$, is $\gamma_{\mathrm{GORZ}}(n)=8\xi(\frac12)\,n!\,a_n$, a constant multiple of our $g_n=n!\,a_n$. Their shifted Jensen polynomials are therefore, up to a constant factor, the polynomials $J_{d,n}(x)=\sum_{j=0}^{d}\binom{d}{j}g_{n+j}x^j$ of Section 20.2.4, the Jensen polynomials of the nth derivative $F^{(n)}(s)$ in the Thorin variable. By Pólya and Schur, the law with Laplace transform $F^{(n)}(0)/F^{(n)}(s)$ is a GGC if and only if $J_{d,n}$ is real rooted for every d.

Corollary 22.11.12. *For every d there is $N(d)$ such that for all $n\ge N(d)$ the derivative laws $F^{(n)}(0)/F^{(n)}(s)$ satisfy the first d rungs of the Pólya Schur ladder [97]. Under RH they are GGCs for every n, by Laguerre's theorem that the Laguerre Pólya class is closed under differentiation.*

The first case is classical: since $F'(s)=\xi'(\frac12+\sqrt{s})/(2\sqrt{s}\,\xi(\frac12))$, the zeros of F' are the squares $s=w^2$ of the zeros $w\neq0$ of $\xi'(\frac12+w)$, so the law $F'(0)/F'(s)$ is a GGC if and only if every zero of ξ' other than $\frac12$ lies on the critical line. That is the Riemann Hypothesis for ξ', which RH implies and which is not known to imply RH; Conrey's theorem [60] gives a positive proportion of its zeros on the line. In general $F^{(n)}(w^2)=(D^n\xi)(\frac12+w)/\xi(\frac12)$ with $D=\frac{1}{2w}\frac{d}{dw}$, so the derivative laws in the Thorin variable correspond to powers of this Bessel type operator rather than of $\frac{d}{dw}$. Numerically $\xi'(\frac12+w)$ has 29 zeros with $1<\operatorname{Im}w<100$ in the box $|\operatorname{Re}w|\le0.6$, all on the imaginary axis.

Numerically every $J_{d,n}$ with $2\le d\le15$ and $0\le n\le25$, 364 polynomials in all, has only real negative zeros, computed at 150 digits from the coefficients of Section 22.4. The refined problem is one n for which all rungs hold. As in Section 20.2.4, and following Farmer [84], large n carries little information about RH, so the useful direction is the opposite one, small n with all d, which is RH itself at $n=0$.

22.11.6 Problem 22.2 again: the compensated condition at almost all heights in the strip

Write $D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) - \frac{\xi'}{\xi}(\alpha)$ and let $N(\sigma,T)$ count zeros with $\beta>\sigma$ and $0<\gamma\le T$.

Lemma 22.11.13. *For $\alpha>\frac12$, unconditionally, $0<b_\alpha=\frac{\xi'}{\xi}(\alpha)\le 2a_1(\alpha-\frac12)<0.04621\,(\alpha-\frac12)$.*

Proof. With $a=\alpha-\frac12$, $b_\alpha=\frac{d}{da}\log F(a^2)=2a\,\psi'(a^2)$, where $\psi=\log F$. By Theorem 8.1.1 ψ is a Bernstein function, so ψ' is completely monotone and in particular $\psi'(a^2)\le\psi'(0)=a_1=0.0231050$. This is part (i) of Theorem 12.1.18, where $V_f=2a_1=0.04620998623$ (Table 12.4) and the bound is proved for every $\alpha>\frac12$. □

Theorem 22.11.14 (Almost all heights). *Fix $\frac12<\alpha\le1$ and put $\sigma=\frac12(\alpha+\frac12)$. There is T_0 such that for $T\ge T_0$ the exceptional set $E_\alpha(T)=\{r\in[T,2T]: D_\alpha(r)\le0\}$ has measure*

$$|E_\alpha(T)|\le C\Big(1+\frac{1}{\alpha-\frac12}\Big)\big(N(\sigma,3T)+1\big) \tag{22.11.10}$$

with an absolute constant C. By Ingham's density theorem [163]

$$|E_\alpha(T)|\ll_\alpha T^{3(1-\sigma)/(2-\sigma)}(\log T)^5=o(T),$$

and under the density hypothesis $|E_\alpha(T)|\ll T^{2(1-\sigma)+\epsilon}$. In particular the compensated clock condition holds at every basepoint in the strip on a set of heights of full density.

Proof. By the Hadamard product, summed symmetrically, $\operatorname{Re}\frac{\xi'}{\xi}(s)=\sum_\rho\operatorname{Re}\frac{1}{s-\rho}$ over all nontrivial zeros, so with $c_\rho=\alpha-\beta_\rho$,

$$D_\alpha(r)=\sum_\rho\left[\frac{c_\rho}{c_\rho^2+(r-\gamma_\rho)^2}-\frac{c_\rho}{c_\rho^2+\gamma_\rho^2}\right],$$

the sum running over zeros of both signs of γ. Call a zero harmful if $\beta>\alpha$, so $c<0$, and good if $|\beta-\frac12|\le\sigma-\frac12$, so $c\in[c_{\min},c_{\max}]$ with $c_{\min}=\alpha-\sigma=\frac12(\alpha-\frac12)$ and $c_{\max}=\alpha-1+\sigma\le\frac34$. Every zero that is not harmful contributes a nonnegative first term, so

$$D_\alpha(r)\ge L(r)-b_\alpha-N_h(r),$$
$$N_h(r)=\sum_{\text{harmful}}\frac{|c|}{c^2+(r-\gamma)^2},$$

where $L(r)$ is the first term of any one good zero; the second terms of all zeros, harmful ones included, sum to b_α.

The local term. By Littlewood's theorem [217] that gaps between consecutive ordinates tend to zero, for $r \geq T_0$ the window $[r-1, r+1]$ contains an ordinate. If one such zero is good, $L(r) \geq \frac{c}{c^2+1} \geq \kappa := \frac{c_{\min}}{c_{\min}^2+1} \geq \frac{4}{5} c_{\min}$, since $c \mapsto c/(c^2+1)$ increases on $(0,1)$. Otherwise r lies within 1 of a zero that is not good, that is one with $\beta > \sigma$ or its mirror with $\beta < 1-\sigma$; the measure of such r in $[T, 2T]$ is at most $4N(\sigma, 2T+1)$.

The harmful sum. Since $\int_{\mathbb{R}} \frac{|c|}{c^2+(r-\gamma)^2} dr = \pi$, the harmful zeros with $\frac{T}{2} \leq \gamma \leq 3T$ contribute at most $2\pi N(\alpha, 3T) \leq 2\pi N(\sigma, 3T)$ to $\int_T^{2T} N_h$, and the remaining harmful zeros contribute $o(1)$ by the trivial bound $|c|/(r-\gamma)^2$ and Ingham's bound for $N(\alpha, t)$. By Markov's inequality, $N_h(r) > \kappa/2$ on a set of measure at most $\frac{2}{\kappa}(2\pi N(\sigma, 3T) + o(1))$.

Outside these two sets, Lemma 22.11.13 gives $b_\alpha \leq 0.04621(\alpha - \frac{1}{2}) = 0.09242\, c_{\min}$, hence $D_\alpha(r) \geq \kappa - \frac{\kappa}{2} - 0.0925\, c_{\min} \geq (0.4 - 0.0925) c_{\min} > 0$. Adding the two exceptional measures and using $\kappa \geq \frac{4}{5} c_{\min}$ gives (22.11.10). □

Theorem 22.11.14 has the same content as Remark 12.1.17, which obtains the comparison from Theorem 12.1.15 with $\sigma_0 = \frac{1}{2} + (\alpha - \frac{1}{2})/\sqrt{2}$ in place of σ and a lower bound growing like $\log \theta$; the proof here is direct, and Theorem 14.3.2 gives a statement of the same kind for the mass of the negative part of the density.

To the right of the strip the inequality holds on the grid numerically (Section 23.1), and at all heights for every $\alpha \geq 1$ by Proposition 12.1.12. Below 1 it holds at all heights, on every line to the right of α_0, under a quasi Riemann hypothesis at α_0 (Theorem 23.1.8).

22.11.7 Problem 22.1 again: coalescing points and the PF_3, PF_4 conditions

Theorem 22.11.6 controls PF_n minors up to an additive error of 1.1×10^{-8} for $n = 3$ and 1.2×10^{-7} for $n = 4$, which is enough wherever the minors of f_{on} are larger than that, but not where they vanish. They vanish at coalescing points and far in the tail. When all x_i and all y_j coalesce at points whose difference is x, the minor divided by the two Vandermonde products tends to a constant multiple of the Hankel determinant $\mathcal{H}_n(x) = \det[f_X^{(i+j)}(x)]_{i,j<n}$, and the PF_n condition becomes $(-1)^{n(n-1)/2}\mathcal{H}_n(x) \geq 0$; for $n = 2$ this is log concavity.

Proposition 22.11.15. *In the tail the coalescing conditions hold, unconditionally, for every n with $\gamma_n < H$: for an exponential sum $f = \sum_k \eta_k e^{-\gamma_k x}$,*

$$\mathcal{H}_n(x) = \sum_{|S|=n} \Big(\prod_{k \in S} \eta_k \Big) \prod_{k<l \in S} (\gamma_l - \gamma_k)^2 \, e^{-x \sum_{k \in S} \gamma_k},$$

and the term of $S = \{1, \dots, n\}$ dominates as $x \to \infty$, with sign $(-1)^{\lfloor n/2 \rfloor} = (-1)^{n(n-1)/2}$ by the alternation of the residues.

Proof. $[f^{(i+j)}(x)]_{i,j<n} = V\,\mathrm{diag}(\eta_k e^{-\gamma_k x})\,V^{\top}$ with $V = [(-\gamma_k)^i]_{i<n,\,k}$, so the Cauchy Binet formula gives the sum over n element subsets S, each weighted by $\det(V_S)^2 = \prod_{k<l\in S}(\gamma_l-\gamma_k)^2$. The residue expansion of Lemma 22.11.3 with T between γ_n and $\min(\gamma_{n+1}, H)$ shows that the remaining terms, including those that contain the remainder E_T and its derivatives, are smaller by a factor of order $e^{-(T-\gamma_n)x}$. □

Numerically, combining quadrature on $[0, 1.6]$ with the residue expansion with fourteen zeros beyond, $-\mathcal{H}_3/f_X^3$ and $\mathcal{H}_4/f_X^4$ are positive at every point of a grid of step 0.02 on $[0,4]$; their smallest values, 1.05×10^{-21} and 2.08×10^{-39}, occur at $x=4$, where Proposition 22.11.15 predicts exponential decay. So PF_3 and PF_4 hold at coalescing points on the grid and, by Proposition 22.11.15, as $x\to\infty$, and the scan of separated minors in Section 22.11 finds none that is negative. A complete proof of PF_3 would need three things that are not proved here: an explicit threshold in x in Proposition 22.11.15, a bound for nearly coalescing points, and a bound for separated points in the tail.

22.11.8 Problem 22.3 again: almost all zeros of the even and odd parts are on the line

Let $B_\alpha(w) = \frac{1}{2i}(\xi(\alpha+w)-\xi(\alpha-w))$ be the odd part, so that $A_\alpha(it) = \mathrm{Re}\,\xi(\alpha+it)$ and $B_\alpha(it) = \mathrm{Im}\,\xi(\alpha+it)$, and let $N_A(T)$, $N_B(T)$ count the zeros with $0<\mathrm{Im}\,w\le T$ and $N_A^0(T)$, $N_B^0(T)$ those on the imaginary axis.

Theorem 22.11.16 (Zero accounting for the even and odd parts). *Let $\frac12<\alpha\le1$ be such that no zero of ξ lies on the line $\mathrm{Re}\,s=\alpha$, which excludes at most countably many α. Then, with no other hypothesis, as $T\to\infty$ through values that are not ordinates of zeros of ξ, A_α or B_α,*

$$\begin{aligned} N_A(T) &= U_\alpha(T) + 2N(\alpha,T) + O(\log T),\\ N_A^0(T) &\ge \left\lfloor U_\alpha(T)+\tfrac12\right\rfloor, \end{aligned} \tag{22.11.11}$$

and the same holds for B_α with $\lfloor U_\alpha(T)\rfloor$ in place of $\lfloor U_\alpha(T)+\frac12\rfloor$. Consequently

$$\begin{aligned} N_A(T)-N_A^0(T) &\le 2N(\alpha,T)+O(\log T),\\ N_B(T)-N_B^0(T) &\le 2N(\alpha,T)+O(\log T), \end{aligned} \tag{22.11.12}$$

so by Ingham's density theorem all but $O(T^{3(1-\alpha)/(2-\alpha)}\log^5 T)$ of the zeros of A_α and of B_α up to height T are purely imaginary, a proportion tending to one.

Proof. For $\mathrm{Re}\,w\ge\alpha$ both $\alpha+w$ and $\alpha-w$, reflected by the functional equation to $1-\alpha+w$, have real part at least 1, where $\log|\xi(\sigma+it)|$ is strictly increasing in σ because every zero has $\beta<1$; as $\alpha+\mathrm{Re}\,w>1-\alpha+\mathrm{Re}\,w$, $|\xi(\alpha+w)|>|\xi(\alpha-w)|$ and $A_\alpha(w)\neq0$. So all zeros lie in $|\mathrm{Re}\,w|<\alpha$. Apply the argument principle to A_α

on the rectangle $|\operatorname{Re} w| \le X$, $0 \le \operatorname{Im} w \le T$ with $X = 2$. On the real axis $A_\alpha > 0$. On the right side $A_\alpha = \frac{1}{2}\xi(\alpha + w)(1 + q)$ with $|q| < 1$, so its argument changes by that of ξ along $\operatorname{Re} s = \alpha + X$ up to $O(1)$, and the left side contributes the same by the symmetries $A_\alpha(-w) = A_\alpha(w)$ and $A_\alpha(\bar{w}) = \overline{A_\alpha(w)}$. The argument principle for ξ on $[\alpha, \alpha + X] \times [0, T]$ converts the change along $\operatorname{Re} s = \alpha + X$ into $\pi U_\alpha(T) + 2\pi N(\alpha, T)$ plus a horizontal variation at height T. The two horizontal variations, of ξ and of A_α, are $O(\log T)$ by Backlund's argument [9] with Jensen's formula, since $\log|\xi|$ and $\log|A_\alpha|$ vary by $O(\log T)$ on discs of bounded radius at height T, by Stirling's formula and polynomial bounds for ζ. This gives the first identity in (22.11.11). For the second, $A_\alpha(it) = |\xi(\alpha + it)| \cos(\pi U_\alpha(t))$ with U_α continuous from $U_\alpha(0) = 0$, so every level $k - \frac{1}{2} \le U_\alpha(T)$ is crossed at least once. The odd part is treated in the same way with sin, except that B_α is odd, with a simple zero at $w = 0$ on the bottom edge, so the contour is indented above 0, which changes the count by $\frac{1}{2}$, inside $O(\log T)$, and $B_\alpha(\bar{w}) = -\overline{B_\alpha(w)}$ replaces $A_\alpha(\bar{w}) = \overline{A_\alpha(w)}$. Subtracting gives (22.11.12). □

Numerically, at $\alpha = 0.55$ the clock reaches $U_{0.55}(300) = 137.921$; $\operatorname{Re}\xi(0.55+it)$ has $138 = \lfloor 137.921 + \frac{1}{2} \rfloor$ zeros on $(0, 300]$ and $\operatorname{Im}\xi(0.55 + it)$ has $137 = \lfloor 137.921 \rfloor$, and the argument principle counts 138.000 zeros of $A_{0.55}$ and 137.0 zeros of $B_{0.55}$ in the box, so the balance is exact there with no off axis zeros.

The odd part family also bears on Problem 22.4: since $B_{\frac{1}{2}+a}(w)/a \to -i\,\xi'(\frac{1}{2} + w)$ locally uniformly as $a \downarrow 0$, the derivative ξ' is the endpoint of the odd part family, just as ξ is the endpoint of the even part family. Under RH every B_α has only imaginary zeros, and Hurwitz's theorem then gives the Riemann Hypothesis for ξ'. Theorem 22.11.16 cannot be pushed to $a \to 0$ directly, because the known bounds for $N(\alpha, T)$ grow to a positive proportion of all zeros as $\alpha \downarrow \frac{1}{2}$, and Hurwitz's theorem needs every zero of B_α on the axis, not almost every one; this is why the Riemann Hypothesis for ξ' remains open.

Chapter 23

Haseo Ki, derivatives and the Keiper Li coefficients

Haseo Ki and his coauthors have studied the objects this book computes with: Fourier transforms of rapidly decaying positive kernels, their zeros under the heat flow and under repeated differentiation, and zeta functions that share the functional equation of ζ but not its Euler product. The de Bruijn Newman flow of Section 4.1 and the Thorin condition of Chapter 1 meet in this work, which we read through the Thorin dictionary: a zero of an even entire function is an atom of the Thorin measure of its reciprocal, and the reciprocal is a generalised gamma convolution (GGC) exactly when the zeros are real.

Four results come out. The theorem of Ki, Kim and Lee on the de Bruijn Newman flow says that after any positive time the reciprocal of the flowed function is a GGC transform up to a finite rational factor, and de Bruijn's strip theorem confines the atoms of that factor to confocal parabolas that shrink with time (Section 23.2). Read together with the theorem of Rodgers and Tao, the flow gives an obstruction: no property that survives a Gaussian tilt of the Pólya density can imply RH (Proposition 23.2.3). Ki's theorem on repeated differentiation is a statement about moment tilts of the Pólya density, and it implies that under RH the reciprocals of the derivatives are GGC transforms whose clocks converge in law to the exit time of Brownian motion from an interval (Section 23.3); a conjecture, supported by computation, gives the rate at which the atoms converge. Ki's approximation theorems for Epstein type zeta functions and the Davenport Heilbronn functions indicate that a Thorin argument for RH must use the Euler product (Section 23.4). Ki's work on the zeros of ζ' concerns the spacing of the zeros (Section 8.3) and gives no route to positivity; his uniqueness theorems in the extended Selberg class and his work in descriptive set theory are not used.

Results of Ki and his coauthors are cited, not reproved.

23.1 The work of Ki and the Thorin dictionary

23.1.1 The strands of the work

Four strands of the work of Ki and his coauthors matter here. The first is the heat flow on Fourier transforms of rapidly decaying kernels: Ki and Kim extend a result of Newman on the zeros of Fourier transforms [122] and answer a question of de Bruijn [123], and Ki, Kim and Lee prove that for every positive time the flowed function has all but finitely many of its zeros real and simple [189]. The second is repeated differentiation: Ki and Kim settle the Fourier Pólya conjecture [188], and Ki studies the Riemann Ξ function under repeated differentiation [118]. The third is approximate zeta functions, which keep the functional equation of a zeta function and lose its Euler product: approximations of the Epstein zeta function [116], modified Epstein zeta functions [117], approximations of the Ramanujan Ξ function [119], Weng's zeta functions [121] and, with Komori and Suzuki, Weng zeta functions for Chevalley groups [124]. The fourth is the zeros of ζ' [120, 85, 95], taken up in Section 23.5.

23.1.2 Zeros as Thorin atoms

The Pólya kernel Φ and $\Xi(\theta) = \xi(\frac{1}{2} + i\theta) = \int_{\mathbb{R}} \Phi(u)\cos(\theta u)\,du$ are those of (2.1.3), and X is the Pólya variable with density $\Phi/\Xi(0)$, so $\Xi(\theta)/\Xi(0) = \mathbb{E}[e^{i\theta X}]$. With $s = -\theta^2$ this reads $\Xi(i\sqrt{s})/\Xi(0) = \mathbb{E}[\cosh(\sqrt{s}\,X)] = G(s)$, the function of Section 2.1, because $\xi(\frac{1}{2} - \sqrt{s}) = \xi(\frac{1}{2} + \sqrt{s})$. The following lemma carries the Thorin reading over to all the real entire functions of this kind that Ki studies.

Lemma 23.1.1. *(a) Let F be an even entire function, real on $\mathbb{R}$, of order less than two, with $F(0) \neq 0$. Let $\pm w_k$ be its zeros, with multiplicity (one sign chosen for each pair, and $w_k \neq 0$), and put $\lambda_k = w_k^2$. Then $s \mapsto F(i\sqrt{s})$ is entire of order less than one and*

$$\frac{F(0)}{F(i\sqrt{s})} = \prod_k \Big(1 + \frac{s}{\lambda_k}\Big)^{-1},$$

locally uniformly off the points $-\lambda_k$. If every w_k is real, this is the Laplace transform of $T = \sum_k \lambda_k^{-1} e_k$, with e_k independent standard exponential variables: a GGC with zero drift and discrete Thorin measure $\sum_k \delta_{\lambda_k}$. If some w_k is not real, then $\lambda_k \notin (0,\infty)$, and $F(0)/F(i\sqrt{s})$ is not the Laplace transform of any GGC.

(b) Let K be even and positive, with $K(u) = O(\exp(-|u|^p))$ for some $p > 2$, and let $F(z) = \int_{\mathbb{R}} K(u)e^{izu}\,du$. Then F satisfies the hypotheses of (a).

Proof. (a) The function $f(s) = F(i\sqrt{s})$ is entire because F is even, and its order is half that of F, so less than one. Such a function has genus zero, and Hadamard's

theorem gives $f(s) = f(0)\prod_k (1+s/\lambda_k)$ with convergent product. When all $\lambda_k > 0$ the product is the Laplace transform of T, since $\mathbb{E}e^{-se_k/\lambda_k} = (1+s/\lambda_k)^{-1}$. A zero $w_k \notin \mathbb{R}$ has $w_k^2 \notin (0,\infty)$. A GGC transform $\exp\{-\int \log(1+s/t)\,\mu(dt)\}$ with $\mu \geq 0$ on $(0,\infty)$ is analytic on $\mathbb{C}\setminus(-\infty,0]$, while $1/f$ has a pole at $-\lambda_k \notin (-\infty,0]$, of the order of the multiplicity of the zero, which no other factor cancels.

(b) $F(0) = \int K > 0$. The bound on K gives $|F(x+iy)| \leq C\int e^{|y||u|-|u|^p}du \leq C'\exp(c_p|y|^{p/(p-1)})$, so F has order at most $p/(p-1) < 2$, and F is even and real. □

For $F = \Xi$ the zeros $\pm\theta$ are $\pm\gamma$ with $\gamma > 0$, and Lemma 23.1.1(a) gives $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s}) = \prod_{\gamma>0}(1+s/\gamma^2)^{-1}$, the clock of Section 2.1. RH is the statement that its Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2}$ is a positive measure on $(0,\infty)$. Special cases of the lemma, and the pole argument, appear in Propositions 2.1.1, 11.1.16 and 14.4.12; the general form is needed for the approximations of Section 23.4. A simple map confines the atoms of a hypothetical counterexample.

Lemma 23.1.2 (Strips become confocal parabolas). *For $\Delta > 0$ the map $w \mapsto w^2$ sends the strip $\{|\operatorname{Im} w| \leq \Delta\}$ onto the closed region to the right of the parabola $\mathcal{P}_\Delta = \{x+iy:\ x = y^2/(4\Delta^2) - \Delta^2\}$. Every $\mathcal{P}_\Delta$ has its focus at the origin, the parabolas are nested, and they collapse onto $[0,\infty)$ as $\Delta \downarrow 0$.*

Proof. The line $w = a + ib$, $a \in \mathbb{R}$, $0 < b \leq \Delta$, goes to $w^2 = (a^2-b^2) + 2iab$, that is, to the parabola $x = y^2/(4b^2) - b^2$. For fixed $y \neq 0$ the right side decreases from $+\infty$ to $y^2/(4\Delta^2) - \Delta^2$ as b increases from 0 to Δ, so these lines fill exactly the points with $x \geq y^2/(4\Delta^2) - \Delta^2$, $y \neq 0$. The real axis goes to $[0,\infty)$, and the points $x \in [-\Delta^2, 0)$ are the images of $w = i\sqrt{-x}$. The focus of $\mathcal{P}_\Delta$ is at $x = -\Delta^2 + \Delta^2 = 0$. For $\Delta' < \Delta$ the parabola $\mathcal{P}_{\Delta'}$ lies to the right of $\mathcal{P}_\Delta$ at every height $y \neq 0$, and for fixed $y \neq 0$ its abscissa tends to $+\infty$ as $\Delta' \downarrow 0$. □

23.2 The heat flow, the barrier and the Gaussian tilt

For real t let Ξ_t be the transform of the damped kernel $e^{tu^2/4}\Phi(u)$ of Section 18.1.6, so $\Xi_t(\theta) = \int_{\mathbb{R}} \Phi(u)e^{tu^2/4}\cos(\theta u)\,du$ and $\Xi_0 = \Xi$. With $D = d/d\theta$, $D^2\cos(\theta u) = -u^2\cos(\theta u)$, hence $\Xi_t = e^{-(t/4)D^2}\Xi$: the backward heat operator is the Gaussian tilt of the Pólya density. Also $\Xi_t(\theta) = 8H_t(2\theta)$ with H_t the function of Section 4.1.1, so the constants of that section carry over. The zeros of Ξ lie in $|\operatorname{Im}\theta| < \frac{1}{2}$ and those of Ξ_t in $(\operatorname{Im}\theta)^2 \leq \max(\frac{1}{4} - \frac{t}{2}, 0)$ by de Bruijn's theorem [73, Theorem 13] (Theorem 8.4.1(c) with $r = 1$, since the zeros of H_0 lie in $|\operatorname{Im} z| \leq 1$); so Ξ_t has only real zeros for $t \geq \frac{1}{2}$. Newman showed that there is a constant Λ such that Ξ_t has only real zeros exactly for $t \geq \Lambda$ [242], and Rodgers and Tao proved $\Lambda \geq 0$ [318]; RH is $\Lambda \leq 0$. By Lemma 23.1.1(b) with $K = \Phi e^{tu^2/4}$, the atoms of the Thorin measure of $1/\Xi_t$ are the squares of the zeros of Ξ_t in the upper half plane and on the positive axis.

23.2.1 The theorem of Ki, Kim and Lee in Thorin form

Ki, Kim and Lee proved that for every $t > 0$ the function Ξ_t has infinitely many zeros, all but finitely many of them real and simple [189]; this gives $\Lambda < \frac{1}{2}$ but no height beyond which the zeros are real (Remark 19.15.1). Lemma 23.1.1 turns it into a statement on the Thorin measure.

Proposition 23.2.1 (Eventually Thorin). *For every $t > 0$ there are a finite, conjugation invariant multiset $E_t \subset \mathbb{C} \setminus (0, \infty)$ and a GGC Laplace transform L_t, with discrete Thorin measure on $(0, \infty)$ whose atoms are simple beyond a finite level, such that*

$$\frac{\Xi_t(0)}{\Xi_t(i\sqrt{s})} = L_t(s) \prod_{\eta \in E_t} \Big(1 + \frac{s}{\eta}\Big)^{-1}.$$

Ξ_t has only real zeros exactly when $E_t = \emptyset$.

Proof. The kernel $\Phi e^{tu^2/4}$ is even and positive and decays like $\exp(-\pi e^{2u}/2)$, so Lemma 23.1.1(b) applies for every p. By the theorem of Ki, Kim and Lee the nonreal zeros w form a finite conjugation invariant set, giving the multiset $E_t = \{w^2\}$, which misses $(0, \infty)$; the real zeros give L_t by Lemma 23.1.1(a), and only finitely many of them are multiple. □

For $t > 0$ the tail of the Thorin measure of the reciprocal is therefore positive, and only a finite rational factor has to be settled by computation. At $t = 0$ the identity would be the clock of ξ with E_0 the multiset of squared off line zeros, and RH is $E_0 = \emptyset$. The barrier of the Polymath project is a separate device, for the zeros that enter the strip from large heights (Chapter 6); with it and an effective version of the theorem the Polymath project proved $\Lambda \le 0.22$ [311], and Platt and Trudgian improved this to $\Lambda \le 0.2$ [277].

De Bruijn's strip theorem says where E_t can be. Ki and Kim [123] proved the form that allows finitely many exceptional zeros: if for every $\varepsilon > 0$ all but finitely many zeros of a real entire function f of genus at most one lie in $|\operatorname{Im} z| \le \Delta + \varepsilon$, then for every $\varepsilon > 0$ all but finitely many zeros of $e^{-\lambda D^2} f$ lie in $|\operatorname{Im} z| \le \sqrt{\max(\Delta^2 - 2\lambda, 0)} + \varepsilon$. This does not locate the finitely many exceptions, so it adds nothing to Proposition 23.2.1; de Bruijn's theorem for all zeros does.

Corollary 23.2.2 (Shrinking parabolas). *For $0 < t < \frac{1}{2}$ every Thorin atom of Ξ_t, in particular every $\eta \in E_t$, lies in the closed region to the right of $\mathcal{P}_{\Delta(t)}$, where $\Delta(t)^2 = \frac{1}{4} - \frac{t}{2}$. The parabolas share the focus 0 and collapse onto $[0, \infty)$ at $t = \frac{1}{2}$, where de Bruijn's theorem takes over.*

Proof. By Theorem 8.4.1(c) with $r = 1$, all zeros of Ξ_t lie in $(\operatorname{Im} \theta)^2 \le \frac{1}{4} - \frac{t}{2} = \Delta(t)^2$, and an atom is the square of a zero. Lemma 23.1.2 gives the region. □

23.2.2 The Gaussian tilt obstruction

Read through Lemma 23.1.1, the theorem of Rodgers and Tao says that for every $t < 0$ the Thorin measure of $1/\Xi_t$ has atoms off $(0, \infty)$. This filters proof strategies.

Proposition 23.2.3 (No slack in the Gaussian direction). *Let $\mathcal{K}$ be a class of positive even kernels closed under $K \mapsto Ke^{-bu^2}$ for every $b > 0$. If every $K \in \mathcal{K}$ has a Fourier transform with only real zeros, then $\Phi \notin \mathcal{K}$.*

Proof. If $\Phi \in \mathcal{K}$, then $\Phi e^{-bu^2} \in \mathcal{K}$ for every $b > 0$, so Ξ_t has only real zeros for every $t < 0$, that is, $\Lambda < 0$, contradicting $\Lambda \geq 0$ [318]. □

This restates Proposition 18.1.16(a) and Proposition 14.4.12; what it adds is the following. Multiplication by e^{-bu^2} adds $-2b$ to $(\log K)''$ and $-bv$ to $\log K(\sqrt{v})$, so every condition that bounds $(\log K)''$ from above is preserved, including log concavity of $K(u)$ and of $K(\sqrt{v})$. No such condition can imply RH, even together with further hypotheses that the tilt also preserves. For log concavity this is Proposition 14.4.12(ii); see also [246]. The same holds for any form of the sine squared lemma (eq. (2.1.9)) whose positivity hypothesis on the Lévy measure survives the corresponding tilt. A Thorin argument for RH must use a property that Φ has and that Φe^{-bu^2} loses for every $b > 0$.

Remark 23.2.4 (Newman's class). Newman [240] characterised the Lee and Yang single spin laws as those whose Fourier transform is in the Laguerre Pólya class (Proposition 11.1.16); the class is preserved by e^{bt^2}, $b > 0$. The law with density $\Phi/\Xi(0)$ belongs to it if and only if RH holds, and by Rodgers and Tao Φe^{-bu^2} does not belong to it for any $b > 0$, so under RH the Pólya law lies on its boundary.

23.3 Derivatives and the Brownian exit time

Ki proved [118] that there are real constants A_n and $\tau_n \to 0$ with

$$A_n \, \Xi^{(2n)}(\tau_n \theta) \to \cos \theta \tag{23.3.1}$$

locally uniformly on $\mathbb{C}$, where $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ is the function of Section 23.1, so no rescaling is needed. Ki also treats the Xi functions of automorphic L functions, and Gunns and Hughes extend the result to the Selberg class [101]. We read (23.3.1) as a statement about moment tilts of the Pólya density.

23.3.1 Moment tilts

Since $\Xi^{(2n)}(\theta) = (-1)^n \int_{\mathbb{R}} u^{2n} \Phi(u) \cos(\theta u) \, du$, the ratio $\Xi^{(2n)}(\theta)/\Xi^{(2n)}(0)$ is the characteristic function of X_n, the symmetric variable with density proportional to $u^{2n}\Phi(u)$. Write $\ell_n(u) = 2n \log u + \log \Phi(u)$ for the log tilted density on $u > 0$. Because $\log \Phi(u) = \frac{9}{2}u - \pi e^{2u} + O(1)$ as $u \to +\infty$, the tilt concentrates $|X_n|$ near

the mode u_n, the root of $2\pi e^{2u} = 2n/u + \frac{9}{2}$, so $u_n \sim \frac{1}{2}\log n$; the convergence is slow (computed: $u_n = 3.4186$ at $n = 10^4$, against $\frac{1}{2}\log n = 4.605$). Setting $\theta = 0$ in (23.3.1) gives $A_n \Xi^{(2n)}(0) \to 1$, so (23.3.1) is equivalent to

$$\mathbb{E}\big[e^{i\theta\tau_n X_n}\big] \to \cos\theta \quad \text{locally uniformly on } \mathbb{C}: \tag{23.3.2}$$

$\tau_n X_n$ converges in law to a symmetric sign, and all exponential moments converge. Put $m_n = \mathbb{E}|X_n|$. The family $|\tau_n X_n|$ is uniformly integrable, its exponential moments being bounded on compact sets by (23.3.2), so $\tau_n m_n \to 1$; since (23.3.2) is stable under multiplying τ_n by a factor tending to one, we may take $\tau_n = 1/m_n$.

Proposition 23.3.1 (Brownian exit time limit). *Let*

$$\psi_n(s) = \frac{\Xi^{(2n)}(0)}{\Xi^{(2n)}(i\sqrt{2s}/m_n)} = \frac{1}{\mathbb{E}[\cosh(\sqrt{2s}\,X_n/m_n)]}.$$

Then $\psi_n(s) \to 1/\cosh\sqrt{2s}$ locally uniformly off the poles of the limit. The limit is $\mathbb{E}[e^{-sT}]$ for T the exit time of standard Brownian motion from $(-1,1)$, a GGC with Thorin measure

$$\sum_{k\geq 1} \delta_{c_k^2/2}, \qquad c_k = \tfrac{(2k-1)\pi}{2}.$$

For each K there is n_K such that for $n \geq n_K$ the K smallest Thorin atoms of ψ_n are positive and simple and converge to $c_k^2/2$. If RH holds, each ψ_n is the Laplace transform of a GGC T_n, and $T_n \to T$ in law.

Proof. Evaluate the locally uniform limit (23.3.2), with $\tau_n = 1/m_n$, at $\theta = i\sqrt{2s}$. The formula $\mathbb{E}e^{-sT} = 1/\cosh\sqrt{2s}$ is classical, and $\cosh\sqrt{2s} = \prod_k (1 + s/(c_k^2/2))$. The statement about atoms is Hurwitz's theorem for the entire functions $1/\psi_n$: the zeros of $\cosh\sqrt{2s}$ are simple and real, so for large n a small disc around each contains exactly one zero of $1/\psi_n$, which is real because $1/\psi_n$ is real on $\mathbb{R}$ and its conjugate lies in the same disc. Under RH, Ξ lies in the Laguerre Pólya class, which is closed under differentiation, so $\Xi^{(2n)}$ has only real zeros; the kernel $u^{2n}\Phi(u)$ is even and positive with the decay required in Lemma 23.1.1(b), and Lemma 23.1.1(a) applies to each ψ_n. Convergence of Laplace transforms on $[0,\infty)$ to a transform that is continuous at 0 gives convergence in law. □

By Kent's theorem (Section 12.2, [184]) a GGC with discrete Thorin measure is the hitting time of a diffusion governed by a Krein string, and the Riemann string is the unknown of the inverse spectral problem. Under RH, Proposition 23.3.1 says that the hitting times of the strings of the scaled high derivatives converge in law to the exit time of Brownian motion from an interval, as the clocks of the de Bruijn Newman flow do (Proposition 4.1.19). Convergence of the strings themselves would need the continuity of Krein's inverse spectral correspondence, which is not used here. The statement about the K smallest atoms is also implied by the local theorem of Ki and Kim recalled in Section 22.3.

23.3.2 The rate of convergence of the atoms

Write $|X_n| = m_n + V_n$, with $\mathbb{E}V_n = 0$, $\sigma_n^2 = \mathbb{E}V_n^2$ and $\kappa_n = \mathbb{E}V_n^3$ (the third central moment, equal to the third cumulant). An exact equation for the zeros leads to a conjecture for the rate. Let $\mathcal{A}_n(\theta) = \mathbb{E}\cos(\theta V_n)$ and $\mathcal{B}_n(\theta) = \mathbb{E}\sin(\theta V_n)$.

Lemma 23.3.2. *A real θ is a zero of $\Xi^{(2n)}$ if and only if*

$$\cos(\theta m_n)\,\mathcal{A}_n(\theta) = \sin(\theta m_n)\,\mathcal{B}_n(\theta).$$

In particular, where $\sin(\theta m_n) \neq 0$ and $\mathcal{A}_n(\theta) \neq 0$, the zeros satisfy $\cot(\theta m_n) = \mathcal{B}_n(\theta)/\mathcal{A}_n(\theta)$.

Proof. By symmetry of X_n, $\mathbb{E}\cos(\theta X_n) = \mathbb{E}\cos(\theta(m_n + V_n)) = \cos(\theta m_n)\mathcal{A}_n(\theta) - \sin(\theta m_n)\mathcal{B}_n(\theta)$. □

Now $\mathcal{A}_n(\theta) = 1 - \frac{1}{2}\theta^2\sigma_n^2 + \dots$ is even in the fluctuation and $\mathcal{B}_n(\theta) = -\frac{1}{6}\theta^3\kappa_n + \dots$ is odd. Near $\theta m_n = c_k$ we have $\cot(c_k + \delta) = -\tan\delta \approx -\delta$, so the shift of the kth zero is $\delta \approx -\mathcal{B}_n/\mathcal{A}_n \approx \kappa_n\theta^3/6$. The variance cancels to leading order: a symmetric broadening of the tilted law does not move the zeros, and to first order only the third central moment does. This heuristic suggests the following statement, supported by the numbers below and not proved.

Remark 23.3.3 (Conjecture: the Edgeworth rate). As $n \to \infty$, for each fixed k the kth positive zero $z_{k,n}$ of $\Xi^{(2n)}$ satisfies

$$z_{k,n}\, m_n = c_k + \frac{\kappa_n\, c_k^3}{6\, m_n^3}\big(1 + o(1)\big),$$

and, with $\tau_n = 1/m_n$,

$$\sup_{0\le s\le S}\Big|\psi_n(s) - \frac{1}{\cosh\sqrt{2s}}\Big| = \Big(\sup_{x\le\sqrt{2S}}\frac{x^2}{2\cosh x}\Big)\frac{\sigma_n^2}{m_n^2}\big(1 + o(1)\big).$$

For $S \ge 2.133$ the constant is $\max_x x^2/(2\cosh x) = 0.53222$, attained at $x = 2.06534$ (computed: the maximiser solves $\tanh x = 2/x$).

The second formula comes from $\mathbb{E}\cosh(x|X_n|/m_n) = \cosh x\,(1 + x^2\sigma_n^2/(2m_n^2)) + O(\kappa_n/m_n^3)$. A saddle point expansion of ℓ_n at u_n gives the orders of the cumulants (heuristic). At the mode, $\ell_n''(u_n) \approx -2n/u_n^2 - 4n/u_n - 9$ and $\ell_n'''(u_n) \approx 4n/u_n^3 - 8n/u_n - 18$, so that

$$\sigma_n^2 \approx -\frac{1}{\ell_n''(u_n)}, \qquad \kappa_n \approx \ell_n'''(u_n)\,\sigma_n^6,$$

and to leading order $\sigma_n^2 \approx u_n/(4n)$ and $\kappa_n \approx -u_n^2/(8n^2)$. The low atoms then move by $O\big(1/(n^2\log n)\big)$ while the Laplace transform converges at rate $O\big(1/(n\log n)\big)$. The leading forms are poor at the computed n (at $n = 10^4$ they exceed the computed σ_n^2 by 15 percent and $|\kappa_n|$ by 57 percent, and the logarithmic factors are not visible), so the numbers below use the full saddle forms.

23.3.3 Numerical evidence

The tilted law is computed, in mpmath at 40 and 50 digits with two windows and steps for each n, by the trapezoid rule on a window around u_n (using $\log\Phi$ to avoid underflow), and so are its cumulants, the first six zeros of $\int u^{2n}\Phi(u)\cos(\theta u)\,du$, and the Laplace transform error on $0 \le s \le 20$. This is done for the normalisation of this book ($n = 10, 10^2, 10^4$) and, for the chapter source's normalisation Φ_{dBN} of Section 4.1.1, ($n = 5$ and $n = 10, 10^2, \dots, 10^6$); the two differ by the scaling $u \mapsto 2u$, which doubles m_n and σ_n, multiplies κ_n by 8, and leaves $z_{k,n}m_n$, σ_n/m_n, κ_n/m_n^3 and the Laplace error unchanged. The rows for $n = 10^3$ and 10^6 of Table 23.1 are those of the second normalisation with m_n and σ_n doubled. The two runs for each n agree to 4×10^{-11} relative at worst ($n = 5$) and to 10^{-26} otherwise.

n	m_n	σ_n	$z_{1,n}m_n - c_1$	Edgeworth	$z_{3,n}m_n - c_3$	Edgeworth
10	0.7832	0.1019	-4.01×10^{-5}	-4.00×10^{-5}	-5.28×10^{-3}	-5.00×10^{-3}
100	1.5319	0.0530	-1.70×10^{-6}	-1.70×10^{-6}	-2.12×10^{-4}	-2.12×10^{-4}
10^3	2.4382	0.0224	-1.72×10^{-8}	-1.72×10^{-8}	-2.15×10^{-6}	-2.15×10^{-6}
10^4	3.4185	0.0086	-1.50×10^{-10}	-1.50×10^{-10}	-1.87×10^{-8}	-1.87×10^{-8}
10^6	5.4844	0.0011	-1.11×10^{-14}	-1.11×10^{-14}	-1.39×10^{-12}	-1.39×10^{-12}

Table 23.1: Computed shifts of the first and third positive zeros of $\Xi^{(2n)}$, rescaled by m_n, against the Edgeworth prediction $\kappa_n c_k^3/(6m_n^3)$ (computed, 40 and 50 digits).

For $100 \le n \le 10^4$ the prediction agrees with the computed shift to within 0.02 percent for every $k \le 6$ (at worst 0.018 percent, at $n = 100$). At $n = 10$ the worst k gives 23 percent, and for small n the higher zeros are not in the regime $c_k^3|\kappa_n| \ll m_n^3$ where the expansion applies. The ratio of the sup of the Laplace transform error on $0 \le s \le 20$ to σ_n^2/m_n^2 rises from 0.514 at $n = 5$ through 0.5307 at $n = 10^2$ and 0.53220 at $n = 10^4$ to 0.53221 at $n = 10^6$, against the predicted 0.53222. At $n = 10^4$ the quadrature gives $\sigma_n^2 = 7.452 \times 10^{-5}$ and $\kappa_n = -9.27 \times 10^{-9}$, and the saddle point formulas above agree with these values.

Two consequences matter for computation. First, the low Thorin atoms of $\Xi^{(2n)}$ are rigid: by $n = 100$ the first six rescaled zeros $z_{k,n}m_n$ lie within 3×10^{-3} of the Brownian values c_k (computed: at most 2.26×10^{-3}), and by $n = 1000$ within 3×10^{-5} (2.29×10^{-5}), and to first order the deviation is governed by κ_n, a functional of the moments of Φ. Second, because the variance drops out of this term, any model of Φ that reproduces the tilted mean and the third central moment reproduces the low atoms of the high derivatives to first order in κ_n/m_n^3; the variance enters at second order through $\mathcal{A}_n$.

Remark 23.3.4 (Scale mixtures survive the tilt). If $\Phi(u) = \int \sigma^{-1} g(u/\sigma)\,F(d\sigma)$ is a positive scale mixture, as in the uniform and arcsine mixtures of Sections 20.4.2

and 14.4.5.2, then

$$u^{2n}\Phi(u) = \int \sigma^{2n}\,\sigma^{-1}\big[(u/\sigma)^{2n} g(u/\sigma)\big] F(d\sigma)$$

is again a positive scale mixture, of the tilted kernel $u^{2n}g(u)$ with mixing measure reweighted by σ^{2n}. Concentration of the rescaled mixture needs concentration of every factor after scaling, since a product of independent positive factors concentrates only if each factor does; Remark 23.3.3 would then say how the third central moment of the reweighted mixture controls the atoms.

23.4 Approximate zeta functions and the Euler product

23.4.1 Ki's approximation theorems

Ki's 2005 paper truncates the Chowla Selberg expansion of an Epstein zeta function [116]. The partial sums keep the functional equation and have no Euler product, and Ki proves that in every strip containing the critical line all but finitely many of their zeros are simple and on the line. Later papers do the same for modified Epstein zeta functions [117] and for approximations of the Ramanujan Ξ function attached to Δ [119]. Lemma 23.1.1(a) turns each into a Thorin statement once the approximation F is even, real on $\mathbb{R}$, of order less than two, has $F(0) \neq 0$, and has all but finitely many of all its zeros (not only those in a fixed strip) real. Positivity of a kernel is not needed; where F is not known to be the Fourier transform of a positive kernel, Lemma 23.1.1(b) does not apply and these hypotheses must be checked. Then $F(0)/F(i\sqrt{s})$ is eventually Thorin in the sense of Proposition 23.2.1: a GGC transform up to a finite rational factor. The truncations and approximations ζ_N of Section 17.1 and the friable approximant of Chapter 18 belong to the same family, and Ki's method, an argument principle count set against a count of real zeros by sign changes, is a template for proving eventual Thorin statements about them.

23.4.2 The Davenport and Heilbronn obstruction

Davenport and Heilbronn [72] showed that the Epstein zeta function of a positive definite binary quadratic form of class number greater than one has zeros off the critical line (in the half plane $\operatorname{Re} s > 1$). Such a function has a functional equation of Riemann type, its completed form has the theta inversion $t \mapsto 1/t$, and by Lemma 23.1.1(a) its reciprocal Xi function is not a GGC.

Remark 23.4.1 (What a Thorin proof must use, heuristic). Any property shared by ξ and the Davenport Heilbronn functions cannot imply RH. The functional equation and the inversion of the theta function are shared. If the Pólya type kernel of such an Epstein zeta function is positive and has the same doubly exponential

decay as Φ, which is not established here, then positivity and decay of the kernel are shared as well. An argument deriving a positive Thorin measure for $1/\Xi$ from these properties alone would then apply equally to those functions and be invalid. This indicates, as a heuristic principle and not a theorem, that a correct argument must use the Euler product or another property that separates ζ from them.

The principle supports placing the Euler product Lévy representation, which writes $\log(1/\zeta)$ through the primes (Section 12.1.4, Section 8.2) and feeds that Lévy measure to the sine squared lemma, at the centre of the programme. Proposition 23.2.3 and Remark 23.4.1 point the same way: the Gaussian tilt keeps the positivity, evenness and decay of the kernel but not the theta inversion, while the passage to Epstein functions keeps the functional equation and the theta inversion, and positivity and decay only under the unproved hypothesis of the remark.

23.4.3 Weng zeta functions as a calibration set

Weng's high rank zeta functions are built from shifted completed zeta values with a functional equation of Riemann type; they have no Euler product. Ki [121], and Ki, Komori and Suzuki [124] for Chevalley groups, proved that their nontrivial zeros lie on the critical line. If that holds, the reciprocals of the corresponding Xi functions have positive Thorin measure by Lemma 23.1.1(a), and they are a test set for the numerical pipeline of this book (Chapter 3): compute the Thorin measure through the sine squared lemma, confirm positivity and measure the margin. Having no Euler product, they test the pipeline, not an argument of the kind Remark 23.4.1 calls for.

23.5 Zeros of the derivative of zeta, open problems and notes

Ki showed, on RH, that $\liminf(\beta' - \frac{1}{2})\log\gamma' \neq 0$ over the zeros $\beta' + i\gamma'$ of ζ' holds exactly when $\zeta'/\zeta(s) = 1/(s-\rho) + O(\log t)$ near the line, with ρ the nearest zero [120]. Farmer and Ki tie zeros of ζ' near the line to small gaps between zeros of ζ and to Landau Siegel zeros [85], and Gonek and Ki study the pair correlation of the zeros of the real and imaginary parts of ζ [95]. In Thorin language ζ'/ζ is the derivative of the Lévy exponent, so Ki's criterion is a local regularity condition on the Lévy measure near the critical line, and in the Farmer and Ki dichotomy small gaps appear as nearly coalescing Thorin atoms. These results concern the spacing of the zeros (Section 8.3); they give no new route to positivity.

For the programme the chapter yields, in order of use: Proposition 23.2.3 (proved), that no property of Φ surviving a Gaussian tilt can carry a proof of RH; Remark 23.4.1 (heuristic), that an argument blind to the Euler product cannot work, with Ki's approximation theorems and the Weng functions as test cases for the numerical pipeline; Proposition 23.3.1 (proved under RH, with Ki's theorem as input) and Remark 23.3.3 (conjecture, computed), that under RH the scaled

clocks of the high derivatives converge to the Brownian exit time and conjecturally their atoms move, to first order, only through the third central moment; and Proposition 23.2.1 and Corollary 23.2.2 (proved, from the theorem of Ki, Kim and Lee and de Bruijn's strip theorem), the Thorin form of that theorem and the confocal parabola picture.

Problem 23.1 (Prove the Edgeworth rate). Prove the conjecture of Remark 23.3.3, with explicit error terms, by a saddle point analysis of ℓ_n and Lemma 23.3.2. The local theorem of Ki and Kim (Section 22.3) gives that for every R the zeros of $\Xi^{(n)}$ in $|\theta| < R$ are real for large n, so that the number K_n of atoms of ψ_n known to be real tends to infinity; determine how large K_n may be unconditionally. The expansion is governed by the condition $c_{K_n}^3 |\kappa_n| / m_n^3 \to 0$, that is $K_n = o\big((n^2 \log n)^{1/3}\big)$; reality of the atoms is a separate question.

Problem 23.2 (Descending the derivative ladder). Find an explicit n_0 and heights $T(n) \to \infty$ such that all zeros of $\Xi^{(2n)}$ below $T(n)$ are real for $n \geq n_0$. This is the effective form of the local theorem of Ki and Kim [188] and of Problem 22.4. It would give a sequence of certified positive Thorin measures computed from the moments of Φ alone, and a derivative analogue of the barrier.

Problem 23.3 (A property that breaks the obstruction). Identify a property of Φ used by the sine squared lemma that is destroyed by $\Phi \mapsto \Phi e^{-bu^2}$ for every $b > 0$ and that fails for the Pólya kernels of Epstein zeta functions of class number greater than one.

Problem 23.4 (Weng calibration). Compute the Thorin measures of reciprocal Weng Xi functions of rank two and three through the sine squared lemma and compare their margin of positivity with that of $\sum_{\gamma > 0} \delta_{\gamma^2}$.

Problem 23.5 (Eventual Thorin for the friable approximant). Adapt Ki's argument principle and sign change method to prove that the reciprocal of the friable approximant of Chapter 18 is eventually Thorin, with an explicit bound on the exceptional set.

Notes. The theorems on the heat flow and on repeated differentiation are due to Ki and his coauthors as cited, to de Bruijn, Newman and to Rodgers and Tao; Proposition 23.2.3 is the theorem of Rodgers and Tao read through the Thorin dictionary, in the form of Proposition 18.1.16(a), and Remark 23.4.1 rests on the Davenport Heilbronn functions. The computations of Section 23.3.3 compute the tilted laws in mpmath and take about two seconds for each n.

23.6 Keiper Li coefficients from the Pólya density

The Pólya density that computes the law of the clock also computes the coefficients of Li's criterion [208], which states that RH holds if and only if

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right] \geq 0 \qquad \text{for every } n \geq 1,$$

the sum over the zeros taken in conjugate pairs. Keiper [182] introduced the coefficients λ_n and computed them before Li proved the criterion, and Bombieri and Lagarias [37] derived an arithmetic expression for them from the explicit formula and related the criterion to Weil's.

23.6.1 Taylor coefficients of $\log \xi$ at $s = 1$

Li [208] expresses the coefficients as derivatives at $s = 1$,

$$\lambda_n = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \Big[s^{n-1} \log \xi(s) \Big]_{s=1}.$$

With $\log \xi(1+w) = \sum_{k \geq 0} a_k w^k$ and $(1+w)^{n-1} = \sum_j \binom{n-1}{j} w^j$, comparing the coefficients of w^n gives

$$\lambda_n = n \sum_{j=0}^{n-1} \binom{n-1}{j} a_{n-j}, \tag{23.6.1}$$

so it suffices to compute the Taylor coefficients of $\log \xi$ at $s = 1$. The zeros nearest to $s = 1$ are $\frac{1}{2} \pm 14.1347i$, so $\log \xi(1+w)$ is analytic in the disc $|w| < R$, where $R = |\rho_1 - 1| = 14.1436$.

The sum (23.6.1) is exact, but its terms can be larger than λ_n. The Hadamard product $\xi(1+w) = \xi(1) \prod_\rho (1 - w/(\rho - 1))$ gives $m a_m = -\sum_\rho (\rho - 1)^{-m}$ for $m \geq 2$, and for large m the two zeros nearest to $s = 1$ dominate, so $|m a_m|$ is at most about $2R^{-m}$. Since $n\binom{n-1}{m-1} = m\binom{n}{m}$, the terms of (23.6.1) are $\binom{n}{m} m a_m$ with $m = n - j$, and for large n their absolute sum is at most about $2(1 + 1/R)^n$. Cancellation therefore costs at most about $\log_{10} \big(2(1 + 1/R)^n / \lambda_n \big)$ digits. This grows like $n \log_{10}(1 + 1/R) \approx 0.03n$, about one digit at $n = 100$, where $2(1 + 1/R)^{100} \approx 1.9 \times 10^3$ and $\lambda_{100} = 118.6$.

We compute the a_k by the trapezoid rule for the Cauchy integral on the circle $|w| = 6$,

$$a_k \approx \frac{1}{N r^k} \sum_{j=0}^{N-1} \log \xi\big(1 + r e^{2\pi i j/N}\big) e^{-2\pi i j k/N}, \qquad r = 6,\ N = 512,$$

which converges geometrically. The rule returns $a_k + a_{k+N} r^N + a_{k+2N} r^{2N} + \cdots$, so aliasing changes a_k by a relative amount of about $(r/R)^N \approx 10^{-191}$. Rounding

errors are amplified in a_k by about $(R/r)^k$, which is 10^{37} at $k = 100$ and is absorbed by the working precision.

Each value $\xi(1+w) = \xi(\frac{1}{2} + (\frac{1}{2} + w))$ comes from the Pólya route at 600 bits; the imaginary part of $\frac{1}{2}+w$ never exceeds 6, so the Pólya sum loses almost nothing to cancellation. The principal branch of the logarithm is the analytic one, because $|\arg \xi(1+w)| < 1$ on the circle: in the Hadamard product (2.1.1) with $s = \frac{1}{2} + w$, so that $|s| \le 6.5$, the factor $1 + s^2/\gamma^2$ of the zeros $\frac{1}{2} \pm i\gamma$ has argument at most $\arcsin(6.5^2/\gamma^2) \le 1.01 \cdot 6.5^2/\gamma^2$, and since $\sum_{\gamma>0} \gamma^{-2} = 0.0231$ these arguments add up to less than 0.99.

The first coefficient has the closed form $\lambda_1 = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, where γ_E is Euler's constant. The Pólya computation gives

$$\lambda_1 = 0.023095708966121033814 31431025,$$

in agreement with the closed form to all twenty five digits. The same number $\lambda_1 = \sum_\rho \rho^{-1} = (\xi'/\xi)(1)$ reappears at the basepoint one (Section 12.1.2): the shifted clock T_1 there, with Laplace transform $\xi(1)/\xi(1+\sqrt{u}) = 1 - \lambda_1\sqrt{u} + O(u)$ near $u = 0$, has a tail of order $x^{-1/2}$, and λ_1/π is its Thorin density at the origin in the chart $\theta = \sqrt{z}$. Table 23.2 lists further values with Keiper's asymptotic $\lambda_n \approx \frac{n}{2}(\log n - 1 - \log 2\pi + \gamma_E)$ [182].

n	λ_n (Pólya)	Keiper asymptotic
1	0.023095708966121033814	−1.13
2	0.092345735228046670386	−1.57
3	0.207638920554324 80379	−1.74
5	0.575542714461177 45243	−1.63
10	2.279339363193157 7437	0.21
20	8.769276872093215 1296	7.35
50	43.531096488374019299	41.28
100	118.60377537679132993	117.23

Table 23.2: Keiper Li coefficients from the Pólya density (all digits shown are correct) and Keiper's leading asymptotic, which becomes accurate only for large n.

23.6.2 Li's criterion as positivity of the Thorin functional

Li's criterion is a statement about the Thorin functional U of Section 8.2, which has a unit atom at $-(\rho - \frac{1}{2})^2$ for each pair $\{\rho, 1-\rho\}$ of zeros, on the line or off it. For a zero ρ put $r = 1 - 1/\rho$. Then $1 - 1/(1-\rho) = 1/r$, so the pair contributes $2 - r^n - r^{-n}$ to λ_n; and if $a = -(\rho - \frac{1}{2})^2$, with the square root chosen so that $\rho - \frac{1}{2} = i\sqrt{a}$, then $r = (2\sqrt{a} + i)/(2\sqrt{a} - i)$. This proves the following.

Proposition 23.6.1. *Without any hypothesis on the zeros,*

$$\lambda_n = \sum_{\{\rho,1-\rho\}} \varphi_n\Big(-\big(\rho-\tfrac12\big)^2\Big) = \langle U, \varphi_n\rangle,$$

where

$$\varphi_n(z) = 2 - \Big(\frac{2\sqrt{z}+i}{2\sqrt{z}-i}\Big)^n - \Big(\frac{2\sqrt{z}-i}{2\sqrt{z}+i}\Big)^n$$

is a rational function of z, and $\varphi_n(z) = 4\sin^2\big(n\arctan\frac{1}{2\sqrt{z}}\big) \ge 0$ for $z>0$.

The two fractions are exchanged when $\sqrt{z}$ changes sign, which is why φ_n is rational in z; it has a pole of order n at $z=-\frac14$. At infinity $\varphi_n(z)\sim n^2/z$, so the sum converges absolutely.

For $z>0$ the two fractions are conjugate numbers of modulus one. On the critical line, for $\rho=\frac12+i\gamma$ with $\gamma>0$, the number $1-1/\rho=(\rho-1)/\rho$ has modulus one and argument $\phi_\gamma = 2\arctan\frac{1}{2\gamma}$, and the conjugate zero gives $-\phi_\gamma$. Adding the contributions of ρ and $\bar\rho$,

$$\lambda_n = \sum_{\gamma>0} 2\big(1-\cos(n\phi_\gamma)\big) = \sum_{\gamma>0} 4\sin^2\Big(n\arctan\frac{1}{2\gamma}\Big) \qquad \text{under RH.}$$

Li's criterion thus says that RH holds if and only if the Thorin functional is nonnegative on the cone spanned by the φ_n.

The test functions are Laplace transforms, so Li's coefficients can be read off the heat trace W of Section 8.1 and off the clock tilted at $u=\frac14$ [145]. Let $q(y)=y/(1-y)^2$ and define polynomials ℓ_n by

$$\sum_{n\ge1}\ell_n(t)\,y^{n-1} = q'(y)\,e^{-tq(y)} = \frac{1+y}{(1-y)^3}\exp\Big(-\frac{ty}{(1-y)^2}\Big),$$

so that $\ell_1=1$, $\ell_2=4-t$, $\ell_3=9-6t+\frac12t^2$, and ℓ_n has degree $n-1$ and leading coefficient $(-1)^{n-1}/(n-1)!$. Let $\widetilde T$ be the Esscher tilt of the clock T at $u=\frac14$ (the variable $\widetilde T$ of Theorem 12.2.2), the law $e^{-x/4}P(T\in dx)/\mathbb{E}e^{-T/4}$, with Laplace transform $G(\frac14)/G(\frac14+u)$, and write $\kappa_j(\widetilde T)$ for its cumulants.

Proposition 23.6.2. *Without any hypothesis on the zeros:*

(i) $\varphi_n(z)=\int_0^\infty e^{-(z+\frac14)t}\ell_n(t)\,dt$ *for* $\operatorname{Re}z>-\frac14$, *and hence*

$$\lambda_n = \int_0^\infty W(t)\,e^{-t/4}\,\ell_n(t)\,dt;$$

(ii) $\kappa_j(\widetilde T)=\int_0^\infty t^{j-1}e^{-t/4}W(t)\,dt=(j-1)!\sum_{\{\rho,1-\rho\}}\big(\rho(1-\rho)\big)^{-j}$ *for* $j\ge1$, *and*

$$\lambda_n = \sum_{j=1}^{n}(-1)^{j+1}\frac{n}{j!}\binom{n+j-1}{2j-1}\kappa_j(\widetilde T), \qquad n\ge1.$$

Proof. (i) For $z>0$ put $r=(2\sqrt{z}+i)/(2\sqrt{z}-i)$, so that $\varphi_n(z)=2-r^n-r^{-n}$ and $r+r^{-1}=2-1/(z+\frac14)$. Then $(1-ry)(1-y/r)=(1-y)^2\big(z+\frac14+q(y)\big)/(z+\frac14)$, and for small $|y|$

$$\sum_{n\ge1}\varphi_n(z)\,y^{n-1}=\frac{d}{dy}\log\frac{(1-ry)(1-y/r)}{(1-y)^2}=\frac{q'(y)}{z+\frac14+q(y)}$$
$$=\int_0^\infty e^{-(z+\frac14)t}\,q'(y)e^{-tq(y)}\,dt.$$

Comparing coefficients gives the formula for $\varphi_n(z)$ when $z>0$, and both sides are analytic in $\operatorname{Re}z>-\frac14$. By Proposition 23.6.1, λ_n is the sum of $\varphi_n(a)$ over $a=-(\rho-\frac12)^2$, one for each pair $\{\rho,1-\rho\}$. For $\rho=\frac12+\delta+i\gamma$ with $|\delta|<\frac12$ one has $\operatorname{Re}a+\frac14>\gamma^2$, and $\int_0^\infty e^{-ct}|\ell_n(t)|\,dt=O(1/c)$ as $c\to\infty$, so the sum of the integrals converges absolutely, and $W(t)=\sum e^{-at}$ gives the formula for λ_n. (ii) Since $1+w=1/(1-y)$ for $w=y/(1-y)$, and $[y^n]w^k=\binom{n-1}{n-k}$, (23.6.1) says $\lambda_n=n[y^n]\log\xi\big(1/(1-y)\big)$. With $\sigma=1/(1-y)$ one has $\sigma(\sigma-1)=q(y)$, so $\sigma=\frac12+\big(\frac14+q(y)\big)^{1/2}$ and $\log\xi(\sigma)=\log\xi(\frac12)+\log G\big(\frac14+q(y)\big)$ near $y=0$. By Section 8.1,

$$\log\frac{G(\frac14+v)}{G(\frac14)}=-\log\mathbb{E}e^{-v\widetilde{T}}=\int_0^\infty\big(1-e^{-vt}\big)e^{-t/4}W(t)\,\frac{dt}{t}=\sum_{j\ge1}(-1)^{j+1}\kappa_j(\widetilde{T})\,\frac{v^j}{j!},$$

which gives the first formula for $\kappa_j(\widetilde{T})$; integrating $W(t)=\sum e^{-at}$ term by term, with $a+\frac14=\rho(1-\rho)$, gives the second. Finally $[y^n]q(y)^j=[y^{n-j}](1-y)^{-2j}=\binom{n+j-1}{2j-1}$. □

In terms of the sums $S_j=\sum\big(\rho(1-\rho)\big)^{-j}$, the relation (ii) is that of Voros [227, Section 8.6.2]; the proposition adds that $(j-1)!\,S_j$ are the cumulants of a probability law, the clock tilted to the point $u=\frac14$, which $\sigma=\frac12+\sqrt{u}$ sends to $\sigma=1$. The relation is triangular with diagonal $(-1)^{n+1}/(n-1)!$, so the λ_n and the cumulants of $\widetilde{T}$ determine each other; in particular $\mathbb{E}\widetilde{T}=\lambda_1$ and $\operatorname{Var}\widetilde{T}=4\lambda_1-\lambda_2=3.71006364\times10^{-5}$. From 100 independently computed cumulants, (ii) reproduces all 20 digits of Table 23.2 (the rows $n=1,2,3,5,10,20,50,100$), and (i), with W from (2.1.5) for $t<0.004$ and from the zeros above, reproduces $\lambda_1,\dots,\lambda_4$ to 30 digits (computed).

Under RH, $W(t)=\sum_{\gamma>0}e^{-\gamma^2t}$, and (i) returns the sum of the nonnegative terms $\varphi_n(\gamma^2)$ displayed above. The functional $f\mapsto\int_0^\infty f(t)e^{-t/4}\ell_n(t)\,dt$ is nonnegative on every completely monotone f for which it converges absolutely, by Bernstein's theorem and $\varphi_n\ge0$ on $[0,\infty)$. It is not nonnegative on nonnegative f, since $\ell_2<0$ on $(4,\infty)$, nor, in general, on k monotone f: $(1-t/20)_+^2$ is 3 monotone, and

$$\int_0^{20}e^{-t/4}\,\ell_3(t)\Big(1-\frac{t}{20}\Big)^2dt=\frac{4-2824e^{-5}}{25}<0.$$

Read on W, Li's criterion asks for the complete monotonicity of Section 8.1, which positivity of W does not give.

The test function φ_n is about n^2/z for $\sqrt{z} \gg n$ and oscillates between 0 and 4 for $\sqrt{z} \lesssim n/\pi$, so λ_n weighs the zeros below height of order n at full size and the higher zeros by about n^2/γ^2 (Figure 23.1). Every computed λ_n is positive, as it must be: every zero below height 3×10^{12} lies on the line [277] and contributes a nonnegative term, while a zero at height $\gamma \gg n$ contributes only about n^2/γ^2 in absolute value, so the zeros above that height cannot change the sign of λ_n for $n \le 100$.

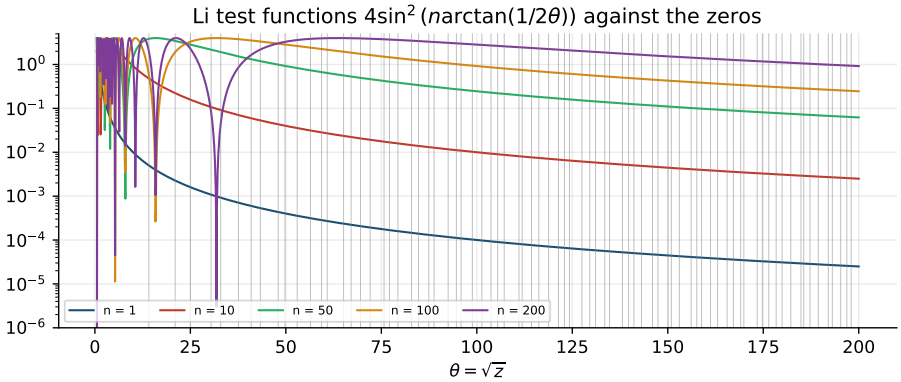


Figure 23.1: The Li test functions $4\sin^2(n \arctan \frac{1}{2\theta})$ for $n = 1, 10, 50, 100, 200$, in the square root chart $\theta = \sqrt{z}$, against the zeros below height 200 (grey vertical lines). Larger n probes higher zeros.

An off line zero makes $|1 - 1/\rho| \neq 1$. For $\rho = \frac{1}{2} - d + i\gamma$ with $d > 0$ one has $|1 - 1/\rho|^2 = 1 + 2d/((\frac{1}{2} - d)^2 + \gamma^2)$, so for large n the term $(1 - 1/\rho)^n$ grows like e^{nd/γ^2}, which would eventually make λ_n negative (Exercise 18.14). The index at which this happens grows with the height of the zero, so finitely many coefficients, like the finite tests of Section 1.4 (Table 1.3), cannot exclude an off line zero at great height.

Summing $4\sin^2(n \arctan \frac{1}{2\gamma})$ over the first 1700 zeros, computed with `mpmath` [237] as in Section 3.1, and adding the tail above γ_{1700} with the smooth zero density $\log(\gamma/2\pi)/2\pi$ gives the Pólya values with relative residuals near 5×10^{-6} for $n \le 20$ and 10^{-5} at $n = 100$. The absolute residual grows like n^2 and comes from the cutoff: at γ_{1700} the zero count exceeds its smooth part by about one half, so the smooth tail counts about half a term, $\frac{1}{2}\varphi_n(\gamma_{1700}^2) \approx n^2/(2\gamma_{1700}^2)$, too many.

As a diagnostic λ_n is poorly localised: it behaves like the heat trace and Bernstein diagnostics of Table 4.2 and unlike the Thorin density, which is local in the ordinate.

Weil's criterion has the same form as Li's, with the functions of the class $\mathcal{T}$ of

Section 8.2 that are nonnegative on $(0,\infty)$ in place of the φ_n (Section 5.1.2), but the two cones are different. The pole of φ_n lies at $-\frac{1}{4}$, the point at which (8.2.1) evaluates its test functions, inside the region where $\mathcal{T}$ requires holomorphy, and φ_n decays only like $1/z$. So φ_n is not in $\mathcal{T}$, $\langle U, \varphi_n \rangle$ is defined through the zeros, and the prime side of λ_n is the regularised arithmetic formula of Bombieri and Lagarias [37].

Part VI

Synthesis

Chapter 24

A route to the Riemann Hypothesis

This chapter assembles the results of the volume into one route to the Riemann Hypothesis, says which steps of it are theorems, and states the one step that is missing. It does not prove RH. Its purpose is to make the missing step as small and as precise as the book allows, and to compare the route with the one setting in which the Riemann Hypothesis is a theorem, curves over finite fields, where the same step is supplied by geometry.

The route is the descent in level of Section 1.4, read in the language of Lee and Yang. Its target has six equivalent forms, each proved equivalent to RH elsewhere in the book (Section 24.1). Its top, the level one or the coupling $\frac{1}{2}$, is a theorem in (T1), (T2), (T3), (T5) and (T6) (Section 24.2). The step from the top to the critical line passes three filters that rule out whole classes of argument (Section 24.3). For curves the step is Hasse's and Weil's positivity, and a computation shows how it decides each curve through finite sections (Section 24.4), and the runs of Section 24.6 carry the same finite sections for ζ, for the Davenport and Heilbronn function and for curves of genus up to 3 much further than the pilot. What is missing for ζ is a positive form built from the primes whose values are those of Weil's functional (Section 24.7). The chapter ends with a work plan of problems ordered by how much of the gap each would close.

24.1 The target in six forms

Let $L(s) = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, the moment generating function of the Pólya law p, and let $u_\theta(v) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\theta+iv)$ be the Thorin density at level θ. The following are equivalent to RH.

(T1) Descent in level. For every $\alpha \in (\frac{1}{2},1)$ and $t \neq 0$, $v_\alpha(t) > b_\alpha$; the minimum of the Thorin density at the origin persists from level one down to $\frac{1}{2}$ (Theorem 12.1.18).

(T2) Deconvolution. The level one density is the Cauchy smoothing of a positive measure at the full width $\frac{1}{2}$; equivalently the backward Cauchy flow

runs from level one to the line through positive measures (Theorem 8.2.2 of Volume II).

(T3) Cauchy mixture of one width. At one basepoint $\alpha \geq 1$, $v_\alpha/\pi = P_{\alpha-1/2} * \nu$ with $\nu \geq 0$ (Theorem 21.3.3).

(T4) Weil positivity. $\sum_\rho \widehat{h}(t_\rho) \geq 0$ for $h = g * \tilde{g}$, g smooth of compact support; it is the canonical measure of the sine squared lemma undamped to the line (Theorem 21.3.4, [357]).

(T5) Ferromagnetism. The Pólya law is ferromagnetic, its characteristic function in the Laguerre Pólya class (Chapter 20).

(T6) Two spin Lee and Yang property, or monotonicity. The Pólya spin coupled to one Ising spin is a Lee and Yang ferromagnet at every coupling $w > 0$; equivalently $|\xi(\sigma + it)|$ is nondecreasing in $\sigma \geq \frac{1}{2}$ for every t (Theorem 21.11.1, [333]).

Each form has a parameter that measures how far from the line the statement is known: the level θ in (T1) and (T2), the width in (T3), the support of the test function in (T4), the tilt w of the law $\cosh(wx)\,p$ in (T5), and the coupling w in (T6). In (T1), (T2), (T3), (T5) and (T6) the parameter at which the statement holds unconditionally corresponds to the line $\operatorname{Re} s = 1$, and in (T2), (T3) and (T6) lowering it is equivalent to a zero free half plane (Theorem 11.9.1 of Volume II, Corollary 21.5.2). The six forms are one statement in six coordinates.

24.2 What is proved at the top

The following hold with no hypothesis, and each is the top of the descent in one of the forms.

1. The Thorin density at level one is a positive Cauchy mixture, with its global minimum at the origin (Proposition 12.1.11); its atoms would be zeros on $\operatorname{Re} s = 1$, of which there are none.

2. The hyperbolic cosine tilted law $\cosh(wx)\,p$ is ferromagnetic for every $w \geq \frac{1}{2}$, and the Pólya spin coupled to an Ising spin is a Lee and Yang ferromagnet at every coupling $w \geq \frac{1}{2}$ (Theorem 21.4.1, Corollary 21.5.2).

3. Ferromagnetic couplings of symmetric Pólya spins never move the zeros of the partition function out of the strip of a single spin (Theorem 21.9.2), by Lieb and Sokal's Proposition 2.2 as restated by Kozitsky.

4. The two spin system built from the truncated Euler product E_N has its Lee and Yang threshold exactly at the coupling $\frac{1}{2}$ (Proposition 21.5.3).

5. At every level $\theta \in (\frac{1}{2}, 1)$ the Thorin measure is positive up to a fraction $O(T^{-(\theta-1/2)/4})$ of its mass on $(0, T]$ (Theorem 13.7.1 of Volume II).

6. The Pólya law lies within total variation 1.932×10^{-6} of a law shown to be ferromagnetic with a computer assisted positivity proof (Proposition 19.1.15 of Volume II).

7. Computations certify the descent in windows: Weil positivity to finite support lengths (Chapter 15 of Volume II), and the positivity of $u_{3/4}$ and $u_{7/8}$ on $[0, 1420]$ together with the zero count $N(T^*) = 1000$ (Chapter 11 of Volume II), extended in Section 24.6 to $[0, 10^4]$ at six levels from $\frac{7}{8}$ down to 0.51 and to $[0, 10^5]$ at $\frac{7}{8}$, $\frac{3}{4}$ and 0.6.

Item 4 is the reason the top sits where it does. The Euler product converges absolutely only on $\operatorname{Re} s > 1$, and its symmetrised truncation E_N of Proposition 21.5.3 has its zeros exactly on $\operatorname{Re} s = 0$ and $\operatorname{Re} s = 1$; heuristically, the unconditional results stop at the coupling $\frac{1}{2}$ because that is as far as finite products of Euler factors reach.

24.3 Three filters on the missing step

Section 1.4 states three constraints. Read as filters on a candidate for the missing step, they say the following.

1. The step cannot be a finite computation, nor any argument that sees only finitely many zeros: for each heat trace test of Theorem 1.4.1 of Volume II there is a perturbation of ξ with the same zeros below a large height that passes it. For curves the corresponding statement is false, since finitely many point counts determine the zeta function; this is the first place where the analogy breaks.

2. The step must use the Euler product, because the Davenport and Heilbronn function shares the functional equation, the Cauchy structure and the positivity of the trace, and has zeros off the line (Theorem 14.4.3 of Volume II).

3. The step must not rest on a property preserved by Gaussian damping (Proposition 23.2.3). In Lee and Yang terms, every general theorem available, from Lee and Yang to Newman and to Lieb and Sokal, preserves the property when ferromagnetic interactions are added; RH asks for one to be removed, and $\Lambda \geq 0$ [318] says that the self interaction of the de Bruijn Newman flow cannot be made negative.

The first filter excludes the computations of the books as proofs, while keeping them as refutations. The third excludes the strip Lee and Yang theorem of item 3

above as a proof: it adds couplings. What survives the three filters is a positivity statement, for all heights at once, that uses the primes.

24.4 The benchmark: curves over finite fields

For an elliptic curve the completed zeta function is a Frobenius spin, and the Hasse bound is its ferromagnetism (Proposition 21.11.2). Hasse's proof supplies the positivity from the degrees of isogenies, and in the language of (T4) it is Weil positivity on two point test functions. For a curve of genus g the same structure has g Frobenius spins, one for each pair of eigenvalues $\alpha_j, \bar{\alpha}_j$, and the analogue of Weil's functional is a Toeplitz form.

Proposition 24.4.1. *Let C be a smooth, projective, geometrically irreducible curve of genus g over $\mathbb{F}_q$ with N_n points over $\mathbb{F}_{q^n}$, and put $s_0 = 2g$ and $s_n = s_{-n} = (q^n + 1 - N_n)/q^{n/2} = \sum_{j=1}^{2g} \alpha_j^n / q^{n/2}$ for $n \geq 1$, the α_j being the $2g$ Frobenius eigenvalues. The Riemann Hypothesis for C, $|\alpha_j| = \sqrt{q}$ for every j, holds if and only if the Toeplitz matrices $(s_{k-l})_{0 \leq k,l < n}$ are positive semidefinite for every n.*

Proof. If $|\alpha_j| = \sqrt{q}$, write $\alpha_j = \sqrt{q}\, e^{i\theta_j}$; then $s_n = \sum_{j=1}^{2g} e^{in\theta_j}$ are the Fourier coefficients of the positive measure $\sum_{j=1}^{2g} \delta_{\theta_j}$ on the circle, and the Toeplitz matrices are positive semidefinite. Conversely, positive semidefinite Toeplitz matrices of every size have entries bounded by s_0 (already the 2×2 principal submatrices give this), so the sequence s_n is bounded; but if some $|\alpha_j| \neq \sqrt{q}$, the functional equation pairs α_j with q/α_j, one of the two has modulus above $\sqrt{q}$, and s_n is unbounded. (By the theorem of Carathéodory and Toeplitz, in its infinite form Herglotz's theorem, the positive semidefinite case is exactly that of the Fourier coefficients of a positive measure.) □

The finite sections of the Toeplitz form play the part of the windows of Weil positivity for ζ. For the curve $y^2 = x^5 + x + 1$ of genus 2 we counted points over $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$ for $p = 11, 13, 17, 19, 29, 31$ (the curve is singular modulo 3, 7 and 23, where the computation does not apply), which determines the numerator $1 + c_1 T + c_2 T^2 + p c_1 T^3 + p^2 T^4$ of the zeta function: for every one of these primes the four Frobenius eigenvalues have modulus $\sqrt{p}$ to machine precision, and the Toeplitz sections of sizes 2 to 40 are positive semidefinite, of rank 4 from size 4 on. A fake numerator with the same c_1 and c_2 moved outside the region the Riemann Hypothesis allows, $c_2 = 29$ or $c_2 = -26$ at $p = 13$, has eigenvalue moduli between $0.80\sqrt{p}$ and $1.26\sqrt{p}$, or $0.69\sqrt{p}$ and $1.46\sqrt{p}$, and its Toeplitz sections already fail at size 3, with smallest eigenvalue -2.2×10^4 and -4.5×10^6 at size 40 (Figure 24.1; computed).

For a curve, three facts make the method work, and each fails for ζ.

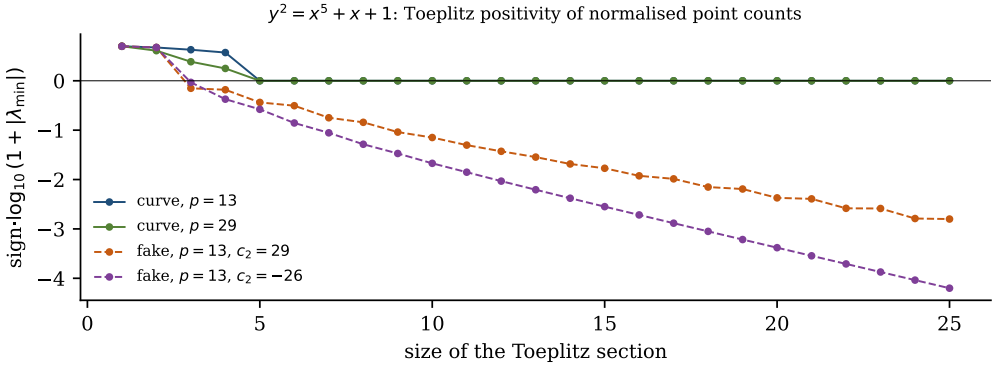


Figure 24.1: Smallest eigenvalue of the Toeplitz sections of the normalised point counts of $y^2 = x^5 + x + 1$ at $p = 13$ and $p = 29$, and of two fake numerators at $p = 13$, against the size of the section, on a signed logarithmic scale. The genuine curve stays at or above 0; the two fakes plotted fail from size 3 (computed).

1. The form is positive for a reason outside the zeta function: Weil's proof uses the Castelnuovo positivity of the intersection form on $C \times C$ [378], and its positivity is a theorem before any zero is located.

2. The form is finite dimensional in effect: the measure has $2g$ atoms, the Toeplitz matrices have rank $2g$, and a fake numerator fails at a small size. For ζ the measure has infinitely many atoms and every finite section is passed by a perturbation whose zeros off the line lie high enough (the first filter).

3. The spins are of exponential type, so the Lee and Yang theory applies to them as they stand. For ζ the Pólya spin has infinite type; Theorem 21.9.2 recovers the strip statement by approximation, but no approximation can supply positivity that the limit does not have.

24.5 Finite sections for ζ, shrinking and bounds

This section takes the benchmark one step further in three directions: how large a finite section must be for ζ, how much a ferromagnetic coupling shrinks the strip, and what the curve proof by bounds along a family would need.

The Hankel analogue of the Toeplitz test For a curve the Toeplitz sections of Proposition 24.4.1 catch the two fake numerators computed above at size 3. For ζ the analogue is the Hankel test of Proposition 19.22.1: with $M_j = \sum_k \gamma_k^{-2j}$, one atom for each pair of zeros, RH holds if and only if $\det[M_{i+j+1}]_{i,j<n} > 0$ and $\det[M_{i+j+2}]_{i,j<n} > 0$ for every order n. The moments are computable from ξ at

$\frac{1}{2}$, without the zeros, so this is a test of the same kind as the curve's. We replaced the ordinates γ_k, γ_{k+1} by the pair $\frac{1}{2}(\gamma_k+\gamma_{k+1}) \pm i\eta$ and recorded the least order at which a determinant fails, with the first 220 ordinates and the tail above them as the positive density $\frac{1}{2\pi}\log\frac{t}{2\pi}\,dt$ (computed, floating point at the precision stated).

k	height	order, $\eta = 0.3$	order, $\eta = 0.05$
1	17.58	4	5
2	23.02	6	7
3	27.72	8	9
5	35.26	11	13
8	45.67	17	19
12	57.90	24	26
17	70.81	32	34
24	88.12	44	47

At 260 digits the genuine moments pass every order up to 34, and from order 35 the test is below the working precision; at 700 digits they pass every order up to 54, and the orders for $k = 12$, 17 and 24 were confirmed or obtained there, including 47 for $k = 24$ and $\eta = 0.05$ (computed at 720 digits). The order needed to see a pair is about twice its index: $4, 6, 8, 11, 17, 24, 32, 44$ for $k = 1, 2, 3, 5, 8, 12, 17, 24$, slightly more when the pair is closer to the line; the runs of Section 24.6 give 1.7 per index to $k = 128$. Since the index of a pair at height T is about $N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi e}$, a Hankel section sees pairs off the line only up to the height where $2N(T)$ equals its order, and the working precision has to grow with the order as well (computed). For a curve a section of size $2g = 4$ catches any fake numerator with distinct roots; for ζ the size must grow without bound with the height, which is the first filter of Section 24.3 in numbers.

Shrinking under a Curie Weiss coupling Problem 24.3 has a quantitative answer when the coupling acts on the magnetisation.

Proposition 24.5.1. *Let $\chi_1, \dots, \chi_N$ be laws of Xi type in $\mathcal{C}_\delta$ whose densities are $O(e^{-|x|^\kappa})$ for some $\kappa > 2$, and couple them with the Curie Weiss weight $e^{\frac{a}{2}S^2}$, $S = \sum_i \sigma_i$, $a \geq 0$. In a common field the partition function $\mathbb{E}e^{\frac{a}{2}S^2+hS}$ has all its zeros in $|\operatorname{Re} h| \leq (\max(\delta^2 - a, 0))^{1/2}$.*

Proof. The moment generating function of S is $\prod_i \varphi_i$, whose zeros lie in the strip of half width δ, and the density of S is again $O(e^{-c|x|^\kappa})$ by convexity of $|x|^\kappa$. The weight $e^{\frac{a}{2}S^2}$ is de Bruijn's multiplier $e^{\lambda t^2}$ with $\lambda = \frac{a}{2}$, which maps a strip of half width Δ into one of half width $(\max(\Delta^2 - 2\lambda, 0))^{1/2}$ [73, Theorem 13], as in Proposition 19.1.2 of Volume II. □

The bound is sharp to first order for one spin with an isolated pair of zeros off the line: under the heat flow $\partial_a = \frac{1}{2}\partial_h^2$ that the multiplier induces on the moment

generating function, a zero moves with velocity $-\sum 1/(r-r')$ over the other zeros r', and for an isolated pair the pull of the mirror zero at distance 2δ gives $\frac{d}{da}\delta^2 = -1$ (heuristic). For N copies the zero has multiplicity N and the pull is N times larger, $\frac{d}{da}\delta^2 = -N$; a computation with two planted spins of strip 0.3 found δ^2 falling from 0.0900 to 0.0860, 0.0704 and 0.0524 at $a = 0.002$, 0.01 and 0.02, that is at rate 2.0, 2.0 and 1.9. The shrinking is a property of the coupled system; RH concerns the single spin, where the coupling would have to be removed.

Bounds along a family There is a second proof of the Riemann Hypothesis for curves, by Stepanov and Bombieri, which goes through bounds and not through positivity of a form. It proves $|N_n - q^n - 1| \le Cq^{n/2}$ for all large n by constructing an auxiliary function that vanishes to high order at the rational points, the fixed points of the Frobenius map $x \mapsto x^q$, and the Riemann Hypothesis follows because bounded normalised power sums s_n force $|\alpha_j| \le \sqrt{q}$, the argument of Proposition 24.4.1. The family is the tower of extensions $\mathbb{F}_{q^n}$. For ζ the family is the cutoff x, the analogue of the bound is $\psi(x) - x = O(x^{1/2}\log^2 x)$, which is equivalent to RH by von Koch's theorem, and the analogue of the growth argument is the passage from the error term to the zeros through the explicit formula. In the language of this book the bound $\psi(x) - x = O(x^{\theta+\varepsilon})$ is the positivity of the Thorin density at the level θ (Section 1.4). What Stepanov uses and ζ lacks is the Frobenius map: a polynomial map whose fixed points are the objects counted, which lets an auxiliary function vanish at all of them with controlled degree. No such map is known on the integers, and this is the same absence, seen through bounds instead of positivity, as that of Section 24.4.

24.6 The runs

The finite sections of the previous sections were computed at pilot size. This section reports four of the five work packages that extend them (the third, the field and the two spin test, is in Section 21.7), each from first principles (the matrices, explicit formulae and certificates were written for this book; none of the pilot code was used). Status words: certified means ball arithmetic throughout, with a sign proved by an LDL^T factorisation of balls or by a cell lemma; computed means floating point or a fit; the laws read off are computed or heuristic and prove nothing.

The Weil ladder The form is the Weil functional on the $N+1$ even cosine modes of $[1/\lambda, \lambda]$, $X = \lambda^2$. It was first checked against the explicit formula: the matrix reproduces $2\sum_k F(\gamma_k)^2$ over the first 1700 ordinates to 10^{-12} relative, and the pilot eigenvalues at fourteen rungs to eight digits (computed). Positivity was then certified on 266 cells (X, N), every one positive: a matrix of balls containing the exact matrix (Gauss and Legendre quadrature with a proved error bound,

prime sums as finite sums of balls), and a ball LDL^T with every pivot positive, with the least eigenvalue bracketed. The certified grid lies in $X \in \{2, \dots, 10^4\}$ against $N \in \{8, \dots, 200\}$ (not every pair), with X to 10^5 at $N \le 200$ and to 10^6 at $N \le 100$ on a sparse set, $N = 300$ and $N = 400$ to $X = 10^5$, and $N = 600$ at $X = 2, 5, 13, 50, 200, 10^3, 10^4, 10^5$. These certify Weil positivity on a finite dimensional subspace, not on all functions supported in $[1/\lambda, \lambda]$. The least eigenvalue ε is tiny, 9.5×10^{-54} at $(13, 40)$ and 9.3×10^{-106} at $(1000, 30)$, and falls by about 17 and 12 digits per doubling of λ on 50 to 200 and on 200 to 1000 (computed). The exponential decay in N seen in the pilot is a law of small N only: $-\log_{10} \varepsilon / N$ falls from 3.83 at $N = 20$ to 2.28 at $N = 200$ for $X = 1000$, at $X = 2$ the eigenvalue saturates at 1.329×10^{-3} already near $N = 100$, at $X = 13$ it still falls by 23 and 10 per cent from $N = 100$ to 200 and from 200 to 400, and at $X = 50$ it is still falling by 37 orders of magnitude from $N = 200$ to 400. So the decay is exponential to about $N = 40$ and no law beyond is claimed. The zero of the transform nearest γ_k is farther than ε, by 4 to 18 digits, and the distance falls like a power 1.05 to 1.17 of ε; a fit $\log_{10} \mathrm{err}_k = 6.03 + 1.168 \log_{10} \varepsilon - 0.289 \gamma_k + 0.604 \gamma_k L$ over all runs has rms residual 3.5 digits (computed). The odd modes decouple: all 36 cells (to $N = 60$, $X = 1000$) are positive (computed, sign certified).

Two replacements of ζ test the second filter of Section 24.3. First, the Davenport and Heilbronn function, with the explicit formula derived for it (no pole term, conductor 5, coefficients of $-f'/f$ on all integers) and validated against its zeros to $T = 200$, 122 on the line and four off it: the matrix and the zero sum agree to 10^{-15} relative on even modes and 2×10^{-12} on odd, and the off line zeros are needed for the agreement. The form is certified positive on every cell with $N \le 52$ and $X \le 2000$ (at $X = 30$ to $N = 80$), and first certified negative at $N = 56$ for $X = 50, 100$, at $N = 60$ for $X = 200, 500$, and at $N = 64$ for $X = 1000, 2000$. Two statements are supported. Proved, given the explicit formula for the Davenport and Heilbronn function: a negative section forces a zero of f off the line, since the line zeros contribute $|F(\gamma)|^2 \ge 0$. Computed, at $X = 50$ and 100: the decomposition over the zeros of f, at the least eigenvector, shows the quadruple of $0.8085 + 85.6993i$ supplying the negative term. At $X \ge 200$ the first negative cells carry no eigenvector data, and at $X = 2000$, $N = 60$ the decomposition does not close, so the mechanism is not shown there; nor does the size at which the form fails follow a simple rule in $\pi N / L$ ($X = 30$ stays positive to $N = 80$). The validation tests the off line terms only weakly: they move the test values by 10^{-14} to 4×10^{-10}, against residuals of 5×10^{-17} to 2×10^{-12}, and it does not reach the range $N \ge 52$ where that zero dominates. Second, cutting the prime sum of ζ at Y: the form is certified negative, with ε of order one for Y well below X and tiny just below the last primes (-1.1×10^{-69} at $Y = 983$, $X = 1000$), At $N = 20$ the largest Y with a negative form and the smallest with a positive one are 7 and 11 at $X = 13$, 47 and 49 at $X = 50$, 193 and 197 at $X = 200$, 983 and 991 at $X = 1000$ (certified signs).

The Thorin level The cell certificate of $u_\alpha > 0$ of Chapter 11 of Volume II, generalised to any level and height range, certifies that ξ has no zero on $\operatorname{Re} s = \alpha$ for $|t| \le 10^4$ at $\alpha = \frac{7}{8}, \frac{3}{4}, 0.6, 0.55, 0.52, 0.51$, and for $|t| \le 10^5$ at $\alpha = \frac{7}{8}, \frac{3}{4}, 0.6$ (no failing cell; 49×10^6 cells at $\frac{3}{4}$ on $[10^4, 10^5]$). The earlier certificate was $[0, 1420]$ at $\frac{3}{4}$ and $\frac{7}{8}$. An independent evaluation with `mpmath` agrees with the ball midpoints to 7×10^{-17} relative (computed). The least certified bound of $v_\alpha = \pi u_\alpha$ on a decade is the margin.

α	$[10^2, 10^3]$	$[10^3, 10^4]$	$[10^4, 10^5]$	ratio, $[10^3, 10^4]$
0.51	0.0094	0.0237	not run	2.377
0.52	0.0189	0.0475	not run	2.377
0.55	0.0474	0.118	not run	2.374
0.6	0.0947	0.236	0.402	2.367
$\frac{3}{4}$	0.234	0.580	0.978	2.321
$\frac{7}{8}$	0.347	0.846	1.413	2.257

The ratio of the margin to $\alpha - \frac{1}{2}$ is almost independent of α (the limit as $\alpha \downarrow \frac{1}{2}$ is the sum of $(v - \gamma)^{-2}$ over the zeros at the worst point, computed, assuming the nearby zeros are on the line), so the margin is linear in $\alpha - \frac{1}{2}$ uniformly in height, with a constant that grows with the height: at $\frac{3}{4}$ it is 0.105, 0.94, 2.32, 3.91 on the decades from $[10, 10^2]$ to $[10^4, 10^5]$, which is faster than $\log T$ and about $(\log T)^2/34$ on the last decades (computed, a description of five decades). The least value is at the same $v = 109.04$, 1069.30, 10677.94 at every level. The minimum over v is at the origin: above v^* (between 0.57 and 1.6 according to the level) every cell bound exceeds $u_\alpha(0)$ (certified), and on $(0, v^*)$ only the midpoints do (computed). The levels 0.55 to 0.51 above 10^4 were not run.

The Hankel order The Hankel test of Section 24.5 was rerun with all orders decided by a ball LDL^T without pivoting, with 2000 ordinates in balls of radius 10^{-719} and the tail above them the positive density, at 3400 and 6800 bits. The pilot orders $4, 6, 8, 11, 17, 24, 32, 44$ at $k = 1, 2, 3, 5, 8, 12, 17, 24$ are reproduced exactly, are the same with 220 or 2000 ordinates and with the ordinates cut to 300 digits, and each failing pivot is a ball below 0 (certified for the model measure; that the zeros above γ_N behave like the tail density is evidence, not proof). The orders continue $57, 86, 113, 168, 222$ at $k = 32, 48, 64, 96, 128$ ($\eta = 0.3$). They are linear in k with slope about 1.7, $n = 1.717k + 2.81$ (rms 0.46), the local slope falling from 2 to 1.69 and no saturation to $k = 128$ (computed); the width $\eta = 0.1$ costs at most three orders more. The genuine moments are positive definite to order 250 for the model measure (certified at 6800 bits), and beyond order about 100 the precision needed exceeds the 700 digits first used (inferred, the 700 digit run was not done).

Curves The Weil ladder was built for curves, by interval arithmetic on the numerator of the zeta function, with the explicit formula of Poisson summation giving the matrix $\ell[2g\,g(0) + 2\sum_m s_m g(m\ell)]$ on cosine modes. Points were counted by brute force on 69 curves of genus $1, 2, 3$ over every prime power $q \le 49$ (the numerators are exact integers and satisfy RH to 10^{-12}), and the ladder was run on 13 of them (genus $1, 2, 3$ at $q = 3, 5, 13, 49$ and $y^2 = x^5 + x + 1$ over $\mathbb{F}_{13}$) at $\tau = 2L/\log q$ from 2 to 32 and N from 2 to 32. Of 819 genuine rungs, 757 are certified positive, 6 positive in floating point without a certificate, and 56 are exact kernels (computed: the least eigenvalue vanishes to the working precision), the supersingular numerators $1 + 49T^2$ (genus 1, $q = 49$) and $1 + 125T^6$ (genus 3, $q = 5$), where the form is semidefinite with a nonzero kernel and strict positivity cannot hold. Of 693 rungs of fake numerators, 626 are certified negative; the first level at which a fake fails is $\tau = 2, 3, 4$ for genus $1, 2, 3$ (at $N = 2$ the genus 2 fakes first fail at $\tau = 4$; see the Toeplitz sizes 2 and 3 of Proposition 24.4.1, and for genus 3 the product of a genuine factor and a genus 2 fake), and the size of the failure grows like ρ^τ, with ρ the largest ratio $|\alpha|/\sqrt{q}$ (computed). Two laws are read off (computed). The smallest eigenvalue ε_N does not depend on N for $\tau \le 2g$ and decays in N for $\tau > 2g$, so the threshold is at $\tau = 2g$. Above it $\log_{10} \varepsilon_N$ is linear in N with a slope c that depends on τ and g only through $x = \tau/g$, not on q: $c = 0.95, 1.53, 2.28, 3.51, 5.23$ at $x = 3, 4, 6, 12, 32$ with spreads of at most 0.06, and $c \approx 4.05 \log_{10} x - 0.83$ for $x \ge 4$. The zeta ladder has the same form (heuristic): replacing the constant density of zeros of a curve, $g \log q/\pi$ per unit height, by $\frac{1}{2\pi} \log \frac{T}{2\pi}$ at the frequency $T_N = \pi N/L$ of the mode, the slopes read off the curve table agree with the computed zeta slopes within a few per cent in the middle of the range and are off by 0.1 to 0.7 at its ends, and the threshold falls at $L \approx 1.2$ as observed. This says the two margin laws have one form, and does not separate a proof for ζ from the curve case, where the margin is exponentially small as well.

What the runs change in the picture is quantitative. The finite sections of Weil's form for ζ are positive far beyond the pilot, with no sign of failure, while the Davenport and Heilbronn form, which lacks only the Euler product, fails at a size at which, in the cases examined, an off line zero supplies the negative term, and the truncated prime sum fails at once. The first of these does not prove RH (first filter), and the second and third suggest that what the positivity uses is the primes in full (heuristic).

24.7 The missing step

Problem 24.1. Construct, from the primes, a positive semidefinite Hermitian form $\mathcal{Q}$ on a space of test functions dense enough for Weil's criterion, such that $\mathcal{Q}(g) = \sum_\rho \widehat{g}(t_\rho)\overline{\widehat{g}(\bar{t}_\rho)}$ for every g in it. Equivalently, in the forms of Section 24.1: prove the monotonicity of $|\xi(\sigma + it)|$ in $\sigma \ge \frac{1}{2}$; or prove that the coupling of the

Pólya spin to an Ising spin can be lowered from $\frac{1}{2}$ to 0 without losing the Lee and Yang property.

For curves, $\mathcal{Q}$ is the intersection form and its positivity is the Hodge index theorem; for ζ, finding $\mathcal{Q}$ is the Hilbert and Pólya problem, and the programmes of Connes and Deninger look for the space on which it would live. Nothing in this book constructs it. What the book does is fix its properties: it must be infinite dimensional in an essential way (first filter), it must use the Euler product beyond its region of absolute convergence (second filter, and item 4 of Section 24.2), and it cannot come from a monotonicity under added interactions (third filter).

24.8 A work plan

The runs of Section 24.6 did the finite computations that this plan first asked for: the Weil ladder and its eigenvalue and zero accuracy, the Davenport and Heilbronn test, the truncated primes, the Thorin level to 10^4 and 10^5, the Hankel orders and the curve ladder. What they left is open and is the first problem below; the others are ordered by how much of the gap each would close, the first two after it would be new theorems with present methods, the last is RH.

Problem 24.2. The margin laws. Prove, at fixed windows X and bounded heights, a lower bound for the margin that the runs only observe: the least eigenvalue of Weil's form on N modes at the window X (exponential in N to $N = 40$, then saturating at small X), the Thorin margin linear in $\alpha - \frac{1}{2}$ with a constant that grows like $(\log T)^2$, and the curve margin $\varepsilon_N \approx (6.8\,x^{-4})^N$ with its threshold at $\tau = 2g$. A bound uniform in X or in the height would imply RH (by Weil's criterion for the form, and for the Thorin margin by the zero free half planes $\operatorname{Re} s > \alpha$ for every $\alpha > \frac{1}{2}$), so only the restricted form is a problem of this plan. Computationally open: the levels 0.51 to 0.55 above 10^4, the Weil grid beyond $N = 600$, and the Hankel orders beyond $k = 128$.

Problem 24.3. Quantitative strip shrinking. Theorem 21.9.2 says that ferromagnetic couplings never widen the strip, and Proposition 24.5.1 gives the shrinking $\delta^2 \mapsto \delta^2 - a$ for a Curie Weiss coupling of strength a. Prove a shrinking bound for pair couplings that do not act through the magnetisation, such as nearest neighbour chains, and prove the multiplicity rate $\delta^2 \mapsto \delta^2 - Na$ of Section 24.5, which is expected only for N identical copies and to first order.

Problem 24.4. The gap law for many pairs. For one pair off the line, Theorem 21.7.1 gives an exact criterion and the law $\eta - w^* = \frac{1}{2}\phi^2\eta^3(1 + O(\eta))$, so the one variable test, ferromagnetism of $\cosh(wx)\,p$, measures the distance of an isolated pair from the line to third order. Theorem 21.8.1 does this for clusters that are small compared with their distance from the other zeros: each cluster is seen through its own field, which includes the other clusters, among simple pairs the widest decide the threshold, and a double pair is missed by the fixed fraction

$1 - \cos\frac{\pi}{4}$ of its distance rather than to third order. What remains is clusters as wide as the spacing of their neighbours, so that a certificate for the tilt in a window says exactly how much of the strip it excludes; by Theorem 21.8.1 such a certificate can exclude at most the zeros with $\operatorname{Re} s > \frac{1}{2} + w/\cos\frac{\pi}{4}$ unless simplicity of the zeros is also certified.

Problem 24.5. A family along the Euler product. For a curve the point counts over all extensions $\mathbb{F}_{q^n}$ are governed by the same $2g$ eigenvalues, and positivity of the Toeplitz form propagates along n. Find a family of laws or spins indexed by an integer, built from the Euler product of ζ, whose Lee and Yang property, or Toeplitz positivity, can be propagated from one member to the next, and whose limit is the Pólya spin. Proposition 21.5.3 shows that finite truncations of the product are not such a family.

Problem 24.6. The first step below the top. Prove the two spin Lee and Yang property for every $w > w_0$, for one $w_0 < \frac{1}{2}$. By Corollary 21.5.2 the two spin form at every $w > w_0$ is the zero free half plane $\operatorname{Re} s > \frac{1}{2} + w_0$, the quasi Riemann hypothesis, announced at $w_0 = \frac{3}{8}$ in October 2026 and not refereed [257]; a proof by the methods of this book would be a proof of positivity from the primes at a level below one.

Problem 24.7. Problem 24.1.

24.9 Is there a path?

There is a route, and the book makes it precise: six equivalent forms of one statement, proved at the top in five of them, filtered by three constraints, and benchmarked against curves, where the same route is a proof, and with its finite sections computed far beyond the pilot size (Section 24.6), which changes the numbers and not the conclusion: they are positive and no finite section can prove RH (first filter). The missing step is the same in every form, a positivity built from the primes that holds for all heights at once. For curves it comes from geometry; for ζ no source is known. What the book adds to Weil's reduction of 1952 is that the step has been restated as that single statement in six forms, that the available theorems of statistical mechanics, which all add interactions, do not supply it (heuristic), and that its first fragment below the top, positivity at one level below one, is already the quasi Riemann hypothesis.

Chapter 25

Conclusions and open problems of the theory

The results of Chapters 1 to 14 can be placed on a single path, which runs along the basepoint from the region where positivity of the Thorin measure is a theorem to the centre, where it is RH.

Chapter 20 follows as a complement on the side of statistical mechanics, where RH is the statement that the Pólya law is ferromagnetic.

25.1 The Thorin approach in summary, and a path to RH

The path is a programme with its gap identified, not a proof.

25.1.1 The path along the basepoint

Figure 25.1 and Table 25.1 set out the path.

Section 15.2.4 draws the two conclusions that fix the shape of the path. At the basepoint one the constructions attached to the reciprocal Xi function are unconditional and explicit (Sections 12.1, 12.2 and 13.1, Table 14.3), with two exceptions: the Krein string representation of Corollary 12.2.3 assumes RH and simple zeros, and Lemma 12.2.4 gives the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$ only for a boundary bounded at infinity, which for the computed boundary is observed, not proved. Below the basepoint one each construction holds, up to a boundary case, exactly when no zero lies to the right of α (Theorems 12.2.5 and 13.1.3, Propositions 12.1.13 and 13.1.12), and the theory of Sato and Bondesson reaches nothing there (Section 14.1.5).

25.1.2 The open steps

By Theorem 2.1.2 ([282, Theorem 3.7 and Remark 3.10]), step 7 is exactly a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < 1$, a quasi Riemann hypothesis. Until 2026 none was known. In October 2026 OpenAI released a proof, produced with one of its

step	statement	status
1	The clock is infinitely divisible and self decomposable; the heat trace is positive and decreasing	proved (Theorem 8.1.1, which is Theorem A of [282]), with the verification of RH below height 1000 as input
2	The heat trace is log convex, its leading Hankel blocks of size up to 3×10^7 are positive definite, derivative signs hold to order 1.16×10^{25}	proved, with verification height as input
3	The reciprocal at basepoint $\alpha \geq 1$ is a GGC with Poisson Thorin density, and at $\alpha = 1$ its Lévy form, hitting time, scale mixture and completeness are explicit; for every $\alpha > \frac{1}{2}$ the shifted clock T_α exists and is an almost GGC	proved (Theorems 2.1.2, 12.2.5 and 14.3.2; Chapters 12 to 14), with the verification of RH below height 1000 as input to Theorem 12.2.5
4	The normalised reciprocal zeta function is infinitely divisible; the canonical function of the floor of the reciprocal Xi function is an arithmetic part plus a completely monotone archimedean part, and RH is the complete monotonicity of the sum	proved (Chapters 10 and 11)
5	The Thorin density is computable at every resolution, with an explicit bound for each source of error, and is positive at every sample point of every window computed; the runs do not yet add these bounds to the ball radii (Section 25.2.2)	computed (Chapters 2 to 8)
6	Zero density $\nu(\frac{1}{2}+\varepsilon, 200) = 0$ for every computed ε, down to $\varepsilon = 0.005$	computed (Section 3.3)
7	For some $\alpha < 1$ the shifted clock T_α is a GGC, equivalently, the half plane $\operatorname{Re} s > \alpha$ is zero free (Theorem 2.1.2); then for every $\alpha' > \alpha$ the measure $U_{\alpha'}$ is the Thorin measure of $T_{\alpha'}$ and is positive at all heights	announced for $\alpha = \frac{7}{8}$ in October 2026, with a formal proof in Lean, not yet refereed (Section 25.1.2); open for every $\alpha < \frac{7}{8}$
8	Positivity of U_α for every $\alpha > \frac{1}{2}$ (Lagarias), equivalently the Thorin condition	open; equivalent to RH

Table 25.1: The path, step by step.

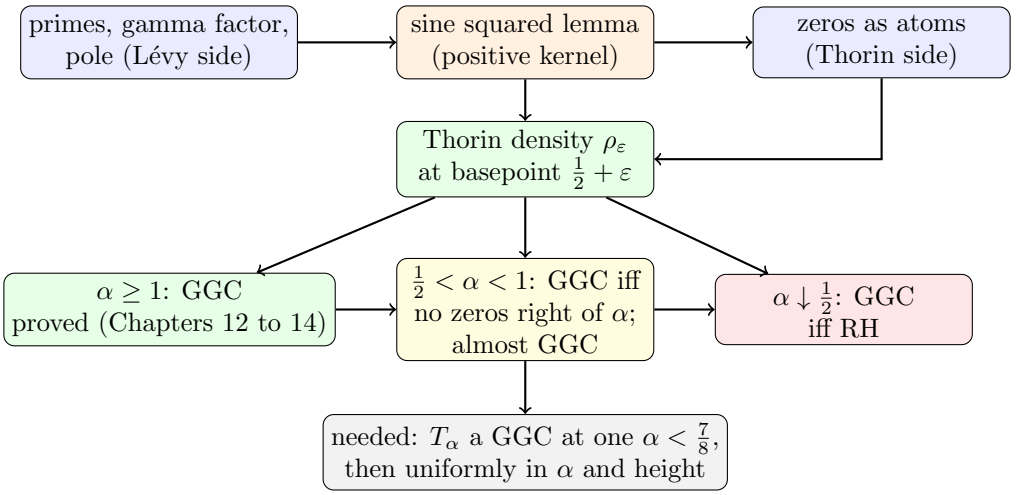


Figure 25.1: The Thorin path to RH. The sine squared lemma connects the arithmetic to the Thorin density. The programme must carry the GGC property across the strip.

AI models, that $\zeta(s) \neq 0$ for $\operatorname{Re} s > \frac{7}{8}$, and the same for every Dirichlet L function [257], together with a formal proof of both statements in the Lean proof assistant and a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$ [258]. The argument proves a power saving for smoothed sums of the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$ and uses nothing from the Thorin picture. It has not yet been refereed. We rebuilt the formal proof from the public repository (commit `adc7f12`, Lean 4.34.1, Mathlib `d13f23b`): the Lean kernel accepts both proofs, which use only the axioms `propext`, `Classical.choice` and `Quot.sound`, and the comparator of the Lean Focused Research Organization, run without its sandbox, confirms that the theorems proved are the published challenge statements. If it stands, step 7 holds with $\alpha = \frac{7}{8}$, and the following constructions of this book follow, conditional on that unrefereed result. By Theorem 2.1.2 the shifted clock T_α is a GGC for every $\alpha \geq \frac{7}{8}$, with Thorin density (2.1.6) and $v_\alpha > 0$ for $\alpha > \frac{7}{8}$; at $\alpha = \frac{7}{8}$ each zero $\frac{7}{8} + i\gamma$ with $\gamma > 0$, which the announcement does not exclude, would add an atom at γ^2 (Theorem 11.1.14); for $\alpha > \frac{7}{8}$ the clock T_α is a Brownian first passage time plus an independent GGC with no drift, because the first two steps of the proof of Proposition 12.1.12 go through for $\frac{7}{8} < \alpha < 1$, where every zero has $0 < \alpha - \beta < \alpha$, with smallest margins 0.0137 and 0.0157 in ball arithmetic; by Proposition 12.1.13 the compensated Lévy form converges for $\frac{7}{8} < \alpha < 1$, with error $O(X^{7/8-\alpha+\delta})$ at prime cutoff X for every $\delta > 0$, and likewise for $L(s, \chi_{12})$ (Section 15.2.2); by Theorem 13.2.7 Riesz's function satisfies $R(x) = O(x^{7/16+\delta})$ for every $\delta > 0$; and by de Bruijn's theorem $\Lambda \leq \frac{9}{32}$ (Section 4.1), weaker than the bound 0.2, and for the cosine tilts of Section 19.1.2 his theorem for cosine factors

gives $\Omega \leq \frac{3}{8}$, against $\frac{1}{2}$ with no hypothesis. Below $\frac{7}{8}$ nothing changes. Positivity of the Thorin density on the single line $\operatorname{Re} s = \alpha$ is a weaker statement than a zero free half plane (Remark 2.1.4). So the first theorem the Thorin path needs beyond the announcement, a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < \frac{7}{8}$, is an open problem in its own right; the Lévy form, the scale mixture, completeness and Riesz's criterion (Sections 12.1, 13.1 and 13.2) restate it and do not get round it. The Thorin formulation offers two tools for it. The sine squared form, compensated as in Proposition 12.1.13, writes the required positivity as domination of the pole piece by the prime and archimedean pieces at basepoint α, a statement about one explicit function of θ for each α; and the Poisson form shows a zero to the right of α as a Cauchy kernel of negative width, whose signature Section 3.2 calibrates exactly.

Step 8 carries the GGC property from a fixed α to every $\alpha > \frac{1}{2}$. Weak limits alone cannot do this (Section 9.1), because the limit of the Cauchy flow is positive whether or not RH holds. Three mechanisms could, each consistent with everything computed in this book:

1. a positive approximating family whose positivity is proved independently, such as zeta functions of Beurling generalised prime systems that provably have only real ordinates, finite Krein strings, or the partial products over verified zeros completed by the Weyl tail; once the first N zeros are verified, RH is equivalent to the GGC property of the tail factor left when their factors are removed from the clock [282, Corollary 18.11];

2. an operator whose spectrum is the set of ordinates, where positivity would follow from self adjointness, the Hilbert Pólya idea in Thorin form (Section 5.2.3);

3. a direct inequality for the variogram pairing that controls the pole by the primes and the gamma factor at every scale, the Weil form of step 8.

Spin chains give finite approximations of another kind: the partition functions of Knauf's chain are Dirichlet polynomials that converge to $\zeta(\beta-1)/\zeta(\beta)$ for $\operatorname{Re}\beta > 2$, and by Remark 20.6.4 the convergence of their corrected form on a strip in $1 < \operatorname{Re}\beta < 2$ would carry their zeros to the zeros $1+\rho$ of $\zeta(\beta-1)$ in it, a consequence that does not decide RH, while its convergence on $\operatorname{Re}\beta > \frac{1}{2}$ without the point 2, where the zeros of $\zeta(\beta)$ are poles of the limit, would imply it (Section 20.6).

25.1.3 What computation contributes

Computation cannot supply the uniformity in the height that steps 7 and 8 require: neither finitely many heat trace tests (Theorem 1.4.1) nor positivity at a fixed resolution (Section 1.4.2) can exclude an off line zero above the verified height.

What it can do (Section 25.2.1) is calibrate the positivity margins, which tells a proof how much room it has: the Thorin density at resolution ε has minimum $0.0147\,\varepsilon$ on the window below height 200 (Section 1.3), a margin that vanishes linearly and must be controlled uniformly. That this margin is the value at the origin is a theorem on the whole verified window, and at every height for every $\varepsilon < \frac{1}{2}$ it is equivalent to RH (Theorem 12.1.18).

The most concrete target for step 8 is the variogram form of the domination inequality (Section 5.1), on which the linear programming formulation of [282, Sections 18.2.8 and 18.2.9] works. Both are posed at the centre, where the variogram form is Weil's criterion and the linear program asks whether the clock $1/G$ is a GGC, so each is equivalent to RH. Section 9.1 and [282, Section 18.2.3] give the closure criterion in its weakest form, the target for approximating families in step 8. Section 14.3 closes with four questions between almost and exact positivity (Section 14.3.7), the last of which is step 7 below $\frac{7}{8}$, the GGC property at a single basepoint $\alpha < \frac{7}{8}$.

25.2 Open problems and lessons for the Thorin programme

The computations below aim at the gap located in Section 25.1, positivity uniform in the height; the lessons that follow are drawn for anyone working on the Thorin route to RH, including the first author's programme [282]. The open problems on the reciprocal zeta function are stated in Section 17.1.4, those on the basepoint one and the critical strip in Section 14.3.7, and those of Pakes's framework in Section 14.4.6.1.

25.2.1 What computation can and cannot do on the path

Computation cannot supply uniformity in height (Section 25.1.3), but it has four aims. It can certify positivity of the Thorin density on explicit windows, at every resolution, with interval arithmetic. It can test intermediate conjectures: the four questions of Section 14.3.7 all have numerically testable consequences. It can calibrate how much room positivity has, against random matrix predictions, so that a proof knows which inequalities are sharp. And it can cross check, computing the same quantities from the primes, through the criteria of Riesz and Salem and the compensated Lévy form, independently of evaluations of ζ.

There are five work packages: twisted Riesz functions as an arithmetic filter bank, the cross check (Section 13.2.1); certified Thorin densities at height (Section 25.2.2); the positivity margins against the circular unitary ensemble (Section 14.3.5); explicit tables of the almost GGC defect (Section 14.3.2); and testing stronger conjectures before pursuing them (Section 25.2.4).

25.2.2 Certified Thorin densities at height

At resolution $\varepsilon = \alpha - \frac{1}{2}$ the Thorin density $\rho_\varepsilon(\theta) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\theta)$ is a smooth function of θ. Its positivity on a window of ordinates is what the absence of zeros with $\beta > \alpha$ predicts there; since an off line zero need not make the density negative (Section 1.4.2), a certificate should also count such zeros. The first of two computations evaluates ρ_ε on a grid with rigorous error bounds, using the ball arithmetic Riemann Siegel and Euler Maclaurin evaluators of Arb [169], exposed in `python-flint`, at moderate height and the Pólya route where it is faster, and bounds the variation between grid points by the derivative. The second counts the zeros to the right of the line $\operatorname{Re} s = \alpha$ through the zero density identity $M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\nu(\frac{1}{2} + \varepsilon, T)$ of Proposition 3.3.2. This is the count of Section 3.3, which at $\varepsilon = \frac{1}{2}$ is Backlund's method. Section 3.3.3 followed the argument in steps of $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) without a bound on its change between grid points, so its counts are computed, not certified; with such a bound the count would certify a whole window. The deliverables are certified positivity of ρ_ε for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ on windows of length 10^4 at heights 10^6, 10^8 and 10^{10}, together with the minimum density on each window, which feeds the margin statistics of Section 14.3.5. Below the verified height 3×10^{12} the positivity itself needs no new computation: an off line pair makes the density negative only within $\frac{1}{2}$ of its ordinate, so $\rho_\varepsilon > 0$ on $[0, 3 \times 10^{12} - \frac{1}{2}]$ for every $\varepsilon > 0$ (Corollary 14.3.7); there the certificates test the evaluators against the verification and measure the margins.

Below height 200 the certificate is within reach. The ball arithmetic of Section 2.1 certifies only the rounding. Adding the discretisation bound of Section 2.1.1 and of Proposition 2.1.1, bounds for the truncations of the theta series and of the u axis, a bound on the change of the argument between grid points, and an explicit bound on the remainder of Proposition 3.1.2, which is $O(\varepsilon^3)$ (Remark 3.1.3), turns positivity of ρ_ε on $0 < z \le 200^2$ at each computed ε, the integer identity of Proposition 3.3.2 and the HCM derivative signs of Section 9.1.2 into theorems on that window.

Above that window the evaluator changes. The tilted contour gives 25 digits at height 5000 and resolves the Lehmer pair near height 7005 (Sections 2.1.3 and 2.2.3.1), but its number of nodes grows linearly with the height. Chirp transform batching of the Pólya route, and the hybrid of Section 2.2.2, a Riemann Siegel or Odlyzko Schönhage evaluator at $\frac{1}{2} + \varepsilon + i\theta$ with Odlyzko's zeros as the RH reference, would give the positivity margin, the matched total variation and the zero density scan near the 10^{12}th [252], 10^{21}st and 10^{22}nd zeros [254, 255]. The same zeros would measure the cell constant of Section 8.3 at large height, whose behaviour is open (Section 8.3.2), and allow a comparison of the cell errors with the prediction of the sine kernel process of the unitary ensemble.

25.2.3 Calibration against known configurations

The diagnostics of Chapters 3 to 8 are calibrated by running them on configurations whose zeros are known exactly. Section 10.8 ran the posterior predictive check of [282, Section 19.3.7] at $\varepsilon = 0.1$ on one window and calibrated it with a displaced zero. Three extensions are open: smaller resolutions, where the signature of a displaced zero is largest and the Pólya route supplies ρ_ε to about 60 digits below height 200 (Section 2.1.1), whereas the deconvolution used double precision; continuous atom positions, which would remove the baseline of the grid; and larger heights. The square root law (3.2.1) of Section 3.2.2, which follows from Proposition 3.2.1, sets what the check can detect.

The de Bruijn Newman flow of Section 4.1 manufactures off line pairs in a controlled way. Running the diagnostics along the flow through the collision of the Lehmer pair near 7005 would calibrate the zero density scan, the matched total variation and the negative mass.

Beurling generalised prime systems whose zeta functions provably have only real ordinates give positive Thorin measures by construction. Measuring how fast their clock transforms approach $1/G$, in the distances of Section 3.1 and along a sequence with an accumulation point as the closure criterion of Corollary 9.1.2 [282, Propositions 18.10 and 18.14] allows, is the computational side of that criterion; the Kent approximants of Section 5.2.3 are the first instance.

Two computations of Chapter 18 remain at their edges. The Brownian boundary of Section 3.1 reproduces the law of the clock in the bulk, to 0.0020 in Kolmogorov distance for the boundary of the Volterra equation and to 0.0059 for the tangent approximation of Roberts and Shortland (Section 3.1.2.2), but the zone of large times t is checked by no simulation, since only about one path in ten million survives past $t = 0.1$. Importance sampling with an exponentially tilted drift, or multilevel splitting, would check that zone and the logarithmic correction of the asymptote of the boundary directly. The Cauchy integral method of Section 23.6 extends to n in the thousands, at a cost of about $0.03n$ digits lost to cancellation. Comparing the Pólya values of the Keiper Li coefficients with the sums $\sum_{\gamma>0} 4\sin^2(n \arctan \frac{1}{2\gamma})$ over the zeros to large n would test how the Li functional sees the zeros near height n.

25.2.4 Testing stronger conjectures before pursuing them

Conrey and Li showed that a positivity condition in de Branges' approach to RH is false even though RH may be true [61], so any intermediate statement proposed on the Thorin path should be tested numerically first. The candidates are a lower bound $\rho_{\alpha-1/2}(\theta) \geq c(\alpha) \log \theta$ outside a sparse set, the first of the four questions of Section 14.3.7; monotonicity or convexity of the margin in α; positivity of the Thorin density for α slightly below one, uniformly on large windows; hyperbolic complete monotonicity of the Laplace transform φ_α of T_α below one; and var-

iogram inequalities at single scales. The first two are partly settled: the lower bound holds for $\rho_{\alpha-1/2}$ itself outside a set of heights of density zero, although not near the zeros where Lemma 14.3.9 would need it (Remark 12.1.17), and the margin divided by $\alpha-\frac{1}{2}$ decreases in α, with the ratio $\rho_{\alpha-1/2}(\theta)/(\alpha-\frac{1}{2})$ nonincreasing in α at every height outside the windows of the zeros off the line (Theorems 12.1.15 and 12.1.18). Each can be evaluated on the computed windows and at the heights of Odlyzko's tables, and discarded if it fails there.

The variogram inequalities can be tested in finite dimension: the reformulation of Section 5.1 poses feasibility problems: is the regularised pole mass dominated by the prime and archimedean mass on the cone of variograms of positive definite functions of a given dimension? Solving them with explicit dual certificates at increasing dimension would measure how the margin behaves.

The cross checks test the zeros themselves. Agreement of the twisted Riesz functions (Proposition 13.2.1) with the explicit formula over the computed zeros is a check that shares no code or method with the computation of the zeros (Remark 13.2.2).

25.2.5 Open problems from free probability

The note [291] closes with eight problems. Three are settled in the book: the positivity of $\widehat{U}$ and the infinite divisibility of k by Theorem 8.1.1 through Theorem 14.2.3, and the translation of the Gamma(2) structure by Propositions 14.2.7 and 14.2.9 with Theorem 14.2.10, up to the residual question of Problem 25.2. Two are taken up by sections of the book, the failure curve of the Hankel test in Section 4.1.3 and the collision brackets in Section 4.1.5, and what remains open there is stated in Problems 25.4 and 25.5. Three are open: Problems 25.1 and 25.3, and the last one, stated after them.

Problem 25.1. Jensen polynomials as finite free products. The multiplicative convolution $\boxtimes_d$ multiplies normalised coefficients and keeps roots real and of one sign (Lemma 14.2.8(i)), and the normalised coefficients of the reversed Jensen polynomial $x^d J_{d,n}(1/x)$ are, up to the signs $(-1)^i$, the numbers $(n+i)!\,b_{n+i}$, $0 \le i \le d$. A factorisation $(n+i)!\,b_{n+i} = \alpha_i \beta_i$ into the normalised coefficients of two polynomials with only real roots of one sign would therefore give real zeros of $J_{d,n}$. Find such factorisations for every shift n at a fixed degree d, or show that none exists with factors of a given kind. The obvious split, a Gaussian multiplier sequence q^{i^2}, $0 < q \le 1$, times a residual close to the coefficients of $(1+x)^d$, fails, because a root of multiplicity d splits into nonreal roots under a generic perturbation; and by Proposition 14.2.9 the factor $1/(a)_i$ can be multiplied in but not divided out. A factorisation has to put real structure into both factors [291].

Problem 25.2. Laguerre and Meixner Pollaczek expansions. Theorem 14.2.10 excludes scale mixtures of gamma laws for $|X|$ and X^2, and Proposition 14.2.9 shows that removing a gamma layer keeps zeros real while adding one does not.

Find a variable in which the Pólya density, or Ξ itself, has an expansion in Laguerre or Meixner Pollaczek polynomials with positive coefficients and truncations with only real zeros [291]. The symmetric Meixner Pollaczek expansions of Section 10.2.3.2 converge slowly, and a real rooted construction from their coefficients is a problem of Section 17.1.4.

Problem 25.3. Jacobi parameters. Under RH, where ν is a positive measure and all its Jacobi parameters exist, prove that $\alpha_n \sim 2\beta_n \sim \frac{1}{2}L_n$, with L_n given by (14.2.13) for the smooth count: a spectral form of the Riemann von Mangoldt formula. Table 14.7 supports it to a few per cent for $n \leq 49$, but the local model of Section 14.2.4 is a heuristic, outside the scaling of [195]. Decide between the rules $n(L_n) = n$ and $n(L'_n) = n - \frac{1}{2}$, which the data do not separate, and explain the scatter of the ratios, about six per cent, through the individual gaps between the zeros, as [291] proposes.

Problem 25.4. The order at which a collision is seen. Section 4.1.3 computes the failure times t_N^* of the Hankel test along the flow to $N = 50$ and explains their steps by collisions; the rule of [291] that the collision of (γ_k, γ_{k+1}) is registered from order $2k + 2$ on fails at $N = 50$ (Table 4.3). Find the order at which a pair is registered as a function of the index k of the pair and of its distance from the axis, and follow the staircase of t_N^* to orders of a few hundred, towards the limit 0 that it has under RH.

Problem 25.5. Unconditional collision brackets. Section 4.1.5 brackets the collision times by quantities along the flow (Corollaries 4.1.13 and 4.1.15) and through the nonreal zeros (Theorems 4.1.12 and 4.1.14), and checks the hypotheses on a grid of times for the 58 collisions of the census. Find the a priori control of the nonreal zeros of H_t, $t < 0$, asked for in Problem 4.1, which would make the brackets statements about H_t computed from H_0 alone.

Problem 25.6. Moving the ladder without new zeros. Theorems 8.1.9 and 8.1.13 prove the derivative signs $(-1)^k W^{(k)} > 0$ on ranges that grow with the height to which RH is verified (Section 8.1.6), and complete monotonicity of W on $(0, \infty)$ is RH. Find a mechanism that extends these ranges with no new information on the zeros, for instance a functional equation for the law of T under Υ, as [291] suggests. Theorem 14.2.3(c) restricts the form such an equation can take: T is not the Upsilon image of a law in BO, and under RH it is the image of the law μ_0, which is neither a GGC nor in BO.

25.2.6 Lessons from the computations

Each lesson is a statement that a computation in the book established or refuted.

Equivalent statements are not equally computable. The Thorin condition has at least eight equivalent faces: positivity of the Thorin measure, complete monotonicity of the heat trace, Stieltjes form of G'/G, Hankel positivity at one time,

HCM of the clock, Lagarias positivity, Weil positivity and Li positivity. They are the same statement mathematically and differ computationally by many orders of magnitude in conditioning. The heat trace at moderate time loses about $87t$ digits to cancellation (Section 8.2); the normalised minors of the even Hankel block at $t = 0.05$ fall to 10^{-44} by size five (Section 8.1); the Keiper Li coefficients weight a zero at height γ by n^2/γ^2 until n reaches γ (Section 23.6); the Thorin density at resolution ε is local in the ordinate and sees each zero at full size $1/\varepsilon$ (Section 3.1). A computational attack should choose its face by conditioning; the local face is the Thorin density.

The primes compute the wrong end: the sine squared lemma is the right way to understand the structure and the wrong way to compute near the line, where k digits at resolution ε need primes up to about $10^{2k/\varepsilon}$ even when no zero lies to the right of the line (Section 5.1.1). Any argument that bounds prime sums to establish positivity near the line must overcome that exponential, and Section 11.1 shows the same obstruction in another form (Proposition 11.1.2).

The heat trace meets the same obstruction. By Proposition 8.1.11, RH is the positivity of the Gaussian smoothing $\mathcal{G}(u,t)$ of the zeros at every height u and every $t > 0$, and (8.1.3) writes the symmetric smoothing $\mathcal{G}(u,t) + \mathcal{G}(-u,t)$ as a polar term plus an archimedean term $A(u,t)$ minus a prime sum $P(u,t)$. Bounding $|\cos(u \log n)|$ by one bounds $P(u,t)$ by $S(t)/\sqrt{2\pi t}$, where $S(t) = \sum_{n \geq 2} \Lambda(n) n^{-1/2} e^{-(\log n)^2/8t}$ is, up to a factor, the prime term of (2.1.5) at time $2t$, which proves $\mathcal{G}(u,t) + \mathcal{G}(-u,t) > 0$ for t below the time $t_0(u)$ at which the bound meets the archimedean and polar terms, which agrees to five digits with the root of $2S(t) = \log(u/2\pi)$ (computed): t_0 is 0.45 at height 10^3, 1.72 at the verified height 3×10^{12} and 3.64 at 10^{50}, to the digits shown (proved in ball arithmetic), and it grows only doubly logarithmically in u, since $S(t)$ is asymptotic to $\sqrt{8\pi t}\, e^{t/2}$. These resolutions are too coarse to see the quadruples of Table 1.2: at their heights t_0 lies between 0.311 and 0.511, while within 8 of each moved pair they make $\mathcal{G}$ negative only for $t > 1.18$ (both proved in ball arithmetic). No bound on P uniform in the height can do better, since by Dirichlet's theorem on simultaneous approximation the phases $u \log n$ of any finite set of prime powers come arbitrarily close to multiples of 2π at arbitrarily large u, where $P(u,t)$ therefore comes arbitrarily close to $S(t)/\sqrt{2\pi t}$. Nor does positivity at a single resolution give a zero free strip: at fixed t the bound $S(t)/\sqrt{2\pi t}$ on P proves $\mathcal{G}(u,t) + \mathcal{G}(-u,t) > 0$ at every height u with $t_0(u) > t$, that is above about $2\pi e^{2S(t)}$, which is 8.1×10^{212} at $t = 6$, and $\mathcal{G}(u,6) > 0$ itself for every $u \geq 10^{213}$ (proved in ball arithmetic, with the bound on $|\mathcal{G}(-u,6)|$ above), whatever the zeros are, so a strip needs t to grow with the height. A proof along the heat trace must use the phases, uniformly in the height.

Limits do not create positivity: it must be proved at every resolution, not in the limit. A proposal that reaches the Thorin condition as a limit must say where the positivity of its approximants comes from.

The perturbation theorem gives a test that a proposed proof of RH must pass. Replace ξ by the function F of Theorem 1.4.1, with R above every height that the proof uses explicitly. A step that holds for F as well as for ξ cannot exclude zeros off the line, because F has four. Such steps include the Riemann von Mangoldt formula with an explicit error, which F satisfies with the error enlarged by one on $[A, B)$, the only heights where the two counts differ; the verified zeros below R; the small time expansion of the heat trace through the constant term, since $W_F - W = \big(\frac{1}{2}(B-A)^2 + \frac{1}{8}\big)t + O(t^2)$; infinite divisibility, self decomposability and log convexity; the Hankel blocks of size N when $R \geq T_N$ and the derivative signs up to order 10^{17} when $R \geq 10^{12}$; and the order, type and functional equation. What separates ξ from F is arithmetic, the Euler product and the explicit formula with the prime weights $\Lambda(n)$, together with local counts of the zeros on the line in windows that contain the height g of the quadruple (Section 1.4.1). A proof through the derivative signs can do without them only below the order at which the signs of F first fail, which is about $t^* R^2$, with t^* between 1.18 and 4.03 in the cases computed below height 2000 and, by the model of Section 1.4.1, of order one or more at larger heights, far beyond 10^{17}. The function F exists unconditionally when $R + 50 \leq 3 \times 10^{12}$; for larger R the test still applies, because a correct proof of RH makes the hypothesis of Theorem 1.4.1(b) true for every R. The criticality filter of Section 14.4.5 is the companion test for properties of the Pólya kernel: a step that holds for $e^{tv^2/4}\Phi(v)$ with some $t < 0$, whose transform has nonreal zeros by the theorem of Rodgers and Tao, cannot be the step that excludes zeros off the line.

Appendix A

Notation

The tables list the notation used in more than one section; the first five cover the book and record second meanings that some letters carry locally. Sums and products over k run over the pairs $\{\rho, 1-\rho\}$.

symbol	meaning	defined in
ξ, Ξ	the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$	Section 2.1
ρ, γ_k	a zero $\rho = \beta + i\gamma$ of ξ, that is a nontrivial zero of ζ; the ordinates $0 < \gamma_1 \le \gamma_2 \le \cdots$ of the zeros with $\gamma > 0$	Section 2.1
a_k	$-(\rho_k - \frac{1}{2})^2$, one for each pair $\{\rho, 1-\rho\}$, equal to γ_k^2 on the line (in Section 23.6, the Taylor coefficients of $\log \xi(1+w)$)	(2.1.1)
Φ, p, X	the Pólya kernel in Titchmarsh's normalisation; the Pólya density $p = \Phi/\Xi(0)$; a random variable with density p (in Section 10.2, p is the Meixner density; in Section 9.3, X is the clock variable; in Sections 12.1 and 15.2, X is the prime cutoff)	(2.1.3), (2.1.4)
S_2, Y, J	$S_2 = \sum_{k\ge1} \frac{2}{\pi^2 k^2}(e_k + e_k')$ with independent standard exponential variables, the sum of two independent copies of the first passage time J of the level 1 by a three dimensional Bessel process started at 0; $Y = \frac{1}{2}\log(\pi S_2/2)$, with $\mathbb{E}e^{sY} = 2\xi(s)$; X is Y tilted by $e^{Y/2}$	Section 2.1, Section 12.1.1
u, z, θ	the variable $u = s^2$ of the clock (also the variable of Φ); the variable of Thorin measures; the chart $\theta = \sqrt{z}$, which is also the ordinate (the Kent approximants of Section 5.2 write w for u)	Section 1.1
$G(u)$	$\xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$; the clock is $1/G$ (in Section 11.1.3, G is a Laguerre Pólya function)	Section 1.1

symbol	meaning	defined in
$\mathcal{G}(u,t)$	the Gaussian smoothing $\sum e^{-2t(\tilde{\gamma}-u)^2}$ of the zeros at resolution $1/(2\sqrt{t})$, with $\tilde{\gamma}=\gamma-i\delta$ for a zero $\frac{1}{2}+\delta+i\gamma$ with $\gamma>0$	Section 8.1.5
T, e_γ	the clock variable, with $\mathbb{E}e^{-uT}=1/G(u)$, which exists with no hypothesis and is the centre clock of Chapters 12 to 14; under RH, $T=\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables e_γ (T is also a height, as in $N(T)$)	Section 2.1, Proposition 2.1.1
U, U_α	the Thorin measure of a GGC H, with $\mathbb{E}e^{-uH}=\exp\{-du-\int\log(1+u/z)\,U(dz)\}$ and drift d; the signed measure $U_\alpha(dz)=v_\alpha(\sqrt{z})\,dz/(2\pi\sqrt{z})$ at a basepoint α; for the clock, the Thorin functional $\sum_k \delta_{a_k}$	Section 2.1, (2.1.6), Section 8.2.1
k	the canonical function of an infinitely divisible law on $[0,\infty)$ with Lévy density $k(t)/t$ ($k(y)$ in Sections 8.1 and 10.2)	Section 4.1

symbol	meaning	defined in
U_{RH}, U_{line}, $U^{\pm}_{\mathrm{RH}}$	$\sum_{\gamma>0}\delta_{\gamma^2}$; the same sum over the zeros on the line; $\sum_{\gamma>0}(\delta_\gamma+\delta_{-\gamma})$	(1.1.1), Proposition 2.1.3
$\mathcal{W}$, $T_{\mathcal{W}}$, $U_{\mathcal{W}}$, U	the principal branch of Lambert's function, $\mathcal{W}(u)e^{\mathcal{W}(u)}=u$ (not the heat trace W); the Lambert law, with Laplace transform $\mathcal{W}(u)/u=e^{-\mathcal{W}(u)}$; its Thorin measure; a variable uniform on $(0,1)$ (not the Thorin functional U)	Sections 14.4.1, 14.4.3
φ_α, T_α	$\xi(\alpha)/\xi(\alpha+\sqrt{u})$, the clock at a basepoint $\alpha\geq\frac{1}{2}$, with $\varphi_{1/2}=1/G$; the variable with $\mathbb{E}e^{-uT_\alpha}=\varphi_\alpha(u)$, which exists for every $\alpha>\frac{1}{2}$ and is a GGC if and only if no zero has $\beta>\alpha$, in particular for every $\alpha\geq 1$; $T_{1/2}=T$	Theorems 2.1.2, 12.2.5
$\widetilde{T}$, $\widetilde{T}_\alpha$, $\sigma^{(\alpha)}$	the centre clock tilted by $e^{-T/4}$, and by $e^{-(\alpha-\frac{1}{2})^2T}$, so that $\widetilde{T}=\widetilde{T}_1$; the subordinator with Laplace exponent $u+2(\alpha-\frac{1}{2})\sqrt{u}$, with $T_\alpha\stackrel{d}{=}\sigma^{(\alpha)}_{\widetilde{T}_\alpha}$	Theorems 12.2.2, 12.2.5
H_a, $H_a^{(c)}$	the first passage time of B_t+ct to the level $a\geq 0$, for a standard Brownian motion B, and $H_a=H_a^{(0)}$; unrelated to H_t	Section 12.2

symbol	meaning	defined in
Y_α, Z	the normal scale mixture $Y_\alpha = \sqrt{2T_\alpha}\, Z$, with Z standard normal and independent of T_α, whose characteristic function is $\xi(\alpha)/\xi(\alpha + \lvert u\rvert)$ (in Section 14.1, Z is the subordinator with $T \stackrel{d}{=} \int_0^\infty e^{-t}\, dZ_t$)	Proposition 13.1.1
$W(t)$, $c_k(t)$	the heat trace $\sum_k e^{-a_k t}$, also written $\int e^{-tz} U(dz)$; the signed derivatives $(-1)^k W^{(k)}(t)$ (in Chapters 10 and 11 the time is x, or y where x is a Lévy variable, as in the proof of Lemma 10.2.5 and from Proposition 10.2.16 to the end of Section 10.2; the numbers $c_k(u)$ of Section 9.1 are local to it)	Sections 2.1, 8.1
m_n, $\mathcal{H}_N$, $\mathcal{H}'_N$, ι_N, ι'_N	the Stieltjes moments $\sum_k a_k^{-n-1}$, with $m_0 = \mathbb{E}T$; the Hankel matrices $[m_{i+j}]_{i,j<N}$ and $[m_{i+j+1}]_{i,j<N}$; the numbers of their negative eigenvalues, which count the zeros off the line (Section 4.1 uses the same symbols for the moments $m_n(t)$ of H_t)	Section 8.1, Theorem 8.1.7
$W^{[j]}$, β_j, B_j	the ladder functions $(-t\frac{d}{dt})^j W$, with $W^{[0]} = W$ (along the de Bruijn Newman flow $W^{[j]}(x;t)$, with x the time of the heat trace and t that of the flow); the polynomials with $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)e^{-at}$; the Touchard polynomials, $\beta_j(u) = (-1)^j B_j(-u)$	Section 14.4.4
L_n, L_∞	Urbanik's classes of multiply self decomposable laws, with L_0 the self decomposable laws; unrelated to the Post Widder functionals $L_n[W]$ of Section 17.1.2	Section 14.4.4.1
$Z^{[j]}$	the jth driving process of the centre clock, $Z^{[1]} = Z$, with Laplace exponent $(u\frac{d}{du})^j \log G$	Lemma 14.4.7
g, g_k, H, λ^*	the candidate generating exponent $\Psi_Z \circ H$ and its Taylor coefficients, with H the inverse of $\log G$; the branch point $\log\lvert G(u^*)\rvert + i\pi$ of H	Section 14.4.4.4
Φ_t, $\widehat{g}$, R_g	the damped Pólya kernel $e^{tv^2/4}\Phi(v)$; the cosine transform $\int g(v)\cos(\theta v)\, dv$ of an even kernel; $\widehat{g}(0)/\widehat{g}(i\sqrt{s})$, with $R_{\Phi_t} = R_t$	Section 14.4.5.1
X_t, p_t, Q_t, ϱ_t, h_t	the variable with density $p_t = \Phi_t/\Xi_t(0)$, $X_0 = X$; the radius of Khinchine's mixture, $X_t = UQ_t$; the radius of the arcsine mixture, $X_t = \varrho_t \cos\Theta$, and its density	Section 14.4.5.2
$\mathcal{B}$, f, $\mathcal{B}_g$, $\mathcal{B}_t$	$\frac{\pi}{2}S_2$, with $\mathbb{E}\mathcal{B}^{s/2} = 2\xi(s)$; its density $b^{-5/4}\Phi(\frac{1}{2}\log b)$; the variable with density proportional to $b^{-5/4}g(\frac{1}{2}\log b)$; $\mathcal{B}_{\Phi_t}$	Section 14.4.5.3
ℓ, $\kappa_k(\lambda)$, $c_k(\lambda)$, σ_λ	a Laplace transform $\mathbb{E}e^{-\lambda\mathcal{V}}$; the cumulants $(-1)^k(\log\ell)^{(k)}(\lambda)$; $\kappa_{k+1}(\lambda)/k!$, the moments of σ_λ for a GGC (not the $c_k(t)$ of the heat trace)	(14.4.9)

symbol	meaning	defined in
$\mathcal{K}_n(\lambda)$, $\mathcal{K}'_n(\lambda)$	the Hankel matrices $[c_{i+j}(\lambda)]_{i,j<n}$ and $[c_{i+j+1}(\lambda)]_{i,j<n}$, positive semidefinite for a GGC	Section 14.4.5.3
ℓ_t, $\dot{\ell}$, $\mathcal{J}$	$\mathbb{E}e^{-\lambda\mathcal{B}_t}$; its derivative in t at $t=0$, up to a term in ℓ_0; the first order change of the Thorin measure in a gap, in the proof of Proposition 14.4.15	Section 14.4.5.3
ϑ_a, $\xi(s,a)$, K_a, $\mathcal{B}_a$	the theta series $\sum n^p a_n e^{-\pi n^2 x/q}$ of a periodic sequence; its completed Dirichlet series; the kernel $2e^{(p+1/2)v}\vartheta_a(e^{2v})$; $\mathcal{B}_{K_a}$ (for $a=\chi_{-4}$, $1/\mathcal{B}_a$ is the variable Z of Remark 15.1.23)	Section 14.4.5.4
χ_d, S_a, Σ_a	the real primitive character of discriminant d ($\chi_{-4}=\chi_4$); the positivity sum of Lemma 14.4.17; the numerator of the reflected transform (14.4.10)	Section 14.4.5.4
f_{DH}, $\varkappa$, f_r	the function of Davenport and Heilbronn; its coefficient 0.28407904…; the self dual family modulo 7	Section 14.4.5.4
k_n	the inverse Laplace transform of Li's φ_n, $\lambda_n = \int k_n W$	Problem 14.4

symbol	meaning	defined in
ε	the resolution; the basepoint is $\frac{1}{2}+\varepsilon$	Section 1.1
$v_\alpha(r)$	$\mathrm{Re}(\xi'/\xi)(\alpha+ir)$, which is π times the density of U_α in the chart $\theta=\sqrt{z}$ (Chapters 12 to 14 write $v_\alpha(\theta)$)	Section 2.1
b_α, Π_α	$b_\alpha = \frac{\xi'}{\xi}(\alpha) = v_\alpha(0)$, the drift of the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty (e^{-sx}-1+sx)\,\Pi_\alpha(dx)$ (for $\alpha\geq 1$, and for $\frac{1}{2}<\alpha<1$ when $\Theta<\alpha$), with $b_1 = \sum_\rho \rho^{-1} = \lambda_1$; its signed Lévy measure, with atoms $\Lambda(n)/(n^\alpha \log n)$ at $\log n$	Proposition 12.1.10, Section 12.1.5
A, A_α, Q, ν_1	the gamma piece $A(\theta) = \frac{1}{2}[\mathrm{Re}\,\psi(\frac{3+i\theta}{2}) - \psi(\frac{3}{2})]$ and the compensated prime piece $Q(\theta) = \lim_{X\to\infty}\int_0^{\log X} 2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ of $v_1 = b_1 + A + Q$; the gamma piece A_α at α; the prime measure $\nu_1 = \sum_n \Lambda(n)n^{-1}\delta_{\log n} - dx$ on $(0,\infty)$, compensated by Lebesgue measure and unrelated to $\nu(\sigma,T)$ (Section 15.2 uses A, A_α and Q for $L(s,\chi_{12})$)	(12.1.6), Proposition 12.1.3

symbol	meaning	defined in
S, T_1°; S_α, T_α°	the positive stable variable of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$; the GGC with no drift, independent of S, whose Thorin density in the chart $\theta = \sqrt{z}$ is $(A+Q)/\pi$, so that $T_1 \stackrel{d}{=} S + T_1^\circ$ (unrelated to $S(t)$ and S_2); the same at a basepoint $\alpha \geq 1$, with b_α and $(v_\alpha - b_\alpha)/\pi$, so that $T_\alpha \stackrel{d}{=} S_\alpha + T_\alpha^\circ$	Propositions 12.1.11, 12.1.12
V_α, V	the variable with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, whose density $e^{(\alpha-\frac{1}{2})y}p(y)\,\xi(\frac{1}{2})/\xi(\alpha)$ is a tilt of the Pólya density, with no hypothesis; $V_{1/2} = X$ and $V = V_1$	Section 2.1, Section 12.1.1
$\rho_\varepsilon(\theta)$	the Thorin density at basepoint $\frac{1}{2}+\varepsilon$ in the chart $\theta = \sqrt{z}$, $\pi^{-1}v_{1/2+\varepsilon}(\theta)$	(2.1.7)
μ_ε	the signed measure $U_{1/2+\varepsilon}$, with density $\rho_\varepsilon(\theta)\,d\theta$ in the chart θ; the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac{1}{2}+\varepsilon > \operatorname{Re}\rho$ for every zero, and called the Thorin measure at resolution ε in general ($\mu(n)$ is the Möbius function)	Section 2.1

symbol	meaning	defined in
P_a	the Poisson (Cauchy) kernel $a/(\pi(a^2+x^2))$, with $P_{-a} = -P_a$	Section 2.1
$d_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * U_{\mathrm{RH}}^{\pm}\|_{\mathrm{TV}}$, the distance to the RH measure smoothed at the same resolution, which vanishes under RH	Section 3.1
$\tilde{d}_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, the matched distance computed against the model $\widetilde{U}$ of the zero list (the first 1700 ordinates and the density $\pi^{-1}\vartheta'$ beyond them, each with its reflection)	Section 3.1
κ_{cell}	the cell constant $\pi^{-1}\sum_k \lvert\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\rvert$ of the 79 cells $(b_{k-1}, b_k]$ below height 200, so that $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$	Section 3.1
$M_\varepsilon(\theta)$	the cumulative mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$, with the branch of the argument that vanishes at $\theta = 0$ (the Mertens function $M(t) = \sum_{d\leq t}\mu(d)$ in Section 13.3)	Section 2.1.2, Section 3.3
$\tau_\varepsilon(T)$	π^{-1} times the change of $\arg\xi$ along the segment from $\frac{1}{2}+\varepsilon+iT$ to $\frac{1}{2}+iT$, so that $M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\nu(\frac{1}{2}+\varepsilon, T)$ ($\tau(n)$ is Ramanujan's function in Section 15.1)	Proposition 3.3.2

symbol	meaning	defined in
$N(T)$, $\nu(\sigma,T)$	the number of zeros with $0<\operatorname{Im}\rho\le T$; the number of those with $\operatorname{Re}\rho>\sigma$ (Sections 3.3 and 14.3 write $N(\sigma,T)$ for the number with $\operatorname{Re}\rho\ge\sigma$)	Section 1.1
$S(t)$, $S_\varepsilon(t)$	$\pi^{-1}\arg\zeta(\frac12+it)$, with the argument continued along $2\to 2+it\to\frac12+it$; $\pi^{-1}\arg\zeta(\frac12+\varepsilon+it)$, continued along the line $\operatorname{Re}s=\frac12+\varepsilon$ from $S_\varepsilon(0+)=-1$ if $0<\varepsilon<\frac12$ and from 0 if $\varepsilon>\frac12$	Sections 2.2.1, 3.3.1
$\vartheta(t)$	the Riemann Siegel theta function $\operatorname{Im}\log\Gamma(\frac14+\frac{it}{2})-\frac{t}{2}\log\pi$	Section 2.2
$Z(t)$	Hardy's function $e^{i\vartheta(t)}\zeta(\frac12+it)$, real for real t	Section 2.2
Θ	a height, the top of a range of ordinates, with $\Theta=200$ for the window of Chapters 2 to 8; in Chapters 10 and 11 and Sections 12.1 and 13.2, $\sup\operatorname{Re}\rho$	Section 2.1.2, Section 10.1
T_0	the height 3×10^{12} below which RH has been verified [277]	Section 1.1
λ_n	the Keiper Li coefficients $\sum_\rho[1-(1-1/\rho)^n]$	Section 23.6
H_t, Λ, Φ_{dBN}	the de Bruijn Newman flow, normalised by $H_0(z)=\frac18\Xi(z/2)$; the de Bruijn Newman constant; the kernel $\Phi_{\mathrm{dBN}}(u)=\frac12\Phi(2u)$ of the flow ($\Lambda(n)$ is von Mangoldt's function)	Section 4.1
$P(\theta)$, ν_n, A_P, K^P	Pólya's Bessel function $P(\theta)=K_{i\theta/2}(2\pi)$; the zeros of $\nu\mapsto K_{i\nu}(2\pi)$, so that P vanishes at $\theta=\pm2\nu_n$; the operator $-\frac{d^2}{dx^2}+e^{2x}$ on $[\log2\pi,\infty)$ with a Dirichlet condition, whose eigenvalues are the ν_n^2; the trace $K^P(y)=\sum_n e^{-2\nu_n y}$	Section 18.1

symbol	meaning	defined in
δ	the displacement of an off line zero $\frac12\pm\delta+i\gamma$, written d in Sections 1.4, 8.2, 8.3, 23.6, 4.1 and 25.2; in Chapter 2 and where its parameters are quoted, the tilt margin, with $0<\delta<\pi/4$; in Section 9.3, the angle of a zero	Propositions 2.1.3, 2.1.1
η	the tilted height $\eta=\pi/4-\delta$ (Dedekind's eta function in Sections 15.1, 9.3 and 15.2; Dirichlet's eta function $\eta(s)=(1-2^{1-s})\zeta(s)$ in Section 13.1 and, as $\eta(1+v)=(1-2^{-v})\zeta(1+v)$, in Chapters 10 and 11)	Proposition 2.1.1
$K_\eta(\varepsilon)$	the kernel constant $\int_{\mathbb{R}}\lvert\Phi(x\pm i\eta)\rvert\,e^{\varepsilon\lvert x\rvert}\,dx$ of the trapezoid bound, with $K_\eta=K_\eta(0)$	Section 2.1.1

symbol	meaning	defined in
ψ	the digamma function Γ'/Γ (Chebyshev's function $\sum_{n\le y}\Lambda(n)$ at real $y \ge 1$ in Sections 11.1, 12.1, 13.2 and 14.3, and $\psi(y,\chi_{12})$ in Section 15.2)	Section 2.1
γ_E	Euler's constant	Section 8.1
$\mathcal{N}$	the standard normal distribution function, with $\overline{\mathcal{N}} = 1 - \mathcal{N}$	Section 3.1
$\xi(s,\chi_{12})$, K	the completed L function $(12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12}) = \int_0^\infty \eta(ix)x^{s/2}\,dx/x$; its kernel $K(u) = 2e^{u/2}\eta(ie^{2u})$, with $\xi(\frac12+s,\chi_{12}) = \int_{\mathbb{R}} K(u)e^{su}\,du$ (a variable Y with density $K/\xi(\frac12,\chi_{12})$ plays the role of X)	Section 15.1.1
$\xi(s,\chi_4)$, K_4, Y_4	the completed Dirichlet beta function $(4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_4)$, equal to $\int_0^\infty \eta(ix)^3 x^{(s+1)/2}\,dx/x$; its kernel $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$, with $\xi(\frac12+s,\chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}\,du$; the variable with density $K_4/\xi(\frac12,\chi_4)$	Section 15.1.7.1
$R(x)$, $R_h(x)$	Riesz's function $x\sum_{n\ge1}\mu(n)n^{-2}e^{-x/n^2}$; its twist $x\sum_{n\ge1}\mu(n)n^{-2}(x/n^2)^{ih}e^{-x/n^2}$, with $R_0 = R$	Sections 13.1.8, 13.2.1
$\xi_K(s)$, C_K	the completed Dedekind zeta function $s(s-1)\lvert d_K\rvert^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$ of a number field K, with $\xi_{\mathbb{Q}} = 2\xi$; the constant of $\sum_{N(\mathfrak{a})\le x}\mu(\mathfrak{a})^2/N(\mathfrak{a})$, with $C = C_{\mathbb{Q}} = 1.7171377$	Sections 13.3.1, 13.3.3
$\lVert\cdot\rVert_{\mathrm{TV}}$	the total variation $\lvert\cdot\rvert(E)$ of a signed measure on the window E	Section 1.1
GGC, HCM, ID, SD, BO, EGGC	generalised gamma convolution; hyperbolically completely monotone; infinitely divisible; self decomposable; the Bondesson class; the extended class of generalised gamma convolutions, the weak limits of convolutions of gamma laws and their reflections	Section 4.1, Section 4.1.2

symbol	meaning	defined in

Sections 9.2 and 9.3 use the following notation.

symbol	meaning	defined in
s, $\varphi(s)$	the variable u of the clock; the clock $\varphi_{1/2}(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$	Section 9.2
W	the Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace	Section 9.2
T_α	Bosch's variable $(Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with independent positive α stable $Z_\alpha, \tilde{Z}_\alpha$; here α is a stability index, not a basepoint	Section 9.2
X, p, ε_h, δ, θ, ϑ	under RH, the clock variable T; an exponent, not the Pólya density; an elasticity, not a resolution; the angle of a zero; angles, not the ordinate and the Riemann Siegel function	Section 9.3
ψ, S_{on}, R_{off}	$-\varphi'/\varphi = \sum_\gamma (s+\gamma^2)^{-1}$, the function of the Stieltjes form; its sums over the real and over the nonreal γ (in Section 14.3.7, the parts of G'/G in the variable u)	Theorems 9.2.4, 9.3.9
D_γ, $\mathcal{E}_{T_0}$, θ_0	for a nonreal zero $\gamma = a+ib$, the closed disc $\lvert s-(b^2-a^2)\rvert \le 2ab$ with diameter from $-\gamma^2$ to $-\bar{\gamma}^2$; the parabolic sliver $\{x+iy:\ x < -T_0^2+T_0+\frac{1}{4},\ y^2 < \frac{1}{2}-x\}$ that contains every D_γ; its angle at the tip, seen from the origin	Theorem 9.3.9

The numerical checks behind Section 9.3 were computed for this book. Chapters 10 and 11 use the following notation.

symbol	meaning	defined in
s, t, x	the variable u of the clock; the variable z of Thorin measures; the time of the heat trace $W(x)$	Section 10.1
τ, z	the ordinate; the argument of ζ and ξ	Section 10.1
G, R, φ	the gamma factor $\Gamma(\frac{1}{4}+\frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$; the reciprocal Xi function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$, the clock $1/G$ of Section 2.1; the normalised reciprocal zeta function $\varphi(s) = \zeta(\frac{1}{2})/\{(1-4s)\zeta(\frac{1}{2}+\sqrt{s})\}$, with $\varphi = RG$	(10.1.4)

symbol	meaning	defined in
R_σ, φ_σ, R_t	$\xi(\sigma)/\xi(\sigma+\sqrt{s})$, which the rest of the book writes φ_σ; the normalised reciprocal zeta function $\zeta(\sigma)/\{(1-s/(1-\sigma)^2)\zeta(\sigma+\sqrt{s})\}$ at σ, with $\varphi_{1/2}=\varphi$, not the φ_σ of the rest of the book; $H_t(0)/H_t(2i\sqrt{s})$, with $R_0=R$ (also in Section 4.1, which writes u for s)	Sections 10.1, 17.1
θ, c, c_n	the Riemann Siegel theta function (ϑ elsewhere); $c=-\theta'(0)=\zeta'(\frac12)/\zeta(\frac12)=2.6860917$ (also quoted in Sections 8.1 and 15.1); $c_n=2n+\frac12$	Section 10.1, (10.1.6)
$W_0(x)$, $w(x)$, $M(t)$	the archimedean part $\frac{1}{\pi}\int_0^\infty e^{-x\tau^2}\theta'(\tau)\,d\tau$ of $W(x)$; the canonical function $w=W-W_0$ of φ; $N(\sqrt{t})-\theta(\sqrt{t})/\pi=1+S(\sqrt{t})$	Section 10.1
X_φ, X_R	random variables with Laplace transforms φ and R, so that X_R has the law of the clock variable T	Section 10.2
κ, Y_κ	a Poisson width; the variable with $\mathbb{E}e^{-sY_\kappa}=\kappa/(\kappa+\sqrt{s})$, whose Thorin density is a Poisson kernel of width κ in $\sqrt{t}$	Section 10.2
$\eta(1+v)$, χ_4, $\mathcal{S}_n$	Dirichlet's eta function; the nontrivial character modulo 4; the truncated lattice variables $\mathcal{S}_n=\sum_{k\le n}k^{-2}(e_k+e'_k)$, with $\mathcal{S}=\frac{\pi^2}{2}S_2$ and tails $\mathcal{T}_n=\mathcal{S}-\mathcal{S}_n$ (also in Section 14.1)	Sections 10.1, 17.1
B_N, h_λ	the weights $B_N=\sum_{n^m=N}d_m a_n$ of the Vere-Jones representation, with $a_n=\Lambda(n)/(\sqrt{n}\log n)$ and $\sum_n d_n n^{-v}=1/\eta(1+v)$, which vanish unless N is a prime power p^n and are then sums of necklace polynomials at $p^{-1/2}$; the heat trace of the sawtooth measure U_λ, a Jacobi theta function	Section 10.1, Theorem 11.1.8

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Bibliography

[1] J. Abate and P. P. Valkó. Multi-precision Laplace transform inversion. *International Journal for Numerical Methods in Engineering* 60 (2004), 979–993.

[2] L. Alpöge and R. Furman. More than two thirds of the zeta zeros are simple and on the critical line. arXiv:2608.13637, 2026.

[3] D. F. Andrews and C. L. Mallows. Scale mixtures of normal distributions. *Journal of the Royal Statistical Society Series B (Methodological)* 36 (1974), 99–102.

[4] S. V. Anulova. On Markov stopping times with a given distribution for a Wiener process. *Theory of Probability and its Applications* 25 (1980), 362–366.

[5] O. Arizmendi, T. Hasebe and N. Sakuma. On the law of free subordinators. *ALEA Latin American Journal of Probability and Mathematical Statistics* 10 (2013), 271–291.

[6] O. Arizmendi and D. Perales. Cumulants for finite free convolution. *Journal of Combinatorial Theory, Series A* 155 (2018), 244–266.

[7] J. V. Armitage. Zeta functions with a zero at $s = \frac{1}{2}$. *Inventiones Mathematicae* 15 (1971), 199–205.

[8] R. Arratia, L. Goldstein and F. Kochman. Size bias for one and all. *Probability Surveys* 16 (2019), 1–61.

[9] R. J. Backlund. Über die Nullstellen der Riemannschen Zetafunktion. *Acta Mathematica* 41 (1918), 345–375.

[10] L. Báez-Duarte. A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis. *Atti della Accademia Nazionale dei Lincei, Rendiconti Lincei, Matematica e Applicazioni* 14 (2003), 5–11.

[11] L. Báez-Duarte. On Maslanka's representation for the Riemann zeta function. *International Journal of Mathematics and Mathematical Sciences* (2010), Article ID 714147. doi:10.1155/2010/714147, arXiv:math/0307214.

[12] R. C. Baker and J. Pintz. The distribution of square-free numbers. *Acta Arithmetica* 46 (1985), 73–79.

[13] A. A. Balkema and L. de Haan. Residual life time at great age. *Annals of Probability* 2 (1974), 792–804.

[14] W. D. Banks. On the OpenAI proof that Landau Siegel zeros do not exist. arXiv:2610.15483, 2026.

[15] R. P. Boas. *Entire Functions*. Academic Press, 1954.

[16] E. Balanzario and J. Sánchez-Ortiz. Zeros of the Davenport–Heilbronn counterexample. *Mathematics of Computation* 76 (2007), 2045–2049.

[17] Á. Baricz and F. Štampach. The Hurwitz-type theorem for the regular Coulomb wave function via Hankel determinants. *Linear Algebra and its Applications* 548 (2018), 259–272.

[18] O. E. Barndorff-Nielsen and D. R. Cox. Edgeworth and saddle-point approximations with statistical applications. *Journal of the Royal Statistical Society B* 41 (1979), 279 ff.

[19] O. E. Barndorff-Nielsen, J. Kent and M. Sørensen. Normal variance-mean mixtures and z distributions. *International Statistical Review* 50 (1982), 145 ff.

[20] O. E. Barndorff-Nielsen, M. Maejima and K. Sato. Some classes of multivariate infinitely divisible distributions admitting stochastic integral representations. *Bernoulli* 12 (2006), 1–33.

[21] O. E. Barndorff-Nielsen and S. Thorbjørnsen. Self-decomposability and Lévy processes in free probability. *Bernoulli* 8 (2002), 323–366.

[22] O. E. Barndorff-Nielsen and S. Thorbjørnsen. A connection between free and classical infinite divisibility. *Infinite Dimensional Analysis, Quantum Probability and Related Topics* 7 (2004), 573–590.

[23] T. Barnet-Lamb, D. Geraghty, M. Harris and R. Taylor. A family of Calabi–Yau varieties and potential automorphy II. *Publications of the Research Institute for Mathematical Sciences* 47 (2011), 29–98.

[24] M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer. Counting zeros of Dirichlet L-functions. *Mathematics of Computation* 90 (2021), 1455–1482.

[25] H. Bercovici and V. Pata. Stable laws and domains of attraction in free probability theory (with an appendix by P. Biane). *Annals of Mathematics* 149 (1999), 1023–1060.

[26] J.-F. Burnol. A lower bound in an approximation problem involving the zeros of the Riemann zeta function. *Advances in Mathematics* 170 (2002), 56–70.

[27] H. Bercovici and D. Voiculescu. Free convolution of measures with unbounded support. *Indiana University Mathematics Journal* 42 (1993), 733–773.

[28] C. Berg. Correction to a paper by A. G. Pakes. *Journal of the Australian Mathematical Society* 76 (2004), 67–74.

[29] C. Berg and G. Forst. Multiply self-decomposable probability measures on $\mathbb{R}_+$ and $\mathbb{Z}_+$. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* 62 (1983), 147–163.

[30] M. V. Berry and J. P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review* 41 (1999), 236–266.

[31] A. Beurling. A closure problem related to the Riemann zeta-function. *Proceedings of the National Academy of Sciences* 41 (1955), 312–314.

[32] P. Biane. Matrix valued Brownian motion and a paper by Pólya. In *Séminaire de Probabilités XLII*, Lecture Notes in Mathematics, Springer, 2009, 171–185.

[33] P. Biane, J. Pitman and M. Yor. Probability laws related to the Jacobi theta and Riemann zeta functions, and Brownian excursions. *Bulletin of the American Mathematical Society* 38 (2001), 435–465.

[34] N. H. Bingham, C. M. Goldie and J. L. Teugels. *Regular Variation*. Encyclopedia of Mathematics and its Applications 27, Cambridge University Press, 1987.

[35] E. Bogomolny, O. Bohigas, P. Leboeuf and A. G. Monastra. On the spacing distribution of the Riemann zeros: corrections to the asymptotic result. *Journal of Physics A: Mathematical and General* 39 (2006), 10743–10754.

[36] H. Bohr and E. Landau. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus de l'Académie des Sciences* 158 (1914), 106–110.

[37] E. Bombieri and J. C. Lagarias. Complements to Li's criterion for the Riemann hypothesis. *Journal of Number Theory* 77 (1999), 274–287.

[38] E. Bombieri. Remarks on Weil's quadratic functional in the theory of prime numbers, I. *Rendiconti Lincei. Matematica e Applicazioni* (9) 11 (2000), 183–233.
[39] L. Bondesson. *Generalized Gamma Convolutions and Related Classes of Distributions and Densities*. Lecture Notes in Statistics 76, Springer, 1992.
[40] L. Bondesson. A class of probability distributions that is closed with respect to addition as well as multiplication of independent random variables. *Journal of Theoretical Probability* 28 (2015), 1063–1081.
[41] P. Bosch. HCM property and the half-Cauchy distribution. *Probability and Mathematical Statistics* 35 (2015), 191–200.
[42] P. Bosch and T. Simon. A proof of Bondesson's conjecture on stable densities. *Arkiv för Matematik* 54 (2016), 31–38.
[43] P. Bourgade, T. Fujita and M. Yor. Euler's formulae for $\zeta(2n)$ and products of Cauchy variables. *Electronic Communications in Probability* 12 (2007), 73–80.
[44] P. Bourgade, C. P. Hughes, A. Nikeghbali and M. Yor. The characteristic polynomial of a random unitary matrix: a probabilistic approach. *Duke Mathematical Journal* 145 (2008), 45–69.
[45] J. Bourgain. On large values estimates for Dirichlet polynomials and the density hypothesis for the Riemann zeta function. *International Mathematics Research Notices* 2000 (2000), 133–146.
[46] C. J. Bouwkamp. A conjectured definite integral (A. H. Nuttall), solution. *SIAM Review* 28 (1986), 568–569.
[47] R. P. Brent. On the zeros of the Riemann zeta function in the critical strip. *Mathematics of Computation* 33 (1979), 1361–1372.
[48] S. Broadbent, H. Kadiri, A. Lumley, N. Ng and K. Wilk. Sharper bounds for the Chebyshev function $\theta(x)$. *Mathematics of Computation* 90 (2021), 2281–2315.
[49] M. Broadie, P. Glasserman and S. Kou. A continuity correction for discrete barrier options. *Mathematical Finance* 7 (1997), 325–349.
[50] D. Bump, K.-K. Choi, P. Kurlberg and J. Vaaler. A local Riemann hypothesis, I. *Mathematische Zeitschrift* 233 (2000), 1–18.
[51] J. Büthe. An analytic method for bounding $\psi(x)$. *Mathematics of Computation* 87 (2018), 1991–2009.
[52] N. Calkin and H. S. Wilf. Recounting the rationals. *The American Mathematical Monthly* 107 (2000), 360–363.
[53] B. Candelpergher. A new expansion of the Riemann zeta function. arXiv:2512.11405, 2025.
[54] D. A. Cardon. Sums of exponential functions having only real zeros. *Manuscripta Mathematica* 113 (2004), 307–317.
[55] D. A. Cardon. Fourier transforms having only real zeros. *Proceedings of the American Mathematical Society* 133 (2005), 1349–1356.
[56] M. A. Chaudhry and A. Qadir. Extension of Hardy's class for Ramanujan's interpolation formula and master theorem with applications. *Journal of Inequalities and Applications* 2012 (2012), paper 52.
[57] X. Chen, L. Cheng, J. Chadam and D. Saunders. Existence and uniqueness of solutions to the inverse boundary crossing problem for diffusions. *Annals of Applied Probability* 21 (2011), 1663–1693.
[58] Z. Ciesielski and S. J. Taylor. First passage times and sojourn times for Brownian motion in space and the exact Hausdorff measure of the sample path. *Transactions of the American Mathematical Society* 103 (1962), 434–450.

[59] M. W. Coffey and G. Csordas. On the log-concavity of a Jacobi theta function. *Mathematics of Computation* 82 (2013), 2265–2272.
[60] J. B. Conrey. Zeros of derivatives of Riemann's ξ-function on the critical line. *Journal of Number Theory* 16 (1983), 49–74.
[61] J. B. Conrey and X.-J. Li. A note on some positivity conditions related to zeta and L-functions. *International Mathematics Research Notices* 2000 (2000), 929–940.
[62] J. B. Conrey, D. W. Farmer, J. P. Keating, M. O. Rubinstein and N. C. Snaith. Integral moments of L-functions. *Proceedings of the London Mathematical Society* 91 (2005), 33–104.
[63] P. Contucci, P. Kleban and A. Knauf. A fully magnetizing phase transition. *Journal of Statistical Physics* 97 (1999), 523–539.
[64] P. Contucci and A. Knauf. The phase transition of the number-theoretical spin chain. *Forum Mathematicum* 9 (1997), 547–568.
[65] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey and D. E. Knuth. On the Lambert W function. *Advances in Computational Mathematics* 5 (1996), 329–359.
[66] H. Cramér. Studien über die Nullstellen der Riemannschen Zetafunktion. *Mathematische Zeitschrift* 4 (1919), 104–130.
[67] H. Cramér. Ein Mittelwertsatz in der Primzahltheorie. *Mathematische Zeitschrift* 12 (1922), 147–153.
[68] T. Craven, G. Csordas and W. Smith. The zeros of derivatives of entire functions and the Pólya-Wiman conjecture. *Annals of Mathematics* 125 (1987), 405–431.
[69] G. Csordas, T. S. Norfolk and R. S. Varga. The Riemann hypothesis and the Turán inequalities. *Transactions of the American Mathematical Society* 296 (1986), 521–541.
[70] H. Davenport. *Multiplicative Number Theory*, third edition, revised by H. L. Montgomery. Graduate Texts in Mathematics 74, Springer, 2000.
[71] N. G. de Bruijn, D. E. Knuth and S. O. Rice. The average height of planted plane trees. In *Graph Theory and Computing*, 15–22. 1972.
[72] H. Davenport and H. Heilbronn. On the zeros of certain Dirichlet series I, II. *Journal of the London Mathematical Society* 11 (1936), 181–185 and 307–312.
[73] N. G. de Bruijn. The roots of trigonometric integrals. *Duke Mathematical Journal* 17 (1950), 197–226.
[74] H. Delange. Une remarque sur la dérivée logarithmique de la fonction zêta de Riemann. *Colloquium Mathematicum* 53 (1987), 333–335.
[75] P. Deligne. La conjecture de Weil. I. *Publications Mathématiques de l'IHÉS* 43 (1974), 273–307.
[76] L. Devroye. *Non-Uniform Random Variate Generation*. Springer, 1986.
[77] A. Dobner. A proof of Newman's conjecture for the extended Selberg class. *Acta Arithmetica* 201 (2021), 29–62.
[78] H. Dym and H. P. McKean. *Gaussian Processes, Function Theory, and the Inverse Spectral Problem*. Probability and Mathematical Statistics 31, Academic Press, 1976.
[79] F. J. Dyson. A Brownian-motion model for the eigenvalues of a random matrix. *Journal of Mathematical Physics* 3 (1962), 1191–1198.
[80] H. M. Edwards. *Riemann's Zeta Function*. Academic Press, 1974.
[81] E. Ekström and S. Janson. The inverse first-passage problem and optimal stopping. *Annals of Applied Probability* 26 (2016), 3154–3177.
[82] R. S. Ellis, J. L. Monroe and C. M. Newman. The GHS and other correlation

inequalities for a class of even ferromagnets. *Communications in Mathematical Physics* 46 (1976), 167–182.

[83] P. Embrechts, C. M. Goldie and N. Veraverbeke. Subexponentiality and infinite divisibility. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 49 (1979), 335–347.

[84] D. W. Farmer. Jensen polynomials are not a plausible route to proving the Riemann Hypothesis. *Advances in Mathematics* 411 (2022), 108781; arXiv:2008.07206.

[85] D. W. Farmer and H. Ki. Landau–Siegel zeros and zeros of the derivative of the Riemann zeta function. *Advances in Mathematics* 230 (2012), 2048–2064.

[86] W. Feller. *An Introduction to Probability Theory and Its Applications*, volume II, second edition. Wiley, 1971.

[87] J. Fiala, P. Kleban and A. Özlük. The phase transition in statistical models defined on Farey fractions. *Journal of Statistical Physics* 110 (2003), 73–86.

[88] S. Fourati. On some analytical properties of stable densities. arXiv:1604.07705, 2016.

[89] S. A. Fulling. Nonuniqueness of canonical field quantization in Riemannian spacetime. *Physical Review D* 7 (1973), 2850–2862.

[90] P. X. Gallagher. Primes and powers of 2. *Inventiones Mathematicae* 29 (1975), 125–142.

[91] G. Gasper. Using integrals of squares of certain real-valued special functions to prove that the Pólya $\Xi^*(z)$ function, the functions $K_{iz}(a)$, $a > 0$, and some other entire functions have only real zeros. arXiv:0801.2996, 2008.

[92] B. V. Gnedenko. Sur la distribution limite du terme maximum d'une série aléatoire. *Annals of Mathematics* 44 (1943), 423–453.

[93] B. V. Gnedenko and A. N. Kolmogorov. *Limit Distributions for Sums of Independent Random Variables*. Translated by K. L. Chung. Addison-Wesley, Cambridge, MA, 1954; revised edition 1968.

[94] C. Goldie. A class of infinitely divisible random variables. *Proceedings of the Cambridge Philosophical Society* 63 (1967), 1141–1143.

[95] S. M. Gonek and H. Ki. Pair correlation of zeros of the real and imaginary parts of the Riemann zeta function. *Journal of Number Theory* 186 (2018), 35–61.

[96] S. M. Gonek and H. L. Montgomery. Zeros of a family of approximations of the Riemann zeta-function. *International Mathematics Research Notices* 2013 (2013), 4712–4733.

[97] M. Griffin, K. Ono, L. Rolen and D. Zagier. Jensen polynomials for the Riemann zeta function and other sequences. *Proceedings of the National Academy of Sciences* 116 (2019), 11103–11110.

[98] R. B. Griffiths, C. A. Hurst and S. Sherman. Concavity of magnetization of an Ising ferromagnet in a positive external field. *Journal of Mathematical Physics* 11 (1970), 790–795.

[99] B. Grigelionis. Processes of Meixner type. *Lithuanian Mathematical Journal* 39 (1999), 33–41.

[100] J. Grommer. Ganze transzendente Funktionen mit lauter reellen Nullstellen. *Journal für die reine und angewandte Mathematik* 144 (1914), 114–166.

[101] J. Gunns and C. Hughes. The effect of repeated differentiation on L-functions. *Journal of Number Theory* 194 (2019), 30–43.

[102] A. Gut. Some remarks on the Riemann zeta distribution. *Revue Roumaine de Mathématiques Pures et Appliquées* 51 (2006).

[103] L. Guth and J. Maynard. New large value estimates for Dirichlet polynomials. *Annals of Mathematics* 203 (2026), 623–675.
[104] J. Haglund. Some conjectures on the zeros of approximates to the Riemann Ξ-function and incomplete gamma functions. arXiv:0910.5228, 2009.
[105] G. H. Hardy and J. E. Littlewood. Contributions to the theory of the Riemann zeta-function and the theory of the distribution of primes. *Acta Mathematica* 41 (1916), 119–196.
[106] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*, sixth edition, revised by D. R. Heath-Brown and J. H. Silverman. Oxford University Press, 2008.
[107] P. Hartman and G. S. Watson. "Normal" distribution functions on spheres and the modified Bessel functions. *Annals of Probability* 2 (1974), 593–607.
[108] E. Hasanalizade, Q. Shen and P.-J. Wong. Counting zeros of the Riemann zeta function. *Journal of Number Theory* 235 (2022), 219–241.
[109] T. Hasebe and S. Thorbjørnsen. Unimodality of the freely selfdecomposable probability laws. *Journal of Theoretical Probability* 29 (2016), 922–940.
[110] C. B. Haselgrove. A disproof of a conjecture of Pólya. *Mathematika* 5 (1958), 141–145.
[111] D. R. Heath-Brown and J. C. Puchta. Integers represented as a sum of primes and powers of two. *Asian Journal of Mathematics* 6 (2002), 535–566.
[112] E. Hecke. Über die Bestimmung Dirichletscher Reihen durch ihre Funktionalgleichung. *Mathematische Annalen* 112 (1936), 664–699.
[113] H. Hedenmalm, P. Lindqvist and K. Seip. A Hilbert space of Dirichlet series and systems of dilated functions in $L^2(0,1)$. *Duke Mathematical Journal* 86 (1997), 1–37.
[114] C. Hermite. Extrait d'une lettre de Mr. Ch. Hermite de Paris à Mr. Borchardt de Berlin sur le nombre des racines d'une équation algébrique comprises entre des limites données. *Journal für die reine und angewandte Mathematik* 52 (1856), 39–51.
[115] I. A. Ibragimov. On the composition of unimodal distributions. *Theory of Probability and its Applications* 1 (1956), 255–260.
[116] H. Ki. All but finitely many nontrivial zeros of the approximations of the Epstein zeta function are simple and on the critical line. *Proceedings of the London Mathematical Society* 90 (2005), 321–344.
[117] H. Ki. On the nontrivial zeros of modified Epstein zeta functions. *Comptes Rendus Mathématique. Académie des Sciences Paris* 342 (2006), 79–81.
[118] H. Ki. The Riemann Ξ function under repeated differentiation. *Journal of Number Theory* 120 (2006), 120–131.
[119] H. Ki. On the zeros of approximations of the Ramanujan Ξ function. *The Ramanujan Journal* 17 (2008), 123–143.
[120] H. Ki. The zeros of the derivative of the Riemann zeta function near the critical line. *International Mathematics Research Notices* (2008), rnn064.
[121] H. Ki. On the zeros of Weng's zeta functions. *International Mathematics Research Notices* (2010), 2367–2393.
[122] H. Ki and Y. O. Kim. A generalization of Newman's result on the zeros of Fourier transforms. *Computational Methods and Function Theory* 2 (2002), 449–467.
[123] H. Ki and Y. O. Kim. De Bruijn's question on the zeros of Fourier transforms. *Journal d'Analyse Mathématique* 91 (2003), 369–387.

[124] H. Ki, Y. Komori and M. Suzuki. On the zeros of Weng zeta functions for Chevalley groups. *Manuscripta Mathematica* 148 (2015), 119–176.
[125] P. Kleban and A. E. Özlük. A Farey fraction spin chain. *Communications in Mathematical Physics* 203 (1999), 635–647.
[126] A. Knauf. On a ferromagnetic spin chain. *Communications in Mathematical Physics* 153 (1993), 77–115.
[127] A. Knauf. The number-theoretical spin chain and the Riemann zeroes. *Communications in Mathematical Physics* 196 (1998), 703–731.
[128] F. B. Knight. Characterization of the Lévy measures of inverse local times of gap diffusion. In *Seminar on Stochastic Processes*, Birkhäuser, 53–78. 1981.
[129] S. Kotani and S. Watanabe. Krein's spectral theory of strings and generalized diffusion processes. *Lecture Notes in Mathematics* 923 (1982), 235–259.
[130] J. C. Lagarias. Hilbert spaces of entire functions and Dirichlet L-functions. In *Frontiers in Number Theory, Physics, and Geometry I*, Springer, 2006, 365–377.
[131] B. Ya. Levin. *Lectures on Entire Functions*. American Mathematical Society, 1996.
[132] K. Maślanka. Hypergeometric-like representation of the zeta-function of Riemann. arXiv:math-ph/0105007, 2001.
[133] N. G. Polson. The Comtet and Majumdar identity, the $2\sin^2$ lemma, and approximating families for the reciprocal Xi function. Manuscript, October 2026.
[134] N. G. Polson. The explicit formula as a deconvolution. Manuscript, October 2026.
[135] N. G. Polson. Extreme values, hitting times and a path to RH. Manuscript, October 2026.
[136] N. G. Polson. The excursion maximum behind the pole cancellation: work on the open problems. Manuscript, October 2026.
[137] N. G. Polson. Ferromagnets and the Riemann zeros. Manuscript, October 2026.
[138] N. G. Polson. The Gamma skeleton, heat flow, and the arithmetic displacement: work on the open problems of the Gumbel mixture path. Manuscript, October 2026.
[139] N. G. Polson. The natural EGGC family for the reciprocal Xi function: heat flow, the $2\sin^2$ lemma, z distributions and a class on $(0, \infty)$. Manuscript, October 2026.
[140] N. G. Polson. Progress on the open problems. Manuscript, October 2026.
[141] N. G. Polson. Hitting times and the reciprocal Xi function. Manuscript, October 2026.
[142] N. G. Polson. How close can a finite approximant get? The lattice bias law for Dyck zeros and the universal Lehmer obstruction. Manuscript, October 2026.
[143] N. G. Polson. The Thorin measure at level one and the backward Cauchy flow. Manuscript, October 2026.
[144] N. G. Polson. The Lee Yang zeta function of a hard wall ferromagnet: Riemann's zeta as the constant weight case, and the rigidity of the critical line. Manuscript, October 2026.
[145] N. G. Polson. The Xi Pólya Gamma law III: self decomposability without RH, Li's coefficients as cumulants, the motion of zeros, and Fejér mixtures. Manuscript, October 2026.
[146] N. G. Polson. Thorin truncation and the zeros of $L(s, \chi_4)$. Manuscript, October 2026.
[147] N. G. Polson. Two uniform spins: the exact ferromagnet behind ξ, Riemann's contour proof as a Lee Yang trace, and the transform RH asks about. Manuscript, October 2026.
[148] N. G. Polson. Weil positivity as the Thorin condition. Manuscript, October 2026.

[149] J. Pickands. Statistical inference using extreme order statistics. *Annals of Statistics* 3 (1975), 119–131.
[150] N. G. Polson. The Euler product of $1/\zeta$ as a Lévy measure, the $2\sin^2$ lemma, and the reciprocal Xi function as a generalized gamma convolution. Note, October 2026.
[151] N. G. Polson. A residue expansion of the reciprocal Xi function and generalized gamma convolution methods. Note, October 2026.
[152] N. G. Polson. The reciprocal Xi function as the Laplace transform of an EGGC: mixing laws, the Euler product, Selberg and Barnes. Note, October 2026.
[153] R. L. Smith, J. A. Tawn and S. G. Coles. Markov chain models for threshold exceedances. *Biometrika* 84 (1997), 249–268.
[154] M. Stern. Ueber eine zahlentheoretische Funktion. *Journal für die reine und angewandte Mathematik* 55 (1858), 193–220.
[155] S. M. Stigler. Fractional order statistics, with applications. *Journal of the American Statistical Association* 72 (1977), 544–550.
[156] G. 't Hooft. On the quantum structure of a black hole. *Nuclear Physics B* 256 (1985), 727–745.
[157] G. Hurst. Computations of the Mertens function and improved bounds on the Mertens conjecture. *Mathematics of Computation* 87 (2018), 1013–1028.
[158] J. I. Hutchinson. On the roots of the Riemann zeta function. *Transactions of the American Mathematical Society* 27 (1925), 49–60.
[159] M. N. Huxley. On the difference between consecutive primes. *Inventiones Mathematicae* 15 (1972), 164–170.
[160] Y. Ihara. On the Euler–Kronecker constants of global fields and primes with small norms. In *Algebraic Geometry and Number Theory*, V. Ginzburg, ed., Progress in Mathematics 253, Birkhäuser, 2006, 407–451.
[161] A. M. Iksanov, Z. J. Jurek and B. M. Schreiber. A new factorization property of the selfdecomposable probability measures. *Annals of Probability* 32 (2004), 1356–1369.
[162] A. E. Ingham. Mean-value theorems in the theory of the Riemann zeta-function. *Proceedings of the London Mathematical Society* (2) 27 (1928), 273–300.
[163] A. E. Ingham. On the estimation of $N(\sigma, T)$. *Quarterly Journal of Mathematics* 11 (1940), 291–292.
[164] A. E. Ingham. On two conjectures in the theory of numbers. *American Journal of Mathematics* 64 (1942), 313–319.
[165] H. Inoue. Expansion of Dirichlet L-function on the critical line in Meixner-Pollaczek polynomials. arXiv:1412.1220, 2014.
[166] K. Itô and H. P. McKean. *Diffusion Processes and Their Sample Paths*. Springer, 1965.
[167] H. Iwaniec and E. Kowalski. *Analytic Number Theory*. American Mathematical Society Colloquium Publications 53, American Mathematical Society, 2004.
[168] L. F. James, B. Roynette and M. Yor. Generalized gamma convolutions, Dirichlet means, Thorin measures, with explicit examples. *Probability Surveys* 5 (2008), 346–415.
[169] F. Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers* 66 (2017), 1281–1292.
[170] D. R. Johnston and A. Yang. Some explicit estimates for the error term in the prime number theorem. *Journal of Mathematical Analysis and Applications* 527 (2023), 127460.
[171] L. Jones and N. O'Connell. Weyl chambers, symmetric spaces and number vari-

ance saturation. *ALEA Latin American Journal of Probability and Mathematical Statistics* 2 (2006), 91–118.

[172] Z. J. Jurek and W. Vervaat. An integral representation for selfdecomposable Banach space valued random variables. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 62 (1983), 247–262.

[173] M. Kac. A remark on Wiener's Tauberian theorem. *Proceedings of the American Mathematical Society* 16 (1965), 1155–1157.

[174] G. A. Kalugin and D. J. Jeffrey. Unimodal sequences show Lambert W is Bernstein. arXiv:1011.5940, 2010.

[175] G. A. Kalugin, D. J. Jeffrey and R. M. Corless. Stieltjes, Poisson and other integral representations for functions of Lambert W. Preprint, arXiv:1103.5640, 2011; published as [176, 177].

[176] G. A. Kalugin, D. J. Jeffrey, R. M. Corless and P. B. Borwein. Stieltjes and other integral representations for functions of Lambert W. *Integral Transforms and Special Functions* 23 (2012), 581–593.

[177] G. A. Kalugin, D. J. Jeffrey and R. M. Corless. Bernstein, Pick, Poisson and related integral expressions for Lambert W. *Integral Transforms and Special Functions* 23 (2012), 817–829.

[178] Y. Kasahara. Tauberian theorems of exponential type. *Journal of Mathematics of Kyoto University* 18 (1978), 209–219.

[179] N. M. Katz and P. Sarnak. Zeroes of zeta functions and symmetry. *Bulletin of the American Mathematical Society* 36 (1999), 1–26.

[180] J. P. Keating and N. C. Snaith. Random matrix theory and $\zeta(1/2 + it)$. *Communications in Mathematical Physics* 214 (2000), 57–89.

[181] B. Kerr. A simpler proof that there are no Landau Siegel zeros. arXiv:2610.21099, 2026.

[182] J. B. Keiper. Power series expansions of Riemann's ξ function. *Mathematics of Computation* 58 (1992), 765–773.

[183] J. T. Kent. Some probabilistic properties of Bessel functions. *Annals of Probability* 6 (1978), 760–770.

[184] J. T. Kent. Eigenvalue expansions for diffusion hitting times. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 52 (1980), 309–319.

[185] J. T. Kent. The spectral decomposition of a diffusion hitting time. *Annals of Probability* 10 (1982), 207–219.

[186] D. Kershaw. Some extensions of W. Gautschi's inequalities for the gamma function. *Mathematics of Computation* 41 (1983), 607–611.

[187] A. Ya. Khinchin. *Limit Laws for Sums of Independent Random Variables* (in Russian). ONTI, Moscow, 1938.

[188] H. Ki and Y. O. Kim. On the number of nonreal zeros of real entire functions and the Fourier-Pólya conjecture. *Duke Mathematical Journal* 104 (2000), 45–73.

[189] H. Ki, Y.-O. Kim and J. Lee. On the de Bruijn–Newman constant. *Advances in Mathematics* 222 (2009), 281–306.

[190] Y. O. Kim. A proof of the Pólya-Wiman conjecture. *Proceedings of the American Mathematical Society* 109 (1990), 1045–1052.

[191] R. Koekoek, P. A. Lesky and R. F. Swarttouw. *Hypergeometric Orthogonal Polynomials and Their q-Analogues*. Springer, 2010.

[192] T. Konstantopoulos, P. Patie and R. Sarkar. A new class of solutions to the van Dantzig problem, the Lee–Yang property, and the Riemann hypothesis. *Annales de*

l'Institut Fourier 74 (2024), 377–421.
[193] J. Korevaar. Distribution proof of Wiener's Tauberian theorem. *Proceedings of the American Mathematical Society* 16 (1965), 353–355.
[194] G. K. Kristiansen. A proof of Steutel's conjecture. *Annals of Probability* 22 (1994), 442–452.
[195] A. B. J. Kuijlaars and W. Van Assche. The asymptotic zero distribution of orthogonal polynomials with varying recurrence coefficients. *Journal of Approximation Theory* 99 (1999), 167–197.
[196] A. Kuznetsov. Integral representations for the Dirichlet L-functions and their expansions in Meixner–Pollaczek polynomials and rising factorials. *Integral Transforms and Special Functions* 18 (2007), 827–835.
[197] A. Kuznetsov. Expansion of the Riemann Ξ function in Meixner–Pollaczek polynomials. *Canadian Mathematical Bulletin* 51 (2008), 561–569.
[198] J. C. Lagarias. On a positivity property of the Riemann ξ-function. *Acta Arithmetica* 89 (1999), 217–234.
[199] J. C. Lagarias. Zero spacing distributions for differenced L-functions. *Acta Arithmetica* 120 (2005), 159–184.
[200] C. D. Lai and D. Vere-Jones. Odd man out: the Meixner hypergeometric distribution. *Australian Journal of Statistics* 21 (1979), 256–265.
[201] J. Lamperti. An occupation time theorem for a class of stochastic processes. *Transactions of the American Mathematical Society* 88 (1958), 380–387.
[202] Y. Lamzouri. A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line. arXiv:2609.02882, 2026.
[203] E. Landau. Über die Nullstellen der Zetafunktion. *Mathematische Annalen* 71 (1912), 548–564.
[204] S. Lang. *Algebraic Number Theory*, second edition. Graduate Texts in Mathematics 110, Springer, 1994.
[205] I. Lazarević. Certaines inégalités où interviennent les fonctions hyperboliques. *Univerzitet u Beogradu, Publikacije Elektrotehničkog Fakulteta, Serija Matematika i Fizika* 159–170 (1966), 41–48.
[206] T. D. Lee and C. N. Yang. Statistical theory of equations of state and phase transitions. II. Lattice gas and Ising model. *Physical Review* 87 (1952), 410–419.
[207] B. Ja. Levin. *Distribution of Zeros of Entire Functions*. Translations of Mathematical Monographs 5, American Mathematical Society, 1964.
[208] X.-J. Li. The positivity of a sequence of numbers and the Riemann hypothesis. *Journal of Number Theory* 65 (1997), 325–333.
[209] H. Z. Li. The number of powers of 2 in a representation of large even integers by sums of such powers and of two primes. *Acta Arithmetica* 92 (2000), 229–237.
[210] H. Z. Li. The number of powers of 2 in a representation of large even integers by sums of such powers and of two primes, II. *Acta Arithmetica* 96 (2001), 369–379.
[211] E. H. Lieb and A. D. Sokal. A general Lee-Yang theorem for one-component and multicomponent ferromagnets. *Communications in Mathematical Physics* 80 (1981), 153–179.
[212] G. D. Lin and C. Y. Hu. The Riemann zeta distribution. *Bernoulli* 7 (2001), 817–828.
[213] A. Lindner, L. Pan and K. Sato. On quasi-infinitely divisible distributions. *Transactions of the American Mathematical Society* 370 (2018), 8483–8520.
[214] Yu. V. Linnik. Prime numbers and powers of two. *Trudy Matematicheskogo Instituta*

imeni V. A. Steklova 38 (1951), 152–169.

[215] Yu. V. Linnik. Addition of prime numbers with powers of one and the same number. *Matematicheskii Sbornik* 32 (1953), 3–60.

[216] Yu. V. Linnik. *Decomposition of Probability Distributions*. Oliver and Boyd, 1964.

[217] J. E. Littlewood. On the zeros of the Riemann zeta-function. *Proceedings of the Cambridge Philosophical Society* 22 (1924), 295–318.

[218] J. Liu, M. C. Liu and T. Wang. The number of powers of 2 in a representation of large even integers, I. *Science in China Series A* 41 (1998), 386–398.

[219] J. Liu, M. C. Liu and T. Wang. On the almost Goldbach problem of Linnik. *Journal de Théorie des Nombres de Bordeaux* 11 (1999), 133–147.

[220] The LMFDB Collaboration. *The L-functions and modular forms database*, `https://www.lmfdb.org`, accessed 27 September 2026.

[221] R. Lugannani and S. Rice. Saddle point approximation for the distribution of the sum of independent random variables. *Advances in Applied Probability* 12 (1980), no. 2, 475–490. doi:10.2307/1426607.

[222] D. Ostrovsky. Theory of Barnes beta distributions. *Electronic Communications in Probability* 18 (2013), paper no. 59, 1–16. doi:10.1214/ECP.v18-2445.

[223] B. Liu. Improved zero-free half-planes for the quasi-Riemann hypothesis. arXiv:2610.12234 [math.NT], submitted 8 October 2026, v2 12 October 2026. Preprint, not refereed.

[224] B. Liu. Zero-free half-planes for Hecke L-functions over number fields. arXiv:2610.26835, submitted 19 October 2026. Preprint, not refereed.

[225] J. Thorner. A quasi Riemann hypothesis for Dedekind zeta functions. Preprint 2610.00200, 2026, proofs attributed to AxiomProver. Unchecked, not refereed.

[226] E. C. Titchmarsh. *The Theory of Functions*, second edition. Oxford University Press, 1939.

[227] A. Voros. *Zeta Functions over Zeros of Zeta Functions*. Lecture Notes of the Unione Matematica Italiana 8, Springer, 2010.

[228] E. Lukacs. Contributions to a problem of D. van Dantzig. *Theory of Probability and its Applications* 13 (1968), 116–127.

[229] S. Mandelbrojt. *Séries de Dirichlet. Principes et méthodes*. Gauthier-Villars, 1969.

[230] A. W. Marcus, D. A. Spielman and N. Srivastava. Interlacing families II: mixed characteristic polynomials and the Kadison Singer problem. *Annals of Mathematics* 182 (2015), 327–350.

[231] A. W. Marcus, D. A. Spielman and N. Srivastava. Finite free convolutions of polynomials. *Probability Theory and Related Fields* 182 (2022), 807–848.

[232] S. Mase. Decomposition of infinitely divisible characteristic functions with absolutely continuous Poisson spectral measure. *Annals of the Institute of Statistical Mathematics* 27 (1975), 289–298.

[233] Math StackExchange question 2063166. Dirichlet series associated to squared Möbius. December 2016, `https://math.stackexchange.com/q/2063166`, accessed 9 October 2026.

[234] N. Metropolis and G.-C. Rota. Witt vectors and the algebra of necklaces. *Advances in Mathematics* 50 (1983), 95–125.

[235] H. L. Montgomery. The pair correlation of zeros of the zeta function. In *Analytic Number Theory*, H. G. Diamond, ed., Proceedings of Symposia in Pure Mathematics 24, American Mathematical Society, 1973, 181–193.

[236] H. L. Montgomery. Zeros of approximations to the zeta function. In *Studies in Pure*

Mathematics: To the Memory of Paul Turán, Birkhäuser, Basel, 1983, 497–506.

[237] The mpmath development team. *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, version 1.3.0, 2023.

[238] T. Nakamura. A complete Riemann zeta distribution and the Riemann hypothesis. *Bernoulli* 21 (2015), 604–617.

[239] T. Nakamura. Zeta distributions generated by Dirichlet series and their (quasi) infinite divisibility. arXiv:2209.13257, 2022.

[240] C. M. Newman. Zeros of the partition function for generalized Ising systems. *Communications on Pure and Applied Mathematics* 27 (1974), 143–159.

[241] C. M. Newman. Inequalities for Ising models and field theories which obey the Lee-Yang theorem. *Communications in Mathematical Physics* 41 (1975), 1–9.

[242] C. M. Newman. Fourier transforms with only real zeros. *Proceedings of the American Mathematical Society* 61 (1976), 245–251.

[243] C. M. Newman. The GHS inequality and the Riemann hypothesis. *Constructive Approximation* 7 (1991), 389–399.

[244] C. M. Newman and W. Wu. Constants of de Bruijn–Newman type in analytic number theory and statistical physics. *Bulletin of the American Mathematical Society* 57 (2020), 595–614.

[245] C. M. Newman and W. Wu. Lee–Yang property and Gaussian multiplicative chaos. *Communications in Mathematical Physics* 369 (2019), 153–170.

[246] T. Newton. Kernel-concavity obstructions and Fourier-side Laguerre bounds for the Riemann Ξ programme. Zenodo, 2026, `https://doi.org/10.5281/zenodo.23461586`, accessed 20 October 2026.

[247] A. Nica and R. Speicher. *Lectures on the Combinatorics of Free Probability*. London Mathematical Society Lecture Note Series 335, Cambridge University Press, 2006.

[248] A. Nikeghbali and M. Yor. The Barnes G function and its relations with sums and products of generalized gamma convolution variables. *Electronic Communications in Probability* 14 (2009), 396–411.

[249] A. A. Novikov. Martingales, Tauberian theorem, and strategies of gambling. *Theory of Probability and its Applications* 41 (1997), 716–729.

[250] A. H. Nuttall. A conjectured definite integral, problem. *SIAM Review* 27 (1985), 573.

[251] B. Nyman. *On the One-Dimensional Translation Group and Semi-Group in Certain Function Spaces*. Thesis, University of Uppsala, 1950.

[252] A. M. Odlyzko. On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation* 48 (1987), 273–308.

[253] A. M. Odlyzko and A. Schönhage. Fast algorithms for multiple evaluations of the Riemann zeta function. *Transactions of the American Mathematical Society* 309 (1988), 797–809.

[254] A. M. Odlyzko. The 10^{22}-nd zero of the Riemann zeta function. In *Dynamical, Spectral, and Arithmetic Zeta Functions*, M. L. Lapidus and M. van Frankenhuysen, eds., Contemporary Mathematics 290, American Mathematical Society, 2001, 139–144.

[255] A. M. Odlyzko. *Tables of zeros of the Riemann zeta function*. Online, `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`, accessed 27 September 2026.

[256] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl and M. A. McClain, eds. *NIST Digital Library of Mathematical Functions*. `https://dlmf.nist.gov/`, Re-

lease 1.2.8 of 2026-09-15.

[257] OpenAI. The quasi-Riemann hypothesis: a zero-free half-plane $\mathrm{Re}(s) > 7/8$. Preprint dated 30 September 2026, released 6 October 2026 with a formalisation in Lean, `https://github.com/openai/math` (family 003), accessed 7 October 2026.

[258] OpenAI. The quasi-Riemann hypothesis. Preprint dated 5 October 2026, released 6 October 2026, `https://github.com/openai/math` (family 003), accessed 7 October 2026.

[259] A. G. Pakes. Length biasing and laws equivalent to the log-normal. *Journal of Mathematical Analysis and Applications* 197 (1996), 825–854.

[260] A. G. Pakes. Remarks on converse Carleman and Krein criteria for the classical moment problem. *Journal of the Australian Mathematical Society* 71 (2001), 81–104.

[261] A. G. Pakes. Structure of Stieltjes classes of moment-equivalent probability laws. *Journal of Mathematical Analysis and Applications* 326 (2007), 1268–1290.

[262] A. G. Pakes. Lambert's W, infinite divisibility and Poisson mixtures. *Journal of Mathematical Analysis and Applications* 378 (2011), 480–492.

[263] A. G. Pakes. On the preservation of infinite divisibility under length-biasing. *Journal of Probability* 2014 (2014), 703697.

[264] A. G. Pakes. The Lambert W function, Nuttall's integral, and the Lambert law. *Statistics and Probability Letters* 139 (2018), 53–60.

[265] A. G. Pakes. Self-decomposable laws from continuous branching processes. *Journal of Theoretical Probability* 33 (2020), 361–395.

[266] A. G. Pakes. Self-decomposability and the Lambert law. *Axioms* 15 (2026), 574.

[267] A. G. Pakes, W.-L. Hung and J.-W. Wu. Criteria for the unique determination of probability distributions by moments. *Australian and New Zealand Journal of Statistics* 43 (2001), 101–111.

[268] A. G. Pakes and R. Khattree. Length-biasing, characterizations of laws and the moment problem. *Australian Journal of Statistics* 34 (1992), 307–322.

[269] A. G. Pakes, T. Sapatinas and E. B. Fosam. Characterizations, length-biasing, and infinite divisibility. *Statistical Papers* 37 (1996), 53–69.

[270] G. Peskir. On integral equations arising in the first-passage problem for Brownian motion. *Journal of Integral Equations and Applications* 14 (2002), 397–423.

[271] J. Pintz and I. Z. Ruzsa. On Linnik's approximation to Goldbach's problem, I. *Acta Arithmetica* 109 (2003), 169–194.

[272] J. Pintz and I. Z. Ruzsa. On Linnik's approximation to Goldbach's problem, II. *Acta Mathematica Hungarica* 161 (2020), 569–582.

[273] J. Pitman and M. Yor. Bessel processes and infinitely divisible laws. In *Stochastic Integrals*, Lecture Notes in Mathematics 851, Springer, 1981, 285–370.

[274] J. Pitman and M. Yor. Infinitely divisible laws associated with hyperbolic functions. *Canadian Journal of Mathematics* 55 (2003), 292–330.

[275] D. J. Platt. Numerical computations concerning the GRH. *Mathematics of Computation* 85 (2016), 3009–3027.

[276] D. J. Platt and T. S. Trudgian. Linnik's approximation to Goldbach's conjecture, and other problems. *Journal of Number Theory* 153 (2015), 54–62.

[277] D. J. Platt and T. S. Trudgian. The Riemann hypothesis is true up to $3 \cdot 10^{12}$. *Bulletin of the London Mathematical Society* 53 (2021), 792–797.

[278] N. G. Polson, J. G. Scott and J. Windle. The Bayesian bridge. *Journal of the Royal Statistical Society, Series B (Statistical Methodology)* 76 (2014), 713; sections and

theorems are cited by the numbering of arXiv:1109.2279v2.
[279] N. G. Polson. On Hilbert's 8th problem. arXiv:1708.02653v24, 2026.
[280] N. G. Polson. On Hilbert's 8th problem. arXiv:1708.02653v5, 2017.
[281] N. G. Polson. Riemann, Thorin, van Dantzig pairs, Wald couples and Hadamard factorisation. arXiv:1804.10043, 2018.
[282] N. G. Polson. *On Hilbert's 8th Problem: the reciprocal Xi clock, Thorin measures, van Dantzig pairs, zero free approximations, hitting times and the reciprocal zeta function*, first edition. Independently published, 2026.
[283] N. G. Polson. The Dirichlet beta function in Thorin form. Manuscript, October 2026.
[284] N. G. Polson. Biane on Pólya: hitting times, Thorin measures and a Wick rotation. Manuscript, October 2026.
[285] N. G. Polson. Thorin measures of the Dedekind eta function. Manuscript, September 2026.
[286] N. G. Polson. Maxima of exponentials, incomplete gamma continuations, and the Gamma(2) structure of L functions. Manuscript, October 2026.
[287] N. G. Polson. The Pólya density as a maximum order statistic and the Thorin measure of the reciprocal Xi function. Manuscript, October 2026.
[288] N. G. Polson. Thorin type representations of the reciprocal zeta function on the critical line. Manuscript, September 2026, now Part IX of the first edition of *On Hilbert's 8th Problem* (2026).
[289] N. G. Polson. The squarefree Dirichlet series $\zeta(s)/\zeta(2s)$: a summary of Math StackExchange 2063166, Pólya densities, Gamma(2) smoothing and Thorin measures of Ξ. Manuscript, October 2026.
[290] N. G. Polson. Gamma, reciprocal Gamma, η^3 and Δ: Thorin forms and the Riemann Ξ function. Manuscript, October 2026.
[291] N. G. Polson. Wigner's semicircle, free probability and the Thorin framework for the reciprocal Xi function. Manuscript, version 7, October 2026.
[292] N. G. Polson. Pakes, the Lambert law and the Thorin condition for the Riemann Xi function. Manuscript, October 2026.
[293] N. G. Polson and V. Sokolov. Computing the Thorin measure of the reciprocal Xi function: a Pólya route and a total variation comparison with the Riemann atoms. Manuscript, September 2026.
[294] N. G. Polson and V. Sokolov. Four attacks on the computational path: a working report. Manuscript, 7 October 2026.
[295] G. Pólya. Über Annäherung durch Polynome mit lauter reellen Wurzeln. *Rendiconti del Circolo Matematico di Palermo* 36 (1913), 279–295.
[296] G. Pólya. Algebraische Untersuchungen über ganze Funktionen vom Geschlechte Null und Eins. *Journal für die reine und angewandte Mathematik* 145 (1915), 224–249.
[297] G. Pólya. Über die Nullstellen gewisser ganzer Funktionen. *Mathematische Zeitschrift* 2 (1918), 352–383.
[298] G. Pólya. Verschiedene Bemerkungen zur Zahlentheorie. *Jahresbericht der Deutschen Mathematiker-Vereinigung* 28 (1919), 31–40.
[299] G. Pólya. Über die Nullstellen sukzessiver Derivierten. *Mathematische Zeitschrift* 12 (1922), 36–60.
[300] G. Pólya. Über die Existenz unendlich vieler singulärer Punkte auf der Konvergenzgeraden gewisser Dirichletscher Reihen. *Sitzungsberichte der Preussischen*

Akademie der Wissenschaften, Physikalisch-mathematische Klasse (1923), 45–50.
[301] G. Pólya. On the zeros of an integral function represented by Fourier's integral. *Messenger of Mathematics* 52 (1923), 185–188.
[302] G. Pólya. Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion. *Acta Mathematica* 48 (1926), 305–317.
[303] G. Pólya. On the zeros of certain trigonometric integrals. *Journal of the London Mathematical Society* 1 (1926), 98–99.
[304] G. Pólya. Über trigonometrische Integrale mit nur reellen Nullstellen. *Journal für die reine und angewandte Mathematik* 158 (1927), 6–18.
[305] G. Pólya. Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen. *Det Kongelige Danske Videnskabernes Selskab, Mathematisk-fysiske Meddelelser* 7 (1927), no. 17, 3–33.
[306] G. Pólya. Untersuchungen über Lücken und Singularitäten von Potenzreihen. *Mathematische Zeitschrift* 29 (1929), 549–640.
[307] G. Pólya. Some problems connected with Fourier's work on transcendental equations. *The Quarterly Journal of Mathematics* (Oxford) 1 (1930), 21–34.
[308] G. Pólya. On the zeros of the derivatives of a function and its analytic character. *Bulletin of the American Mathematical Society* 49 (1943), 178–191.
[309] G. Pólya. *Collected Papers, Vol. II: Location of Zeros*, edited by R. P. Boas. Mathematicians of Our Time 8, MIT Press, 1974.
[310] G. Pólya and J. Schur. Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen. *Journal für die reine und angewandte Mathematik* 144 (1914), 89–113.
[311] D. H. J. Polymath. Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant. *Research in the Mathematical Sciences* 6 (2019), paper 31.
[312] K. Pratt, N. Robles, A. Zaharescu and D. Zeindler. More than five-twelfths of the zeros of ζ are on the critical line. *Research in the Mathematical Sciences* 7 (2020), article 2.
[313] R. Remmert. *Classical Topics in Complex Function Theory*, translated by L. Kay. Graduate Texts in Mathematics 172, Springer, 1998.
[314] M. Riesz. Sur l'hypothèse de Riemann. *Acta Mathematica* 40 (1916), 185–190.
[315] H. Robbins. A remark on Stirling's formula. *American Mathematical Monthly* 62 (1955), 26–29.
[316] G. O. Roberts. Asymptotic approximations for Brownian motion boundary hitting times. *Annals of Probability* 19 (1991), 1689–1731.
[317] G. O. Roberts and C. F. Shortland. The hazard rate tangent approximation for boundary hitting times. *Annals of Applied Probability* 5 (1995), 446–460.
[318] B. Rodgers and T. Tao. The de Bruijn–Newman constant is non-negative. *Forum of Mathematics, Pi* 8 (2020), e6.
[319] D. Romik. Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases. *Acta Arithmetica* 200 (2021), 259–329.
[320] J. B. Rosser and L. Schoenfeld. Approximate formulas for some functions of prime numbers. *Illinois Journal of Mathematics* 6 (1962), 64–94.
[321] R. Rumely. Numerical computations concerning the ERH. *Mathematics of Computation* 61 (1993), 415–440.
[322] R. Salem. Sur une proposition équivalente à l'hypothèse de Riemann. *Comptes*

Rendus de l'Académie des Sciences 236 (1953), 1127–1128.
[323] P. Salminen and P. Vallois. On maximum increase and decrease of Brownian motion. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 43 (2007), 655–676.
[324] K. Sato. Class L of multivariate distributions and its subclasses. *Journal of Multivariate Analysis* 10 (1980), 207–232.
[325] K. Sato. *Lévy Processes and Infinitely Divisible Distributions*. Cambridge Studies in Advanced Mathematics 68, Cambridge University Press, 1999.
[326] A. Saumard and J. A. Wellner. Log-concavity and strong log-concavity: a review. *Statistics Surveys* 8 (2014), 45–114.
[327] R. L. Schilling, R. Song and Z. Vondraček. *Bernstein Functions: Theory and Applications*, second edition. Walter de Gruyter, 2012.
[328] I. J. Schoenberg. On Pólya frequency functions. I. The totally positive functions and their Laplace transforms. *Journal d'Analyse Mathématique* 1 (1951), 331–374.
[329] W. Schoutens and J. L. Teugels. Lévy processes, polynomials and martingales. *Communications in Statistics. Stochastic Models* 14 (1998), 335–349.
[330] A. Selberg. Contributions to the theory of the Riemann zeta-function. *Archiv for Mathematik og Naturvidenskab* 48 (1946), 89–155.
[331] G. Sierra. The Riemann zeros as energy levels of a Dirac fermion in a potential built from the prime numbers in Rindler spacetime. *Journal of Physics A: Mathematical and Theoretical* 47 (2014), 325204.
[332] B. Simon and R. B. Griffiths. The $(\phi^4)_2$ field theory as a classical Ising model. *Communications in Mathematical Physics* 33 (1973), 145–164.
[333] J. Sondow and C. Dumitrescu. A monotonicity property of Riemann's xi function and a reformulation of the Riemann hypothesis. *Periodica Mathematica Hungarica* 60 (2010), 37–40.
[334] R. Spira. Zeros of approximate functional approximations. *Mathematics of Computation* 21 (1967), 41–48.
[335] R. Spira. Some zeros of the Titchmarsh counterexample. *Mathematics of Computation* 63 (1994), 747–748.
[336] L. A. Stefanski. A normal scale mixture representation of the logistic distribution. *Statistics & Probability Letters* 11 (1991), 69–70.
[337] F. W. Steutel. Note on the infinite divisibility of exponential mixtures. *Annals of Mathematical Statistics* 38 (1967), 1303–1305.
[338] F. W. Steutel. Some recent results in infinite divisibility. *Stochastic Processes and their Applications* 1 (1973), 125–143.
[339] F. W. Steutel and J. G. F. Thiemann. On the independence of integer and fractional parts. *Statistica Neerlandica* 43 (1989), 53–59.
[340] F. W. Steutel and K. van Harn. *Infinite Divisibility of Probability Distributions on the Real Line*. Marcel Dekker, 2004.
[341] A. Talbot. The accurate numerical inversion of Laplace transforms. *Journal of the Institute of Mathematics and its Applications* 23 (1979), 97–120.
[342] M. Tanaka. A numerical investigation on cumulative sum of the Liouville function. *Tokyo Journal of Mathematics* 3 (1980), 187–189.
[343] J. T. Tate. Fourier analysis in number fields and Hecke's zeta-functions. In *Algebraic Number Theory*, J. W. S. Cassels and A. Fröhlich, eds., Academic Press, 1967, 305–347.
[344] O. Thorin. On the infinite divisibility of the Pareto distribution. *Scandinavian Ac-*

tuarial Journal (1977), 31–40.
[345] O. Thorin. On the infinite divisibility of the lognormal distribution. *Scandinavian Actuarial Journal* (1977), 121–148.
[346] O. Thorin. An extension of the notion of a generalized Γ-convolution. *Scandinavian Actuarial Journal* (1978), 141–149.
[347] E. C. Titchmarsh. *Eigenfunction Expansions Associated with Second-Order Differential Equations*. Clarendon Press, Oxford, 1946.
[348] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*, second edition, revised by D. R. Heath-Brown. Oxford University Press, 1986.
[349] P. Turán. On some approximative Dirichlet polynomials in the theory of the zeta function of Riemann. *Danske Vid. Selsk. Mat.-Fys. Medd.* 24 (1948).
[350] T. S. Trudgian. Improvements to Turing's method. *Mathematics of Computation* 80 (2011), 2259–2279.
[351] T. S. Trudgian. An improved upper bound for the argument of the Riemann zeta-function on the critical line II. *Journal of Number Theory* 134 (2014), 280–292.
[352] K.-M. Tsang. Some Ω-theorems for the Riemann zeta-function. *Acta Arithmetica* 46 (1986), 369–395.
[353] N. Tschebotareff. Über die Realität von Nullstellen ganzer transzendenter Funktionen. *Mathematische Annalen* 99 (1928), 660–686.
[354] K. Urbanik. Limit laws for sequences of normed sums satisfying some stability conditions. In *Multivariate Analysis III*, P. R. Krishnaiah, ed., Academic Press, 1973, 225–237.
[355] D. Voiculescu. Addition of certain non-commuting random variables. *Journal of Functional Analysis* 66 (1986), 323–346.
[356] T. Wang. On Linnik's almost Goldbach theorem. *Science in China Series A* 42 (1999), 1155–1172.
[357] A. Weil. Sur les "formules explicites" de la théorie des nombres premiers. *Communications du Séminaire Mathématique de l'Université de Lund*, tome supplémentaire (1952), 252–265.
[358] G. C. Wick. Properties of Bethe-Salpeter wave functions. *Physical Review* 96 (1954), 1124–1134.
[359] D. V. Widder. *The Laplace Transform*. Princeton Mathematical Series 6, Princeton University Press, 1941.
[360] N. Wiener. Tauberian theorems. *Annals of Mathematics* 33 (1932), 1–100.
[361] N. Wiener. *The Fourier Integral and Certain of Its Applications*. Cambridge University Press, 1933.
[362] E. P. Wigner. On the distribution of the roots of certain symmetric matrices. *Annals of Mathematics* 67 (1958), 325–327.
[363] D. Williams. Brownian motion and the Riemann zeta-function. In *Disorder in Physical Systems*, G. R. Grimmett and D. J. A. Welsh, eds., Oxford University Press, 1990, 361–372.
[364] R. E. Williamson. Multiply monotone functions and their Laplace transforms. *Duke Mathematical Journal* 23 (1956), 189–207.
[365] S. J. Wolfe. On a continuous analogue of the stochastic difference equation $X_n = \rho X_{n-1} + B_n$. *Stochastic Processes and their Applications* 12 (1982), 301–312.
[366] M. Yamazato. Unimodality of infinitely divisible distribution functions of class L. *Annals of Probability* 6 (1978), 523–531.
[367] A. C. Yao and D. E. Knuth. Analysis of the subtractive algorithm for greatest

common divisors. *Proceedings of the National Academy of Sciences of the United States of America* 72 (1975), 4720–4722.

[368] M. Yor. Loi de l'indice du lacet brownien, et distribution de Hartman-Watson. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 53 (1980), 71–95.

[369] X. Zhu. Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law. arXiv:2608.24827v2, 2 September 2026.

[370] B. Riemann. Über die Anzahl der Primzahlen unter einer gegebenen Grösse. *Monatsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin* (1859), 671–680. English translation in [80], Appendix.

[371] W. Jedidi and T. Simon. Further examples of GGC and HCM densities. *Bernoulli* 19 (2013), 1818–1838.

[372] T. Sjödin. On mixtures of gamma distributions, distributions with hyperbolically monotone densities and generalized gamma convolutions (GGC). arXiv:1806.03926, version of 20 February 2019.

[373] T. Sjödin. The power problem for generalized gamma convolutions (GGC) and related questions. arXiv:2601.04038, submitted 7 January 2026. Preprint.

[374] M. Wang and S. Yin. Powers of generalized gamma convolutions. arXiv:2609.26558, 2026. Preprint.

[375] P. J. McLane. Mathematical considerations of the modified normal distribution. *Bulletin Géodésique* 83 (1967), 9–20.

[376] C. Platt and Z. Paprzycki. About some properties of the Romanowski distributions. *Manuscripta Geodaetica* 20 (1995), 408–412.

[377] Y. Kozitsky. On the Lee Yang property of some ferromagnets. arXiv:2503.03510, March 2025.

[378] A. Weil. *Sur les courbes algébriques et les variétés qui s'en déduisent.* Actualités scientifiques et industrielles 1041, Hermann, Paris, 1948.